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DIAGNOSTIC PATHOLOGY

Normal Histology

SECOND EDITION

LINDBERG | LAMPS

QUICK • MCKENNEY • MCHUGH • CHANG • COX • MILLER • POLYDORIDES
CASSARINO • WALLENTINE • FOUCAR • NELSON • FISHER • HICKS



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Normal Histology

SECOND EDITION

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Dedication

*To my beautiful wife, Rani, for her love, patience, and support.
I would also like to thank the hardworking team of editors,
coordinators, and managers at Elsevier for their tireless
dedication to making the many, many wonderful books within
the Diagnostic Pathology series a reality.*

MRL





Preface

A thorough knowledge of normal histology, in all of its wonderful variation, is essential to the study of disease. *Diagnostic Pathology: Normal Histology*, second edition is designed for practicing pathologists, pathologists in training, medical students, and students of histology in any setting as they strive to learn about the spectrum of normal.

This reference provides clear, concise information on the normal histology of every organ system. It also features nearly 1,800 images, including photomicrographs, gross images, electron micrographs, and medical illustrations. In this second edition, a number of chapters have been expanded with new histologic images showing additional artifacts, mimics, and pitfalls often encountered in normal tissues, and a new chapter on synovium is introduced. In addition to the existing introductory chapters on electron microscopy, immunofluorescence, immunohistochemistry, histochemistry, the cell, and the basic organization of tissues, a new chapter detailing histologic artifacts has also been included.

On behalf of all of the authors of this book, we hope that *Diagnostic Pathology: Normal Histology*, second edition becomes a mainstay of your library, and that it is a favorite reference to which you return on a regular basis.

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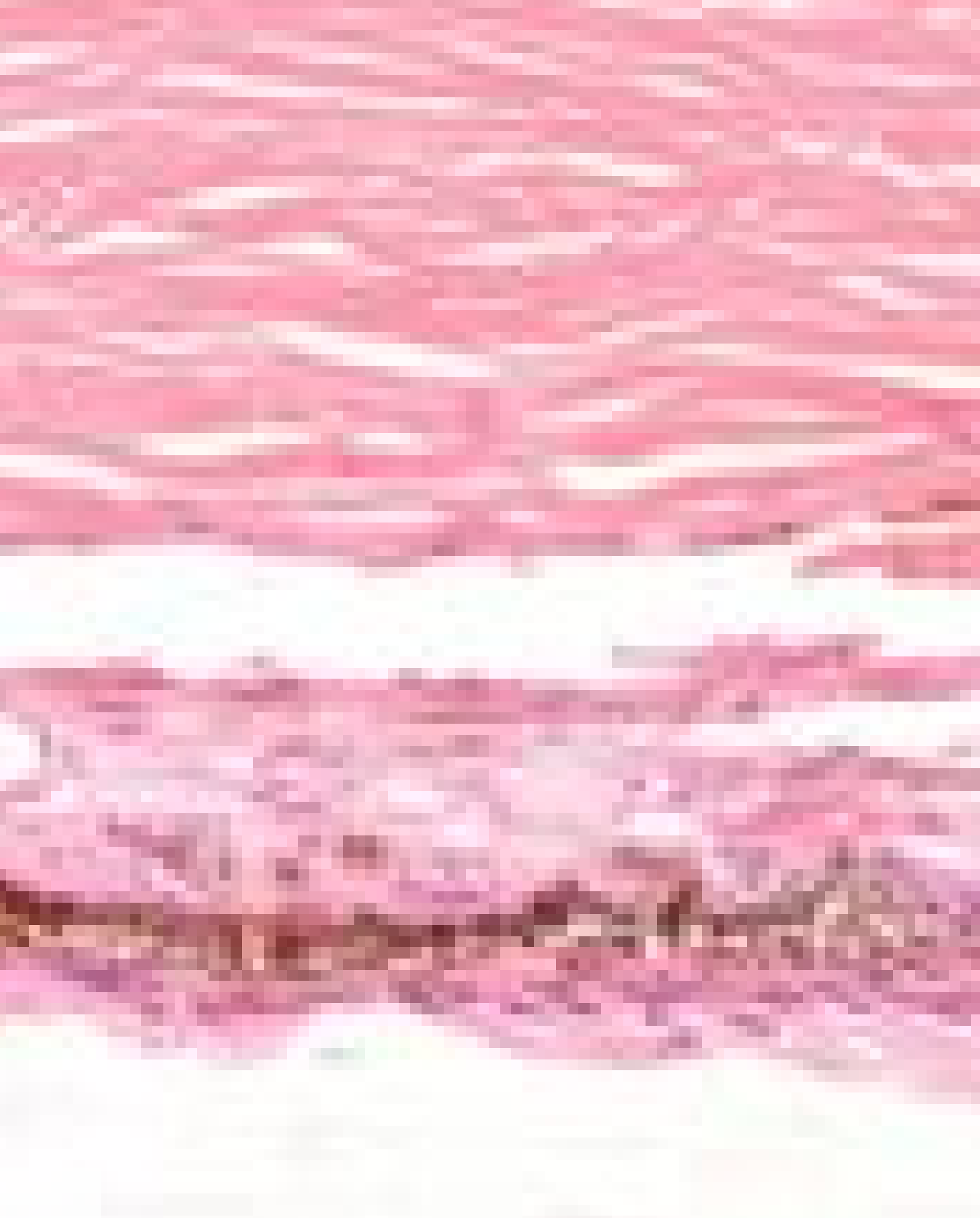
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Introduction to Normal Histology and Basic Histological Techniques



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TERMINOLOGY

Definitions

- Cell
 - Basic structural unit of all tissue
- Organelles
 - Intracellular heterogeneous interactive substructures

OVERVIEW OF CELL ORGANIZATION

Membrane

- Major unit of organization
- Membranes appear as single linear structure by light microscopy and low-magnification electron microscopy
- Phospholipid bilayer
 - Hydrophilic surface, hydrophobic center with interspersed variable-sized protein globules

Membrane-Bound Organelles

- Nucleus
- Plasma (cell) membrane
- Endoplasmic reticulum (ER)
 - Rough and smooth types
- Golgi apparatus
- Mitochondria
- Lysosomes
- Cell membrane specializations
 - Microvilli, pinocytotic vesicles
- Various types of secretory/neuroendocrine granules

Nonmembrane-Bound Organelles

- Nuclear chromatin, nucleolus
- Intracytoplasmic contractile and anchoring filaments
- Cytoskeletal tubules
- Free ribosomes
- Glycogen, fat globules
- Desmosomes, attachment plaques
- Centrioles, basal bodies

Extracellular Elements

- Basal lamina, collagen, matrix proteoglycans

NUCLEUS

Chromatin

- Electron-dense aggregates representing DNA
 - Heterochromatin: Clumped, electron-dense DNA, biologically inactive
 - Euchromatin: Dispersed DNA, biologically active
- Individual chromosomes not apparent without cell culture

Chromosomes

- 46 total chromosomes per human cell
 - 22 pairs of identical chromosomes (or autosomes) with either 2 X chromosomes for females or 1 X and 1 Y chromosome for males
- 2 chromatids join together at centromere to form chromosome
- Telomeres are located at end of each chromosome

Nucleolus

- Focus of variable prominence in nucleus
- Filamentous and granular nucleolonema composed of histones, ribonucleic acid (RNA), and regulatory cell cycle proteins

Cytoplasmic Interactions

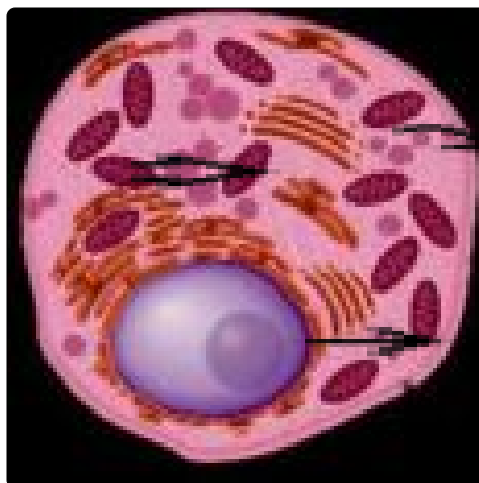
- Via
 - Nuclear pores
 - Nuclear blebs
 - Cisternal space contiguous with ER
 - Dissolution of nuclear membrane during mitosis

CYTOPLASM

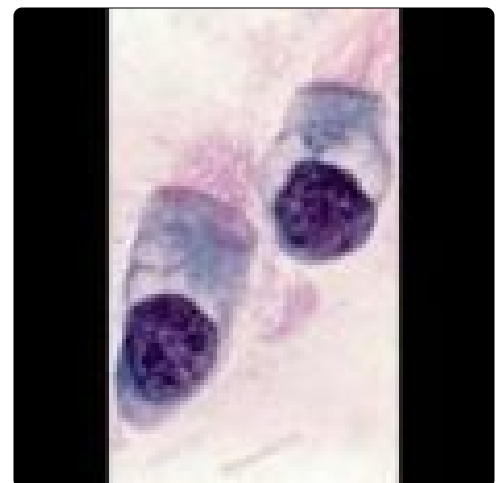
Endoplasmic Reticulum

- Definition
 - Parallel arrangement of membranes adjacent to and contiguous with nucleus
- Rough ER (rER)
 - Ribosomes are free, small granular aggregates of cytoplasmic RNA, which, when attached to ER membranes, constitute rER

Cellular Components



Cellular Organization



(Left) This depiction of organelle distribution within a cell shows a basally placed nucleus [1], mitochondria [2], endoplasmic reticulum, and variably sized granules [3] (mucin or other proteins). (Right) Endocervical cells from a Pap smear show typical cellular organization (basal nucleus, variable chromatin clumping, terminal bar [4] with cilia, granular cytoplasm, and vacuoles, Golgi [5] with mucin droplet formation).

- Messenger RNA (mRNA) attaches at junction of 2 ribosomal regions
- Transfer RNA sequentially attaches one end to mRNA with opposite end bearing amino acid
- Protein synthesis accumulates in cistern of rER for cell membrane maintenance or for secretory export
- Smooth ER
 - Same membrane complex but no associated ribosomes
 - Often contiguous with rER
 - Involved in detoxification, synthesis of steroids, lipid absorption, and metabolism
 - a.k.a. sarcoplasmic reticulum in cardiac and skeletal muscle

Golgi Apparatus

- Receives proteins from rER for posttranslational modification and packaging
- Delivers enzymes to lysosomes
- Histologic appearance is clear area adjacent to nucleus (hof in typical plasma cell)

Mitochondria

- Complex of internal membranes (cristae) bound by external membrane
- Energy source for cell through adenosine triphosphate production
- High numbers impart granular eosinophilic cytoplasm, e.g., oncocytes

Lysosomes

- Membrane bound, round, protein-containing bodies
- May contain hydrolytic enzymes
- Involved in cell maintenance by degrading membranes and obsolescent organelles
- End stage is "residual body"

Cell Membrane Specializations

- Microvilli
 - Apical foldings protruding into gland lumen
 - Increase surface area of cell for absorption from gland lumen
- Pinocytotic vesicles
 - Tiny plasma membrane invaginations
 - Facilitate exchange or movement of fluid
 - Prominent in endothelial cells and smooth muscle cells

CYTOSKELETON

Thin Filaments (Actin)

- Actin biochemically ubiquitous
- In muscle cells, organized into short, wavy bundles held together by "dense bodies"
- Best seen in smooth muscle, myoepithelial cells, myofibroblasts
- Contractile function

Thick Filaments

- Predominantly myosin, most prominent in skeletal and cardiac muscle
- Organized with actin filaments and Z bands giving striated appearance by light microscopy
- Contractile function

Microtubules

- Long, hollow cylindrical structure
- Consists of tubulin
- Anchored to centrosome

Intermediate Filaments

- Cytokeratin
 - Short, wavy bundles often inserting on desmosomes or plasma membrane plaques
 - Best seen in squamous epithelium
- Vimentin
 - Widespread in mesenchymal cells
- Desmin
 - Contractile protein in muscle cells
- Glial fibrillary acidic protein
 - Predominant in glia of CNS
- Neurofilament
 - Predominant in CNS, peripheral nervous system, and neuroendocrine cells

Centrioles

- Basal bodies
 - Anchoring structures, usually at apex of epithelial cells
- Cilia
 - Tubular doublets arranged 9 (peripheral) + 2 (central)

CELL ADHESION

Desmosomes (Macula Adherens)

- Attachment of epithelial cell to another epithelial cell
- Zonula occludens
- Zonula adherens
- Terminal bar
 - Light microscopic structure composed of desmosomes linked by horizontal filaments near apex beneath numerous microvilli

Hemidesmosomes

- Attachment of epithelial cell to connective tissue

EXTRACELLULAR SUBSTANCES

Basal Lamina

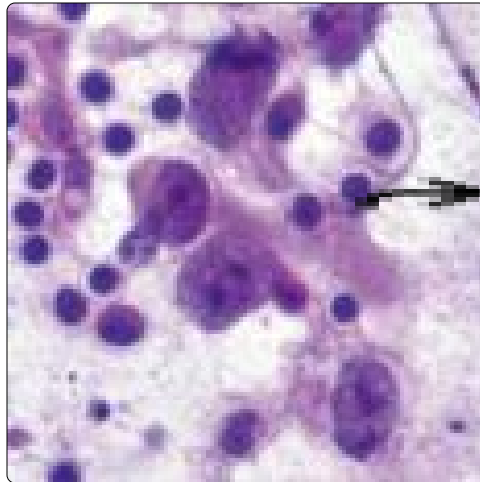
- Cell product forming interface between cell and underlying connective tissue
- Epithelial cells (principally glandular), myoepithelial cells, endothelial cells, pericytes, smooth muscle, Schwann cells
- Composed of type IV collagen
- Basement membrane
 - Light microscopic structure composed of basal lamina and underlying polymerized collagen fibers, best seen by PAS staining

Collagen

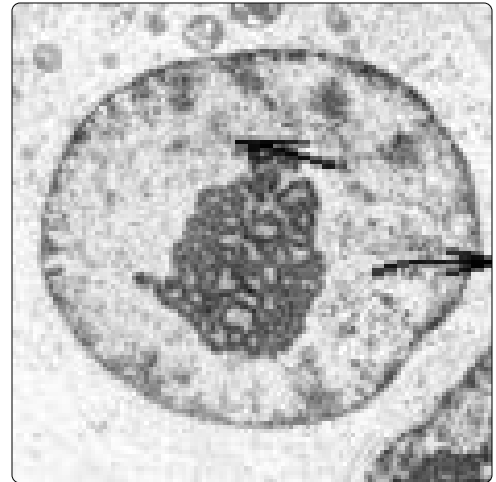
- Major protein product of some cells
- Organized as fibrils with periodicity of 64 nm

Nucleus and Nucleoli

(Left) This touch imprint from a seminoma of the testis with a cluster of tumor cells and background lymphocytes demonstrates well-defined, enlarged nuclei, each with at least 1 prominent nucleolus. (Right) The nuclear membrane appears as a single linear structure on this EM. There is both heterochromatin (clumped along the nuclear membrane) and euchromatin (dispersed), the latter reflecting metabolic activity. The central prominent nucleolus is composed of filamentous nucleolonema.

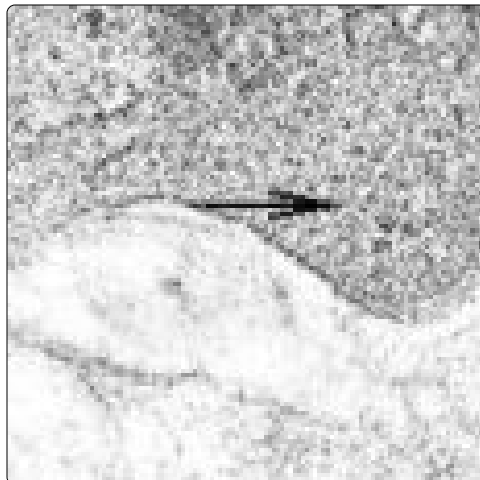


Nucleus, Nucleolus, and Unit Membrane

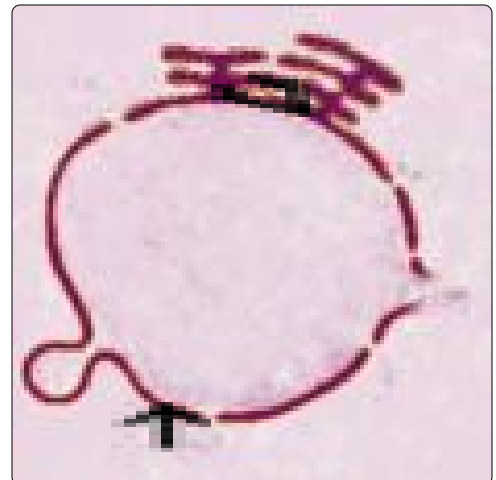


Unit Membrane

(Left) This high-magnification EM of nuclear membrane demonstrates a trilaminar structure with 2 fine parallel densities separated by a lucent region. The top 1/2 of the image is represented by heterochromatin. (Right) This diagram highlights the nuclear-cytoplasmic communication that occurs through the following structures: (1) Nuclear blebs, (2) nuclear pores, (3) communication between perinuclear space and rough endoplasmic reticulum, and (4) dissolution of the nuclear membrane during mitosis.

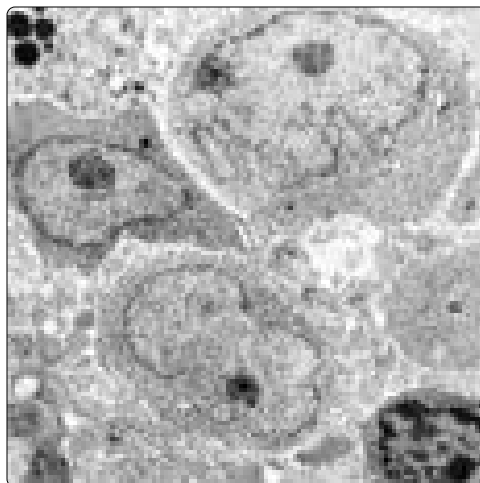


Nuclear-Cytoplasmic Interactions



Nuclear Blebs: EM

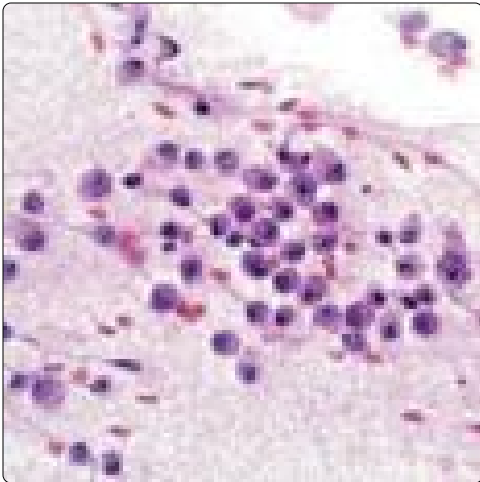
(Left) EM shows that there are several lymphoid cells in various stages of activation. A mature lymphocyte is in the lower right. Each of the remaining nuclei has a prominent nucleolus. The nuclear membranes of the 2 center cells are convoluted and show occasional thin circular extensions entrapping cytoplasm. (Right) Apposing membranes of the endoplasmic reticulum create a flattened compartment for protein synthesis. The attached external granules are ribosomes.



Rough Endoplasmic Reticulum



Plasma Cells

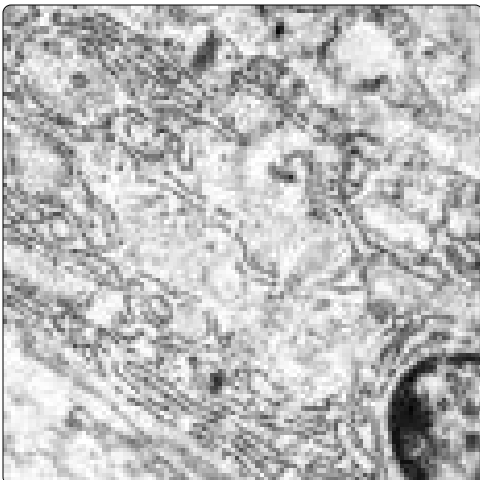


Plasma Cell Mitochondria

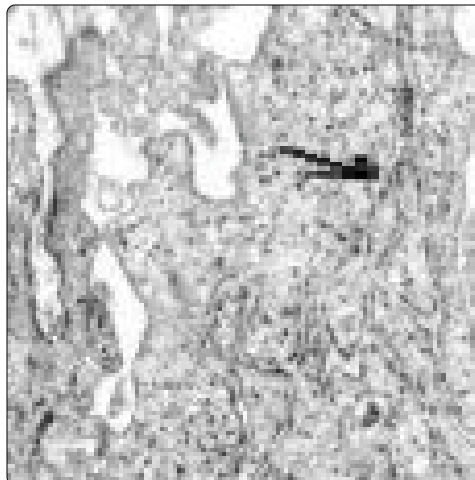


(Left) Touch imprint of a cluster of reactive plasma cells shows the displaced, clockface nucleus adjacent to a cleared area of cytoplasm (hof) that corresponds ultrastructurally to the location of the Golgi apparatus, mitochondria, and assorted organelles. Basophilic cytoplasm corresponds to distribution of rough endoplasmic reticulum. **(Right)** In this EM of mature plasma cell, area next to the nucleus is occupied by mitochondria. Most of the cytoplasm is occupied by flat cisterns of endoplasmic reticulum.

Plasma Cell Golgi Apparatus

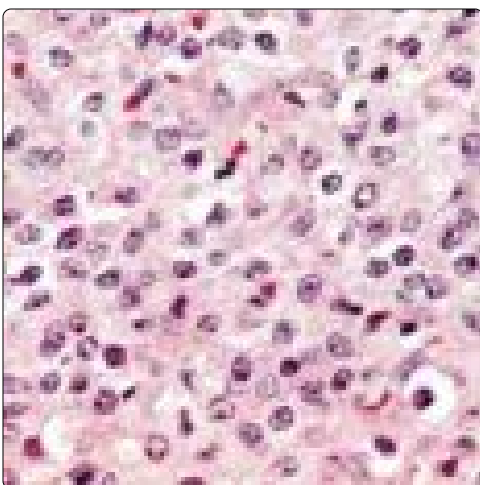


Free Ribosomes and Rough Endoplasmic Reticulum

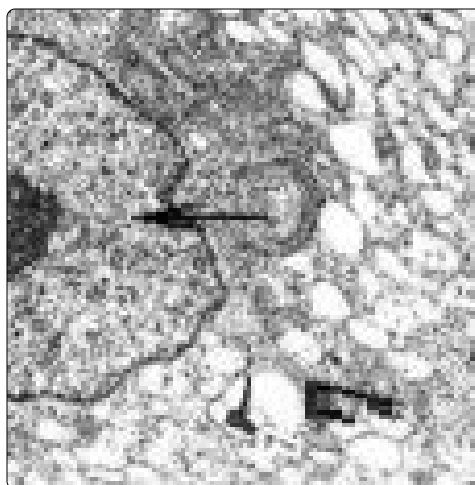


(Left) EM shows that in the perinuclear area of this plasma cell, the nucleus is apposed to flat cisterns of the Golgi apparatus, all of which are surrounded by flattened rough endoplasmic reticulum. **(Right)** This portion of a cell periphery on EM shows convoluted plasma membranes and scattered intracytoplasmic ribosomes, some of which attach to rough endoplasmic reticulum.

Smooth Endoplasmic Reticulum



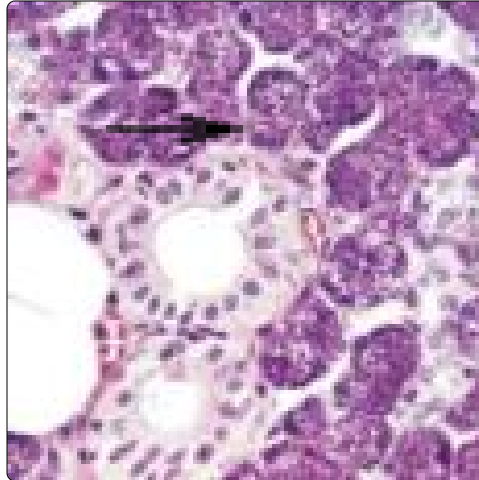
Smooth Endoplasmic Reticulum



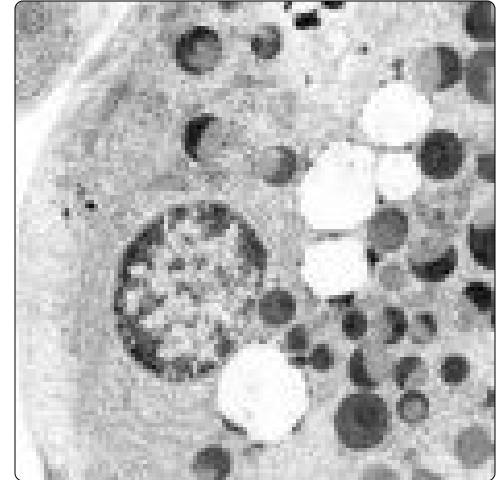
(Left) In this portion of an adrenal cortical adenoma, the tumor cells have abundant foamy cytoplasm reflective of prominent smooth endoplasmic reticulum. **(Right)** EM shows a portion of an adrenal cortical tumor cell with cytoplasm filled with rounded, clear spaces of smooth endoplasmic reticulum. Note the large nucleus with dispersed chromatin and part of a prominent nucleolus. The prominent smooth endoplasmic reticulum correlates with steroid production.

Salivary Gland Ducts and Acini

(Left) A salivary gland lobule with 2 ducts is surrounded by acini with cytoplasm filled with enzymatic granules, which is indicative of an active Golgi system. (Right) EM shows an acinar cell with a basal nucleus and apical variably sized granules of heterogeneous electron density. Flat, rough endoplasmic reticulum surrounds the bottom and sides of the nucleus. Basally, there are numerous folds of plasma membrane abutting the stroma and outlined by a basal lamina.

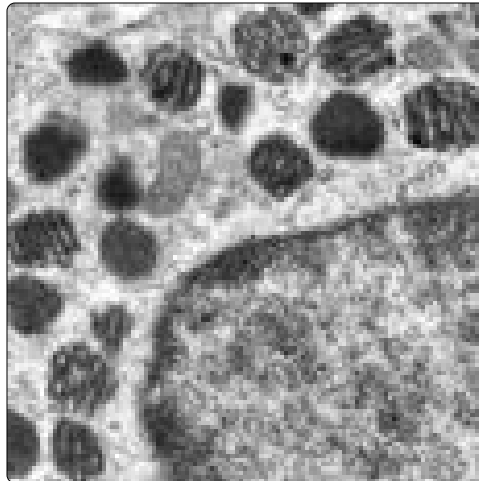


Acinar Cell

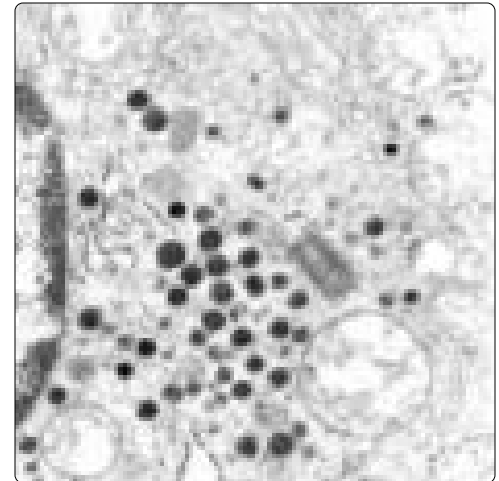


Mast Cell Granules

(Left) Secretory granules may contain mucin or digestive enzymes, be lysosomal as in lipofuscin, or be specialized. These granules have a scroll-like structure and are characteristic of mast cells. (Right) This juxtanuclear collection of neuroendocrine granules is from a paraganglioma. These small granules contain mixtures of peptides and amines. There is a centriole adjacent to the cluster of granules.

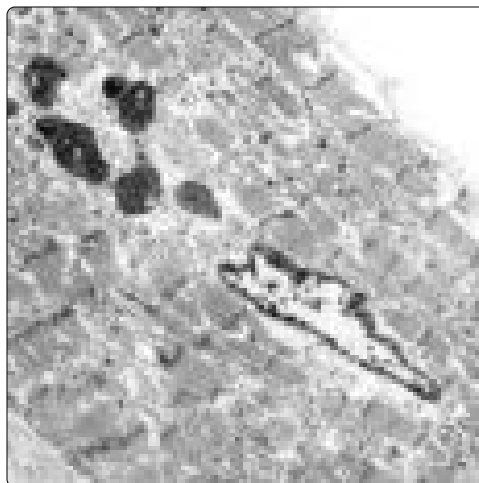


Neuroendocrine Granules and Centriole

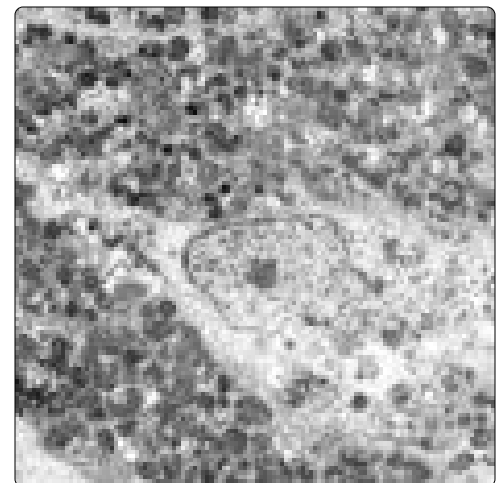


Lipofuscin Granules

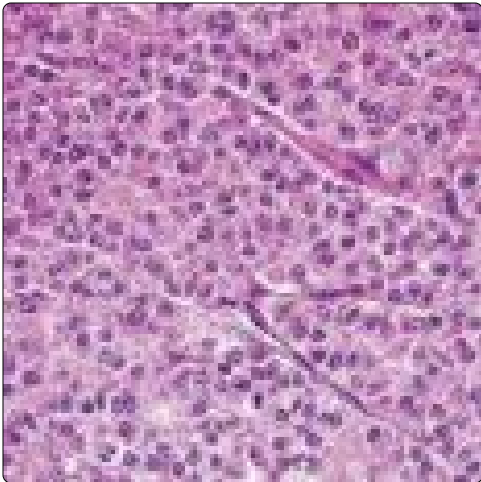
(Left) EM shows this portion of a cardiac muscle cell contains paranuclear clusters of variably sized, dark, membrane-bound granules. Histologically, these granules are brown and similar to those seen in the liver. (Right) EM shows several tumor cells from a granular cell tumor containing intracytoplasmic granules, which are of variable size, density, and internal complexity. These are characteristic of lysosomes.



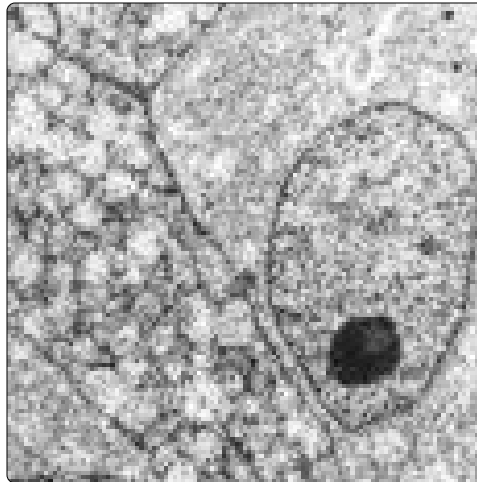
Lysosomes



Oncocytes

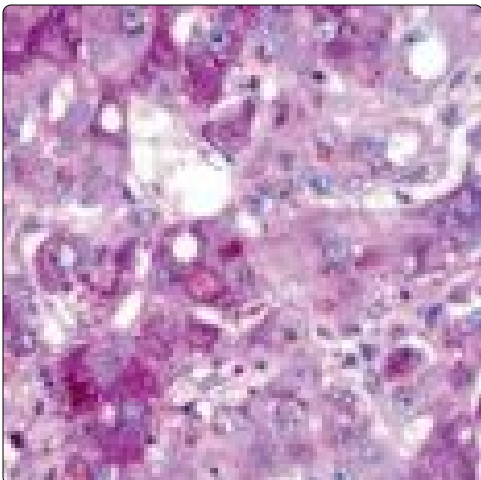


Oncocytoma

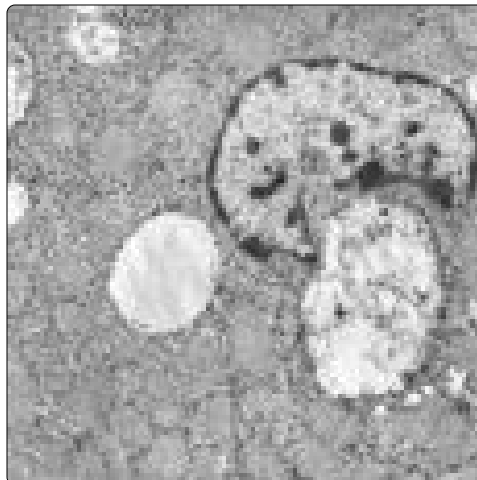


(Left) The cells of an oncocytoma have granular, eosinophilic cytoplasm reflecting numerous intracytoplasmic mitochondria. **(Right)** EM shows the cytoplasm is filled with mitochondria, to the exclusion of most other organelles. The internal substructure of cristae is clearly visible.

Fat, Glycogen, and Lipofuscin

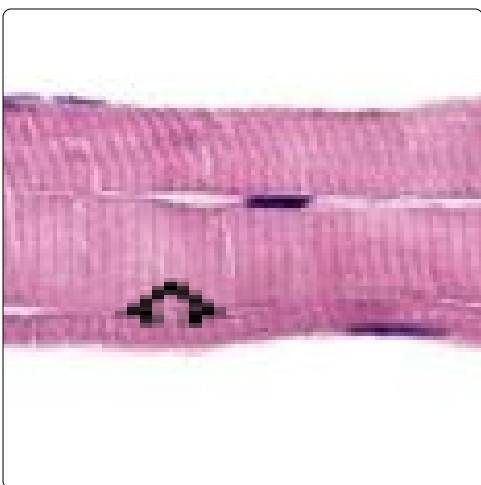


Glycogen and Fat

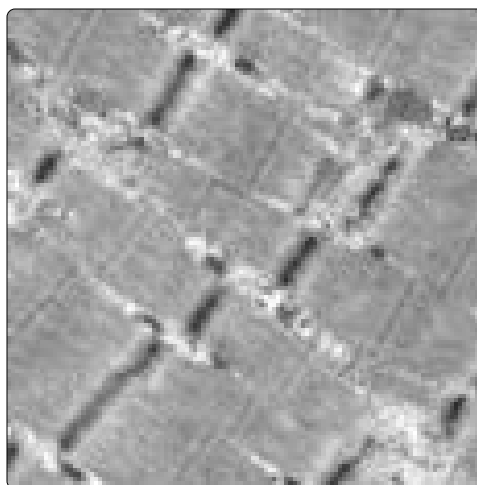


(Left) A section of liver stained with PAS before diastase digestion reveals scattered fat globules and intracytoplasmic brown pigment (lipofuscin) coexisting with numerous dark-staining glycogen granules. **(Right)** EM shows this liver cell contains numerous mitochondria, dispersed granules of glycogen, and scattered globules of fat. Glycogen granules may occasionally be confused with free ribosomes.

Skeletal Muscle



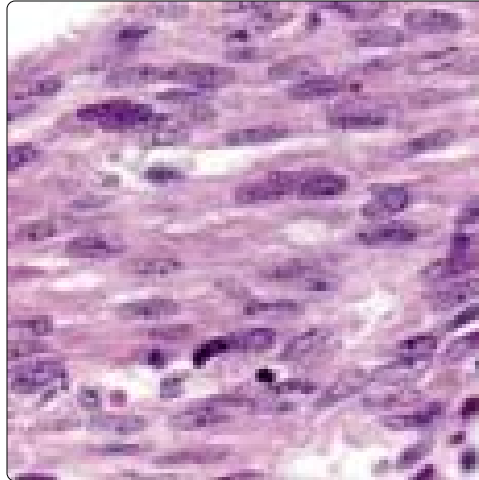
Skeletal Muscle Contractile Filaments



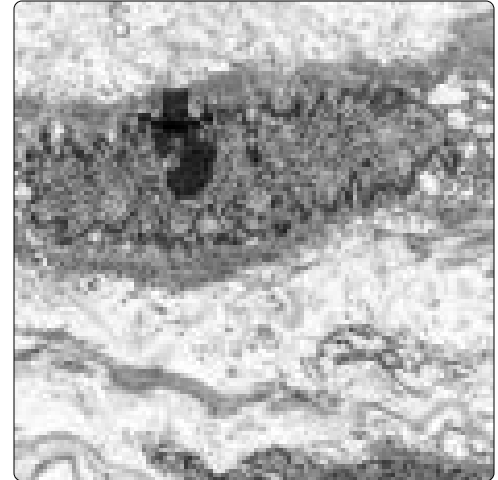
(Left) Normal skeletal muscle demonstrates peripheral elongated nuclei. Typical cross striations of the cytoplasm are present. **(Right)** EM shows that the skeletal muscle cytoplasm is organized into dark Z bands, which receive thick and thin myofilaments responsible for contractile function. These correlate directly with the light microscopic striations.

Smooth Muscle

(Left) Smooth muscle cells demonstrate elongated nuclei that are blunt at the ends and have occasional transverse lines perpendicular to the long axis , reflecting deep nuclear folds. The cytoplasm is fibrillar and eosinophilic. The cell borders are distinct. **(Right)** EM shows a smooth muscle cell with a corrugated nuclear contour  corresponding to transverse lines histologically. The cytoplasm is filled with fine actin filaments and dense bodies. The collagen fibers are extracellular.



Smooth Muscle Cell

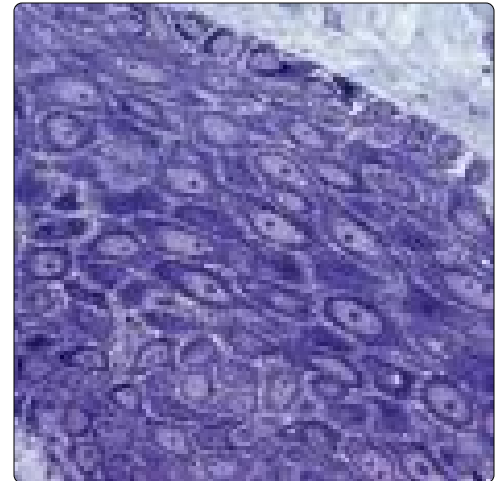


Smooth Muscle Actin Filaments

(Left) On EM, the edge of a smooth muscle cell shows actin filaments parallel to and inserting on the plasma membrane  from which pinocytotic vesicles invaginate . Thin basal lamina parallels the plasma membrane. **(Right)** A toluidine blue-stained tangential section through the mid epidermis demonstrates stratified squamous epithelium with numerous intercellular bridges.

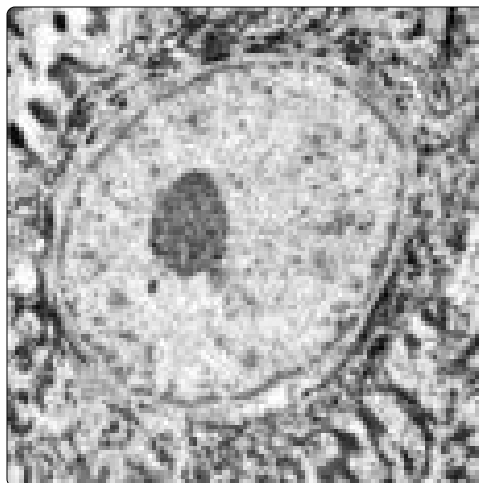


Intercellular Bridges in Squamous Epithelium

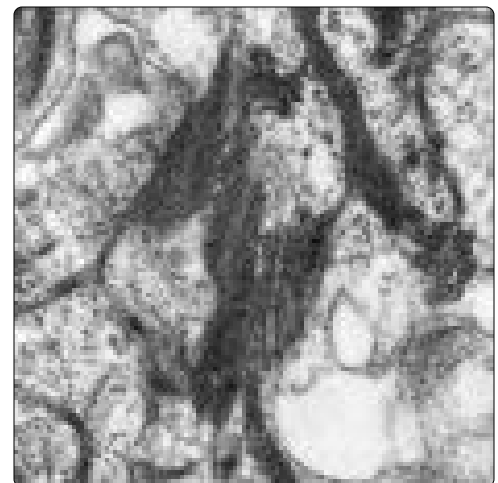


Intercellular Bridges, Tonofilaments, and Squamous Epithelium

(Left) EM shows the cytoplasm of this squamous epithelial cell filled with wavy fascicles of tonofilaments inserting on desmosomes. Cytoplasmic extensions of these complexes form the intercellular bridges. **(Right)** This high-magnification EM shows an intercellular bridge consisting of a central desmosome  (thickened apposing segments of plasma membrane and a central intervening linear density) onto which wavy tonofilaments insert.



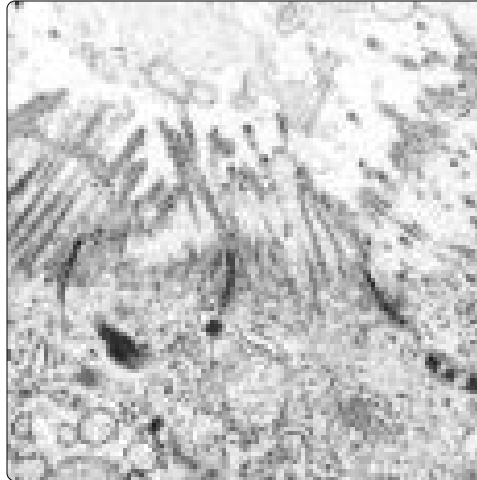
Tonofilaments Inserting on Desmosomes



Small Bowel Epithelium

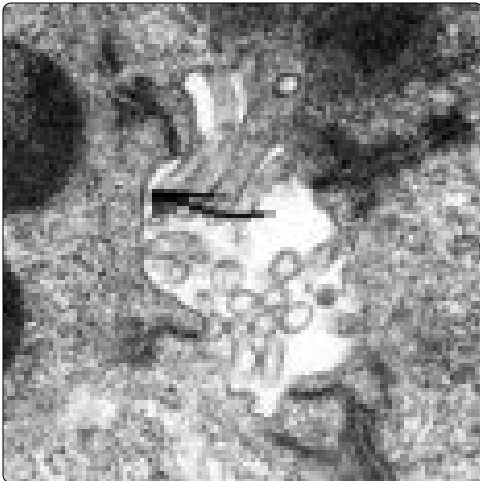


Small Bowel Apex

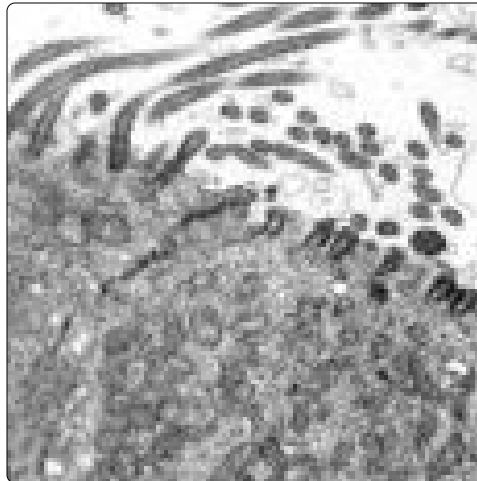


(Left) Toluidine blue stain shows glandular epithelium of the small bowel demonstrating mucinous vacuolization [X] and a dense apical terminal bar [X]. (Right) The apical region of several adjacent glandular cells of the small bowel is visualized by EM. The terminal bar by light microscopy is represented by numerous microvilli anchored into the apical cytoplasm by thin filaments. The cells are joined by desmosomal complexes.

Glandular Space

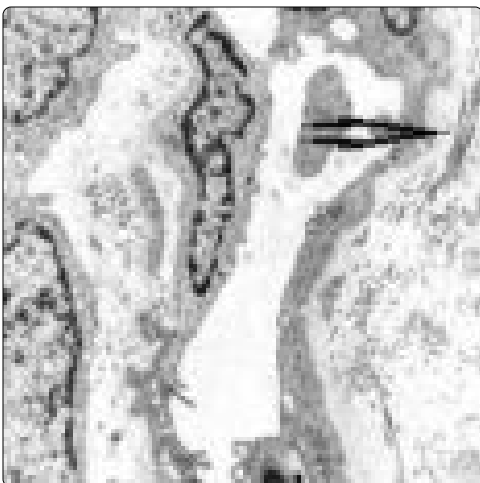


Cilia



(Left) EM shows a glandular lumen [X] formed by several salivary gland acinar cells and filled with microvilli. A secretory granule is present [X]. The apical portion of the cells is held together by junctional complexes [X]. (Right) EM shows a portion of 2 ciliated respiratory epithelial cells. The cilia are modified centrioles with the same tubular substructure. They are anchored into the cytoplasm by angled striated blepharoplasts [X].

Capillary, Basal Lamina, and Collagen



Extracellular Collagen



(Left) EM shows a thin-walled capillary lined by endothelial cells. A thin single layer of basal lamina [X] surrounds and runs parallel to the outer plasma membrane. Collagen fibers [X] in varying stages of cross section are in the stroma. (Right) Extracellular collagen fibers exhibit the characteristic periodicity of 64 nm, which is visualized by EM.

TERMINOLOGY

Histology

- Study of tissues
 - All tissues are composed of cells and extracellular matrix
 - Tissue types are grouped together to form organs
 - Organs are grouped together to form organ systems
- In humans, term histology is used synonymously with microscopic anatomy
- 4 basic tissue types
 - Epithelium
 - Connective tissue
 - Muscle
 - Nerve

TYPES OF TISSUE

Epithelium

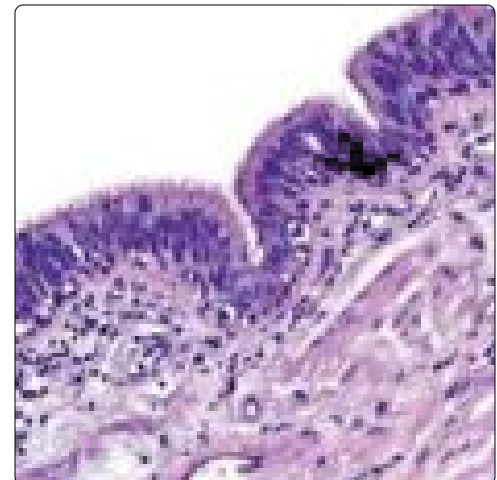
- Sheets of tightly bound contiguous cells
 - Cover external surface of body and line internal surfaces
 - Epithelial cells are bound together by junctional complexes
 - Little intercellular space between cells
- Derived from all 3 embryonic germ layers (ectoderm, mesoderm, and endoderm)
 - Although mostly from ectoderm and endoderm
- Separated from underlying connective tissue by extracellular matrix called basal lamina
 - Extracellular matrix is made by epithelial cells
- Epithelium is avascular
 - Derives nourishment and oxygen from underlying connective tissue
- Functions of epithelium
 - Protection
 - Lining of surfaces
 - Transport of molecules across epithelial layers and between body compartments
 - Absorption
 - Detection/sensation
 - Secretion

- Classification of epithelium is by number of cell layers as well as morphology
 - Number of cell layers between basal lamina and free surface
 - **Simple epithelium**
 - ◻ Composed of single layer of tightly packed cells
 - ◻ Functions as lining or limiting membrane as well as for fluid transport, gas exchange, lubrication, absorption
 - **Stratified epithelium**
 - ◻ Composed of > 1 cell layer
 - ◻ Used for protection, absorption, secretion
 - **Pseudostratified epithelium**
 - ◻ Appears stratified but is actually composed of single layer of cells
 - ◻ Examples: Trachea, primary bronchi, auditory canal, nasal cavity, epididymis, lacrimal sac
 - Morphology
 - **Squamous**
 - ◻ Flat to polygonal cells
 - **Simple squamous**
 - ◻ Thin, plate-like cells arranged in single layer
 - ◻ Examples: Pulmonary alveolar lining, loop of Henle, mesothelium
 - **Stratified squamous**
 - ◻ Layers of squamous epithelium that provide toughness
 - ◻ May be keratinized or nonkeratinized
 - ◻ Examples: Skin, esophagus, larynx, vagina
 - **Cuboidal**
 - ◻ Short and square cells
 - ◻ Centrally placed, rounded nuclei
 - **Simple cuboidal**
 - ◻ Lining of ducts, external surface of ovary, some renal tubules
 - **Stratified cuboidal**
 - ◻ Lines sweat gland ducts
 - **Columnar**
 - ◻ Tall and rectangular with ovoid, usually basally located nuclei

Epithelium



Cilia



(Left) Epithelium consists of sheets of tightly bound contiguous cells that cover the external surfaces of the body and line the internal surfaces. Epithelial cells are bound to the underlying connective tissue by a basement membrane. **(Right)** Cilia are motile, hair-like projections on the surfaces of some epithelial cells that are used to propel substances along the lumen, as seen here in the bronchus.

- **Simple columnar**
 - Examples: Lining of gut, some large ducts; may have cilia or microvilli at luminal border
- **Stratified columnar**
 - Examples: Conjunctiva, some large ducts, portions of male urethra
- **Transitional**
 - Originally believed to be intermediate between stratified squamous and stratified columnar, now known to be special distensible epithelium that lines bladder and much of urinary tract
- o Polarity and specialized structures
 - Most epithelial cells have apical domain and basolateral domain that differ according to cell function
 - Apical domain faces lumen
 - Microvilli
 - Finger-like projections of cytoplasm extending from free surface of cell into lumen
 - Critical for absorption
 - Glycocalyx
 - Carbohydrate coating overlying luminal surface of microvilli
 - Used for protection, cell recognition
 - Combination of glycocalyx and microvilli make up brush border of intestines
 - Stereocilia
 - Long microvilli found only in epididymis and on sensory hairs cells of cochlea
 - Used to increase surface area (epididymis) and aid in signal generation (cochlea)
 - Cilia
 - Motile, hair-like projections on surfaces of some epithelial cells
 - Used to propel substances along lumen and for sensation
 - Basolateral domain
 - In contact with basal lamina
 - Lateral membrane specializations hold cells together
 - Terminal bars, junctional complexes
 - Basal membrane specializations anchor cell to underlying connective tissue
 - Basal lamina, plasma membrane foldings, hemidesmosomes

Glands

- Epithelial structures that originate from invaginated epithelial cells
- Secretory units and accompanying ducts make up parenchyma of gland
- Underlying supporting connective tissue is known as stroma
 - o In some larger glands, connective tissue subdivides gland into lobules
 - o Connective tissue lobules contain vessels, lymphatics, nerves
- Formation
 - o Epithelial cells proliferate and invade underlying stroma
 - If contact with surface maintained: Exocrine gland
 - No contact with surface: Endocrine gland
 - o Differentiate further into specific type of gland
- Function
 - o Secrete mucus, hormones, enzymes, etc.
 - o Manufacture products intracellularly
 - o Products usually stored in vesicles called secretory granules
- **Exocrine** glands
 - o Secrete their products onto external or internal surface from which they originated, via ducts
 - Mucous glands
 - Secrete mucinogens that ultimately become mucus
 - Examples: Goblet cells of GI tract, minor salivary glands, mucous glands of stomach, genital tract, respiratory tract
 - Serous glands
 - Secrete watery fluid rich in enzymes
 - Example: Exocrine pancreas
 - Mixed serous and mucinous glands
 - Produce both mucous and serous secretions
 - Example: Minor salivary glands
 - o 3 different mechanisms for releasing secretory products
 - Merocrine
 - Released by exocytosis; cell membranes/cytoplasm not part of secretion
 - Example: Parotid gland
 - Apocrine
 - Small portion of apical cytoplasm released with product
 - Example: Lactating mammary gland
 - Ironically, apocrine sweat glands are believed to use merocrine mechanism of secretion release
 - Holocrine
 - Mature secretory cells die and become secretory product
 - Example: Sebaceous gland
 - o Unicellular
 - Simplest type of exocrine gland
 - Isolated cells within epithelium
 - Example: Goblet cells of GI tract
 - o Multicellular
 - Organized clusters of secretory cells
 - Majority of glands are multicellular
 - Simple
 - Consist of 1 unbranched duct
 - Compound
 - Consists of branching ducts
 - Further classified according to morphology
 - Tubular
 - Acinar (or alveolar)
 - Tubuloacinar (tubuloalveolar)
 - Multicellular glands typically have capsule
- o Myoepithelium
 - Cells with features of both smooth muscle and epithelium
 - Located between basal lamina and basal pole of duct or secretory cell of many multicellular glands
 - Help express secretions
 - Examples: Sweat, mammary, lacrimal, salivary glands
 - Can be identified by myoepithelial immunohistochemical markers
 - SMA, calponin, p63, others

- **Endocrine glands**
 - Secrete their products into blood or lymphatics
 - Secrete hormones, including peptides, proteins, glycoproteins, steroids
 - Do not have ducts that connect them to originating epithelium
 - Contact with originating epithelium lost during formation
 - Major endocrine glands
 - Adrenal
 - Pituitary
 - Thyroid
 - Parathyroid
 - Pineal
 - 2 major structural forms
 - Cords
 - Anastomosing cords around capillaries or sinusoids
 - Examples: Adrenal, parathyroid
 - Follicles
 - Glandular cells surround cavity that receives hormone
 - Example: Thyroid
 - Some other organs also have significant endocrine component
 - Ovary
 - Granulosa cells
 - Thecal cells
 - Interstitial cells
 - Hilus cells
 - Testes
 - Sertoli cells
 - Leydig cells
 - Pancreas
 - Islets of Langerhans
 - Placenta
 - Syncytiotrophoblast
 - Decidua
- Some glands are **mixed endocrine and exocrine**
 - Example: Pancreas has both exocrine acini and endocrine Islets of Langerhans

Connective Tissue

- Connective tissue proper
 - 4 types
 - Loose
 - Right beneath skin, below mesothelium, around blood vessels, around glands, lamina propria
 - Dense
 - Dermis, nerve sheaths, organ capsules, tendons, ligaments, aponeuroses, in blood vessels
 - Reticular
 - Framework of hepatic sinusoids, bone marrow, lymph nodes, spleen, islets of Langerhans
 - Adipose
 - White or brown fat
- Specialized connective tissue
 - Cartilage
 - Hyaline, elastic, or fibrocartilage
 - Bone
- Composed of cells and extracellular matrix

- Functions
 - Structural support
 - Medium for exchange
 - Fat storage
 - Defense/protection
- Cell types
 - Fibroblasts
 - Adipocytes
 - Pericytes
 - White blood cells
 - Mast cell
 - Macrophages

Muscle

- Striated
 - Skeletal
 - Cardiac
- Smooth
 - Walls of blood vessels and viscera; dermis of skin

Nerve

- Central nervous system
 - Brain
 - Spinal cord
- Peripheral nervous system
 - Cranial nerves
 - Spinal nerves
 - Ganglia

Extracellular Matrix

- Complex of nonliving macromolecules manufactured by cells and exported into extracellular space
 - Ground substance
 - Amorphous, gel-like material composed of glycoproteins, proteoglycans, glycosaminoglycans
 - Specialized fibers provide strength and elasticity
 - Collagen
 - Elastin
 - Reticulin
 - Basement membrane
 - Interface between epithelium and underlying connective tissue
- Integrins and dystroglycans
 - Transmembrane proteins that serve as receptors and facilitate formation of basal lamina

ORGAN SYSTEMS

Blood

- Red blood cells
- White blood cells
- Plasma

Circulatory System

- Cardiovascular system
 - Arteries
 - Capillaries
 - Veins
 - Heart
 - Cardiac valves
- Lymphatic System

Lymphoid (Immune) System

- Lymph nodes
- Bone marrow
- Thymus
- Spleen
- Mucosa-associated lymphoid tissue
- Tonsils

Endocrine System

- Glands
 - Pituitary
 - Thyroid
 - Parathyroid
 - Adrenal
 - Pineal
- Dispersed neuroendocrine cells
- Paraganglia

Integumentary System

- Epidermis
- Dermis
- Skin appendages
- Fingernails

Breast

- Nipple
- Glandular tissue
- Adipose tissue

Respiratory System

- Nasal cavity
- Sinuses
- Nasopharynx
- Larynx
- Trachea
- Lungs
- Bronchial tree

Oral Cavity

- Oral mucosa
- Tongue
- Gingiva
- Teeth
- Palate
- Minor salivary glands

Digestive System: Alimentary Canal

- Esophagus
- Stomach
- Small bowel
- Appendix
- Large bowel
- Anus

Digestive System: Viscera/Glands

- Major salivary glands
- Pancreas
- Liver
- Gallbladder
- Biliary tree

Genitourinary System

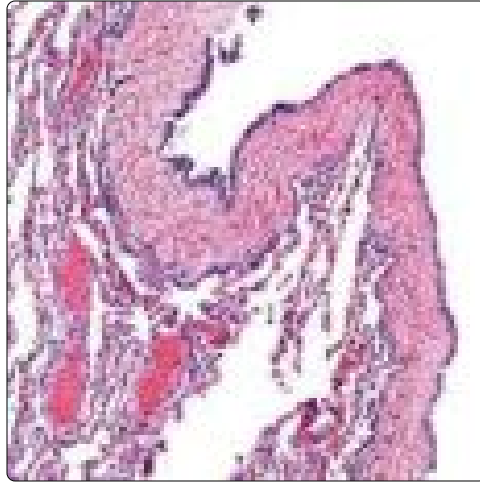
- Kidneys
- Urinary bladder
- Ureters
- Urethra
- Prostate
- Ovaries
- Fallopian tubes
- Uterus
- Cervix
- Vagina
- Vulva
- External genitalia
- Prostate
- Testes
- Penis
- Genital ducts and glands

Special Sensory Organs

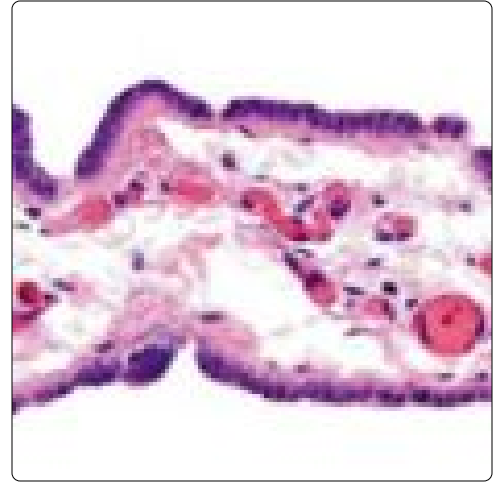
- Eye
- Nose
- Ear
- Tongue/taste buds

Simple Squamous

(Left) The mesothelium  consists of a single layer of squamous epithelial cells. **(Right)** The epithelial cells are anchored to underlying stroma by the basement membrane , as seen in this high-power view of mesothelium. Epithelium is avascular, so it derives nourishment and oxygen from underlying connective tissue.



Simple Squamous (High Power)

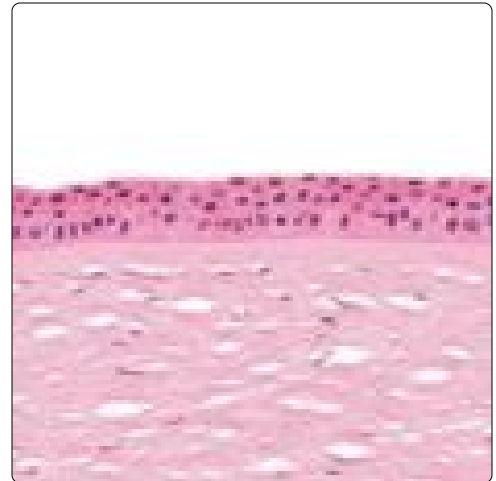


Stratified Squamous

(Left) Stratified squamous epithelium consists of multiple layers of polygonal squamous cells, as seen here in the esophagus. This stratified squamous epithelium is nonkeratinizing because the top layers have nuclei. **(Right)** This example of nonkeratinizing squamous epithelium is from the conjunctiva; it consists of multiple layers of polyhedral squamous cells. The nuclei extend to the surface. This type of epithelium is usually moist and found on mucous membranes.

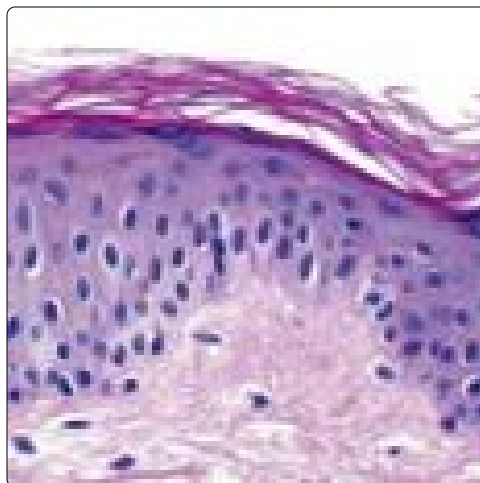


Stratified Squamous, Nonkeratinizing

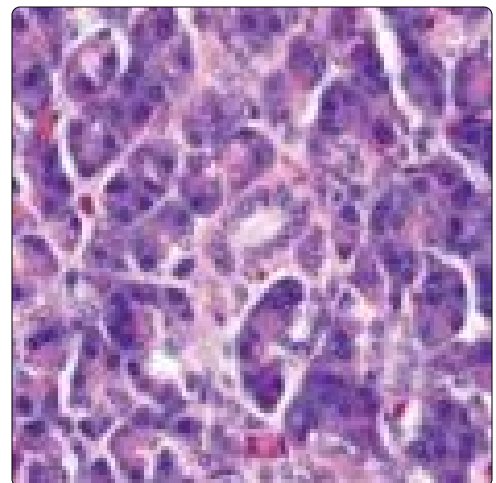


Stratified Squamous, Keratinizing

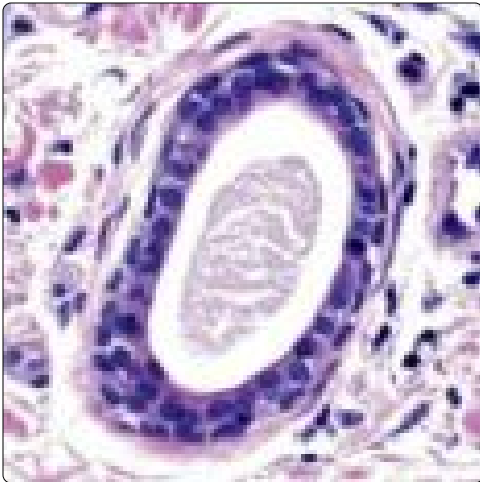
(Left) In keratinizing stratified squamous epithelium, the superficial layers  are composed of dead cells that have been replaced with keratin, providing toughness, as in the skin. **(Right)** Cuboidal epithelial cells  are short and square with centrally placed round nuclei. This example is a small pancreatic duct. Examples of simple cuboidal epithelium include the linings of ducts, the external surface of the ovary, and some renal tubules.



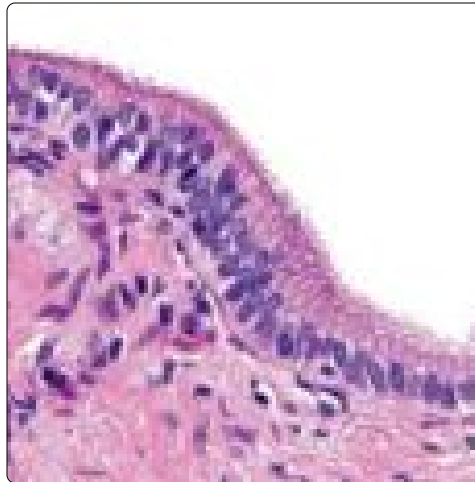
Simple Cuboidal



Stratified Cuboidal

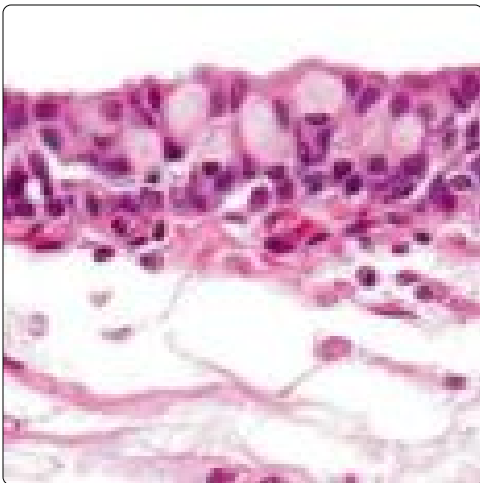


Simple Columnar

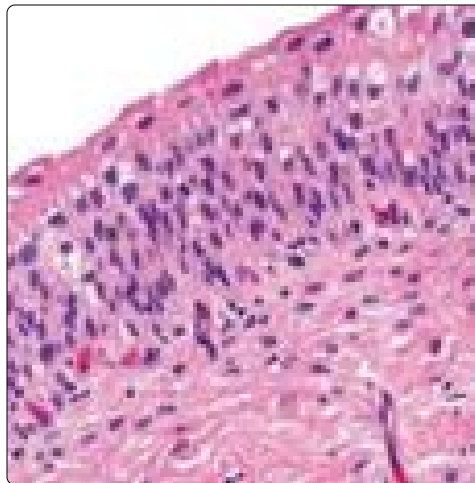


(Left) This sweat gland duct is lined by multiple layers of cuboidal epithelium. Stratified cuboidal epithelium is rare. **(Right)** Simple columnar epithelium consists of tall, rectangular cells with ovoid, basally located nuclei. This example is from the pancreatobiliary tree. Other examples include the lining of the gut and other large ducts. Cilia or microvilli may be present at the apical border.

Stratified Columnar

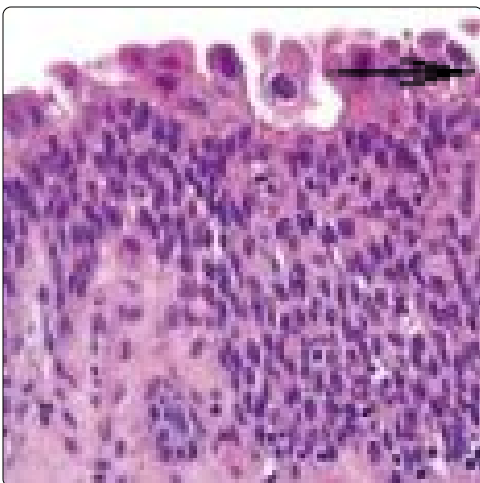


Transitional

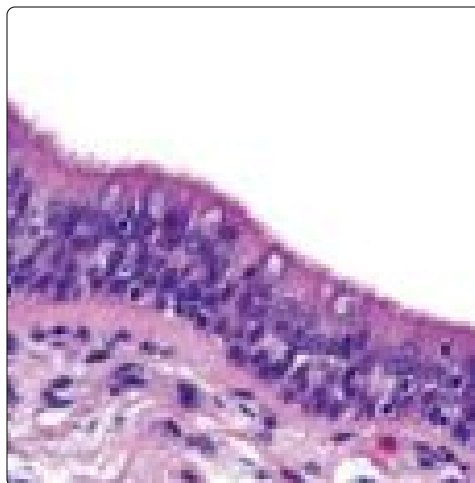


(Left) Stratified columnar epithelium is seen in some portions of the conjunctiva as well as in some large ducts and portions of the male urethra. **(Right)** Transitional epithelium is a specialized multilayered epithelium with basal low columnar or cuboidal cells, overlying intermediate cells, and large, sometimes dome-shaped surface or umbrella cells. It allows structures of the urinary tract to expand when distended with urine and collapse when empty.

Transitional (Urothelium)



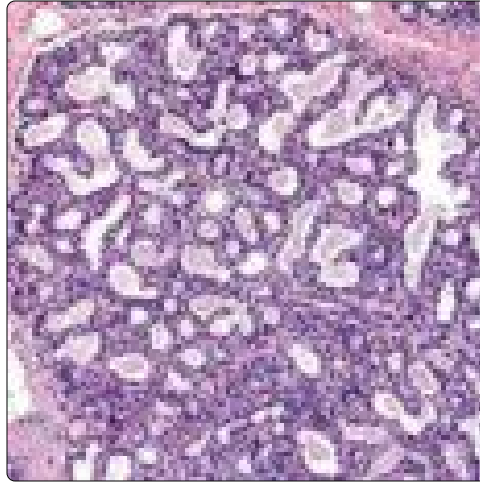
Pseudostratified



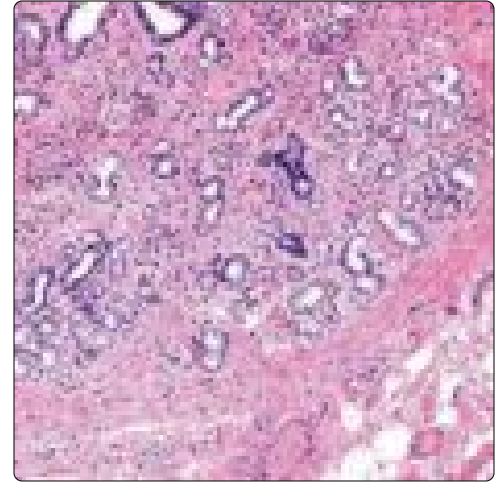
(Left) The umbrella cells  in urothelium may be binucleate and may bulge into the luminal space. **(Right)** This example of pseudostratified epithelium is from the trachea. All cells rest on the basement membrane but not all reach the surface. Since the nuclei are at different heights, it imparts the appearance of stratification. Most pseudostratified epithelia are ciliated, as seen here . Functions of this type of epithelium include secretion, absorption, lubrication, protection, and transportation.

Glands

(Left) The secretory units and their ducts make up the parenchyma of a gland (as shown here by this example of lactating breast). **(Right)** The connective tissue surrounding a gland constitutes the supporting stroma. Here, peribiliary mucous glands are surrounded by lobular connective tissue stroma.

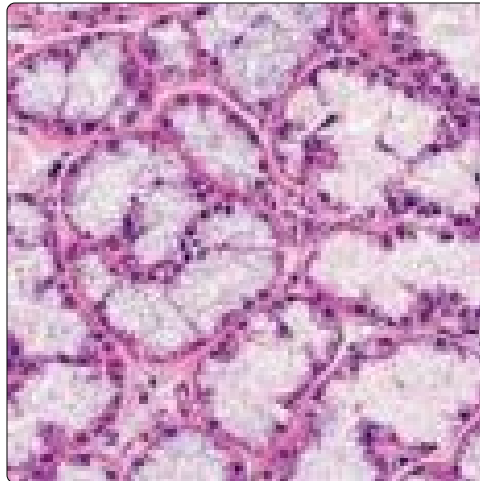


Glands and Stroma

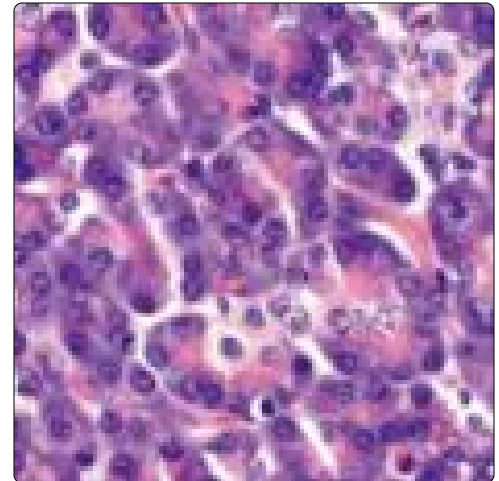


Mucous Glands

(Left) Mucous glands, as seen here in this example from a minor salivary gland, are exocrine glands that secrete mucinogens, which ultimately become mucus. Note that the secretory unit is surrounded by connective tissue stroma. **(Right)** Serous glands, such as the exocrine pancreatic acini shown here, are exocrine glands that secrete a watery, enzyme-rich fluid. The products of glands are often stored in secretory granules.

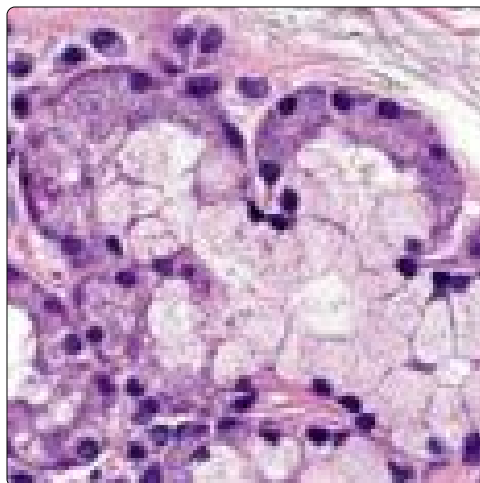


Serous Glands

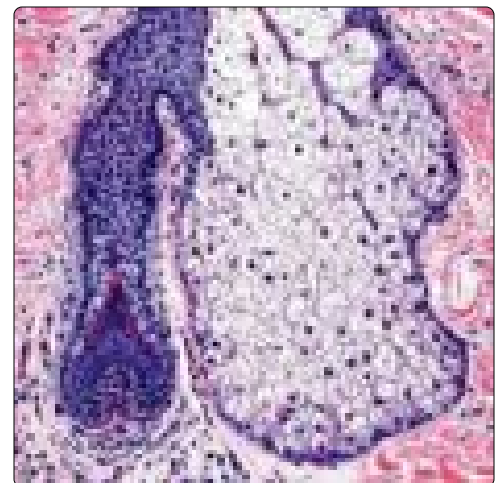


Mixed Mucinous and Serous Gland

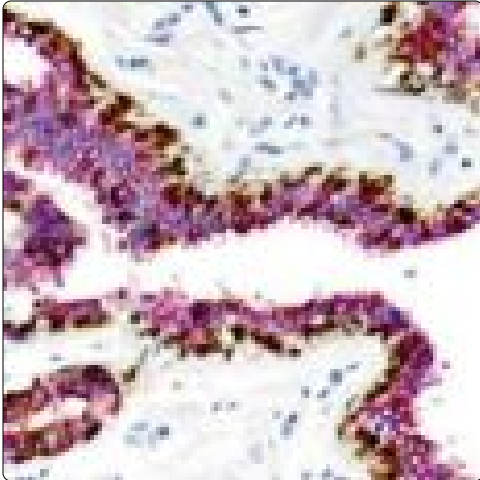
(Left) Mixed serous and mucinous glands, illustrated here in a minor salivary gland, are exocrine glands that produce both mucous and serous secretions. The distinct mucous cells [X] are filled with basophilic mucin with overlying granular serous demilunes [Y]. **(Right)** Sebaceous glands have abundant clear-staining cytoplasm due to abundant lipid. This sebaceous gland is attached to an adjacent hair follicle and empties its contents into the follicle via the sebaceous duct.



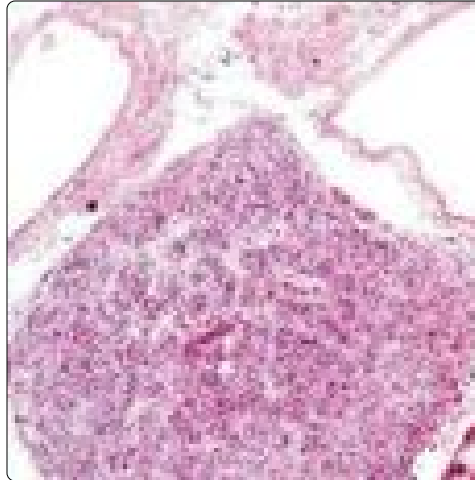
Sebaceous Exocrine Gland



Myoepithelium

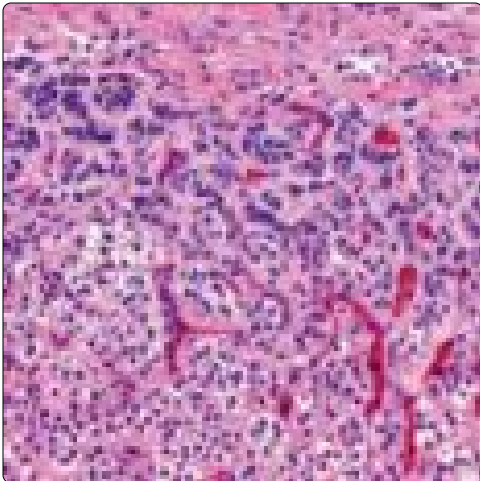


Endocrine Gland (Pineal)



(Left) Myoepithelial cells have features of both smooth muscle and epithelium, and they help express secretions. In this section of breast, the epithelial cells mark with keratin (red chromagen), and the myoepithelial cells mark with calponin (brown chromagen). **(Right)** Endocrine glands (such as the pineal gland shown here) have no ducts, as they have lost the connection to the originating epithelium during formation. They secrete their products directly into blood or lymphatics.

Endocrine Gland (Adrenal)

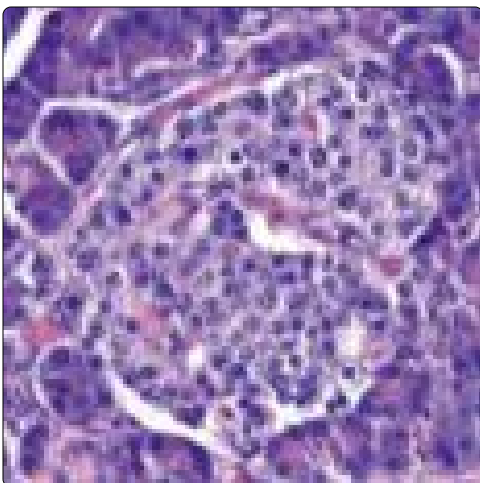


Endocrine Gland (Thyroid)

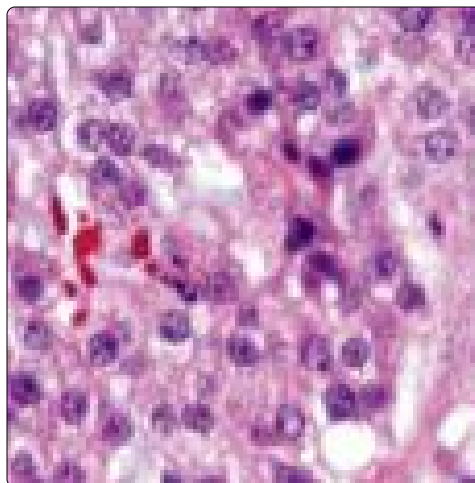


(Left) The adrenal gland is an endocrine gland arranged in cords of endocrine cells surrounding capillaries. **(Right)** Some endocrine glands are arranged in follicles, or glandular cells that surround a cavity, which receive the hormone. (Courtesy J. Hunt, MD.)

Mixed Glands



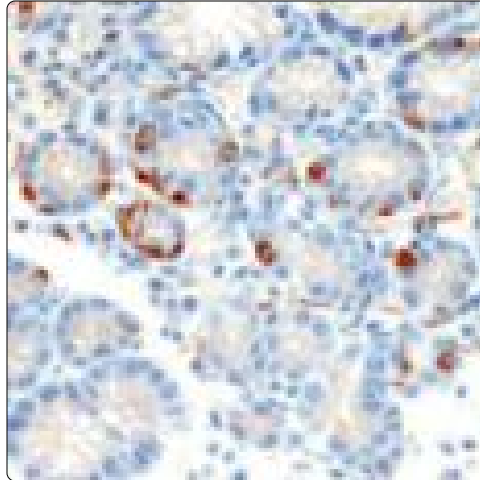
Endocrine Cells



(Left) Some glands, such as the pancreas, are mixed endocrine and exocrine. This photomicrograph of pancreas shows the endocrine islet of Langerhans surrounded by exocrine acinar parenchyma. **(Right)** The granulosa cells of the ovary, shown here, are an example of a collection of endocrine secretory cells that exist within an organ that contains multiple tissue types. The testis, pancreas, and placenta are other examples.

Diffuse Neuroendocrine System

(Left) The diffuse neuroendocrine system is present throughout the digestive tract and respiratory tract. These cells make many different types of hormones. They are also known as APUD cells (amine precursor uptake and decarboxylation). These neuroendocrine cells in the stomach are highlighted by CD56. **(Right)** Loose connective tissue consists of a loose arrangement of fibers with abundant ground substance and extracellular fluid. It also contains scattered collagen, reticulin, and elastic fibers.

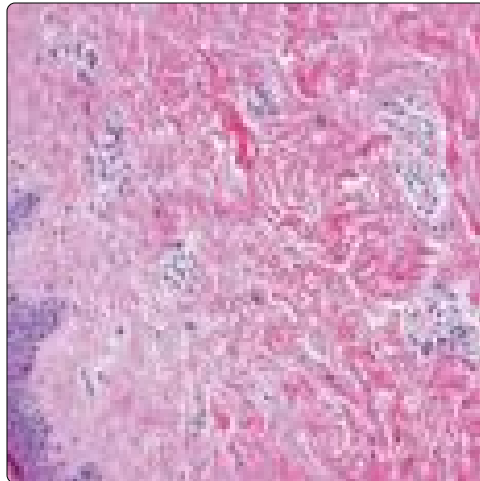


Loose Connective Tissue

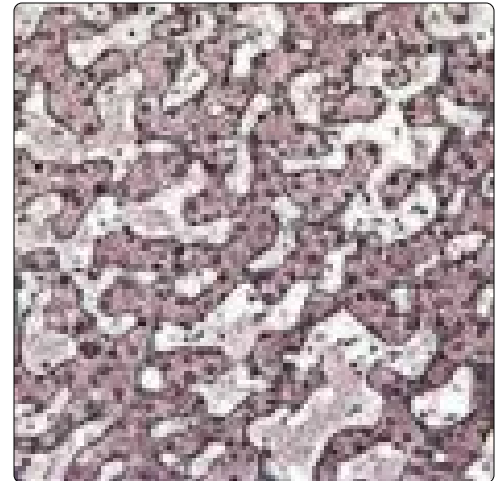


Dense Connective Tissue

(Left) Dense connective tissue, seen here in the dermis, contains more fibers than loose connective tissue. Note the dense eosinophilic collagen fibers. **(Right)** Reticular connective tissue is composed of type III collagen, which forms a network that makes the framework for many organs and glands. A reticulin stain shows the reticular network in the liver, which supports the sinusoids and hepatic cell plates.

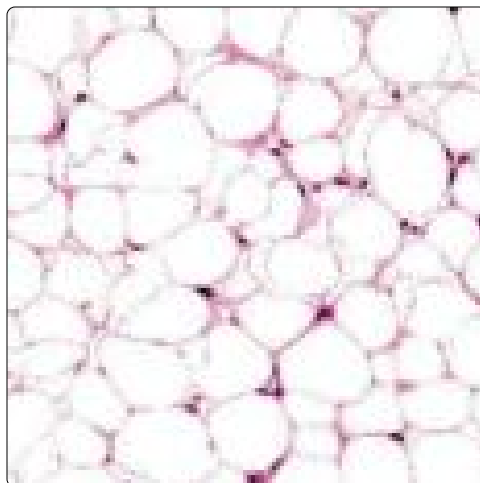


Reticular Connective Tissue



Adipose Tissue

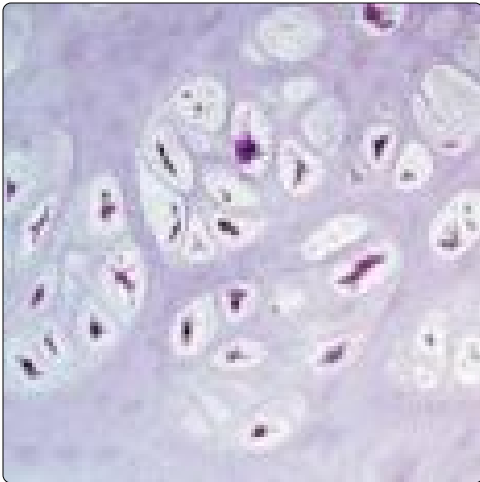
(Left) Adipose tissue is a type of connective tissue that stores fat. It also contains receptors for many substances, including glucocorticoids, growth hormone, and insulin. It is present in the subcutaneous tissue, omentum, and mesentery. **(Right)** Bone is a type of specialized connective tissue that comprises the primary structural framework for the body (skeleton). Its extracellular matrix is calcified and surrounds the cells that secreted it.



Bone



Cartilage



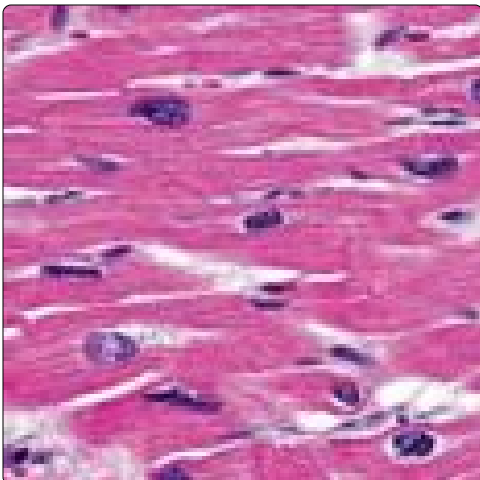
Skeletal Muscle



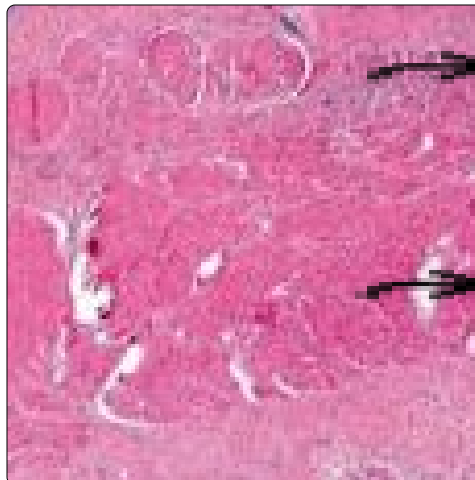
(Left) Cartilage is a specialized connective tissue that is very flexible. It lines the articulating surfaces of the bones, serves as a template for endochondral bone formation, and is part of many organs, such as the ear, larynx, and intervertebral discs.

(Right) Skeletal muscle is a type of striated muscle that comprises the majority of the voluntary muscle mass in the body. The cells are long and cylindrical (shown here in cross section) and have peripheral nuclei [\[2\]](#).

Cardiac Muscle



Smooth Muscle



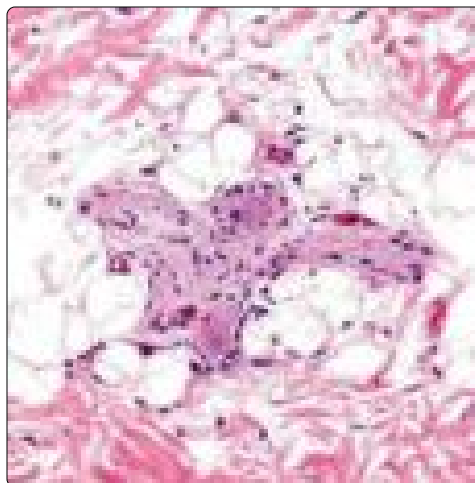
(Left) Cardiac muscle is a nonvoluntary, striated muscle that is present only in the heart and proximal segments of the pulmonary veins where they join the heart. The muscle fibers are striated, and the nuclei are centrally located.

(Right) Smooth muscle [\[2\]](#) is present in the walls of the viscera, such as in this example of the wall of the stomach.

Peripheral Nerve



Ganglia



(Left) Peripheral nerves are composed of multiple bundles (or fascicles) of nerve cells, each enclosed by connective tissue known as perineurium. The peripheral nervous system includes cranial and spinal nerves. (Courtesy P. Burger, MD.) **(Right)** Ganglia are located throughout the body. They are aggregates of neurons within the peripheral nervous system that contain the cell bodies of sensory and postganglionic autonomic neurons.

TERMINOLOGY

Histologic Artifacts

- Substances, structures, or alterations formed by artificial means that are not naturally present in given tissue
- May be introduced at any time during course of specimen from patient to pathologist
 - Collection
 - Fixation
 - Embedding
 - Processing
 - Microtome sectioning
 - Staining
 - Coverslipping
- Issues
 - Can limit satisfactory histologic evaluation of tissue
 - Can be misinterpreted as significant findings

COMMON ARTIFACTS

Cauterization (Fulguration)

- Occurs during specimen collection
- Electrocautery results in distortion of tissue
 - Nuclear spindling and pseudostratification
 - Connective tissue, fat, and muscle appear dense and amorphous
 - Separation of epithelium from basement membrane

Crushing

- Occurs during specimen collection
- Excess pressure upon biopsy forceps results in distortion of tissue
 - Destruction and compression of nuclei
 - Extrusion of cytoplasmic contents
 - Can also lead to telescoping (gland-in-gland) appearance

Ice Crystals

- Occurs during specimen freezing
- Prominent gaps and holes in tissue from crystal formation

Air Bubbles

- Occurs during coverslipping
- Introduction of small collections of air between glass slide and coverslip

Nicks, Tears, and Fragmentation

- Occurs during microtome sectioning
- Leads to tissue holes &/or linear streaks
- May result in thin parallel lines (known as chatter)

Pigments and Precipitates

- Occurs during processing or staining
- Formalin, mercury, chromate
- Appear as dark brown, yellow-brown, or black granules/clusters

Contaminants

- May occur during collection, processing, staining, or coverslipping
- Flecks and fragments of exogenous material
 - Squamous epithelial cells (touching slide with fingers)
 - Pollen grains
 - Small fibers
 - Tissue from different specimen

Underdecalcification

- Occurs during gross examination or processing
- Large pieces of tissue containing bone must be sufficiently demineralized (softened) in order to cut with microtome blade
- Underdecalcified bone appears dark purple and gritty

Improper Fixation

- Occurs during specimen collection
- Tissue collected fresh must be placed in formalin or it will begin to autolyze (degenerate)
- Closed specimens that are resected (e.g., bowel, gallbladder) must be opened to allow fixation of inner surface
- Use of alcohol instead of formalin as fixative can distort and shrink tissue

Tissue Folds

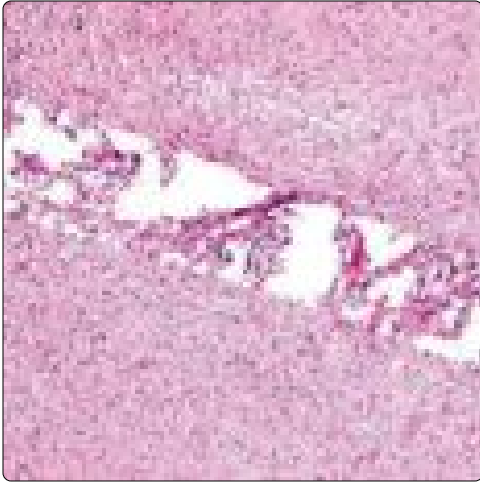
(Left) Tissue folds and curls are a very common artifact in histology. Although large folds can be prevented with careful technique, some small folds are unavoidable, particularly in tissues with hard components (e.g., bone) or edematous/myxoid change. **(Right)** Air may get trapped between the slide and coverslip during preparation. This artifact appears histologically as sharply demarcated glassy zones and often prevents clear visualization of the underlying tissue.



Air Bubbles



Tissue Nicks and Tears

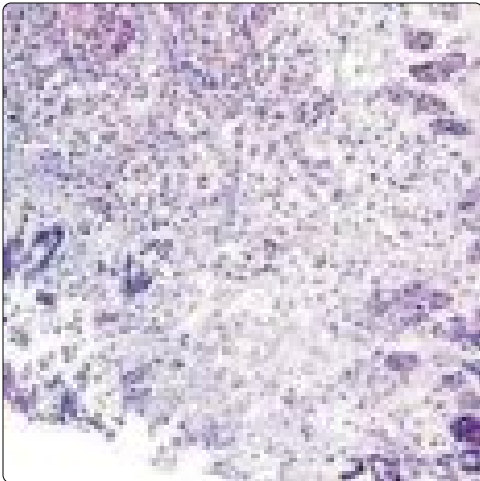


Tissue Fragmentation

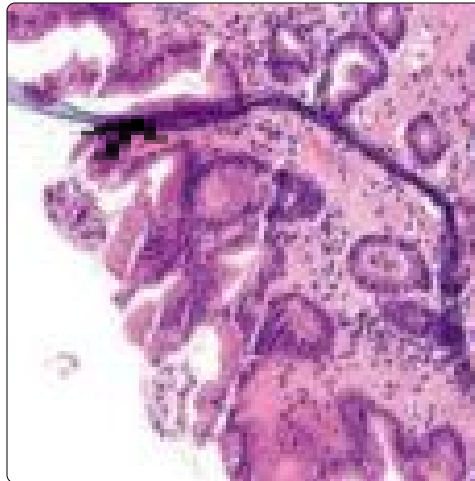


(Left) Linear breaks [\[E\]](#) or tears may be identified in tissue sections and are the result of blade damage during microtome sectioning of the paraffin block. Sometimes these tears have a repetitive, parallel appearance and have been described as "chatter." **(Right)** Fragmentation of tissue during sectioning and processing can limit or prevent adequate histologic evaluation, particularly in small samples.

Autolysis

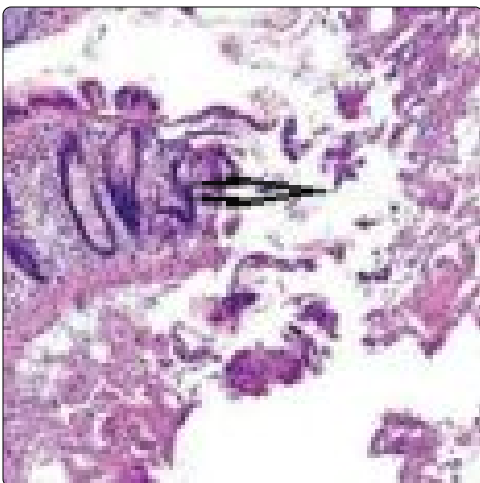


Contaminants

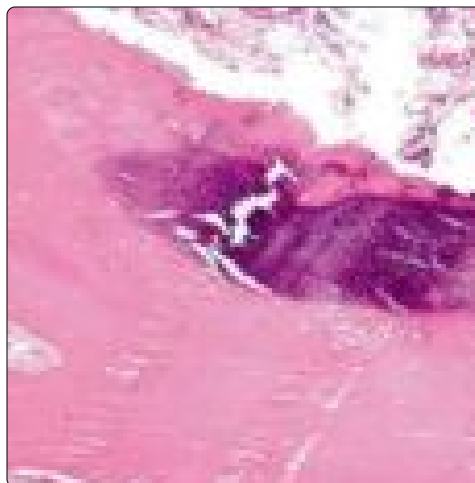


(Left) Fresh tissue removed from the body will begin degenerating (autolyzing) if not placed in fixative. Autolyzed tissue is characterized by loss of cytoplasmic and nuclear detail, cell shrinkage, and overall pale staining. These findings may be misinterpreted as necrosis. **(Right)** A variety of exogenous contaminants may be identified in histologic sections, including pollen, small fibers, and other material [\[E\]](#). Occasional bits of tissue from other specimens may also be seen ("floaters").

Cauterization



Underdecalcification



(Left) Tissue removed via electrocautery or similar means shows characteristic artifactual changes, including nuclear spindling and pseudostratification [\[E\]](#). Affected connective tissue [\[E\]](#) often appears denser and amorphous. **(Right)** Tissue containing grossly identifiable bone must be decalcified prior to processing so that satisfactory microtome sectioning can be performed. Underdecalcified bone is mineralized and appears gritty and purple [\[E\]](#). Other artifacts, such as folds and tears, are also common in this setting.

Histochemistry (Special Stains)

TERMINOLOGY

Fixation

- Alteration of proteins in tissue so that tissue is made resistant to further changes

Processing

- Process of preparing tissue so that microscopic sections can be made
 - Dehydration, clearing, infiltrating tissue with wax, embedding, cutting tissue sections for microscopic slides

Histochemical Staining

- Use of chemical reactions to highlight or differentiate various components on slide

NUCLEAR AND CYTOPLASMIC STAINING

Hematoxylin and Eosin

- Routine stain used most commonly in examination of tissues
 - Stains nuclei blue/purple
 - Stains cytoplasm pink

Feulgen Stain

- Stains nuclear DNA reddish purple

Methyl Green-Pyronin Y

- Differentiates between DNA and RNA
 - Stains nuclear DNA blue to green
 - Stains cytoplasmic RNA pink to rose
- Used to detect plasma cells and immunoblasts

Romanowsky

- Includes several different methods
 - Wright stain
 - Giemsa stain
 - Others
- Differentiates between hematopoietic cells based on staining properties

CARBOHYDRATE STAINS

Periodic Acid-Schiff

- Identifies polysaccharides, neutral mucins, basement membranes
 - Bright pink staining

Periodic Acid-Schiff With Diastase

- Can be used to demonstrate presence of glycogen
 - Bright pink staining
- Diastase used to digest glycogen; subsequent loss of staining

Mucicarmine

- Identifies acid mucins (typically in epithelia)
 - Stains bright pink
- Yellow counterstain typically added

Alcian Blue (pH 2.5)

- Identifies acid mucopolysaccharides, including sulfated and carboxylated acid mucopolysaccharides and sulfated and carboxylated sialomucins
 - Stains bright blue to blue/green

Alcian Blue (pH 1.0)

- Identifies sulfated mucins
 - Stains bright blue to blue/green

Alcian Blue With Hyaluronidase

- Differentiates epithelial and connective tissue mucins
 - Hyaluronic acid rich in connective tissues
- Acid mucopolysaccharides and sialomucins stain deep blue
- Substances containing hyaluronic acid and chondroitin sulfates A and C lose staining with digestion by hyaluronidase

Alcian Blue With Periodic Acid-Schiff

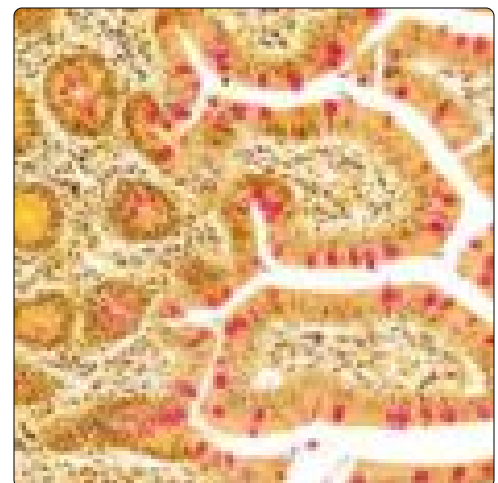
- Differentiates between neutral and acidic mucins
 - Acid mucosubstances stain blue
 - Neutral polysaccharides stain magenta

Hematoxylin and Eosin

(Left) The hematoxylin and eosin stain is the standard histochemical stain used in routine histopathologic evaluation. The hematoxylin stains nuclei dark blue/purple, and the eosin stains cytoplasm pink. (Right) The mucicarmine stain (sometimes referred to as mucin stain) highlights acid mucins with a bright pink stain , as seen in goblet cells of the small bowel epithelium. Other mucins and background tissues stain yellow.



Mucicarmine



Colloidal Iron

- Identifies carboxylated and sulfated mucopolysaccharides and glycoproteins
 - Stains deep blue

CONNECTIVE TISSUE STAINS

Masson Trichrome

- Differentiates between collagen/connective tissue and muscle
 - Collagen and connective tissue stains blue to blue-green
 - Muscle stains red
- Also stains nuclei black

Elastic Stain

- Demonstrates elastic fibers
 - Stains black or dark purple depending on method of staining
 - Found in connective tissues, skin, lung/pleura, arteries, veins, elastic cartilage

Pentachrome Stain (Movat)

- Demonstrates 5 tissue types
 - Collagen stains yellow
 - Muscle stains red
 - Elastic fibers stain black
 - Mucins stain blue
 - Fibrin stains intense red
- Generally utilized in connective tissues, heart, blood vessels

Reticulin Stain

- Demonstrates reticulin fibers
 - Stains black
 - Found in connective tissues of liver, bone marrow, lymphoreticular organs

MUSCLE STAINS

Phosphotungstic Acid-Hematoxylin

- Highlights muscle cross striations, fibrin, and glial fibers
 - Stains blue
- Also stains elastic fibers purple and connective tissue orange-pink to brown-red

FAT STAINS

Oil Red O

- Highlights neutral lipids
 - Stains red
- Used on frozen section tissue

Sudan Black B

- Highlights neutral lipids
 - Stains black
- Used on frozen section tissue

NERVE STAINS

Silver Nitrate

- Highlights nerve fibers and neurofibrils
 - Stains black

Mallory Phosphotungstic Acid-Hematoxylin

- Highlights glial fibers, neurons, and myelin
 - Stains glial fibers blue
 - Stains neurons salmon pink
 - Stains myelin blue

Cajal Stain

- Highlights astrocytes
 - Stains black

Weil Stain

- Highlights myelin
 - Stains blue/black

Luxol Fast Blue

- Highlights myelin
 - Stains blue
- Also stains neuropil pink

PIGMENTS, MINERALS, AND GRANULES

Prussian Blue

- Highlights ferric iron
 - Stains blue

Fontana-Masson

- Highlights melanin
 - Stains black

von Kossa

- Identifies calcium and calcified structures
 - Stains black

Rhodanine

- Identifies copper
 - Stains rust brown

MISCELLANEOUS STAINS

Toluidine Blue

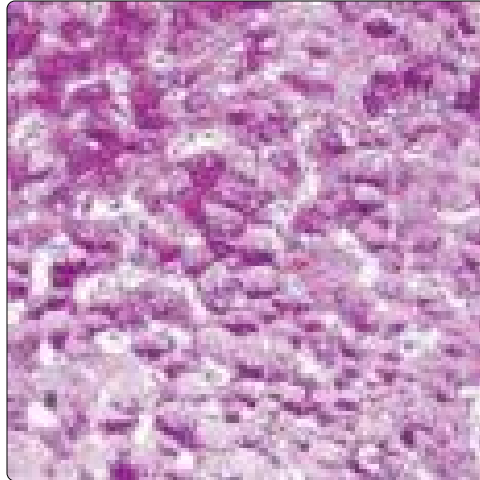
- Highlights mast cells
 - Stains deep violet

Jones Methenamine Silver

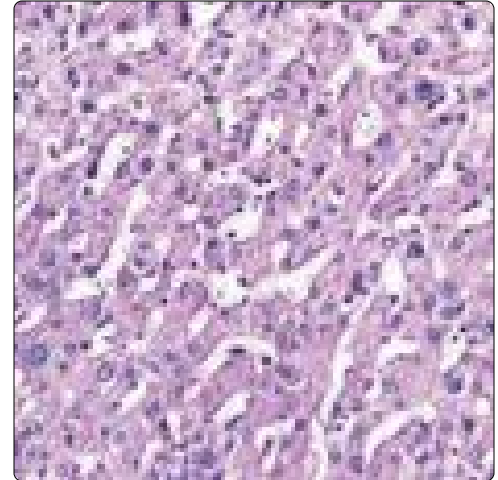
- Highlights glomerular basement membranes
 - Stains black
- Combination of methenamine silver and periodic acid-Schiff stains

Periodic Acid-Schiff

(Left) The periodic acid-Schiff stain is used to highlight a variety of substances from neutral mucins to polysaccharides (glycogen). This image shows intracytoplasmic glycogen (bright pink) in normal liver tissue. **(Right)** Pretreatment of tissue sections with diastase causes the digestion of glycogen, removing the bright pink periodic acid-Schiff staining from tissues that normally contain the polysaccharide.



Periodic Acid-Schiff With Diastase

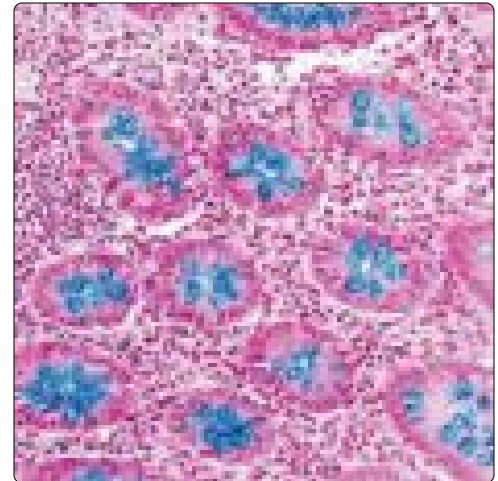


PAS on Mucinous Epithelia

(Left) The periodic acid-Schiff stain highlights neutral mucins, such as those present in the foveolar lining of the stomach (depicted). As with other tissues, the staining is bright pink. **(Right)** The alcian blue stain highlights acidic mucopolysaccharides and sulfated mucins, staining them blue/green, as seen in this section from normal intestine.

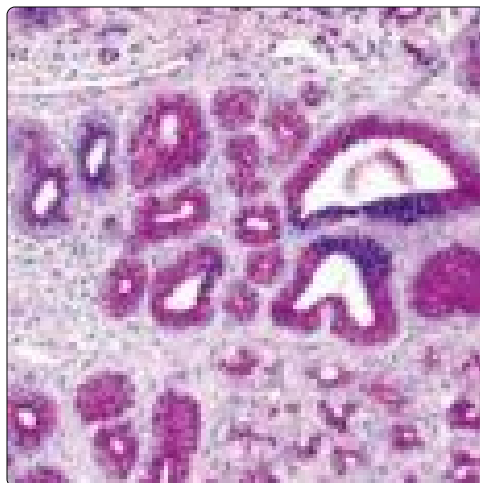


Alcian Blue

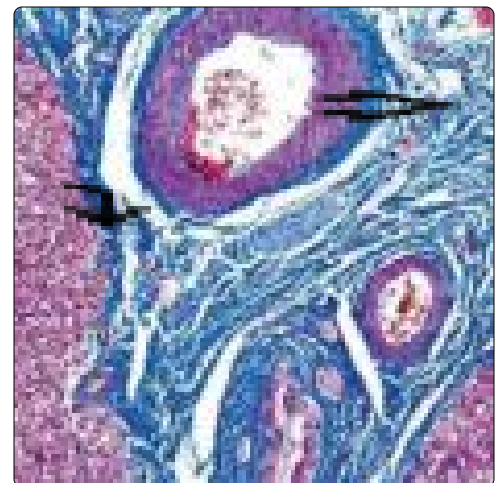


Alcian Blue and Periodic Acid-Schiff

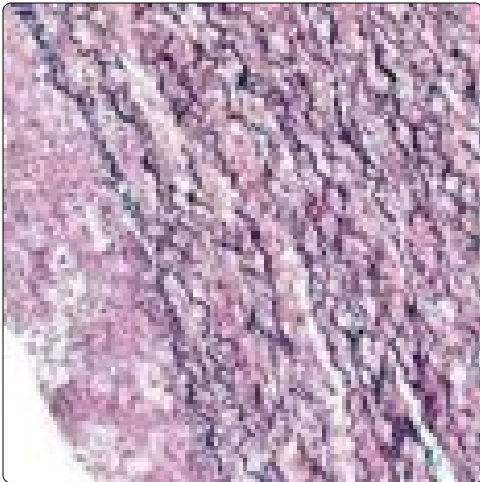
(Left) A combined stain containing alcian blue and periodic acid-Schiff can be used to distinguish acid mucins (blue/green) from neutral mucins (bright pink). **(Right)** The trichrome stain differentiates connective tissue and collagen from muscle tissue. Connective tissues stain blue to blue/green, while muscle (as well as parenchyma) stains red/pink.



Masson Trichrome



Elastic Stain

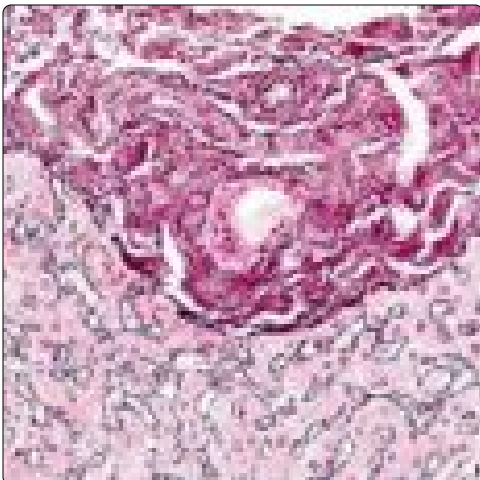


Elastic Stain in Pleura

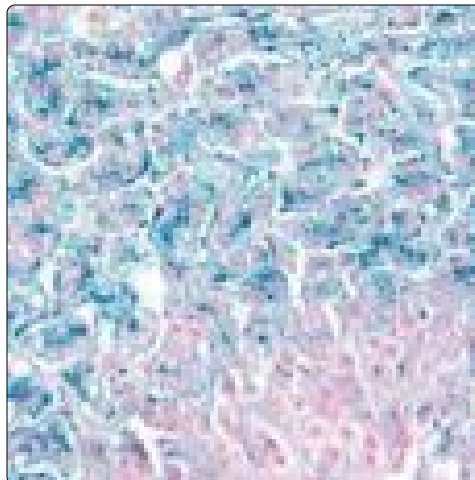


(Left) As the name implies, an elastic stain highlights elastin fibers, which appear black or dark purple, depending upon the exact histochemical technique. These fibers are common in connective tissue. **(Right)** An elastic stain can be used to identify the elastic lamina that delineates the visceral pleura from the periphery of the lung parenchyma. This structure may be transgressed in some lung tumors.

Reticulin

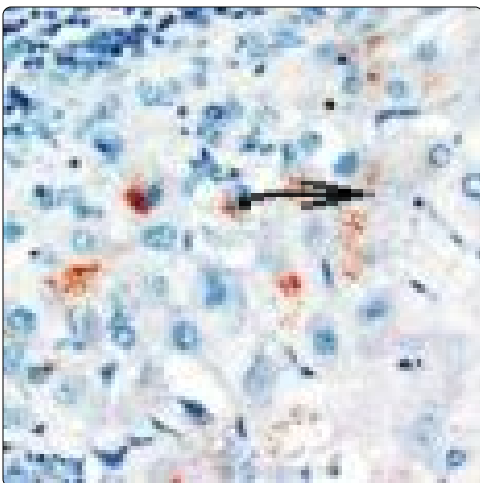


Prussian Blue

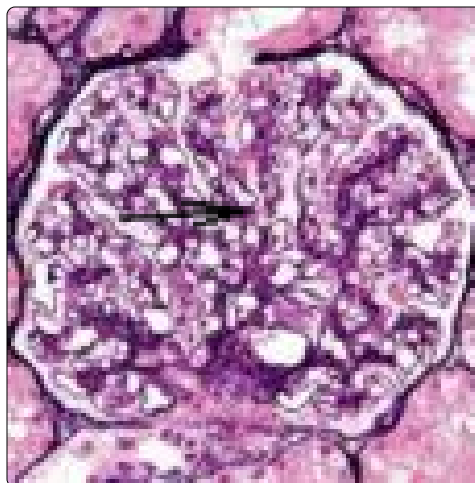


(Left) Reticulin stain highlights reticulin fibers in black in this image of normal liver. Reticulin fibers can also be demonstrated in bone marrow and lymphoreticular organs. **(Right)** The Prussian blue stain highlights ferric iron when present in tissue (liver in this image). Iron appears as small blue granules. Depending on the tissue, this finding may or may not be significant.

Rhodanine (Copper)



Jones Stain



(Left) The rhodanine stain (often referred to as copper stain) highlights cytoplasmic accumulation of copper. This pigment appears a distinctive rust brown color. **(Right)** The Jones methenamine silver stain highlights the basement membrane of the glomerulus. In this tissue, the basement membrane stains black.

Immunohistochemistry of Normal Tissues

TERMINOLOGY

Fixation

- Alteration of proteins in tissue so that tissue is made resistant to further changes

Processing

- Process of preparing tissue so that microscopic sections can be made

Immunohistochemical Staining

- Use of labeled antibody of known specificity to detect antigen in tissue
 - Antibody is labeled with chromogen so that positive reaction is identified by visualizing color
 - Usually brown or red
- Types of staining
 - Nuclear
 - Antigen expressed only in nucleus
 - Generally clean, crisp stain with little nonspecific background staining
 - Cytoplasmic
 - Antigen expressed predominantly in cytoplasm ± staining of cytoplasmic membrane
 - Membranous
 - Antigen expressed predominantly in cytoplasmic membrane
 - Mixed
 - Antigen expressed in both nucleus and cytoplasm
- Technical considerations
 - Most antibodies suitable for use in formalin-fixed, paraffin-embedded routinely processed tissue
 - Fixation may impair antigenicity due to protein cross linking
 - Epitope retrieval may be required to unmask antigens (antigen retrieval) using heat or enzymes

GENERAL CATEGORIES OF IMMUNOHISTOCHEMICAL ANTIBODIES

Epithelial Markers

- Broad spectrum keratins (e.g., cytokeratin AE1/AE3)
- CK7 (lung, breast, thyroid, salivary gland, other)
- CK20 (GI tract, other)
- EMA
- Calretinin

Muscle Markers

- MSA
- SMA
- Desmin
- Myogenin
- MYOD1

Myoepithelial Markers

- High molecular weight cytokeratins (HMWKs) (e.g., CK5/6)
- Calponin
- p63

Vascular Markers

- CD31
- CD34
- ERG
- D2-40 (lymphatic endothelium)

Neural Markers

- S100 protein
- SOX10
- CD57
- NFP
- GFAP

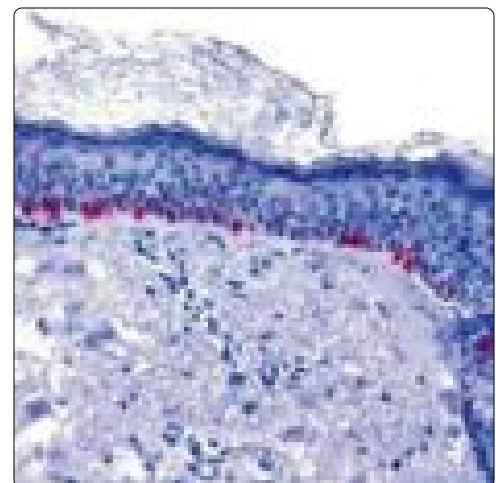
Melanocyte Markers

- S100 protein
- SOX10
- HMB-45
- Melan-A
- Tyrosinase

Brown Chromagen



Red Chromagen



(Left) The most common chromogen color used in immunohistochemistry is brown. The degree of expression (staining) depends heavily on the distribution and quantity of a particular antigen. In this image, a pancytokeratin immunostain highlights squamous epithelium. **(Right)** Red chromogen is most often utilized in immunohistochemical stains that are performed on specimens, such as skin, that can feature dark pigments (e.g., melanin).

Neuroendocrine Markers

- CD56
- Synaptophysin
- Chromogranin

Hematopoietic/Lymphoid Markers

- CD45 (leukocyte common antigen)
- CD20 (B cells)
- CD3, CD4, CD8 (T cells)
- CD68, CD163 (histiocytes)
- CD138 (plasma cells)

Hormones

- ACTH
- Parathyroid hormone
- Insulin
- HCG
- Glucagon
- Somatostatin
- Gastrin
- FSH
- LH

Enzymes

- Chymotrypsin
- Trypsin

Mucins

- MUC2
- MUC3 (intestinal)
- MUC6 (gastric)

ANTIGENIC PROPERTIES OF SELECTED ORGANS AND TISSUES

Adipose Tissue

- S100 protein
- Calretinin

Adrenal Cortex

- Calretinin
- Inhibin
- D2-40
- Synaptophysin

Adrenal Medulla

- Synaptophysin
- Chromogranin
- CD56

Bladder (Epithelium)

- p63
- HMWKs (e.g., 34 β E12)
- GATA3

Brain and Spinal Cord

- GFAP (astrocytes)
- Synaptophysin (neuronal cells)

Breast

- GATA3
- p63, SMA, calponin, CK5/6 (myoepithelial cells)

Cartilage

- S100 protein

Connective Tissues

- SMA
- Calponin
- Desmin
- CD34

Liver

- Hep-Par1 (hepatocytes)

Lung

- TTF-1 (pneumocytes)

Mesothelium

- Calretinin
- HMWKs (e.g., CK5/6)
- D2-40
- WT-1

Nerve

- S100 protein
- NFP
- EMA (perineurium)

Pancreas

- Synaptophysin, chromogranin, CD56, pax-8 (islets of Langerhans)

Pineal Gland

- Synaptophysin

Pituitary Gland

- Various hormonal antigens (e.g., GH, ACTH)
- GFAP (pituicytes)

Prostate

- PSA
- PSAP
- p63, HMWKs (basal cells)

Skeletal Muscle

- Myogenin
- Desmin
- MYOD1
- MSA

Smooth Muscle

- SMA
- Desmin
- MSA
- Calponin

Thyroid

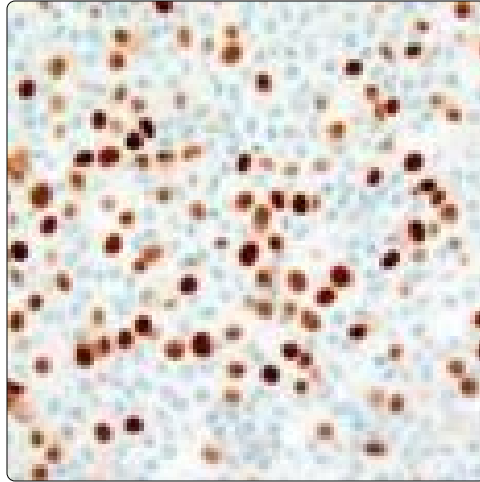
- TTF-1
- pax-8
- Thyroglobulin
- Calcitonin (C cells)

Uterus (Endometrium)

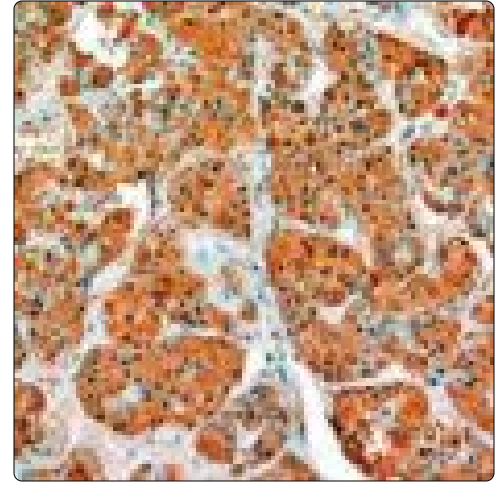
- pax-8

(Left) Immunostains with nuclear expression are often clean of background staining and easy to interpret. Examples include TTF-1, pax-8, myogenin, p63, and GATA3. Cytoplasmic staining with these stains is aberrant and should be ignored. **(Right)** Many immunostains identify antigens that are present only in the cytoplasm of cells. Examples of stains from this group include synaptophysin, chromogranin, desmin, and SMA.

Nuclear Stain

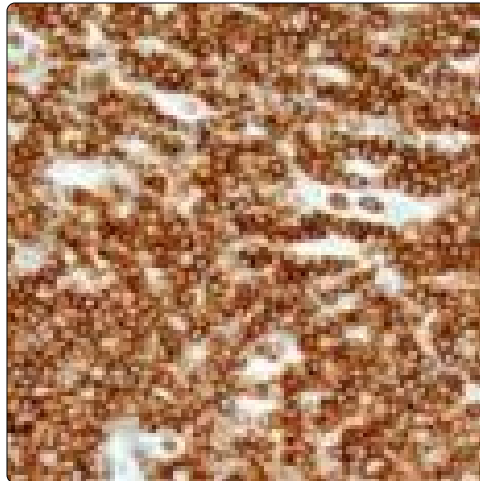


Cytoplasmic Stain

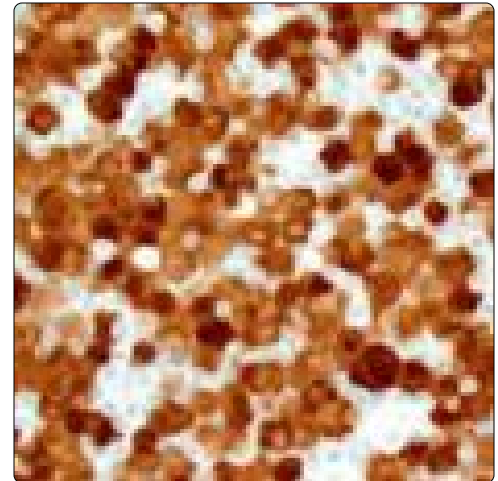


(Left) Some immunostains only highlight antigens on the surface or cell membrane of cells. CD markers are classic examples and include CD99, CD3, and CD20 among many others. **(Right)** A few immunostains can highlight antigens in both the nucleus and cytoplasm. Examples include calretinin (shown) in mesothelial cells and S100 protein in Schwann cells.

Membranous Stain

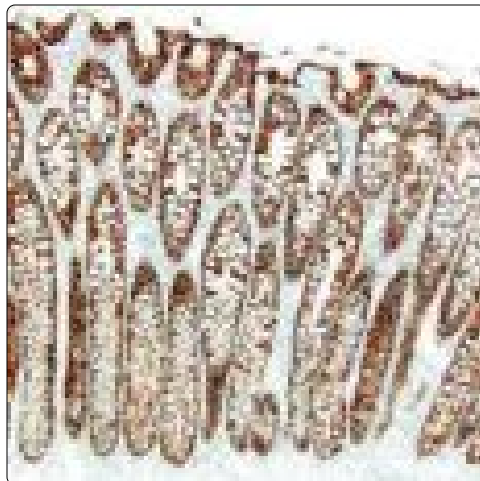


Nuclear and Cytoplasmic Stain

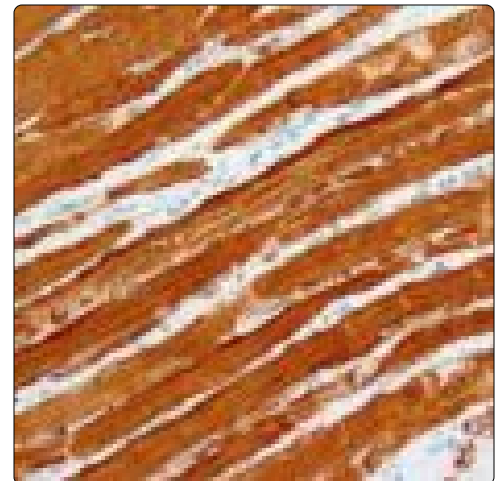


(Left) Epithelia from nearly all sites in the body is positive for broad-spectrum keratin, such as pancytokeratin and cytokeratin AE1/AE3. Staining is usually strong, as depicted in this section of a normal colon. **(Right)** Muscle fibers show strong expression of desmin and MSA. Smooth muscle fibers also show expression of SMA and calponin. Skeletal muscle shows nuclear expression of myogenin and MYOD1.

Epithelium



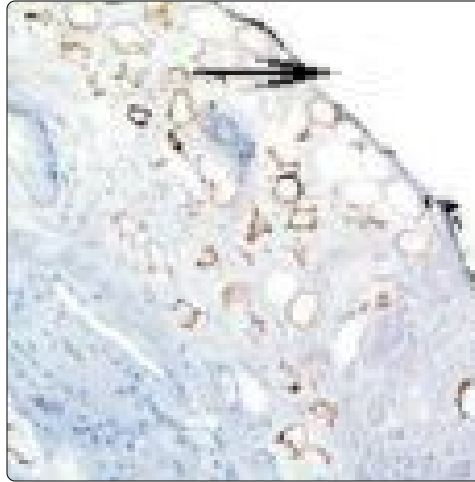
Muscle



Nerve

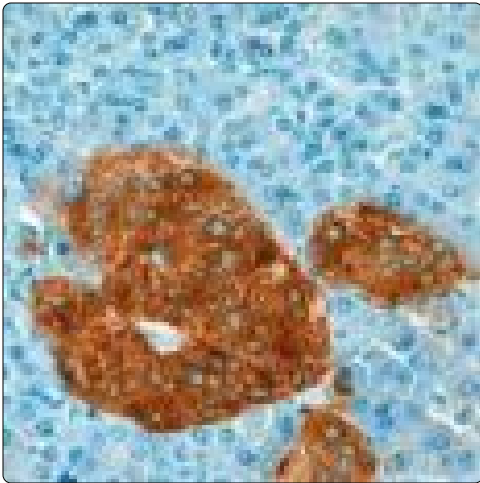


Adipose Tissue

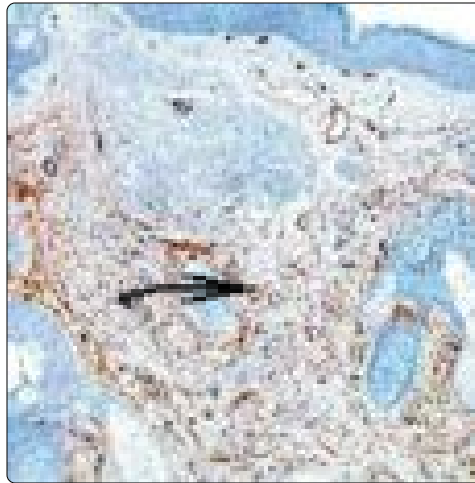


(Left) S100 protein is expressed in a variety of normal tissue, including nerve fibers. **(Right)** S100 protein expression is also typical of normal mature adipose tissue. The staining appears membranous rather than cytoplasmic due to the fact that much of the individual cells are composed of lipid, which pushes the actual cytoplasm to the periphery.

Hormone Stains



Connective Tissue

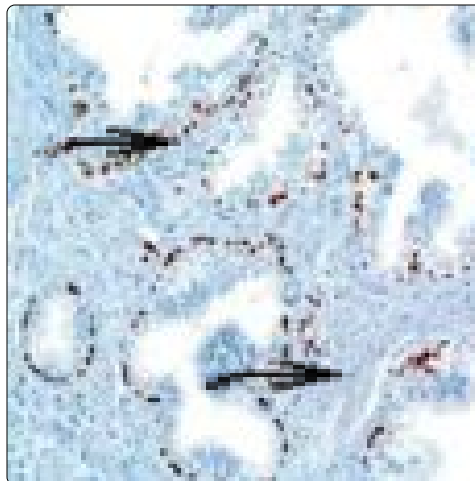


(Left) Hormone immunostains include a wide variety of antigens, including ACTH, FSH, LH, GH, and somatostatin. This image shows insulin highlighting the cells of an islet of Langerhans in a normal pancreas. **(Right)** Connective tissue features a variety of fibroblastic cells, some of which express SMA and others that express CD34. The latter is commonly expressed in dermal fibroblasts. CD34 is also expressed in endothelial cells.

Myoepithelial Markers (CK5/6)



Myoepithelial Markers (p63)



(Left) Markers such as CK5/6 (shown), calponin, and SMA are useful for highlighting the myoepithelial or basal cells surrounding normal ducts in the breast and normal acini in the prostate. Loss of these cells in these particular tissues is often seen in carcinoma. **(Right)** Myoepithelial or basal markers may also show nuclear expression. In this example, p63 highlights basal cells in normal prostatic acinar tissue, whereas the luminal cells are negative.

TERMINOLOGY

Abbreviations

- Immunofluorescence (IF)

Definitions

- IF uses fluorescently labeled antibodies to detect antigens of interest with fluorescence microscope
 - Direct IF
 - Primary antibody directly conjugated with fluorescent dye for detection using fluorescence microscope
 - Typically fluorescein isothiocyanate (FITC)
 - Indirect IF
 - Primary antibody not conjugated with fluorescent dye
 - Requires adding secondary antibody conjugated with fluorescent dye for detection using fluorescence microscope

METHODOLOGY

Transport Media

- Michel or Zeus
 - These media must be kept refrigerated until use
 - Once placed in media, tissue can be kept at room temperature for up to 5 days prior to processing
 - These are transport media and not fixatives

Fresh Tissue

- Embed and freeze tissue in optimal cutting temperature medium
- Generally more sensitive than IF or immunohistochemistry that is performed using formalin-fixed tissue sections

Formalin-Fixed, Paraffin-Embedded Tissue

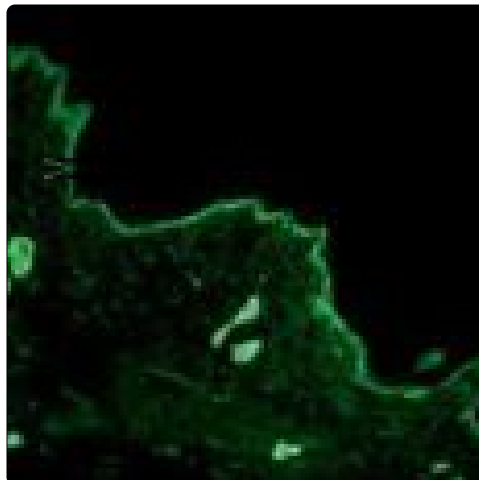
- Important salvage technique when fresh frozen tissue is not available for IF
 - In general, formalin fixation reduces sensitivity of antigen detection by IF compared with fresh frozen tissue
- Requires digestion of tissue for antigen retrieval
- Digestion enzymes include

- Pronase
- Proteinase XXIV
- Trypsin
- Pepsin
- Protease
- Proteinase K

Antibodies

- Primary antibodies often directly conjugated with FITC
- Antibodies used in clinical evaluation of nonneoplastic biopsies (particularly kidney) include
 - IgG
 - IgG₁
 - IgG₂
 - IgG₃
 - IgG₄
 - IgG subclass testing **not** needed on routine basis
 - IgA
 - IgM
 - Complement component C3
 - Normal synthesis in proximal tubular epithelial cell
 - Complement component C1q
 - Fibrinogen or fibrin
 - κ -light chain
 - λ -light chain
 - Albumin
 - Complement component C4d
 - Tested in transplant allografts, typically kidney or heart
 - Strong and diffuse capillary endothelial staining is suggestive of antibody-mediated rejection
 - Normal granular staining of glomerular mesangial cells
 - α -3 and α -5 chains of collagen IV
- Slide storage
 - Ambient light causes photobleaching
 - Store slides in dark

Skin Biopsy



(Left) The α -1 chain of collagen IV shows normal, strong, continuous staining of the epidermal basement membrane and capillary basement membranes in this skin biopsy. (Right) The α -5 chain of collagen IV shows normal, strong, and continuous linear staining of the glomerular and distal tubular basement membrane and Bowman capsule.

Kidney Biopsy



ASSESSMENT OF CLINICAL SPECIMENS

Kidney

- Glomerulus
 - Normal glomeruli show no immunoglobulin or complement component staining
- Mesangium
 - Granular C4d staining **not** considered pathologic
 - C4d IF tested on renal allograft specimens
- Podocyte (visceral epithelial cell)
 - Normal podocytes show no immunoglobulin or complement component staining
 - Protein resorption droplets can be prominent in some proteinuric states
 - Globular appearance and **not** to be mistaken for immune complexes, which are granular
- Glomerular basement membrane
 - Normal linear staining of α -3 and α -5 chains of collagen IV
 - **Absence** of staining for α -3 and α -5 chains of collagen IV
 - Associated with X-linked inheritance pattern of Alport syndrome in **males**
 - **Segmental** staining for α -3 and α -5 chains of collagen IV
 - Associated with X-linked inheritance pattern of Alport syndrome in **females**
- Bowman capsule
 - Normal expression of α -5 and α -6 chains of collagen IV
- Tubule
 - C3 synthesized in proximal tubular epithelial cell cytoplasm
 - κ - and λ -light chains and albumin highlight protein reabsorption droplets in proximal tubular epithelial cell cytoplasm
 - May be pronounced in proteinuric states
 - Monoclonal light chain staining observed in light chain proximal tubulopathy
- Tubular basement membrane
 - Normal linear staining of α -1 chain of collagen IV
 - Normal linear staining of α -3 and α -5 chains of collagen IV in distal tubules
- Arterioles
 - Normal granular C3 staining of endothelial cells due to cell synthesis in cytoplasm
- Interstitial inflammatory cells
 - κ - &/or λ -light chain cytoplasmic staining can demonstrate monoclonal or polyclonal nature

Skin

- Normal skin shows no significant immunoglobulin or complement component staining
 - Abnormal staining in various vesiculobullous skin diseases
- Epidermal basement membrane
 - Normal linear staining for α -1 and α -5 chains of collagen IV
 - **Absence** of staining for α -5 chain of collagen IV
 - Associated with X-linked inheritance pattern of Alport syndrome in **males**
 - **Segmental** staining for α -5 chain of collagen IV

- Associated with X-linked inheritance pattern of Alport syndrome in **females**

- α -3 chain of collagen IV **not** expressed
 - Autosomal inheritance pattern of Alport syndrome (α -3 or α -4 chain of collagen IV mutations) **cannot** be excluded using skin specimen

Oral Mucosa

- Normal oral mucosa shows no significant immunoglobulin or complement component staining
 - Abnormal staining in various vesiculobullous diseases

Lung

- Normal basement membranes show no significant immunoglobulin or complement component staining
 - Strong linear staining for IgG and κ - and λ -light chains observed in pulmonary involvement by antglomerular basement membrane disease (Goodpasture syndrome)

Transplant Allograft

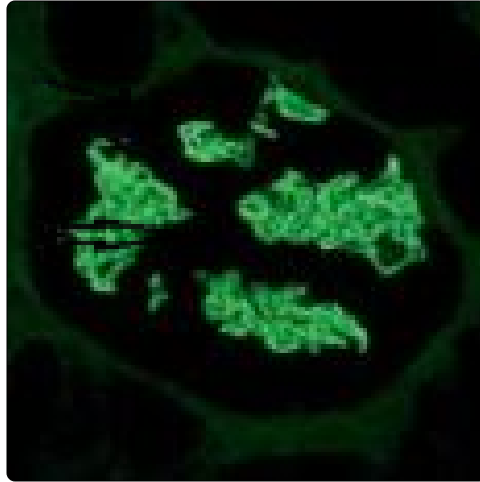
- C4d capillary staining is suggestive of antibody-mediated rejection
 - Typically tested in heart and renal allografts
 - Interstitial C4d staining is nonspecific finding in heart allografts
 - Tubular basement membrane C4d staining of atrophic tubules is nonspecific and can mimic peritubular capillaries

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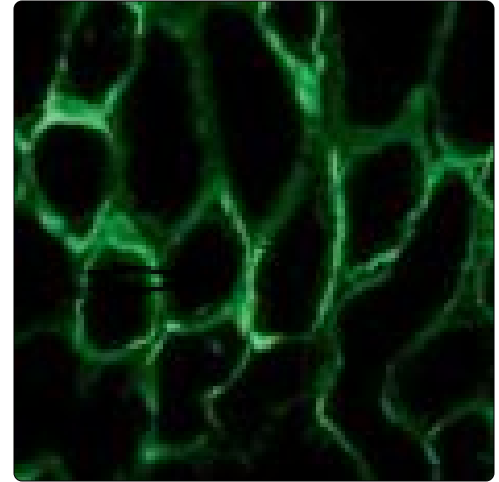
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Normal Glomerulus

(Left) This glomerulus shows normal linear staining of the glomerular basement membranes  for the α -3 chain of collagen IV. Bowman capsule  consists of only α -5 and α -6 chains of collagen IV and shows no staining. The adjacent proximal tubular basement membranes  show no staining for the α -3 chain of collagen IV. (Right) The distal tubules in this medullary ray demonstrate strong linear staining of the tubular basement membranes  for α -3 chain of collagen IV.

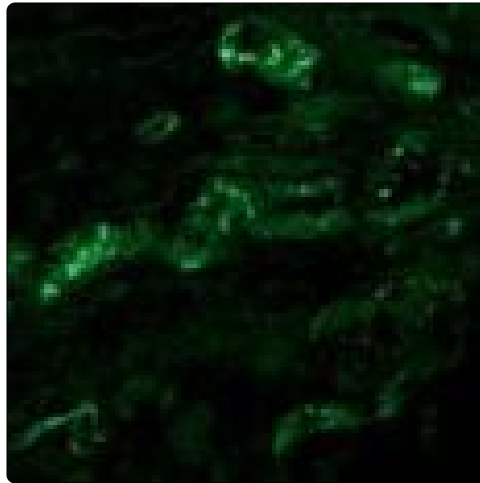


Normal Distal Tubules

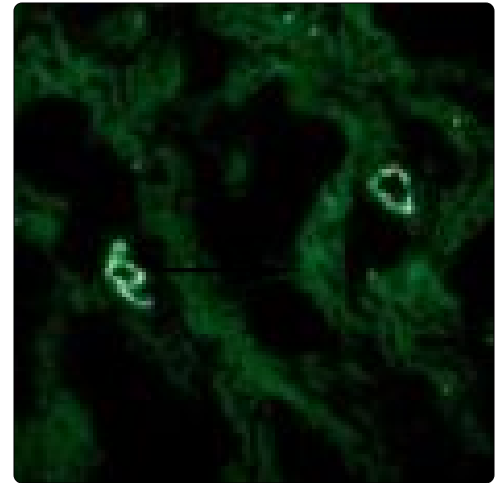


Normal C3 Synthesis in Tubular Epithelial Cells

(Left) Immunofluorescence (IF) for C3 reveals this protein within the tubular epithelial cell cytoplasm , which is due to normal synthesis. Linear C3 staining of tubular basement membranes (not shown) may be present and is of uncertain significance. (Right) C3 is synthesized in endothelial cells  of several arterioles, and this is a useful internal positive control. Granular C3 staining of the vessel walls along with C1q and other immunoglobulins can be observed in lupus nephritis and would be consistent with immune complexes.

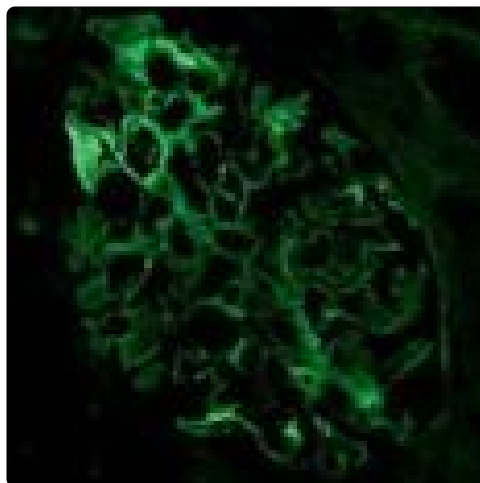


C3 in Endothelial Cells

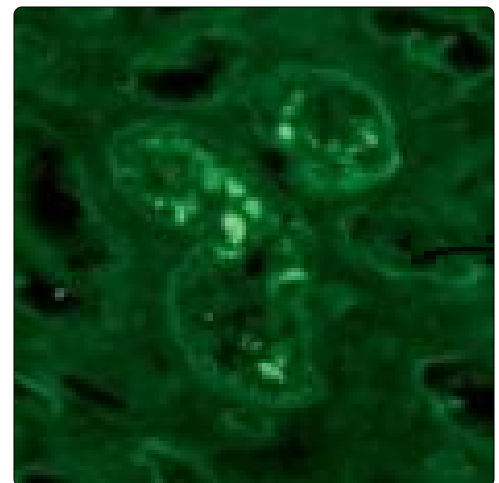


C4d in Glomerulus

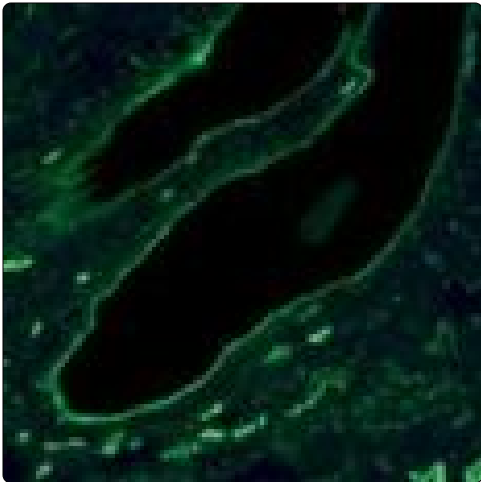
(Left) Granular mesangial  C4d staining is a useful internal control and not considered a pathologic finding. The presence of C4d deposition of peritubular capillaries (not shown) in a kidney transplant biopsy would be suggestive of antibody-mediated rejection. (Right) IF reveals normal reabsorption for albumin within the proximal tubular epithelial cell cytoplasm . Tubular resorption droplets may be increased in proteinuric states. κ - and λ -light chains often stain these droplets.



Albumin in Tubular Epithelial Cells



Normal Skin

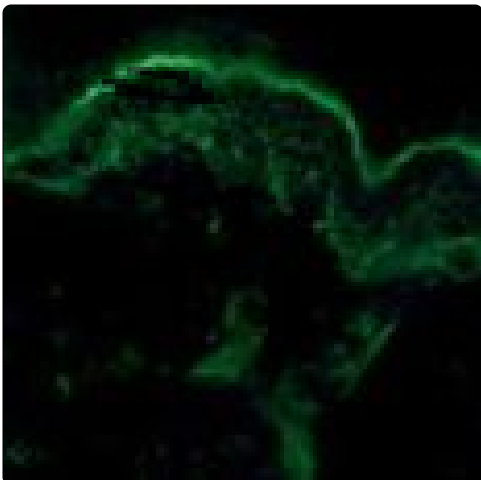


Normal Skin

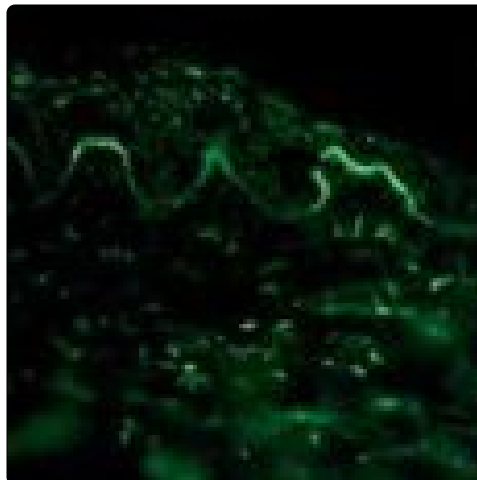


(Left) The epidermal and capillary basement membranes from this normal skin biopsy demonstrate strong linear staining for the α -1 chain of collagen IV. (Right) The epidermal basement membrane from this normal skin biopsy reveals strong linear staining for the α -5 chain of collagen IV. Significant background staining of the epidermis (not shown) may be present in some specimens and should not be mistaken for the epidermal basement membrane.

X-Linked Alport Syndrome Skin Biopsy in Male Patient

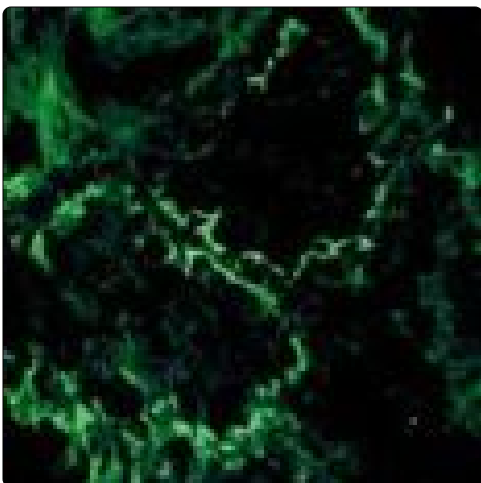


X-Linked Alport Syndrome Skin Biopsy in Female Patient

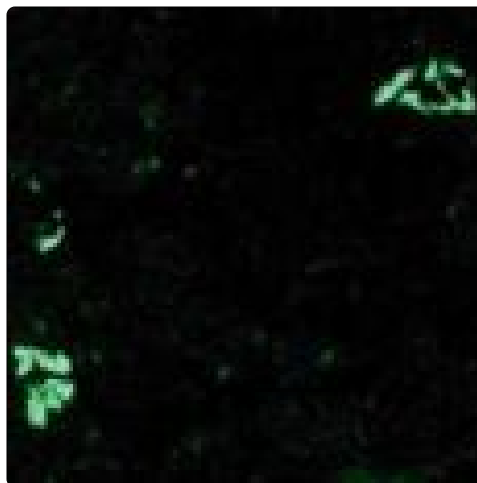


(Left) The α -5 chain of collagen IV shows no staining of the epidermal basement membrane, which is diagnostic of the X-linked inheritance pattern of Alport syndrome (hereditary nephritis). There is a blush of staining in the overlying cornified layer (stratum corneum) that should not be mistaken as a positive result. (Right) Segmental staining of the epidermal basement membrane for the α -5 chain of collagen IV is diagnostic of Alport syndrome in a female patient.

Pulmonary Basement Membranes



C4d in Heart Allograft



(Left) IgG linear staining of the basement membranes is observed in a patient with pulmonary hemorrhage (Goodpasture syndrome) and antglomerular basement membrane nephritis. The epithelial and capillary basement membranes are normally irregularly fused as shown in these alveoli, but a normal lung would not reveal any IgG deposition. (Right) C4d stains several necrotic myocytes, but the capillaries are negative, which argues against antibody-mediated rejection.

Electron Microscopy

TERMINOLOGY

Abbreviations

- Electron microscopy (EM)

Definitions

- Transmission EM
 - Resolution of 0.2 nm compared to 200 nm for light microscope
 - Clinically useful in certain settings
- Scanning EM
 - Electron beam scans surface of (not through) examined specimen
 - Provides 3D image of surfaces and topography

SPECIMEN HANDLING

Tissue Handling

- Place tissue in fixative immediately

Fixatives

- Karnovsky
 - 2.0% formaldehyde and 2.5% glutaraldehyde
 - Fix tissue for at least 2 hours
- Glutaraldehyde
 - Cross links and preserves proteins
- 10% buffered formalin
 - Tissue may not have been initially submitted for EM after gross examination
 - Tissue fixed in formalin may be subsequently placed in glutaraldehyde or Karnovsky media for EM processing
 - Processing formalin-fixed, paraffin-embedded tissue results in poor preservation of fine cellular structures

Post Fixation

- Osmium tetroxide
 - Heavy metal that reacts with phospholipids to increase electron density for imaging

Microwave Techniques

- Substantially reduces processing time

SPECIMEN PROCESSING

Embedding Media

- Methacrylate/acrylate
- Epoxy resin/EPON

Sectioning Blocks and Staining Sections

- 1- μ m thick (or semithin) sections are generated from plastic-embedded tissue
 - Light microscopic examination with following special stains
 - Toluidine blue
 - Methylene blue
 - Thin sections (~ 100 nm-thickness) are placed on special grids for definitive EM evaluation
 - Thicker sections (> 100 nm) may be visualized with more recently developed EMs
 - Grids stained with uranyl acetate and lead citrate

Immunolabeling

- Primary or secondary antibodies labeled with particles (e.g., gold) of uniform size may detect antigens of interest

ELECTRON MICROSCOPY ASSESSMENT OF CLINICAL SAMPLES

Kidney

- Glomerular basement membrane
 - Consists of lamina rara externa (region between podocyte and lamina densa), lamina densa, and lamina rara interna (region between lamina densa and endothelial cell)
 - Range for 9 years or older: 264-460 nm
 - < 264 nm considered thin basement membrane disease by World Health Organization
 - Lamellation or basketweave appearance characteristic of Alport syndrome (hereditary nephritis)
- Podocyte (visceral epithelial cell)
 - Located on outer aspect of glomerular basement membranes
 - Intact foot processes

Electron Microscopy Preparation

(Left) The tissue  measures 1 mm in width at the tip of the epoxy resin block  with the specimen label. The block is subsequently placed in the microtome holder  for sectioning. (Courtesy J. Brainer, MS.) (Right) The EPON block  is in the microtome. The knife trough  collects the tissue sections as they come off the diamond knife . These sections are placed on grids for electron microscopy evaluation. (Courtesy J. Brainer, MS.)



Electron Microscopy Tissue Sectioning



- Effacement of foot processes correlates with proteinuria
- Absence of inclusions or crystals
 - Crystals present in light chain Fanconi syndrome
 - Myelin bodies present in Fabry disease
- Endothelial cell (fenestrated)
 - Swelling of cytoplasm (endotheliosis) in preeclampsia
 - Tubuloreticular inclusions
 - Observed in states of high interferon, including systemic lupus erythematosus and human immunodeficiency virus infection
- Mesangial matrix
 - Increased in diabetic nephropathy
- Mesangial cell
- Proximal tubular epithelial cell
 - Brush border or microvilli at apical surface
 - Loss or blunted in acute tubular injury
 - Basolateral interdigitations
 - Loss in acute tubular injury
- Peritubular capillary
 - Multilayering of basement membrane in renal allografts with chronic antibody-mediated rejection

Muscle

- Syncytial multinucleated myofibers develop through fusion of mononucleated myoblasts
- Cell membrane (sarcolemma) surrounded by continuous basement membrane
- Neuromuscular junction
 - Ultrastructural changes with myasthenia gravis and other myasthenic syndromes
- T tubules
 - Tubular invaginations of sarcolemma allowing uniform spread of depolarization throughout myofibers
 - Sandwiched between endoplasmic reticulum (sarcoplasmic reticulum) as "triad"
 - Triad essential for electromechanical coupling
 - Uniform calcium release in response to depolarization triggering contraction
- Sarcomeres
 - Contractile filaments are arranged into myofibrils
 - Serially arranged sarcomeres; each ~ 1 µm in diameter and 2-3 µm in length
 - Z-discs anchor thin actin filaments
 - M line marks anchorage of thick myosin filaments
 - Disorganized or disrupted arrangement of sarcomere elements, e.g., intensive care unit myopathy or nemaline myopathy
- Organelles
 - Peripheral subsarcolemmal nuclei
 - Mitochondria, glycogen, and other organelles between myofibrils
 - Increased numbers of and morphologically abnormal mitochondria with mitochondrial disease
 - Abnormal accumulations of glycogen or lipid with some storage diseases
- Blood vessels between myofibers may show endothelial cell inclusions (tubuloreticular inclusions) with autoimmune conditions like dermatomyositis

Nerve

- Axons arranged into fascicles
- Fascicles surrounded by perineurial cells (long thin processes, continuous basement membrane, and prominent pinocytotic vesicles)
- Myelinated axons
 - 1 Schwann cell myelinates single internode of single axon
 - Myelin thickness proportional to axonal diameter
- Unmyelinated axons
 - 1 Schwann cell surrounds segments of several axons
- Axonal density decreased with axonal disease
- Thinned myelin sheath with demyelinating disease or axonal regeneration

Sinonasal Mucosa

- Cilia
 - Normal 9+2 structure of microtubules
 - Absence of dynein arms in Kartagener syndrome or primary ciliary dyskinesia

Heart

- Adriamycin (doxorubicin) toxicity
 - Swelling of myocyte sarcoplasmic reticulum
 - Disruption or loss of myofibrils
 - Possible focal involvement

Small Intestinal Mucosa

- Intact microvilli
 - Microvillus inclusion disease
 - Absence of apical microvilli
 - Accumulation of secretory granules
 - Tufting enteropathy
 - Increased length and number of desmosomes between small intestinal epithelial cells

Neoplasm

- EM may reveal ultrastructural clues to cell type, such as
 - Premelanosomes and melanosomes in melanocytes
 - Weibel-Palade bodies in endothelial cells
 - Birbeck granules in Langerhans cells
 - Neuroendocrine granules in neuroendocrine tumors
 - Fibrillary background in neuroblastoma
 - Muscle filaments and Z-band material in rhabdomyosarcoma
 - Immunohistochemistry has largely supplanted EM for cellular characterization

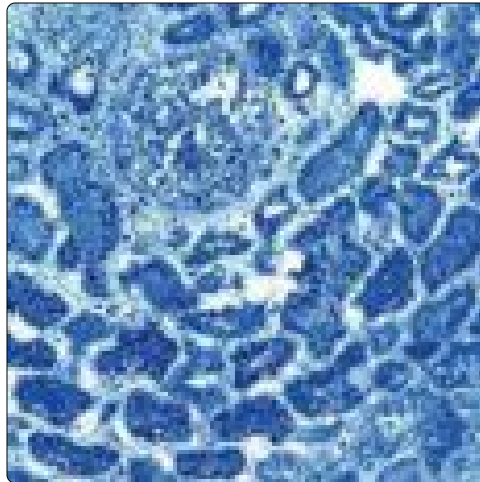
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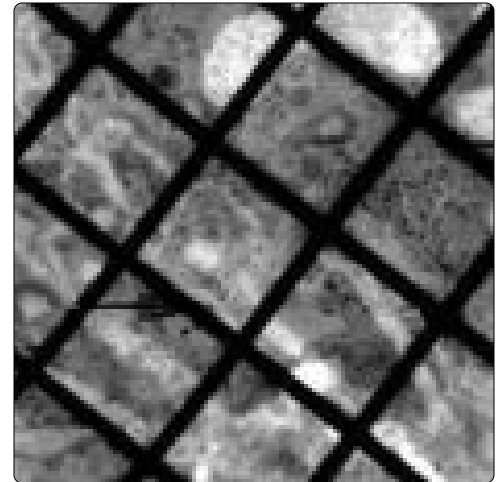
Electron Microscopy

Toluidine Blue

(Left) Semithin or thick (1- μ m) sections are evaluated to assess if additional thin sections should be obtained for ultrastructural evaluation. Significant histologic information can be obtained by light microscopic examination. (Right) This low-magnification photograph (150x) demonstrates the grid upon which the thin tissue section sits. Occasionally, the gridlines may obstruct a specific structure of interest, but several tissue sections are often available on the grid for ultrastructural evaluation.

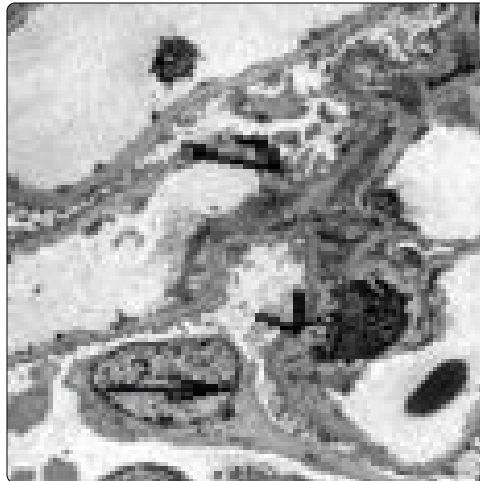


Electron Microscopy Grid

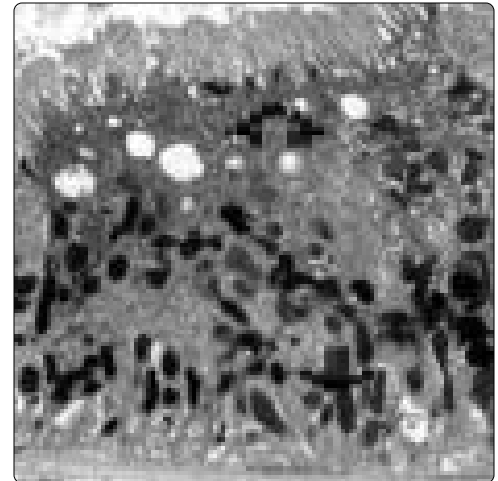


Normal Glomerulus

(Left) The glomerular basement membranes of several capillaries are of normal thickness and are covered by podocytes with intact foot processes. A mesangial cell nucleus is present. (Right) This normal proximal tubular epithelial cell contains many mitochondria. There is an intact brush border with microvilli at the apical surface with basolateral interdigitations. In acute tubular injury, the microvilli are blunted, and there is loss of the basolateral interdigitations.

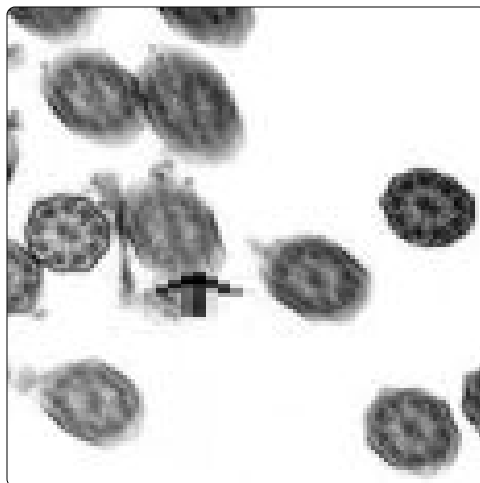


Normal Proximal Tubular Epithelial Cells

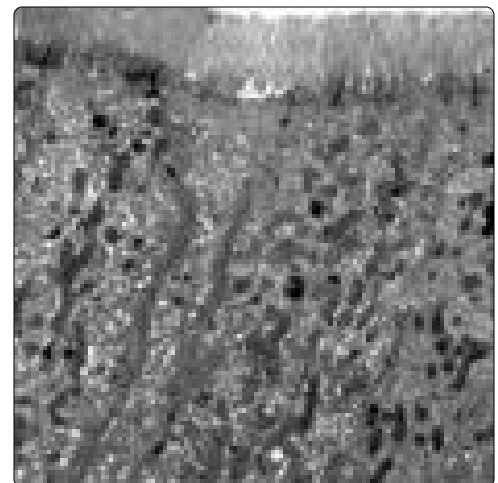


Normal Cilia

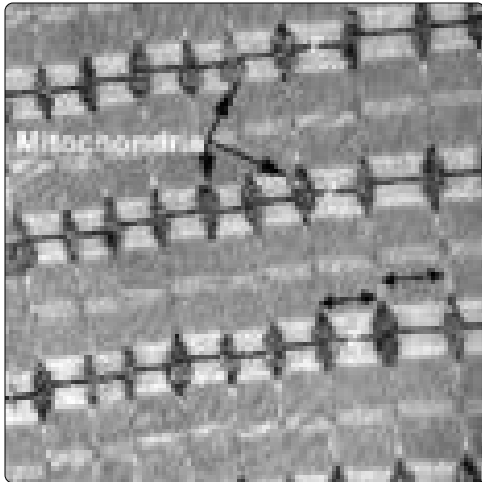
(Left) The cilia in this sinonasal biopsy, captured in the correct plane, demonstrate the normal 9+2 configuration of microtubules with dynein arms, which excludes the diagnosis of Kartagener syndrome. (Courtesy E. Sengupta, MD.) (Right) Numerous villi are present at the apical surface of this normal small intestinal epithelial cell from a pediatric patient with diarrhea, which excludes the diagnosis of microvillus inclusion disease. (Courtesy E. Sengupta, MD.)



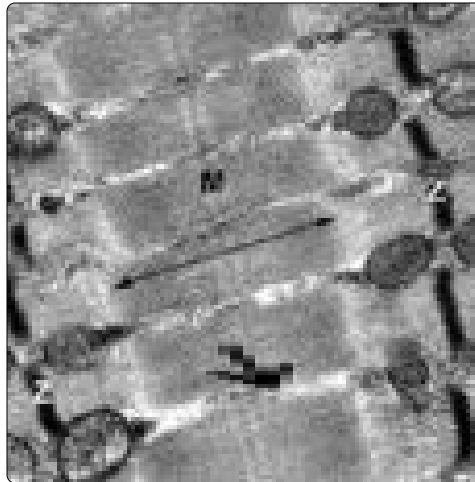
Normal Small Intestinal Epithelial Cell



Normal Skeletal Muscle



Normal Skeletal Muscle: Sarcomere



(Left) The characteristic arrangement of contractile filaments and organelles appear as cross striations on light microscopy. Myofibrils consist of sarcomeres, defined as the area between adjacent Z-discs (Z). Double arrows mark the diameter of myofibrils. Mitochondria are found between myofibrils. (Right) A sarcomere is bordered by 2 Z-discs (Z), which also are the I band midline. The M line (M) anchors myosin filaments at the sarcomere midline. The A band (double arrow) is separated by I bands.

Normal Skeletal Muscle: T Tubule

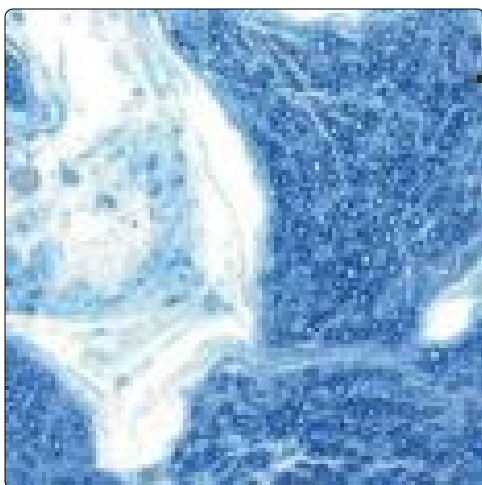


Malignant Neoplasm

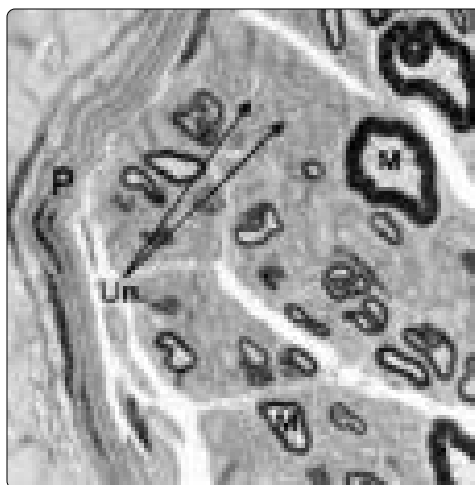


(Left) The T tubule is the membranous structure running perpendicular to the orientation of the sarcomere. A Z-disc (Z) and a mitochondrion (Mi) are present. (Right) Skeletal muscle differentiation is established based on the distinct sarcomeres that are identified within the cell cytoplasm of this neoplasm. This finding in the setting of a malignant peripheral nerve sheath tumor indicates a triton tumor, which often has an aggressive clinical course.

Normal Nerve



Normal

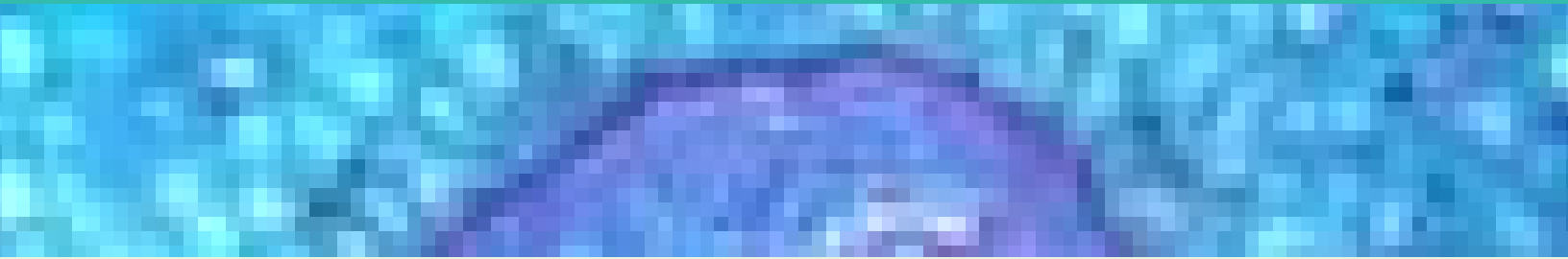


(Left) The embedding process nicely preserves the lipid-rich myelin layer around axons in this normal nerve, which is better visualized with the toluidine blue stain than on a paraffin section. (Right) Cross section of peripheral nerve shows perineurium marking the outer border of a fascicle (P). Myelinated axons of varying diameter are seen (M). The thickness of the myelin sheaths correlates with the diameters of the axons. Thin, unmyelinated axons are found in the background (Un.).

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SECTION 2

Integument



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Epidermis (Including Keratinocytes and Melanocytes)

EPIDERMIS

Macroscopic Anatomy

- Thin, most superficial layer of skin
- Appears white-gray or brown grossly
 - Exact color depends on amount of intraepidermal melanin pigment (ethnic variations)
- Exact thickness varies based on anatomic location
 - Thinnest on eyelids and thickest on palms/soles

Microscopic Anatomy

- Composed of stratified squamous epithelium
- Multiple microanatomic levels (superficial to deep)
 - Stratum corneum
 - Outermost (surface) layer
 - Composed of multiple layers of anucleate keratinocytes that often intertwine (basketweave appearance)
 - Stratum granulosum
 - 1-3 layers of flattened cells oriented parallel to surface
 - Basophilic keratohyalin granules
 - Stratum spinosum
 - 5-10 layers of keratinocytes with desmosomes
 - Stratum basalis
 - Innermost (basal) layer
 - Proliferating keratinocytes and melanocytes, attached to basement membrane by desmosomes
 - Basement membrane separates epidermis from dermis
 - Composed of collagens and laminins
- Types of cells
 - Keratinocytes
 - Main cell of stratified squamous epithelium
 - Round to oval nuclei
 - Cell-to-cell bridges (desmosomes)
 - IHC: CK5/6(+), p63(+)
 - Melanocytes
 - Melanin-producing cells with cleared-out cytoplasm
 - Hyperchromatic round to oval nuclei
 - IHC: S100(+), HMB-45(+), Melan-A(+), tyrosinase (+), MITF(+)

- Langerhans cells
 - Intraepidermal dendritic cells with reniform nuclei
 - Usually present in stratum spinosum
 - IHC: CD1a(+), S100(+), langerin (+)
- Merkel cells
 - Indistinct ovoid cells in stratum basalis
 - Associated with underlying sensory nerve endings
 - IHC: CK20(+) (required to visualize histologically)
- Blood vessels, nerves not normally found in epidermis
- Rete ridges
 - Epidermal areas extending downward between dermal papillae
- Acrosyringium
 - Intraepidermal ductal portion of eccrine gland
 - Spirals through epidermis in corkscrew pattern
- Follicular ostia
 - Intraepidermal hair follicle opening with lamellated keratin
- Melanin pigment
 - Accumulated in basilar keratinocytes
 - Increased accumulation in darker skinned individuals
- Acral skin
 - Thickened stratum corneum; prominent granular layer
 - Overall thicker epidermis than seen at other locations

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Increased melanocytes and Langerhans cells in children
 - Sun-damaged skin in adults can have increased numbers of junctional melanocytes

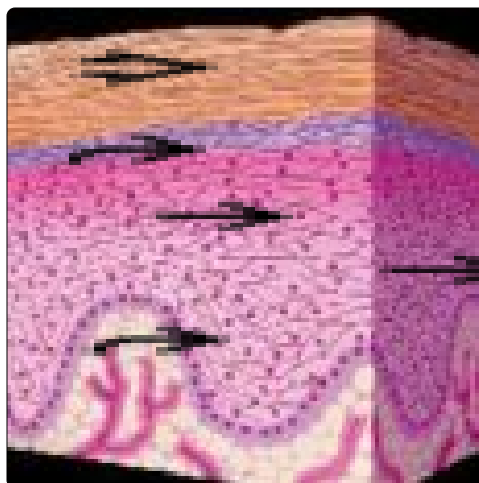
Hyperplasia

- Common; can occur in any layer of skin
 - In stratum corneum, parakeratosis is retention of nuclei in keratinocytes

Pitfalls and Artifacts

- Increased melanin pigment can be seen in both darker skinned individuals and in lentigines
- Reactive keratinocytic atypia in inflammatory conditions

Skin

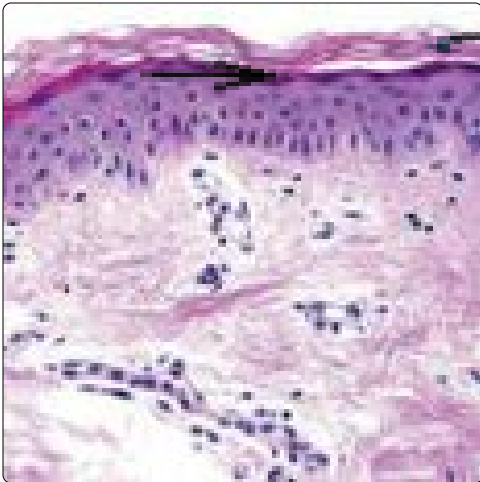


(Left) This graphic of normal thick skin (palm or sole) shows the layers of the epidermis, including the stratum basalis [red box], stratum spinosum [blue box], stratum granulosum [green box], and a thickened stratum corneum [yellow box]. (Right) The keratinocytes in normal epidermis form the normal stratum basalis [red box], stratum spinosum [blue box], stratum granulosum [green box], and stratum corneum [yellow box], which has associated orthokeratosis (anuclear keratin layer).

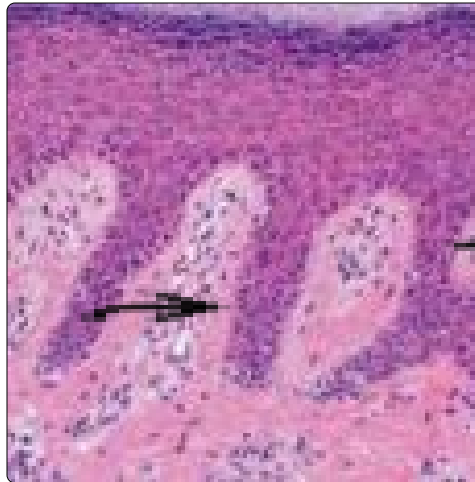
Normal Epidermis



Normal Epidermis and Superficial Dermis



Rete Ridges

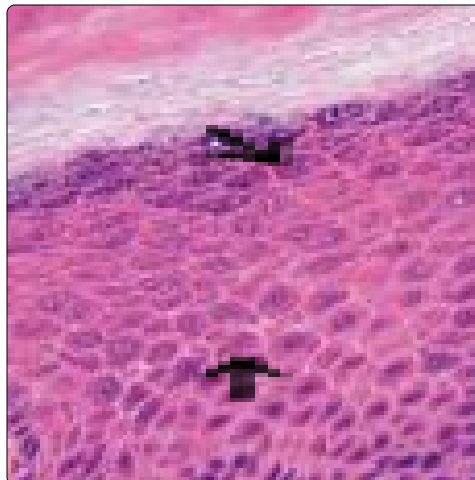


(Left) Normal thin skin on the arm features epidermis composed of 4-5 layers of bland-appearing keratinocytes with round to oval, hyperchromatic nuclei. The granular layer is well formed, and there is overlying basketweave stratum corneum. The superficial dermis is underneath the epidermis. (Right) Rete ridges (or rete pegs) are downward extensions of the epidermis between dermal papillae. These ridges are variably formed and may appear absent in some cases.

Normal Epidermis

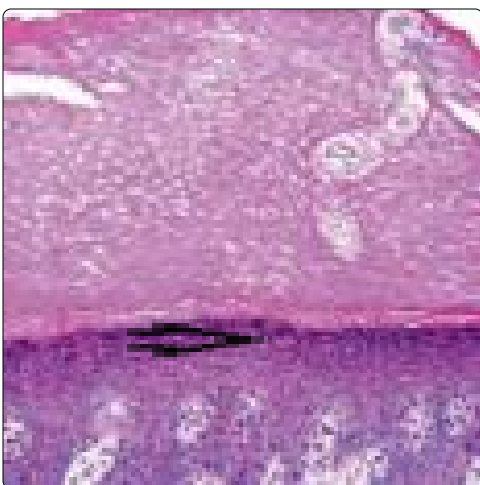


Spinosum and Granulosum

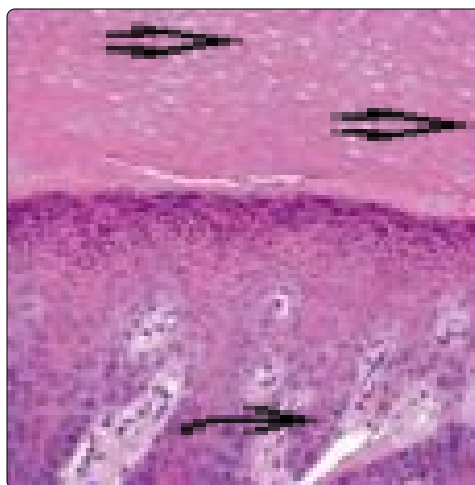


(Left) This photograph shows fused rete ridges, which can be seen occasionally in normal skin. Focal cytoplasmic melanin pigment is also seen in a keratinocyte. (Right) At high magnification, characteristic hair-like intercellular bridges (desmosomes) can often be identified between keratinocytes of the stratum spinosum. Basophilic keratohyalin granules of the stratum granulosum can also be appreciated in this image.

Acral Skin



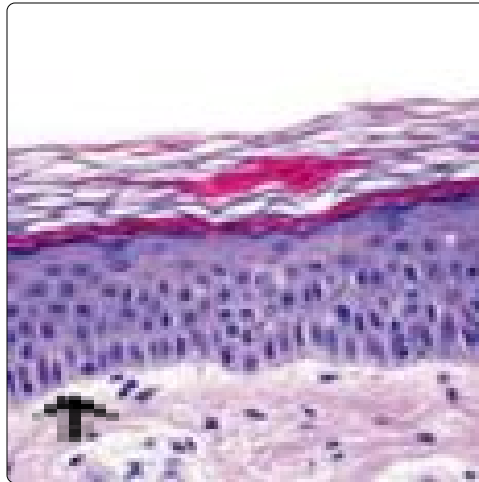
Acral Skin



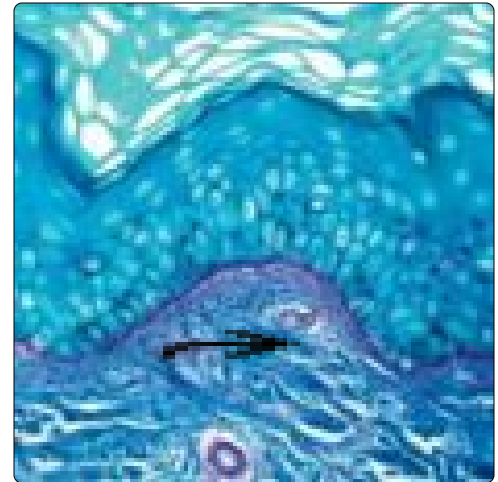
(Left) Acral-type skin contains prominent, dense keratin in the stratum corneum with an intracorneal acrosyringium. The epidermis also has a prominent granular layer. (Right) Higher magnification of acral skin shows a thickened stratum corneum. The epidermis is also thicker than at other sites and the dermis displays mild dermal fibrosis.

Epidermis With Focal Parakeratosis

(Left) This high magnification of the epidermis in a case of resolving spongiotic dermatitis (which can be mistaken for normal skin) shows focal parakeratosis [E], rare lymphocytes [E] in the basilar layer, and focal apoptotic bodies [E]. These mild reactive changes indicate a previous inflammatory insult. (Right) PAS stain will highlight the normal basement membrane [E]. It can be thickened or reduplicated in pathologic processes, such as lupus and lichen sclerosus et atrophicus.



PAS Stain in Normal Epidermis

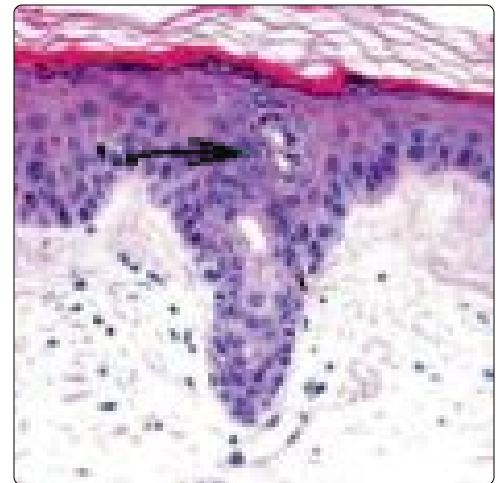


Epidermis With Follicular Ostia

(Left) The follicular ostia (intraepidermal opening of a hair follicle) is composed of an invagination of the epidermis containing laminated keratin [E]. The adjacent stratum corneum shows basketweave orthokeratosis [E]. (Right) Histologic section of an intraepidermal eccrine duct (acrosyringium) shows a small ductal lumen [E] within the epidermis. Melanin pigment is seen in the cytoplasm of basilar keratinocytes [E] and a melanocyte [E].

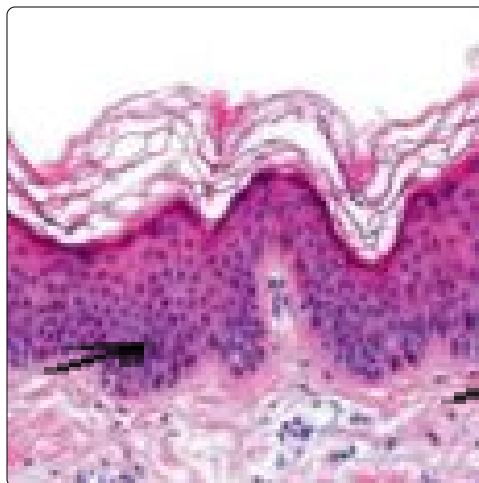


Acrosyringium

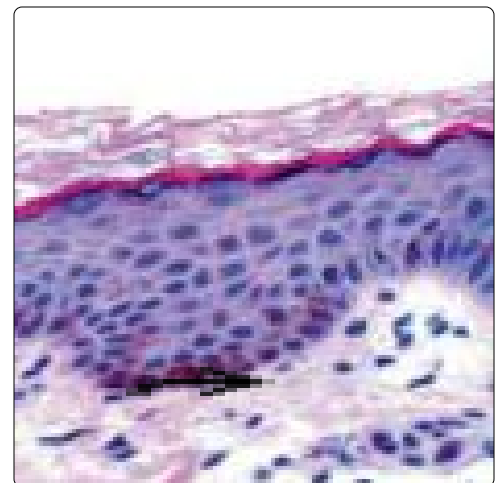


Moderately Pigmented Skin

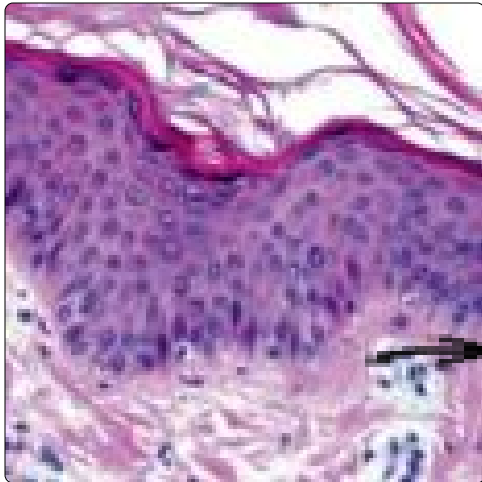
(Left) The normal epidermis in a darker skinned (i.e., Hispanic or Southeast Asian) patient shows increased melanin pigment in the basilar keratinocytes [E]. (Right) Normal epidermis from a dark-skinned (i.e., African origin) patient shows prominent basilar keratinocytic pigmentation [E]. The increased pigment is due to greater accumulation in keratinocytes, not an increased number of melanocytes.



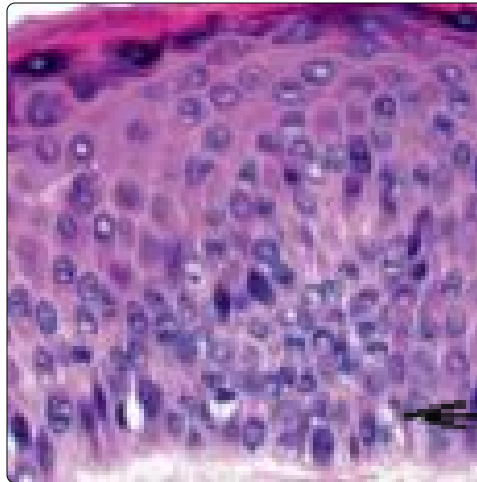
Darkly Pigmented Skin



Normal Epidermis With Melanocytes

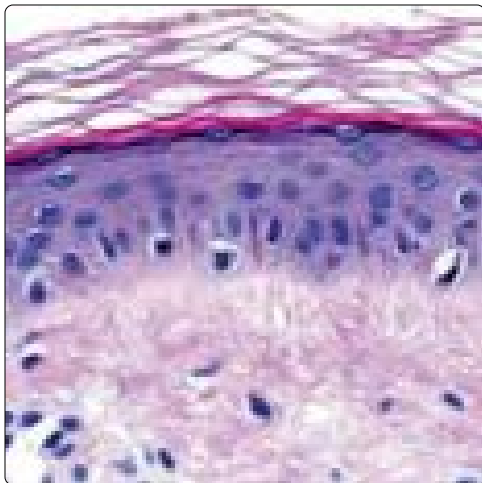


Normal Epidermis With Melanocytes

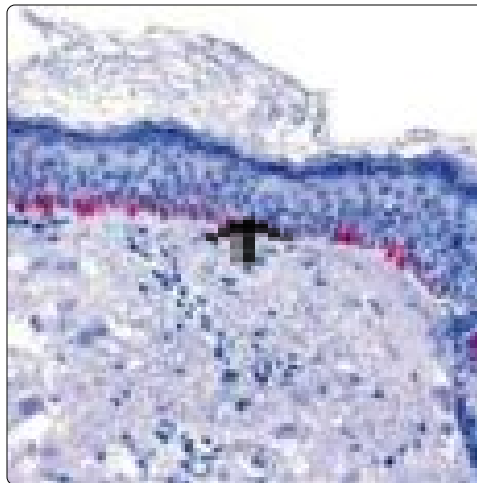


(Left) High magnification of the epidermis in a Caucasian patient shows a few basilar melanocytes with clear cytoplasm. **(Right)** Higher magnification section of the epidermis in a Caucasian patient shows several basilar melanocytes, which display hyperchromatic nuclei and cleared-out cytoplasm.

Junctional Melanocytes in Sun-Damaged Skin

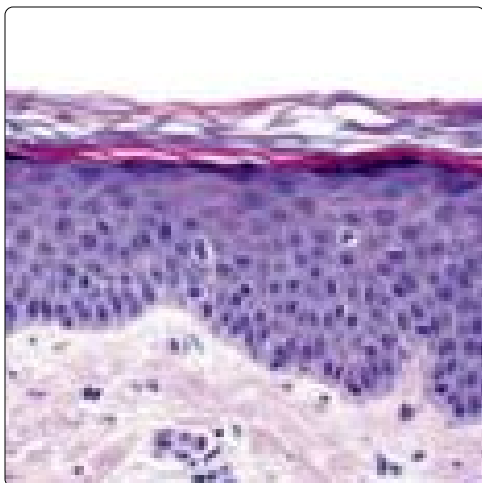


Melan-A Immunohistochemistry in Sun-Damaged Skin

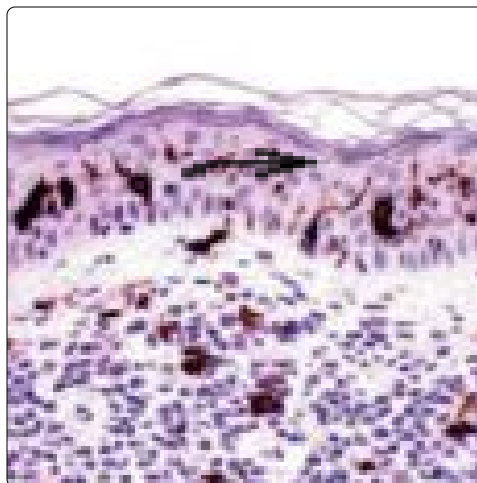


(Left) This epidermis from sun-damaged skin shows several junctional melanocytes with clear spaces separating them from the keratinocytes. While the nuclei are hyperchromatic, there is no significant cytologic atypia or pagetoid spread. **(Right)** This MART-1/Melan-A immunohistochemical-stained section shows mildly increased numbers of small, normal-appearing junctional melanocytes in the basilar epidermis and focally in the superficial portion of a hair follicle.

Intraepidermal Langerhans Cells



CD1a Immunohistochemical Stain for Langerhans Cells



(Left) The normal epidermis contains scattered Langerhans cells with somewhat curved to kidney bean-shaped nuclei. There is overlying orthokeratosis present as well. **(Right)** CD1a immunohistochemical stain for Langerhans cells shows very strong cytoplasmic staining of multiple intraepidermal cells with elongated, branching cytoplasmic processes typical of Langerhans cells.

DERMIS**Macroscopic Anatomy**

- Firm, white-yellow appearance

Microscopic Anatomy

- Papillary dermis
 - Loose, thin collagen fibers (mostly type I, some type III), with associated mucin
 - Elastic fibers typically sparse, thin, and branching
 - Small capillary-type blood vessels
 - Numerous vessels present at papillary/reticular dermis interface (superficial plexus)
 - Capillary beds also form beneath epidermis
 - Small nerve fibers
 - Meissner corpuscles present in acral skin
 - Encapsulated nerve endings (touch receptors)
 - Dermal papillae
 - Finger-like projections of dermis pushing into overlying epidermal basement membrane
 - Surrounded by epidermal rete ridges
- Reticular dermis
 - Dense collagen fibers (mostly type I) with less mucin
 - Much more dense and sclerotic-appearing on back
 - Elastic fibers thicker and more numerous
 - Small and medium-sized blood vessels (arterioles and venules, lymphatics) form deep plexus
 - Nerves typically associated with vessels (neurovascular bundles)
 - Pacinian corpuscles
 - Encapsulated nerve endings surrounded by circumferential lamellae
 - Function as pressure receptors in acral skin
 - Pilar muscles
 - Small muscle bundles associated with hair follicles
 - Insert below level of sebaceous glands
 - Smooth muscle bundles present in genitalia and areola
 - Adnexal structures
 - Based in deep dermis or subcutis (i.e., anagen hairs)
 - Folliculosebaceous units lacking on acral sites

Stromal cells

- Fibroblasts and myofibroblasts
 - Normal component; sparsely distributed
 - Bland, spindle-shaped hyperchromatic nuclei with inconspicuous nucleoli
- Mast cells (tissue basophils)
 - Normal component; sparsely distributed
 - Central round to oval nuclei; abundant basophilic-staining granular cytoplasm
- Lymphocytes
 - At least a few small perivascular lymphocytes physiologically present
- Plasma cells
 - Normally a few present in mucosal sites and often seen in scalp skin
- Histiocytes (tissue macrophages) and dendritic cells
 - Normal component; sparsely distributed
 - Indented nuclei with vesicular chromatin, abundant lightly staining cytoplasm
- Melanophages
 - Pigment-laden macrophages in superficial dermis
 - Often seen after inflammatory processes affecting the epidermis

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

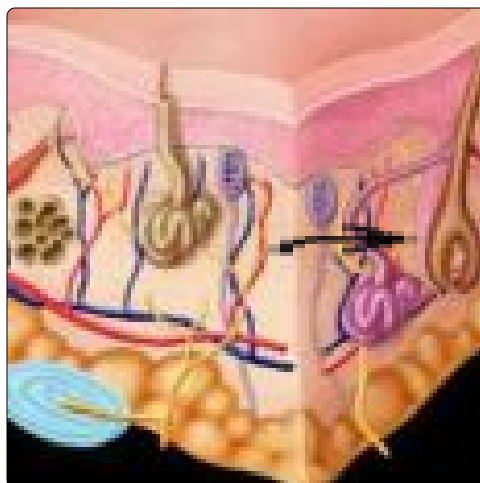
- Dermis is less cellular in infants and children
- Dermal atrophy can be seen with aging, due to loss of mucin and stromal cells
- Solar elastosis often present in older adults
 - Affects nonsun-protected sites
- Decreased fibroblasts, dendritic cells, mast cells in adults
- Hemorrhage commonly seen in elderly patients' biopsies, due to fragility of vessels

Pitfalls and Artifacts

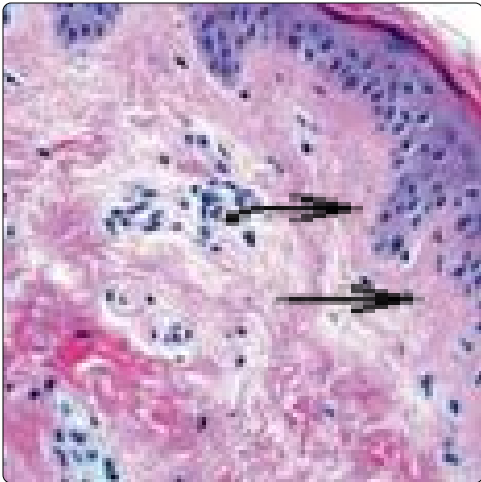
- Suboptimal staining due to poor fixation
- Artifactual clefts in dermis from processing
- Tissue cautery leads to homogenization of collagen fibers and streaming effects on nuclei

Dermis

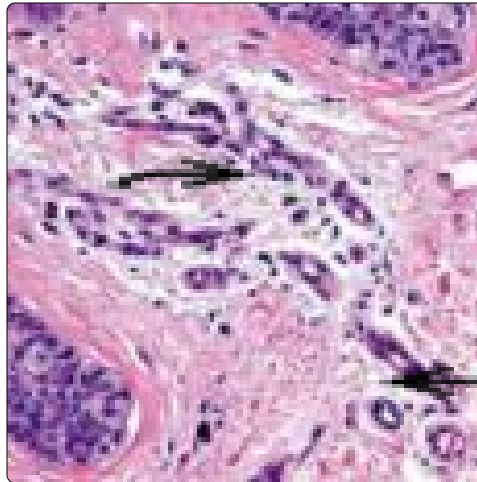
(Left) The dermis is composed of connective tissue, blood vessels, and nerves. Multiple adnexal structures, including eccrine glands, apocrine glands, pacinian corpuscles (blue oval), and hair follicles are present. (Right) Normal skin shows epidermis, papillary dermis with thin, lightly eosinophilic-staining collagen, and reticular dermis with thicker, denser collagen bundles.

**Papillary and Reticular Dermis**

Papillary Dermis

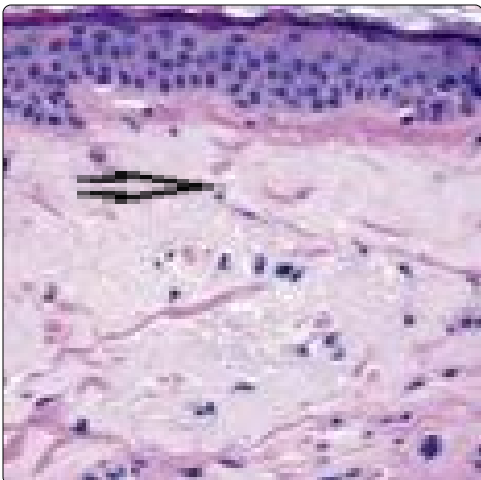


Papillary Dermis

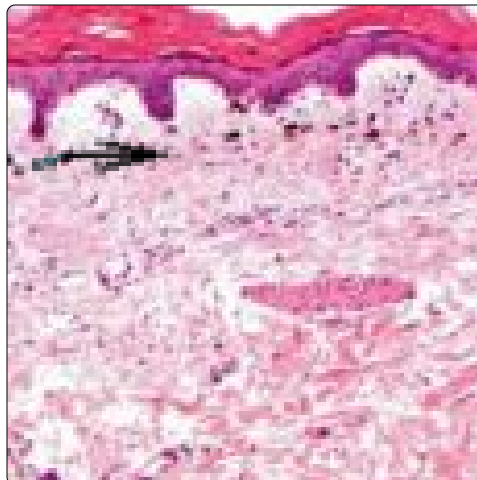


(Left) High-magnification H&E of the papillary dermis shows loose collagen and multiple small, capillary-type blood vessels. A few stromal lymphocytes are normally present. **(Right)** The papillary dermis between rete ridges normally contains multiple small, capillary-type blood vessels and a loose collagen matrix with mucinous material.

Severe Solar Elastosis in Superficial Dermis

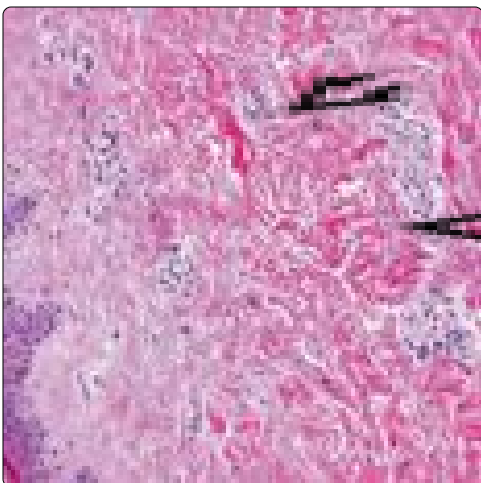


Papillary Dermal Melanin Pigment

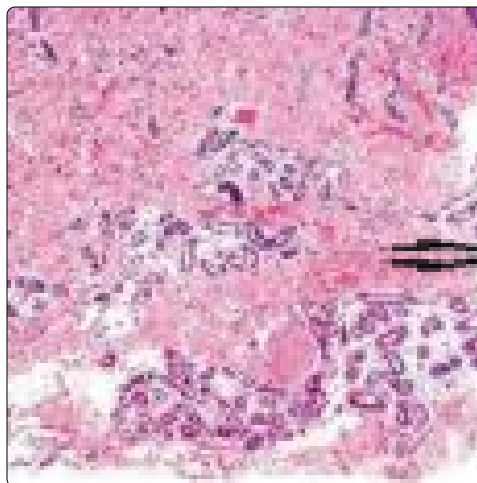


(Left) Solar elastosis is a common finding in sun-damaged skin of the elderly. It consists of degenerating elastic fibers in the papillary dermis and superficial reticular dermis that stain blue-gray. Note the few scattered mast cells and fibroblasts. **(Right)** Pigment is often seen in the dermis in skin that has previously been inflamed. In this case, there is no residual inflammation, but prominent melanin pigment deposits are seen in the papillary dermis.

Interface Between Papillary and Reticular Dermis



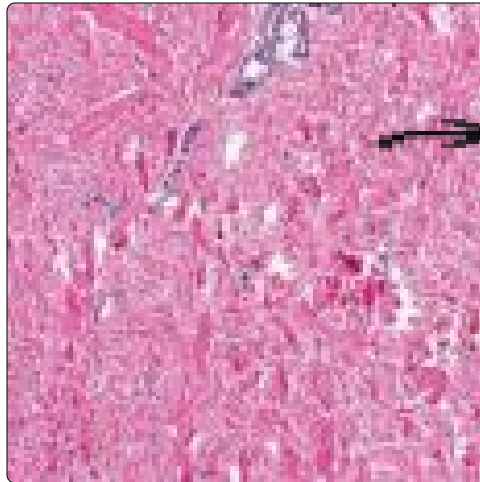
Reticular Dermis



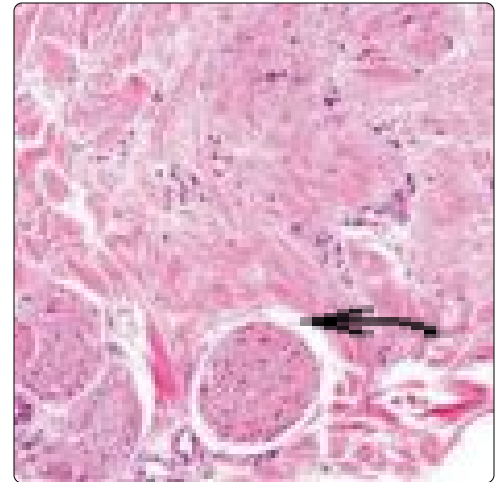
(Left) In this low-magnification H&E of the papillary dermal/reticular dermal interface, the papillary dermis on the left is composed of thinner, lighter staining collagen bundles, whereas the reticular dermis on the right is composed of thicker, more eosinophilic-appearing collagen bundles. **(Right)** The reticular dermis contains multiple adnexal structures, including eccrine ducts, eccrine and apocrine glands, and a pilar muscle in the deeper dermis.

Reticular Dermis

(Left) The reticular dermis from the back shows a dense arrangement of thickened eosinophilic collagen bundles. Note the eccrine ducts at the top of the field . **(Right)** The deep reticular dermis contains dense, sclerotic-appearing, hyalinized collagen bundles. This finding is usually abnormal except for the skin of the back, which typically has this appearance. Note the cluster of deep pilar muscles .

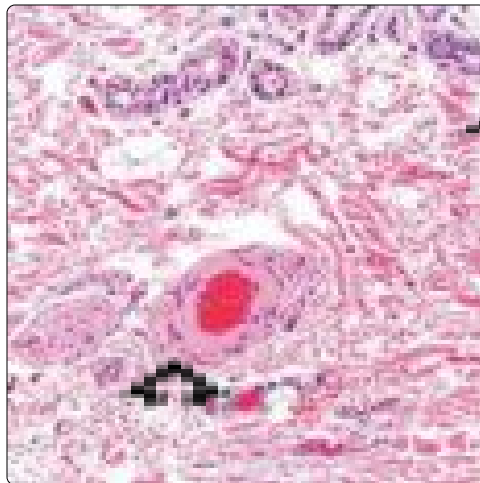


Deep Dermis

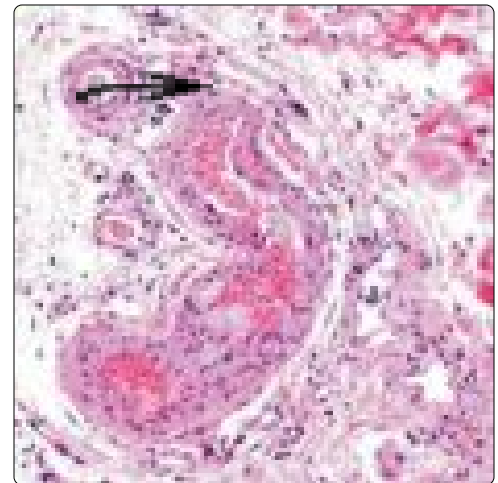


Neurovascular Bundle and Apocrine Glands

(Left) This H&E illustrates a neurovascular bundle and a small muscular vessel  on the right and a small nerve on the left . A more superficial cluster of apocrine glands is also seen . **(Right)** This H&E shows a deep dermal blood vessel with a thin muscular lining, consistent with a venule. Note the small adjacent arteriole  and lymphatics .

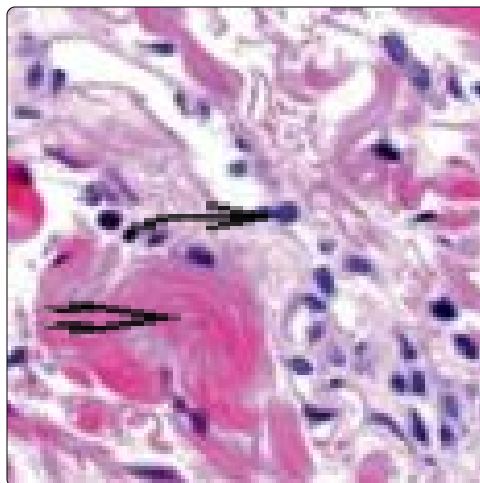


Dermal Blood Vessels and Lymphatics

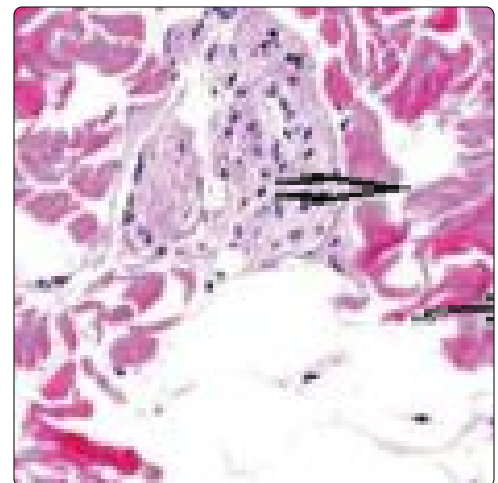


Blood Vessels and Collagen Bundles

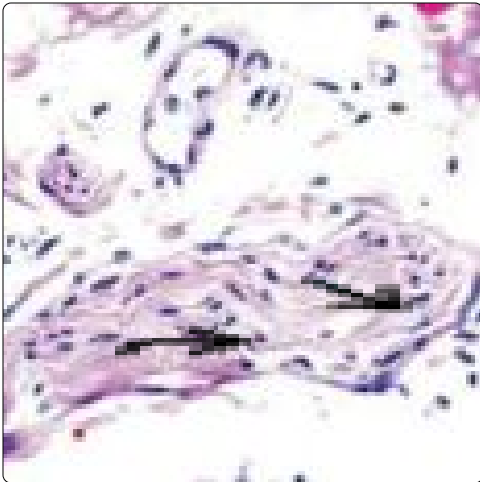
(Left) This high-magnification H&E shows small blood vessels and prominent, thickened collagen bundles . A few lymphocytes and a mast cell  are seen in the stroma. **(Right)** This deep neurovascular bundle at the dermal/subcutaneous junction  consists of a small nerve on the left  and the vessels on the right .



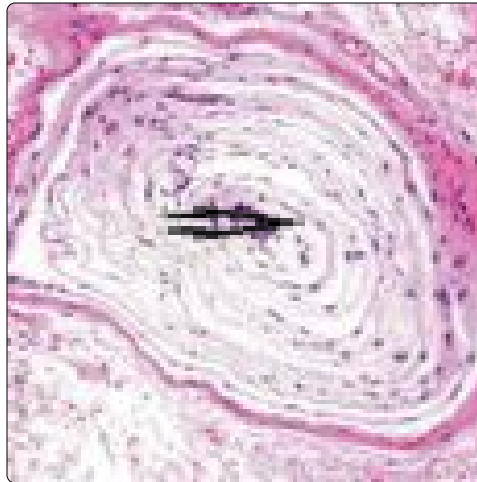
Deep Neurovascular Bundle



Deep Dermal Nerve

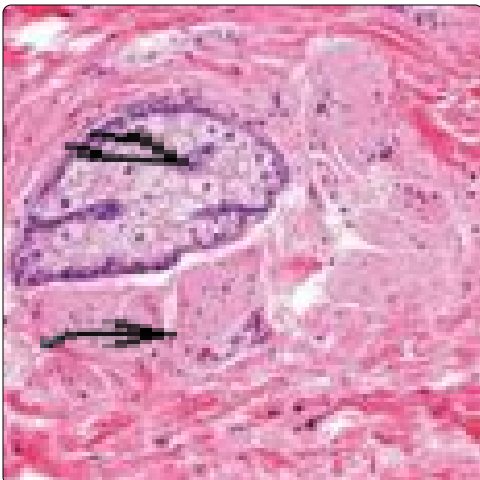


Pacinian Corpuscle

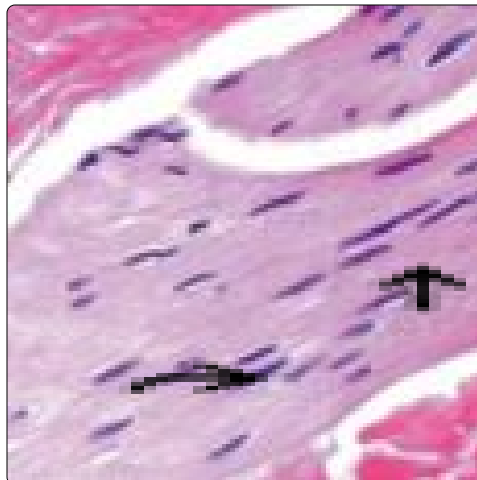


(Left) A medium-sized nerve in the deep dermis is composed of a collection of spindle-shaped cells with angulated nuclei and wispy, lightly eosinophilic-staining cytoplasm. A few small blood vessels are also present superior to the nerve.
(Right) This H&E of a pacinian corpuscle on acral skin illustrates the distinctive concentric laminated pattern of nerve sheath fibers.

Sebaceous Gland and Pilar Muscles

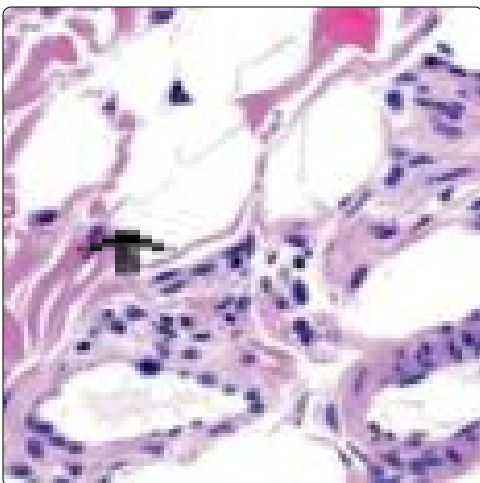


Pilar Muscle

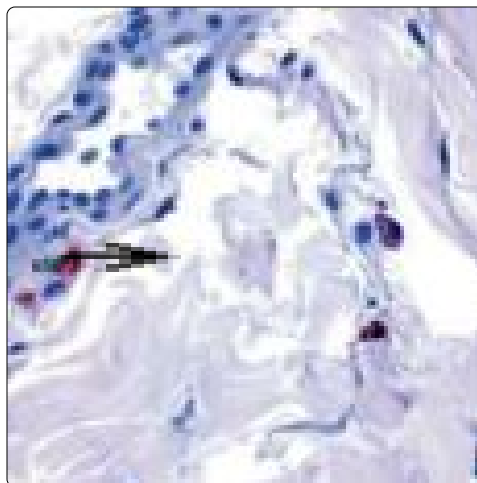


(Left) This section shows a sebaceous gland and surrounding pilar muscles in the reticular dermis. (Right) This pilar muscle at high magnification features elongated, spindle-shaped nuclei with a cigar-shaped (or blunt-ended) appearance and abundant eosinophilic-staining cytoplasm. Note the perinuclear clearing in some cells.

Mast Cells at Deep Dermis/Subcutaneous Junction



Mast Cells



(Left) This H&E of the deep dermis/subcutaneous junction shows multiple mast cells that are normally present in the perivascular stroma. The mast cells show hyperchromatic-staining oval nuclei and abundant basophilic-staining granular cytoplasm. (Right) CD117 immunohistochemistry prominently highlights the cytoplasmic membranes of mast cells in the deep dermis.

Adnexal Structures

ADNEXAL STRUCTURES

Macroscopic Anatomy

- Only hair follicles typically visible grossly
 - Hairs may be fine/vellus (arms, legs, face, back) or coarse (scalp, genital)

Microscopic Anatomy

- Hair follicles
 - Bulb
 - Base of follicle, composed of small basaloid cells (matrix), often with cytoplasmic melanin pigmentation
 - Dermal papilla (papillary mesenchymal body)
 - Invagination of cellular stroma into base of bulb
 - Inner root sheath
 - Eosinophilic-staining cells around hair shaft in suprabulbar region (includes Huxley and Henle layers)
 - Outer root sheath
 - Clear-staining cells (due to cytoplasmic glycogen) surrounding inner root sheath in suprabulbar region
 - Isthmus
 - Above suprabulbar region; extends from bulge up to insertion of sebaceous duct
 - Keratinization without granular layer
 - Bulge region
 - Insertion of pilar muscle
 - Proposed origin of follicular stem cells
 - Infundibulum
 - Uppermost region of follicle (above insertion of sebaceous duct)
 - Keratinization with granular layer
 - Ostium
 - Opening of follicle at epidermal surface
 - Often contains keratin and small collections of bacterial organisms
- Sebaceous glands
 - More abundant in facial and eyelid areas
 - Lobular collections of large, central clear cells with abundant multivacuolated cytoplasm and small, vesicular nuclei with indentations

- Peripheral layer of small, cuboidal basaloid cells
- Glands secrete lipids and entire necrotic cells into sebaceous duct, which attaches to follicular lumen
- Apocrine glands
 - More abundant in axillary, groin, breast, and eyelid areas
 - Clusters of glands in deep dermis or superficial subcutis
 - Composed of large epithelial cells with eosinophilic cytoplasm; may show apical snouts and secretions
 - Apocrine duct connects to follicle; ductal cells appear identical to eccrine ductal cells
- Eccrine glands
 - Present everywhere but concentrated on acral surfaces
 - Clusters of glands in deep dermis or superficial subcutis
 - Composed mostly of cuboidal cells with pale eosinophilic to clear cytoplasm (clear cells)
 - Glands also contain dark cells, which line lumina and contain secretory granules
 - Peripheral layer of myoepithelial cells present
 - Eccrine ducts
 - Composed of proximal convoluted duct, straight dermal duct, and spiral acrosyringium
 - Connect directly to epidermis (acrosyringium is intraepidermal duct)

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Adnexal structures are hypoplastic in infancy
- Hair follicles and sebaceous glands mature during puberty, especially in axillary, facial, and genital areas

Hyperplasia

- Sebaceous hyperplasia common, usually on face in elderly

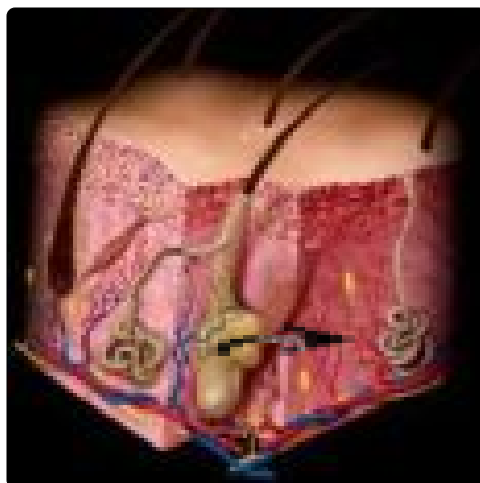
Metaplasia

- Squamous metaplasia may be seen in adnexal ducts

Pitfalls and Artifacts

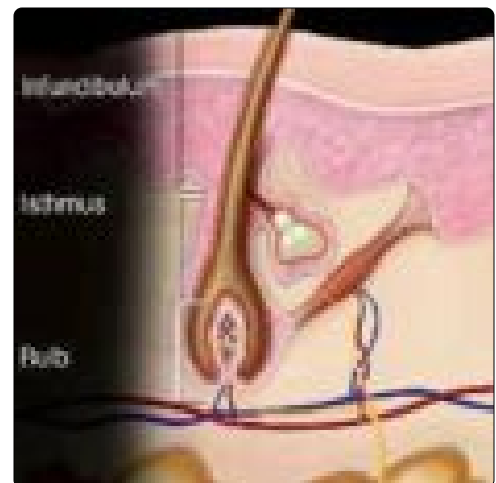
- Fewer follicles present in patients of African descent
- Tangentially sectioned hair bulbs can be mistaken for basal cell carcinoma

Adnexal Structures

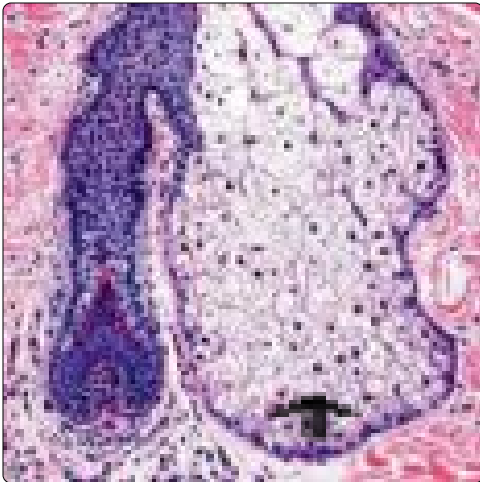


(Left) Normal cutaneous adnexal structures include hair follicles with associated apocrine and sebaceous glands. The eccrine gland attaches directly to the epidermis. (Right) The isthmus of the hair follicle extends from the bulge region (insertion of pilar muscle) up to the insertion of the sebaceous duct. The infundibulum extends from this point up to the ostium at the surface. Also note the dermal papilla at the base of the bulb.

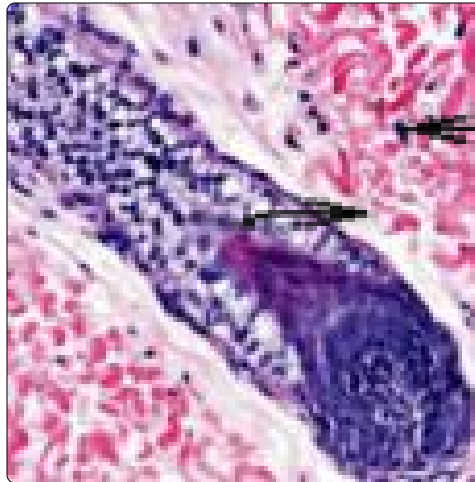
Hair Follicle



Hair Follicle

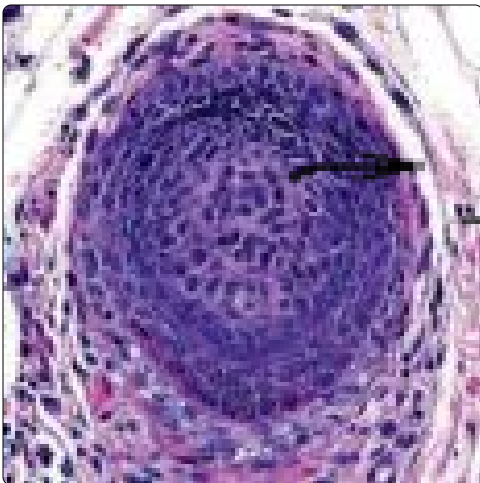


Bulb and Suprabulbar Region of Vellus Follicle

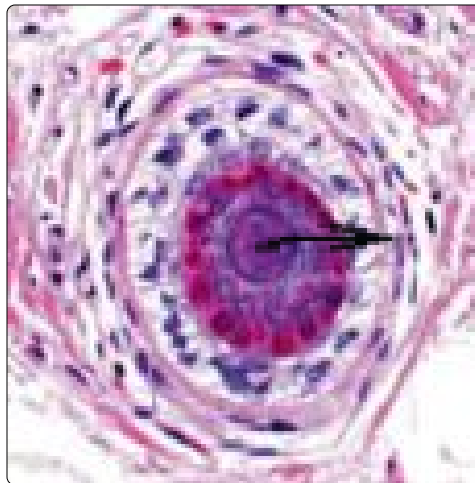


(Left) This is a small (vellus) hair follicle and sebaceous gland. The follicle shows the dermal papilla (papillary mesenchymal body) [arrow] invaginating into the bulb. The sebaceous gland is composed of large clear cells. (Right) H&E shows a hair follicle at the bulbar region with the dermal papilla [arrow] surrounded by small basaloid cells [arrow] with scant amounts of cytoplasm. Note the more superficial outer root sheath cells with clear cytoplasm [arrow].

Cross Section of Hair Bulb

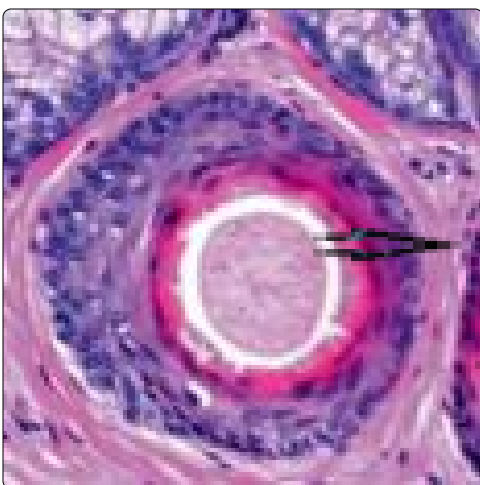


Cross Section of Suprabulbar Region

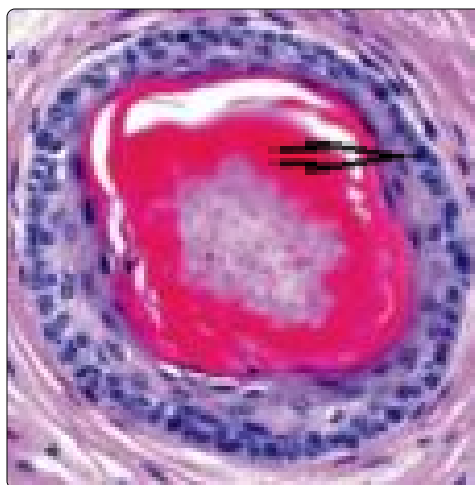


(Left) Cross section of a hair follicle at the bulbar region shows the papillary mesenchymal body [arrow] surrounded by small basaloid cells [arrow] with scant amounts of cytoplasm. (Right) Cross section of an anagen (growth phase) follicle in the suprabulbar region shows the follicle with clear outer root sheath cells [arrow] surrounding eosinophilic-staining inner root sheath cells [arrow] and the central hair matrix [arrow].

Cross Section of Anagen-Phase Hair Follicle



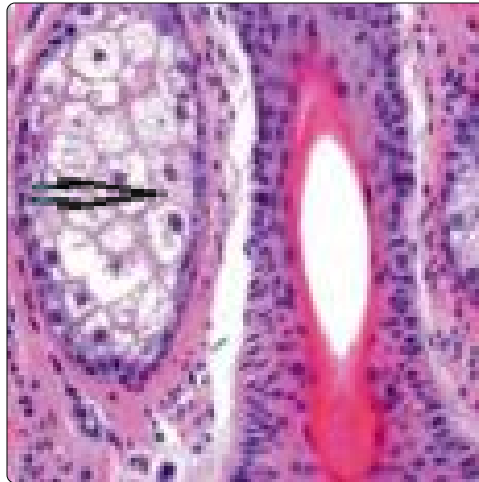
Cross Section of Telogen-Phase Hair Follicle



(Left) This cross section of an anagen-phase follicle at the level of the isthmus shows a central hair shaft, presence of the inner root sheath [arrow], and keratinization without a granular layer [arrow]. (Right) This cross section of a telogen-phase follicle (resting phase) at the level of the isthmus shows absence of the inner root sheath and brightly eosinophilic-staining keratinous material [arrow]. Note the absence of a granular layer (which is not present until the level of the infundibulum).

Isthmus of Hair Follicle With Sebaceous Glands

(Left) This longitudinal section of a telogen hair follicle at the level of the isthmus shows keratinization without a granular layer. Adjacent sebaceous glands are present. (Right) This cross section of a vellus (fine, light colored) hair follicle at the lower level of the infundibulum shows the miniaturized hair shaft and adjacent small sebaceous glands. Note the focal presence of a granular layer.

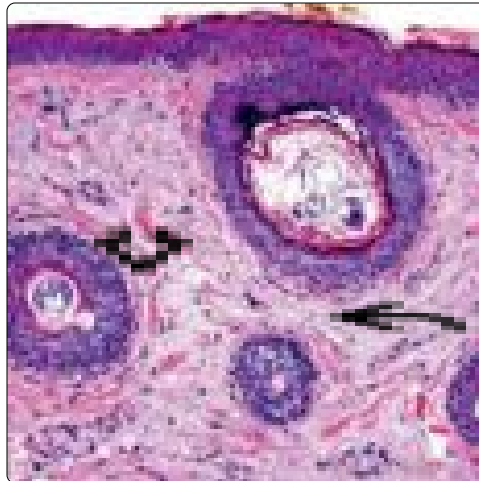


Vellus Follicle With Sebaceous Glands



Infundibular Region of Follicles With Demodex

(Left) This section of superficial dermis contains multiple follicles (including a terminal hair and several vellus hairs) with multiple intrafollicular Demodex organisms present (these are common in adult skin). (Right) The follicular ostia (intraepidermal opening of the hair follicle) is composed of an invagination of the epidermis containing laminated keratin and the hair shaft. The adjacent stratum corneum shows basket-weave orthokeratosis.

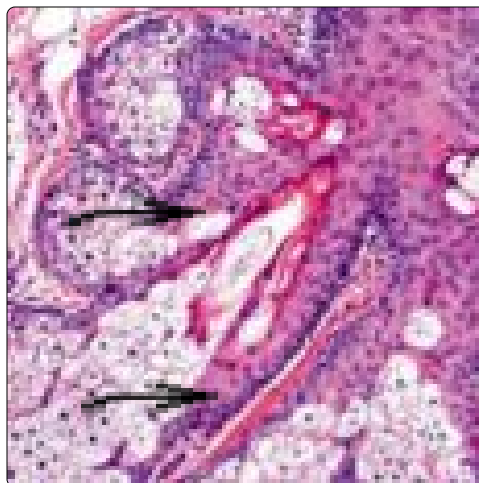


Epidermis With Follicular Ostia

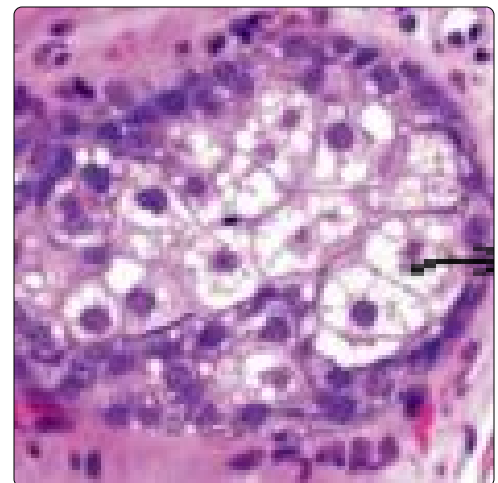


Sebaceous Glands Attached to Hair Follicle

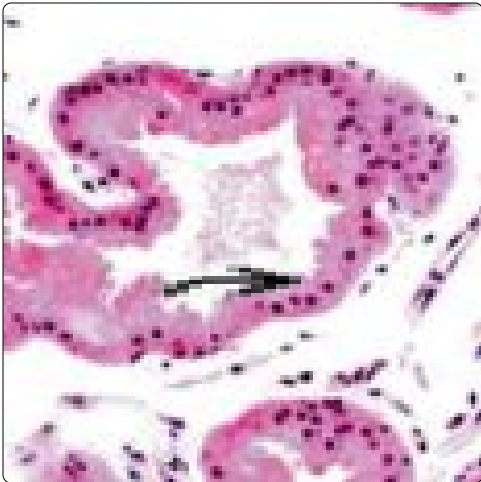
(Left) Vertical section of a hair follicle shows multiple attached mature sebaceous glands, which empty directly into the follicle via sebaceous ducts. (Right) High-magnification cross section of a sebaceous gland shows cells with abundant, multivacuolated, clear-staining cytoplasm. The nuclei are round to oval with vesicular to hyperchromatic-appearing chromatin. Some show indentations due to the cytoplasmic lipid.



Sebaceous Gland at High Magnification



Apocrine Glands

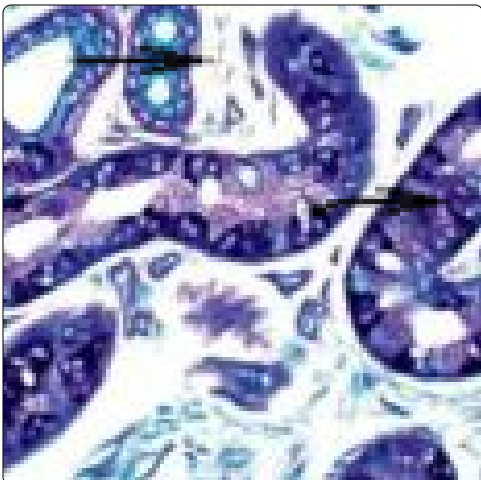


High Magnification of Apocrine Glands

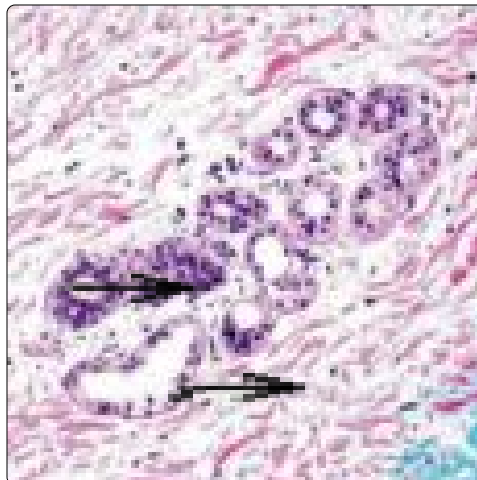


(Left) Apocrine glands are characterized by large epithelial cells with abundant eosinophilic-staining cytoplasm and numerous apical cytoplasmic snouts and secretions. Focal lipofuscin is also present in some of the cells. (Right) High-magnification view of apocrine glands shows a population of large columnar cells with hyperchromatic nuclei, abundant eosinophilic-staining cytoplasm, and protuberant apical snouts.

PAS Stain in Apocrine Glands and Ducts

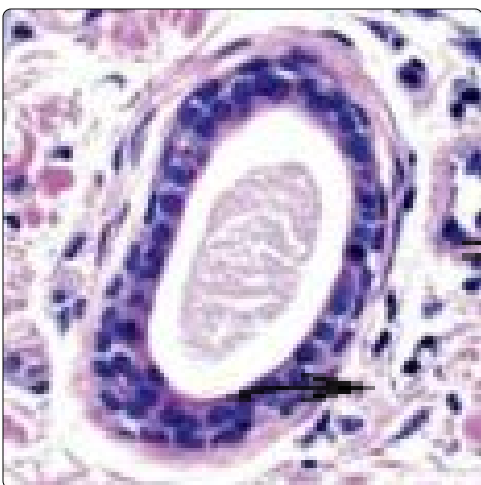


Eccrine Glands and Ducts



(Left) PAS stain in apocrine glands strongly stains the cytoplasm of most of the cells and highlights small granules in many of the luminal cells. Note that the apocrine ducts do not show strong staining. (Right) Eccrine glands and ducts, here in the deep dermis, are characterized by a proliferation of small, bland-appearing epithelial cells with abundant clear cytoplasm surrounding central lumina. The ducts show smaller lumina with focal intraluminal secretions.

Eccrine Duct



Superficial Dermal Eccrine Duct



(Left) This cross section of an eccrine duct in the dermis shows a well-formed lumen with proteinaceous debris surrounded by a 2- to 3-cell-thick layer of small, cuboidal-shaped epithelial cells with eosinophilic-staining cytoplasm. (Right) Histologic section of an intradermal eccrine duct entering into the epidermis (which contains the acrosyringium) shows a small ductal lumen within the dermal portion.

NAIL**Macroscopic Anatomy**

- Keratin plate and associated epithelium covering dorsal parts of terminal phalanges of fingers and toes
 - Consists of convex to flat, rectangular, white/pink hard nail plate, which lies on nail bed
 - Translucent but appears pink from underlying vasculature
- Part of nail that grows is at proximal end of nail, beneath epidermis
- Protects fingertips and toes from injury and enhances sensation of fingertips
- Onychodermal band
 - 0.5- to 1.5-mm distal transverse band (glassy/gray color) proximal to free margin of nail
 - Forms seal between nail plate and hyponychium

Microscopic Anatomy

- Nail plate
 - Consists of nail proper or body of nail
 - Composed of dense layers of interdigitating polyhedral keratinocytes that lack nuclei or organelles
 - Free margin is distal edge or anterior margin of nail plate (cutting edge of nail)
- Proximal nail fold
 - Wedge-shaped skin fold on dorsum of digit that overlies nail plate
 - Made of 2 layers of epidermis (dorsal and ventral)
 - Dorsal proximal nail fold
 - Contiguous with epidermis on digit
 - Cuticle is part of dorsal proximal nail fold and protects nail base/matrix
 - Ventral proximal nail fold
 - Overlies newly formed base of nail
 - Eponychium is epithelium of ventral part of proximal nail fold
 - Eponychium and cuticle form protective seal over proximal nail

- Perionych is projecting edge of eponychium covering proximal lunula

- Matrix
 - Tissue upon which proximal nail rests
 - Produces cells that form nail plate
 - Consists of acanthotic squamous epithelium without granular layer, under proximal nail fold
 - Basaloid cells progress to larger squamoid keratinocytes, which form nail plate after flattening, losing nuclei, and condensing cytoplasm
 - Matrix contains small numbers of normal melanocytes, Merkel cells, and Langerhans cells
 - Lunula is distal portion of matrix: Whitish, crescent-shaped base of visible nail
 - Nail bed
 - Flat epithelium underlying remainder of nail plate
 - ~ 3 cell layers thick without melanocytes
 - Hyponychium
 - Distal nail bed epithelium under nail plate between free edge and skin of finger tip
 - Thick keratin layer, acanthotic squamous epithelium, and transverse papillae that forms seal that protects nail bed

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

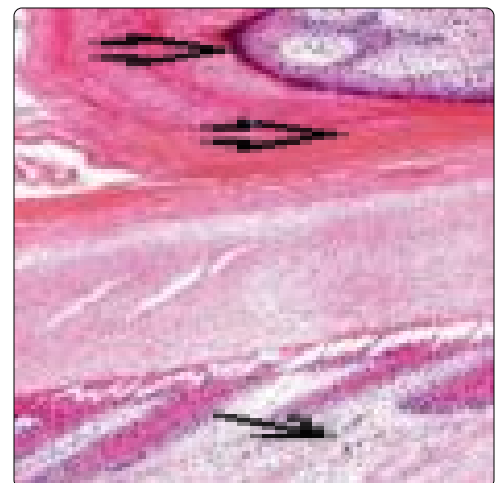
- Nail increases in thickness with advancing age
- Adults have larger dorsal nail keratinocytes compared to children

Pitfalls and Artifacts

- Fixation and processing artifacts common
 - Due to dense keratin material of nail
- Artifactual staining or entrapped air bubbles may make it difficult to assess for pigment (melanin or hemosiderin) deposition

Proximal Nail Fold and Cuticle

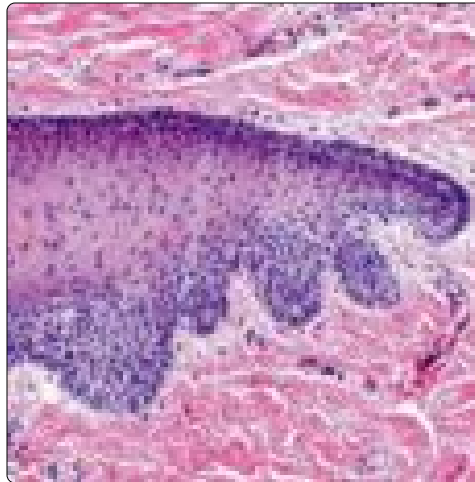
(Left) This histologic section shows the proximal nail fold [black arrow] and cuticle [blue arrow] overlying the proximal nail plate [green arrow] and the nail matrix [red arrow] in an infant. Note the underlying immature bone [red arrow]. **(Right)** Higher magnification of the proximal nail fold shows keratinizing eponychium [blue arrow] as well as the cuticle [blue arrow] overlying the nail plate. The nail matrix shows elongated rete ridges [red arrow] composed of bland squamous cells.

Proximal Nail Fold

Proximal Nail Matrix

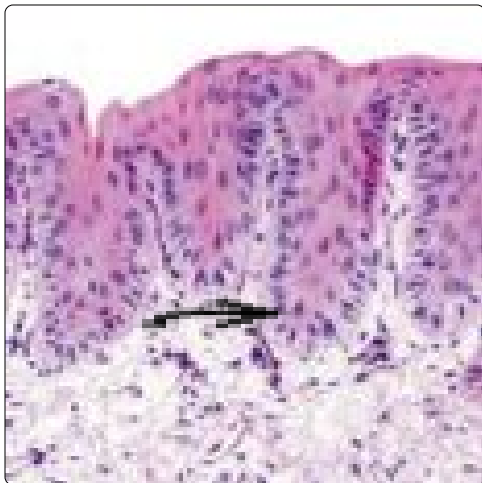


Matrix Angle Epithelium

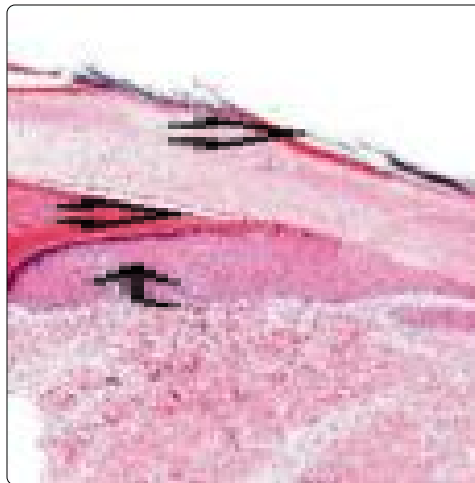


(Left) The proximal nail matrix (both dorsal and ventral/intermediate) shows keratinizing squamous epithelium forming the proximal nail plate. Note the focal elongated rete ridges of the most proximal area of the ventral matrix. (Right) This H&E illustrates the most proximal portion of the matrix (matrix angle/dorsal and intermediate matrix). The matrix in this region is formed by proliferating basaloid cells that lead to the production of the nail plate.

Higher Magnification of Intermediate Matrix



Hyponychium

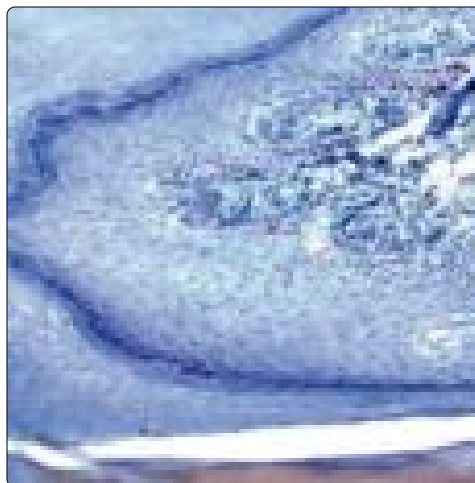


(Left) The intermediate matrix shows an acanthotic epithelium composed of bland-appearing squamous cells with eosinophilic cytoplasm and rare intraepithelial lymphocytes. No increase in melanocytes or melanin pigment is present. (Right) This is a view of the hyponychium, which is the distal nail bed epithelium under the nail plate between the free edge and the skin of the finger tip. Note the very dense keratin layer and prominent granular layer.

Higher Magnification of Hyponychium



Melanocytes in Proximal Nail Fold



(Left) The hyponychium at higher magnification shows epithelial acanthosis with a thickened layer of mature squamous keratinocytes. There is marked overlying hypergranulosis and dense hyperkeratosis. (Right) Immunohistochemical stain for MART-1/Melan-A highlights scattered normal melanocytes in the proximal nail fold and the eponychium. Normally, only a few scattered single melanocytes should be present in this region.

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SECTION 3

Musculoskeletal System



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Bone and Cartilage

BONE

Macroscopic Anatomy

- Bones classified into 3 general groups: Flat, cuboid, and tubular
- Tubular bones have 3 regions: Epiphysis, metaphysis, and diaphysis
 - Epiphysis: Extends from articular surface to growth plate
 - Metaphysis: Extends from growth plate to diaphysis
 - Diaphysis (shaft): Narrowest portion of bone
- Individual bones have unique apophyses (bumps, protuberances) for tendinous/ligamentous attachment

Microscopic Anatomy

- Periosteum: Lines outer surface of cortical bone
- Endosteum: Lines inner surfaces of cortical bone and cancellous bone
- 2 basic types of bone, which can be present in any location
 - Woven bone (immature bone): Irregular collagen fibrils, hypercellular, large osteocytes and lacunae in haphazard distribution
 - Lamellar bone (mature bone): Parallel collagen fibrils, hypocellular, small osteocytes and lacunae in organized distribution
- Cortex: Composed of cortical (compact) bone
 - 3 architectural patterns: Circumferential (outermost layer of cortex), concentric (around osteons), and interstitial (everything else)
- Haversian system (osteon): Composed of haversian canals filled with blood vessels, nerves, and mesenchymal cells and surrounded by concentric lamellar bone
 - Canal of Volkmann: Connects haversian canals to other haversian canals as well as to external cortical surface and medullary cavity
- Medullary cavity: Contains cancellous (trabecular) bone and bone marrow
 - Cancellous or spongy bone: Interconnecting trabeculae of bone within medullary cavity
 - Bone marrow: Contains mixture of hematopoietic elements and adipose tissue

- Osteoblasts: Elongated mononuclear cells with eccentric nuclei, prominent nucleoli, and perinuclear halos; often form thin line around trabecular bone
- Osteocytes: Small, mature osteoblasts with high nuclear:cytoplasmic ratio; found within lacunae
- Osteoclasts: Large, multinucleated cells, often found in Howship lacunae

CARTILAGE

Macroscopic Anatomy

- 3 types of cartilage: Hyaline, elastic, and fibrocartilage
- Hyaline cartilage is most common and covers articular surfaces of all long bones
 - Glassy and blue to gray-white in color
 - Also found in growth plate (physis) during active growth

Microscopic Anatomy

- Hyaline cartilage: Abundant glassy extracellular matrix containing chondrocytes in variable cellularity
- Tidemark: Sharp demarcation between cartilage and bone that represents site of previous growth plate
- Elastic cartilage: Matrix contains abundant elastin fibers
- Fibrocartilage: Matrix contains abundant collagen fibers
- Chondrocytes: Generally small cells with eosinophilic cytoplasm and dark, round nuclei; reside in lacunae

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

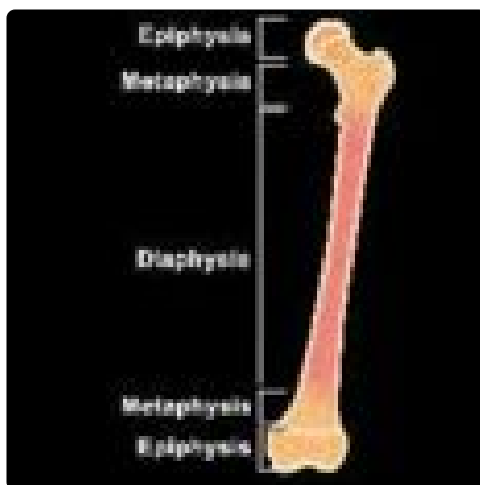
- Bony trabeculae often become thinner as result of age- or hormone-related osteopenia (clinical osteoporosis)
- Growth plate (physis) disappears after long bone growth ceases

Pitfalls and Artifacts

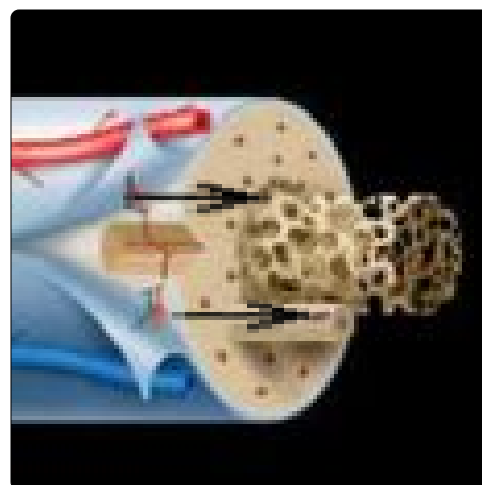
- Irregular fragments of woven bone with prominent osteoblastic rimming can mimic osteosarcoma
- Markedly increased cellularity at growth plate (physis) can be mistaken for chondroblastic neoplasm
- Enchondral ossification can be mistaken for cartilaginous tumor

Anatomic Regions of Bone

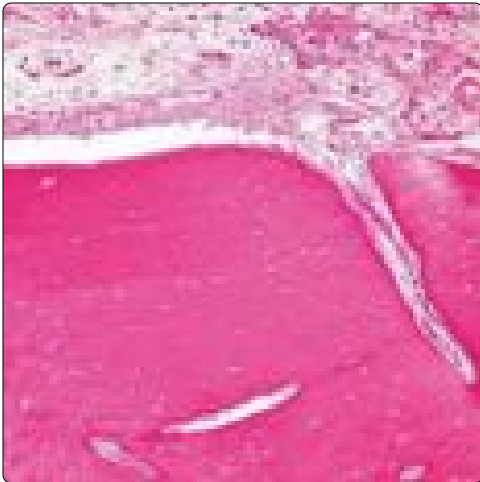
(Left) This graphic demonstrates the 3 regions of a long bone: The epiphysis, metaphysis, and diaphysis. Also highlighted are the cortex, medullary cavity, and articular surfaces. (Right) This graphic illustrates the cross-sectional anatomy of a long bone. Note the peeled back periosteal layer as well as the haversian system (osteon) containing a central haversian canal.



Microanatomy of Bone



Periosteum



Cortical (Compact) Bone

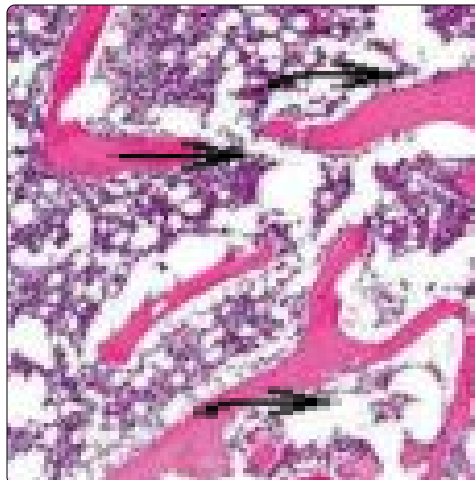


(Left) The periosteum is a thin layer of connective tissue that is present on the outer surface of cortical bone. Note the presence of a periosteal vessel, which is seen penetrating the outer cortex. These small vessels are thought to provide nutrients to the outer cortical bone. **(Right)** Cortical bone is very dense and lacks bone marrow. Haversian canals are circular or tubular structures within the bone that contain nutrient vessels. Note the periosteal layer.

Cortical (Compact) Bone

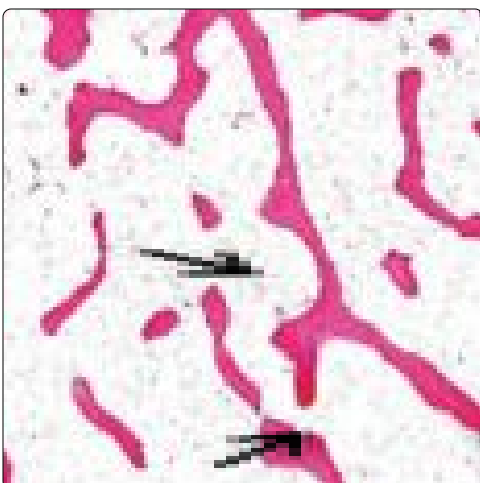


Cancellous (Trabecular) Bone

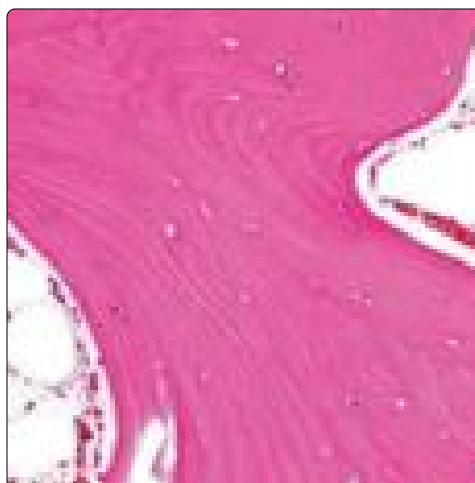


(Left) Haversian canals are easily identified as round or tubular spaces within otherwise solid-appearing cortical bone. Note the small blood vessels seen within some of the canals. **(Right)** Cancellous (or trabecular) bone is found in the medullary cavity of long bones and appears as interanastomosing trabeculae. Unlike cortical bone, there is marrow and fat present in cancellous bone.

Cancellous (Trabecular) Bone



Lamellar Bone



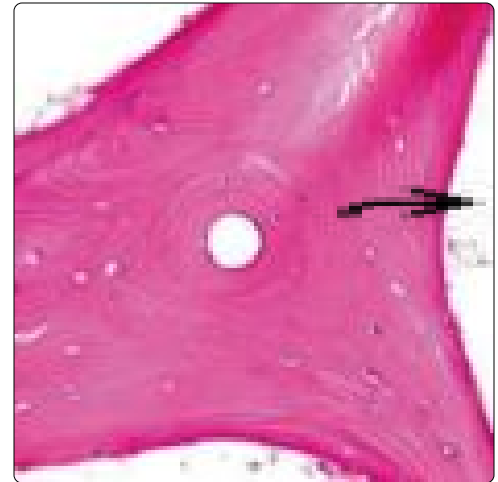
(Left) In elderly patients, particularly those with osteoporosis, it is common to see thinning of the bony trabeculae, also referred to as osteopenia. As intertrabecular adipose tissue tends to increase with age, this trabecular thinning also leads to the illusion that some trabeculae are "floating" in adipose tissue. **(Right)** Lamellar (or mature) bone is composed of an orderly, parallel arrangement of type I collagen fibers, giving it a topographic or plate-like appearance. It is seen in both cortical and medullary bone.

Parallel Arrays

(Left) The parallel arrays of collagen fibers in lamellar bone are readily apparent on routine H&E sections. This type of lamellar architecture is also referred to as interstitial lamellar bone. **(Right)** One of the 3 architectural patterns of lamellar bone, concentric, is displayed in this image. Note the concentric arrays of collagen  and localization around a haversian canal . The remaining bone architecture seen elsewhere in this image is interstitial lamellar bone.



Concentric Arrays



Lamellar Bone (Polarized)

(Left) Under polarized light, the parallel collagen arrays of lamellar bone are brilliantly depicted. Note the concentric lamellae around the small haversian canal . **(Right)** This image shows 2 tubular haversian canals  running lengthwise, each containing nutrient vessels. The smaller channel  that runs perpendicularly and connects the 2 haversian canals is known as a Volkmann canal.



Haversian System

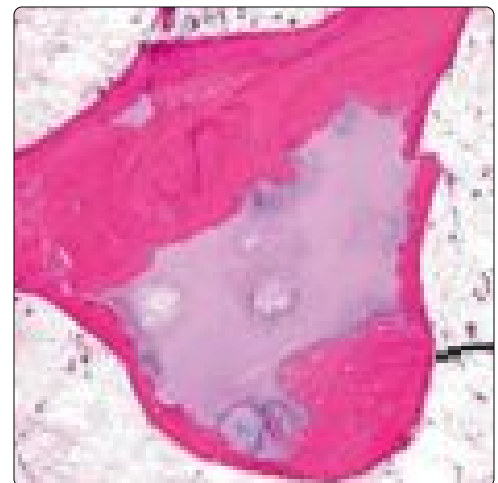


Lamellar Bone Mineralization

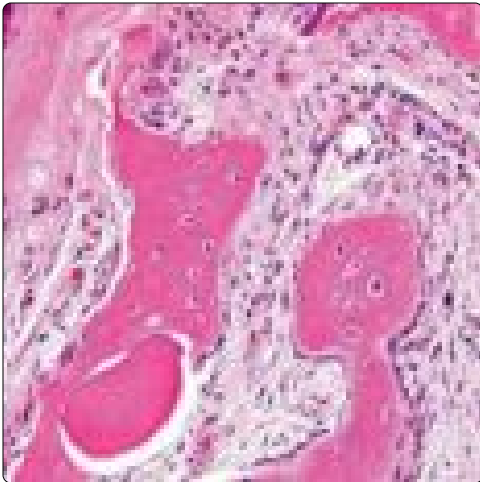
(Left) Often, lamellar bone can take on a purple hue  in routine histologic preparations. This is due to the fact that the bone is mineralized and has not been decalcified completely. This finding usually varies from focal to diffuse. **(Right)** During the process of enchondral ossification (a part of bone growth), bone is laid down on top of a cartilage framework . This can lead to the appearance of cartilage trapped in bone and should not be confused with a cartilaginous tumor.



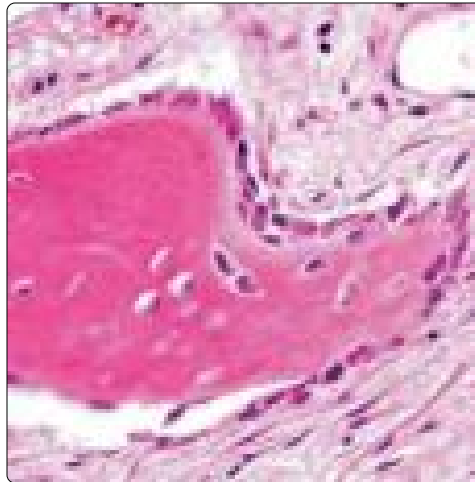
Enchondral Ossification



Woven Bone

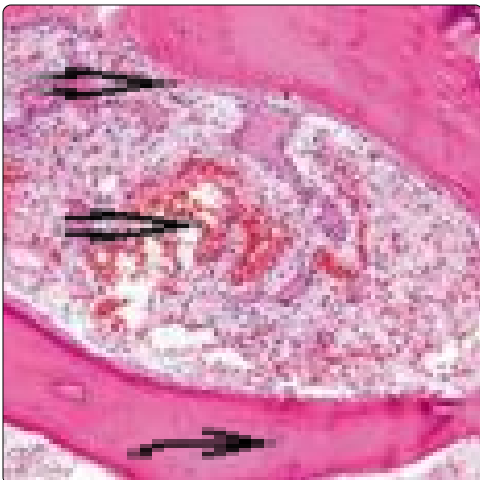


Woven Bone and Osteoblasts

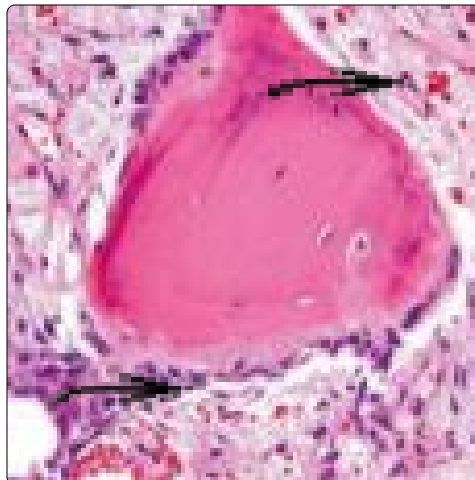


(Left) Compared to lamellar bone, woven (immature) bone is more irregularly shaped, less organized, and contains more osteocytes per unit area. Osteoblasts and osteoclasts are commonly seen near or around the periphery. (Right) Woven bone often shows a prominent lining of activated osteoblasts, which are responsible for producing the bone. Osteoblasts can become trapped in the bony matrix and mature into osteocytes.

Woven Bone in Fracture Site

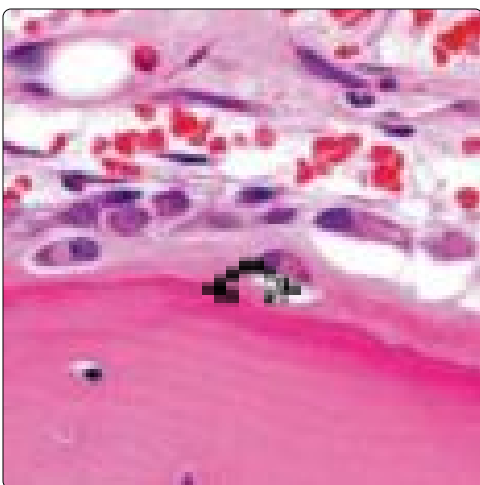


Osteoblasts

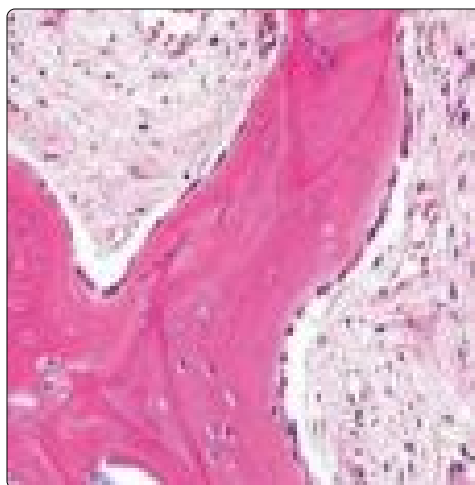


(Left) Following a fracture, it is common to see irregular tongues of woven bone being formed between larger, mature bony trabeculae. This appearance can be alarming and may be confused with an osteosarcoma; however, the osteoblasts in normal woven bone formation are not cytologically abnormal. (Right) Osteoblasts are small, elongated cells with pink cytoplasm and single nuclei. They are readily identified around woven bone but can also be seen around lamellar bone.

Osteoblast Cytology



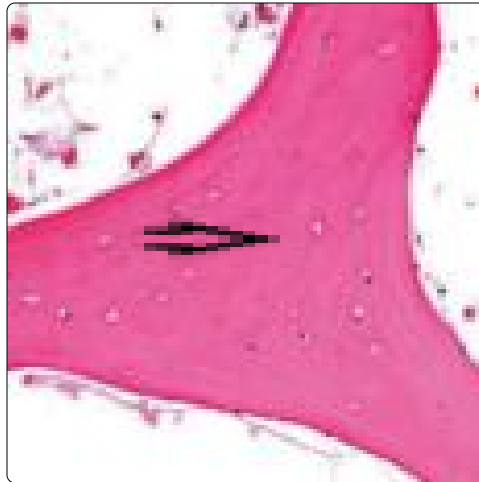
Osteoblastic Lining



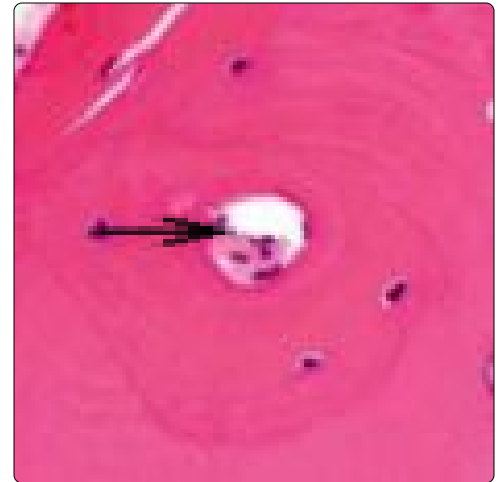
(Left) Osteoblasts characteristically contain a single nucleus that is eccentrically located at one pole of the cell. Note the paler cytoplasm near the nucleus. This region contains the Golgi apparatus. Activated osteoblasts can show 1 or 2 prominent nucleoli within the nucleus. (Right) Although more commonly associated with woven bone, osteoblasts can be found lining lamellar bone. However, this layer is usually more subtle, and the osteoblasts are smaller.

Osteocytes

(Left) Osteocytes are mature osteoblasts that have become enveloped by the bony matrix. Each one resides in a small cavity called a lacuna. Note that not all lacunae contain an osteocyte in any given section, which is often a function of viewing a 3D object in a 2D plane. (Right) Due to the small size of the osteocyte, the nucleus is often the only visible part of the cell. Nuclear detail cannot be appreciated. Note the haversian canal.

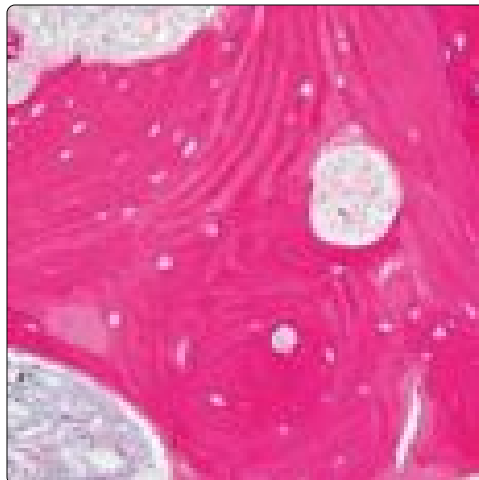


Osteocyte Cytology

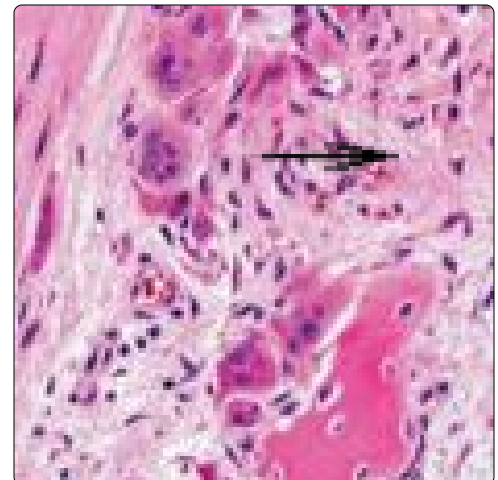


Loss of Osteocytes

(Left) Empty lacunae from loss of osteocytes can be seen in dead (nonviable) bone or as an artifact in otherwise normal bone occurring during specimen procurement. In some cases, overdecalcification of the tissue can lead to pale osteocyte nuclei that are difficult to appreciate. (Right) Osteoclasts are large, multinucleated cells with eosinophilic cytoplasm and variably prominent nucleoli. They are often found clustered together and are responsible for bone resorption. Note the woven (immature) bone.

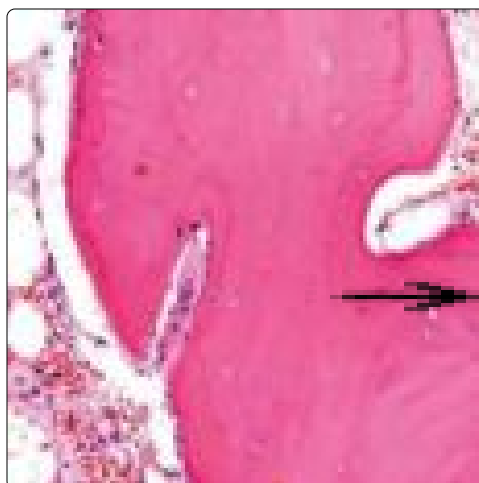


Osteoclasts

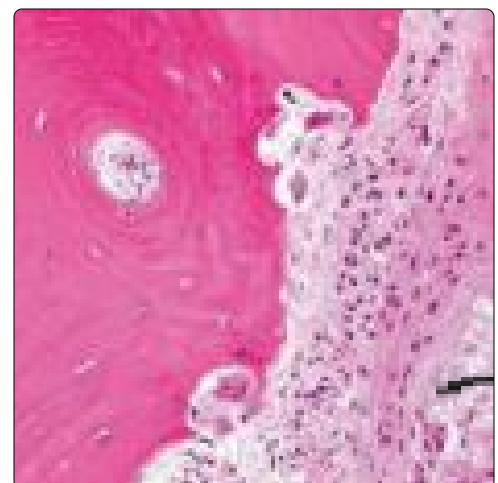


Osteoclast Tunneling

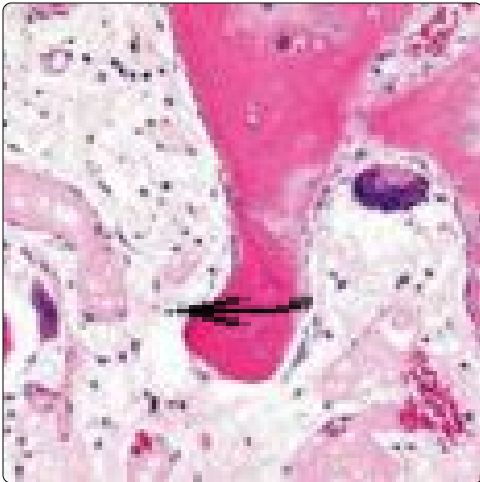
(Left) Osteoclasts can sometimes be seen "chewing" or "burrowing" through bony trabeculae. This tunneling is what ultimately leads to the formation of haversian canals. (Right) During the resorption process, osteoclasts often cluster together in a concave space known as a Howship lacuna. These lacunae form as a result of bony matrix digestion.



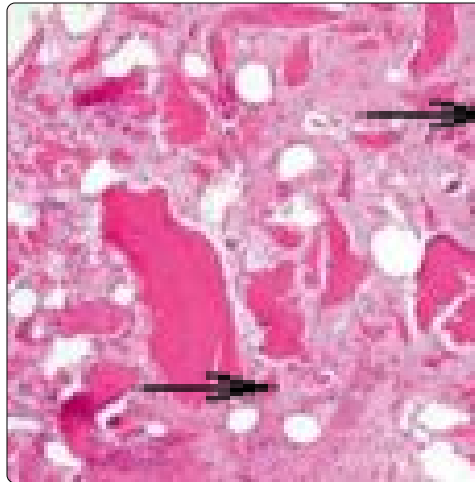
Osteoclastic Resorption



Large Osteoclasts

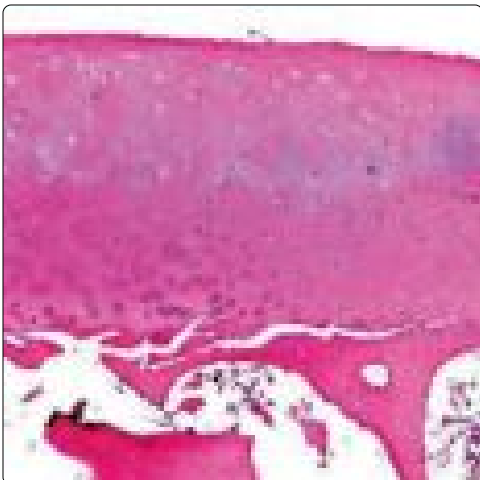


Procedure-Related Disruption

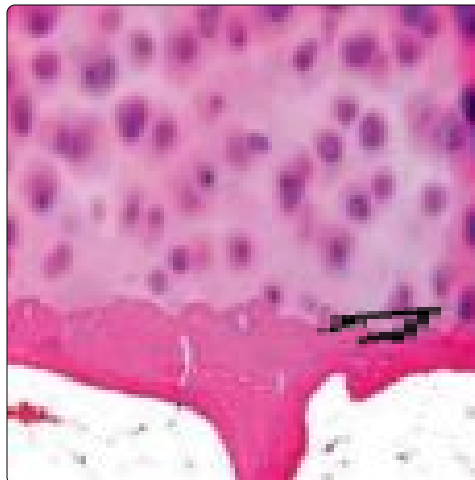


(Left) Some osteoclasts can grow quite large and feature a substantial number of nuclei. Note the smaller, more typical example. (Right) Bone specimens are often procured through curettage or scraping/scooping of tissue. As a result, small, irregular fragments of bone are commonly seen in these specimens.

Articular Cartilage

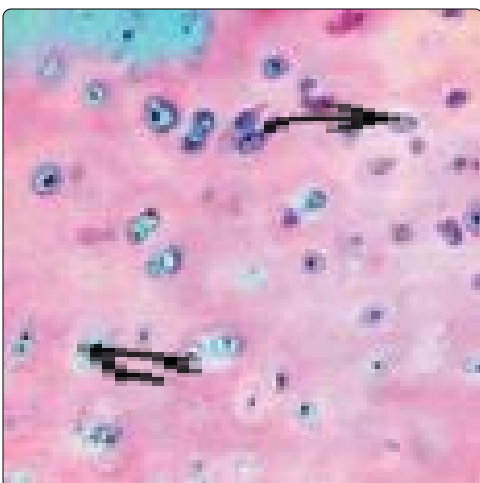


Tidemark

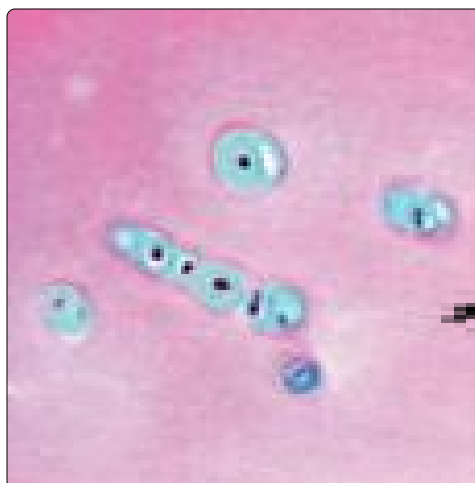


(Left) Hyaline cartilage is the most common type of cartilage in the human body. It is found on the articular (joint) surfaces of all long bones. It has a glassy appearance and ranges from blue to pink in color, depending on stain preparations. (Right) At the base of the articular cartilage is a conspicuous undulating line known as a tidemark. The tidemark separates unmineralized articular cartilage from the underlying subchondral lamellar bone. It is a remnant of a previously active growth plate (physis).

Chondrocytes



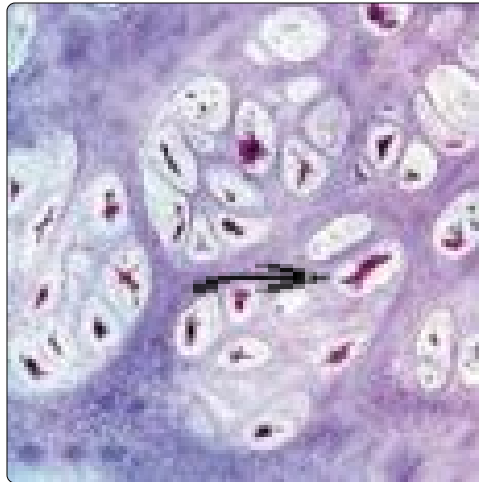
Chondrocyte Cytology



(Left) Chondrocytes are found within spaces known as lacunae and are often evenly distributed throughout the extracellular matrix of hyaline cartilage. At times, an occasional lacuna may appear to contain more than 1 chondrocyte. (Right) Chondrocytes are typically small and cytologically bland. In fact, they often appear much smaller than the lacunae in which they reside. They can be seen in clusters, as single cells, or in linear arrays.

Stellate Chondrocytes

(Left) This image shows clusters of chondrocytes. Sometimes chondrocytes can appear larger and more stellate than usual (a processing artifact). However, even though the cytoplasm is larger, the nuclei are still small and bland. **(Right)** Hyaline cartilage characteristically features a prominent glassy extracellular matrix. Variations in color and staining are common in cartilage, such as in this image, where the matrix around the chondrocytes stains darkly.

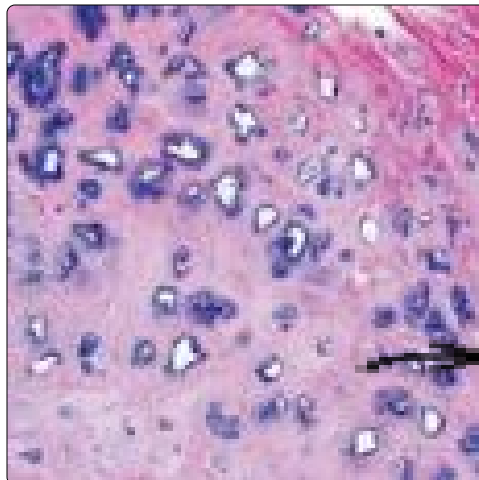


Hyaline Cartilage

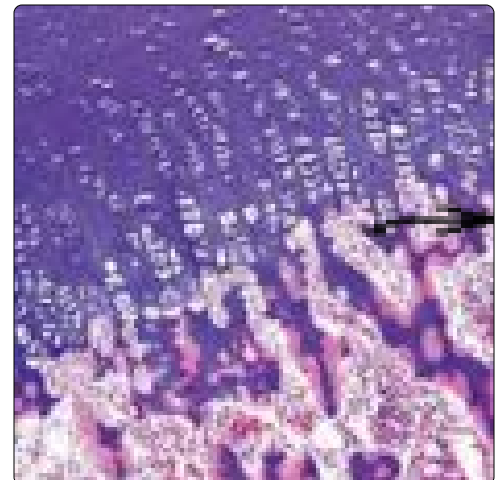


Chondrocyte Drop-Out

(Left) Empty lacunae from loss of chondrocytes are rare and are likely due to tissue disruption. **(Right)** At the normal growth plate (also known as the physis), bone is produced upon cartilage (enchondral ossification) and results in the longitudinal growth of bone. At the interface between the cartilage and bone formation, chondrocytes are classically arranged in columns.

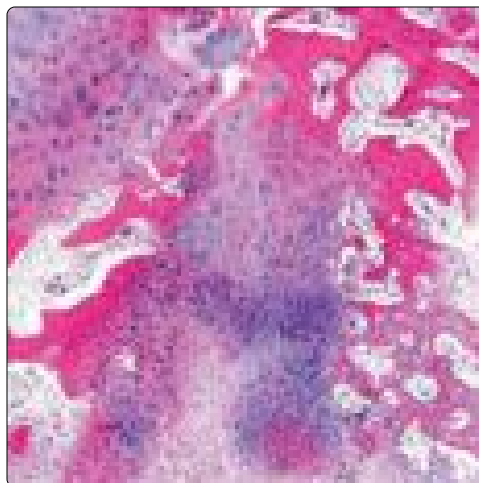


Growth Plate (Physis)

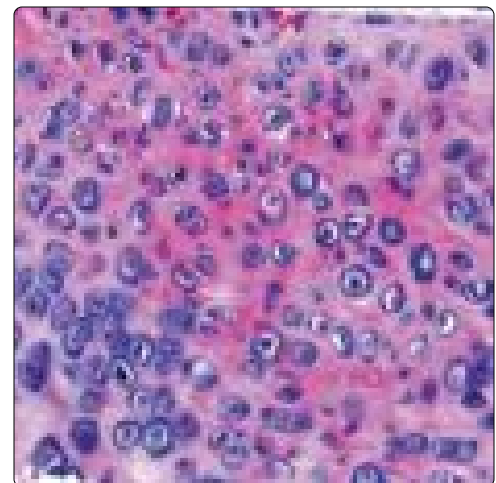


Fracture Callus

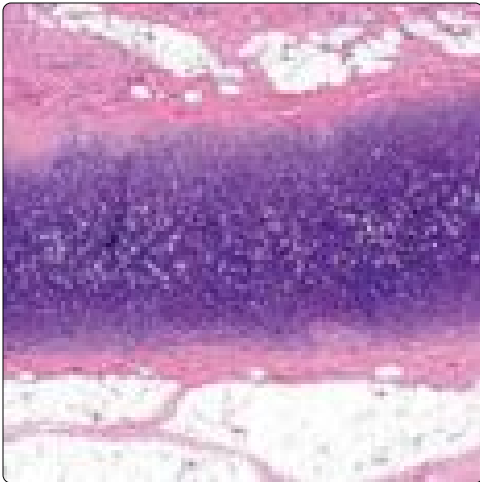
(Left) A fracture callus is formed to repair breaks or disruptions in bone and often aligns along the line of fracture. Irregular nodules of cellular cartilage are commonly seen within the callus as well as woven (immature) bone formation and a fibroblastic proliferation. **(Right)** The cellularity of the cartilage within a fracture callus may be alarming and should not be mistaken for chondrosarcoma.



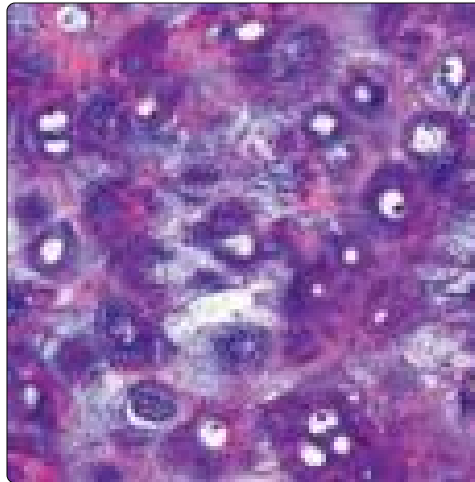
Fracture Callus Cartilage



Elastic Cartilage

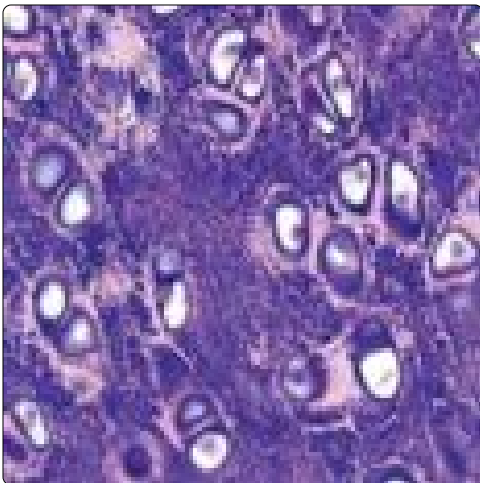


Elastin Fibers

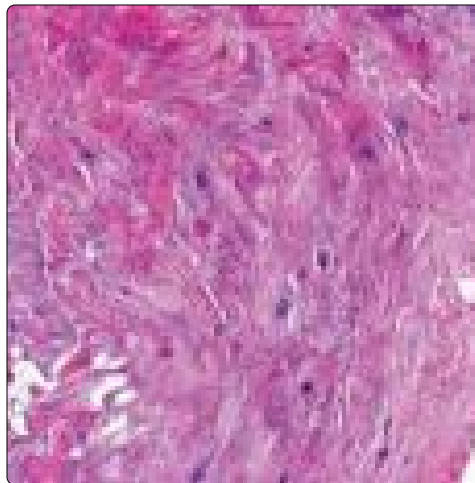


(Left) Elastic cartilage is much less common than hyaline cartilage and is characteristically found within the external ear (shown), epiglottis, and ligamentum flavum. **(Right)** Elastic cartilage contains an abundance of elastin fibers within its matrix, lending greater elasticity to the tissue. These fibers  may or may not be identifiable on H&E and in some cases require a special elastin histochemical stain.

Elastin Fibers

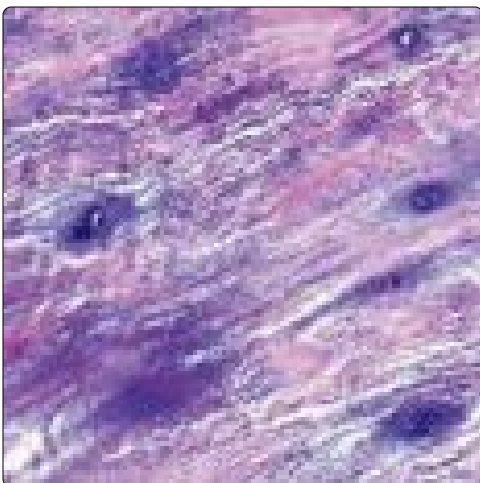


Fibrocartilage

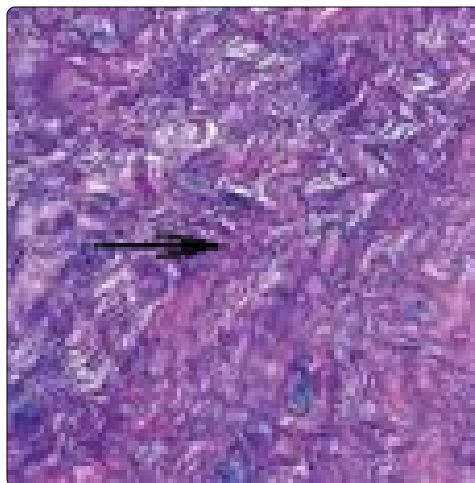


(Left) This H&E stain of elastic cartilage shows elastic fibers appearing as "cracks" within the extracellular matrix. **(Right)** Fibrocartilage is also much less common than hyaline cartilage and is found in the intervertebral discs (annulus fibrosus), pubic symphysis, menisci, as well as tendinous and ligamentous insertion points.

Abundant Collagenous Matrix



Wavy Collagen Fibers



(Left) Fibrocartilage features scattered chondrocytes  within a matrix of abundant collagen fibers, often arranged in parallel arrays . The collagenous matrix may be loose or dense. **(Right)** In some cases, the arrays of collagen fibers in fibrocartilage may appear conspicuously wavy, somewhat resembling tendon tissue. Note the sparsely scattered chondrocytes .

Synovial Membrane

SYNOVIAL MEMBRANE

Macroscopic Anatomy

- Lines inner surface of joints; also tendon sheaths and bursae
- Smooth, glistening appearance
 - Occasional small folds or projections (villi)
- Functions of synovial cells
 - Secrete synovial fluid
 - Rich in hyaluronate
 - Phagocytosis
 - Lubrication and nourishment of cartilage

Microscopic Anatomy

- Intima
 - Simple lining of closely packed cells (synoviocytes) with elongated nuclei
 - Often 1-2 cells in thickness
 - 2 types of lining cells have been defined by electron microscopy and IHC
 - Type A synoviocytes are related to macrophages
 - IHC: CD68(+), CD163(+), NSE(+)
 - Type B synoviocytes are related to fibroblasts
 - Predominant cell type
 - Produce hyaluronate
 - IHC: CD55(+); (-) for CD68, CD163, and keratins
 - Not true epithelial lining
 - No basal lamina or basement membrane
- Subintima
 - Loose extracellular matrix
 - Fibroblasts and collagen
 - Rich capillary vasculature network and lymphatics
 - Adipose tissue (variable)
 - Small nerves
 - Few inflammatory cells
 - Usually macrophages and mast cells
 - Rare lymphocytes (mostly T cells)

Electron Microscopy

- Type A intimal cells

- Minor cell type (25% of cells)
- Macrophage lineage
- Cytoplasmic vacuoles and vesicles
- Golgi apparatus
- Type B intimal cells
 - Major cell type (75% of cells)
 - Fibroblastic lineage
 - Abundant rough endoplasmic reticulum
 - Pinocytic vesicles
- Cells with features of both type A and type B have been described as intermediate cells

CONSIDERATIONS AND VARIATIONS

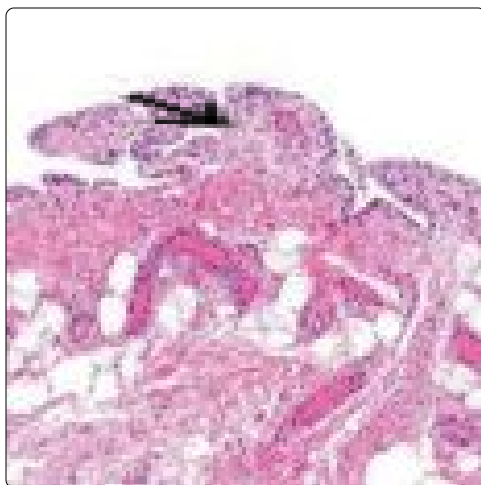
Hyperplasia

- Common in setting of inflammatory arthritides
 - Rheumatoid arthritis, ankylosing spondylitis
- Disproportionate increase in number of macrophages vs. fibroblastic cells
- Often associated with increased subintimal inflammation and fibrosis

Pitfalls and Artifacts

- Intimal layer may not be visible on histologic section
- Oblique sectioning of synovial lining may resemble glands or ducts
- Normal adipose tissue may mimic synovial lipomatosis
 - Arthroscopic demonstration of intraarticular proliferation in true lipomatosis
- Reactive synovium may be confused for pigmented villonodular synovitis (diffuse-type giant cell tumor)
 - Absence of nodular growth of epithelioid tumor cells in reactive setting
- Thickened reactive synovium can resemble true epithelial lining

Synovial Membrane

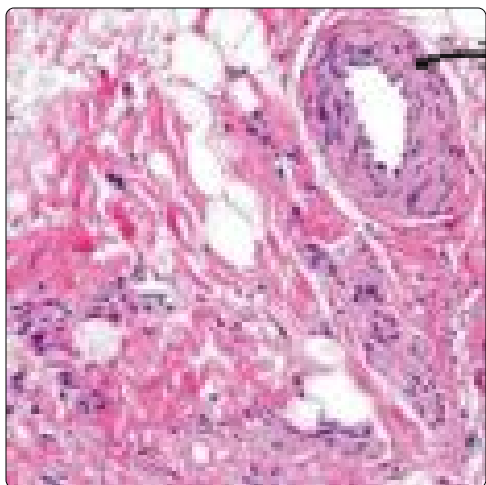


Intima

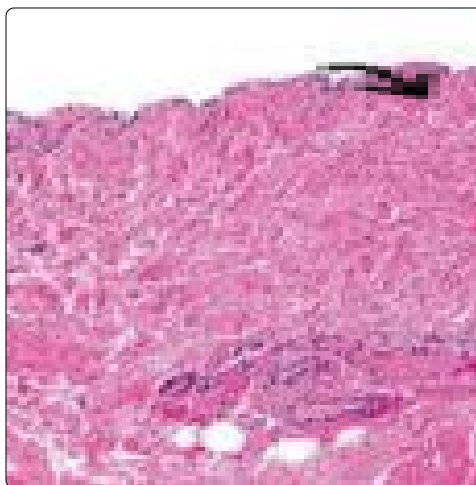


(Left) Synovium is a thin glistening layer of connective tissue that lines joint cavities, bursa, and tendon sheath. It is composed of 2 layers: The superficial intima and underlying subintima. The intima is often flat and smooth but can also feature small papillary projections known as villi. **(Right)** The very thin intimal layer of the synovial membrane is composed of a mixture of cells (synoviocytes) that feature properties of either fibroblasts or macrophages. This layer is generally inconspicuous in normal, nonreactive tissue.

Subintima

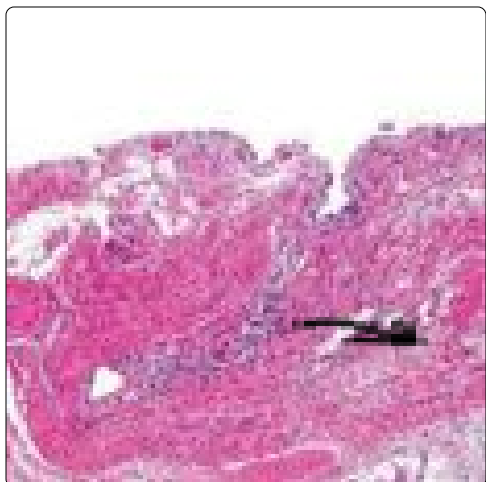


Fibrous Synovial Membrane

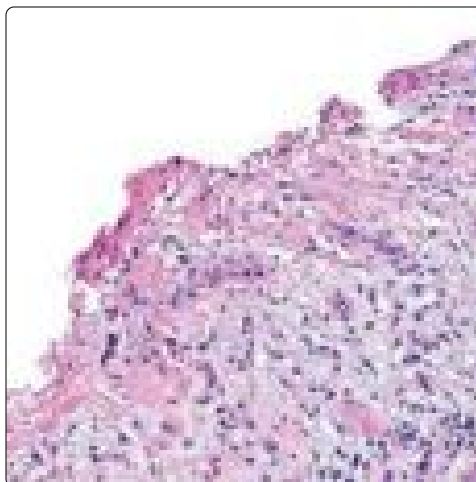


(Left) The subintimal layer is more abundant than the overlying intima. It is generally composed of loose connective tissue and contains a variable mixture of collagen, adipose tissue, and small nerve fibers. A rich vascular network is also identified. (Right) In some sites, the connective tissue of the synovial subintimal is denser and more fibrocollagenous, as depicted, and is not considered pathologic. Note the thin intimal layer at the surface.

Oblique Sectioning Artifact

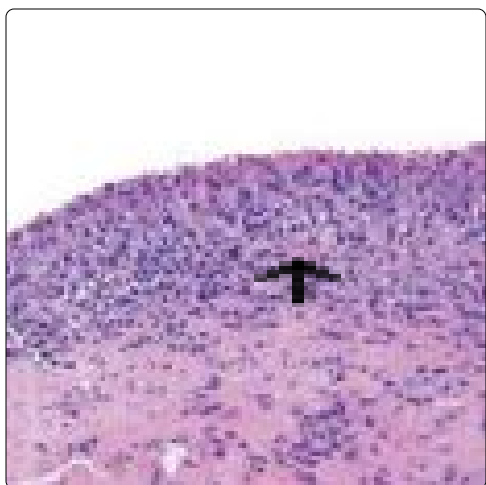


Absence of Intimal Layer

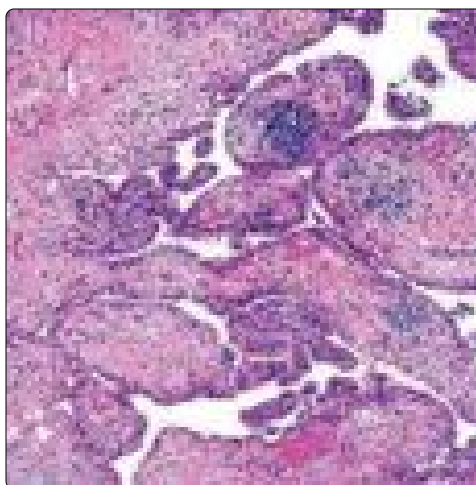


(Left) Oblique or tangential sectioning of synovial tissue may lead to the misidentification of normal surface intima as epithelial elements, such as glands or ducts, particularly in patients with a history of carcinoma. (Right) The intima may not always be identifiable in histologic sections, particularly in tissue taken from a site of trauma or prior surgical procedure. This artifact can make it difficult to identify the tissue as synovium.

Reactive Synovium



Reactive Papillary Hyperplasia



(Left) The synovial intima often appears dramatically thickened in reactive conditions, such as rheumatoid arthritis. More of the cells are macrophages. Resemblance to an epithelial lining or even a neoplastic process is more likely in this setting. (Right) Synovial reactive changes can also take the form of papillary hyperplasia, which can be striking, as depicted. Unlike pigmented villonodular synovitis, however, nodules and sheets of epithelioid tumor cells do not fill the papillae.

Connective Tissue

CONNECTIVE TISSUE**Microscopic Anatomy**

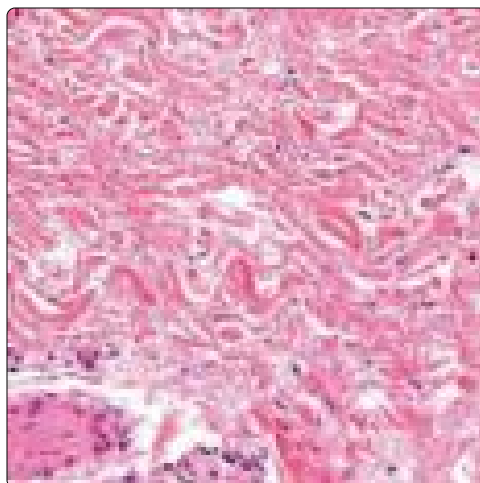
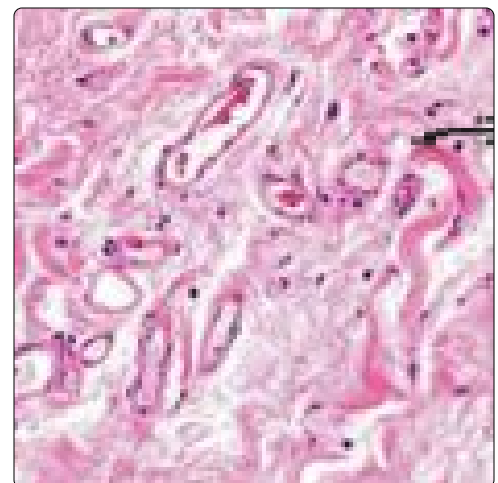
- Variable composition consisting of fibroblasts/myofibroblasts, collagen fibers, elastic fibers, and scant mononuclear inflammatory cells
 - Fibroblasts and myofibroblasts
 - Spindled to stellate cells
 - Amphophilic to eosinophilic cytoplasm
 - Single elongated nucleus
 - Small nuclear indentation or "notch" often present
 - May show tiny intranuclear pseudoinclusions
 - 1 or 2 small nucleoli
 - Often prominent in reactive states
 - Myofibroblasts more commonly seen in setting of tissue injury, repair, and response
 - Collagen
 - Eosinophilic, acellular fibrillar bundles
 - Variable thickness and distribution
 - Fibers may be abundant and dense with few associated fibroblasts
 - Elastic fibers
 - Very thin, small fibers
 - Provide tissue flexibility
 - Elastin histochemical stain needed to properly visualize
- Exact composition of connective tissue depends upon site and physiologic state
 - Often also contains vessels, nerves, &/or fat
- Variations in certain conditions
 - Granulation tissue
 - Reactive myofibroblasts
 - Prominent capillary proliferation, fibrin, and acute and chronic inflammatory cells
 - Myxoid change
 - Decreased cellularity with light blue tinge to stroma
 - Desmoplasia
 - Myofibroblasts within loose blue-gray matrix
 - Usually stromal response to invasive neoplasm
- Fibroblasts and myofibroblasts are morphologically indistinguishable and require immunohistochemistry or electron microscopy
 - Immunohistochemistry
 - Myofibroblasts express smooth muscle actin in characteristic wispy, peripheral cytoplasmic (tram-track) pattern
 - Focal desmin expression can also be seen
 - Vimentin (+) but extremely nonspecific
 - Electron microscopy
 - Most useful in making distinction between fibroblasts and myofibroblasts
 - Fibroblast
 - Well-developed Golgi apparatus and rough endoplasmic reticulum
 - Small number of microfilaments
 - No intercellular junctions
 - Fibronexus absent
 - Myofibroblast
 - Numerous bundles of microfilaments (stress fibers) and dense bodies
 - Well-developed fibronexus
 - Intercellular intermediate (adherens) junctions and gap junctions
 - Nuclear indentations
 - Collagen
 - Extracellular fibrils with periodicity of 64 nm

CONSIDERATIONS AND VARIATIONS**Pitfalls and Artifacts**

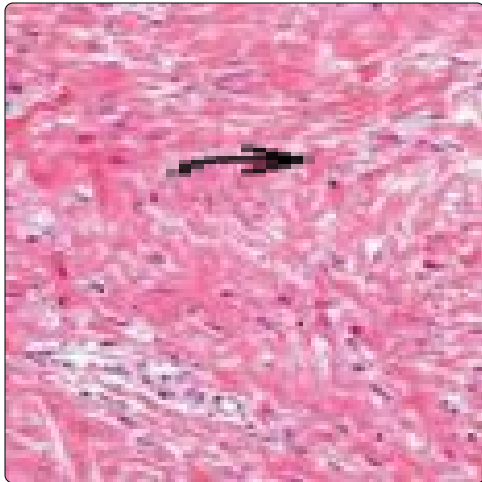
- Cracking artifact in areas of dense, thick collagen bundles
- Reactive myofibroblasts often have prominent nucleoli
- Exuberant granulation tissue can mimic neoplasm
 - Reactive myofibroblasts can show marked atypia
- Myofibroblasts can show high level of mitotic activity in reactive states
 - Mitotic figures are not abnormal

Collagenous Connective Tissue

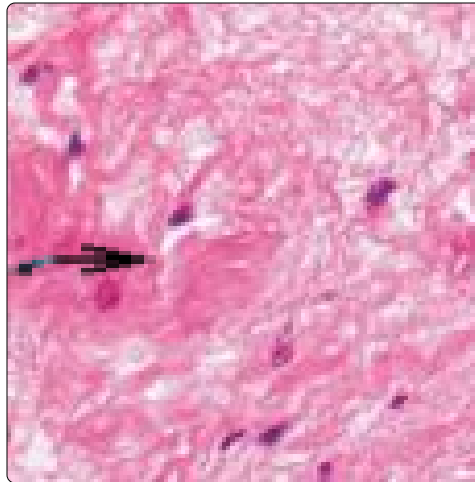
(Left) The composition of connective tissue varies depending upon the site and physiologic state of the tissue. This H&E shows dense connective tissue with an abundance of pink collagen  and scattered myofibroblasts identifiable by their dark nuclei . **(Right)** This H&E shows loose connective tissue with more extracellular matrix between the collagen fibers . Note the myofibroblastic nuclei  and blood vessels .

**Loose Connective Tissue**

Dense Connective Tissue

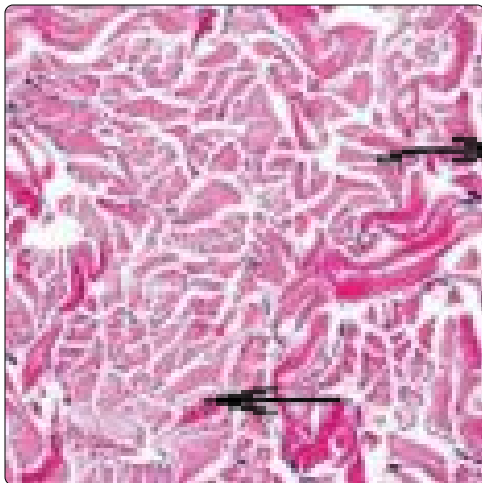


Thin and Thick Collagen Fibers

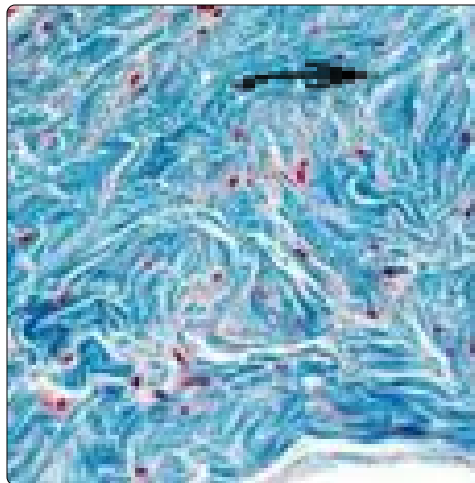


(Left) Collagen is an eosinophilic acellular protein that constitutes a variable proportion of the connective tissue or stroma. Upon closer inspection, the individual fibers that make up the collagen bundles are often wavy. **Note the myofibroblasts.** **(Right)** Collagen fibers are often fine and wispy but also regularly form bundles of varying thickness. A wavy morphology to the fibers is also common.

Collagen Cracking Artifact

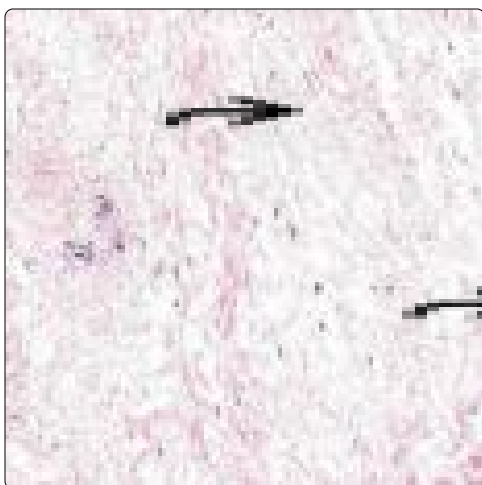


Trichrome Stain

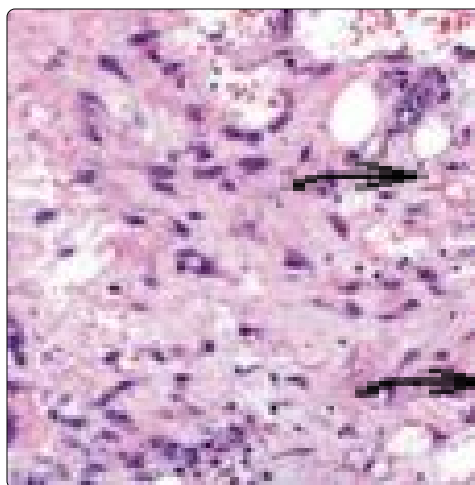


(Left) In some regions of the body, such as the reticular dermis of the back, the collagen fibers are thickened, densely packed, and brightly eosinophilic. Note the artifactual appearance of "cracks" or clear spaces between the fibers. Fibroblast nuclei are sparse. **(Right)** A standard trichrome histochemical stain highlights collagen in blue, which can aid in the visualization of fibrosis (thickening or scarring of tissues). Note the fibroblast nuclei, which stain red.

Edematous Connective Tissue



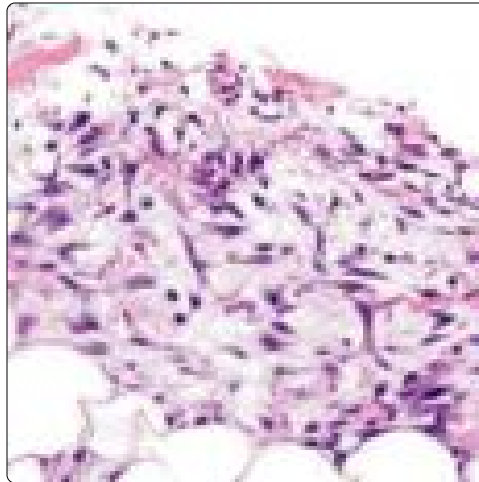
Fibroblasts and Myofibroblasts



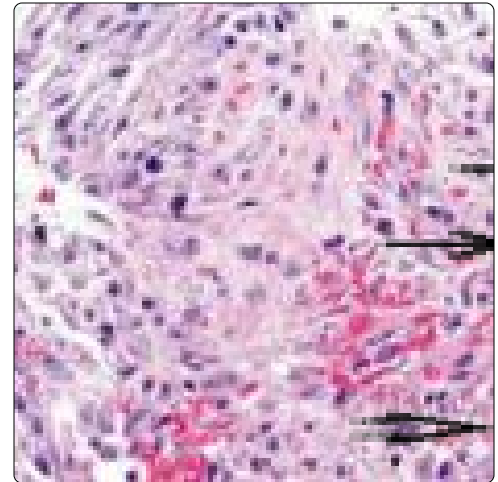
(Left) Lymphovascular obstruction can lead to increased fluid within the extracellular connective tissue matrix, making the tissue appear pale and hypocellular. Note the splayed out collagen fibers. **(Right)** Both fibroblasts and myofibroblasts often show amphophilic, spindled to stellate cytoplasm and prominent nuclei. In reactive conditions, these cells can show more eosinophilic cytoplasm and prominent central nucleoli. Without immunohistochemical stains or electron microscopy, these 2 cells are indistinguishable.

Reactive Myofibroblasts

(Left) This H&E shows reactive myofibroblasts in a traumatized tissue containing fat. Note the spindled to stellate shape of the cells as well as the loose, myxoid stroma. **(Right)** Myofibroblasts often demonstrate 1 or a few central nucleoli, which may be prominent in exuberant reactive states. This appearance can look atypical and may result in misidentification as a neoplastic process. Note the extravasated blood cells, a common finding in reactive states.

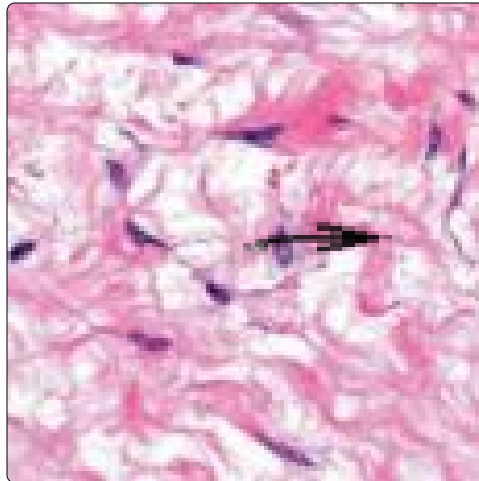


Reactive Myofibroblasts

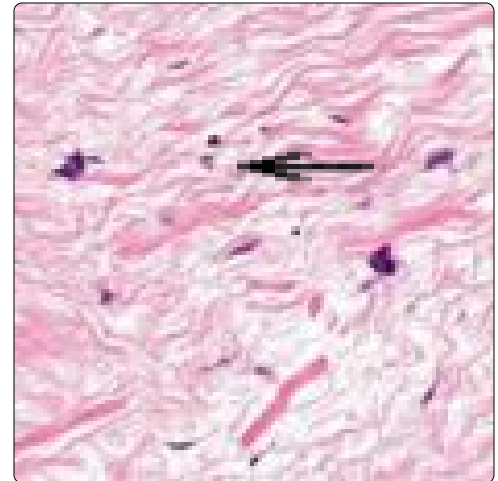


Intranuclear Pseudoinclusions

(Left) Fibroblasts may feature small, round nuclear pseudoinclusions, which represent focal intranuclear invaginations of the cytoplasm. A small "notch" or irregularity in the nuclear membrane is also a common finding. **(Right)** Multinucleated fibroblasts may be seen in nonneoplastic and edematous tissues, such as in this H&E from a case of massive localized lymphedema. However, they may also be encountered in true neoplasms.

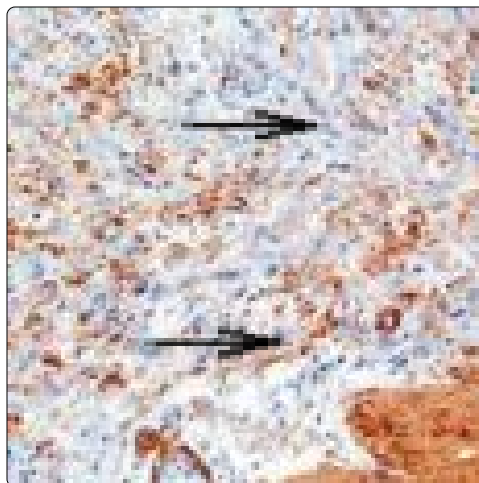


Multinucleation



Myofibroblastic Immunophenotype

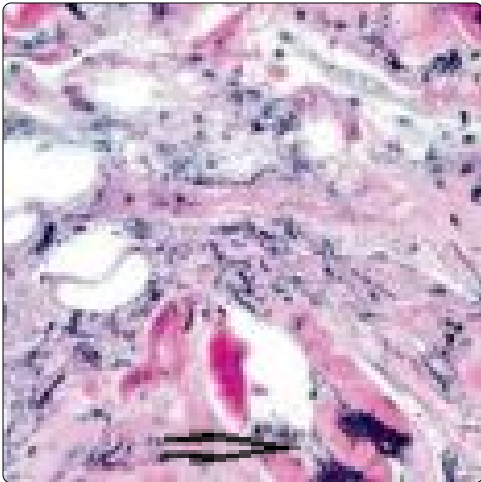
(Left) Myofibroblasts can be distinguished from fibroblasts by smooth muscle actin, which is characteristically expressed in a fine peripheral cytoplasmic distribution in myofibroblasts. Compare to diffuse cytoplasmic staining in true smooth muscle. **(Right)** This transmission electron microscope image shows collagen fibers that have been produced by fibroblasts within the extracellular connective tissue matrix.



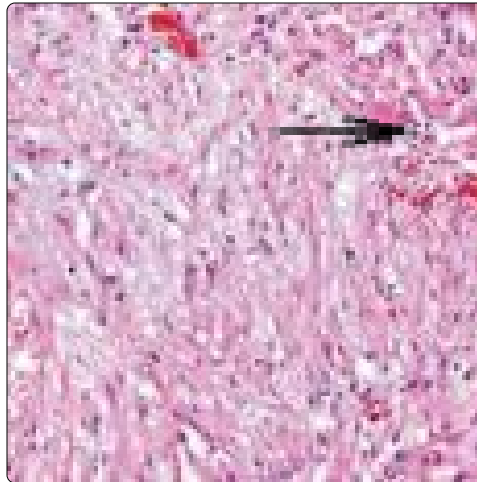
Collagen Fibers (Electron Microscopy)



Elastic Fibers

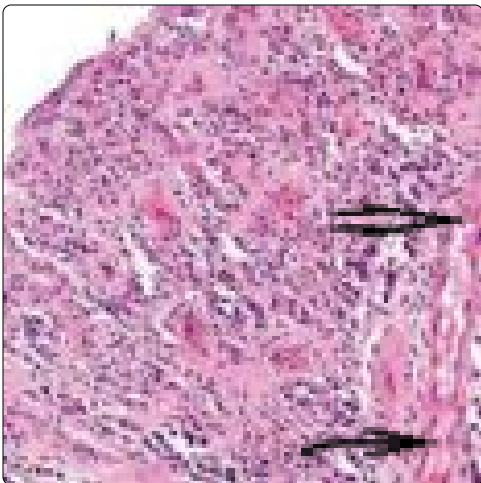


Myxoid Stromal Change

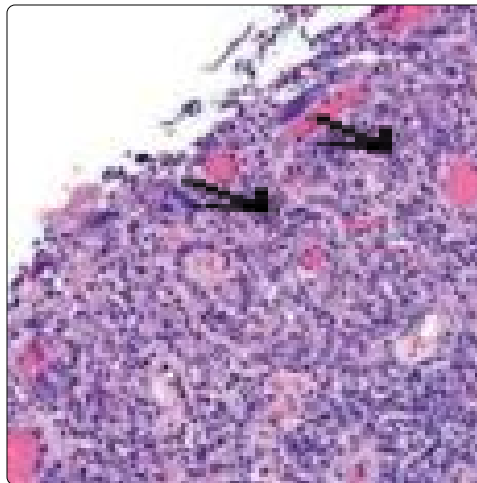


(Left) Wavy elastic fibers are often difficult to see with the naked eye, but they can be highlighted by an elastin-specific histochemical stain (elastic fibers appear black). These fibers provide flexibility and support to the connective tissue. (Right) Myxoid stromal change can be seen in either reactive or neoplastic states and imparts a light blue or gray tinge to the connective tissue, sometimes with a bubbly appearance. Note the plump reactive myofibroblasts.

Granulation Tissue

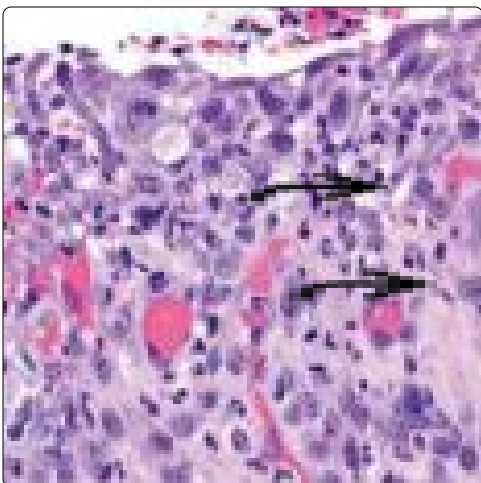


Myofibroblast Atypia

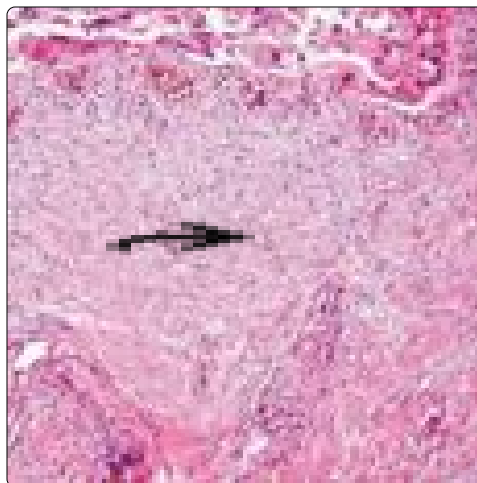


(Left) Granulation tissue is often seen in areas of ulceration. It is composed of a reactive capillary proliferation associated with numerous acute and chronic inflammatory cells and scattered reactive myofibroblasts. (Right) Reactive myofibroblasts can show prominent nuclear atypia, particularly within the setting of epithelial ulceration with underlying granulation tissue formation. Out of context, the atypia may suggest malignancy.

Mitotic Activity



Desmoplasia



(Left) Normal mitoses are commonly seen in reactive myofibroblasts (particularly granulation tissue) and should not immediately suggest a neoplastic process. Bizarre or abnormal forms, however, are generally absent. (Right) Desmoplasia is a reactive stromal (host) myofibroblastic proliferation in response to invasive epithelial malignancy (carcinoma). It often has a blue or pink myxoid appearance and contains a cellular population of myofibroblasts.

Adipose Tissue

ADIPOSE TISSUE

Macroscopic Anatomy

- White fat
 - Yellow soft tissue
 - Lobulated by fine fibrous septa
 - Widely distributed
 - Mainly subcutis, intraabdominal sites
 - Component of several organs in health or disease
 - Has insulation, energy storage, and endocrine function; produces leptin, adiponectin, resistin, and cytokines
 - Can replace normal muscle in pathological processes
- Brown fat
 - Pale brown lobulated tissue
 - In neck, mediastinum, paravertebral, paraaortic, and other retroperitoneal locations
 - Cells are thermogenic; uncoupling protein on sympathetic stimulation uncouples fatty oxidation from ATP production, producing heat
 - Subset in brown fat areas derive from MYF5(+) precursors, possibly sharing origin with muscle cells
 - MYF5(-) brown adipocytes occur in white fat

Microscopic Anatomy

- White fat
 - Lobular architecture with fatty lobules separated by thin collagenous septa with blood vessels
 - Mature adipocyte is polygonal spherical
 - Intranuclear vacuole (Lochkern) can be present
 - Cytoplasm forms very thin peripheral layer
 - Fatty vacuole fills most of cell
 - Mitotic activity absent
 - Nuclei and cytoplasm are S100 protein (+)
 - Blood vessels scanty
- Brown fat
 - Lobular architecture
 - Cells smaller than mature white adipocytes
 - Nucleus rounded, often centrally located
 - Variable numbers of lipid vacuoles

- Cytoplasm between vacuoles is granular due to numerous mitochondria
- Mitotic activity absent
- S100 protein (+) and CD31(+)
- More vascular than white fat and has nerves
- Univacuolated white adipocytes can be admixed

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- White fat
 - Cells enlarge in first 6 months of life
 - Number of cells increases in childhood
 - Number and size of cell increases during puberty
- Brown fat
 - Maximal in childhood
 - Gradual shrinkage and loss with age with replacement by white fat
 - Persists mostly in neck, mediastinum, and around aorta

Hyperplasia

- Size of fat cells increases (hypertrophy) in obesity
- Number of fat cells increases (hyperplasia) in obesity
 - Mitotic activity not seen
- Number of fat cells can increase in fatty infiltration of organs

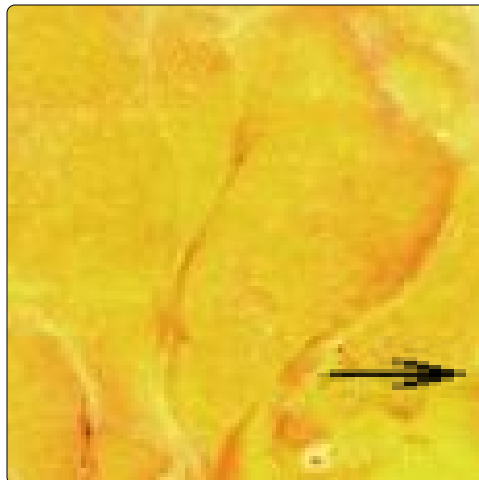
Metaplasia

- Fat can replace variety of normal tissues with advancing age and pathologic conditions
 - Thymus, parathyroid, pancreas, liver
 - Bone marrow, lymph nodes
 - Skeletal muscle, heart, cardiac valve

Pitfalls and Artifacts

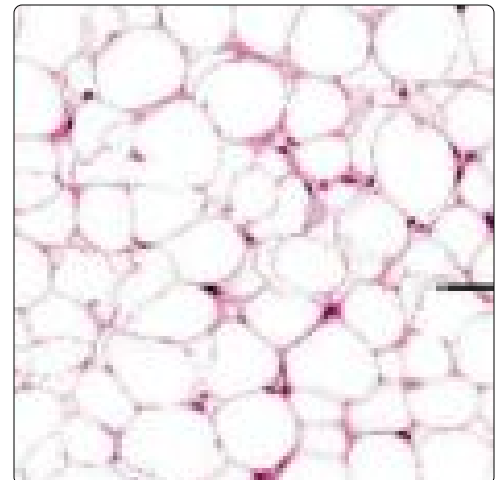
- Intranuclear vacuole (Lochkern) can mimic lipoblasts
- Fat atrophy
- Fat necrosis

Yellow Fat

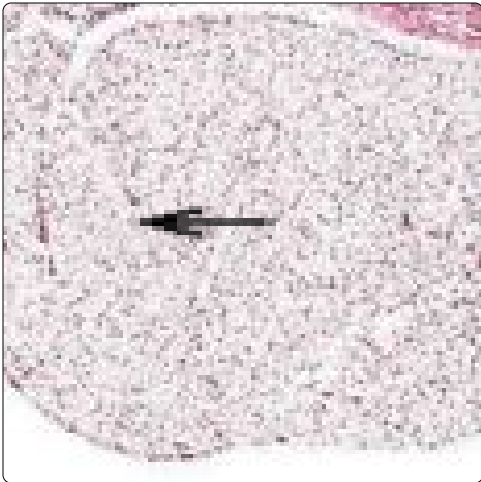


(Left) Normal subcutaneous adipose tissue is lobulated by slender fibrous septa . There is a finely nodular pattern, and the fat is largely homogeneous in appearance and color. **(Right)** Lipocytes vary in size and appear empty because lipid is removed during processing. Nuclei  are inconspicuous; peripheral cell membranes are sharply defined. Some cells have a thin layer of cytoplasm.

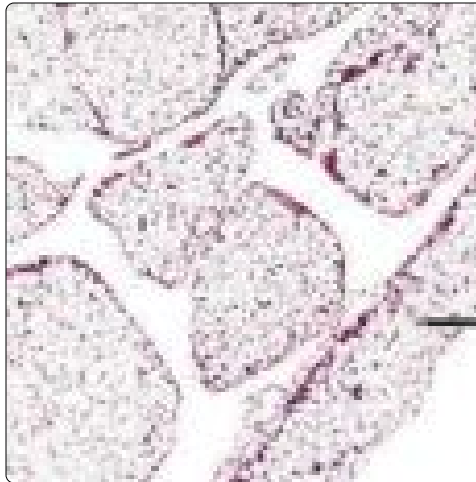
Adipocytes



Normal White Adipose Tissue

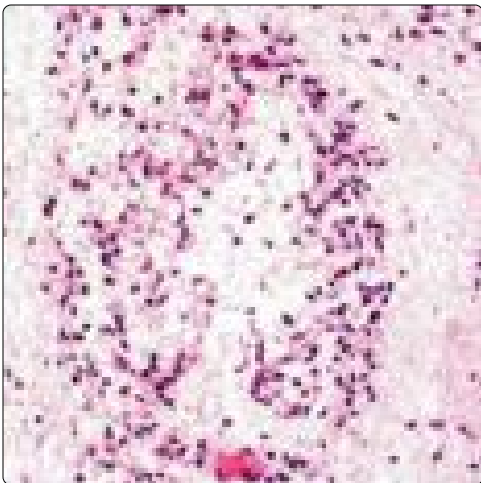


White Fat in Omentum

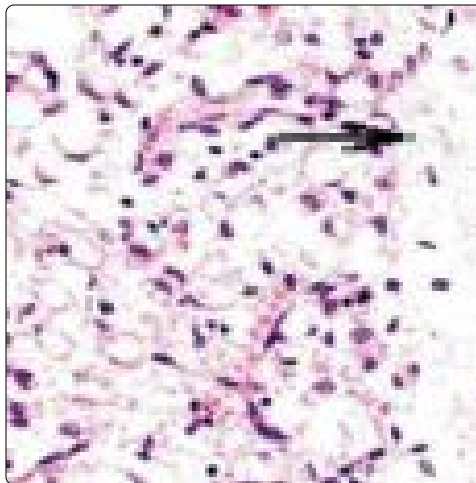


(Left) Normal white fat in the subcutis forms lobules divided by delicate fibrovascular septa. The adipocytes and nuclei are uniform. Nuclear atypia is not seen in normal fat and if seen on low magnification suggests an atypical lipomatous tumor. Fat is relatively hypovascular, but occasional small-caliber blood vessels are seen [1]. (Right) The omentum is one of the storage sites of normal white fat. It forms variably sized lobules within the mesothelial surface layer [2].

Fetal Fat

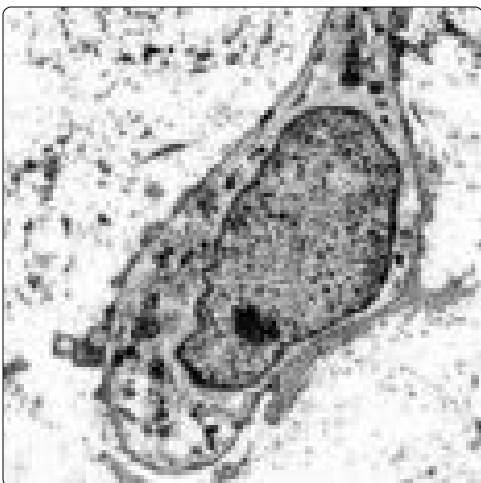


Fetal Fat



(Left) Primordial fat lobules consist of myxoid stroma and pericyte-derived, S100(+) spindle cells [1]. These adipocyte precursors (preadipocytes) condense around lobules of proliferating capillaries [2] by the 14th gestational week. (Right) Multilocular adipocytes predominate at 1st and then become unilocular. Continued proliferation results in vacuolated fat cells [3] within a rich capillary network [4]. At 24 weeks, the lobules are enclosed within fibrous septa.

Fetal Fat (Electron Micrograph)



S100 Protein in White Fat



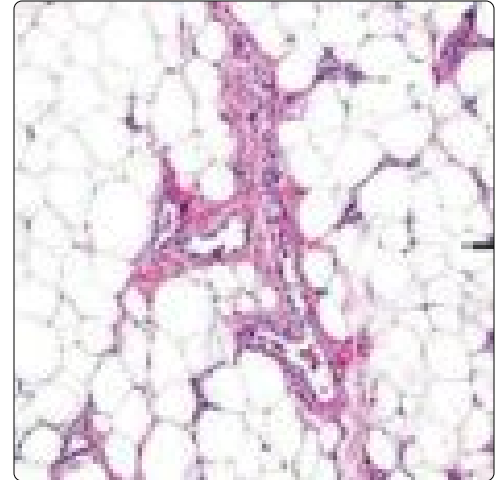
(Left) Preadipocytes are spindle cells with incomplete basement membranes [1]. As the cell matures, small paranuclear lipid droplets [2] appear, enlarge, and coalesce as a vacuole. CHOP gene controls differentiation of fibroblasts into adipocytes. (Right) Immunostaining for S100 protein is strong and diffuse in adipocytic nuclei [3] and also highlights the rim of cytoplasm [4] in some cells. Demonstration of this antigen can sometimes be useful in the diagnosis of liposarcoma.

Lochkern

(Left) Normal adipocytes in white fat can have 1 or more rounded, clear vacuoles within the nucleus, termed Lochkern. These cells should not be mistaken for lipoblasts, which have cytoplasmic vacuoles that push the nucleus to 1 side and indent it with a scalloped appearance. **(Right)** Scattered blood vessels are commonly seen within white fat and serve to provide nutrients to the tissue. Note that these vessels are unlike the capillary channels with thrombi seen in angiolipoma.

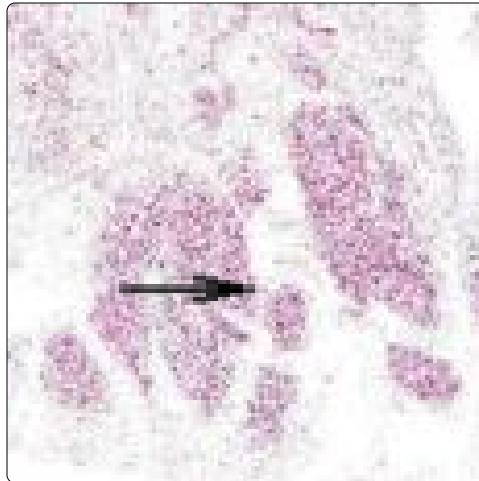


Vasculature

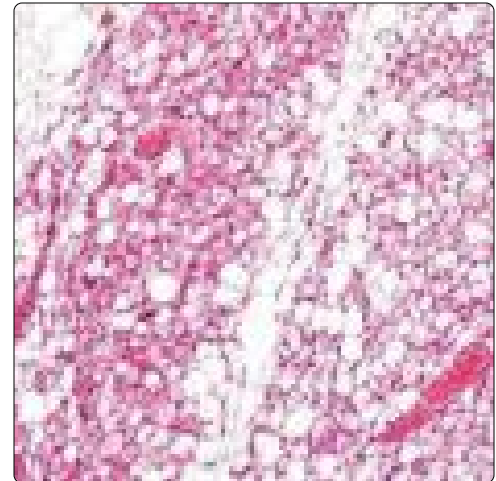


Brown Fat

(Left) In adults, brown fat exists as small islands in supraclavicular, mediastinal, and retroperitoneal sites, especially in the paraaortic, paravertebral, and periaxillary areas. The variably sized lobules are separated by loose fibrous tissue. **(Right)** There are variable numbers of brown fat cells and white fat cells showing mild variation in diameter. The 2 types are admixed within the cellular lobules. Thin-walled blood vessels are congested but sparse.

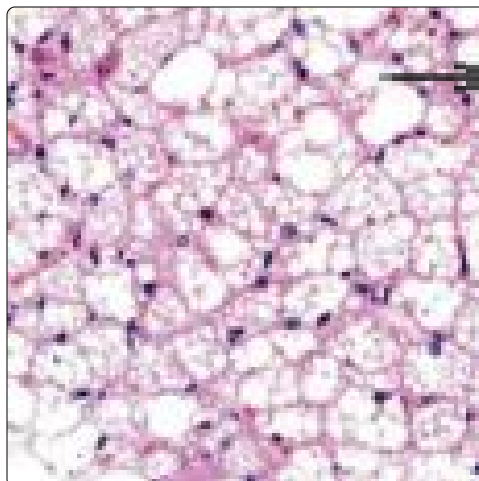


Brown Fat Cells

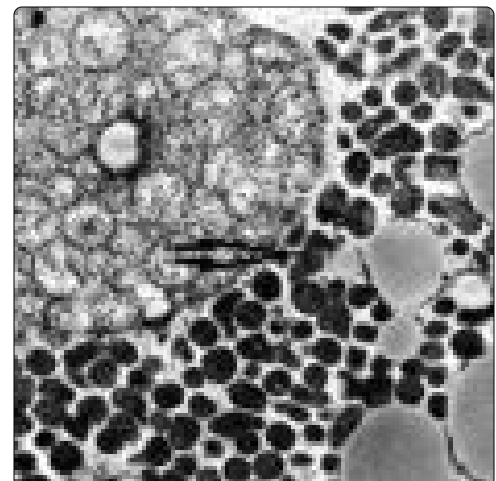


Multivacuolated Cells

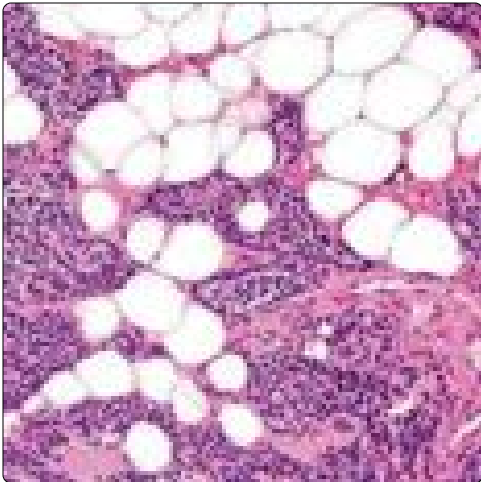
(Left) Brown fat is composed of closely packed rounded cells with numerous small intracytoplasmic vacuoles. The nucleus is rounded and located centrally or eccentrically. Nuclear atypia and mitoses are absent. An occasional white fat cell is seen. **(Right)** An electron micrograph of the cytoplasm in a brown fat cell in a hibernoma shows numerous mitochondria displaying prominent transverse cristae as well as electron-dense forms and lipid droplets.



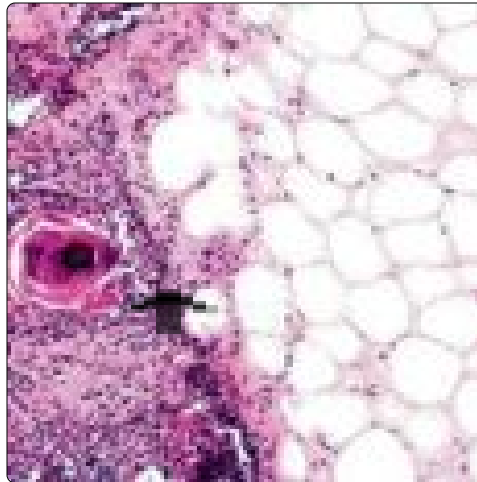
Brown Fat (Electron Micrograph)



Fatty Replacement in Parathyroid Gland

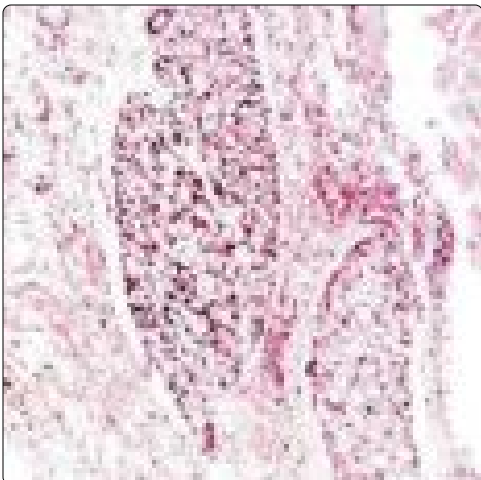


Fatty Replacement in Thymus

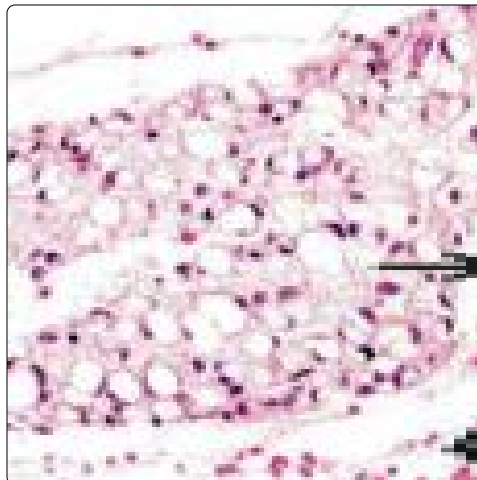


(Left) Islands of parathyroid tissue [] are admixed with fat. With advancing age, parathyroid glandular tissue is gradually replaced by normal adipose tissue. Fatty replacement usually reaches a maximum between 40 and 60 years of age. The amount of fat can be different in each of the 4 parathyroid glands. (Right) With increasing age, the thymus undergoes involution, a process in which the gland is replaced by normal white adipose tissue. Residual thymic tissue, including a Hassall corpuscle [], is seen.

Atrophy of White Fat

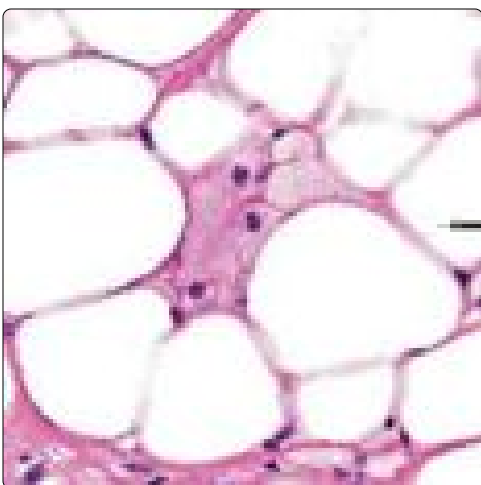


Atrophy of White Fat

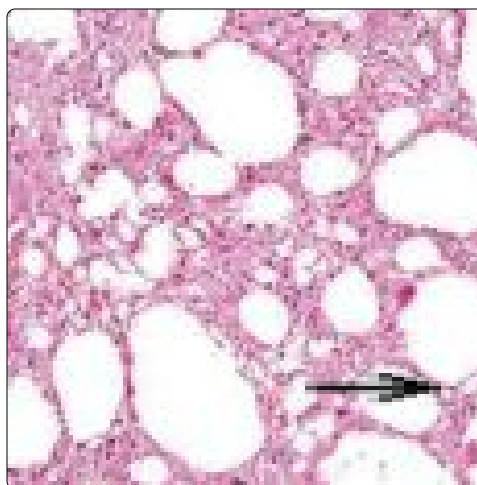


(Left) White adipose tissue can undergo atrophy in starvation and severe systemic illness. The fatty tissue shrinks so that lobules [] become smaller and discrete. The intervening tissue shows myxoid change. (Right) Individual adipocytic cells are smaller than normal [] due to reduction of intracytoplasmic lipid, resulting in a more prominent appearance of nuclei []. The lobular configuration is maintained, and nuclear atypia is absent. Lipofuscin pigment is sometimes seen.

Fat Necrosis



Fat Necrosis



(Left) In fat necrosis, fatty spaces become enlarged when the cell membranes break down and macrophages phagocytose the escaped lipid substance, resulting in "foamy" cells with finely vacuolated cytoplasm []. (Right) In later stages, there is inflammation, fibrosis, and focal calcification. No significant nuclear atypia is seen, which helps to exclude sclerosing and inflammatory variants of well-differentiated liposarcoma. Note the large empty spaces [] due to coalescence of cells.

Skeletal Muscle

SKELETAL MUSCLE

Macroscopic Anatomy

- Over 400 distinct skeletal muscles in human body
 - Skeletal muscle is voluntary (can be controlled willfully)
 - Wide range in size, shape, and function
 - Types include fusiform (parallel), simple pennate, bipennate
- With few exceptions, skeletal muscles connect to bone via tendinous insertions

Microscopic Anatomy

- Muscles are composed of bundles of fibers within groups called fascicles
 - Each fiber is elongated tube with polygonal cross section 25-90 μm in diameter
 - Contains myofibrils, composed of actin (thin) and myosin (thick) myofilaments
 - Fibers are larger in male patients and in proximal muscles
 - 4-6 nuclei per cell (cross section); peripheral beneath sarcolemma
 - 2 fiber types (can be distinguished by histochemistry)
 - Type 1 (aerobic)
 - a.k.a. slow-twitch or red fibers
 - Sustained activity
 - High oxidative enzyme and lipid content
 - Low glycogen and phosphorylase content
 - Type 2 (anaerobic)
 - a.k.a. fast-twitch or white fibers
 - Rapid, unsustained activity
 - Low oxidative enzyme and lipid content
 - High glycogen and phosphorylase content
- Cytoplasm has cross striations on light microscopy
 - Striations due to alignment of actin (thin) and myosin (thick) filaments and related structures
 - Highlighted by immunostaining for desmin
 - A band: Dark thick and thin filaments; has lighter H zone and central M line linked to myosin
 - I band: Light thin filaments; has Z bands (discs) delineating sarcomere

- Muscles are microanatomically compartmentalized by sheaths of connective tissue
 - Endomysium: Thin fibrous layer around single fibers
 - Perimysium: Encloses several fibers to form fascicles
 - Contains nerves and blood vessels
 - Epimysium: Encloses entire muscle
- Special histochemical stains for evaluation of muscle
 - ATPase (at varying pH), esterase
 - Gomori modified trichrome
 - NADH-TR, cytochrome oxidase (COX), succinate dehydrogenase (SDH)
 - PAS, Oil red O, phosphorylase
- Tendons join most skeletal muscles to bone
 - Dense collagen, often wavy in longitudinal sections
 - At interface with tendon, muscle fibers often show variation in size and shape
- Function of skeletal muscle
 - In contraction, calcium binds to troponin, exposing actin site that interacts with myosin head
 - Thick and thin filaments slide past each other, contracting myofibril
 - Adenosine triphosphate releases myosin from actin, relaxing muscle

VARIATIONS AND CONSIDERATIONS

Age-Related Variations

- Muscle fibers are smaller in children and elderly
- Loss of muscle volume with age (partly neurogenic)

Hyperplasia

- Muscle fibers can enlarge in width after exercise

Metaplasia

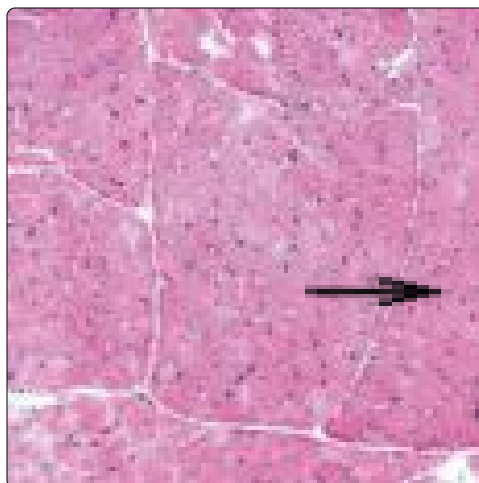
- Adipose tissue can replace muscle fibers

Pitfalls and Artifacts

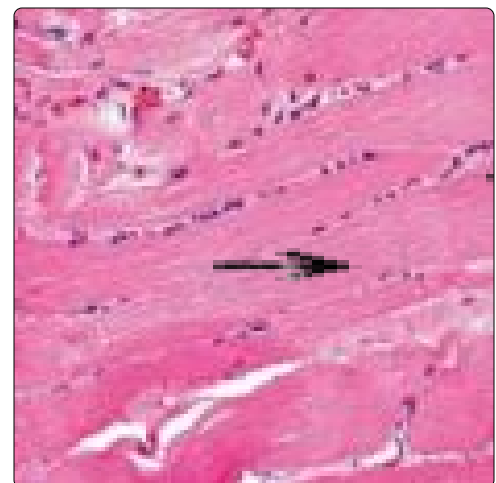
- Regenerating or atrophic muscle fibers
- Vacuolar sarcoplasmic change
- Contraction, wrinkling, folding artifacts

(Left) All skeletal muscles are composed of fascicles containing muscle fibers. These units can be delineated by the different connective tissue sheaths: Endomysium  (around individual muscle fibers), perimysium  (around fascicles), and epimysium (around whole muscle). **(Right)** Skeletal muscle fibers  are long and tube-shaped and contain multiple nuclei  that are characteristically located subadjacent to the sarcolemmal membrane (periphery of the cell).

Skeletal Muscle



Longitudinal Section of Skeletal Muscle



Transverse Section of Skeletal Muscle



Cytoplasmic Cross Striations



(Left) In transverse (cross) section, there is little variation in skeletal muscle fiber diameter. This image also demonstrates the characteristic peripherally located nuclei [X]. Endomysium [Y] is located around individual muscle fibers. **(Right)** On H&E stain, cytoplasmic cross striations [Z] can often be appreciated in skeletal muscle, especially in longitudinal section. These striations represent the overlapping actin and myosin myofilaments within sarcomeres.

Cross Striations (Desmin)



Cross Striations (Desmin)

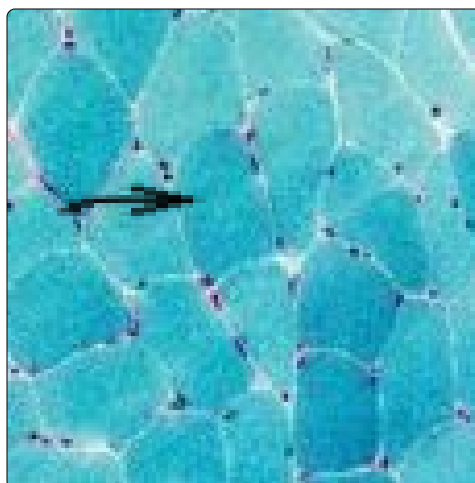


(Left) Immunostaining for desmin nicely shows cross striations [X] spaced regularly within each skeletal muscle fiber. **(Right)** Another desmin immunostain shows striations in longitudinal section. This feature is diagnostic of skeletal muscle and can also be seen on H&E stain, special stains such as phosphotungstic acid hematoxylin and periodic acid-Schiff (PAS), and p63 immunohistochemistry.

Tendon



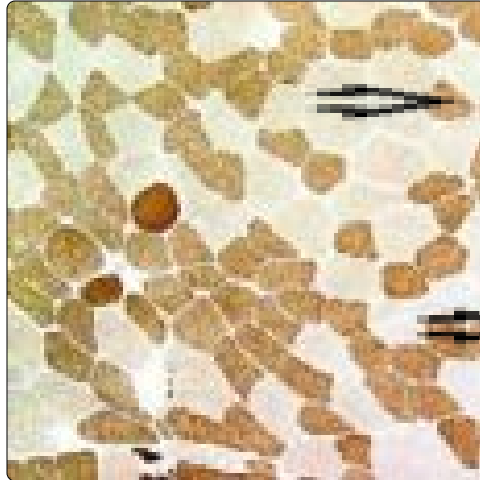
Modified Gomori Trichrome



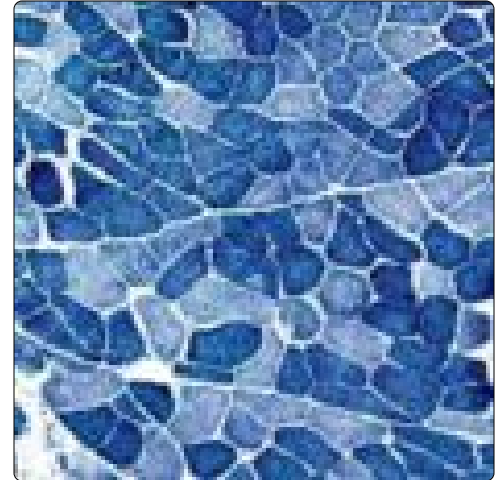
(Left) Most skeletal muscles attach to bone via tendons, which are composed of dense collagenous fibroconnective tissue. As shown here, the collagen is arranged in parallel arrays and often shows a prominent wavy contour. **(Right)** On modified Gomori trichrome stain, mitochondria within skeletal muscle fibers appear as tiny red dots [X] and are typically accentuated [Y] under the sarcolemmal membrane at the periphery of each individual fiber.

ATPase Reaction

(Left) ATPase enzymatic stains can be utilized to distinguish type 1 fibers from type 2 fibers and evaluate their distribution. When conducted at a pH of 9.4, type 2 fibers show dark staining and type 1 fibers show pale staining. (Right) Similar to other enzymatic stains, NADH-TR can distinguish between type 1 and type 2 fibers. It does so by highlighting intracellular mitochondria, which are more numerous in type 1 fibers and thus stain darkly.

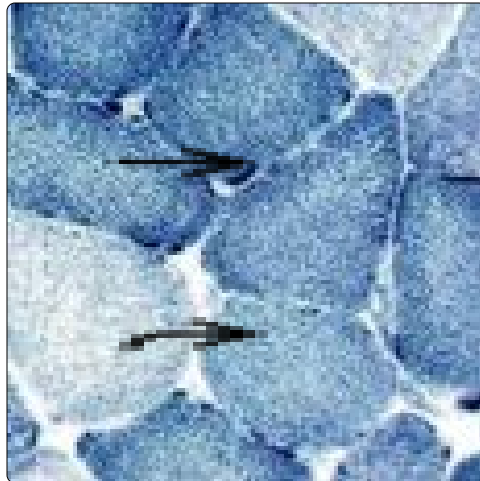


NADH-Tetrazolium Reductase Stain

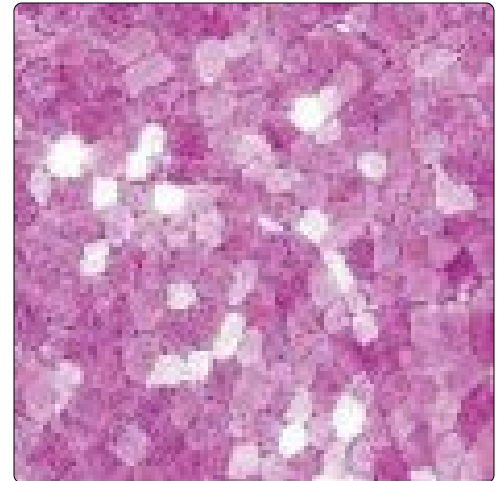


NADH-Tetrazolium Reductase Stain

(Left) At high magnification, NADH-TR stain highlights both mitochondria and the sarcoplasmic reticulum of the muscle fibers. Type 1 fibers have more mitochondria than type 2 fibers. (Right) As in other tissues, a PAS stain can also be utilized in sections of skeletal muscle to demonstrate glycogen. Type 2 fibers have a higher glycogen content than type 1 fibers and therefore show darker staining.



Periodic Acid-Schiff (PAS)



Perimysium and Epimysium

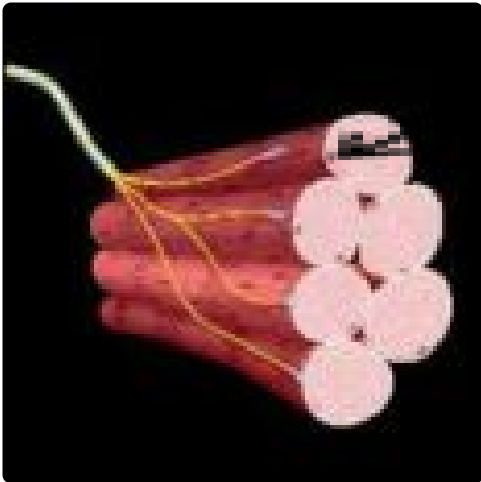
(Left) The endomysium, a thin layer of collagen stained green in this trichrome preparation, encloses individual muscle fibers. Several fibers enclosed within a thicker layer of connective tissue, termed perimysium, form a fascicle. The fibrous layer enclosing the entire muscle is termed epimysium. (Right) The perimysium encloses several muscle fibers and also contains intramuscular nerve fibers and blood vessels. The connective tissue stains green in this modified trichrome preparation.



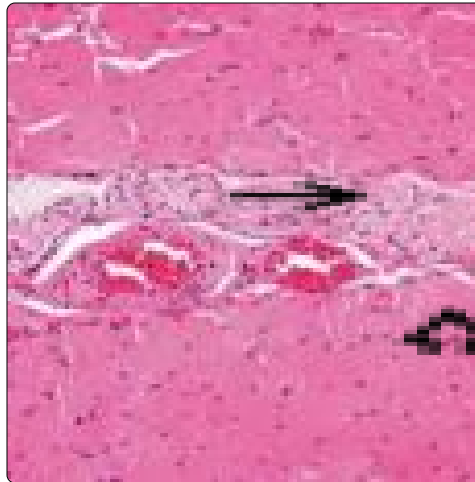
Perimysium



Innervation of Skeletal Muscle



Innervation of Skeletal Muscle

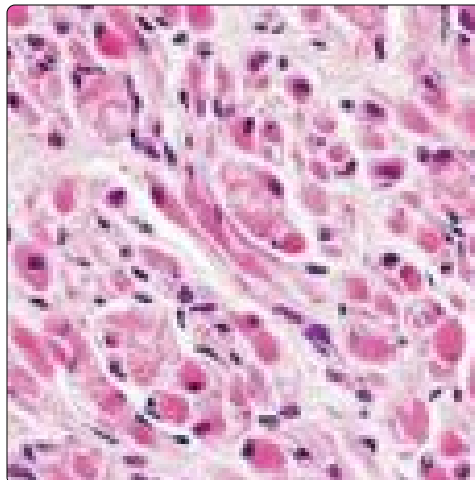


(Left) Nerves branch [] to supply individual muscle fibers. A motor unit, the functional unit of muscle contraction, comprises a motor neuron and all of the skeletal muscle fibers it innervates. The size of the motor unit relates to the muscle's functional requirements. (Right) A nerve branch [], along with an artery [] and a vein [], lies between muscle fascicles [] in the connective tissue forming the perimysium. Mechanoreceptors are also located in this plane.

Skeletal Muscle Fiber Atrophy

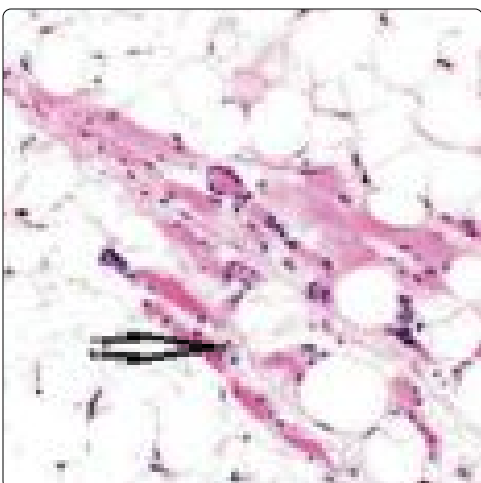


Atrophic Muscle Cells

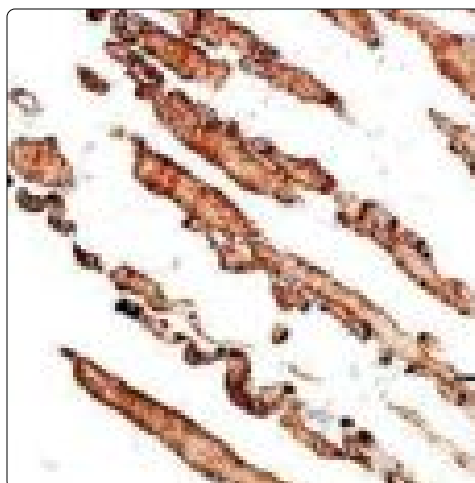


(Left) Atrophy in skeletal muscle fibers is represented by a significant decrease in size relative to normal muscle fibers of the region. In addition to a decrease in size, nuclei [] often overlap and appear hyperchromatic. (Right) This skeletal muscle has shrunk muscle fibers with hyperchromatic nuclei (some central) and eosinophilic cytoplasm. There is interstitial fibrosis [] that can result in scarring.

Fatty Replacement of Muscle



Fatty Replacement of Muscle

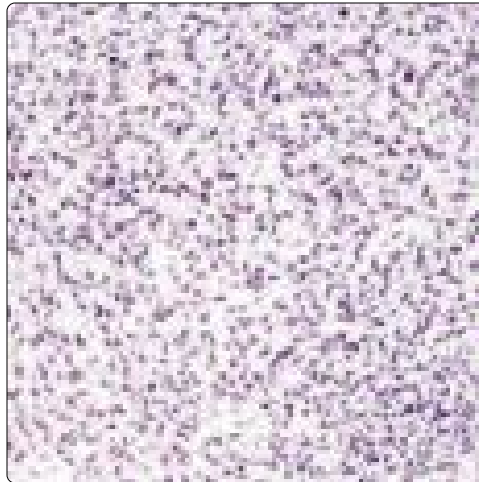


(Left) Atrophic muscle is often replaced by fat. This skeletal muscle, located in the lower limb close to a degenerative hip joint, was atrophied due to inactivity. The skeletal muscle fibers have been extensively replaced by mature fatty tissue []. The residual muscle fibers are smaller than normal and exhibit condensation of nuclei []. (Right) Immunohistochemistry for desmin shows atrophic muscle fibers of variable diameter but mostly thinner than normal and sometimes wavy []. The usual pattern of cross striations is indistinct.

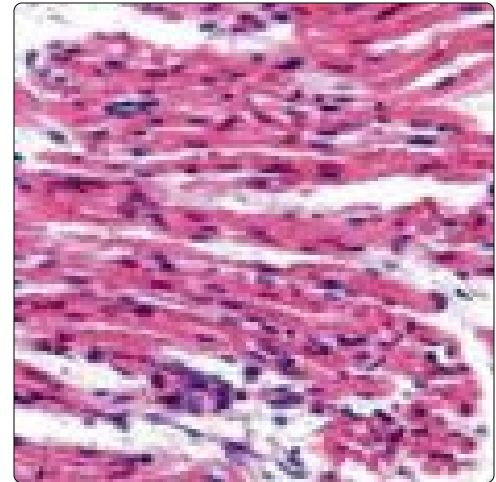
Skeletal Muscle

Embryonic Skeletal Muscle

(Left) During embryogenesis, skeletal muscle develops within the paraxial mesoderm of somites as a proliferation of primitive spindle and stellate mesenchymal cells within myxoid stroma. Myogenesis is controlled in stages by myogenic regulatory factors, including MYOD1, myogenin, and pax-7. (Right) Myoblasts proliferate and then form into multinucleated myotubes with central and later peripheral nuclei. Myofilaments appear in the 9th week, then develop cross striations.

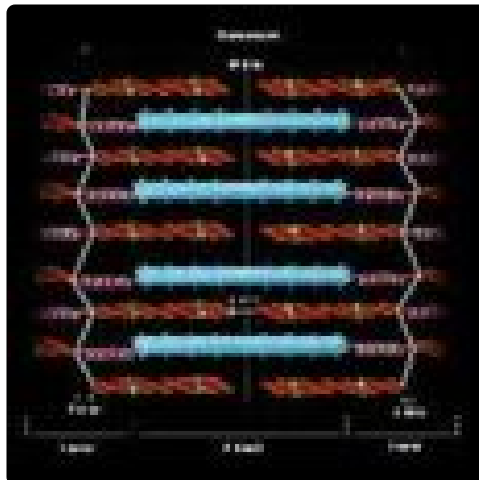


Fetal Skeletal Muscle

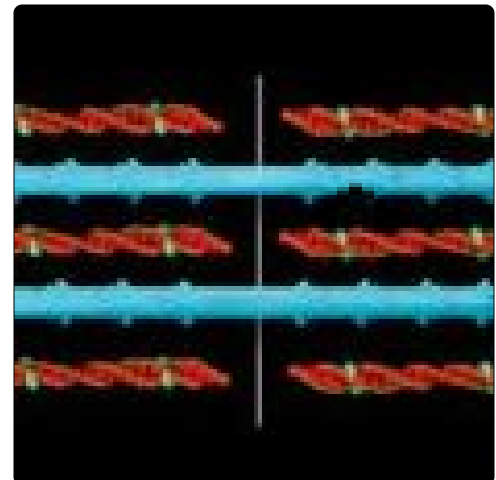


Sarcomere Structure

(Left) The basic contractile unit of the skeletal muscle fibers is the sarcomere, defined as the length between adjacent Z discs. The red bars denote the actin (thin) contractile proteins. The blue bars denote myosin (thick). In light microscopy, A bands are darker than I bands. The lightest area in the A band is the H zone. The center of each H zone is a dark M line. Z discs bisect the I bands. (Right) This graphic shows an A band that is dark because both thin (actin) and thick (myosin) myofilaments overlap.

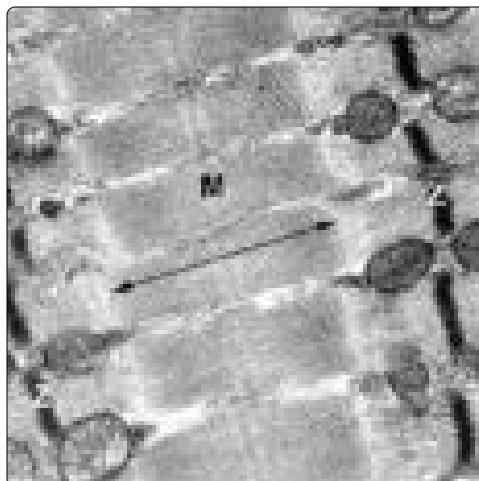


Structure of Sarcomeric A Band

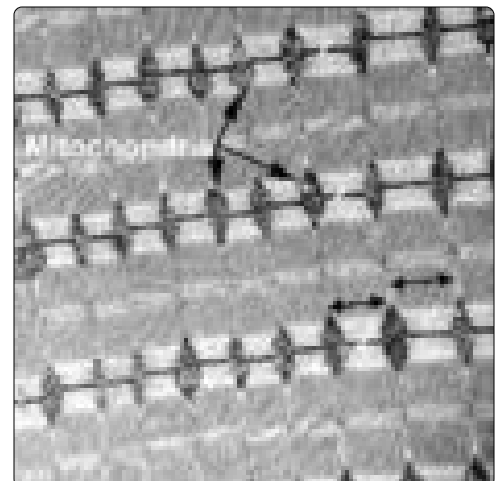


Ultrastructure of Sarcomere

(Left) A sarcomere is bordered by 2 Z discs (Z). The M line (M) is a thin midline band that anchors myosin filaments. The double arrow marks the width of an A band. (Courtesy P. Pytel, MD.) (Right) This electron micrograph shows several sarcomeres arranged in consistent bands. The sarcomere is the area between adjacent Z discs (Z). Double-pointed arrows mark the diameter of myofibrils. Mitochondria lie between myofibrils. (Courtesy P. Pytel, MD.)



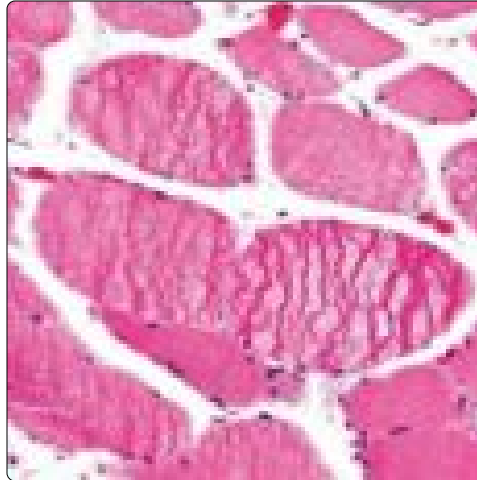
Ultrastructure



Wavy Fibers

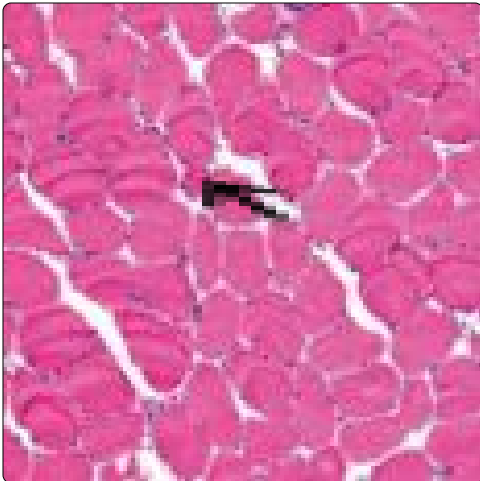


Contraction Artifact

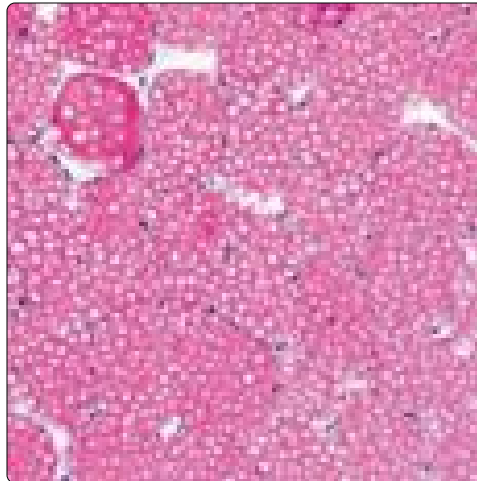


(Left) In this image, skeletal muscle fibers show a wavy contour, likely due to specimen disruption. **(Right)** In some muscle specimens, dark contraction bands  can be seen separated by pale zones of sarcomeric disruption. This artifact is most common in specimens collected without the use of isometric clamping.

Wrinkling Artifact

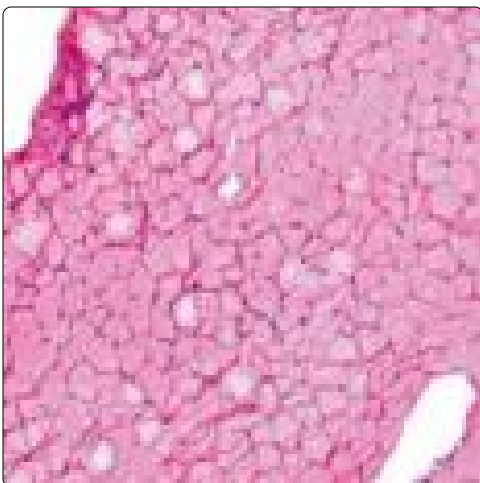


Vacuolization Artifact



(Left) Folding or wrinkling  of muscle fiber cytoplasm (sarcoplasm) may be encountered, particularly on frozen sections. This finding is sometimes referred to as "pie crust" artifact. **(Right)** Vacuolization of the sarcoplasm results from improper freezing of specimen tissue and is due to formation of ice crystals.

Folding and Overlapping



Indistinct Fibers



(Left) This biopsy of skeletal muscle was not properly transferred to the slide and shows prominent folding and overlapping of tissue. **(Right)** In this formalin-fixed tissue section, compression of the tissue makes it difficult to identify individual muscle fibers, and the distinction between skeletal muscle cytoplasm and collagen may be challenging.

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SECTION 4

Circulatory System



Heart	84
Cardiac Valves	88
Cardiac Conduction System	90
Arteries	94
Capillaries, Veins, and Lymphatics	98

HEART

Macroscopic Anatomy

- Cardiac chamber structure reflects pressure and volume physiology
 - Right atrium (2-6 mm Hg) receives systemic venous drainage and provides small atrial "kick" to help fill right ventricle; 2- to 3-mm wall thickness
 - Right ventricle (15-30 mm Hg) pumps blood through pulmonary artery against pulmonary vascular resistance; 4- to 5-mm wall thickness
 - Left atrium (4-12 mm Hg) receives pulmonary venous drainage and provides small atrial "kick" to help fill left ventricle; 2- to 3-mm wall thickness
 - Left ventricle (100-140 mm Hg) pumps blood through aorta against systemic vascular resistance; 13- to 15-mm wall thickness
- Atrial "sidedness" landmarks
 - Right appendage larger and pyramid-shaped, left is finger-like
 - Right atrial pectinate muscles prominent, left smooth except for appendage
 - Right atrium has coronary sinus ostium, limbus of oval fossa, terminal crest, sinoatrial node, atrioventricular node
- Ventricular "sidedness" landmarks
 - Right septal wall trabecular, left septal wall smooth
 - Tricuspid cordal attachments to right septum; no septal cordal attachments on left
 - Muscle separates tricuspid valve from pulmonary valves on right; mitral valve and aortic valves in continuity on left
- Pericardium envelopes and encircles heart
 - Epicardium (visceral pericardium) is delicate serous layer adhering to heart, whereas pericardium proper (parietal pericardium) forms thick fibrous-walled sac surrounding heart
 - Space between epicardium and pericardium contains ~ 30 mL of cushioning fluid

Microscopic Anatomy

- Chamber walls composed of 3 layers
 - Endocardium: Inner single endothelial cell lining with basement membrane and thin layer of fibroelastic connective tissue
 - Myocardium: Middle layer of stratified and interconnected myocytes with associated vessels and stromal cells
 - Epicardium: Outer layer of adipose tissue through which epicardial nerves, arteries, veins, and lymphatics course; bounded by serous epicardial lining
- Interventricular septum contains only myocardium, whereas interatrial septum may contain epicardium separating 2 myocardial layers

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Brown atrophy: Lipofuscin pigment is present in normal heart and increases with age
- Proportion of sarcoplasm to nuclei in infant myocardium is much lower than in adults

Hypertrophy

- Hypertrophy: Increased myocardial mass
 - Nuclear features: Binucleation, enlargement, hyperchromasia, less ovoid and more rectangular contours (even angulated and bizarre when severe)
 - Myocyte size: Less reliable due to tangential sectioning, variable tissue shrinkage during fixation, and distortion during microtomy

Pitfalls and Artifacts

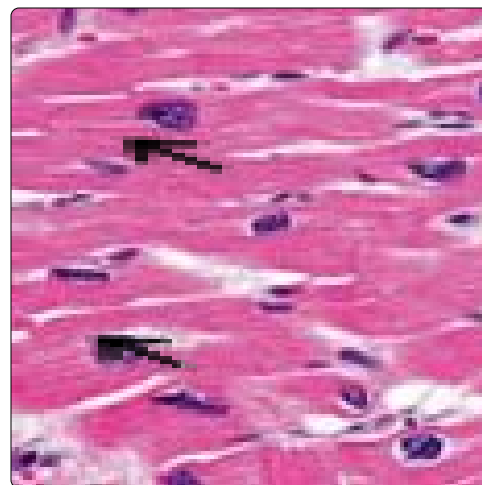
- Contraction bands (in endomyocardial biopsy) result from compression, stretching, and tearing of myocytes
- Perivascular collagen: Normal amount of dense collagenous tissue surrounds intramyocardial vessels; this is not interstitial fibrosis
- Papillary muscle tip: Dense collagen normally; not fibrosis
- Atrioventricular valve annulus: Dense collagen normally; not fibrosis

Heart Anatomy

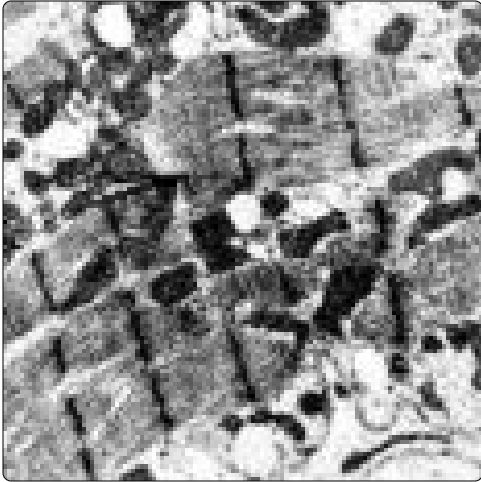


(Left) This view of a cadaver heart illustrates the right atrium [A], right ventricle [B], epicardium (visceral pericardium) [C], parietal pericardium [D], and myocardium [E] of the "infundibulum." (Right) Cardiomyocytes contain sarcoplasmic cross striations [F], centrally placed nuclei [G], and interconnections between cells [H] that facilitate cellular electrical "coupling."

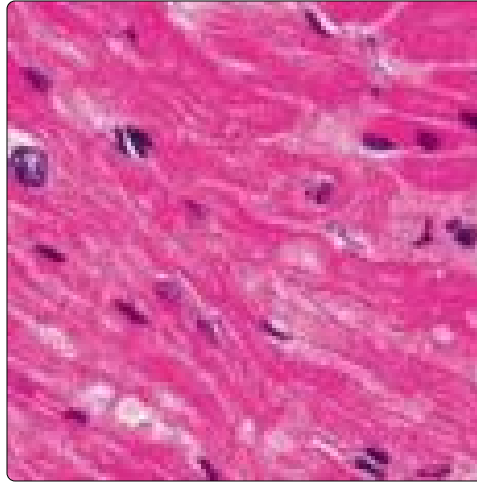
Cardiomyocytes



Sarcomere Ultrastructure

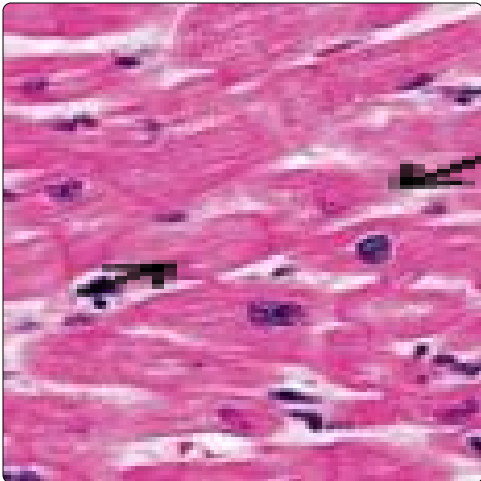


Cardiomyocytes: Cross Striations

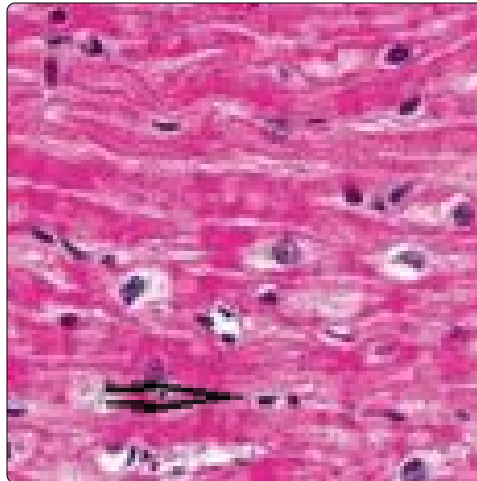


(Left) This EM view (x 14,000) shows individual myofilaments within a cardiomyocyte. The Z bands anchor the thin actin filaments. M lines connect thick myosin filaments and represent the center of the A band (between 2 Z lines). Mitochondria are abundant. (Right) Parallel stripes running perpendicular to the long axis of the myocytes are referred to as cross striations. These result from alignment of hundreds of myofilament Z bands across the width of the cell.

Cardiomyocytes: Intercalated Discs

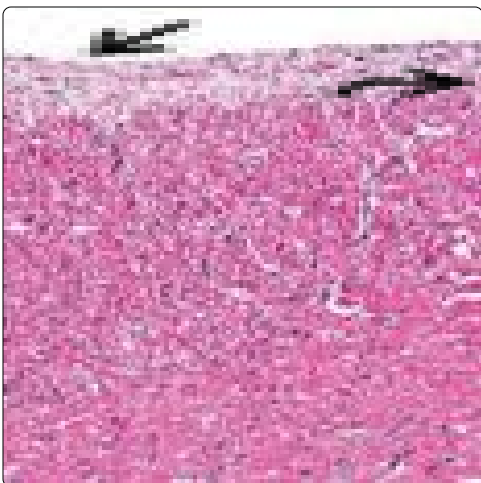


Lipofuscin (Brown Atrophy)

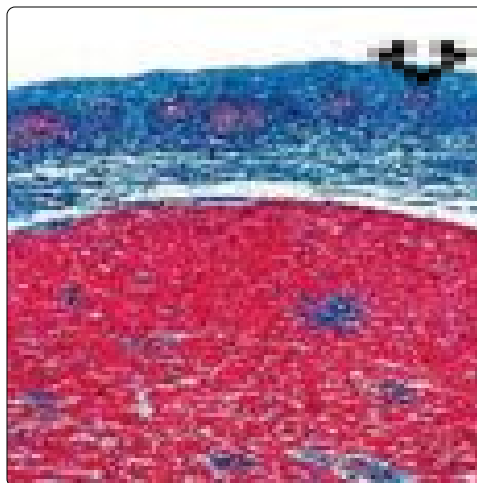


(Left) At the junctions between interconnecting cells, desmosome-rich intercalated discs are seen as dense, irregular eosinophilic lines. (Right) Brown atrophy is an age-related phenomenon and consists of the accumulation of granular gold-brown lipofuscin (or lipochrome) pigment around cardiomyocyte nuclei.

Endocardium



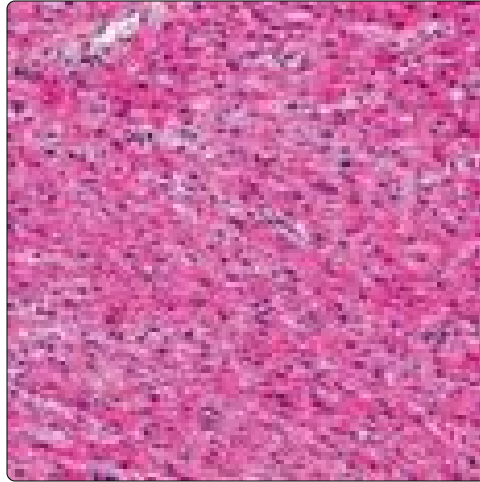
Endocardial Thickening



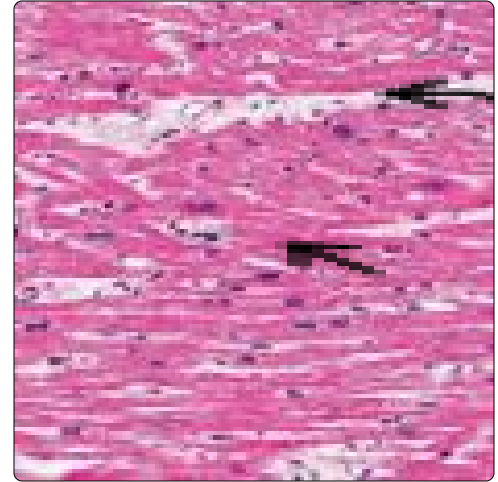
(Left) Ventricular endocardium is comprised of a single layer of endothelial cells (with flattened inconspicuous nuclei) and varying amounts of subendothelial connective tissue on the ventricular luminal surface. (Right) Thickened endocardium is seen normally in the left atrium and subvalvular portions of the ventricles. There is increased collagen fibrosis and elastic tissue. Endocardium also contains a thin smooth muscle layer throughout much of the heart.

Infant Myocardium

(Left) The proportion of sarcoplasm to nuclei in infant myocardium is much lower than in adults, consistent with the dogma that myocardial growth occurs via hypertrophy and not hyperplasia. (Right) This adult myocardium (note the increased sarcoplasm relative to infants) shows early features of (abnormal or pathologic) myocyte hypertrophy, including frequent binucleation of myocytes  and nuclear darkening and enlargement .

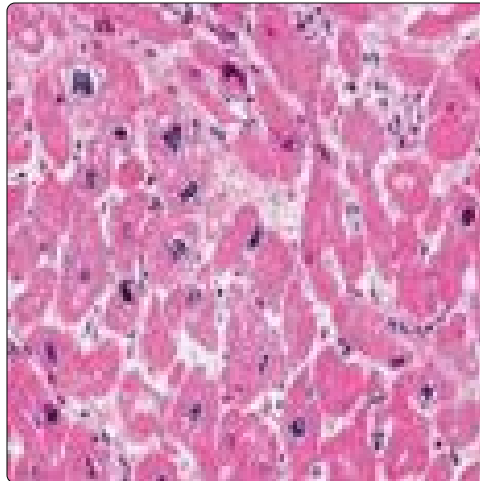


Early Myocyte Hypertrophy

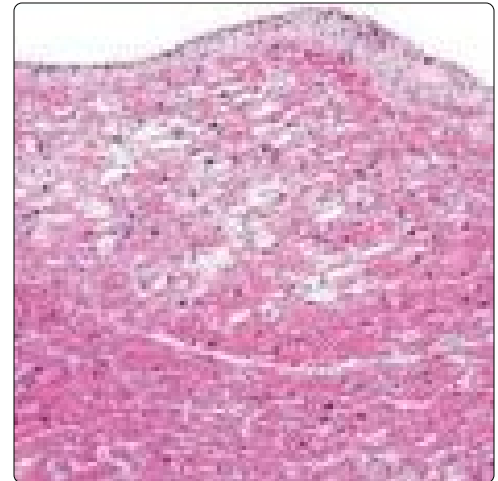


Severe Myocyte Hypertrophy

(Left) In severe (abnormal) hypertrophy, the nuclei are dramatically enlarged, hyperchromatic, and irregular. Note the absence of mitoses. Nuclear features are a more reliable measure of hypertrophy than other cell width measurements due to the complex 3D architecture of cardiomyocytes. (Right) Subendocardial myocardium is furthest from coronary blood flow but too deep to receive oxygen and nutrients from luminal blood. In chronic ischemia, starved cells show vacuolation.

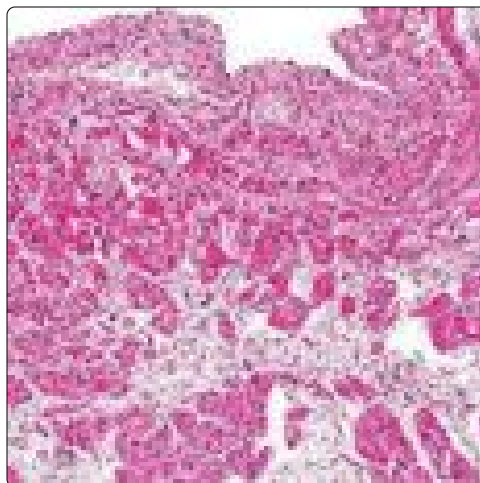


Subendocardial (Ischemic) Vacuolation

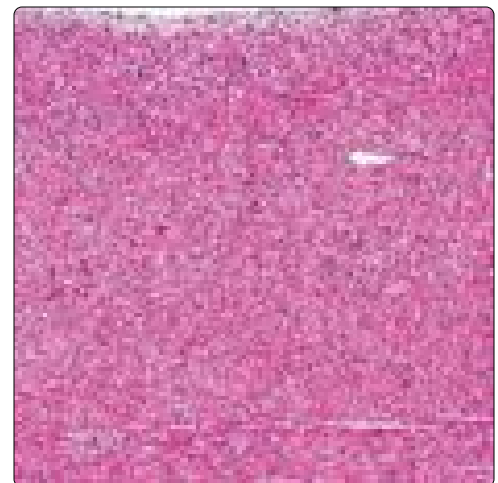


Atrial Myocardium

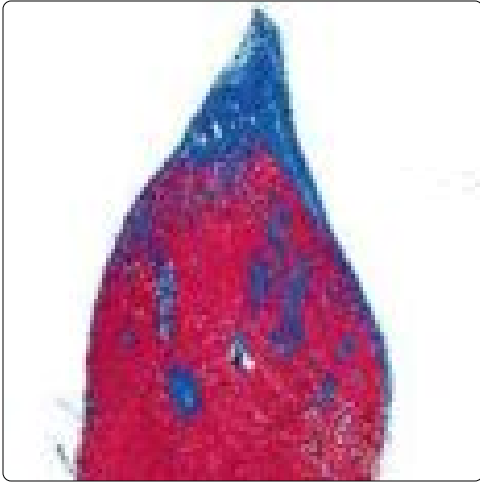
(Left) The atrial walls are thinner than the ventricles but still contain all 3 tissue layers. The atrial endocardium  is typically thicker than in the ventricles. The myocardial architecture is also more disorganized and haphazard. Adipose infiltration in the atria . (Right) Aside from being thicker, the ventricular myocardium is also more compact and composed of layers of myocyte "fascicles," successively oriented in alternating directions.



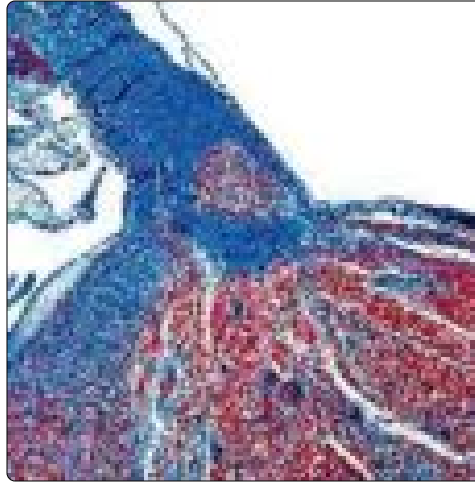
Ventricular Myocardium



Papillary Muscle Tip

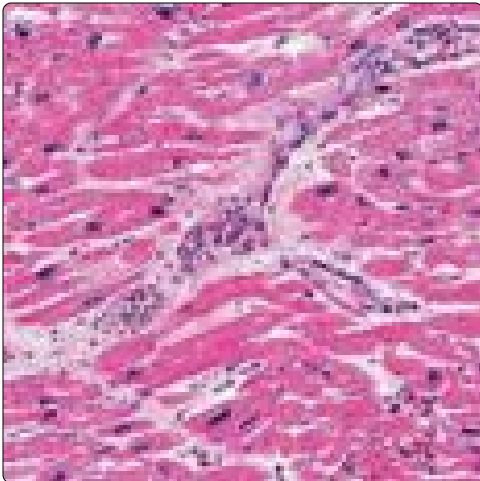


Tricuspid Valve Annulus

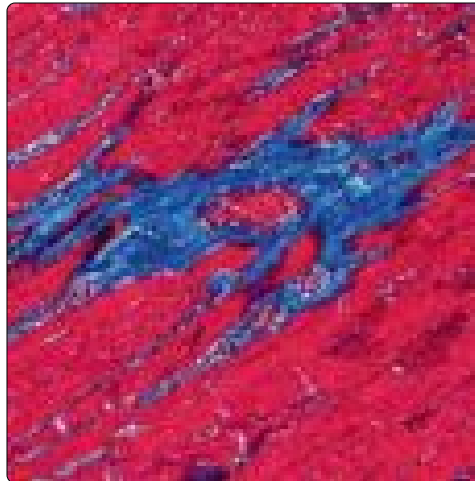


(Left) The tip of a papillary muscle is stained with trichrome to highlight the interdigitation of myocardium and dense collagenous valve apparatus (tendinous cord) tissue . Cross sections through the tips of the papillary muscle may falsely appear to have increased interstitial fibrosis, though this is a normal anatomic phenomenon. (Right) Similar to the papillary muscle tip, the dense collagenous interventricular valve annulus also shows some interdigitation with myocardium.

Intramyocardial Vessels

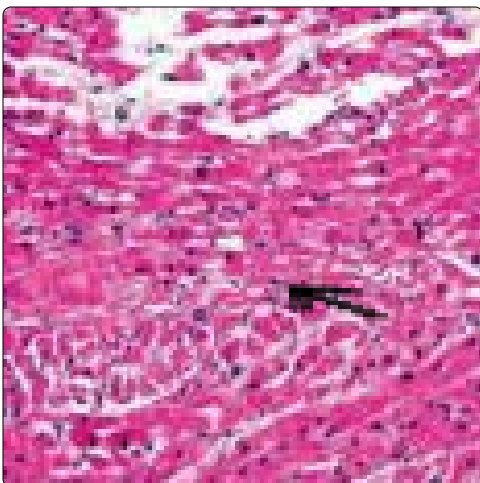


Intramyocardial Vessels

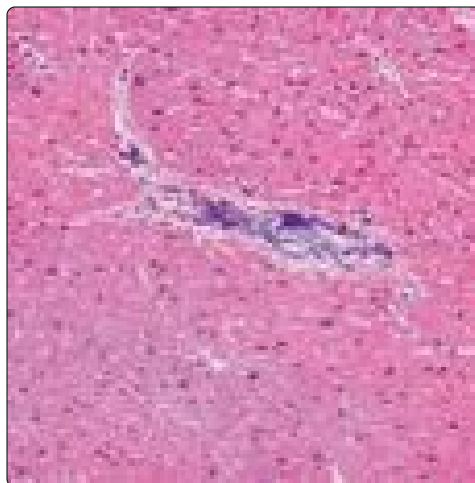


(Left) As penetrating arteries and other intramyocardial vessels course through the myocardium, they are surrounded by a layer of connective tissue. This is normal and should not be confused with pathologic interstitial fibrosis. (Right) Trichrome staining highlights the collagenous connective tissue surrounding intramyocardial vessels. Note the absence of collagenous fibrosis between myocytes, where no vessels are seen.

Contraction Band Artifact



Hydrophilic Polymer Gel Embolus



(Left) Contraction bands are thick perpendicular lines  (much thicker than cross striations or intercalated discs) within myocytes caused by clumping together of the contractile elements. In full-thickness sections of the heart, these indicate infarction, but in biopsies, they do not (rather due to biopsy-related trauma). (Right) Hydrophilic polymer coatings  are common on endovascular catheters and may embolize to coronary arteries. This is an artifact that is not typically pathologic.

Cardiac Valves

CARDIAC VALVES

Macroscopic Anatomy

- Semilunar (ventriculoarterial) valves
 - Aortic and pulmonary valves; anatomy is similar
 - 3 shirt-pocket-like cusps or leaflets arranged radially
 - Retrograde blood flow from aorta during diastole fills cusp pockets, occluding valve orifice
 - Outer "sheath" of these valves varies in composition at different levels
 - Below leaflets (pulmonary): Ventricular septum and right ventricular outflow tract (infundibulum)
 - Below leaflets (aortic): Ventricular septum and anterior mitral leaflet-aortic valve fibrous continuity
 - Behind leaflets: Bulbar or "sinus of Valsalva" portion of aorta or pulmonary artery (and coronary artery ostia in aorta); sinuses are more dilated in aortic valve
 - Above leaflets: Sinotubular junction or supraaortic/pulmonary ridge
 - Line of fine fibrosis develops toward top of aortic cusps along line of closure/coaptation; above this line there is redundant and fenestrated tissue called lunula
 - At cusp edge midpoint is pyramidal focus of fibrosis (nodule of Arantius), from which often emanate fine strands of fibrous tissue (Lambl excrescences)
- Atrioventricular valves
 - Mitral and tricuspid valves; anatomy is similar
 - Each is suspended from fibrous annulus that separates atria above from ventricle below
 - Each is tethered from below (preventing prolapse into atria during systole) by tendinous cords attached to papillary muscle (outcroppings from ventricular muscular wall)
 - Mitral valve has 2 leaflets (anterior and posterior) supported by 2 papillary muscles (1 at each commissure)
 - Tricuspid valve has 3 leaflets (anterior, posterior, and septal) and 3 papillary muscles

Microscopic Anatomy

- Semilunar and atrioventricular valves share same trilaminar histoarchitecture

- Thickness and proportional contribution of each layer vary
- Fibrosa
 - Dense collagenous layer in continuity with annular connective tissue (and for semilunar valves, the arterial wall)
 - Bounded on either side by thin layer of elastic fibers (more prominent beneath endothelium)
 - Elastic layers diminish toward free edge of leaflets
 - This layer continues as tendinous cords of atrioventricular valves
- Spongiosa
 - Central layer, composed of loose connective tissue, fibroblasts, and valvular interstitial cells
 - Most prominent near annulus
 - In semilunar valves, atrial myocardium, nerve bundles, and lymphatics may infiltrate spongiosa near annulus
 - This layer also rises to surface distally in semilunar valves to become "auricularis" layer
- Ventricularis
 - Most elastic fiber-rich layer with multiple layers of elastic tissue
 - Lines ventricular surfaces of both semilunar and atrioventricular valves
 - Also coats tendinous cords

CONSIDERATIONS AND VARIATIONS

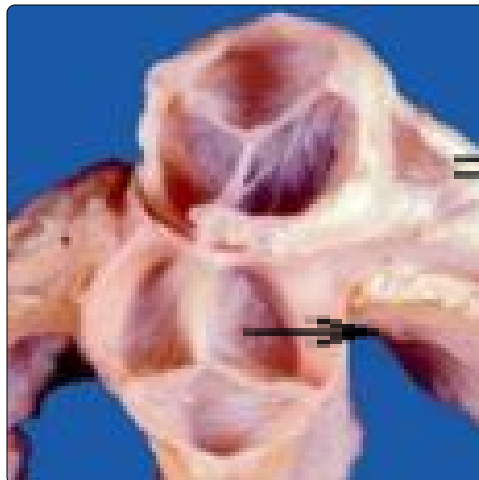
Age-Related Variations

- Diffuse rubbery thickening, accentuation of noduli Arantii, and mitral annular and sinotubular junction (supravalvular ridge) calcification all occur with aging
- Calcification and fibrosis in fibrosa in valve sclerosis and expansion of loose connective tissue in spongiosa also occur with aging

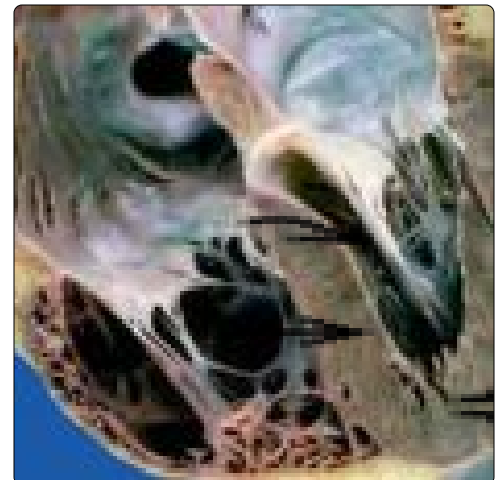
Pitfalls and Artifacts

- Tangential sectioning (when blocking tissue or at embedding)
- Presence of vessels within valve tissue is abnormal, signifies postinflammatory process, such as rheumatic valve disease

Semilunar Valves

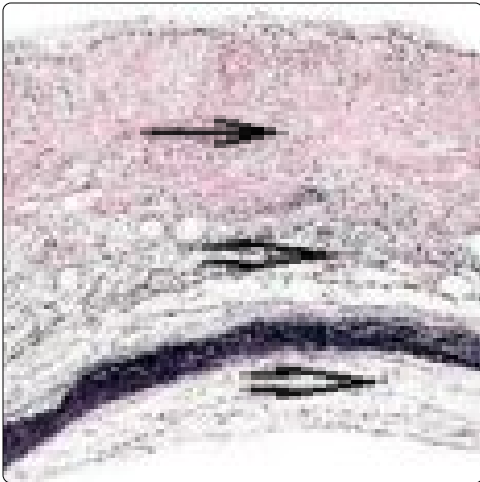


Atrioventricular Valves

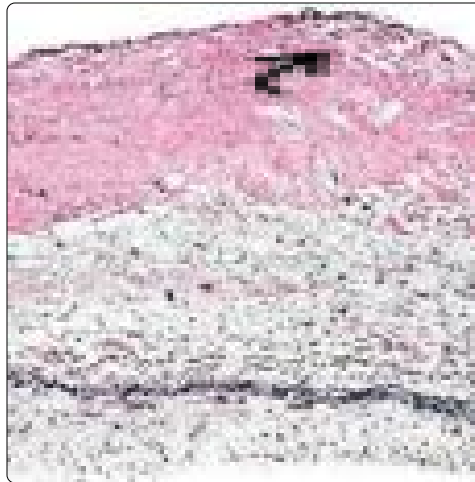


(Left) This photo shows the semilunar valves after removal of the great arteries. The pulmonary artery is anterior and the aorta posterior with the right coronary artery branching from the aorta. (Right) The atrioventricular valves (tricuspid and mitral) are shown in this 4-chamber view of the heart. The valves are suspended from annular rings and tethered to papillary muscles by tendinous cords.

Semilunar Valve

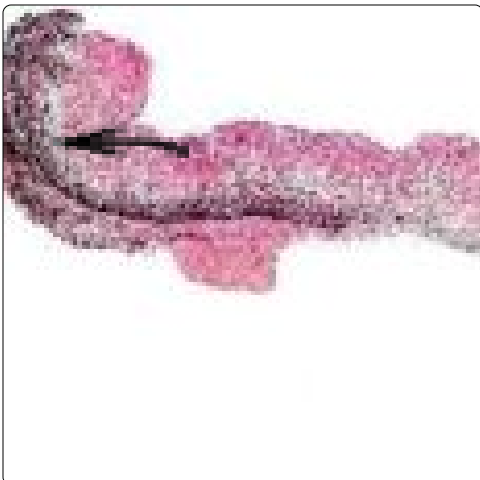


Atrioventricular Valve



(Left) The 3 distinct valvular layers are the fibrosa, spongiosa, and ventricularis. A fine line of elastic tissue is seen beneath the endocardium overlying the fibrosa layer. The ventricularis layer is the richest in elastic tissue, as shown by the heavy black band on this elastic stain. (Right) This photo shows the 3 distinct valvular layers: Fibrosa, spongiosa, and ventricularis, with the subendocardial elastic layer.

Semilunar Valve (Free Aspect)



Semilunar Valve (Free Aspect)

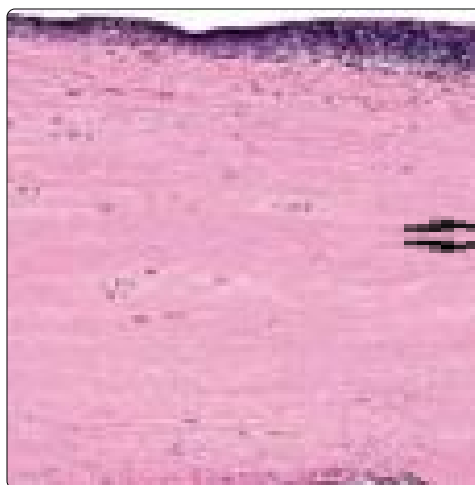


(Left) This photo demonstrates continuity between the aortic valve and the elastic wall of the aortic root along with the gradual diminution of the ventricularis layer and fibrosa layer toward the free edge of the cusp. (Right) There is gradual diminution of the ventricularis layer and fibrosa layer with progression toward the free edge of the cusp shown here. The spongiosa layer expands to become most prominent at the free edge.

Atrioventricular Valve Near Annulus



Tendinous Cord



(Left) The myocardium extends (along with lymphatics, fat, and nerves) into the proximal portion of the valve substance. The heavy band of elastic tissue in the ventricularis layer is also prominent. (Right) This section shows a thickened portion of tendinous cord. The core is dense collagen that is continuous with the valve fibrosa layer. There is a coating of prominent elastic tissue covering the cords as well.

Cardiac Conduction System

CARDIAC CONDUCTION SYSTEM

Macroscopic Anatomy

- Sinoatrial (SA) node
 - a.k.a. sinus node
 - Specialized "pacemaker" myocyte collection located in right atrium
 - Subepicardial structure near superior vena cava, overlying terminal crest (vertical crest on interior wall of right atrium that separates sinus of vena cava from rest of right atrium)
 - Found at union of smooth-walled "sinus venosus" portion and trabecular portion of right atrium
 - Supplied by sinus nodal artery
 - Usually branch from right coronary
- Atrioventricular (AV) node
 - Specialized conducting myocytes within tricuspid annulus near AV (membranous) septum
 - Subendocardial structure found within "Koch triangle"
 - Koch triangle: Anatomic area defined by 3 vertices
 - Membranous septum
 - Roof of coronary sinus ostium
 - Tricuspid annulus at point directly below coronary sinus ostium
 - Supplied by AV nodal artery
 - Usually from posterior descending artery

Microscopic Anatomy

- SA node
 - Compact and polyhedral myocytes surrounded by dense collagenous tissue
 - Sarcoplasm and cross striations less prominent than typical myocytes
 - Sinus nodal artery courses through sinus node
 - Autonomic nerve fibers and ganglia seen in vicinity
- AV node
 - Compact "bulb" of smaller polyhedral myocytes merging with larger stellate to spindled myocytes with vacuolar sarcoplasm toward bundle of His

- Mesothelial-like cells and cystic structures occasionally intermixed
- Arterioventricular nodal artery courses through AV node
- Connective tissue surrounds node, "insulating" adjacent myocardium
- Autonomic nerves and lymphatics seen in vicinity
- Bundle of His and bundle branches
 - Constituent cells are mostly smaller than myocardial myocytes and feature cytoplasm that is more vacuolated
 - Right bundle branch is smaller and cord-like
 - Left bundle branch is larger and splays out over leftward ventricular septum
- Purkinje cell
 - This term is applied to cells in left bundle branch and distal right bundle branch because they are larger than myocardial myocytes and feature cytoplasm that is more vacuolated
 - As with all conduction system cells, Purkinje cells are myocytes
 - Contain myofibrils by electron microscopy
 - Differences from normal cardiomyocytes
 - Absence of T tubules
 - Striking abundance of cell:cell junctions

CONSIDERATIONS AND VARIATIONS

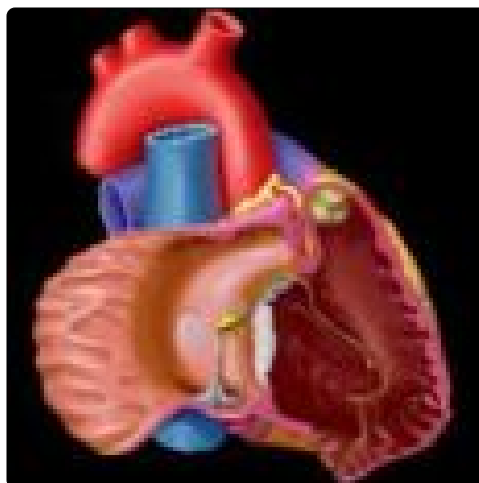
Age-Related Variations

- With advancing age, increased dense collagen:myocytes ratio in SA node reported by some authors

Pitfalls and Artifacts

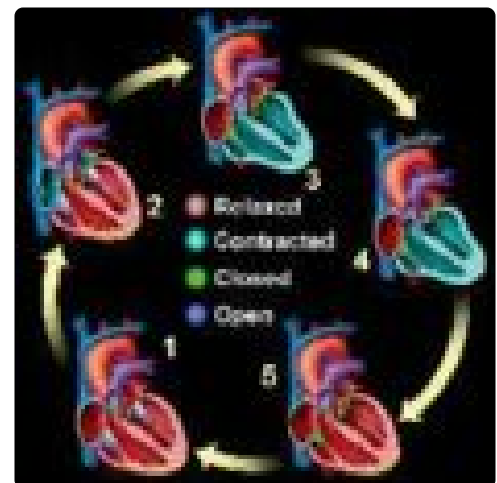
- Normally occurring smooth muscle bundles in endocardium should not be mistaken for Purkinje cells or conduction system tracts

Conduction System of Heart

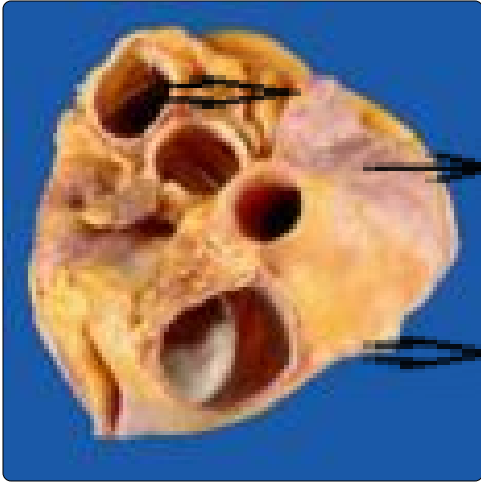


(Left) This illustration shows the approximate locations of the sinus node (subepicardial), atrioventricular (AV) node (subendocardial), and His bundle as well as the bundle branches (subendocardial). (Right) The conduction system coordinates each cardiac contraction. Every beat originates in the sinus node and propagates through the atrium to the AV node. The AV node, His bundle, and bundle branches control contraction of the ventricles.

Conduction Sequence



Heart, Superior Oblique View



Heart, Superior Oblique View (Sinoatrial Node Block Removed)

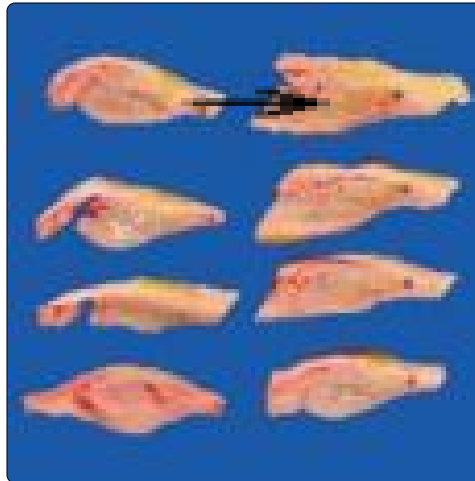


(Left) This heart removed at autopsy illustrates the pulmonary artery, aorta, superior vena cava, and left pulmonary vein confluence when viewed from above. (Right) A block of tissue along the lateral border near the superior vena cava connection, containing the sinus node, has been removed from this heart. Internally, a segment of the terminal crest in the right atrium is removed with this block.

Excised Tissue Block Containing Sinus Node

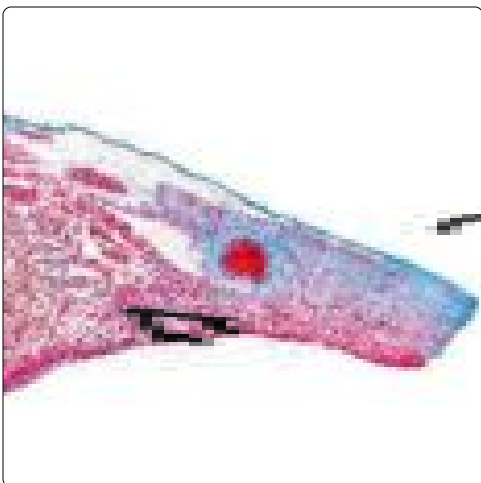


Sinus Node Tissue Block (Cross Sections)



(Left) Viewed from the endocardial aspect, this excised tissue block consists of a portion of terminal crest with a few pectinate muscles attached. The overlying epicardium (not seen, facing down) is intact. (Right) After serial sectioning, the belly of the terminal crest is now seen in cross section in each piece. The pinpoint sinus nodal artery can also be seen in many of the sections, identifying the locus of the node.

Sinoatrial Node



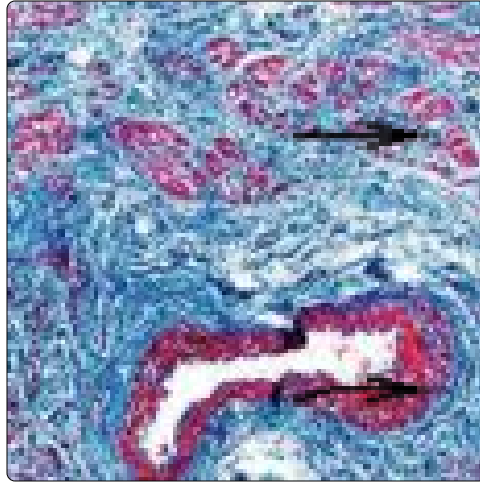
Sinoatrial Node



(Left) This trichrome-stained section at low magnification shows the atrial (terminal crest) myocardium, sinus node artery, and sinoatrial node itself characterized by intermixed dense collagen and small polygonal myocytes. (Right) This cross section illustrates the dense connective tissue-rich sinoatrial node tissue surrounding the sinus node artery.

Sinus Node, High Magnification

(Left) This trichrome-stained section shows a portion of normal sinoatrial node with accompanying sinus node artery [A]. The nodal myocytes show prominent vacuolization [B]. (Right) The right side of the heart has been opened through the tricuspid valve. Koch triangle can be visualized as imaginary lines connecting the roof of the coronary sinus ostium [C], the tricuspid annulus at a point directly below the coronary sinus ostium [D], and the membranous septum [E].



Right Atrium and Ventricle (Opened)

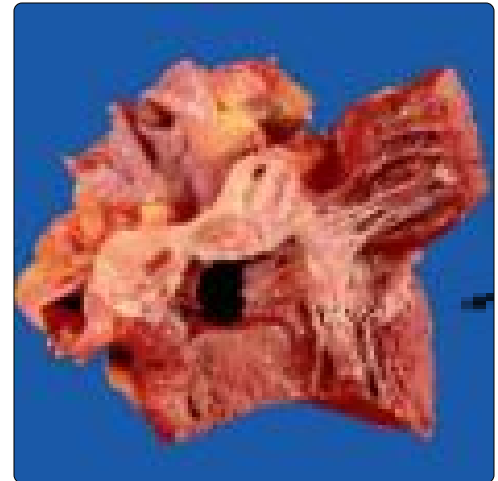


Right Atrium and Ventricle, Opened (AV Node Tissue Block Removed)

(Left) This view shows the right side of the heart, opened through the tricuspid valve, after the tissue block containing Koch triangle (including the AV node) has been removed [A]. (Right) This view demonstrates that the excised tissue block [B] included the interventricular septum and part of the aortic valve and aortomitral fibrous continuity.



Left Ventricular Outflow Tract, Opened (AV Node Tissue Block Removed)

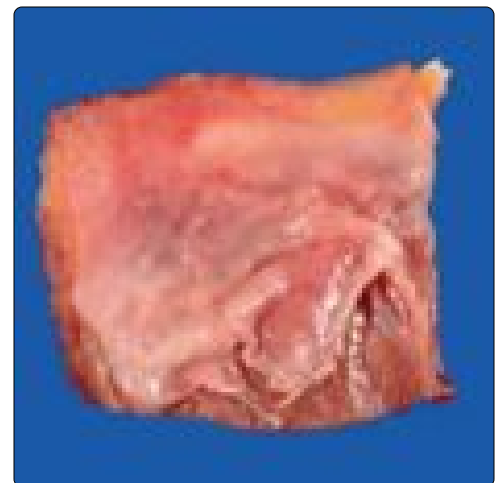


Left Atrium and Ventricle, Opened (AV Node Tissue Block Removed)

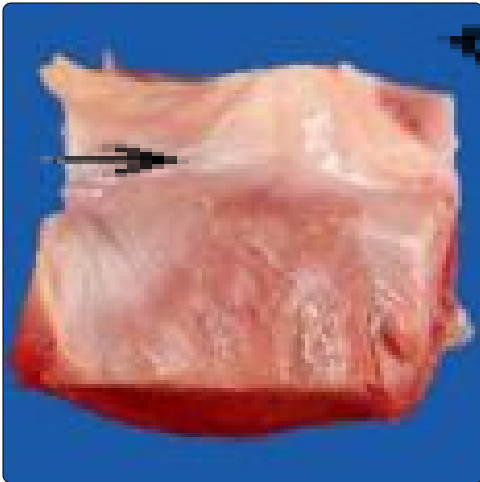
(Left) This view demonstrates that the excised tissue block [A] also included the portion of anterior mitral valve leaflet that forms the aortomitral fibrous continuity. (Right) The AV node tissue block is shown with a portion of posterior tricuspid leaflet [B] that traverses the rightward aspect diagonally. The membranous (AV) septum is also seen [C].



AV Node Tissue Block (Rightward Aspect)



AV Node Tissue Block (Leftward Aspect)

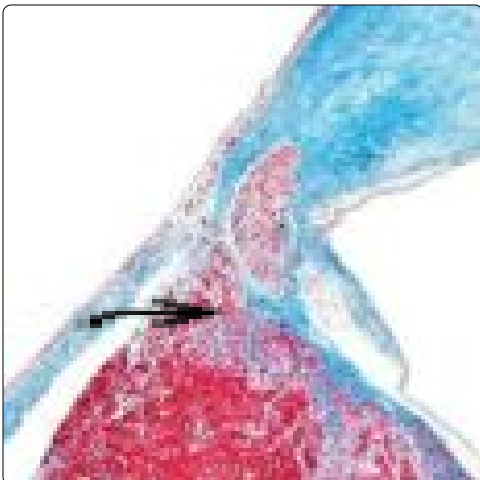


Cross Sections Through AV Node Tissue Block

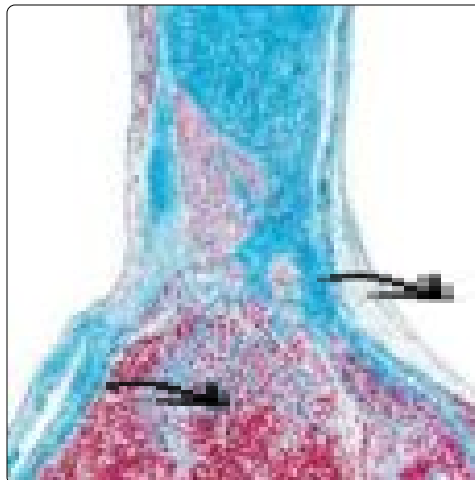


(Left) The left-facing aspect of the AV node tissue block is shown here with membranous septum [X] and pockets of 2 aortic valve cusps [X]. (Right) Serial cross sections through the excised AV node tissue block illustrate portions of tricuspid valve leaflet [X], membranous (AV) septum [X], and aortic valve cusp [X].

AV Node: Trichrome

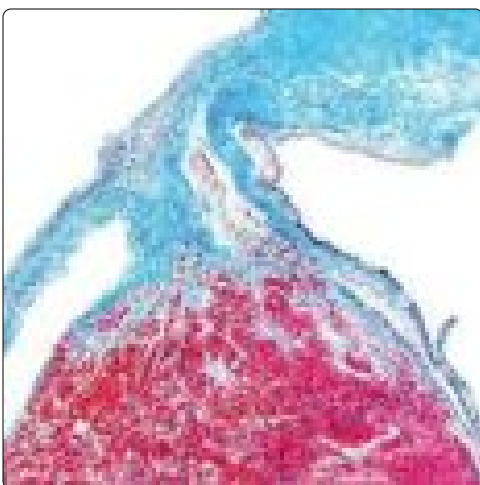


His Bundle and Right Bundle Branch: Trichrome

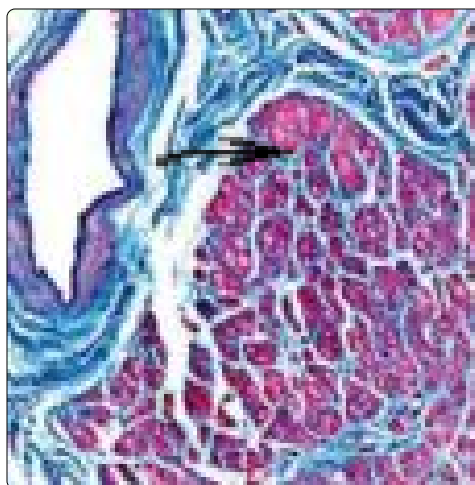


(Left) Intermixed dense collagen and small myocytes comprise the AV node [X], situated in the annular connective tissue, insulated from the atrial and ventricular myocardium. A portion of tricuspid valve leaflet is also seen [X]. (Right) The right bundle branch [X] consists of a thin, cord-like wisp of conduction system myocytes arising from the main His bundle [X]. Because of its small size, step sections through the paraffin block are typically required to see it well.

Left Bundle Branch: Trichrome



AV Node, High Magnification: Trichrome



(Left) The left bundle branch [X] is larger and broadly splays out beneath the left ventricular outflow tract endocardium. The specialized myocytes are smaller than the adjacent normal myocardial myocytes and may also be more vacuolated. (Right) This trichrome-stained section shows a portion of the AV node with accompanying AV nodal artery [X]. The conduction system myocytes are compact and show vacuolization of the sarcoplasm [X]. There is also dense collagenous connective tissue in the node [X].

ARTERIES

Macroscopic Anatomy

- Elastic arteries
 - 1st order branches of aorta (carotid, subclavian, proximal renal, and iliac; also internal mammary arteries)
 - Walls composed of multielastic-layer concentric strata
- Muscular arteries
 - Arterial segments between elastic arteries and arterioles
 - Distinct internal and external elastic lamina
- Arterioles
 - Smooth muscle only; no definite elastic layer

Microscopic Anatomy

- Intimal layer (tunica intima)
 - Single layer of endothelial cells with subendothelial loose connective tissue and smooth muscle cells
- Medial layer (tunica media)
 - Multiple layers of smooth muscle cells with associated elastic tissue

- Adventitial layer (tunica adventitia)
 - Dense connective tissue

CONSIDERATIONS AND VARIATIONS

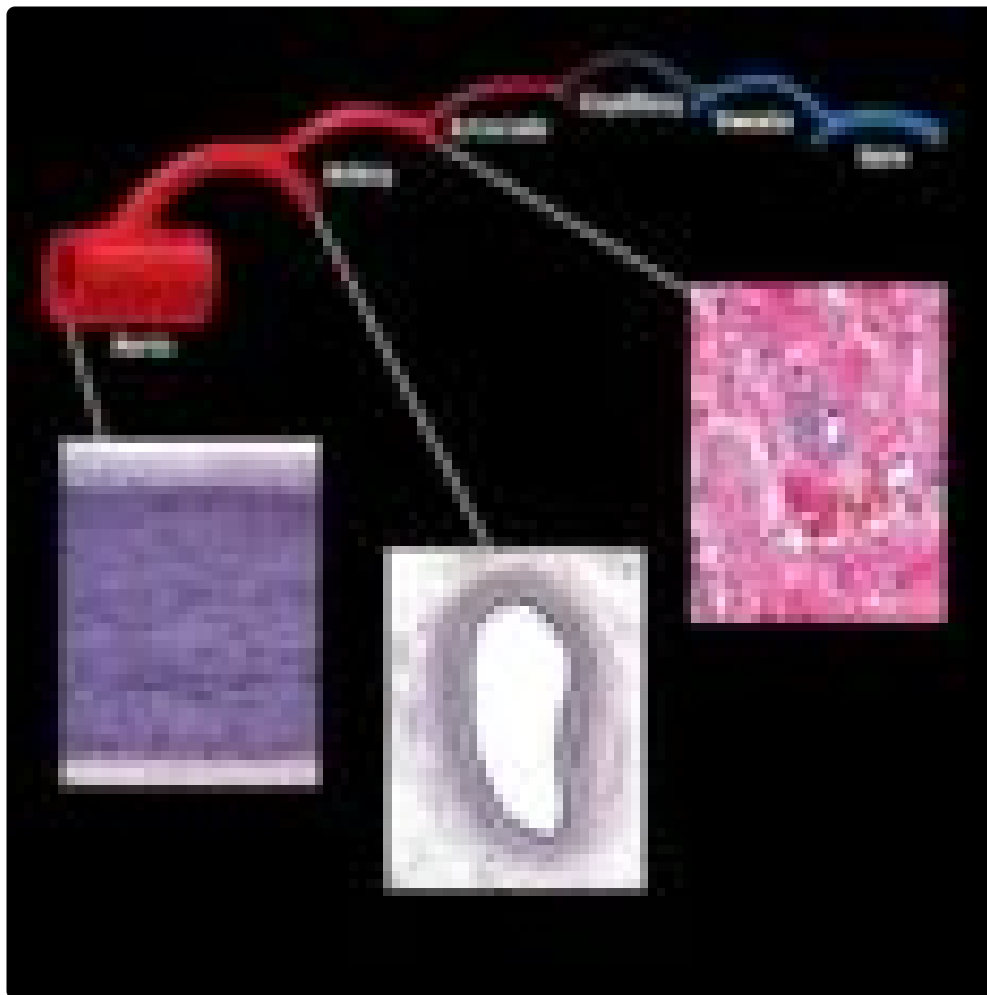
Pitfalls and Artifacts

- Tangential sectioning of tortuous vessels can result in asymmetric luminal contours and wall thicknesses
- En face sectioning of elastic layers in aortic sections (when not embedded "on edge")

Age-Related Variations

- Intimal thickness increases with age in manner independent of atherosclerosis
- Mild medial elastic fiber fragmentation in large elastic arteries also can be seen in elderly
- Calcification of internal elastic lamina in muscular arteries (Mönckeberg-type medial calcific sclerosis) is another nonpathologic, age-related phenomenon

Normal Vascular Network



This schematic shows the spectrum of vascular segments from elastic artery to capillaries and back again to the great veins. For each arterial segment, the histologic structure reflects its function. The aorta is rich in elastic tissue, which helps provide recoil to sustain blood flow during diastole. The muscular arteries have some elastic properties but also substantial smooth muscle that contracts and relaxes to alter luminal flow. Arterioles are almost entirely smooth muscle and serve as the main locus of control for peripheral vascular resistance.

Muscular Artery (Cross Section)

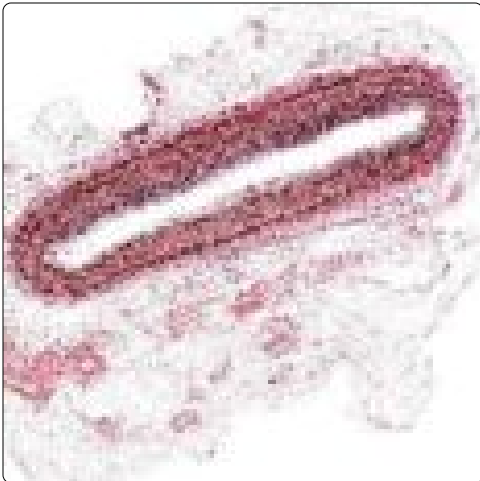


Muscular Artery (Cross Section)



(Left) Photomicrograph of a muscular artery shows the normal configuration and appearance of the intima, media, and adventitia layers. (Right) Elastic-stained section of a muscular artery illustrates the normal configuration and appearance of the intima, media, and adventitia layers. The stain also demonstrates the internal and external elastic laminae.

Muscular Artery (Compressed)

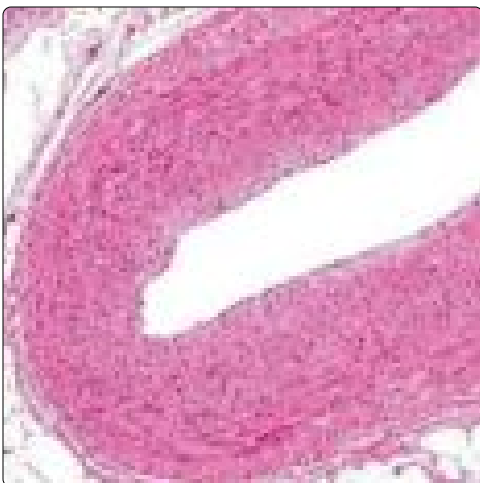


Muscular Artery Branching

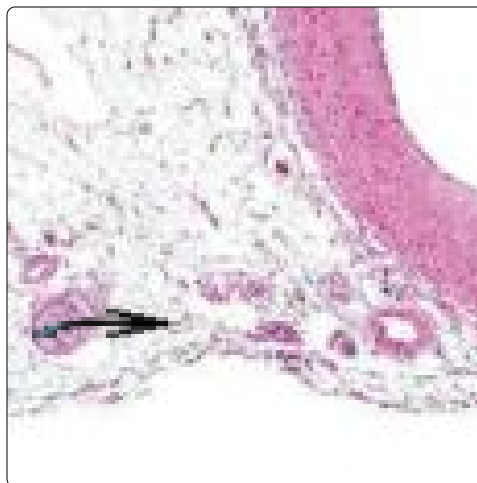


(Left) The lumen of this artery is elongated and narrowed, rather than the usual circular appearance, but this reflects an artifact of compression during tissue fixation. (Right) A perpendicular branch arises from a muscular artery. Note the difference in caliber between the main artery (cross section) and branch (longitudinal section). Note also that the tissue layers and elastic laminae are continuous from the main artery to the branch with thickened intima at the junction.

Muscular Artery



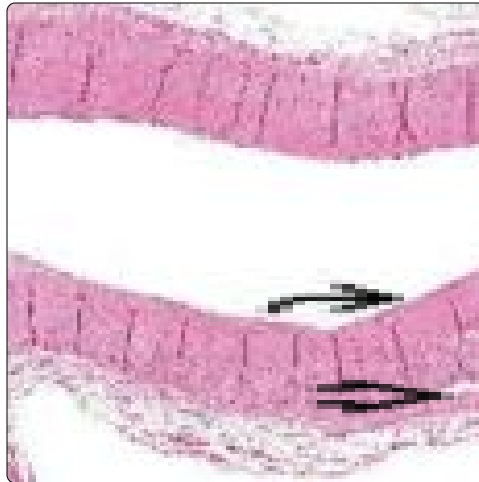
Adventitia and Periadventitial Tissue



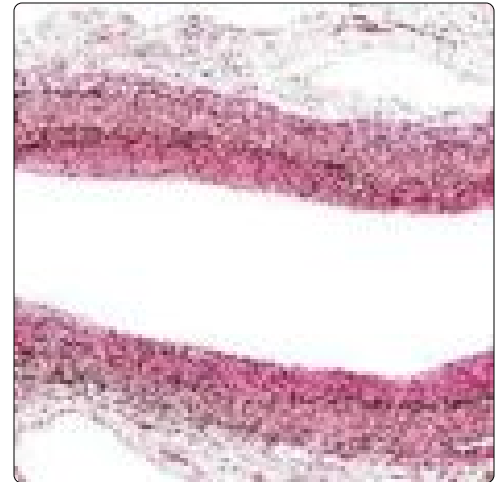
(Left) Photomicrograph of a muscular artery in cross section demonstrates that elastic staining is not always needed to see the wavy, refractile elastic fibers of the internal elastic lamina. (Right) Beyond the adventitia, there is adipose tissue in which nerve fibers and small vessels of the vasa vasorum can be found. The vasa vasorum run parallel to the artery but give rise to perpendicular branches that penetrate into the media to supply the smooth muscle cells.

Muscular Artery (Longitudinal)

(Left) Longitudinal section of an arterial segment (as opposed to cross section) shows the same trilayered architecture with intima, media, and adventitia. **(Right)** Elastic stain of a longitudinally sectioned muscular artery demonstrates the internal and external elastic laminae.



Muscular Artery (Longitudinal)



Elastic Artery (Aorta)

(Left) Photograph compares H&E and elastic staining of the aortic wall. Note the relatively thin intimal layer; nearly the entire thickness of the wall is comprised of media with alternating elastic fiber and intervening smooth muscle layers. The adventitia is also prominent with large vasa vasorum readily apparent. **(Right)** Normal aortic wall is composed of multilayered strata of wavy elastic fibers.

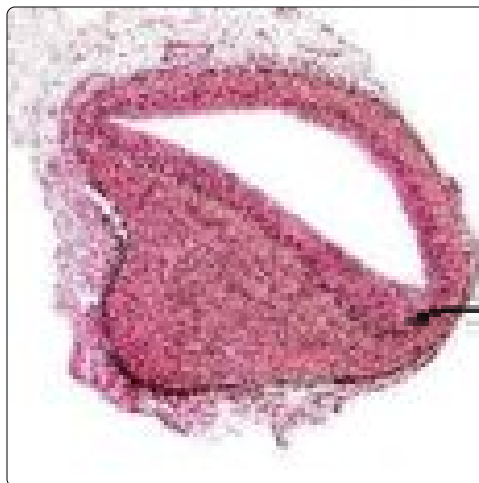


Elastic Artery (Aorta)

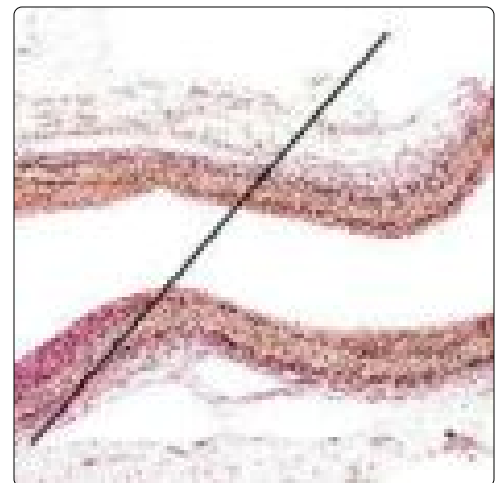


Tangential Sectioning Artifact

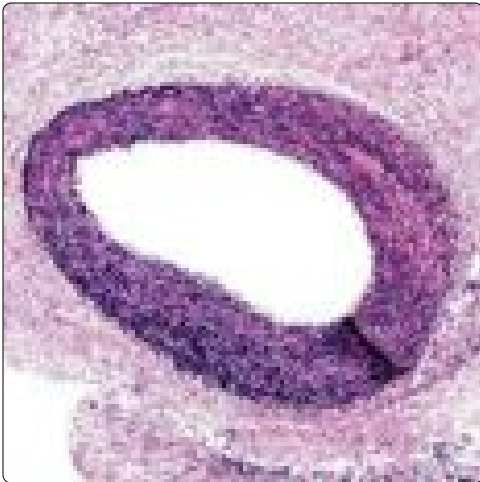
(Left) An asymmetric bulging is seen on one side of this vessel (otherwise in cross section). Elastic staining shows that this is composed of medial layer. The orientation of the medial smooth muscle is altered relative to the cross-sectional areas (another clue to tangential sectioning). **(Right)** A slightly tortuous segment of muscular artery is shown with a line representing a tangential plane that could give rise to an artificial bulge secondary to sectioning artifact.



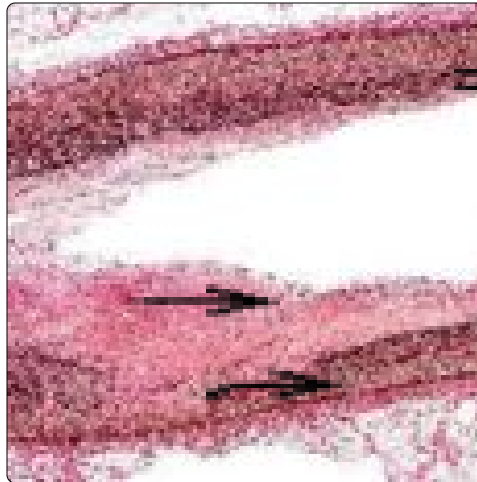
Tangential Sectioning Artifact



Elastic Artery (Internal Mammary Artery)

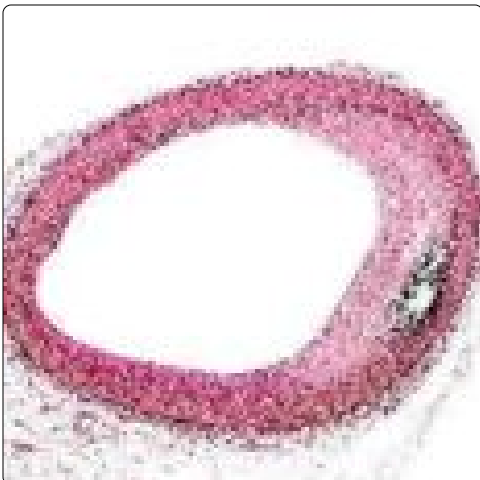


Atherosclerotic Plaque (Longitudinal)



(Left) The internal mammary arteries are unique as the smallest caliber elastic arteries in the arterial tree. These are used as conduits in coronary bypass surgery, so this feature is helpful distinguishing conduit vessels from the native target vessel histologically. (Right) This artery contains an atherosclerotic plaque. Note the dissolution of the internal elastic lamina and atrophy of the media underlying the plaque (compared to the medial thickness opposite).

Atherosclerotic Plaque (Cross Section)

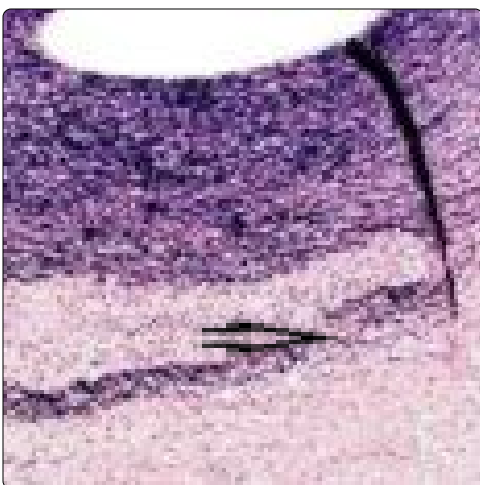


Medial Calcific (Mönckeberg-Type) Sclerosis

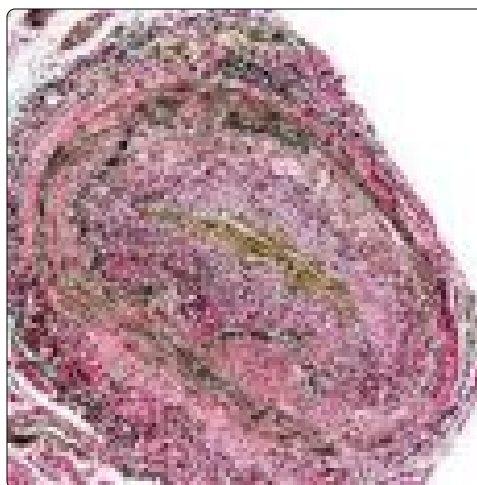


(Left) Elastic staining of this muscular artery shows atherosclerotic plaque in the intima (inside the internal elastic lamina). Atrophy of the underlying media and focal calcification are also seen. (Right) Paraffin section of muscular artery shows extensive calcification centered on the internal elastic lamina. This is largely an age-related phenomenon and of no major clinical significance.

Chronic Medial Dissection



Vasculitis: Disruption of Internal Elastic Lamina



(Left) Photomicrograph shows an elastic artery with a chronic medial dissection plane. A new cavity (false lumen) has developed in a dissection plane within the media filled with old, organized thrombus. (Right) This is a markedly abnormal vessel showing an important hallmark of inflammatory damage caused by vasculitis. The inflammation causes discrete breaks in the internal elastic lamina, and there is replacement of smooth muscle with collagen and fibroblasts (fibrosis).

Capillaries, Veins, and Lymphatics

CAPILLARIES

Microscopic Anatomy

- Lack muscular media
- Lack elastic fibers
- Composed only of endothelial cells with basement membrane and supporting pericyte(s)
- Smallest diameter (~ 5-40 μm) but greatest aggregate surface area in circulatory system
- Capillary density is proportionate to metabolic activity of organ/tissue
- 3 primary types
 - Fenestrated
 - Pores (60-90 nm) perforating endothelial cytoplasm, forming direct communication to extracellular space (kidney, GI tract, endocrine organs)
 - Continuous
 - Absent fenestrations; endothelium forms barrier to extracellular space (muscle, nervous tissue)
 - Sinusoid
 - Complex 3D endothelial-lined cavity (liver, spleen, bone marrow)
- Portal system: Capillary bed bound by venous drainage on either end (pituitary, liver)

VEINS

Macroscopic Anatomy

- Thinner wall and larger caliber compared to accompanying arteries
- Valves cause periodic widening within larger veins
- Venous circulation is high volume but low flow

Microscopic Anatomy

- Venules
 - Transition from capillaries somewhat arbitrary, based on size
 - Pericytes still present
 - More subendothelial connective tissue than capillaries
 - At most, single smooth muscle medial layer (often absent)

- High endothelial venules: Specialized venular segment within lymph nodes; site of leukocyte migration
- Veins
 - Intima: Endothelium and connective tissue; absent internal elastic lamina
 - Media: Variable thickness; greatest in lower extremities, mesentery, uterus, umbilicus, and nearly absent in CNS, retina, medullary bone, penis
 - Adventitia: Most prominent layer; predominantly longitudinally oriented bundles of dense collagen &/or smooth muscle with coarse elastic fibers
 - Valves are present in most veins, composed of paired infoldings from intimal layer

LYMPHATICS

Macroscopic Anatomy

- Difficult to see macroscopically
- Visualization during surgery aided by preoperative ingestion of lipids (chylomicrons taken up by lymphatics)
- Lymphatic mapping in vivo performed by injecting fluorescent dyes or radioactive isotopes (lymphoscintigraphy)
- Larger lymphatics may have beads on string appearance due to lymphatic valves

Microscopic Anatomy

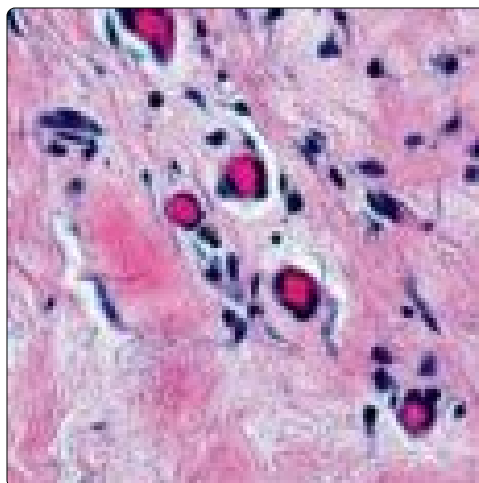
- Smallest lymphatics resemble capillaries but with larger lumina and less rounded contours
- Larger lymphatics have thin wisps of muscular media and intimal valves
- Lack elastic tissue
- Negligible basement membrane layer
- Endothelial cells express podoplanin (D2-40) and LYVE-1

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Phlebosclerosis
 - Age-related medial and adventitial fibrosis with less frequent intimal thickening and calcification

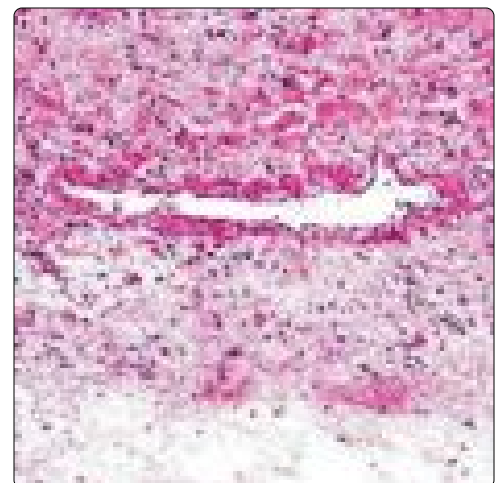
Capillaries



(Left) This photomicrograph shows capillaries within organizing granulation tissue. Capillaries consist of endothelial cells with a basement membrane and supporting pericytes.

(Right) This is a small vein. The intimal layer consists of only endothelium and loose collagen. The media is essentially absent. Longitudinal smooth muscle bundles comprise the adventitia.

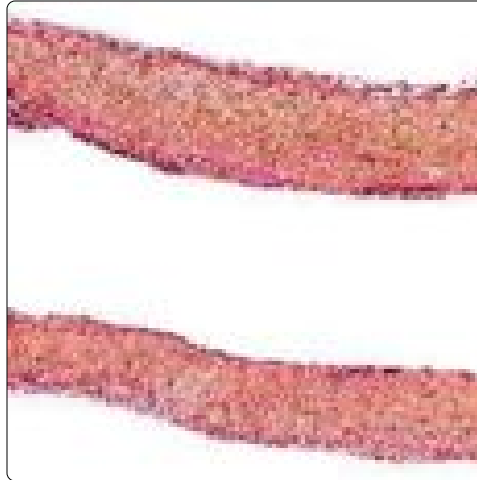
Small Vein



Medium-Sized Vein in Cross Section

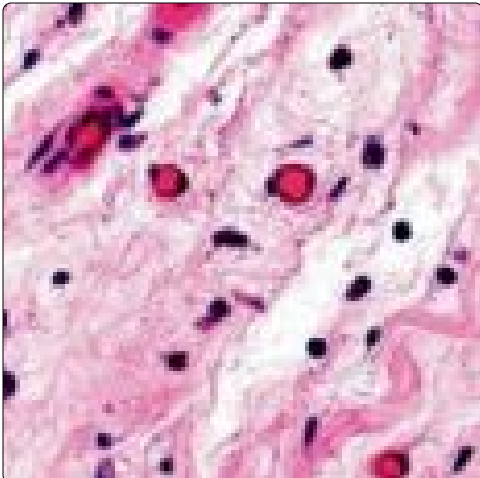


Medium-Sized Vein in Longitudinal Section

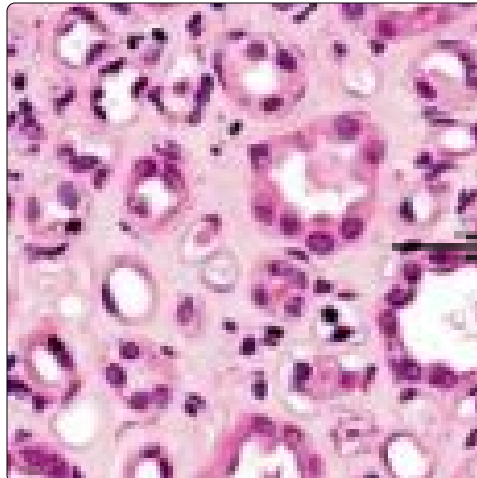


(Left) Veins lack a well-developed internal elastic layer beneath the intima. This small saphenous vein branch has substantial muscular media. There is abundant adventitial elastic tissue, but it does not form a discrete lamina. The adventitia is stratified with longitudinal and circumferential bands of smooth muscle and collagen. **(Right)** This longitudinal section of a saphenous vein branch shows that most of the adventitial bundles run parallel to the vessel length.

Organizing Granulation Tissue, Capillaries

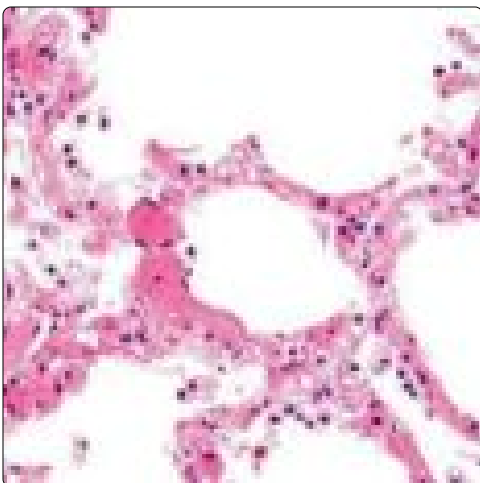


Capillaries, Renal Medulla

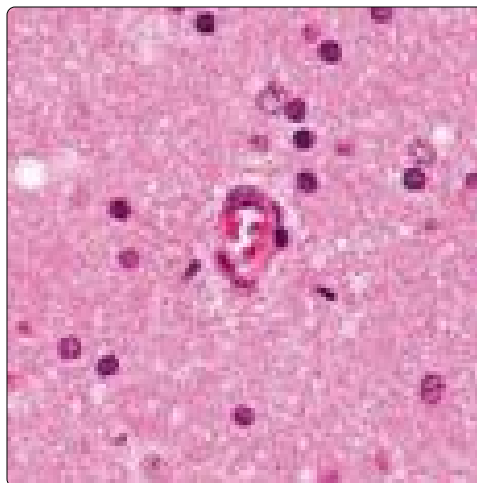


(Left) Capillaries surrounded by a loose connective tissue matrix are a component of the classic wound healing response. By light microscopy, the endothelial cells and supporting pericytes are clearly seen. **(Right)** These capillaries are composed of a single endothelial cell circumscribing the lumen. They run alongside renal tubules and form part of the vasa recta, running parallel to Henle loops in the renal papillae.

Pulmonary Alveolar Capillaries



Cerebral Cortical Capillary



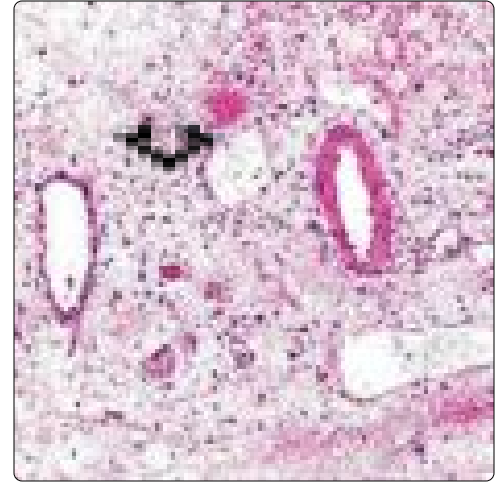
(Left) Contiguous capillaries comprise the bulk of the alveolar septal structure, allowing maximum surface area exposure for gas exchange. Several capillaries with open lumina can be seen in this section. **(Right)** Continuous-type capillaries account for the blood-brain barrier, which limits solute exchange in CNS circulation. As with other capillaries, endothelial cells and supporting pericytes are seen by light microscopy.

Fenestrations in Glomerular Capillaries

(Left) This electron micrograph shows glomerular capillary fenestrated endothelium. Endothelial cytoplasm forms a discontinuous "dashed" line to the left of the glomerular basement membrane. The clear spaces (pores) between the cytoplasm allow free communication with the extracellular compartment. **(Right)** This photograph of adjacent vein, lymphatic, and artery in cross section highlights their structural differences.



Comparison of Vein, Lymphatic, and Artery

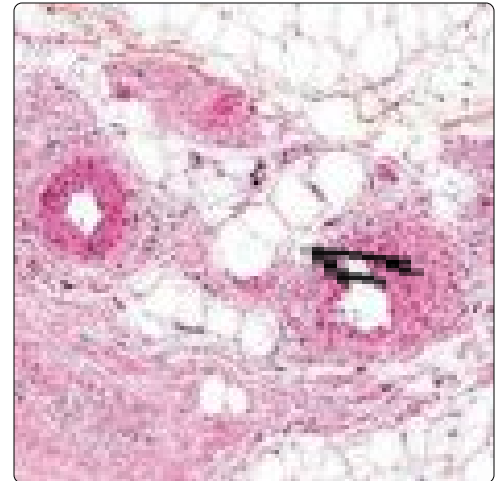


Medium-Sized Vein and Artery

(Left) This section contains a medium-sized vein and accompanying artery. Veins lack an internal elastic lamina and have thinner, less orderly medial smooth muscle. The vein adventitia is the most prominent layer, and most adventitial bundles are oriented longitudinally. **(Right)** This vein has an irregular lumen profile and prominent longitudinal adventitial collagen bundles. An artery is next to it for comparison. A nerve is also seen in this "neurovascular bundle".

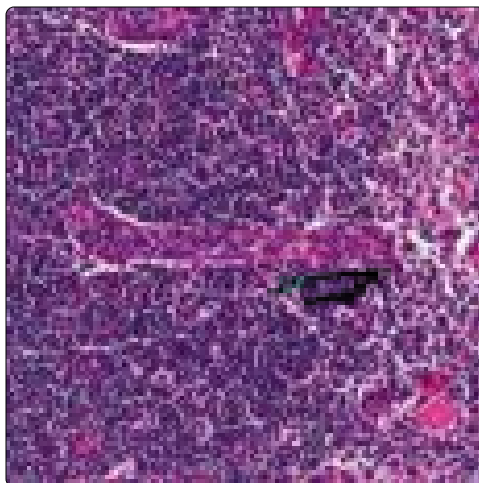


Small Vein and Muscular Artery

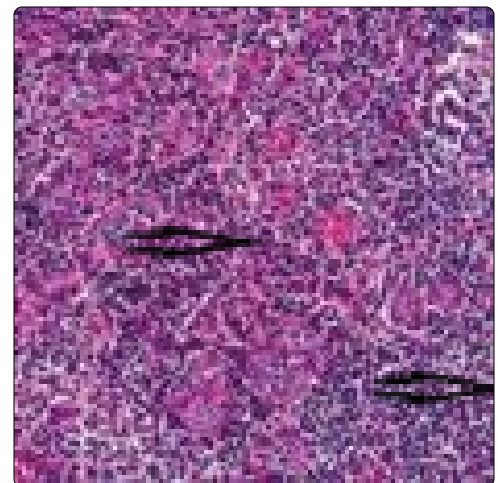


High Endothelial Venule (Longitudinal Section)

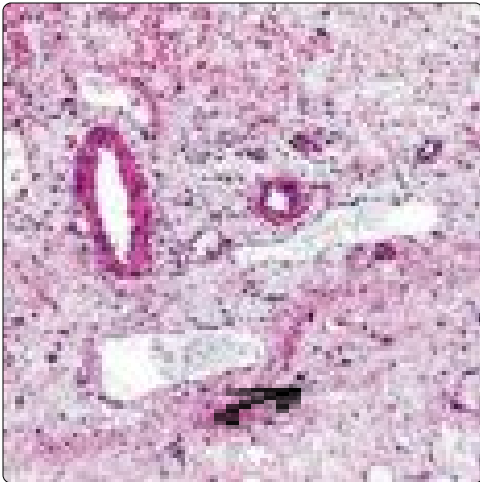
(Left) High endothelial venules within lymph nodes are specialized venule segments that serve as an active site of leukocyte homing and migration. The endothelial cells are cuboidal rather than flattened. Small lymphocytes are seen filling the lumen and within the vessel walls. **(Right)** The endothelial cells are cuboidal rather than flattened within these high endothelial venules in a lymph node. Small lymphocytes are seen filling the lumen and within the vessel walls.



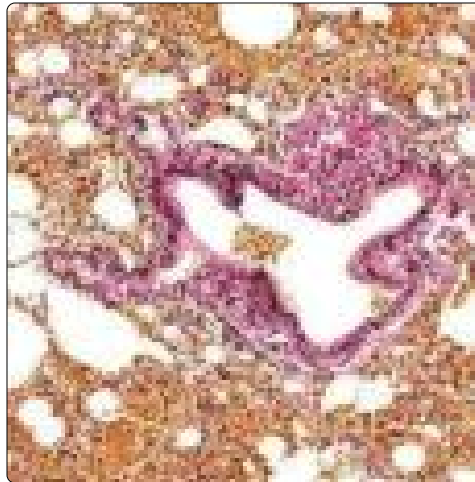
High Endothelial Venules (Cross Section)



Lymphatic Vessels

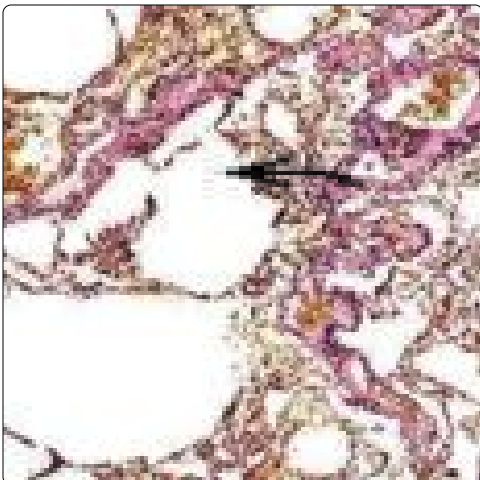


Pulmonary Vein

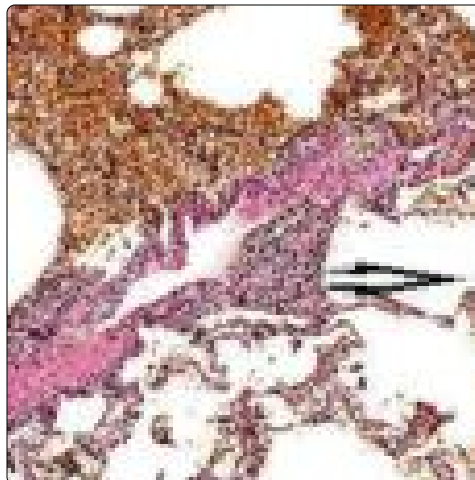


(Left) Two dilated lymphatic vessels are seen within organizing granulation tissue. Lymphatics are composed of a single layer of endothelial cells with minimal supporting connective tissue. There is little to no muscular medial wall. They often have irregular luminal contours. (Right) Veins in the lung are recognized by the absence of an accompanying airway. No internal elastic lamina is seen. The adventitial elastic fibers are coarse and poorly organized.

Pulmonary Venules and Lymphatics



Pulmonary Lymphatics

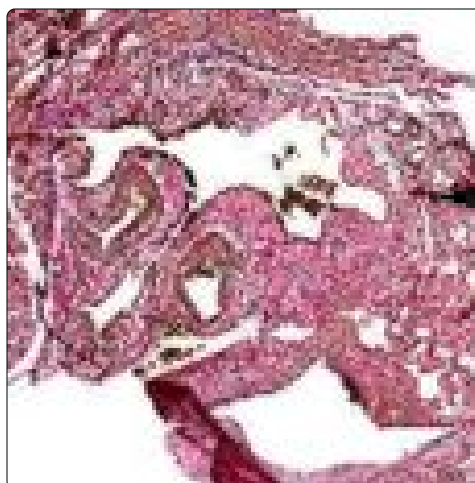


(Left) Venules and lymphatics in the lung travel in the interlobular septa. Venules have a better developed elastic adventitia and contain blood. Lymphatics show more irregular luminal contours, thinner walls, and no luminal blood cells. (Right) This lymphatic channel shows complex luminal branching and some associated extranodal lymphoid tissue. A small vein is also present for comparison.

Arteriovenous Malformation



Arteriovenous Malformation

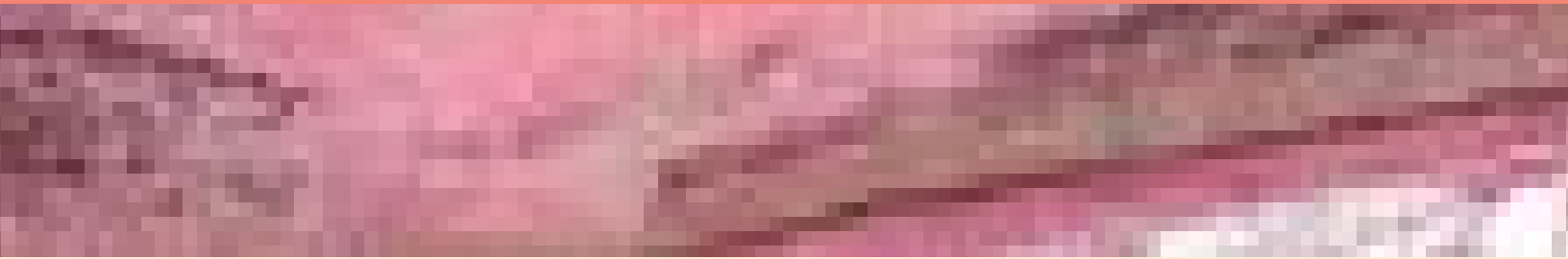


(Left) Hybrid vessel walls with arterial (internal elastic lamina) and venous (absent internal elastic lamina and prominent adventitial collagen and longitudinal smooth muscle) features are seen in this arteriovenous malformation (AVM). (Right) Irregular cavernous lumina and vessel walls showing venous features (absent internal elastic lamina and prominent adventitial bundles) are seen in this AVM. Disproportionately thick-walled vessels are also embedded in the walls of larger vessels.

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SECTION 5

Nervous System



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Peripheral Nervous System

PERIPHERAL NERVOUS SYSTEM

Macroscopic Anatomy

- Nerve trunks: Glistening, grayish-white cords
- 12 cranial nerves and 31 spinal nerves
- Spinal nerves connected to spinal cord (SC) via 2 roots: Anterior (efferent/motor), posterior (afferent/sensory)
- Sensory cell bodies: In posterior (dorsal) root ganglion
- Motor cell bodies: SC anterior horn, brainstem nuclei
- Distally spinal nerves divide: Anterior + posterior rami
 - Supply anterior and posterior parts of body
 - Anterior rami-supplying limbs form nerve plexuses
 - Upper: Cervical, brachial; lower: Lumbar, sacral

Microscopic Anatomy

- Neuron/ganglion cell: Basic anatomic/functional unit
 - Highly specialized with complex morphology
 - Perikaryon (soma): Large, spherical neuron cell body
 - Large euchromatic pale nuclei, prominent nucleoli
 - Nissl substance: Basophilic material in cytoplasm
 - Granular endoplasmic reticulum and ribosomes
 - Neurofibrils: Silver + neurofilaments, neurotubules
 - Dendrites: Multiple branching short processes
 - Receive impulses and conduct to soma
 - Axon: Single, slender, long; conducts away from soma
 - Axoplasmic flow and retrograde transport system
- Types of neurons
 - Based on axon/dendrite configuration (polarity)
 - Multipolar: Several branching dendrites, 1 axon, polygonal (stellate-pyramidal) soma
 - Unipolar (pseudounipolar): Single T-shaped process branches into axon/dendrites; rare (in ganglia)
 - Bipolar: Single dendrite and single axon at opposite poles of spindle-shaped soma; rare (retina, cochlea)
 - Based on axon length (Golgi types)
 - Type I: Extensive dendrites, long axon forms nerve
 - Type II: Short axons; interneurons in CNS pathways
 - Based on stimulus direction (motor vs. sensory)
 - Motor: Control effector organs (muscles, glands)
 - Sensory: Receive stimuli from environment

- Nonneuronal (satellite) cells
 - Schwann cells: Neural crest derived; S100(+)
 - Separated from endoneurium by basal lamina
 - Perineurial cells: Fibroblast derived; EMA(+)
 - Endoneurial capillaries from vasa nervorum
 - Endothelium tight junctions (blood-nerve barrier)
- Connective tissue sheaths
 - Endoneurium: Delicate connective tissue, few cells
 - Surrounds each individual nerve fiber
 - Perineurium: Multilayered concentric cells, collagen
 - Encloses each fiber bundle/fascicle
 - Fuses to form sensory endings, motor endplate
 - Blends with piaarachnoid
 - Epineurium: Dense fibrous connective tissue
 - Envelops main nerve trunk (several bundles)
 - Merges with surrounding adipose tissue
 - Continuous with dura mater

Organization of Peripheral Nervous System

- Nerve fiber: Long process of neuron (usually axon)
 - Grouped into fascicles by connective tissue sheaths
- Fibers myelinated (2-18 μm) or unmyelinated (0.2-3.0 μm)
 - Myelin: Lipoprotein complex; insulating sheath
 - Pale, unstained halo around axon on H&E stains
- Nodes of Ranvier: Junctions (gaps) in myelination
 - Internodal segment myelinated by 1 Schwann cell
 - Variable number of fibers enveloped (5-20 axons)
- Synapse: Specialized membranous contact (neurons)
 - Usually: Axon of 1 to dendrites of another
- Ganglion: Collection of neuron cell bodies, glial cells
 - Autonomic or sensory

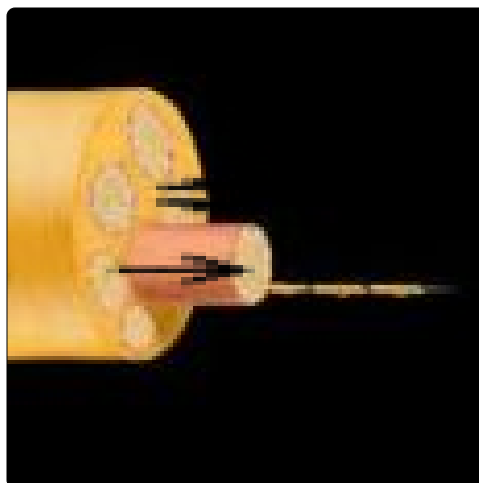
CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

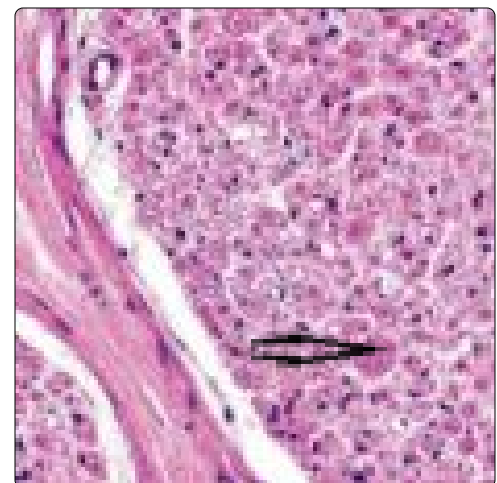
- Crush injury/artifact resembles axonal degeneration
- Electrocautery/thermal artifact may mimic infarction
- Crushed vessels mistaken for healed/fibrotic vasculitis
- Fixation in hyperosmolar solution causes shrinkage
 - Empty space around neurons mistaken for atrophy

Anatomy of Peripheral Nerve

(Left) Each axon  is surrounded by myelin-producing Schwann cells  and covered by endoneurium . Multiple axons are grouped into fascicles , each covered by perineurium . Several fascicles together are bound by epineurium , forming the main nerve trunk. (Right) High-power view of a peripheral nerve cross section shows individual fibers (axons)  in parallel arrays, surrounded by endoneurium  and bundled into fascicles covered by perineurium . (Courtesy P. Burger, MD.)



Peripheral Nerve



Peripheral Nerve Cross Section



Nerve Composed of Bundles and Fascicles

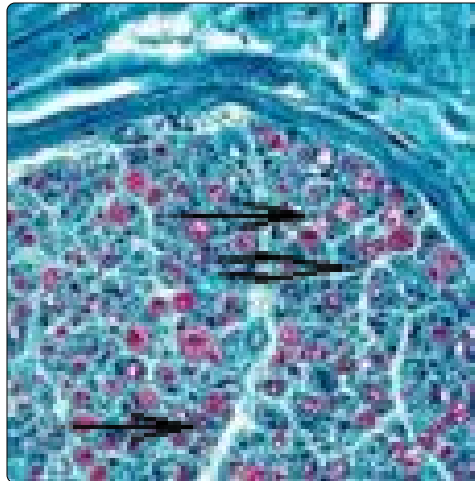


(Left) Each main nerve trunk is enveloped by epineurium, the outer layer of dense fibrous connective tissue that merges with surrounding fibroadipose tissue. Blood vessels (vasa nervorum) pass through this epineurium to reach the nerves. (Courtesy P. Burger, MD.) (Right) Peripheral nerves are composed of multiple bundles (or fascicles), each enclosed by perineurium.

Peripheral Nerve (Trichrome)

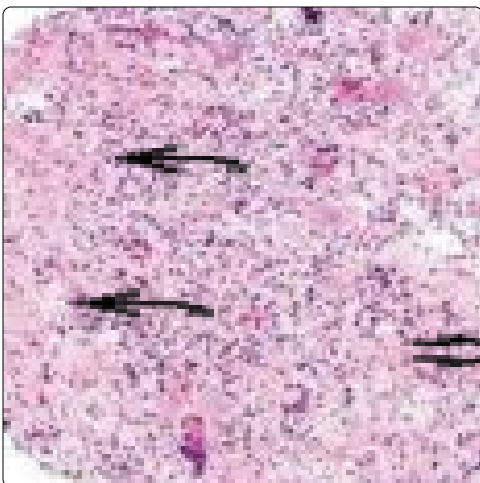


Endoneurium (Trichrome)

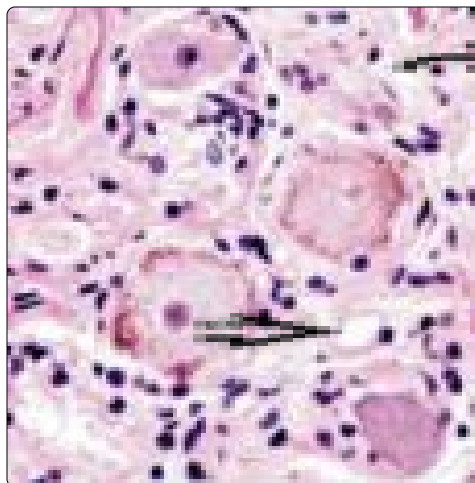


(Left) Peripheral nerve axons are red on trichrome stain, and connective tissue is blue-green. Perineurium, composed of multiple concentric layers of flattened cells and collagen, surrounds each bundle/fascicle. (Courtesy P. Burger, MD.) (Right) The individual nerve fibers stain red, and the endoneurium is blue-green. Endoneurium is composed of satellite cells and fine connective tissue that extends around and between each individual nerve fiber. (Courtesy P. Burger, MD.)

Dorsal Spinal Root (Sensory) Ganglion



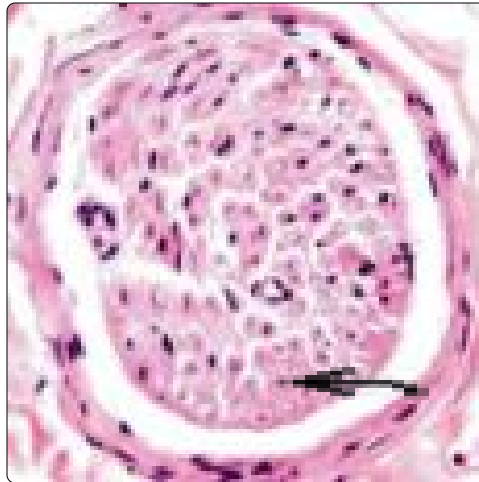
Sensory Ganglion



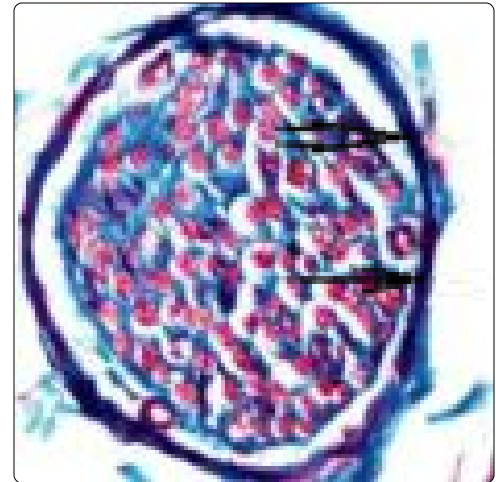
(Left) A section from the dorsal root ganglion (T10 level) shows clusters of neuronal cell bodies. The surrounding smaller, darker nuclei belong to satellite cells. The longitudinal bundles of nerve fibers belong to the spinal nerve associated with this ganglion. (Right) This high-power view shows large spherical perikarya with large, pale, centrally located nuclei and prominent nucleoli. Brown-blue Nissl substance can be seen in the cytoplasm.

Perineurium

(Left) The perineurium around this single, small nerve fascicle is composed of collagen and spindle-shaped perineurial cells. The empty space is due to fixation artifact and should not be mistaken for atrophy. Nuclei of nonneuronal satellite cells in the endoneurium mostly belong to Schwann cells. (Right) A trichrome-stained section of a nerve fascicle shows axons and myelin sheaths in red and the delicate connective tissue of endoneurium in blue.

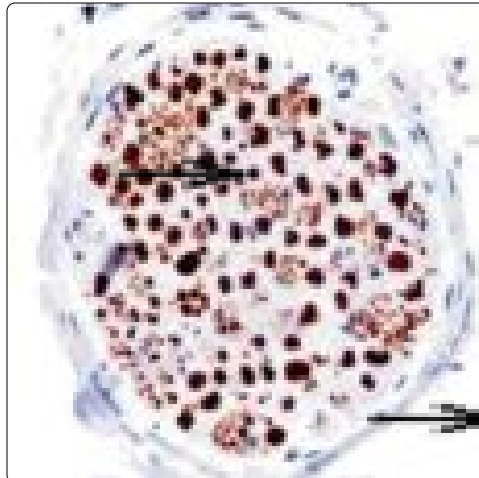


Myelin Sheath (Trichrome)

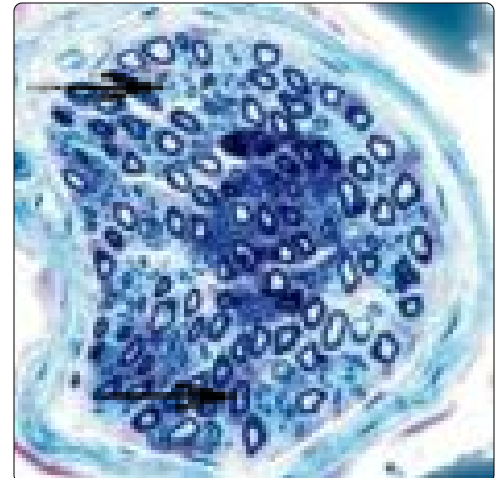


Neurofilament Protein

(Left) Immunohistochemical stain for neurofilament protein shows large, myelinated axons that are strongly positive. The variable diameters of the staining axons reflect different amounts of myelin surrounding each nerve fiber/axon. Unmyelinated axons are very narrow and difficult to detect. (Right) The myelin stains dark blue in this toluidine blue-stained plastic section. A mixture of large- and small-diameter myelinated nerve fibers can be seen.

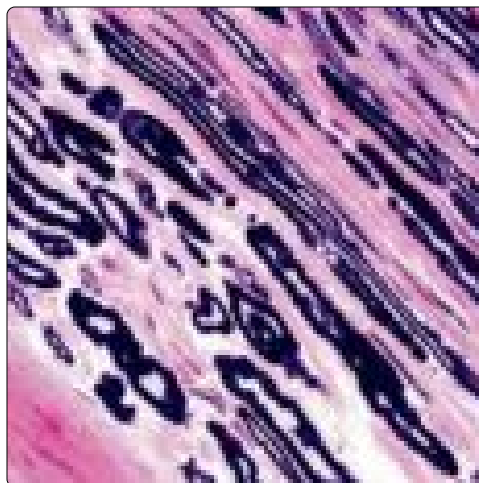


Myelin (Toluidine Blue)

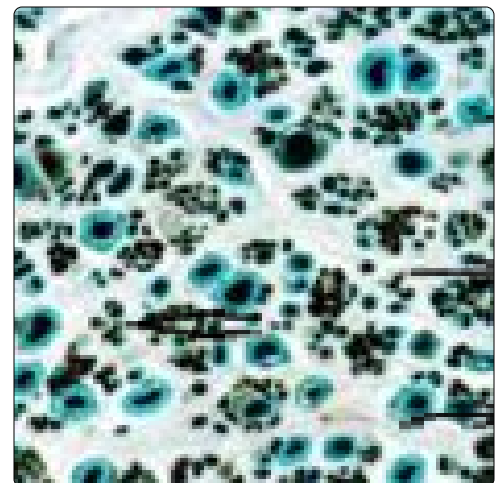


Nodes of Ranvier

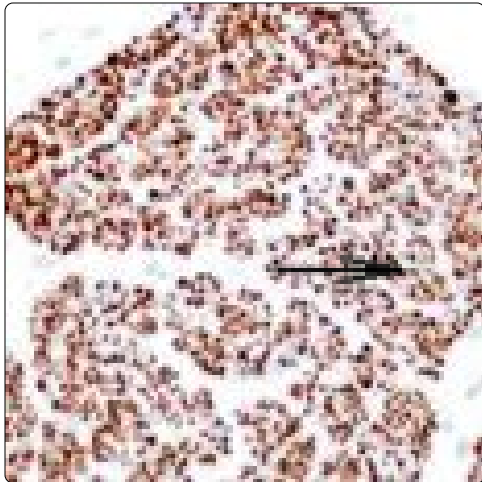
(Left) This toluidine blue longitudinal section shows nodes of Ranvier seen as gaps in myelination. The internodal segment between is myelinated by a single Schwann cell. (Courtesy B. Scheithauer, MD.) (Right) Luxol fast blue stains myelin blue-cyan (seen here around large-diameter axons), which are stained black with a Bielschowsky stain. Small-diameter axons are unmyelinated. This can identify demyelinating/remyelinating processes. (Courtesy B. Scheithauer, MD.)



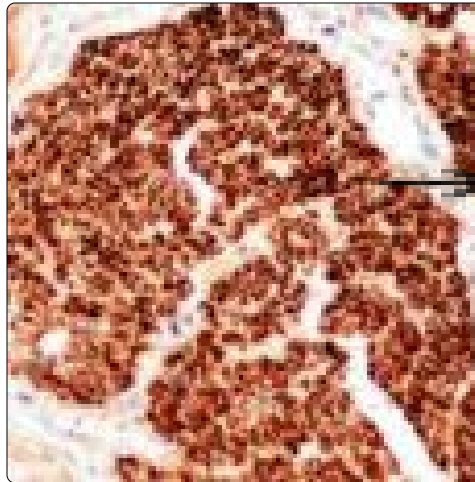
Myelin (Luxol Fast Blue)



Neurofilament Protein in Nerve



Schwann Cells

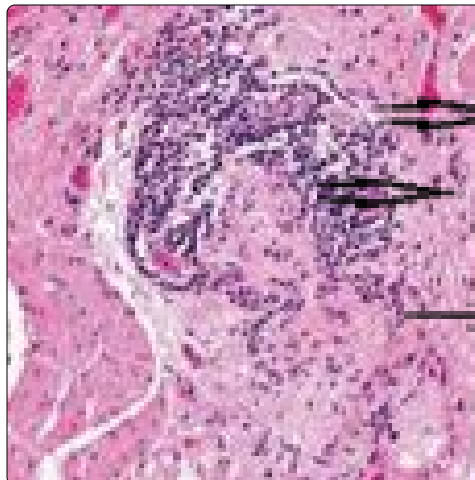


(Left) This medium-power neurofilament protein stain shows a cross section of peripheral nerve with strongly positive fascicles of nerve fibers. The space between fascicles represents perineurium. (Right) S100 protein shows Schwann cells staining strongly positive in the endoneurium, while surrounding nerve fibers are negative.

Axons and Schwann Cells



Myenteric Ganglion



(Left) Axons are surrounded by myelin sheaths produced by Schwann cells. Schwann cell nuclei and cytoplasm with abundant endoplasmic reticulum are also seen. (Courtesy P. Boyer, MD, PhD.) (Right) This section from the colon wall shows Auerbach (myenteric) plexus, a collection of ganglion cells in the enteric nervous system. Increased numbers of lymphocytes, as seen here, are abnormal, but the cause is not always clear.

Peripheral Nerve Tumor



Peripheral Nerve Tumor (Neurofilament Protein)



(Left) This cross section from a peripheral nerve shows a schwannoma in the center of the nerve trunk displacing normal nerve fibers to the periphery. (Right) Neurofilament protein staining shows a schwannoma that does not stain, whereas the peripherally displaced normal nerve fibers as well as a neighboring nerve are both positive.

CENTRAL NERVOUS SYSTEM

Macroscopic Anatomy

- Brain
 - Supratentorial components (over tentorium cerebelli)
 - Cerebrum
 - Frontal, parietal, temporal, occipital lobes
 - Cerebral cortex has 6 layers
 - Molecular, external granular, external pyramidal, internal granular, internal pyramidal, polymorphic/multiform
 - Subcortical white matter
 - Projection fibers and commissural fibers
 - Fissures/sulci separate gyri/lobes
 - Sylvian fissure/lateral sulcus (temporal from frontal/parietal), longitudinal cerebral fissure (left/right hemispheres)
 - Deep basal ganglia nuclei (caudate, putamen, globus pallidus); diencephalon (thalamus, hypothalamus)
 - Infratentorial components (posterior fossa contents)
 - Cerebellum
 - Superior, middle, and inferior peduncles
 - Cerebellar cortex has 3 layers
 - Outer hypocellular molecular, central Purkinje cell, inner/deep granular
 - Brainstem (midbrain, pons, medulla)
 - Cranial nerve motor nuclei, sensory nuclei, and autonomic nuclei
 - Ventricular system: Hollow chambers/connections that contain CSF (2 lateral, 3rd, 4th ventricles)
- Spinal cord (SC)
 - Central, H-shaped gray matter horns
 - Posterior/dorsal: Sensory input (dorsal root ganglia)
 - Anterior/ventral: Motor neurons to spinal nerves
 - Lateral autonomic (intermediolateral cell columns)
 - Surrounding white matter fiber bundles (funiculi)
 - Thickest in cervical SC (ascending/descending fibers)
 - Terminus (filum terminale): Mostly meningeal tissue
 - Central canal: Ependyma-lined neural tube remnant

Microscopic Anatomy

- Gray matter: Rich in neuronal and glial cell bodies
- Neuron (nerve cell): Anatomical/functional unit
- Cell body (perikaryon): Spherical, ovoid, or angular
 - Size: 5-10 μm (granular) to 50-100 μm (Purkinje cells)
 - Nissl substance: Granular basophilic areas (rough ER)
- Axon: Single long process; conducts downstream
 - Short (interneurons) to very long (upper motor)
- Dendrites: Multiple short processes; receive stimuli
- Neuropil: Hypocellular, finely fibrillar meshwork of unmyelinated neuronal and astrocytic cell processes
- White matter: Compact bundles (tracts, fascicles) of myelinated axons and supporting oligodendrocytes
- Astrocytes: Most common; outnumber neurons 10:1
 - Protoplasmic: Many short processes; in gray matter
 - Fibrous: Few long processes; mostly in white matter
 - Round-oval nuclei (10 μm); even, pale chromatin
- Oligodendroglia (oligodendrocytes)
 - Small round nucleus (8 μm), dense chromatin
 - Perinuclear "halo": Artifactual rim of clear cytoplasm
 - Accumulate around neurons ("neuronal satellitosis")
- Pigmented catecholaminergic neurons (substantia nigra, locus coeruleus): Cytoplasmic neuromelanin
- Microglia: Mesoderm-derived resident macrophages
 - Small elongated cells with short processes, thin rod nuclei (5-10 μm); express lysosomal marker CD68
- Ependyma: Epithelioid cuboidal/low columnar cells
 - Line ventricular system (form brain-CSF barrier)
 - Pale vesiculated nucleus (8 μm) on abluminal side
 - Ciliated/microvillous border on ventricular side

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

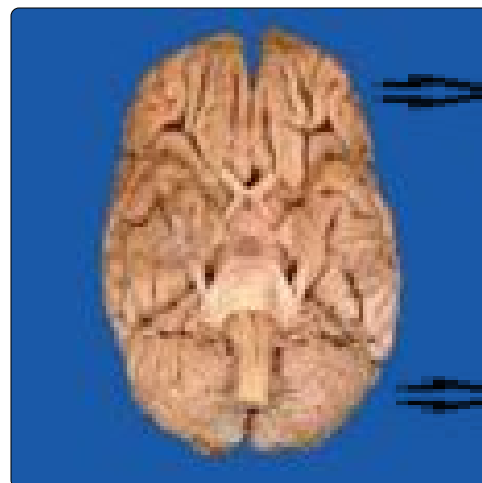
- Neurons accumulate substances with increasing age
 - Yellow-brown intracytoplasmic lipofuscin
 - Neurofibrillary tangles, plaques, Hirano bodies
 - Marinesco bodies: Bright red intranuclear inclusions
- Astrocytes accumulate corpora amylacea in processes

Brain: Sagittal Section

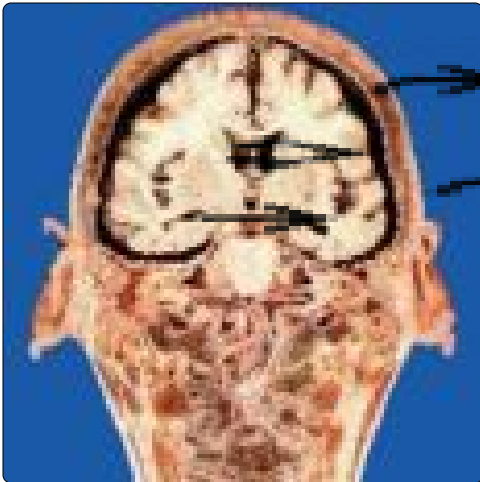


(Left) Midline sagittal brain section (medial view) shows the frontal, parietal, and occipital lobes of the cerebral hemispheres, as well as the cerebellum, pons, and medulla. (Courtesy M. Nielsen, MS.) (Right) Inferior view shows the 2 cerebellar hemispheres, beginning of spinal cord, pons, and optic chiasm. The longitudinal cerebral fissure separates left from right cerebral hemispheres, here at the level of frontal lobes. (Courtesy M. Nielsen, MS.)

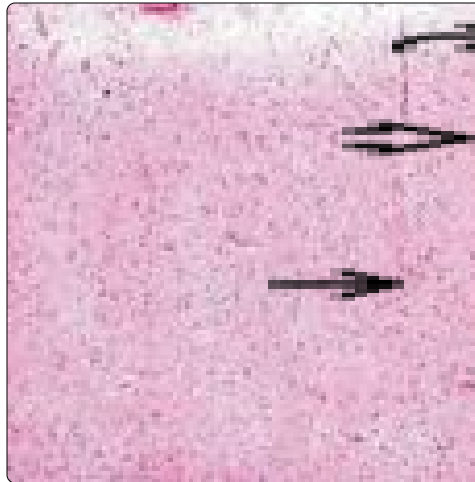
Brain: Inferior View



Coronal Brain Section

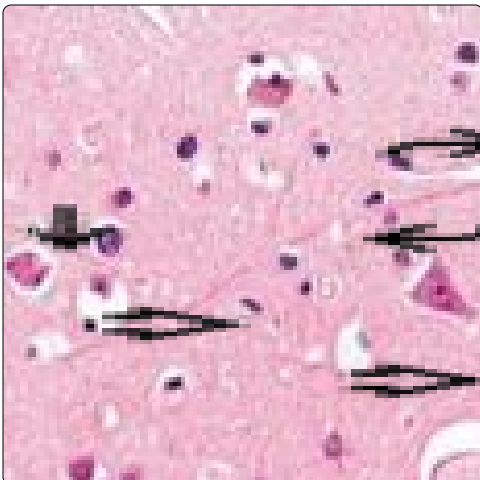


Cerebral Cortex Layers



(Left) Anterior view of coronal head section shows the sylvian fissure or lateral sulcus separating temporal from frontal lobes. Also seen are the paired lateral ventricles, deep nuclei of basal ganglia, and pons. (Courtesy M. Nielsen, MS.) (Right) This H&E shows the 6 layers of the cerebral cortex between the pia mater (top) and subcortical white matter (bottom left): Molecular, external granular, external pyramidal, internal granular, internal pyramidal, and polymorphic or multiform.

Cerebral Cortex Cells

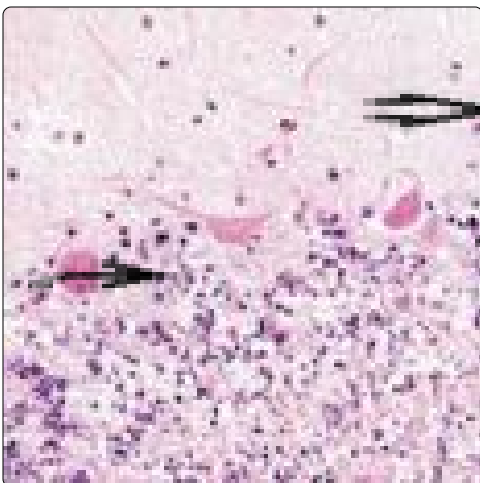


Cerebellar Cortex Layers

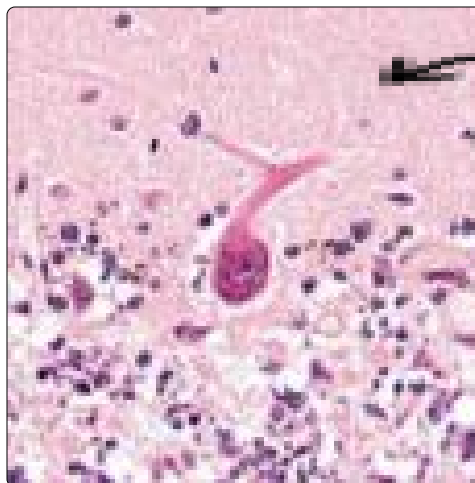


(Left) The cellular components of the cerebral cortex (and all of the CNS) include neurons, astrocytes, oligodendrocytes, and microglia. (Right) This view shows the 3 layers of the cerebellar cortex between surface pia mater and the white matter of cerebellar medulla: Outer hypocellular molecular layer, middle/central Purkinje cell layer, and inner/deep hypercellular granular cell layer. The small cells of the latter are difficult to recognize histologically as classic neurons.

Purkinje Cell Layer



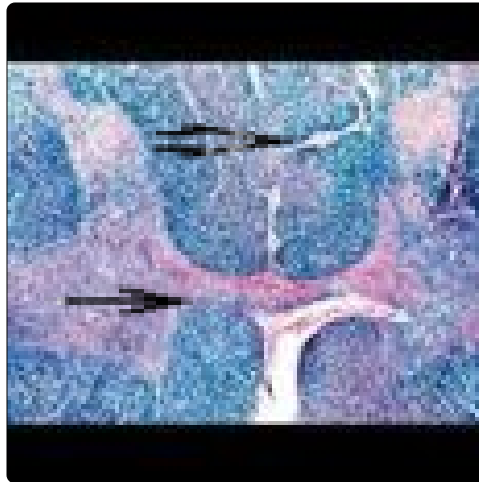
Purkinje Cell



(Left) The intermediate lamina of the cerebellar cortex consists of a single cell layer of Purkinje cells, sitting between the outer, relatively hypocellular molecular layer and the inner, densely populated granular cell layer. (Right) Purkinje cells in the cerebellar cortex are composed of a large (50 μ m) nerve cell body (perikaryon) with a prominent vesicular nucleus, granular basophilic Nissl bodies/substance, and a dendritic arbor that extends deep into the molecular layer.

Cervical Spinal Cord

(Left) Cross section of cervical spinal cord shows large anterior/ventral horns of gray matter containing motor neurons that innervate upper extremities. Posterior/dorsal horns get sensory input from dorsal root ganglia. Funiculi of white matter fibers are thickest here (largest number of ascending and descending tracts). (Right) Thoracic cord has smaller anterior/posterior horns but also contains the lateral horns of the intermediolateral cell column (sympathetic autonomic) neurons.



Thoracic Spinal Cord

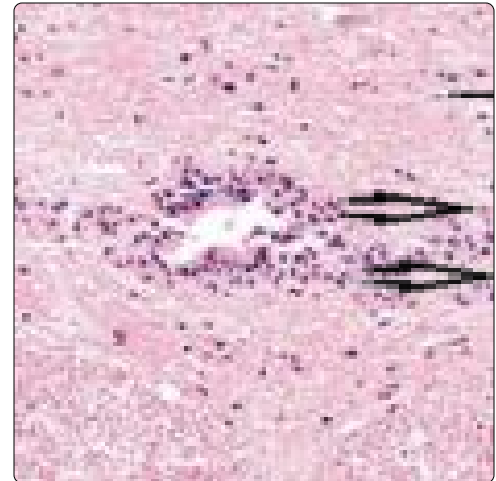


Lumbosacral Spinal Cord

(Left) Lumbosacral spinal cord has very large anterior/ventral gray matter horns that house motor neurons supplying the lower extremities. White matter tracts are thinner here, as small numbers of ascending tracts have joined. (Right) The central canal of the spinal cord is lined (as is the ventricular system) by ciliated cuboidal/columnar ependyma cells. The lumen, often obliterated in adults, contains CSF. Surrounding gray matter is part of the gray commissure.

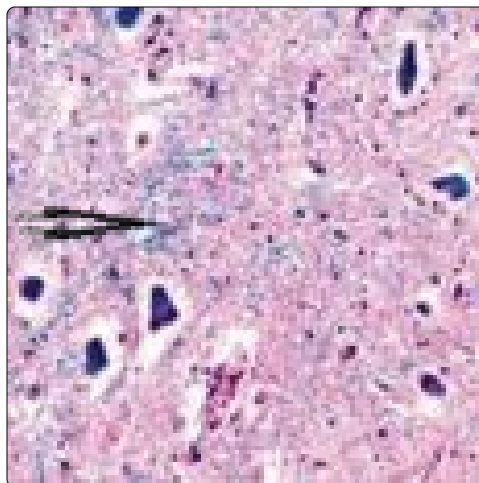


Spinal Cord: Central Canal

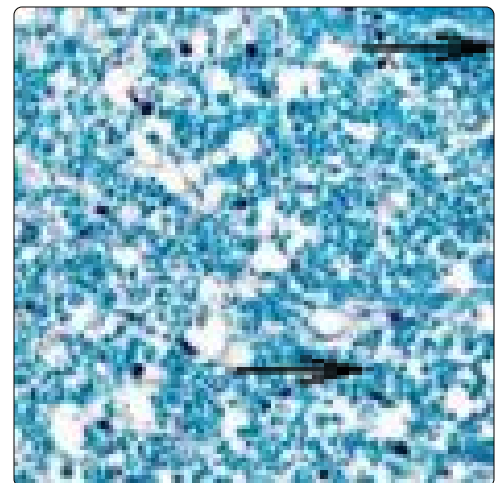


Spinal Cord Gray Matter

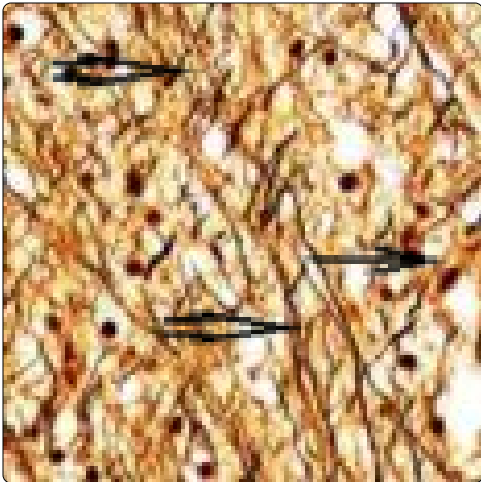
(Left) Spinal cord gray matter contains multiple neuronal cell bodies and nuclei of various supporting glial cells amid the finely fibrillar background of the neuropil meshwork. The latter is composed of unmyelinated (negative in this Luxol fast blue stain) neuritic and astrocytic cell processes. (Right) Luxol fast blue stain highlights the blue myelinated axons of neurons in the white matter of the cord (and CNS) and the small dense round nuclei of oligodendrocytes that produce this myelin sheath.



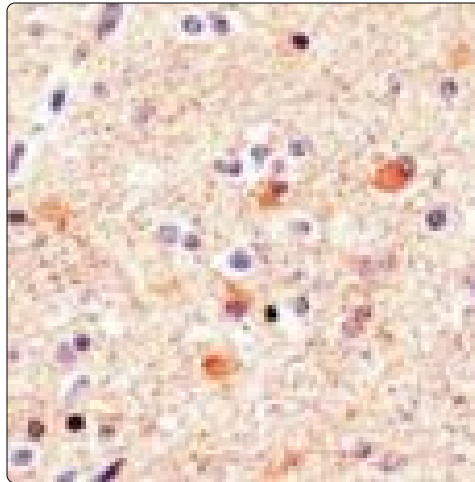
Spinal Cord White Matter



White Matter Axons

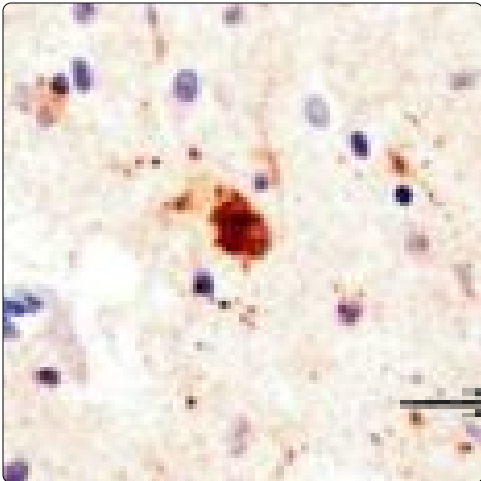


GFAP-Positive Astrocyte

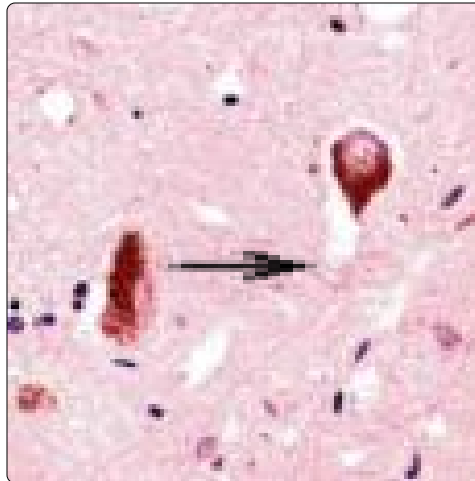


(Left) High-power view of a Bielschowsky silver stain highlights the argyrophilic (silver-positive) long axonal processes of neurons in largely uniform white matter peppered by the nuclei of myelinating oligodendroglia. **(Right)** High-power view of an immunohistochemical stain for GFAP highlights immunoreactive astrocytes with star-shaped cytoplasm, round pale nuclei, and multiple processes.

Synaptophysin-Positive Neuron

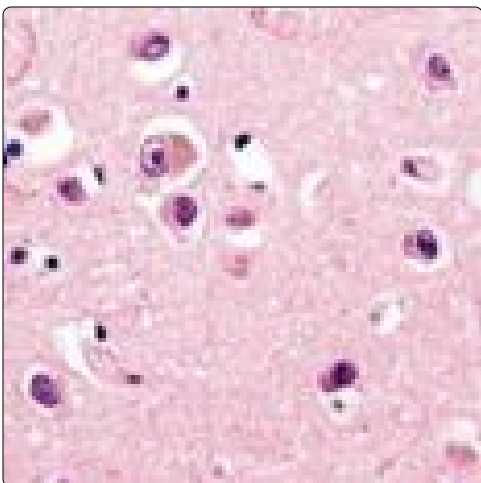


Substantia Nigra Neurons

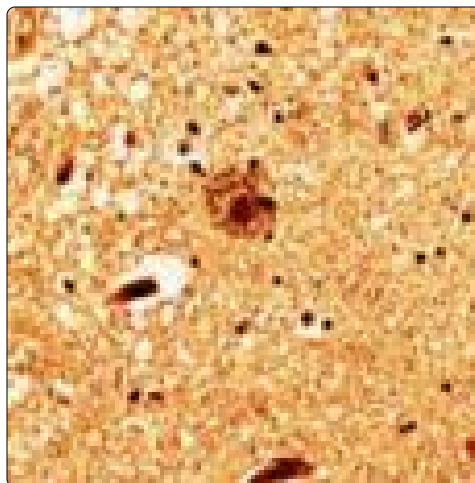


(Left) Synaptophysin is a widely used marker of neuronal differentiation that is found in synaptic vesicles and, thus, highlights the nerve cell body and proximal dendrites of a large neuron in a punctate pattern. In the background, neuropil, seen as diffuse, finely granular staining indicates synaptic contacts. **(Right)** Pigmented catecholaminergic neurons of the brainstem (substantia nigra in midbrain, locus coeruleus in the pons) contain neuromelanin, seen as coarse dark brown cytoplasmic granules.

Lipofuscin (Lipochrome) Accumulation



Senile (Neuritic) Plaque



(Left) Lipofuscin is a pale, yellow-brown cytoplasmic accumulation seen with age in neurons and often peripherally displaces nucleus and organelles. **(Right)** Bielschowsky silver stain shows a senile plaque in the aging brain as an ill-defined round cluster of extracellular granular material (abnormal neurites) that alters the smooth fibrillarity of background neuropil and causes nearby axons to disperse. Mature plaques are visible on H&E and acquire a central eosinophilic core that is positive for amyloid.

Meninges

MENINGES

Macroscopic Anatomy

- Meninges consist of 3 separate layers of connective tissue membranes that envelop central nervous system (CNS)
 - Dura mater, arachnoid mater, and pia mater
 - Arachnoid and pia mater constitute leptomeninges
- Dura mater (pachymeninx, meninx fibrosa)
 - Tough outermost layer
 - Composed of 2 tightly juxtaposed or fused layers of dense fibrous connective tissue that separate to form dural venous sinuses (major venous channels)
 - Outer (external) layer of dura is adherent to inner skull and functions as periosteum of cranium
 - Inner dura layer joins arachnoid to form infoldings that separate parts of brain (falx cerebri, falx cerebelli, tentorium cerebelli, diaphragm sellae)
 - Penetrated by arachnoid villi, draining veins
 - Potential spaces
 - Epidural (between dura and cranium/vertebral canal)
 - Subdural (between dura and leptomeninges)
- Arachnoid mater
 - Delicate, thin, continuous, and transparent membrane closely apposed to dura mater above
 - Wispy, web-like connective fibers (arachnoid trabeculae) adjoin arachnoid to underlying pia
 - Subarachnoid space (between pia and arachnoid)
 - Subarachnoid cisterns
 - Enlarged areas of subarachnoid space (e.g., cisterna magna) may contain additional cranial nerves
- Pia mater
 - Internal delicate membrane covers brain surface, follows contours (gyri and sulci)
 - 2 closely attached layers
 - Outer layer (epipia)
 - Covers CNS surface, coats all penetrating arterioles/venules until they become capillaries
 - Inner layer (intima pia)
 - Closely apposed to peripheral astrocytic end processes (pia-glial limiting membrane)

- Potential space
 - Subpial (between pia and underlying brain/spinal cord)

Microscopic Anatomy

- Dura mater
 - Elastic and dense collagen fibers arranged in laminae
 - Contains numerous blood vessels
 - Cerebral venous sinuses lined by endothelium
- Arachnoid mater
 - Composed of 2 layers (subdural and inner) of mesothelial cells
 - Compact central layer: Polygonal cells, round nuclei, pale cytoplasm
 - Desmosomes and intercellular tight junctions
 - Granulations: Small aggregates/clusters near intracranial venous sinuses
- Pia mater
 - Loose connective tissue with collagen and elastic fibers
 - Thin sheet of flattened mesothelial cells with desmosomes and gap junctions

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Dura can fuse with skull in older individuals
- Nonspecific calcifications in dura increase with age
- Arachnoid mater and granulations increase in size, thicken with advancing age due to collagen deposition

Pitfalls and Artifacts

- Arachnoiditis ossificans
 - White plaques covering dorsal leptomeninges of spinal cord (thoracic, lumbar, sacral)
 - Composed of laminated and hyalinized fibrous tissue
 - Incidental autopsy finding, no relevant clinical history
 - True mineralization occurs with prior symptomatic inflammation or trauma
- Leptomeningeal melanocytes
 - Normal cellular components of pia and arachnoid
 - Impart brown discoloration grossly, especially over ventral midbrain

(Left) This coronal view shows the brain and cranial dura mater. The 2 layers (outer and inner) of the dura mater separate to form the superior sagittal sinus. The inner dural layer joins the underlying arachnoid to form the falx cerebri. **(Right)** The dense fibrous dura mater overlies the delicate translucent arachnoid, which is closely attached to it. Web-like connective tissue fibers (arachnoid trabeculae) join the arachnoid to the underlying pia mater.

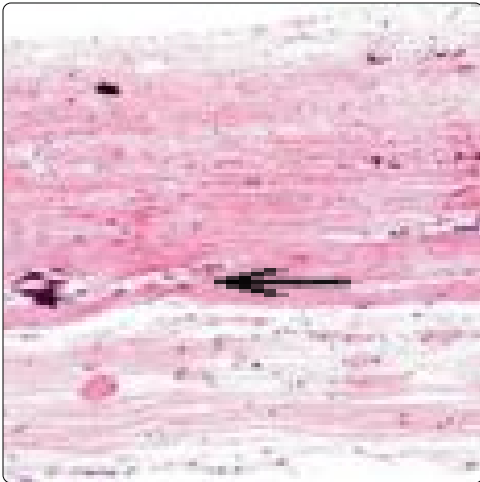
Meninges



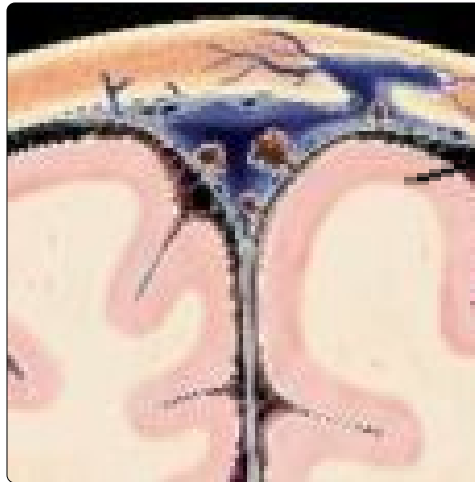
Dura and Arachnoid



Dura Mater

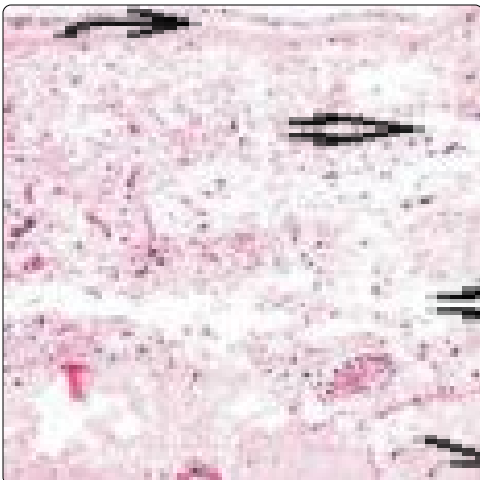


Superior Sagittal Sinus



(Left) The dura mater is composed of dense fibrous connective tissue. The outer and inner layers are often fused and difficult to distinguish. Nonspecific calcifications can be seen, especially with increasing age. (Right) The superior sagittal sinus is a venous channel formed by the separation of the outer (periosteal) and inner (meningeal) dural layers. Arachnoid granulations are projections of the arachnoid-covered, CSF-containing subarachnoid space into the blood-filled venous sinus.

Arachnoid and Subarachnoid Space



Leptomeninges

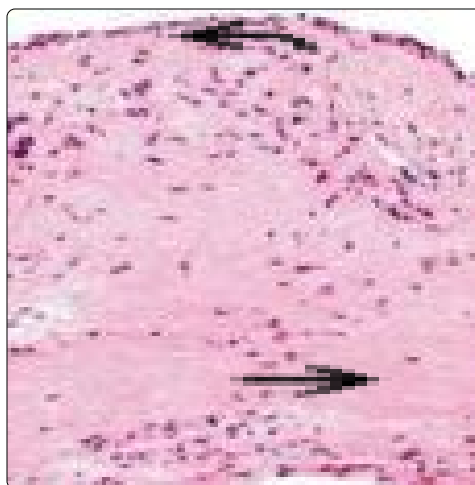


(Left) The subarachnoid space is a potential space found below the arachnoid membrane and occupied by CSF, superficial cerebral vessels, and arachnoid trabeculae (connective tissue fibers between arachnoid and underlying pia). (Right) This low-power view illustrates the relationships between the arachnoid (thin layer of mesothelial cells), subarachnoid space, and pia mater (delicate membrane covering the brain, adherent to gyri and sulci).

Pia Mater



Arachnoid Cyst



(Left) The pia mater is a delicate membrane that coats the brain surface as well as penetrating cerebral vessels. The potential space between the pia mater and CNS surface is called the subpial space. (Right) Arachnoid cysts are usually congenital and asymptomatic. The arachnoid is replaced by dense fibrosis, but the characteristic overlying single cell layer of mesothelium is still visible.

Choroid Plexus

CHOROID PLEXUS

Macroscopic Anatomy

- Specialized vascular organ in ventricles of CNS
- Site of cerebrospinal fluid (CSF) production
- Locations
 - Lateral ventricles: Body, atrium, and temporal horns
 - Roof of 3rd and 4th ventricles, foramina of Monro
 - Cerebellopontine angle cisterns (in subarachnoid space)

Microscopic Anatomy

- Invaginated pia mater folds (vascular leptomeninges)
 - Papillary, frond-like processes project into ventricles
 - Contain loose tissue core containing collagen, dilated blood vessels
 - Small nests of meningotheial (arachnoid) cells
- Covered by single layer of cuboidal/low columnar cells
 - Modified ependyma cells: Larger, hobnailed surface
 - Greatly increased surface area due to villi, microvilli
 - Tight intermediate junctions (blood-CSF barrier)

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Mineral salt deposition in connective tissue cores
 - Seen with increasing age; accounts for radiodensity
- Cytoplasmic vacuoles, cystic xanthomatous change
 - Common incidental finding with increasing age
- Choroid plexus cysts
 - Common, often in fetuses
 - Some associated with aneuploidy (trisomy 18, 21)

Pitfalls and Artifacts

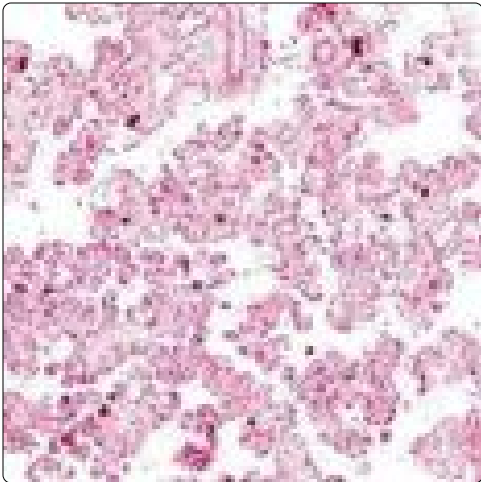
- Biopsy tissue compression can mimic hypercellularity
 - Misinterpretation as choroid plexus papilloma
 - Normal choroid cells retain hobnail appearance
 - True papilloma is contrast-enhancing mass
- Identification of choroid plexus in unusual location or as tissue "floater" by carryover
 - May be misinterpreted as metastatic carcinoma
 - Normal choroid plexus is not cytologically atypical

Choroid Plexus

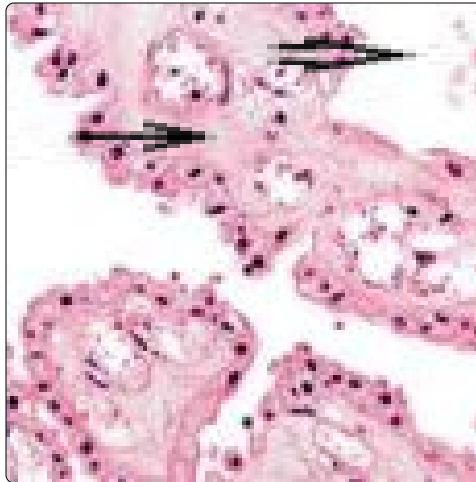


This graphic shows the sites of choroid plexus in the brain, including the roof of the 3rd  and 4th  ventricles.

Frond-Like Processes

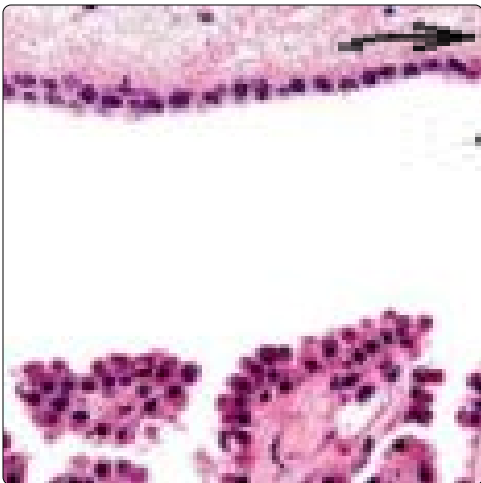


Connective Tissue Cores

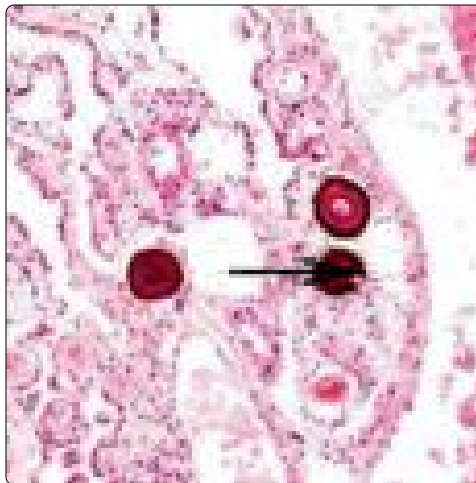


(Left) This low-power view demonstrates the papillary architecture of the choroid plexus, characterized by frond-like processes of vascular leptomeningeal tissue that project into the ventricular lumen. (Right) The cores of loose connective tissue and dilated blood vessels in the choroid plexus are covered by a single cell layer of modified ependyma cells.

Choroid Plexus Epithelium

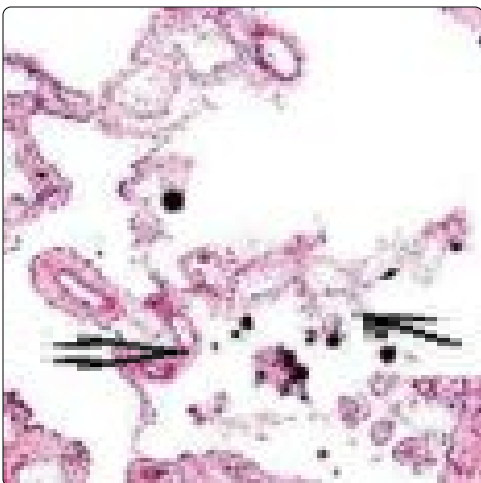


Psammomatous Calcification

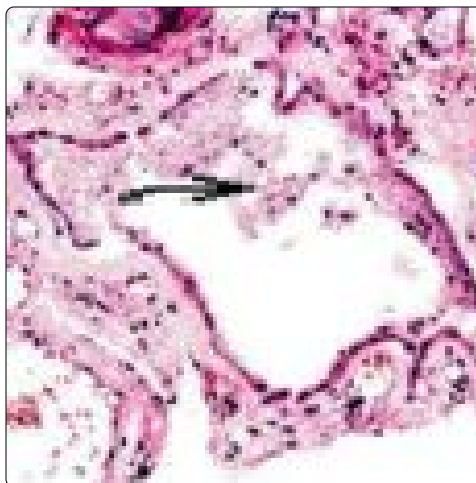


(Left) This high-power view shows the choroid plexus and ependyma cells covering the surface of the ventricle wall. In contrast to the ependyma cells, from which they are derived and with which they are continuous, choroid epithelial cells are larger, with a more hobnailed or tombstone appearance. (Right) The calcifications within the connective tissue cores of choroid plexus may assume a laminated, concentric appearance (psammoma bodies).

Choroid Plexus Cyst



Cystic Xanthomatous Change

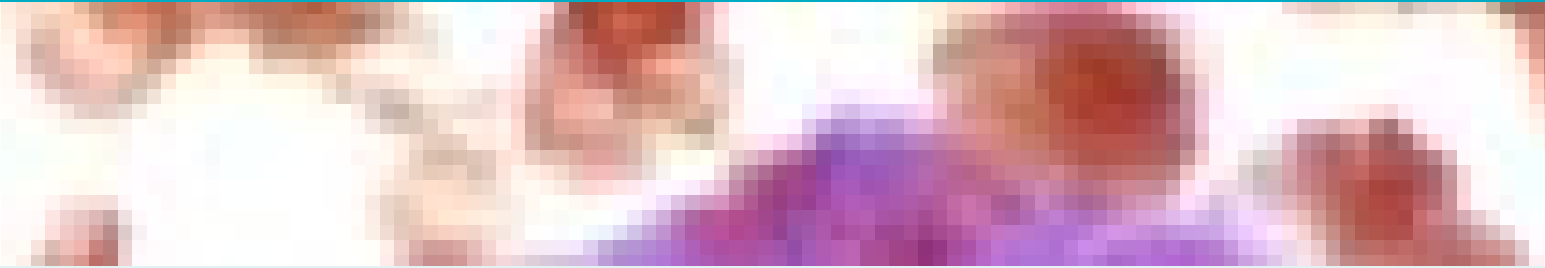


(Left) Choroid plexus cysts are a common benign lesion often seen in fetuses. Vascular connective tissue fronds of choroid plexus can be seen surrounding the lumen of the cyst. (Right) This photograph of cystic xanthomatous change shows a cystic lumen filled with xanthoma cells (foamy macrophages) and surrounded by attenuated choroid epithelial cells. This is a common incidental finding, seen with older age, but it has no specific pathologic significance.

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SECTION 6

Hematopoietic and Immune Systems



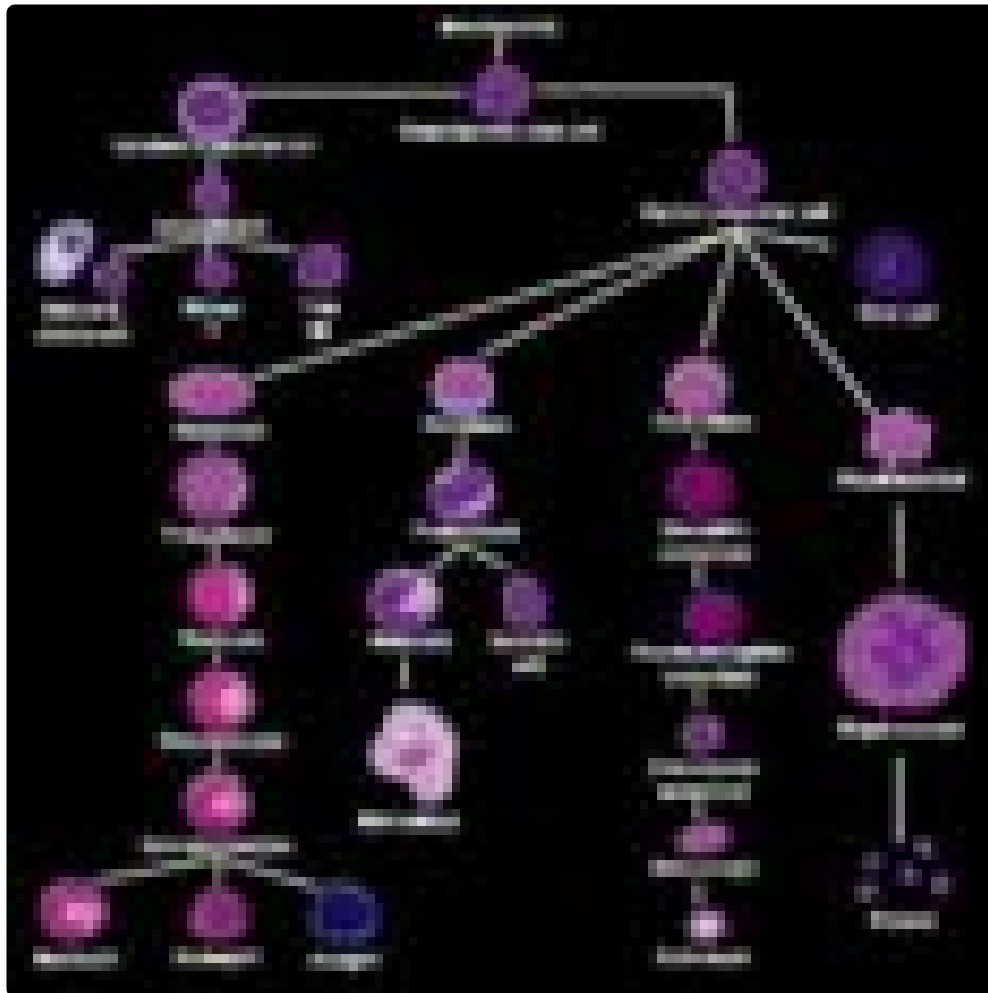
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Spleen	126
Bone Marrow	130
Peripheral Blood	138
Thymus	142

TERMINOLOGY

Definitions

- Conceptually divided into innate and adaptive immunity
- **Innate immunity**
 - Rapid and stereotyped response
 - Consists of physical, chemical, and biological barriers
 - Resistance is not acquired through interaction with antigen
 - Recognize foreign carbohydrates on surface of microorganisms
 - Carried out by specialized cells
 - Macrophages/monocytes, neutrophils, dendritic cells, natural killer cells, mast cells, eosinophils, and basophils
 - Main mechanisms include
 - Phagocytosis
 - Release of inflammatory mediators
 - Activation of complement system proteins
- Synthesis of acute-phase proteins, cytokines, and chemokines
- Present in all individuals
- Does not change or adapt following contact; nonspecific
- **Adaptive immunity**
 - Induced following contact with foreign antigens
 - Dependent on activation of specialized cells and soluble molecules produced by lymphocytes
 - NK/T, T, and B lymphocytes (TLs and BLs), dendritic cells, or antigen-presenting cells
 - Soluble molecules include antibodies, cytokines, and chemokines
 - Exhibits diversity, memory, and specificity
 - Involves 2 main components
 - Cell mediated (e.g., cytotoxic TLs)
 - Antibody mediated (antibody-producing BLs)
 - Specificity and diversity of recognition are key

Hematopoietic Cell Lineages



This schematic illustrates general features of lymphoid and myeloid cell derivation from pluripotent stem cells. Both myeloid and lymphoid lineage cells are derived from a common precursor cell. The general maturation stages, from immature to fully differentiated hematopoietic cells, are listed.

COMPONENTS OF INNATE IMMUNE SYSTEM

Dendritic Cells

- Engulf and present antigens to lymphocytes
 - Antigens are presented bound to major histocompatibility complex (MHC) molecules
- Act as bridge from innate to adaptive immune system
- Originate in bone marrow and migrate to peripheral tissues where they reside (skin, liver, intestine)
 - Migrate to regional lymph nodes upon activation

Neutrophils

- Most abundant leukocytes in peripheral blood
- Constitute 40-60% of circulating leukocytes
- Activated by bacterial products, complement proteins (e.g., C5a), immune complex, chemokines, and cytokines
- Involved in phagocytosis
- Contain primary, secondary, and tertiary granules

Macrophages

- Referred to as monocytes in peripheral blood
- Constitute 3-8% of circulating leukocytes
- Monocytes give rise to macrophages and myeloid dendritic cells in tissues
- Engulf pathogens and cellular debris; efficient phagocytes
 - Process and present antigens via MHC molecules to lymphocytes

Natural Killer Cells

- Originate from common progenitor cells in bone marrow
- Constitute 5-20% of circulating leukocytes
- Important line of nonspecific defense
 - Recognize and lyse virus-infected cells, bacteria, protozoa, and tumor cells
 - Lysis mediated by perforins and granzymes
 - Virus-infected cells and tumor cells are susceptible due to their low expression of MHC class I proteins
 - Recruit neutrophils and macrophages
 - Activate dendritic cells and T and B lymphocytes
 - Secrete proinflammatory cytokines

Mast Cells

- Derived from hematopoietic progenitor cells in bone marrow
 - Migrate to peripheral tissues where differentiation and maturation occur
 - Distributed along blood vessels, nerves, and under epithelial surface of skin and mucous membranes
 - Abundant in areas of environmental contact
 - Typically not observed in circulation
- Play key role in acute inflammatory reactions
 - Type I hypersensitivity reactions
- Upon activation, both preformed and newly formed mediators are released
 - Preformed mediators: Vasoactive amines, proteases, heparin, IL-4, TNF- α , and granulocyte-macrophage colony-stimulating factor (GM-CSF)
 - Newly formed mediators: Platelet activating factor, arachidonic acid derivatives, cytokines

- Release of mediators results in: Inflammatory cell migration, increased vascular permeability, mucus secretion, increased gastrointestinal motility, and bronchoconstriction

Basophils

- Derived from hematopoietic progenitor cells in bone marrow
- Constitute < 1% of circulating leukocytes
- Cytoplasmic granules contain mediators similar to mast cells
- May contribute to immediate hypersensitivity reactions

Eosinophils

- Derived from hematopoietic progenitor cells in bone marrow
- Constitute $\leq$ 5% of circulating leukocytes
- Found in greater numbers in mucosal regions and in patients with allergies &/or asthma
- Important infection-fighting cells
- Strong antiparasitic action
 - Adhere to parasites and pathogens coated with IgE or IgA
 - Release granular contents upon binding and activation
- Important components in allergic reactions and asthma
- Produce and store various proteolytic granules
- Recruited by adhesion molecules and chemokines

Complement System

- 3 forms of complement system activation: Classic, alternative, and via mannose-binding lectin
 - **Classic activation:** Stimulated by antigen-antibody reactions
 - **Alternative activation:** Stimulated by polysaccharides from yeasts and gram-negative bacteria
 - **Mannose-binding lectin activation:** Stimulated by mannose-containing proteins and carbohydrates on microbes (e.g., viruses and yeasts)
- Consists of ~ 20 plasma glycoproteins
- Upon activation, glycoproteins acquire proteolytic activity and activate subsequent components in cascade effect
 - Mediators
 - Alter vascular permeability
 - Contribute to inflammatory response
 - End result is formation of membrane attack complex
 - Promotes osmotic lysis and removal of target cell/infectious agent
 - Other antiinfectious functions include
 - Opsonic action of C3b
 - Release of soluble C3a and C5a (anaphylatoxins)
 - Inflammatory infiltrate induced by C5a

Major Histocompatibility Complex

- HLA genes reside within genetic complex referred to as MHC
- **Class I MHC proteins**
 - Glycoproteins found on surfaces of nearly all nucleated cells
 - HLA-A, HLA-B, and HLA-C loci encode numerous different proteins that make up group of MHC class I proteins

- **Class II MHC proteins**
 - Highly polymorphic glycoproteins found on surface of certain cells; macrophages, B cells, dendritic cells of spleen, and Langerhans cells of skin
 - HLA-D loci encode for class II MHC proteins (i.e., DP, DQ, DR)
 - Composed of 2 polypeptides; both encoded by MHC locus
 - Hypervariable region (provides polymorphism) and constant region (site of CD4 T-cell binding)
- Association of antigen with MHC class proteins allows recognition by T cells
 - CD8-positive cytotoxic T cells recognize antigen complexed with class I MHC
 - CD4-positive helper T cells recognize antigen complexed with class II MHC

COMPONENTS OF ADAPTIVE IMMUNE SYSTEM

B Lymphocytes

- Lymphoid progenitors arise from pluripotent stem cells within bone marrow and give rise to BLs
- BLs undergo maturation within bone marrow
 - Pro B cell → pre B cell I → pre B cell II → immature B cell → mature B cell → plasmocyte and memory cell
- Mature BLs migrate to secondary lymphoid organs (e.g., lymph nodes and spleen)
- Igs
 - Igs consist of 2 heavy chains (α , γ , δ , ϵ , μ) and 2 light chains (κ , λ)
 - Membrane immunoglobulins (IgM, IgD) recognize antigens
 - During differentiation, changes in constant portion of heavy chain allow for class switching (e.g., IgD or IgM → IgG, IgE, or IgA)

T Lymphocytes

- Lymphoid progenitors arise from pluripotent stem cells within bone marrow and give rise to TLs
- TL differentiation, maturation, and selection take place within thymus
 - Maturation process involves expression of functional T-cell receptor and coreceptors CD4 &/or CD8
 - Positive selection: Selects TLs capable of interacting with MHC I or II; positively selected TLs migrate from thymic cortex toward medulla; TLs that do not interact with MHC I or II undergo apoptosis
 - Negative selection: TLs that have survived positive selection but interact too strongly with "self" peptides are removed by way of apoptosis; process allows for self tolerance
- TLs recognize processed antigens presented by MHC molecules on surface of antigen-presenting cell
- **T-cell receptors**
 - Formed by 2 peptide chains of immunoglobulin superfamily
 - Variable and constant regions
 - Undergo recombination process similarly to BL immunoglobulins
 - Formed by α and β chains in 95% of circulating TLs ($\alpha\beta$ -TLs)

- Formed by γ and δ chains in 5% of circulating TLs ($\gamma\delta$ -TLs)
- **CD4 TLs**
 - Recognize antigens presented by class II MHC molecules
 - Responsible for orchestration of other cells involved in immune response
 - Activation of BLs, macrophages, and CD8 TLs
 - T-helper (Th) lymphocytes
 - Subdivided by cytokine production patterns during activation and effector patterns
 - **Th1 lymphocytes**: Produce IFN- γ and IL-2; function in phagocyte activation and opsonizing antibody production
 - **Th2 lymphocytes**: Produce IL-4, IL-5, IL-6, IL-10, and IL-13; function in BL activation, BL differentiation, antibody production, and eosinophil activation
 - **Th17 lymphocytes**: Produce IL-17, IL-22, IL-26, M-CSF, GM-CSF; function in myeloid expansion, chemokine production, and inflammatory cytokine production
- **CD8 TLs** (cytotoxic TLs)
 - Recognize antigens presented by class I MHC molecules
 - Recognize viral infected and tumor cells
 - Induce apoptosis in target cell via perforins, granzymes, and expression of FasL receptor (CD95)
- **Regulatory TLs**
 - Function in immune system regulation and autotolerance
 - Produce immunosuppressive cytokines (e.g., IL-4, IL-10, TGF- β)
 - Express high levels of CD25

Lymphoid Tissues

- Naïve (nonantigen-stimulated) BLs and TLs populate lymph nodes, spleen, tonsils, and mucosa-associated lymphoid tissue
- Provide microenvironment for naïve BLs and TLs to encounter specific antigen
- Contain antigen-presenting cells that produce necessary cytokines to maintain BLs and TLs
- Communicate with tissues via lymphatics and blood vessels

CLINICAL IMPLICATIONS

MHC and Transplantation

- Success of transplantation requires compatibility of donor and recipient MHC class proteins
- Class II MHC plays major role
- Severity and rapidity of rejection depend on degree of difference between donor and recipient MHC class proteins
 - **Hyperacute allograft rejection**: Occurs within minutes of engraftment due to large amounts of preformed antibody (e.g., anti-ABO antibodies)
 - **Acute allograft rejection**: Occurs within 11-14 days of engraftment; marked decrease in vascular circulation with increased mononuclear infiltrate and eventual necrosis; T-cell mediated
 - **Chronic allograft rejection**: Occurs months to years following engraftment; thought to be related to incompatibility of minor histocompatibility antigens and immunosuppressive drug side effects; results in atherosclerosis of vasculature
- Allografts require some degree of immunosuppression

Basic Characteristics of Immunoglobulin Classes

Immunoglobulin Class	Structure	Characteristics
IgD	Monomeric	Membrane-bound immunoglobulin; part of naïve B-lymphocyte membrane receptor
IgM	Monomeric; pentameric	Monomeric form present on surface of nearly all B lymphocytes; produced early in primary immune response to antigen; pentameric form present in serum; does not cross placenta
IgG	Monomeric	Main immunoglobulin of acquired immunity; has capacity to cross placenta
IgE	Monomeric	Involved in response to allergic and parasitic processes; mediator of immediate hypersensitivity; interaction with basophils and mast cells results in histamine release
IgA	Monomeric and dimeric	Present in gastrointestinal, respiratory, and urogenital tract mucosa; prevents colonization of mucosa by bacteria and viruses

Selection of HLA Disease Associations

HLA Molecule	Disease Association	Comments
HLA-B27	Ankylosing spondylitis	Found in > 95% of ankylosing spondylitis patients; absence of HLA-B27 essentially excludes ankylosing spondylitis
HLA-DQ2, HLA-DQ8	Celiac disease	
HLA-DRB1*01:03	Ulcerative colitis	
HLA-DRB1*07	Crohn disease	
HLA-DQB1*06:02, HLA-DR2	Narcolepsy	Absence of HLA-DQB1*06:02 tends to exclude narcolepsy
HLA-DQ2, HLA-DQ3	Insulin-dependent diabetes mellitus	No value in routine testing of patients
HLA-DRB1	Rheumatoid arthritis	DRB1*04:01, 04:04, 04:05, and 04:08 are main susceptibility subtypes

- **Graft-vs.-host reaction**
 - Graft T cells proliferate in irradiated immunocompromised host
 - Graft T cells reject cells with foreign MHC class proteins
 - May occur in perfectly MHC classes I and II matched patients due to differences in minor antigens
 - Results in severe organ dysfunction
 - Common symptoms: Maculopapular rash, diarrhea, hepatosplenomegaly, and jaundice
 - Reaction can be reduced by treating allograft with antithymocyte globulin or monoclonal antibodies to reduce mature T cells in allograft
- **MHC and autoimmune disease**
 - Many autoimmune diseases occur in individuals with expression of certain MHC genes (HLA associations)
 - Associations have strong negative predictive value but weak positive predictive value (e.g., presence of certain HLA type/subtype does not predict disease, but its absence may be helpful in excluding disease)

LYMPHOID ORGANS

Lymph Node

- Small, oval-shaped organ of immune system
- Distributed throughout body

- Linked by lymphatic vessels that drain into left subclavian vein via thoracic duct

Spleen

- Located in upper left quadrant of abdomen
- Largest secondary immune organ in body
- Functions as mechanical filter of old, damaged, or defective red blood cells, reserve of monocytes, and site of active immune response through innate and adaptive pathways
- Initiates immune reactions to blood-borne antigens

BONE MARROW

Hematopoiesis

- Contains hematopoietic stem cells, which give rise to 3 types of blood cells
 - Red blood cells (erythrocytes)
 - Leukocytes (lymphocytes and granulocytes)
 - Megakaryocytes/platelets (thrombocytes)

THYMUS

T-Cell Maturation

- Specialized organ of immune system involved in T-cell education, selection, and maturation

Lymph Nodes

LYMPH NODES

Macroscopic Anatomy

- Round or reniform shape
- Normally do not exceed 1 cm in diameter
 - Larger during stimulation (2-3 cm in diameter)
- Tan-pink homogeneous cut surface
- Blood supply
 - Arteries and veins enter and exit at hilus, respectively
- Lymphatics
 - Afferent lymphatics enter in subcapsular sinus
 - Branching network of sinuses drains into efferent lymphatic vessels
 - Efferent lymphatics exit at hilus

Microscopic Anatomy

- Composed of 4 compartments
- **Cortex**
 - Primary follicle
 - Homogeneous nodules of small, darkly staining, naïve inactivated B lymphocytes
 - Secondary follicle (due to antigenic stimulation)
 - Shows features such as germinal center (GC) formation, polarization, tingible body macrophages
 - GC composed of centroblasts, centrocytes, small lymphocytes, tingible body macrophages, dendritic reticulum cells, and few small T cells
 - Zonal pattern of staining on H&E
 - Related to direction of antigen processing and process of B-cell maturation from centroblasts to centrocytes
 - Dark zone is predominantly centroblasts, oriented toward center of lymph node
 - Light zone is predominantly centrocytes, oriented toward periphery
 - Lack expression of Bcl-2
 - Follicles may show infiltration by small lymphocytes (progressive transformation of GC)
 - Mantle zone

- Small, dark-staining, tightly packed B cells that surround GC
- Similar characteristics as primary follicle
- Marginal zone
 - Less compact B cells, more abundant cytoplasm, located along outer layer of mantle zone
- **Paracortex**
 - Interfollicular area composed predominantly of T cells with post capillary/high endothelial venules, and interdigitating cells (IDC)
 - T cells are mostly small and naïve, may become activated and change into larger immunoblasts upon stimulation
 - IDCs involved in antigen presentation, when present in large numbers, impart mottled appearance
- **Medullary region**
 - Cords of cells, including lymphocytes, plasmacytoid lymphocytes, mature plasma cells, and plasmablasts
 - Cords are separated by medullary sinuses
- **Sinuses**
 - Primarily in medullary region, but do extend into cortex
 - Subcapsular sinus located directly beneath capsule
 - Lined by thin, pale-staining endothelial cells; acquire lining of macrophages within hilum
 - Contain macrophages, lymphocytes, plasma cells, immunoblasts, and occasional neutrophils

CONSIDERATIONS AND VARIATIONS

Hyperplasia

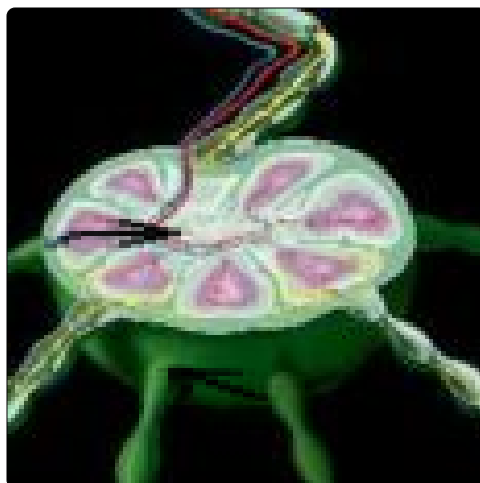
- Enlarged reactive lymph nodes with prominent follicular hyperplasia more common in younger age groups
 - Considered atypical in elderly and should prompt close evaluation

Pitfalls and Artifacts

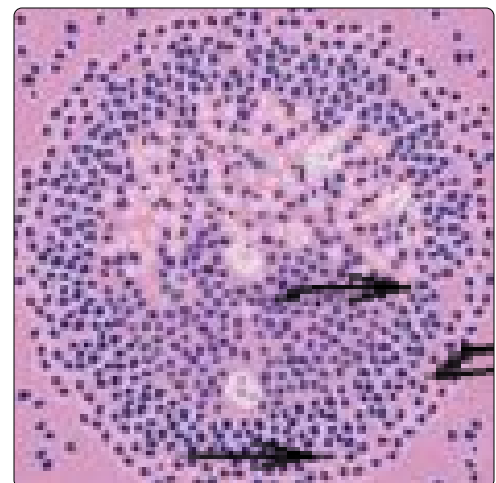
- Thoracic/pulmonary lymph nodes show abundant anthracotic pigment and histiocytes
- Paraaortic/parailiac lymph nodes may show sclerotic or hyaline deposits
- Inguinal lymph nodes commonly appear fibrotic
- Reactive lymph nodes may mimic follicular lymphoma

Lymph Node

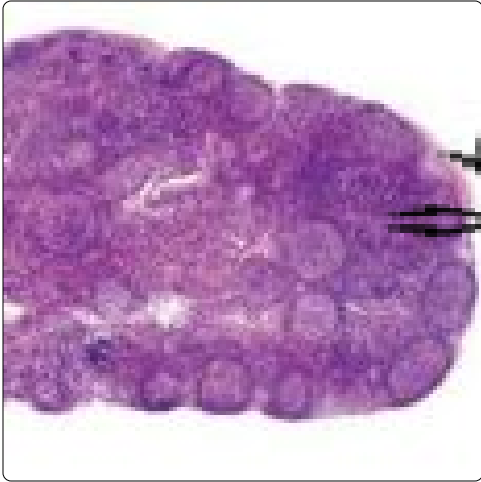
(Left) This lymph node graphic shows the usual arrangement of cortical-based follicles. The directional flow of lymph through afferent and efferent lymphatics is indicated by yellow arrows. (Right) This illustration of a secondary follicle shows many of the key histologic features: Light and dark zonation, tingible body macrophages, follicular dendritic cells, mantle zone, and marginal zone.



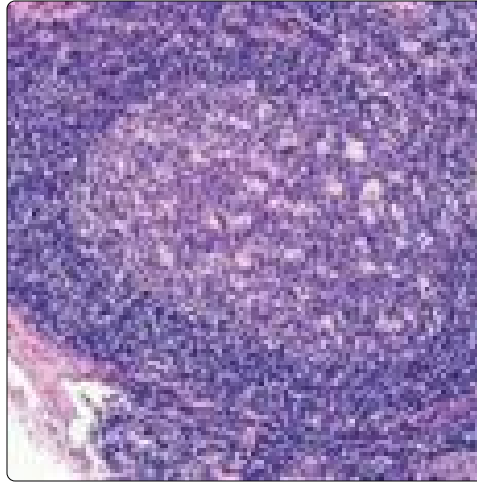
Follicle Features



Lymph Node (Low Power)

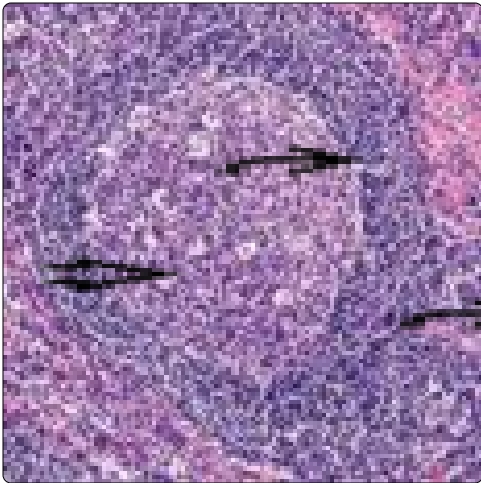


Lymph Node Follicle

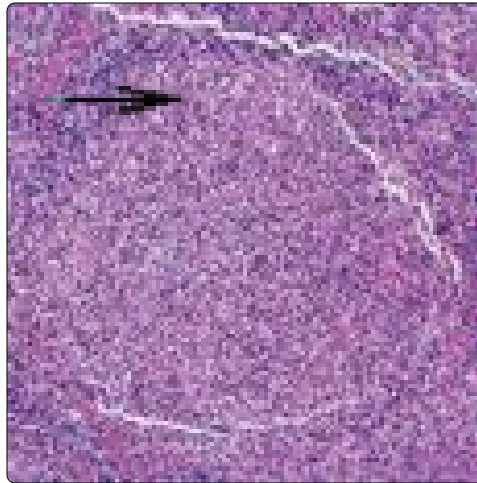


(Left) Cortically based secondary follicles, paracortex, medullary region, and sinuses are visualized in this reactive-appearing lymph node. **(Right)** Polarization (light and dark zones) and prominent tingible body macrophages impart a starry-sky pattern in this secondary follicle. The dark zone is composed predominantly of centroblasts that, following antigenic stimulation, mature into centrocytes located primarily within the light zone.

Secondary Follicle

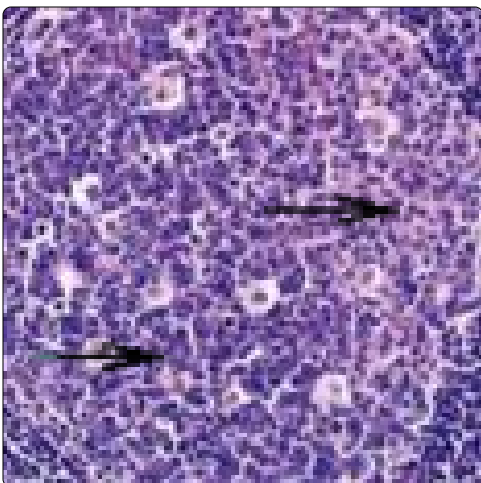


Neoplastic Follicle

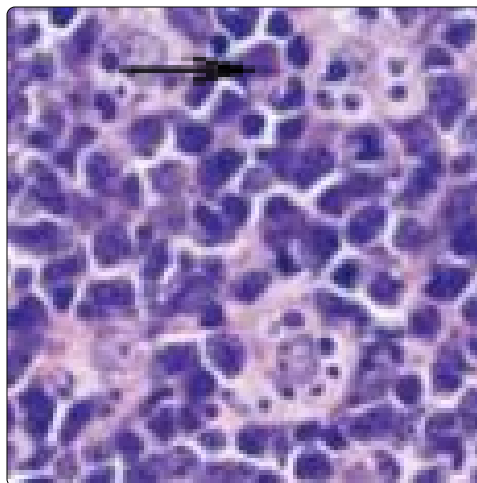


(Left) A secondary follicle with dark and light zonation and a distinct surrounding mantle zone is shown. The dark zone consists predominantly of immature centroblasts and a few immunoblasts, while the light zone consists of more mature centrocytes. The surrounding mantle zone consists of naïve B cells. **(Right)** This neoplastic follicle from a follicular lymphoma lacks polarization and tingible body macrophages and has a diminished mantle zone.

Germinal Center



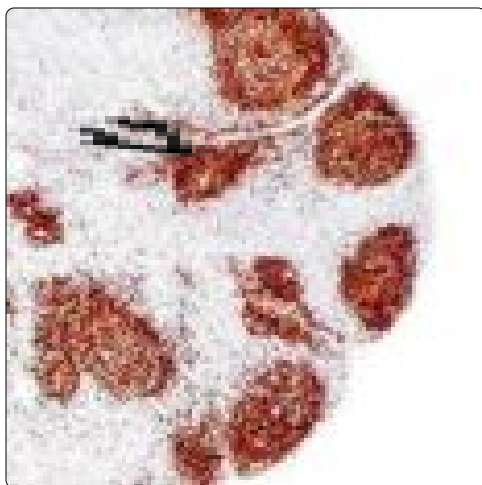
Tingible Body Macrophages



(Left) The predominance of centroblasts in the dark zone and centrocytes in the light zone of this germinal center is illustrated here at high power. **(Right)** The dark zone of a germinal center contains several tingible body macrophages, shown here at high power. Note the cytoplasmic debris. This finding is common among antigenically activated secondary follicles. The centroblasts show large oval nuclei that often have multiple nucleoli.

CD20 Immunostain, Lymph Node

(Left) CD20 immunostaining highlights B cells of the cortical-based follicles in this lymph node. Note the paucity of CD20(+) B cells in the paracortical and medullary regions of the lymph node. (Right) Medium-power view shows a cortical-based lymph node follicle that is composed primarily of CD20(+) B cells.

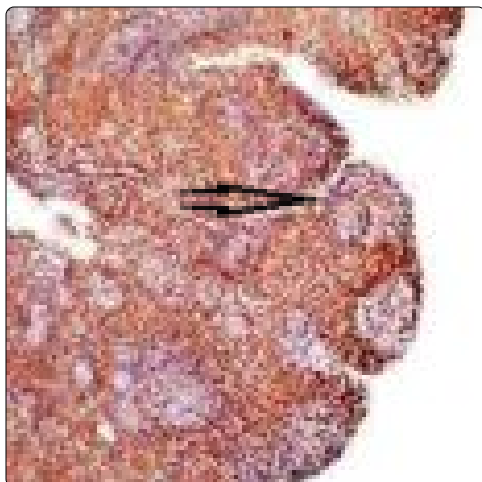


CD20 Immunostain, Follicle

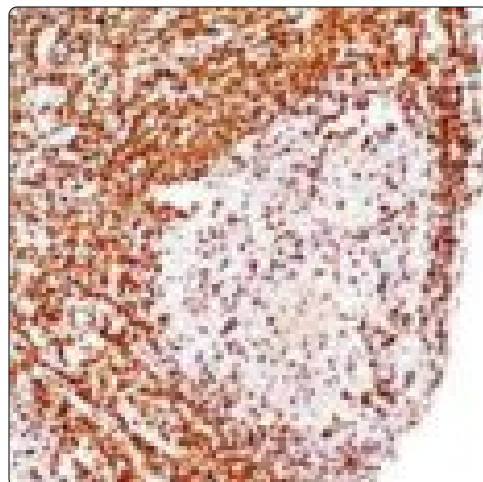


CD3 Immunostain, Lymph Node

(Left) CD3 immunostain highlights the general distribution of the CD3(+) T cells within the paracortical and medullary regions. Note the few scattered T cells (follicular helper T cells) seen within the B-cell-predominant follicles. (Right) CD3 immunostain at medium power highlights the scattered T cells within the germinal center (follicular helper T cells) and the predominance of T cells within the paracortical region.

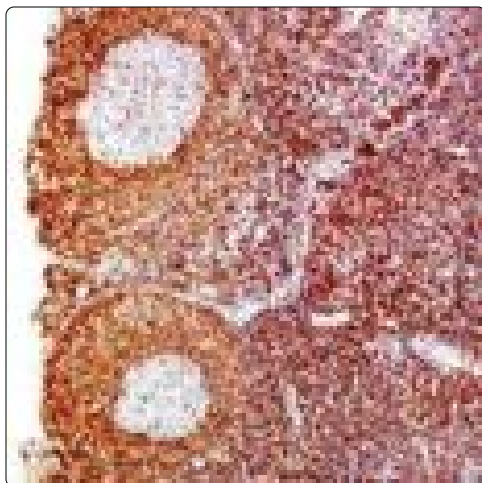


CD3 Immunostain, Follicle

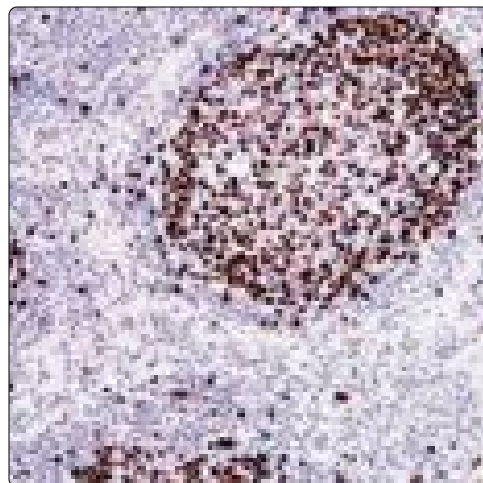


Bcl-2 Immunostain, Lymph Node

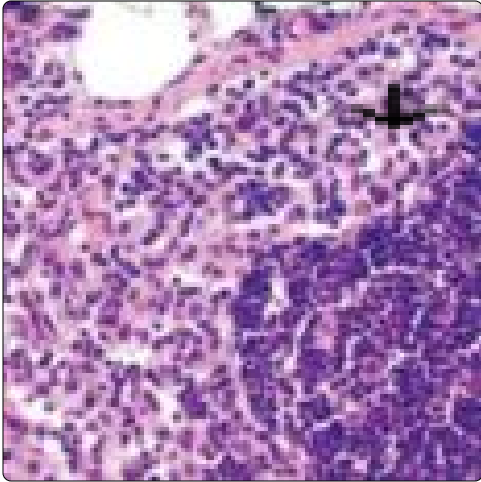
(Left) Bcl-2 immunostaining shows the typical staining pattern of a reactive lymph node: Negative staining in the germinal center and positive staining in all other areas. Bcl-2 has antiapoptotic activity, so the lack of Bcl-2 expression in B cells undergoing antigen-driven selection in the germinal center allows these cells to undergo apoptosis if necessary. (Right) Ki-67 immunostaining shows a polarized appearance (dark and light zones), which is typical of a benign reactive follicle.



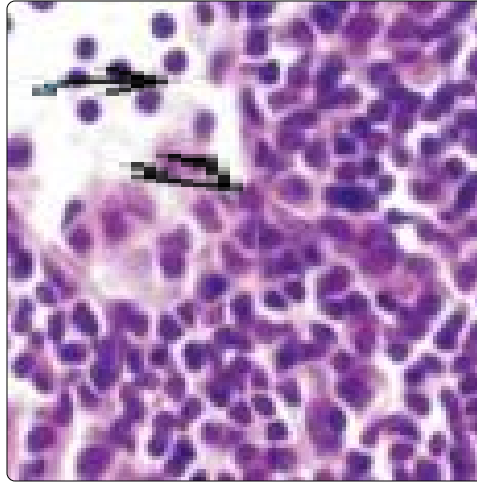
Ki-67 Immunostain, Follicle



Lymph Node Subcapsular Sinus

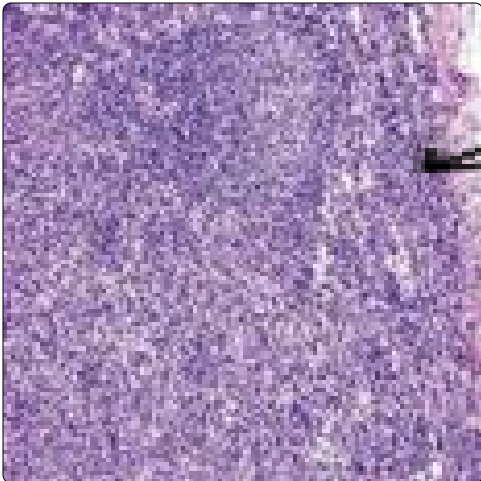


Lymph Node Medullary Region

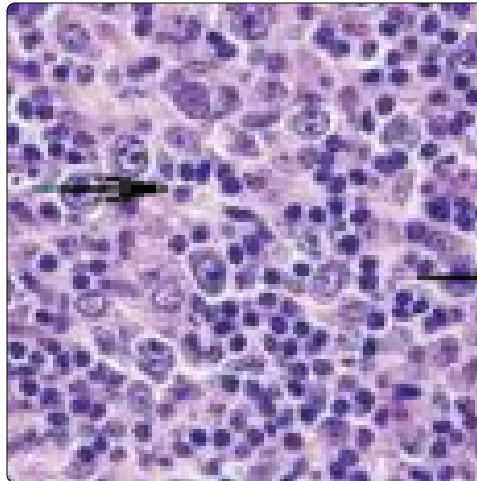


(Left) A dilated subcapsular sinus with numerous histiocytes (round nuclei, abundant eosinophilic cytoplasm) and scattered small lymphocytes is shown at medium power. (Right) A small portion of a medullary cord with mature plasma cells is shown at high power. Adjacent is a dilated sinus with scattered histiocytes and small lymphocytes.

Paracortical Hyperplasia

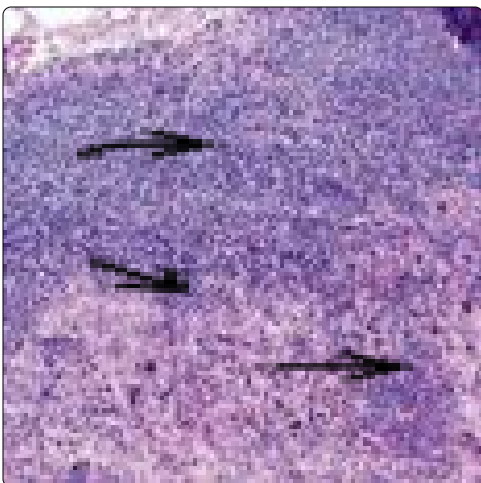


Paracortical Hyperplasia

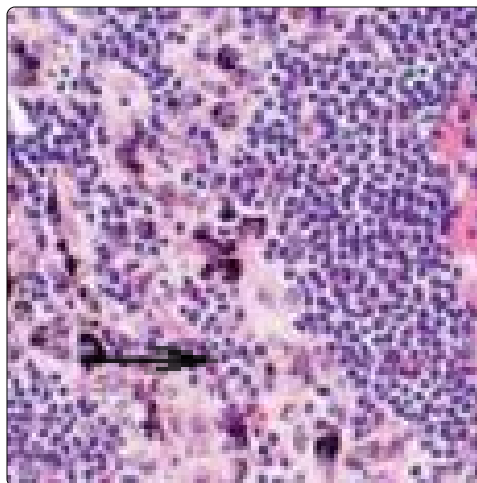


(Left) This reactive lymph node has marked paracortical (interfollicular) expansion. A residual follicle is at the top of the field. This feature is commonly associated with viral-mediated lymphadenopathies. (Right) A hyperplastic paracortex with a heterogeneous cell population is shown. Note the large immunoblasts with prominent centrally located nucleoli admixed with small lymphocytes and histiocytes. These features are commonly associated with viral infections.

Thoracic Lymph Node



Thoracic Lymph Node



(Left) Thoracic lymph nodes often show abundant anthracotic pigment that is typically associated with histiocytic infiltrates and distortion of the usual lymph node architecture. (Right) The abundant anthracotic pigment and scattered histiocytes are shown at high power. These features are common among thoracic/pulmonary lymph nodes.

SPLEEN**Macroscopic Anatomy**

- Bean-shaped organ covered by smooth capsule
- Blood supply via splenic artery
 - Splenic artery enters at hilus and branches in spleen

Microscopic Anatomy

- Splenic vasculature
 - Blood enters spleen via splenic artery, which then branches into trabecular arteries
 - Accompanied by veins and lymphatic vessels, branching trabecular arteries surrounded by dense connective tissue comprise splenic cuff
 - Trabecular arteries emerge from connective tissue and become arterioles of white pulp (central arterioles), which are surrounded by periaarteriolar lymphatic sheath
 - Central arterioles continue into follicles (follicular arterioles)
 - Follicular arterioles become smaller and terminate in marginal zone or form vascular tuft of capillaries in red pulp
 - Capillaries in red pulp end as sheathed capillaries
 - Sheathed capillaries lack direct communication to sinuses
 - Sheathed capillaries are lined by concentrically arranged macrophages and reticular fibers that become continuous with reticular network (stroma) of red pulp (cords)
 - Sheathed capillaries in conjunction with cords function as filtering unit of spleen
 - Red cells enter adjacent sinuses via sheathed capillaries and cords
- Red pulp
 - Loose reticular network of capillaries, penetrating venous sinuses, and cords
 - Venous sinuses are large leaky vessels lined by discontinuous layer of cuboidal littoral cells
 - Littoral cell
 - Type of endothelial cell that expresses both histiocytic and endothelial immunohistochemical markers

- Splenic cords (cords of Billroth) represent tissue between venous sinuses; contain reticular cells, macrophages, and plasma cells
- White pulp
 - Consists of B- and T-cell compartments
 - B-cell compartments may be seen as primary follicles (unstimulated) or secondary follicles (antigen stimulated with germinal center present)
 - ◻ Follicles are surrounded by rim of mantle zone cells and outer rim of marginal zone cells
 - T-cell compartment is known as periaarteriolar lymphatic sheath
 - ◻ Lies adjacent to arterioles
 - ◻ Irregular areas composed primarily of CD4(+) T cells
- Perifollicular zone
 - Area adjacent to follicles and T-cell compartments
 - Indicated by numerous erythrocytes directly adjacent to lymphoid cells
 - Capillaries and sheathed capillaries are also present in this zone

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

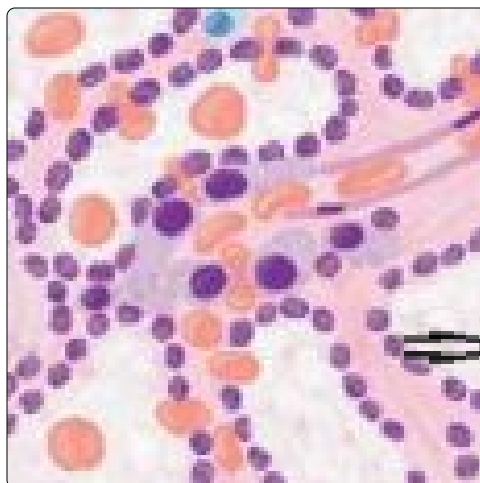
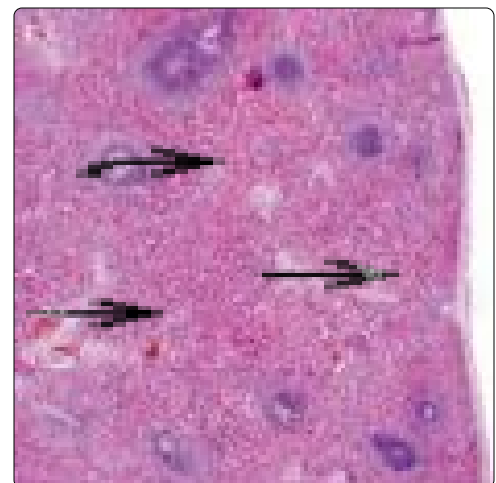
- White pulp does not contain well-formed follicles until birth
- Maturing hematopoietic precursors are commonly seen in fetal spleen
- Secondary follicles are more common in patients < 20 years of age
- Patients > 20 years of age typically have fewer secondary follicles
- Hyalinization of vessels is common in both old and young patients
- Extramedullary hematopoiesis in adult spleen is associated with pathologic conditions

Pitfalls and Artifacts

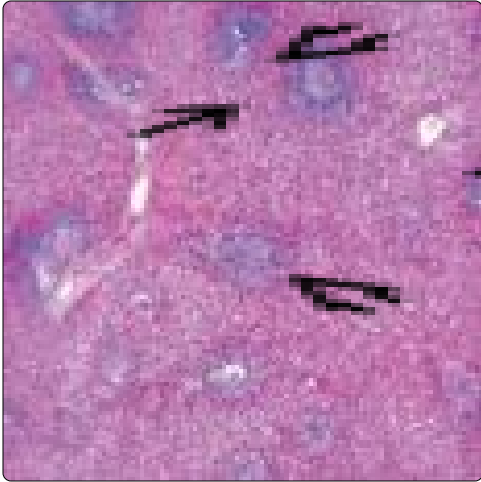
- Extremely vulnerable to autolysis
- Additional stains may be helpful to visualize splenic architecture (e.g., reticulin, periodic acid-Schiff)

Spleen

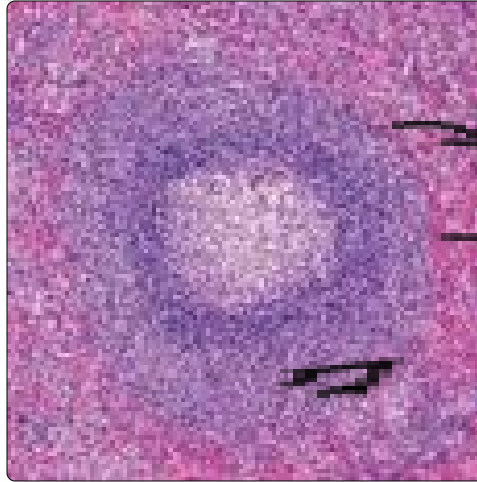
(Left) The functional filtering unit of the spleen is illustrated. Red cells pass from the sheathed capillaries lined by concentrically arranged macrophages into the cords and then into the sinuses. Old &/or abnormal red cells will be removed during this process. **(Right)** White and red pulp are viewed at low power in this section of normal spleen. A thin splenic capsule with slivers of branching trabeculae is also noted.

**Normal Spleen**

Spleen: Red and White Pulp

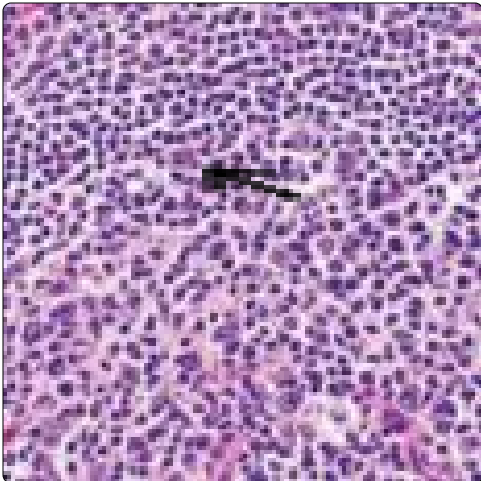


White Pulp: Secondary Follicle

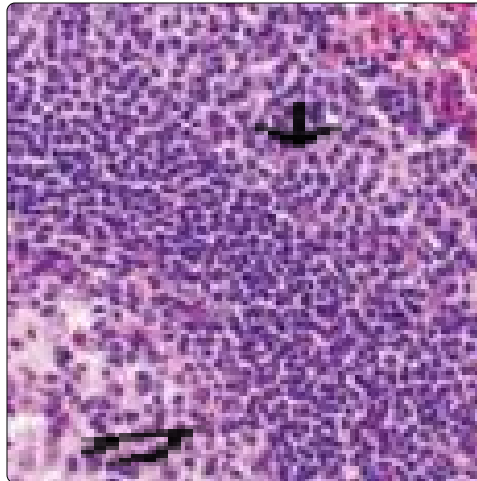


(Left) Nearly all splenic compartments are illustrated in this image: White pulp (B-cell and T-cell compartments), red pulp, and splenic cuff. (Right) A B-cell follicle is shown at medium power. The follicle shows features of activation, as noted by the germinal center. Note the surrounding rim of dense small lymphocytes and the less dense and more expanded rim of lymphocytes, which represent the mantle and marginal zones, respectively.

White Pulp: Follicle

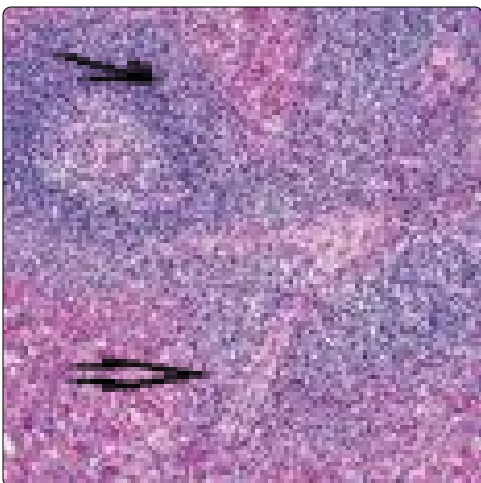


White Pulp: Follicle

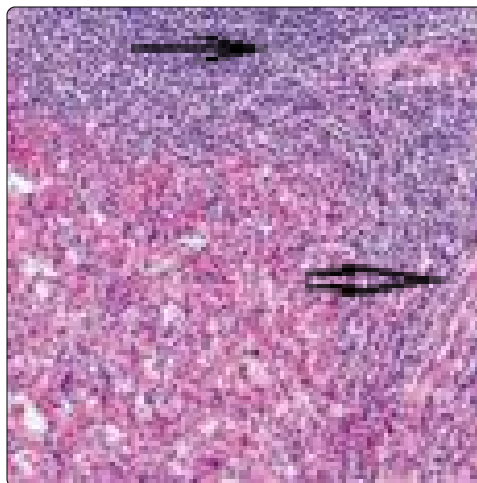


(Left) Splenic follicles show many of the same features as those of lymph nodes. In this secondary (antigen-stimulated) follicle, a tingible body macrophage is seen within the germinal center, and the surrounding mantle zone is shown. (Right) This high-power view illustrates the mantle zone (dense rim of small B cells with scant cytoplasm), the outer marginal zone (less dense rim of B cells with more abundant cytoplasm), and a small section of germinal center.

White Pulp: Follicle and Periarteriole Lymphatic Sheath

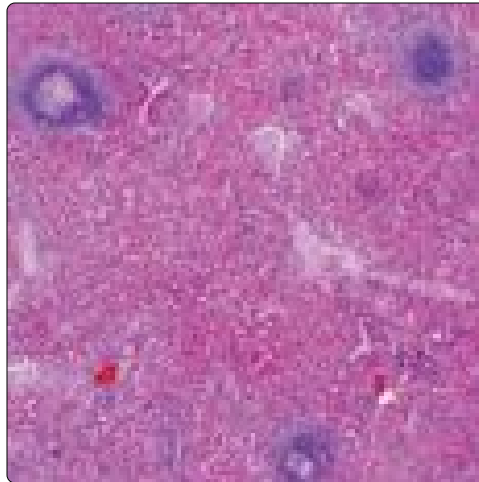


Spleen: Perifollicular Zone



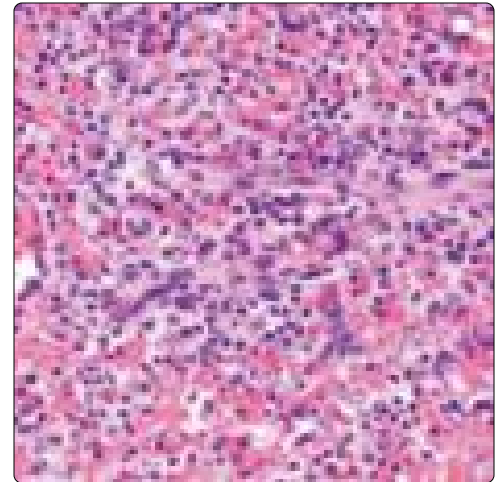
(Left) The T-cell compartment (periarteriole lymphatic sheath), B-cell compartment (follicle), and perifollicular zone are shown. The T-cell and B-cell compartments together make up the white pulp. (Right) The perifollicular zone lies adjacent to the follicle and T-cell compartment (periarteriole lymphatic sheath) and contains numerous erythrocytes in addition to capillaries and sheathed capillaries (not shown).

Spleen: Red Pulp

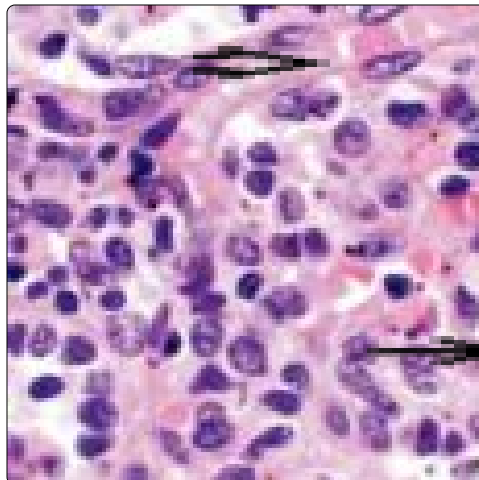


(Left) Red pulp consists of capillaries, venous sinuses, and splenic cords (cords of Billroth). Several follicles along the edges are also present. **(Right)** A branching capillary lined by flattened endothelial cells is shown. In the spleen, arterioles lead to capillaries, which then lead to sheathed capillaries that are lined by concentrically arranged macrophages.

Red Pulp: Splenic Capillary

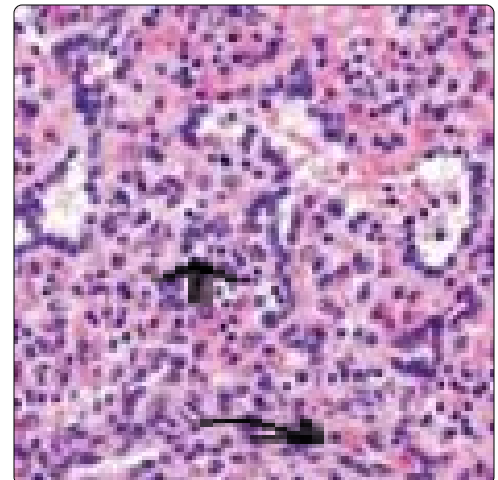


Red Pulp: Splenic Capillary

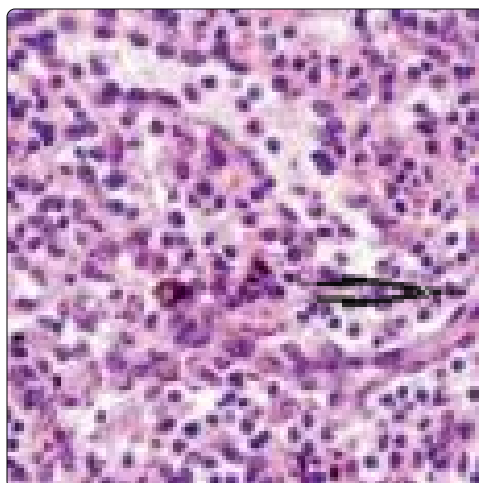


(Left) A splenic capillary lined by flat endothelial cells precedes the sheathed capillary, which is lined by concentrically arranged macrophages and a network of reticular cells and fibers. **(Right)** Sinuses lined by littoral cells, a type of cuboidal endothelial cell that stains for both endothelial and histiocytic markers, are shown. Several small capillaries are also noted along the bottom. Note the flatter endothelial cells that line the capillaries.

Red Pulp: Sinuses

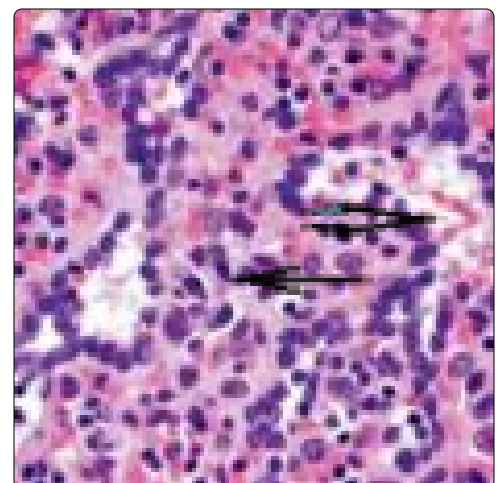


Red Pulp: Capillaries and Sinus

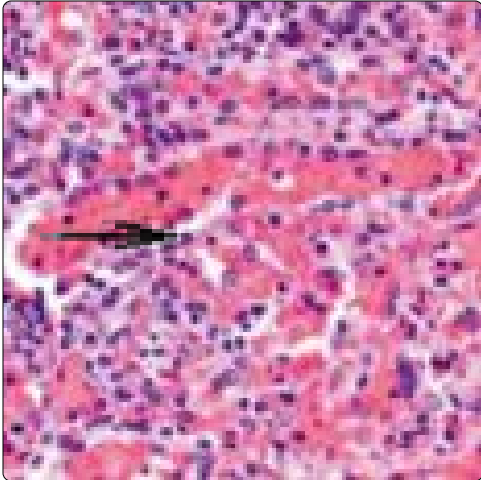


(Left) Capillaries, sheathed capillaries, and a sinus are shown. Note the abundant hemosiderin in the macrophages that are part of the sheathed capillaries and cords. **(Right)** Splenic cords that contain macrophages, reticular cells, and plasma cells represent the tissue that lies between the sinuses. Note the individual red cells that are passing from the cords into the sinuses. Old &/or damaged red cells that cannot squeeze through will be removed by the spleen.

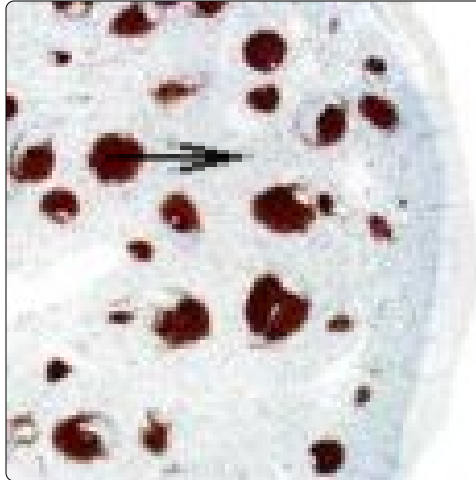
Red Pulp: Cords and Sinuses



Splenic Congestion

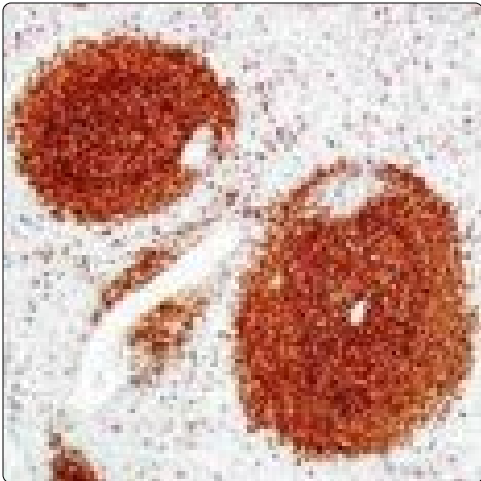


Spleen: CD20 Immunostain

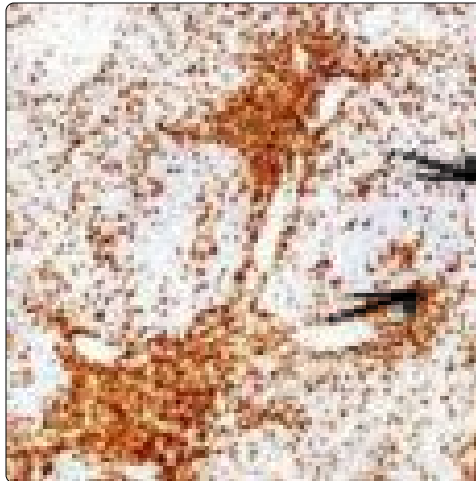


(Left) Sinuses filled with packed red blood cells [] are seen in this image from a congested spleen. (Right) B-cell follicles [] are prominent in this spleen that has undergone follicular hyperplasia and are highlighted by a CD20 immunohistochemical stain.

Spleen: CD20 Immunostain

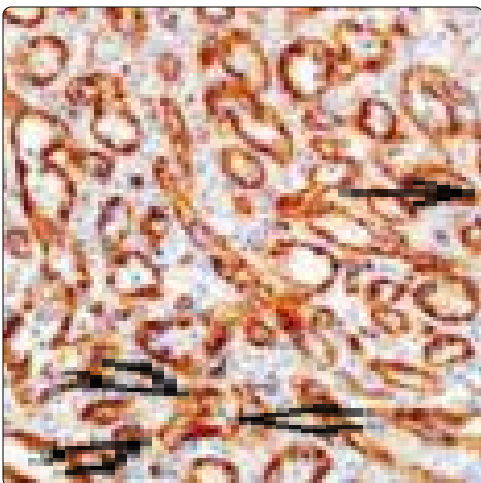


Spleen: CD3 Immunostain

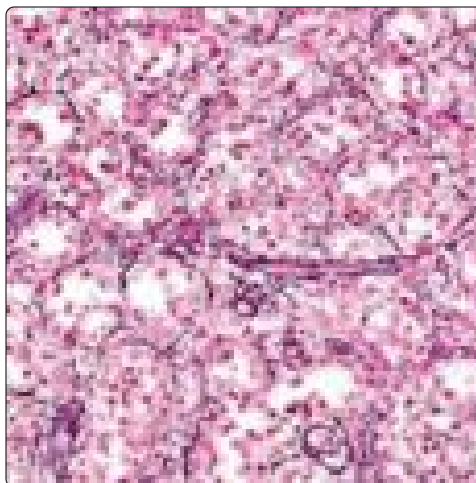


(Left) CD20 immunostain highlights the B-cell follicles of this spleen. (Right) The T-cell compartment (periarteriolar lymphatic sheath) is highlighted by CD3. Note the negative areas of staining [] that represent the B-cell follicles. Together, these 2 compartments make up the white pulp of the spleen.

Spleen: Factor VIII Immunostain



Red Pulp: Reticulin Stain



(Left) The splenic sinuses [] and capillaries [] are highlighted by factor VIII. The space between the sinuses represents the splenic cords (cords of Billroth) []. After passing through the cords into the sinuses, the red cells then progress to venules and veins. (Right) The reticulin network of the sinuses and capillaries is highlighted. The more discontinuous reticulin network of the sinuses allows for the passage of cells from the cords into the sinuses.

BONE MARROW

Hematopoiesis

- Cell production in bone marrow (BM) is highly regulated; both inducers and suppressors in microenvironment
- Both hematopoietic (HP) and mesenchymal stem cells delineated by functional properties or immunophenotype
- Cell types produced by HP stem cells include erythrocytes (RBCs), neutrophils, eosinophils, basophils, monocytes/macrophages/dendritic cells, mast cells, lymphocytes, natural killer cells, osteoclasts, and platelets
- Microenvironment specifies each lineage production site at submicroscopic level
- Mesenchymal stem cells produce fat cells, osteoblasts, vessels, and stromal cells
- Blast is term used for earliest identifiable cell in each HP lineage
- Unique features of HP include endomitosis of megakaryocytic lineage cells and enucleation of RBCs

Macroscopic Anatomy

- Specimens that can be assessed include BM core biopsies, imprint preparations, clot sections, and BM aspirate smears
- Anticoagulated aspirate samples can be used for numerous specialized studies, including flow cytometry, cytogenetics, and molecular assays

Microscopic Anatomy

- Granulopoiesis
 - Myeloblast to neutrophil maturation process characterized by progressive increase in cytoplasmic granulation essential for cell function
 - Immature granulocytic cells reside adjacent to bony trabeculae and around blood vessels
 - Both enzymatic (cytochemical) stains and immunophenotyping can be used to delineate lineage and stage of maturation of granulocytic cells
- Erythropoiesis
 - Erythroblast to RBC maturation process characterized by RNA loss, hemoglobin gain, nuclear shrinkage, and ultimately, nuclear extrusion

- Erythroid cells form colonies by adhering to central macrophages
- Erythroid cells are best identified by morphology and immunophenotyping for CD71, glycophorin A, and hemoglobin A
- Iron in erythroid cells and macrophages can be assessed by Prussian blue staining
- Megakaryopoiesis
 - Megakaryoblast to platelet maturation process characterized by progressive doubling of DNA without mitosis, creating very large cell with lobulated nuclei and abundant cytoplasm
 - Megakaryocytes reside adjacent to sinuses to facilitate platelet release into circulation by shedding cytoplasm
 - Megakaryocytic cells, especially immature and small forms, can be delineated by immunophenotyping for CD61, CD42b, CD41, and factor VIII
- Fat cells and macrophages are key components of the BM microenvironment
- Macrophages play role in regulation of hematopoiesis and in iron storage
- Plasma cells reside adjacent to blood vessels and are generally infrequent
- Mast cells reside within bone marrow particles and can be highlighted by CD117 and tryptase immunostains

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Distinctive age-based variation in cellularity, bone features, number of lymphocytes, lymphoid immunophenotype
 - Both osteoclasts and osteoblasts are seen along bony trabeculae in children and adolescents
 - Overall cellularity declines with age

Pitfalls and Artifacts

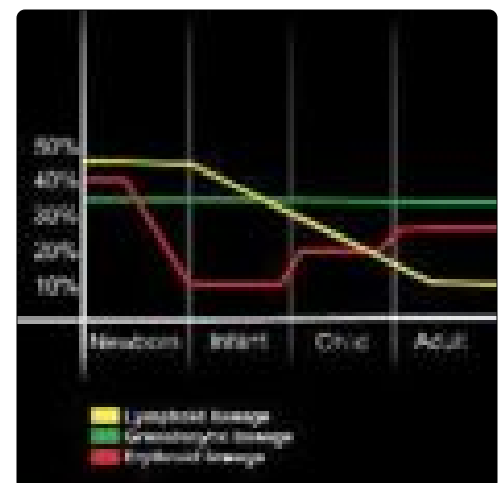
- Poor specimen quality is major source of diagnostic problems
- Aspiration of medullary space is common in core biopsies where aspiration is performed first, then biopsy is done close to aspiration site

Hematopoietic Cells

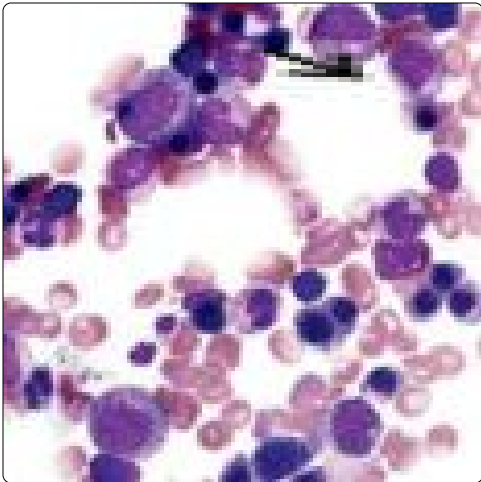
(Left) Schematic illustrates likely hematopoietic and mesenchymal stem cells, which reside in distinct niches. Osteoblasts and macrophages play key roles in hematopoietic regulation. (Right) Graph displays age-related variations in the proportion of erythroid, granulocytic, and lymphoid lineages within bone marrow. Note the age-related variation of erythroid and lymphoid lineage cells, while granulocytic lineage cell levels are fairly stable throughout life.



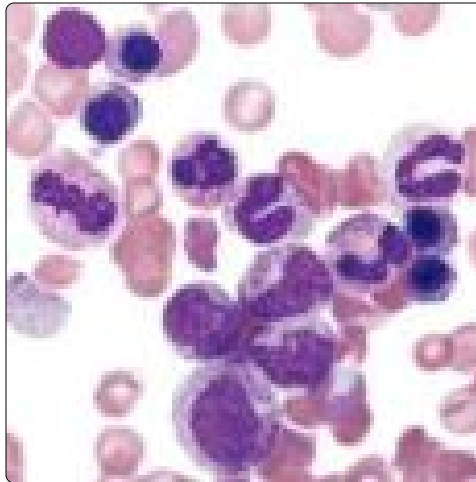
Blood Cell Lineages



Bone Marrow Aspirate

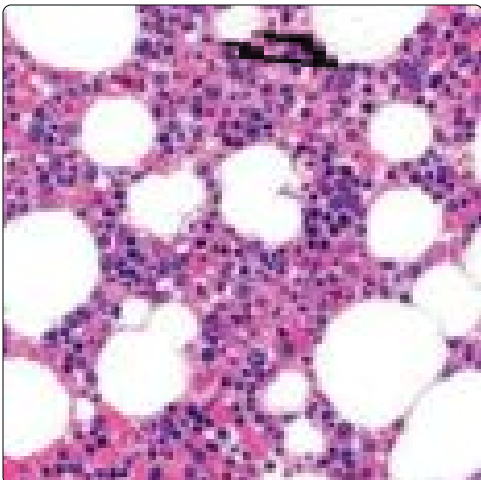


Bone Marrow Aspirate

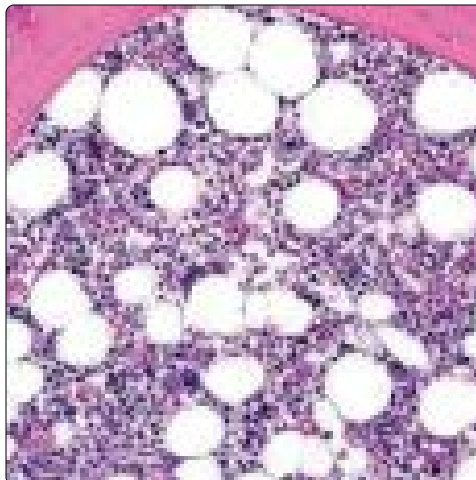


(Left) Bone marrow aspirate smear at intermediate magnification shows an admixture of developing myeloid and erythroid cells. Note that the immature myeloid cells are least numerous [1], and the maturing forms predominate with intact nuclear and cytoplasmic maturation. Erythroid cells [2] are generally less numerous than granulocytic cells. (Right) The morphologic features of the nuclei and cytoplasm of developing hematopoietic cells are best appreciated at high magnification.

Bone Marrow Clot Section

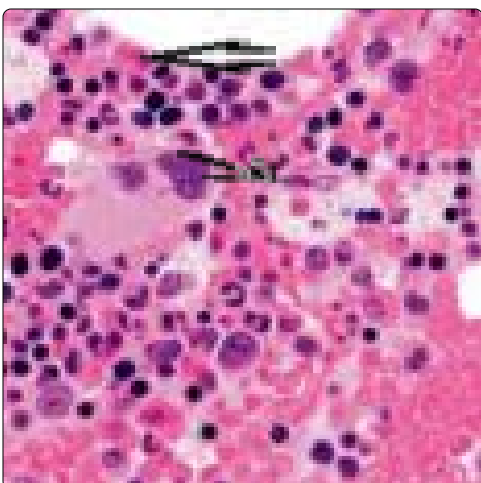


Bone Marrow Core Biopsy

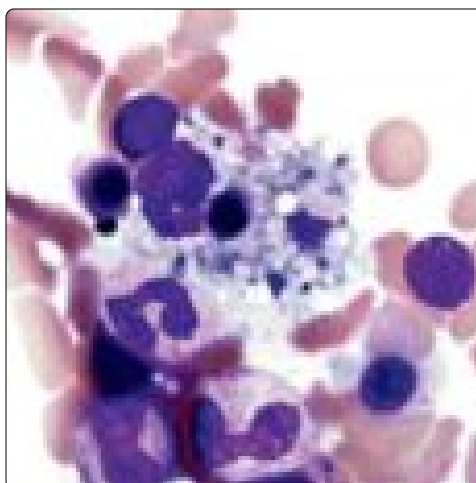


(Left) On clot sections, the overall cellularity can be assessed (70% in this image). In addition, hematopoietic cells, including erythroid colonies, are also evident [1]. Note hemosiderin in the individually dispersed macrophages [2]. (Right) On bone marrow core biopsies, bony trabeculae are evaluated in addition to hematopoietic and stromal cells. Bony trabeculae in normal adult patients do not show evidence of remodeling. Note dispersed megakaryocytes [3] and erythroid colonies [4].

Bone Marrow Clot Section



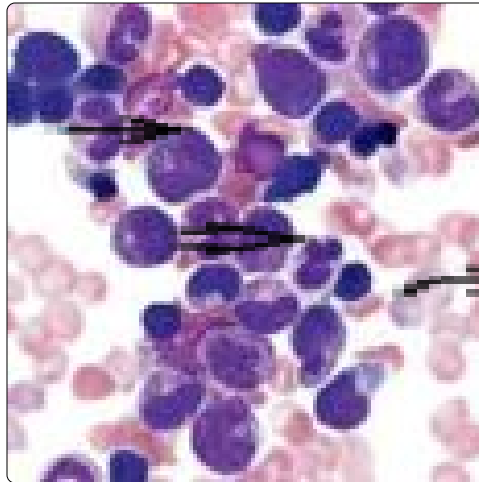
Macrophage, Aspirate Smear



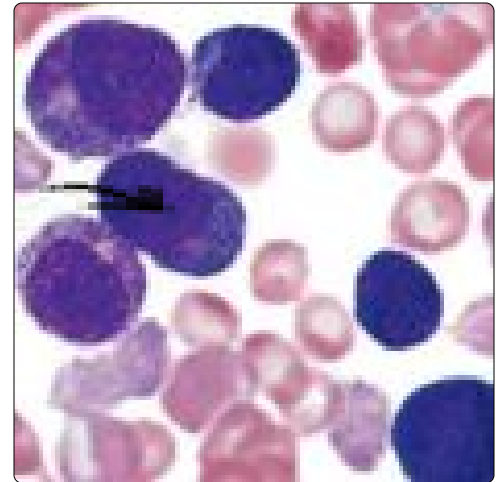
(Left) This bone marrow clot section shows an individual megakaryocyte [1] and multiple foamy macrophages [2]. Macrophages are typically inconspicuous on clot and core biopsy sections. (Right) Macrophages can be assessed on bone marrow aspirate smears and should be evaluated for nuclear features and cytoplasmic contents. Note the ingested nucleus, likely from an erythroid cell [3]. The adjacent hematopoietic cells are unremarkable.

Granulocytic Lineage Cells

(Left) Bone marrow aspirate smear shows numerous maturing granulocytic lineage cells, including promyelocytes [A], myelocytes [B], metamyelocytes [C], band neutrophils [D], and a mature segmented neutrophil [E]. (Right) Immature granulocytic cells, such as this central promyelocyte [A], have basophilic cytoplasm with sparse granules, while myelocytes [B] and successive granulocytic lineage cells have eosinophilic cytoplasm due to secondary granules.

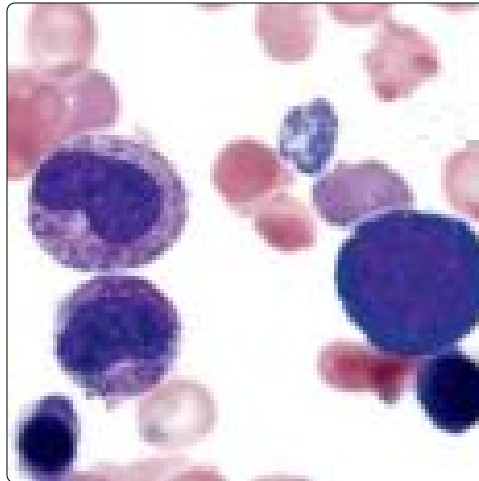


Immature Granulocytic Cells

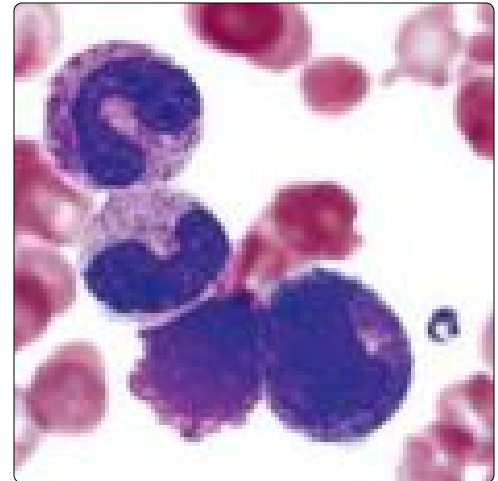


Maturing Granulocytic Cells

(Left) Nuclear indentation distinguishes a metamyelocyte [A] from a myelocyte [B], although maturation is a biologic continuum. Note that darker primary granules are more conspicuous in the myelocyte. (Right) Band neutrophils [A] are distinguished from segmented neutrophils [B] by nuclear features. Band neutrophils have uniform C-shaped nuclei without constriction. Terminal maturation is characterized by progressive nuclear lobulation, just beginning in the neutrophil shown here.

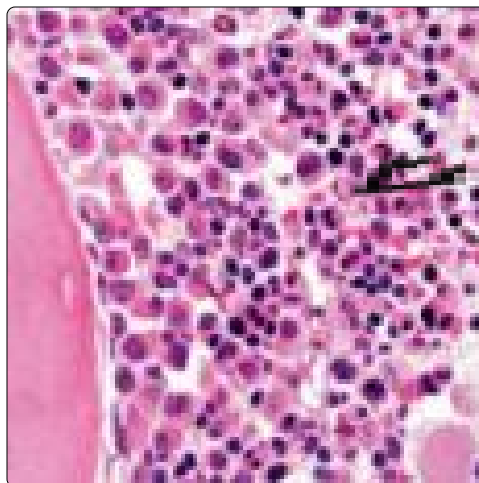


Stages of Granulocytic Maturation

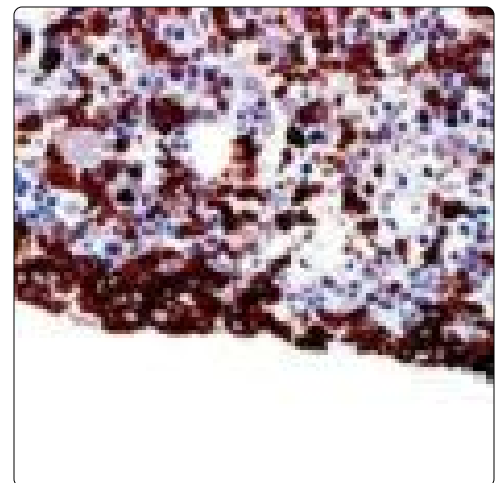


Paratrabecular Granulocytic Precursor Cells

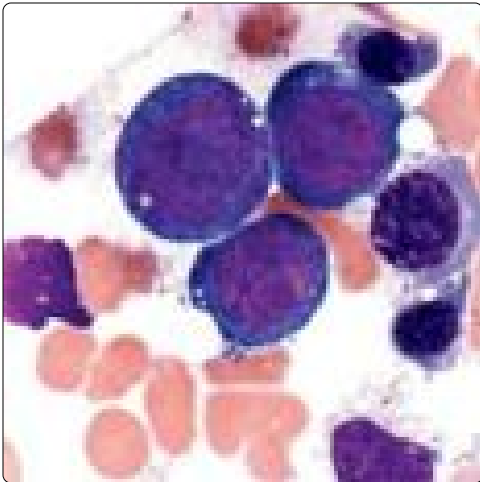
(Left) The normal paratrabecular localization of immature granulocytic lineage cells [A] is evident in this high-magnification image of a bone marrow core biopsy. More mature granulocytic lineage cells [B] are localized in the central regions of the core biopsy and are thus readily available for release into circulation. (Right) The lower edge of this biopsy shows the paratrabecular localization of normal immature granulocytic lineage cells that are highlighted by myeloperoxidase staining [A].



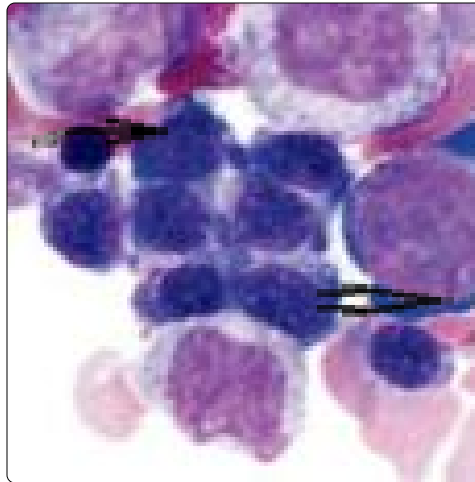
Paratrabecular Granulocytic Precursors



Erythroblasts

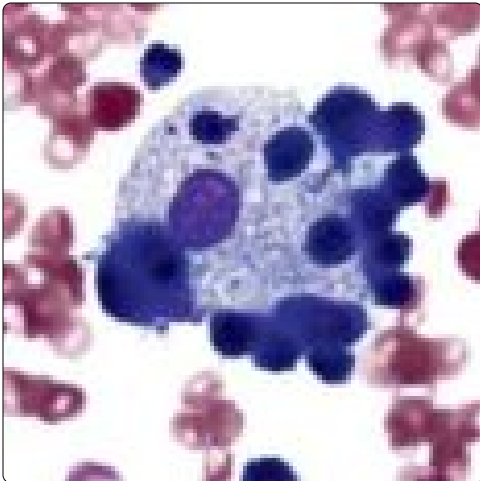


Erythroid Colony

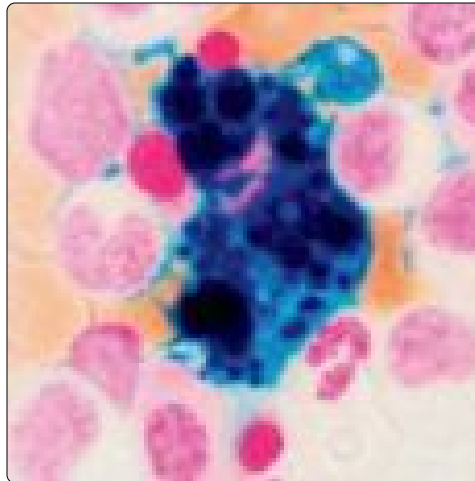


(Left) Erythroblasts (center) are characterized by a high nuclear:cytoplasmic ratio and deeply basophilic cytoplasm. Dramatic nuclear and cytoplasmic changes characterize maturation. (Right) More mature erythroid lineage cells show gray to eosinophilic cytoplasm reflecting hemoglobin production. There is dramatic nuclear shrinkage and pyknosis with ultimate extrusion of the pyknotic nucleus. Erythroid cells tend to aggregate in colonies.

Macrophage With Adherent Erythroids

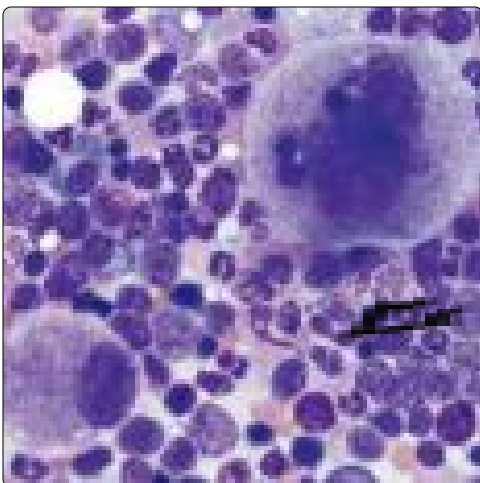


Storage Iron in Macrophage

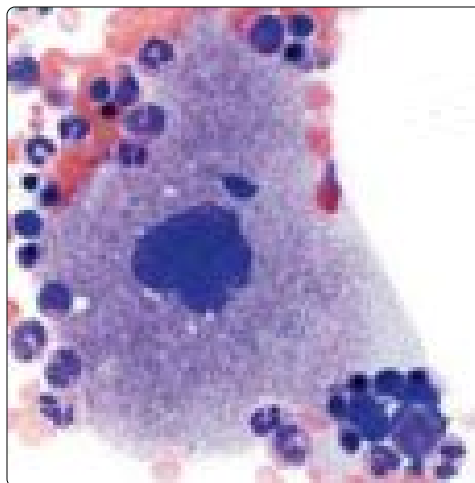


(Left) Erythroid cells selectively attach to macrophages, producing a so-called nurse cell. This adherence is critical for iron homeostasis. The macrophage is also immediately available to ingest extruded pyknotic erythroid cell nuclei in the terminal stages of erythrocyte maturation. (Right) This macrophage, seen on an aspirate smear, contains abundant storage iron as seen in this iron stain.

Megakaryocytes



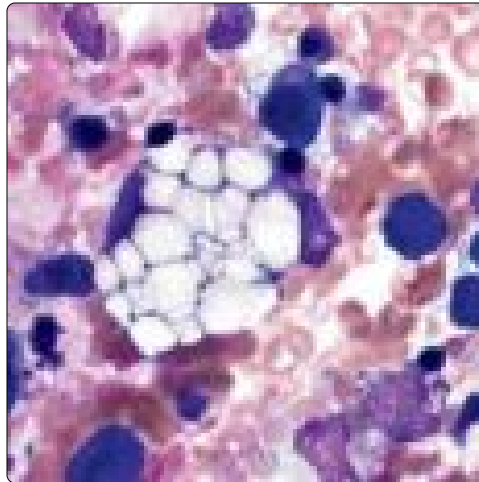
Mature Megakaryocyte



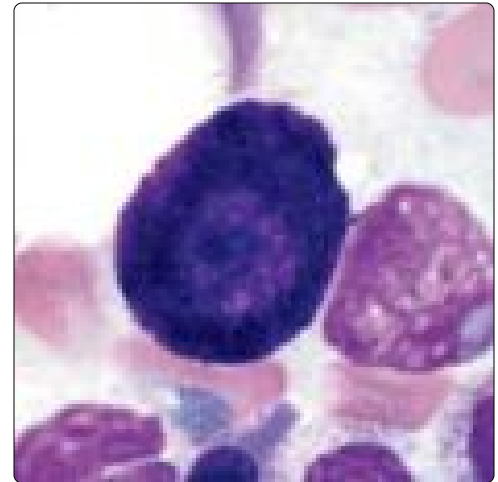
(Left) An immature megakaryocyte with a nonlobate nucleus is evident compared to a hyperlobated mature megakaryocyte. Megakaryocyte maturation is characterized by progressive doubling of DNA without cell division (endomitosis). (Right) This mature megakaryocyte has voluminous amounts of cytoplasm, which will eventually shed off directly into the circulation as platelets. Neutrophils may pass through megakaryocyte cytoplasm.

Fat Cell

(Left) Adipose (fat) cells are a normal constituent of the bone marrow microenvironment. These cells can have multiple vesicles, especially during regeneration following bone marrow suppression. (Right) Mast cells are an inconspicuous cell in bone marrow aspirate smears, especially within the darkest staining regions of bone marrow particles. The dark purple granules of mast cells obscure the round nucleus.

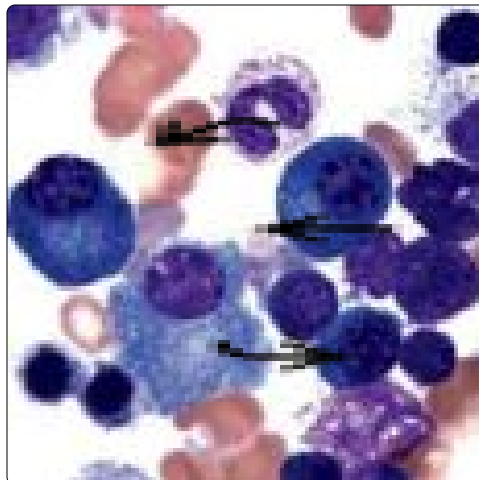


Mast Cell

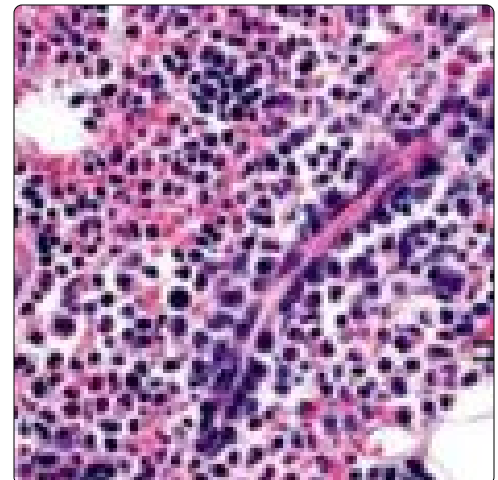


Plasma Cells

(Left) Plasma cells are evident on bone marrow aspirate smears and can be identified by their eccentric nuclei and basophilic cytoplasm. A pale paranuclear region is evident. Plasma cells may be confused with erythroid precursor cells because both cell types have deeply basophilic cytoplasm. (Right) Plasma cells are present in a perivascular location on bone marrow core biopsy sections.

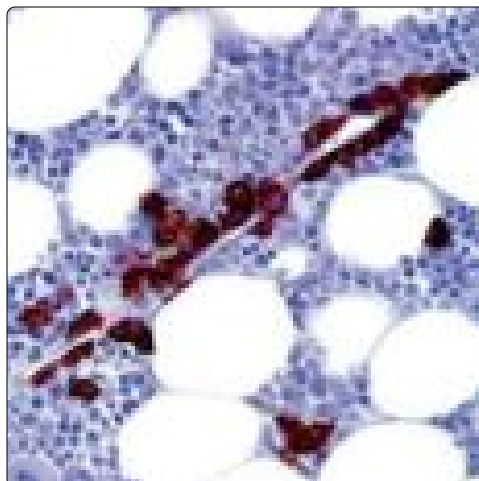


Perivascular Plasma Cells

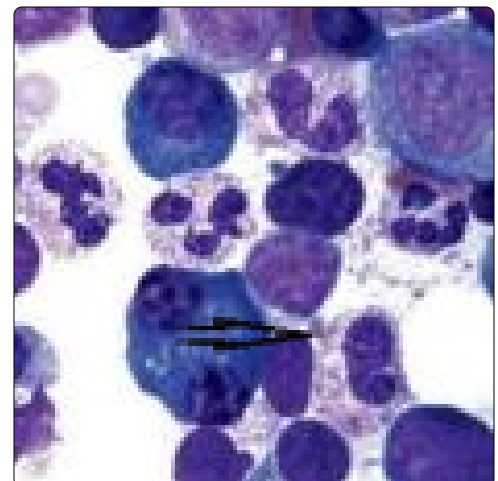


Perivascular Plasma Cells

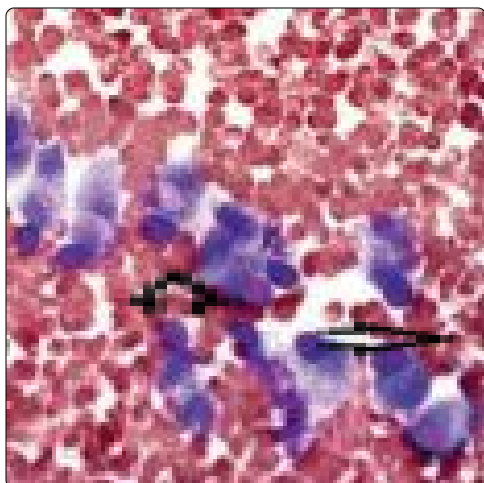
(Left) Immunohistochemical staining for CD138 can highlight the number, distribution, and location of plasma cells on bone marrow core biopsy sections. Note that the CD138(+) plasma cells form a collar around small blood vessels. (Right) Binucleated plasma cells may be seen on bone marrow aspirate smears. Binucleation is not indicative of a neoplastic plasma cell disorder and can be seen in many reactive processes.



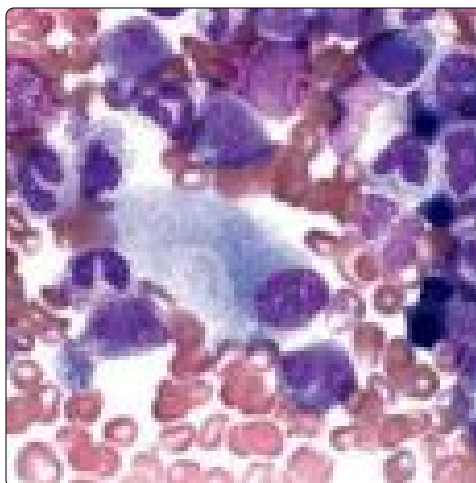
Binucleate Plasma Cell



Osteoblasts

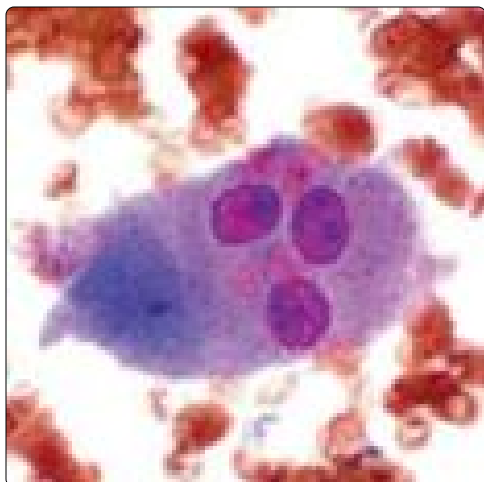


Osteoblast

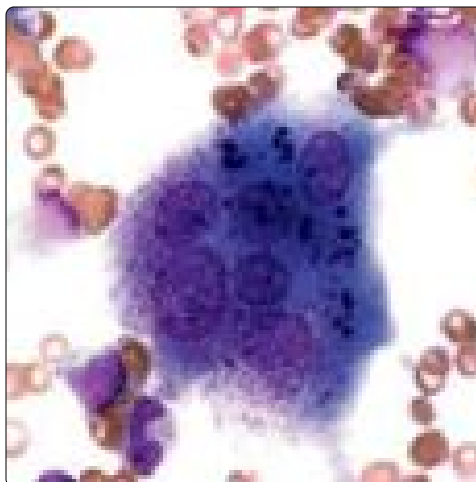


(Left) Osteoblasts are bone-forming cells that can be seen on bone marrow aspirate smears, especially in pediatric patients. They can be numerous in children with physiologic active bone growth and remodeling. Osteoblasts have markedly eccentric nuclei that appear to be separating from the cytoplasm . **(Right)** On high magnification, an osteoblast is characterized by an eccentric nucleus and a hof (cytoplasmic pale area), which, unlike the plasma cell, is separated from the nucleus .

Osteoclast



Osteoclast With Bone Sand

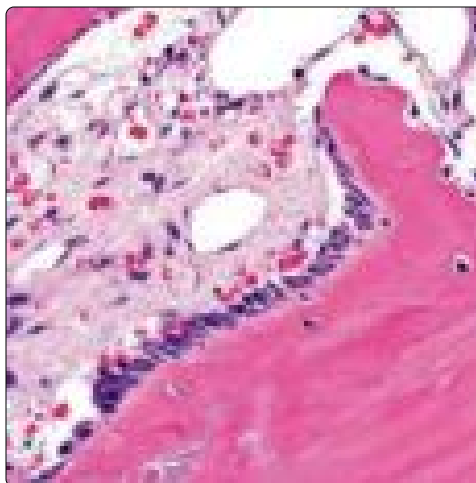


(Left) Osteoclasts can also be seen on bone marrow aspirate specimens in pediatric patients. These cells are characterized by multiple separate nuclei rather than the interconnected lobulation of megakaryocytes. Osteoclasts are derived from hematopoietic stem cells and function in bone resorption. **(Right)** More mature osteoclasts may contain coarse cytoplasmic bone sand. Osteoclasts are very large cells that may be confused with either megakaryocytes or multinucleated histiocytes.

Osteoclast in Lacunar Space



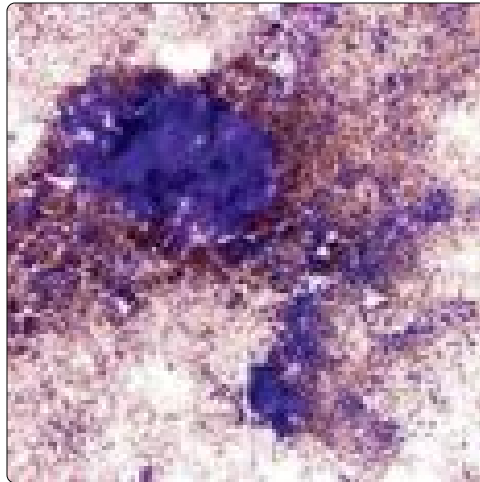
Osteoblasts Rimming Bone



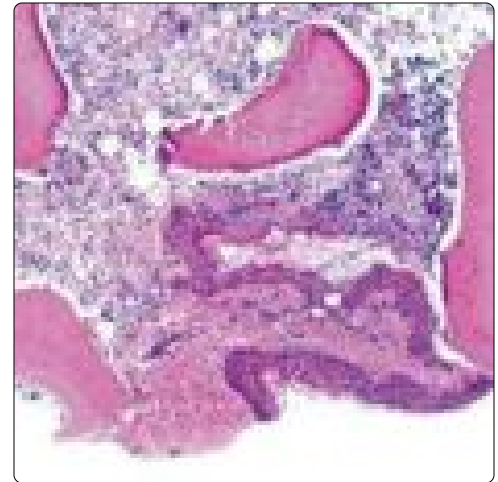
(Left) Bony remodeling is a normal feature in bone marrow core biopsy specimens from children. Osteoclasts reside in lacunar spaces , whereas osteoblasts rim the bony trabeculae . Osteoclasts are derived from hematopoietic stem cells, whereas osteoblasts are derived from mesenchymal stem cells. **(Right)** Osteoblast rimming is evident adjacent to the bony trabeculae in this bone marrow core biopsy from a teenager. Osteoblasts are bone-forming cells and are normally present in children and adolescents.

Clumping on Aspirate Smear

(Left) Prominent clumping is present on this aspirate smear, reflecting early specimen clotting. This phenomenon may greatly skew differential cell counts and distort cytologic assessment. **(Right)** Skin can sometimes be forced through the trocar needle during the bone marrow core biopsy procedure, as shown in this bone marrow core biopsy section. This has become more common due to concerns regarding scarring of the skin from lancing the entry site prior to inserting the trocar needle.

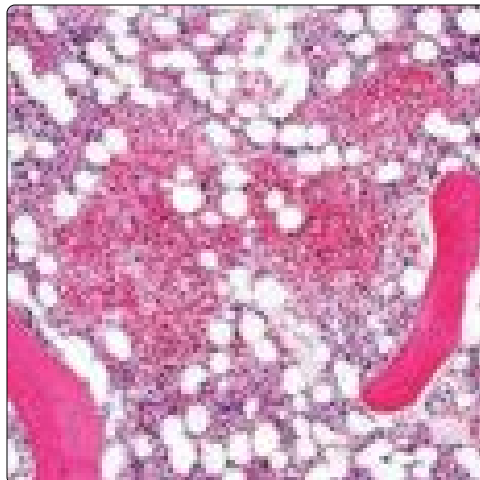


Skin in Core Biopsy Specimen



Aspiration Artifact

(Left) Partial bone marrow aspiration artifact is evident in the center of this bone marrow core biopsy section, showing an area in the biopsy devoid of marrow. Aspiration artifact is caused by performing the biopsy too close to the prior aspiration site. Aspiration artifact can be avoided by performing the biopsy first. **(Right)** This composite shows a sizable, intact, technically excellent bone marrow core biopsy adjacent to an inadequate core biopsy showing crushed bone without hematopoietic cells.

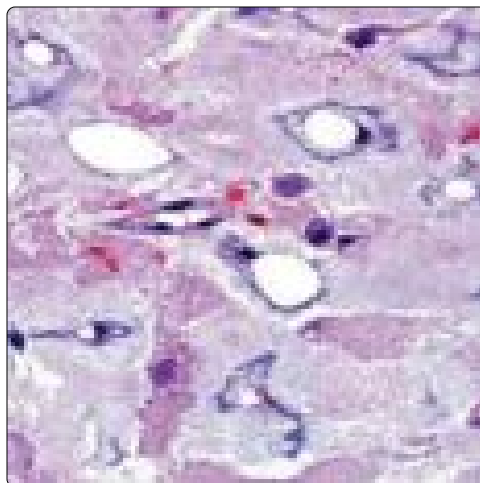


Adequate and Inadequate Core Biopsies

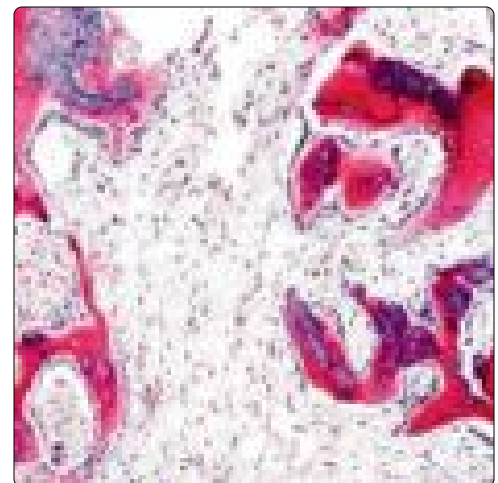


Gelatinous Transformation

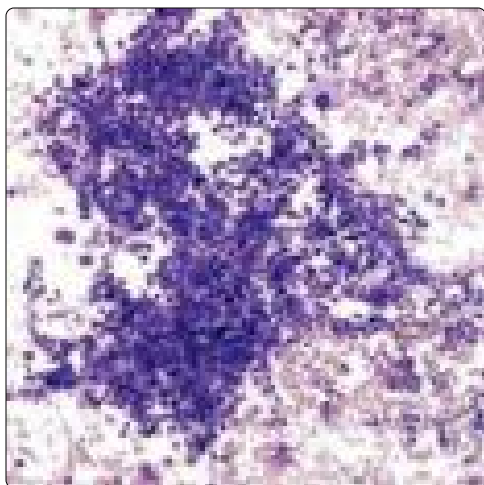
(Left) Gelatinous transformation, a change in the marrow secondary to cachexia, can mimic bone marrow aspirate artifact. RBCs are infrequent in areas of gelatinous transformation whereas numerous RBCs are present with aspiration artifact. **(Right)** This bone marrow core biopsy section is taken from a previous core biopsy site. Note fibrosis and prominent bone remodeling that reflects the repair process from the previous biopsy.



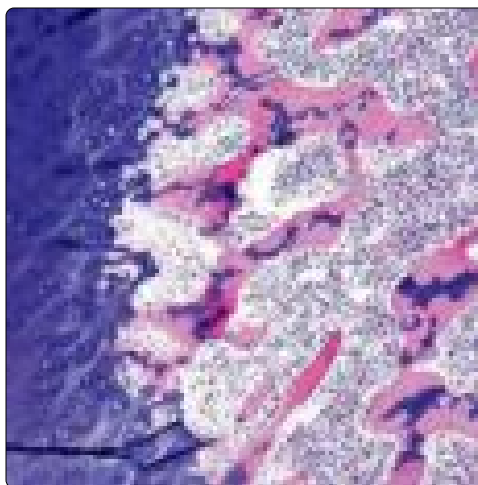
Previous Biopsy Site



Bone Marrow Aspirate, Newborn

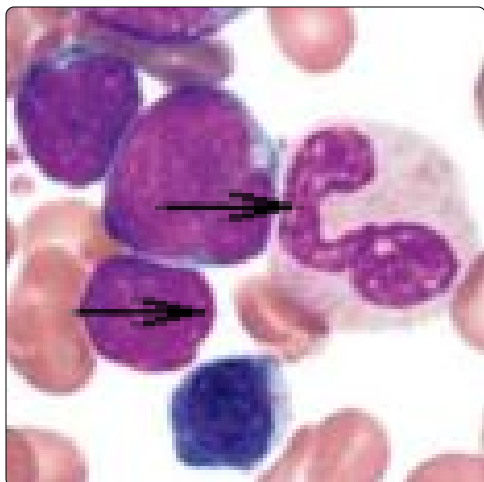


Bone Marrow Core Biopsy, Infant

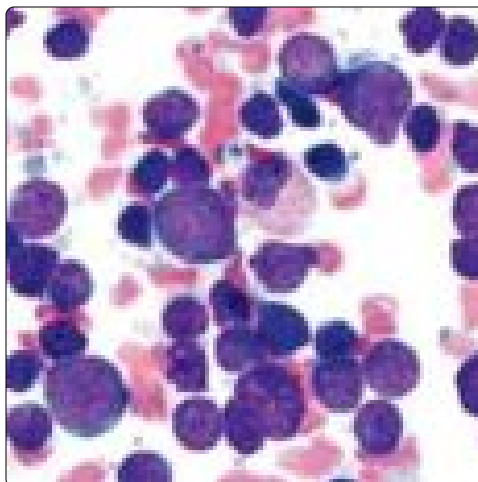


(Left) Bone marrow aspirate smear from a newborn shows ~ 100% cellularity since few, if any, fat cells are present. This high level of cellularity is physiologic in newborns and infants. **(Right)** Bone marrow core biopsy from a 21-month-old infant shows prominent bony remodeling and mineralization, which is physiologic for patient age. Note that cellularity approaches 100%, which is also physiologic for age.

Hematogones



Prominent Hematogones, Young Patient



(Left) Hematogones are benign lymphoid precursor cells that show a range in overall cell size. On high magnification, hematogones show dense nuclear chromatin with inconspicuous nucleoli and scant cytoplasm. **(Right)** Lymphoid cells are especially prominent on bone marrow aspirate smears from young patients. These cells often have morphologic features of hematogones (benign lymphoid precursor cells). Specimens with physiologically increased hematogones can be misdiagnosed as leukemia.

Bone Marrow Core, Adolescent Male



Bone Marrow Core Biopsy, Middle-Aged Woman



(Left) Bone marrow core biopsy from a 17-year-old boy shows prominent bone remodeling and thick bony trabeculae, which is physiologic for age. Note that overall cellularity is ~ 75%, normal for age. **(Right)** Bone marrow core biopsy from a 52-year-old woman shows ~ 50% fat cells and 50% hematopoietic cells. Increasing amounts of bone marrow fat cells with decreasing hematopoietic cells is typical with aging. Bony trabeculae are also thinner than in younger patients.

Peripheral Blood

PERIPHERAL BLOOD

Macroscopic Anatomy

- Total blood volume varies by body size
- Blood comprises ~ 7% of total body weight
- Adult blood volume is ~ 5 liters
- Blood consists of cells [predominantly red blood cells (RBCs)] and plasma
- By centrifugation, blood can be separated into packed RBCs (hematocrit), plasma, and white blood cells (WBCs) (called buffy coat)

Microscopic Anatomy

- All normal circulating cells are bone marrow derived
- Blood is assessed by automated cell counting and morphologic review
- Clinical and laboratory integration is essential
- Numerous RBC parameters are assessed to gauge number, size, shape, hemoglobin content, and uniformity
- WBCs enumerated, differentiated into specific types
- Automated WBC differentials include neutrophils, lymphocytes, monocytes, eosinophils, and basophils
- Absolute WBC counts are more useful than percentages
- In blood, ~ 1/2 of neutrophils are circulating, and other 1/2 have attached to endothelium (marginated cells)
- Only circulating pool of neutrophils is included in WBC count
- Migration to tissues is key for WBC function
- Platelets are enumerated and sized

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Highly specific age- and sex-related normal ranges have been established
- Residence at high altitude impacts normal ranges for RBC parameters
- Term neonates exhibit higher RBC parameters with nucleated RBCs and higher WBCs than other ages

- Blood in preterm neonates has lower RBC count, higher mean corpuscular volume (MCV), and more nucleated red blood cells (NRBCs) compared to term infants
- Blood smear review of normal neonate shows polychromasia, macrocytosis, NRBCs, and left shift
- WBC count is typically above 15,000 per μL in term neonates due to high numbers of neutrophils
- WBC count declines rapidly after birth, due to neutrophil reductions; RBC parameters gradually decline; NRBCs absent after 1st week
- By ~ 2 weeks of age, lymphocytes predominate; this finding persists throughout early childhood
- After puberty, RBC parameters are higher in males than females
- Normal ranges for RBC parameters in advanced elderly are not clear cut, especially definition of anemia

Hyperplasia

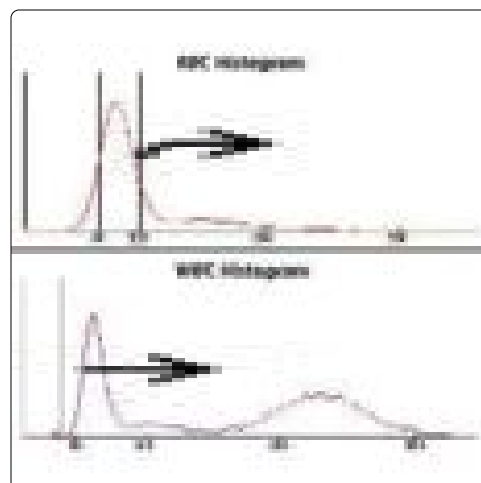
- Expansions in neutrophils, lymphocytes, eosinophils, monocytes, and platelets may occur in reactive conditions
- Cause of these reactive conditions can usually be determined by clinical correlation
- Significant absolute basophilia is uncommon in reactive conditions and is generally linked to myeloid neoplasms

Pitfalls and Artifacts

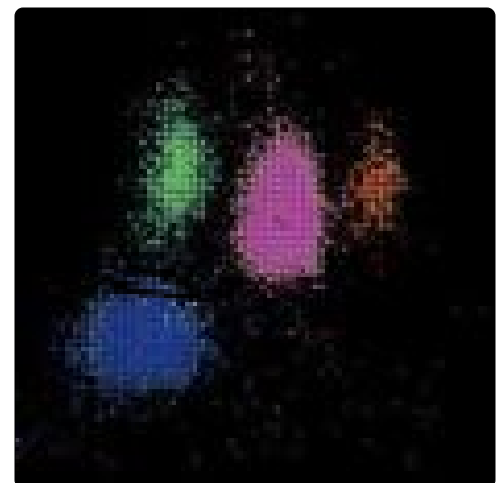
- Automated cell counting may be inaccurate in aged specimens or specimens exposed to high temperatures
- Cryoglobulins, cold agglutinins, and other substances can interfere with automated cell counting
- Large cells and microfilarial worms are selectively pulled to feather edge and may not be conspicuous in region of slide where morphology assessed
- Poor stain quality may interfere with cell identification and morphologic assessment
- Platelet clumping and satellitism cause spurious counts
- Degranulated basophils may be misidentified
- Lymphocytes in neonates and young children may have immature nuclear features
- Normal ranges for RBC parameters/absolute neutrophil counts are lower in individuals of African ancestry

(Left) By automated cell counting techniques, cells pass single file through an aperture and are counted, sized, and interrogated. These 2 graphs illustrate the size profile (i.e., mean corpuscular volume [MCV]) of normal red blood cells (RBCs). The 2 most prominent peaks on the white blood cell (WBC) histogram illustrate lymphocytes and granulocytes. (Right) Based on size and cytoplasmic granularity, differential counts are performed. Lymphocytes are small and agranular, whereas neutrophils are larger and granular.

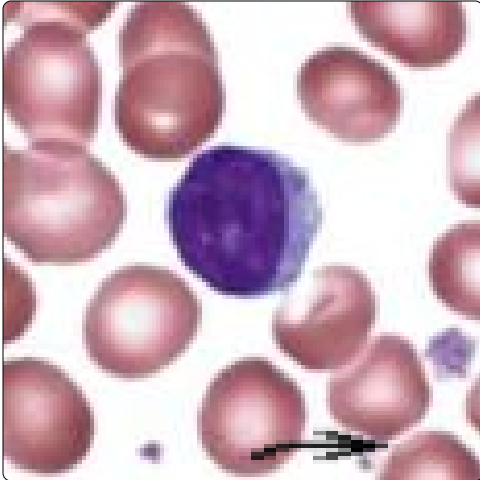
Peripheral Blood



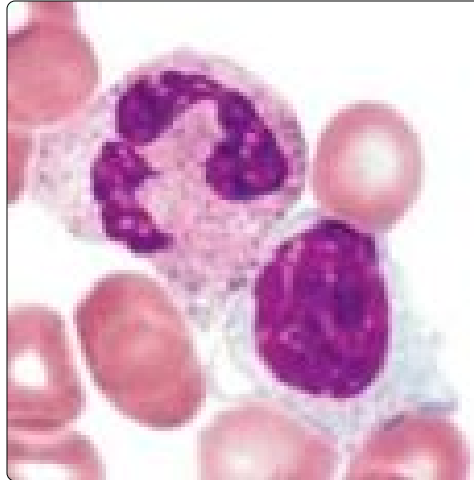
Cell Differential



Normal Blood (Adult)

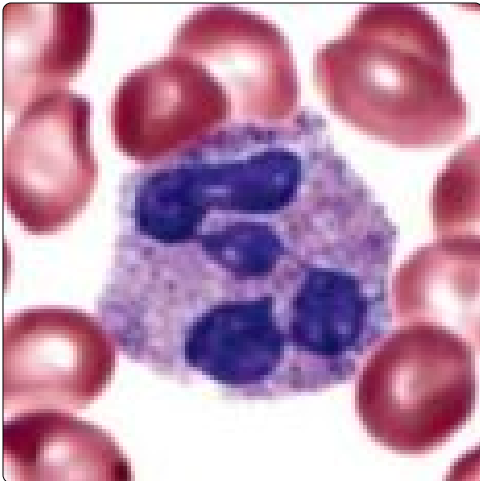


Normal White Blood Cells

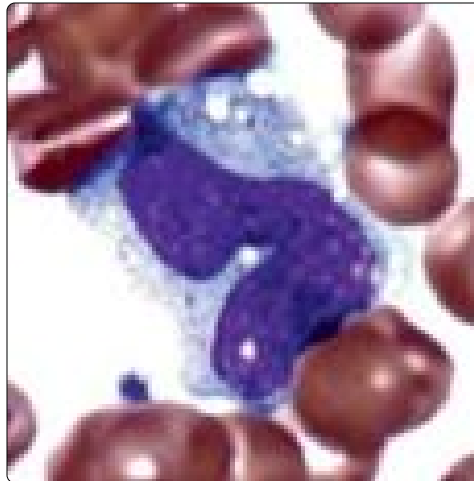


(Left) This peripheral blood smear from a normal adult woman shows uniform erythrocytes with a normal central pallor, normal platelets, and a nonactivated lymphocyte. There is some variation in platelet size , but all are well granulated. **(Right)** A normal neutrophil and a large granular lymphocyte are evident at high magnification on this blood smear from an adult. Large granular lymphocytes are present in low numbers in blood; cells with this morphology are either NK cells or cytotoxic/suppressor T cells.

Neutrophil

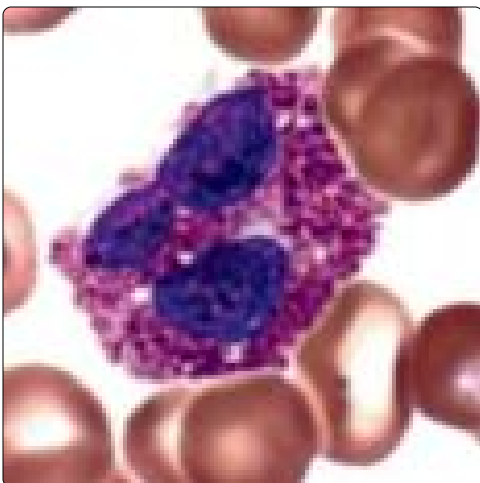


Monocyte

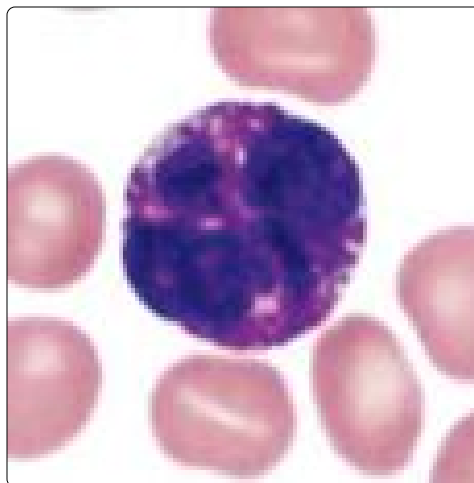


(Left) Normal neutrophil nuclear segmentation and normal granulation of the cytoplasm are evident in this circulating neutrophil. Neutrophils typically have 3-5 nuclear lobes with a thin strand of chromatin connecting these lobes. The cytoplasm has a pinkish tint from secondary granules. **(Right)** A monocyte is characterized by large size, blue-gray cytoplasm with occasional granules and vacuoles, and a somewhat folded nucleus. The nuclear chromatin has a hills and valleys appearance.

Eosinophil



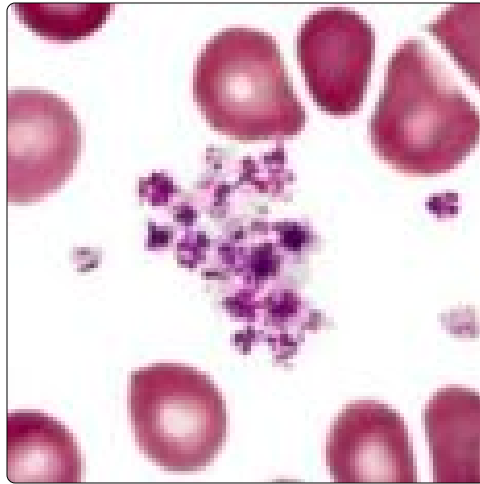
Basophil



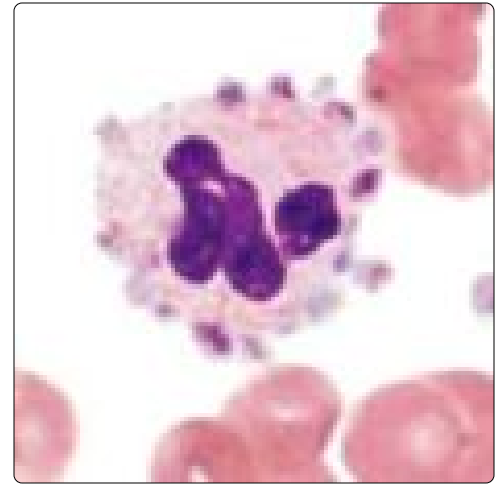
(Left) Eosinophils are recognized by the distinctive eosinophilic, refractile appearance of the secondary granules as shown on the blood smear. Eosinophils typically have 2 nuclear lobes, but sometimes 3, as illustrated here. **(Right)** Basophils are the least numerous WBC in the blood. The secondary granules of basophils are dark and coarse. These granules often overlie the nucleus, which typically shows 3-4 nuclear lobes. Note the adjacent normal RBCs.

Clumped Platelets

(Left) Platelet clumping is a common artifact in peripheral blood and may interfere with accurate platelet counts. Manual scanning for platelet clumps is recommended on blood smears when thrombocytopenia is detected by automated counting. **(Right)** Platelet satellitism is an artifactual phenomenon in which platelets adhere to neutrophils. A spuriously low platelet count can result from satellitism. Repeat testing utilizing a different anticoagulant is recommended.

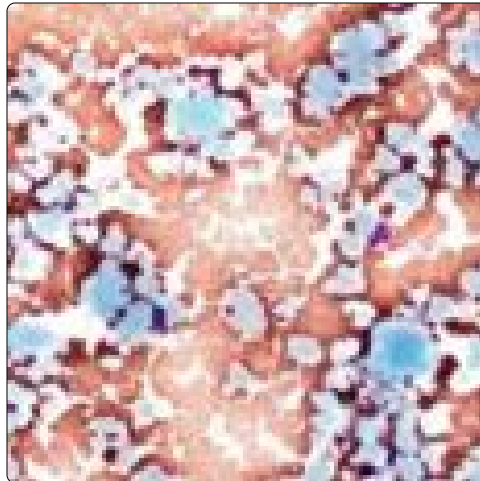


Platelet Satellitism

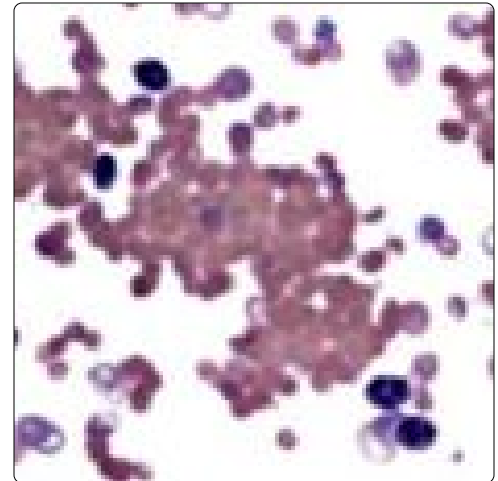


Cryoglobulin

(Left) The extracellular amorphous blue globules shown here are cryoglobulin. They can precipitate in peripheral blood, interfering with automated cell counting. **(Right)** Erythrocyte agglutination is common in patients with a cold agglutinin. Agglutination often results with cooling of the specimen during transport after phlebotomy. Warming the specimen eliminates the agglutination so that accurate automated cell counts can occur. Note the clumps of erythrocytes in this photograph.

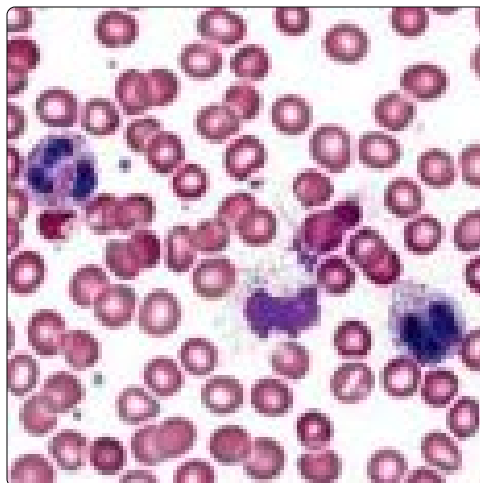


Agglutination

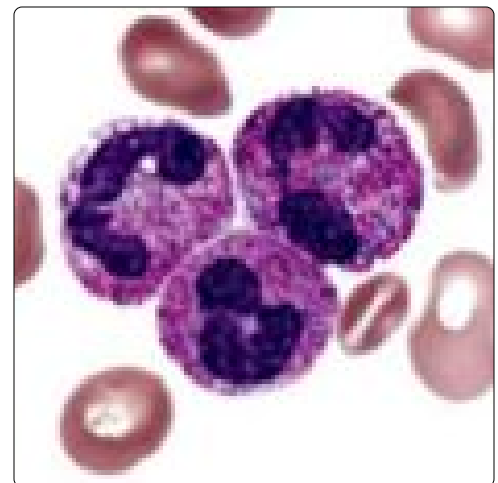


Partial Degranulation of Basophil

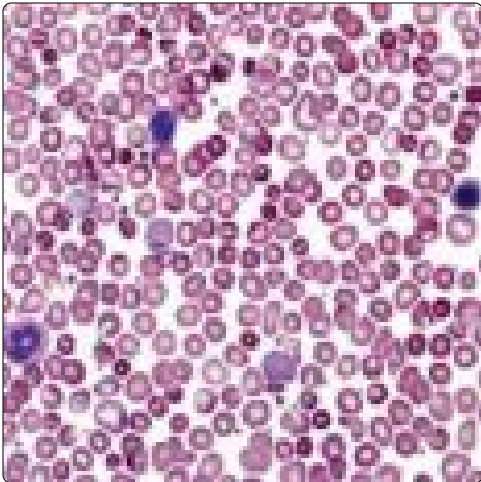
(Left) This photo compares a partially degranulated basophil (lower right) with a normal neutrophil (upper left). Note that basophil granules still overlie the nucleus. Note the broken neutrophil , a so-called basket cell. **(Right)** Toxic neutrophils, as illustrated here, can mimic basophils because of the prominent basophilic granulation that characterizes an activated neutrophil. This appearance is due to staining changes in the activated secondary granules.



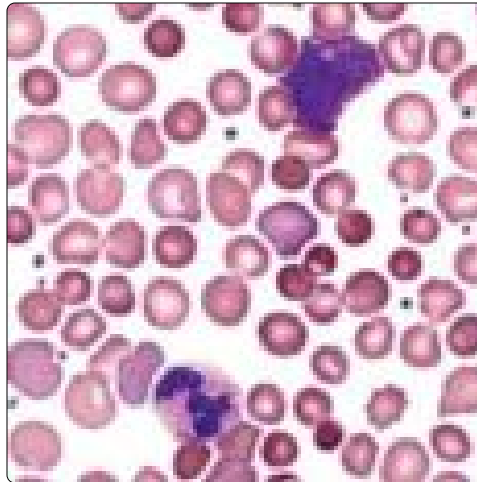
Toxic Neutrophils



Newborn Blood

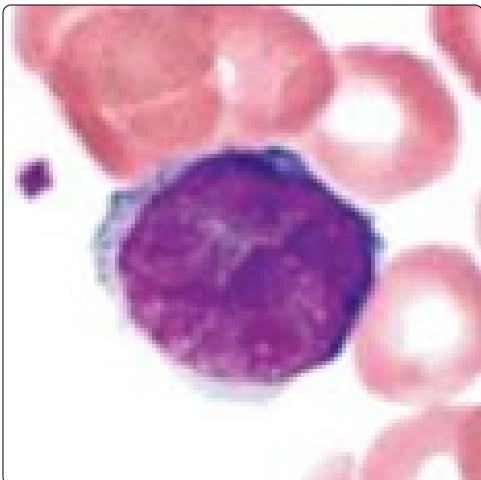


Normal Term Newborn

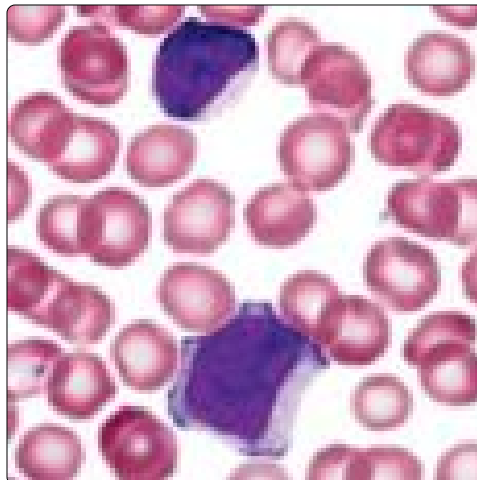


(Left) This blood smear from a term newborn shows a high number of erythrocytes and many reticulocytes, which is physiologic for this age. A leukocytosis with left shift is also physiologic for age, as are nucleated red blood cells . The leukocytosis, polychromasia, and NRBCs should all decline shortly after birth in healthy babies. **(Right)** Features that are physiologic in the newborn period are illustrated in this peripheral blood smear, including prominent polychromasia , left shift, and leukocytosis.

Normal Immature Lymphocyte in Infant

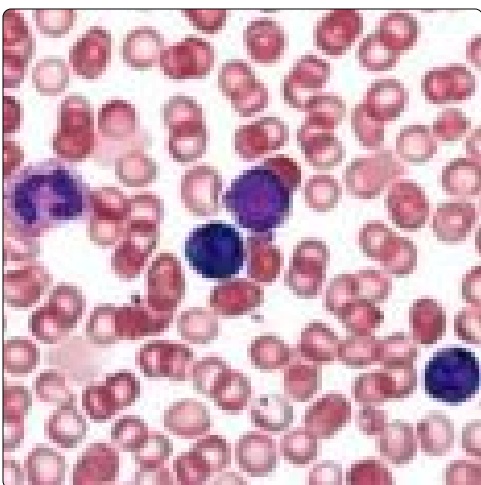


Normal Lymphocyte Spectrum in Infant



(Left) Immature lymphocytes are physiologic in infants and young children, even though they are reminiscent of lymphoblasts. These cells have dispersed chromatin and scant cytoplasm. Hematopoietic parameters are normal, a finding useful in the distinction from leukemia. **(Right)** This peripheral blood smear from a 4-month-old infant shows a more normal-appearing lymphocyte (top) in conjunction with a larger lymphocyte that exhibits some nuclear immaturity. The size and nuclear features are physiologic for this age.

Normal Blood, 4-Month-Old Infant



Feather Edge



(Left) This blood smear from a 4-month-old boy shows a predominance of lymphoid cells (absolute lymphocyte count of $10.8 \times 10^9/L$), which is physiologic for age. **(Right)** It is important to scan the feather edge of a smear during manual morphologic review to detect cells and even microfilarial worms that have been "dragged" to this portion of the blood smear. Larger cells, such as this immunoblast , are often pulled to the feather edge in the preparation of blood smears as well.

THYMUS

Macroscopic Anatomy

- Located in midline anterosuperior mediastinum
 - Anterior to great vessels
 - Posterior to sternum
 - Often extends from lower neck down to upper heart, along pericardium
 - Parietal pleura forms lateral borders
- Composed of 2 lobes
 - Each lobe is composed of lobules
- Size and weight vary with age and from person to person
 - Grows in size through puberty and then involutes
 - 10-20 grams at or around birth
 - Growth up to 25-50 grams
 - Despite involution/atrophy, overall size and shape of thymus remains same
- Function: Responsible for T lymphocyte development

Microscopic Anatomy

- Outer fibrous capsule
 - Surrounds lobes and lobules
- Gland is composed primarily of mixture of lymphoid and epithelial cells
 - Epithelial cells
 - Plump with abundant, pale eosinophilic cytoplasm
 - IHC: Cytokeratin (+), p63(+)
 - Thymocytes
 - Immature T lymphocytes
 - Mostly small with dark, dense nuclei
 - Minor population of larger lymphoblasts
 - ◻ Most in outer cortex
 - IHC: CD45(+), CD3(+), Tdt(+)
- Each lobule is composed of cortex and medulla
 - Cortex
 - Outer region of lobule
 - Contains numerous T lymphocytes
 - Epithelial cells are sparse
 - Medulla
 - Inner region of lobule

- Contains larger proportion of epithelial cells
- Fewer lymphocytes, more mature
- Hassall corpuscles
 - ◻ Most distinctive feature of thymus gland
 - ◻ Only found in medulla
 - ◻ Structures formed by concentric layers of keratin
 - ◻ Morphologic variations include calcification, cystic change, inflammatory infiltrates, goblet cells
- Additional constituent cells of thymus include B lymphocytes, Langerhans cells, neuroendocrine cells, mast cells, histiocytes
 - B cells form follicles; may contain germinal centers
- Physiologic involution (atrophy)
 - Process begins after puberty
 - Thymic lymphoid parenchyma shrinks and is gradually replaced by mature adipose tissue
 - Epithelial elements remain and often appear as small nests within lymphoid aggregates or islands within fat
 - May also form small rosette-like structures

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Size and weight of thymus increase from birth to puberty
- Prepubertal thymus is overtly cellular with prominent dual epithelial/lymphocyte population
 - Epithelial cells are larger and more prominent
- Adult thymus shows significantly decreased cellularity, epithelial cell atrophy, paucity of lymphocytes, and mature fat infiltration

Hyperplasia

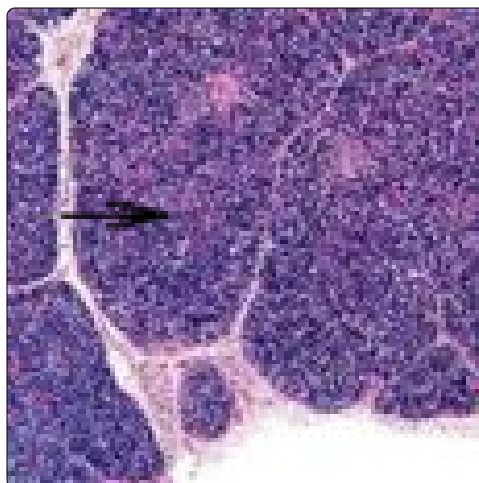
- True thymic hyperplasia is increase in size of thymus
- Lymphoid hyperplasia is due to influx of reactive B cells

Pitfalls and Artifacts

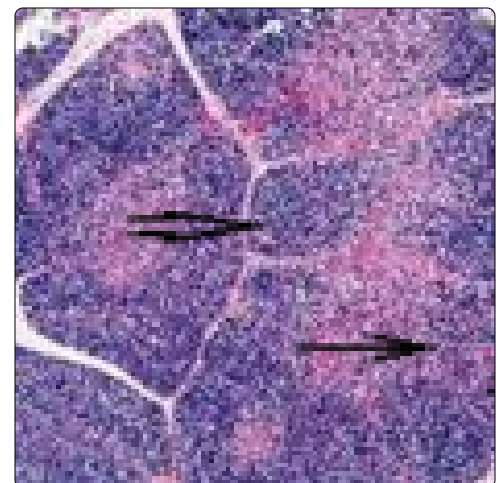
- Atrophied thymic remnants can mimic neuroendocrine proliferation
- Regional brown fat can be misinterpreted as hibernoma or liposarcoma
- Caulerization, processing, and traumatic changes

Pediatric Thymus

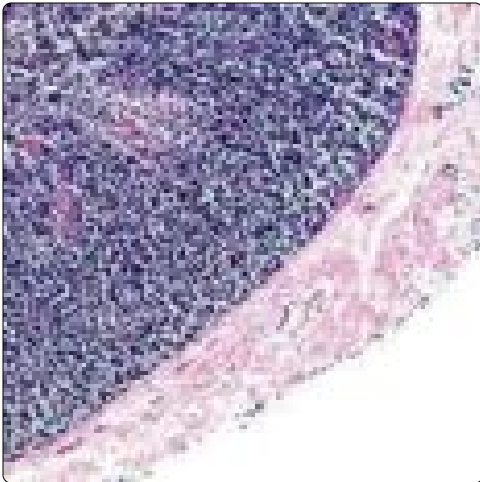
(Left) The pediatric thymus is markedly compartmentalized with lobules of lymphoepithelial tissue separated by wispy fibrous interstitium . Hassall corpuscles can often be seen at low magnification . **(Right)** Within each lobule are 2 compartments: A dark outer cortex  and a pale inner medulla . This disparity in color is due to the presence of significantly more lymphocytes in the cortex than in the medulla.



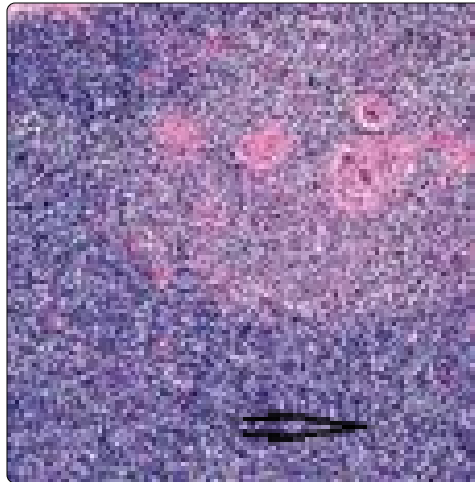
2 Microanatomic Compartments



Fibrous Capsule

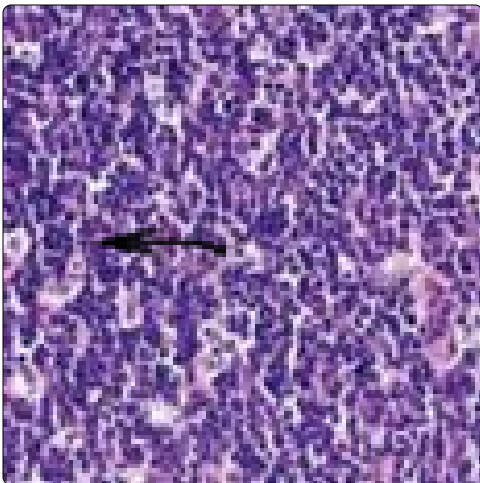


Cortex and Medulla

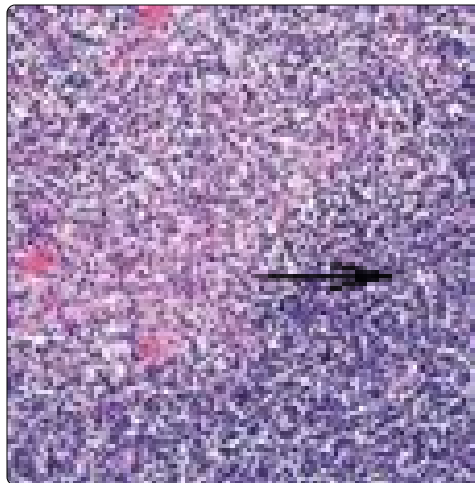


(Left) There is usually a connective tissue capsule at the periphery of the thymus. This capsule varies in size from thin and wispy to thick and fibrous. It is contiguous with the fibrous interstitium between the thymic lobules. (Right) The cortex is composed predominantly of immature T lymphocytes and a few epithelial cells, giving it a darker appearance than the adjacent medulla. The pink squamoid Hassall's corpuscles are easily visible against the background of lymphocytes.

Thymic Cortex

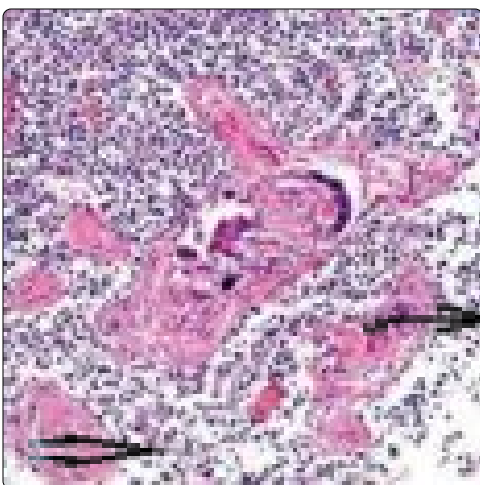


Thymic Medulla

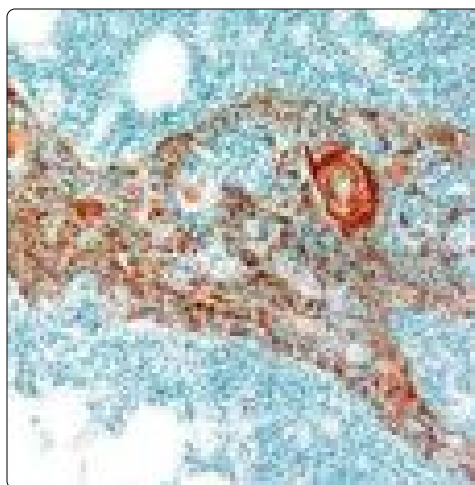


(Left) The immature T lymphocytes found in the thymic cortex have very little cytoplasm and are closely packed. The larger thymic epithelial cells are interspersed between the lymphocytes and can be identified by their more abundant cytoplasm. (Right) Compared to the cortex, the thymic medulla contains a much larger population of pink, thymic epithelial cells and a smaller proportion of dark T lymphocytes.

Hassall's Corpuscles



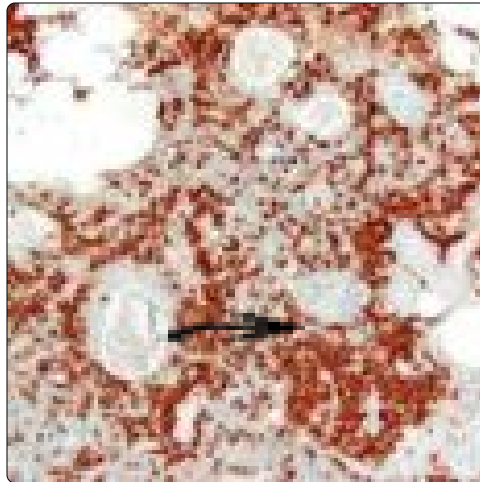
Cytokeratin Immunohistochemistry



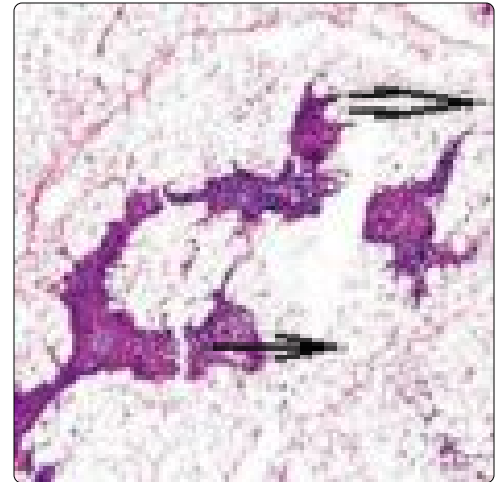
(Left) Hassall's corpuscles are composed of keratinized epithelial cells arranged in concentrically oriented nests, which often contain keratin debris. These unique structures are easily seen in the medulla even at low magnification and are very useful in identifying thymic tissue. (Right) A pancytokeratin immunostain highlights the thymic epithelial cells, while the thymic lymphocytes (thymocytes) are negative. Note also the positivity of the keratinized Hassall's corpuscle.

CD3 Immunohistochemistry

(Left) Thymic T lymphocytes can be highlighted by a variety of immunostains, including CD3 (shown). Note that epithelial cells and the Hassall corpuscles are negative. **(Right)** After puberty, the thymus begins the process of physiologic involution, or atrophy, during which the thymic lymphoepithelial tissue is gradually replaced by mature fat.

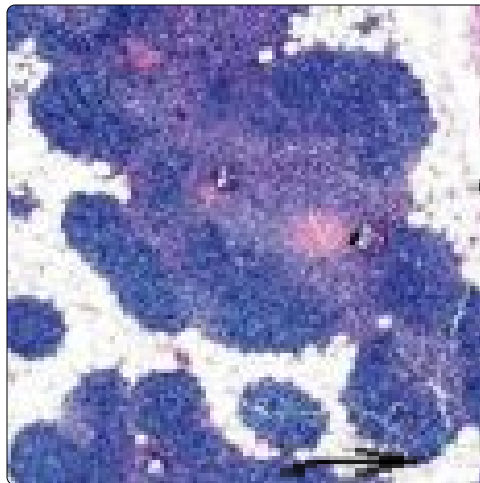


Physiologic Involution

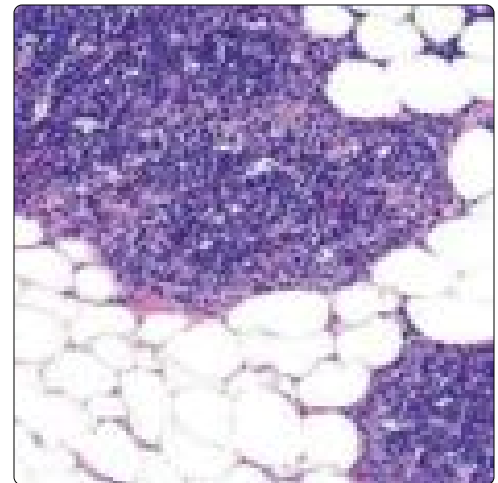


Calcification

(Left) Calcification of the Hassall corpuscles is often seen at least focally in thymic tissue undergoing involution. Other alterations to these structures include cystic change and inflammatory infiltrates. **(Right)** In some areas of atrophic thymic tissue, the epithelial cells appear to form a layer around the periphery of the thymocyte aggregates and can be distinguished by their more abundant pale eosinophilic cytoplasm.

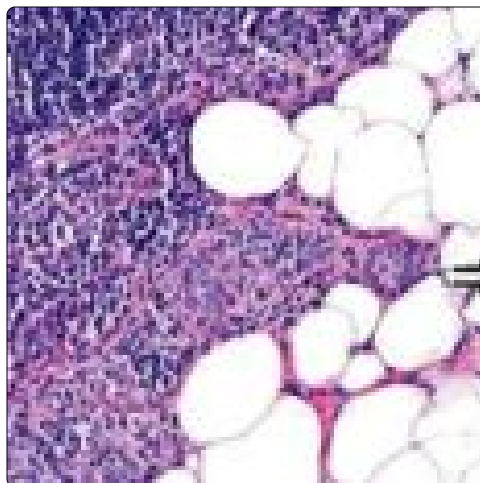


Peripheral Epithelial Cells

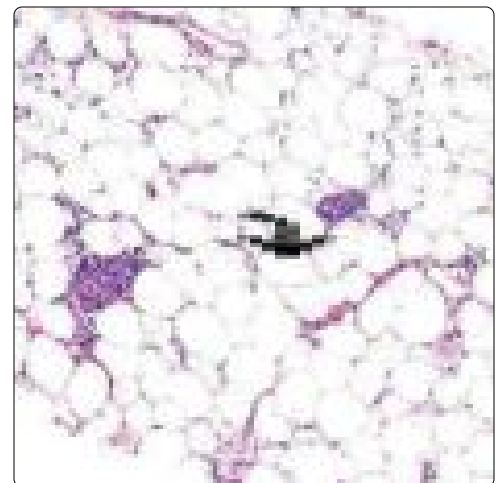


Atrophic Epithelial Elements

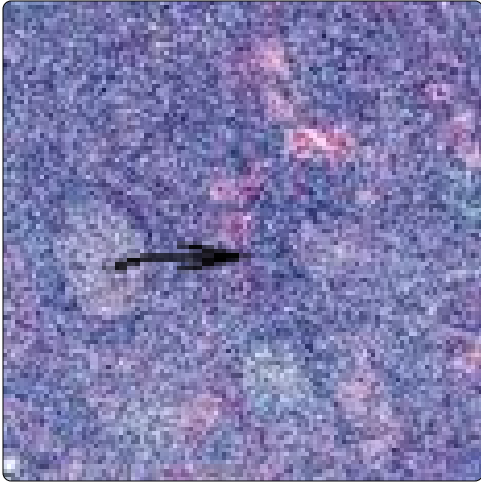
(Left) During the process of involution, thymic atrophy can result in small nests or aggregates of atrophied epithelial cells that are reminiscent of neuroendocrine proliferations, such as a carcinoid tumor. **(Right)** In older age groups, the thymic lymphocyte parenchyma may show marked atrophy and fatty replacement. However, epithelial elements remain.



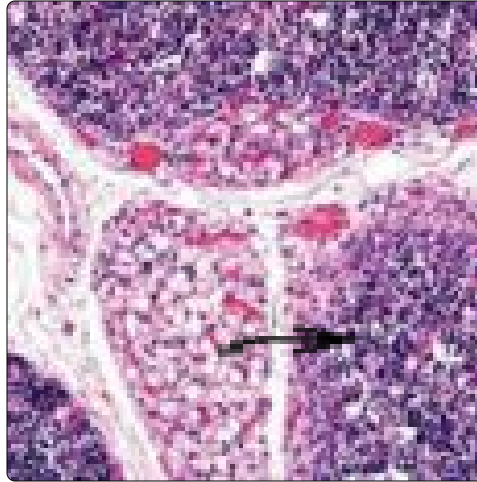
Residual Epithelial Elements



Germinal Center Formation

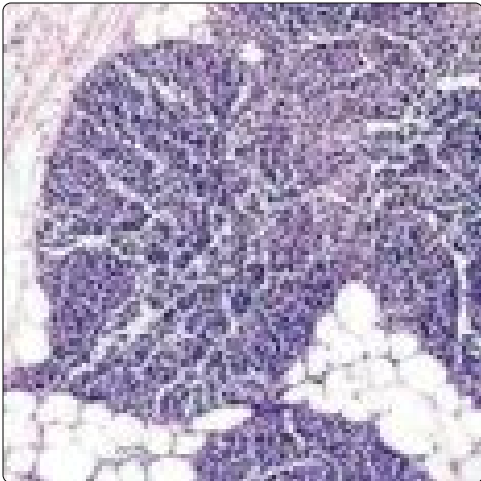


Brown Fat

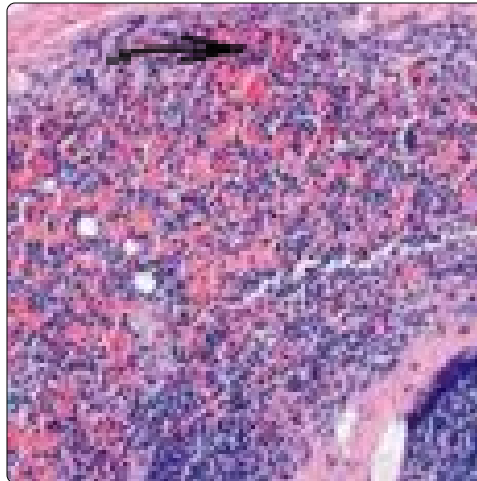


(Left) Reactive germinal center formation [\[2\]](#) can be seen in otherwise normal thymus glands, particularly in the pediatric age group. However, they are also characteristically prominent in immune-mediated conditions such as myasthenia gravis. (Right) In the pediatric population, the mediastinum contains a prominent component of brown fat [\[2\]](#). It is not unusual to see nests or sheets of these multivacuolated, often pink granular cells adjacent to the thymic lobules [\[2\]](#).

Cracking Artifact

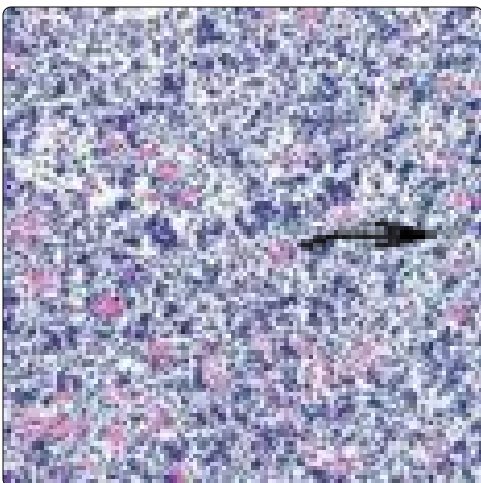


Artifactual Hemorrhage



(Left) Artifactual cracking or clefts may be present in some thymic specimens, as depicted, and are related to tissue processing. (Right) Intraparenchymal blood may be identified in some thymus specimens, secondary to surgical manipulation. It can also be seen in postmortem specimens. In this image, both thymic epithelial cells [\[2\]](#) and lymphocytes [\[2\]](#) are visible.

Clumping Morphology



Cautery Artifact



(Left) In some cases, thymic lymphocytes may appear to clump together in small aggregates [\[2\]](#). (Right) Cauterization during surgery is a common cause of histologic artifact, particularly elongation or spindling of both thymic epithelial cells and lymphocytes. This morphology can lead to a mistaken impression of a neoplasm, such as thymoma or sarcoma. Crush artifact may also occur (not shown), resembling a morphology often seen in small cell carcinoma.

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SECTION 7

Head and Neck



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Eye and Ocular Adnexa

EYE

Macroscopic Anatomy

- Globe
 - Spherical, hollow body of eyeball
- Cornea
 - Transparent outermost structure
 - Refracts incoming light
- Iris
 - Membranous diaphragm with central opening (pupil)
- Lens
 - Biconvex crystalline structure
 - Focuses light on retina
- Anterior chamber
 - Area between posterior cornea and anterior iris
- Posterior chamber
 - Area between vitreous body and posterior aspect of iris
- Aqueous body
 - Clear fluid filling anterior and posterior chambers
- Vitreous body
 - Gel-like material filling posterior cavity of globe
- Sclera
 - White, tough outer covering of globe
- Retina
 - Complex organization of nerve cells and processes
 - Translates light into electrical impulses to brain via optic nerve

Microscopic Anatomy

- Schlemm canal
 - Drainage channel for aqueous
- Descemet membrane
 - Basement membrane lining of posterior cornea
- Schwalbe ring
 - Termination of Descemet membrane
- Ciliary body
 - Supports lens via zonular fibers
- Choroid
 - Vascular layer between sclera and retina

- Retina
 - Consists of 8-10 layers
 - Nerve fiber layer (innermost layer)
 - Ganglion cell layer
 - Inner plexiform layer
 - Inner nuclear layer
 - Outer plexiform layer
 - Outer nuclear layer
 - Rods and cones
 - Retinal pigment epithelium (outermost layer)
 - Internal limiting membrane is boundary between nerve fiber layer and vitreous body
 - External limiting membrane demarcates outer nuclear layer from rods and cones

OCULAR ADNEXA

Macroscopic Anatomy

- Eyelid
 - Skin and conjunctival covering of eye
- Optic nerve
 - Cranial nerve attached to posterior aspect of globe
 - Conveys signals from retina to brain
- Orbital soft tissue
 - Supportive fat, muscle, and collagen

Microscopic Anatomy

- Lacrimal glands
 - Serous glands that produce tears
- Meibomian gland
 - Sebaceous glands in tarsus
- Glands of Moll
 - Eyelash follicle glands composed of apocrine cells
- Glands of Zeis
 - Sebaceous glands that empty into eyelash follicle

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

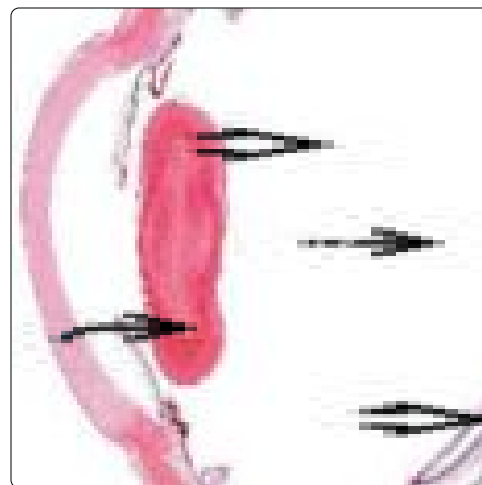
- Thickening and hyalinization of pars plicata and ciliary processes

Anatomy of Globe



(Left) This graphic illustrates the globe in a sagittal section. The globe is supported by muscle and adipose tissue and connects to the brain via the optic nerve. (Right) In this sagittal section, the cornea, iris, lens, vitreous body, and ciliary body can be identified. The anterior chamber is the space between the anterior iris and posterior cornea, and the posterior chamber is comprised of the space between the vitreous and the posterior aspect of the iris. (Courtesy H. Brown, MD.)

Anterior Eye Anatomy



Posterior Eye Anatomy



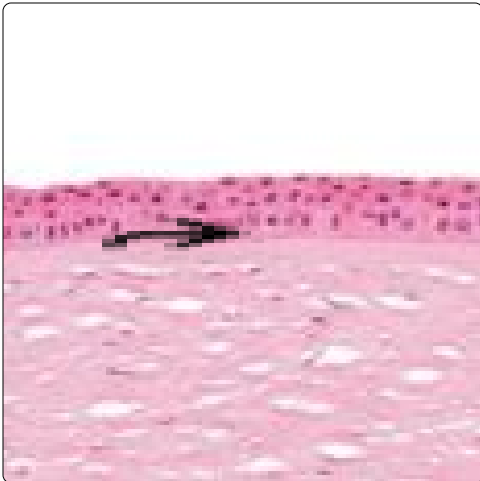
Anterior Chamber



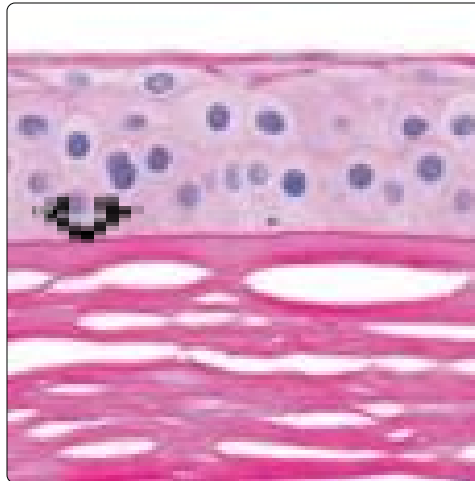
(Left) In this low-power field of the posterior globe, the head of the optic nerve, retina, choroid, and the sclera can be identified. (Courtesy H. Brown, MD.)

(Right) The anterior chamber of the globe is situated between the posterior cornea and the anterior iris. At low power, the lens is composed of eosinophilic material. The iris sphincter constricts and dilates to adjust the amount of light that the lens is allowed to focus on the retina. (Courtesy H. Brown, MD.)

Cornea: Stratified Squamous Epithelium

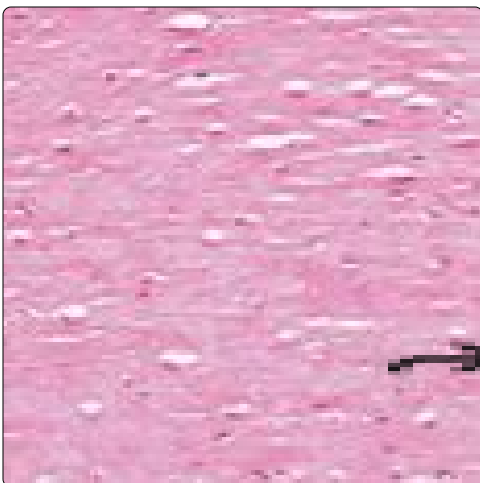


Cornea: Basal Lamina

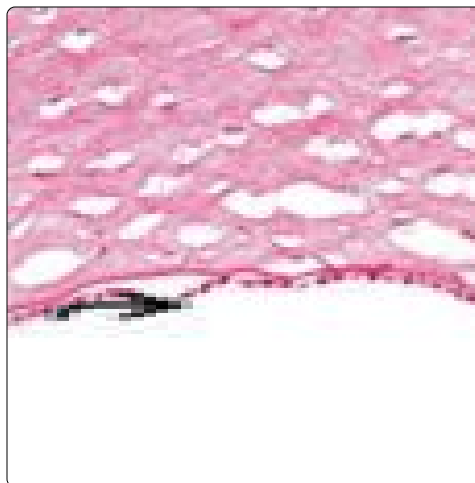


(Left) Corneal epithelium is a nonkeratinizing, stratified, squamous epithelium with nuclei present in all layers. The basal layer of the epithelium is composed of larger, more columnar cells, and as the cells rise to the corneal surface, they become flattened. (Courtesy H. Brown, MD.) (Right) The corneal epithelium rests upon a thin basal lamina, stained bright pink in this periodic acid-Schiff preparation. The acellular band beneath the basal lamina is the Bowman layer. (Courtesy H. Brown, MD.)

Cornea: Stroma



Cornea: Descemet Membrane



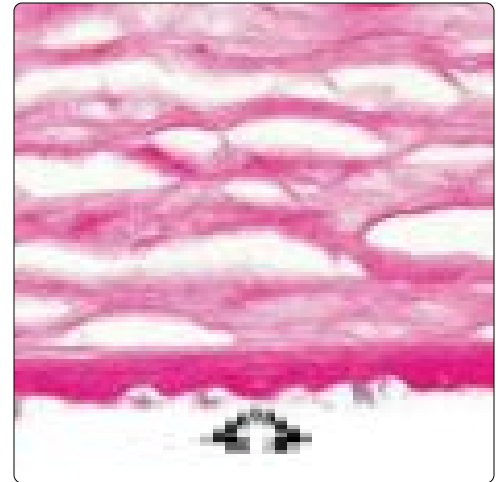
(Left) The corneal stroma is composed of regularly spaced collagen fibers embedded in an extracellular matrix. Of note, no blood vessels or lymphatics are present in the corneal stroma. The cleft-like spaces are secondary to tissue fixation/processing. (Courtesy H. Brown, MD.) (Right) The posterior cornea is lined by a single layer of cuboidal "endothelium" (a misnomer as it does not line a vessel). The Descemet membrane is the basement membrane of the "endothelium." (Courtesy H. Brown, MD.)

Corneoscleral Limbus

(Left) The junction of the cornea and the sclera is known as the corneoscleral limbus, seen here. The corneoscleral limbus is an important surgical landmark and posteriorly contains several structures that are vital to the flow of aqueous fluid. (Courtesy H. Brown, MD.) (Right) Normal, age-related changes are seen in the Descemet membrane consisting of focal thickenings (Hassall-Henle warts) that appear to alternate with the posterior "endothelium." (Courtesy H. Brown, MD.)



Cornea: Descemet Membrane



Corneoscleral Limbus

(Left) The corneal stroma and iris can be seen forming the anterior chamber angle. The canal of Schlemm is anterior to the iris. As the corneal stroma blends into the sclera, delicate blood vessels (absent in the corneal stroma) can be identified. (Courtesy H. Brown, MD.) (Right) The episclera composes the outermost layer of the sclera and contains more numerous blood vessels admixed with collagen and fibroblasts.

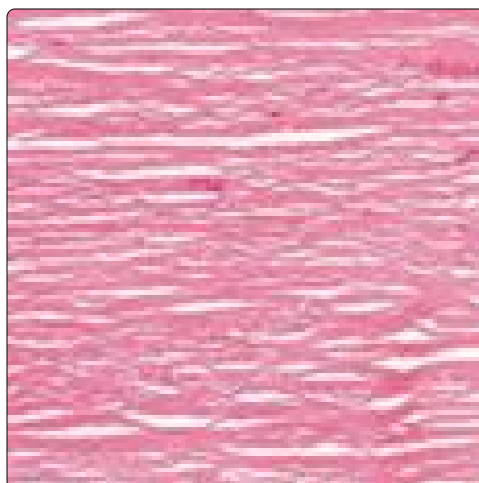


Sclera: Episclera

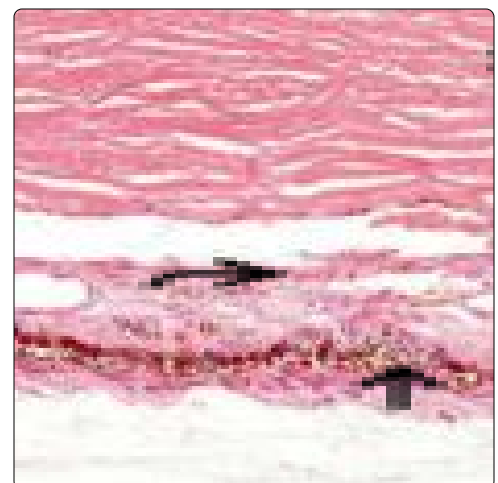


Sclera: Stroma

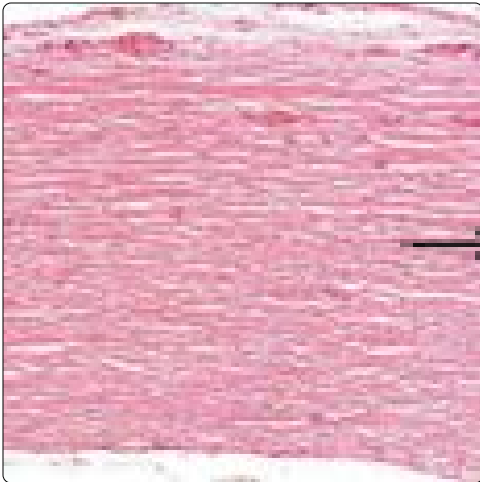
(Left) The middle portion, and the bulk of the sclera, is the scleral stroma. The stroma is composed of varying sizes of collagen fibers admixed with elastic fibers. Within the stroma, rare fibroblasts and blood vessels may be identified. (Courtesy H. Brown, MD.) (Right) The lamina fusca, seen here under the scleral stroma, composes the innermost layer of the sclera. The lamina fusca is composed of melanocytes, collagen, and fibroblasts.



Sclera: Lamina Fusca



Sclera: Vessels

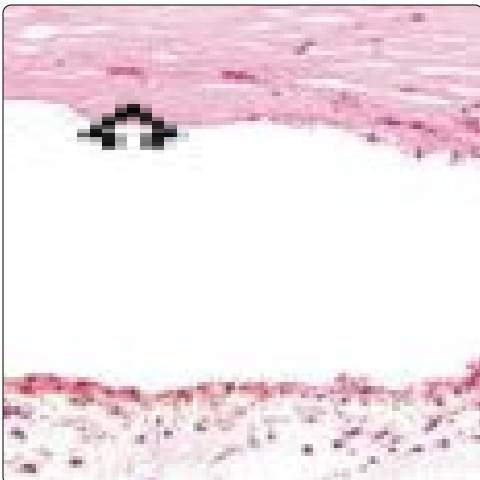


Canal of Schlemm

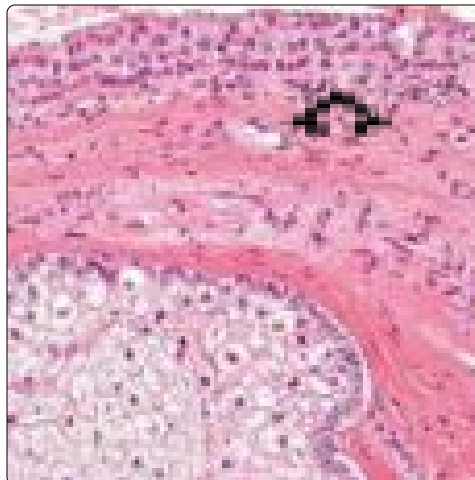


(Left) Blood vessels and fibroblasts are present within the sclera. The episclera is located at the top of the image. (Courtesy H. Brown, MD.) (Right) Aqueous fluid from the anterior chamber circulates through the trabecular meshwork and drains into the canal of Schlemm. From there, it flows into the episcleral venous plexus via aqueous veins. Blockage of the aqueous flow at any of these points may lead to increased intraocular pressure and glaucoma. (Courtesy H. Brown, MD.)

Schwalbe Ring

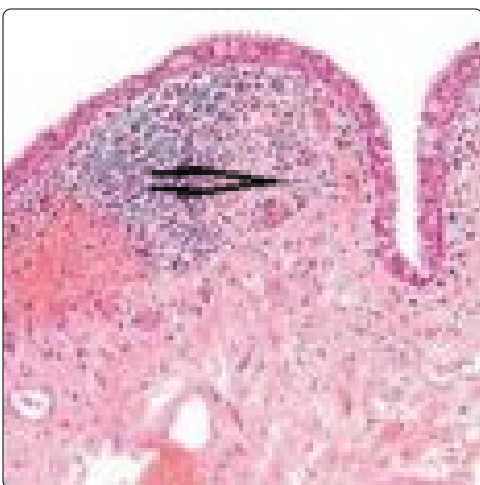


Conjunctiva: Palpebral Conjunctiva

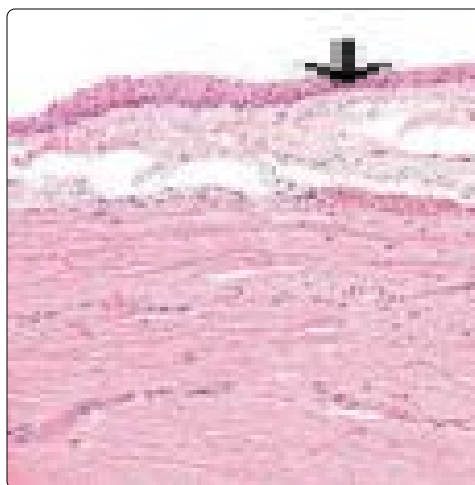


(Left) Schwalbe ring identifies the termination of Descemet membrane peripherally. The trabecular meshwork begins immediately beyond the Schwalbe ring. (Courtesy H. Brown, MD.) (Right) The conjunctiva is the thin, transparent lining relatively covering the inner surface of the eyelid (palpebral conjunctiva) and reflected onto the anterior surface of the globe (bulbar conjunctiva). Its function is to allow the eyelid to slide smoothly over the globe. (Courtesy H. Brown, MD.)

Conjunctiva: Lymphoid Aggregates



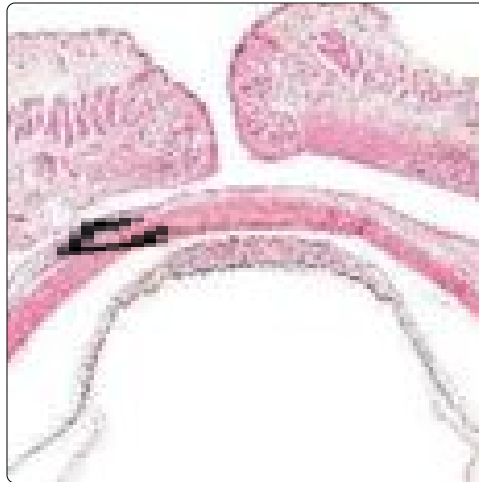
Conjunctiva: Bulbar



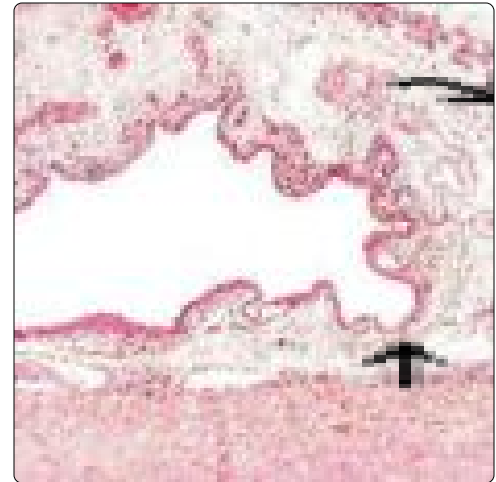
(Left) Occasional aggregates of lymphocytes may be identified within the conjunctiva. Lymphoid aggregates tend to be more common in the area of the superior and inferior fornices. They are not a sign of chronic infection. (Courtesy H. Brown, MD.) (Right) The portion of the conjunctiva overlying the sclera is known as the bulbar conjunctiva. The bulbar conjunctival stroma is composed of loose connective tissue. (Courtesy H. Brown, MD.)

Conjunctiva: Superior and Inferior Fornix Openings

(Left) The conjunctiva terminates anteriorly at the beginning of the squamous lining of the eyelids. The junction of the palpebral conjunctiva and the bulbar conjunctiva occurs in the superior and inferior fornix, the openings of which can be seen here. (Courtesy H. Brown, MD.) (Right) The palpebral conjunctiva and the bulbar conjunctiva meet in the fornix. It is in this region that lymphoid aggregates and abundant epithelial goblet cells can be identified. (Courtesy H. Brown, MD.)

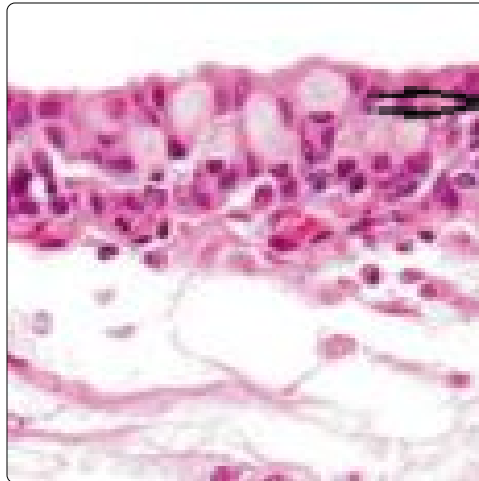


Conjunctiva: Fornix

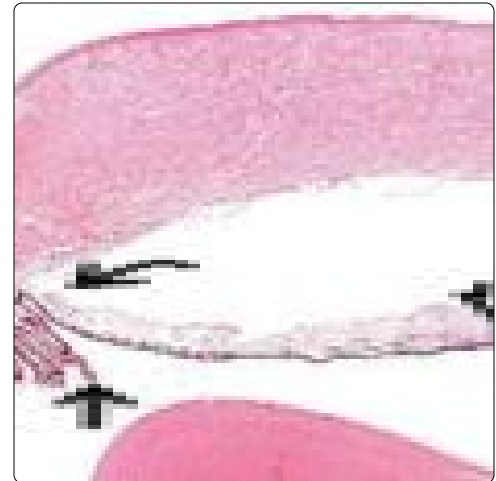


Conjunctiva: Goblet Cells

(Left) Goblet cells are seen in the forniceal conjunctiva. Goblet cells are large mucin-containing cells with gray-blue cytoplasm and basal nuclei. This mucin helps to lubricate the eyelid and conjunctiva. (Courtesy H. Brown, MD.) (Right) The iris can be seen in the middle of this H&E. The origin of the iris (iris root) is from the anterior ciliary face. Smooth muscle fibers in the iris cause the pupil to constrict and dilate, regulating the amount of light entering the lens. (Courtesy H. Brown, MD.)

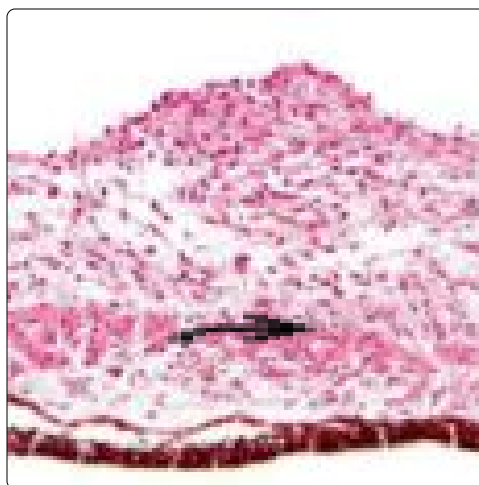


Iris



Iris: Stroma

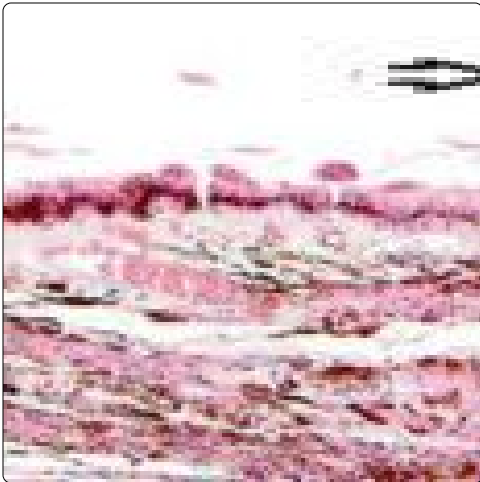
(Left) The iris is composed of connective tissue, smooth muscle, blood vessels, nerve cells, and melanocytes. The number of stromal melanocytes determines the color of the iris. The posterior epithelial lining is heavily pigmented. (Courtesy H. Brown, MD.) (Right) The epithelium that lines the ciliary body is composed of 2 distinct layers. The innermost layer is nonpigmented and abuts the posterior chamber, while the outer layer is pigmented and adjacent to the ciliary body stroma. (Courtesy H. Brown, MD.)



Ciliary Body: Epithelium



Ciliary Body: Zonular Fibers

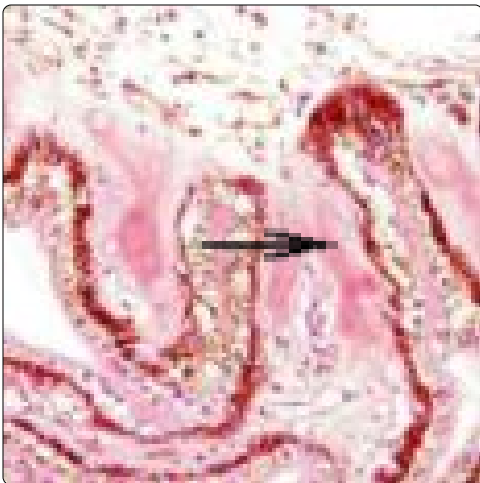


Ciliary Body: Pars Plicata



(Left) Thin, acellular, eosinophilic zonular fibers may be identified extending from the pars plana of the ciliary body to the lens equator. The zonular fibers anchor the lens to the ciliary body, and decreases in zonular tension allow alteration in lens curvature. (Courtesy H. Brown, MD.) (Right) The fold-like pars plicata can be seen here. The fibrovascular cores of the pars plicata produce the aqueous humor. (Courtesy H. Brown, MD.)

Ciliary Body: Pars Plicata Aging Changes

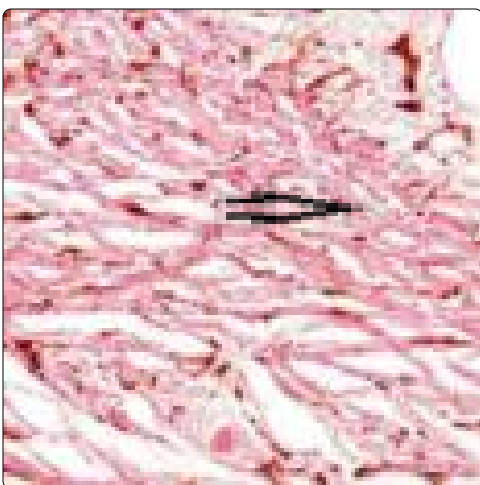


Ciliary Body: Smooth Muscle

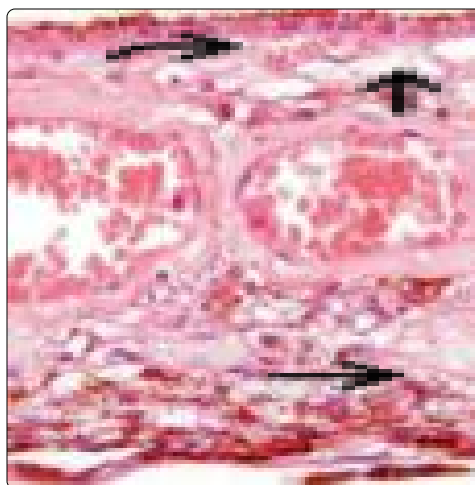


(Left) The amount of stroma present in the pars plicata increases with age. In this section, the stroma is thickened and hyalinized. (Courtesy H. Brown, MD.) (Right) The ciliary muscle (smooth muscle) forms the bulk of the ciliary body. Although histologically difficult to distinguish, 3 layers of muscle (inner circular, middle oblique, and outer longitudinal) are present. Contraction relaxes the zonular fibers and allows the lens to change shape. (Courtesy H. Brown, MD.)

Ciliary Body: Pigment Within Smooth Muscle



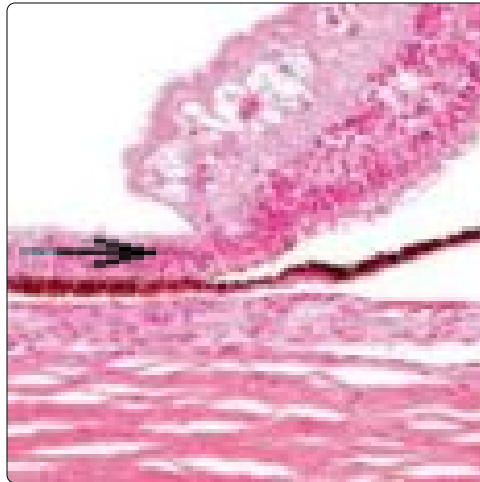
Choroid



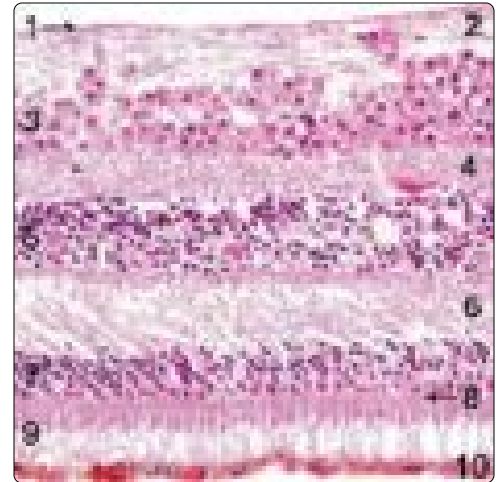
(Left) Interspersed within the smooth muscle of the ciliary body, numerous pigmented melanocytes may be identified. (Courtesy H. Brown, MD.) (Right) The choroid begins posterior to the ciliary body and terminates at the optic nerve, lying between the retina and the sclera. The choroid is composed of vessels with interspersed melanocytes and fibroblasts. Bruch membrane is the basement membrane for both the choriocapillaris and pigmented retinal epithelium. (Courtesy H. Brown, MD.)

Retina: Ora Serrata

(Left) The ora serrata is the anterior termination of the retina. The neurosensory retina is contiguous with the nonpigmented ciliary epithelium. (Courtesy H. Brown, MD.) (Right) The 10 retinal layers are as follows: (1) Internal limiting membrane, (2) nerve fiber layer, (3) ganglion cell layer, (4) inner plexiform layer, (5) inner nuclear layer, (6) outer plexiform layer, (7) outer nuclear layer, (8) external limiting membrane, (9) rods and cones, and (10) retinal pigment epithelium. (Courtesy H. Brown, MD.)

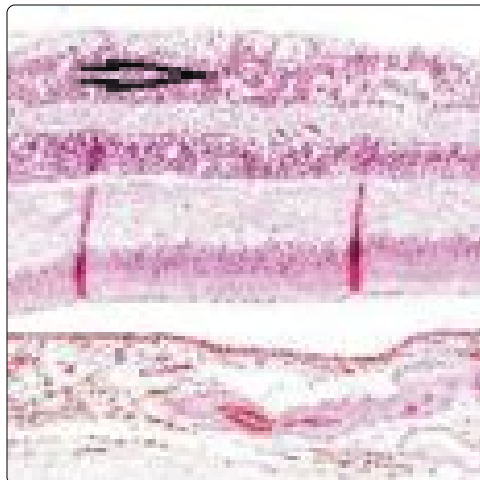


Retina: Multilayered Epithelium

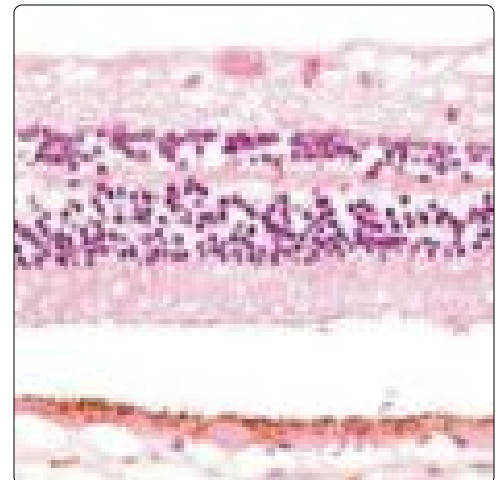


Retina: Macula

(Left) Grossly, the macula lutea appears as a yellow spot within the retina. The macula is a region of the retina that can be identified histologically when the ganglion cell layer is > 1 cell thick, as seen here. (Courtesy H. Brown, MD.) (Right) Away from the macula, the ganglion cell layer thins. This image of the peripheral retina shows only rare ganglion cells. (Courtesy H. Brown, MD.)



Retina: Nonmacular

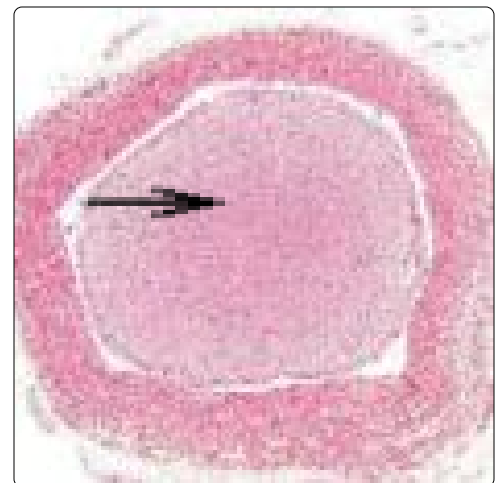


Optic Nerve: Head

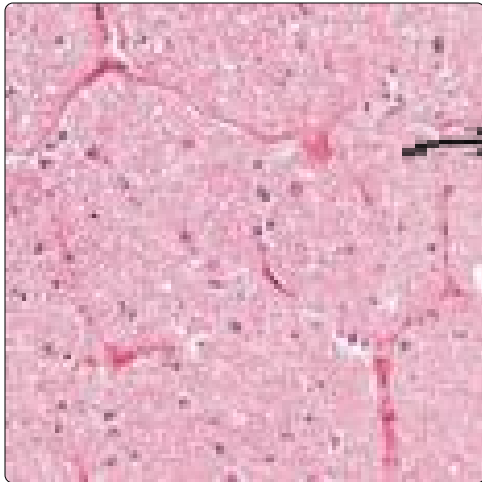
(Left) The optic nerve exits the posterior aspect of the globe. Note the absence of nuclear layers of the retina overlying the most anterior aspect of the optic nerve, leading to the presence of a physiological "blind spot." (Courtesy H. Brown, MD.) (Right) A cross section of the optic nerve shows the nerve centrally, surrounded by a thin pia mater, arachnoid, and the much thicker dura mater. (Courtesy H. Brown, MD.)



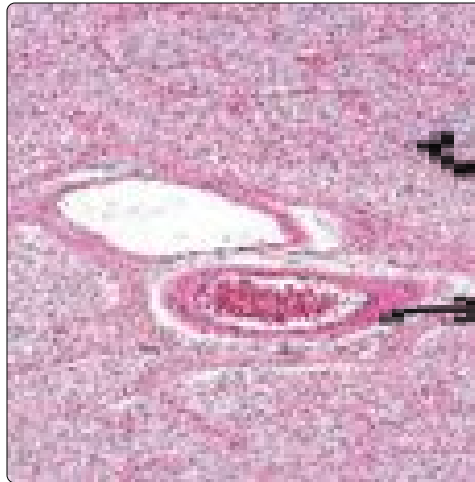
Optic Nerve



Optic Nerve

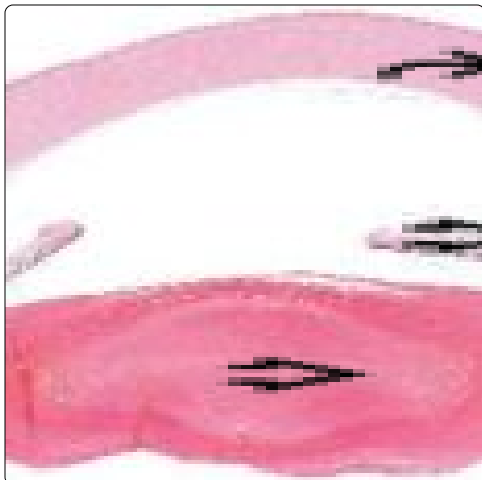


Optic Nerve: Central Artery and Vein



(Left) High-power examination of the central portion of the optic nerve reveals glial cells and nerve bundles divided into fascicles by delicate fibrovascular septa. (Courtesy H. Brown, MD.) **(Right)** This high-power cross section shows the central artery and vein. These structures can be identified in the middle of the optic nerve in its most anterior aspect (extending to the head of the nerve), entering and exiting the nerve 1.0-1.5 cm posterior to the globe. (Courtesy H. Brown, MD.)

Lens

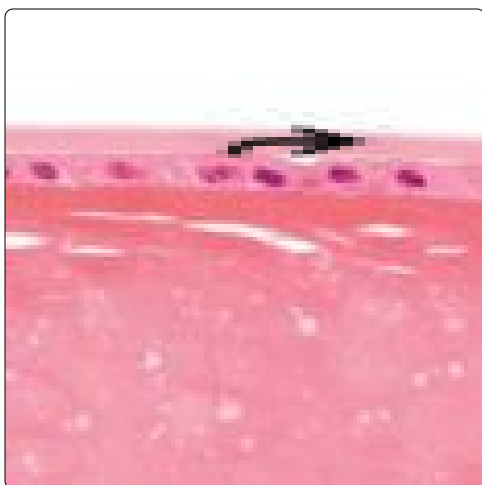


Lens: Equator

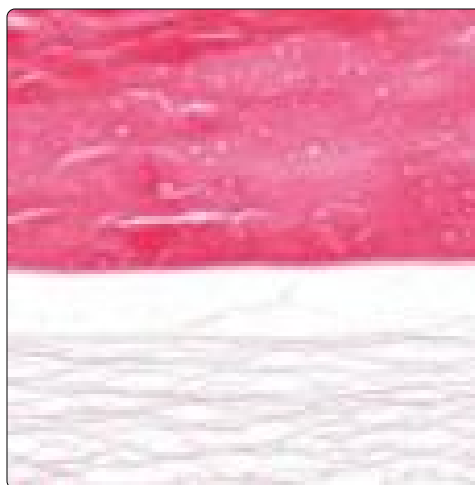


(Left) This low-magnification H&E of the anterior globe illustrates the lens lying posterior to the iris, anterior chamber, and cornea. The pupil consists of the opening within the center of the iris. (Courtesy H. Brown, MD.) **(Right)** The equator of the lens is located peripherally. In this area, the monolayer lens epithelium lining the anterior lens elongates to form lens fibers. Small clefts in the lens stroma are secondary to specimen processing. (Courtesy H. Brown, MD.)

Lens: Anterior



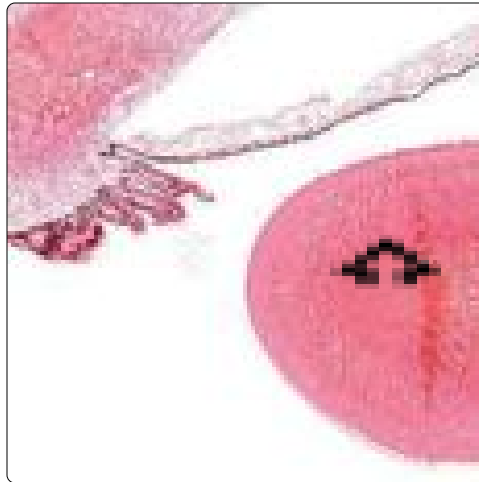
Lens: Posterior



(Left) A thin, carbohydrate-rich lens capsule encircles the lens, and the equator is the site of attachment for the zonular fibers that connect the lens to the ciliary body. The lens capsule is the basement membrane of the cuboidal epithelial cells. (Courtesy H. Brown, MD.) **(Right)** The posterior aspect of the lens lacks an epithelial layer. Since only anterior epithelium is present postnatally, the anterior capsule thickens with age, while the posterior does not. (Courtesy H. Brown, MD.)

Lens and Ciliary Body

(Left) The equator of the lens is in close approximation to the ciliary body and held in place by zonular fibers (invisible at this magnification). The iris can be seen anterior to the lens. (Courtesy H. Brown, MD.) **(Right)** The nearly acellular, transparent vitreous fills the area posterior to the lens, extending circumferentially to the multilayered retinal epithelium. The vitreous is a gel-like substance principally composed of water and hyaluronic acid. (Courtesy H. Brown, MD.)

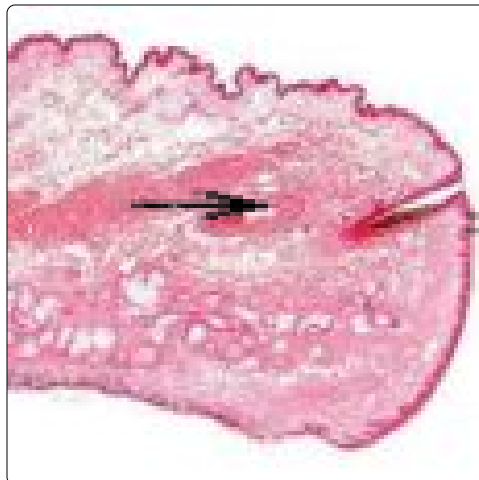


Vitreous



Eyelid

(Left) This section of the eyelid demonstrates the palpebral conjunctiva merging with epidermis of the anterior eyelid. The eyelid contains many structures, including orbicularis muscle, glands, and the cilia (eyelashes). (Courtesy H. Brown, MD.) **(Right)** One of the most abundant types of glands is the sebaceous meibomian glands that are found in the tarsus in the posterior portion of the eyelid. Here, acini of meibomian glands are seen in association with a central duct. (Courtesy H. Brown, MD.)



Eyelid: Meibomian Glands



Eyelid: Glands of Moll and Zeis

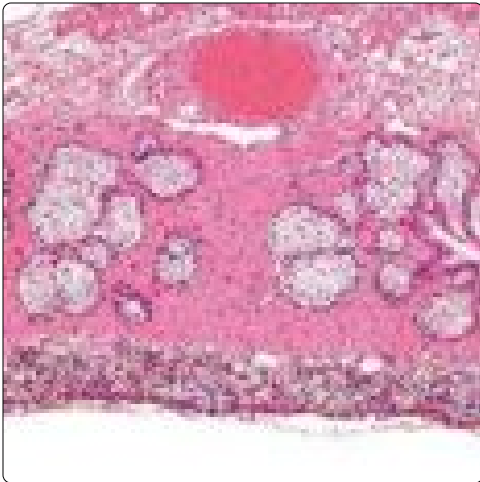
(Left) The glands of Moll are apocrine glands that drain via a duct into the hair follicles that make up the eyelashes. The glands of Zeis are sebaceous glands that may be found in the same area and drain into the hair follicle as well. (Courtesy H. Brown, MD.) **(Right)** Accessory lacrimal glands may be identified within the most superior and inferior portions of the eyelids at the bases of the tarsal plates and in the superior and inferior fornices of the conjunctiva. (Courtesy H. Brown, MD.)



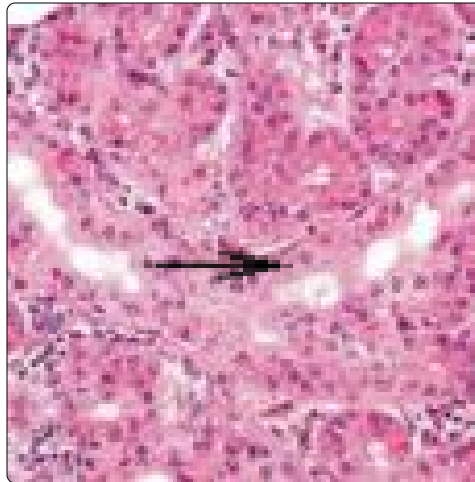
Eyelid: Accessory Lacrimal Glands



Eyelid: Tarsus



Lacrimal Glands and Duct

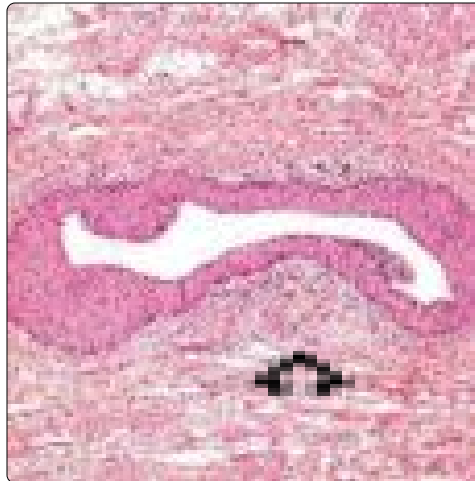


(Left) The tarsus (or tarsal plate) is composed of dense collagen [1] and is present in the posterior aspect of the eyelid. It contains the meibomian glands [2]. (Courtesy H. Brown, MD.)
(Right) The lacrimal glands are composed of serous-type cells [1] that drain into a central duct [2]. The glands are found within the superior lateral orbit. (Courtesy H. Brown, MD.)

Eyelid: Punctum Canaliculus

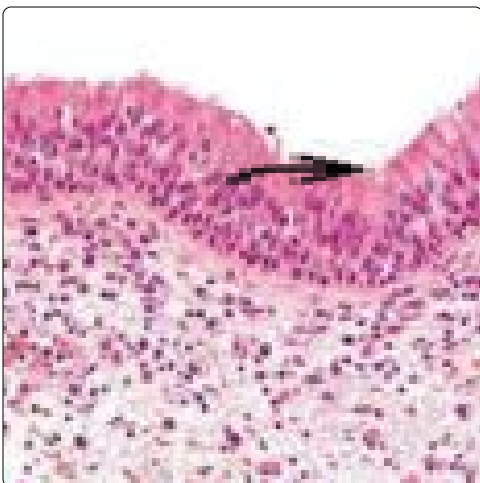


Lacrimal Canaliculus

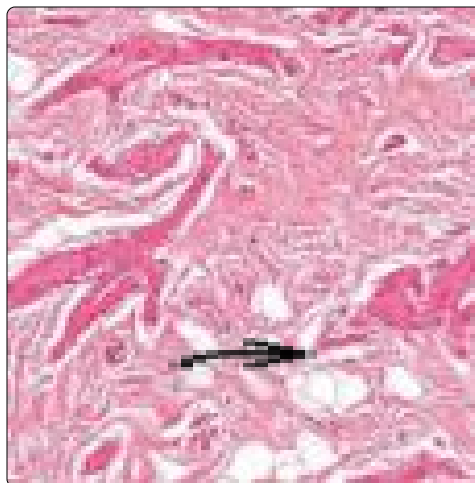


(Left) Tears drain from the lacrimal ducts into the fornices of the conjunctiva and lubricate the anterior globe surface. The tears drain into the punctum canaliculus (opening [1]), which is situated in the medial aspect of each eyelid. (Courtesy H. Brown, MD.)
(Right) Tears drain from the punctum canaliculus into the lacrimal canaliculi [1]. The paired canaliculi (1 superior and 1 inferior) are located in the medial portion of each eye. The lining is composed of stratified nonkeratinizing squamous epithelium. (Courtesy H. Brown, MD.)

Lacrimal Sac



Orbital Soft Tissue



(Left) From the lacrimal canaliculi, the tears drain into the lacrimal sacs, where they collect and drain via the nasolacrimal duct into the sinuses. The lining of the nasolacrimal duct consists of a pseudostratified, ciliated, columnar epithelium [1] with occasional goblet cells [2]. (Courtesy H. Brown, MD.)
(Right) The orbital soft tissue is composed of mature adipose tissue [1] and supporting collagenous tissue [2]. Small blood vessels are also present [3]. (Courtesy H. Brown, MD.)

Oral Mucosae

ORAL MUCOSAE

Macroscopic Anatomy

- Pink, red, or brown in color
- Moist
- Smooth surface
- Lack appendages seen in skin

Microscopic Anatomy

- Epithelium
 - Stratified squamous type
 - Orthokeratinized in areas exposed to high friction, like palate
 - Nonkeratinized in most other areas of oral cavity
 - Parakeratinized as mucosa transitions to skin on lip
 - Rete ridges are present
 - Pigmentation
 - Melanin (endogenous)
 - Foreign (exogenous)

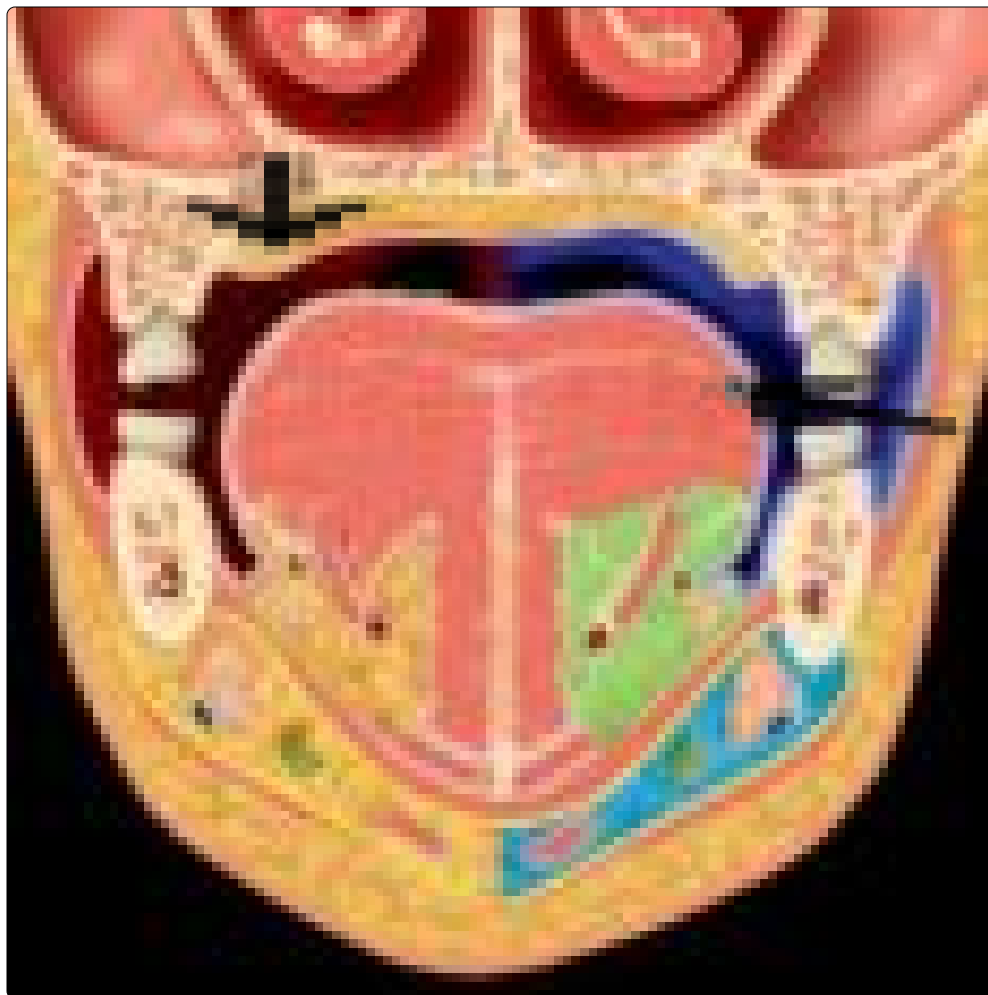
- Nonkeratinocytes: Melanocytes, Langerhans cells, Merkel cells, and lymphocytes
- Lamina propria
 - Composed of dense connective tissue
 - Sebaceous glands (called Fordyce granules) are commonly found in lip and buccal mucosa
 - Inflammatory cells variably present
- Submucosa
 - Dense when overlying bone/periosteum
 - Loose when overlying muscle
 - Contains blood vessels, nerves
 - Minor salivary glands and ducts are present

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

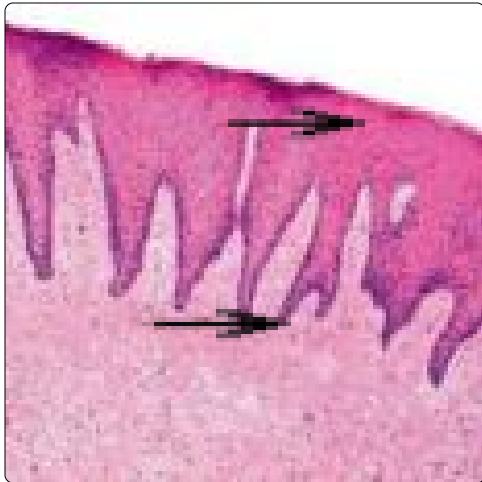
- Epithelium usually becomes thinner and more fragile with aging
- Fordyce granules increase

Oral Cavity Anatomy

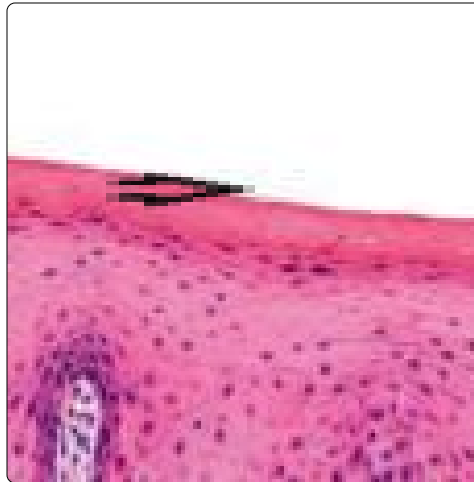


The oral cavity is lined with oral mucosae that are involved in secretion as well as sensations, such as touch, temperature, and taste on the tongue. This protective lining transitions from skin to mucosae at the lips and is continuous with the mucosae of the digestive system posteriorly. Their protective functions are essential, and the mucosae are also a 1st-line defense against microorganisms and infection.

Mucosa



Epithelium

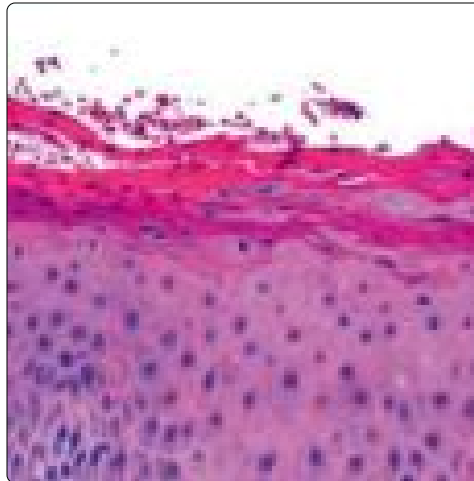


(Left) The oral mucosae are composed of 3 layers. Epithelium [1], with its distinctive rete ridges, is easily discernible from the underlying lamina propria [2]. The distinction between the lamina propria and submucosa is not always as obvious. The submucosa may contain fat, glandular tissue, nerves, and blood vessels. (Right) The keratinization of oral epithelium varies with location and degree of friction. This image shows epithelium from the palate, surfaced by orthokeratin (flat eosinophilic cells with no nuclei [3]).

Epithelium

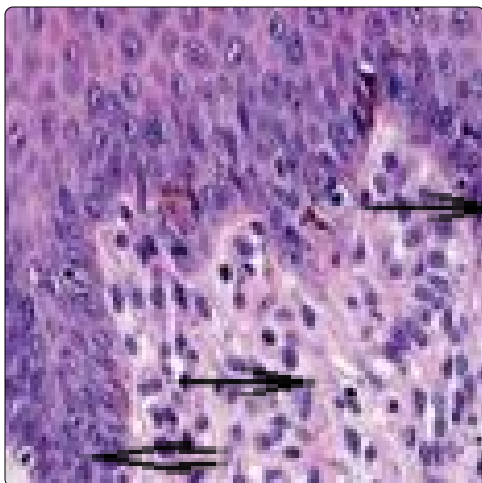


Epithelium



(Left) This image shows an example of hyperorthokeratotic epithelium seen on the tongue. This appearance is the result of exposure to strong mechanical forces. Note the adherent bacterial colonies [4]. (Right) Areas with increased friction or areas like the epithelium near the lips may show areas of parakeratinization, characterized by flat eosinophilic cells with retained pyknotic nuclei [5].

Epithelium



Submucosa



(Left) The color of mucosae varies among individuals, much like the color of skin. This H&E shows deposition of melanin [6] along the basal layer of epithelium. The number of melanocytes [7] found in the epithelium is generally consistent from one person to the next, despite color variation. (Right) This H&E shows blood vessels [8], a relatively large salivary gland duct [9], and a minor salivary gland [10] within the submucosa. Deep to these structures, adipose tissue [11] and muscle [12] are seen.

GINGIVAE**Macroscopic Anatomy**

- Mucosae that cover alveolar ridge, most closely associated with teeth (masticatory mucosae)
- Moist and pink, red, or brown with fine stippling
- May be attached (adherent to associated bone) or free (forms cuffs around neck/cervix of tooth)
- Gingival crevice
 - Space between free gingiva and tooth
 - Extends from tip of free gingiva (papilla) to cemento-enamel junction of tooth

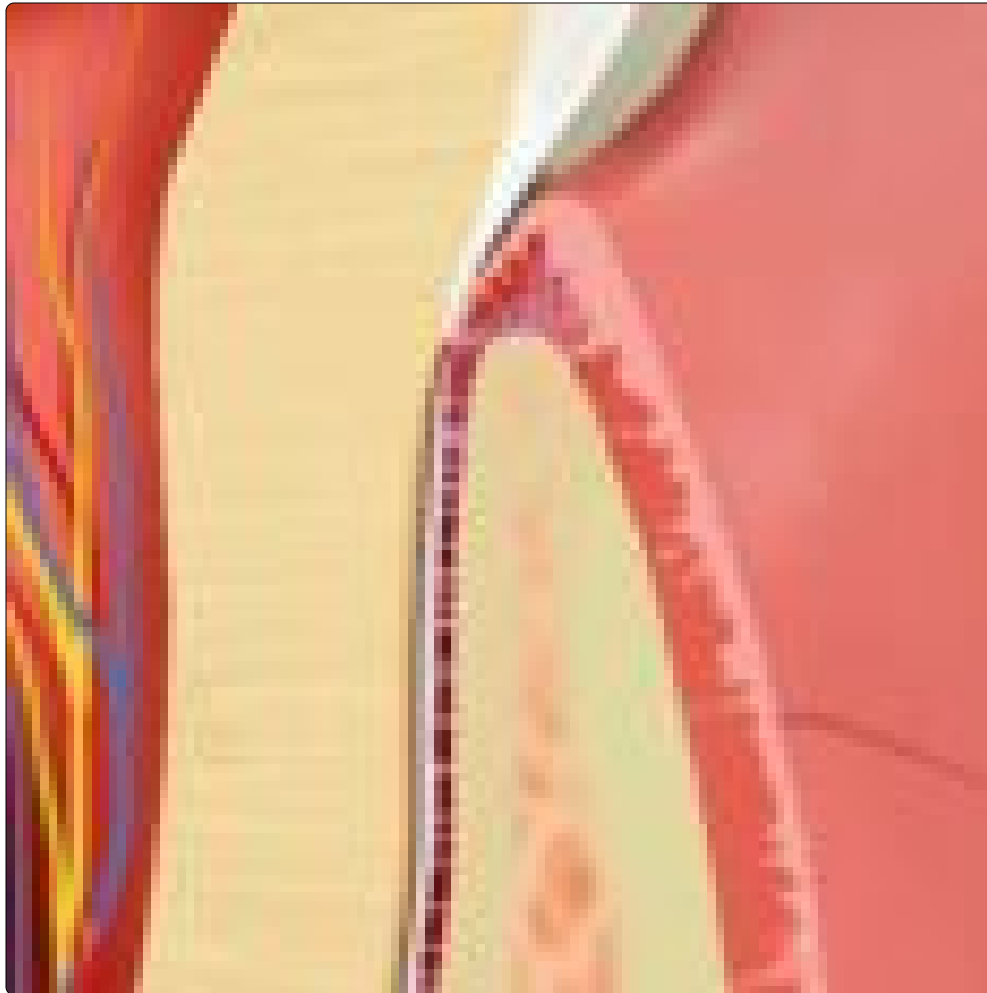
Microscopic Anatomy

- Epithelium
 - Stratified squamous type
 - Parakeratinization or orthokeratinization may be present
 - Epithelium thins in gingival crevice
 - Rete ridges are long and sometimes narrow
 - Melanin pigmentation varies with population

- Nonkeratinocytes: Melanocytes, Langerhans cells, Merkel cells, and lymphocytes
- Lamina propria
 - Composed of dense connective tissue collagen fibers
 - Sebaceous glands (Fordyce granules) are seen
 - Less frequent than in other oral mucosal sites
 - Inflammatory cells variably present
 - Capillaries and small nerve twigs
 - Odontogenic epithelial rests
- Mucoperiosteum
 - Provides firm and direct attachment to underlying periosteum of bone of maxilla and mandible
 - No submucosal layer in gingiva

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

- Epithelium becomes thinner and more fragile with age

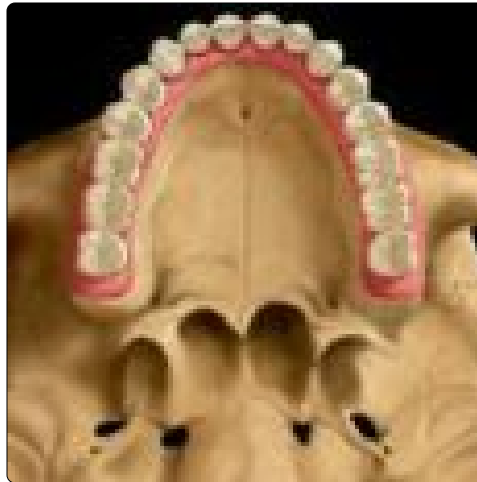
Gingival Anatomy

This graphic shows the relationship of the gingivae to the tooth and the alveolar bone. The gingiva intimately associated with the bone is referred to as the attached gingiva, while the gingiva that forms a cuff around the tooth is designated as the free gingiva. The space, or potential space, that is created from the tip of the free gingiva to the attachment of the gingiva to the tooth, at the cemento-enamel junction, is called the gingival crevice. This space is an excellent environment for pathogens, and the presence of an inflammatory infiltrate is not uncommon. This space is also the area generally targeted during dental cleanings.

Gingivae: Gross



Gingivae: Gross

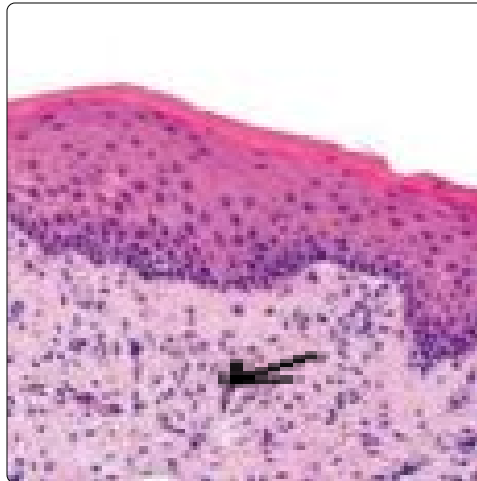


(Left) The attached gingiva provides a protective layer for the underlying bone (mandible in this image). Macroscopically, this tissue is firm and immobile with a pink stippled surface. (Right) The forces involved in biting and chewing food expose the gingivae to mechanical forces, such as compression and shearing. Additionally, the tissues are exposed to numerous microorganisms found in the normal oral flora. The gingivae help provide a barrier to infection.

Gingivae: Epithelium

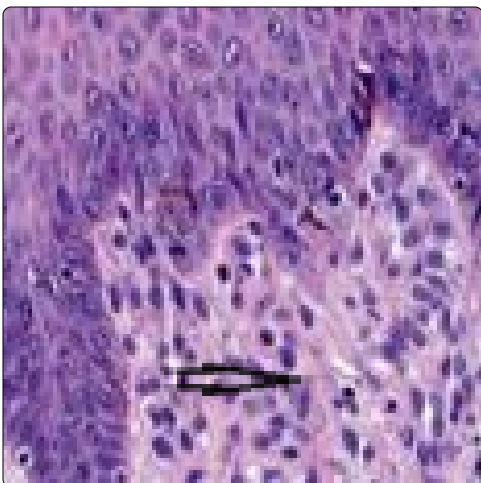


Gingivae: Epithelium

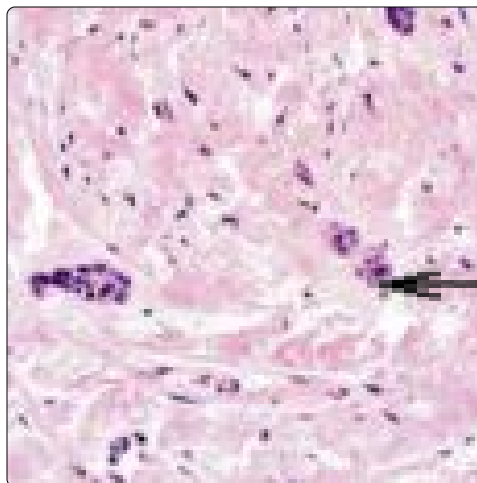


(Left) H&E-stained tissue section shows gingiva. The epithelium in this particular location has a parakeratinized surface. The rete ridges are long and, in some areas, narrow. (Right) This section of gingiva shows fewer rete ridges and has an orthokeratinized surface. The underlying lamina propria contains small capillary loops. Sparse inflammatory cells and nerve twigs are sometimes seen in the lamina propria.

Melanin Pigment



Odontogenic Epithelial Rests



(Left) The pigmentation of gingivae within a population may vary as much as skin color; however, there tends to be a relationship between the amount of pigmentation seen in the skin and the gingivae. The melanin pigmentation seen here is produced by melanocytes. Melanocytes lack desmosomes and may have visible dendritic processes that extend between the cells of the epithelium. (Right) Odontogenic epithelial rests can be seen, especially in the posterior areas of gingiva.

TEETH**Macroscopic Anatomy**

- Crown
 - Part of tooth easily seen clinically
 - Surfaced by hard, brittle, translucent enamel supported by underlying dentin
 - Pulp forms central chamber of tooth that supplies nutrition and contains nerve supply
- Root
 - Anchors tooth to alveolar bone
 - Root number varies depending on tooth
 - Apical foramen is opening through which blood and nerve supply enter and exit tooth

Microscopic Anatomy

- Enamel
 - Acellular, highly mineralized, 96% inorganic material made up of hydroxyapatite crystals
 - Crystals are aligned to create rods
 - Immature enamel has basophilic, fish scale appearance on H&E sections
- Dentin
 - 70% mineralized tissue
 - Eosinophilic, closely packed tubules
 - Odontoblasts are responsible for production of dentin, and their processes occupy tubules
 - Reparative dentin deposited at site of injury shows distorted tubular pattern
- Pulp
 - Loose connective tissue that resembles primitive mesenchyme
 - Odontoblasts line pulp chamber
 - Stellate fibroblasts
 - Small blood vessels
- Periodontal ligament
 - Thin fibrous attachment between root and surrounding bone
 - Permits slight movement of teeth
 - Seldom preserved or seen on routine processing

- Cementum
 - Mineralized organic material that resembles bone
 - Cementocytes are responsible for production of cementum and occupy its lacunae
 - Location where enamel and cementum meet (or sometimes overlap) is referred to as cements enamel junction

CONSIDERATIONS AND VARIATIONS**Pitfalls and Artifacts**

- Teeth need to be decalcified for routine H&E slides
 - Mature enamel is lost to processing
- Often, teeth that are submitted are surgically sectioned, losing normal anatomy/histology
- Dental restorations may alter or replace normal histology

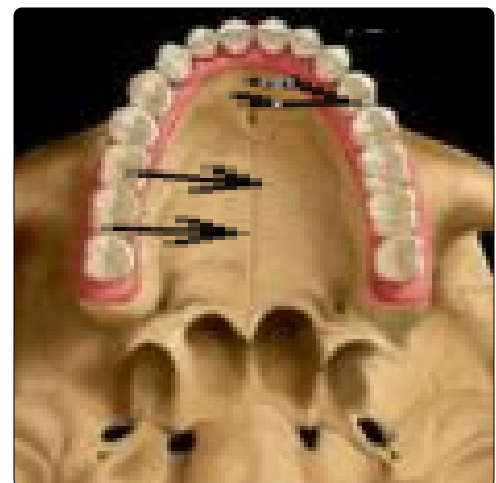
Age-Related Variations

- Primary (deciduous)
 - 20 teeth: Smaller in size, erupt between 6-30 months
 - 8 incisors, 4 canines, 8 molars
- Secondary (permanent)
 - 32 teeth: Larger in size, erupt starting at ~ 6 years of age
 - 8 incisors, 4 canines, 8 premolars, 12 molars
- Changes found in both primary and secondary teeth
 - Hyperdontia (supernumerary teeth)
 - Hypodontia
 - Dentin continues to be laid down with age, reducing or obliterating size of dentin tubules and shrinking pulp chamber
 - Attrition: Loss of tooth structure due to tooth-to-tooth contact
 - Abrasion: Loss of tooth structure due to tooth-to-nontooth contact (i.e., contact with tooth brush)
 - Erosion: Loss of tooth structure due to chemical process
 - Caries: Loss of tooth structure due to bacterial decay
 - Teeth generally darken with age
 - Teeth may be lost due to disease or trauma

Teeth: Mandible

(Left) This graphic shows the mandibular permanent teeth: Incisors [light blue], canines [dark blue], premolars [green], and molars [yellow]. The teeth occupy ~ 1/5 of the surface area of the mouth.

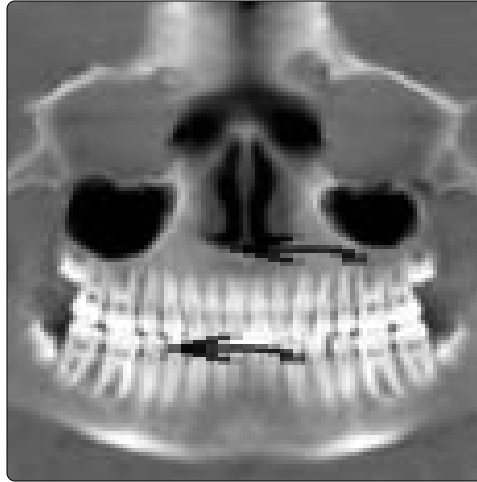
(Right) This graphic shows the maxillary permanent teeth: Incisors [light blue], canines [dark blue], premolars [green], and molars [yellow]. The maxillary and mandibular teeth work together for proper eating and speech.

Teeth: Maxilla

Normal Full Dentition: Child

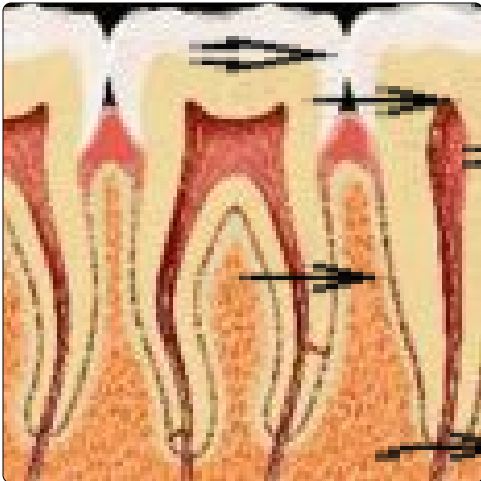


Normal Full Dentition: Young Adult

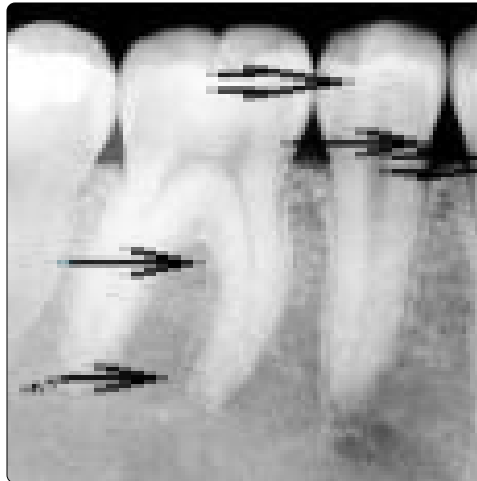


(Left) This radiographic image shows the normal full dentition of a child, likely around 5 years of age. All primary teeth are erupted, and the developing permanent teeth can be seen within the jaws. (Right) This radiographic image shows the normal full dentition of a young adult. Note that the 3rd molars are not yet erupted. The eruption of the 1st permanent teeth begins around age 6 with the 1st molars and generally ends in the early 20s with the eruption of the 3rd molars.

Tooth: Major Components



Tooth: Radiographic Image

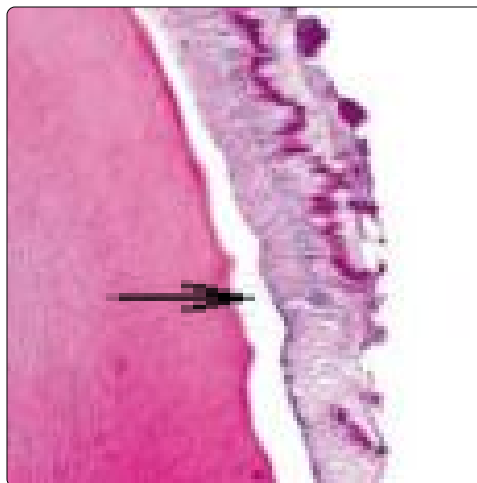


(Left) This graphic shows the major components of a tooth. Enamel surfaces the crown and is supported by underlying dentin. The pulp is found centrally and communicates with surrounding bone via the foramen at the apex. The periodontal ligament (PDL) is in close association with the root. (Right) In this radiographic image, the components of the tooth are readily seen: Enamel, dentin, pulp, apex of the tooth, and PDL.

Cusp of Canine Tooth



Enamel Matrix



(Left) This H&E shows the cusp of a canine tooth. Due to the demineralization process required for routine H&E preparations, no enamel is present. Enamel is an inorganic material made up primarily of hydroxyapatite crystallites. (Right) Enamel matrix, or immature enamel, is less mineralized and can be seen on a routine H&E. Mature teeth do not have enamel matrix; however, this specimen is from an odontoma. Note the underlying dentin.

Root of Tooth: Apex

(Left) H&E shows the root of the tooth near the apex. The apex of the root is the area where the main foramen is seen; however, smaller accessory foramina may be found anywhere on the root surface. (Right) H&E shows a cross section of the pulp chamber as it narrows toward the foramen. The foramen allows for entry and exit of vessels and nerves. Infection of the pulp may create swelling, resulting in the disruption of the blood supply and tooth death.

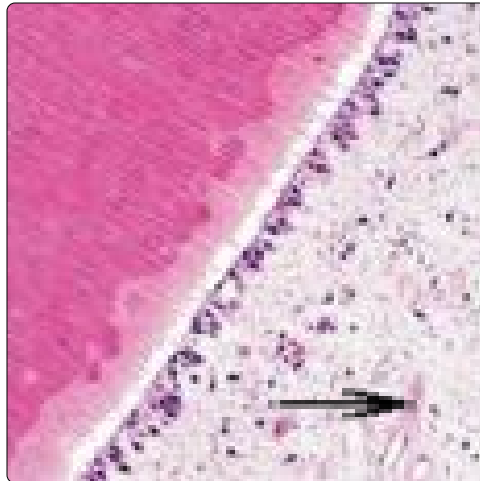


Pulp Chamber

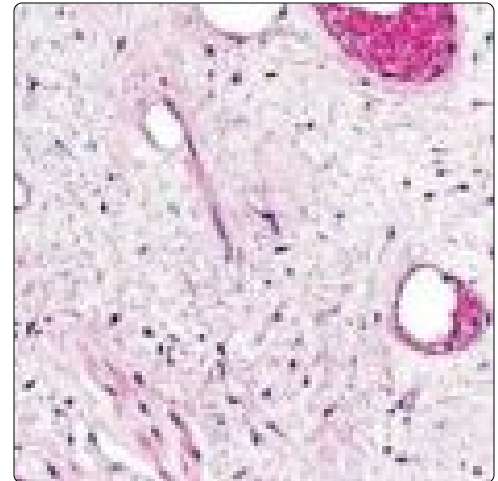


Dentin-Pulp Complex

(Left) High-power H&E shows the dentin-pulp complex. The odontoblasts form a single layer along the periphery of the pulp and have processes that extend into the tubules. It is the presence of these cells and their processes that makes dentin a sensitive tissue, unlike enamel. (Right) High-power view of the pulp shows its loose matrix, which is made up of fibroblasts, collagen, and ground substance. Note the presence of small vessels with red blood cells.

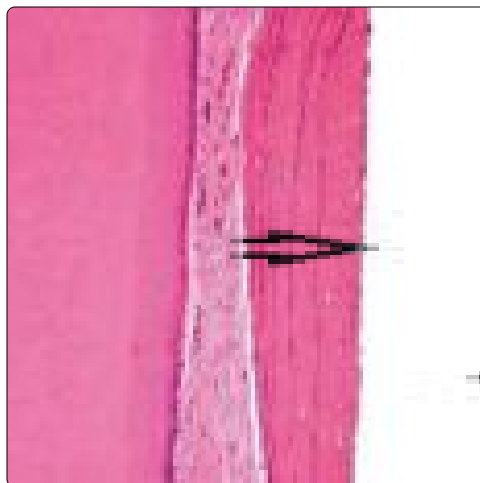


Pulp



Periodontal Ligament

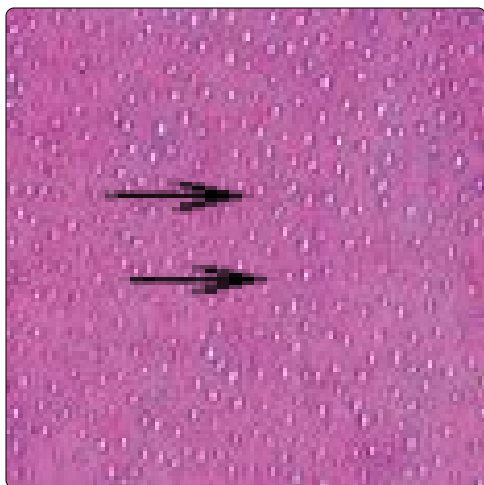
(Left) The PDL provides attachment between the tooth root and the surrounding alveolar bone. The PDL is made up of fibrous connective tissue that allows for slight movement during normal eating and biting. It is also the PDL that allows for the correction of malocclusion by orthodontic movements. (Right) High-power H&E shows the relationship of the tooth to the PDL. Note the transition of dentin into cementum.



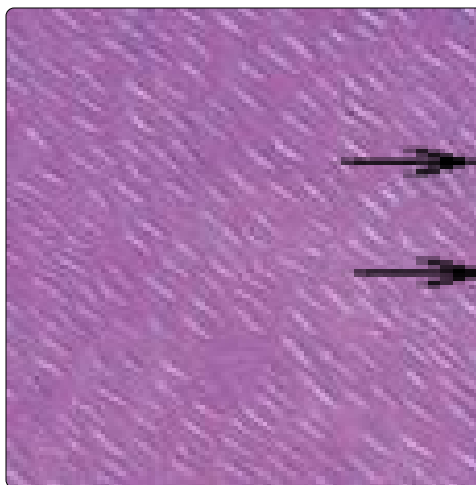
Periodontal Ligament



Dentin



Dentin

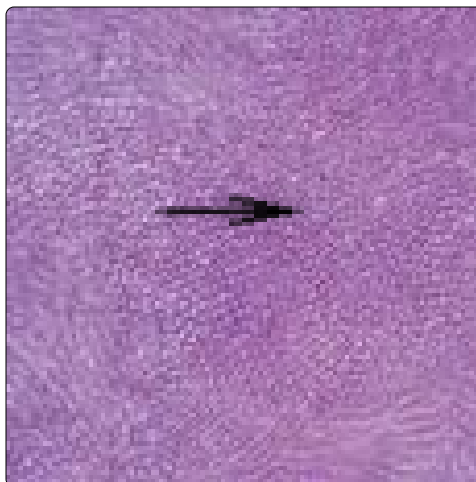


(Left) High-power cross section of demineralized dentin shows characteristic small tubules. (Right) High-power longitudinal section of dentin shows the slight curvature of the tubules. These tubules are filled with fluid and odontoblast processes. The tubules, however, make the dentin vulnerable to invasion of bacteria, which may result in dental caries. Severe infection may result in the infection of the pulp in addition to decay of tooth structure.

Reparative Dentin



Enamel Matrix

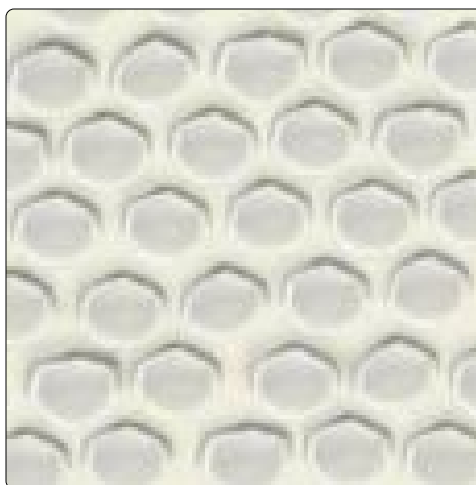


(Left) Insult to the tooth due to caries or mechanical trauma results in the deposition of reparative dentin, seen here adjacent to normal dentin. Reparative dentin has fewer and more irregular tubules; this may be due to the fact that it is likely rapidly laid down in an attempt to protect the pulp. (Right) High-power view shows a classic basophilic, fish scale appearance of enamel matrix. Mature enamel, unlike matrix, is destroyed by routine tissue processing.

Enamel Matrix



SEM Image of Enamel



(Left) H&E illustrates enamel matrix. Mature enamel is 96% inorganic and acellular. Unlike dentin, it is nonvital and not sensitive when affected by caries or trauma. Enamel cannot be regenerated or replaced and is, unfortunately, vulnerable to demineralization by acids in the environment because it is so highly mineralized. (Right) This graphic depicts a scanning electron microscope (SEM) image of enamel. SEM images are primarily used for research purposes.

TONGUE

Macroscopic Anatomy

- Anterior 2/3 (mobile tongue) and posterior 1/3 (fixed tongue) separated by sulcus terminalis
- Mobile tongue has dorsal and ventral surfaces
 - Dorsal surface covered by various types of papillae
 - Ventral surface smooth, lacks papillae

Microscopic Anatomy

- Epithelium is modified stratified squamous mucosa with various types of papillae
 - Ventral surface lined by thin, nonkeratinizing, stratified squamous epithelium with blunt rete pegs
- Filiform papillae (most numerous)
 - Line majority of dorsal tongue
 - 2-3 mm long, conical-shaped, and curved slightly posteriorly
 - Arranged in vague rows parallel to sulcus terminalis
 - Heavily keratinized and often have oral flora bacterial colonization
- Fungiform papillae
 - Scattered throughout tongue
 - 0.5-1.0 mm wide, dome-shaped, rise higher than filiform papillae
 - Thin, nonkeratinized epithelium with underlying vascular stroma (appear red-pink in situ)
 - May contain rare taste buds
- Foliate papillae
 - Only found in lateral tongue posteriorly
 - Parallel ridges lined by nonkeratinized stratified squamous mucosa
 - Rudimentary in humans with only rare taste buds
- Circumvallate papillae
 - Arranged in V-shaped (6-14 papillae) orientation anterior to sulcus terminalis
 - 2-3 mm wide, dome-shaped, surrounded by small circular furrow where most taste buds reside
 - Serosus (von Ebner) glands secrete into furrow base and serve to rinse out furrow contents

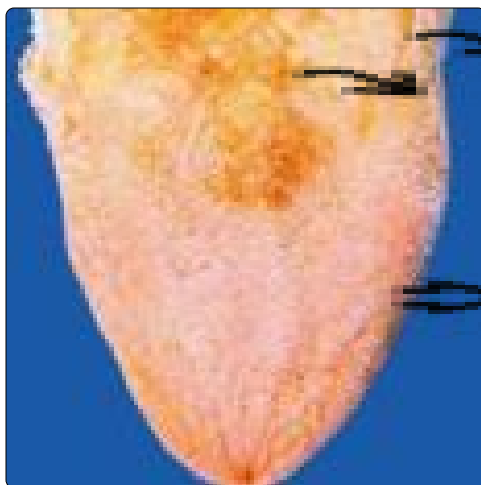
- Peripheral to furrow and papilla is circular mucosal elevation (vallum) that may have taste buds
- Taste buds
 - Sensory taste receptors that communicate with surface via gustatory pore
 - Oval-shaped, comprised of 3 cell types: Gustatory (taste), supporting (sustentacular), and basal (regenerative)
 - Gustatory (taste) cells are crescent-shaped, simple epithelial cells with pale cytoplasm innervated by nonmyelinated nerves
 - Nonmyelinated nerve fibers communicate with myelinated fibers in papillae connective tissue core, forming subgemmal nerve plexus
 - Besides papillae, taste buds can be seen in glossopalatine arch, soft palate, lingual epiglottis, and posterior pharynx
- Majority of tongue parenchyma comprised of interlacing striated muscle (intrinsic) bundles arranged horizontally, vertically, and longitudinally
 - Varying amounts of mature adipose tissue located between skeletal muscle bundles
 - Extrinsic muscles also present; have origin outside of tongue proper
 - Richly vascular with numerous anastomoses and rich nerve supply
- Small amount of loose connective tissue located between epithelial surface and underlying muscle
- Mucous minor salivary gland lobules present in muscle in posterior portion of mobile tongue
- Serosus minor salivary glands (von Ebner glands) present only underlying circumvallate papillae

CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

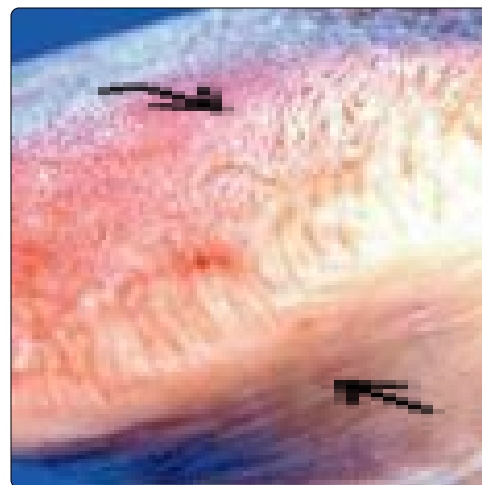
- Salivary gland tissue may be intimately associated with nerves, mimicking perineural invasion
- Neural tissue in connective tissue cores of papillae may be confused with neuromas
- Ectopic tonsil tissue can mimic neoplasms or inflammatory processes

Dorsal View of Tongue

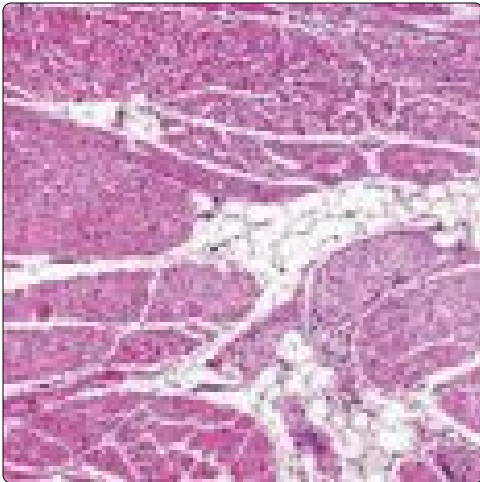


(Left) The posterior border of the mobile tongue consists of a V-shaped row of circumvallate papillae. The dorsal tongue has numerous filiform papillae and scattered fungiform papillae. (Right) The linear foliate papillae are found on the posterior lateral tongue. Note the numerous filiform papillae on the dorsal tongue. The ventral tongue is smooth, lacking papillae.

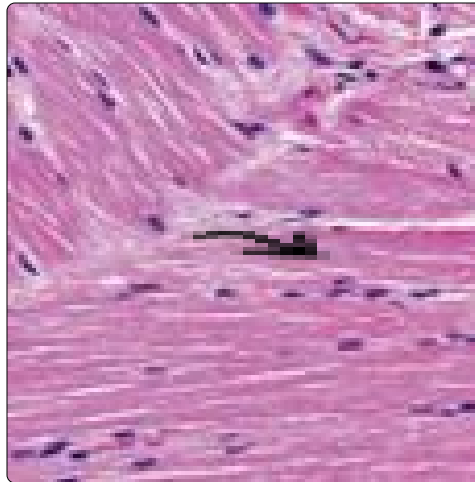
Lateral View of Tongue



Tongue Parenchyma

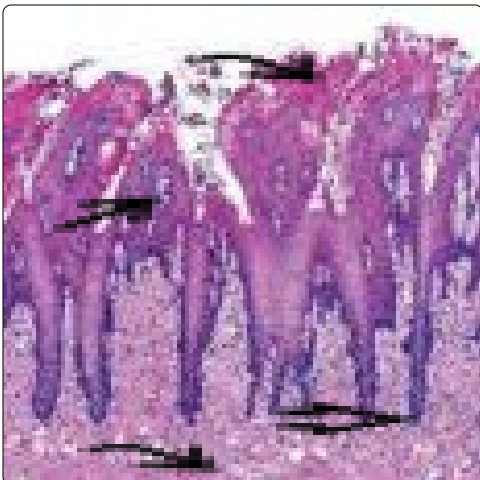


Tongue Muscle

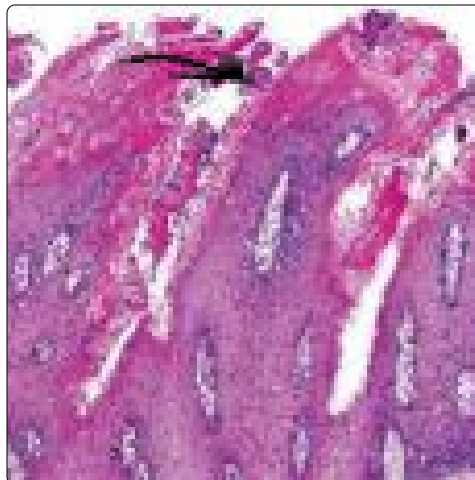


(Left) The bulk of the tongue parenchyma consists of muscle bundles arranged in a 3D configuration to allow for maximal mobility and functionality. Varying amounts of mature adipose tissue [1] are associated with the muscle bundles. The tongue is richly innervated [2] and contains many blood vessels [3], replete with vascular anastomoses. (Right) Interlacing striated muscle bundles [4] allow for such essential functions as mastication, phonation, swallowing, etc.

Filiform Papillae

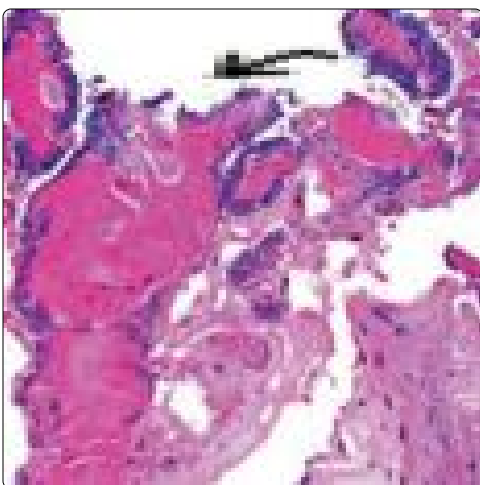


Keratinization of Filiform Papillae



(Left) The dorsal tongue surface is carpeted with conical-shaped, 2-3 mm long, filiform papillae [1]. In contrast to other papillae, the filiform papillae are heavily keratinized [2], especially on the tips. Throughout the tongue, only a small amount of loose connective tissue [3] separates the mucosa from the underlying striated tongue muscle [4]. (Right) The tips of the filiform papillae are heavily keratinized [5]. The filiform papillae generally lack taste buds.

Bacterial Colonization of Filiform Papilla



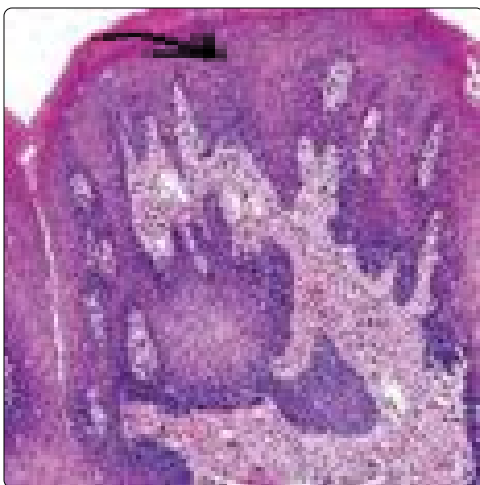
Fungiform Papilla



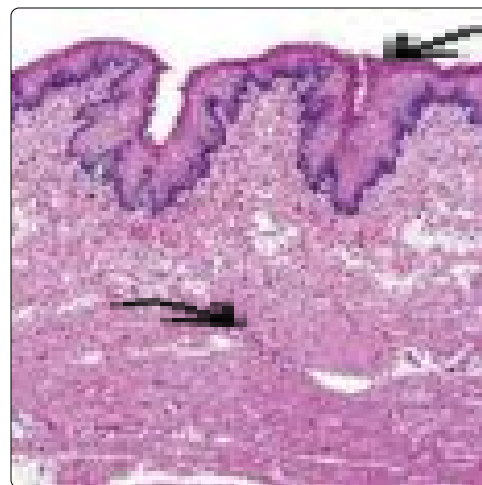
(Left) Bacterial colonization by oral flora bacteria [1] can be seen to varying degrees in the keratin of filiform papillae. With extensive bacterial colonization, the tongue can appear brown-black macroscopically. (Right) Fungiform papillae [2] are the 2nd most numerous papillae and are irregularly distributed throughout the tongue between the numerous filiform papillae [3]. These are 0.5-1.0 mm wide, dome-shaped, and slightly taller than the filiform papillae.

Fungiform Papilla

(Left) Fungiform papillae have a thin epithelial surface with limited parakeratinization and underlying stroma rich in small blood vessels making them look red or pink grossly. The connective tissue stroma has small nerves that innervate the rare taste buds found in some fungiform papillae. (Right) Unlike in some mammals, foliate papillae are rudimentary in humans. Note the small amount of loose connective tissue separating the mucosa from the striated muscle.



Foliate Papilla

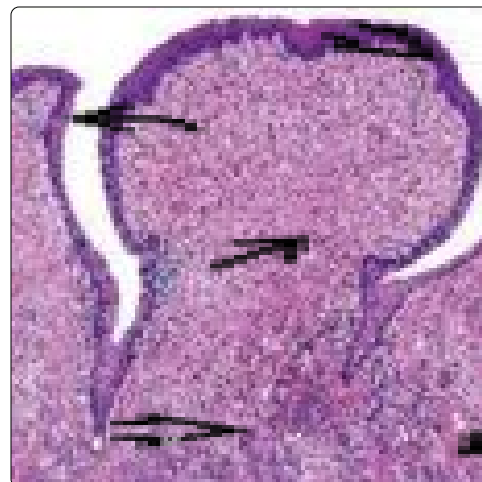


Foliate Papilla

(Left) Foliate papillae can have rare taste buds and are lined by a thin stratified squamous epithelium with minimal parakeratinization. The connective tissue stroma contains many small blood vessels and nerves. (Right) The dome-shaped circumvallate papillae are surrounded by a circumferential furrow and a circular mucosal ridge (vallum) lateral to that. von Ebner serous glands empty into the base of the furrow and are thought to flush out its contents.

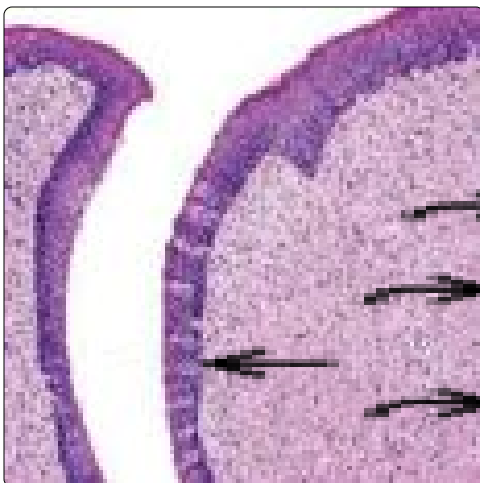


Circumvallate Papilla



Circumvallate Papilla

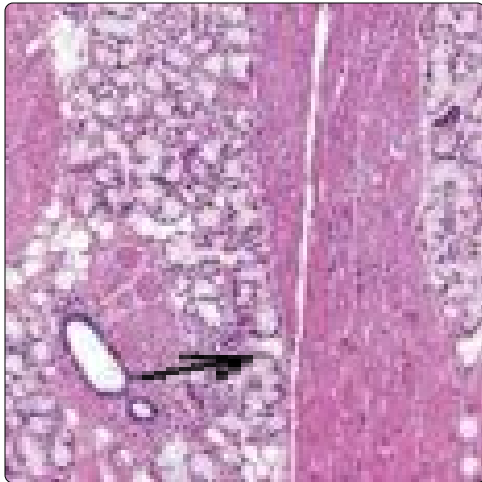
(Left) Circumvallate papillae have a connective tissue core rich in haphazardly arranged small nerves that innervate the numerous intraepithelial taste buds. Taste buds can also be seen in the vallum. Papillae nerve tissue can be mistaken for neuromas. (Right) Taste buds are oval, clear-staining structures comprised of crescent-shaped taste, supporting, and basal cells. A taste pore transmits tastes via receptors on microvilli to the underlying neurites.



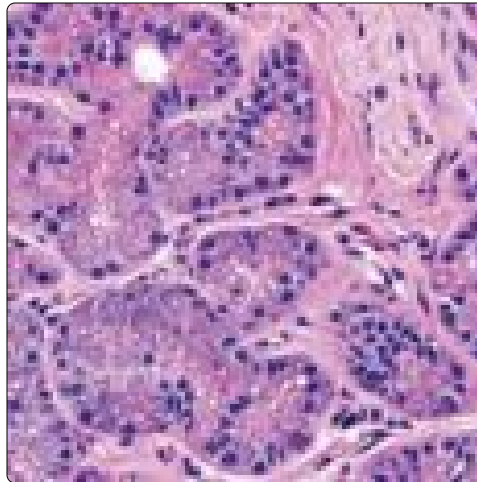
Taste Buds



Mucous Glands

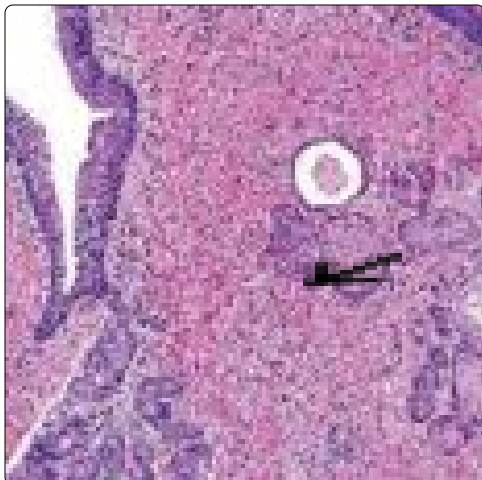


von Ebner Glands

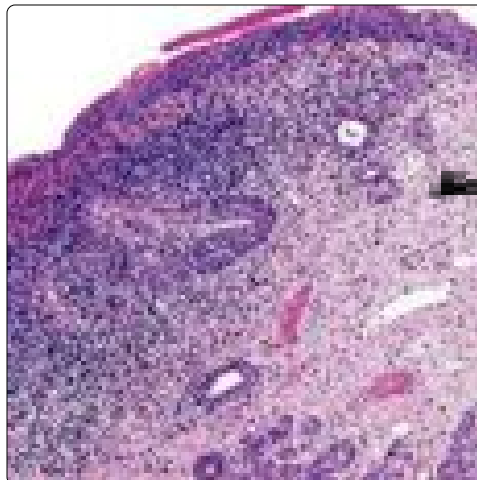


(Left) Salivary gland tissue is present in the posterior portion of the tongue, and these consist mostly of mucous gland lobules embedded in striated muscle. The lobules have a central salivary duct [1] that drains to the ventral tongue. (Right) von Ebner glands are serous glands located only below the circumvallate papillae. Their ducts drain into the furrows to flush out its contents. Both serous and mucous glands in the tongue are intimately associated with nerves [2].

Squamous Metaplasia

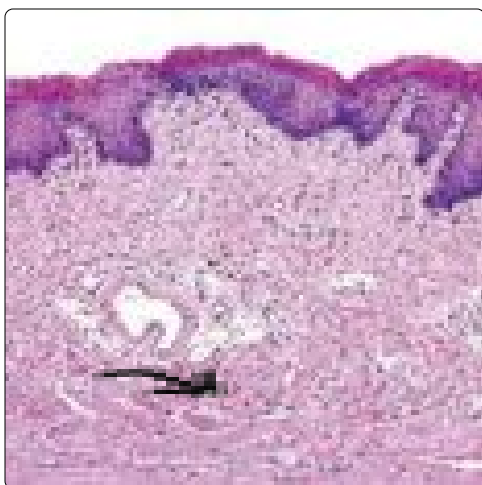


Ectopic Tonsil

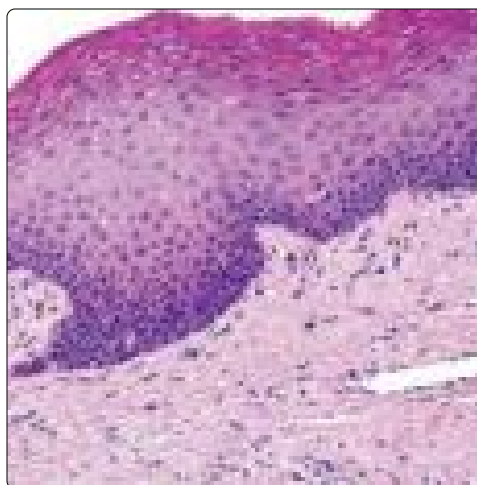


(Left) Serous von Ebner glands [3] drain into the furrow [4] separating the circumvallate papillae from the adjacent furrow. This is thought to flush out the furrow contents in order to allow for new tastes to be better discriminated. Squamous metaplasia of the salivary gland ducts [5] can mimic squamous cell carcinoma on biopsy or at frozen section. (Right) Ectopic tonsil tissue [6] can be seen anterior to the sulcus terminalis and can mimic neoplasms or lesions clinically.

Ventral Tongue



Ventral Tongue



(Left) The ventral tongue lacks papillae, giving it a smooth, glistening appearance macroscopically. Note the small amount of loose connective tissue [7] separating the mucosa from the tongue muscle [8]. (Right) The ventral tongue is lined by a thin, nonkeratinizing, stratified squamous mucosa that lacks the papillae seen throughout the dorsal tongue.

TONSILS/ADENOIDS**Macroscopic Anatomy**

- Lymphoid aggregates covered by epithelium that form majority of Waldeyer ring
 - 3 major tonsils: Palatine, lingual, and pharyngeal
- Palatine tonsil (a.k.a. faucial tonsil, tonsil)
 - Paired tonsils located in lateral oropharynx between palatoglossal (anterior pillar) and palatopharyngeal (posterior pillar) arches
 - Surface is smooth with 15-20 invaginations (crypts) into tonsil parenchyma
 - Yellow (sulfur) granules often seen within crypts and may be calcified (tonsilloliths)
- Lingual tonsil (a.k.a. base of tongue)
 - Covers surface of posterior 1/3 of tongue (fixed tongue) between sulcus terminalis and valleculae
 - Separated into halves by glossoepiglottic ligament
 - Consists of mucosa-covered nodules with surface invaginations into tonsil parenchyma (crypts)
- Pharyngeal tonsil (a.k.a. tonsil of Luschka, adenoid)
 - Pyramid-shaped, mucosa-covered nodular tissue located in superior midline of nasopharynx
 - Unlike other tonsils, do not have crypts but have elongated folds of surface mucosa
 - When enlarged clinically, referred to as adenoids

Microscopic Anatomy

- Palatine tonsils
 - Surface is nonkeratinizing stratified squamous epithelium
 - Crypts represent extension of surface mucosa into underlying lymphoid stroma; most are simple, but may be branched, and they extend to connective tissue pseudocapsule
 - Crypt epithelium contains intraepithelial lymphocytes (both B and T cells) and plasma cells, resulting in distortion of normal architecture with reticulated appearance (lymphoepithelium)
 - Most crypt epithelial cells have basaloid appearance with layer of differentiated keratinocytes at surface

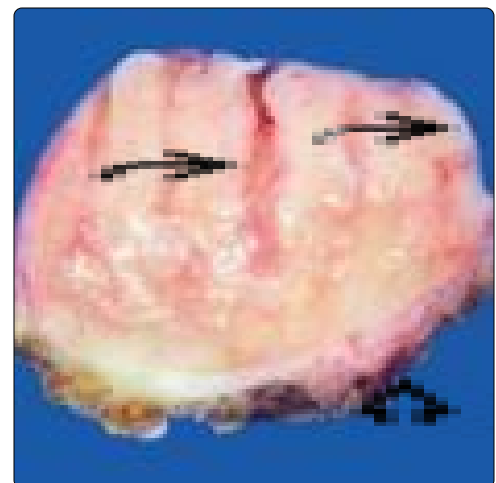
- Sulfur granules (aggregates of *Actinomyces* and other nonpathogenic oral flora) common in crypt lumina
- Deep crypts often contain desquamated keratin debris mixed with oral bacteria and inflammatory cells
- Lymphoid tissue is composed of B-cell-rich follicles, many with germinal centers and T-cell-rich interfollicular zones
- Reactive follicles polarize toward antigenic exposure similar to lymph nodes; unlike lymph nodes, there is no capsule, trabeculae, or sinuses
- Small mucous salivary glands are present in connective tissue peripheral to tonsil; ducts typically drain to surface
- Small islands of ectopic cartilage can be present in adjacent connective tissue
- Lingual tonsils
 - Histologically very similar to palatine tonsils
 - Crypts are shorter than those in palatine tonsils
 - Sulfur granules less common than in palatine tonsils
 - Mucous minor salivary glands present, underlying lymphoid tissue
- Pharyngeal tonsil
 - No crypts but elongated mucosal folds containing lymphoid tissue
 - Surface mucosa is respiratory-type epithelium with variably present foci of squamous epithelium
 - Deep portions of epithelial folds contain intraepithelial lymphocytes (lymphoepithelium) similar to tonsillar crypts
 - Lymphoid component similar to other tonsils
 - Seromucous minor salivary glands present adjacent to tonsillar tissue; not usually sampled

CONSIDERATIONS AND VARIATIONS**Age-Related Variation**

- Tonsils largest in childhood, progressively atrophy following puberty

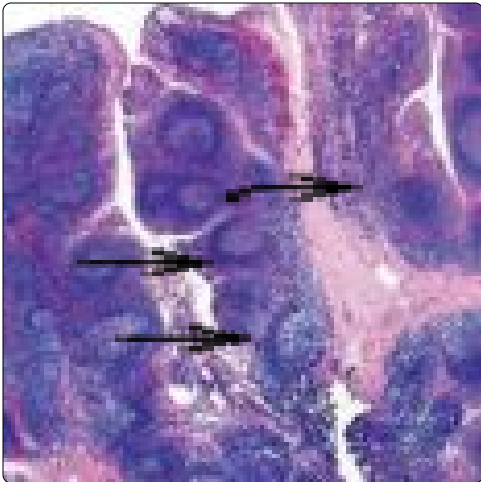
Pitfalls and Artifacts

- Deep crypts may be mistaken for invasive squamous cell carcinoma, especially on frozen section

Tonsil Mucosal Surface**Tonsil Cross Section**

(Left) The mucosal surface of the palatine tonsil is mainly smooth with ~ 15-20 surface invaginations. Tonsils are largest in childhood and gradually atrophy by adulthood. **(Right)** The cut surface shows that the surface invaginations correspond to crypts that extend deeply into the lymphoid stroma close to the underlying compressed connective tissue pseudocapsule.

Palatine Tonsil Crypts

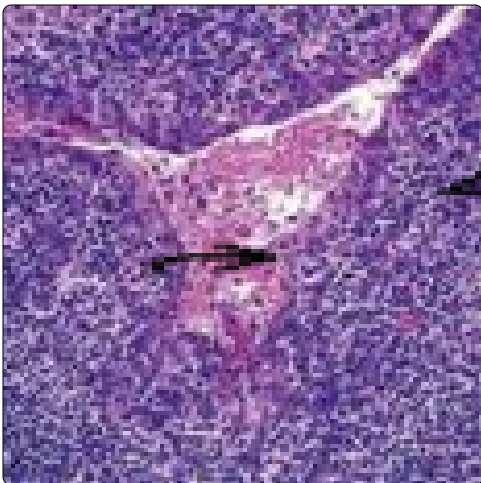


Palatine Tonsil Epithelium

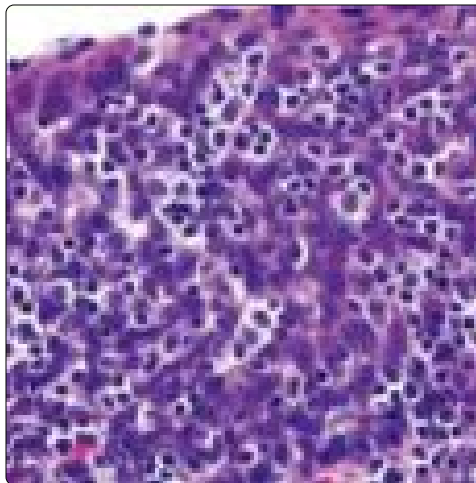


(Left) Tonsillar crypts represent invaginations of the surface mucosa into the lymphoid stroma. Unlike lymph nodes, tonsils lack capsules, sinuses, and trabeculae but do have connective tissue bands that extend to the surface. The stroma is mainly composed of lymphoid aggregates and interfollicular small lymphocytes. (Right) The surface epithelium is nonkeratinizing stratified squamous mucosa, which becomes a specialized lymphoepithelium in the crypt zones.

Tonsillar Crypts: Palatine Tonsil

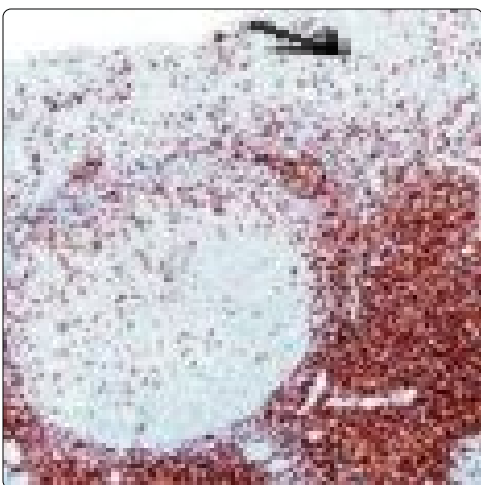


Lymphoepithelium

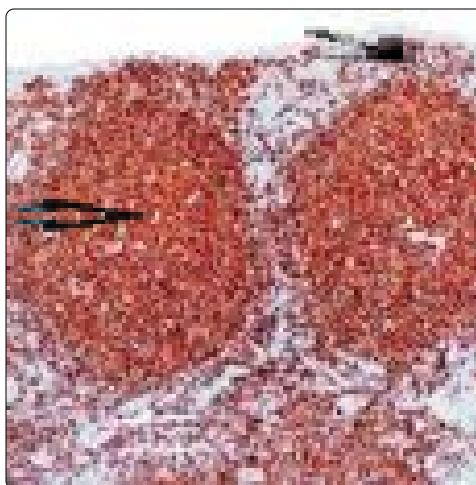


(Left) Crypt epithelium is distorted by infiltrating lymphocytes and plasma cells, which impart a reticular appearance. The squamous cells are predominantly basaloid. The surface has a thin layer of differentiated keratinocytes. Although nonkeratinizing, desquamated squamous cells often accumulate within the crypts. (Right) The lymphoepithelium consists of basaloid keratinocytes with infiltrating lymphocytes and plasma cells, imparting a reticulated appearance.

CD3 Immunohistochemistry



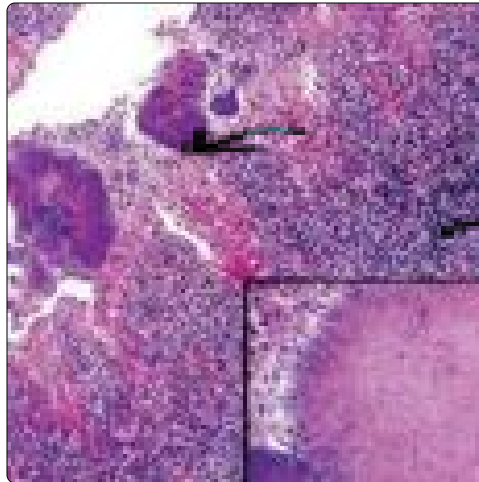
CD20 Immunohistochemistry



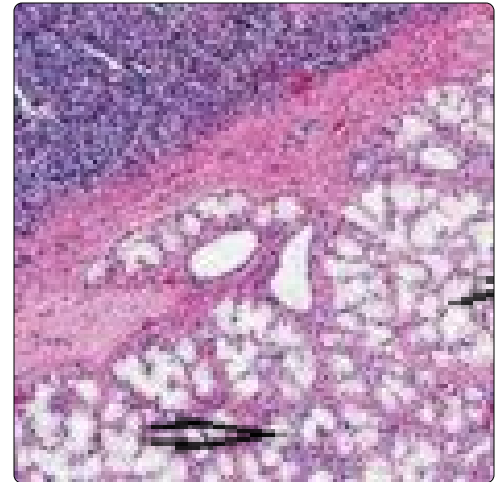
(Left) CD3 immunohistochemical staining highlights interfollicular T-cell-rich zones in a normal tonsil. Note the presence of scattered T cells within the lymphoepithelium. (Right) CD20 immunohistochemical staining highlights B-cell-rich follicles in a normal tonsil. Many B cells are also present in the T-cell-rich interfollicular zones. Note the presence of more abundant B cells within the lymphoepithelium.

Sulfur Granules

(Left) Tonsil crypts are blind-ending and typically accumulate luminal debris, such as desquamated keratin [1] and aggregates of oral bacteria [2]. This likely contributes to the development of chronic tonsillitis clinically. The inset demonstrates a sulfur granule composed of numerous actinomycotic organisms mixed with other oral flora. (Right) The connective tissue pseudocapsule contains scattered small mucous glands [3] with ducts [4] draining to the surface.

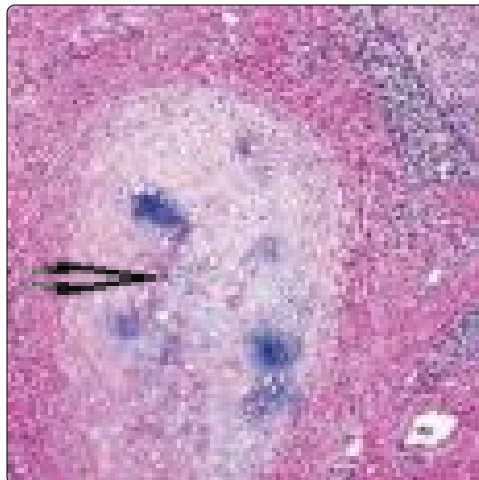


Mucous Glands: Palatine Tonsil

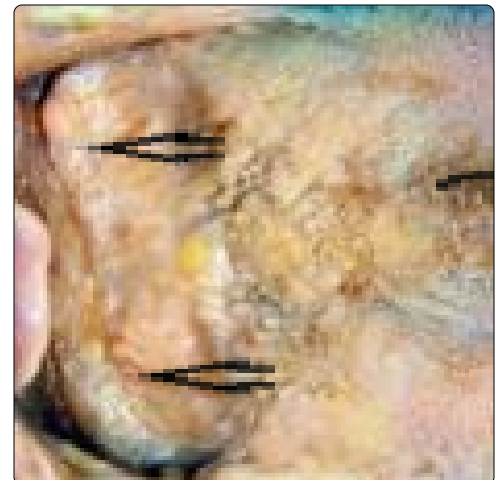


Ectopic Cartilage

(Left) Small nodules of ectopic cartilage [1] may be seen in the tonsillar connective tissue pseudocapsule. These are often associated with mucous glands and likely represent 2nd branchial cleft developmental abnormalities. (Right) The lingual tonsil (base of tongue) consists of irregular nodular tissue [2] corresponding to mucosa-covered lymphoid tissue with surface invaginations. It is located posterior to the sulcus terminalis [3] and anterior to the valleculae [4].

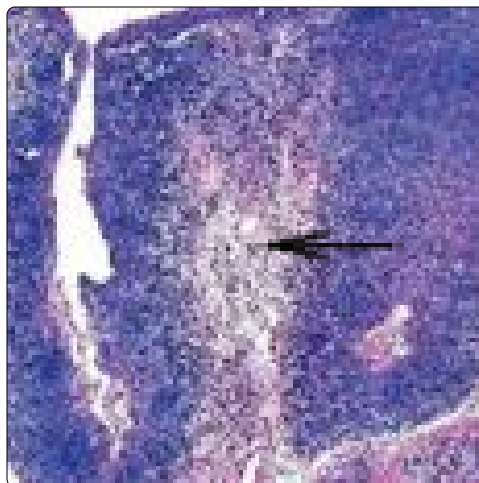


Lingual Tonsil

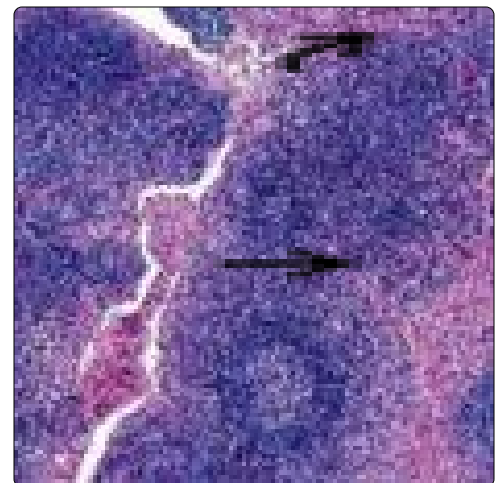


Lingual Tonsil Crypts

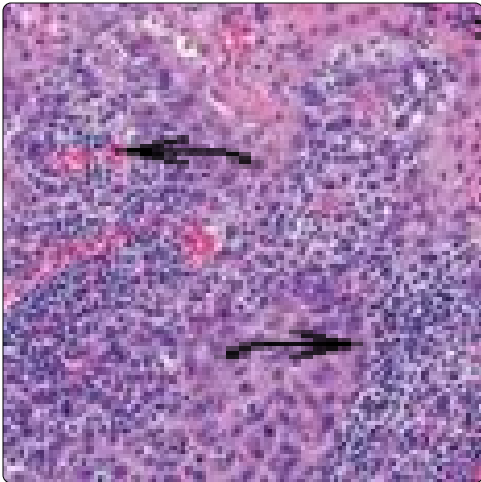
(Left) Similar to the palatine tonsils, lingual tonsillar crypts [1] represent invaginations of the surface mucosa into the lymphoid stroma. They are shorter than those seen in the palatine tonsil and often extend downward to underlying skeletal muscle and salivary glands [2]. (Right) The surface epithelium is nonkeratinizing stratified squamous mucosa [3], which becomes a specialized lymphoepithelium in the crypt zones [4].



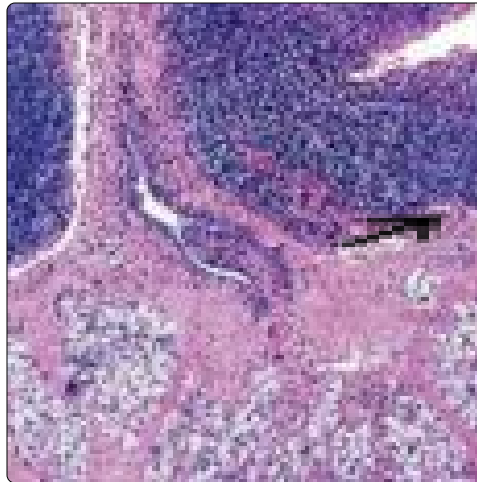
Lingual Tonsil Epithelium



Lymphoepithelium: Lingual Tonsil



Mucous Glands: Lingual Tonsil

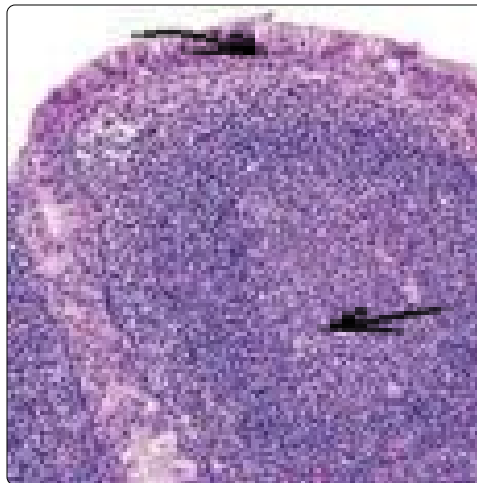


(Left) The lingual tonsil crypt epithelium is similar to that of the palatine tonsil. Basaloid keratinocytes are infiltrated by lymphocytes and plasma cells, which distort the mucosa. The surface has a thin layer of differentiated keratinocytes. Desquamated keratin debris can accumulate in the lumen. **(Right)** Identical to the palatine tonsils, scattered small, pure mucous glands with ducts draining to the surface are seen in the connective tissue pseudocapsule.

Pharyngeal Tonsil

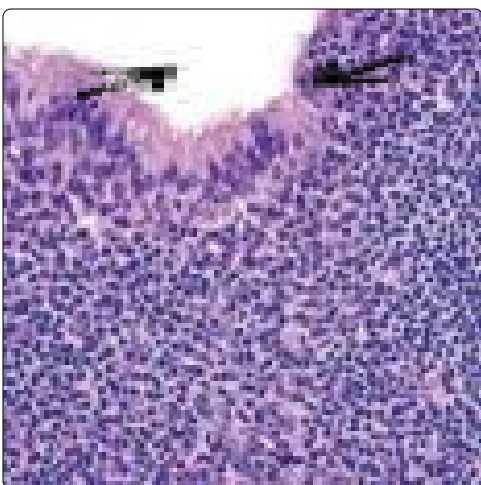


Pharyngeal Tonsil Folds

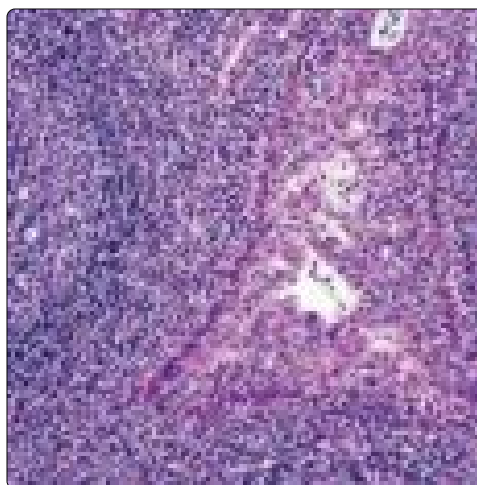


(Left) Unlike other tonsils, the pharyngeal tonsil does not have true crypts but instead has variably elongated mucosal folds that contain lymphoid tissue composed of follicles and interfollicular lymphocytes. **(Right)** Primary lymphoid follicles and germinal centers are intimately associated with the nasopharyngeal surface mucosa and mucosal folds. Intraepithelial lymphocytes are present, similar to other tonsils.

Pharyngeal Tonsil Epithelium



Deep Pharyngeal Tonsil Folds



(Left) Unlike other tonsils, the mucosa of the pharyngeal tonsil is a ciliated, pseudostratified, columnar epithelium with scattered goblet cells identical to respiratory-type epithelium. Note areas of intraepithelial lymphocytes and plasma cells. **(Right)** At the deeper aspects of the mucosal folds, the epithelium contains more intraepithelial lymphocytes and plasma cells, resulting in a characteristic lymphoepithelium similar to other tonsils.

EAR

Macroscopic Anatomy

- Composed of 3 regions
 - External ear
 - Composed of auricle (pinna), external auditory canal, and outer surface of tympanic membrane
 - Function: Sound conduction
 - Middle ear
 - Composed of tympanic cavity, inner surface of tympanic membrane, ossicles, mastoid air cells, and internal auditory canal (IAC) (eustachian tube)
 - Function: Sound conduction for auditory portion of inner ear
 - Inner ear
 - Comprised of membranous labyrinth embedded within medial petrous temporal bone (osseous labyrinth) and consists of cochlea, semicircular canals, vestibule (sacculle and utricle), and endolymphatic sac organ
 - Function: Sensory reception for hearing and balance

Microscopic Anatomy

- External ear
 - Auricle
 - Modified skin structure with cartilaginous skeleton
 - Contains normal skin adnexal structures
 - Outer 1/3 of external auditory canal
 - Supported by cartilage that is continuous with auricular elastic cartilage
 - Has modified apocrine glands (ceruminous glands) instead of eccrine glands
 - Inner 2/3 of external auditory canal
 - Epidermal lining devoid of adnexal structures/glands
 - Supported by portions of temporal bone instead of cartilage
 - Tympanic membrane
 - Covered by epidermis externally and nonsquamous cuboidal epithelium internally with connective tissue framework

- Middle ear
 - Tympanic cavity
 - Lined by modified respiratory epithelium consisting of single layer of cuboidal epithelium
 - No glands or squamous epithelium present under normal circumstances but may have patches of ciliated columnar epithelium
 - Ossicles
 - Composed of compact lamellar bone; incudomalleal and incudostapedial joints are diarthroses with synovial lining
 - Incudomalleal and incudostapedial joints are diarthroses with synovial lining
 - Mastoid air cells
 - Lined by flat cuboidal modified respiratory epithelium adherent to underlying periosteum
 - IAC
 - lined by ciliated columnar epithelium
 - Goblet cells and seromucous glands can be seen mainly in cartilaginous component
 - Proximal 1/3 of IAC is intraosseous, while distal 2/3 is surrounded by hyaline cartilage
 - Lymphoid aggregates (tubal tonsil) can be seen in IAC, especially in children
- Inner ear
 - Semicircular canal, utricle, and sacculle
 - Lined by specialized epithelia with sensory hair cells
 - Endolymphatic sac organ
 - Lined by low cuboidal to columnar epithelium, which may have papillary structures in sac portion
 - Cochlea
 - Highly specialized, complex sensory organ

CONSIDERATIONS AND VARIATIONS

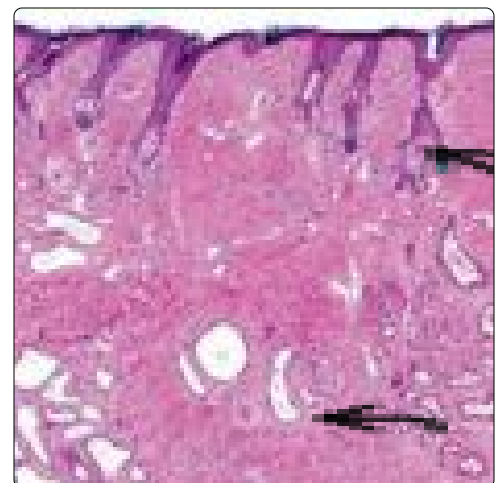
Metaplasia

- Squamous metaplasia in middle ear cavity
- Chronic otitis media can result in glandular "metaplasia"

External Ear

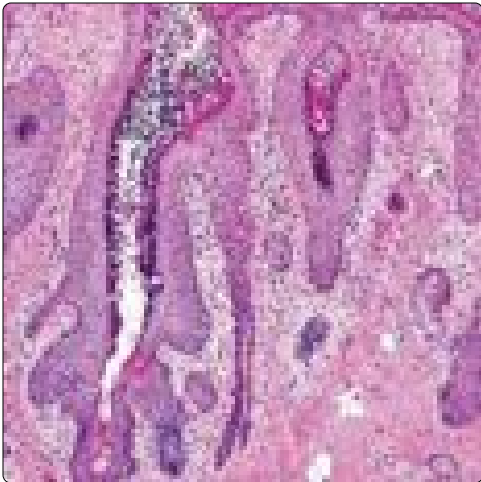


External Auditory Canal

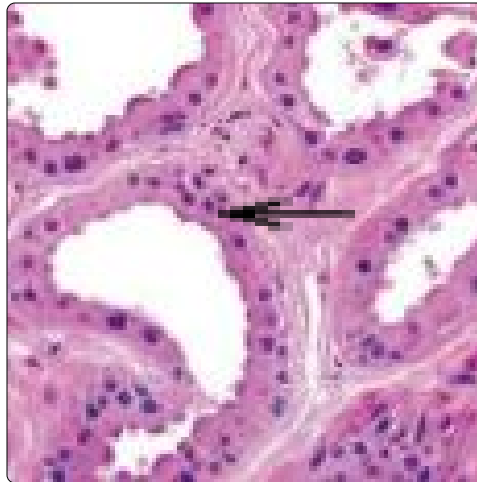


(Left) The external ear auricular cartilaginous framework is in continuity with the external auditory canal at the concha. The outer 1/3 is supported by cartilage, and the inner 2/3 is supported by bone. (Right) The outer 1/3 of the external auditory canal is lined by squamous epithelium with numerous associated hair follicles, sebaceous glands, and modified apocrine sweat glands known as ceruminous glands.

External Auditory Canal

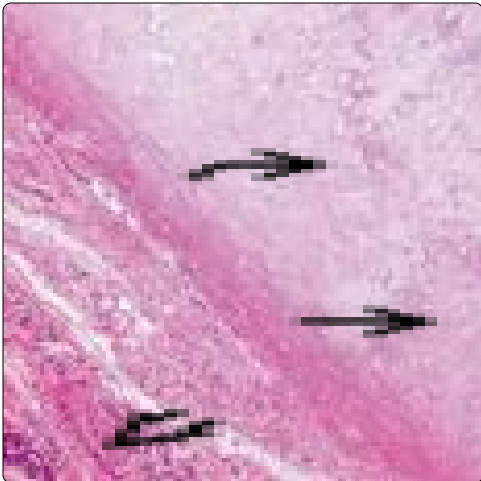


Ceruminous Glands

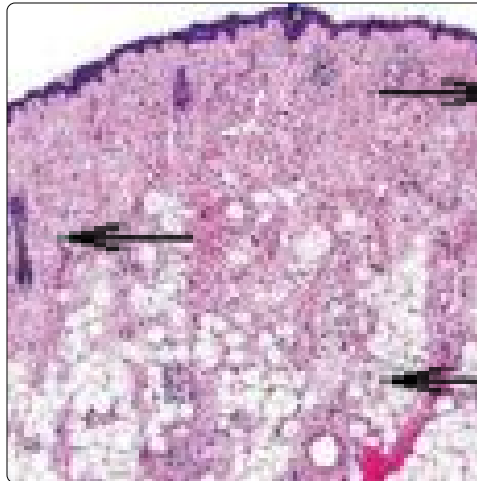


(Left) Cutaneous adnexal structures are numerous in the outer 1/3 of the canal, but are typically absent in the inner 2/3, which is lined by epithelium adherent to bone. Ceruminous gland ducts may drain to the surface or empty into the pilosebaceous unit as shown here. (Right) Ceruminous glands are modified apocrine glands characterized by cells with abundant granular eosinophilic cytoplasm, apical snouts, apocrine-type secretion, and yellow cytoplasmic granules.

Auricular Cartilage

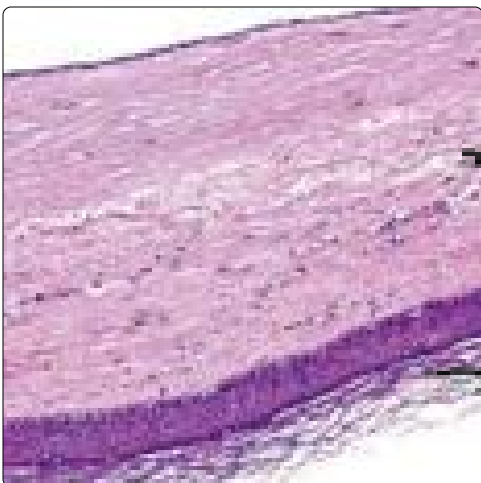


Ear Lobule

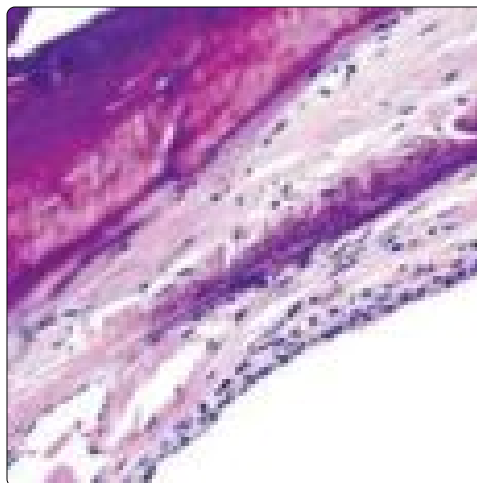


(Left) The auricular and external auditory canal cartilages are elastic-type cartilages. The cartilage is separated from the cutaneous adnexal structures and ceruminous glands by loose connective tissue and a thin perichondrium. (Right) The ear lobule is similar to the rest of the auricle; it is lined by squamous epithelium with adnexal structures. In contrast, no cartilage framework is present, but the stroma does contain abundant mature adipose tissue.

Tympanic Membrane



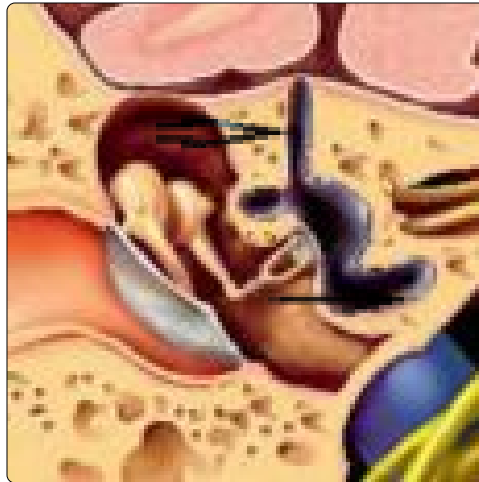
Tympanosclerosis



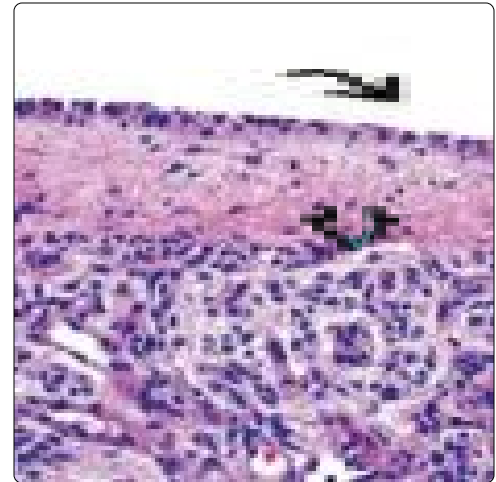
(Left) The tympanic membrane consists of a thin connective tissue plate lined externally by squamous epithelium devoid of adnexal structures and internally by modified respiratory epithelium. This tympanic membrane shows thickened, sclerotic connective tissue stroma due to chronic irritation. (Right) Chronic injury to the tympanic membrane can result in subepithelial sclerosis with secondary dystrophic calcification, known as tympanosclerosis.

Middle Ear

(Left) The tympanic cavity contains the 3 ossicles (malleus, incus, and stapes from lateral to medial) and is delimited laterally by the tympanic membrane. The tympanic cavity is in continuity with the internal auditory canal, which extends to the lateral nasopharynx. (Right) Tympanic cavity epithelium typically consists of flattened respiratory-type epithelium lacking underlying glands. Here it is seen adjacent to a paraganglioma.

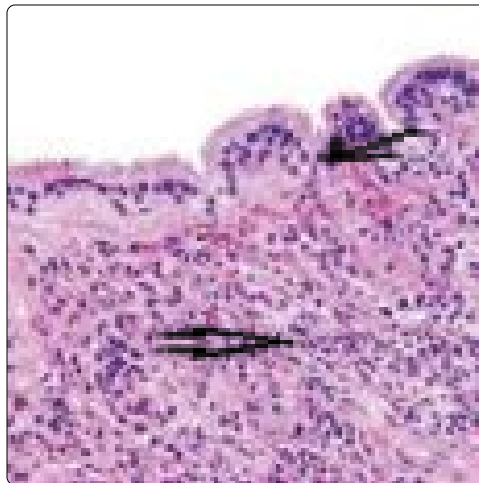


Tympanic Cavity

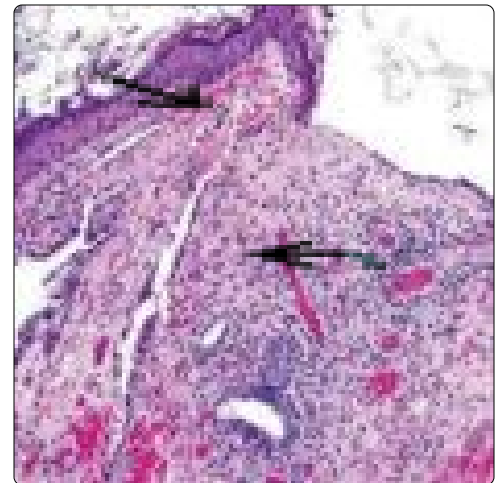


Tympanic Cavity Epithelium

(Left) While normally lined by cuboidal respiratory-type epithelium, patches of columnar respiratory epithelium complete with cilia are present to varying degrees lining this jugulotympanic paraganglioma. (Right) Squamous epithelium is not present under normal circumstances in the middle ear. Its presence indicates a cholesteatoma as seen here. These are often associated with chronic inflammation and glandular metaplasia, indicating chronic otitis.

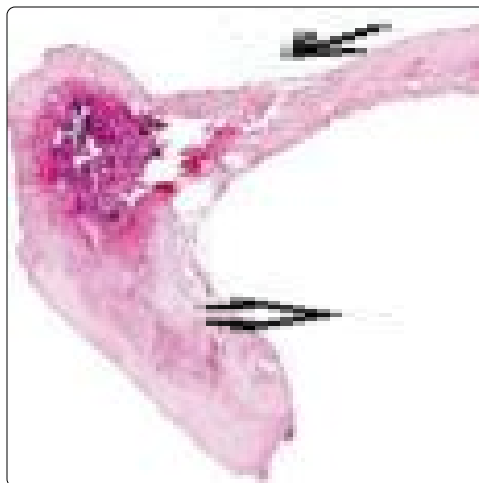


Cholesteatoma and Chronic Otitis Media

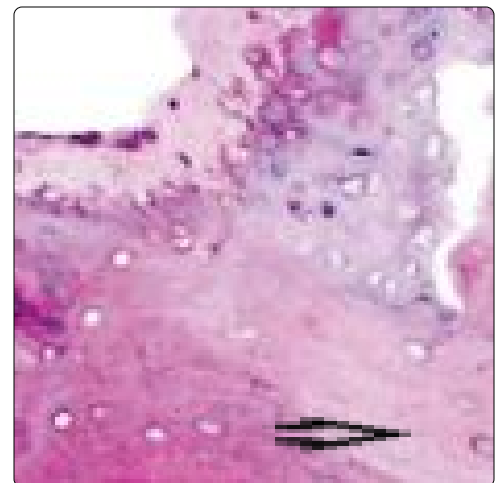


Middle Ear Ossicle (Incus)

(Left) The ossicles arise from membranous ossification and consist of compact lamellar bone. Islands of persistent cartilage can be seen at the ends of most of the ossicles. (Right) The ossicles predominantly consist of compact lamellar bone with osteocytes set within lacunae as well as varying numbers of haversian systems. Islands of cartilage can be seen at the periphery of most ossicles.



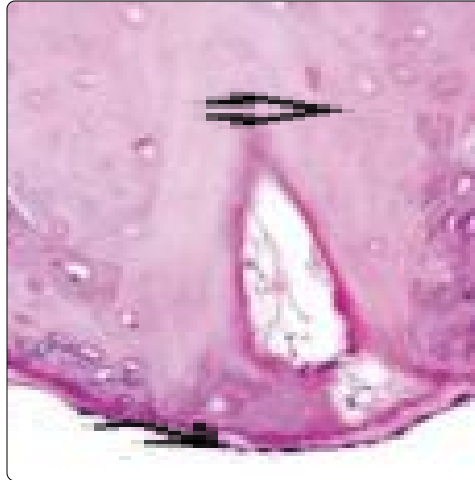
Ossicles



Ossicle (Stapes)

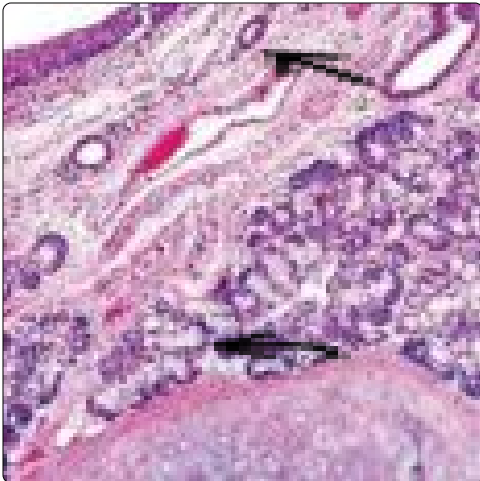


Mastoid Air Cells



(Left) This section from a stapes bone demonstrates compact lamellar bone. Tympanic cavity epithelium typically lines the periosteal surface of the ossicles. The epithelium is identical to the tympanic cavity epithelium except that the presence of cilia is uncommon. (Right) Mastoid air cells consist of air-filled cystic spaces within the dense compact lamellar temporal bone. The lining is identical to the tympanic cavity and is tightly adherent to bone.

Internal Auditory Canal

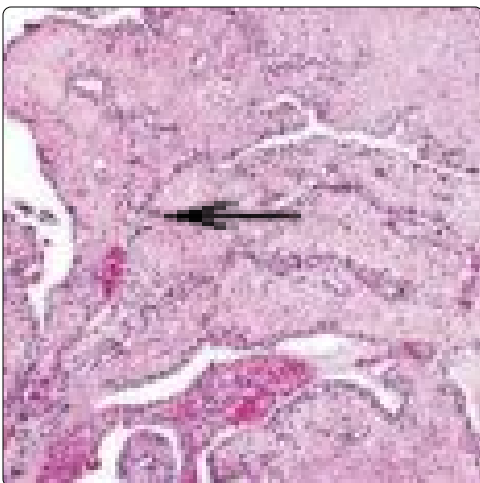


Internal Auditory Canal

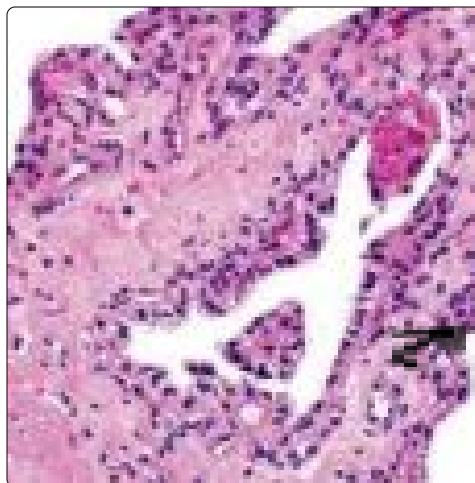


(Left) The internal auditory canal is lined by respiratory-type epithelium. It is present within bone proximally, but a hyaline cartilage tube supports it distally. Submucosal seromucous glands are present in the distal 2/3 (cartilaginous portion). (Right) The internal auditory canal is lined by respiratory-type epithelium, but goblet cells and seromucous glands can be seen, primarily in the distal 2/3. Lymphoid aggregates (tubal tonsil) may be seen as well.

Endolymphatic Sac



Endolymphatic Sac Epithelium



(Left) While not typically seen as a surgical specimen, portions of the endolymphatic sac can be seen in tumor resection specimens. These appear as epithelial-lined cystic spaces embedded within loose connective tissue or bone. (Right) The endolymphatic sac is lined by flat cuboidal to columnar epithelium and may have small papillae. The cells have small central nuclei and clear to pale eosinophilic cytoplasm and rest on a loose vascular connective tissue stroma.

NOSE

Macroscopic Anatomy

- Paired nasal cavities extend anteriorly from nares (nostrils) and posteriorly to choanae and are separated by nasal septum
 - Consist of 3 distinct regions: Vestibule, respiratory, and olfactory
- Vestibule communicates with external environment and contains hairs that act as coarse filter
- Respiratory region is most voluminous portion of nasal cavity
 - Medial wall is smooth and formed by nasal septum
 - Lateral wall contains 3 coiled bony projections called turbinates, which increase surface area and induce turbulence to inhaled air
 - Floor is smooth and sits on hard palate and anterior maxilla
- Olfactory region occupies majority of roof of nose and extends medially onto superior nasal septum and laterally onto superior portion of superior turbinate

Microscopic Anatomy

- External nose is covered by epidermis with high concentration of sebaceous glands and fine hairs
- Vestibule consists of epidermis that is continuous with surface skin and contains sebaceous glands, sweat glands, and coarse hairs
- Majority of nasal cavity is lined by ciliated pseudostratified columnar respiratory epithelium
 - Epithelium is ectodermally derived and is referred to as schneiderian membrane
 - There is short transition of nonciliated columnar or transitional epithelium between vestibule and ciliated epithelium
 - 3 cell types are identifiable: Basal cells, goblet cells, and ciliated columnar cells
- Lamina propria contains numerous seromucous glands, which drain to surface via small ducts
- Scattered melanocytes are present in surface mucosa, glands, and lamina propria

- Turbinates are lined by same schneiderian membrane but contain more prominent, specialized vascular component
 - Vessels form dense network of variably sized spaces resembling erectile tissue
 - Vessels have thick, muscular walls and contract and dilate to regulate temperature and secretions
 - Osseous core consists of thin plates of lamellar bone with intraosseous vessels
- Nasal septum is composed of hyaline cartilage anteriorly with remainder being plates of lamellar bone
 - Focal collection of ectatic thin-walled vessels can be seen on anterior cartilaginous portion (known as Kiesselbach area or Little area)
- Olfactory mucosa is specialized sensory epithelium lined by ciliated pseudostratified columnar epithelium
 - 3 cell types are identifiable: Basal cells, ciliated columnar supporting/sustentacular cells, and olfactory neural cells
 - No goblet cells are present in olfactory region
 - Sustentacular cells may have lipofuscin pigment in older individuals
 - Olfactory neural cells are bipolar spindle cells with olfactory sense receptor cilia present on surface and axons at basal surface
 - Olfactory neural cells collect in lamina propria to form myelinated nerves that traverse cribriform plate to join olfactory nerve
 - Specialized serous secretory glands (of Bowman) present in lamina propria and empty to surface via small ducts

PARANASAL SINUSES

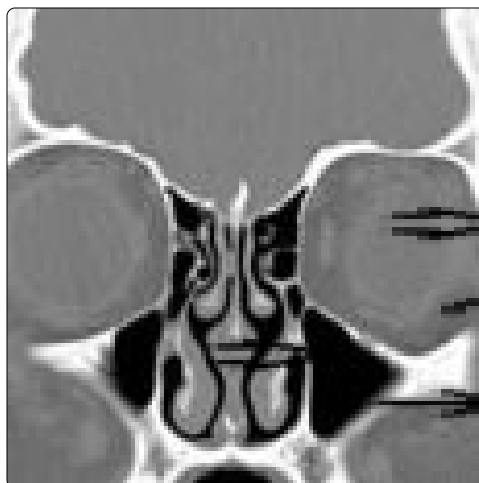
Macroscopic Anatomy

- Paired air-filled paranasal sinuses include maxillary, frontal, ethmoid, and sphenoid sinuses
 - They communicate with, and are directly connected to, nasal cavity via several ostia

Microscopic Anatomy

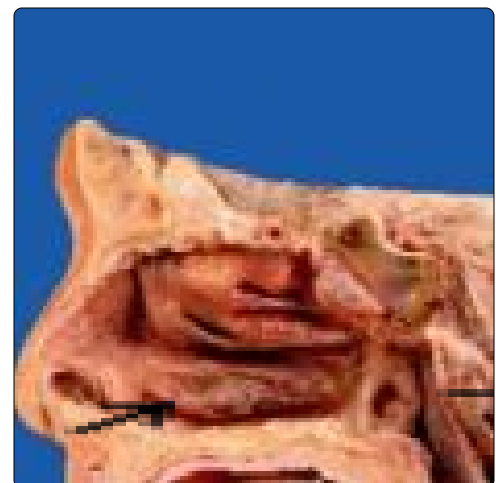
- Sinuses are lined by schneiderian membrane, but it is thinner, less vascular, and contains fewer seromucous glands

Nasal Cavities

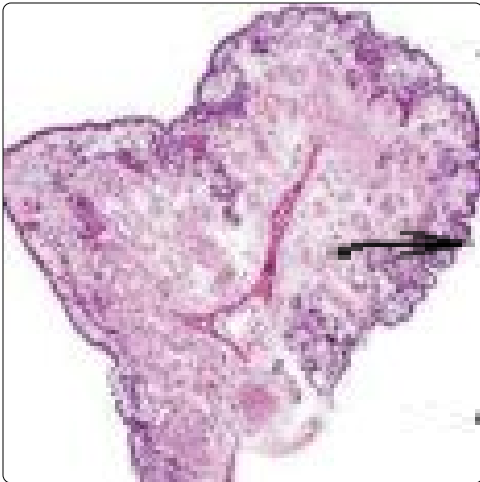


(Left) This coronal CT scan illustrates the nasal cavity with inferior and middle turbinates and paranasal sinuses. The cribriform plate forms the roof, and the floor sits on the hard palate. (Right) The nasal vestibule communicates with the environment and contains coarse hairs. The lateral nasal wall contains superior, middle, and inferior turbinates. (Courtesy M. Nielsen, MS.)

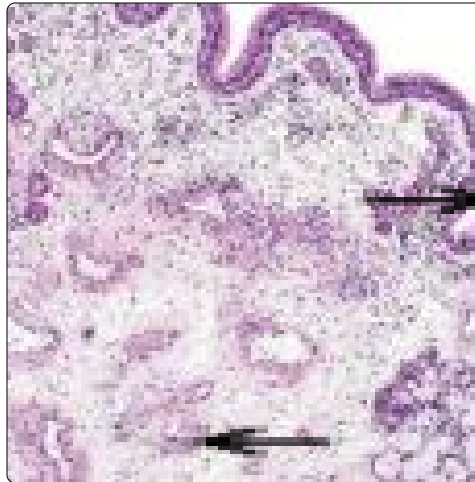
Nasal Vestibule



Turbinate



Turbinate Lamina Propria

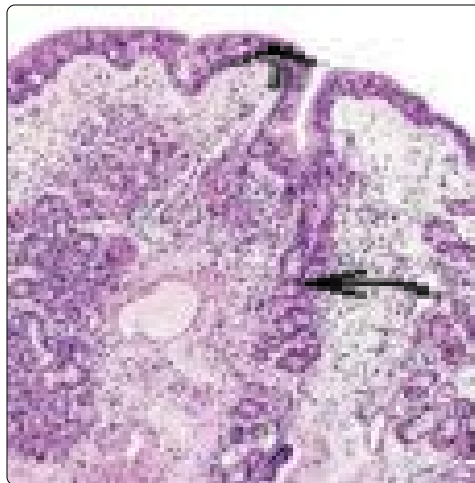


(Left) This cross section of a middle turbinate shows the delicate, curved bone core surrounded by thick lamina propria, rich in variably sized vessels and covered by schneiderian mucosa. **(Right)** The lamina propria of the nasal mucosa, especially the turbinates and inferior septum, contains a network of variably sized venous sinuses with thick, muscular walls. These sinuses can rapidly constrict/dilate in response to various stimuli and histologically resemble erectile tissue.

Schneiderian Mucosa

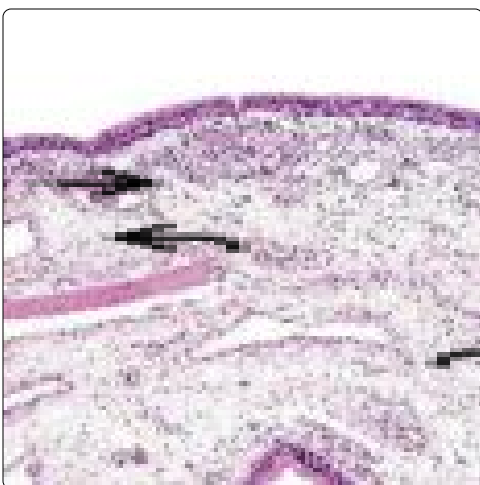


Distal Turbinate



(Left) Similar to the rest of the respiratory region, the turbinates are lined by ciliated, pseudostratified columnar epithelium with scattered mucous (goblet) cells. Varying numbers of seromucous glands are present in the lamina propria and drain directly to the surface via small ducts. **(Right)** There are regional differences in turbinate histology. The most distal region, which directly confronts inhaled air, is relatively rich in mucous cells and salivary glands.

Proximal Turbinate



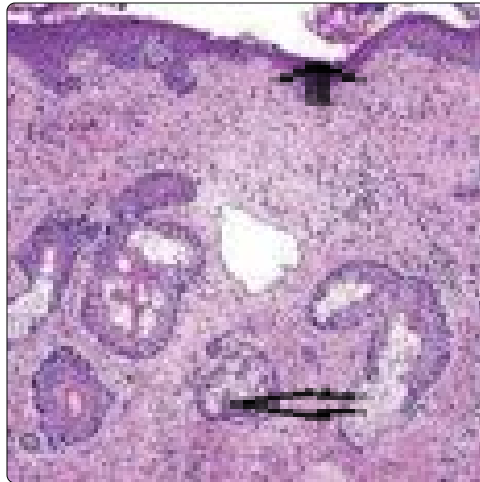
Turbinate Bone



(Left) The turbinate stalk has a thinner lamina propria with fewer vessels and salivary glands. The epithelium has fewer mucous cells as well. **(Right)** Turbinate bone is composed of thin trabeculae of mature lamellar bone. Similar to the lamina propria, there are numerous muscular veins, dilated vessels, and arteries that may mimic vascular neoplasms or malformations. Although there may be adipose tissue, no hematopoietic elements are present.

Nasal Vestibule

(Left) The nasal vestibule is lined by skin with stratified squamous epithelium containing all layers of normal epidermis. There are numerous hairs with associated sebaceous glands. The coarse hairs and secretions act to trap inhaled particulate matter. **(Right)** The transition from vestibule to respiratory mucosa is gradual with thinning of the epidermis, loss of pilosebaceous units, and increased venous vasculature with occasional salivary glands and ducts.

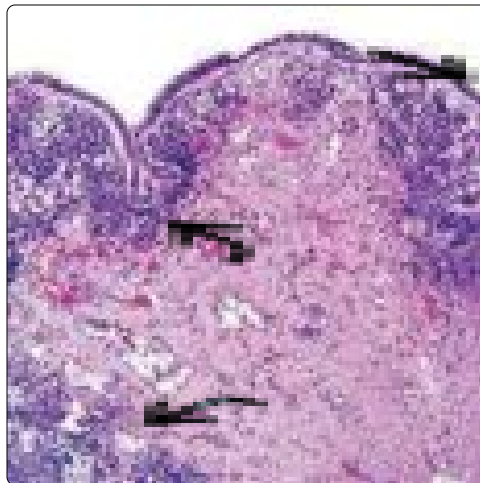


Vestibule: Respiratory Transition



Respiratory Region

(Left) Similar to the turbinates, the remainder of the respiratory portion of the nasal cavity is lined by ciliated pseudostratified columnar epithelium with numerous seromucous minor salivary glands. The lamina propria also contains numerous venous sinuses that warm inhaled air. **(Right)** Mucoserous glands drain to the surface via small excretory ducts. The schneiderian epithelium is pseudostratified and columnar, containing cilia and scattered mucous (goblet) cells.

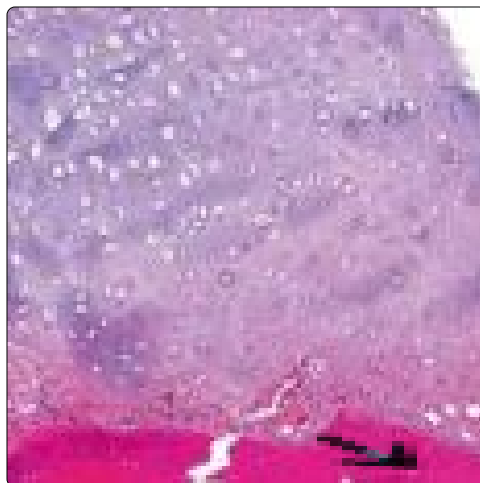


Respiratory Region

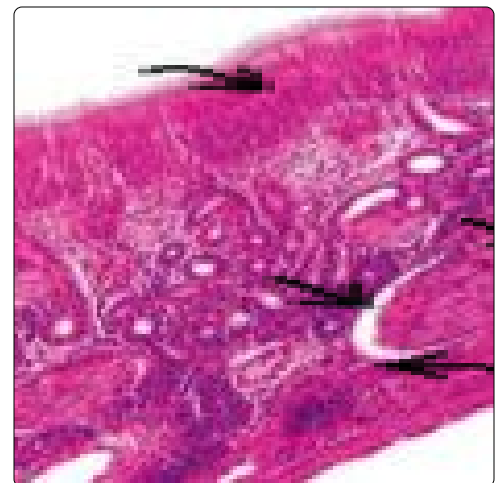


Septal Cartilage

(Left) The nasal septum consists of a plate of hyaline cartilage associated with 4 smaller bony plates fused with fibrous sutures. **(Right)** Olfactory mucosa is a modified ciliated pseudostratified columnar mucosa with specialized intraepithelial olfactory nerve cells. These neurons fuse to form olfactory nerve bundles in the lamina propria. Specialized olfactory glands (Bowman glands) are also present in the lamina propria and are unique to olfactory mucosa.



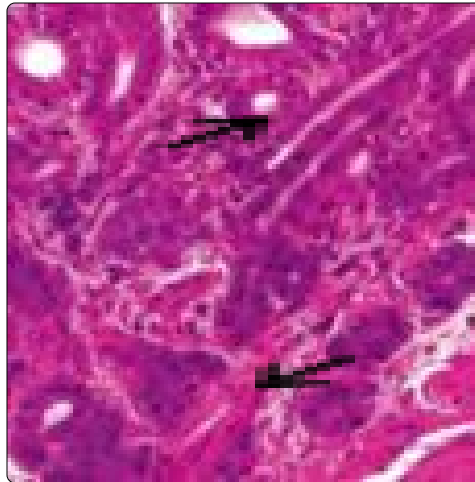
Olfactory Region



Olfactory Epithelium



Glands of Bowman

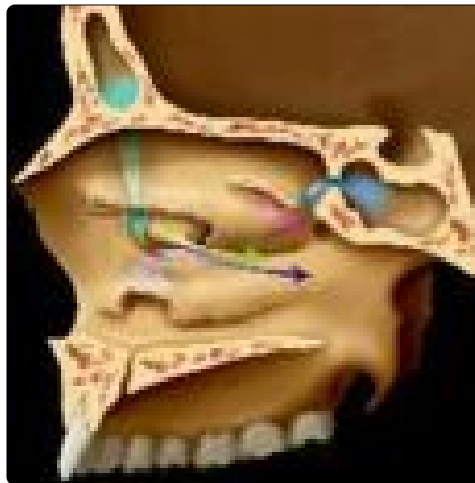


(Left) Olfactory epithelium contains bipolar olfactory nerve cells with nuclei stratified between the more uniform superficial nuclei of the ciliated columnar supporting (sustentacular) cells and the progenitor basal cells. No mucous cells are present in the olfactory mucosa. (Right) Olfactory glands are unique to the olfactory region. They are tubulovillous serous glands that may have lipofuscin pigment in the cytoplasm. They drain into a small central duct.

Secretory Duct

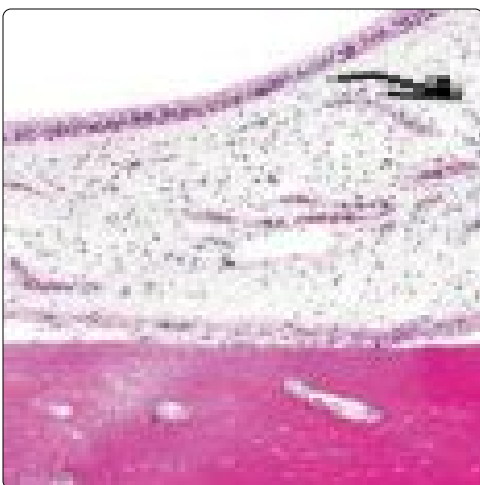


Paranasal Sinus Ostia

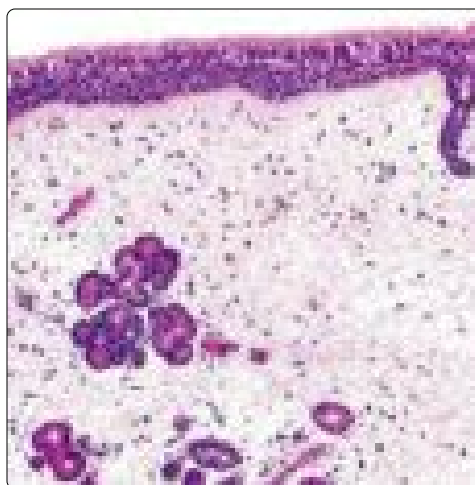


(Left) Olfactory glands drain secretions directly to the surface via short ducts. The secretions are thought to trap and dissolve odoriferous substances, allowing new substances to be recognized. (Right) The paranasal sinuses are air-filled spaces in the bones adjacent to the nasal cavity and are named after the bones in which they are situated. They communicate with the nose via small openings (ostia) as follows: Blue = sphenoid, pink and green = ethmoid, aqua = frontal, and purple = maxillary.

Paranasal Sinus



Paranasal Sinus



(Left) The paranasal sinuses are lined with ciliated, pseudostratified (Schneiderian) mucosa with mucous cells, which is in continuity with the nasal mucosa via ostia. The mucosa is thinner and attaches to the periosteum of the various sinus bones. (Right) In contrast to nasal mucosa, seromucous glands are less numerous and more randomly distributed. These drain to the surface via small ducts. Also, the stroma does not contain a prominent vascular component.

Minor Salivary Glands

MINOR SALIVARY GLANDS

Macroscopic Anatomy

- 500-1,000 lobules of minor salivary glands
- Arise outside of parotid, submandibular, and sublingual glands (i.e., major salivary glands)
- Sites
 - Buccal mucosa
 - Lips
 - Floor of mouth
 - Tonsillar pillars
 - Glossopalatine arch
 - Hard palate
 - Soft palate
 - Uvula
 - Tongue
 - Blandin and Nunn glands
 - von Ebner glands
 - Nasal and paranasal regions

Microscopic Anatomy

- 3 types
 - Mucous
 - Hard palate, soft palate, tonsillar pillars, glossopalatine arch
 - Base of tongue
 - Serous
 - von Ebner glands (tongue)
 - Mixed seromucinous
 - Lips, buccal mucosa, floor of mouth
 - Hard palate, soft palate, uvula
 - Blandin and Nunn glands (anterior tongue)
 - Nasal and paranasal regions
- Acinus
 - Rounded collection of mucous or serous cells surrounding central lumen
- Mucous cells
 - Large, polygonal shape
 - Pale, basophilic, mucin-filled cytoplasm

- Dense, basoloid nuclei flattened against basement membrane
- Positive histochemical stains
 - PAS
 - Alcian blue
 - Mucicarmine
- Serous cells
 - Polygonal shape
 - Cytoplasm has dense, strongly staining eosinophilic secretory granules
 - PAS(+), diastase-resistant
 - Round, open nuclei
 - Present in central or basal 1/3 of cell
 - Serous demilunes
 - Collections of serous cells surrounding terminal part of mucous acini
 - Found in glands where mucous cells predominate
- Myoepithelial cells
 - May rarely be seen as spindle cells associated with acini
- Ducts
 - Intercalated ducts
 - Smallest type
 - Lined by low cuboidal cells with scant cytoplasm and round nuclei
 - Striated ducts
 - Larger than intercalated ducts
 - Lined by columnar and eosinophilic cells with uniform round nuclei
 - Excretory
 - Largest type
 - Lined by pseudostratified columnar epithelium

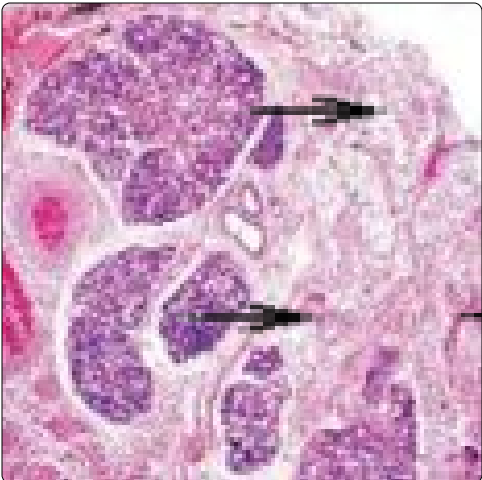
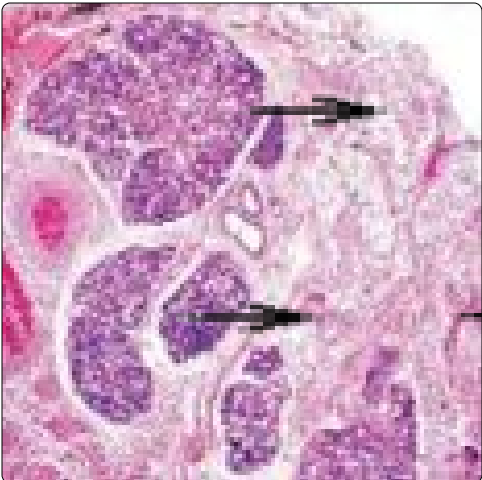
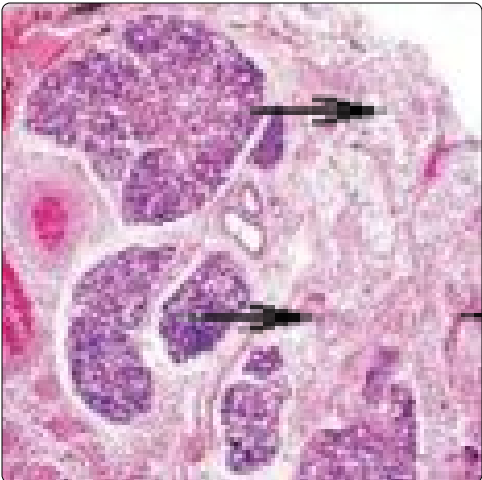
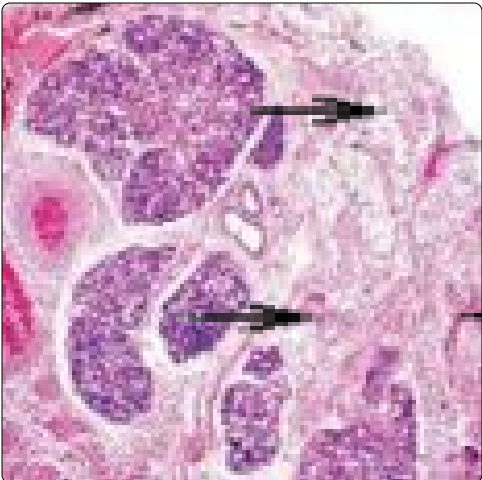
CONSIDERATIONS AND VARIATIONS

Metaplasia

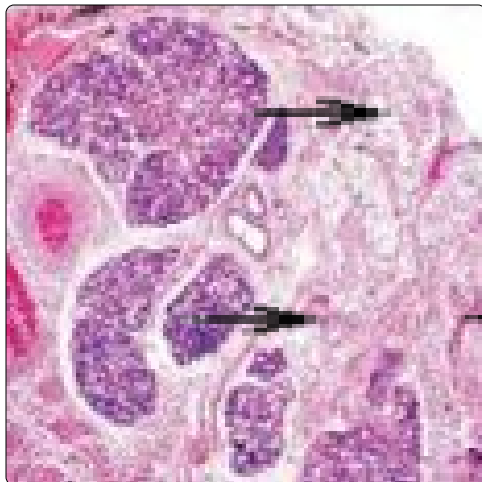
- Squamous, oncocytic

Pitfalls and Artifacts

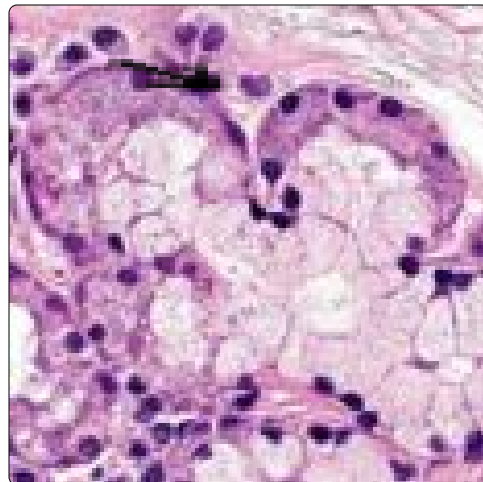
- Irradiated normal salivary glands and ducts may simulate carcinoma

(Left) Low-power view of seromucous minor salivary glands  shows the unencapsulated relationship the glands have with the surrounding tissue. These glands, from the buccal mucosa, are intimately associated with the nearby connective and adipose tissue. Small excretory  ducts are seen between lobules. **(Right)** High-power view of a seromucous gland shows the distinct mucous cells  filled with basophilic mucin and associated serous, granular demilunes .

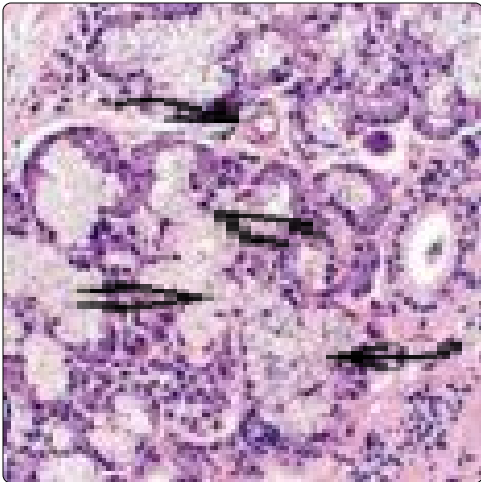
Minor Salivary Glands



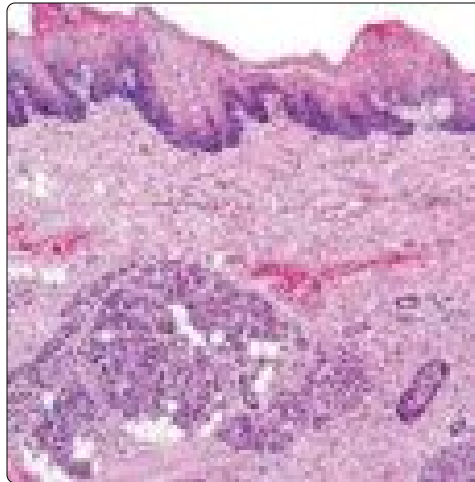
Mixed Seromucinous Minor Salivary Gland



Seromucinous Gland

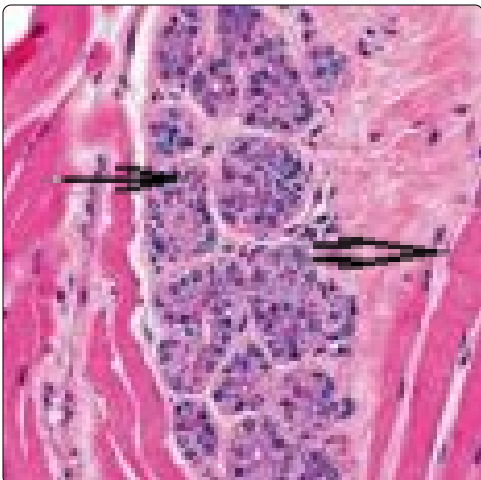


Serous Minor Salivary Gland

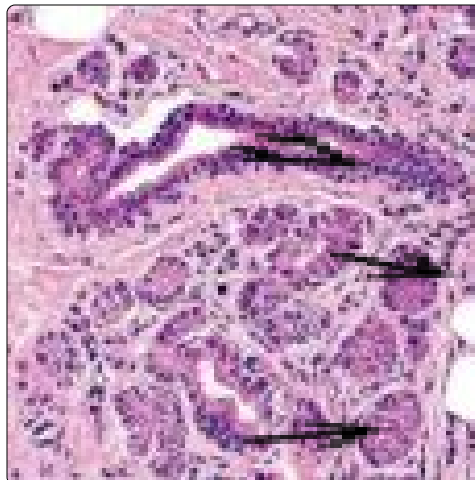


(Left) Medium-power H&E of a seromucous minor salivary gland shows several seromucous units with mucous acini capped by serous demilunes. Secretions are collected from the secretory units by ducts. Note the striated duct lined by columnar cells with round, uniform nuclei. (Right) Low-power H&E shows the von Ebner minor salivary glands. These glands are found in association with the circumvallate papillae found on the posterior dorsal tongue.

von Ebner Glands

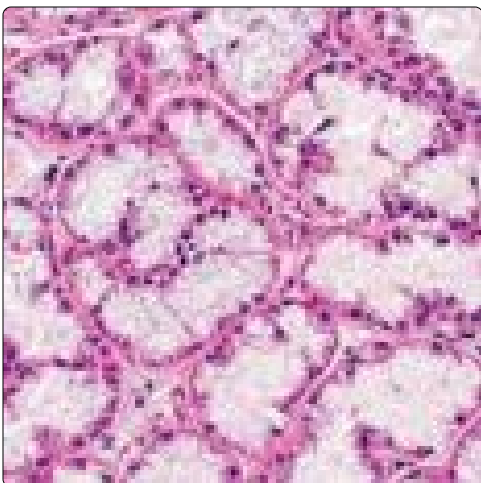


Excretory Duct

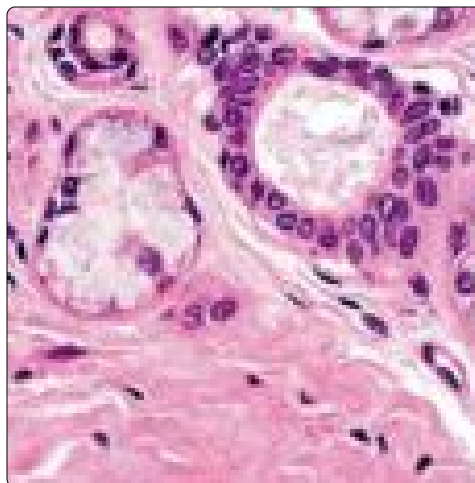


(Left) H&E of the serous von Ebner glands shows the close association to the surrounding skeletal muscle. When minor salivary glands develop neoplasms, this intimate association can make determining invasion difficult. (Right) H&E shows serous acini with an adjacent excretory duct, usually shorter than those in major glands. This type of duct is lined by pseudostratified columnar epithelium, supported by surrounding fibrous connective tissue.

Mucous Minor Salivary Gland



Mucous Acinus and Intercalated Duct



(Left) High-power view shows the Blandin and Nunn glands. This type of gland consists entirely of mucous acini with each acinus surrounded by thin connective tissue septa. (Right) High-power H&E shows a mucous acinus adjacent to an intercalated duct. The small size of this type of duct makes them very inconspicuous in routine H&E slides. Intercalated ducts are lined by cuboidal, eosinophilic cells with centrally placed nuclei.

Major Salivary Glands

MAJOR SALIVARY GLANDS

Macroscopic Anatomy

- Paired parotid, submandibular, and sublingual glands
 - Surrounded by delicate fibrous capsule with intraglandular septa dividing them into lobules
- Parotid glands (15-30 g) are largest
 - Located subcutaneously anterior to ear
 - Divided by facial nerve into superficial and deep lobes
 - Superficial lobe is larger, flat, and quadrangular
 - Deep lobe is wedge-shaped and extends into parapharyngeal space
 - Main (Stensen) duct (~ 7 cm) traverses through masseter and buccinator muscles and drains to oral cavity opposite 2nd maxillary molar
 - Only gland with intraparenchymal lymph nodes
- Submandibular (submaxillary) glands (~ 10 g)
 - Located in deep posterior floor of mouth behind mandible
 - Main (Wharton) duct (~ 5 cm) traverses floor of mouth and drains to oral cavity at sublingual caruncle, just lateral to frenulum
- Sublingual glands (2-4 g)
 - Located in anterior floor of mouth behind mandible
 - Main (Bartholin) duct drains into submandibular duct near its termination
 - Numerous small (Rivinus) ducts drain directly to floor of mouth along plica sublingualis

Microscopic Anatomy

- Functional unit consists of acinus and draining duct system
- Acini can be serous, mucous, or mixed
 - Parotid is purely serous
 - Submandibular is mixed with serous predominance
 - Sublingual is mixed with mucous predominance
- Serous acini
 - Spherical with pyramid-shaped cells containing large basophilic zymogen granules
- Mucous acini
 - Tubular with round cells containing sialomucin granules

- Mixed acini
 - Mucous acini with crescent-shaped caps of serous cells (serous demilunes)
- Acinar lumina drain into branching duct system that modifies saliva and delivers it to oral cavity
- Intercalated ducts are intralobular and are located between acini and striated ducts
 - Lined by single layer of cuboidal cells
 - Longest in parotid, shorter in submandibular, and very short in sublingual gland
- Striated ducts are intralobular and deliver saliva to larger excretory ducts
 - Lined by columnar cells with eosinophilic cytoplasm and parallel striations at basal aspect
 - Longest in submandibular, shorter in parotid, and very short in sublingual gland
- Interlobular and main excretory ducts are located in interlobular septa and outside gland, respectively
 - Lined by pseudostratified columnar epithelium with rare goblet cells; become stratified and squamous near their termination in oral cavity
- Contractile myoepithelial cells are flat cells with numerous cytoplasmic processes (basket cells)
 - Present between epithelial cells and basal lamina in acini, intercalated ducts, and proximal striated ducts
- Noncontractile basal cells are present in striated and excretory ducts
- Parotid parenchyma contains abundant mature adipose tissue; submandibular gland has less
 - Large branches of facial nerve are seen in parotid interlobular septa
- Rare sebaceous cells or lobules can be seen in parotid

CONSIDERATIONS AND VARIATIONS

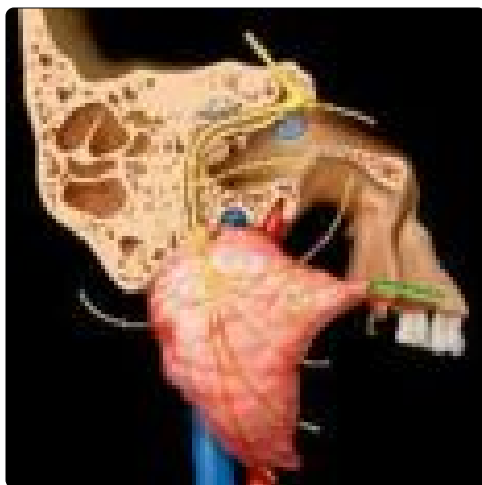
Age-Related Variations

- Atrophy and oncocytic metaplasia are more common with age

Pitfalls and Artifacts

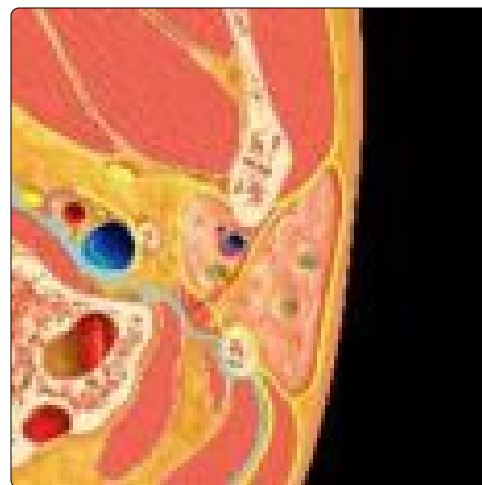
- Intranasal epithelial inclusions may mimic metastases

Parotid Gland



(Left) Branches of the facial nerve artificially divide the quadrangular superficial parotid lobe from the deep lobe. The main duct drains into the oral cavity opposite the 2nd maxillary molar. **(Right)** The deep parotid lobe is wedge-shaped and extends into the parapharyngeal space. Note the intraparotid lymph nodes and the facial nerve branch separating the deep lobe from the superficial lobe.

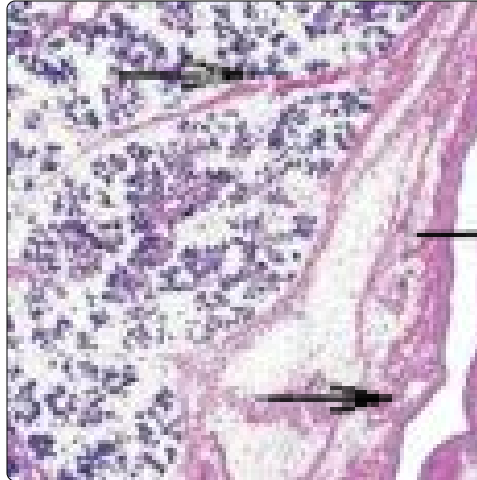
Superficial and Deep Lobes



Submandibular and Sublingual Glands

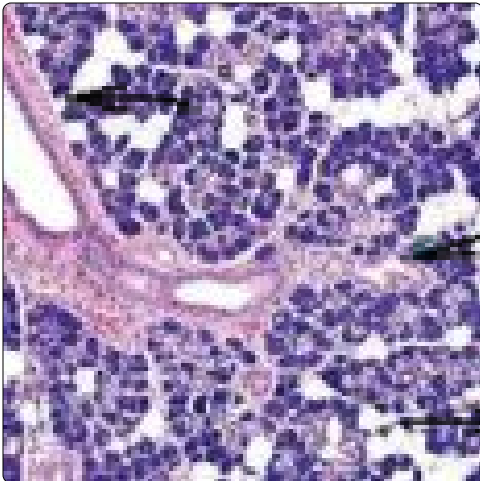


Parotid Lobules

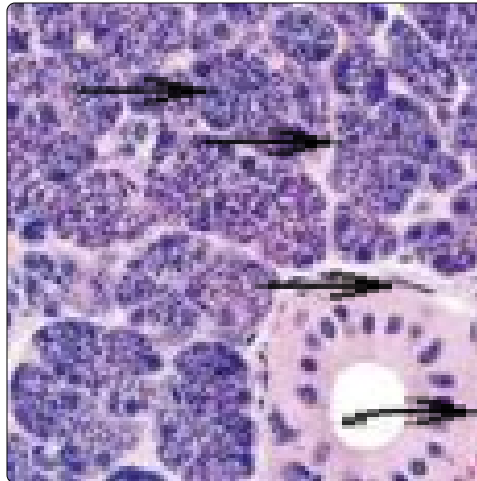


(Left) Submandibular glands sit below the mylohyoid muscles. A small portion extends above the muscle, and the main duct extends medially to drain into the oral cavity at the sublingual caruncle. The sublingual gland is above the muscle, and its main duct drains into the submandibular duct. (Right) Fibrous septa extend from the capsule to divide the parotid into indistinct lobules. This parotid, from an older individual, contains abundant adipose tissue.

Parotid Excretory Ducts

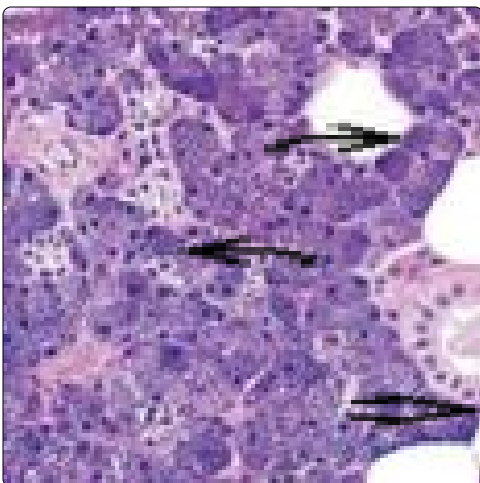


Parotid Acini

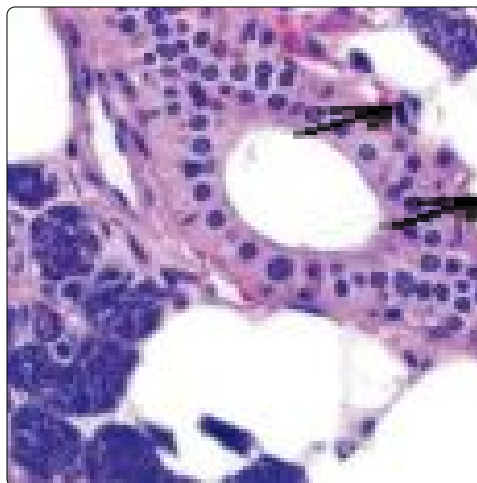


(Left) Intercalated ducts receive secretions from acini and are easily recognizable in parotid glands. They deliver secretions to intralobular striated ducts, which drain to progressively larger intralobar and interlobar excretory ducts. (Right) Serous acini are spherical and drain into intercalated ducts. The secretory cells are pyramidal with basal nuclei and cytoplasm containing numerous coarse, basophilic secretory (zymogen) granules.

Parotid Intercalated Ducts



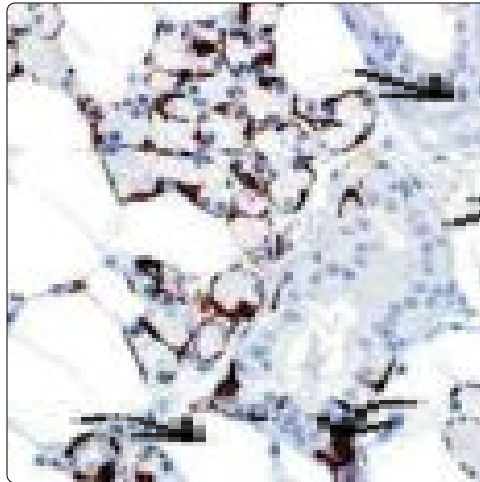
Parotid Striated Duct



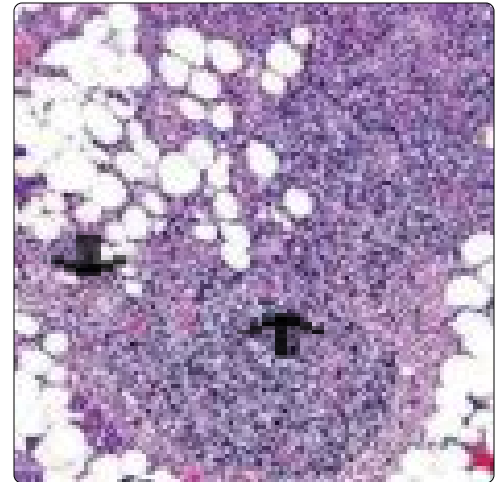
(Left) Intercalated ducts are lined by simple cuboidal epithelium with pale cytoplasm. Some intercalated duct cells may contain residual zymogen granules. (Right) Striated ducts are simple columnar intralobular ducts that consist of cells with abundant eosinophilic cytoplasm and round central nuclei. Invaginations of the basal cell membrane account for the striations seen in histologic sections. The cells have numerous mitochondria, which cause the cytoplasmic eosinophilia.

Myoepithelial Cells

(Left) Contractile myoepithelial cells are difficult to discern on H&E but can be appreciated using immunohistochemistry (calponin). They are located around acini and intercalated ducts and focally in some proximal striated ducts. **(Right)** Late embryologic encapsulation accounts for the 5-20 intraparotid lymph nodes that represent 1st-echelon drainage for malignancies of the face, scalp, and ear. They often contain salivary gland inclusions, which may mimic metastases.

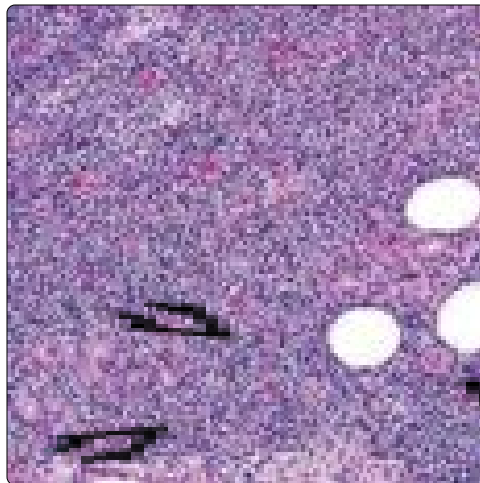


Intraparotid Lymph Node

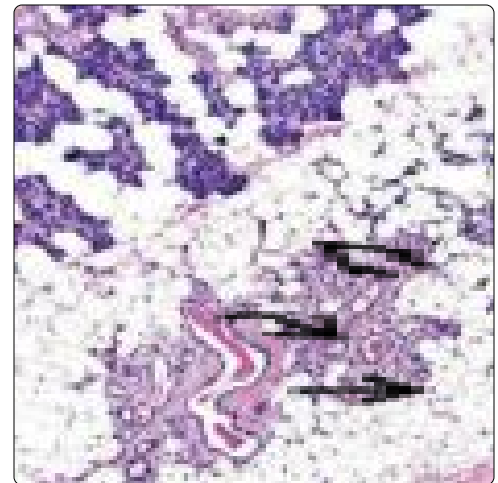


Periparotid Lymph Node

(Left) Similar to intraparotid lymph nodes, periparotid lymph nodes also may contain salivary gland inclusions, likely due to late embryologic encapsulation as well. These may mimic metastatic disease and can also be a source of extraparotid salivary gland neoplasms. **(Right)** Focal areas of atrophy are common in excised parotid glands. These result from excretory duct obstruction and may mimic neoplasms. Acini are lost, and only ducts with periductal fibrosis persist.

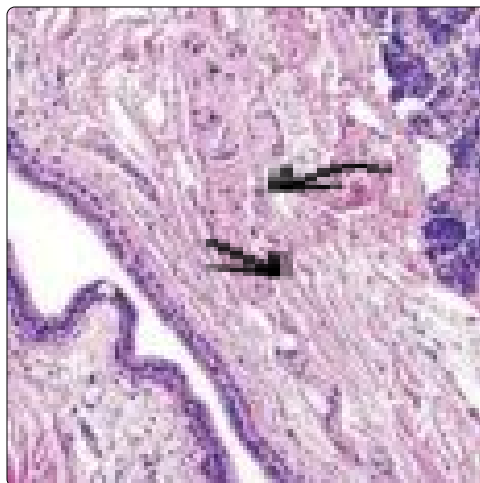


Parotid Atrophy

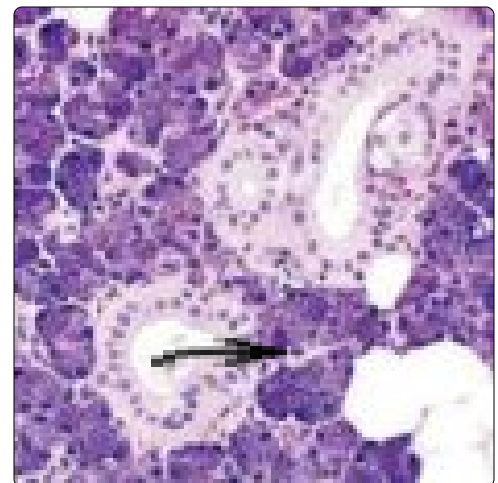


Main Parotid (Stensen) Duct

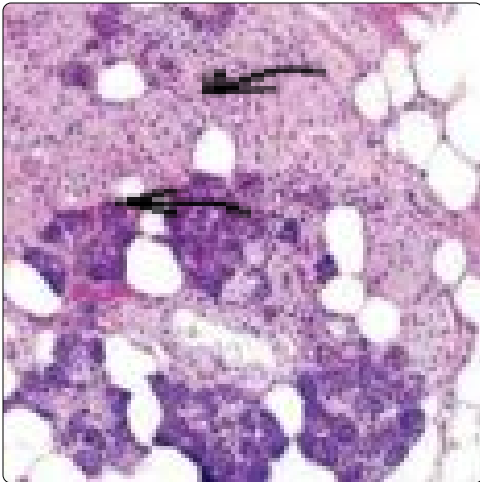
(Left) The main parotid ducts traverse connective tissue and muscle to reach their termination in the buccal mucosae. They are lined by pseudostratified or stratified columnar epithelium and may contain goblet cells and cilia. Accessory parotid tissue can be seen in ~ 20-50% of the population. **(Right)** Focal sebaceous cells or lobules are not uncommon in parotid gland duct epithelium. They can be seen in the majority of parotid glands with careful examination.



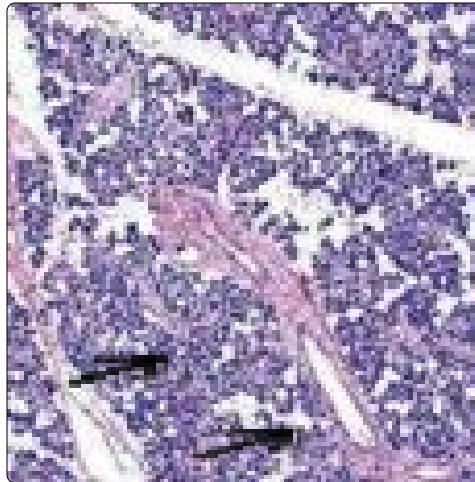
Sebaceous Glands



Parotid Gland Oncocytic Metaplasia

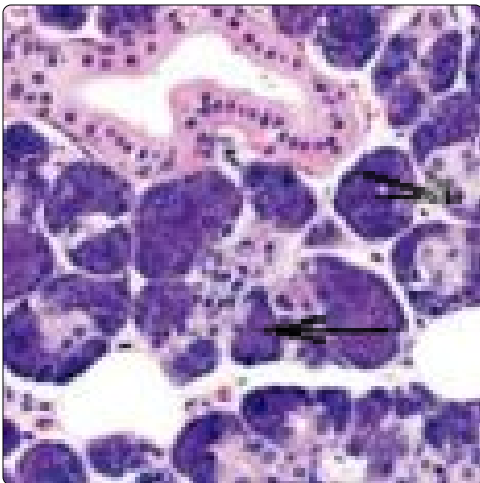


Submandibular Gland

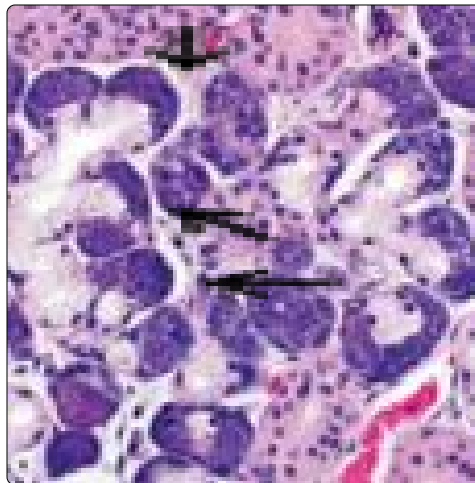


(Left) Oncocytes are metaplastic cells with granular eosinophilic cytoplasm containing numerous irregular mitochondria. They are often seen in the parotid glands of those > 50 years of age but are less common in submandibular glands. They resemble striated duct cells superficially. (Right) Lobulation is more pronounced in submandibular glands, and there is less intraparenchymal adipose tissue than in parotid glands. A central intralobular excretory duct is present.

Submandibular Ducts

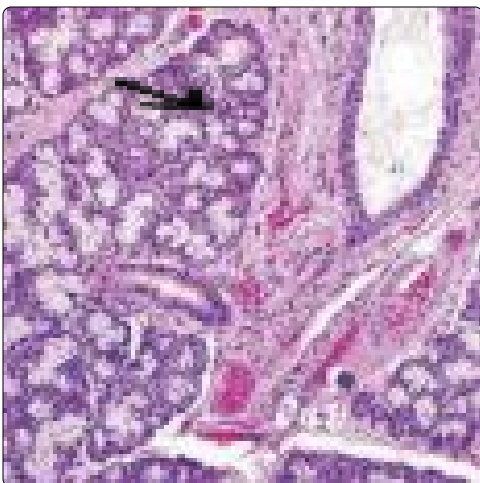


Serous Demilunes

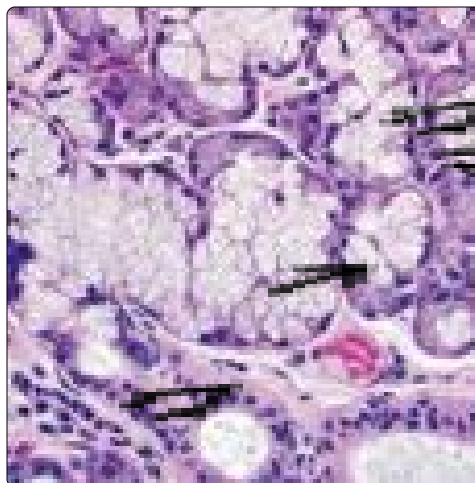


(Left) The submandibular gland is a mixed gland predominantly composed of serous acini but also containing a minor mucous acinar component. Striated ducts are longest in the submandibular gland and are easily recognizable. Intercalated ducts are shorter than in the parotid gland but are easily seen. (Right) Serous demilunes consist of tubular mucous acini capped by a crescent-shaped collection of serous cells. These have been shown to be an artifact of fixation.

Sublingual Gland



Sublingual Acini



(Left) The sublingual gland is arranged into lobules but lacks adipose tissue, unlike the other major salivary glands. Intralobular excretory ducts empty into large interlobular ducts, which are lined by a pseudostratified columnar epithelium. (Right) The sublingual gland is mixed with a predominance of tubular mucous acini, many of which are capped by serous acini. They have very short intercalated ducts that communicate with short striated ducts.

PHARYNX

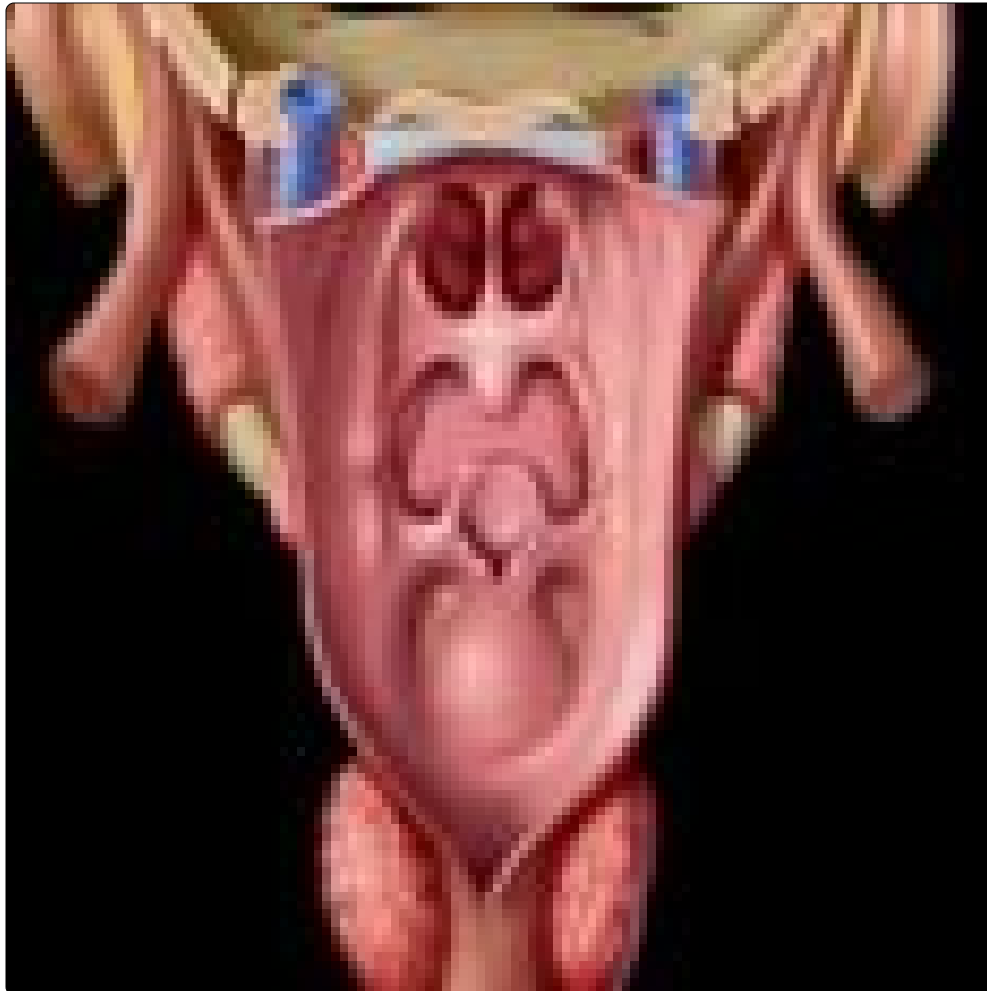
Macroscopic Anatomy

- 3 functionally and anatomically distinct subsites
 - Nasopharynx, oropharynx, and hypopharynx
- Nasopharynx
 - Extends from choanae to superior (nasal) soft palate border
 - Includes pharyngeal tonsils and posterolateral walls (with eustachian tube openings)
- Oropharynx
 - Extends from superior soft palate border to hyoid bone/valleculae
 - Includes palatine tonsils, tonsillar pillars, base of tongue, soft palate, uvula, posterolateral walls
- Hypopharynx
 - Extends from hyoid bone/valleculae to inferior cricoid cartilage border
 - Includes pyriform sinuses, postcricoid region, posterolateral walls

Microscopic Anatomy

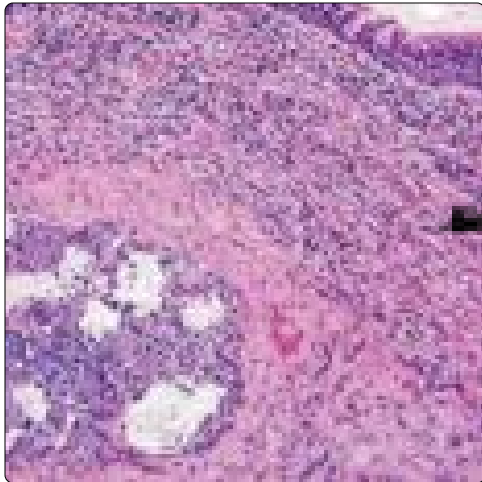
- Nasopharynx
 - Lined by both ciliated respiratory-type epithelium and stratified squamous epithelium
 - Pharyngeal tonsil consists of surface invaginations extending into lymphoid stroma
 - Scattered, less prominent lymphoid aggregates can be seen in nontonsil mucosa
 - Mixed seromucous minor salivary glands in submucosa
- Oropharynx
 - Lined by stratified squamous epithelium
 - Palatine and lingual (base of tongue) tonsils consist of lymphoepithelial crypts extending into lymphoid stroma
 - Pure mucous and scattered mixed minor salivary glands are present in submucosa
- Hypopharynx
 - Lined by stratified squamous epithelium
 - Pure mucous and scattered mixed minor salivary glands are present in submucosa
- No muscularis mucosa in any of subsites

Pharyngeal Anatomy



The nasopharynx extends from the choanae [1] to the superior soft palate [2]. The oropharynx extends from the superior soft palate to the valleculae [3]. The hypopharynx extends from the valleculae to the inferior cricoid cartilage [4] and includes the pyriform sinuses [5].

Nasopharynx

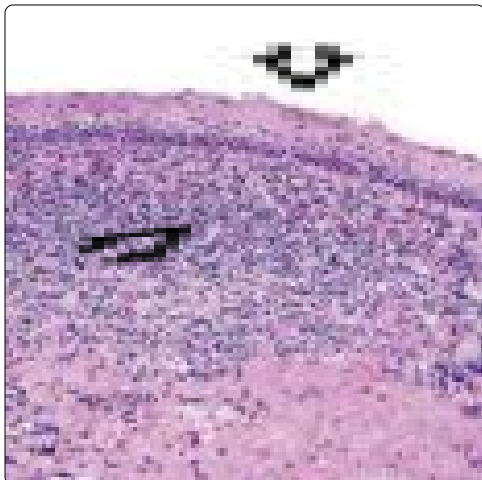


Lymphoid Aggregates



(Left) The nasopharynx is lined by both ciliated respiratory-type epithelium and stratified squamous epithelium. The pharyngeal tonsil, roof, and anterior regions are lined by respiratory-type epithelium. Numerous submucosal mixed seromucous minor salivary glands are present. (Right) Scattered submucosal (nontonsillar) lymphoid aggregates are present throughout the nasopharynx. The overlying mucosa consists of stratified squamous epithelium alternating with respiratory-type epithelium.

Oropharyngeal Mucosa

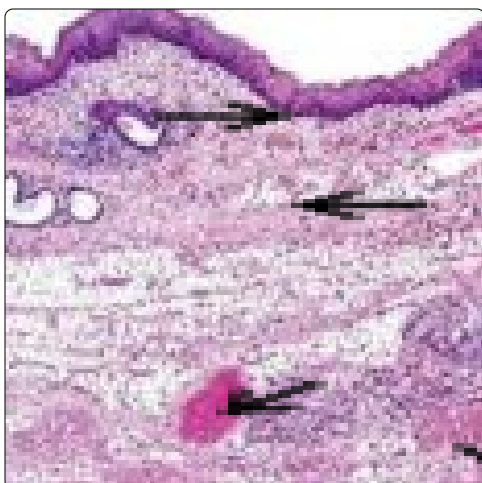


Uvula

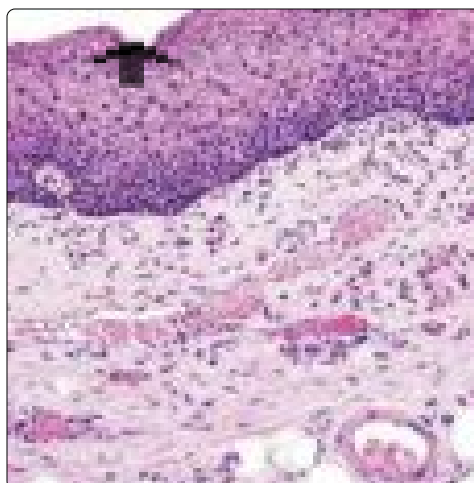


(Left) The oropharynx is lined purely by nonkeratinizing stratified squamous epithelium. Scattered submucosal (nontonsillar) lymphoid aggregates are present throughout the oropharynx. (Right) The uvula is lined by stratified squamous epithelium and contains numerous mucous salivary gland lobules with ducts. Thin bundles of skeletal muscle are present in the uvula stroma.

Hypopharynx



Hypopharyngeal Mucosa



(Left) Similar to the oropharynx, the hypopharynx is lined entirely by nonkeratinizing stratified squamous epithelium. Scattered, predominantly mucous, minor salivary gland lobules are present in the submucosa with ducts draining to the surface. Pharyngeal constrictor skeletal muscle is present below the submucosa. (Right) Although the surface typically consists of nonkeratinizing squamous mucosa, exposure to irritants, such as tobacco smoke, can result in surface keratosis.

LARYNX

Macroscopic Anatomy

- Extends from epiglottic tip superiorly to inferior border of cricoid cartilage inferiorly
 - Main functions are phonation and preventing aspiration of ingested or inhaled particles
- Composed of cartilaginous framework supported by ligaments as well as intrinsic and extrinsic muscle
 - Epiglottic, thyroid, cricoid, and arytenoid cartilages
- Covered by mucosa
- Composed of 3 regions
 - Supraglottis
 - Extends from epiglottis to superior portion of true vocal cords
 - Includes epiglottis, aryepiglottic folds, false vocal cords, and ventricles
 - Glottis
 - Extends from superior portion of true vocal cords inferiorly for 1 cm
 - Includes true vocal cords (folds), anterior commissure, and posterior commissure
 - Subglottis
 - Extends from inferior glottis to inferior border of cricoid cartilage
- Anterior commissure tendon (Broyles ligament)
 - Where vocal cord elastic tissue attaches to thyroid cartilage
 - Represents "weak" point where carcinomas can spread beyond larynx
- Preepiglottic space
 - Adipose and loose connective tissue anterior to epiglottis
- Paraglottic space
 - Adipose and loose connective tissue deep to true and false vocal cords
 - Bounded by cricovocal membrane, thyroid cartilage, quadrangular membrane, and pyriform sinus

Microscopic Anatomy

- Lined by both ciliated columnar epithelium with mucous (goblet) cells and nonkeratinized stratified squamous epithelium
 - Majority of supraglottic and glottic larynx are covered by squamous epithelium with occasional patches of ciliated columnar epithelium in supraglottis
 - Superior portion of supraglottic larynx is always squamous, whereas lower 1/2 is more likely to have patches of ciliated columnar epithelium
 - Ventricles and subglottis are always lined by ciliated columnar epithelium
- Submucosal seromucous glands are present throughout with exception of vocal cords
- Submucosa of true vocal cords is loose connective tissue without lymphatics or salivary glands and is known as Reinke space
 - True vocal cords have band of elastic tissue (vocal ligament), which sits adjacent to vocalis (skeletal) muscle
- Epiglottis is elastic cartilage and has varying numbers of fenestrations, whereas remaining laryngeal cartilages are hyaline cartilages
 - Epiglottic fenestrations and anterior commissure are weak points where laryngeal carcinomas can spread beyond larynx

CONSIDERATIONS AND VARIATIONS

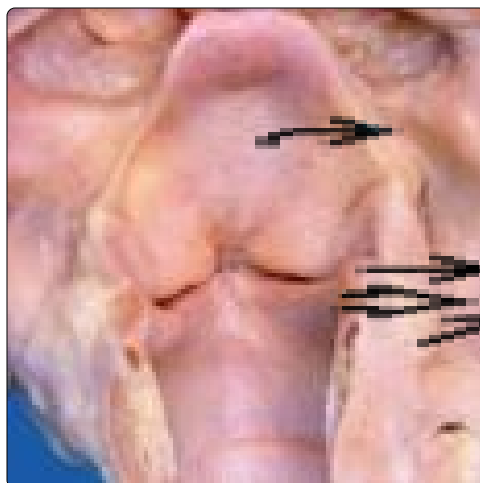
Age-Related Variations

- At birth, larynx is lined entirely by ciliated pseudostratified respiratory mucosa
- With age, it is gradually replaced by nonkeratinizing stratified squamous mucosa except for ventricle, subglottis, and rare microscopic patches in supraglottis
- Variable ossification of thyroid and cricoid cartilages

Metaplasia

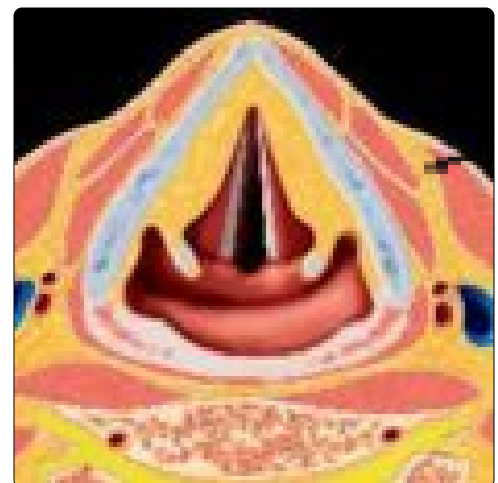
- Squamous metaplasia (common in smokers)
- Chondroid metaplasia of vocal cord ligament
- Oncocytic metaplasia of seromucous gland ducts is common over age 50

Laryngeal Anatomy



(Left) The glottis contains the true vocal cords, which are just below the ventricles. The supraglottis is also composed of the epiglottis, false vocal cords, and aryepiglottic folds. (Right) The vocal cord ligaments, as well as their attachment at the anterior commissure, can be seen in this graphic. The paraglottic space is between the thyroid cartilage and vocal cords.

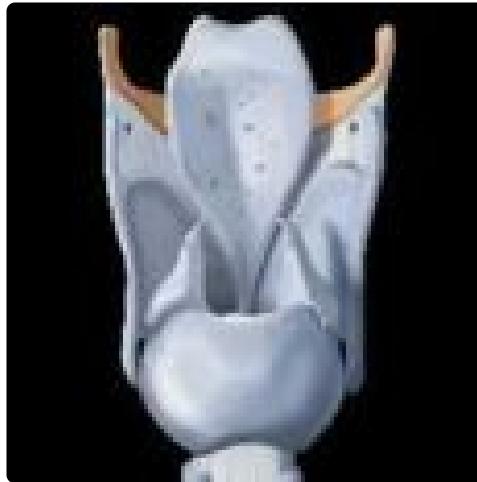
Vocal Cords and Paraglottic Space



Laryngeal Cartilages (Anterior)

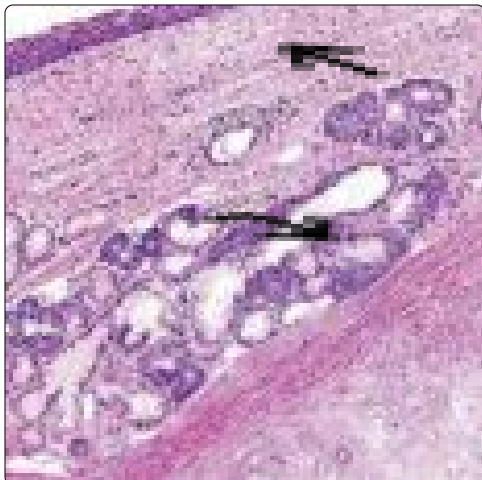


Laryngeal Cartilages (Posterior)

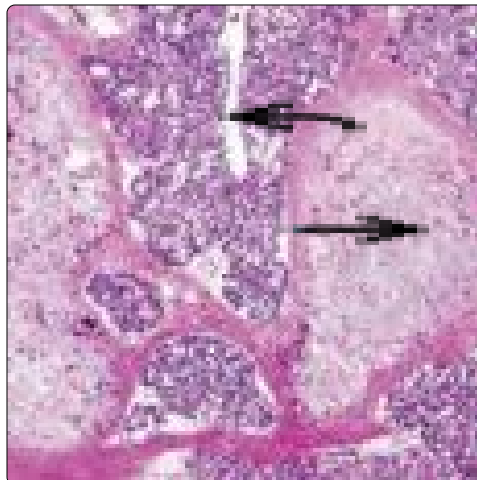


(Left) The main laryngeal cartilages are the epiglottic, thyroid, and cricoid cartilages. The thyroid cartilage is broad anteriorly but does not form a complete ring. There are superior and inferior horns and a notch just above the laryngeal prominence ("Adam's apple"). The cricoid cartilage forms a complete ring, being narrower anteriorly. (Right) The epiglottis has small holes (fenestrations), and the arytenoids sit on the posterior cricoid cartilage.

Epiglottis

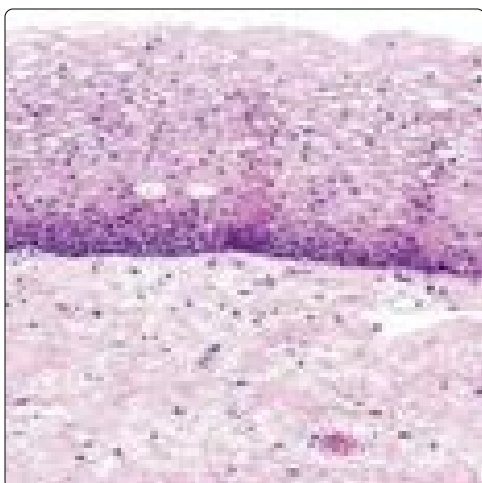


Epiglottic Fenestrations

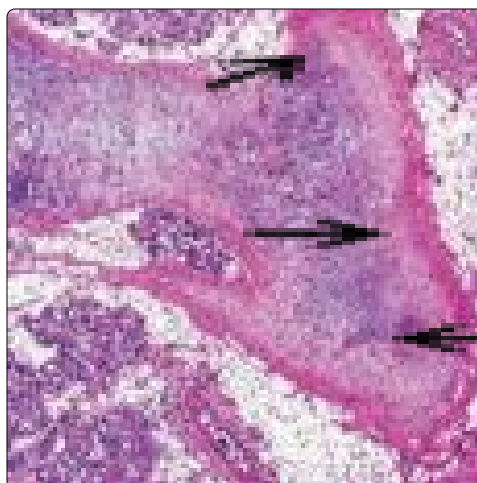


(Left) The epiglottis in adults is typically lined by stratified squamous epithelium, but patches of ciliated columnar epithelium may be present as well. Numerous submucosal seromucous glands are present in the supraglottic larynx and typically are situated near the epiglottic cartilage. (Right) The epiglottis is elastic cartilage and has fenestrations, which usually contain seromucous glands, and may serve as a conduit for carcinomas to enter the preepiglottic space.

Aryepiglottic Fold



Arytenoid Cartilage



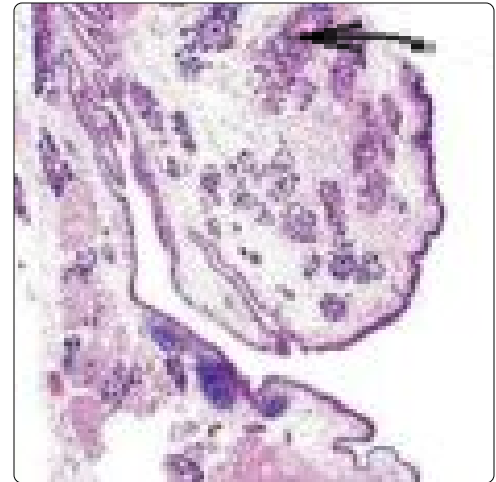
(Left) The aryepiglottic folds are part of the supraglottic larynx and extend from the epiglottis to the arytenoid cartilages. They are usually lined by stratified squamous epithelium without keratinization and have less numerous minor salivary glands in the subepithelial stroma. They represent one of the mucosal margins in laryngectomy specimens. (Right) The arytenoid cartilages are small hyaline cartilages associated with mucoserous glands and are important in phonation.

Vocal Cords

(Left) The true vocal cord is lined by squamous epithelium and is devoid of mucoserous glands. The false vocal cord is lined by squamous or respiratory epithelium and has mucoserous glands. The ventricle is lined by respiratory epithelium and also contains mucoserous glands. (Right) The anterior portion of the ventricle contains the saccule, a segment that extends caudally and can result in cystic dilation if obstructed.



Ventricle and Saccule

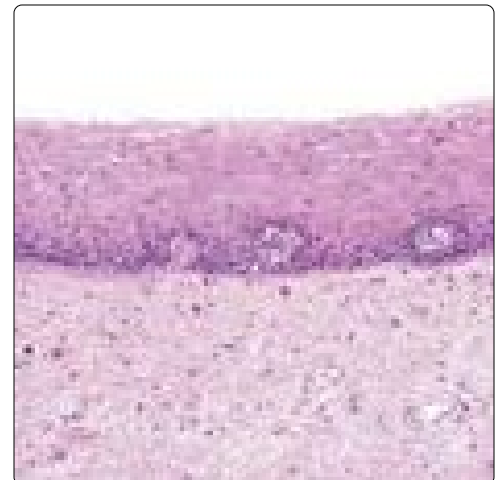


Reinke Space

(Left) True vocal cords contain the vocalis muscle. The lamina propria between the vocalis muscle and surface epithelium (Reinke space) contains relatively few capillaries and lacks lymphatic channels. This explains the better prognosis of low-stage glottic carcinomas and the etiology of vocal cord nodules. (Right) The true vocal cords are lined by nonkeratinizing squamous epithelium, ~ 5-20 cells thick. Voice abuse and irritant exposure can lead to keratosis.

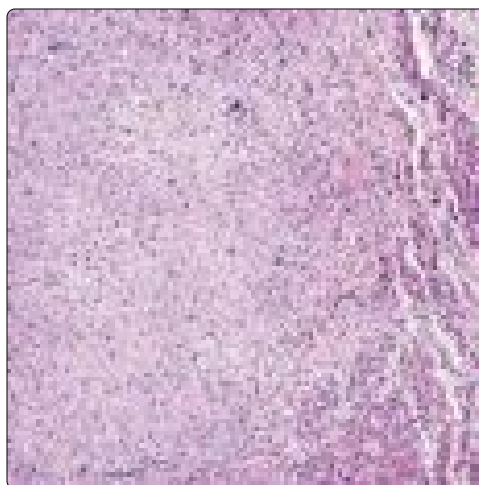


True Vocal Cord

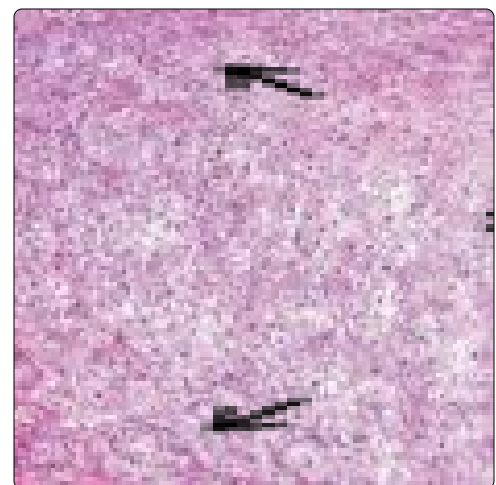


Vocal Cord Ligament

(Left) The vocalis muscle consists of a band of connective tissue rich in elastic fibers and often can be seen adjacent to the vocal ligament. (Right) Chondroid metaplasia of the larynx is not uncommon and usually occurs in the posterior 2/3 of the vocal cord ligament. In contrast to true cartilage, the periphery is not sharply demarcated, and elastic fibers can be seen extending into the metaplastic focus. Chondroid metaplasia can be seen in other parts of the larynx as well.



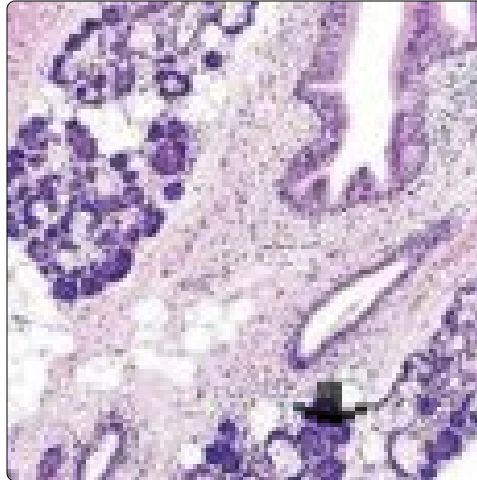
Cartilaginous Metaplasia of Vocal Cord Ligament



Lymphoid Aggregate

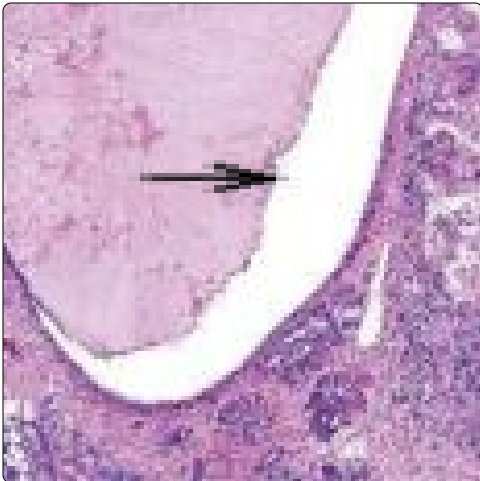


Sacculle

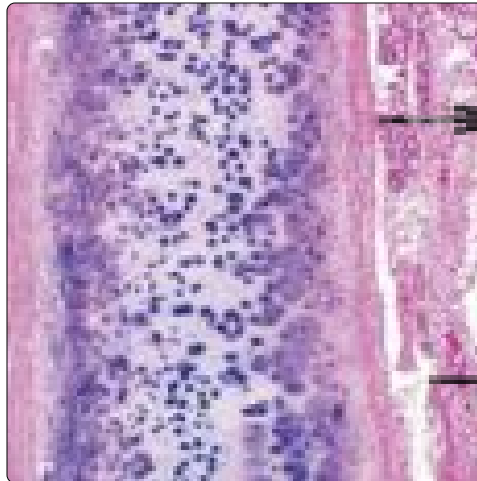


(Left) Lymphoid aggregates can be seen in the larynx but are uncommon. When present, they are typically seen in the supraglottic larynx. (Right) The laryngeal saccules represent caudal extensions of the ventricles and are normal. They are lined by respiratory epithelium and contain numerous mucoserous glands that drain to the mucosal surface via small ducts. Both the saccule and the ducts can become obstructed, leading to saccular and ductal cysts, respectively.

Saccular Cyst



Thyroid Cartilage

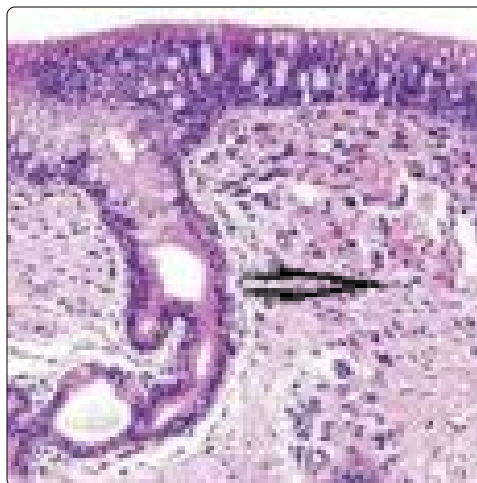


(Left) Saccular cysts represent cystic dilation of the laryngeal saccule; they present in the supraglottis and histologically are respiratory epithelium-lined cysts containing mucin. They usually result from obstruction by invasive squamous cell carcinomas. (Right) In contrast to the epiglottis, most of the other laryngeal cartilages are composed of hyaline cartilage with an adjacent perichondrium. The paraglottic space can be seen adjacent to this thyroid cartilage.

Ossification of Thyroid Cartilage



Subglottis



(Left) The thyroid and cricoid cartilages of most adult larynges undergo extensive ossification by adulthood. This is an important oncologic point, as nonossified cartilage is typically an excellent barrier to invasion by carcinomas. Once ossified, it is easier for carcinomas to breach the laryngeal cartilages and subsequently spread beyond the larynx. (Right) The subglottis is lined by respiratory mucosa with mucoserous glands that drain to the surface via small ducts.

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SECTION 8

Respiratory System



Trachea	196
Lung	198
Pleura and Pericardium	206

Trachea

TRACHEA

Macroscopic Anatomy

- Tubular structure designed solely for air conduction
 - a.k.a. windpipe
 - ~ 10-12 cm long
 - ~ 2.0-2.5 cm in diameter
- Extends from cricoid cartilage to carina
 - Superior border contiguous with larynx
 - Branches distally at carina to become mainstem bronchi
- Runs directly anterior to esophagus within mediastinum
- Protected anteriorly by rings of flexible cartilage
 - 15-20 rings
 - C- or U-shaped
 - Prevents collapse or obstruction of airway
- Posterior aspect is flat and devoid of cartilage
 - Instead contains trachealis muscle and membranous fibroelastic connective tissue
 - Flexible connective tissue allows for expansion of adjacent esophagus during consumption of food and liquid

Microscopic Anatomy

- 4 main layers (superficial to deep)
 - Mucosa
 - Respiratory epithelium
 - Basement membrane
 - Lamina propria
 - Connective tissue
 - Elastin fibers
 - Submucosa
 - Serous and mucous glands
 - Similar to those seen in major and minor salivary glands and esophagus
 - Serous glands have basophilic, granular cytoplasm
 - Mucous glands have clear, granular cytoplasm
 - Blood vessels
 - Nerve fibers
 - Cartilaginous rings
 - Composed of hyaline cartilage

- Adventitia
 - Loose connective tissue sheath
 - Binds trachea to adjacent structures in mediastinum
- Respiratory epithelium is composed of pseudostratified columnar epithelium
 - Characteristically ciliated
 - Goblet cells are common
- Seromucous glands connect to luminal surface of trachea via ducts lined by cuboidal epithelium
 - Glands are more numerous in posterior trachea
- Function of epithelium and seromucous glands is to warm and filter conducted air
 - Mucus from goblet cells and glands traps particles
 - No gas exchange occurs
- Smooth (nonstriated) muscle of trachealis present in posterior trachea

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Hyaline cartilage rings may calcify in elderly

Metaplasia

- Squamous metaplasia of surface epithelium
 - Associated with smoking, chronic irritation
- Squamous metaplasia of seromucinous ducts and acini
- Oncocytic metaplasia of seromucinous ducts and acini
 - More common in older individuals

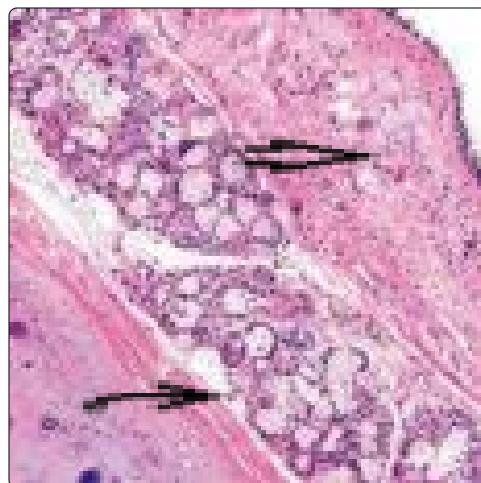
Pitfalls and Artifacts

- Cartilaginous rings can appear highly cellular
- Numerous seromucous glands in posterior trachea can mimic neoplasm

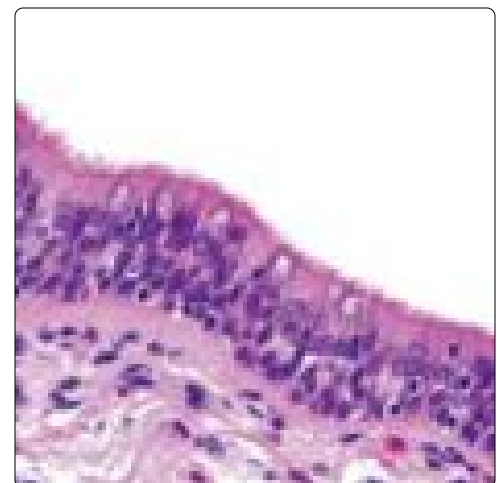
Tracheal Anatomic Layers

(Left) This image shows the surface epithelium [E], underlying lamina propria, submucosal seromucous glands [G], and cartilage ring of the trachea [C]. The outer adventitia is not shown.

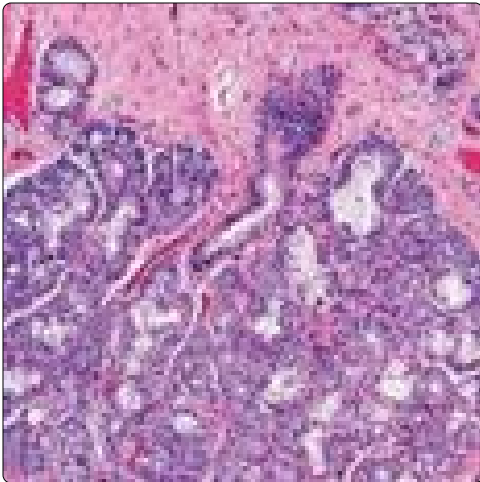
(Right) The trachea is lined by a pseudostratified columnar epithelium containing numerous apical cilia [C] and scattered goblet cells [G]. This epithelium is also referred to as respiratory-type epithelium. The pink basement membrane [E] separates the epithelium from the underlying lamina propria.



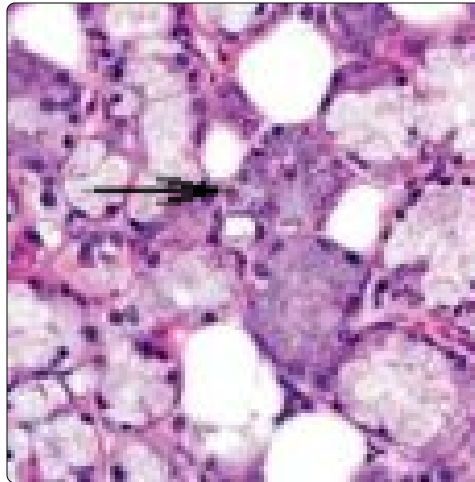
Pseudostratified Ciliated Columnar Epithelium



Seromucous Glands and Ducts

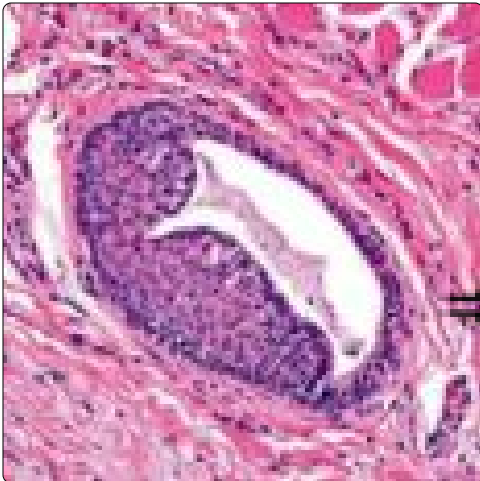


Seromucous Glands

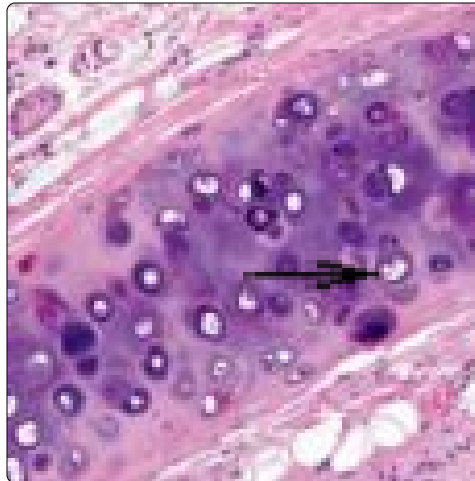


(Left) The submucosal glands of the trachea are very similar to those found in the normal salivary gland. The glands are arranged in lobules, and each lobule is associated with a duct that conveys the glandular secretions to the luminal surface. **(Right)** Within a lobule, each acinus (cluster of cells) is composed of a variable mixture of serous cells and mucous cells. The serous cells are usually easily identified by their basophilic, granular cytoplasm.

Squamous Metaplasia

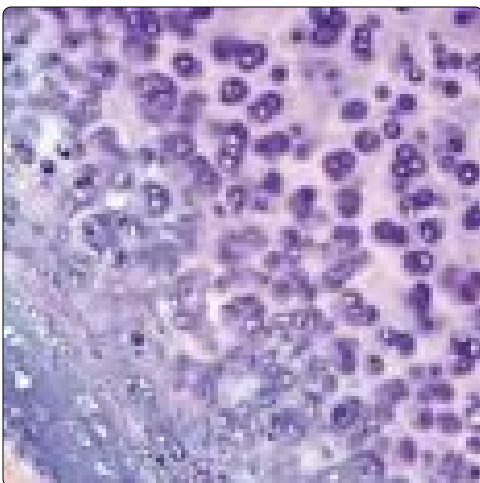


Cartilage Ring

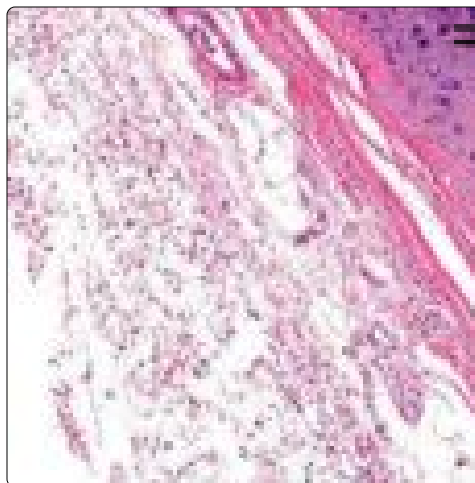


(Left) Similar to the columnar surface epithelium, the epithelial lining (flat cuboidal) of the glandular ducts will occasionally show features of squamous metaplasia, particularly in the setting of chronic irritation. **(Right)** The cartilage rings that protect the trachea anteriorly are found immediately deep to the submucosa. Each demonstrates a fibrous perichondrium and contains numerous chondrocytes. The rings are absent posteriorly where they are bridged by fibroelastic connective tissue.

Hyaline Cartilage



Adventitia



(Left) The cartilage rings are composed of hyaline cartilage, which is the most common type of cartilage in the body. On a routine H&E stain, this hyaline cartilage can appear blue, purple, or even pink. **(Right)** The adventitia is loose connective tissue that invests the external surface of the trachea. Similar to the submucosa, it contains fibroblasts, nerve twigs, adipose tissue, and small vessels. There is no mesothelial lining.

LUNG

Macroscopic Anatomy

- Airways
 - Bronchi (> 1 cm in diameter) connect trachea to bronchioles within hilum of lung
 - Bronchioles (usually < 1 mm in diameter) with extensive distal dichotomous branching
 - Terminal bronchioles (< 0.5 mm in diameter) connect proximal bronchioles with respiratory bronchioles
 - Respiratory bronchioles are connected directly to alveolar ducts, alveolar sacs, and alveoli
 - Alveoli: Sites of gas exchange
- Right lung
 - 3 lobes: Upper, middle, and lower
 - Each lobe divided into 10 bronchopulmonary segments (lobules)
- Left lung
 - 2 lobes: Upper and lower
 - Presence of lingula (no true middle lobe)
 - Each lobe divided into 9 bronchopulmonary segments (lobules)
- Vasculature
 - Pulmonary arteries provide deoxygenated venous blood from systemic circulation to alveoli for oxygenation
 - Pulmonary veins transport oxygenated blood from alveoli to systemic circulation
 - Bronchial arteries provide oxygenated systemic blood to parenchyma of lung and visceral pleura
 - Bronchial veins remove deoxygenated blood from parenchyma of lung and visceral pleura
- Pleura
 - Visceral pleura invests periphery of each pulmonary lobe
 - Parietal pleura lines inner chest wall
- Lymphatics
 - Rich lymphatic network drains toward hilum

Microscopic Anatomy

- Airways
 - Ciliated pseudocolumnar epithelium with goblet cells

- Submucosal seromucous glands
- Smooth muscle present in walls of airways
- Bronchi have protective anterior cartilaginous rings
- Lobar and terminal bronchioles lack cartilaginous rings
- Respiratory bronchioles and alveolar structures are not lined by pseudocolumnar epithelium
- Parenchyma
 - Bronchopulmonary segments contain central lobar bronchiole and are delineated from adjacent segments by interlobular septa
 - Alveolar tissue is composed of type 1 pneumocytes, type 2 pneumocytes, and alveolar capillaries
 - Alveolar macrophages remove alveolar debris
- Vasculature
 - Lobar bronchioles are paired with pulmonary arteries and contain bronchial arteries in their walls
 - Pulmonary vein branches are in interlobular septa and are not paired with arteries or lobar bronchi
 - Lymphatic channels can be found adjacent to lobar arterioles as well as interlobular septal venules
 - In general, arteries contain double- (inner and outer) elastic lamina, and veins contain single outer lamina
- Visceral pleura
 - Composed of connective tissue and elastic fibers and lined by layer of mesothelial cells

PITFALLS/ARTIFACTS

Age-Related Variations

- Bronchial cartilage may calcify in old age

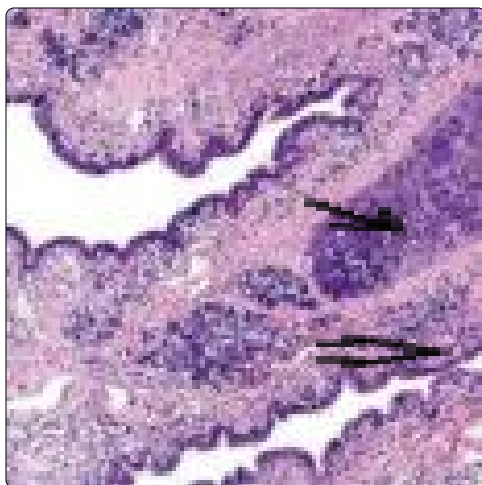
Metaplasia

- Peribronchial metaplasia
- Metaplastic ossification

Pitfalls and Artifacts

- Hemorrhage related to surgical procedure
- Insufficient inflation prior to tissue processing
- Carbon pigment (anthracosis)
- Corpora amylacea, megakaryocytes, bone marrow emboli
- Minute meningothelial-like nodules

Bronchus

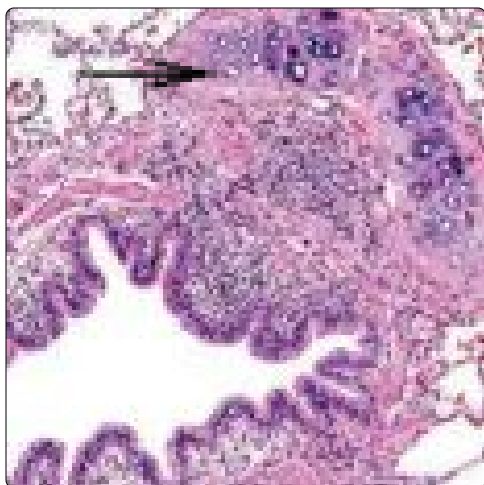


Bronchi and Bronchioles

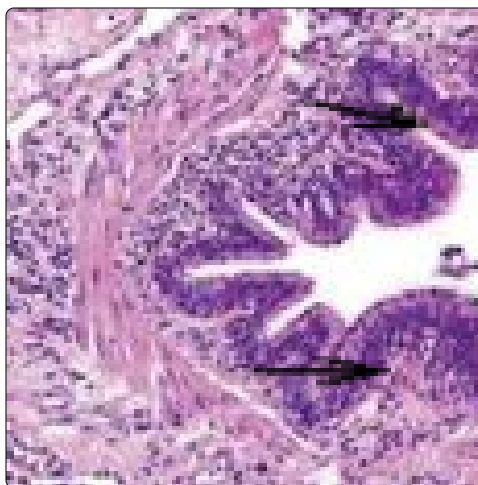


(Left) Akin to the trachea, a bronchus is composed of surface epithelium [E] with underlying seromucous glands (for lubrication) [F] and C-shaped rings of hyaline cartilage (for structural support) [G]. **(Right)** The bronchi and bronchioles also contain a thin layer of smooth muscle [H] that allows for bronchoconstriction. An artery [I] can also be identified in this field.

Main Bronchi

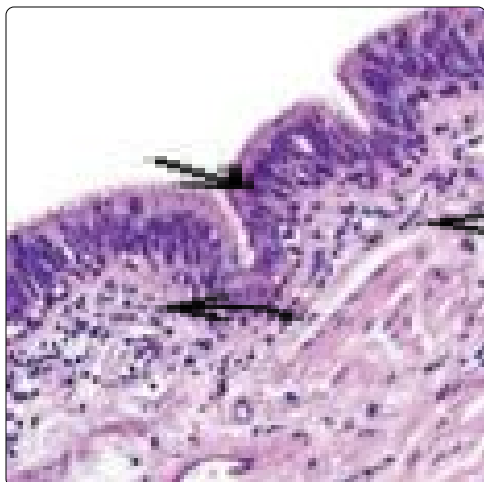


Lobar Bronchioles

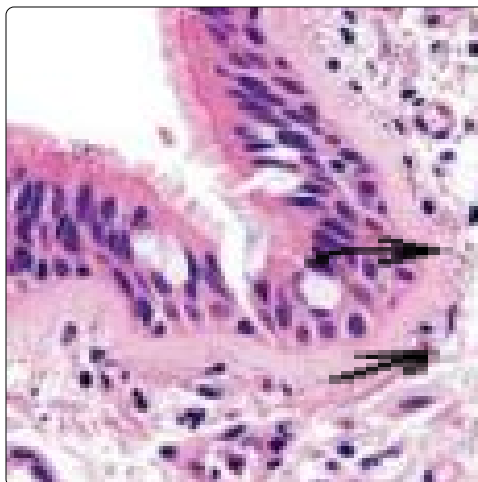


(Left) This image shows a main bronchus as it enters the lung parenchyma (note the surrounding alveoli [1]). It also serves to demonstrate how the protective cartilaginous rings [2] are not completely circumferential and only protect the anterior aspect of the large airways. (Right) Unlike the large main bronchi, the lobar bronchi do not contain cartilaginous rings. They do, however, contain a ring of smooth muscle [3] that helps constrict the airway for protective purposes.

Ciliated Pseudostratified Columnar Epithelium

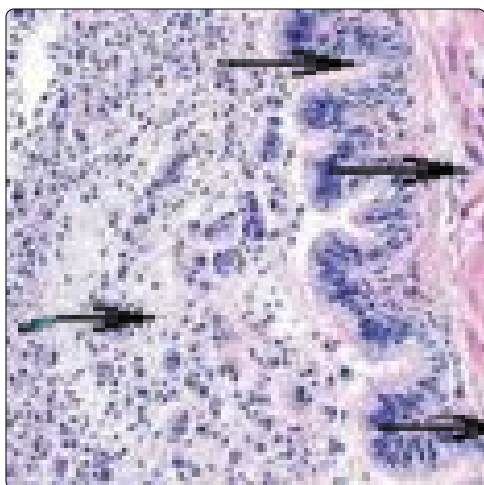


Asthma

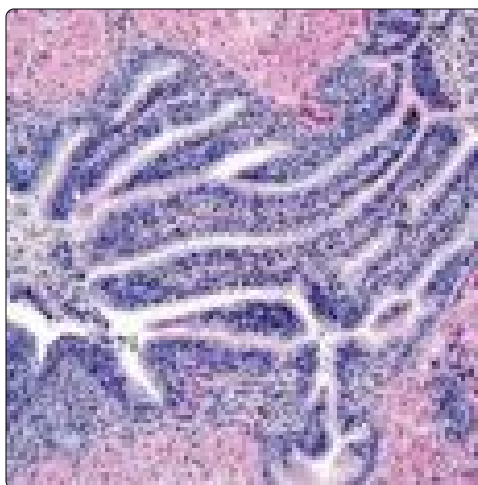


(Left) The respiratory lining is composed of pseudostratified columnar epithelium with prominent apical cilia [1] and goblet cells [2]. This epithelium rests on a thin, pink basement membrane [3], which separates the epithelium from the underlying lamina propria. (Right) In patients with asthma, the basement membrane [4] is often markedly thickened. Other changes include increased numbers of goblet cells [5] as well as mucosal eosinophils [6].

Denuded Respiratory Epithelial Cells



Tangential Section of Bronchus

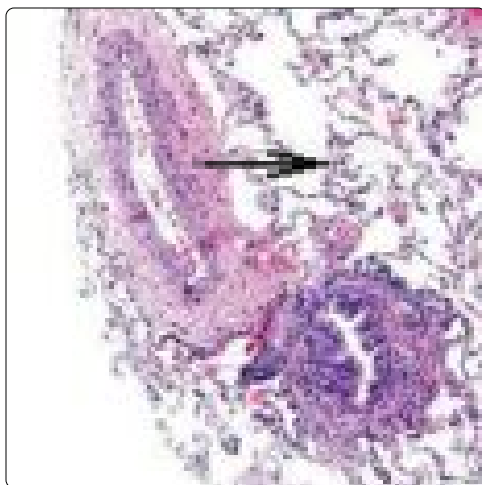


(Left) Otherwise normal ciliated epithelial cells [1] may be shed out into the bronchial lumina secondary to trauma or rough specimen handling. These denuded cells are often associated with mucin [2] or mucus. (Right) A normal bronchus (or bronchiole) can show unusual patterns when sectioned tangentially, including a stratified papillary appearance, as seen in this image.

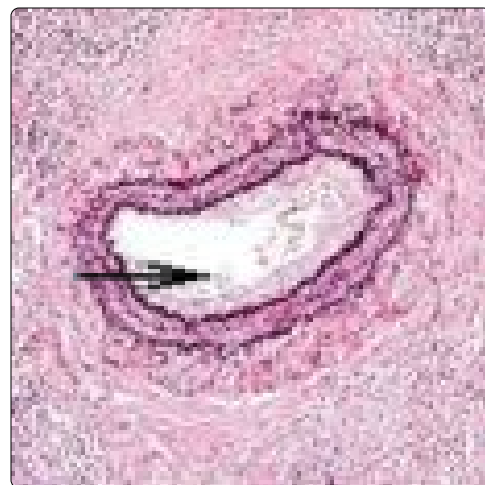
Lung

Lobar Bronchioles and Arteries

(Left) Lobar bronchioles are paired with an adjacent pulmonary arteriole. These medium-sized vessels further ramify within the lung parenchyma and show up as small isolated vessels unaccompanied by an airway. It can be difficult to tell a small venule from an arteriole. **(Right)** Both arteries and veins contain elastic lamina within their walls; however, the larger arteries show both inner and outer lamina, as highlighted in black by an elastin stain.

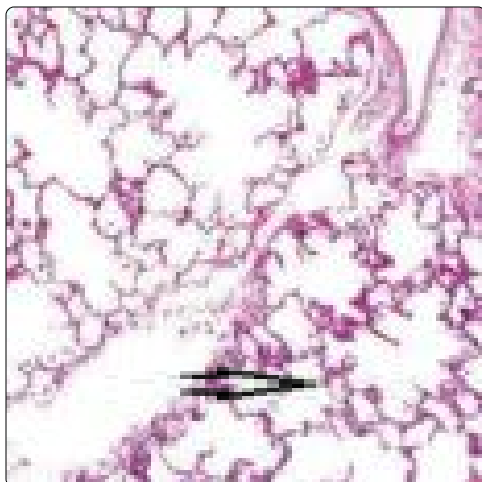


Vasculature

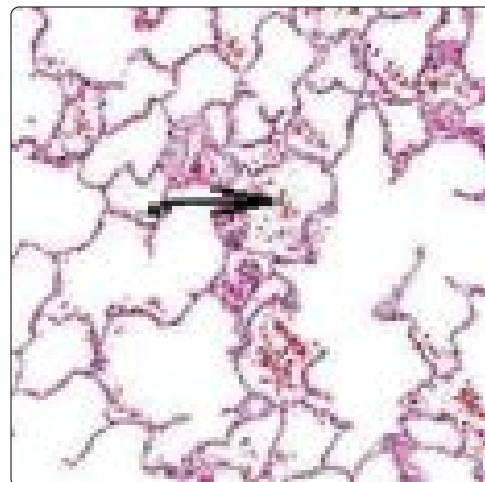


Interlobular Septa

(Left) The interlobular septa are composed of loose connective tissue and define the anatomic boundary between lobules. Pulmonary veins and lymphatic channels course within these septa. **(Right)** Respiratory bronchioles transfer air into elongated alveolar ducts, which are in direct communication with multiple alveoli. Note that in the normal lung, the alveolar walls are very thin and form a relatively intact mesh-like network.

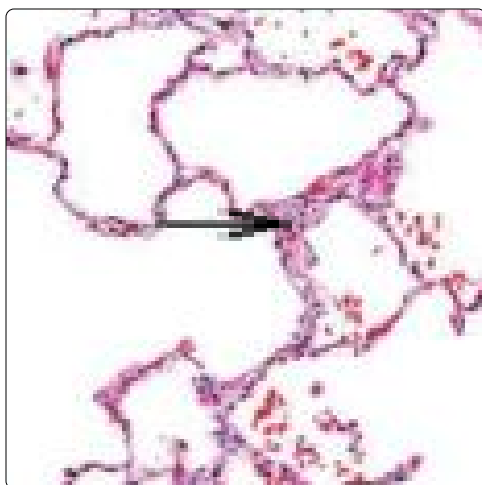


Alveoli

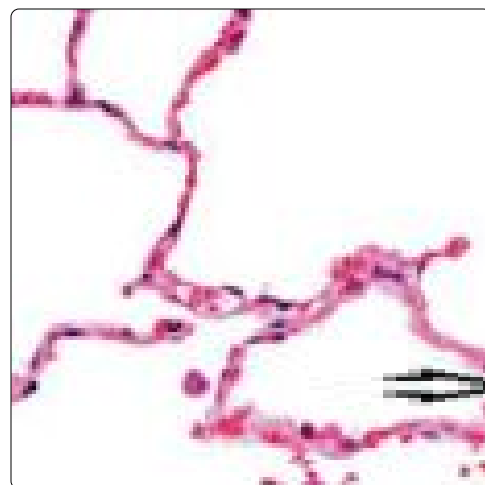


Alveolar Epithelium

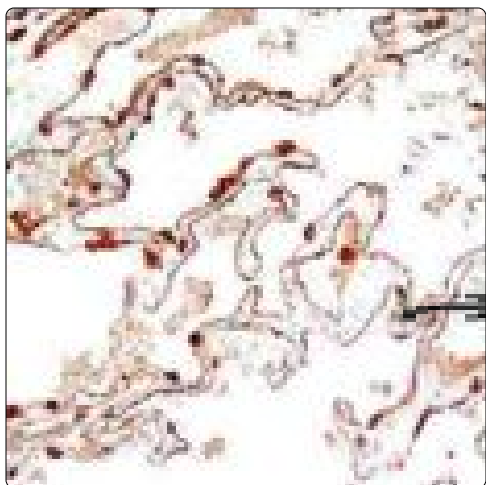
(Left) Alveolar epithelium is composed of capillaries (identified by their lumina) and types 1 and 2 pneumocytes, which can be difficult to see on routine H&E stain. Tiny alveolar macrophages are not uncommon. **(Right)** Even at very high magnification, it can be difficult to identify pneumocytes and distinguish them from endothelial cells. Capillaries, however, are usually easily identified by their open lumina. Note the single alveolar macrophage.



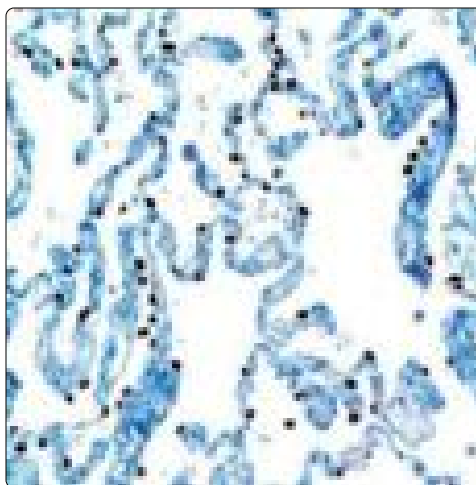
Alveolar Epithelium



Pneumocytes

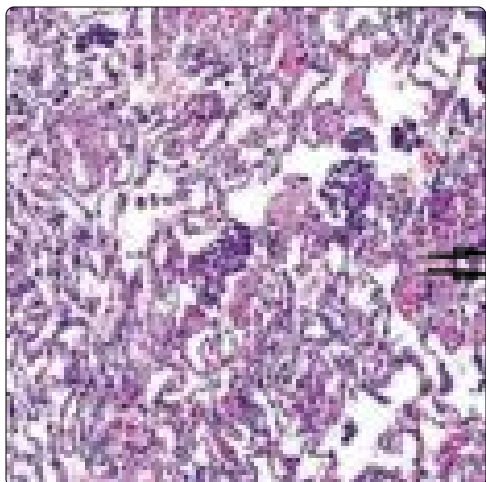


Pneumocytes

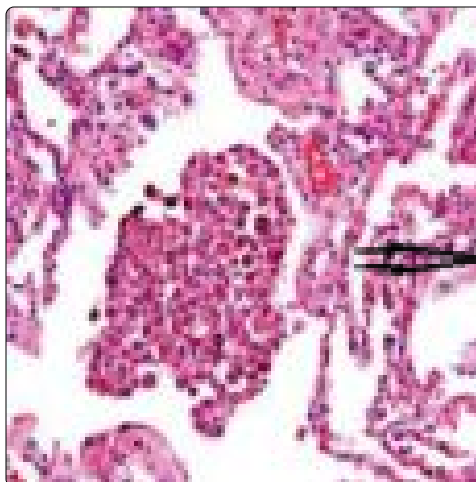


(Left) Pancytokeratin immunostain nicely highlights the pneumocytes [1] lining the pulmonary alveoli. Note that the pulmonary macrophages [2] are negative for this marker. **(Right)** TTF-1 immunostain highlights the nuclei [3] of the alveolar pneumocytes. The endothelial cells of the capillaries [4] and the alveolar macrophages [5] do not stain.

Alveolar Macrophages

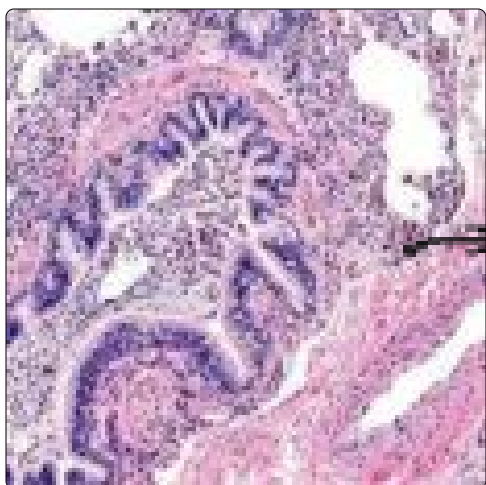


Pigmented Macrophages

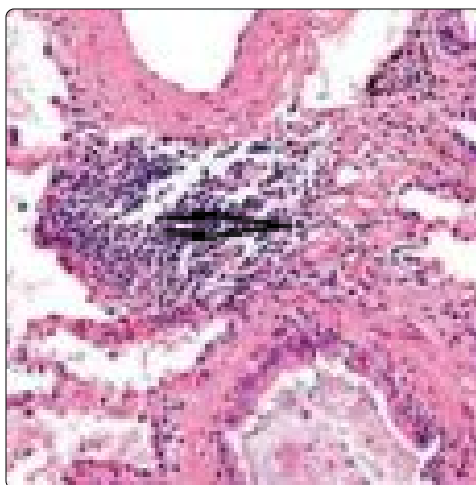


(Left) Although alveolar macrophages are typically scant in normal lung, they can be increased [1] in patients who smoke. Under these circumstances, they often form large aggregates that fill the alveoli. Sometimes they can appear multinucleated. **(Right)** In smokers, the aggregates of macrophages often show a fine, light brown pigmentation [2]. In contrast, iron (hemosiderin) pigment, such as that seen in true alveolar hemorrhage, is bright golden-brown and chunky.

Respiratory Bronchiolitis



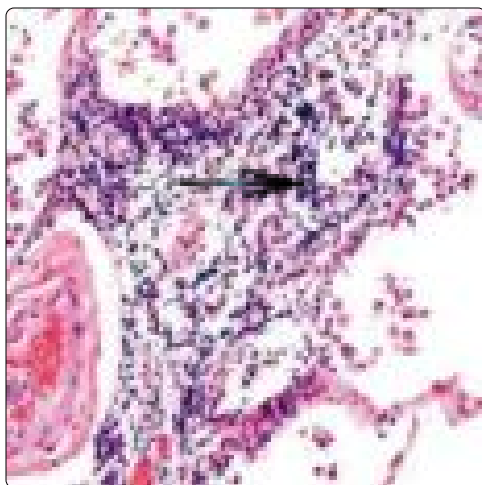
Bronchial-Associated Lymphoid Tissue



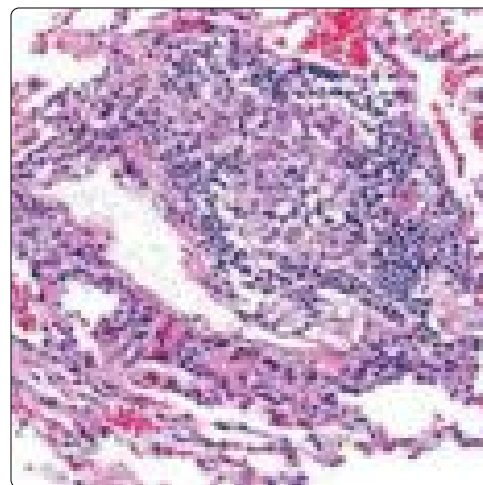
(Left) Respiratory bronchiolitis is characterized by pigmented macrophages within bronchiolar lumina [1], often with extension into peribronchiolar alveoli [2]. Although not a normal finding per se, it is a common finding in lung specimens and is associated with smoking. **(Right)** Similar to the GI tract, the lung has small aggregates of lymphoid tissue [1] associated with the bronchioles [2]. They can be enlarged in infectious or inflammatory states and may feature reactive germinal centers.

Bronchial-Associated Lymphoid Tissue Crush Artifact

(Left) Lymphocytes are somewhat fragile and are easily crushed during a biopsy procedure. This can give them a spindled (elongated) appearance that may be mistaken for a neoplasm or fungus. **(Right)** Small clusters or aggregates of benign, reactive histiocytes may be seen within bronchial-associated lymphoid tissue, often within the setting of lung neoplasia. In isolation, this finding should not be mistaken for sarcoidosis or hypersensitivity pneumonia.

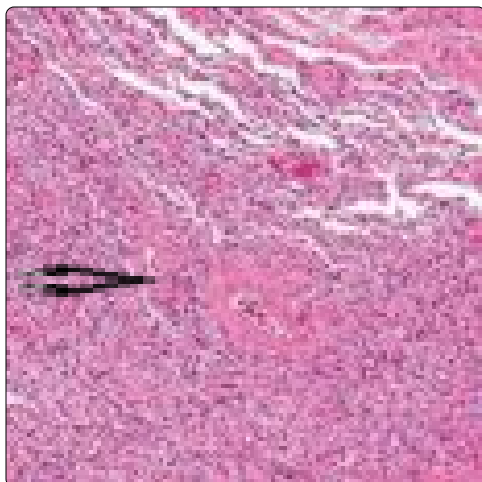


Small Histiocyte Clusters

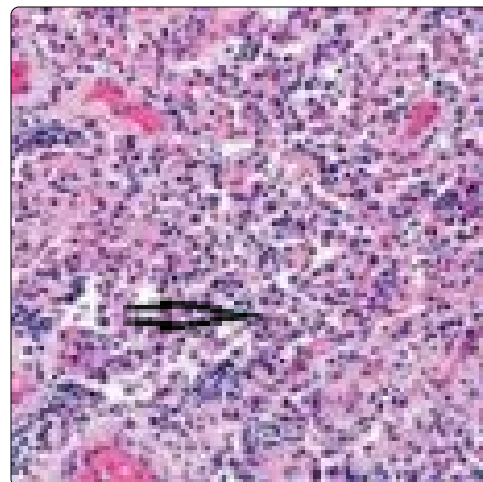


Atelectasis

(Left) Atelectasis is a collapse of the lung parenchyma. It can be real or artifactual, but it is most commonly due to insufficient inflation of the lung specimen prior to fixation. This impression of increased cellularity can lead to the mistaken notion of fibrotic lung disease. **(Right)** Even when artifactual atelectasis is significant, the normal lung architecture, although compressed, can often still be appreciated. Note the macrophages in the alveoli.

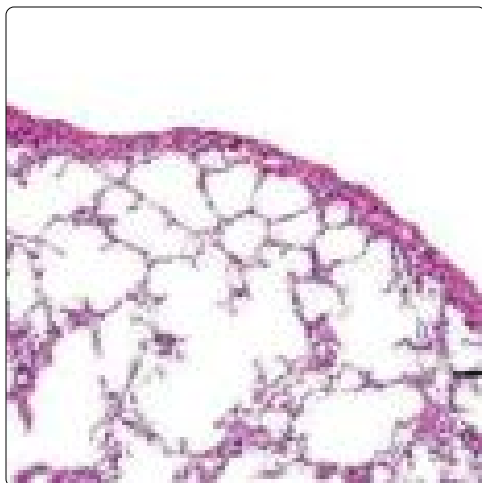


Atelectasis

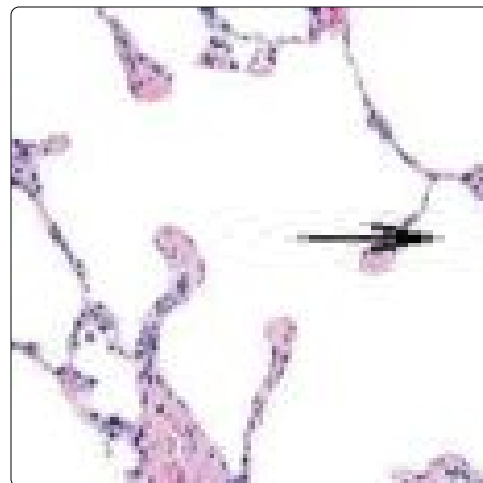


Peripheral Emphysematous Changes

(Left) A common finding in smokers is emphysema, which is characterized by alveolar wall destruction and gives the impression of enlarging alveolar spaces. It is most common at the periphery of the lung, near the pleura. **(Right)** The appearance of enlarged alveolar spaces is due to loss of alveolar walls. Formerly intact walls often show small fibrous nubs at the damaged end.



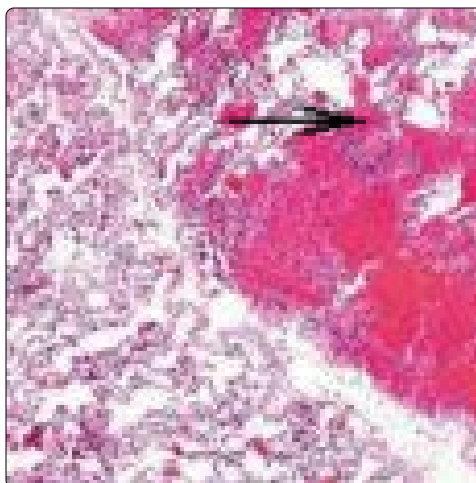
Emphysema



Visceral Pleura

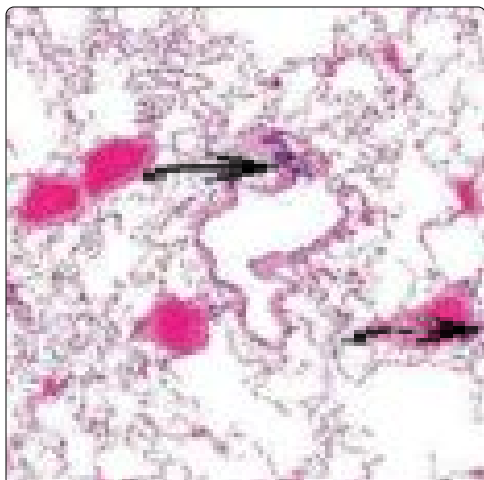


Artifactual Hemorrhage

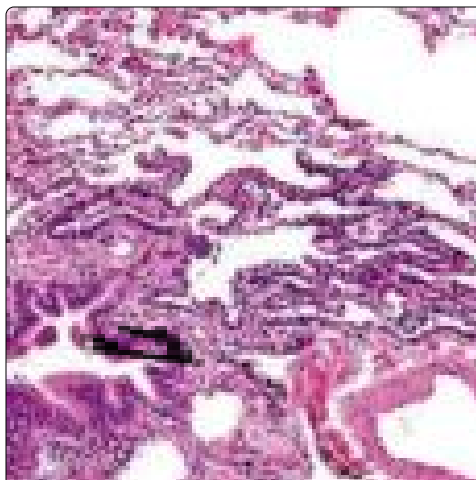


(Left) The periphery of each lobe is invested in a rind of fibroconnective tissue and elastin, called the visceral pleura. The external surface is lined by a thin layer of mesothelial cells; however, this lining is not always entirely visible in routine histologic sections. **(Right)** Blood is often seen if the lung is injured during the biopsy/resection process. Unlike true hemorrhage, artifactual (procedure-related) bleeding is patchy and usually does not cross interlobular septa.

Congested Blood Vessels

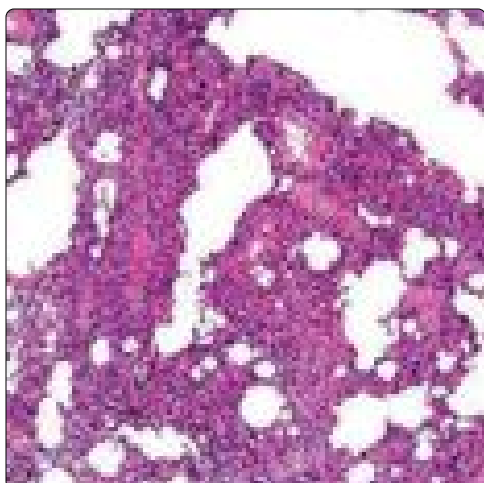


Peribronchiolar Metaplasia

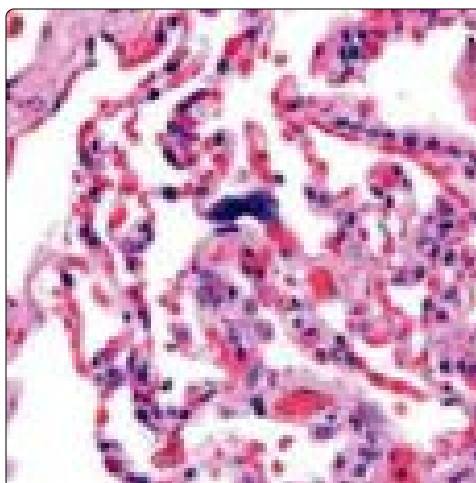


(Left) Blood vessels may appear congested in lung excision or resections, particularly within the postmortem setting. Importantly, there is no fibrin or organization within these vessels that would suggest a thrombus or embolus. **(Right)** Peribronchiolar metaplasia is due to chronic bronchiolar irritation, such as with cigarette smoking. It is characterized by metaplastic cuboidal cells that replace the normally flat alveolar epithelium that is immediately adjacent to bronchioles.

Pseudolipoid Artifact



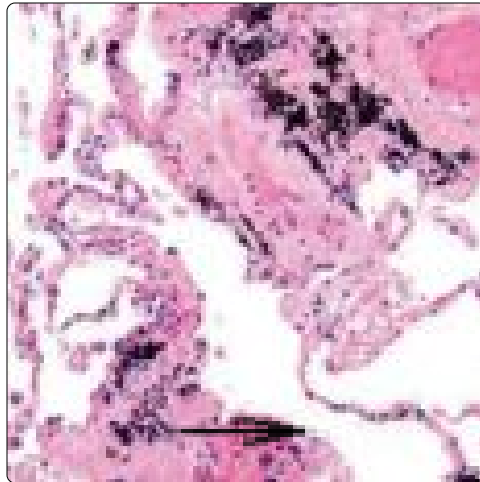
Megakaryocytes



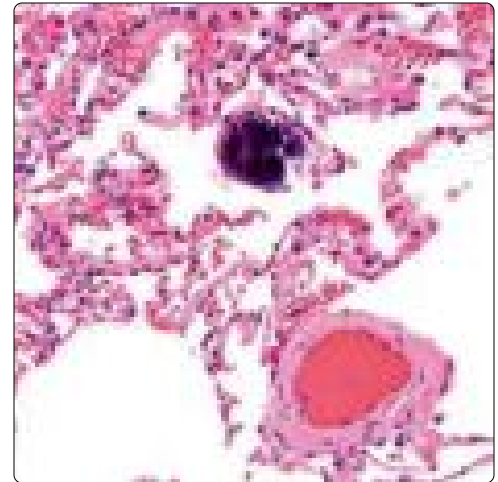
(Left) Compression of air within a lung specimen may lead to the artifactual formation of large, round spaces resembling fat within the tissue. Unlike in cases of lipid aspiration, this artifact shows an absence of multinucleated giant cells around the spaces. **(Right)** Megakaryocytes within alveolar capillary spaces can be an incidental finding. They are typically large, dark, elongated, and irregularly shaped. They should not be confused with neoplastic cells.

Anthraxosis

(Left) A very common finding in lung specimens is the presence of dark carbon (anthracotic) pigment within the lung parenchyma, histiocytes, and lymph node sinuses. It is usually most prominent in smokers but can be seen in any city dweller. **(Right)** Foci of calcification can be seen in pathologic conditions or with increasing age and may or may not be a significant finding. They are commonly associated with old hyalinized granulomas.

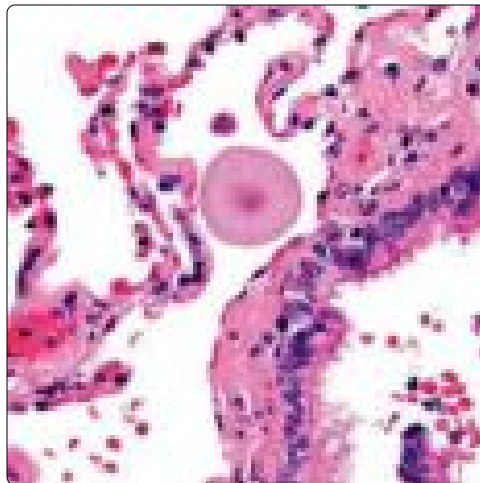


Calcification

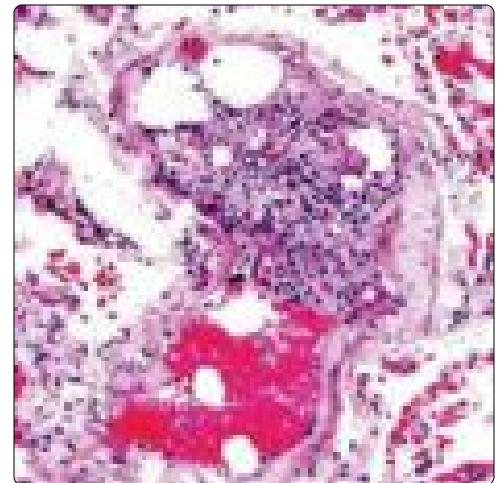


Corpora Amylacea

(Left) Similar to what is seen in the prostate, corpora amylacea in the lung are concentrically lamellated pink structures occasionally found within alveoli, particularly in older adults. Their etiology is unclear, and they have no clinical significance. **(Right)** Occasionally, bone marrow tissue may be identified within a pulmonary blood vessel. This finding is common at autopsy and suggests prior bone manipulation (e.g., chest compressions).

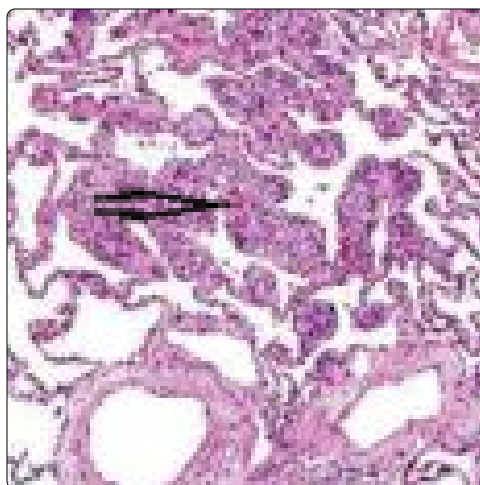


Bone Marrow Embolus

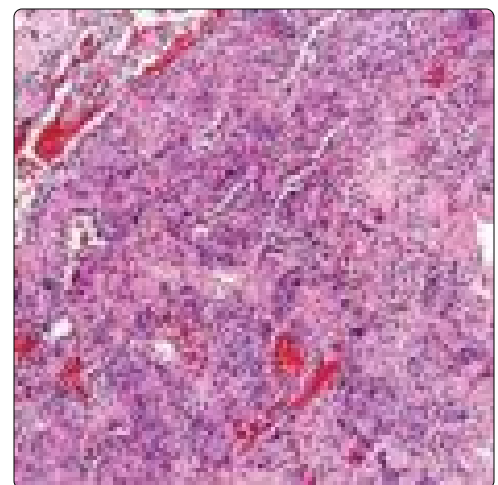


Minute Meningothelial-Like Nodule

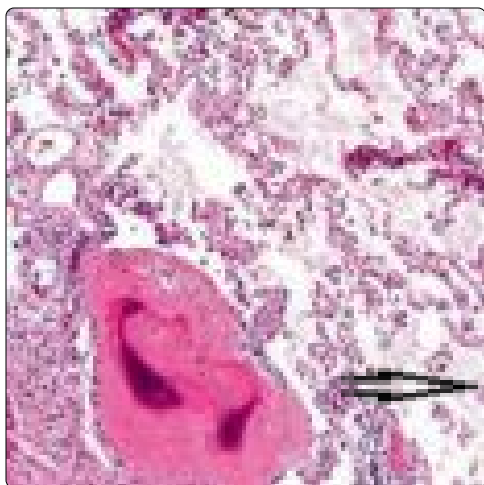
(Left) Meningothelial-like nodules (MLNs) in the lung are small, septal, and whorled proliferations of benign cells. They are often incidental findings and are of unknown etiology, though they are widely considered to be reactive in nature. The cells show a similar immunophenotype to true meningeal cells. **(Right)** Rare benign MLNs can appear larger and more complex than others and even simulate metastases on imaging. Of note, MLNs are believed to give rise to the very rare pulmonary meningiomas.



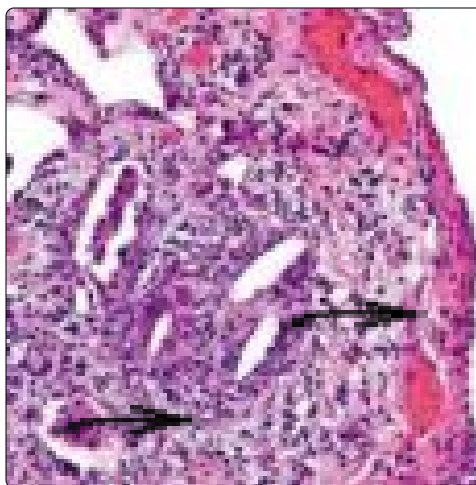
Meningothelial-Like Nodule



Metaplastic Ossification

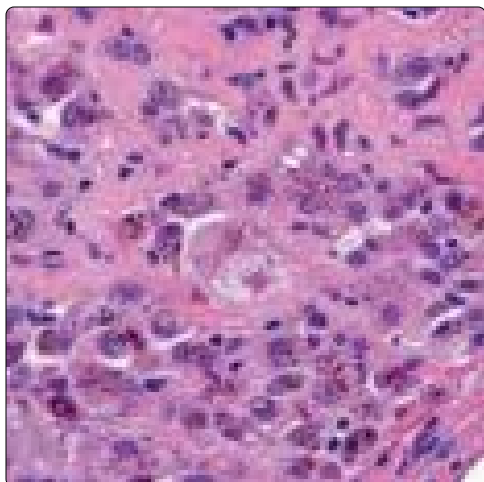


Focal Foreign Body Giant Cell Reaction

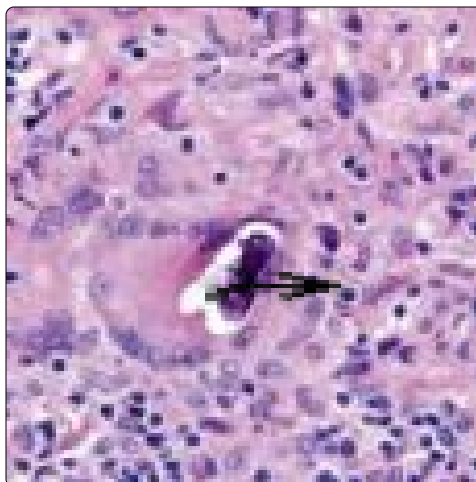


(Left) Mature bone formation [E] can rarely be seen in alveoli, particularly in sites of prior inflammation or lung injury, and it can be focal or patchy in distribution. It is also occasionally seen associated with diffuse interstitial (fibrosing) lung disease. (Right) Focal foreign body giant cells [E] can occasionally be seen reacting to aspirated food particles or talc and other substances. Depending on its composition, a particle [E] may appear crystalline and glassy unless polarized light is utilized.

Asteroid Bodies

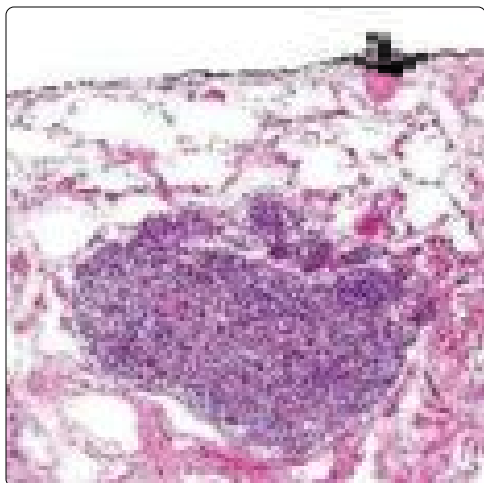


Schaumann Bodies



(Left) Unusual stellate or star-shaped structures (known as asteroid bodies [E]) are occasionally seen in the cytoplasm of multinucleated histiocytes/giant cells. Though once believed to be indicative of sarcoidosis, it is now recognized that any giant cell reaction can produce these structures. (Right) Similar to asteroid bodies, Schaumann bodies [E] are occasionally present within multinucleated histiocytes/giant cells [E]. These calcified structures can be seen in any florid granulomatous reaction.

Intrapulmonary Lymph Node



Anthraxis of Lymph Node



(Left) Rarely, an otherwise normal lymph node arises within the parenchyma of the lung, well away from other lymph node chains. Note the visceral pleura [E] present in this section. (Right) Similar to hilar lymph nodes, intrapulmonary lymph nodes often contain at least focal black anthracotic pigment within histiocytes. Extensive pigment is often associated with abundant histiocytes.

Pleura and Pericardium

PLEURA AND PERICARDIUM

Macroscopic Anatomy

- Pleura
 - Thin, continuous membranous lining within thoracic cavity
 - Visceral pleura invests lungs
 - Also extends along lobar fissures
 - Parietal pleura lines outer surface of thoracic cavity
 - Extends along inner chest wall (including ribs and sternum) and mediastinal structures
 - Superior extent (cervical pleura)
 - Inferior extent (diaphragmatic pleura)
 - Parietal pleura reflects over parietal pericardium anteriorly
- Pericardium
 - Thin, continuous layers surrounding heart and great vessels
 - Visceral pericardium
 - a.k.a. serous pericardium or epicardium
 - Contacts myocardium
 - Parietal pericardium
 - a.k.a. fibrous pericardium
 - Potential space exists between visceral and parietal pericardial layers
 - Contains minimal amounts of clear fluid
 - Visceral and parietal layers only make contact at site of pericardial reflections

Microscopic Anatomy

- Thin layer of mesothelial cells upon thin basement membrane overlying fibroelastic connective tissue
 - Mesothelial cell lining faces cavities and potential spaces
- Normal mesothelial cells
 - Small, flat to cuboidal cells
 - Well-defined cell borders
 - Single central nucleus
 - Homogeneous chromatin
 - No nucleolus
- Reactive mesothelial cells

- Larger, more conspicuous cells
- Enlarged nucleus
- Prominent central nucleoli
- May show multinucleation
- May show limited papillary growth
- Submesothelium
 - Spindled fibroblastic cells, collagen, elastin
 - Parietal pericardium predominantly collagenous
 - Often thickened and more cellular in reactive conditions
 - Inflammatory cells, reactive myofibroblasts, capillaries
- Mesothelial immunophenotype
 - Positive markers
 - Keratins
 - Including AE1/AE3, CK7, CK 5/6, CK 8/18
 - Calretinin
 - WT1
 - D2-40 (podoplanin)
 - HBME-1
 - Thrombomodulin
 - Desmin (in reactive states)
 - Negative markers
 - CK20
 - MOC-31
 - BER-EP4
 - CEA
 - CD15
 - TTF-1
 - In reactive states, submesothelial stromal cells express keratins

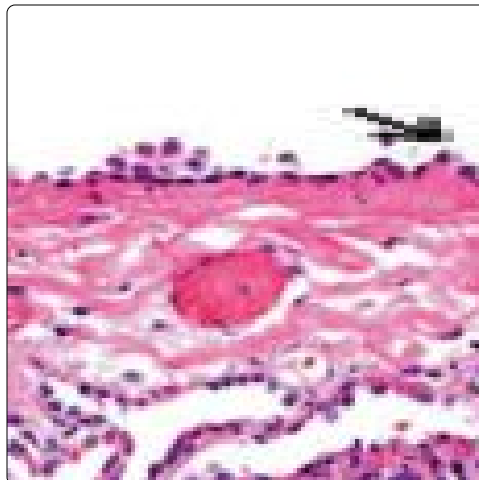
CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

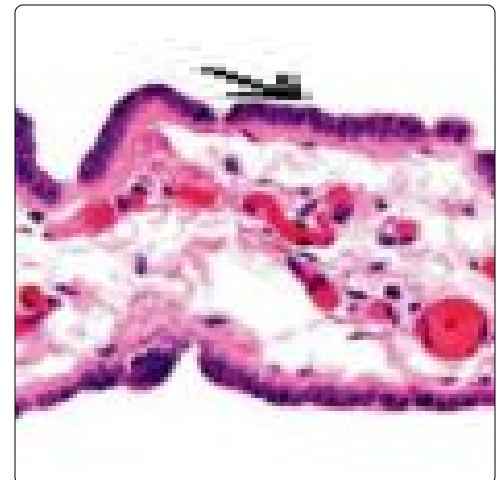
- Normal, nonreactive mesothelium often not visible (also easily denuded)
- Tangential sectioning can lead to unusual mesothelial cell architectures
- Mesothelial cells and submesothelial stromal cells express epithelial antigens (e.g., keratins), which may lead to confusion with carcinoma or mesothelioma

Pleura

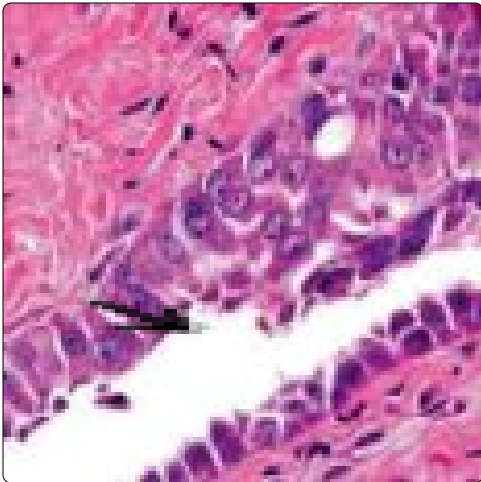
(Left) H&E stain of the pleura shows a single layer of normal, nonreactive mesothelial cells [E]. They appear as small cuboidal or flat cells with eosinophilic cytoplasm and indistinct nuclear features. At times, they are difficult to identify in histologic sections and may appear to be absent. **(Right)** H&E stain shows cuboidal mesothelial cells [E], which are easier to identify than flat mesothelial cells but still show a similar bland cytomorphology.



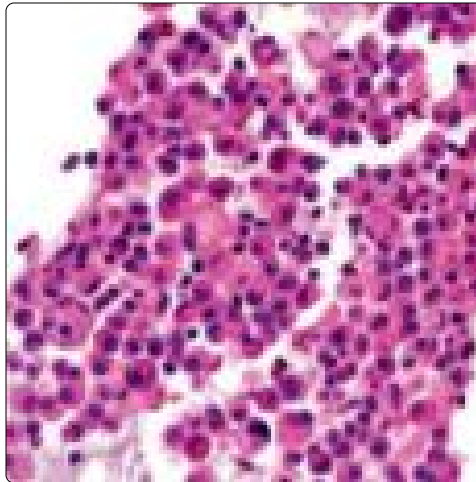
Mesothelial Cells



Reactive Mesothelial Cells

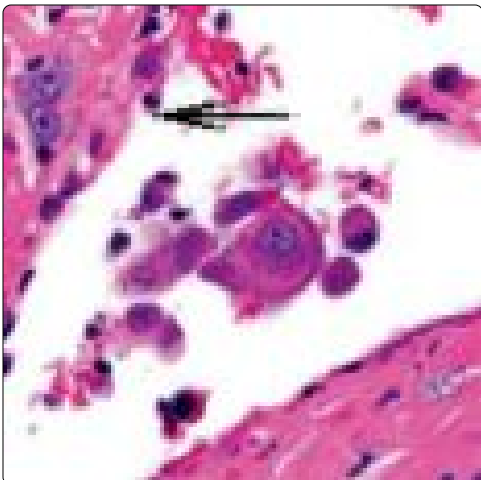


Sloughed Mesothelial Cells

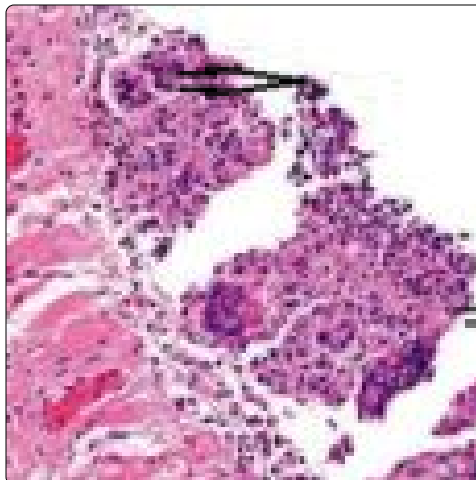


(Left) A distinctive feature of reactive mesothelial cells is an open chromatin pattern with 1 to a few small but prominent basophilic nucleoli. This image also shows the characteristic hobnailing or tombstoning of these cells. **(Right)** Reactive mesothelial cells may slough off and appear as nests or sheets of eosinophilic epithelioid cells. They are not cytologically malignant, however, which helps distinguish them from an epithelial malignancy (carcinoma) or mesothelioma.

Mesothelial "Window"



Reactive Mesothelial Cells

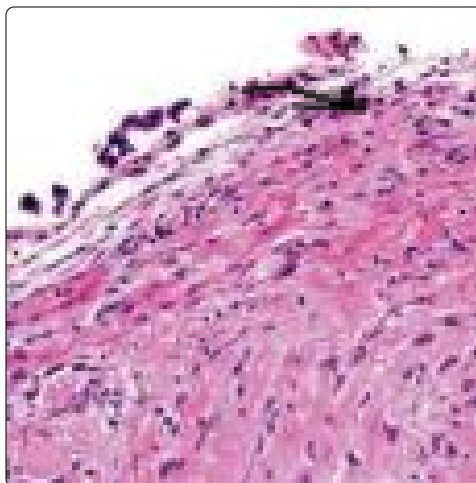


(Left) A unique finding in reactive mesothelial cells is the mesothelial "window," which is a thin, pale zone that forms between neighboring cells. Note the prominent reactive nucleoli. **(Right)** Occasionally, reactive mesothelial cells will show a marked increase in the number of nuclei, leading to confusion with the multinucleated giant cells of granulomatous infections or foreign body giant cell reactions.

Reactive Papillary Hyperplasia



Submesothelial Fibroblasts

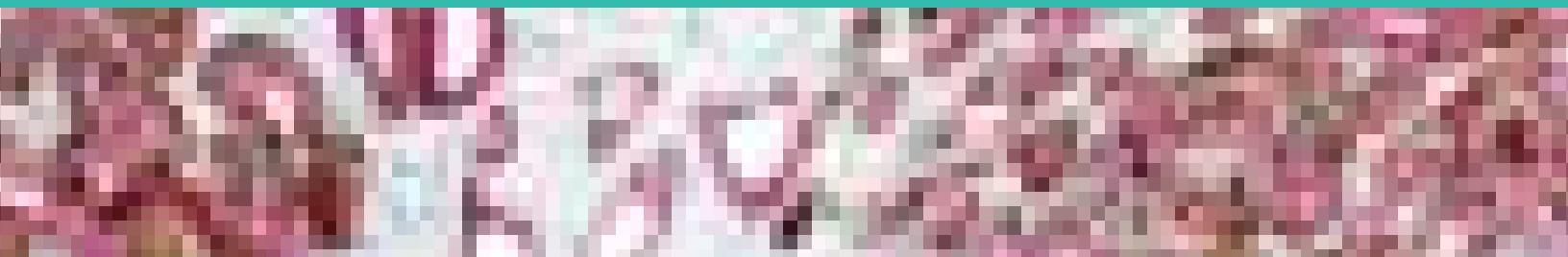


(Left) Reactive mesothelial cells can form small papillary structures that project out into the pleural space. When these changes are diffuse, distinction from a well-differentiated papillary mesothelioma can be challenging. **(Right)** Submesothelial stromal cells exist in the connective tissue underlying the mesothelium and are generally inconspicuous. However, they are more visible in reactive conditions, as their cytoplasm becomes more eosinophilic and stellate (angulated).

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SECTION 9

Breast



Breast

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BREAST

Macroscopic Anatomy

- Distinctive conical protuberance extending from 2nd to 6th rib in midclavicular line
 - Breast tissue spreads from lateral edge of sternum to anterior axillary line and often extends into axilla
 - Suspensory (Cooper) ligaments are fascial attachments from skin to chest wall
 - Provide support and allow for mobility
- Adult female breast
 - Modified specialized cutaneous glandular structure
 - Located in superficial fascia of anterior chest wall overlying pectoralis major and minor muscles
- Nipple
 - Projects from anterior surface of breast, positioned slightly medial and inferior to center
 - Consists of dense fibrous tissue and smooth muscle bundles covered by hyperpigmented skin
 - Smooth muscle makes nipple "erectile"
 - Assists with milk expression
 - Milk secretion occurs through 10-15 major duct orifices opening onto nipple surface
- Breast parenchyma consists of branching duct system
 - 15-20 lobes radiate from nipple, bound together by interlobular connective tissue
 - Ducts are continuous with terminal ductal lobular units (TDLU), responsible for milk production
 - TDLUs are embedded in specialized, hormonally responsive intralobular stroma
 - Glandular tissue is admixed with varying amounts of adipose tissue, responsible for variability in breast size

Microscopic Anatomy

- Nipple
 - Epidermis has thick keratin layer
 - Protects against trauma and drying from nursing
 - Some cells feature pyknotic nuclei and clear cytoplasm
 - May be keratinocytes or Toker cells (mammary ductal epithelial cells)

- Lactiferous duct is lined by squamous epithelium where duct traverses epidermis of nipple
- Lining cells transition from squamous to columnar epithelium distal to lactiferous sinus
- Breast epithelium
 - Composed of simple bilayer of cells lining entire mammary duct/lobular system
 - Inner luminal cells are required for milk production
 - IHC: CK7(+), CK8/18(+), GATA3(+), CK20(-)
 - Outer myoepithelial cells assist with milk ejection
 - IHC: S100(+), CK 5/6(+), SMA(+), p63(+), calponin (+), SMHC(+)
 - Stem cells that give rise to both cell types are present but require special techniques for recognition

Physiologic Function

- Secretes milk for nourishment and immunologic protection of young during postpartum lactation
 - Target organ for variety of hormones that regulate development and physiologic function
 - Hormonal and growth factor effects result in ongoing lifelong changes in response to menarche, pregnancy, lactation, and menopause
 - These cyclical changes over life cycle result in broad range of nonpathologic breast histology
 - Some histologic patterns seen in specimens submitted for pathologic examination represent variants of physiologic changes

CONSIDERATIONS AND VARIATIONS

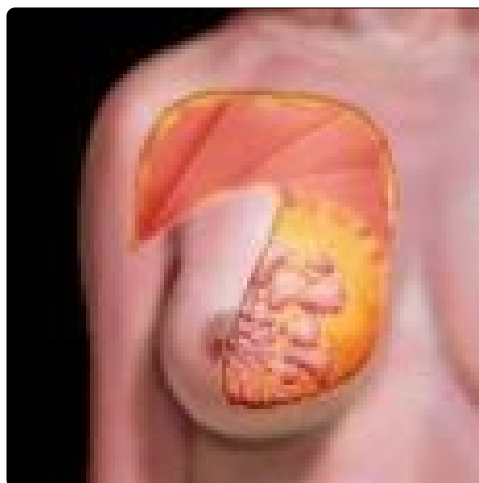
Age-Related Variations

- Histology of breast tissue undergoes substantial change during period from early adolescence to menopause
- Estrogen mediates development of ductal tissue; progesterone facilitates TDLU development; prolactin regulates milk production

Pitfalls and Artifacts

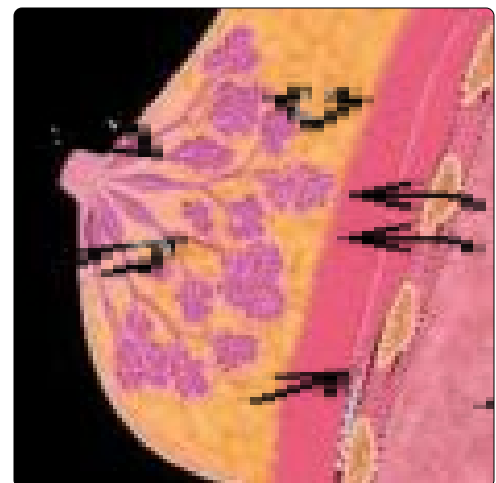
- Lactational changes in breast
- Clear cells in nipple epithelium

Adult Female Breast

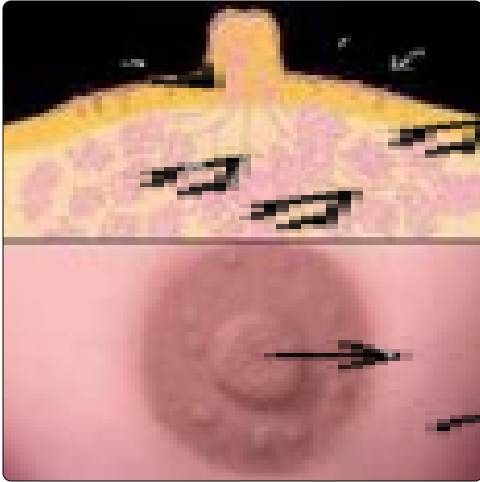


(Left) The adult female breast is located on the anterior chest wall, overlying the pectoralis muscles. The nipple-areolar complex is located slightly inferior to center and connects the overlying skin to the mammary parenchyma. (Right) The major breast structures are the nipple-areolar complex, the large duct system, and the terminal duct lobular units (TDLUs). Glandular breast tissue is supported by connective tissue (interlobular and intralobular stroma and adipose tissue).

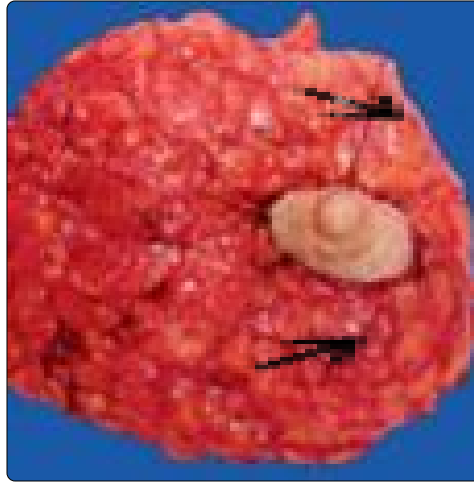
Anatomic Structures



Nipple/Areolar Complex



Prophylactic Subcutaneous Mastectomy

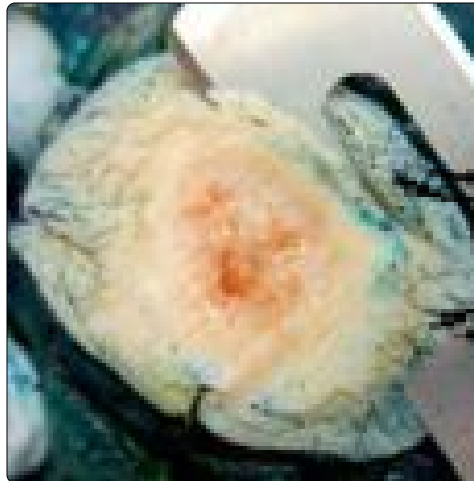


(Left) The nipple-areolar complex is covered by a pigmented squamous epithelium [A] and contains sebaceous glands (Montgomery tubercles [B]). During lactation, milk is produced in the lobules [C] and is transported via ducts [D] that lead to the nipple orifices [E]. (Right) A mastectomy may be performed to reduce risk of subsequent cancer in some high-risk patients. In this case, the nipple [F] and breast tissue were removed, sparing the skin. The anterior breast tissue [G] corresponds to the skin-flap margins.

Nipple, Gross Evaluation



Nipple, Gross Evaluation

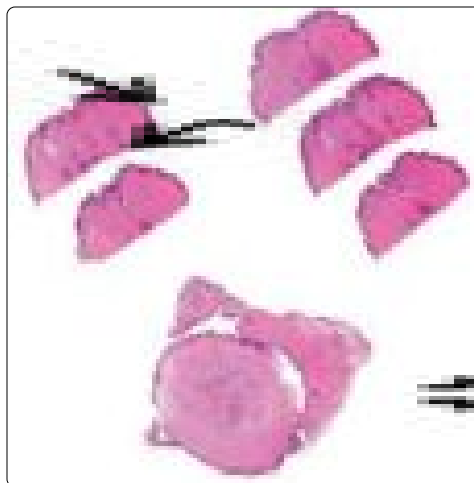


(Left) Microscopic examination of the nipple is important to exclude nipple or skin involvement by disease. The nipple is 1st removed by carefully cutting across the base [A], parallel to the skin surface. (Right) After the nipple has been removed, an en face section of the nipple can be taken for microscopic examination of the base of the nipple [B]. This tissue will show cross sections of the major ducts and lactiferous sinuses as they extend to the nipple surface.

Nipple, Gross Evaluation



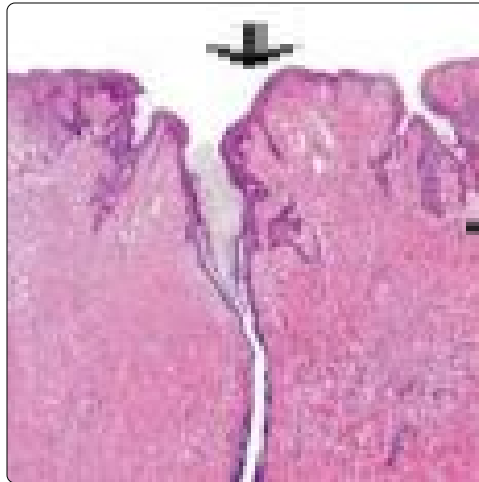
Nipple, Sections for Microscopy



(Left) Perpendicular sections [A] of the nipple allow for examination of the overlying nipple epidermis and the lactiferous ducts as they extend to the nipple orifice. (Right) Microscopic examination of perpendicular sections of the nipple show the epidermis [B] and the major lactiferous ducts [C]. Microscopic examination of the base of the nipple shows the lactiferous sinuses [D].

Nipple and Lactiferous Duct

(Left) Squamous epithelium covers the nipple and areola. The orifices of the lactiferous ducts are also squamous lined where the duct traverses the epidermis. Keratin plugs may be present in the nipple orifices in the nonlactating breast. (Right) Lactiferous ducts in the nipple are surrounded by bundles of smooth muscle, which are embedded in a collagenous and elastic stroma. Longitudinal muscle bundles along the lactiferous ducts are responsible for nipple erection.



Calponin Immunohistochemistry

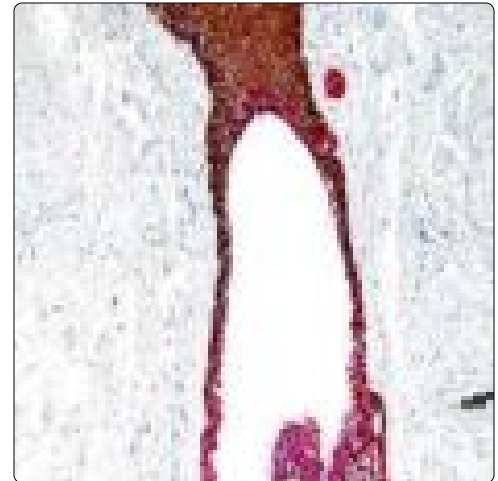


Squamous/Columnar Junction

(Left) The squamous epithelium of the epidermis dips into the nipple orifice for 1-3 mm, where it joins the columnar epithelium to form an abrupt squamocolumnar junction distal to the lactiferous sinus. (Right) Immunohistochemistry can be used to highlight the different cell types seen in the squamocolumnar junction of the lactiferous ducts. CK7 stains the luminal cells (red chromogen), while CK5/6 and p63 stain the squamous epithelium (brown chromogen).

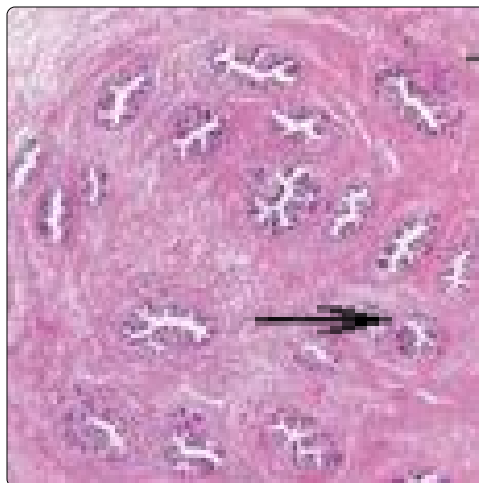


CK7 and CK5/6 With p63 Immunohistochemistry

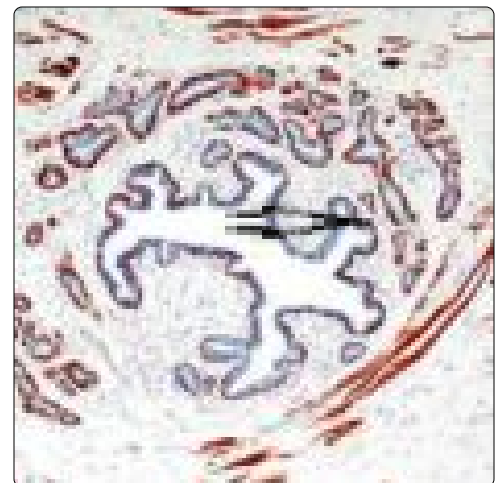


Lactiferous Sinuses

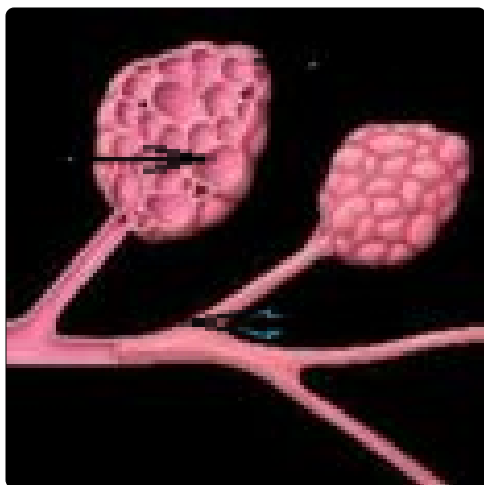
(Left) Immediately deep to the areola, each lactiferous duct dilates to form the lactiferous sinus or collecting ducts in which milk accumulates during lactation. Note the typical undulating outline of these ducts, which are easily distended. Epithelial and myoepithelial cells line the duct space. (Right) The nipple nodule is composed of bundles of smooth muscle and elastic tissue traversed by the lactiferous ducts. Radially arranged bundles of smooth muscle assist with milk expression during lactation.



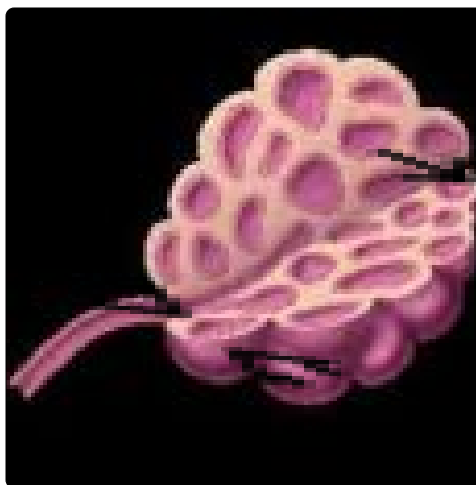
Calponin Immunohistochemistry



Mammary Duct System

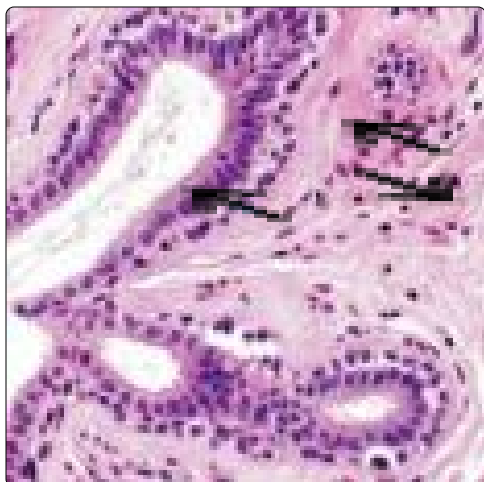


Terminal Ductal Lobular Units

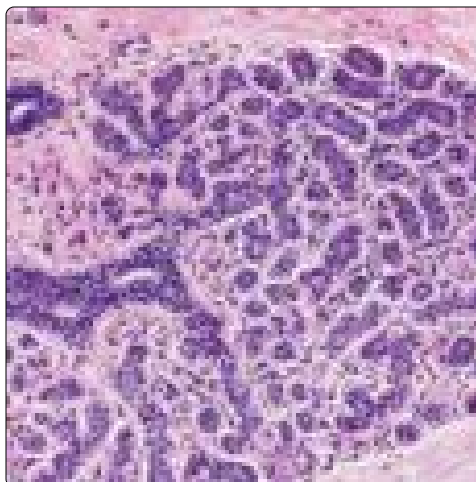


(Left) The epithelial components of the breast consist of a branching system of large and intermediate interlobular ducts [E] that connect the structural and functional units of the breast (TDLUs [F]) to the nipple. (Right) The terminal ductules [E] end in lobular/glandular units called the TDLUs [F]. Lobule formation is apparent within 1 to 2 years after the onset of the menses. Thereafter, glandular development is variable. Full differentiation of the mammary gland is attained during pregnancy.

Interlobular Mammary Ducts



Terminal Ductal Lobular Units

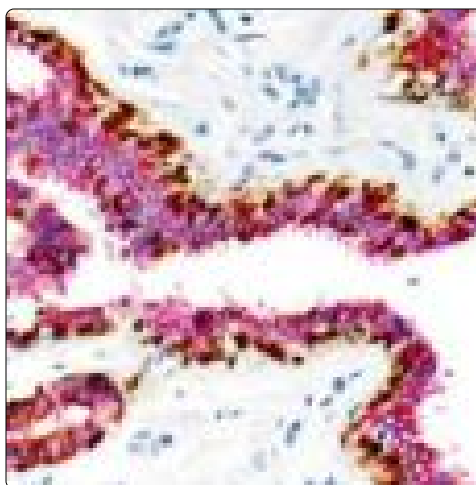


(Left) The mammary ductal/lobular system is composed of a bilayer of cells, extending from the major lactiferous duct to the TDLUs. The luminal cells form a columnar layer [E] in the interlobular ducts and are surrounded by a layer of myoepithelial cells [F]. (Right) The normal histology of the lobules is not constant in the mature female breast because of changes associated with fluctuations in hormonal levels (during menstrual cycle, pregnancy, lactation, and menopause).

CK7 and CK5/6 With p63 Immunohistochemistry



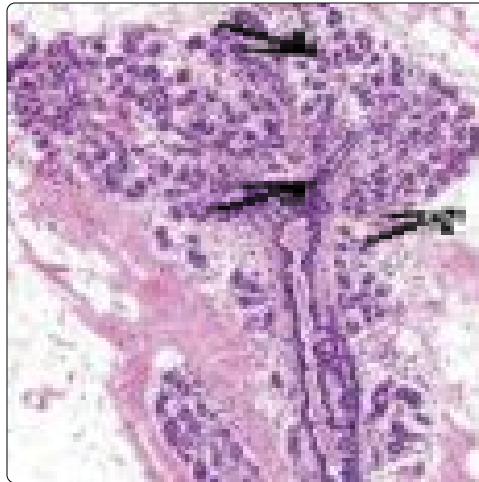
CK7 and CK5/6 With p63 Immunohistochemistry



(Left) Luminal and myoepithelial cells differentially express low and high molecular weight cytokeratins, respectively. Luminal cells express low molecular weight luminal cytokeratins (CK7, red chromogen). Myoepithelial cells express high molecular weight basal cytokeratins (CK5/14/17) and other distinguishing markers such as p63 (brown chromogen). (Right) High-power photo shows luminal cells expressing low molecular weight CK7 [E] and myoepithelial cells expressing p63 [F].

Terminal Ductal Lobular Units

(Left) TDLUs have an architectural organization similar to a tree if the trunk and branches were hollow. The terminal duct ("trunk") opens into the acini ("branches") of the lobule. This lobulocentric organization is a key microscopic feature in recognizing normal breast structure. (Right) There are 2 types of epithelial cells. Luminal cells express keratins 7, 8, and 18 (red cytoplasmic chromogen). Myoepithelial cells express p63 (brown nuclear chromogen).

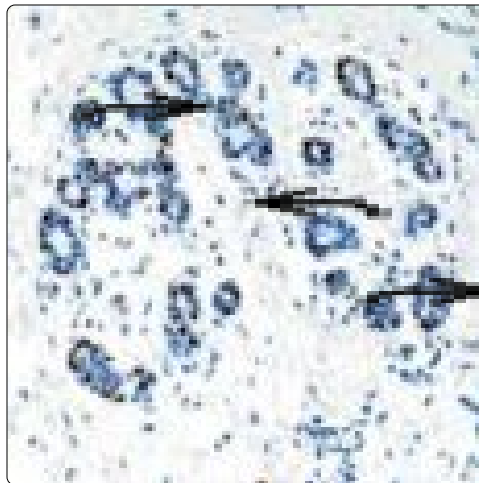


CK7/p63 Immunohistochemistry

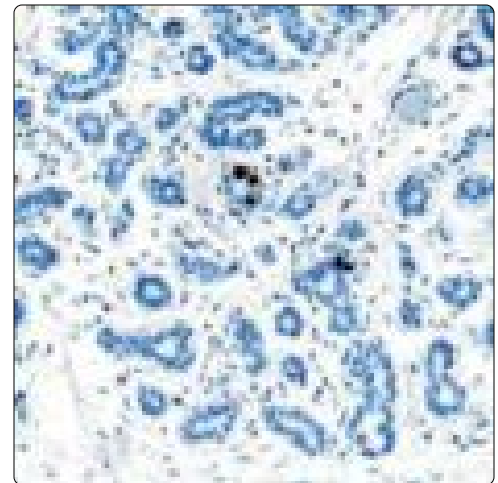


Estrogen Receptor Immunohistochemistry

(Left) Not all luminal cells express ERP. The content of ERP and PRP in the normal luminal cells varies with the degree of lobular development, in parallel with cell proliferation. (Right) Proliferation is an important part of the growth and development of breast tissue. Scattered Ki-67(+) cells (a marker of proliferation) can be found in the normal TDLU; however, these cells are negative for ERP. Different populations of luminal cells within the breast may interact in a paracrine fashion.



Ki-67 Immunohistochemistry

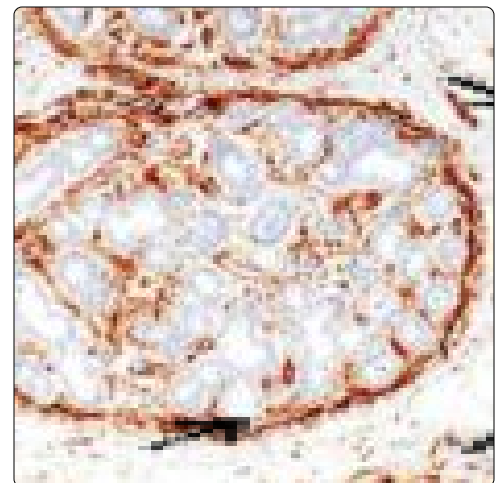


Terminal Ductal Lobular Units

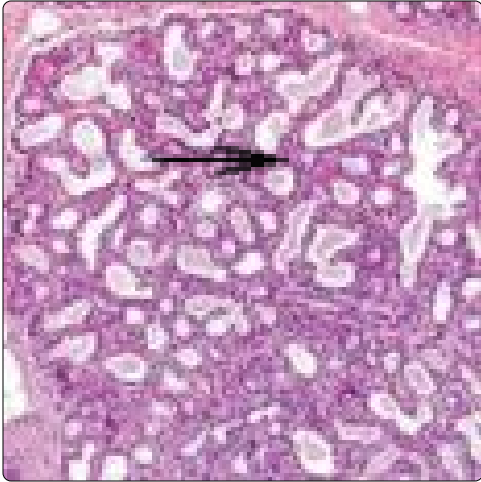
(Left) The basement membrane surrounding the ducts and lobules contains type IV collagen and laminin and appears as a bright pink layer on Alcian blue/PAS staining. This basement membrane is contiguous with that of the large ducts and skin. (Right) The TDLU has a specialized intralobular stroma enveloping the acini and consisting of CD34(+) stromal fibroblasts and capillaries. Lesions arising from this stroma (fibroadenomas and phyllodes tumors) are biphasic.



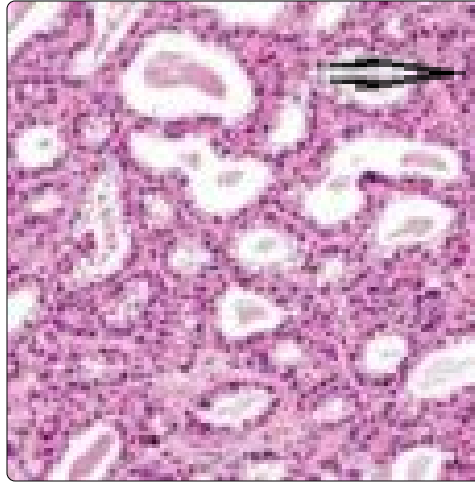
CD34 Immunohistochemistry



Pregnancy and Lactation

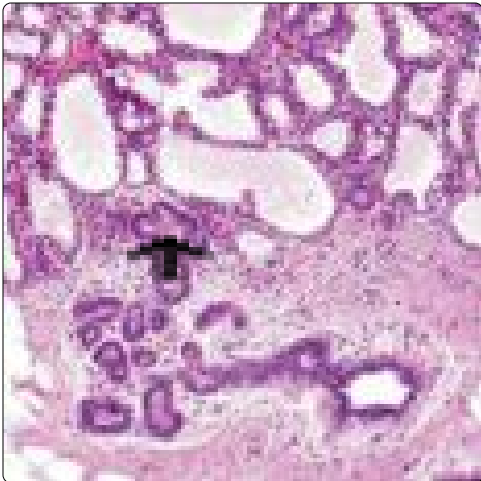


Pregnancy and Lactation

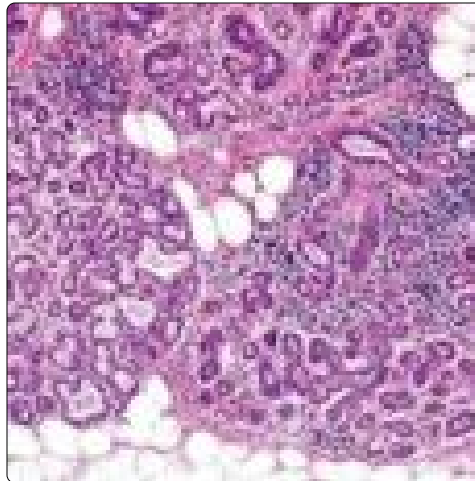


(Left) During early pregnancy, epithelial cells rapidly proliferate, with expansion of both the size and number of lobules due to the influence of placental hormones. The lobular acini dilate and undergo secretory changes in preparation for lactation. **(Right)** The luminal cells of the TDLU become vacuolated and show nuclear enlargement with prominent nucleoli during pregnancy and lactation. The acinar lumina become progressively dilated and distended by the accumulation of secretory material or colostrum.

Pregnancy and Lactation

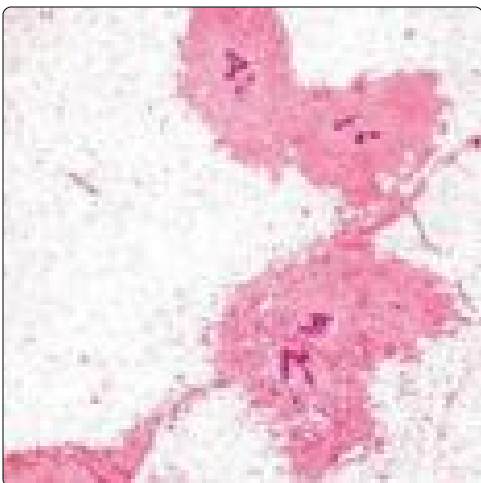


Postlactation Involution

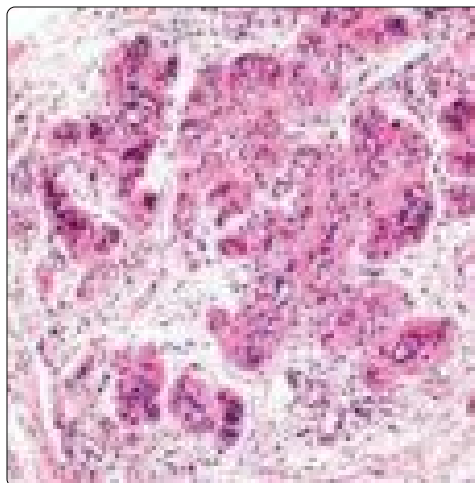


(Left) The large duct system in lactational breast can be distinguished from the TDLUs by its absence of secretory changes. There is an abrupt transition from the terminal duct to the TDLU. **(Right)** The postlactational breast requires both lactogenic hormone deprivation and local signals to undergo glandular involution. A prominent lymphocytic infiltrate surrounds the lobules, which do not regress to their prepregnancy state but maintain a more differentiated lobular organization.

Menopause



Menopause



(Left) The breast undergoes involution following menopause and more of the breast consists of stroma and fatty tissue. During involution, the lobules and the specialized lobular stroma begin to disappear, leaving only small terminal ducts. The interlobular stroma becomes more fatty, making the breast more radiolucent. **(Right)** In an atrophic lobule, myoepithelial cells and basement membranes become more prominent and luminal cells regress. The myoepithelial cells sometimes appear spindled in shape.

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SECTION 10

Tubular Gut and Peritoneum



Esophagus	218
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Small Intestine	232
Large Intestine	238
Appendix	246
Anus and Anal Canal	252
Peritoneum	256

ESOPHAGUS

Macroscopic Anatomy

- Muscular tube ~ 25 cm in length
 - Prominent longitudinal folds when not insufflated
- Begins at level of cricoid cartilage
- Extends down within posterior mediastinum and through diaphragm
- Connects to stomach at gastroesophageal junction (GEJ)
 - Z-line: Macroscopically defines GEJ
 - Lower esophageal sphincter: Site of increased muscular pressure near GEJ
- 3 anatomic regions
 - Cervical
 - Thoracic
 - Abdominal
- Natural points of constriction
 - Cricoid cartilage
 - Aortic arch, left main bronchus, left atrium
 - Esophageal hiatus (in diaphragm)

Microscopic Anatomy

- Mucosa
 - Nonkeratinizing, stratified squamous epithelium
 - p63(+), cytokeratin 5/6 (+)
 - Few scattered intraepithelial lymphocytes and Langerhans cells
 - Lymphocytes often have irregular contour ("squiggle cells")
 - Lamina propria
 - Located under epithelium and above muscularis mucosa
 - Contains blood vessels, inflammatory cells, and mucous glands
 - Minority of people have melanocytes at epithelial-stromal interface
 - Muscularis mucosae
 - Thin, longitudinal bands of smooth muscle
 - Thicker at GEJ
 - Demarcates mucosa from submucosa

- Submucosa
 - Loose connective tissue containing blood vessels, lymphatics
 - Meissner plexus: Nerve fibers, ganglion cells
 - Mucous (salivary-type) glands
 - More prominent in upper and lower 1/3
- Muscularis propria
 - Thick smooth muscle bundles (inner circular and outer longitudinal layers)
 - Auerbach (myenteric) plexus and interstitial cells of Cajal between layers
- Adventitia
 - Most of esophagus lined by loose connective fascial sheath
 - Lacks true serosa for most of its length
- Squamocolumnar junction
 - Transition from stratified squamous epithelium of esophagus to glandular epithelium of stomach
 - Junctional glandular epithelium may be mix of mucous and oxyntic glands or exclusively mucous glands

VARIATIONS AND CONSIDERATIONS

Heterotopia

- Gastric mucosa in upper 1/3 of esophagus
 - a.k.a. "inlet patch"
- Sebaceous glands

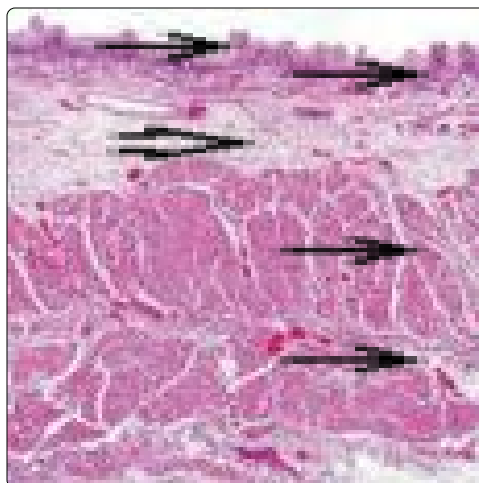
Metaplasia

- Intestinal metaplasia (Barrett esophagus)
- Pancreatic acinar metaplasia

Pitfalls and Artifacts

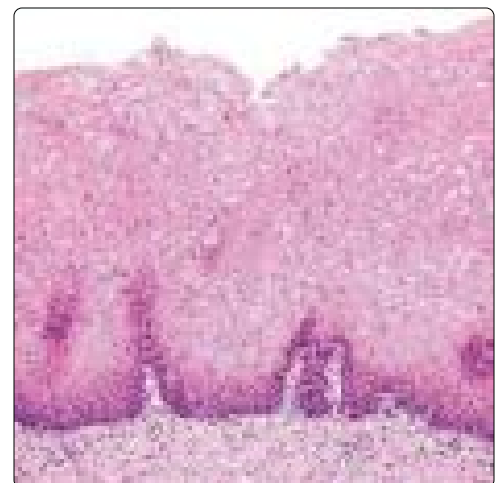
- Glycogenic acanthosis can mimic candidiasis macroscopically
- Contamination by normal oral flora (bacteria and yeast) can mimic true infection

Esophagus at Low Magnification

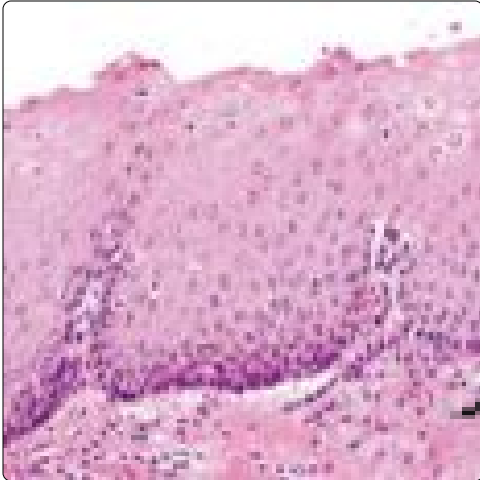


(Left) From superficial (luminal) to deep, the anatomical layers of the esophagus are: Stratified squamous epithelium, lamina propria, muscularis mucosa, submucosa, muscularis propria, and the adventitial connective tissue. (Right) The esophagus is lined by nonkeratinized stratified squamous epithelium. The basal layer contains actively proliferating basal cells that give rise to the squamous cells above them, which mature as they progress toward the surface where they are ultimately sloughed off.

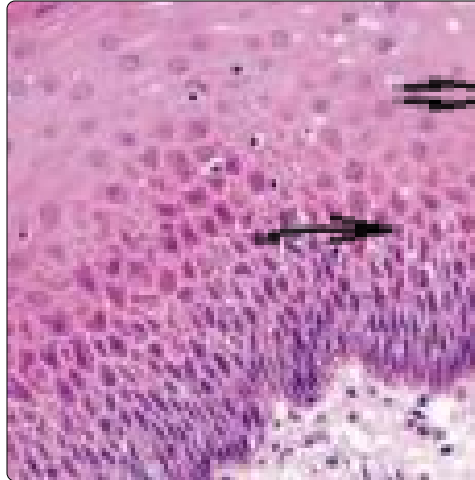
Squamous Epithelium



Squamous Epithelium

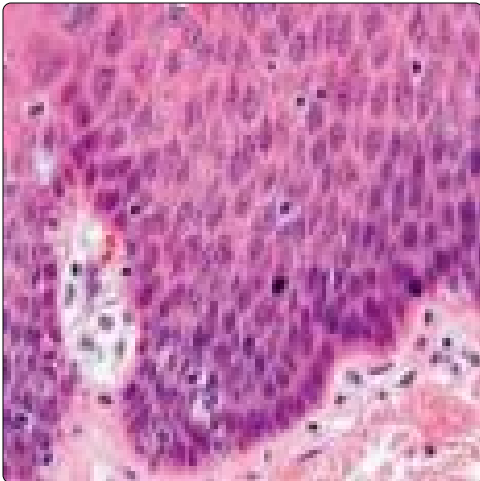


Squamous Epithelium



(Left) The basal cell layer is readily identified due to its higher nuclear density. (Right) In various reactive conditions (e.g., reflux), the basal 1/2 of the epithelium often displays prominent intercellular edema, which highlights the desmosomes connecting the squamous cells. Scattered intraepithelial lymphocytes are common normally, especially in the basal area.

Squamous Epithelium

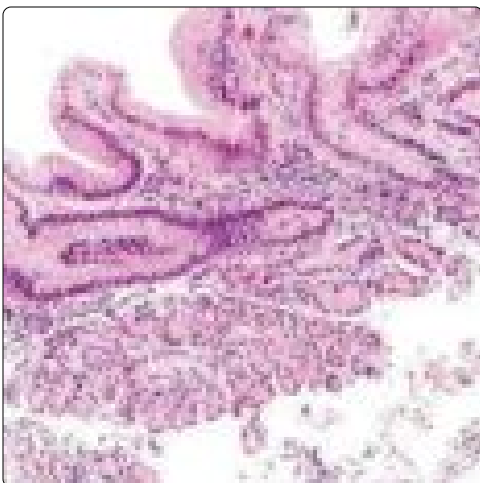


Squamous Epithelium, Reactive

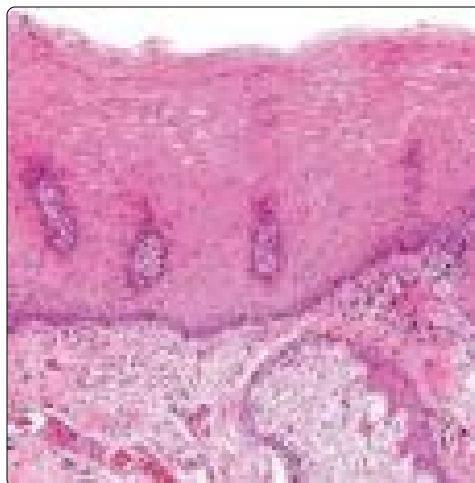


(Left) Since it is a site of active proliferation, mitotic figures can often be identified in the normal basal layer. They are larger than the intraepithelial lymphocytes. (Right) In reactive conditions, elongation of the tongue-like epithelial rete pegs is common. Compare the significant elongation in this case of reflux to images of the normal esophagus.

Inlet Patch



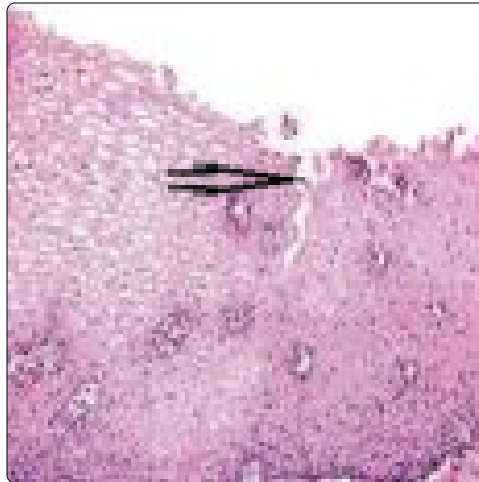
Heterotopia



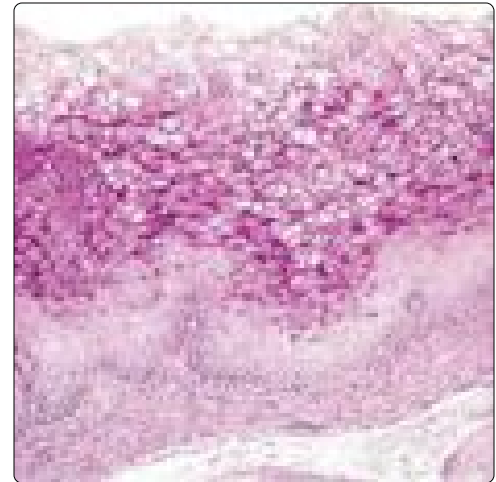
(Left) On occasion, normal mucosal tissue not typically found in the esophagus (such as gastric, shown here) will appear as small nodules in the proximal esophagus. This is considered a normal variant finding (heterotopia) and is not neoplastic. Heterotopic gastric tissue in the esophagus is also known as an "inlet patch." (Right) Rarely, heterotopic sebaceous tissue may arise in the proximal esophagus, similar to an inlet patch.

Glycogenic Acanthosis

(Left) All maturing squamous cells contain some intracytoplasmic glycogen (which gives them a clear appearance), but occasionally, the glycogen is prominent and grossly nodular, known as glycogenic acanthosis. This is a normal variant but can mimic candidiasis grossly. **(Right)** A PAS stain highlights the intracytoplasmic glycogen in glycogenic acanthosis.

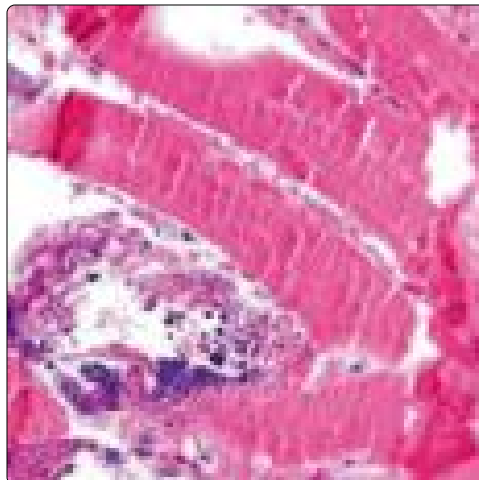


Glycogenic Acanthosis

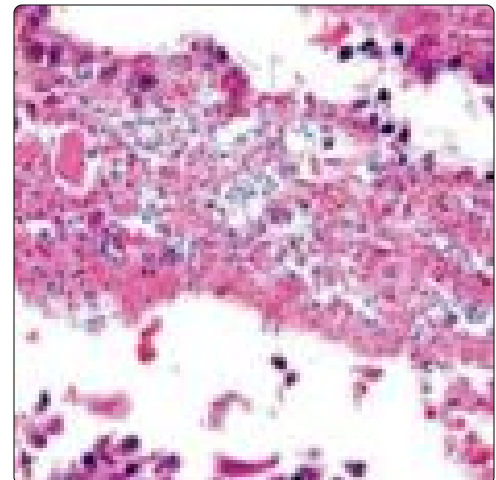


Food Matter

(Left) Food matter is an incidental finding, particularly near esophageal strictures. Meat products appear as red striated skeletal muscle. Accompanying blue clusters of bacteria are often present. **(Right)** In addition to bacteria, the oral cavity normally contains fungi (e.g., *Candida*). These organisms are frequently swept up with denuded oral epithelial cells or food and swallowed and can appear in esophageal biopsies. This should not be mistaken for true infection.

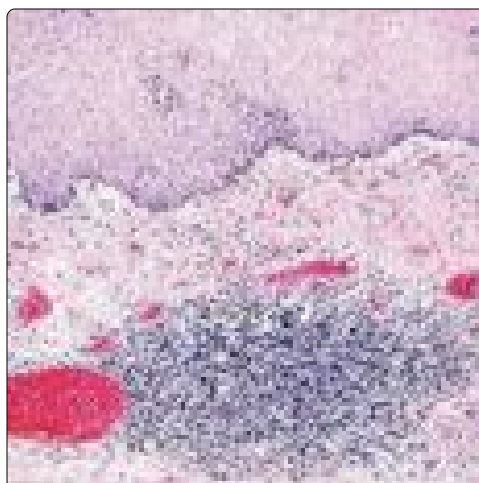


Oral Flora

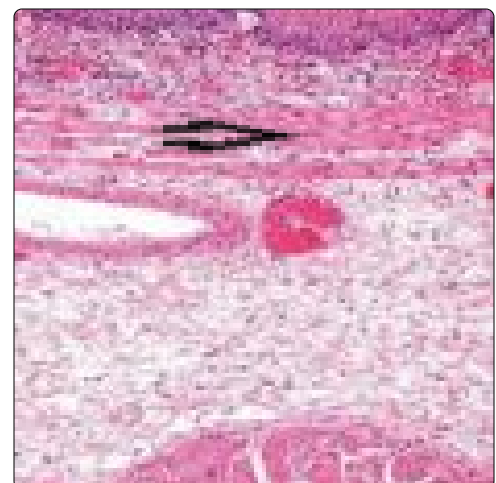


Lamina Propria

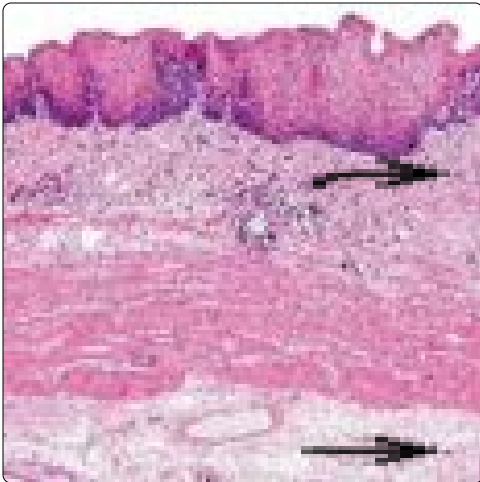
(Left) The lamina propria is directly underneath the epithelium. It contains connective tissue with vessels and nerves. Occasional benign lymphoid aggregates are seen as well. **(Right)** The muscularis mucosa is typically a thin layer of smooth muscle that demarcates the overlying mucosa from the underlying submucosa. It lies directly under the lamina propria.



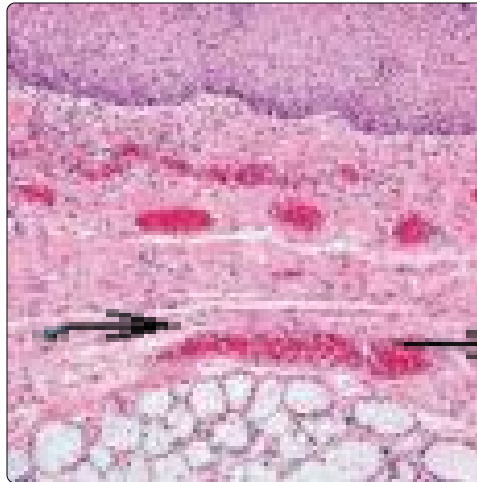
Muscularis Mucosa and Submucosa



Muscularis Mucosa

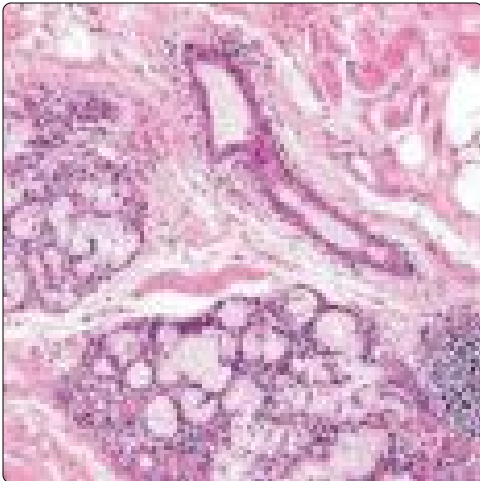


Submucosa

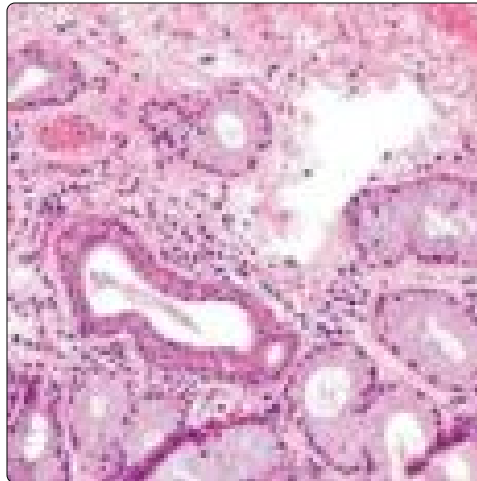


(Left) The muscularis mucosa is thicker in the distal esophagus and in some diseases (e.g., Barrett esophagus) sometimes leads to confusion with the muscularis propria. The lamina propria and submucosa can help orient the tissue. **(Right)** The submucosa contains loose connective tissue, blood vessels, lymphatics, inflammatory cells, and submucosal mucous glands. Note that the muscularis mucosa is very thin and difficult to identify in this image.

Submucosal Glands

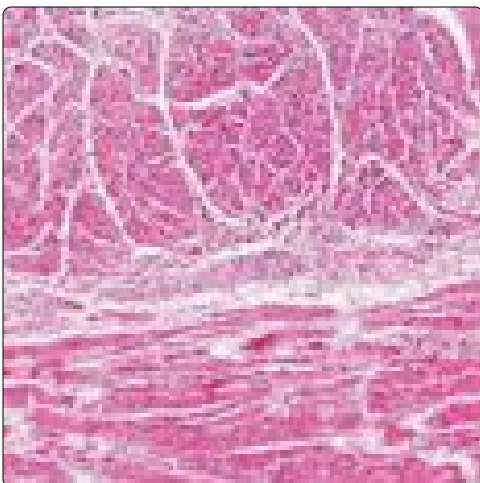


Submucosal Glands



(Left) Submucosal glands are lobular mucous glands associated with a simple duct. The ducts deliver gland secretions to the surface for lubrication and protection. **(Right)** Individual acinar units of the submucosal glands, like those in the salivary gland, consist of mucous cells arranged radially around a central lumen. The lumen connects to the duct, which is lined by a double layer of cuboidal cells. A mild lymphoid infiltrate associated with the duct is common.

Muscularis Propria

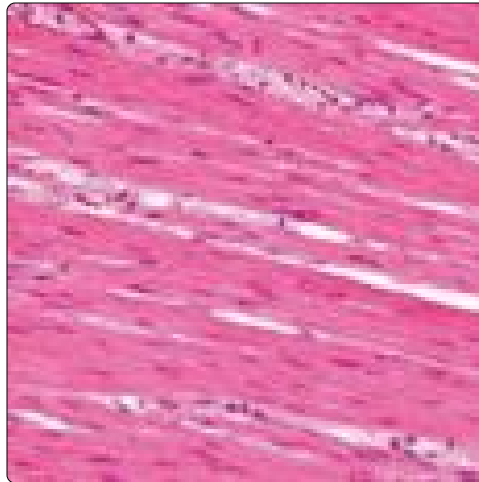


Muscularis Propria

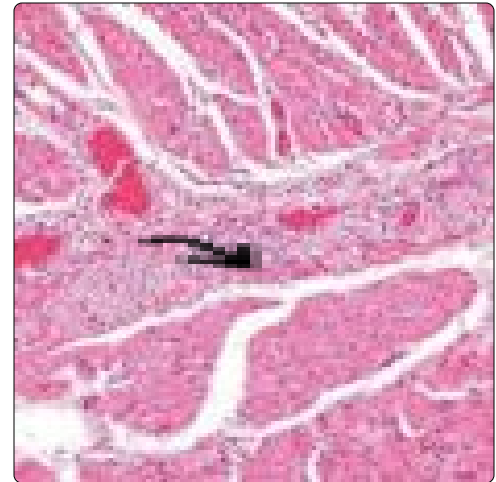


(Left) The muscularis propria is divided into an inner circular (or circumferential) layer and an outer longitudinal layer. In between lies the myenteric (Auerbach) neural plexus. **(Right)** The inner circular (circumferential) layer of the muscularis propria is composed of smooth muscle that encircles the tubular esophagus. As a result of this orientation, the muscle fibers are often cut in cross section and appear as lobular bundles rather than long fascicles.

Muscularis Propria



Auerbach Plexus

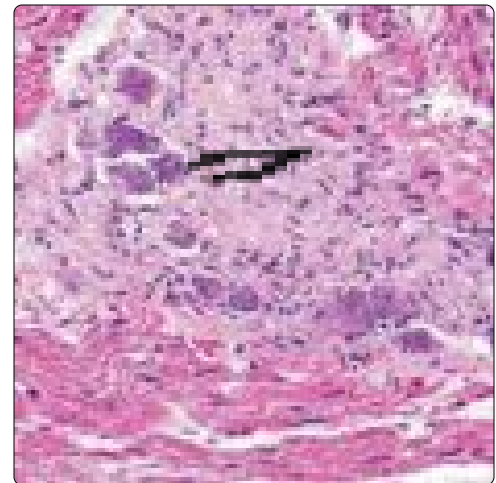


(Left) In contrast to the inner layer, the outer (longitudinal) layer of the muscularis propria is oriented to run along the length of the esophagus. This creates a streaming fascicular pattern to the smooth muscle cells. **(Right)** Auerbach (myenteric) plexus is composed of a network of nerve fibers that run the length of the gastrointestinal tract. These nerve fibers branch out to innervate both layers of the muscularis propria.

Interstitial Cells of Cajal

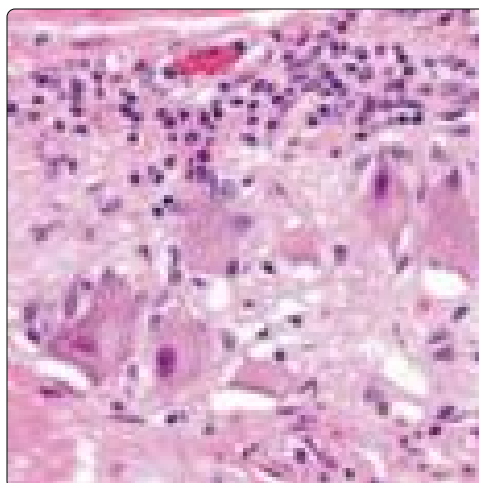


Ganglion Cells of Auerbach Plexus

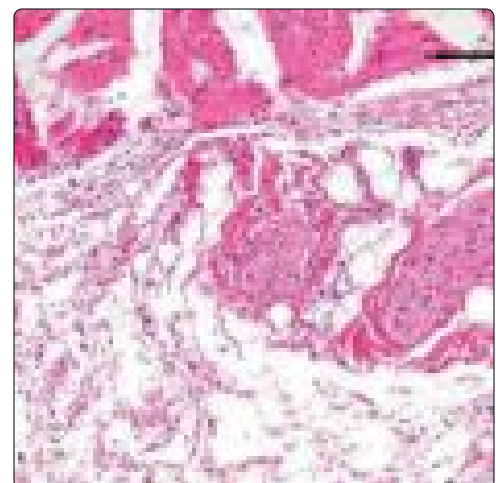


(Left) Interstitial cells of Cajal regulate motility in the gastrointestinal tract and vary in number depending on the site. They are difficult to identify on routine stain, but a CD117 immunostain highlights them. Note that the smooth muscle cells of the muscularis propria in the background do not stain. **(Right)** Ganglion cells, either individually or in clusters, are present within the network of neural fibers and are present in varying numbers. They are large, round to polygonal cells, often with prominent nucleoli.

Ganglion Cells

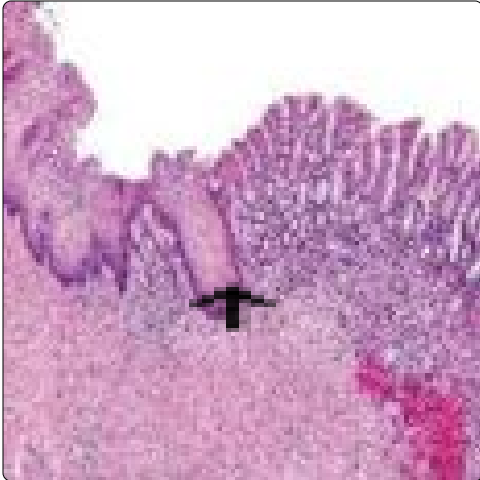


Adventitia

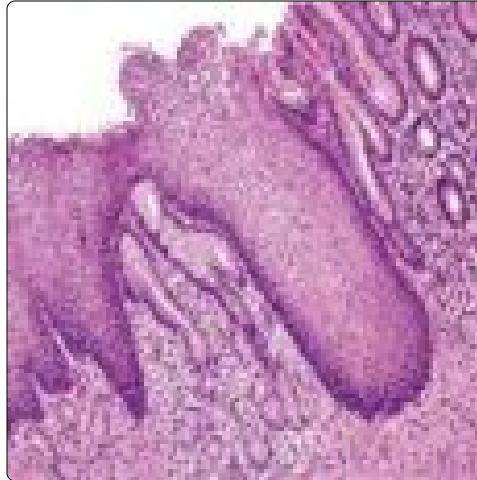


(Left) Ganglion cells have basophilic (blue) cytoplasm with large nuclei and prominent nucleoli. **(Right)** Beneath the muscularis propria, the adventitia of the esophagus is a loose connective tissue fascia that forms a sheath around most of the esophagus. It contains fibroblasts, adipose tissue, and neurovascular elements. The majority of the esophagus lacks an outer mesothelial lining, or serosa.

Squamocolumnar Junction



Squamocolumnar Junction

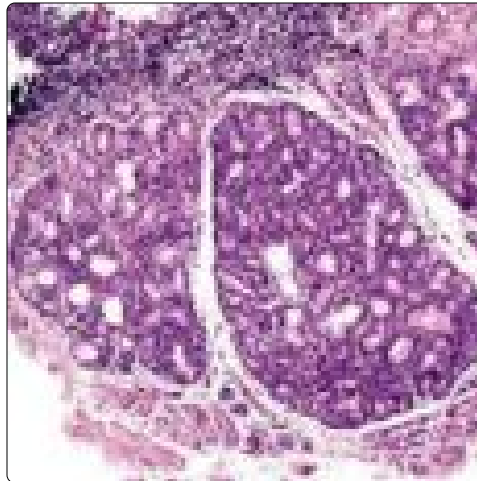


(Left) This photomicrograph illustrates the squamocolumnar junction, or the transition from squamous epithelium to gastric glandular epithelium. **(Right)** The gastroesophageal junction, a.k.a. the squamocolumnar junction, is a gradual transition from the stratified squamous epithelium of the esophagus to the simple columnar epithelium of the stomach. Due to the gradual change, it is not uncommon to see glandular epithelium admixed with squamous epithelium.

Squamocolumnar Junction

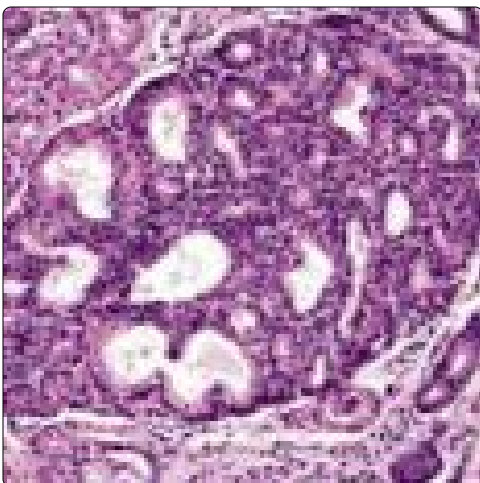


Pancreatic Acinar Metaplasia



(Left) The glandular epithelium at the squamocolumnar junction may be a mix of oxyntic and mucous glands or consist primarily of mucous glands. **(Right)** Small aggregates of pancreatic acinar-type glands are found in up to 5% of biopsies of the gastroesophageal junction, usually in the deeper part of the mucosa. This finding is known as pancreatic acinar metaplasia, and the origin (true metaplasia vs. heterotopia) is unknown. It has no known clinical significance.

Pancreatic Acinar Metaplasia



Intestinal Metaplasia



(Left) Pancreatic acinar metaplasia is characterized by groups of pancreatic acinar-like cells in a lobular configuration. Like normal pancreatic acini, many of the cells have supranuclear granular eosinophilic cytoplasm and subnuclear basophilic cytoplasm. **(Right)** Intestinal metaplasia is characterized by large, barrel-shaped goblet cells within the epithelium (like those seen in the bowel). In the appropriate endoscopic setting, this is known as Barrett esophagus.

STOMACH

Macroscopic Anatomy

- Sac-like, J-shaped organ in central left upper abdomen
 - Can accommodate up to 1.5 L of food and gastric juice in normal adults
 - Lined by rugae or prominent folds
 - Mucosal surface is finely granular or nodular; nodules are known as *areae gastricae*
- Cardia: Uppermost portion where esophagus joins stomach
 - Extends 1-2 cm or less
 - Immediately distal to lower esophageal sphincter musculature
- Body: Extends distally from cardia to angularis (bend in lesser curvature near pylorus)
 - Greater curvature on left, inferolateral margin
 - Lesser curvature on right, superomedial margin
 - Angularis roughly divides body from antrum
- Fundus: Cephalad part of body that forms bulge on upper left of stomach
- Antrum: Distal to angularis; terminates in pylorus
 - Pylorus overlies pyloric sphincter, which is dividing line between stomach and duodenum
- External surface almost entirely covered by peritoneum
 - Greater curvature attached to omentum and mesentery of transverse colon
 - Lesser curvature attached to liver by gastrohepatic ligament

Microscopic Anatomy

- Mucosa consists of epithelium, lamina propria, muscularis mucosa at base
 - Types of mucosa do not exactly correlate with gross anatomic divisions
 - Mucosa has 2 compartments: Superficial pit (or foveolar) and deep glands
 - Surface/pit epithelium is same throughout stomach
 - CK 20 (+), CK 7 (-), CDX-2 (-)
 - Gland type differs with region
 - Cardiac and antral/pyloric glands are mucinous

- Body and fundus glands contain specialized chief and parietal cells (oxyntic mucosa)
- Transitional zones can share features of both types of mucosae
- Endocrine cells are present throughout mucosa but type differs with region
- Lamina propria
 - Scant fibromuscular stroma between glands
 - Contains arterioles, venules, and lymphatics
 - Scattered mononuclear cells are normal
- Muscularis mucosa
 - Thin double layer of smooth muscle that separates mucosa from submucosa
- Submucosa
 - Loose connective tissue containing vessels, lymphatics, nerves, ganglion cells
- Muscularis propria
 - 3 muscle layers: Inner oblique, middle circular, and outer longitudinal
- Subserosa/serosa
 - Thin covering of collagen and single layer of flat mesothelium
 - Present except where stomach is attached to omentum, mesocolon, and ligaments

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Antral/body transitional mucosa extends more proximally with age

Metaplasia

- Intestinal and pyloric-type metaplasia may be seen in variety of inflammatory conditions

Pitfalls and Artifacts

- Detached glandular cells can mimic signet ring cells
- Hypertrophic muscularis mucosa can be seen in numerous inflammatory and neoplastic conditions
- Proton pump inhibitor therapy may cause vacuolization, hypertrophy, dilated lumina in oxyntic glands

Anatomy of Stomach

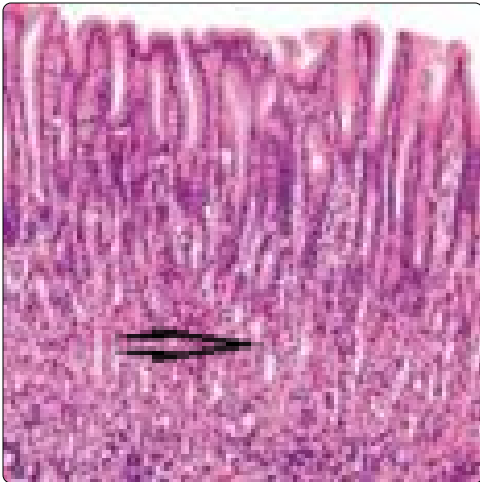
(Left) The anatomic regions of the stomach include the cardia (blue), the fundus (green), the body (yellow), and the pylorus (purple). The duodenum is red and the esophagus is orange in this graphic. **(Right)** The superficial gastric mucosa consists of pits or foveolae, which are invaginations of the surface epithelium. The pits are a conduit to the lumen for glandular secretions. The depth of the pits varies from short in the body (1/4 of total mucosal thickness) to longer in the cardia and antrum (~ 1/2 of total thickness).



Pits



Glands



Pit/Surface Epithelium

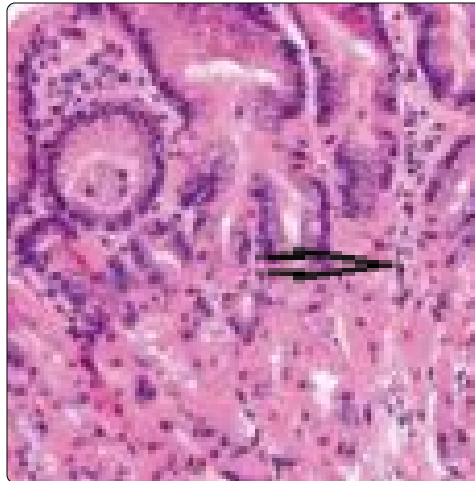


(Left) The deeper layer of the mucosa consists of coiled glands that empty into the bases of the pits. The glands differ in both structure and function in the various regions of the stomach; however, the types of mucosa do not exactly correlate with the gross anatomic division of the stomach. (Right) The surface/pit epithelium is mucinous and tall columnar throughout the stomach, regardless of region. Nuclei are small and basally located.

Pit/Surface Epithelium

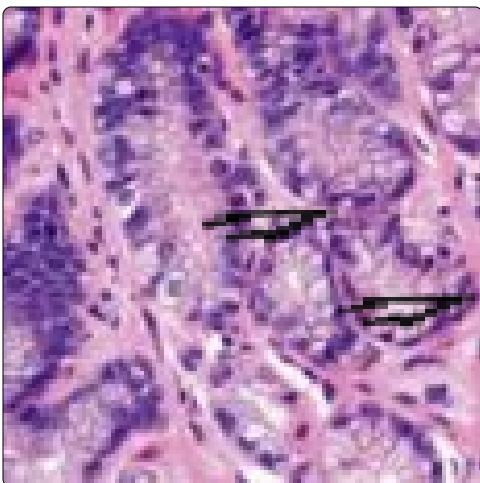


Mucous Neck Cells

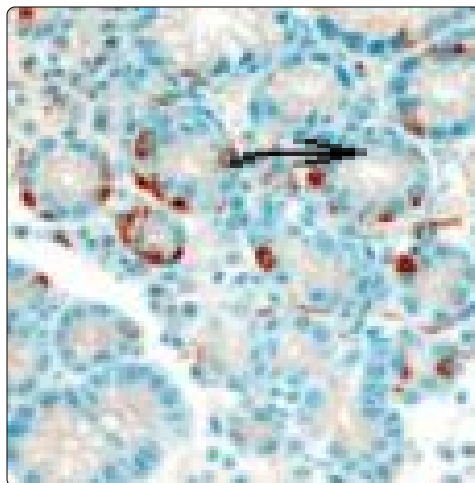


(Left) The surface/pit epithelium often appears to have pale pink apical mucin vacuoles that contain neutral mucins. (Right) The area between the pits and the glands is the neck. Mucous neck cells are mucinous, and the nuclei of the neck region may be slightly larger than those in the pits or the glands. Mucous neck cells are believed to be the proliferative zone for the gastric mucosa.

Endocrine Cells



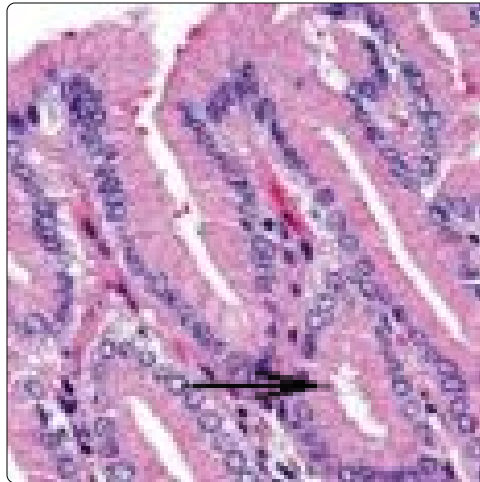
Endocrine Cells



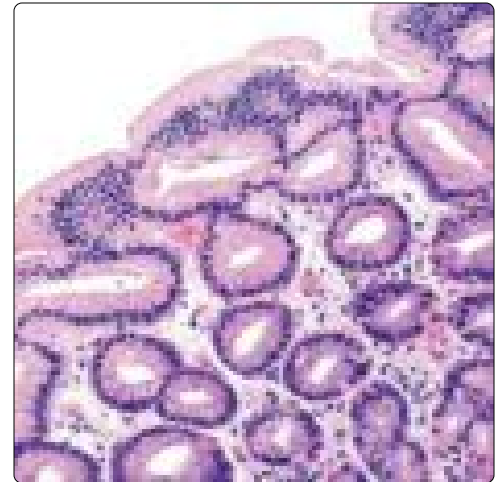
(Left) The type of endocrine cell differs with the region of the stomach. In the antrum, the pale, ovoid cells with gray cytoplasm are the gastrin-producing or G cells. The body contains ECL cells, which produce histamine, in the lower 1/3 of the mucosa near the chief cells. (Right) Endocrine cells are present throughout the gastric mucosa but may be difficult to appreciate on H&E. This CD56 stain highlights neuroendocrine cells within the glands.

Surface and Foveolar Epithelium

(Left) In this image, the cytoplasm of the epithelial cells shows a more granular eosinophilic quality due to numerous small mucin vacuoles. (Right) The lamina propria consists of relatively sparse stroma, which normally contains scattered macrophages, lymphocytes, mast cells, and plasma cells. Lamina propria is more abundant in the superficial mucosa between the foveolae.



Lamina Propria

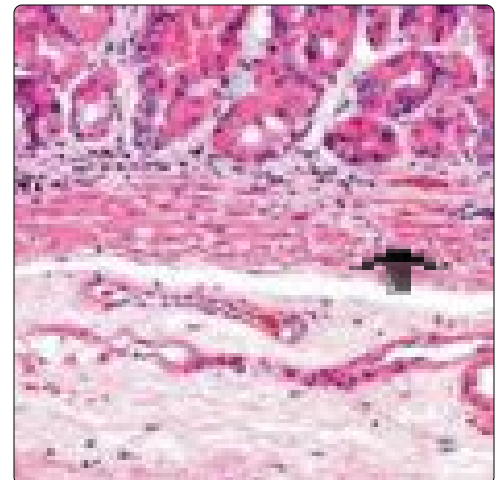


Lamina Propria

(Left) The lamina propria is composed of smooth muscle and loose collagen, and it contains arterioles and venules. In the normal stomach, the lymphatics are present in the lamina propria at the base of the mucosa, not at the surface. (Right) The muscularis mucosa consists of a thin double layer of smooth muscle that defines the base of the mucosa and separates it from the submucosa. Muscle fibers can occasionally extend into the base of the mucosa.

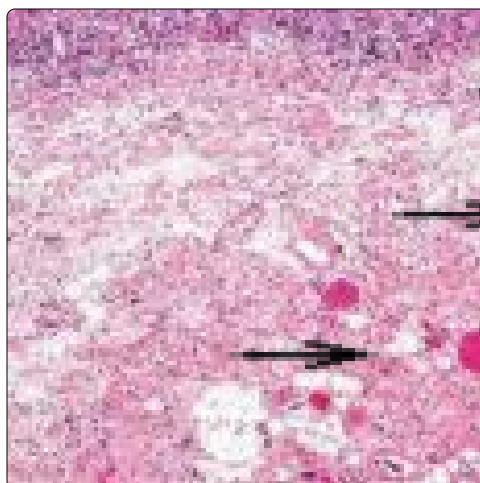


Muscularis Mucosa



Muscularis Mucosa

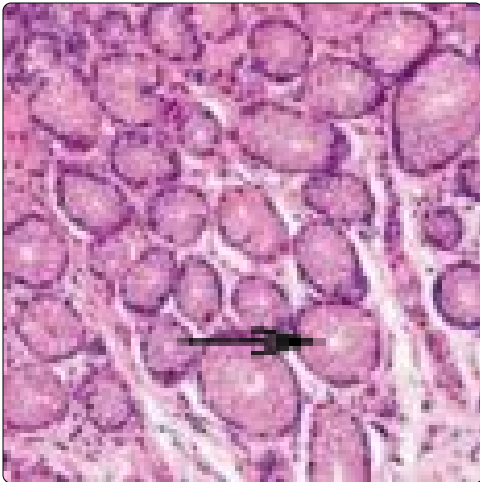
(Left) The muscularis mucosa becomes hypertrophic in a wide variety of reactive and neoplastic processes. A hypertrophic muscularis mucosa should not be confused with muscularis propria. (Right) Cardiac glands are mucinous and loosely aggregated. Occasional cystic glands are normal. The mucosa at the gastroesophageal junction may be mixed mucinous and oxyntic, as seen here, entirely mucinous, or rarely, entirely oxyntic.



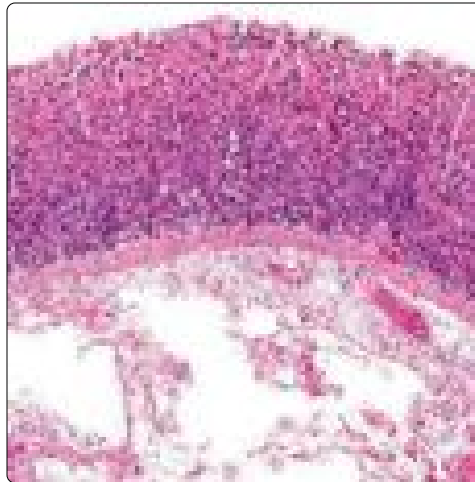
Cardia



Cardia and Antrum

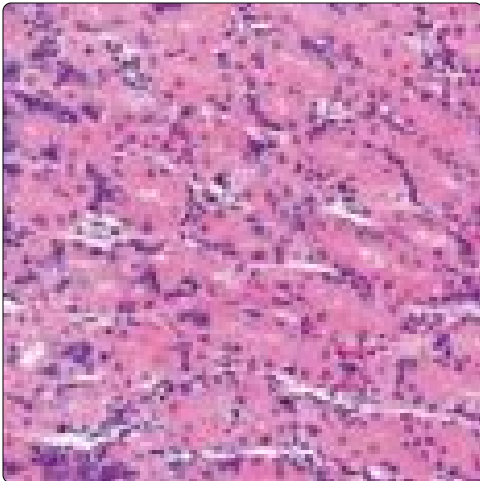


Oxyntic Mucosa

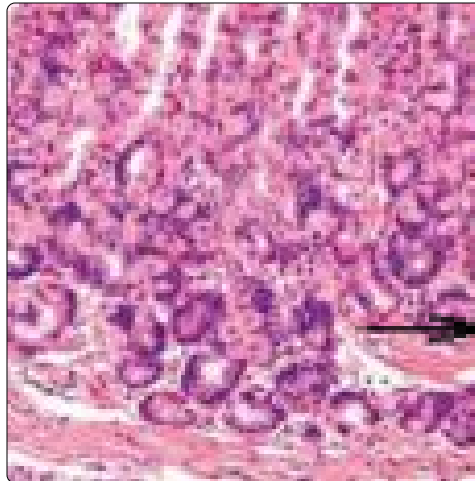


(Left) The cardia and antrum typically have more stroma than the fundus and body. The mucinous glands of the cardia and antrum may be separated into loose clusters by fine strands of collagen and smooth muscle . **(Right)** The fundus and body contain the specialized oxyntic glands. These glands consist of parietal cells, which produce acid, and chief cells, which produce digestive enzymes. The glands in the oxyntic mucosa are tightly packed and straight rather than coiled.

Parietal Cells

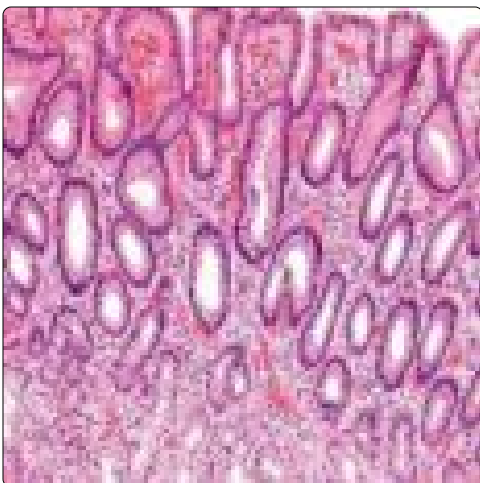


Parietal and Chief Cells



(Left) Parietal cells are pink with centrally placed nuclei. Parietal cells secrete acid and intrinsic factor. **(Right)** Parietal cells  are more prominent in the upper part of the mucosa, whereas the darker purple chief cells  are more prominent at the base. Chief cells are cuboidal to columnar with basally located nuclei, and they secrete digestive enzymes, such as pepsinogen.

Antrum



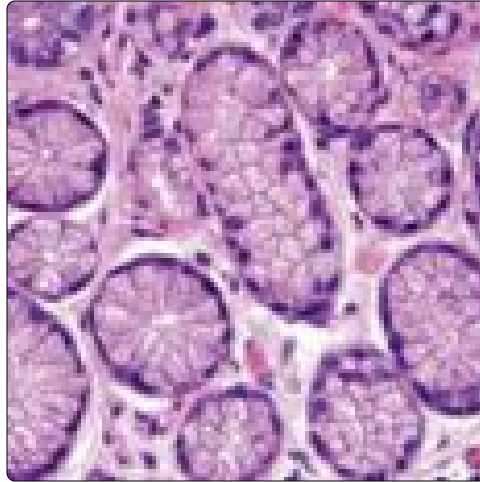
Intraepithelial Lymphocytes



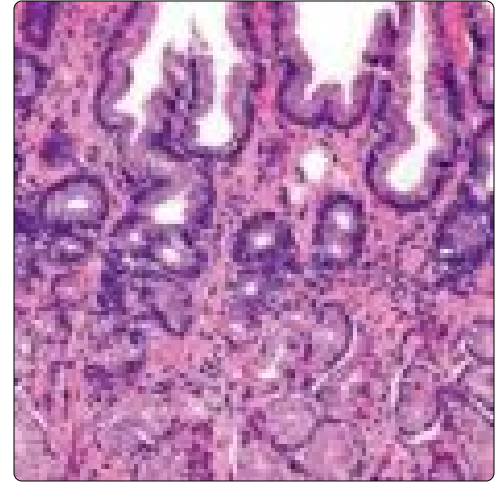
(Left) The antral and pyloric mucosa is similar to the cardiac mucosa, consisting of mucinous glands with abundant stroma. **(Right)** Scattered intraepithelial lymphocytes (T cells) within the gastric foveolar epithelium are not uncommon and can be identified by their artifactual clear halo .

Antrum

(Left) The mucinous antral glands have bubbly cytoplasm resembling Brunner glands and small, basally located nuclei. **(Right)** Transitional mucosa, such as that at the interface between the body and antrum, may contain both oxyntic and mucinous glands. This particular transition migrates proximally with age.

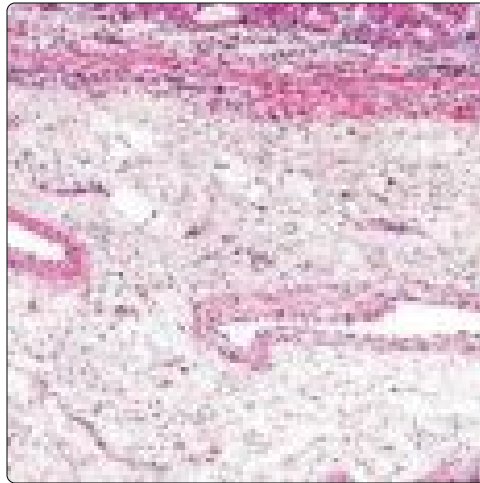


Transitional Mucosa

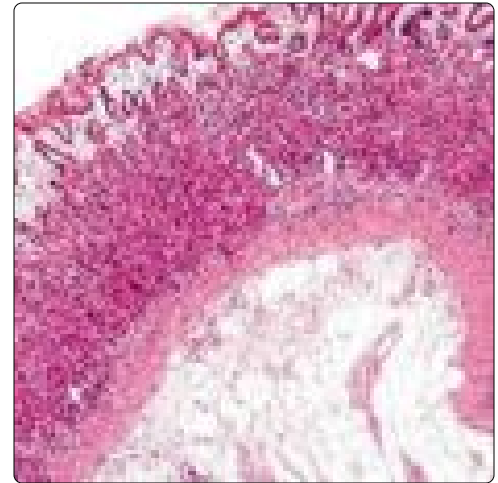


Submucosa

(Left) The submucosa lies beneath the muscularis mucosa and consists of loose connective tissue admixed with smooth muscle and variable amounts of adipose tissue. It contains vessels, lymphatics, nerves, ganglion cells, and scattered mononuclear cells as well. **(Right)** The amount of submucosal adipose tissue can be quite prominent in some individuals, as seen here. Note the submucosal vessels.

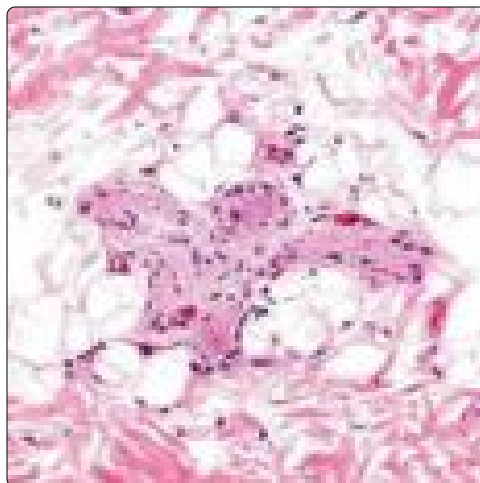


Submucosa

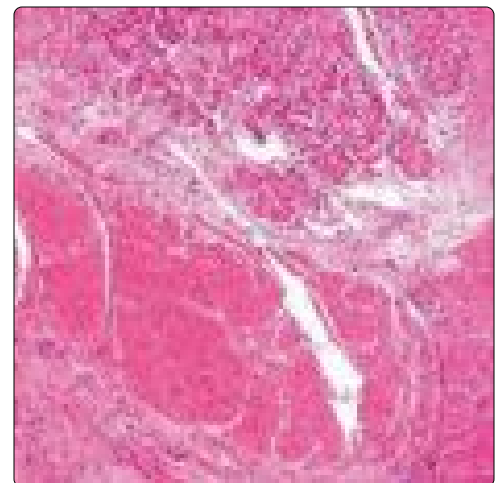


Submucosa

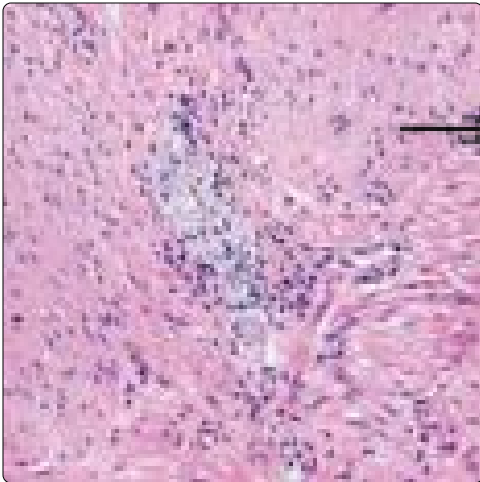
(Left) The network of nerves and ganglion cells present within the gastric submucosa is known as Meissner plexus. **(Right)** Unlike the rest of the tubular gut, the gastric muscularis propria consists of 3 muscle layers: Inner oblique, middle circular, and outer longitudinal.



Muscularis Propria



Auerbach Plexus

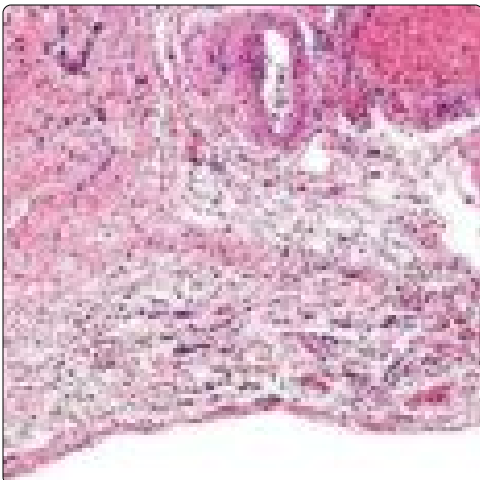


Interstitial Cells of Cajal

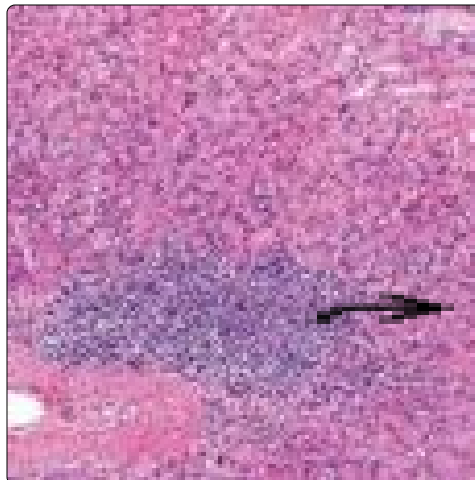


(Left) The nerves and ganglion cells of Auerbach (myenteric) plexus are present between the outer 2 layers of the muscularis propria. (Right) As in other regions of the gastrointestinal tract, the muscularis propria of the stomach contains a network of interstitial cells of Cajal that regulate gut motility. They can only be visualized on special stain (here, CD117 immunostain).

Subserosa

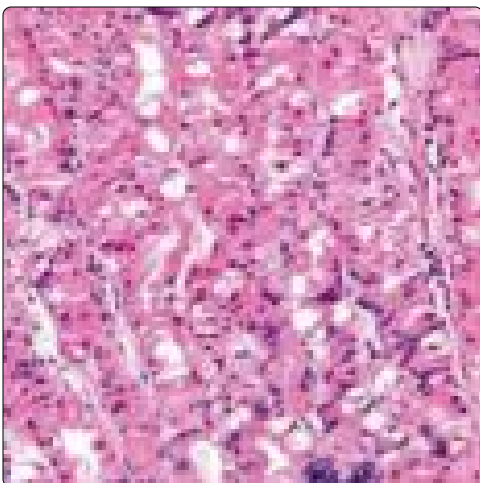


Lymphoid Tissue

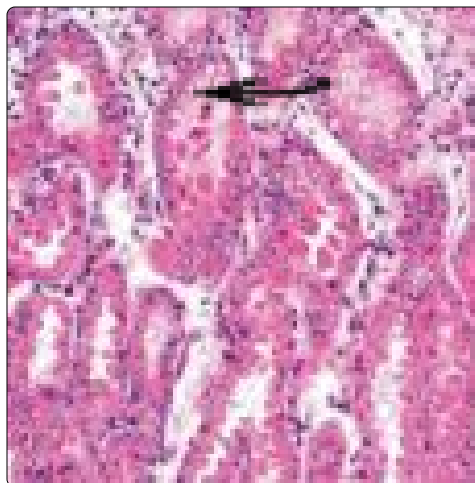


(Left) The subserosa consists of a thin layer of collagen that is present on the external surface of the stomach except where it is attached to the omentum, mesocolon, and ligaments. The subserosa is covered by a single layer of flat mesothelium, the serosa proper, which is part of the visceral peritoneum. (Right) Small lymphoid aggregates lacking reactive germinal centers are often seen in otherwise normal gastric biopsies and do not constitute a de facto diagnosis of chronic gastritis.

Proton Pump Inhibitor Effect



Proton Pump Inhibitor Effect



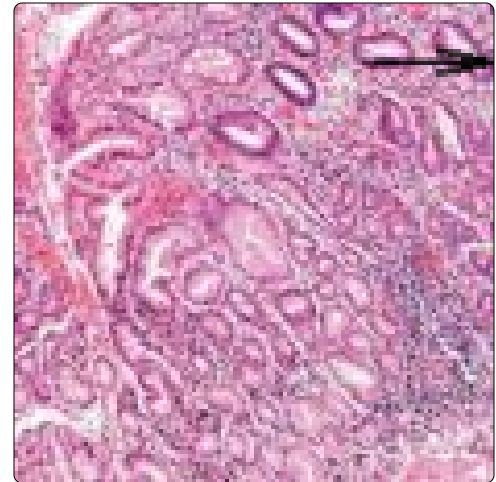
(Left) Proton pump inhibitor (PPI) use can cause numerous changes in parietal cells, including cytoplasmic vacuolization, increased glandular tortuosity, dilated glandular lumina, and protrusion of cytoplasm into the gland lumina creating a tufted, irregular appearance. (Right) This image shows prominent intraluminal projection of gastric parietal cells in a patient on chronic PPI, imparting a hobnail appearance to the epithelium.

Pylorus

(Left) Foci of goblet cells are normally found in the transition between antral and duodenal mucosa, and their presence in this location does not mean that the patient has intestinal metaplasia. **(Right)** Both intestinal and pyloric metaplasia are seen in this fundic biopsy of chronic atrophic gastritis. The intestinal and pyloric glands have replaced the oxyntic mucosa.

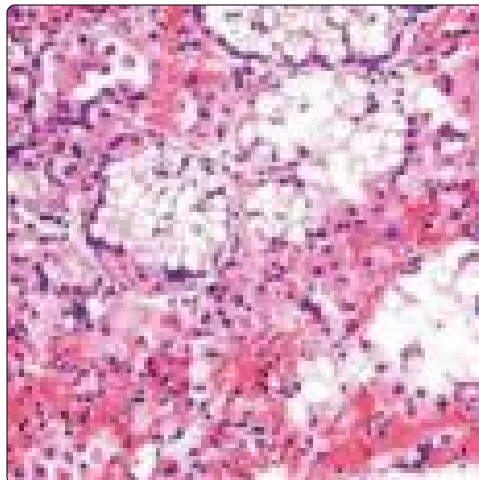


Metaplasia

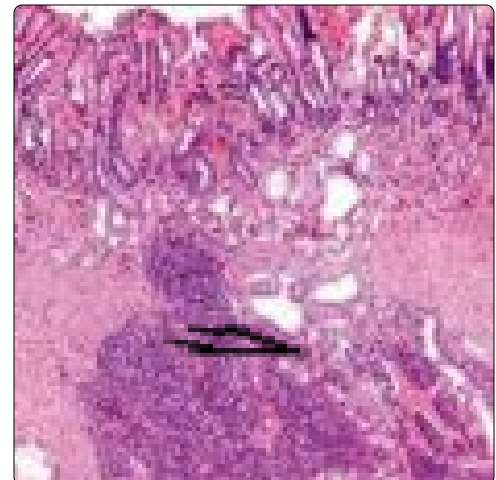


Pseudosignet Ring Cells

(Left) Detached clusters of signet ring-like cells can be seen in a number of inflammatory conditions. These detached clusters should not be mistaken for true signet ring cell carcinoma. **(Right)** Heterotopic pancreatic tissue is commonly found within the stomach. This example shows pancreatic acinar and ductal tissue at the distal antrum/pylorus.



Heterotopic Pancreas

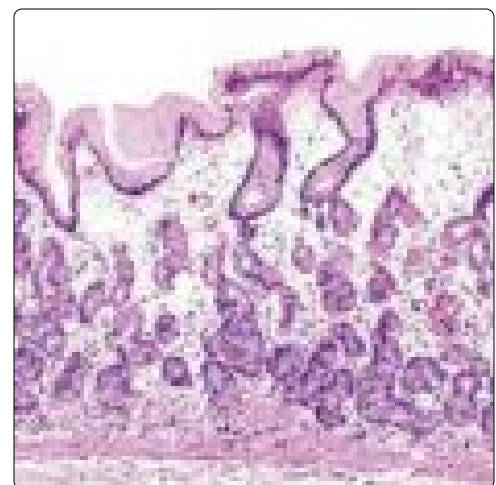


Surface Vacuolization Artifact

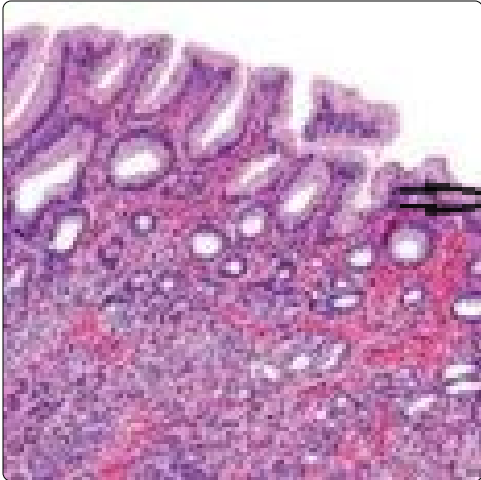
(Left) Artifactual cytoplasmic vacuolization is commonly seen as a focal finding in normal gastric mucosa and tends to manifest on the mucosal surface. **(Right)** Stretching of the gastric mucosa, either from the endoscopic procedure or from an underlying mass lesion, can splay apart the mucosal glandular elements, mimicking edema.



Stretching Artifact



Procedure-Related Mucosal Hemorrhage

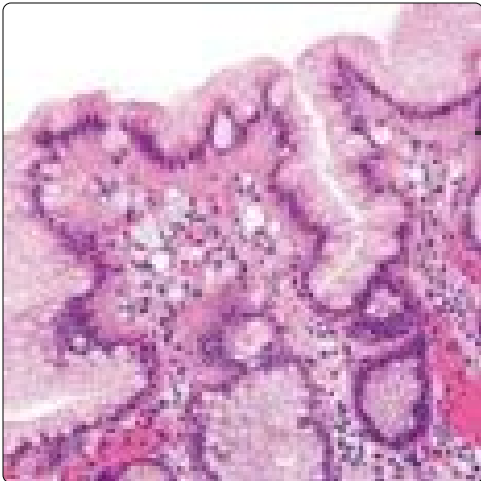


Surface Epithelial Disruption

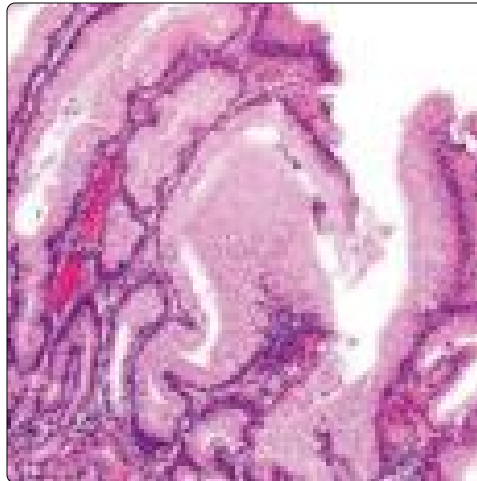


(Left) Artfactual hemorrhage  into the lamina propria is not uncommon and is usually related to specimen procurement during the biopsy procedure. Importantly, other features of true hemorrhagic gastritis are absent. **(Right)** Disruption of the surface epithelium  may occur during the biopsy procedure and may histologically mimic a true erosion; however, the latter often features some degree of acute inflammation.

Pseudogoblet Cells

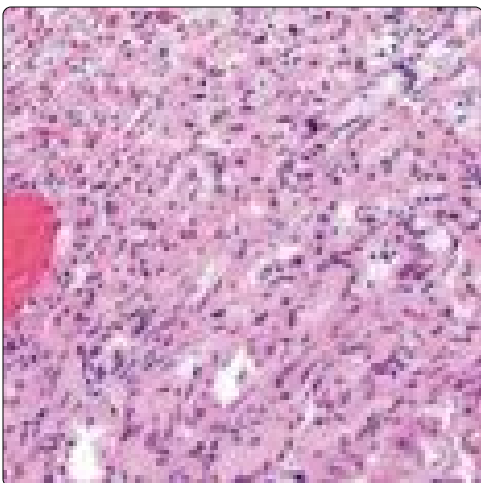


Tangential Sectioning

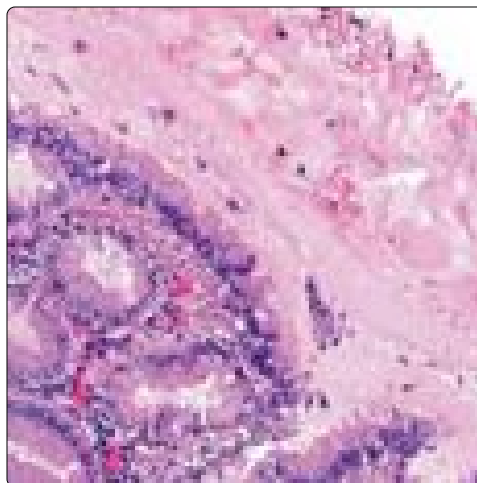


(Left) Some surface and foveolar epithelial cells  may become so distended by mucin that they resemble goblet cells; however, the latter often has blue-gray cytoplasm and stains intensely blue rather than magenta on alcian blue/PAS combined stain. **(Right)** Tangential sectioning of the gastric mucosa can lead to unusual pit and gland architectures as well as the impression of multilayered epithelium.

Crush and Cautery Artifacts



Surface Mucin



(Left) During the biopsy procedure, tissue may be crushed or even distorted by electrocautery, leading to interpretive difficulties. **(Right)** Acellular surface mucin may be conspicuous in some gastric biopsies or resection specimens and is related to disruption of mucin-containing epithelial cells on the surface.

SMALL INTESTINE

Macroscopic Anatomy

- Coiled tubular organ extending from pylorus of stomach to ileocecal valve (ICV)
 - Average length in adults is 6-7 m
 - Primary function is absorption of nutrients
 - Plicae circulares (circular mucosal folds)
- Duodenum
 - Most proximal portion; extends from pylorus to duodenojejunal flexure
- Jejunum
 - Middle portion, primarily lies within upper abdomen; no distinct anatomic transition to ileum
- Ileum
 - Most distal portion, primarily lies within lower abdomen and pelvis

Microscopic Anatomy

- Mucosa
 - Epithelium, lamina propria, muscularis mucosa
 - 2 epithelial compartments: Villi and crypts
 - Villous compartment
 - Absorptive cells, goblet cells, endocrine cells
 - Crypt compartment (crypts of Lieberkühn)
 - Crypts are depressions of surface epithelium
 - Absorptive cells, goblet cells, endocrine cells, Paneth cells
 - Mitoses are frequent (1-12 per crypt)
 - Normal villous to crypt ratio is 3-5:1
 - Epithelium is single layer composed of multiple cell types
 - CK 20 (+), CK 7 (-), CDX-2(+)
 - Scattered intraepithelial lymphocytes (T cells) normal
 - Delicate basement membrane separates epithelium from lamina propria
 - Lamina propria
 - Forms cores of villi
 - Mixed mononuclear inflammatory cells are present
 - Contain vessels and central blind-ending lymphatic channel or lacteal

- Thin strips of smooth muscle (muscularis mucosa) at interface with submucosa
- Submucosa
 - Loose connective tissue that contains inflammatory cells, adipose tissue, vessels, lymphatics, nerves, ganglion cells (Meissner plexus)
- Muscularis propria
 - 2 layers: Outer longitudinal and inner circular
 - Auerbach (myenteric) plexus lies between layers, as do interstitial cells of Cajal
- Serosa
 - Mesothelial-lined thin layer of connective tissue external to muscularis propria; vessels, nerves in subserosa
- Site-specific features
 - Duodenum
 - Abundant mononuclear cells in lamina propria
 - Submucosal Brunner glands
 - Villi shorter than in jejunum and ileum
 - Jejunum
 - Taller villi with club-shaped tips
 - No Brunner glands or Peyer patches
 - Ileum
 - ICV and submucosa may have very prominent fat
 - Ileum has most goblet cells
 - Peyer patches

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Peyer patches increase until puberty and regress with age but are present throughout life

Environment-Related Variations

- Persons from tropics may have shorter, broader villi, but villus to crypt ratio is maintained

Pitfalls and Artifacts

- Artifactual shortening and blunting of villi
- Tangential sectioning of tissue
- Pigment deposition within lamina propria or Peyer patches

Small Intestine Plicae

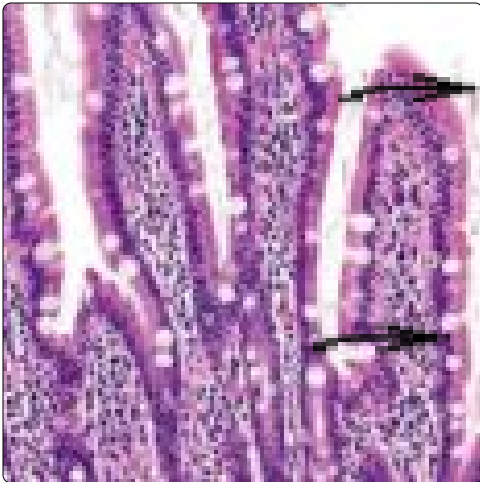


(Left) The plicae circulares, or circular folds, of the small intestine are represented histologically as long, finger-like projections of mucosa and underlying submucosa. The mucosa, in turn, has numerous finger-like projections of its own (villi). (Right) The mucosa consists of the villi and crypt compartments. Each villus is a long, slender projection covered by epithelium; the lamina propria forms the cores of the villi. The crypts lie between and beneath the villi. Villi cover the entire luminal surface of the small bowel to enhance the surface area.

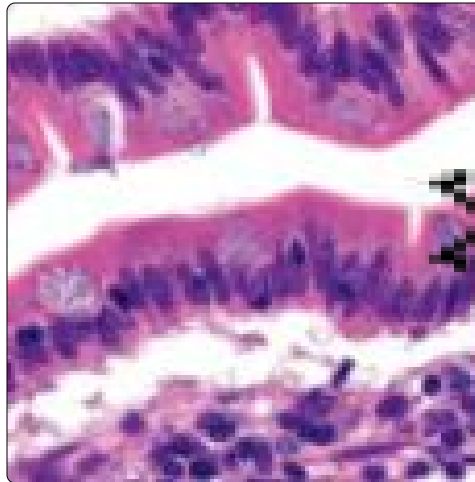
Small Intestine Mucosa



Epithelium

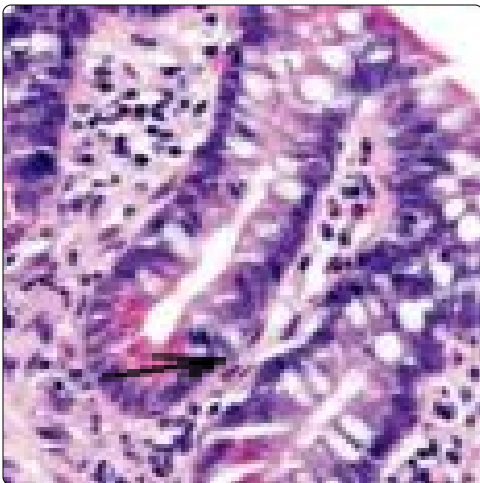


Brush Border

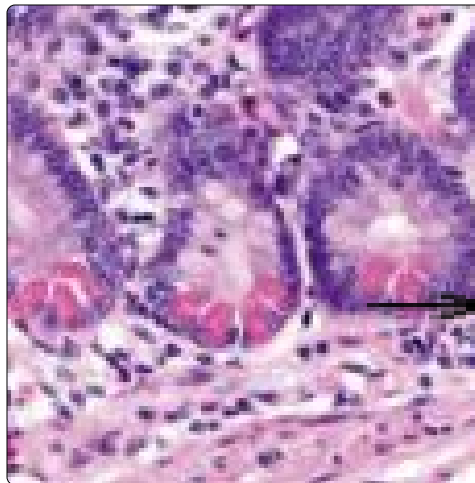


(Left) The absorptive cells are simple columnar cells with round to oval, basally located nuclei and eosinophilic cytoplasm. Goblet cells are interspersed among the absorptive cells. These barrel-shaped cells with characteristic mucin droplets secrete mucin. (Right) The thin brush border of the small intestine epithelium is composed of microvilli (best seen on electron microscopy) and the glycocalyx. This complex increases the surface area of the small intestine and houses important digestive enzymes.

Crypts

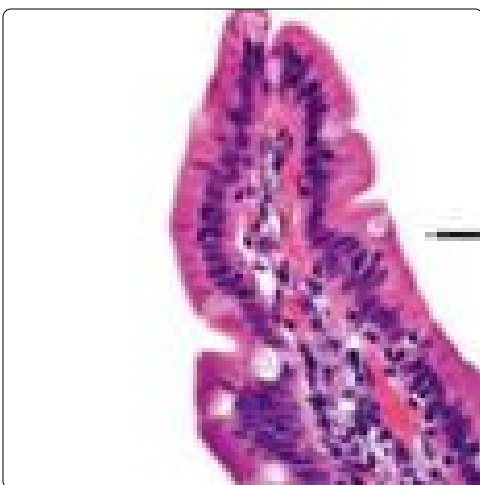


Paneth Cells



(Left) The epithelium rests on a delicate basement membrane separating it from the lamina propria. Crypts contain Paneth cells, goblet cells, absorptive cells, neuroendocrine cells, and occasional intraepithelial lymphocytes. Crypts are the regenerative compartment of the mucosa and thus contain mitoses. (Right) Paneth cells, present throughout the small intestine, are pyramidal cells with their apices toward the lumina of the crypts. The supranuclear eosinophilic granules contain various enzymes, such as lysozyme.

Intraepithelial Lymphocytes



Lacteals



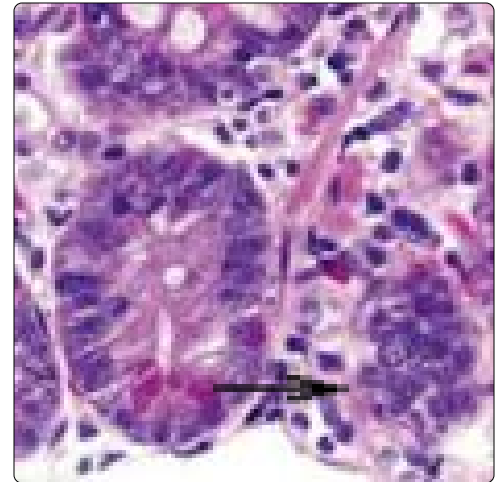
(Left) Intraepithelial T lymphocytes are normally present in the small intestine and are usually located right above the basement membrane of the epithelium. Approximately 1 lymphocyte per 4-5 epithelial cells is considered normal. In the duodenum, the lymphocytes often extend along the sides of the villi and spare the tips. (Right) Villi contain a central blind-ending lacteal or lymphatic channel.

Lamina Propria

(Left) The lamina propria of the small intestine normally contains abundant plasma cells, lymphocytes, mast cells, histiocytes, and scattered eosinophils. Neutrophils are not a normal component of the inflammatory milieu. **(Right)** Endocrine cells in the small intestine are small, pyramid-shaped, and have small dark pink or red granules. In contrast to Paneth cells, which have supranuclear granules, endocrine cells have subnuclear granules.

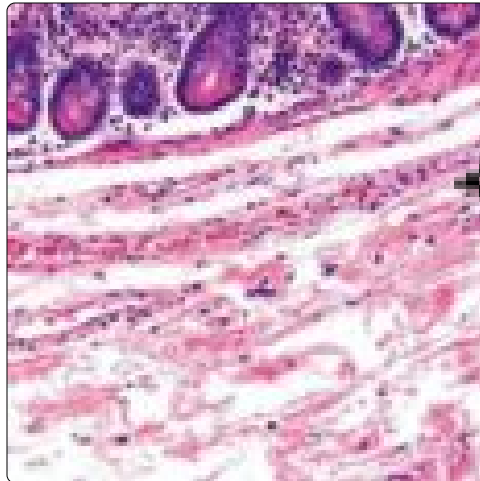


Endocrine Cells

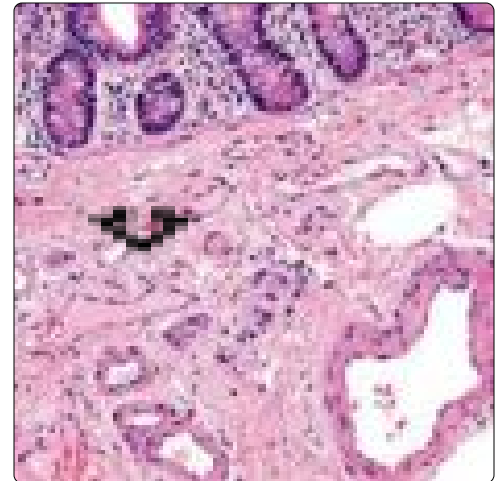


Muscularis Mucosa

(Left) The muscularis mucosa is a thin layer of muscle that separates the mucosa, above, from the underlying submucosa. It anchors the mucosa and provides structural support. **(Right)** The submucosa consists of loose connective tissue and contains nerves, vessels, lymphatics, and fat. The Meissner plexus is also present, represented by clusters of ganglion cells.

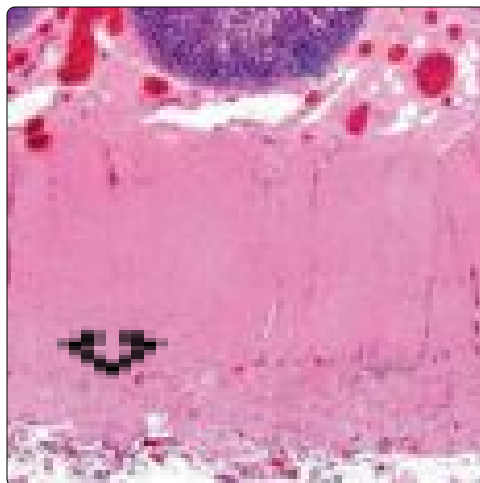


Submucosa

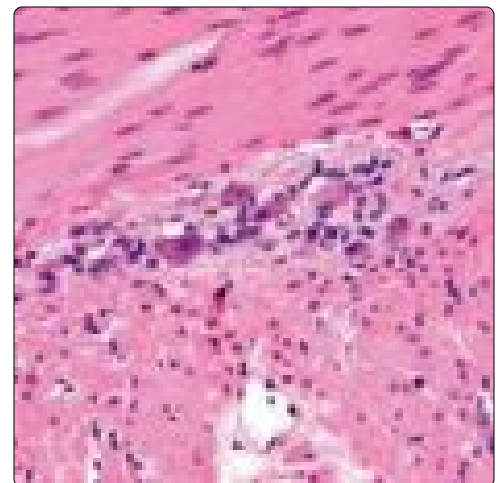


Muscularis Propria

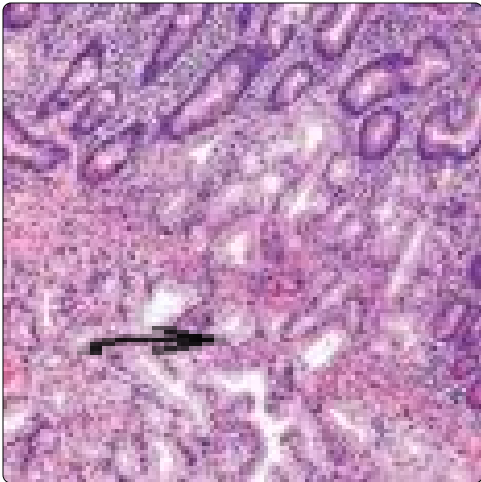
(Left) The muscularis propria consists of 2 layers: The inner circular layer and the outer longitudinal layer. Auerbach (or myenteric) nerve plexus can be seen in between these layers. **(Right)** The Auerbach plexus lies between the 2 layers of the muscularis propria and contains both nerves and ganglion cells.



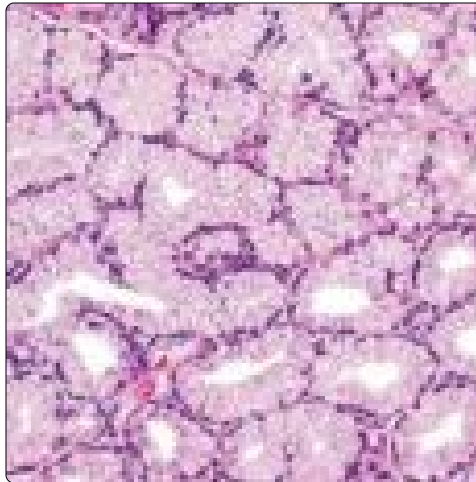
Auerbach (Myenteric) Plexus



Brunner Glands

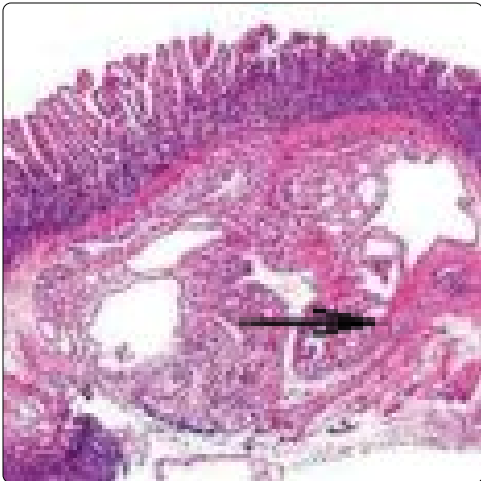


Brunner Glands



(Left) Brunner glands are a characteristic feature of the duodenum. These tubuloalveolar glands form lobules of variable sizes and are located primarily in the submucosa but may also extend into the mucosa, as depicted. (Right) Brunner glands contain cuboidal/columnar cells with small basal nuclei and abundant pale cytoplasm filled with mucin. The lobules are drained to the crypt epithelium by elongated ducts (not shown).

Brunner Gland Hyperplasia/Hamartoma



Jejunum

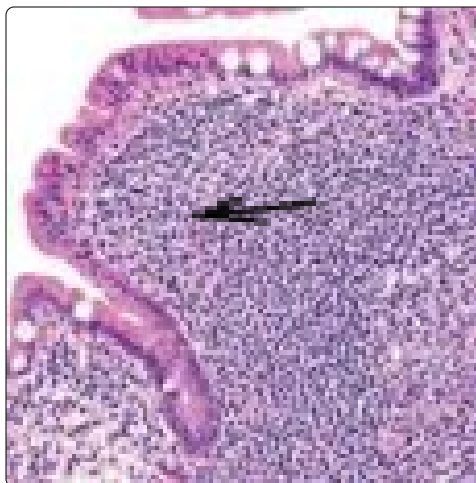


(Left) Enlarged nodules of prominent or hyperplastic Brunner glands can be seen in the duodenum and appear as a mass or polyp macroscopically. In some cases, the glands are cystically dilated and have been described as Brunner gland hamartoma. (Right) The jejunal villi are tall and slender but may have broader or club-shaped tips. The jejunum lacks both the Brunner glands of the duodenum and the Peyer patches of the ileum.

Ileum



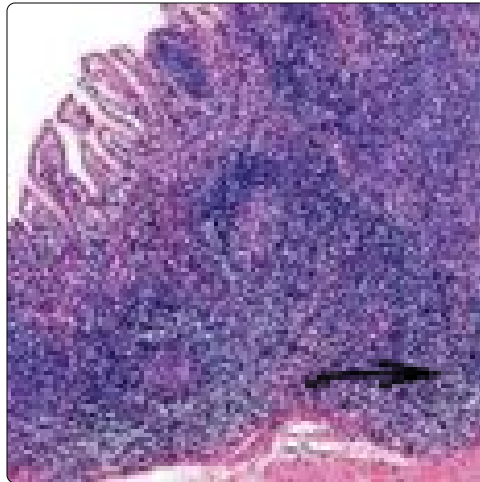
Peyer Patches



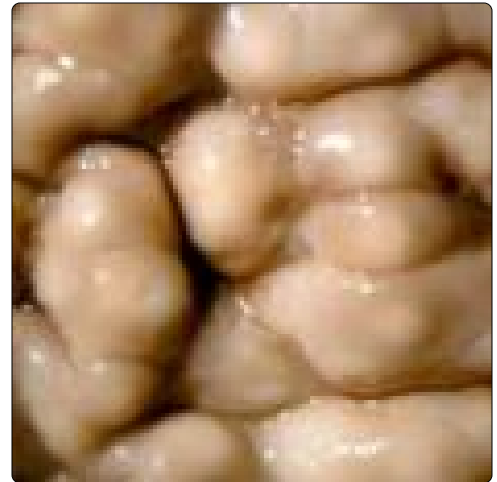
(Left) The ileum contains more goblet cells and fewer absorptive cells than the more proximal segments of the small intestine. (Right) Peyer patches are large aggregates of lymphoid tissue that are unique to the ileum. They involve both the mucosa and submucosa and usually have germinal centers. The overlying follicle-associated epithelium lacks goblet cells and generally shows a loss of developed villous architecture. Small lymphoid aggregates (non-Peyer) may be found anywhere in the small intestine, however.

Peyer Patches

(Left) In some cases, Peyer patches may be quite large and floridly hyperplastic, as depicted. Note the reactive germinal centers . Occasionally, brown-black pigment deposits (not shown) can be identified within the basal portions of the Peyer patches. (Right) Peyer patches may be very prominent in children and visible to the naked eye as white nodules . (Courtesy G. Gray, MD.)

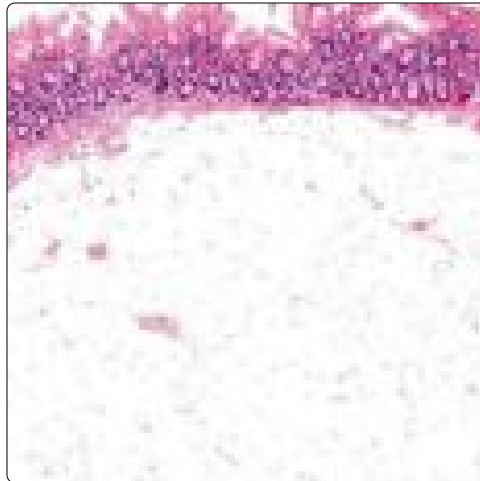


Peyer Patch



Ileocecal Valve

(Left) The ileocecal valve, as well as the ileocecal area in general, may contain remarkable amounts of adipose tissue that may even form a mass. (Right) When the mucosa is stripped off of the muscularis mucosa during a biopsy procedure, the tissue spreads laterally due to the loss of structural support, and the normal villous architecture may be lost. Note, however, that there is no abnormal intraepithelial lymphocytosis. Artfactual blunting can also be seen in areas overlying prominent Brunner glands (not shown).

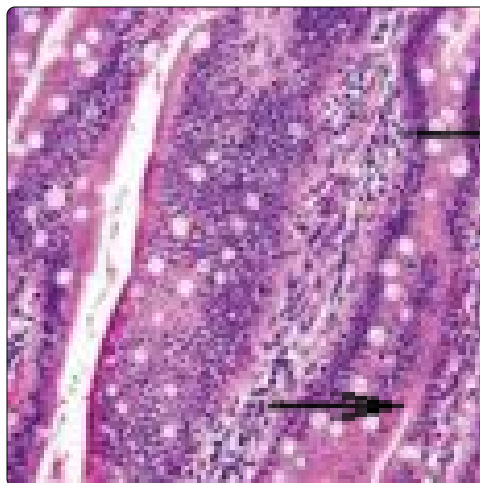


Artfactual Villous Blunting

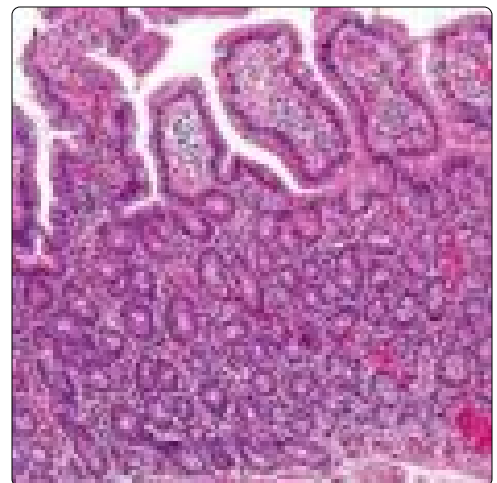


Tangential Orientation

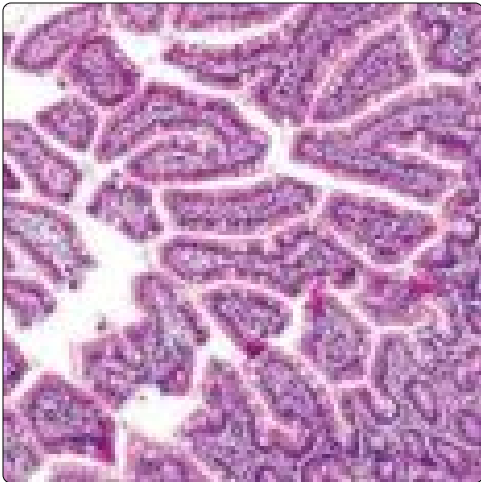
(Left) Tangential or oblique sectioning of maloriented small intestinal mucosa can lead to the appearance of increased cellularity or stratification  of the epithelium. (Right) Tangential sectioning of the small intestine can also lead to the false appearance of shortened or blunted villi; however, a useful clue to identifying this artifact is the appearance of multiple round to oblong crypts in the lower half of the mucosa rather than a single layer of elongated crypts.



Tangential Orientation



Tangential Orientation

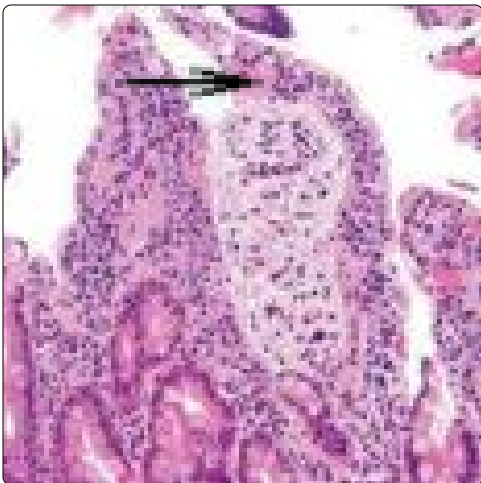


Intraluminal Mucin and Debris



(Left) Tangential sectioning of malorientated small intestine mucosa can create an appearance of interlocking villi floating within the lumina. **(Right)** Small intraluminal mucin flecks and other forms of debris are commonly identified in small intestine histologic preparations. This finding may potentially be mistaken for *Giardia* infection.

Crushed Brunner Glands

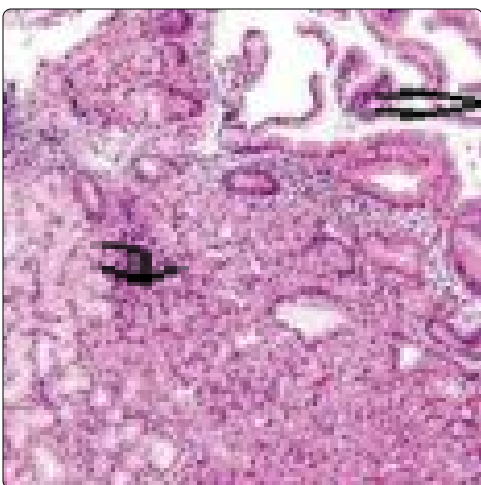


Pseudomelanosis

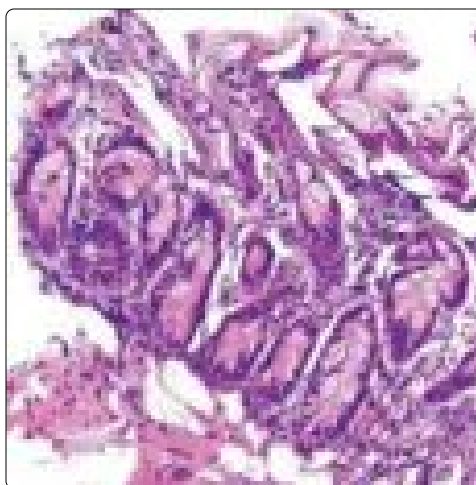


(Left) During the biopsy process, small intestine tissue can suffer several artifacts, including damage to surface epithelium and Brunner glands. The latter, when localized, can mimic the appearance of a small mesenchymal or neuroendocrine proliferation. **(Right)** Occasionally, various pigments, such as hemosiderin, can be identified within the lamina propria of the duodenum. This finding has been described as pseudomelanosis duodeni. (From DP: GI Endoscopy.)

Heterotopic Gastric Mucosa



Cautery Artifact



(Left) Heterotopic tissue in the small intestine is relatively common and is usually composed of either gastric mucosa or pancreatic acinar &/or ductal tissue. Note the presence of Brunner glands and small intestinal surface epithelium in this image. **(Right)** As in other sites in the tubular gut, cautery artifact from the biopsy procedure can lead to difficulty in the histologic evaluation of the small intestine, particularly in the assessment of dysplasia.

LARGE INTESTINE (COLON)

Macroscopic Anatomy

- Right colon
 - Ileocecal valve, cecum, ascending colon, hepatic flexure, and proximal transverse colon
- Left colon
 - Distal transverse colon, splenic flexure, descending colon, sigmoid colon, rectum
- Ascending, descending, sigmoid, and rectum are retroperitoneal and have no serosa/peritoneal lining

Microscopic Anatomy

- Mucosa
 - Simple columnar epithelium with apical microvilli, numerous goblet cells, and underlying basement membrane
 - CK20(+), CK7(-), CDX-2(+)
 - Paneth cells in right side of colon only
 - Lacks villi

- Lamina propria containing mixed mononuclear inflammatory cells
- Thin strips of smooth muscle (muscularis mucosa) at interface with submucosa
- Submucosa
 - Loose connective tissue containing fat, vessels, nerves
- Muscularis propria
 - 2 layers: Inner circular and outer longitudinal
- Serosa and subserosa
 - Mature fibroadipose tissue ± mesothelial (peritoneal) lining

CONSIDERATIONS AND VARIATIONS

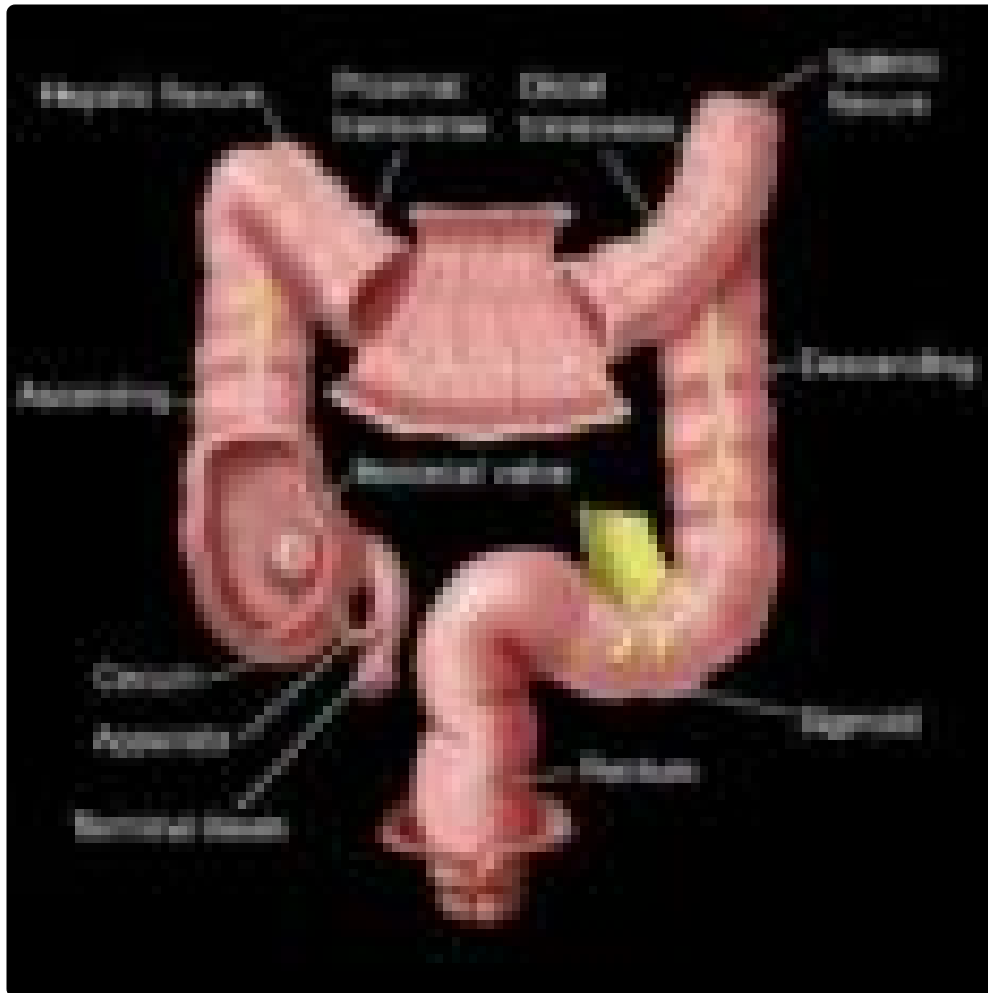
Metaplasia

- Paneth cell metaplasia (left side of colon)

Pitfalls and Artifacts

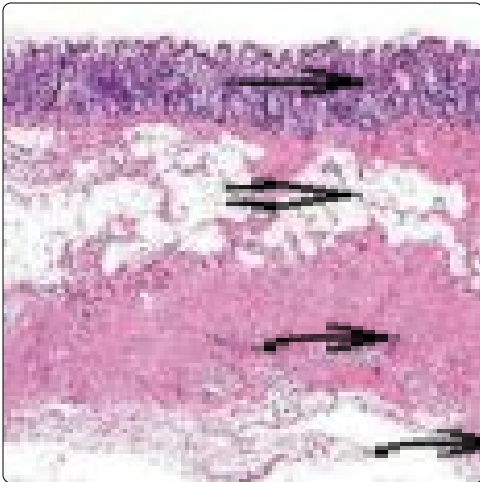
- Tangential sectioning of tissue
- Endoscopy-related trauma
- Vegetable (fecal) matter

Large Intestine



This illustration shows the major regions of the large intestine from proximal (cecum and ileocecal valve) to distal (rectum). They are sometimes referred to by collective designations: Right colon (cecum, ascending colon, hepatic flexure, proximal transverse colon) and left colon (distal transverse, splenic flexure, descending colon, sigmoid, rectum).

Anatomic Layers



Mucosa

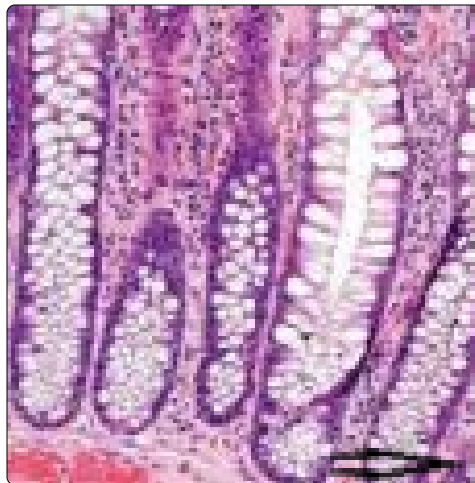


(Left) This scanning magnification H&E shows the microanatomic layers of the large intestine (colon), including the mucosa [arrow], submucosa [arrow], muscularis propria [arrow], and serosa/subserosa [arrow]. (Right) The mucosa of the large intestine extends from the luminal epithelium at the surface [arrow] down to the muscularis mucosa [arrow]. The crypts may appear round instead of tube-like if tissue is oriented obliquely. Note the mucosa-associated lymphoid tissue (MALT) [arrow].

Columnar Epithelium

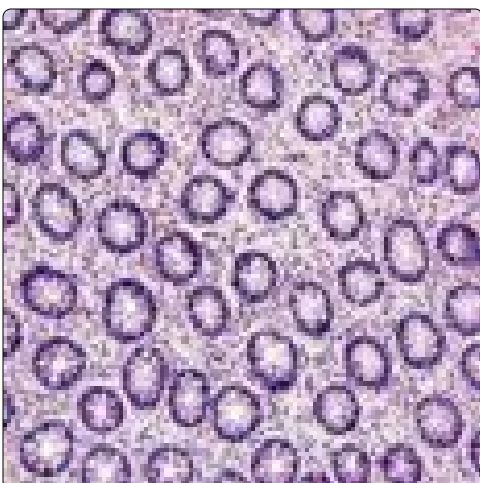


Colonic Crypts

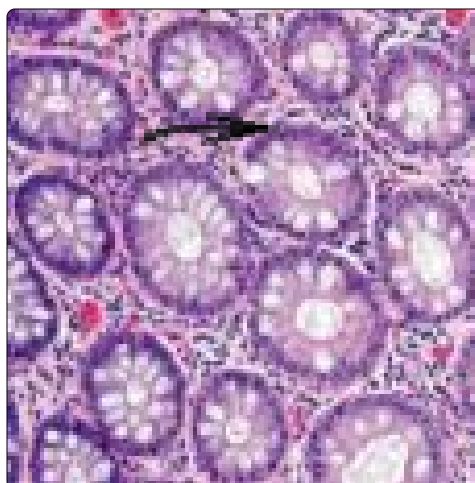


(Left) The simple columnar epithelium of the colon has a fine brush border containing innumerable microvilli [arrow]. Interspersed goblet cells [arrow] are responsible for secreting mucin. (Right) The crypts have a rack of test tubes configuration when viewed longitudinally. They are relatively evenly distributed and extend from the luminal surface down to the muscularis mucosa [arrow]. They generally do not show branching or distortion.

Colonic Crypts



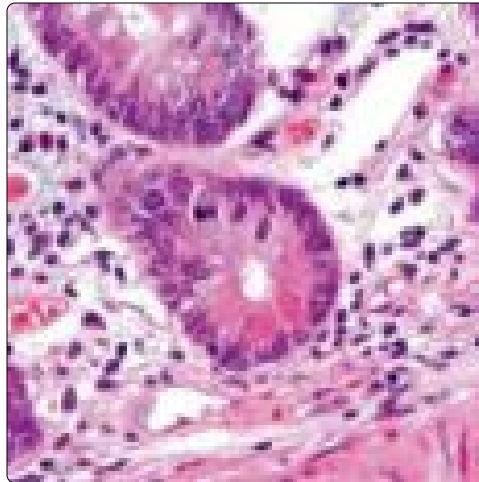
Lamina Propria



(Left) When viewed from the top down, the colon appears to be a sea of glands with regularly spaced crypts. The intervening lamina propria is normally filled with a mixed chronic inflammatory infiltrate. For this reason, a diagnosis of "chronic inflammation" in the colon is inappropriate. (Right) The mixed inflammatory infiltrate of the lamina propria includes lymphocytes, plasma cells, and eosinophils [arrow]. Scattered normal mitotic figures [arrow] are not an uncommon finding in the crypt epithelium.

Paneth Cells

(Left) On the right side of the colon (proximal to the midtransverse colon), specialized immune cells with prominent pink/red granules (Paneth cells) are common. Distal to the midtransverse colon, Paneth cells are metaplastic and indicate chronic injury. (Right) Endocrine cells can also be seen scattered throughout the colon and are identified by their small, basally located red granules. Unlike Paneth cells, these granules do not face the lumen of the crypt.

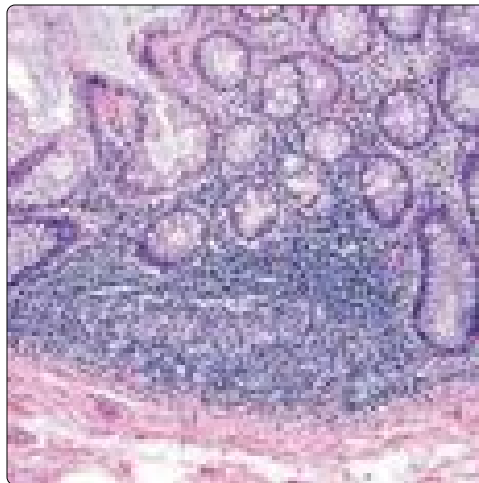


Endocrine Cells

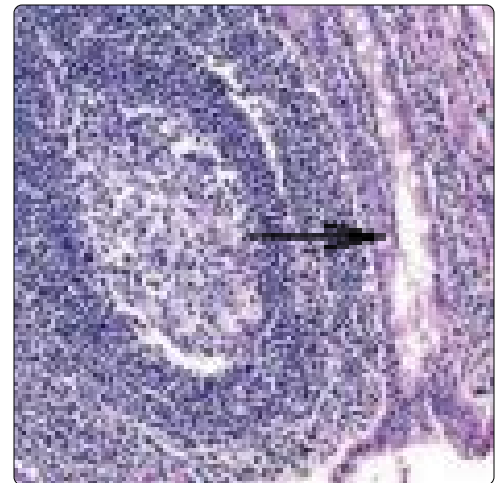


Mucosa-Associated Lymphoid Tissue

(Left) MALT is common throughout the colon and represents the colonic component of the immune system. These aggregates function similar to the tonsils or lymph nodes, are composed mostly of T cells, and vary widely in size from small to very large. (Right) Some aggregates show a central, reactive germinal center that is composed of B cells. These can be seen in any reactive, infectious, or inflammatory conditions in the colon.

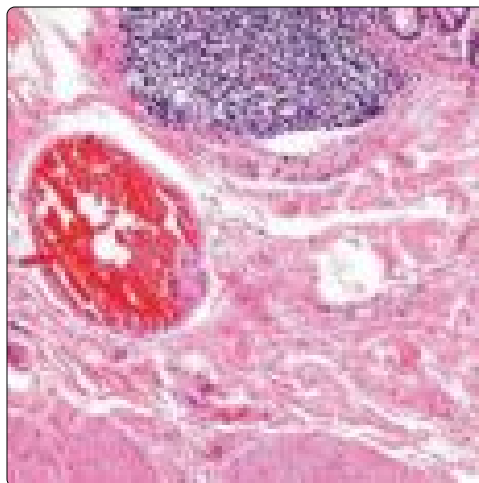


Reactive Mucosa-Associated Lymphoid Tissue

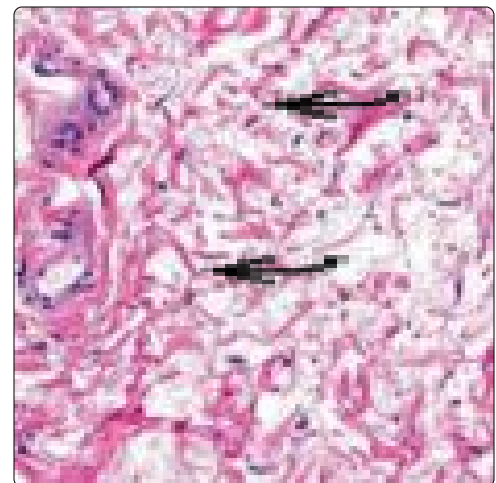


Submucosa

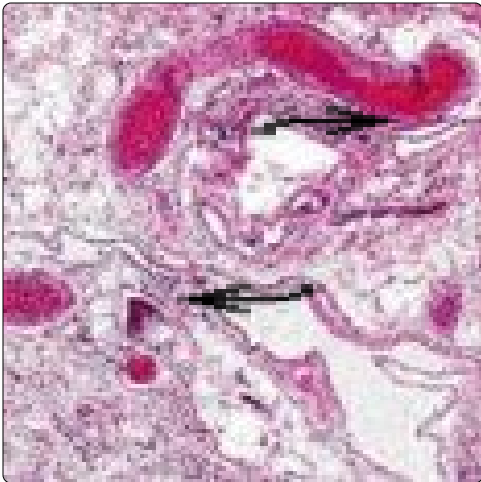
(Left) As is seen in other segments of the gastrointestinal tract, the colonic submucosa contains blood vessels, nerves and ganglion plexi, fibrocollagenous stroma, and downward extensions of the MALT. (Right) Sometimes the submucosal tissue is looser and more edematous than dense and fibrous, and the collagen takes on a shredded and splayed appearance. This image also shows blood vessels.



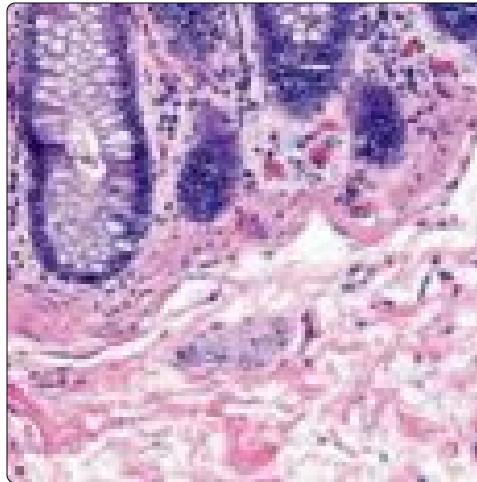
Loose or Edematous Submucosa



Prominent Vascularity in Submucosa

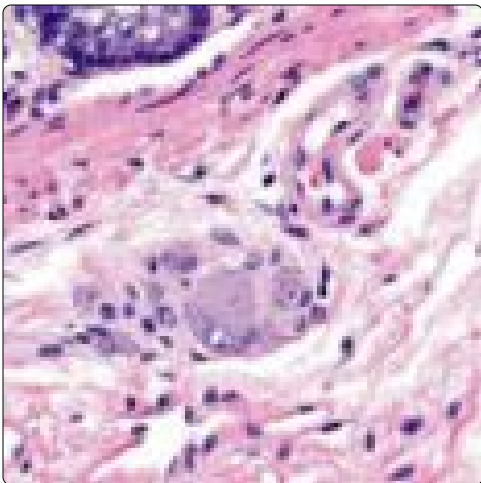


Meissner Plexus

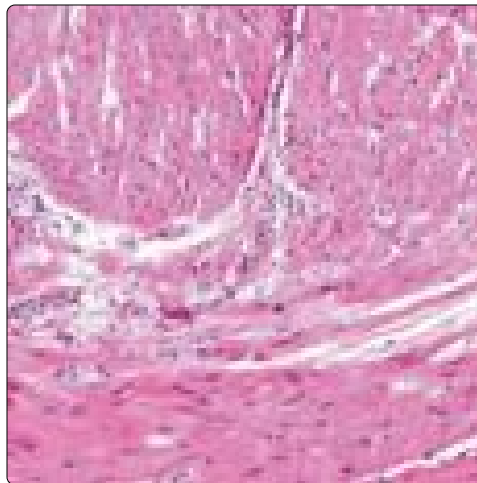


(Left) The vasculature of the colonic mucosa is often prominent in resection specimens with tortuous or serpentine contours. Arterioles  often appear filled with blood. **(Right)** The network of ganglion cells  and nerve twigs in the submucosa is collectively known as Meissner plexus.

Ganglion Cell

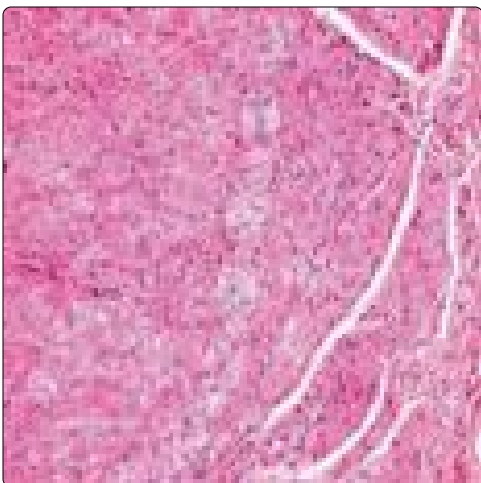


Muscularis Propria

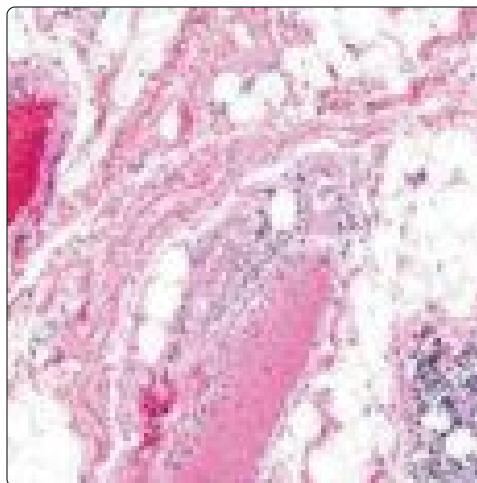


(Left) Ganglion cells are often arranged in clusters and feature abundant basophilic cytoplasm. Nuclei are generally round, and prominent nucleoli  are common. **(Right)** The muscularis propria is composed of 2 layers of mature smooth muscle: An inner circular layer and an outer longitudinal layer. The longitudinal layer runs the length of the colon (parallel), and the circular layer encircles the colon.

Myenteric Plexus With Ganglion Cells



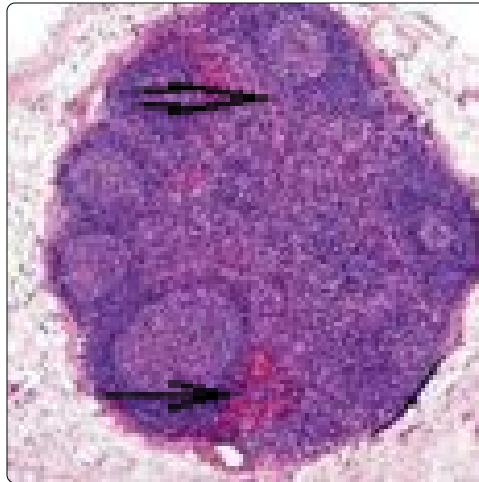
Serosa and Subserosa



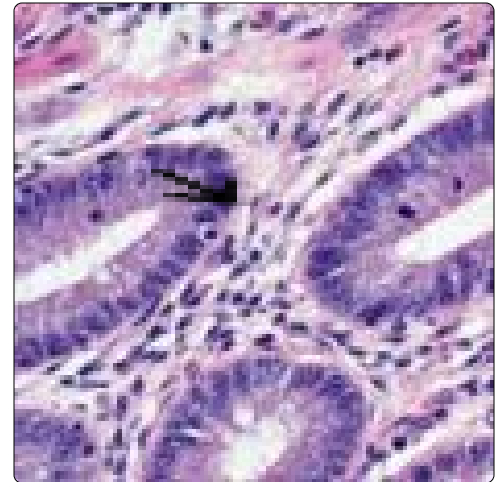
(Left) In between the 2 layers of the muscularis propria lies the myenteric (Auerbach) nerve plexus , which contains both nerves and ganglion cells and helps coordinate muscular contractions. **(Right)** The subserosa contains fat, vessels, collagen, fibroblasts, and chronic inflammatory cells. Intraperitoneal segments of colon are lined by mesothelium (known as serosa). Retroperitoneal segments of the large bowel lack a true peritoneal lining; this is important to know for cancer staging purposes.

(Left) Lymph nodes are commonly found within serosal fat. They are distinguished from lymphoid aggregates by the presence of a pink outer capsule  and subcapsular sinuses . **(Right)** Scattered mitoses  are a common finding in the colonic epithelium and are often increased in reactive conditions. Atypical forms, however, are not seen.

Lymph Node



Mitotic Figures

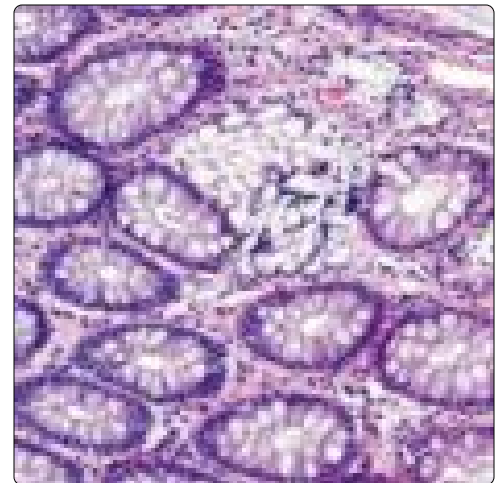


(Left) Small blebs of mucin  may extend off the luminal surface of the colonic epithelium, mimicking infectious organisms, such as *Cryptosporidium*. **(Right)** Histologic artifacts introduced by colonoscopy trauma are common. This image shows a crushed and distorted colonic crypt .

Luminal Mucin Blebs

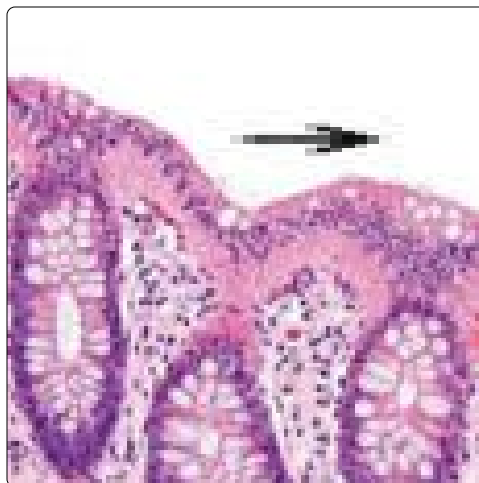


Crushed Colonic Crypts

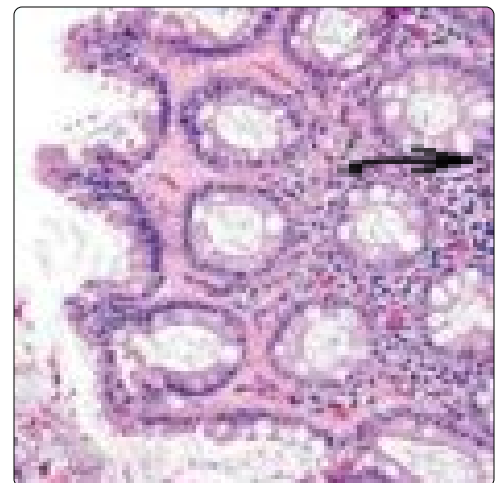


(Left) A common finding in colon biopsies is artificial thickening of the epithelium  and underlying basement membrane  due to oblique or tangential sectioning of the tissue. This artifact can generate the impression of nuclear stratification, such as that seen in dysplasia, or a thickened subepithelial collagen band, such as that seen in collagenous colitis. **(Right)** Oblique sectioning in this case gives the impression of an abnormal subepithelial collagen band  mimicking collagenous colitis; however, there is no colitis present.

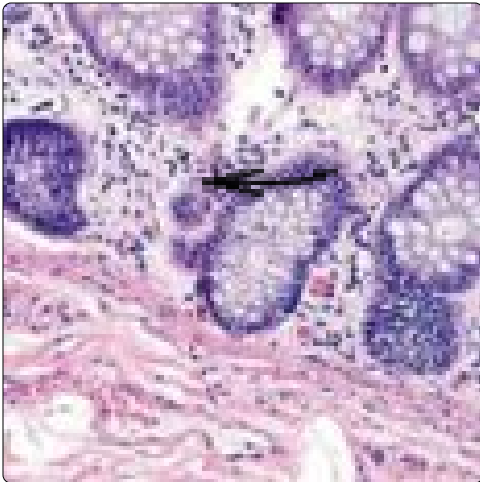
Tangential Sectioning



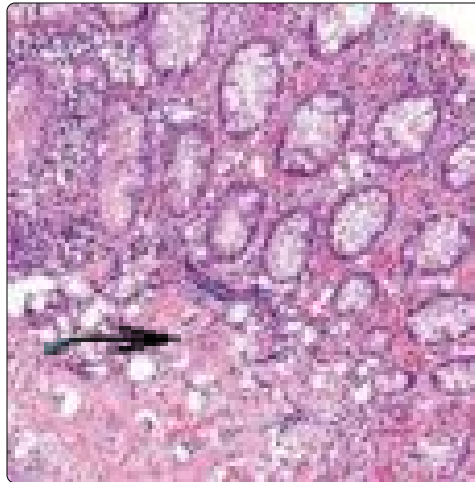
Tangential Sectioning



Tangential Sectioning

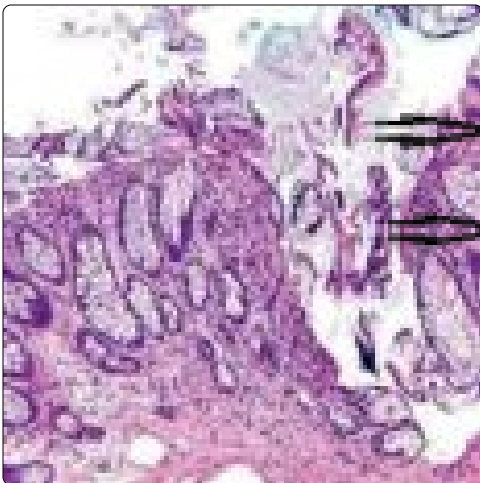


Pseudolipomatosis

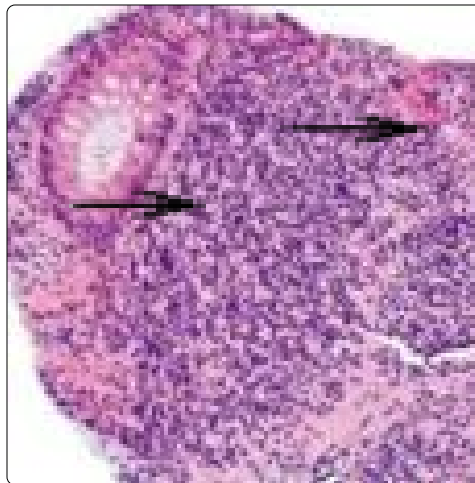


(Left) Tangential or oblique sectioning of colonic mucosal tissue can lead to crypts that appear more cellular than normal and can lead to the simulated appearance of a neuroendocrine proliferation. (Right) Mild endoscope trauma can lead to air entering the lamina propria and submucosa, an artifact known as pseudolipomatosis.

Tissue Cauterization

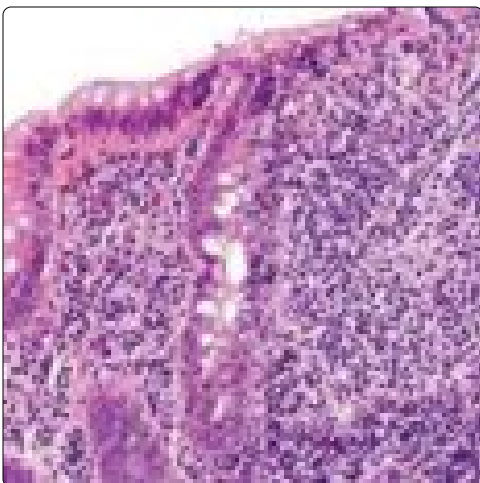


Intraepithelial Lymphocytes

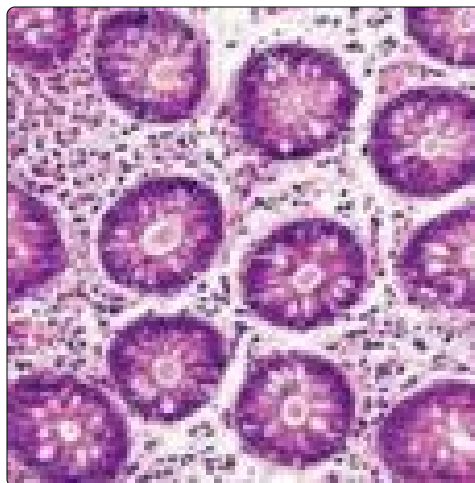


(Left) If the endoscopist uses cautery when collecting tissue, it is very common to see at least some degree of tissue cautery or burning, which gives all affected cells and their nuclei a markedly stretched out appearance. (Right) The epithelium overlying a lymphoid aggregate (MALT) usually shows an increased number of intraepithelial lymphocytes. This normal finding can lead to a misdiagnosis of lymphocytic colitis, particularly if lymphoid aggregates are numerous.

Intraepithelial Lymphocytes



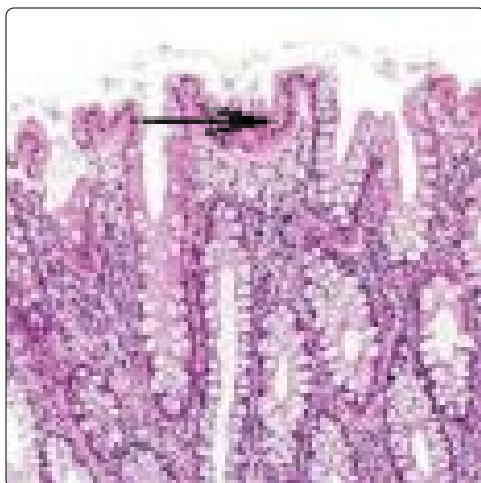
Luminal Debris



(Left) This image shows a substantial increase in intraepithelial lymphocytes adjacent to a reactive lymphoid aggregate. Occasionally, intraepithelial neutrophils can also be seen next to lymphoid aggregates as well (not shown). (Right) Biologic matter (such as vegetable matter), admixed with gland secretions, can be seen within the crypt lumina as pink fluffy or chunky material and is of no particular significance. It should not be confused with parasitic organisms.

Hyperplastic Mucosal Changes

(Left) Hyperplastic mucosal changes are typically seen within the context of mucosal irritation/friction, such as exposure to the fecal stream. It is identified as splitting or branching of the luminal surface epithelium. Note how there is no distortion at the base of the crypts, which is often a more significant finding. **(Right)** In the rectum, it is common to find aggregates of benign foamy histiocytes known as muciphages. These aggregates vary in size.

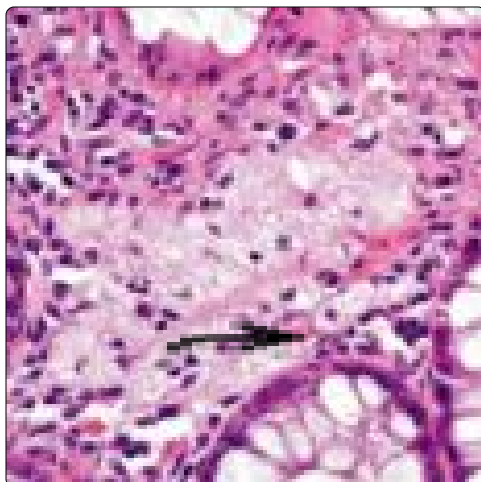


Rectal Histiocytes (Muciphages)

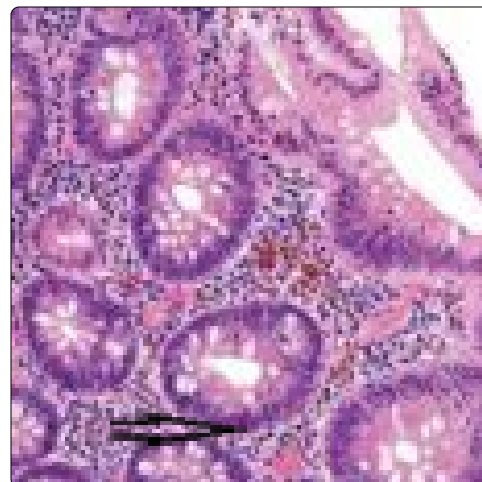


Muciphages

(Left) At high magnification, the muciphages show abundant foamy blue-white cytoplasm and small, dark nuclei. The lack of nuclear atypia is an important feature and helps distinguish them from the signet ring cells present in some variants of adenocarcinoma. **(Right)** Melanosis coli is seen as brown/black pigmented macrophages predominantly within the colonic mucosa. It is a common finding in patients with chronic laxative usage.

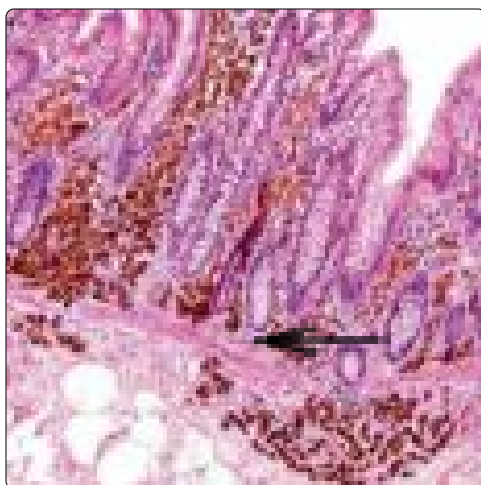


Melanosis Coli

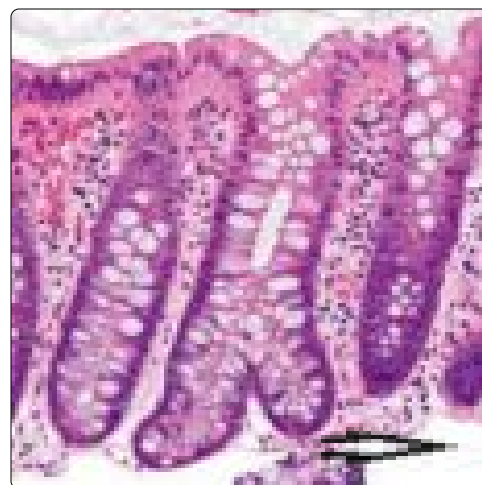


Melanosis Coli

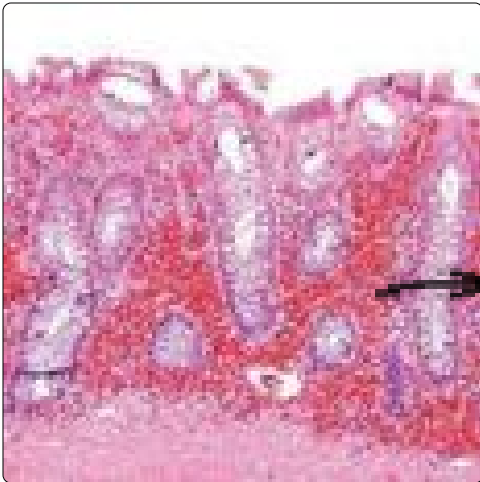
(Left) In some examples of melanosis coli, the pigmented macrophages are very prominent and even extend into the submucosa. If extensive, this finding may lead to consideration of malignant melanoma. **(Right)** A bifid crypt is a normal variant of a colonic crypt in which the basal aspect demonstrates a fork-like split appearance. This finding should not be confused with true architectural distortion seen within the context of chronic colitis and inflammatory bowel disease.



Bifid Crypt



Artifactual Mucosal Hemorrhage



Artifactual Epithelial Telescoping

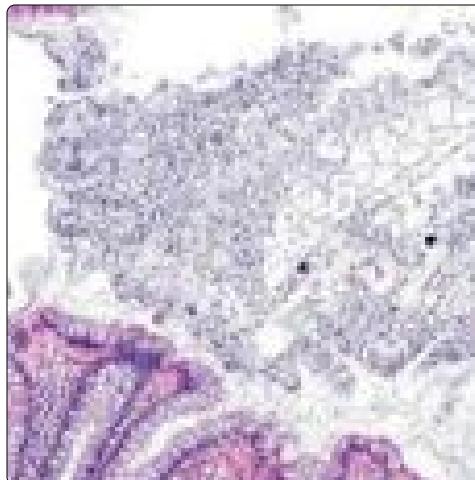


(Left) Nonpathologic hemorrhage localized within the lamina propria can be seen as a result of trauma related to the surgical or biopsy procedure or to bowel prep. It is often patchy and confined to the mucosa but without hemosiderin-laden macrophages (a feature of chronic hemorrhage). (Right) As a consequence of aggressive biopsy techniques and tissue compression, colonic crypts can "telescope" within themselves and mimic the appearance of an intraluminal organism.

Endoscope Trauma

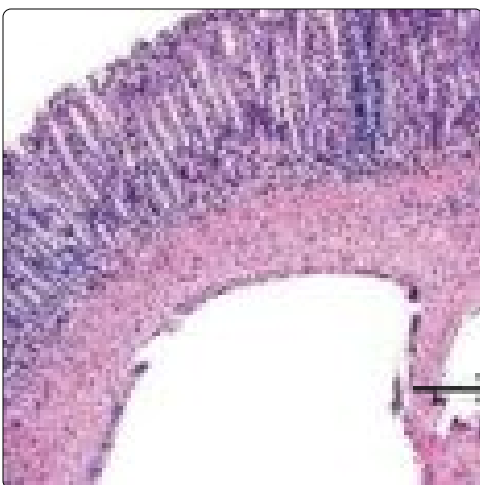


Vegetable Matter

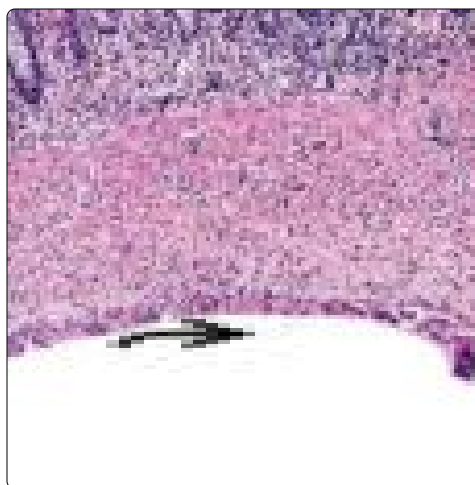


(Left) Aggressive biopsy techniques can lead to disruption of the tissue and can manifest as crypt "dropout" and/or "peeling off" of the epithelial surface. These specimens often show additional features of scope trauma, including telescoping, hemorrhage, or cautery. (Right) Fragments of vegetable matter (feces) within the bowel lumen are a common finding in colectomy specimens in which the patient does not undergo adequate preoperative evacuative measures.

Pneumatosis



Pneumatosis



(Left) Pneumatosis is caused by gas entering the submucosa or subserosa of the bowel and is related to various etiologies, including infection and mechanical trauma. Compared to pseudolipomatosis, pneumatosis can be seen endoscopically and is not an artifact. (Right) A higher magnification of pneumatosis shows the characteristic finding of foreign body multinucleated giant cells rimming the space filled with gas. This finding is not seen in artifactual pseudolipomatosis.

APPENDIX

Macroscopic Anatomy

- Tubular organ arising from cecum in right lower quadrant of abdomen
- Can range in length from < 5 cm to > 9 cm
- Mesoappendix is contiguous with mesentery of bowel

Microscopic Anatomy

- Layers are very similar to large bowel
- Mucin &/or fecal material are commonly seen in lumen
- Mucosa
 - Simple columnar epithelium, basically colonic in type
 - CK20 (+), CDX-2(+)
 - CK7 expression variable, often focal
 - Crypts are more unevenly distributed and irregularly aligned than in colon
 - Paneth cells, goblet cells, and neuroendocrine cells
 - Synaptophysin (+), chromogranin (+) neuroendocrine cells
 - Large component of lymphoid tissue in children, adolescents, and young adults
 - CD20(+) B cells in germinal center, marginal zone
 - CD3(+) T cells in parafollicular zone
 - Epithelium over lymphoid tissue may be less columnar, and crypts may be absent
 - Lamina propria contains mononuclear cells, scattered eosinophils, and neuroendocrine cells
 - Muscularis mucosa is variably developed and may appear discontinuous
 - Can appear focally absent where large lymphoid aggregates are present
- Submucosa
 - Loose connective tissue containing adipose tissue, vessels, nerves, and scattered inflammatory cells
 - Lymphoid tissue may also be present within submucosa
- Muscularis propria
 - Inner circular layer, outer longitudinal layer
 - Longitudinal layer becomes continuous with taenia coli at base of appendix

- Nerves and ganglion cells are present irregularly throughout muscular wall
- Serosa and subserosa
 - Mature fibroconnective tissue with mesothelial lining (peritonealized)
- Mesoappendix
 - Mature fibroadipose tissue contiguous with mesentery of small intestine

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

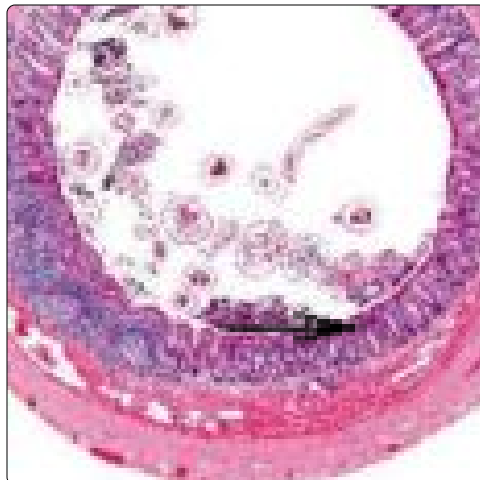
- Lymphoid tissue minimal to absent in neonates
- Mucosal lymphoid aggregates become smaller and less confluent with increasing age
- Entire appendix becomes narrower in middle age, with increasing fibrosis
- Fibrous obliteration (neuroma) of appendix more common with increasing age

Pitfalls and Artifacts

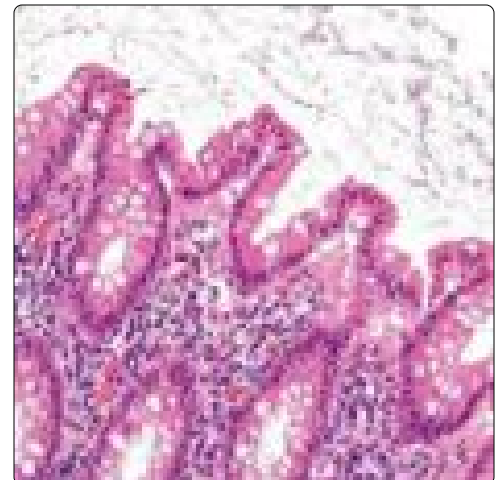
- Tangential sectioning
- Surface hyperplastic changes can mimic serrated polyps
- Irregular distribution and uneven alignment of crypts can raise possibility of chronic idiopathic inflammatory bowel disease
 - Lack of architectural derangement that is much less common in normal appendix will help in making distinction
- Appendiceal diverticula
 - Ruptured diverticula can spill mucin, mimicking low-grade appendiceal mucinous neoplasm (LGAMN); however, neoplastic epithelium is absent
- Luminal mucin
 - Abundant mucin can mimic LGAMN; however, underlying epithelium is normal
- Müllerian elements: Endometriosis, endocervicosis, and endosalpingiosis are common
 - May mimic metastatic adenocarcinoma on frozen section

Appendix Cross Section

(Left) This low-power view of the appendix demonstrates the epithelial layer [1] and prominent lymphoid tissue [2] of the mucosa, as well as the underlying submucosa [3], muscularis propria [4], and the serosa and subserosal connective tissue. Pinworms are visible in the lumen. **(Right)** The epithelium of the appendix resembles the colon with columnar cells and interspersed goblet cells.



Epithelium



Epithelium



Lamina Propria

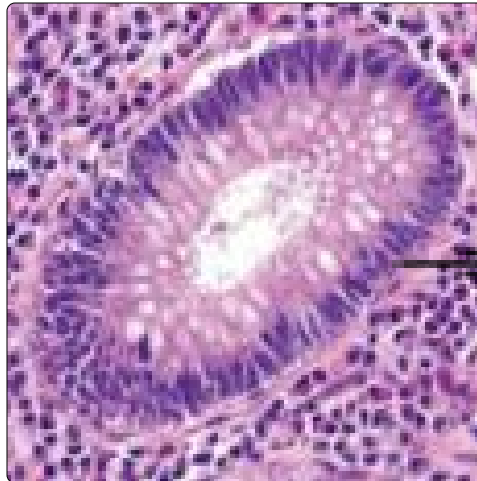


(Left) The appendiceal epithelium is a mucinous columnar lining with goblet cells, similar to the colon. The crypts in the appendix are much less orderly and more irregularly aligned than those in the colon. Occasional intraepithelial lymphocytes and crypt mitoses can also be identified. (Right) Similar to the colon, the lamina propria of the appendix contains mononuclear cells and occasional eosinophils.

Paneth Cells

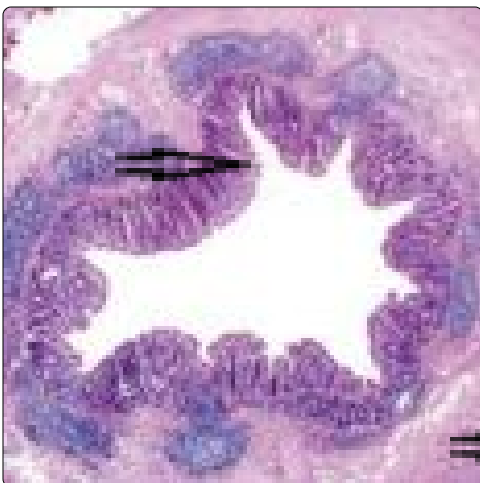


Luminal Mucin Blebs

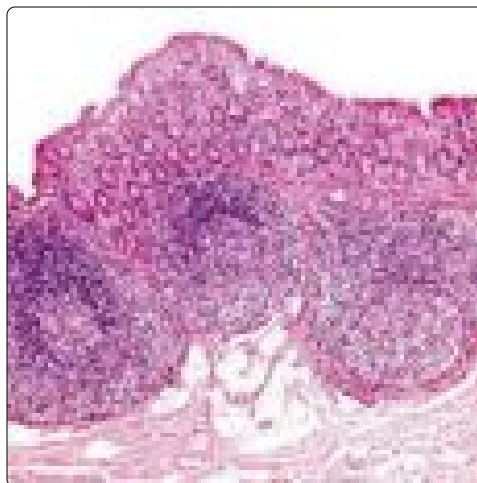


(Left) Paneth cells are commonly present in the appendix and are localized to the bases of the crypts. They can be easily identified by their brightly eosinophilic supranuclear granules. (Right) In some cases, small mucin blebs can be identified on the luminal surface of the appendiceal crypts and should not be confused for infectious organisms.

Lymphoid Tissue



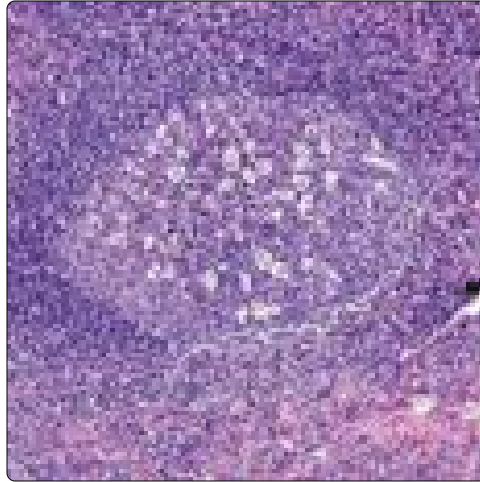
Lymphoid Tissue



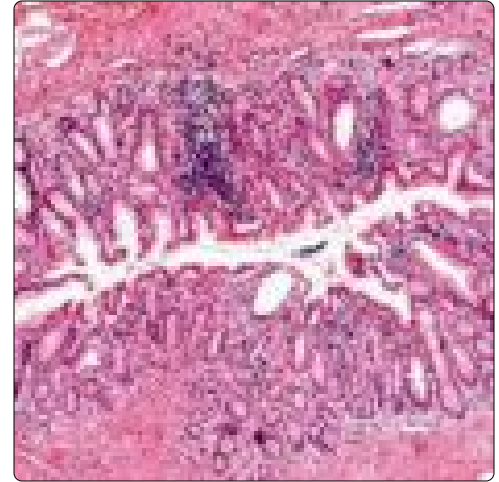
(Left) The conspicuous (and often confluent) lymphoid aggregates are a characteristic feature of the appendix. The prominent lymphoid aggregates can involve the mucosa and the submucosa. (Right) This appendix from a child shows hyperplastic lymphoid follicles with prominent germinal centers. Note that the large lymphoid follicles interrupt the muscularis mucosa.

Reactive Germinal Centers

(Left) As in other lymphoid sites, reactive germinal centers in the appendix often feature variable numbers of histiocytes containing intracytoplasmic cellular debris (a.k.a. tingible body macrophages [2]). (Right) Cross section of an appendix from an adult shows marked atrophy of the lymphoid tissue, which is a common change with increasing age.

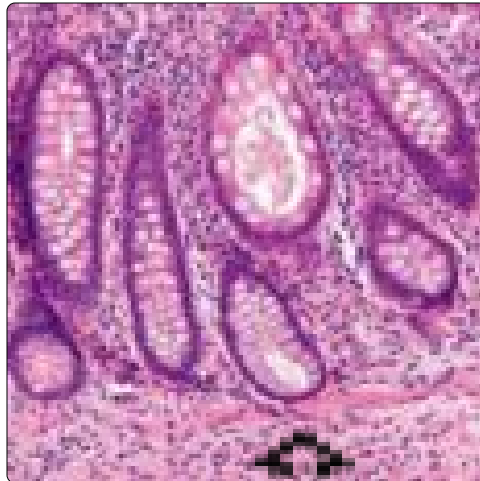


Lymphoid Tissue

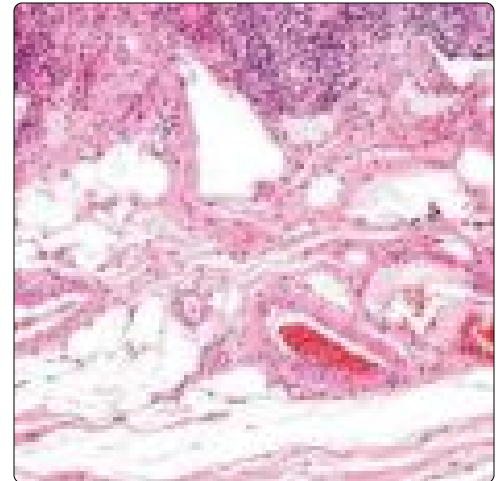


Muscularis Mucosa

(Left) In this image, the muscularis mucosa consists of tiny, discontinuous wisps of muscle [2] and demarcates the lamina propria above from the submucosa below. (Right) The submucosa consists of adipose tissue, vessels, lymphatics, and nerves. Lymphoid tissue may also be present.

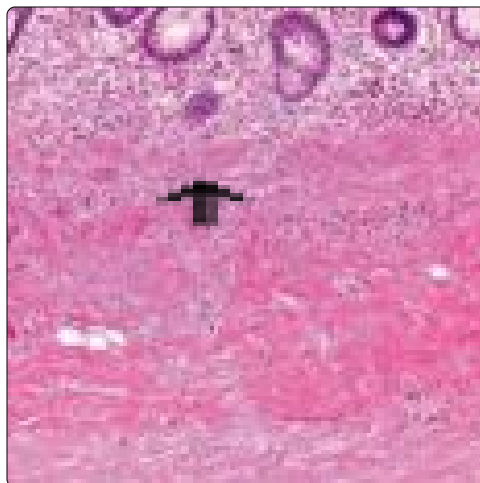


Submucosa

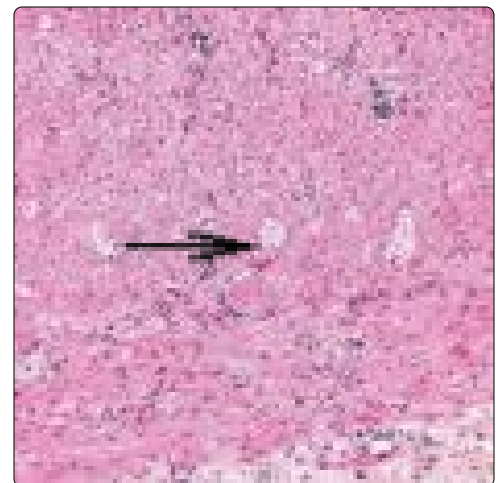


Muscularis Mucosa

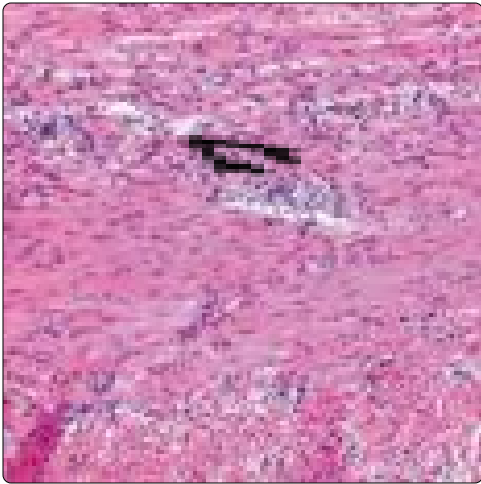
(Left) This image shows a well-developed muscularis mucosa [2]. The epithelium is visible above, and the dense connective tissue of the submucosa is visible below. The thickened smooth muscle wall of the muscularis propria is also seen [2]. (Right) The inner circular [2] and outer longitudinal layers [2] of the muscularis propria are illustrated here. Scattered neural elements are also visible [2].



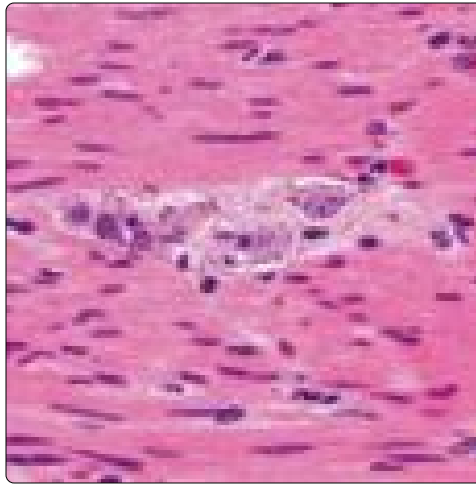
Muscularis Propria



Neural Elements

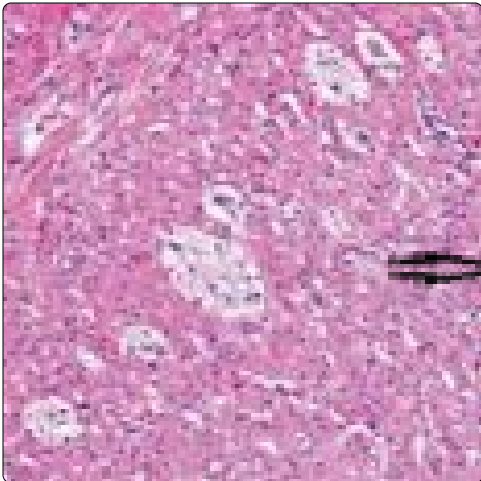


Ganglion Cells

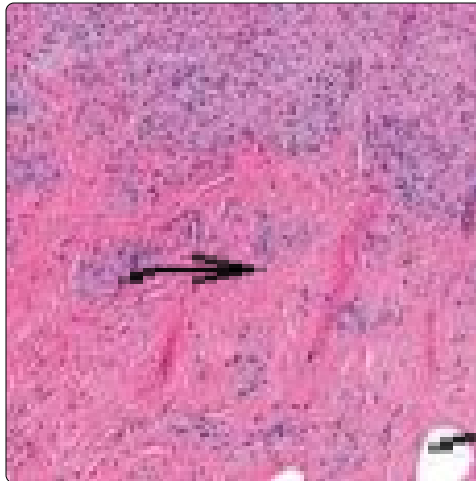


(Left) Nerves and ganglion cells  are often easily identified within the inner and outer layers of the muscularis propria. Similar to those at other sites, they have prominent basophilic cytoplasm and large nucleoli. **(Right)** This high-power view shows the eccentrically located nucleus and prominent nucleolus of mural ganglion cells.

Ganglion Cell Artifact

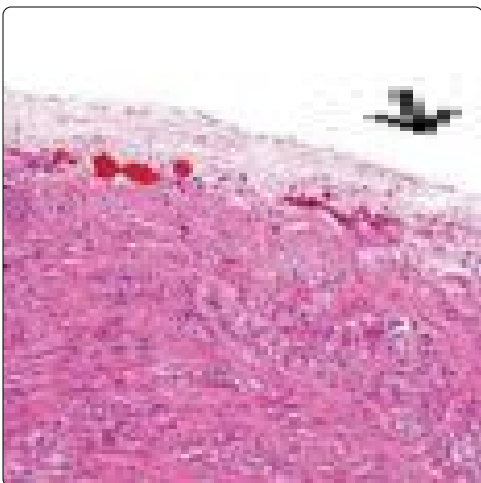


Submucosa-Muscularis Propria Interface

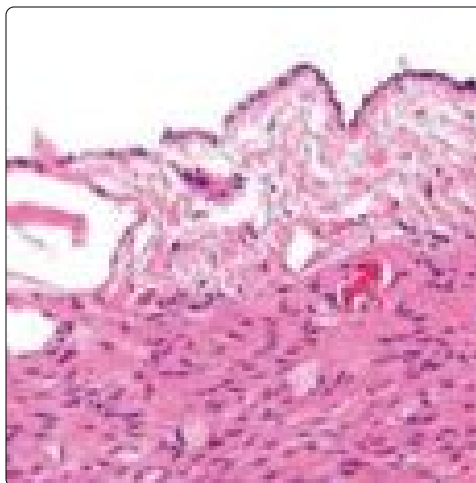


(Left) Ganglion cells  may appear to have clear or foamy cytoplasm. Together with the nested or packeted arrangement of these cells, this artifact can simulate the appearance of a goblet cell carcinoid. **(Right)** In this image, small bundles of smooth muscle  from the inner muscularis propria appear to extend between the dense collagen of the deep submucosa, simulating a neural proliferation or granular cell tumor.

Subserosa



Serosal Lining

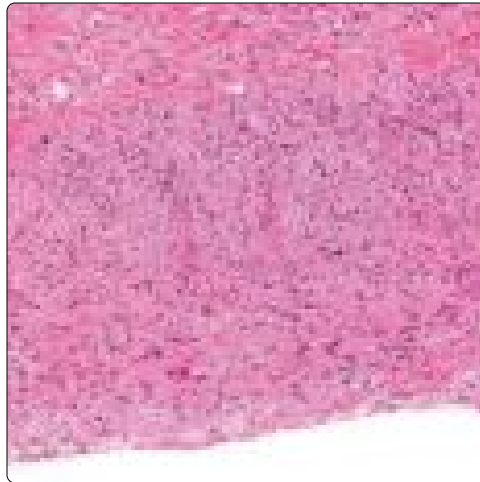


(Left) The subserosal region  deep to the muscularis propria contains blood vessels, lymphatics, and nerve fibers. External to the subserosa is an outer layer of mesothelial cells called the serosa. **(Right)** The serosa is composed of a single layer of mesothelial cells . These cells are often small and inconspicuous under normal conditions but can be enlarged in reactive states. Note the subserosa  and outer layer of muscularis propria .

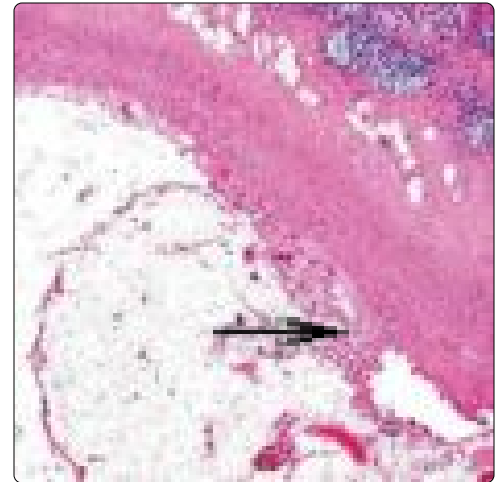
Subtle Serosa

(Left) The serosa and subserosa may be extremely subtle or seemingly absent in some histologic sections.

(Right) The mesoappendix runs most of the length of the external appendix and has a peritoneal mesothelial lining. It is contiguous with the mesentery of the small intestine.



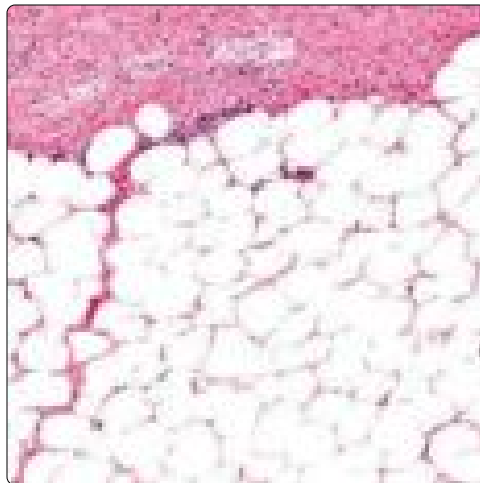
Mesoappendix



Mesoappendix Adipose Tissue

(Left) The mesoappendix is composed of mature fibrofatty tissue and contains the blood vessels that feed the appendix.

(Right) Benign neural proliferations are occasionally seen within the lamina propria or at the tip of the appendix (sometimes referred to as fibrous obliteration of the tip). They are typically of no clinical significance.

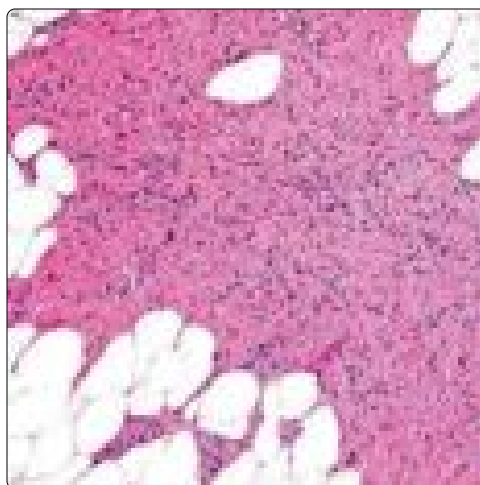


Fibrous Obliteration of Tip



Fibrous Obliteration

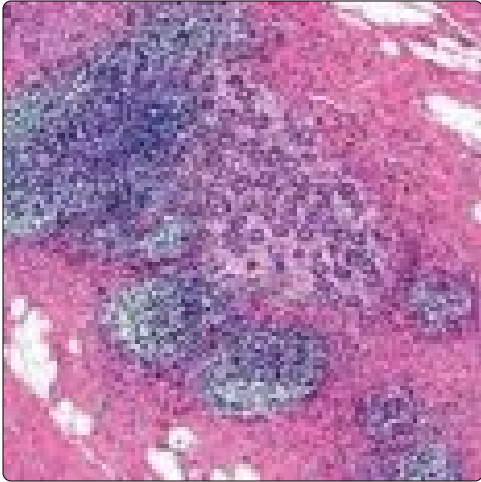
(Left) Fibrous obliteration (neuroma) is most common in the distal aspect of the appendix; however, it can also appear to involve the entire lumen. The proliferation often has a fibroblastic or neural appearance and often expresses neural markers (e.g., S100 protein). **(Right)** Benign neural proliferations may also arise focally within the appendiceal mucosa and are not associated with luminal obliteration. They are also of no clinical significance.



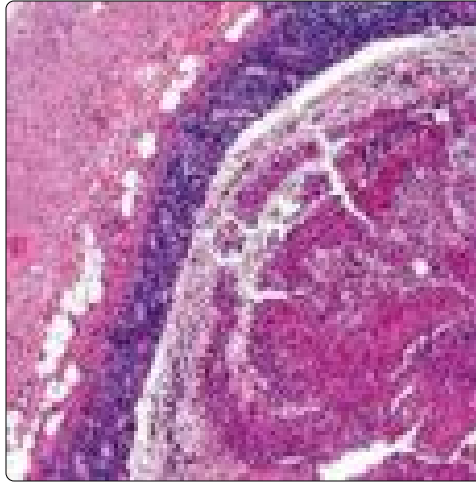
Intramucosal Neuroma



Tangential Sectioning

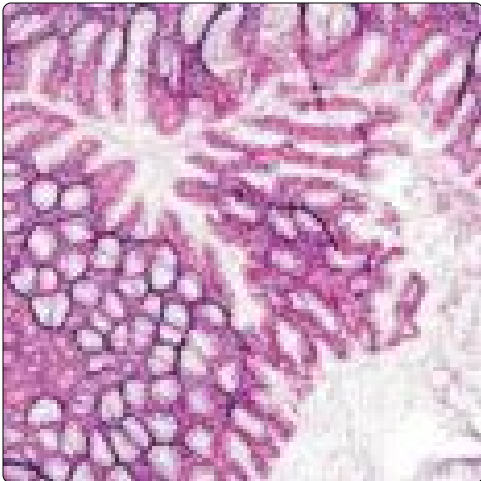


Intraluminal Fecalith



(Left) Tangential sectioning of the appendix can lead to a variety of unusual appearances, including clusters of ovoid crypts and isolated lymphoid patches. It can also lead to the mistaken impression of fibrous obliteration. (Right) Fecal matter can become entrapped or lodged within the appendiceal lumen, as depicted in this image, often described as fecaliths. Large fecaliths can cause dilatation of the appendix and attenuation of the mucosa.

Surface Hyperplastic Changes



Diverticulum



(Left) Mucosal irritants, such as fecaliths, can lead to hyperplastic changes in the epithelium, which may superficially resemble a serrated polyp or adenoma. Compared to these 2 entities, however, the hyperplastic change is not focal, has less architectural complexity, and does not show cytologic dysplasia. (Right) In states of increased luminal pressure, a portion of the appendiceal mucosa may herniate through the muscularis propria, creating a diverticulum, which may become secondarily inflamed or even rupture.

Intraluminal Mucin



Endometriosis



(Left) Abundant mucin may be produced secondary to trauma or inflammation. Large amounts of mucin can raise the possibility of a low-grade appendiceal mucinous neoplasm, but the underlying epithelium is not neoplastic. (Right) In women, müllerian elements may be identified in the muscular wall or serosa. This image shows endometrial glands and stroma (endometriosis), but endocervical glands (endocervicosis) and fallopian tube epithelium (endosalpingiosis) can be seen as well.

ANUS AND ANAL CANAL

Macroscopic Anatomy

- Complex anatomy with controversial nomenclature
 - Extent of zones is variable, and delineations are irregular
 - Macroscopic and microscopic zones may not correspond
- Perianal skin
- Anal canal: Squamous zone, transition zone, colorectal zone
- Dentate line lies within anal canal
 - Includes anal valves and sinuses, and bases of anal columns
 - Anal columns are vertical folds extending from mid to upper anal canal connected at base by small semilunar anal valves, which may appear as small papillae; anal sinuses or crypts are behind valves
- Anal cushions: Pads of tissue outlined by Y-shaped anal lumen; hemorrhoids represent prolapse of anal cushions
- Anal musculature: External sphincter, intersphincteric longitudinal muscle, internal sphincter, musculus submucosae ani

Microscopic Anatomy

- Colorectal zone: Continuation of rectal glandular mucosa
 - CK20 (+); CK7 (-)
- Transition zone: Variable types of epithelium
 - IHC: CK20(-); variable CK7(+)
 - Anal glands open here
- Squamous zone: Nonkeratinized squamous epithelium lacking hair, appendages
- Perianal skin: Keratinizing epithelium with hair, appendages

CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

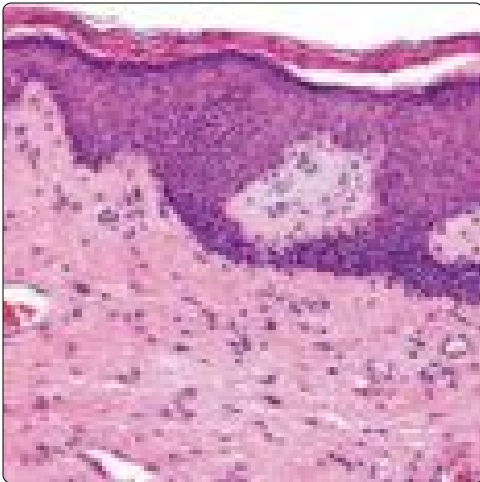
- Colorectal zone may have irregular crypts
- Anal transition zone epithelium may mimic high-grade anal intraepithelial neoplasia
 - Smaller cells with high nuclear:cytoplasmic ratio
- Ganglion cells normally absent or sparse in 1st cm above dentate line

Anal Canal

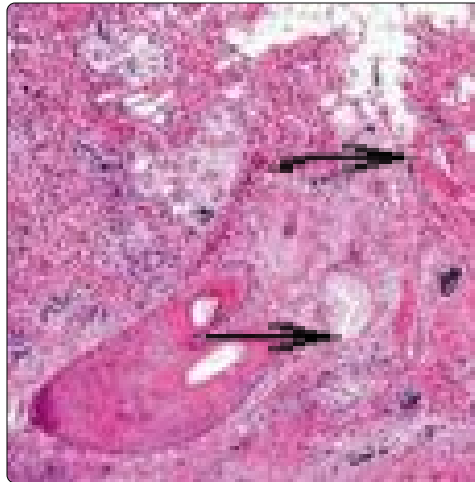


The proximal zone of the anal canal is the colorectal zone, which is lined by uninterrupted glandular mucosa. The distal zone is the squamous zone, lined by uninterrupted squamous mucosa. The transition zone lies in between, and contains a number of types of epithelia. The dentate line lies within the anal canal and includes the anal valves, anal sinuses, and bases of anal columns. The histologic anal canal is defined by the extent of the specialized mucosa, i.e., the transitional zone and squamous zone down to the perianal skin. The dentate line is an important anatomical landmark with respect to orientation of biopsies from this area.

Perianal Skin

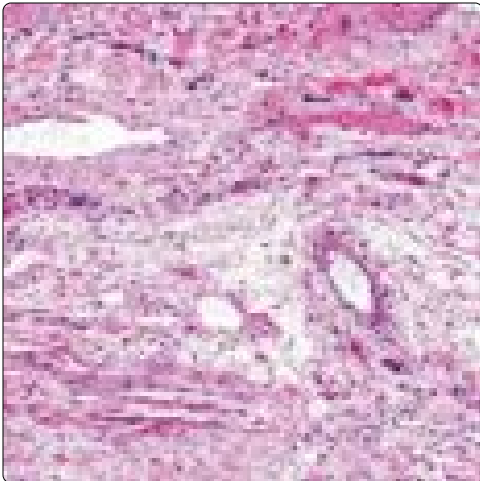


Perianal Skin and Appendages

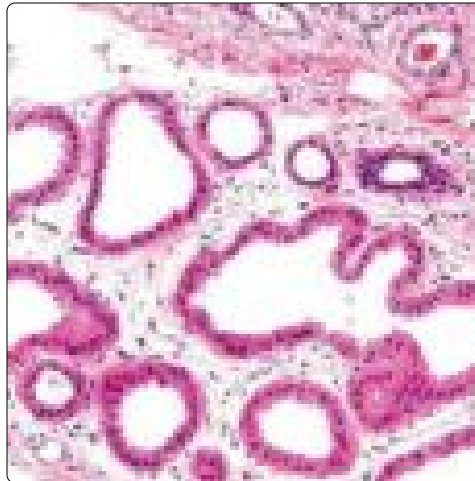


(Left) Perianal skin is keratinized, unlike the nonkeratinized squamous zone of the anal canal. (Right) Perianal skin contains hair follicles, apocrine glands, eccrine glands, and sebaceous glands. The squamous zone of the anal canal lacks hair, glands, and appendages.

Lamina Propria



Apocrine Glands



(Left) Anal canal lamina propria consists of loose connective tissue with scattered lymphocytes and mast cells. It contains vessels, lymphatics, and nerves. Ganglion cells are normally absent or sparse within 1 cm of the dentate line, which is important for evaluating patients for Hirschsprung disease. (Right) Apocrine glands, which can be identified by their bright pink epithelium, are commonly found in perianal skin. (Courtesy S. Shalin, MD.)

Eccrine Glands

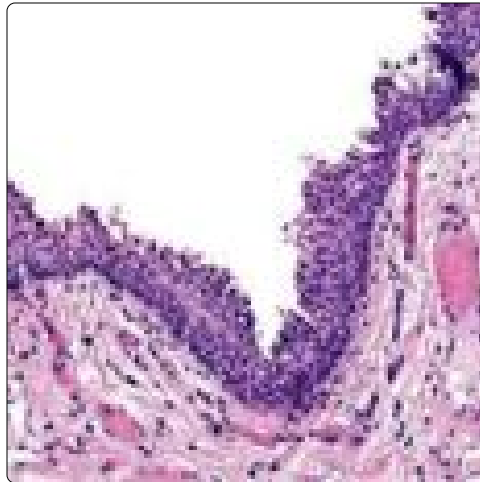


Anogenital Sweat Glands

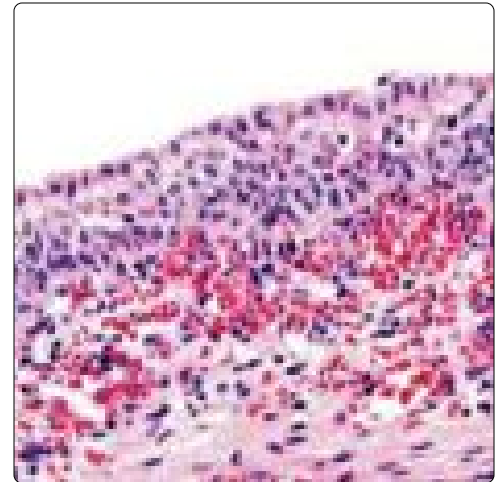


(Left) Eccrine glands are commonly found in perianal skin. They form small well-circumscribed clusters and have low cuboidal epithelium. A variably distinct myoepithelial layer is present. (Right) Anogenital sweat glands are another type of gland found in perianal skin. They consist of columnar epithelium with cytoplasmic "snouts" protruding into the lumen of the gland.

Transition Zone

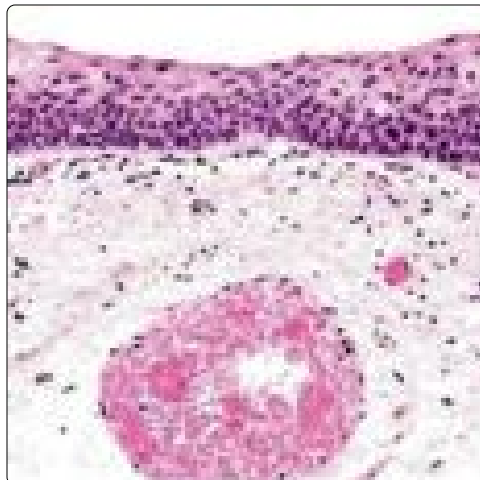


Transition Zone



(Left) Transition zone lies between squamous & colorectal zones and contains various types of epithelium, including anal transitional epithelium, which may resemble urothelium or immature squamous epithelium. Surface cells may be flat or resemble urothelial umbrella cells. (Right) Anal transition zone epithelium consists of basal cells with overlying columnar, cuboidal, or polygonal cells. Note cuboidal or polygonal cells on the surface. Melanocytes may be present in transition zone.

Transition Zone



Colorectal and Transition Zones



(Left) This example of transition zone epithelium more closely resembles squamous epithelium and has flattened surface epithelial cells. Scattered mononuclear cells are normally present in the lamina propria. (Right) The colorectal zone is a continuation of the rectal mucosa and consists of uninterrupted glandular epithelium. The muscularis mucosa extends from the rectum into the colorectal zone and upper transition zone.

Colorectal and Transition Zones

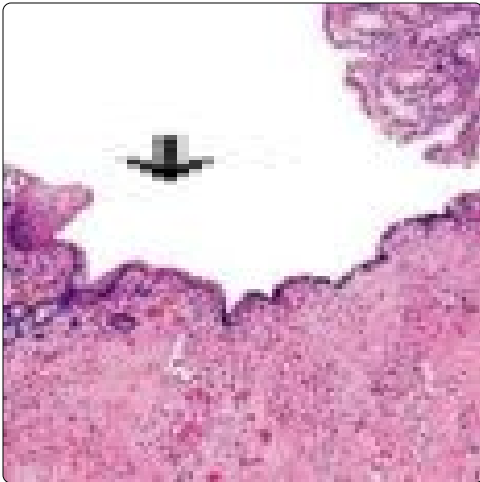


Squamous Zone

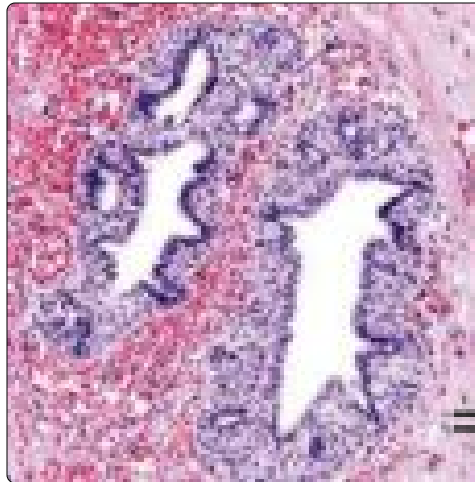


(Left) The mucosa of the colorectal zone may have shorter, more irregular crypts; this should not be mistaken for evidence of chronic inflammatory changes. (Right) Squamous zone consists of uninterrupted nonkeratinizing epithelium with few or no rete pegs. It lacks hair and skin appendages. Melanocytes, lymphocytes, histiocytes, and Merkel cells are present. Anal canal stroma may contain large multilobated stromal cells that should not be mistaken for a neoplastic process.

Transition Zone

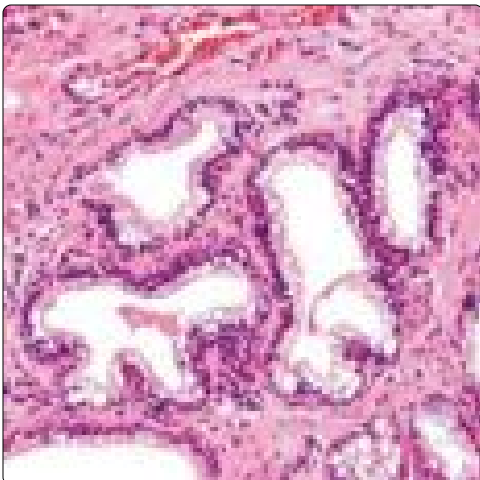


Anal Glands

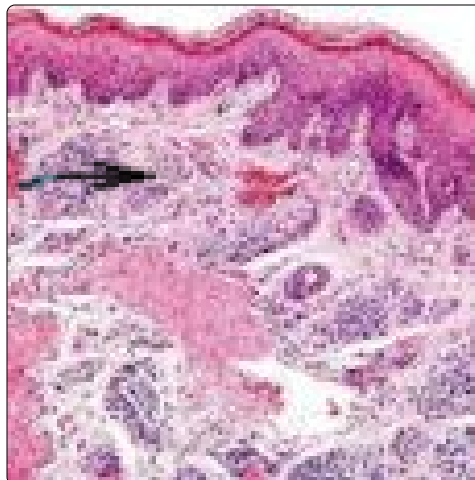


(Left) Mature squamous epithelium [1], colorectal mucosa, and simple columnar epithelium [2] may also be present in the transition zone. (Right) Anal glands open into the transition zone. The epithelium resembles transition zone epithelium, and microcyst formation [3] is common. Some are present in the submucosa, but others extend into the anal musculature. The anal glands have a similar cytokeratin staining pattern to the transition zone mucosa.

Anal Glands



Perianal Nevus



(Left) Goblet cell metaplasia is a frequent finding in anal glands. (Right) The melanocytes within the transition zone and perianal skin may give rise to nevi, as seen here [4], as well as to anal melanoma.

Anal Intraepithelial Neoplasia in Transition Zone



Anal Intraepithelial Neoplasia in Transition Zone



(Left) Occasionally, transition zone epithelium may be difficult to distinguish from high-grade anal intraepithelial neoplasia (AIN), which is seen here in the transition zone. (Right) p16 immunostaining may help distinguish high-grade AIN (full-thickness staining, on the left) from transition zone epithelium (patchy staining, on the right).

PERITONEUM

Macroscopic Anatomy

- Thin, continuous membranous lining of abdominal cavity
- Smooth, glistening surface
- 2 layers: Visceral and parietal
 - Visceral peritoneum lines intraabdominal viscera
 - Stomach, large and small intestines (only 1st part of duodenum), upper 1/3 of rectum
 - Liver, spleen
 - Uterus, fallopian tubes, ovaries
 - Parietal peritoneum lines outer surfaces of abdominal cavity
 - Abdominal wall, diaphragm, retroperitoneum, upper pelvis
 - In setting of increased intraabdominal pressure, sacs of parietal peritoneum may extend through abdominal wall musculature (herniations)
- Visceral and parietal layers meet at mesentery
 - Mesentery contains blood vessels, nerves, lymphatics, and lymph nodes
- Omentum is mature adipose tissue sandwiched between 2 double layers of visceral peritoneum
- Tunica vaginalis
 - Parietal peritoneal lining of scrotum

Microscopic Anatomy

- Composed of thin layer of cuboidal mesothelial cells upon thin basement membrane overlying fibroelastic connective tissue
- Normal mesothelial cells
 - Difficult to see in routine histologic sections
 - Small, flat to cuboidal cells
 - Well-defined cell borders
 - Single central nucleus
 - Homogeneous chromatin
 - Nucleolus is typically absent
- Reactive mesothelial cells
 - Larger and more conspicuous
 - Enlarged nucleus

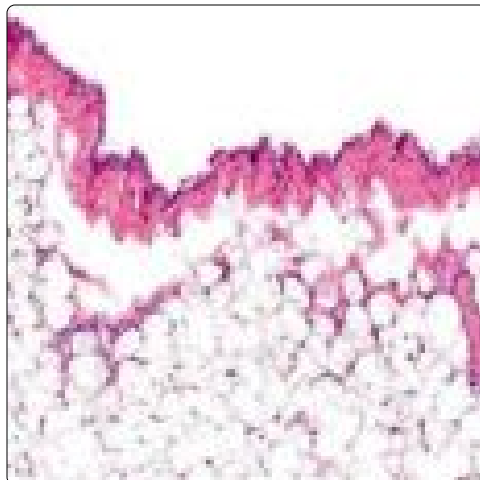
- Prominent central nucleoli
- May show multinucleation
- May form small papillary structures
- Submesothelium
 - Spindled fibroblastic cells, collagen, elastin
 - Often thickened and more cellular in reactive conditions
 - Inflammatory cells, reactive myofibroblasts, capillaries
- Mesothelial Immunophenotype
 - Positive markers
 - Keratins
 - Including AE1/AE3, CK7, CK5/6, CK8/18
 - Calretinin
 - WT-1
 - D2-40 (podoplanin)
 - HBME-1
 - Thrombomodulin
 - Desmin (in reactive states)
 - Negative markers
 - CK20
 - MOC-31
 - BER-EP4
 - CEA
 - CD15
 - TTF-1
 - pax-8
 - In reactive states, submesothelial stromal cells express keratins

CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

- Tangential sectioning can lead to unusual mesothelial cell architectures
- Reactive papillary mesothelial hyperplasia
- Mesothelial and reactive submesothelial stromal cells express epithelial antigens (e.g., keratins), which may lead to confusion with carcinoma or mesothelioma
- Endometriosis
- Endosalpingiosis
- Deciduosis

Peritoneum

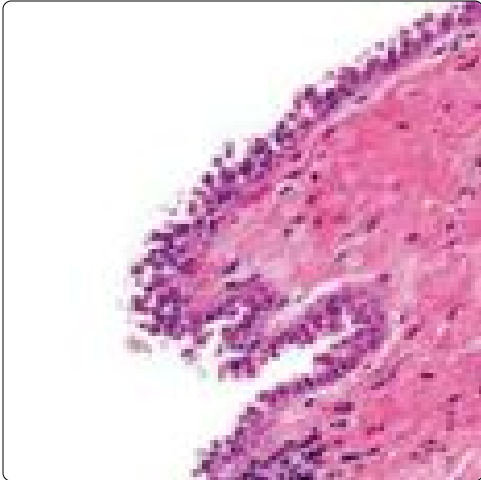


(Left) When visible, the peritoneal lining appears as a relatively thin layer of small eosinophilic cells (mesothelial cells) lining the surface of mature adipose (omental) or fibroconnective tissue. **(Right)** Mesothelial cells are small, cuboidal eosinophilic cells that demonstrate a single central nucleus and usually lack a nucleolus. The elongated to spindled cells in the underlying connective tissue are submesothelial fibroblasts.

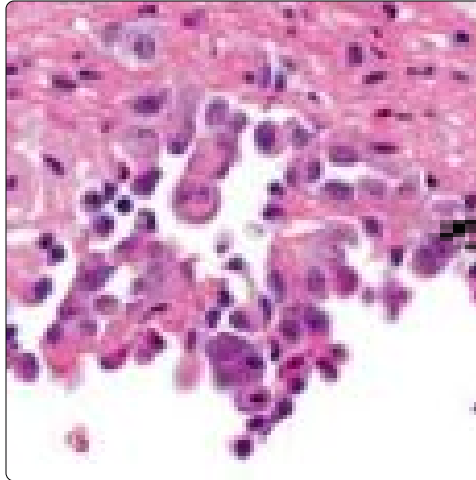
Mesothelial Cells



Reactive Mesothelial Cells

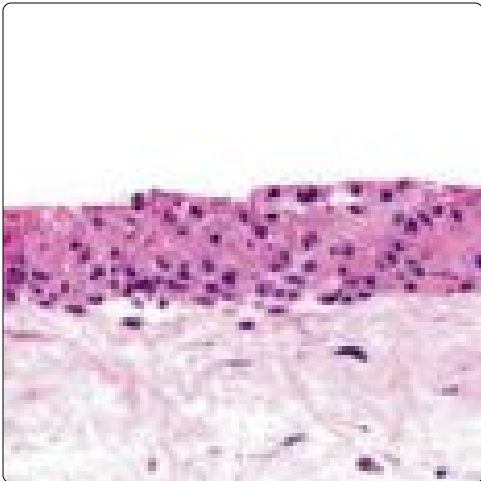


Reactive Mesothelial Cells

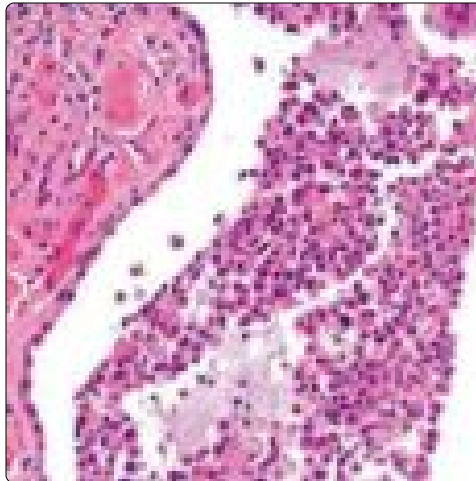


(Left) Under reactive conditions, mesothelial cells commonly adopt a plump, hobnail morphology  that is reminiscent of a row of tombstones or matchsticks. The lining is often also more cellular and disorganized, making it look very "busy." **(Right)** Cytologically, reactive mesothelial cells commonly show cellular and nuclear enlargement with the formation of conspicuous nucleoli . Multinucleation  and mitotic division figures  can also be seen.

Reactive Peritoneal Thickening

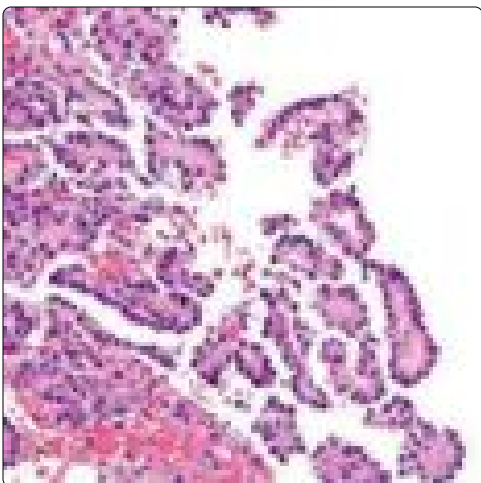


Sloughed Mesothelial Cells

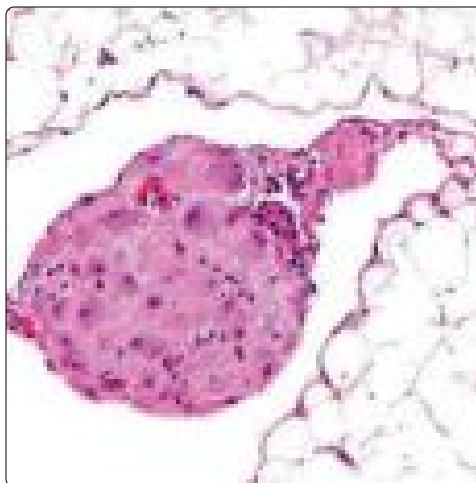


(Left) Under reactive conditions, the peritoneal lining can also become more thickened and less hobnailed. Importantly, despite their number, cells are limited to the surface and never show destructive invasion of the underlying fibroconnective tissue. **(Right)** A reactive lining can slough off of the underlying submesothelium, sometimes mimicking a neoplastic process; however, these cells are usually loosely cohesive, contain admixed inflammatory cells, and lack features of malignancy.

Reactive Papillary Mesothelial Hyperplasia



Ectopic Decidua



(Left) The peritoneal lining is capable of forming simple papillary structures (cells surrounding a central fibrovascular core) . When these structures are prominent and diffuse, a well-differentiated mesothelioma must be considered. **(Right)** Ectopic decidua (deciduosis) is a rare, incidental finding in omental, mesenteric, or peritoneal tissue of postpartum women and is of no clinical significance. It can, however, morphologically mimic mesothelial cells.

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SECTION 11

Hepatobiliary Tract and Pancreas

Liver	260
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Extrahepatic Biliary Tract	272
Vaterian System	276
Pancreas	278

LIVER**Macroscopic Anatomy**

- Dual blood supply: Hepatic artery and portal vein
 - Vessels enter liver at porta hepatis (fissure at central inferior border of liver)
 - Venous drainage: Right, middle, and left hepatic veins drain into inferior vena cava
- Bile flow
 - Bile canaliculi → interlobular bile ducts → right and left hepatic ducts → common hepatic duct

Microscopic Anatomy

- 2 competing conceptions of subunits: Lobule and acinus
 - Lobule: Central (hepatic) vein at center, portal tracts at periphery
 - Acinus: Portal tract is central axis, central veins at periphery; defined by liver microcirculation
- Hepatocytes
 - Polygonal cells with abundant cytoplasm
 - Nuclei are centrally placed, round to oval, with delicate chromatin and 1 or more nucleoli
 - Positive IHC: CK8, CK18, Hep-Par1
 - Negative IHC: Vimentin, CK7, CK19
- Portal tracts
 - Contain interlobular bile duct and bile ductules, branch of hepatic artery, branch of portal vein
 - Connective tissue framework has variably present lymphocytes and mast cells; limiting plate delineates interface with parenchyma
- Bile ducts
 - Interlobular ducts lined by cuboidal or low columnar epithelium
 - Larger septal ducts (> 100 μ m in diameter) lined by tall columnar epithelium; have denser surrounding collagen
 - 70-90% of hepatic arteries should have attendant bile duct; unaccompanied arteries signal biliary disease
 - Nourished by hepatic arteries via peribiliary capillary plexus
 - Positive IHC: CK7, CK19, CK8/18, CK AE1/AE3

- Bile ductules
 - Present at periphery of portal tracts, not accompanied by hepatic arteriole
 - Bile ducts drain bile from canaliculi via bile ductules and canals of Hering
 - Rarely seen in normal liver but proliferate in wide variety of pathological conditions
 - Same immunophenotype as bile ducts
- Bile canaliculi
 - Formed by apposition of specialized grooves in surfaces of contributing liver cells
 - Continuous conduit that extends along hepatic plates and ultimately empties into biliary tree
 - Positive IHC: CD10, polyclonal CEA
- Sinusoids
 - Narrow vascular channels between radiating hepatic plates
 - Lined by Kupffer cells and endothelial cells
 - Supported by delicate reticulin framework
 - Space of Disse: Not normally visible
 - Stellate cells: Store fat and vitamin A
- Central veins (terminal hepatic veins)

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

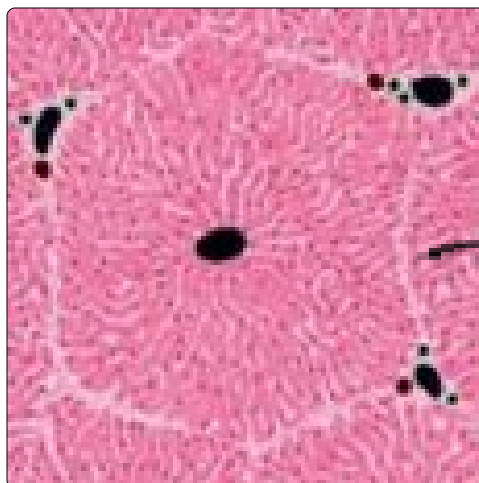
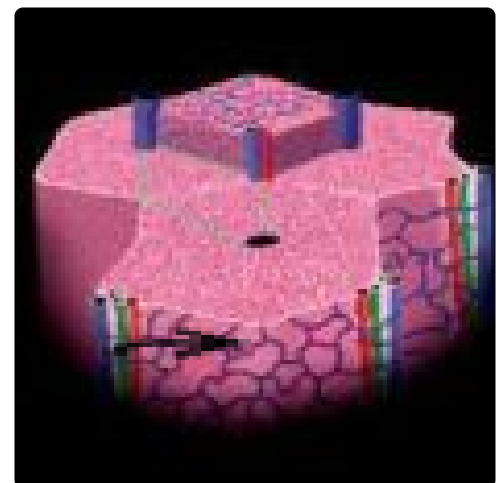
- Infants: Extramedullary hematopoiesis common; increased stainable iron and copper
- Young children: Glycogenated nuclei common; cell plates usually 2 cells thick
- Elderly persons: Increased variation in nuclear size and hepatocyte size; prominent lipofuscin; increased portal tract hyalinization, inflammation

Pitfalls and Artifacts

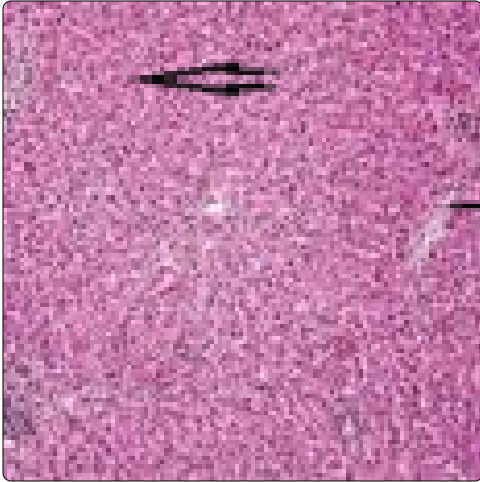
- Subcapsular zone in shallow wedge biopsies
 - Elastin can cause contraction of portal areas, imitating cirrhosis
- Longitudinally cut large portal tracts may mimic fibrous septa

Lobule

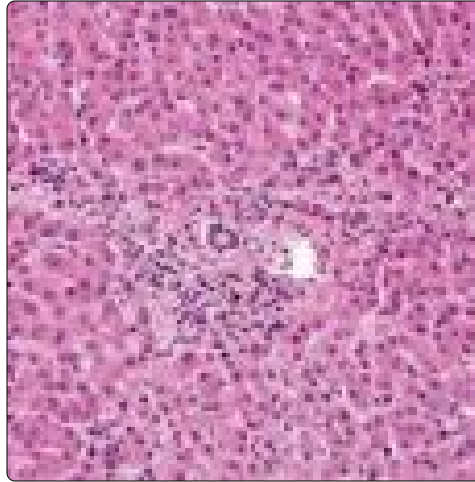
(Left) The classic liver lobule consists of a hexagonal structure with portal tracts  at the periphery and the central venule  at the center. (Right) The acinus is defined by the microcirculation of the liver. The central axis consists of the portal tract , and the hepatic plates radiate outward to the central (hepatic) venules .

**Acinus**

Normal Lobule



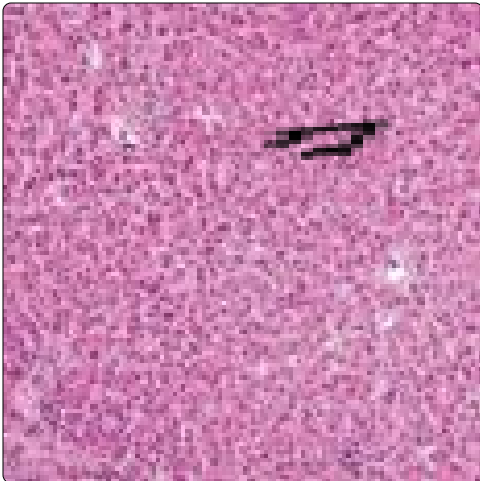
Periportal Zone/Zone 1



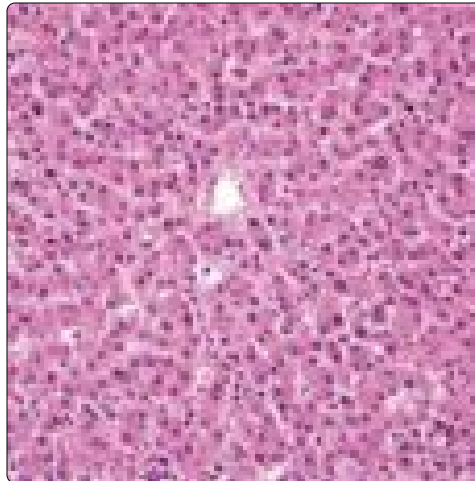
(Left) The normal lobule, which is a 2D structure, is easy to conceptualize microscopically. The central vein is the hub of the hexagon, with the portal tracts at the periphery.

(Right) The periportal zone in the lobule is comparable to zone 1 in the acinus. Zone 1 is best supplied with oxygen and nutrients because of the blood flow and is thus least susceptible to ischemia and toxic injury.

Midlobular Zone/Zone 2



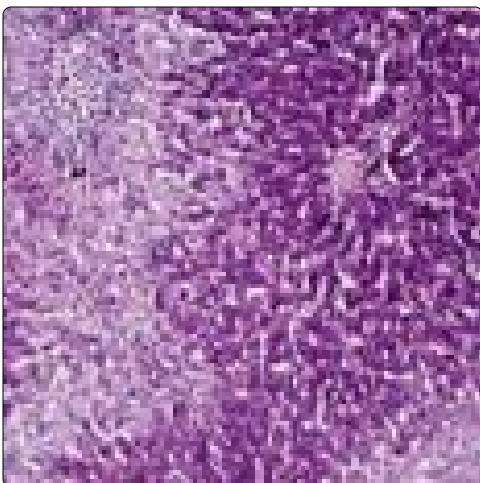
Centrilobular Zone/Zone 3



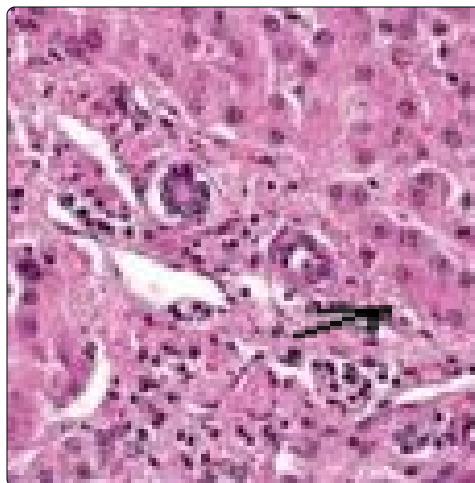
(Left) The midlobular zone, which is intermediate between the periportal (zone 1) and centrilobular (zone 3) areas, is comparable to zone 2 in the acinus. Note the portal tract and the central venule.

(Right) The centrilobular zone around the central venule is comparable to zone 3 in the acinus. This area is the most devoid of oxygen and nutrients and is thus more susceptible to ischemia and toxic injury.

Hepatocyte Glycogen



Portal Tract

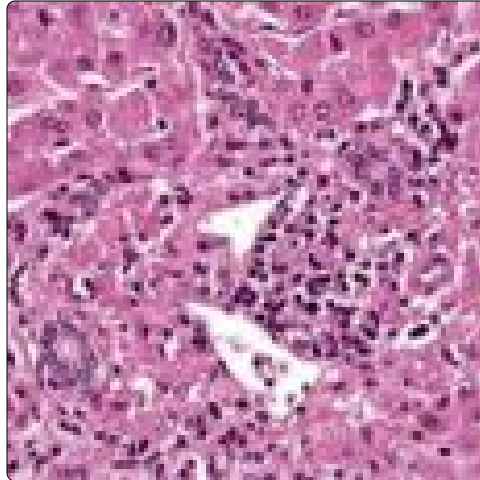


(Left) The hepatocytes contain abundant cytoplasmic glycogen (dark pink color), highlighted by a PAS stain. An irregular distribution pattern is common, as in this picture; it has no clinical significance.

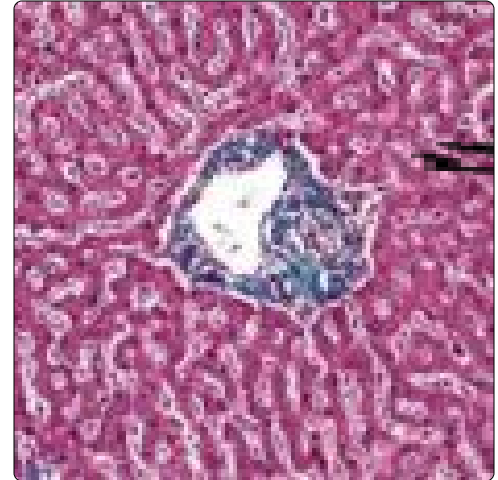
(Right) Portal tracts contain a bile duct, an accompanying small branch of the hepatic artery, and a portal vein branch. Occasionally, more than 1 bile duct is present per portal tract.

(Left) Portal tracts normally contain a variable number of lymphocytes and mast cells, and this should not be interpreted as abnormal inflammation. Neutrophils, eosinophils, and plasma cells are not usually seen in the normal liver. **(Right)** Portal tracts are supported by connective tissue stroma that becomes more coarse and dense with age. The normal interface between the connective tissue of the portal tract and the hepatic parenchyma is smooth.

Portal Tract Cells

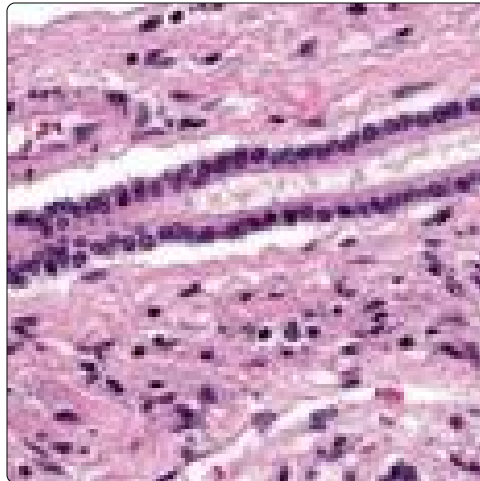


Portal Tract (Trichrome)

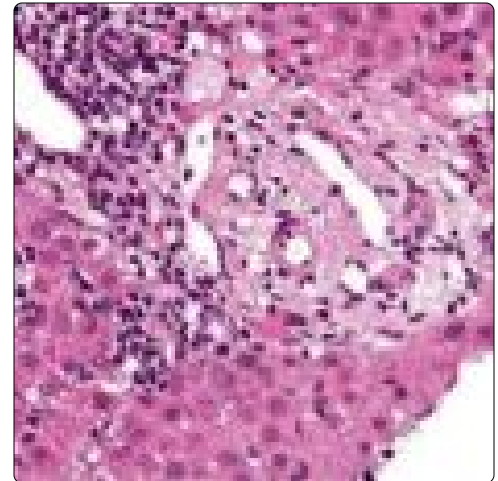


(Left) Interlobular bile ducts are lined by cuboidal or low columnar epithelium. **(Right)** 70-90% of hepatic arterioles should be accompanied by a bile duct. When a significant percentage of arterioles lack an accompanying bile duct, loss of ducts should be suspected, as in this case of primary biliary cirrhosis.

Interlobular Bile Duct

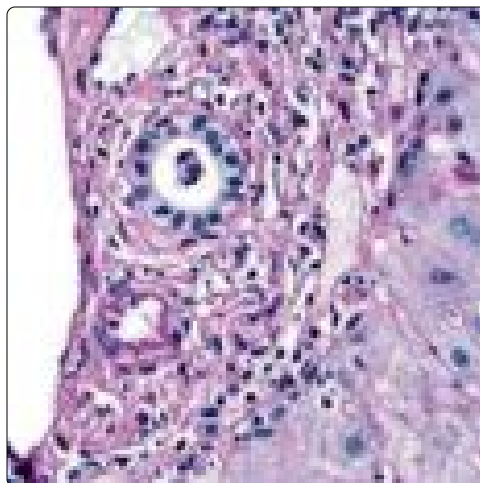


Interlobular Bile Duct Loss

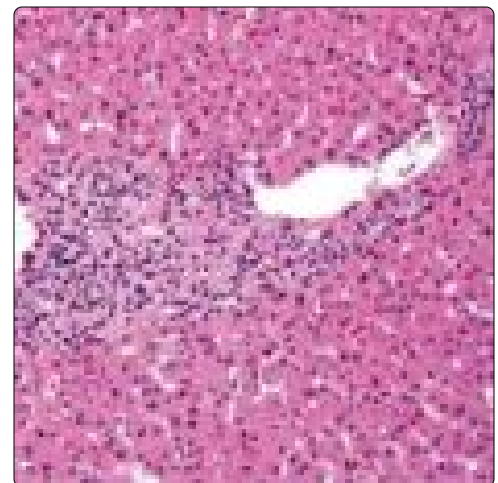


(Left) A PAS stain highlights the normal delicate basement membrane of bile ducts. Disruption of this basement membrane can be seen in bile duct damage. **(Right)** Bile ductules (cholangioles) drain bile from the canaliculi to the interlobular ducts. They are located at the periphery of the portal tracts and lack an accompanying hepatic artery. They are seldom seen in the normal liver but proliferate in many diseases.

Bile Duct Basement Membrane



Bile Ductules



Bile Duct

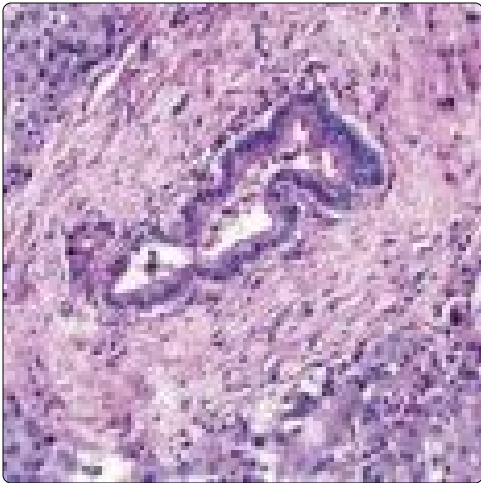


Bile Duct Collagen

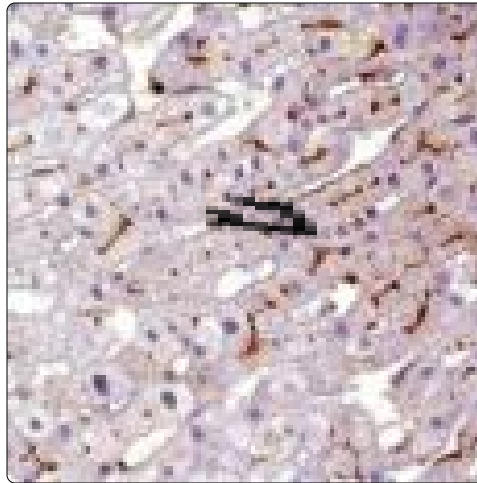


(Left) Normal larger bile ducts are lined by tall columnar epithelium. The surrounding collagen is more dense than that which surrounds the smaller interlobular bile ducts. **(Right)** The dense concentric collagen that surrounds larger bile ducts should not be mistaken for the "onion skin" lesion of primary sclerosing cholangitis.

Bile Duct (PAS)

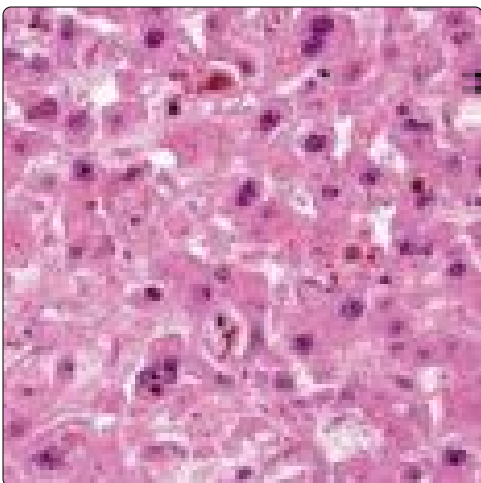


Bile Canaliculi

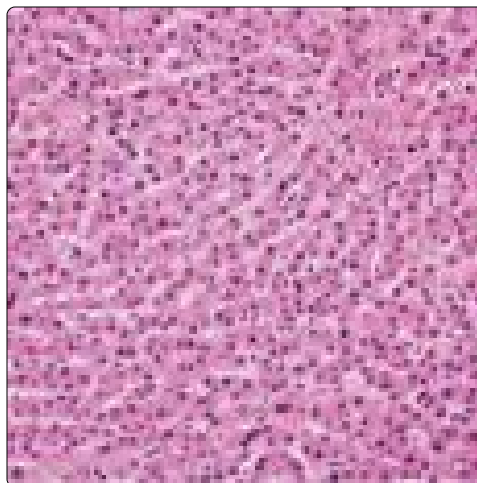


(Left) PAS stain highlights the tall columnar epithelium of larger bile ducts as well as the dense supporting stroma. **(Right)** Polyclonal CEA immunostain illustrates bile canaliculi in the normal liver. The canaliculi are created from grooves in hepatocytes that form a continuous conduit along the hepatic plates, which empties into the bile ductules at the edges of portal tracts. The canals of Hering link the canaliculi to the biliary tree and are not visible on routine stains.

Bile Plugs Within Canaliculi



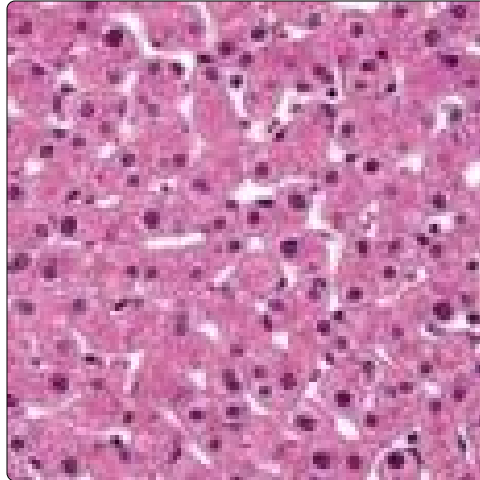
Hepatocytes



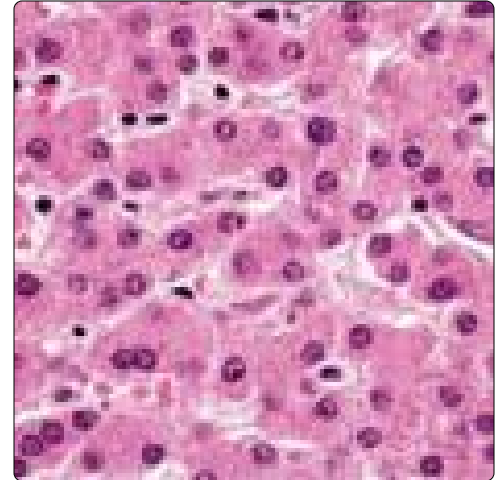
(Left) Bile canaliculi are not visible in the normal liver. When bile plugs are present in cholestatic liver disease, however, they delineate the bile canaliculi. Bile may also accumulate in hepatocytes and must be distinguished from lipofuscin. Intracellular bile is less granular than iron and lipofuscin. **(Right)** Hepatocytes measure 25–30 μm . Normal hepatocyte cell plates are 1–2 cells thick; they may be 2 cells thick in children younger than 6 years.

(Left) Hepatocytes are polygonal cells with abundant eosinophilic, granular cytoplasm. Improper fixation or processing can cause false hepatocyte swelling or shrinking and can produce inconsistent cytoplasmic staining. **(Right)** Hepatocyte nuclei are round, centrally placed, and frequently have a prominent nucleolus. Double nuclei are common. Variation in nuclear size increases with age. Mitotic activity is rare in the normal liver.

Hepatocyte Cytoplasm

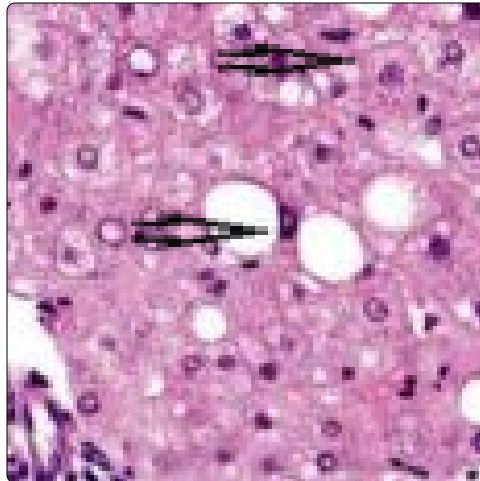


Hepatocyte Nuclei

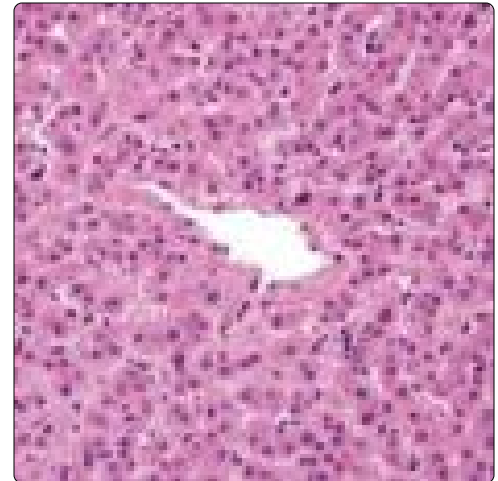


(Left) Glycogenated nuclei  are common in children and young adults but may also be seen in pathological conditions, such as fatty liver disease or Wilson disease. The enlarged nucleus has a clear, vacuolated appearance. **(Right)** Central veins (also called terminal hepatic veins) are the smallest component of venous outflow tract.

Glycogenated Hepatocyte Nuclei

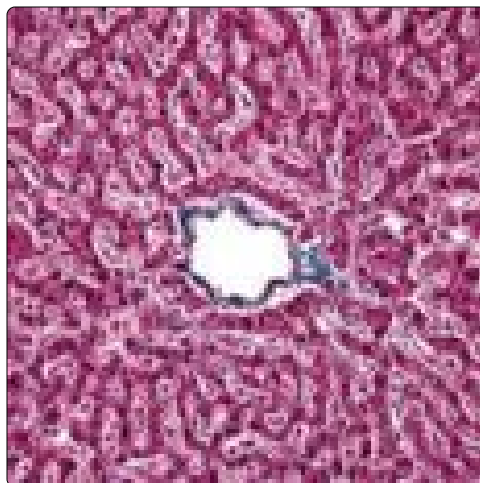


Central Vein

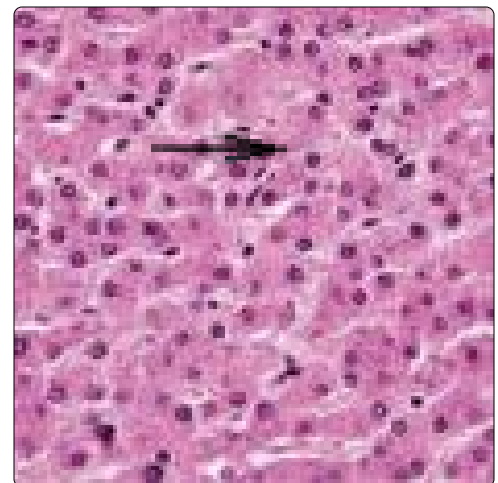


(Left) Central veins are lined by endothelium and supported by a thin layer of collagen. They join to form the sublobular hepatic veins, and then the hepatic veins, which drain into the inferior vena cava. **(Right)** The sinusoids  are the narrow vascular channels between the radiating hepatic plates. They are lined by Kupffer cells  and endothelial cells. Typically, the sinusoids are empty except for a few erythrocytes.

Central Vein (Trichrome)



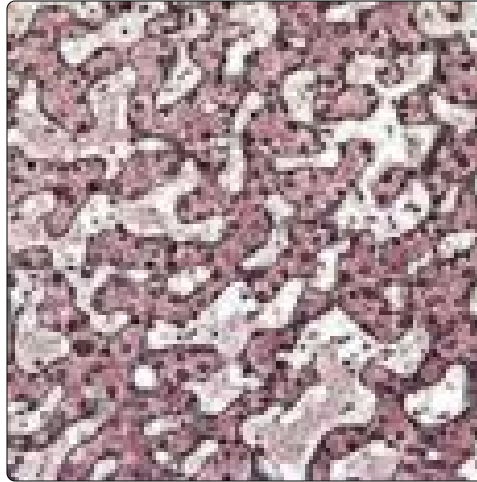
Sinusoids



Reticulin Framework

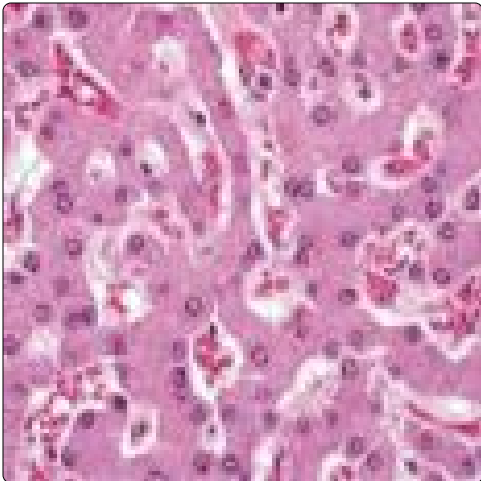


Sinusoids and Space of Disse

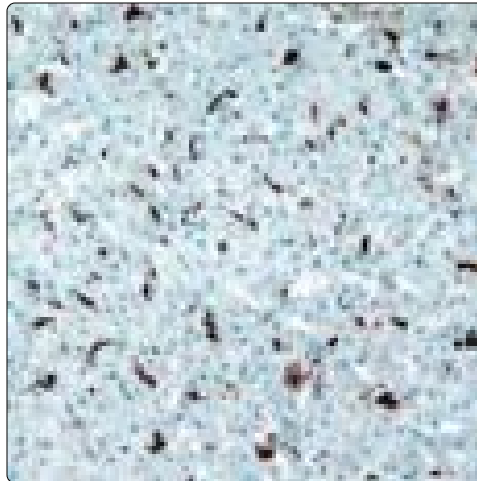


(Left) The reticulin stain highlights the normal reticulin framework of the liver. The cell plates and sinusoids form a 3D network that directs blood from portal tracts to central venules. (Right) The space of Disse is the zone between the endothelial cells and the hepatocytes. It is not normally visible, and it contains the reticulin framework of the liver (a continuous network of type III collagen fibers that outline the hepatic plates, as seen here).

Space of Disse

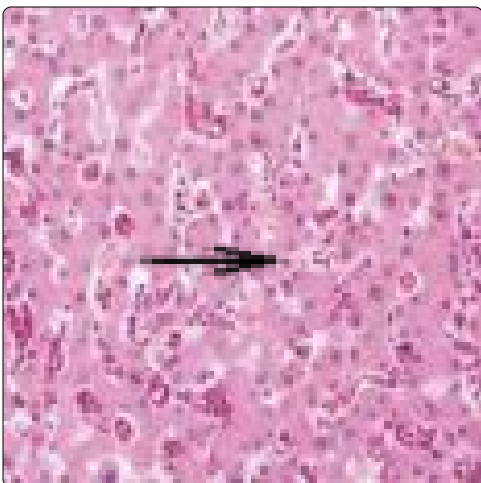


Kupffer Cells

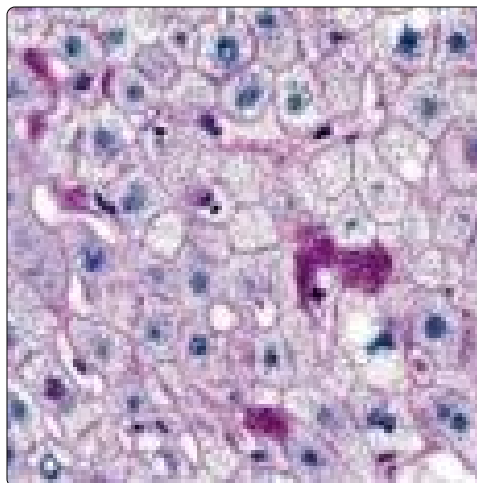


(Left) The space of Disse is the zone between the endothelial cells and the hepatocytes. It is not normally visible in routinely fixed and processed specimens, but it can be seen in suboptimally fixed autopsy material. (Right) A CD68 immunostain highlights the numerous Kupffer cells that normally line the sinusoids. Kupffer cells are highly phagocytic macrophages which, in response to hepatocyte damage, proliferate and enlarge.

Ceroid Pigment Within Kupffer Cells



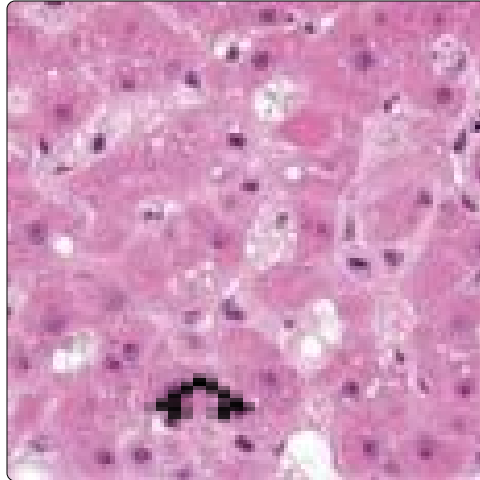
Kupffer Cells (PAS)



(Left) Kupffer cells may accumulate ceroid pigment, a brown-yellow pigment that is composed of cellular debris. (Right) Ceroid pigment is strongly PAS positive, and, thus, this stain is useful in highlighting activated Kupffer cells.

Stellate Cells

(Left) The space of Disse contains the stellate cells (also known as Ito cells, lipocytes, or interstitial fat storing cells) , which are modified fibroblasts that store fat and vitamin A. They also have the capacity for collagen synthesis. When fat laden, they appear as multivesicular cells with cytoplasmic striations. **(Right)** Lipofuscin  is a yellow-brown pigment that accumulates in hepatocytes, most prominently in the centrilobular area.

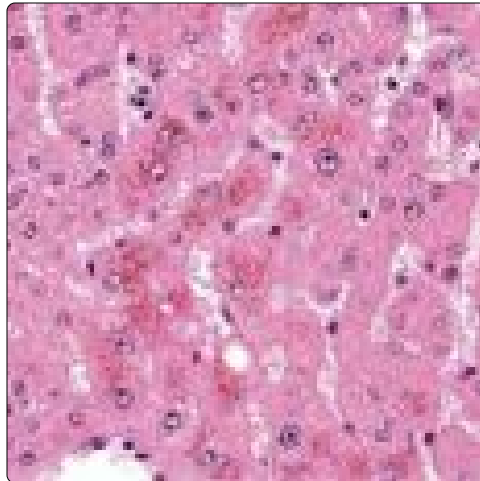


Lipofuscin Pigment

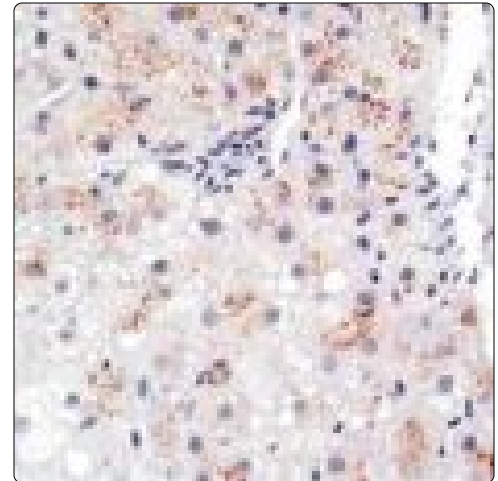


Lipofuscin

(Left) This high-power photomicrograph illustrates the yellow-brown color of lipofuscin. It is common in varying amounts in the adult liver and increases with age. It may be acid fast and PAS/diastase positive and must be distinguished from hemosiderin and hepatocellular bile. **(Right)** Copper  is abundant in the cytoplasm of hepatocytes in infants. It gradually disappears and should be absent by 6-9 months.

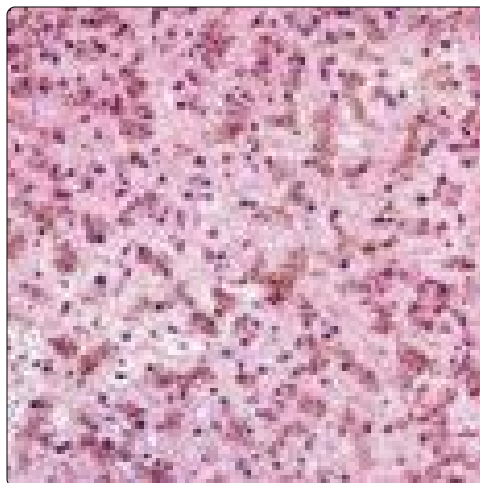


Copper

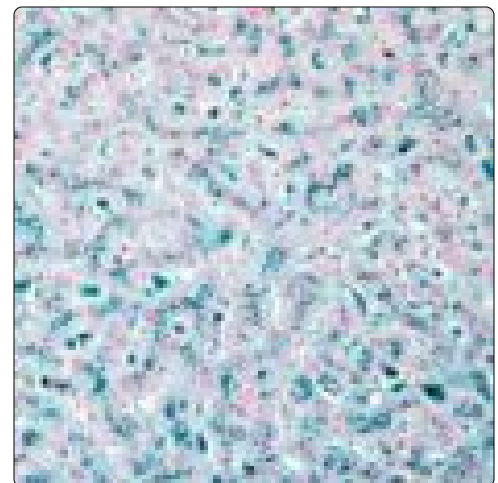


Hemosiderin

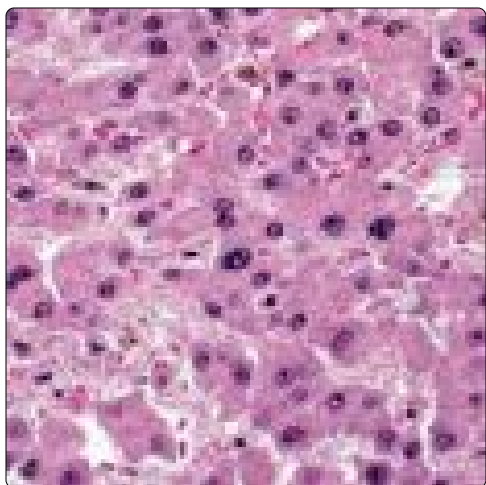
(Left) Hemosiderin is abundant in the livers of newborns and decreases over the 1st few months of life. Only minor quantities of iron are normally found in adults. Iron is most commonly found in periportal hepatocytes, where it consists of a coarse, golden-brown, refractile material. **(Right)** The Prussian blue stain can be used to highlight iron pigment within hepatocytes and Kupffer cells.



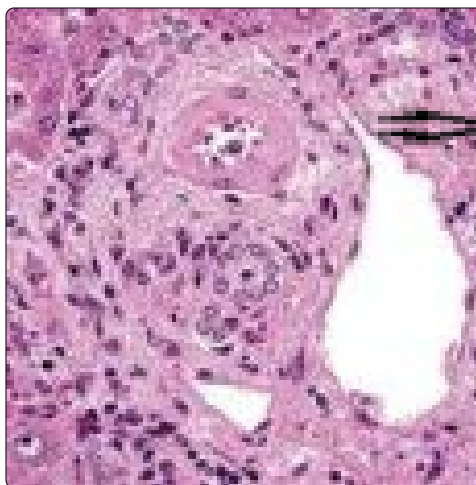
Hemosiderin (Prussian Blue)



Age-Related Changes

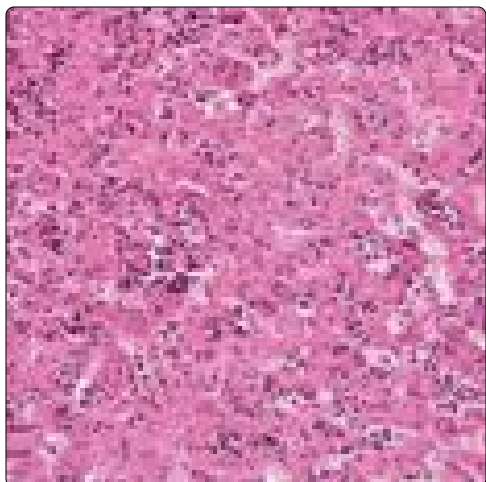


Age-Related Changes

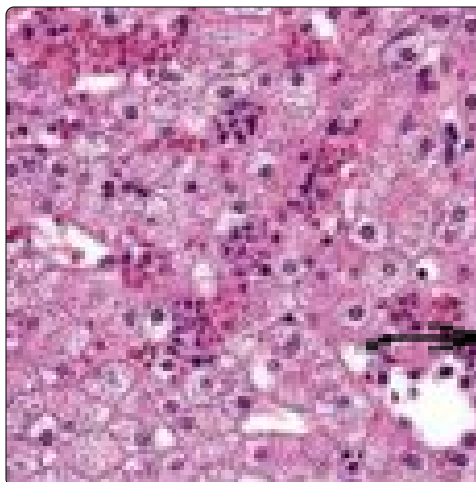


(Left) Increased variation in nuclear size and shape are commonly seen in the aging liver. (Right) Hepatic arterioles may become thickened with age, as shown here. These aging changes can mimic similar changes seen in the context of diabetes.

Extramedullary Hematopoiesis



Surgical Hepatitis

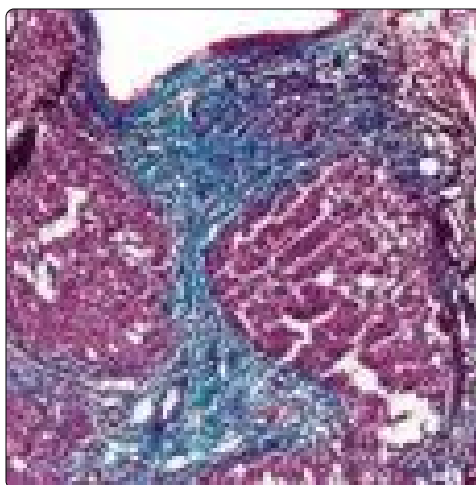


(Left) Extramedullary hematopoiesis is common in newborns but abates within a few weeks after birth. It is largely erythropoietic, but granulocytes, megakaryocytes, and monocytes may also be seen. (Right) Small aggregates of neutrophils may be found within the liver in intraoperative biopsy specimens, usually those taken late during surgery. This must be distinguished from alcoholic hepatitis and other causes of pathologic neutrophilic inflammation.

Bridging Fibrosis Mimic



Subcapsular Artifact



(Left) When cut at a certain angle, large portal tracts may resemble bridging fibrous septa. The comparably sized vessels and ducts are clues that this is not a pathological change. (Right) Portal tracts in the subcapsular zone can appear unusually fibrotic. When wedge biopsies are taken, the capsular contraction may also form artifactual septa that isolate islands of parenchyma under the capsule and mimic cirrhosis. These changes rarely extend > 2 mm into the liver parenchyma, however.

GALLBLADDER

Macroscopic Anatomy

- Located in shallow depression on inferior surface of liver
 - Composed of fundus, body, and neck
 - Neck empties into cystic duct, connecting gallbladder to biliary tree
 - Cystic duct contains spiral valves of Heister, which are mucosal folds that prevent collapse
- Up to 10 cm long by 4 cm wide in adults
 - Normal wall: 1-2 mm thick
- Peritoneum reflected from liver covers free side; subserosal connective tissue of gallbladder merges with interlobular connective tissue of liver on hepatic side

Microscopic Anatomy

- Mucosa (surface epithelium plus lamina propria)
 - Cores of lamina propria lined by single layer of columnar epithelium
 - Branching of mucosal surface is common
 - Epithelial cells have eosinophilic cytoplasm with variably present small apical vacuoles
 - IHC: CK7(+), CK19(+), CK20(-), CDX-2(-)
 - Nuclei are at base of cell or central; oval and uniform; fine chromatin and smooth nuclear membranes
 - Nucleoli are inconspicuous or absent
 - Additional scattered basal cells and narrower, darker penciloid cells admixed with biliary epithelial cells
 - Few scattered mucosal lymphocytes, plasma cells, mast cells, and histiocytes are normal
 - Lamina propria is composed of loose connective tissue and contains blood vessels, lymphatics, and elastic fibers
- Muscular wall
 - Loosely arranged bundles of circular, longitudinal, and oblique muscle fibers
 - Lacks well-formed layers seen in luminal gut
 - Fibrovascular tissue separates muscle bundles
 - Gallbladder lacks muscularis mucosa and submucosa; thus, lamina propria abuts directly onto muscularis propria

- Perimuscular connective tissue
 - Typically contains blood vessels, nerves, lymphatics, and fat
 - Occasionally contains lymph nodes, paraganglia
- Serosa
 - Peritoneum covers part of gallbladder not connected to liver (~ 60%)
- Mucous glands
 - Located in neck of gallbladder
 - Cuboidal or low columnar epithelium with clear to lightly basophilic cytoplasm
 - Round, uniform, basally located nuclei
- Ducts of Luschka: Small accessory bile ducts in perimuscular connective tissue, most often on hepatic side of gallbladder
 - Supported by distinctive ring of connective tissue
 - Epithelium is similar to biliary epithelium
 - Occasionally, accessory bile ducts can be found in gallbladder bed; may leak after cholecystectomy
- Ganglion cells: Present in perimuscular connective tissue, lamina propria, muscular wall
- Ectopic liver, pancreas, adrenal, stomach, thyroid have been reported in gallbladder

CONSIDERATIONS AND VARIATIONS

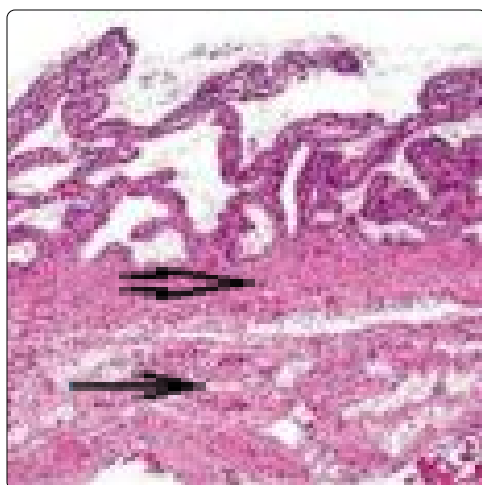
Metaplasia

- Pseudopyloric and intestinal metaplasia can be seen
 - Rare outside context of chronic cholecystitis

Pitfalls and Artifacts

- Ducts of Luschka can be mistaken for carcinoma on frozen section
 - Look for lobular architecture
- Rokitansky-Aschoff sinuses found in almost 1/2 of normal gallbladders
 - Do not indicate chronic cholecystitis by themselves
 - Ruptured Rokitansky-Aschoff sinuses with extruded mucin can mimic cancer, as can Rokitansky-Aschoff sinuses that contain dysplastic epithelium

Gallbladder Layers



(Left) This low-power photomicrograph shows all layers of the gallbladder wall, including the mucosa (epithelium plus lamina propria), muscularis propria, and perimuscular connective tissue. (Right) The nuclei of the gallbladder epithelium are uniform with fine chromatin and smooth nuclear membranes. They are located in the centers or at the bases of the cells.

Epithelium



Mucosa

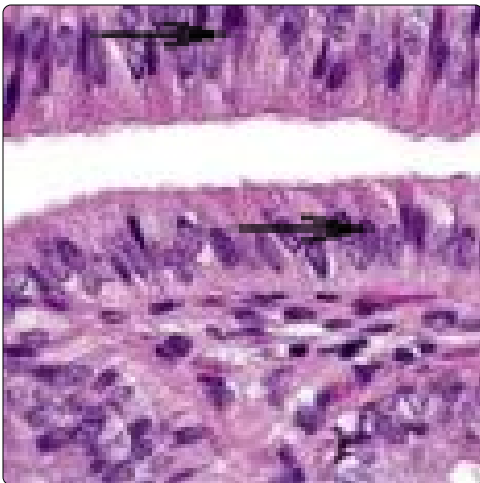


Epithelial Cells



(Left) The gallbladder mucosa consists of cores of lamina propria lined by a single layer of columnar epithelium [E]. Mucosal branching is common in the normal gallbladder, as seen here. (Right) Gallbladder epithelial cells have eosinophilic cytoplasm with variably present small apical vacuoles [E]. The nuclei are round to oval with smooth nuclear membranes. Nucleoli are inconspicuous or absent.

Penciloid Cells



Basal Cells

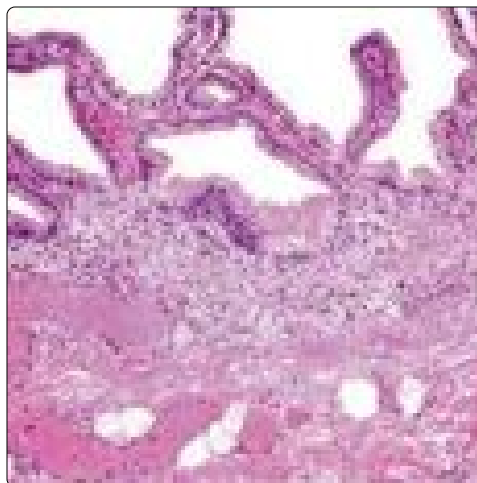


(Left) Dark, thin, elongated penciloid cells [E], the function of which is unknown, are admixed with the biliary epithelial cells in the gallbladder mucosa. (Right) The epithelium also contains scattered basal cells [E], which should not be mistaken for lymphocytes.

Lamina Propria



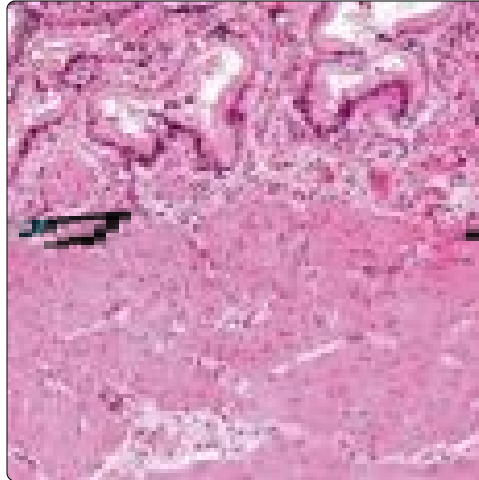
Scattered Inflammatory Cells



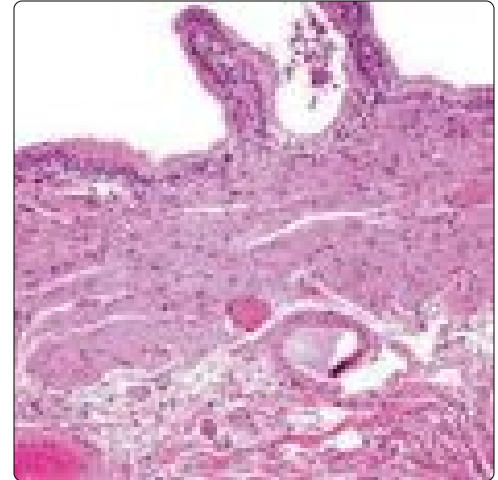
(Left) The lamina propria cores are composed of loose connective tissue and contain blood vessels, lymphatics, and elastic fibers. (Right) A few scattered mucosal lymphocytes, plasma cells, mast cells, and histiocytes are normally present in the gallbladder. This sparse inflammatory infiltrate alone is not diagnostic of chronic cholecystitis.

Muscularis Propria

(Left) Since the gallbladder lacks a true muscularis mucosa and submucosa, the lamina propria abuts directly onto the muscle bundles of the muscularis propria. The muscle bundles may actually extend up into the lamina propria. **(Right)** The muscular wall consists of loosely arranged bundles of circular, longitudinal, and oblique muscle fibers. It lacks the well-developed longitudinal and circumferential layers seen in the intestine.

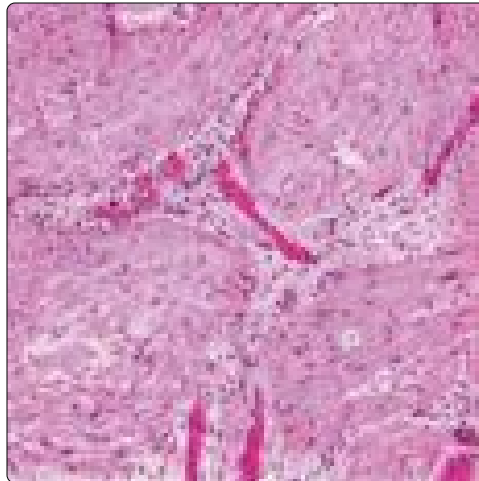


Muscularis Propria

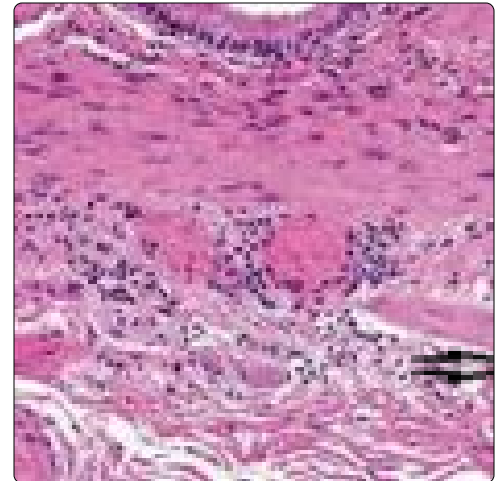


Muscularis Propria

(Left) Fibrovascular tissue separates the muscle bundles in the muscularis propria. **(Right)** Ganglion cells are present in the perimuscular connective tissue, lamina propria, and muscularis propria.

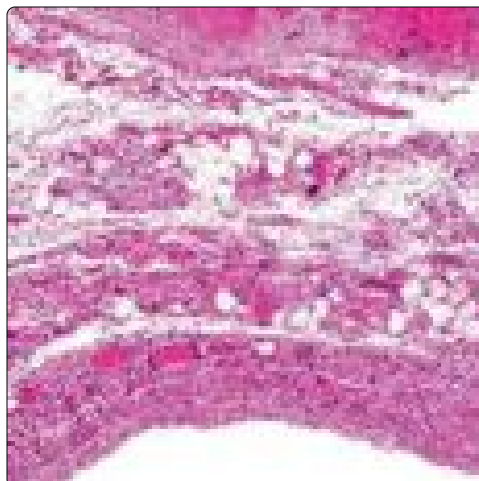


Ganglion Cells

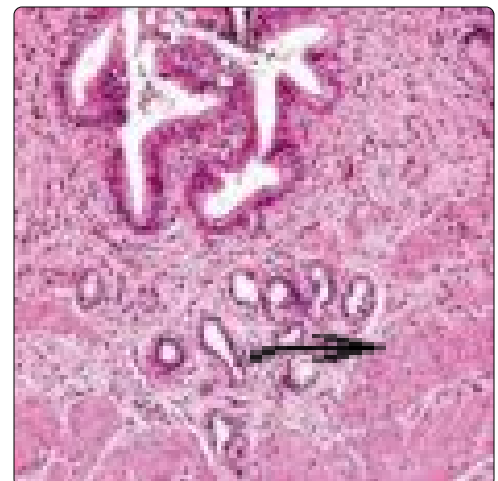


Perimuscular Connective Tissue

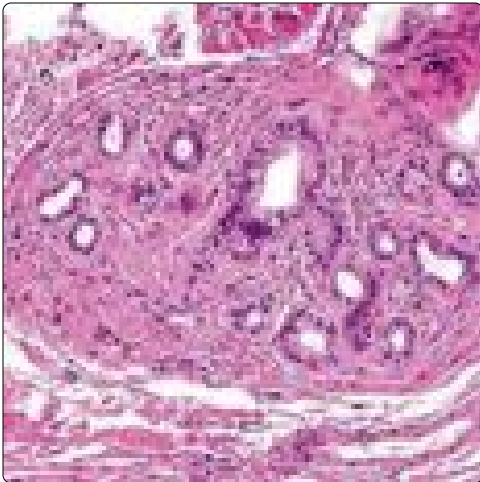
(Left) The perimuscular connective tissue contains blood vessels, nerves, lymphatics, and fat. It occasionally contains lymph nodes and paraganglia. Peritoneum covers the portion of the gallbladder that is not connected to the liver (~60%). **(Right)** Mucous neck glands are located in the neck of the gallbladder. They have cuboidal or low columnar epithelium with clear to lightly basophilic cytoplasm and round, uniform, basally located nuclei.



Mucous Neck Glands



Ducts of Luschka

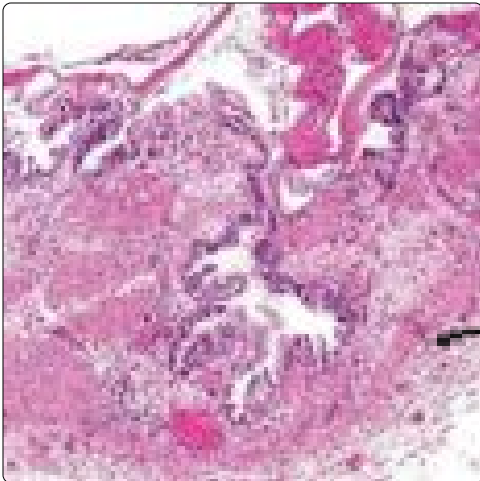


Accessory Bile Ducts



(Left) The ducts of Luschka are small accessory bile ducts in the perimuscular connective tissue, most often on the hepatic side. Note the distinctive ring of connective tissue surrounding the glands. These should not be mistaken for carcinoma on frozen sections. (Right) Occasionally, larger accessory bile ducts can be found in the perimuscular soft tissue in the gallbladder bed. These are significant because they may leak after cholecystectomy.

Rokitansky-Aschoff Sinuses

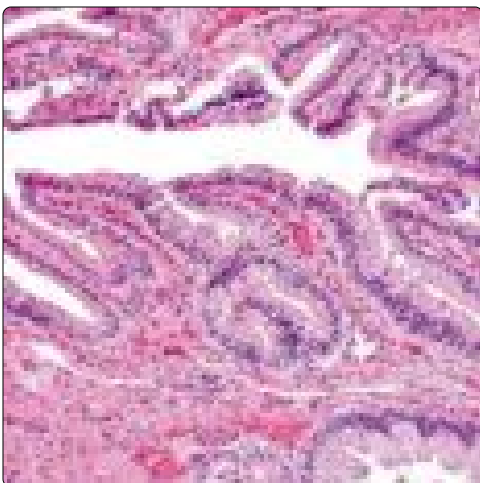


Gastric Metaplasia

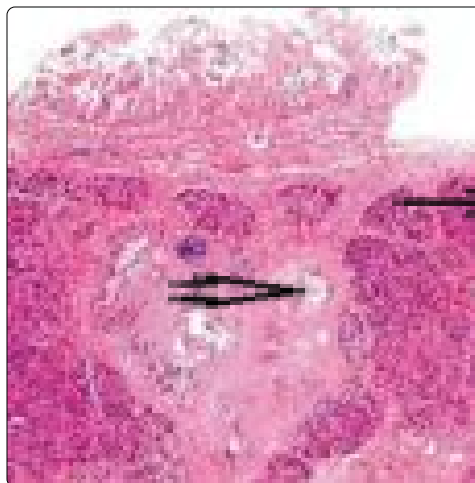


(Left) Rokitansky-Aschoff sinuses [\[1\]](#) can be found in almost 1/2 of normal gallbladders and do not by themselves indicate chronic cholecystitis. (Right) Gastric-type or pseudopyloric metaplasia [\[2\]](#), similar to that seen in the duodenum, is a common finding in chronic cholecystitis.

Intestinal Metaplasia



Ectopic Tissues



(Left) Intestinal metaplasia [\[5\]](#) is also a common finding in chronic cholecystitis. (Courtesy L. Yerian, MD.) (Right) Ectopic liver, pancreas, adrenal, stomach, and thyroid have all been reported in the gallbladder. This photograph illustrates an example of ectopic pancreas with acinar tissue [\[3\]](#), ducts [\[4\]](#), and smooth muscle. (Courtesy L. Yerian, MD.)

Extrahepatic Biliary Tract

EXTRAHEPATIC BILIARY TREE

Macroscopic Anatomy

- Extrahepatic biliary tree is composed of left and right hepatic ducts, common hepatic duct (CHD), cystic duct, and common bile duct (CBD); serves as conduit for bile flow
 - Left and right hepatic ducts exit liver, extend for ~ 1 cm, and fuse to become CHD
 - CHD extends for ~ 2 cm before joining cystic duct; fusion of these 2 ducts forms CBD
 - Cystic duct is ~ 3 cm in length; anatomic variability in way cystic duct joins CHD has implications for biliary surgery
 - On average, CBD is 0.4-1.3 cm in diameter and 5 cm long
 - Lower end of CBD runs alongside and usually unites with main pancreatic duct before entering duodenum at papilla

Microscopic Anatomy

- Epithelium: Single layer of columnar epithelium
 - IHC: CK7(+), CK19(+), CK20(-), CDX2(-)

- Dense connective tissue layer (lacks smooth muscle except for CBD and cystic duct at junction with gallbladder)
 - Upper portion of CBD contains occasional strands of muscle; thick smooth muscles invests lower 1/3
- Peribiliary connective tissue

CONSIDERATIONS AND VARIATIONS

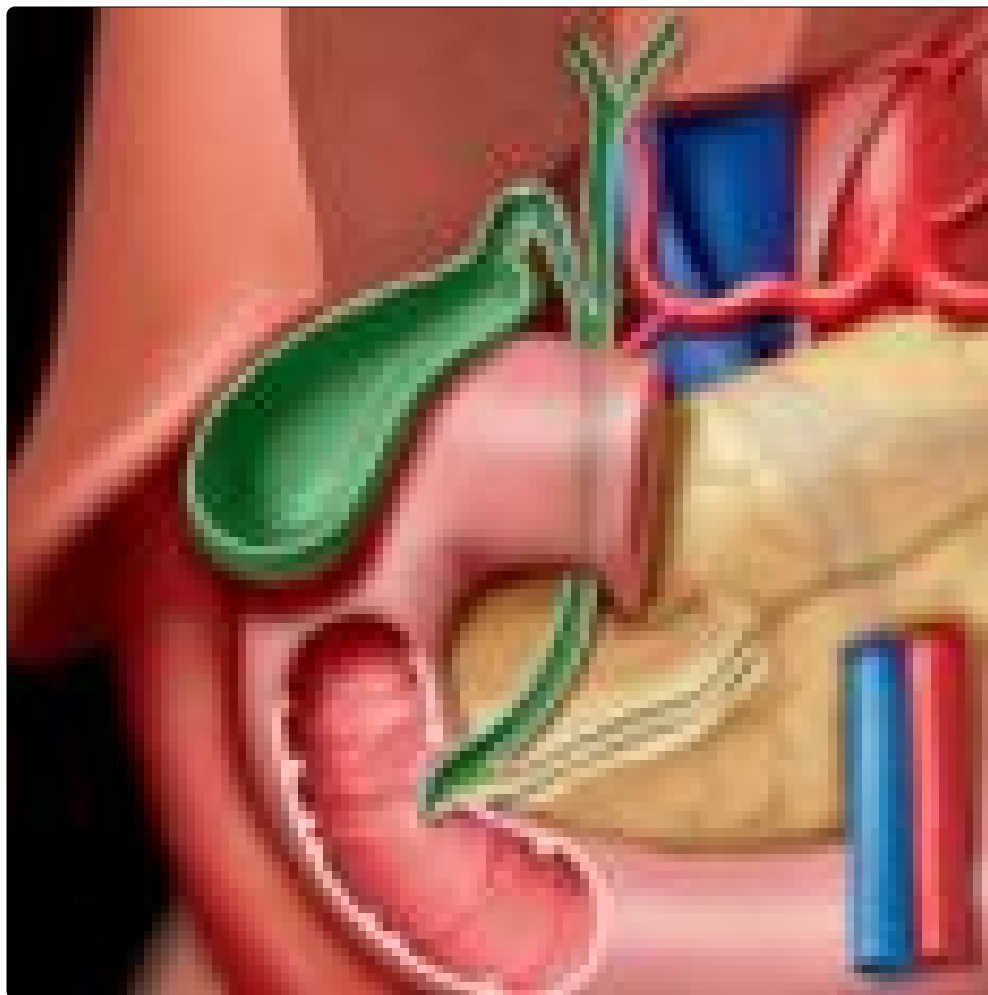
Metaplasia

- Gastric
- Intestinal

Pitfalls and Artifacts

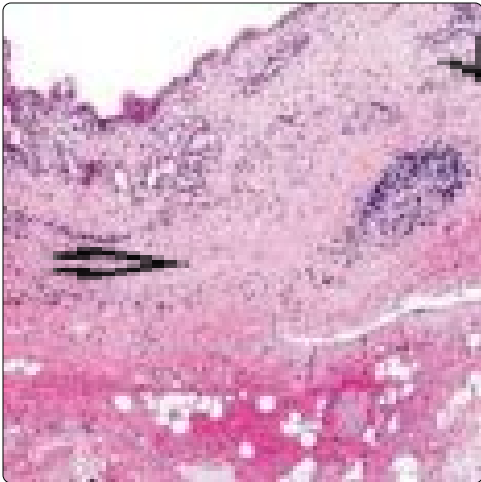
- Peribiliary mucous glands may be mistaken for carcinoma on frozen section
 - Perineural "invasion" by these glands has rarely been reported
 - Glands retain their lobular architecture and dense stroma; lack atypia and mitoses

Extrahepatic Biliary Tract

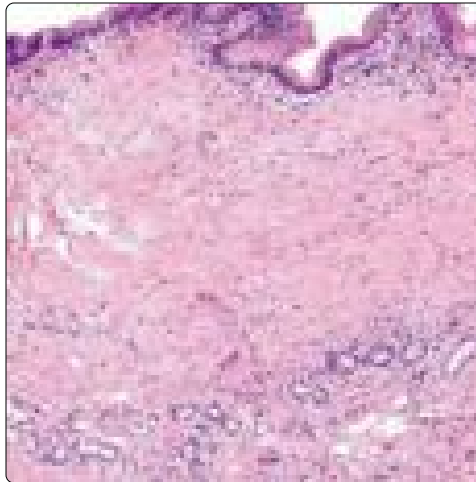


The left and right hepatic ducts join to become the common hepatic duct (CHD), which extends for ~ 2 cm before joining the cystic duct. The common bile duct (CBD) is formed by the fusion of the cystic duct and CHD. The lower end of the CBD runs beside the pancreas and usually unites with the main pancreatic duct before entering the duodenum at the papilla.

Extrahepatic Bile Ducts: Layers

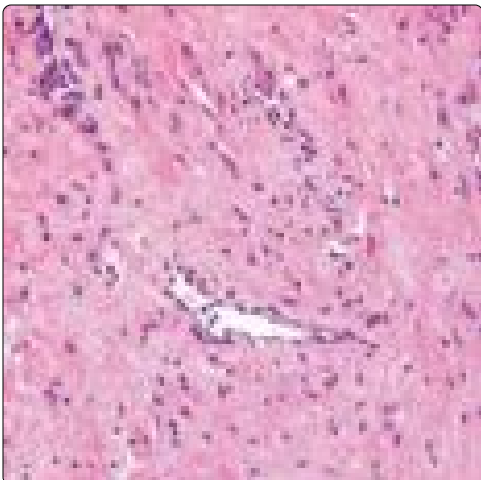


Fibrous Stroma

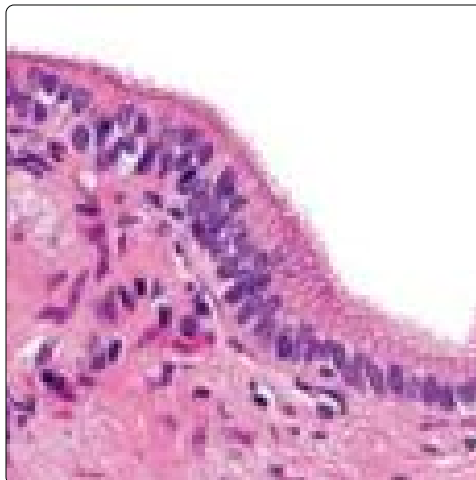


(Left) The ducts in the extrahepatic biliary tree are composed of an epithelial layer [1], subepithelial stroma [2], and the peribiliary connective tissue [3]. The peribiliary connective tissue contains nerves, lymphatics, and vessels. **(Right)** The periepithelial fibrous stroma [4] is a distinctive feature of the extrahepatic biliary tree and is composed of benign fibroblasts and dense collagen.

Stromal Layer

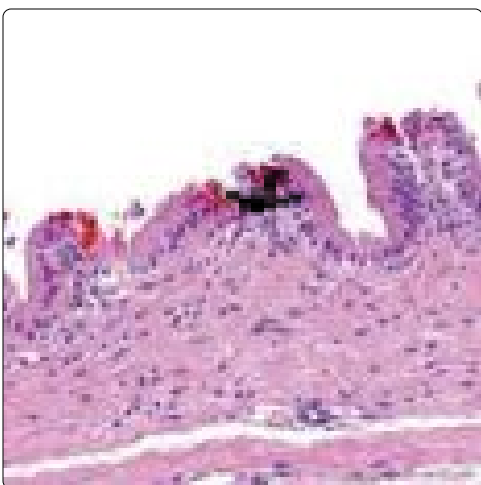


Biliary Epithelium

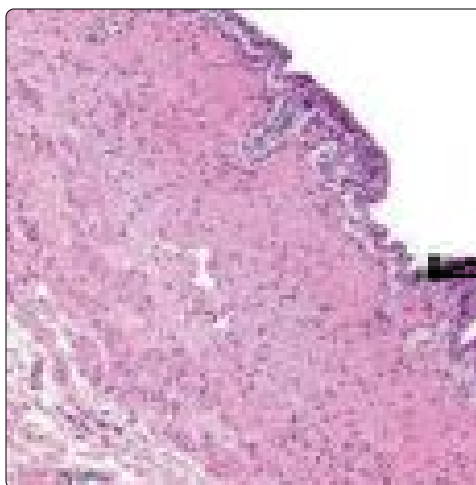


(Left) The subepithelial stroma is composed of dense collagen, elastic fibers, and small vessels. Inflammatory cells are sparse. **(Right)** Extrahepatic bile ducts are lined by a single layer of columnar epithelium. The nuclei are round and basally located with evenly distributed chromatin and inconspicuous nucleoli. The cytoplasm contains small amounts of mucin. Goblet cells are normally absent. Scattered neuroendocrine cells are present throughout.

Bile Pigment



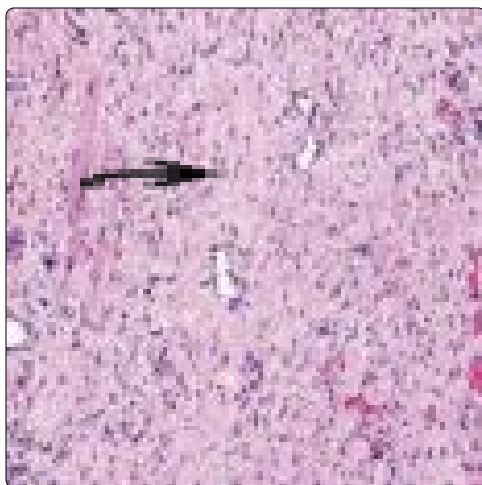
Patchy Smooth Muscle



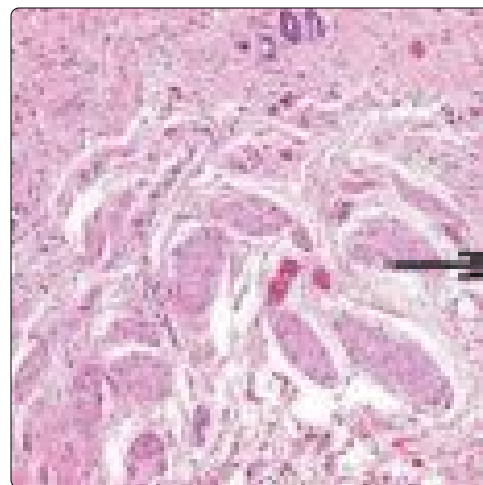
(Left) Anywhere along its length, the extrahepatic biliary tree may show bile pigment [1] within or along the surface of the epithelium. **(Right)** Smooth muscle fibers [2] are present within the fibrous stromal layer along much of the length of the CBD and at the junction of the cystic duct and gallbladder. In the CBD, the smooth muscle is more prominent in the distal third of its length.

Subtle Smooth Muscle

(Left) Smooth muscle fibers  are sparse in the proximal CBD and appear as subtle strands within the dense fibrous stroma. **(Right)** Smooth muscle fibers  are more prominent in the distal CBD. Smooth muscle is also commonly seen at the junction between the cystic duct and gallbladder (known as spiral valve of Heister).

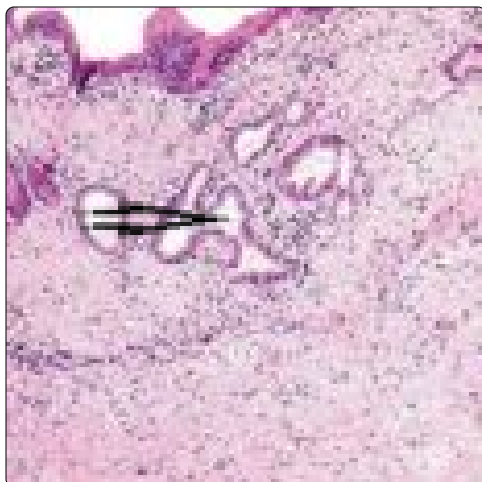


Prominent Smooth Muscle



Sacculi of Beale

(Left) The hepatic and CBD surface epithelia contain numerous regular depressions or invaginations known as the sacculi of Beale . They may appear to be isolated from the surface in tangential sections and thus may mimic carcinoma. **(Right)** Numerous lobular mucous glands , known as peribiliary glands, biliary glands, or periductal glands, surround the sacculi  and empty into them. The more peripheral lobular mucous glands are invested by dense stroma.

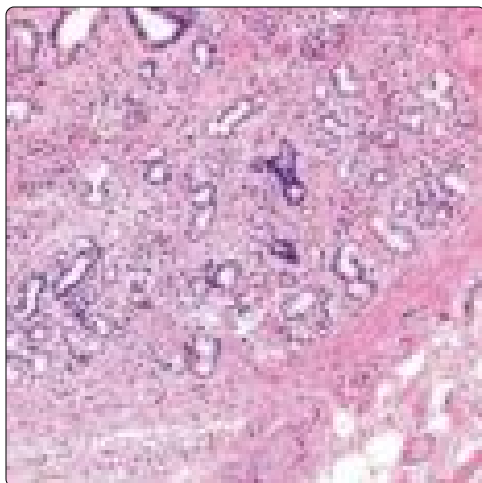


Lobular Mucous Glands

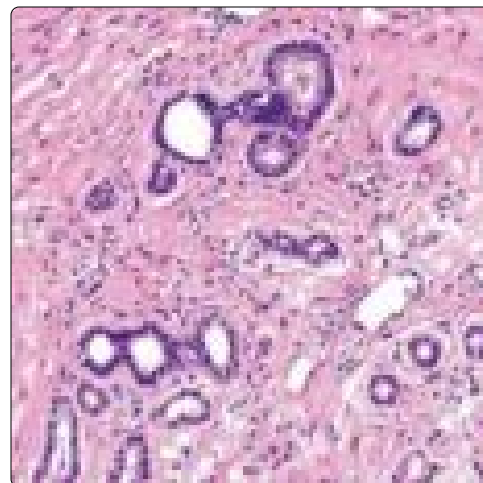


Lobular Mucous Glands

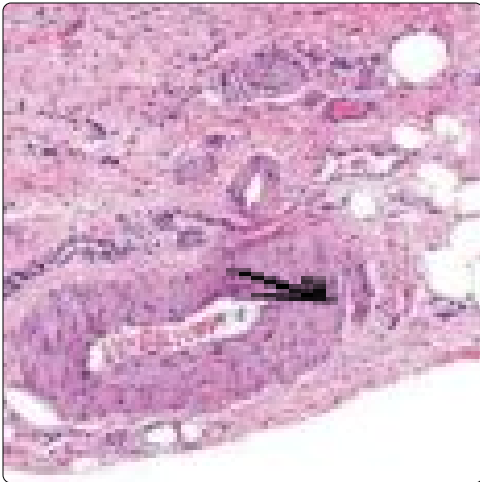
(Left) These glands are lined by low columnar or cuboidal epithelium with prominent mucin. Note the thick lobular stroma that encircles the glands; it can help differentiate normal mucous glands from an infiltrating carcinoma. **(Right)** Peribiliary glands can appear angulated and irregular in some cases, mimicking invasive adenocarcinoma, particularly on a frozen section. The lobular arrangement and dense stroma of the glands can aid in identifying them as benign.



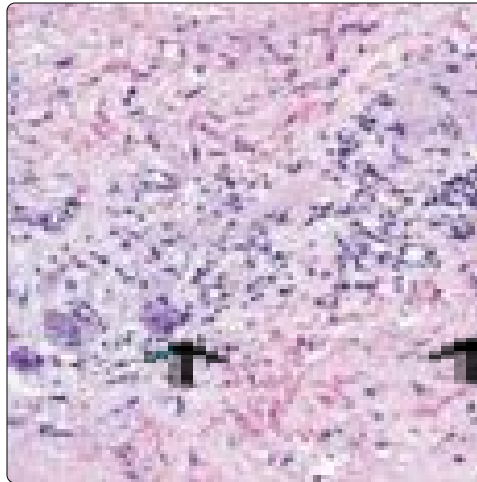
Lobular Mucous Glands



Peribiliary Connective Tissue

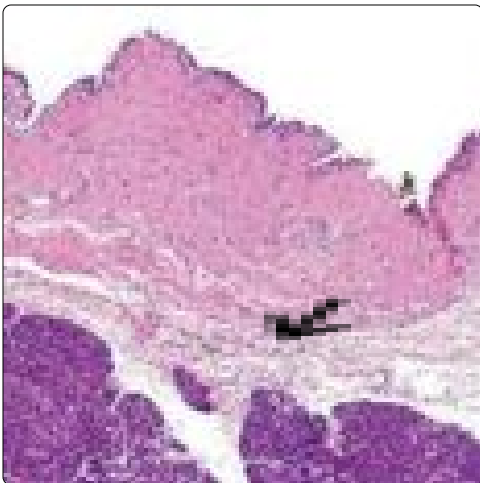


Ganglion Cells

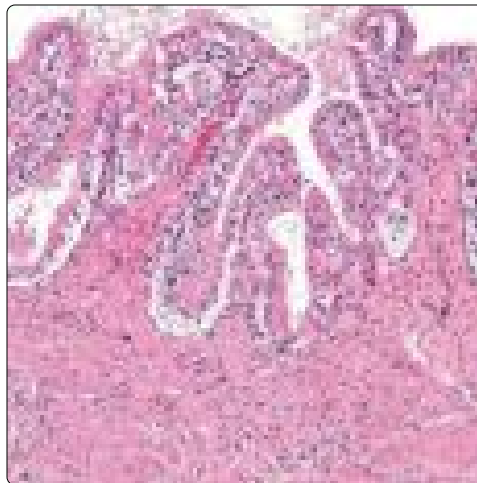


(Left) The peribiliary connective tissue consists of adipose tissue that contains vessels [1], lymphatics, and nerves [2]. (Right) Ganglion cells [3] can be identified in the peribiliary stromal connective tissue and are often associated with nerve fibers [4].

Common Bile Duct

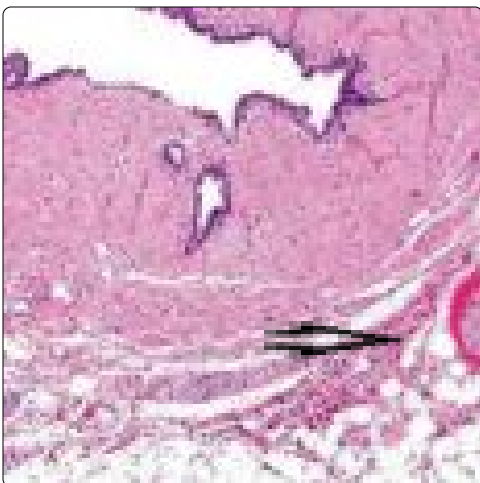


Cystic Duct

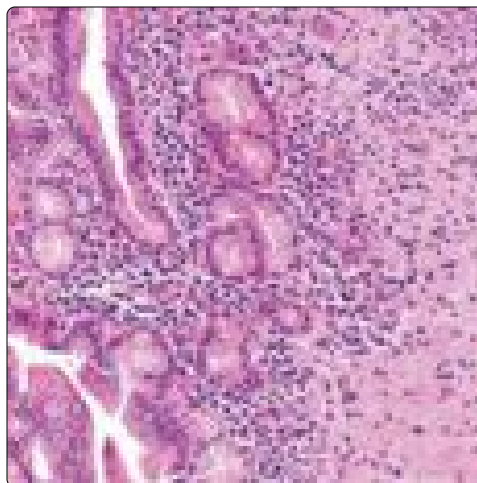


(Left) The distal portion of the CBD [1] runs alongside the pancreas [2] and eventually joins up with the main pancreatic duct before entering the duodenum at the major papilla. (Right) The cystic duct lining epithelium is virtually identical to that of the gallbladder and is thrown into pleats or folds. Mucous glands may be present in the lamina propria (not shown here).

Cystic Duct



Pyloric Metaplasia



(Left) Cystic duct stroma contains muscle fibers [1] admixed with dense connective tissue, particularly at the junction with the gallbladder (spiral valve of Heister). (Right) Both pyloric-type [2] and intestinal metaplasia can be seen in the extrahepatic bile ducts, usually in association with inflammation.

Vaterian System

VATERIAN SYSTEM

Macroscopic Anatomy

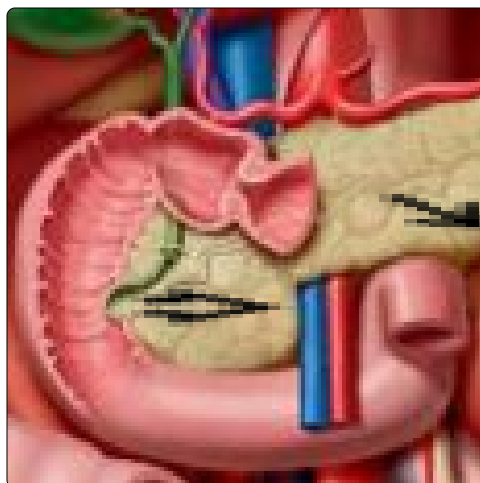
- Vaterian system: Distal segments of common bile duct (CBD) and main pancreatic duct, major papilla and ampulla, minor papilla, and sphincter muscle (sphincter of Oddi)
- Complex and variable anatomy
 - CBD and main pancreatic duct join to form common channel before entering duodenum in 74% of people
 - 19% of people have ducts running in parallel with adjacent but separate openings into duodenum (bile duct opening cranial and medial to duct of Wirsung); these people usually do not have well-developed ampulla
 - 7% of people have ducts that unite before entering duodenum, but lumina separated by connective tissue septum
- Major papilla
 - Protuberance located 8-10 cm distal to pylorus
 - Projects into 2nd portion of duodenum
- Ampulla of Vater
 - Strictly defined, ampulla of Vater is dilated conduit present at major papilla (which is the protuberance)
 - Ampulla results from union of CBD and main pancreatic duct where they open into duodenal lumen
 - Mean length: 11 mm; mean diameter: 5 mm
 - Mucosal folds or flaps are present covering ampulla in majority of people; may serve as valves
 - Only 25% of people with combined distal channel have well-developed ampulla
- Minor papilla
 - May be difficult to appreciate macroscopically
 - Duct of Santorini usually opens into minor papilla
- Sphincter of Oddi
 - Circular and longitudinal musculature that regulates flow of bile and pancreatic enzymes into duodenum
 - Muscles held in place by muscularis propria of duodenal wall
 - Sphincter muscle fibers believed to extend up into CBD and main pancreatic duct

- Pancreatic ducts
 - Major pancreatic duct of Wirsung drains small ducts from tail of pancreas to duodenum
 - Minor pancreatic duct of Santorini joins major duct at variety of intrapancreatic locations
- CBD
 - Formed by joining of cystic duct and common hepatic duct
 - Distal CBD either runs alongside or joins with duct of Wirsung before entering duodenum at major papilla

Microscopic Anatomy

- CBD and main pancreatic ducts have essentially identical epithelial linings
 - Columnar mucinous epithelium
 - No goblet cells
- Ampullary epithelium
 - Epithelium of ampulla and distal ducts consist of folds or papillae
 - Fibrovascular cores that contain sparse inflammation
 - Epithelium similar to ducts (columnar), but scattered goblet cells are common
- Associated ducts and glands
 - Numerous mucous glands present around terminal portions of ducts and ampulla
 - Morphologically similar to peribiliary mucous glands, but often larger
 - Accessory pancreatic ducts also drain into ampulla
 - Pancreatic acini may also be present
- Sphincter of Oddi
 - Muscle fibers become apparent in distal ducts (CBD and pancreatic) proximal to their entrance into duodenum, and muscle fibers may extend up into epithelial folds at ampulla
- Duodenal villi are few or absent at papilla
- Scattered neuroendocrine cells sometimes present around pancreatic ducts and accessory glands

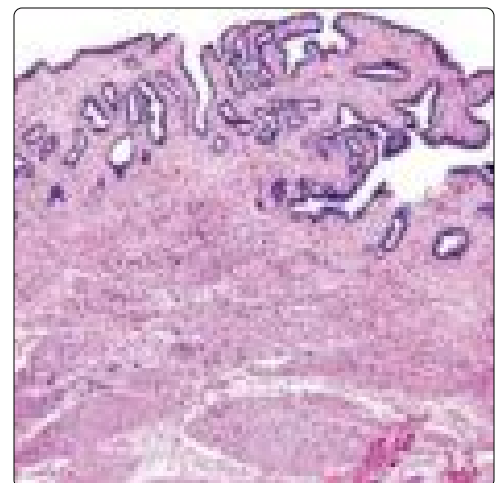
Vaterian System



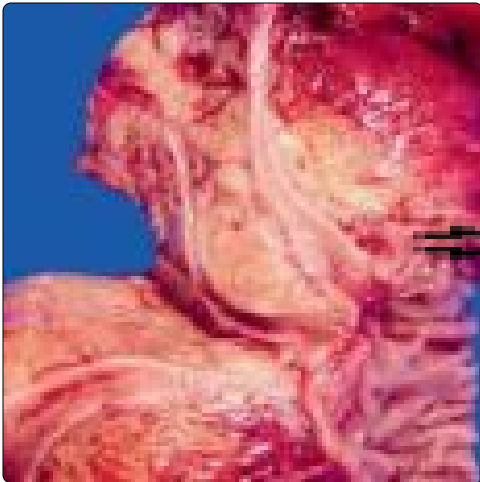
(Left) The Vaterian system consists of the common bile duct and main pancreatic duct that typically join proximal to the duodenum and enter at the major papilla.

(Right) The epithelium of the ampulla and distal ducts that converge there consists of prominent folds or papillary fronds, which are supported by connective tissue stroma.

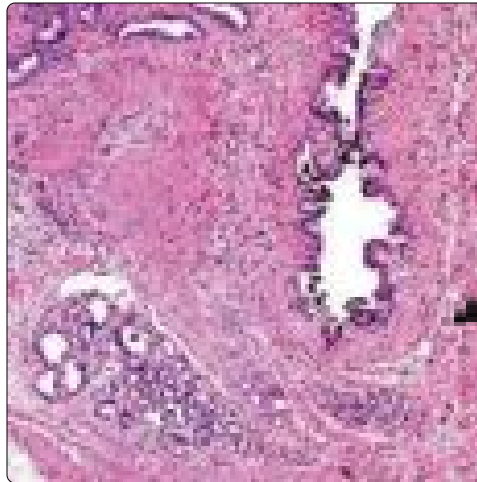
Ampullary Epithelium



Vaterian System (Gross)

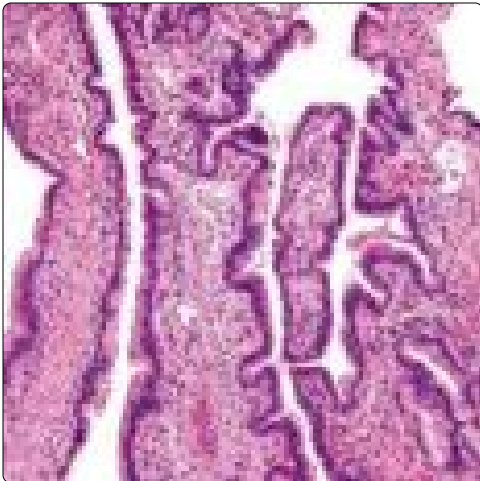


Vaterian System (Low Power)

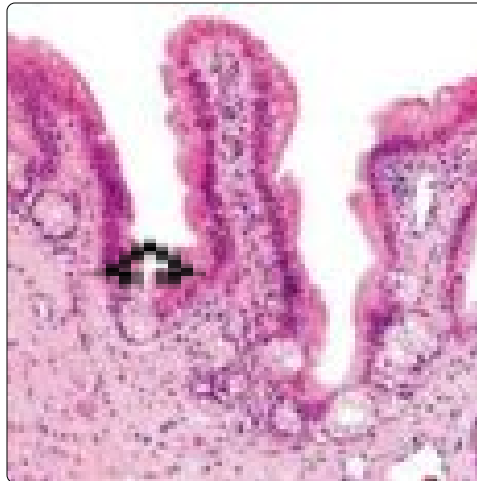


(Left) This photo of a bivalved pancreas illustrates the pancreatic duct and the common bile duct entering the duodenum together at the ampulla. (Right) This photomicrograph shows the common bile duct entering the duodenum at the ampulla. Note the numerous lobular collections of mucous glands in this area.

Ampullary Epithelium

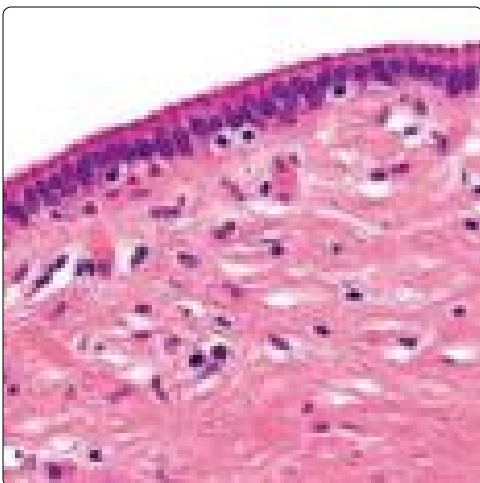


Ampullary Epithelium

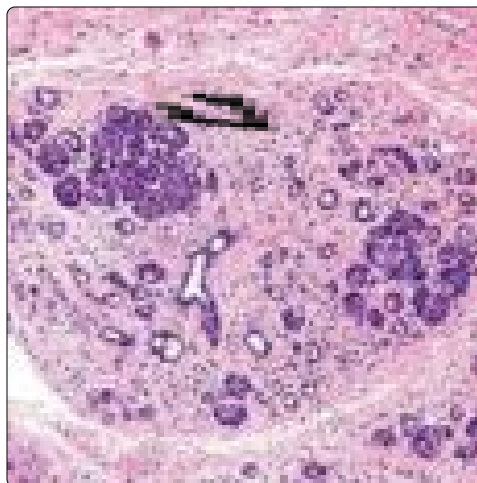


(Left) The folds or fronds of the ampullary epithelium are supported by fibrovascular cores. These cores normally contain sparse chronic inflammatory cells. (Right) Unlike the epithelium of the pancreatic and bile ducts, the ampullary epithelium contains scattered goblet cells. Note also the mucous glands at the base of the epithelial fronds.

Ductal Epithelium



Glands and Acini



(Left) The epithelium of the terminal portions of the main pancreatic ducts and the common bile duct, which terminate in the ampulla, is a simple columnar mucinous epithelium without goblet cells. (Right) Numerous mucous glands are present around the ampulla and the terminal portions of the ducts. In addition, accessory pancreatic glands and pancreatic acini are often present in this area.

PANCREAS

Macroscopic Anatomy

- Lobular gland located in retroperitoneum behind stomach and transverse colon, and in front of aorta and vena cava
- Anterior surface is peritonealized
- 15-20 cm in length; 41-182 grams on average
- No discrete capsule

Microscopic Anatomy

- Acini
 - Exocrine secretory component, comprises > 80% of pancreas
 - Spherical or tubular glands with small central lumens, surrounded by triangular secretory cells
 - Cytoplasm packed with organelles, imparting granular appearance
 - Number of granules varies depending on secretory activity
 - Some cells have Golgi zone adjacent to nucleus
 - Variant: Acinar ectasia (focal dilatation with flattening of epithelium and inspissated secretions) is associated with obstruction, dehydration, sepsis, uremia
- Islets of Langerhans
 - Endocrine component of pancreas is present mostly within islets and comprises 1-2% of gland in adult but 20% in neonates
 - 2 types of islets: Compact and diffuse
 - Scattered single endocrine cells are present in ducts (mostly larger ducts) and acini, but these constitute < 10% of total number of endocrine cells in pancreas
 - 4 types of endocrine cells: β cells, α cells, Δ cells, PP cells
 - β cells (60-70% of islet cells) secrete insulin; at center of islets with occasional markedly enlarged nuclei
 - α cells (15-20% of islet cells) secrete glucagon
 - Δ cells (< 10% of islet cells) secrete somatostatin
 - PP cells are very rare, produce pancreatic polypeptide; mostly present in diffuse islets where they are majority of cells

- Ducts: Main ducts, interlobular ducts, intralobular ducts, intercalated ducts, centroacinar cells
 - Main ducts lined by simple columnar mucinous epithelium except in ampulla, which contains more papillary epithelium
 - Interlobular ducts are also lined by low columnar mucinous epithelium with round nuclei
 - Main and interlobular ducts are surrounded by thick collagen layer that may contain lobules of mucous glands within wall, similar to extrahepatic bile ducts
 - Intralobular ducts have epithelium identical to intercalated ducts and have only scant collagen
 - Intercalated ducts lined by single layer of flattened cells with oval nuclei and nonsecretory cytoplasm; fuse to form intralobular ducts
 - Centroacinar cells are inconspicuous flat to cuboidal cells with pale cytoplasm in middle of acini; attach to exocrine acinar cells by tight junctions
- Connective tissue
 - Lobules separated by thin strands of loose connective tissue containing nerves and vessels
 - Adipose tissue (3-20% of gland); also present in peripancreatic soft tissue
- Pancreatic stellate cells: Similar to hepatic stellate cells, store vitamin A

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Neonates have larger volume of endocrine cells and mesenchymal tissue (up to 30%)
- Ratio of hormone-producing cells in islets varies with age

Metaplasia

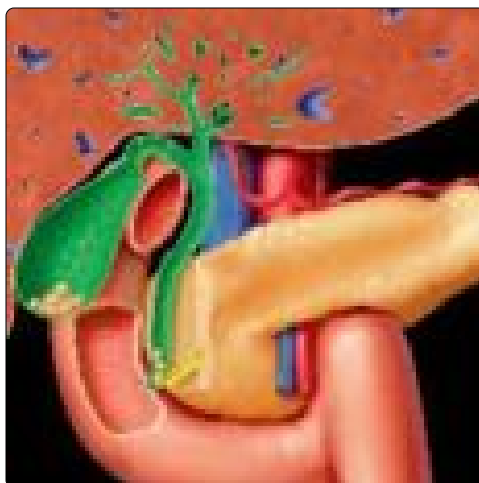
- Ducts: Squamous, oncocytic, acinar, intestinal/goblet cell
- Acini: Squamous, mucinous, centroacinar/ductal

Pitfalls and Artifacts

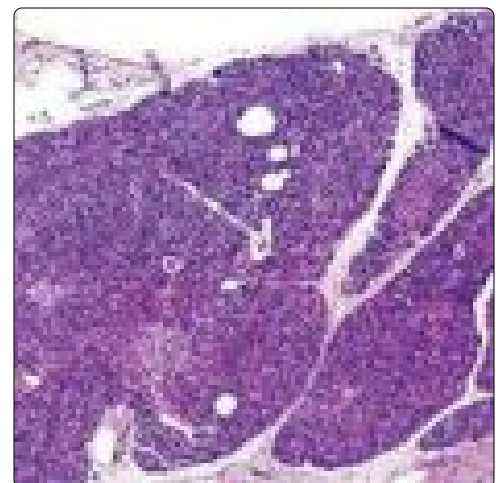
- Focal acinar transformation may mimic tumor
- Diffuse islets may be mistaken for infiltrative tumor given irregular borders

Pancreas

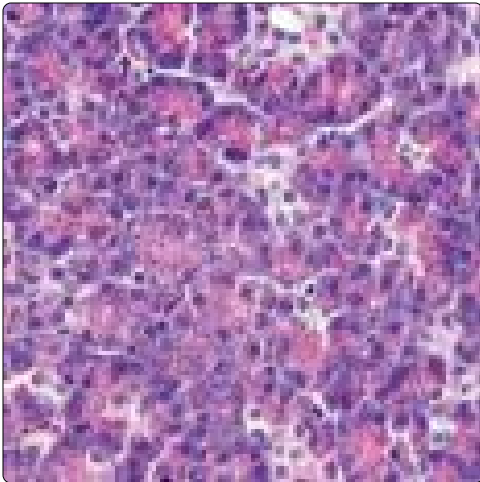
(Left) The pancreas is a lobular gland located on the posterior wall of the abdomen. It lies behind the stomach and transverse colon and in front of the aorta and vena cava. The pancreas is divided into the head, neck, body, and tail, but the demarcations are indistinct. The head lies within the curve of the duodenum, and the tail reaches to the hilum of the spleen. **(Right)** Delicate connective tissue septa divide the pancreatic parenchyma into lobules.



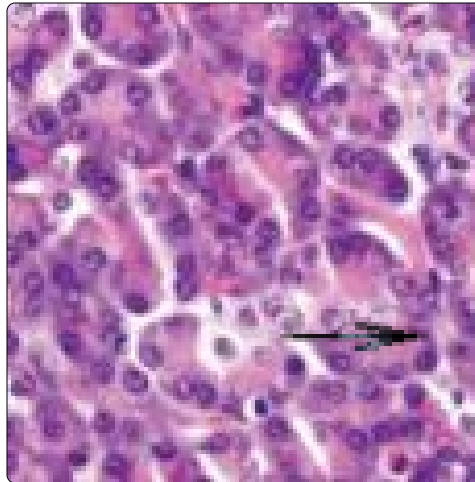
Lobular Organization



Acinar Parenchyma

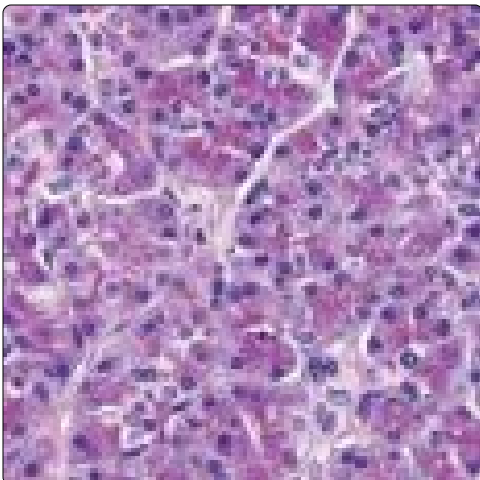


Acinar Cells

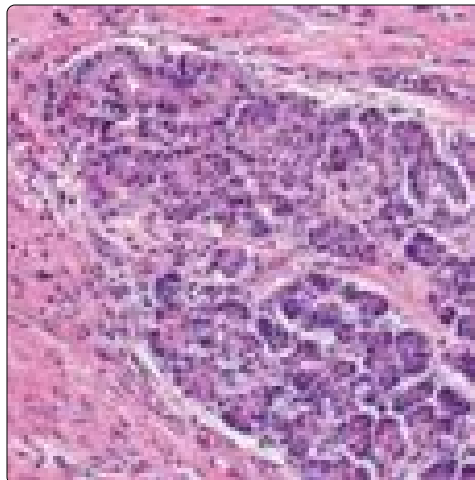


(Left) The exocrine acinar parenchyma consists of spherical or tubular glands with small central lumina, surrounded by a single layer of triangular secretory cells. There is no myoepithelial layer. Nuclei are round and basally located with even chromatin and small nucleoli. **(Right)** Secretory cells have basophilic basal granules [icon] and eosinophilic apical granules [icon]. A centroacinar cell, which represents the beginning of the ductal system, is visible in a lumen [icon].

Acinar Cytoplasmic Granules

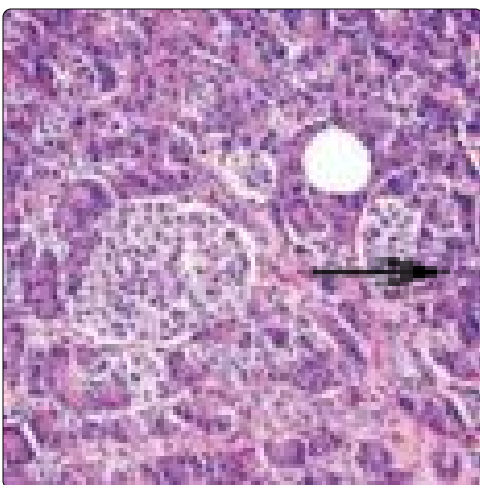


Metaplasia

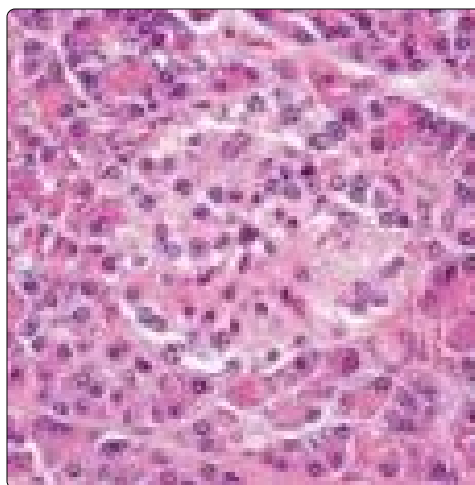


(Left) Zymogen granules in secretory cells are PAS(+) and diastase resistant, and negative for mucin and Alcian blue because they do not contain mucin. Immunohistochemically, they are positive for pancreatic enzymes (amylase, lipase, trypsin, chymotrypsin), as well as CAM 5.2. They are negative for AE1/AE3, CK7, CK19, and CK20. **(Right)** Rarely, acinar cells can show focal metaplastic changes, including mucinous, squamous, and ductal/centroacinar types.

Islets of Langerhans



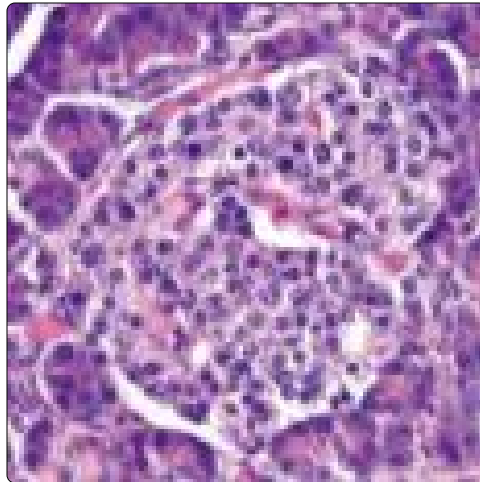
Islets (Compact)



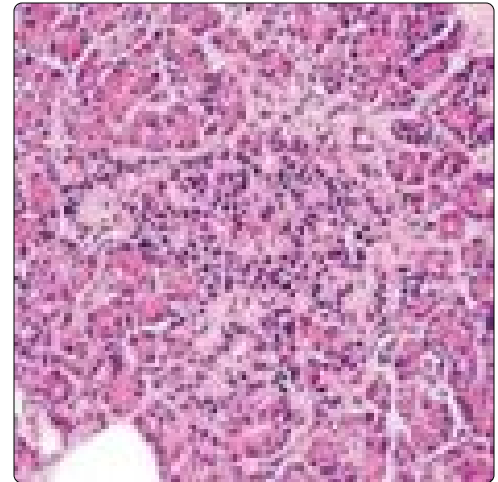
(Left) Islets of Langerhans [icon] are present throughout the pancreas but are randomly distributed with some lobules having more than others. The vast majority of the endocrine cells in the pancreas are in the islets. **(Right)** Compact islets account for 90% of the total and occur in the body and tail of the pancreas as well as the superior portion of the head. They have well-defined borders.

Islet Cytology

(Left) The nuclei of compact islets have coarsely clumped or speckled chromatin. **(Right)** Diffuse islets are found in the inferior portion of the head of the pancreas, and they are much less numerous than compact islets. They have ill-defined borders, are often larger than compact islets, and have more hyperchromatic nuclei.

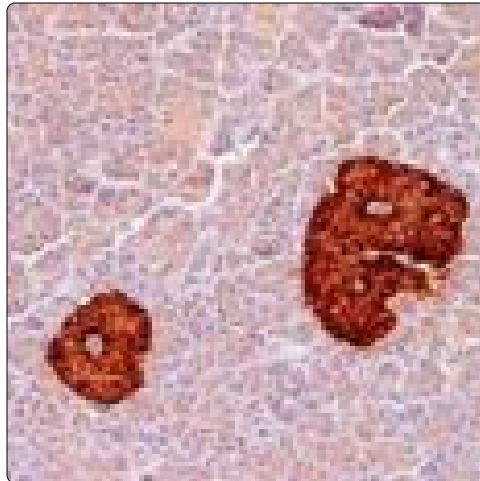


Islets (Diffuse)

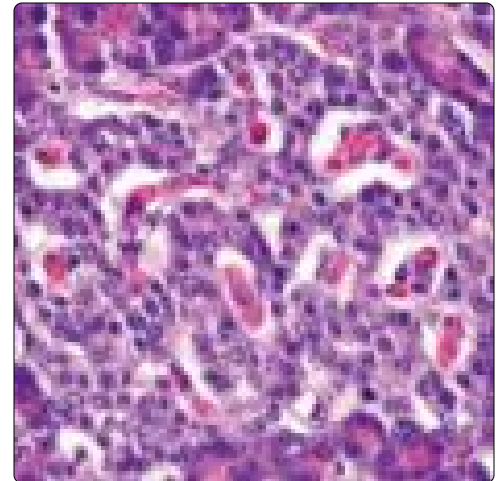


Islet Immunophenotype

(Left) Synaptophysin immunostain highlights the islets of Langerhans. All islets produce general endocrine markers such as CD56, synaptophysin, and chromogranin. Islets are typically keratin negative. **(Right)** Islets of Langerhans are not encapsulated and have a rich capillary network within them.

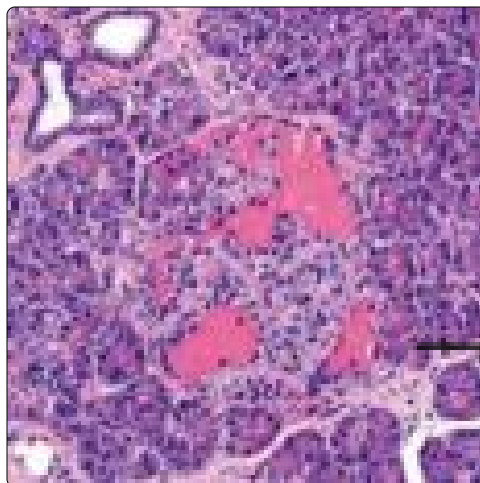


Capillary Network

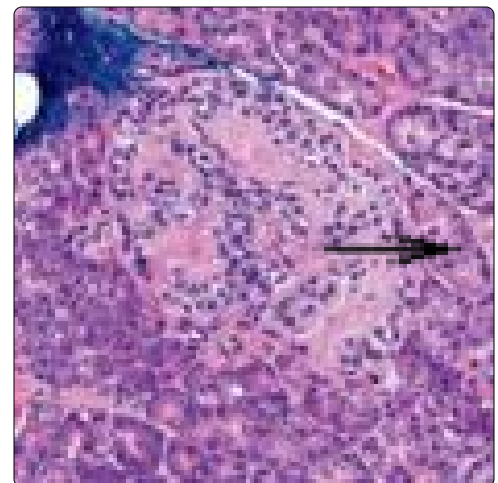


Peliosis Insulis

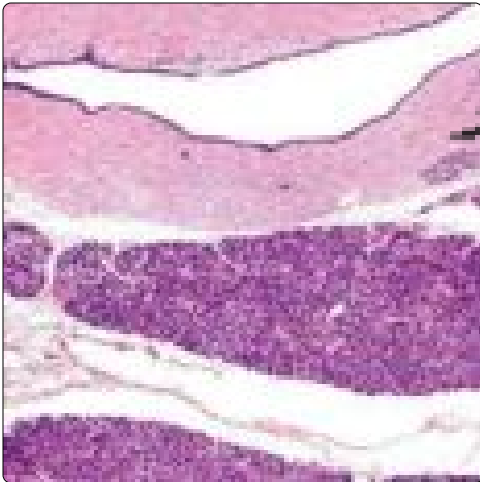
(Left) Dilated blood-filled spaces may be seen within the islets. This finding has been referred to as peliosis insulis and is believed to be of no clinical significance. **(Right)** Deposition of amyloid or amyloid-like material around the small blood vessels within pancreatic islets is an uncommon finding, but is more likely to be seen in older patients, particularly in association with diabetes mellitus. This finding is not related to systemic amyloidosis.



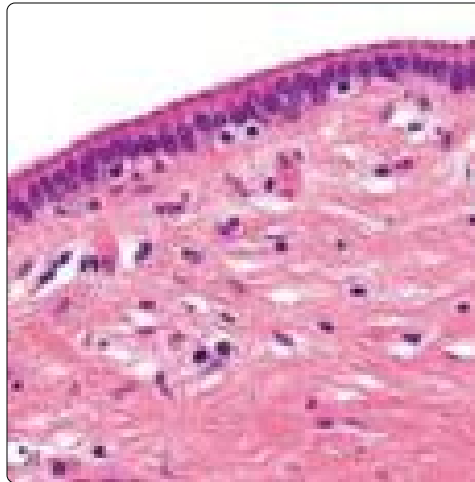
Perivascular Amyloid



Main Pancreatic Ducts

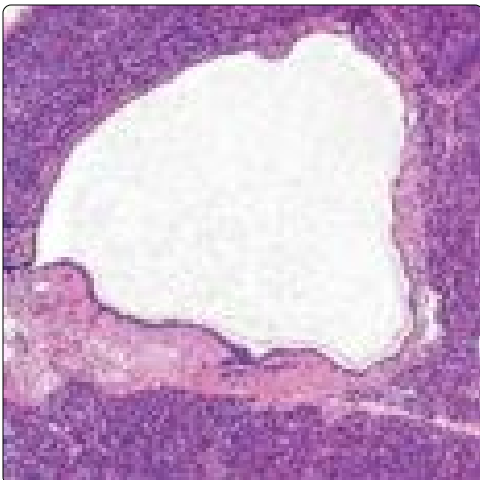


Pancreatic Ducts

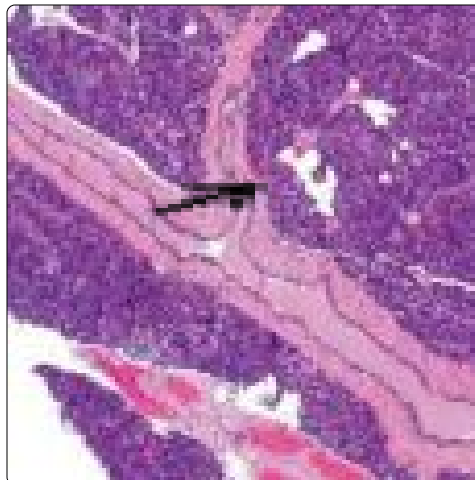


(Left) The main pancreatic ducts (also known as ducts of Wirsung and Santorini) and the interlobular ducts are surrounded by a thick collagen layer. The pancreatic ductal system transports pancreatic secretions to the duodenum. **(Right)** The main pancreatic ducts are lined by a simple columnar mucinous epithelium. The nuclei are round and basally located. Ductal cells are positive for keratins AE1/AE3, CAM5.2, CK7, and CK19 and are CK20 negative.

Duct Ectasia

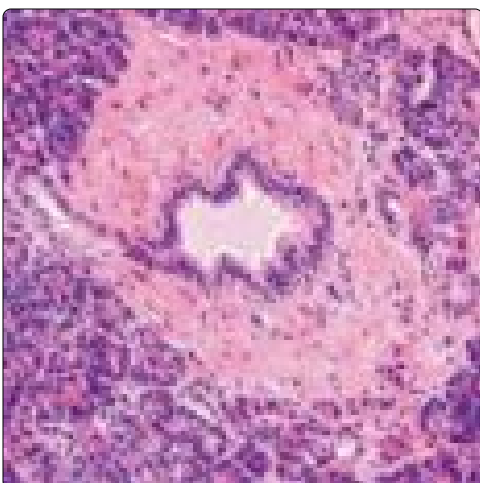


Interlobular Ducts



(Left) Pancreatic duct ectasia is characterized by prominent dilatation of the large ductal system. The epithelial lining is often compressed and appears cuboidal rather than columnar. Although usually seen within the setting of chronic pancreatitis, duct ectasia can also be an incidental finding in otherwise normal pancreatic tissue. **(Right)** Interlobular ducts connect the main pancreatic ducts to the small intralobular ducts. These structures are characteristically invested by a rim of collagenous tissue, which varies in thickness.

Interlobular Duct



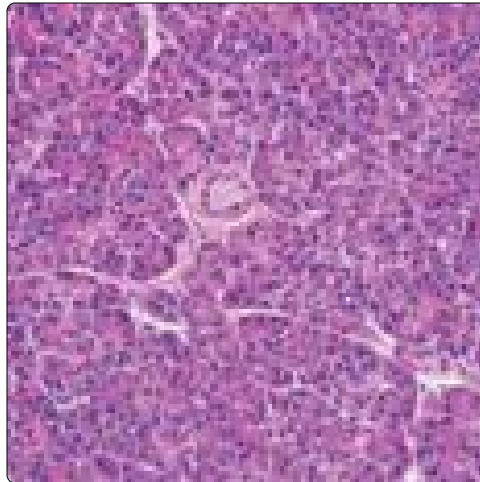
Intercalated Ducts



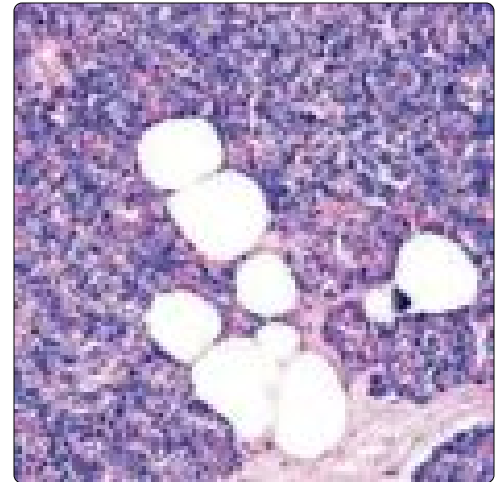
(Left) This cross section of a smaller interlobular duct shows the surrounding band of supporting collagen and also contains low columnar mucinous epithelium. **(Right)** The intercalated ducts are the smallest ducts outside the acini. They are lined by cuboidal epithelium that does not contain mucin and have minimal surrounding connective tissue stroma. The intercalated ducts fuse to form the intralobular ducts.

Intraductal Secretions

(Left) Secretions from the acinar cells are carried through the pancreatic ductal system to the duodenum. These secretions can often be identified as dense eosinophilic material within the ductal lumina. **(Right)** Most pancreata contain a small amount of intrapancreatic fat. The proportion of the gland composed of fat varies with age and nutritional status.

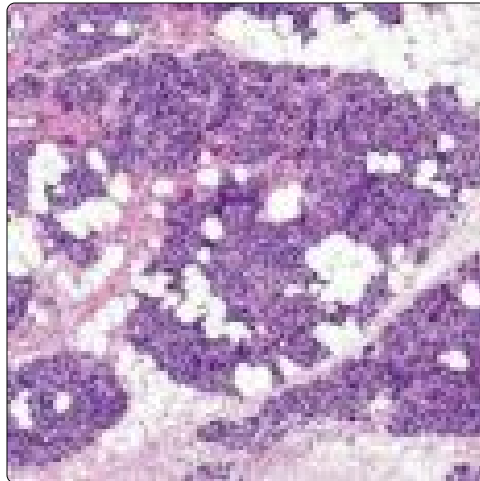


Intrapancreatic Fat

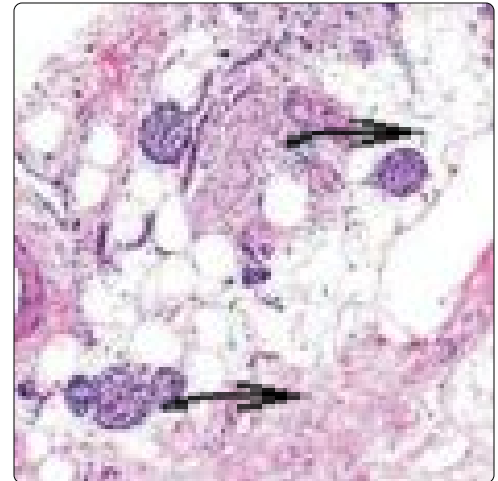


Intrapancreatic Fat

(Left) Mature adipose tissue may become more prominent in the pancreas of older individuals, particularly in the setting of parenchymal atrophy secondary to chronic duct obstruction. **(Right)** In cases of severe parenchymal atrophy, the exocrine component of the pancreas vanishes, and residual islets appear to lie within fibroadipose tissue.

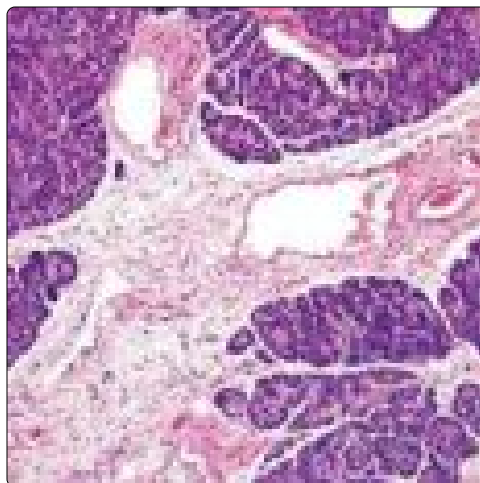


Exocrine Atrophy

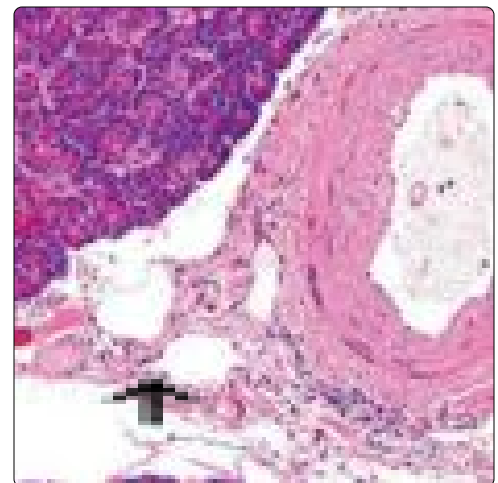


Connective Tissue

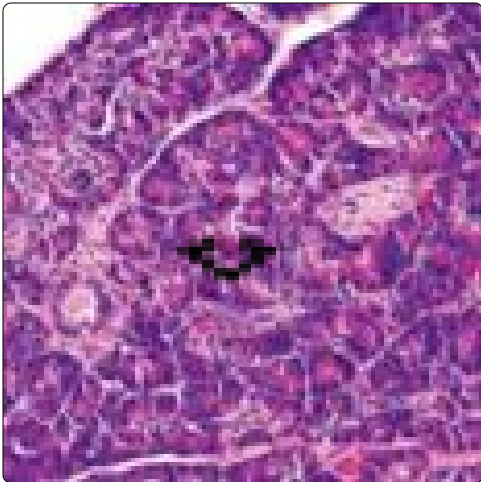
(Left) The connective tissue septa between the pancreatic lobules contain vessels and nerves. **(Right)** A large artery, lymphatics, veins, and a nerve are seen within this connective tissue interlobular septum. Of note, these muscular arteries are always separated from the ducts within the pancreatic lobules and never course alongside them. This information can be helpful when trying to distinguish benign from malignant glands within the setting of chronic pancreatitis.



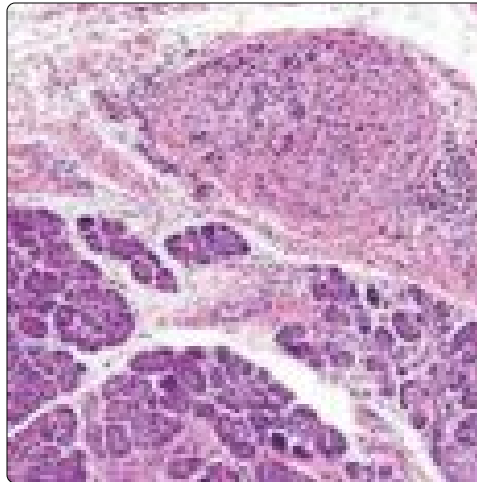
Connective Tissue



Nerves

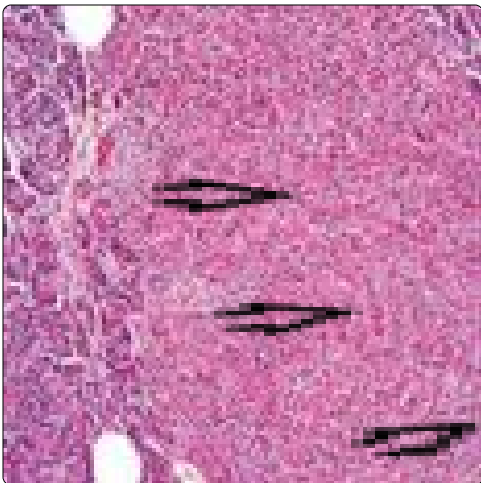


Ganglion

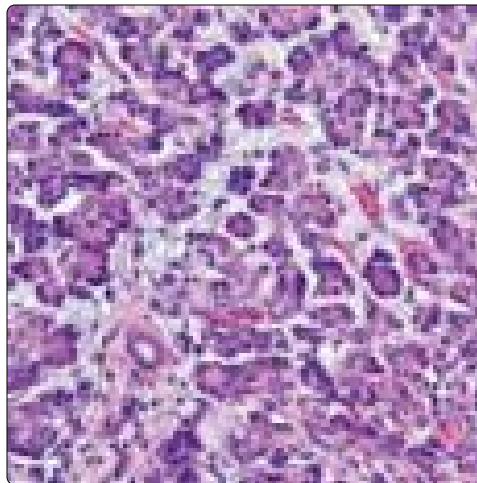


(Left) Most nerves and vessels are in the septa, but some small vessels and nerves may be present within the lobular parenchyma. Note the small duct profile at the left. (Right) Autonomic ganglia may be identified in some sections of pancreatic tissue. In some cases, small ganglia may be present between acini.

Focal Acinar Transformation

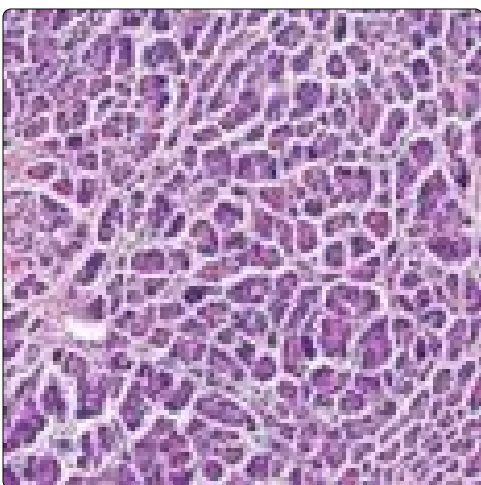


Focal Myxoid Change

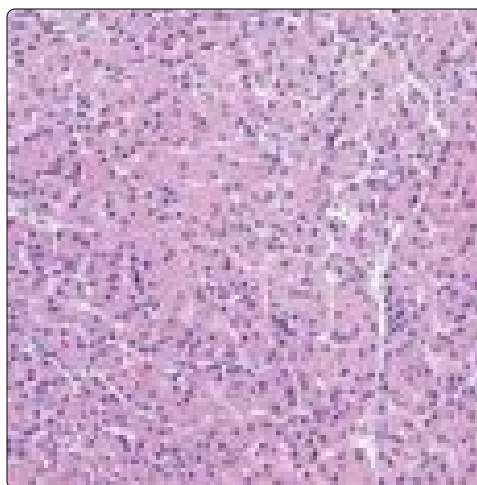


(Left) Focal acinar transformation consists of a pale eosinophilic nodular area that often involves an entire lobule. This change is secondary to dilatation of rough endoplasmic reticulum. The cause is unknown, but it has no pathologic significance. (Right) Myxoid change within the lobules of acinar cells is most often seen in the setting of chronic pancreatitis; however, it can be seen focally in otherwise normal pancreatic tissue.

Retraction Artifact



Cautery Artifact



(Left) Although a tubular pattern is a normal finding in some pancreatic acini, extensive retraction artifact due to poor fixation can accentuate this pattern and may simulate a neuroendocrine tumor or micropapillary carcinoma. (Right) Cauterization used during surgical resection can distort pancreatic acinar tissue, giving the cells a dense, pale pink appearance.

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SECTION 12

Genitourinary and Male Genital Tract



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Testis and Associated Excretory Ducts	336

KIDNEY**Macroscopic Anatomy**

- ~ 1 million nephrons per kidney (range: 200,000 to > 2.5 million)
- Composed of 11-14 lobes
 - Each lobe consists of medullary pyramid and surrounding cortical parenchyma
- Cortex
 - Cortical labyrinth
 - 10-14 generations of glomeruli
 - Medullary rays contain straight portion of proximal/distal tubules and collecting ducts
- Medulla
 - Divided into pyramids
 - Outer medulla: Outer and inner stripe
 - Inner medulla: Papilla
- Renal pelvis
 - Expansion of proximal ureter
 - Features major and minor calyces
- Renal sinus
 - Region containing renal pelvis and neurovasculature
- Renal hilum vasculature
 - Renal artery branches → interlobar → arcuate → interlobular arteries → afferent arterioles → efferent arterioles → peritubular capillaries or vasa recta
 - Renal vein

Microscopic Anatomy

- Glomeruli
 - Glomerular filtration barrier consists of visceral epithelial cells (podocytes), glomerular basement membrane, and fenestrated endothelial cells
 - Parietal epithelial cells and podocytes line Bowman capsules
 - Mesangial cells comprise stalk into which glomerular capillaries are anchored
- Juxtaglomerular apparatus
 - Consists of portions of afferent and efferent arterioles, Goormaghtigh cells, macula densa

- Granular cells most abundant in afferent arteriolar wall
- Proximal tubules
 - Cells have eosinophilic cytoplasm (high mitochondria content) and brush border
- Loop of Henle
- Distal tubules
- Collecting ducts
 - Principal cells reabsorb sodium and secrete potassium
 - Intercalated cells have ↑ mitochondria content and cortical density > medulla
- Arteries: Adventitia, media, and intima lined by nonfenestrated endothelial cells
- Arterioles (afferent and efferent) and vasa recta
- Peritubular capillary network
- Venules, lymphatics, dendritic cell network, nerves

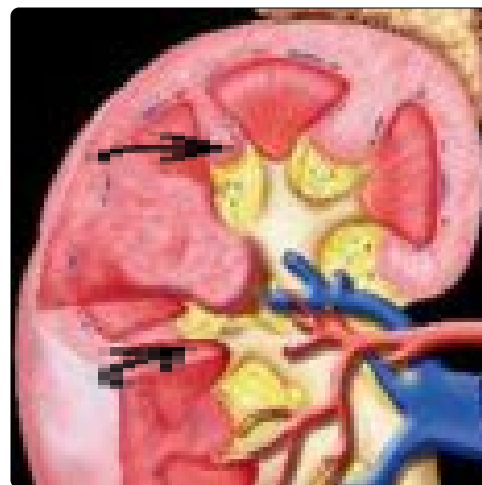
CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

- Infants and young children
 - Kidney weight: 13-44 g
 - Renal lobulations more prominent
 - Smaller glomeruli and podocytes can be quite prominent
 - Low-birth-weight infants have fewer glomeruli/nephrons
- Adults
 - Kidney weight: 115-175 g: generally larger in men
 - Fetal lobulations often disappear
- Global glomerulosclerosis (nonpathologic)
 - < 40 years of age
 - Usually < 10% of glomeruli (mean: 2.5%) are sclerotic
 - > 40 years of age
 - 1/2 of person's age minus 10 = percentage of global glomerulosclerosis considered normal

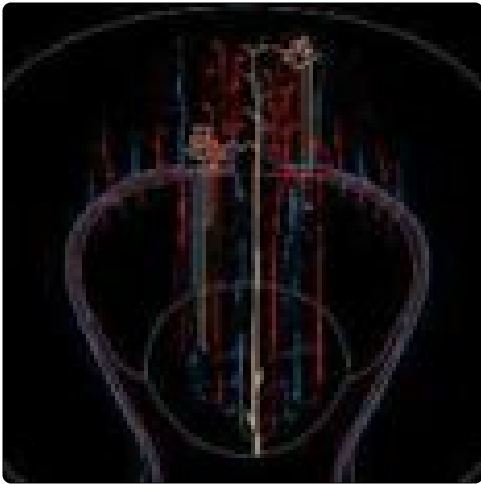
Pitfalls and Artifacts

- Interstitium more prominent in renal medulla
 - Difficult to assess interstitial fibrosis in this region
- Prominent podocytes of pediatric glomeruli
- Increased extracellular matrix in perivascular regions can be mistaken for interstitial fibrosis

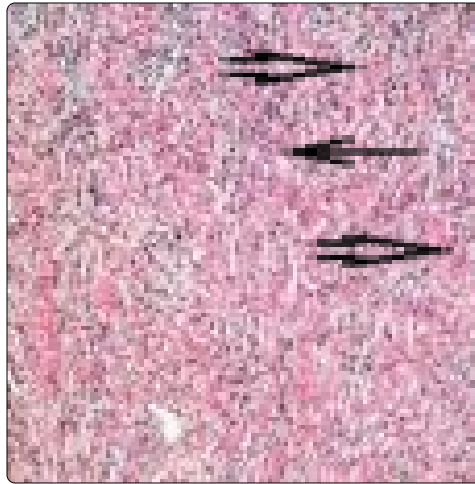
(Left) Every day, 1,800 L of blood (roughly 25% of the cardiac output) flow into the kidneys through the renal artery and exit the renal vein. Up to 1 L of urine is produced, which exits through the ureter into the bladder. **(Right)** The cortex runs along the outer rim and extends as columns between the medullary pyramids. Urine drains into the minor calyces through the papillae. A capsule is not present to contain tumors from gaining access to lymphatics in the sinus fat region.

Kidney**Cortex and Medulla**

Schematic of 2 Nephrons With Blood Supply



Normal Cortex



(Left) Juxtamedullary glomeruli/nephrons have a longer loop of Henle compared with cortical glomeruli/nephrons. The efferent arterioles of cortical nephrons supply the peritubular capillary network, while the efferent arterioles of juxtamedullary nephrons may supply the vasa recta. (Right) Glomeruli are present only in the cortex. Medullary rays are part of the cortex and run up toward the capsule and down toward the medulla in this photomicrograph. The tubules are closely packed in the cortex.

Schematic of Nephron

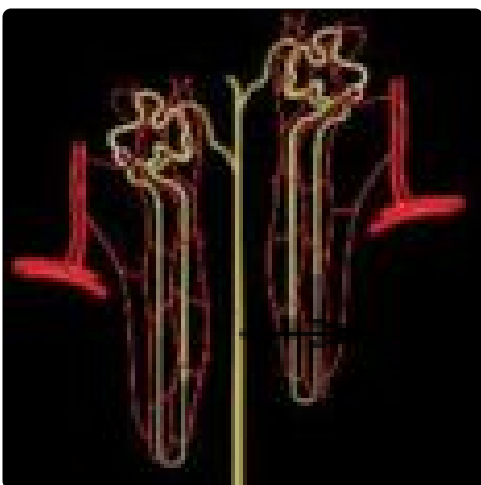


Renal Medulla

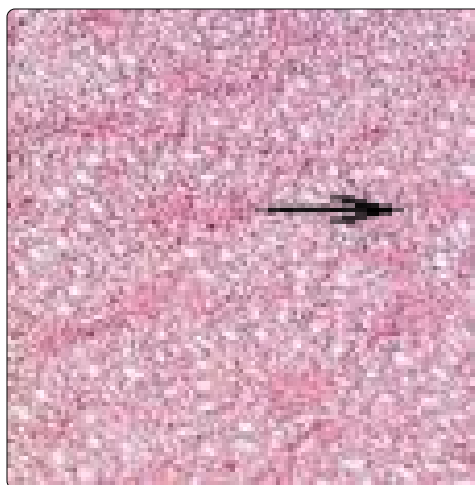


(Left) The renal corpuscle consists of a Bowman capsule and its included glomerulus. The proximal tubule joins the renal corpuscle at the urinary pole. The diagram also shows the descending portion of the loop of Henle, the distal tubule, and the macula densa (light brown), which is in continuity with the juxtaglomerular apparatus. (Right) The renal medulla consists of distal tubules, loops of Henle, and collecting ducts. The tubules are not as closely packed as those in the cortex. No glomeruli are present.

Schematic of 2 Nephrons



Medulla



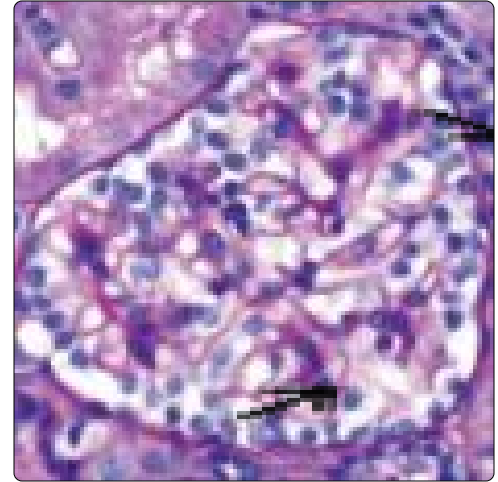
(Left) The kidney consists of ~1 million nephrons, and its complex vasculature is demonstrated in this schematic of 2 nephrons. The elaborate peritubular capillary network is supplied by blood from the efferent arterioles that exit the cortical glomeruli. Therefore, primary glomerular diseases will lead to secondary interstitial fibrosis and tubular atrophy when significant glomerular injury or scarring occurs. (Right) Efferent arterioles of juxtamedullary glomeruli become the vasa recta and form bundles.

Schematic of Normal Glomerulus

(Left) The glomerulus consists of mesangial cells [A], fenestrated endothelial cells [B], and podocytes [C] overlying the glomerular basement membrane. The macula densa [D] is adjacent to the juxtaglomerular apparatus. It is a group of densely staining cells in the distal tubule that may function as receptors that feed information to the juxtaglomerular cells. (Right) Pediatric glomeruli are smaller than adult glomeruli and have prominent podocytes [E] (visceral epithelial cells).

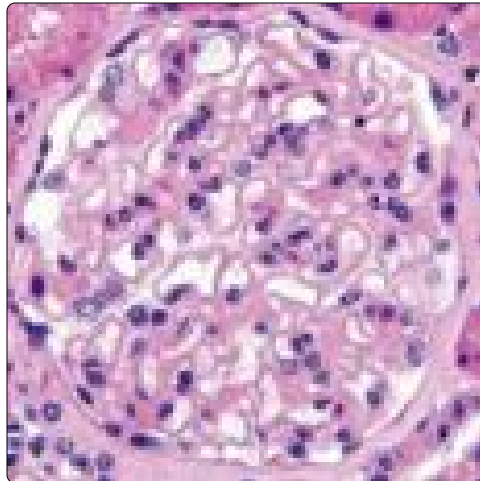


Normal Pediatric Glomerulus

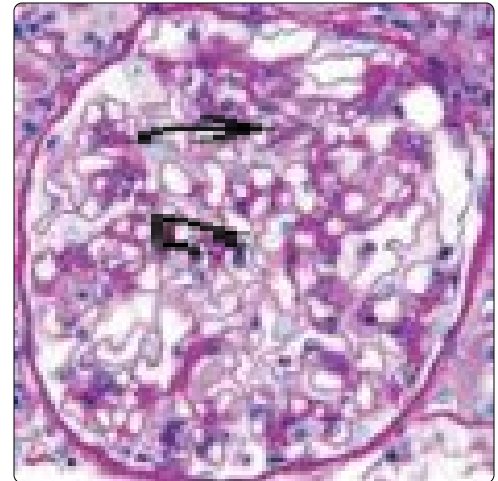


Normal Glomerulus: H&E

(Left) Hematoxylin and eosin is not the ideal histochemical stain for assessing structural abnormalities of the glomerulus. (Right) Periodic acid-Schiff nicely outlines the glomerular basement membrane [A], thereby facilitating assessment of the glomerulus. Mesangial areas [B] normally have < 3 nuclei per region in a 2- μ m thick section. An incidental circulating neutrophil [C] is present in a glomerular capillary.

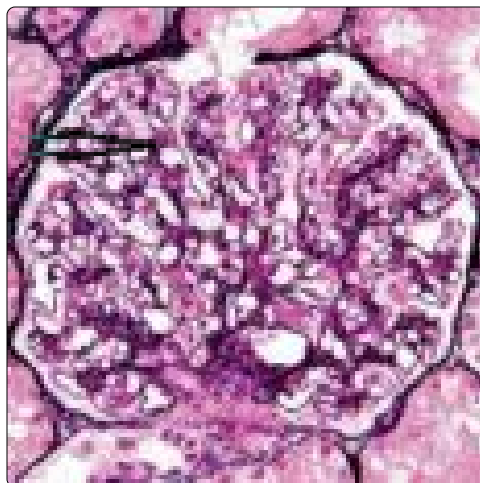


Normal Glomerulus: PAS

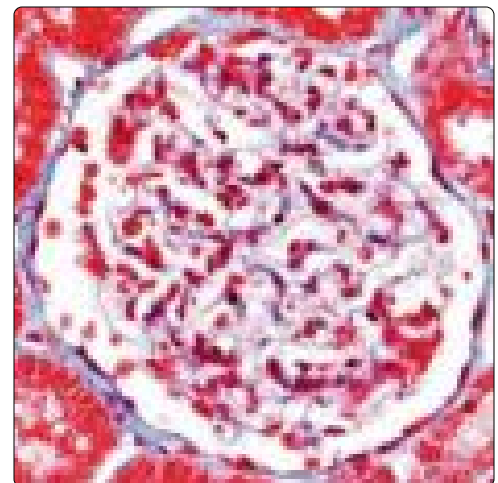


Normal Glomerulus: Jones Methenamine Silver

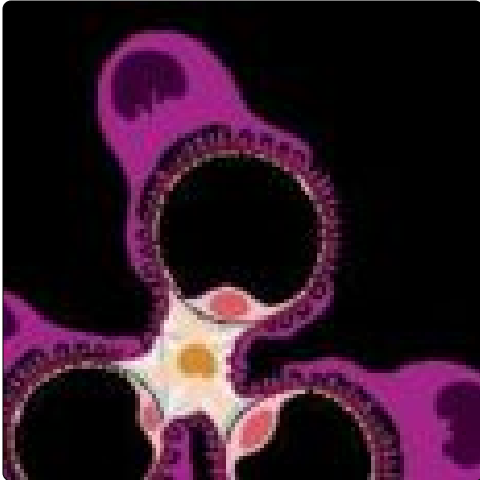
(Left) Jones methenamine silver stain allows optimal visualization of the normal delicate glomerular basement membranes [A] and normal mesangial cellularity. The juxtaglomerular apparatus [B] is next to the hilar arteriole at the vascular pole. The tip of the glomerulus empties into the urinary pole [C] at the most proximal portion of the proximal tubule. (Right) Masson trichrome stains the Bowman capsule and the glomerular and tubular basement membranes blue.



Normal Glomerulus: Masson Trichrome



Schematic of Glomerular Capillaries

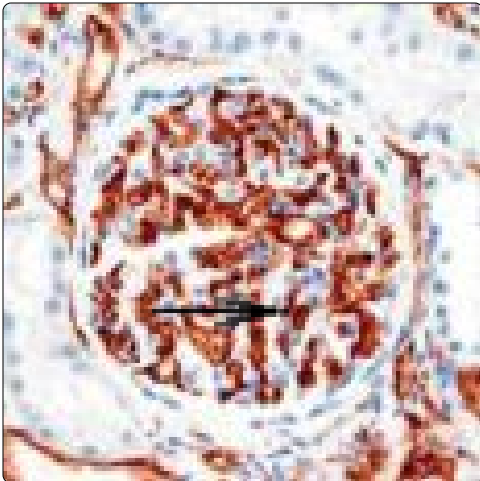


Normal Glomerular Capillaries: EM

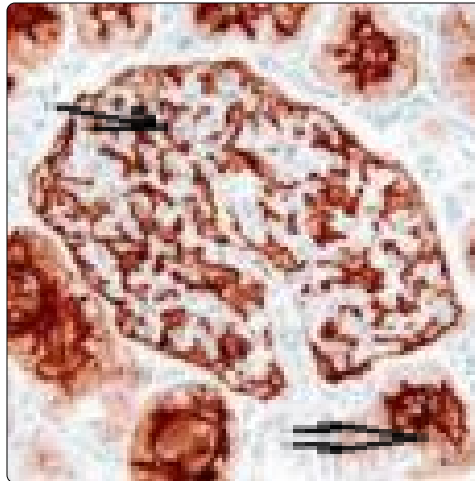


(Left) The podocytes and their foot processes  cover the glomerular basement membranes. The fenestrated endothelial cells  also contribute to the glomerular filtration barrier. The glomerular basement membranes are anchored to the mesangium . (Right) The glomerular capillary basement membrane anchors in the mesangium . It is lined by fenestrated endothelial cells  and covered by podocytes and their foot processes . Normally, 2 mesangial cells  are present per mesangial region.

Glomerular Capillaries: CD31

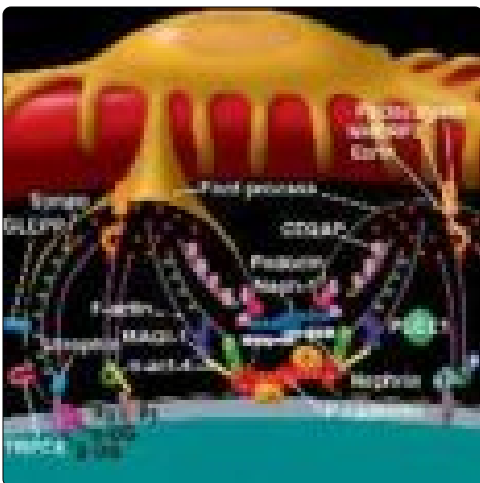


Glomerular Podocytes: CD10

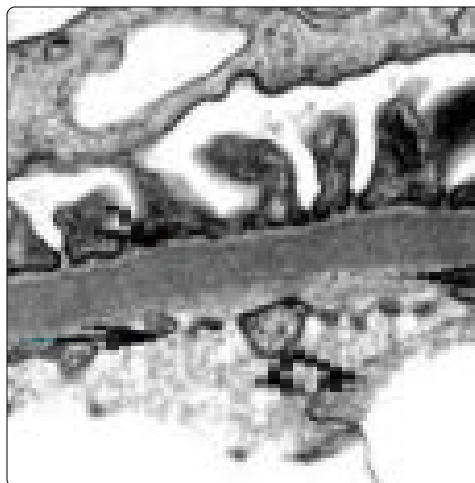


(Left) The endothelial cells within glomerular capillaries  and peritubular capillaries  are highlighted by CD31. Podocytes  are located on the outside of the glomerular capillaries and are CD31 negative. (Right) CD10 (neutral endopeptidase) is expressed on podocytes (visceral glomerular epithelial cells)  and the brush border  of proximal tubular epithelial cells. The podocytes intricately cover the glomerular capillary surface.

Podocyte Foot Process and Slit Diaphragm Complex



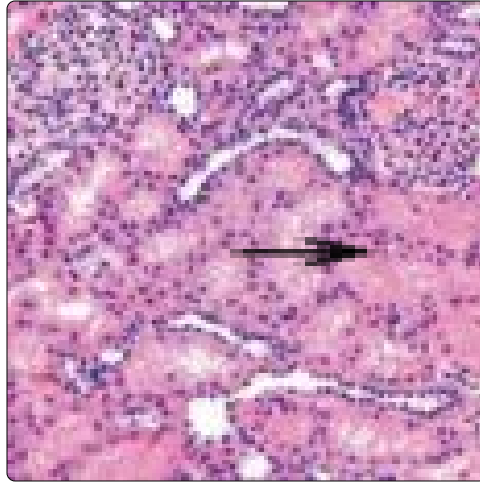
Glomerular Filtration Barrier: EM



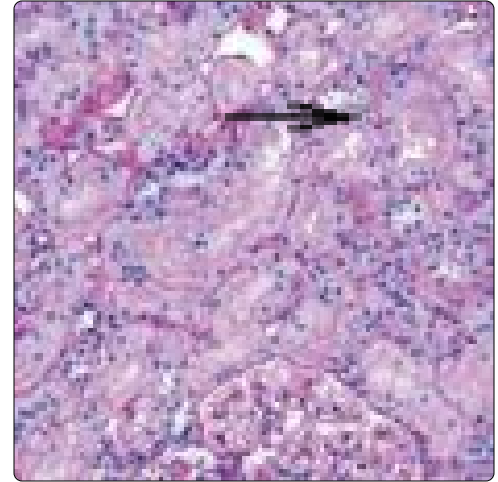
(Left) The slit diaphragm complex  is a highly specialized cell junction connecting the foot processes of adjacent podocytes. It is composed of proteins (e.g., nephrin, podocin, CD2AP) expressed by the podocytes. (Right) The glomerular basement membrane consists of the lamina densa , lamina rara externa , and lamina rara interna . Slit diaphragms  are noted. Endothelial fenestrae  are 70-100 nm in diameter.

Renal Cortex: Proximal Tubules

(Left) The convoluted proximal tubules [1] comprise the majority of tubules seen in the renal cortex. They are characterized by abundant eosinophilic cytoplasm, in contrast with the distal nephron segments [2] that have much smaller cells and thus more nuclei per tubule length. (Right) Periodic acid-Schiff stain shows the abundant cytoplasm of the proximal tubular epithelial cells [1]. No significant space is present between tubules except for the network of peritubular capillaries [2].

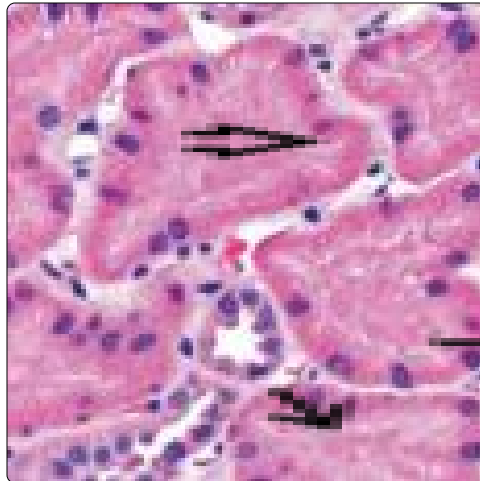


Proximal Tubules: PAS

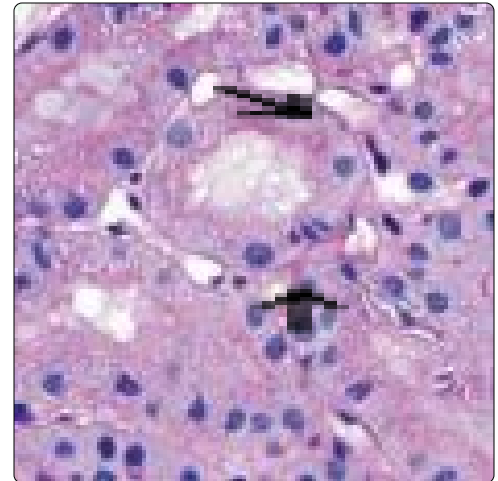


Proximal Tubules: H&E

(Left) Proximal tubular cells [1] have abundant eosinophilic cytoplasm (due to high mitochondrial content) compared with distal tubules [2]. 60% (~ 110 L) of the total glomerular filtrate (180 L) is reabsorbed by the proximal tubules daily. The ratio of proximal tubular profiles to distal nephron segments is ~ 3:1 in the cortex. (Right) The brush border [1] at the apical aspect of the proximal tubules is best visualized with periodic acid-Schiff and may be attenuated in acute tubular injury/necrosis.

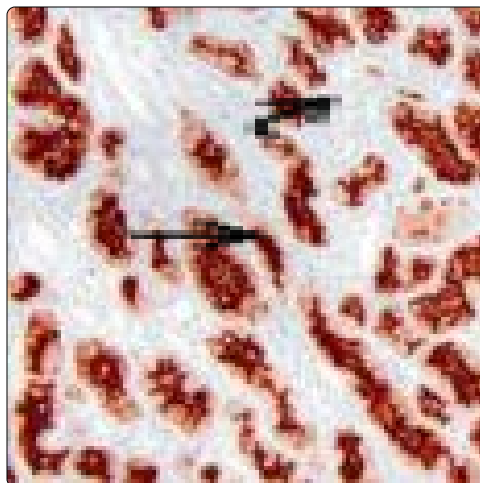


Brush Border: PAS

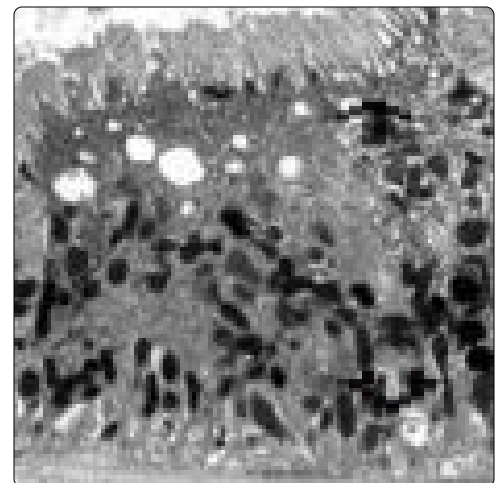


Proximal Tubules: CD10

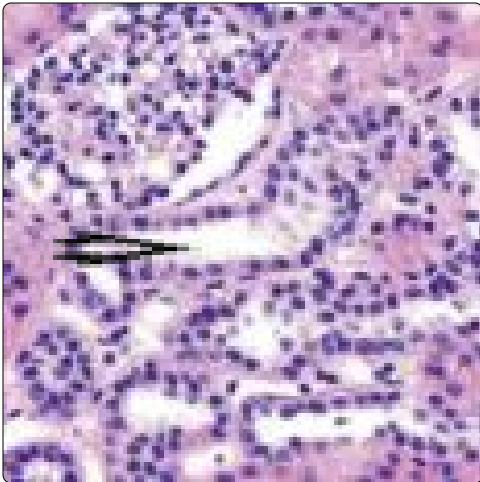
(Left) CD10 strongly stains the brush borders [1] on the apical aspect of the proximal tubular epithelial cells. The distal nephron segments [2] are negative. (Right) This electron micrograph shows the brush border [1] along the apical surface that is characteristic of proximal tubular epithelial cells. Many mitochondria [2] are associated with the basolateral interdigitations [3] along the tubular basement membrane.



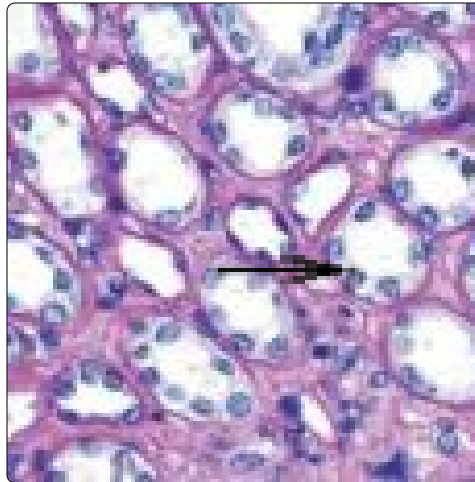
Proximal Tubular Epithelial Cell: EM



Distal Convolved Tubules

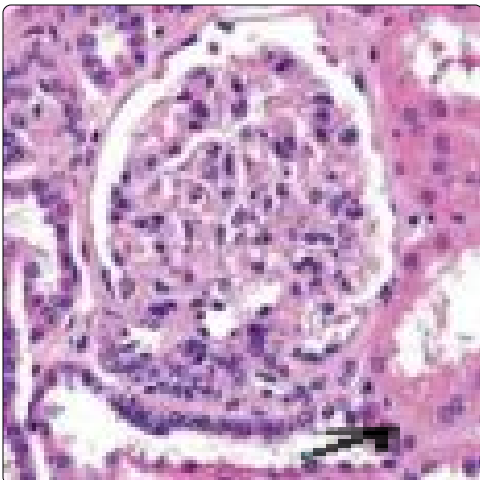


Loops of Henle

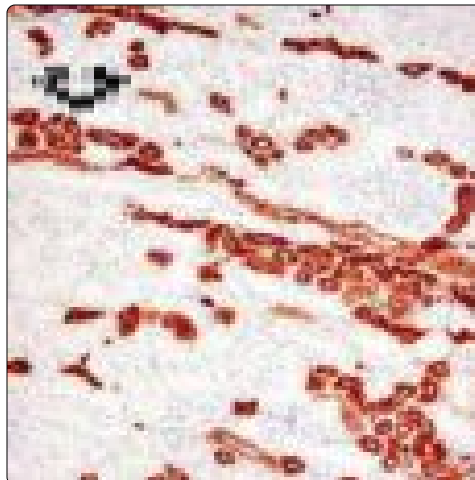


(Left) The distal tubules  are easily distinguished from the larger proximal tubular epithelial cells  as they have many more cells and nuclei per tubule length and each cell has much less cytoplasm (and thus a less eosinophilic appearance). (Right) Cross sections of the loops of Henle  show their nuclei protruding into the tubular lumina with attenuated cell cytoplasm. Adjacent peritubular capillaries  have a smaller caliber.

Macula Densa

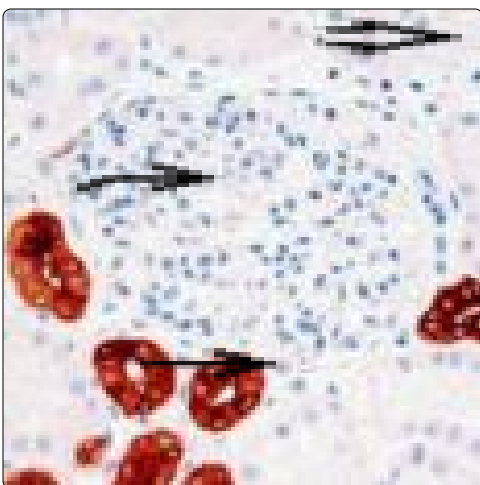


Distal Tubules and Collecting Ducts: EMA

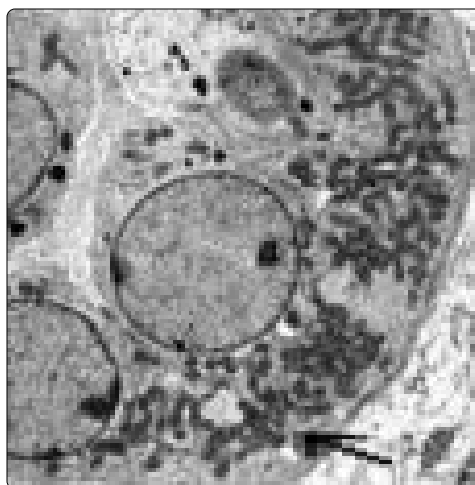


(Left) The cells of the macula densa  are columnar; they represent a specialized portion of the distal tubule that is adjacent to the juxtaglomerular apparatus and should not be mistaken for a cellular crescent. (Right) EMA is expressed in the distal tubules  but not in glomeruli or proximal tubules. The principal cells of the collecting ducts  demonstrate less intense cytoplasmic staining, whereas the scattered intercalated cells  have strong staining.

Distal Tubules: EMA



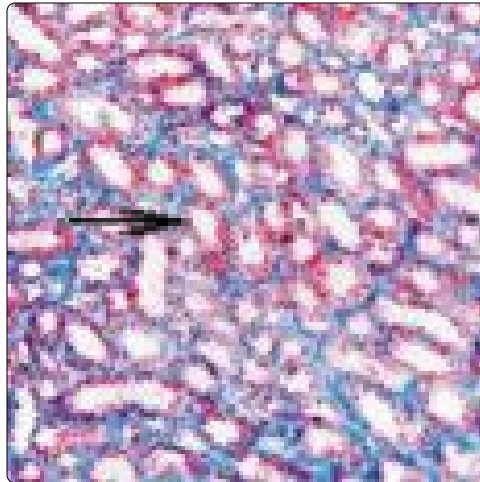
Distal Tubular Epithelial Cell: EM



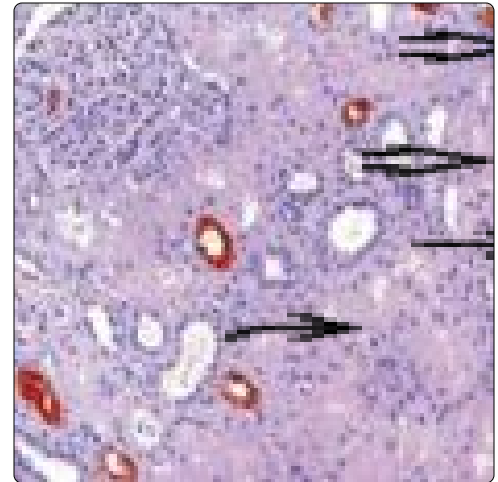
(Left) Epithelial membrane antigen highlights distal tubules , but there is no expression within the adjacent proximal tubules  or the glomerulus . (Right) The nuclei  of the distal tubular epithelial cells are close to the apical surface or tubular lumen. The cells are much smaller in size compared with proximal tubular epithelial cells, so many more nuclei can be seen in a cross section. No brush border is present at the apical surface, but short, blunted villi may be noted.

Renal Medulla: Masson Trichrome

(Left) Masson trichrome stains tubular basement membranes and interstitial fibrosis blue [1]. The tubules have more space between them in the medulla, where assessing the extent of interstitial fibrosis and tubular atrophy is much more difficult than in the cortex. (Right) Collecting ducts [2] are highlighted by immunohistochemistry for the lectin *Dolichos biflorus*. Adjacent proximal tubules [3] and distal tubules [4] show no staining.

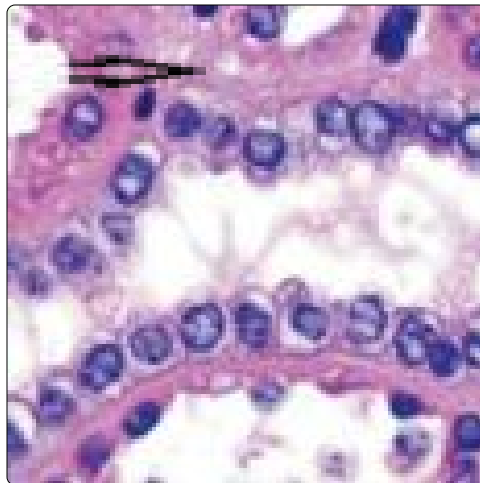


Dolichos biflorus

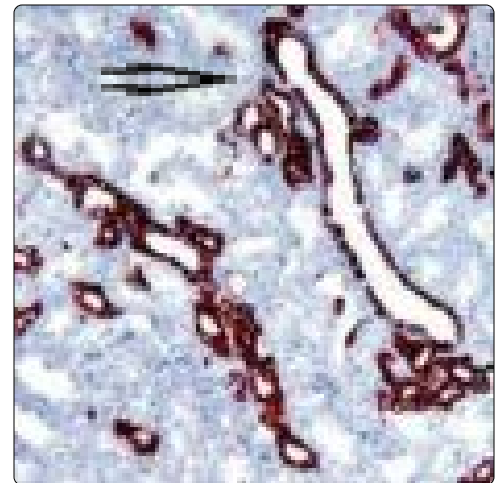


Collecting Duct: H&E

(Left) The lateral cell borders [1] of the collecting ducts are often distinct, in contrast to the adjacent proximal tubular epithelial cells [2] that lack this feature. (Right) Cytokeratin 7 highlights the distal tubules [3] and collecting ducts [4], but there is no expression in the proximal tubules [5]. The principal cells within the collecting ducts express cytokeratin 7, while the intercalated cells [6] show no expression.

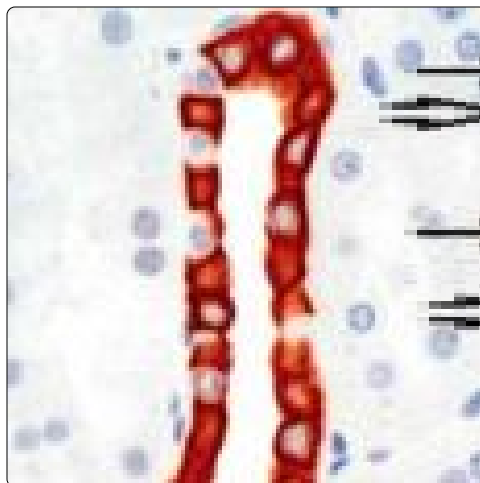


Distal Tubules and Collecting Ducts: Cytokeratin 7

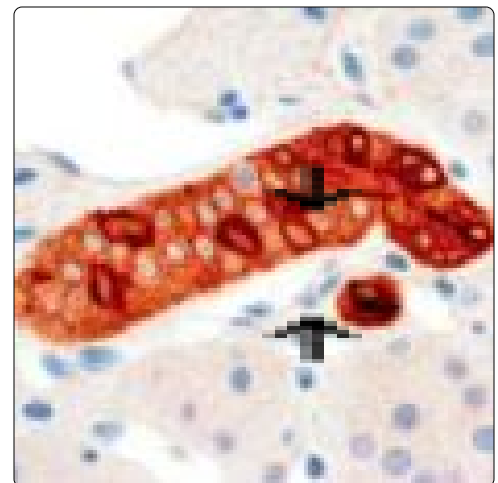


Collecting Duct: AE1/AE3

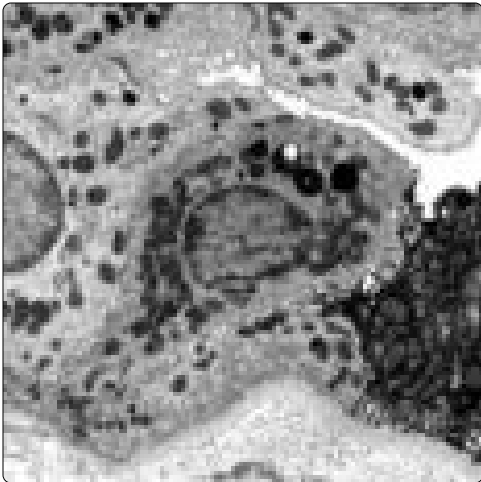
(Left) AE1/AE3 immunohistochemistry strongly stains the cytoplasm of principal cells [1] of the collecting duct, while intercalated cells [2] are negative. The adjacent proximal tubules [3] demonstrate no staining. (Right) The intercalated cells [4] show strong staining for EMA, while there is a cytoplasmic blush in the principal cells [5] of the collecting duct. Intercalated cells are specialized cells that assist in acid-base homeostasis.



Collecting Duct: EMA



Principal Cell: EM

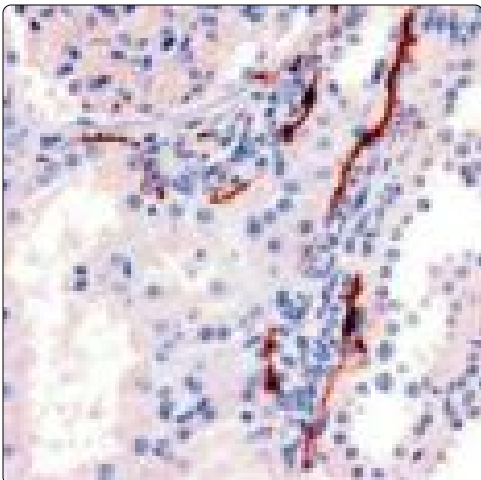


Intercalated Cell: EM

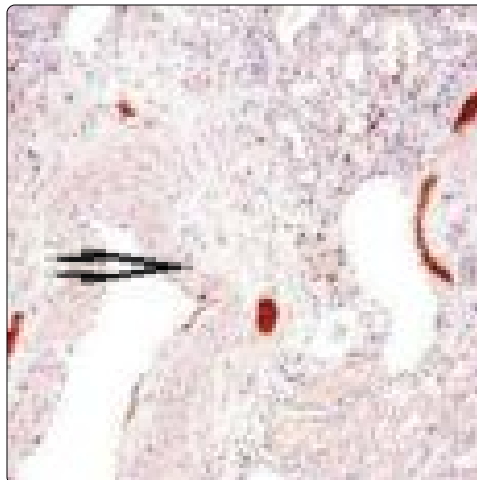


(Left) The principal cell contains fewer mitochondria and other organelles and has a distinctly lighter cytoplasmic color compared with the adjacent intercalated cell. (Right) Abundant mitochondria are present in the cytoplasm of this intercalated cell and give the darker cytoplasmic hue seen by light microscopy. Two principal cells are adjacent in this collecting duct. Collagen fibrils are present between the tubular basement membrane and the peritubular capillary basement membrane.

Nerves: S100

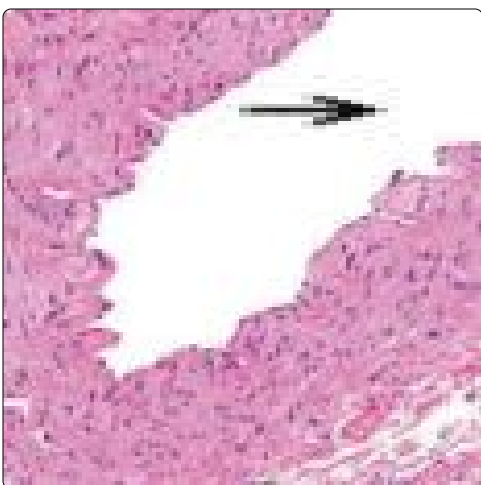


Nerves: S100

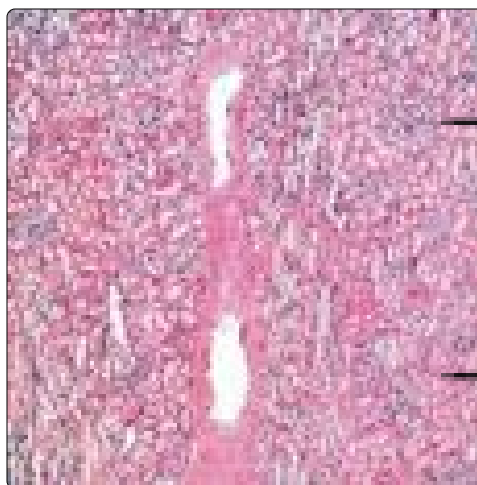


(Left) Nerves are not generally abundant in the renal cortex, but S100 highlights small nerves that course along the arterioles. Renal denervation is currently gaining attention as a potential safe and effective treatment for refractory hypertension. (Right) S100 immunohistochemistry reveals the larger nerves that accompany large-caliber arteries and venules in the medulla and hilum of the kidney.

Renal Artery



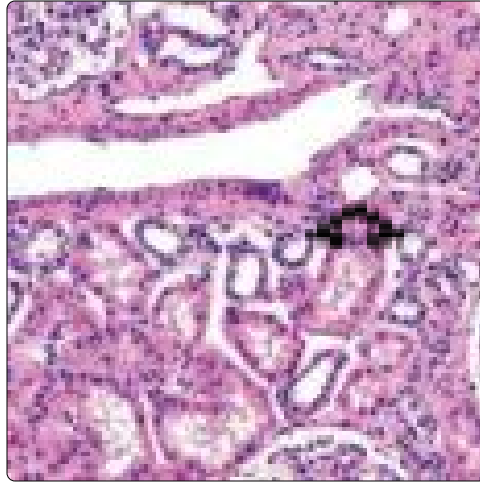
Interlobular Artery



(Left) The internal elastic lamina has a refractile quality that is best appreciated when viewed under the microscope while focusing the slide up and down. This structure is not present in the veins. (Right) This interlobular artery extends from the arcuate artery (not shown) into the cortex adjacent to a medullary ray as afferent arterioles branch off into glomeruli.

Artery and Arteriole

(Left) This interlobular artery branches into an afferent arteriole before entering a glomerulus (not shown). The arteriole wall thickness normally consists of ≤ 2 layers of smooth muscle cells. (Right) Jones methenamine silver shows an afferent arteriole, which often has a thicker vessel wall and larger caliber than the efferent arteriole. A fortuitous section to visualize both arterioles simultaneously may be necessary to distinguish one from the other.

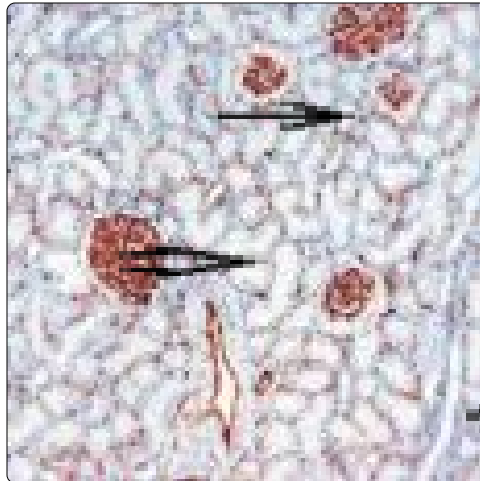


Afferent and Efferent Arterioles: Silver Stain

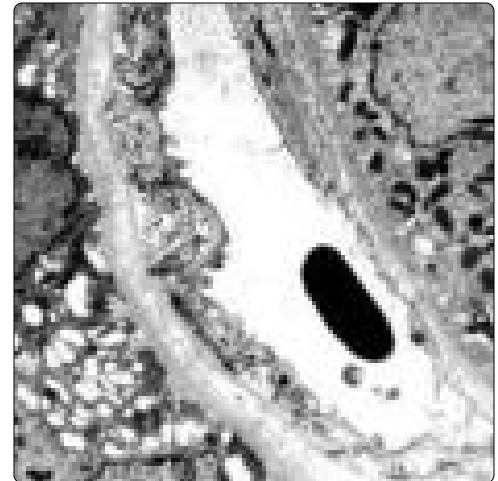


Peritubular Capillary Network: CD31

(Left) CD31 highlights the intricate peritubular capillary network as well as the glomerular capillaries and 1 artery. Pathologic glomerular injury can decrease blood flow to the peritubular capillaries, which results in tubular atrophy and interstitial fibrosis. (Right) This peritubular capillary contains a red blood cell. The endothelial cell sits atop a single basement membrane layer, which may be multilayered in chronic antibody-mediated rejection in renal allografts.

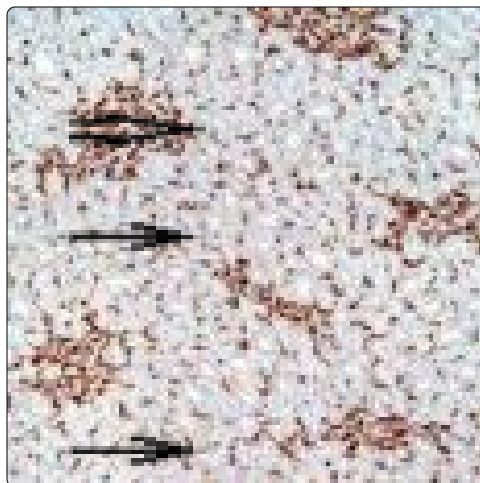


Peritubular Capillary: EM

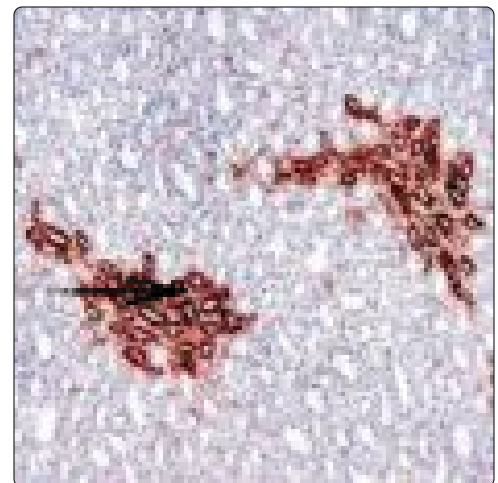


Renal Medulla: CD31

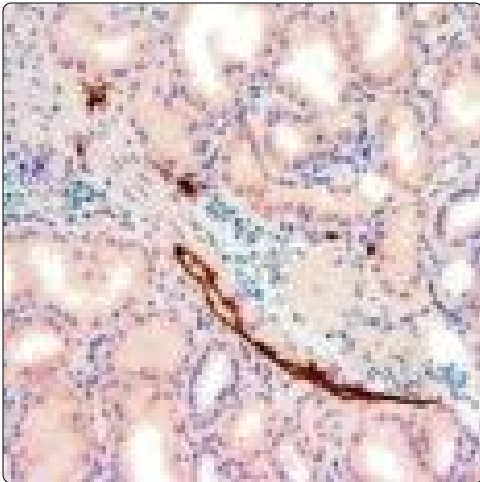
(Left) CD31 highlights the vascular bundles and individual peritubular capillaries that course through the renal medulla. (Right) α -smooth muscle actin highlights the vasa recta, which travel in bundles through the renal medulla. Note that the peritubular capillary network that is present between the tubules is not highlighted by this immunostain.



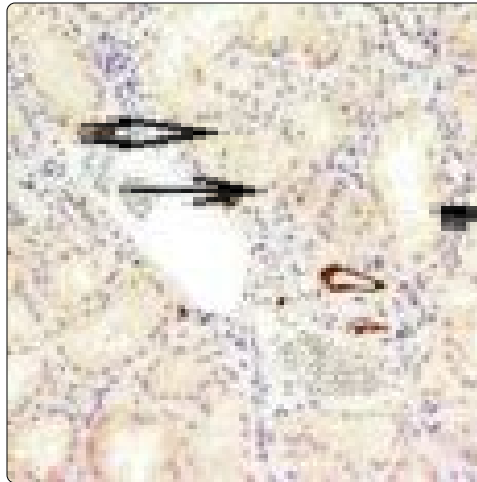
Vascular Bundle: SMAa



Lymphatic Vessels: D2-40

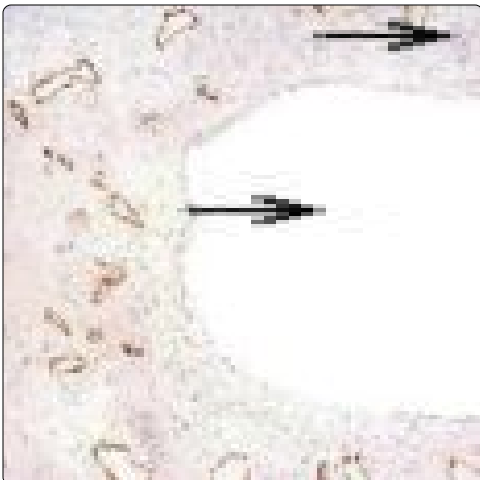


Lymphatics in Renal Cortex: D2-40

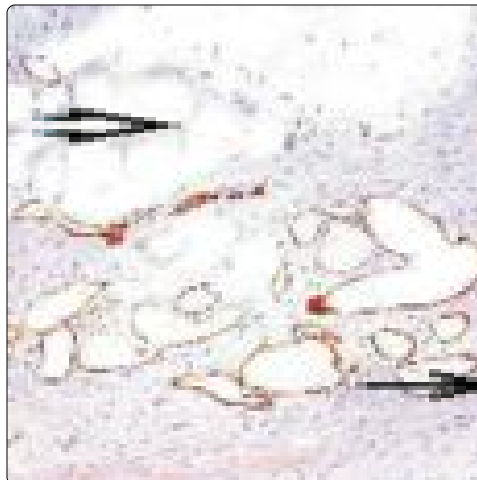


(Left) Lymphatic vessels are sparse in the renal cortex as demonstrated by D2-40, but they are relatively more abundant when in close proximity to renal veins and venules. Lymphatics are not present in the medulla. (Right) Lymphatic vessels are sparse in the normal renal cortex as demonstrated by D2-40 but are usually present in association with renal venules (as noted by the numerous red blood cells in their lumina). Lymphatics are not present in the medulla.

Lymphatics: D2-40

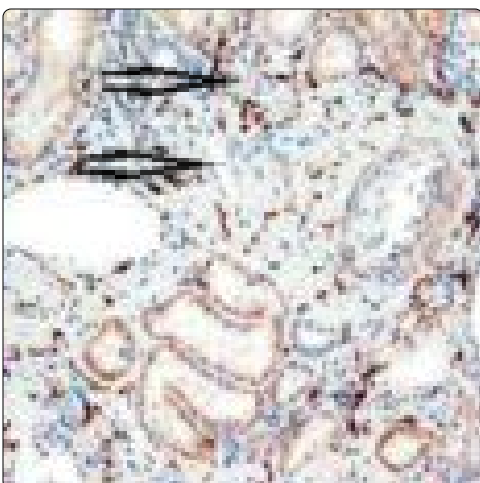


Lymphatics in Renal Sinus: D2-40

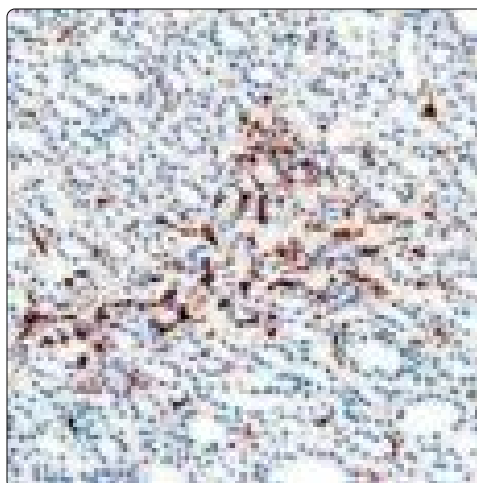


(Left) D2-40 outlines the numerous lymphatic channels that are present in the wall of this renal vein. (Right) D2-40 highlights the many lymphatic vessels that are present in close proximity to the renal sinus fat. In contrast to the outer aspect of the kidney, a capsule does not exist between the medulla and renal sinus. Therefore, large malignant renal neoplasms can extend into this region and easily gain access to these lymphatic vessels.

Resident Dendritic Cells: CD163



Resident Dendritic Cells: CD163



(Left) CD163 reveals numerous dendritic cells in the normal renal cortex, which are closely approximated with venules, peritubular capillaries, and lymphatics. They may have a role in immune surveillance. (Right) CD163 highlights the resident dendritic cells that are present in greater density along the vascular bundles that course through the medulla.

Ureter and Renal Pelvis

URETER

Macroscopic Anatomy

- Conduit for urine from kidney (renal pelvis) to urinary bladder through retroperitoneum and pelvis
- Becomes intravesicular at base of bladder
- Empties into bladder lumen at ureteral orifices

Microscopic Anatomy

- Epithelium comprised of urothelium
 - Previously termed "transitional" epithelium
 - Features umbrella, intermediate, and basal cells
- Umbrella cells
 - Superficial layer in contact with urinary space
 - Abundant eosinophilic cytoplasm
 - Larger, sometimes multilobated nuclei
- Intermediate cells
 - Comprise majority of urothelium
 - Usually 3-7 cell layers thick
 - Oval in shape with fine chromatin and conspicuous intranuclear grooves
- Basal cells
 - Form thin layer that lines basement membrane
 - Small and cuboidal
- Lamina propria
 - Subepithelial connective tissue of variable density
 - Contains small blood vessels
- Muscularis propria is comprised of smooth muscle cells, often with indistinct layers
- Surrounding soft tissue is comprised of adipose tissue, blood vessels, and small peripheral nerves

RENAL PELVIS

Macroscopic Anatomy

- Collects urine from major calyces and drains to ureter
- Minor calyces (calyx: Cup-shaped cavity)
 - Usually 7-14 per kidney
 - Urine from collecting ducts of renal medulla drain into minor calyces at tips of renal papilla

- Major calyces
 - Usually 2-3 per kidney
 - Collects urine from several individual minor calyces

Microscopic Anatomy

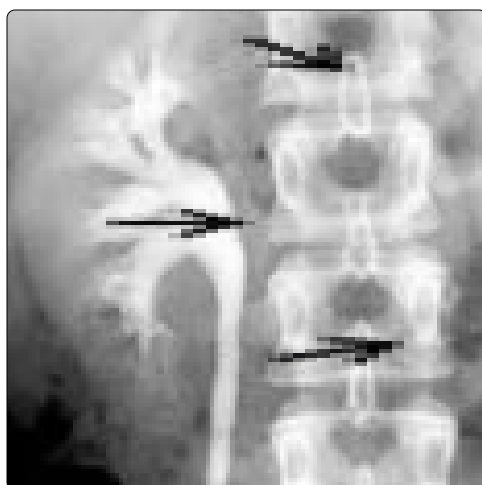
- Renal pelvis and major calyces
 - Urothelial, lamina propria, and muscularis propria layers of variable thickness
- Minor calyces
 - Renal papillae are lined by urothelium
 - No underlying lamina propria or muscularis is present
 - Walls of minor calyces that abut renal sinus fat are similar in histology to those of renal pelvis and major calyces but not as thick

CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

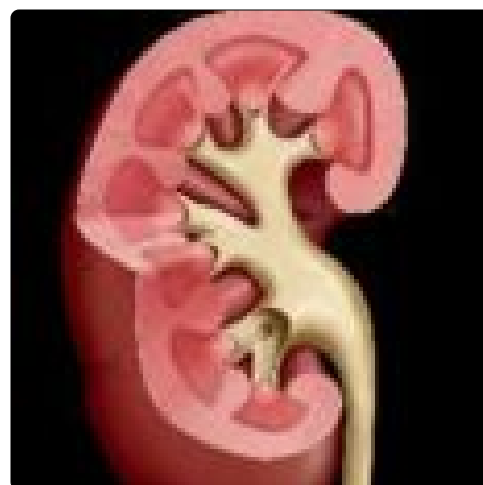
- von Brunn nests in ureter may mimic invasive urothelial carcinoma
 - Urothelial nests are smaller, more confluent, and more disorganized than those of urinary bladder
 - Extend to uniform level in lamina propria, which is distinct from invasive urothelial carcinoma (nested type)
 - Occasionally have cytoplasmic clearing that may mimic renal cell carcinoma
- Cystitis cystica/cystitis glandularis
 - Similar to von Brunn nests but with central luminal space
 - With cystitis glandularis, luminal cells have apically oriented cytoplasm and columnar appearance
- Cancer staging pitfall in invasive urothelial carcinoma
 - At junction with kidney in minor calyces, urothelium transitions from renal pelvis-type wall to that of renal papillae
 - Invasive urothelial carcinoma present at renal papilla will be staged higher than that of adjacent wall because "renal parenchyma" directly underlies urothelium in this anatomic location

Ureter and Renal Pelvis

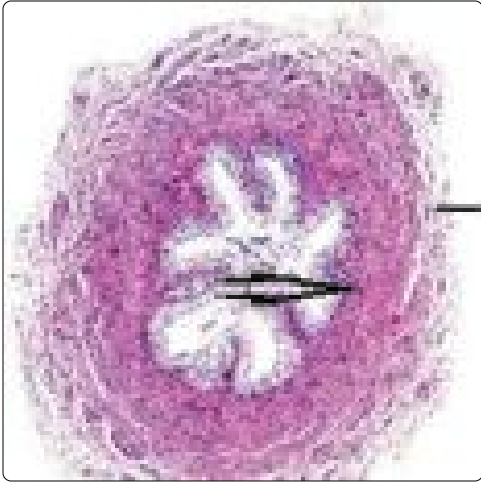


(Left) In this intravenous pyelogram, the course of the ureter and the organization of the collecting system of the kidney (renal pelvis and calyces) are highlighted. (Right) This schematic demonstrates the renal calyces draining into the renal pelvis. The junction of the calyces and the renal medulla has important histologic considerations.

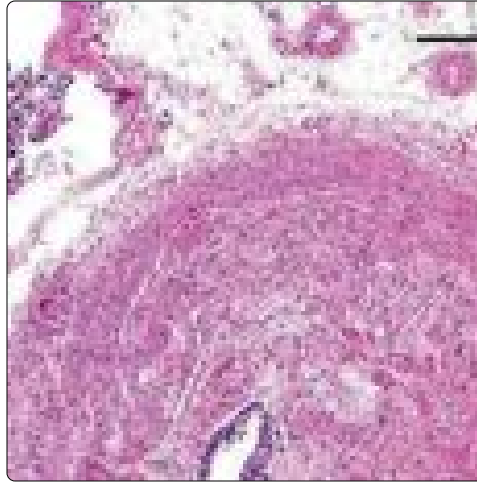
Renal Pelvis



Ureter



Ureter Soft Tissue

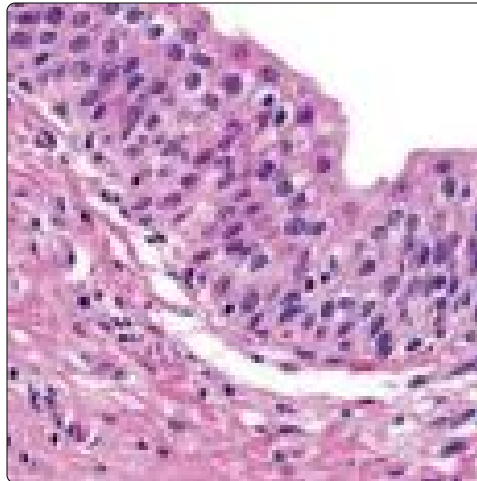


(Left) The lumen of the ureter [1], which is lined by urothelium, typically has complex infolding near the mid portion. The lumen is surrounded by a layer of fibrous connective tissue [2] (lamina propria/subepithelial connective tissue) and an outer layer of smooth muscle [3] (muscularis propria). (Right) This section of ureter shows the surrounding soft tissues, which are often comprised of adipose tissue [4], blood vessels [5], and small peripheral nerves [6].

Ureter Layers

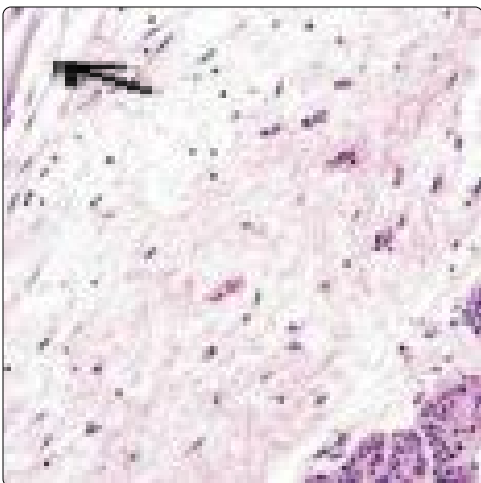


Urothelium

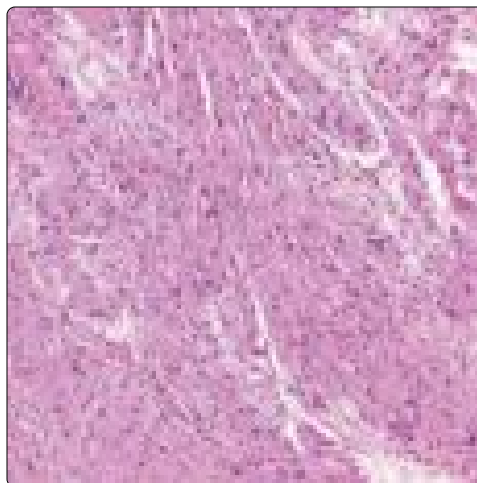


(Left) The ureter has urothelial [1], connective tissue (lamina propria) [2], and smooth muscle (muscularis propria) [3] layers. (Right) The urothelium has a stratified layer of predominantly intermediate cells, organized perpendicular to the basement membrane, with fine nuclear chromatin and intranuclear grooves [4]. The superficial (umbrella) layer has more eosinophilic cytoplasm [5]. A basal cell layer, which lines the basement membrane, is often inconspicuous.

Subepithelial Connective Tissue (Lamina Propria)



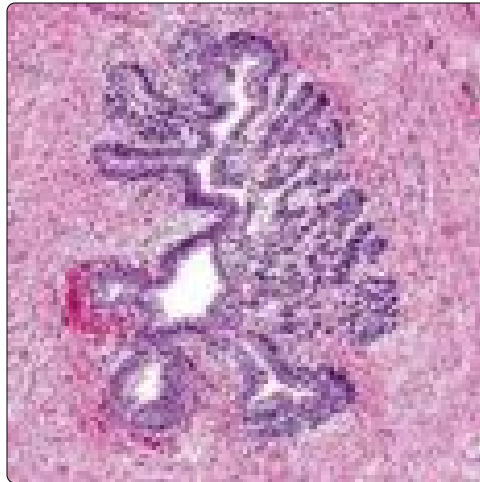
Ureter Muscularis Propria



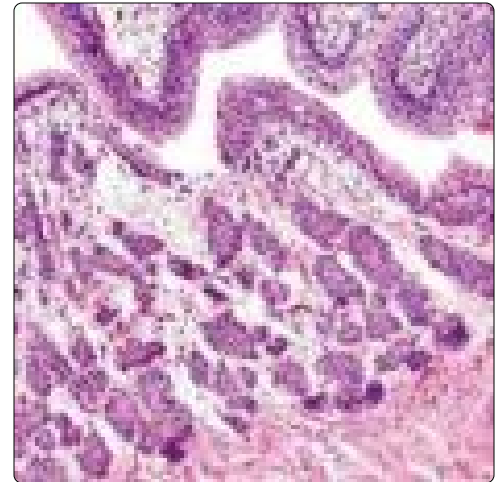
(Left) The tissue between the basement membrane of the urothelium [1] and the smooth muscle cells of the muscularis propria [2] is referred to as the subepithelial connective tissue (or lamina propria). In the ureter and renal pelvis, it consists of connective tissue of variable density and small blood vessels [3]. (Right) The muscularis propria of the ureter is characterized by compact aggregates of smooth muscle.

Ureter With von Brunn Nests

(Left) In the ureter, von Brunn nests are more numerous and somewhat more disorganized than typically seen in the urinary bladder; however, the nests extend to a relatively uniform level in the lamina propria. **(Right)** These invaginated urothelial nests (von Brunn nests) are smaller and more confluent than those typically seen in the bladder; therefore, they may closely mimic invasive urothelial carcinoma (nested type) on biopsy.

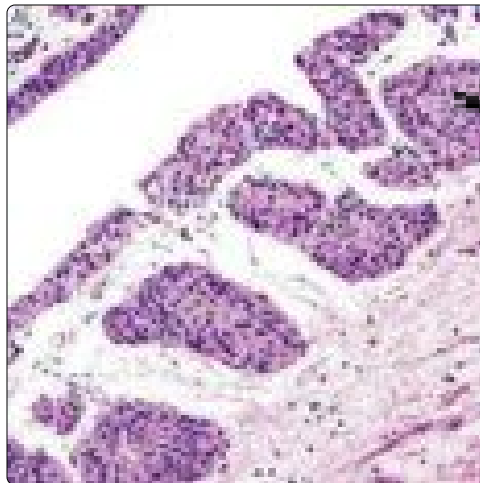


von Brunn Nests

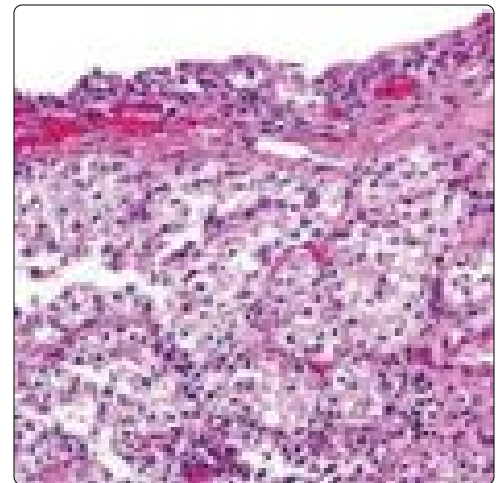


von Brunn Nest Cytology

(Left) The urothelial cells within the von Brunn nests have cytologic features that are similar to those of the overlying urothelium. **(Right)** The von Brunn nests in the ureter &/or renal pelvis occasionally have more abundant intracytoplasmic glycogen, which may histologically mimic a clear cell renal cell carcinoma in radical nephrectomy specimens; however, renal cell carcinoma does not typically extend down the wall of the ureter.

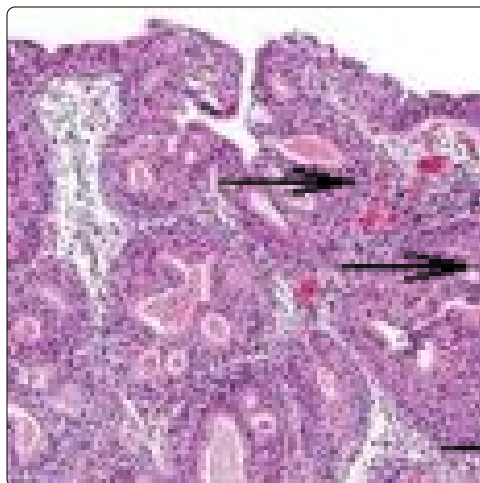


Glycogenated von Brunn Nests



Cystitis Glandularis

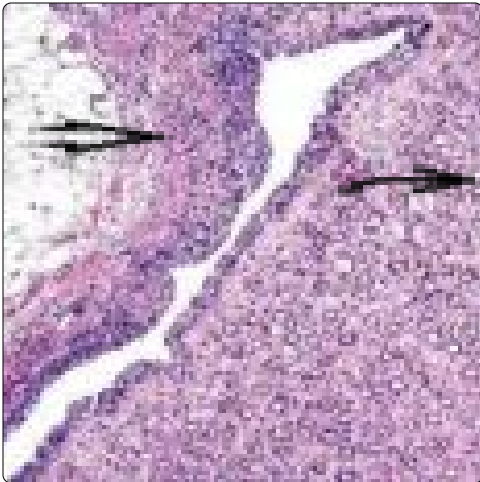
(Left) The von Brunn nests may have central cystic spaces with a more columnar luminal layer of cells and apically oriented cytoplasm (a change that is termed cystitis glandularis). This is a benign lesion that is thought to represent a reactive process secondary to prior injury or inflammation. **(Right)** The renal papillae bulge into the minor calyces as demonstrated in this example. The wall of the calyx is very thin in this region.



Renal Papilla



Renal Calyx



Renal Pelvis



(Left) The junction of the renal papilla with the calyx does not contain lamina propria or muscularis, which is present in the wall of the calyx that is adjacent to the renal sinus fat. This has important implications for staging urothelial carcinoma. **(Right)** The renal pelvis has urothelial, lamina propria, and muscularis layers, similar to the ureter. The wall becomes thinner in the regions of the major, and particularly the minor, calyces.

Renal Papilla

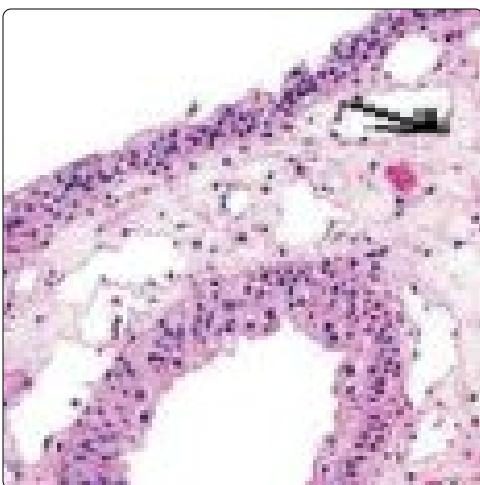


Minor Calyx Over Renal Papillae



(Left) The renal papilla region of the minor calyces has distinct features (e.g., it lacks a lamina propria layer underlying the urothelium over the renal papillae). **(Right)** There is no lamina propria layer underlying the urothelium over the renal papillae. The underlying tissue contains urothelial-lined collecting ducts, renal tubules, and supporting stroma. This is important for staging urothelial carcinomas in this region, as even superficial invasion will be classified as renal parenchymal involvement.

Minor Calyx Urothelium



Renal Pelvis



(Left) The urothelium overlying the renal papilla may be very thin. **(Right)** The urothelial lining of the renal pelvis is similar to that of the urinary bladder. The intermediate cell layer is oriented with the long axis perpendicular to the basement membrane. The nuclear chromatin is fine and pinpoint nucleoli may be seen. Intranuclear grooves are also common. As in other sites, the superficial (umbrella cell) layer has variable densely eosinophilic cytoplasm.

BLADDER

Macroscopic Anatomy

- Inverted pyramidal-shaped organ within anterior pelvis
 - Connected to left and right ureters (dorsolaterally) and urethra distally (at bladder neck)
- Posterior surface of bladder a.k.a. "base"
 - Also contains trigone region
 - Triangular region bordered by bladder neck and both ureteral orifices
- Anterosuperior region of bladder a.k.a. "apex" or "dome"
 - Site of urachal remnant (median umbilical ligament)
- Adjacent structures
 - Female
 - Cervix and superior vagina are posterior to bladder
 - Male
 - Seminal vesicles and vas deferens are posterior to bladder
 - Levator ani muscles are lateral to bladder

Microscopic Anatomy

- Urothelium
 - Previously known as transitional epithelium
 - Most superficial (innermost) layer of mucosa
 - Composed of basal cells, intermediate cells, and umbrella cells
 - Basal cells
 - ◻ Small and cuboidal
 - ◻ Often appear absent in distended bladders
 - ◻ Basal cell layer overlies basement membrane
 - Intermediate cells
 - ◻ Compose majority of urothelium
 - ◻ Often show nuclear grooves
 - ◻ Display polarity (longitudinal axis of cell is perpendicular to basement membrane)
 - Umbrella cells
 - ◻ Larger and contain abundant eosinophilic cytoplasm
 - ◻ May show multinucleation

- ◻ Often flattened and may appear absent in distended bladders
- Thickness
 - 3-6 cell layers in contracted (nondistended) state
 - 2-3 cell layers in distended state
- Lamina propria
 - Loose connective tissue containing vasculature, lymphatics, and nerve fibers
 - Separated from overlying urothelium by basement membrane
 - Muscularis mucosa may be present
 - Discontinuous layer of small fascicles and wisps of smooth muscle
- Muscularis propria
 - Composed of thick bundles of smooth muscle
- Adventitia
 - Outermost layer
 - Composed of fibroadipose tissue

CONSIDERATIONS AND VARIATIONS

Metaplasia

- Glycogenated squamous metaplasia
 - Often seen in trigone of female bladder
- Glandular metaplasia (cystitis glandularis)
- Intestinal metaplasia in background of cystitis glandularis

Pitfalls and Artifacts

- Tangential sectioning of urothelium can give false impression of urothelial hyperplasia
- Poor orientation in transurethral resection of bladder tumor specimens can make it difficult to distinguish between muscularis mucosae and muscularis propria
- Adipose tissue within lamina propria and muscularis propria can be mistaken for perivesicular fat
- Various nonneoplastic proliferations can mimic neoplasia
 - von Brunn nests, cystitis cystica, and cystitis glandularis

Urinary Bladder (Male)

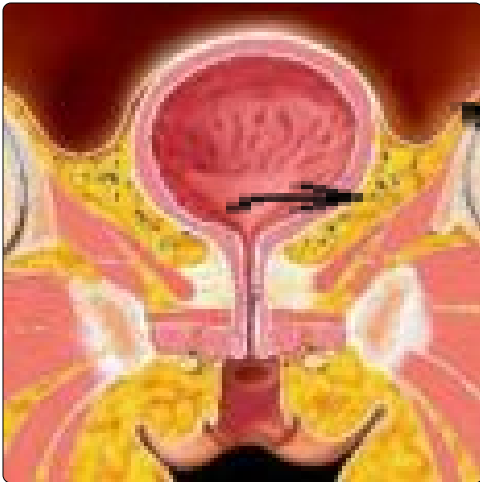


(Left) The urinary bladder  lies below the peritoneum within the pelvis, and the surface of the dome is covered by pelvic parietal peritoneum. The bladder is attached at the bladder neck with the remainder free to expand and contract as it fills with urine and empties. The seminal vesicles and vas deferens lie posterior to the bladder in the male. (Right) In females, the urinary bladder  is separated from the rectum by the cervix and superior portion of the vagina.

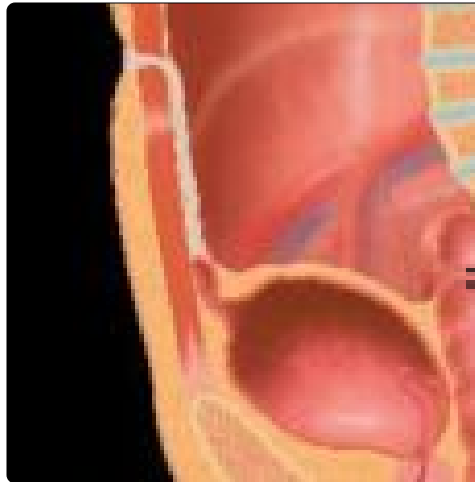
Urinary Bladder (Female)



Layers of Bladder



Urachal Remnants

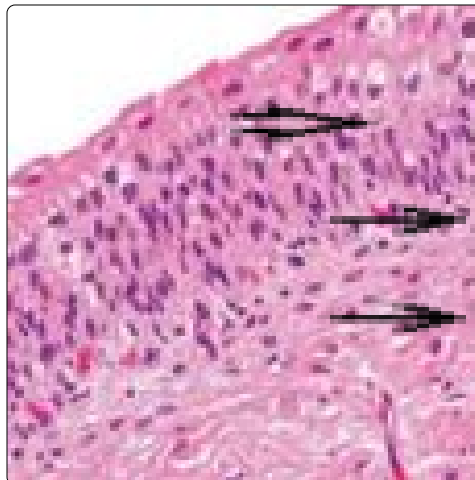


(Left) The 4 layers of the bladder can be visualized on cross section. The innermost layer is composed of the urothelium with the underlying lamina propria separated from the urothelium by the basement membrane. Below the lamina propria lies the smooth muscle of the muscularis propria. The outermost layer of the bladder consists of the perivesical fat. (Right) Urachal remnants, which have a urothelial or glandular lining, may be present at the apex of the bladder wall.

Normal Urothelium

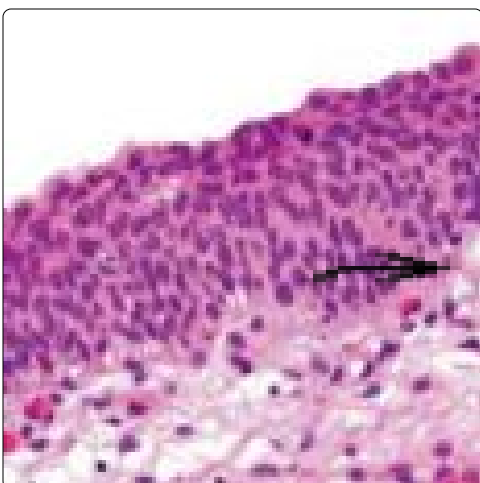


Basal, Intermediate, and Umbrella Cells

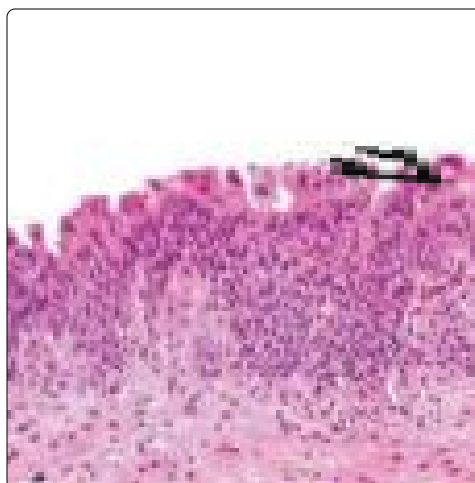


(Left) The thickness of the urothelium varies depending on the degree of distention of the bladder. The average thickness in the contracted bladder is 3-6 cell layers, while the distended bladder may only be 2-3 cell layers thick. The superficial cell layer is composed of umbrella cells. The intermediate and basal cell layers lie underneath the umbrella cells. (Right) Higher power shows the intermediate and basal cell layers and the superficial umbrella cells.

Intermediate Cells



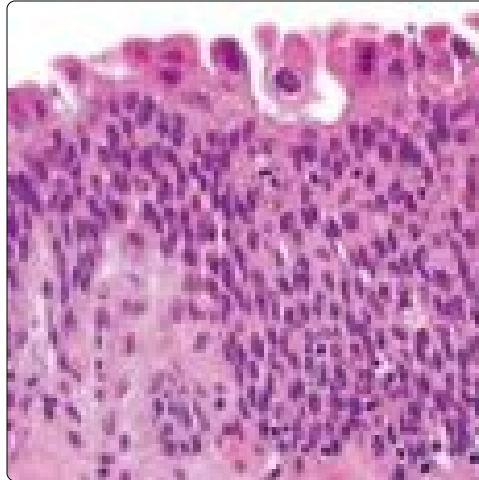
Umbrella Cells



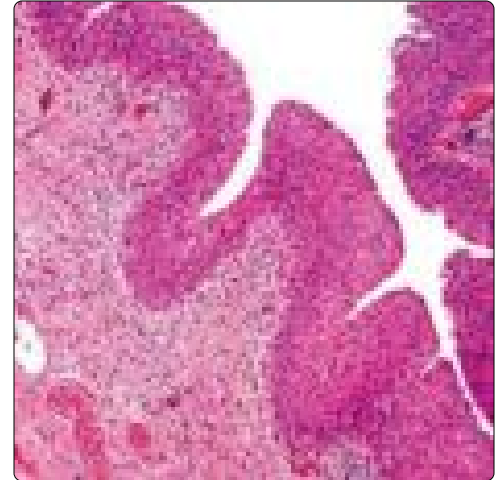
(Left) The intermediate cells may have longitudinal nuclear grooves. The nuclei are ovoid with indistinct nucleoli and display polarity with their longitudinal axis arranged perpendicular to the basement membrane. (Right) The umbrella cells are large with abundant eosinophilic cytoplasm and a rounded contour at the luminal surface. Invaginations of the luminal cell membrane and underlying cytoplasmic vesicles allow the cells to stretch with bladder distention.

Umbrella Cell Binucleation

(Left) The umbrella cells may be binucleate . Although the nuclei of these cells may appear large, the cells have abundant cytoplasm, maintaining a low nuclear:cytoplasmic ratio. The presence of the umbrella cells should not be used in determining the presence of neoplasia, as in situ lesions can have an intact superficial cell layer. (Right) Tangential sectioning  may give the false impression of a thickened urothelium and should not be misdiagnosed as hyperplasia.



Tangential Sectioning

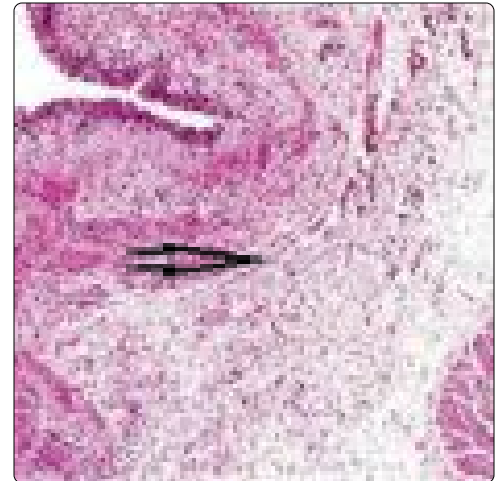


Lamina Propria

(Left) The vascular plexus  is identified in the mid portion of the lamina propria and is composed of medium-sized vessels. The muscularis mucosae, when present, is often seen in association with the vascular plexus. (Right) The muscularis mucosae  is a thin layer of smooth muscle often associated with the vascular plexus in the lamina propria. It is composed of wisps and small fascicles of smooth muscle as opposed to the thick muscle bundles of the muscularis propria .

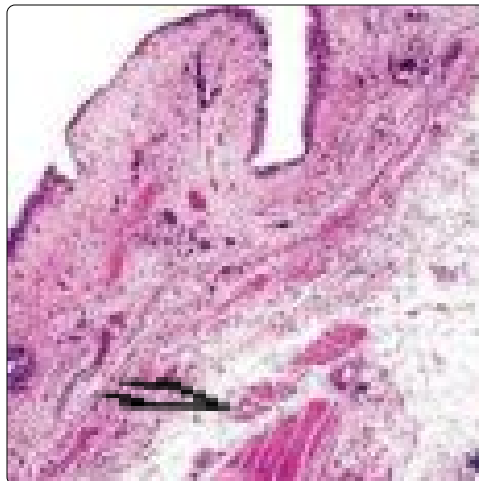


Muscularis Mucosae

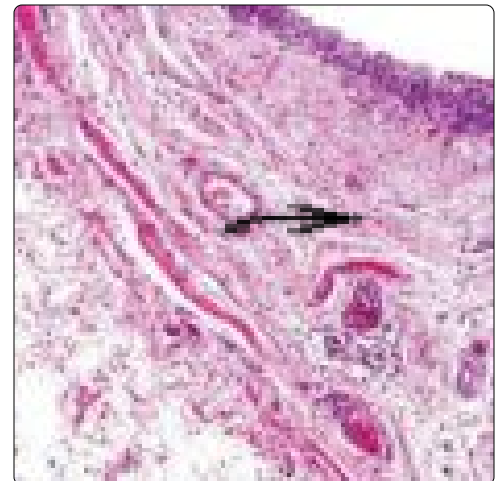


Discontinuous Layer

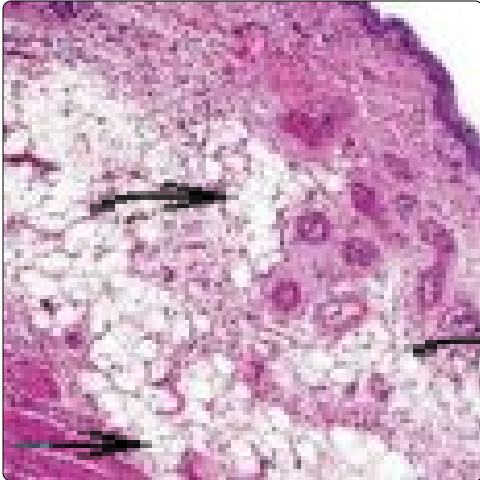
(Left) The muscularis mucosae is composed of wisps and small fascicles of smooth muscle and when present often forms a discontinuous layer within the lamina propria . The wisps and small bundles should not be confused with the large bundles of the muscularis propria when evaluating depth of invasion of carcinoma into the bladder wall, as this will upstage the patient from a pT1 lesion to pT2. (Right) Wisps of smooth muscle form the muscularis mucosae  in the lamina propria.



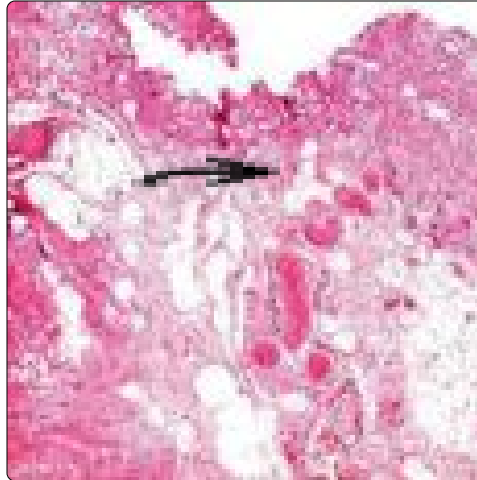
Wisps of Smooth Muscle



Adipose Tissue Within Lamina Propria

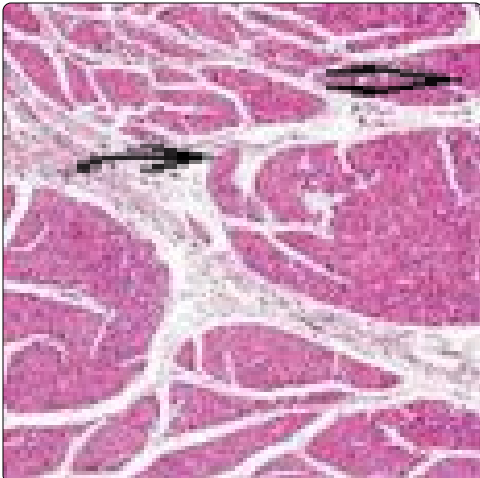


Transurethral Resection of Bladder Tumor

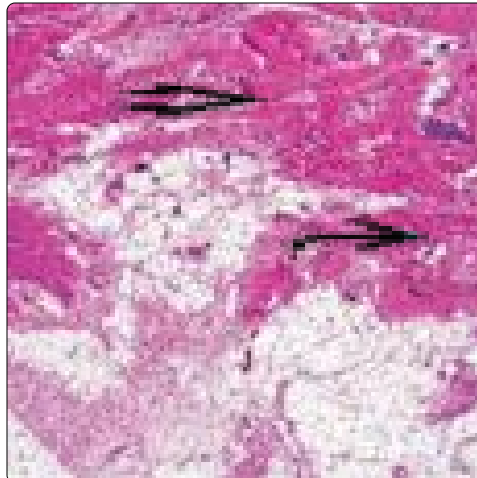


(Left) Mature adipose tissue [A] can be present within the lamina propria [B] and muscularis propria [C] and should not be mistaken for perivesicular fat. (Right) This transurethral resection of bladder tumor section shows an example of mature adipose tissue [A] intermixed within the lamina propria and muscularis propria layers. The presence of this adipose tissue should not be confused as perivesical fat when evaluating for depth of tumor invasion.

Muscularis Propria

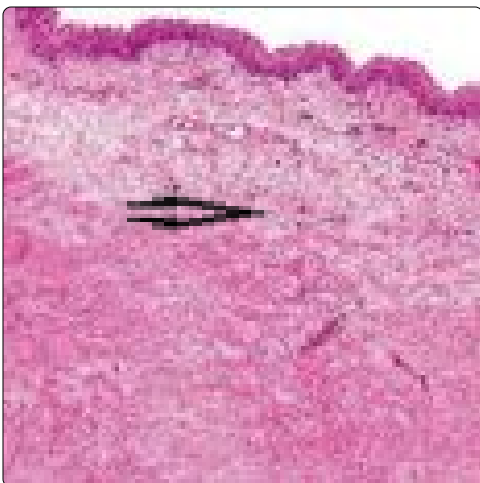


Adventitia

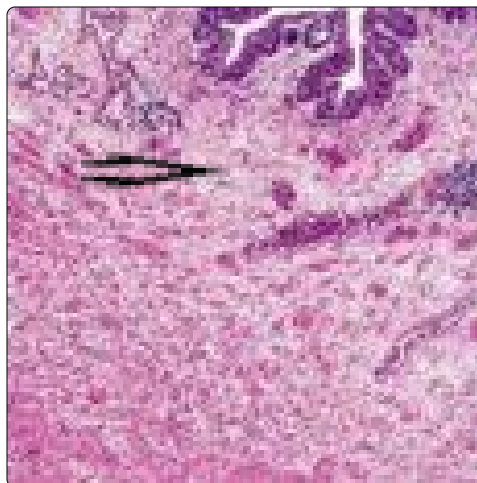


(Left) The muscularis propria is composed of large fascicles [A] of smooth muscle and connective tissue [B] containing blood vessels, lymphatic spaces, and nerves. (Right) The adventitia of the bladder is composed of fibroadipose tissue [A], as seen here at the interface of the adventitia with the muscularis propria [B]. The dome of the bladder is covered by the pelvic parietal peritoneum, while other areas of the bladder lack a serosal surface.

Muscularis Propria (Trigone and Bladder Neck)



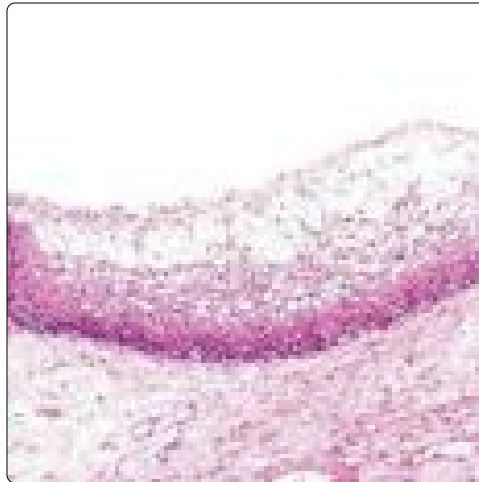
Trigone and Bladder Neck



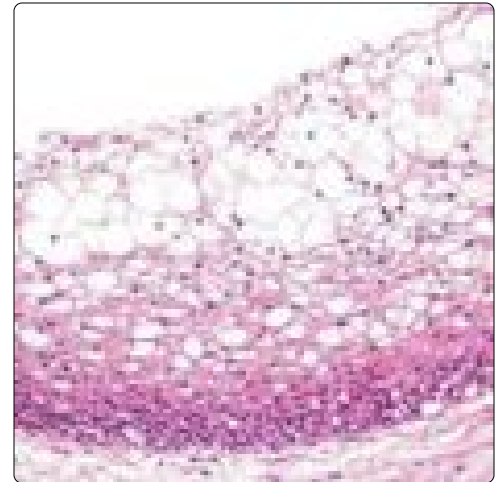
(Left) The muscularis propria in the trigone and bladder neck has unique morphological features. The muscle fascicles gradually decrease in size as they become more superficial in location [A] in the muscularis propria. (Right) The lamina propria is thinner in the trigone and bladder neck as the muscularis propria muscle bundles extend to a more superficial location. The muscle fascicles [B] become smaller and more disorganized in the superficial portions of the muscularis propria.

Glycogenated Squamous Epithelium

(Left) Glycogenated squamous epithelium is commonly found in the trigone area in female patients and can be seen in male patients undergoing estrogen therapy for prostatic carcinoma. **(Right)** Glycogenated squamous epithelium found in the trigone area of the bladder is morphologically similar to the squamous epithelium of the cervix and vagina. The cytoplasm is abundant and clear due to the accumulation of glycogen. No nuclear atypia is present, and there is no keratinization.

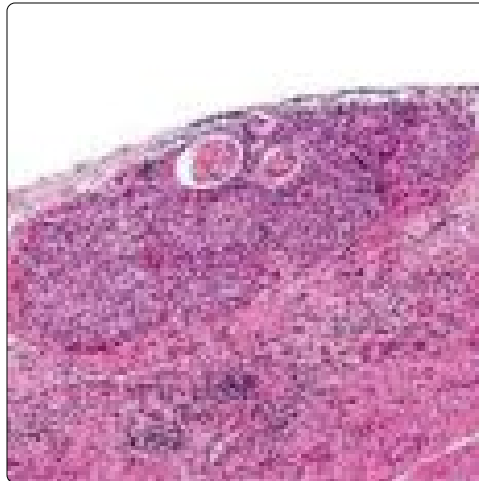


Glycogenated Squamous Epithelium

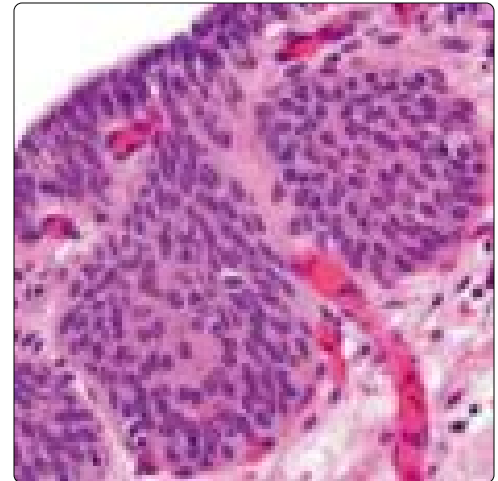


von Brunn Nests

(Left) von Brunn nests are a common normal variation present in the urothelium; they represent invaginations of the urothelium as nests into the underlying lamina propria. These lesions may occasionally be florid, causing a visible nodule on cystoscopy. The architecture shows nests with smooth contours, an orderly arrangement, and a sharp border with the lamina propria below. **(Right)** The cytologic features of von Brunn nests are bland with no mitoses or atypia.

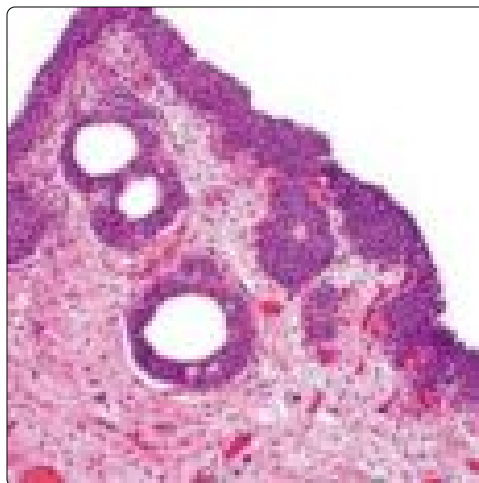


von Brunn Nest Cytology

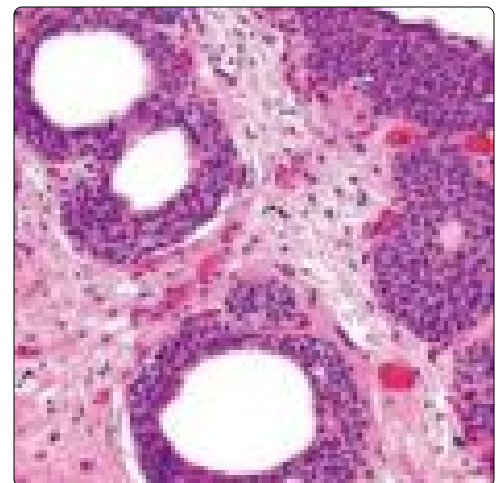


Cystitis Cystica

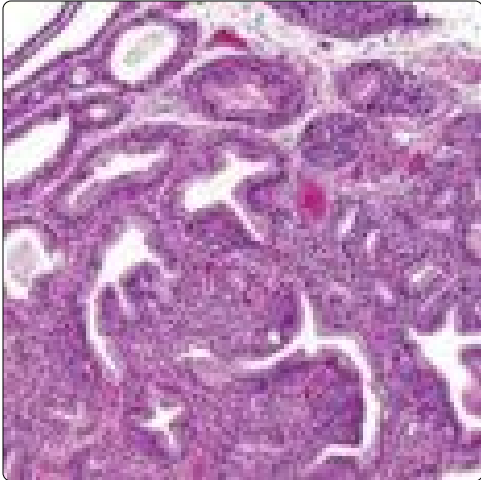
(Left) Cystitis cystica refers to von Brunn nests that have acquired a cystic lumen. The overall architecture of the nests is the same as that of von Brunn nests with an orderly overall appearance and a sharp linear border with the underlying lamina propria. **(Right)** The epithelium lining the cysts in cystitis cystica is composed of cuboidal or flattened urothelium. The nuclei are small and lack atypia, sharing a similar morphology with the overlying urothelium.



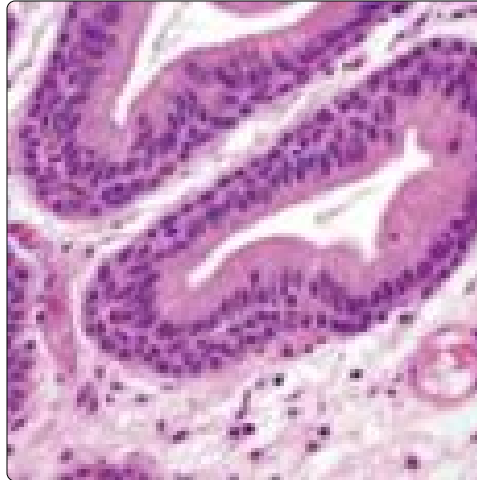
Cystitis Cystica Cytology



Cystitis Glandularis



Cystitis Glandularis Cytology



(Left) Cystitis glandularis refers to proliferations of nests of urothelial cells where the lining has undergone glandular metaplasia. The cells lining the cysts are cuboidal to columnar in shape. The overall architecture shows smooth, contoured, round to branching glands with a distinct border with the lamina propria. **(Right)** The cells of cystitis glandularis lack atypia and are composed of cuboidal to columnar epithelium, which may contain mucin within the cytoplasm.

Cystitis Glandularis With Intestinal Metaplasia

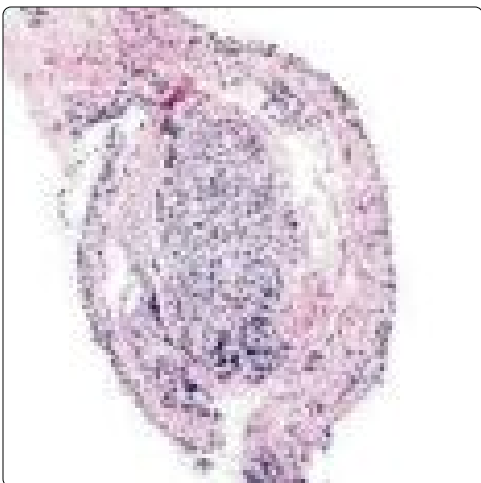


Cystitis Glandularis With Intestinal Metaplasia Cytology

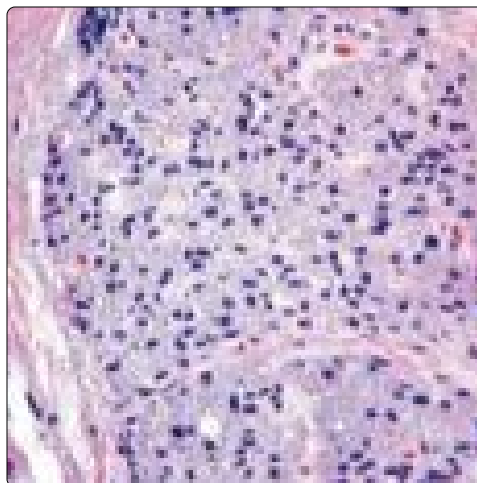


(Left) The invaginations of nests of cystitis glandularis can acquire intestinal-type goblet cells, similar to colonic epithelium. When this occurs, the proliferation is called cystitis glandularis with intestinal metaplasia. **(Right)** At higher power, the bland nuclear features of cystitis glandularis with intestinal metaplasia are apparent. No mitotic figures are present, and the nuclei are small, lacking atypia.

Paraganglia



Paraganglia Cytology



(Left) Paraganglia may be seen within the lamina propria, where they may be confused for invasive carcinoma or paraganglioma. The paraganglia are often seen in association with nerves and small-caliber vessels. **(Right)** A higher power view of the paraganglia shows an aggregate of cells with basophilic cytoplasm and bland-appearing, small to medium-sized nuclei. Immunohistochemistry will show positive staining for chromogranin or synaptophysin and negative cytokeratin staining.

URETHRA**Macroscopic Anatomy**

- Male urethra (15-20 cm)
 - Preprostatic: Intramural segment (within wall of urinary bladder)
 - Prostatic: Receives secretions from prostatic and ejaculatory ducts at verumontanum
 - Membranous (intermediate): Passes through pelvic and urogenital diaphragm
 - Penile or spongy (subdivided: Bulbar, pendulous, fossa navicularis)
- Female urethra (4 cm)
 - Proximal: Extends from bladder to mid urethra
 - Distal: Exits body between clitoris and vagina
- Paraurethral/periurethral glands also present
 - Female: Skene
 - Male: Cowper (membranous) and Littre (penile)

Microscopic Anatomy

- Male
 - Preprostatic, prostatic, and membranous urethra lined by stratified urothelium with superficial umbrella cell layer
 - Bulbar and pendulous regions of penile urethra lined by distinctive stratified columnar layer
 - Most distal aspect (fossa navicularis) lined by nonkeratinizing squamous epithelium
- Female
 - Proximal 1/3 lined by stratified urothelium that transitions to nonkeratinizing squamous epithelium distally
- Paraurethral and periurethral glands
 - Ducts lined by stratified columnar/cuboidal epithelial layer
 - Acini comprised of cells with intracytoplasmic mucin
 - Prostatic-type glands may also be seen
- Intraepithelial glandular structures
 - Small collections of intraepithelial cells with intracytoplasmic mucin

CONSIDERATIONS AND VARIATIONS**Hyperplasia**

- "Transitional" cell hyperplasia
 - Prominent thickened urothelial layer within periurethral ducts and glands
- Lobular hyperplasia of Cowper gland

Metaplasia

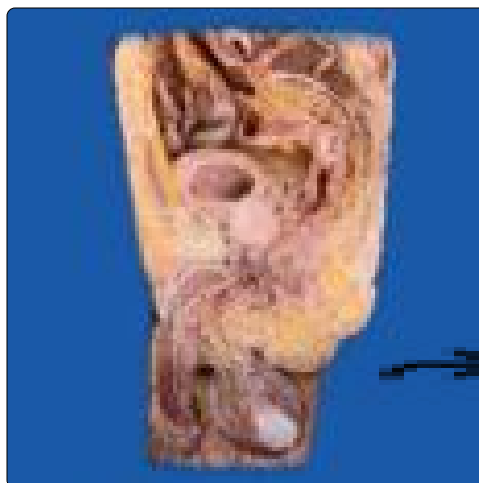
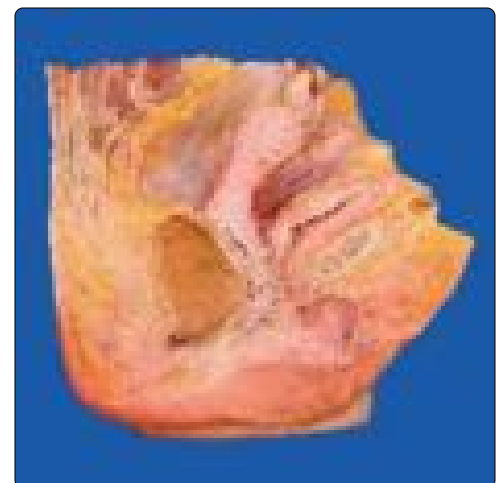
- Nonkeratinizing glycogenated squamous metaplasia
 - Common in proximal urethra and bladder trigone of women
- Glandular metaplasia
 - May be seen with chronic irritative conditions

Pitfalls and Artifacts

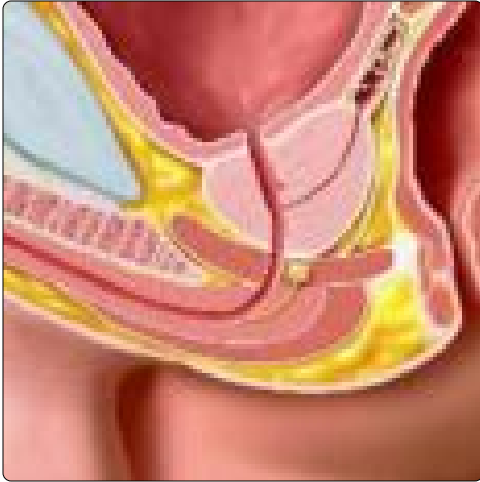
- Florid von Brunn nests
 - Numerous rounded invaginations of benign urothelium into underlying connective tissue
 - Invaginations extend to uniform depth
- Polypoid/papillary urethritis
 - Papillae and polypoid excrescences with broad base
 - No complex hierarchical branching
- Nephrogenic adenoma
 - Generally considered regenerative/repairative lesion
 - Papillary, tubular, cystic, solid/diffuse, or flat patterns
- Urethral caruncle
 - Inflammatory lesion at urethral meatus, usually in postmenopausal women
 - Consists of inflammatory cells, blood vessels, and varying edematous to fibrotic stroma
 - Lining epithelium may be urothelial or squamous
- Fibroepithelial polyp
 - Rare, but may occur in children
 - Most common in posterior urethra
 - Loose fibrous stroma lined by urothelium, sometimes with reactive epithelial atypia
- Prostatic-type polyp
 - Polypoid lesion lined by admixture of benign urothelial and prostatic secretory-type cells

Urethra (Male)

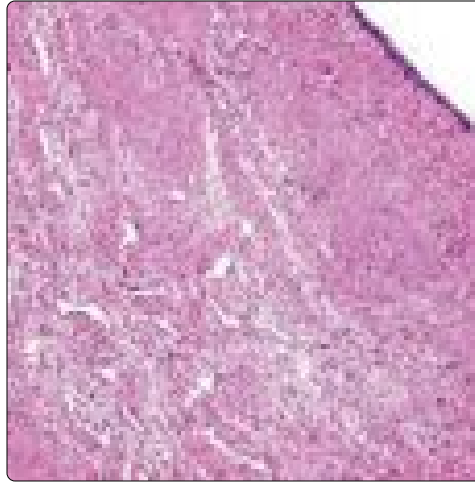
(Left) In this dissection of the male pelvis (midsagittal view), the lumen of the urethra is not present at every level, but the anatomic distribution of its course is highlighted: Bladder , intramural , prostatic , membranous , and penile . (Right) In this dissection of the female pelvis (midsagittal view), the lumen of the urethra  is seen in continuity with the bladder . Compared to the male urethra (15-20 cm), the female urethra is relatively short in length (4 cm).

**Urethra (Female)**

Urethral Anatomy (Male)

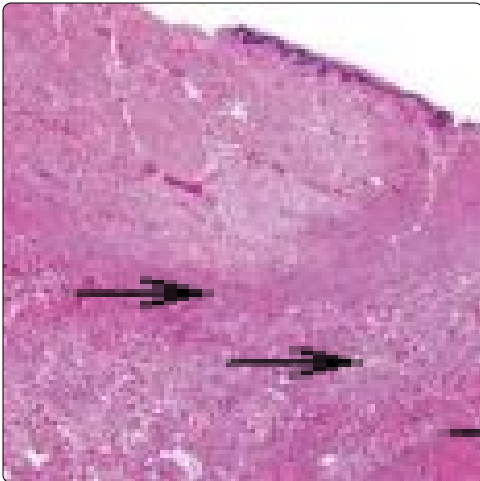


Membranous Urethra (Male)

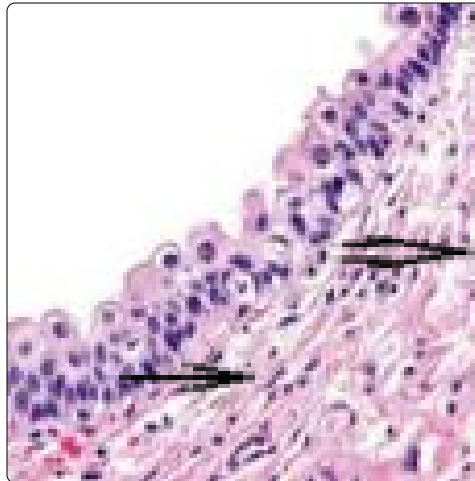


(Left) Graphic of the male pelvis highlights the location of the verumontanum [1], where the ejaculatory ducts [2] empty into the prostatic urethra. The membranous urethra [3] (1-2 cm) courses from the apex of the prostate to the perineal membrane at the bulb of the penis within the deep perineal space. (Right) The membranous urethra has an epithelial lining [4], an underlying lamina propria [5], and a region rich in vascular channels [6].

Membranous Urethra (Male)

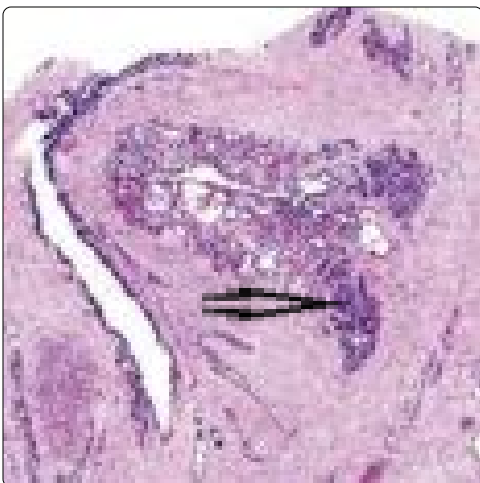


Urothelium



(Left) Skeletal muscle [1] may be seen deep in the wall of the membranous urethra because it traverses the pelvic floor musculature, which is composed of muscle fibers of the levator ani and the coccygeus, just distal to the apex of the prostate gland. (Right) The proximal portions of the urethra may have urothelium (transitional epithelium) identical to that of the bladder. The elongated intermediate cells with intranuclear grooves [2] are the predominant type, but an umbrella cell layer is also seen [3].

Prostatic Urethra (Verumontanum)

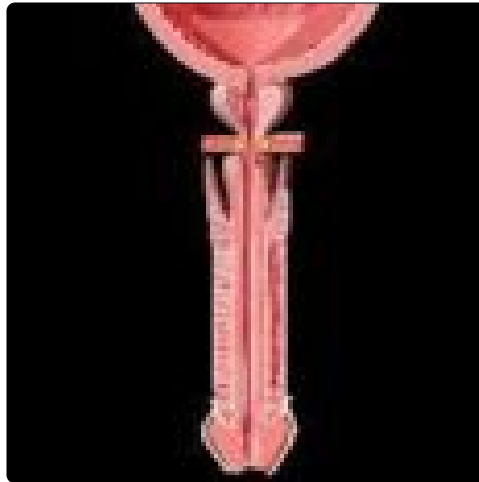


Prostatic Urethra



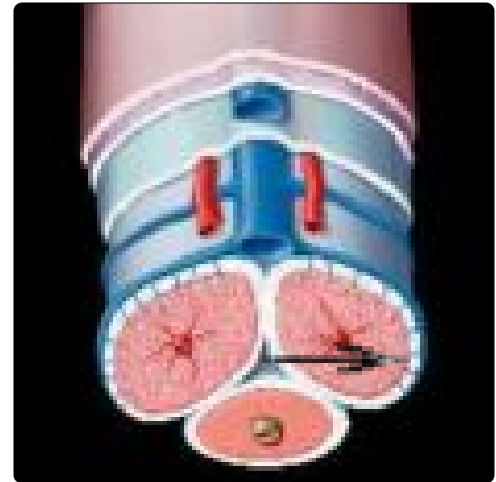
(Left) The verumontanum (or colliculus seminalis) is a posterior protuberance into the prostatic urethral lumen [1] that represents the site where the prostatic and ejaculatory ducts [2] enter the urethra. (Right) The normal prostatic urethra has an irregular surface with small polypoid projections [3] that may mimic a neoplastic process. The lining epithelial cells of the prostatic urethra are an admixture of urothelial [4] and prostatic secretory [5] cells.

Urethra (Male)

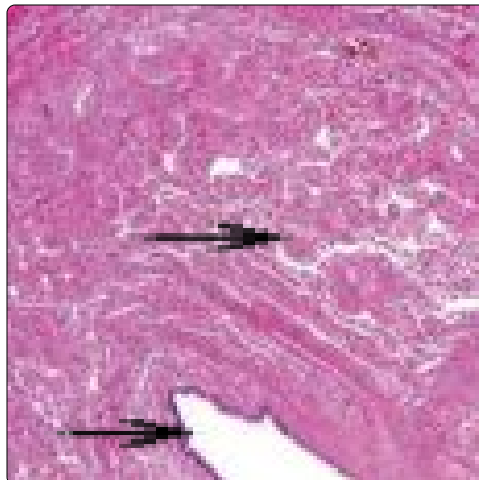


(Left) Graphic highlights the anatomic regions of the male urethra: Intramural [1], prostatic [2], membranous [3], and penile (including bulbar [4] and pendulous [5]). (Right) Graphic shows a cross section of the penis and highlights the urethra [6], which travels through the corpus spongiosum [7]. The location of the corpora cavernosa [8] and deep dorsal vein [9] are also highlighted for comparison.

Penile Urethra



Penile Urethra



(Left) The lumen of the penile urethra [1] is surrounded by a thin layer of fibrous connective tissue (the lamina propria) [2] and then by the periurethral corpus spongiosum [3]. (Right) The corpus spongiosum [4], which underlies the lamina propria [2] and the epithelial layer [1] of the penile urethra, consists of interconnected, branching vascular spaces that are surrounded by smooth muscle and connective tissue.

Penile Urethra

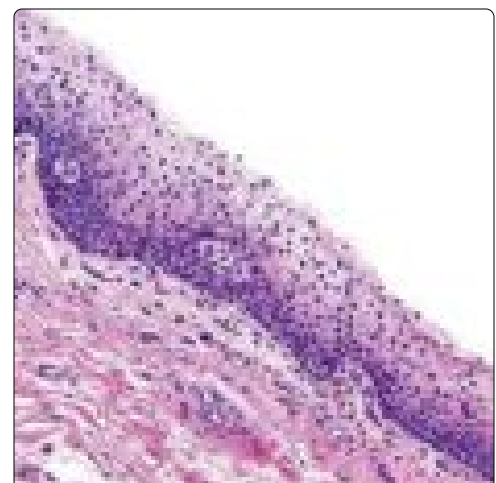


Penile Urethra (Distal/Pendulous)

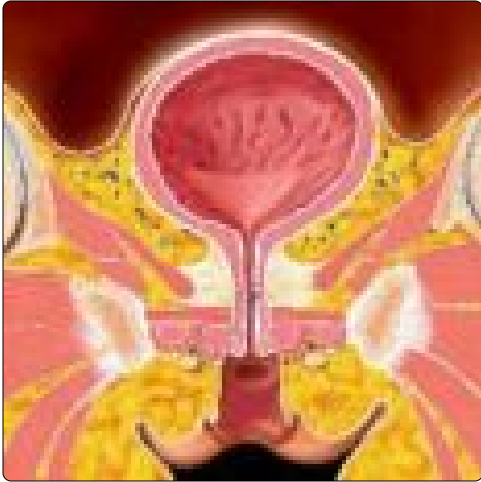


(Left) The epithelium of the penile urethra has been referred to as a stratified or pseudostratified columnar layer and ranges from ~ 4-15 layers of cells. (Right) The most distal portion of the male urethra, a saccular dilatation called the fossa navicularis, is lined by nonkeratinizing squamous epithelium that becomes continuous with the epithelium of the glans penis.

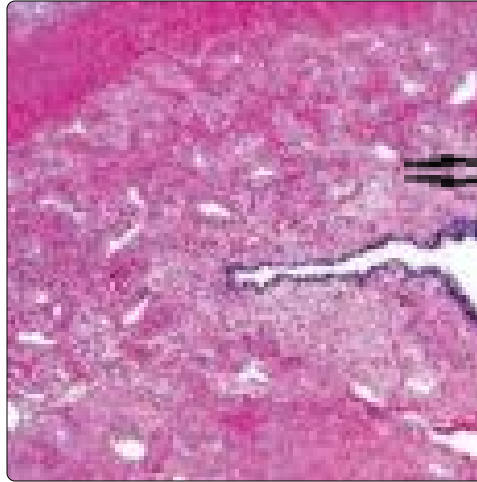
Penile Urethra (Fossa Navicularis)



Urethral Anatomy (Female)

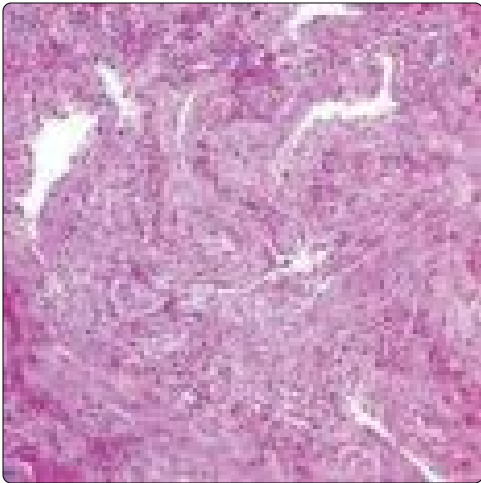


Urethra (Female)

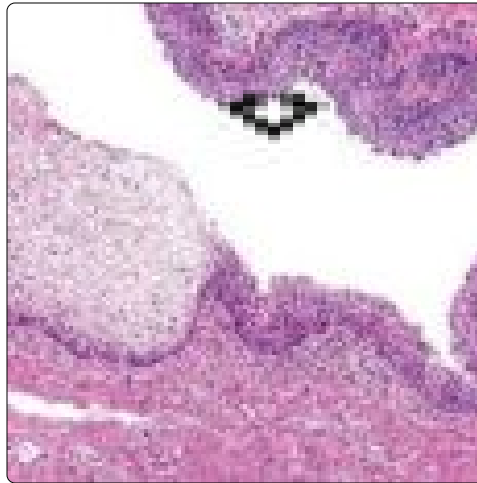


(Left) The female urethra has a much shorter course compared to that of the male. It is divided simply into the proximal and distal regions. The internal and external sphincters are also shown. (Right) The female urethra is composed of an epithelial layer, underlying fibrous connective tissue with small capillary-sized vessels (lamina propria), and surrounding larger "erectile-type" vascular spaces.

Urethra (Female)

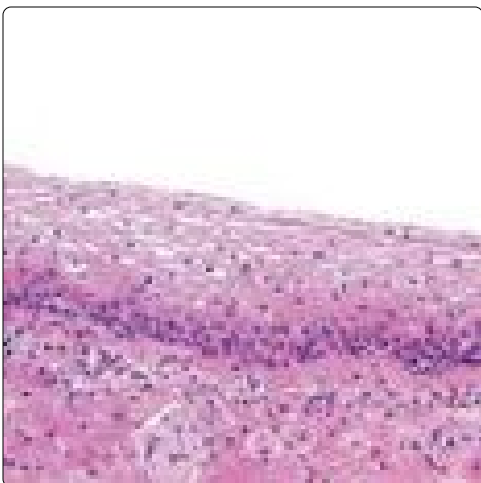


Squamous Metaplasia (Female)

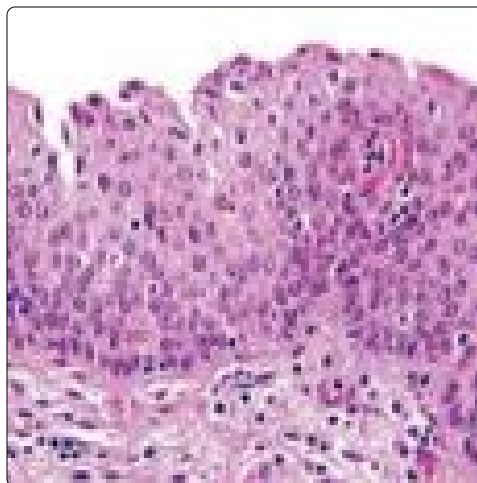


(Left) These large, venous-like spaces with surrounding smooth muscle stroma are typically seen beneath the lamina propria in the female urethra. (Right) Nonkeratinizing squamous epithelium is commonly seen in the proximal urethra and trigone of females, and is typically considered normal. It typically contains abundant glycogen, lending the squamous cells a pale appearance. Note the adjacent urothelium.

Distal Urethra (Female)



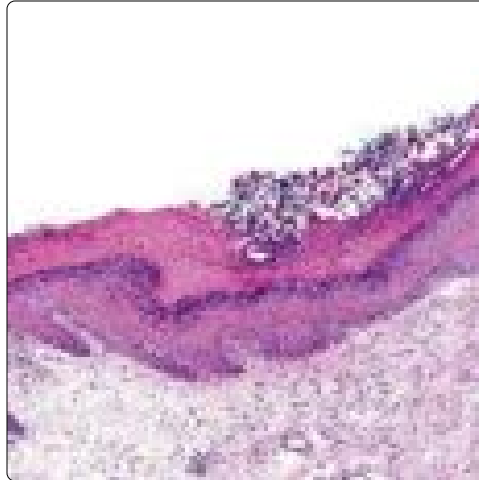
Squamous Metaplasia (Female)



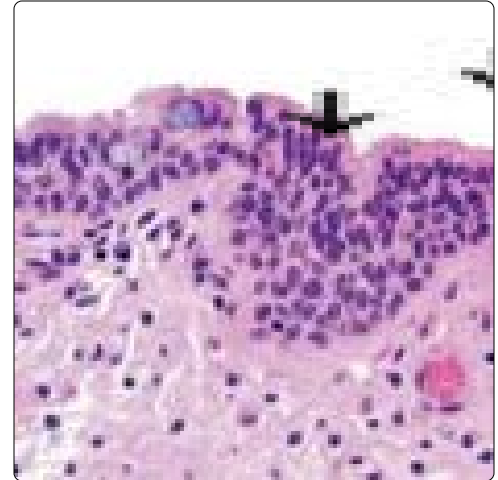
(Left) The distal region of the female urethra is typically lined by nonkeratinizing squamous epithelium, as shown here. (Right) A urothelial lining is typically present in the proximal portions of the female urethra, but this may be replaced by metaplastic epithelium, especially nonkeratinizing glycogenated squamous epithelium in regions near the urinary bladder.

Keratinizing Squamous Metaplasia

(Left) Keratinizing squamous metaplasia is an abnormal process that is most commonly associated with states of chronic irritation secondary to a variety of etiologies. (Right) The urothelium may occasionally show individual cells with mucinous features, especially in states of chronic irritation or inflammation. These metaplastic cells are characterized by intracytoplasmic intestinal-type mucin. This is not generally regarded as a preneoplastic condition.

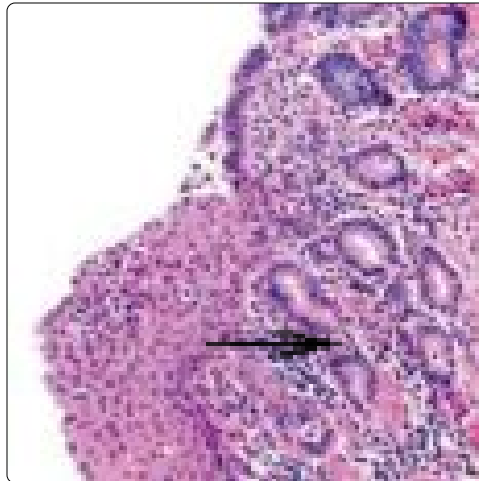


Mucinous Intraepithelial Cells

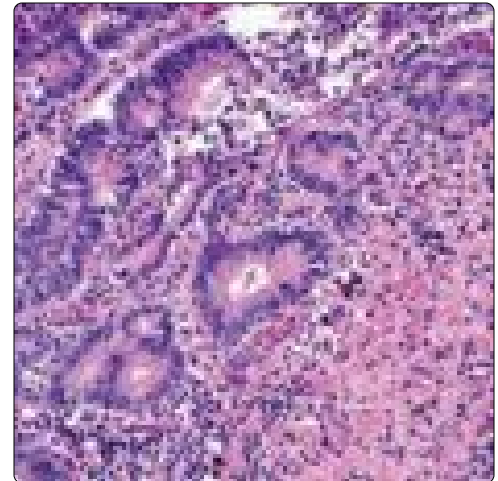


Intestinal Metaplasia

(Left) In this section of urethral epithelium in a patient with a long history of urethral diverticula and infections, there is well-developed intestinal metaplasia, which may be contrasted with the adjacent urothelium. (Right) Intestinal metaplasia of the urethra has a histologic appearance (and immunophenotype) that is identical to that of enteric-type mucosa from any anatomic site.

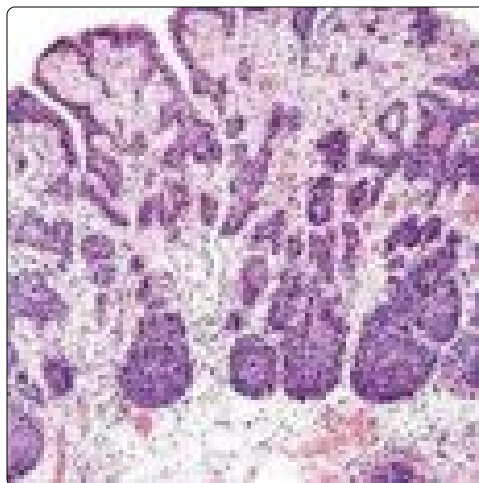


Intestinal Metaplasia

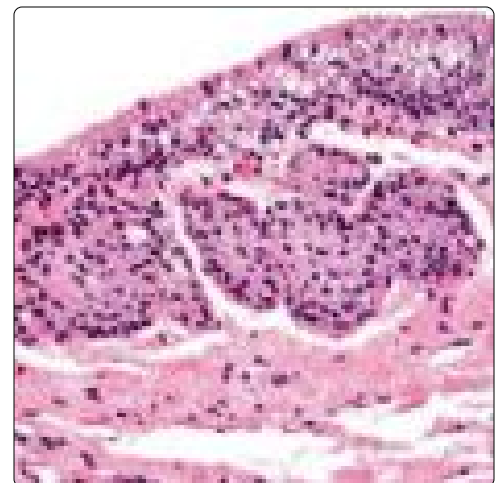


von Brunn Nests

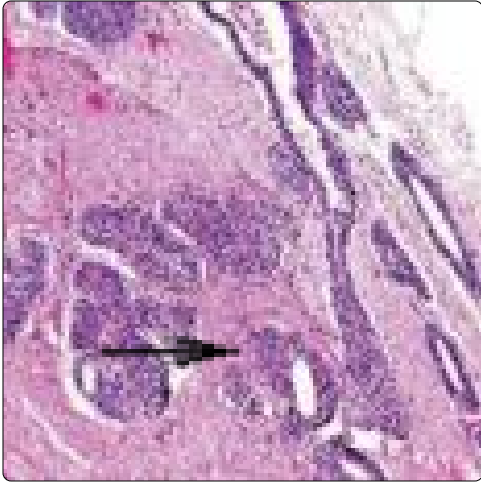
(Left) von Brunn nests are invaginations of benign urothelium into the lamina propria. Unlike cytologically bland forms of invasive urothelial carcinoma, such as nested type, these benign urothelial nests extend to a relatively uniform level within the lamina propria. (Right) The urothelial cells within von Brunn nests are cytologically bland with histologic features similar to those in the overlying flat urothelium.



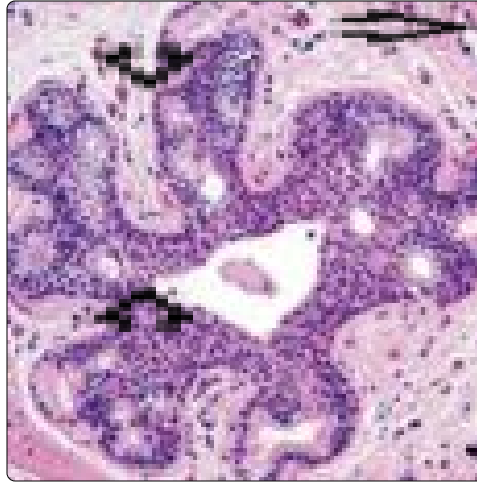
von Brunn Nests



Urethral Ducts

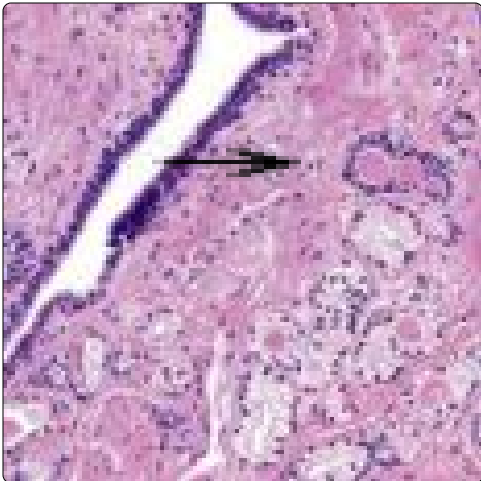


Periurethral/Paraurethral Glands

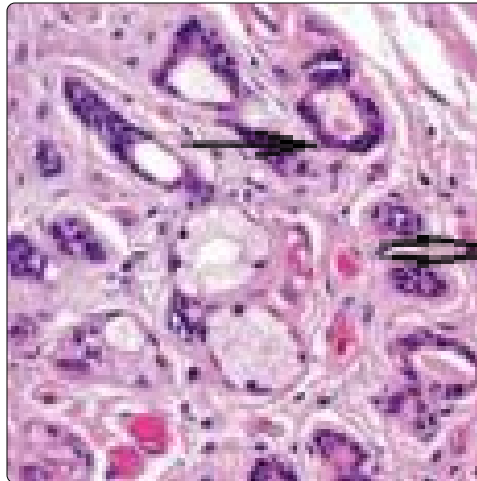


(Left) Numerous ducts of varying caliber are frequently identified within the wall of the urethra; they connect the surrounding glands to the urethral lumen and are typically lined by cuboidal or urothelial-type epithelium. (Right) This section from the penile urethra shows periurethral glands with a tubuloacinar arrangement, consisting of columnar epithelium and peripherally placed mucinous-type epithelium.

Periurethral/Paraurethral Glands

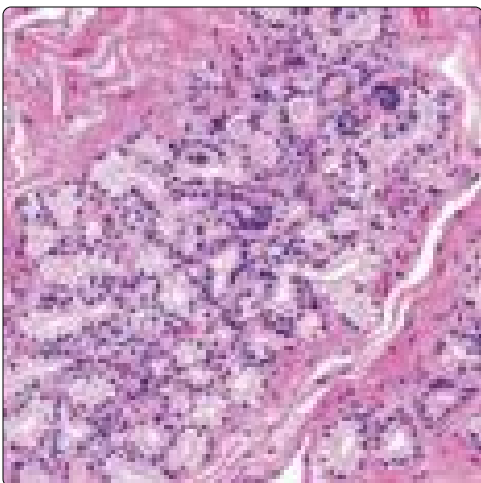


Periurethral/Paraurethral Glands

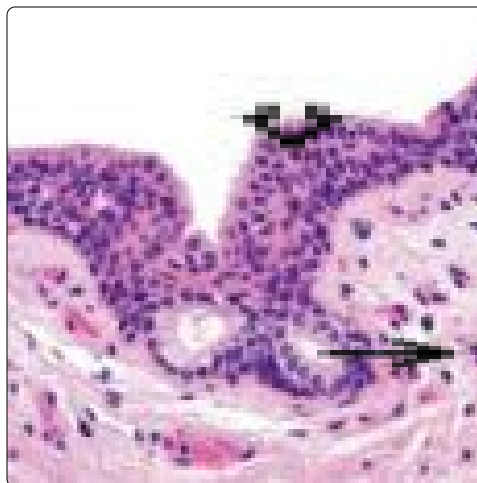


(Left) This deeper, large-caliber duct is present adjacent to a collection of mucinous acini. This architectural arrangement is typical of the benign glandular structures that are frequently identified adjacent to the urethra. (Right) The accessory glands that are present within the wall of the urethra or in surrounding tissues may consist of an admixture of small acini lined by both mucinous and cuboidal epithelium.

Periurethral/Paraurethral Glands



Intraepithelial Glandular Cells



(Left) This collection of acini with mucinous epithelium is typical of the glands identified within the wall of the urethra. The lobular/circumscribed architectural arrangement aids in their recognition as a normal benign structure. (Right) Intraepithelial glandular cells may also be seen within the otherwise normal epithelium that lines the urethra, as in this example. These are usually seen within the vicinity of underlying ducts and accessory glands.

PROSTATE

Macroscopic Anatomy

- Prostate divided into regional zones of McNeal
 - Transition zone surrounds prostatic urethra
 - Central zone extends from bladder base to mid prostate (inverted pyramidal shape)
 - Peripheral zone extends from superior prostate to apex and surrounds both central and transition zones

Microscopic Anatomy

- Peripheral zone
 - Glands show variable caliber and often follow contour of prostate gland surface
- Transition zone
 - More nodular aggregates of prostate glands
- Central zone
 - Distinct histology with intraluminal epithelial bridging and papillae
 - Often have more eosinophilic cytoplasm

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Benign prostatic hyperplasia may distort normal anatomy in older adults
 - Transition zone may extend into apex, and stromal hyperplasia may extend to bladder base

Hyperplasia

- Benign prostatic hyperplasia

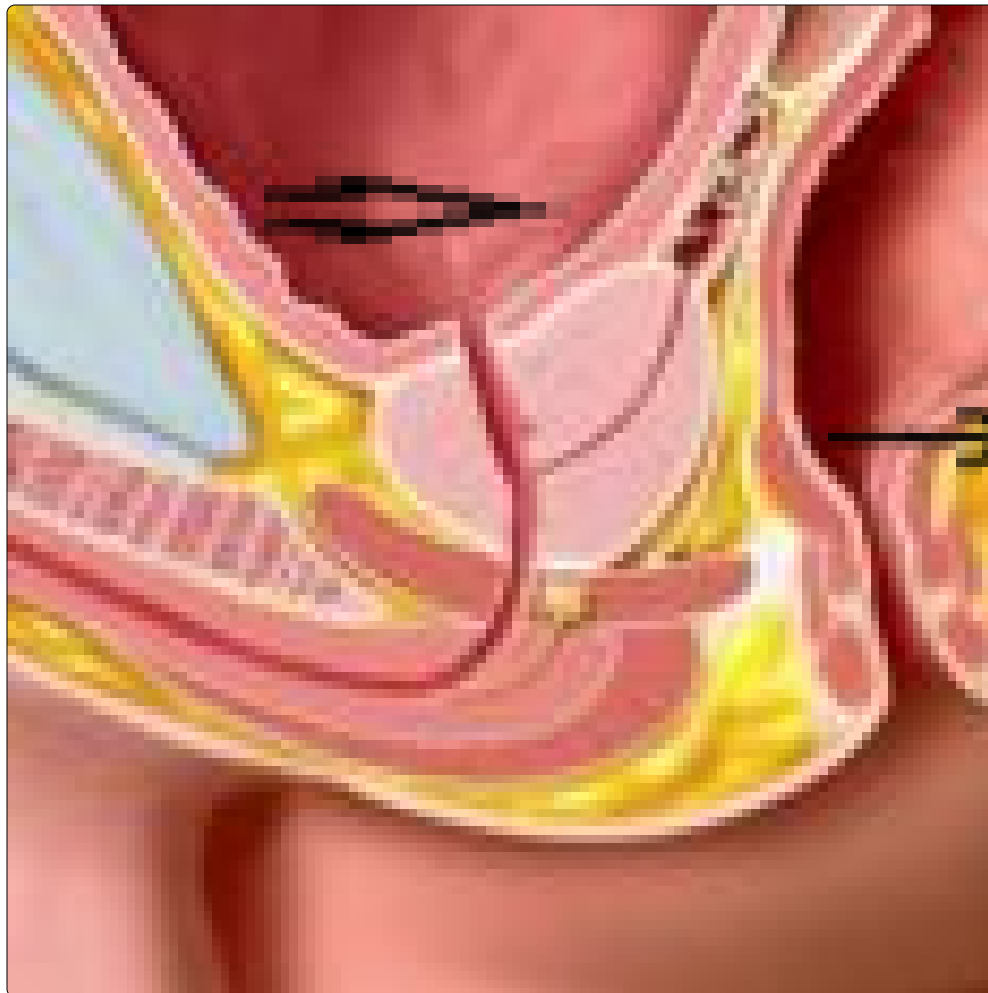
Metaplasia

- Urothelial (transitional) metaplasia
 - Common in periurethral region, rarely extends into peripheral zone

Pitfalls and Artifacts

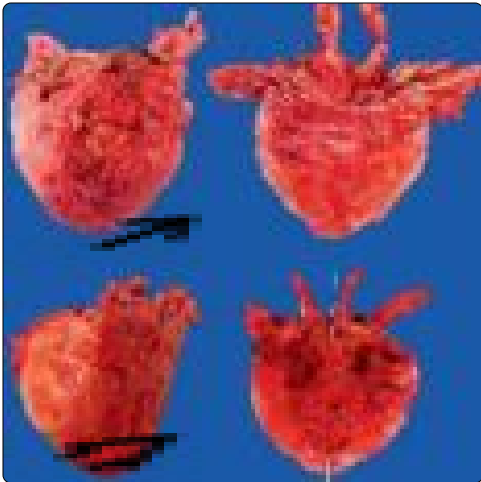
- Normal anatomic variation may mimic adenocarcinoma histologically
 - Central zone glands
 - Urothelium within prostate glands

Prostate Gland

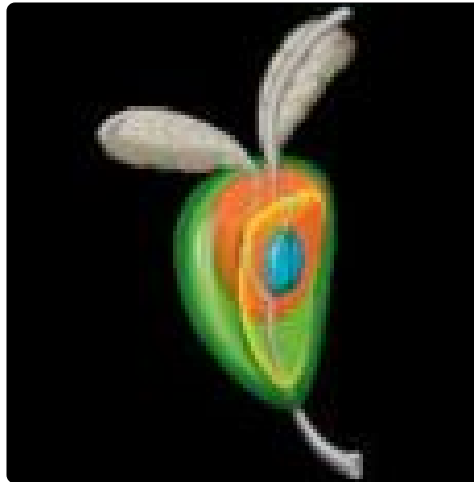


The prostate gland  is inferior to the bladder , superior to the pelvic floor musculature , and anterior to the rectum . Urine flows to the prostatic urethra, starting at the bladder neck .

Gross Specimen

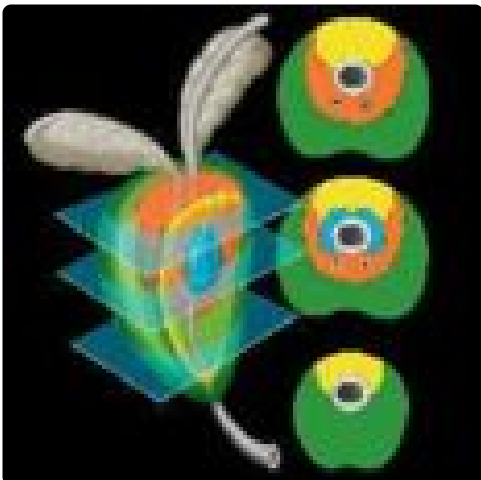


Anatomic Zones of Prostate

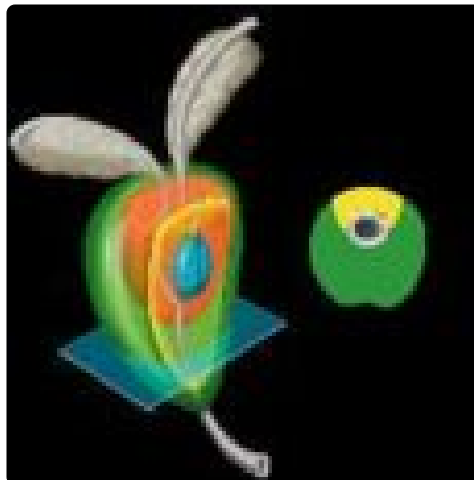


(Left) This gross radical prostatectomy specimen shows anterior, posterior, and posterolateral views. The paired seminal vesicles and vas deferentia are seen posteriorly and superiorly (probe through prostatic urethra). (Courtesy G. Paner, MD.) (Right) The central zone is an inverted pyramid shape at the superior aspect (orange), the transition zone is in the periurethral region (blue), and the surrounding tissue is the peripheral zone (green). Yellow represents the anterior fibromuscular stroma.

Anatomic Zones of Prostate

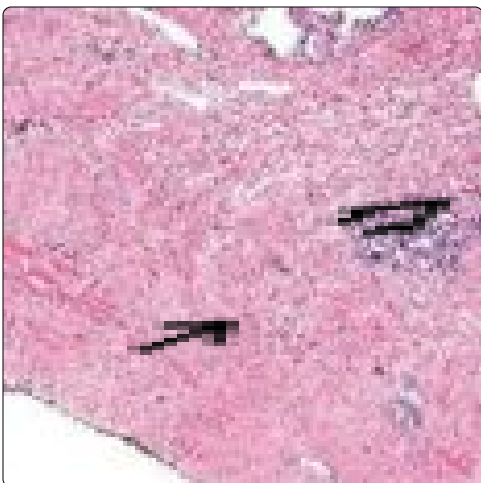


Apical Cross Section of Prostate

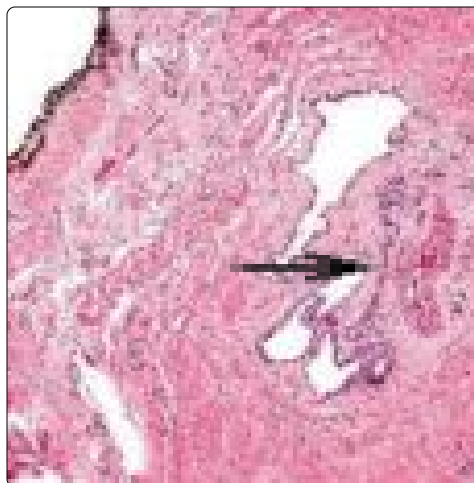


(Left) The microscopic anatomy of the prostate varies from superior to inferior, depending on the anatomic zones that are present in a given cross section level. (Right) Apical cross sections of the prostate gland typically show peripheral zone prostatic tissue and anterior fibromuscular stroma. In severe benign prostatic hyperplasia, the transition zone glands and stroma may extend inferiorly into the apical tissues.

Apical Prostate



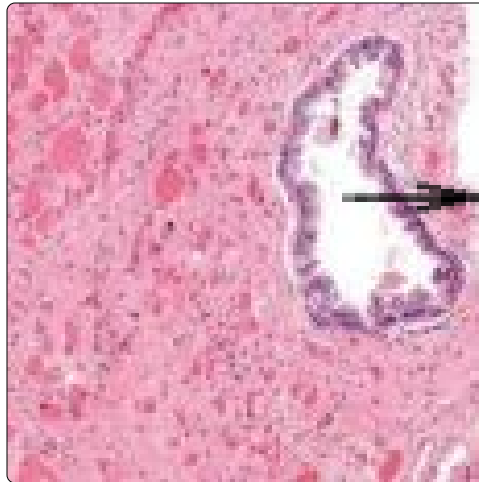
Skeletal Muscle in Apical Prostate



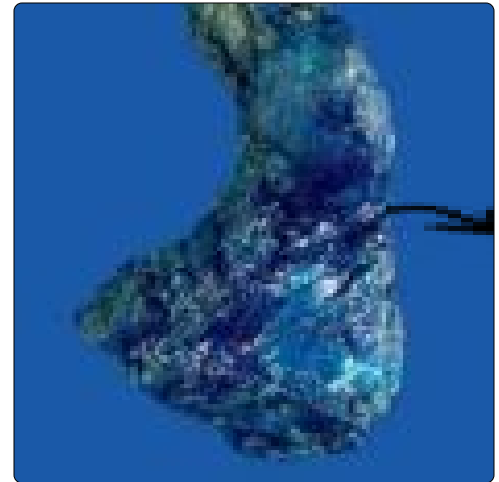
(Left) The apical prostate typically consists of peripheral zone prostatic glands, fibromuscular stroma, and admixed skeletal muscle from the pelvic floor &/or anterior soft tissues. (Right) The benign prostate glands of the apex are often intermixed with scattered skeletal muscle fibers; therefore, skeletal muscle fibers should not be considered to be extraprostatic tissue in the apex.

Apical Prostate Glands

(Left) This image shows the benign prostate gland  completely surrounded by skeletal muscle . This underscores the fact that skeletal muscle is often intraprostatic in histologic sections taken from the prostatic apex. (Right) This lateral view of an inked gross prostate specimen can be oriented by the following landmarks: Posterior seminal vesicles , base , and apex . When a thick section of the apex is amputated, the most anterior aspect  may contain adipose tissue from the anterior stroma.

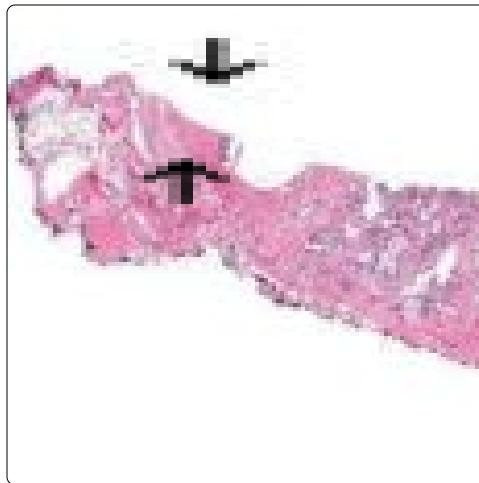


Anteriormost Aspect of Apex

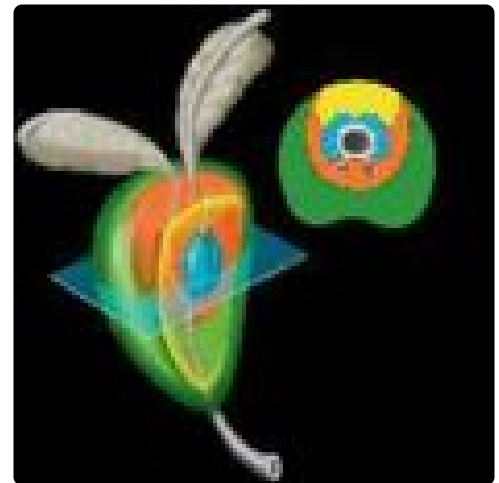


Prostatic Apex

(Left) In this single section from the prostatic apex, which was serially sectioned in an anterior/posterior plane after amputation, adipose tissue is identified anteriorly . (Right) A mid level macroscopic cross section of the prostate gland typically contains peripheral zone (green) and transition zone (blue) tissue, but the central zone (orange) may also be seen depending on the exact level of the section and anatomic variation between patients.

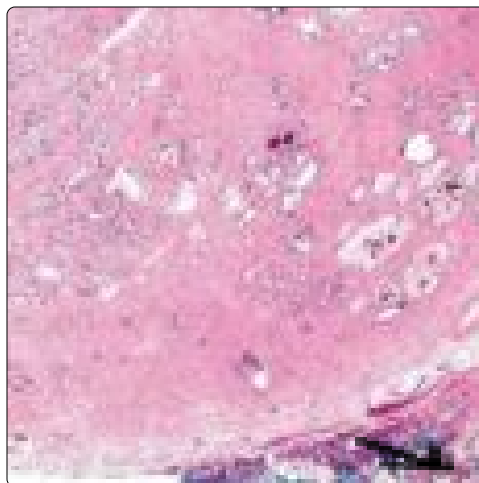


Mid Level Section of Prostate

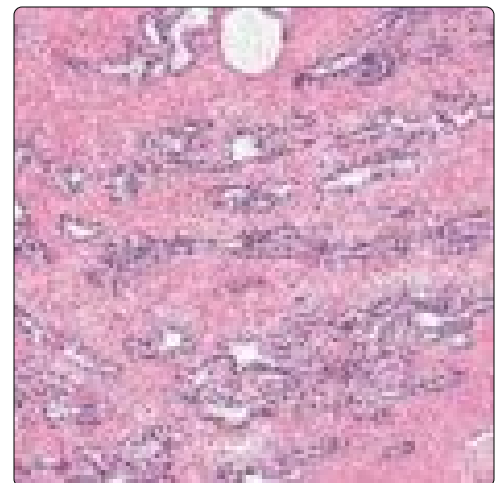


Mid Prostate Peripheral Zone

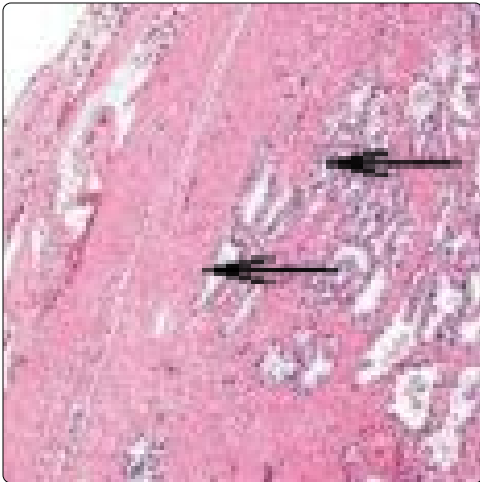
(Left) This normal posterolateral quarter section from the mid prostate shows a typical low-power distribution of the prostate glands. The outer contour of the prostate is highlighted . (Right) The benign prostate glands of the peripheral zone at the mid level have a variable orientation. A linear arrangement of the glands  that runs parallel to the outer contour of the prostate gland is common.



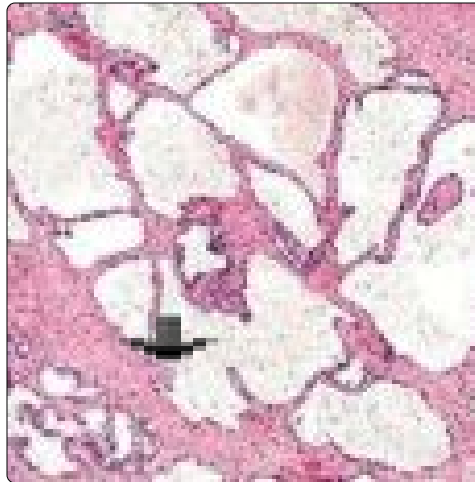
Mid Prostate Peripheral Zone Glands



Mid Prostate Peripheral Zone Boundary

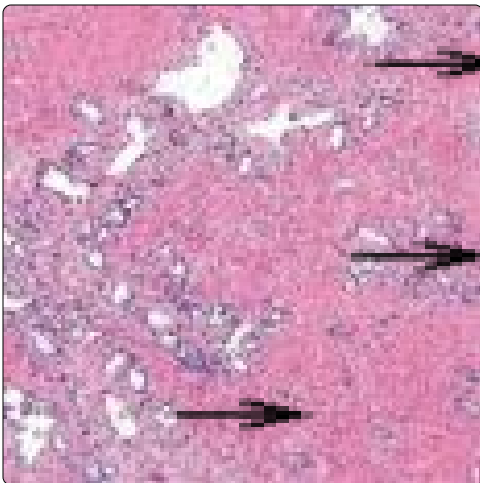


Mid Prostate Peripheral Zone Atrophy

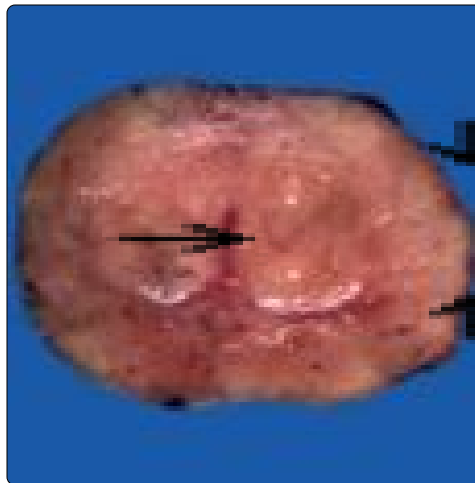


(Left) This mid level section shows the boundary of the extraprostatic tissue at the posterolateral aspect, which is defined as the level of the adipose tissue [1]. The benign glands show a typical caliber and distribution. (Right) Normal prostate glands may show marked dilatation and a nearly complete loss of cytoplasm (referred to as cystic atrophy). The large size of the glands is more appreciable by comparison to adjacent normal-sized glands [2]. In some patients, this cystic change may fill the peripheral zone.

Mid Prostate Peripheral Zone Variation

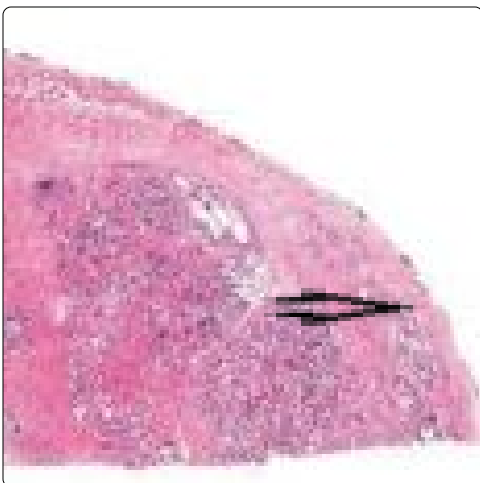


Mid Prostate Transition Zone

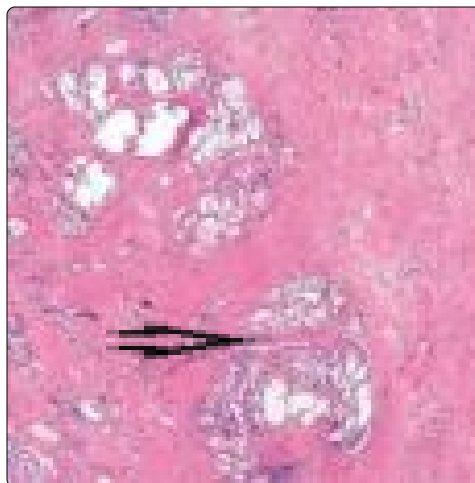


(Left) In this example of the mid peripheral zone, there is more heterogeneous gland morphology with an admixture of both dilated cystic [1] and smaller round glands [2]. (Right) This gross macrosection of the mid prostate is from a patient with benign prostatic hyperplasia. The transition zone, which surrounds the urethra [3], is prominent [4].

Transition Zone Boundary With Peripheral Zone



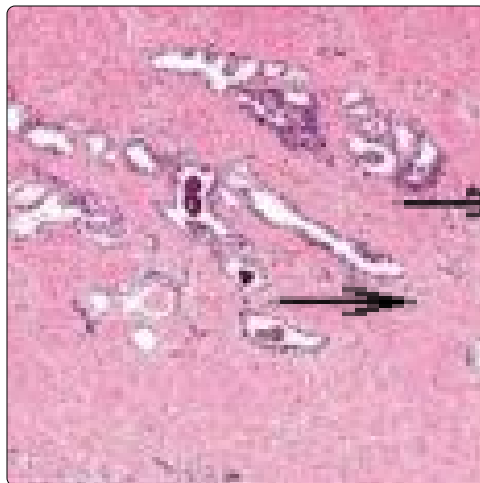
Transition Zone



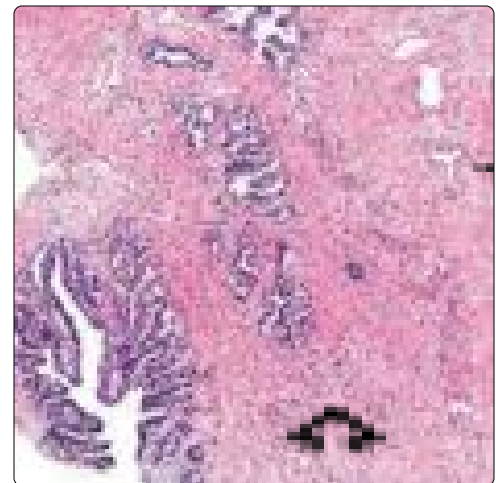
(Left) A lobular arrangement of the prostate glands [1] is typical of the transition zone. The boundary [2] with the peripheral zone is fairly distinct in this section. The peripheral zone glands [3] have a more linear arrangement that follows the outer contour of the prostate gland [4]. (Right) The transition zone is commonly composed of circumscribed nodules of glands with an increased epithelial:stromal ratio [5]. There is architectural variation with other foci showing fewer glands of less pronounced organization [6].

(Left) In the transition zone, the density of the glands and the nodular architecture vary considerably depending on the presence of benign prostatic hyperplasia. Intraluminal corpora amylacea are common. **(Right)** The prostatic glands directly underlying the prostatic urethra are less lobular in architecture and are more commonly lined by an admixture of prostatic epithelium and urothelium.

Transition Zone Glands

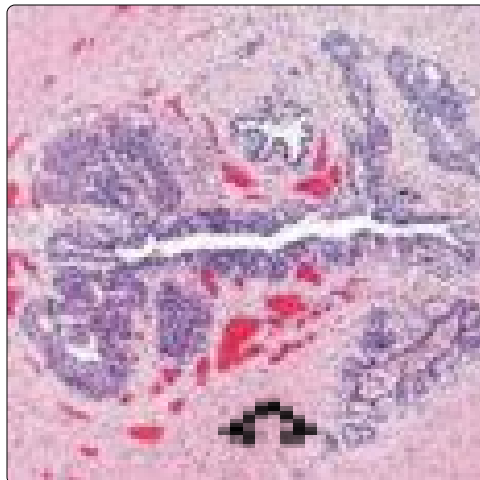


Periurethral Prostate

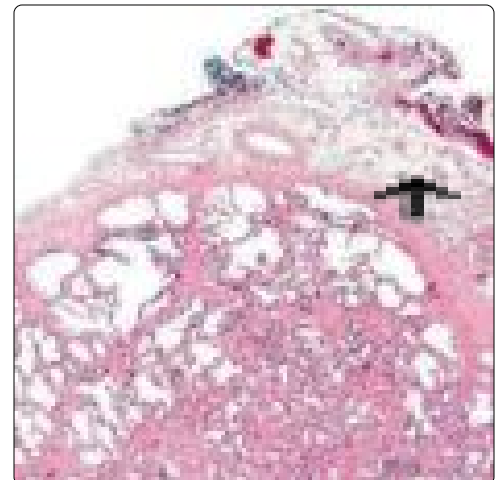


(Left) The lumen of the prostatic urethra is lined predominantly by urothelium. The surrounding prostatic glands are variably lined by a stratified urothelium, a feature that has been referred to as transitional (urothelial) metaplasia. **(Right)** Often, there is a distinct junction between the prostate and the extraprostatic tissue at the posterolateral aspect of the prostate, which is the location of the neurovascular bundles. Even at low power, the sharp line at the start of the extraprostatic adipose tissue is evident.

Periurethral Area

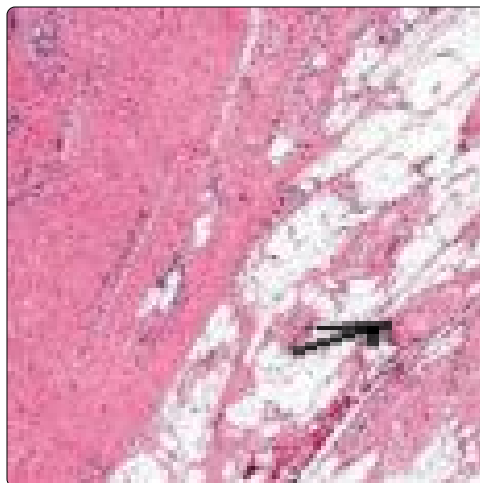


Extraprostatic Tissue: Posterolateral

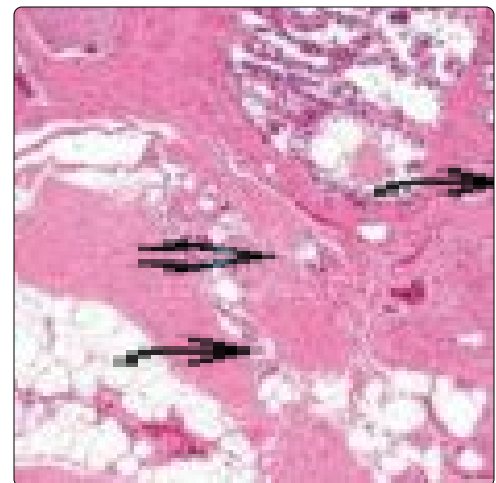


(Left) The extraprostatic tissue begins at the plane with the 1st identifiable adipose tissue. This is the most reproducible landmark for the prostatic/extraprostatic junction. **(Right)** An admixture of fibrous tissue, adipocytes, large-caliber blood vessels, and peripheral nerves are commonly seen in the extraprostatic tissues, particularly in the posterolateral region.

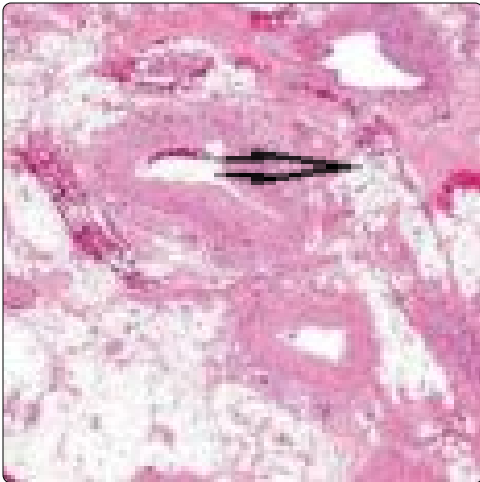
Extraprostatic Tissue: Adipose



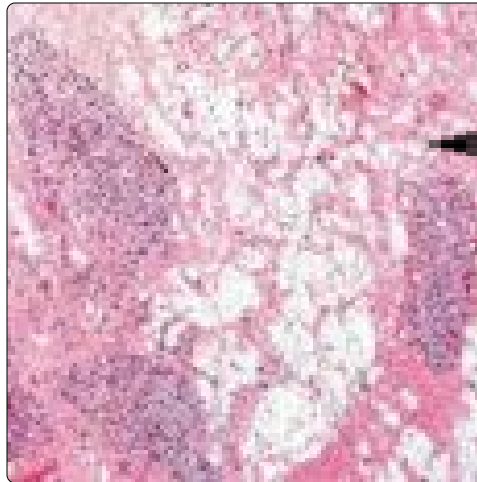
Extraprostatic Tissue Composition



Extraprostatic Tissue: Vessels and Nerves

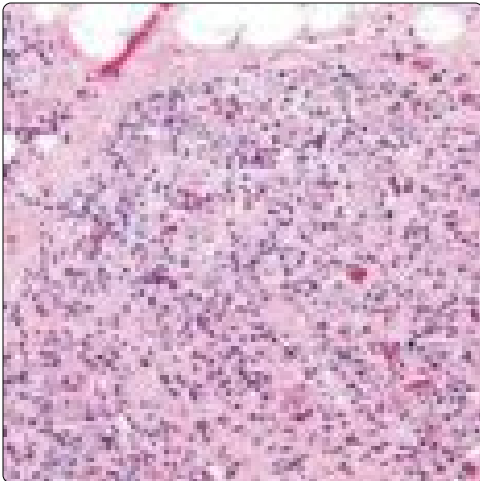


Extraprostatic Tissue: Ganglia



(Left) Large-caliber blood vessels and peripheral nerves are often prominent in the posterolateral extraprostatic soft tissue when the neurovascular bundle is not surgically spared. **(Right)** Collections of ganglion cells are commonly found admixed with adipose tissue in the extraprostatic tissues. They may mimic high-grade prostatic carcinoma because of the sheet-like arrangement and the presence of nucleoli at higher power magnification.

Ganglion

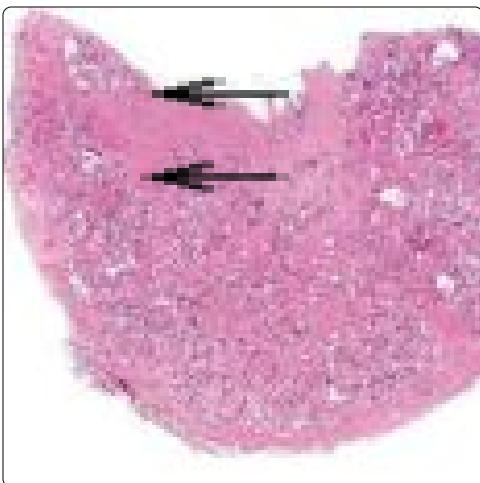


Anterior Soft Tissue

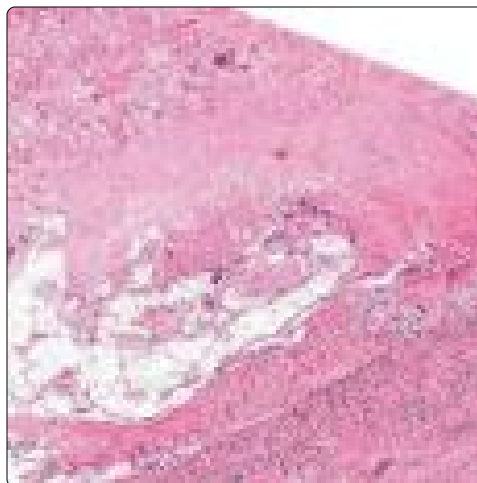


(Left) Ganglia are common in the periprostatic tissues. They are composed of collections of ganglion cells, which are characterized by large nuclei, prominent nucleoli, and cytoplasmic granules. **(Right)** This macroscopic cross section from a radical prostatectomy specimen is oriented with the anterior soft tissue facing upward. Knowledge of the composition of these anterior tissues and their boundary with the prostatic tissues can greatly facilitate cancer staging. (Courtesy G. Paner, MD.)

Anterior Prostate



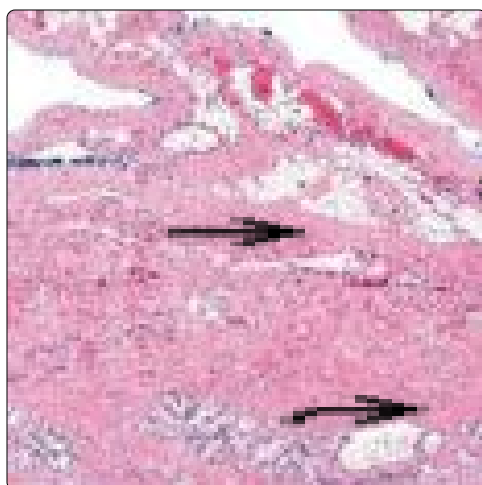
Anterior Prostate Boundary



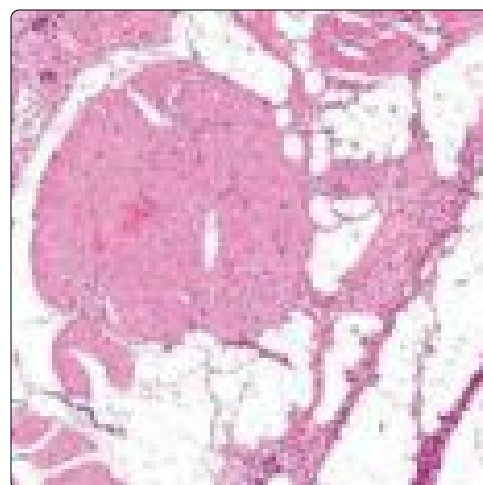
(Left) This image of a 1/2 macrosection from a radical prostatectomy specimen shows the anterior stromal tissues, which are identified by skeletal muscle fibers. **(Right)** In the anterior zone of the prostate, the extraprostatic tissue is defined by the plane in which the adipose tissue is first seen. Skeletal muscle may be seen on both the intra- and the extraprostatic sides of this line.

Anterior Prostate: Skeletal Muscle

(Left) This example of the anterior region highlights the presence of intraprostatic skeletal muscle  admixed with benign glands . The extraprostatic tissue would begin at the adipose tissue  in this section. (Right) Large, compact smooth muscle bundles within the anterior extraprostatic soft tissue may become continuous with the bladder neck musculature as one moves superiorly. Large amputations of the bladder base that extend too anteriorly may contain these bundles and, occasionally, some skeletal muscle fibers.

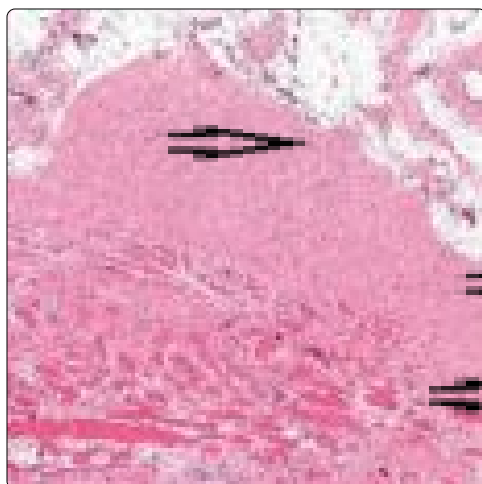


Anterior Prostate: Smooth Muscle

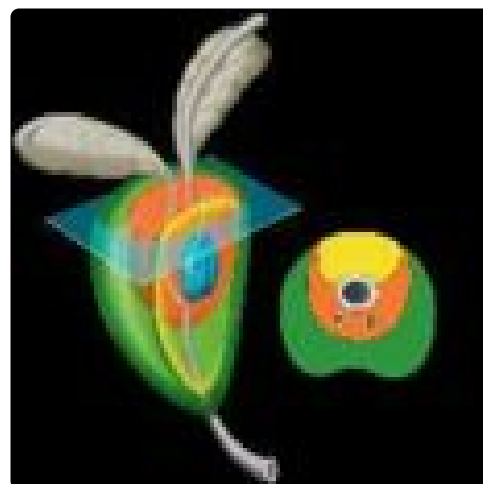


Anterior Prostate: Mixture of Muscle Fibers

(Left) In the more superior regions of the prostate, the anterior soft tissue contains large compact bundles of smooth muscle  that may be intermixed with skeletal muscle . (Right) The central zone glands are typically seen in sections from the bladder base. The central zone comprises a larger proportion of each macrosection as one moves superiorly. In this schematic, the paired ejaculatory ducts , which course toward the urethra , are also seen within the central zone.

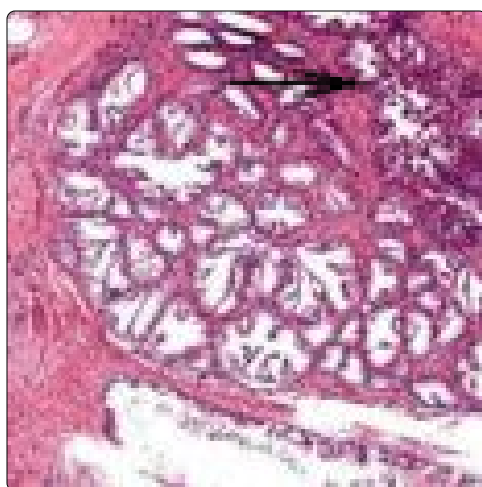


Superior Prostate Cross Section

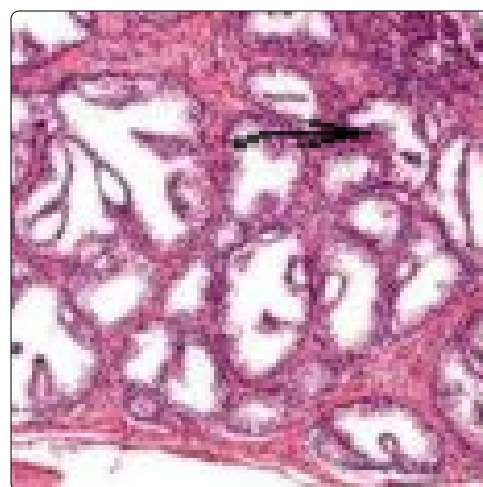


Central Zone

(Left) This section shows the unique morphology of the central zone glands, which are characterized by epithelial bridging  and intraluminal papillary tufts . (Right) This example of central zone glands shows the characteristic epithelial bridging that forms so-called lacunar spaces  and the increased number of elongated intraluminal papillary excrescences . These features may be mistaken for prostatic neoplasia due to the degree of epithelial complexity.



Central Zone Glands



Central Zone: Cribriform Morphology

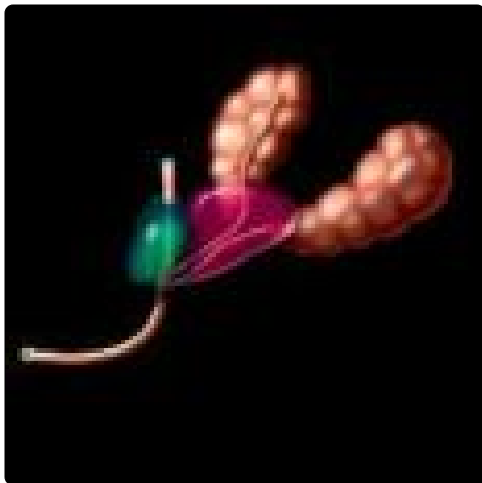


Central Zone: Papillary Infoldings

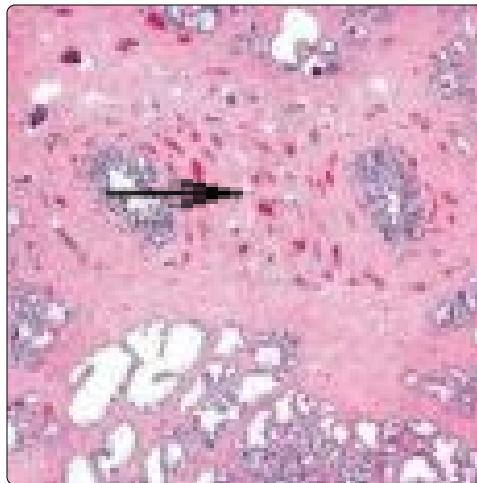


(Left) The central zone of the prostate gland may occasionally have a focal cribriform appearance that may mimic high-grade prostatic intraepithelial neoplasia or other intraductal proliferations. Typical epithelial bridging with lacunae is also seen. (Right) This benign prostate gland was seen in a transrectal needle core biopsy labeled "bladder base." The papillary excrescences and lacunae are typical histologic features of the central zone.

Course of Ejaculatory Ducts

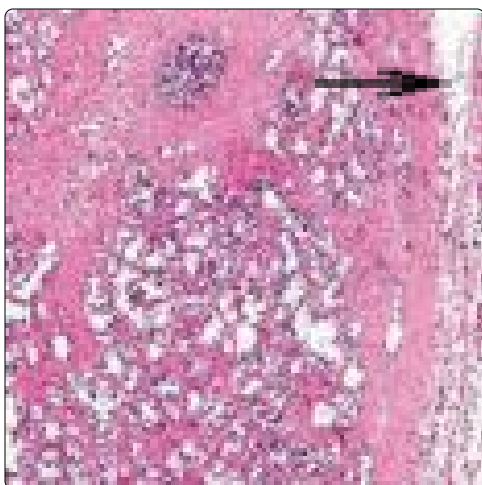


Paired Ejaculatory Ducts



(Left) This schematic shows the relationship of the seminal vesicles to the ejaculatory ducts and urethra. The paired ejaculatory ducts course through the central zone to the urethra, where the ejaculatory contents are emptied. (Right) The paired ejaculatory ducts are seen in this section from a radical prostatectomy specimen. These ducts are typically surrounded by loose fibrous stroma containing numerous thin-walled blood vessels.

Ejaculatory Duct



Ejaculatory Duct



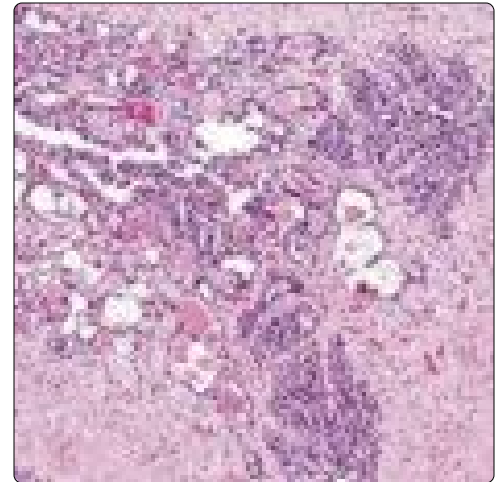
(Left) The ejaculatory duct is readily appreciable on low-power evaluation due to the central lumen with complex intraluminal epithelial infolding, which is very similar to that of the seminal vesicle. Note the surrounding rim of loose fibrous stroma. (Right) As shown here, each ejaculatory duct typically has a complex intraluminal papillary architecture. The rim of fibrous tissue with associated thin-walled blood vessels is also a characteristic feature.

(Left) In this low-power photomicrograph of the prostatic urethra near the verumontanum, the paired ejaculatory ducts end their course through the central zone and begin to connect with the prostatic urethral lumen, where their ejaculatory products are emptied. **(Right)** At higher power, the typical complex papillary infolding of the ejaculatory ducts is appreciated. The ejaculatory duct epithelium gradually admixes with prostatic secretory and urothelial epithelium.

Paired Ejaculatory Ducts Near Urethra

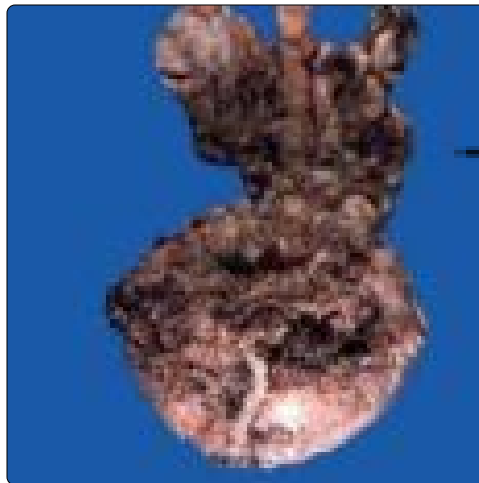


Paired Ejaculatory Ducts

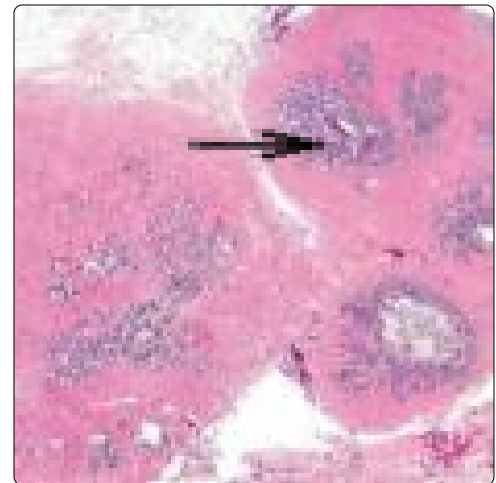


(Left) This prostate gland gross specimen may be oriented by the seminal vesicles, the urethral opening at the bladder base, and the apex. The paired seminal vesicles are attached to the prostate at the superior and posterior-most aspect of the gland. **(Right)** On low-power magnification, the tortuosity of the seminal vesicle lumen may be appreciated. The surrounding thick layer of smooth muscle is a characteristic feature of the seminal vesicle as well.

Seminal Vesicles Attached to Prostate



Seminal Vesicle

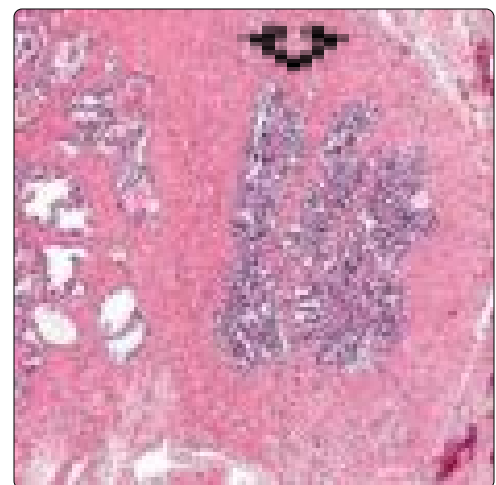


(Left) Invaginations of seminal vesicle epithelium may form small glandular structures surrounding the central lumen. This finding may occasionally mimic prostatic carcinoma glands. High-power examination will confirm seminal vesicle-type epithelium. **(Right)** This is the normal junction between the seminal vesicle and the prostate gland. Because of the very complex infolding of the seminal vesicle epithelium, the central lumen may be difficult to appreciate in some sections.

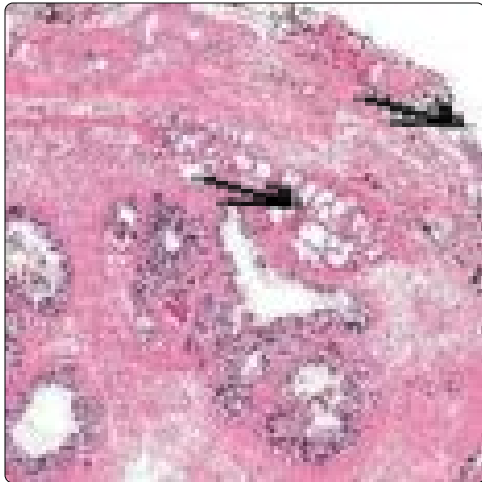
Seminal Vesicle Glands



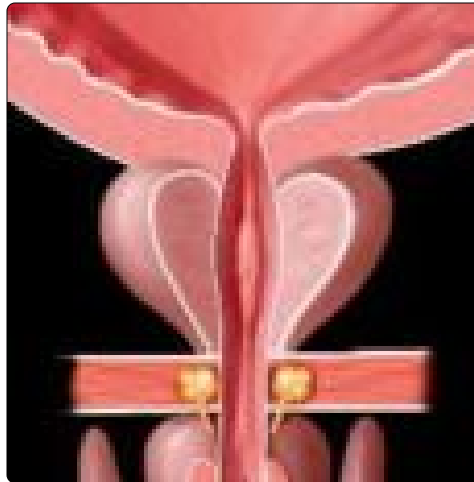
Seminal Vesicle Adjacent to Prostate



Seminal Vesicle With Prostatic Tissue in Wall

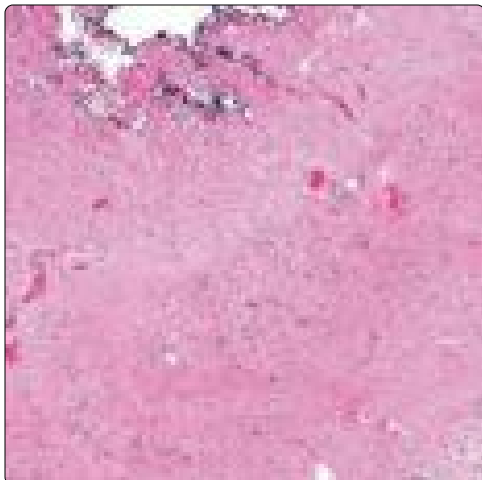


Bladder Neck

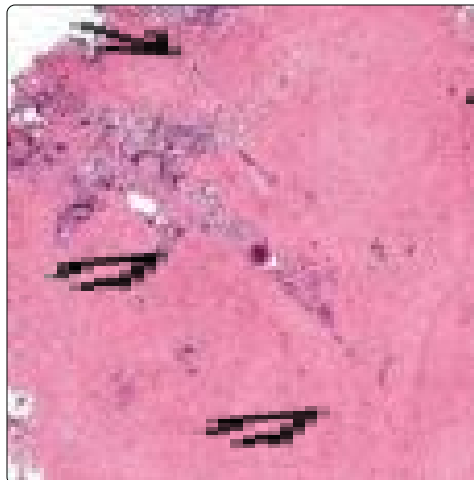


(Left) In some prostates, collections of benign prostate glands [red box] may be found in the wall of the seminal vesicle [blue box]. This may be due to the proposed shared embryogenesis of the seminal vesicle and central zone of the prostate. (Right) The resection plane of a prostatectomy procedure determines the type of tissue seen in the bladder base margin. A plane through the true bladder neck [blue box] will produce smooth muscle, while a more inferior plane [orange box] yields prostate glands. This plane may be distorted by benign prostatic hyperplasia.

True Bladder Neck

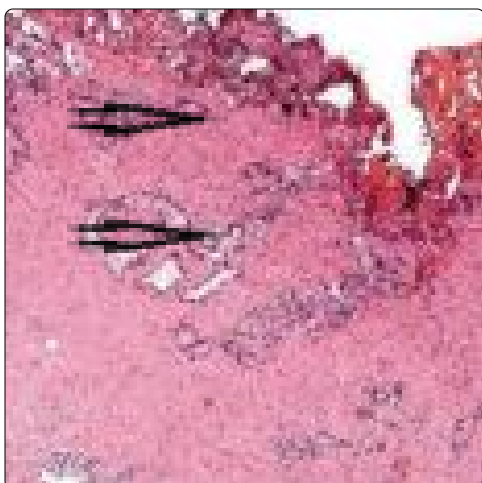


Bladder Base

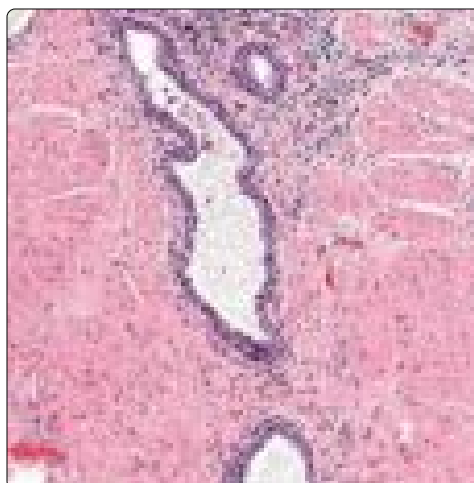


(Left) This tissue taken as a margin from the bladder base of a prostatectomy specimen shows thick smooth muscle without prostate glands. This histology is diagnostic of microscopic bladder neck tissue. (Right) Taken at the bladder base region (superior aspect) of a radical prostatectomy, this tissue should not be regarded as bladder neck for staging purposes, because of the admixed benign prostate glands [red box] (the true margin is inked [blue box]).

Prostate Tissue Near Bladder Base



Prostate Glands Near Bladder Base



(Left) This tissue from the bladder base of a radical prostatectomy specimen also contains admixed benign prostate glands [red box] and should therefore not be regarded as bladder neck musculature. (Right) On high-power magnification, these glands in the bladder base margin tissue are prostatic in appearance. This represents prostatic tissue in the bladder base due to a resection plane inferior to the true bladder neck. Carcinoma involving such tissue should be staged as intraprostatic.

Prostate: Benign Glandular and Stromal Histology

PROSTATE

Macroscopic Anatomy

- Prostate gland
 - Located inferior to urinary bladder, superior to pelvic floor, and anterior to rectum
- Seminal vesicles
 - Paired glandular structures located in superior/posterior region of prostate
 - Extend superiorly along posterior aspect of bladder
- Cowper (bulbourethral) glands
 - Paired glandular structures located just distal to prostate gland in periurethral region

Microscopic Anatomy

- Prostate glands are composed of 2 cell layers
 - Secretory cells
 - Luminal-facing cells with clear to eosinophilic, sometimes vacuolated, cytoplasm
 - IHC: PSA(+), PSAP(+), p63(-), high molecular weight (HMW) cytokeratin (-)
 - Basal cells
 - Located adjacent to underlying stroma
 - High nuclear:cytoplasmic ratios and small nucleoli
 - IHC: p63(+), HMW cytokeratin (+), PSA(-), PSAP(-)
- Prostatic stroma
 - Normal stroma has fibromuscular appearance
 - Cytologically bland spindled cells with varying collagenous to myxoid stroma and associated small blood vessels in transition zone with hyperplasia
- Periurethral tissue
 - Complex glandular tissue with admixture of prostatic secretory epithelium and urothelium
- Seminal vesicle and ejaculatory duct
 - Both feature scattered pleomorphic nuclei and intracytoplasmic lipochrome pigment
 - Seminal vesicle is surrounded by dense smooth muscle; ejaculatory duct is surrounded by loose collagenous stroma

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Benign prostatic hyperplasia and glandular atrophy are more common with increasing age

Hyperplasia

- Benign prostatic glandular and stromal hyperplasia in periurethral transition zone
 - Stromal hyperplasia with cellular nodules of benign spindled cells
 - Lobular glandular hyperplasia with increased gland density
- Basal cell hyperplasia
 - Cellular aggregates of basophilic cells at periphery of prostate glands
 - Basal cells may entirely fill glands
 - Fairly common in transition zone
- Clear cell cribriform hyperplasia

Metaplasia

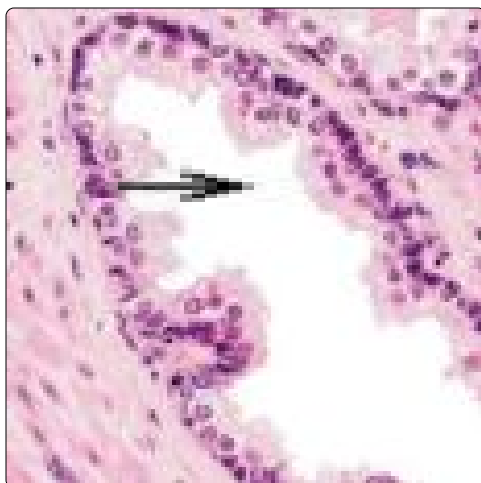
- Transitional (urothelial)
 - Extension of urothelium into prostatic glands, typically of transition zone
 - Stratified epithelium with longitudinal nuclear grooves
- Mucinous metaplasia
 - Rare cells with intracytoplasmic mucin may be present

Pitfalls and Artifacts

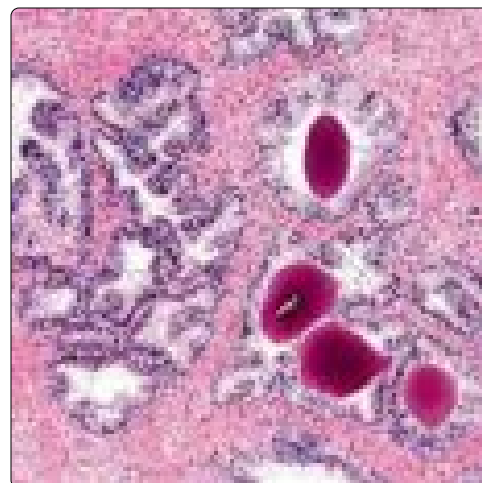
- Normal variations in prostate may mimic malignancy
 - Glandular atrophy
 - Includes simple, cystic, partial, and sclerotic types
 - Adenosis, sclerosing adenosis
 - Inflammatory changes
 - Mesonephric remnants
 - Nephrogenic adenoma
- Radiation atypia in benign glands
 - Glands with scattered pleomorphic cells
 - Atrophic eosinophilic cytoplasm
- Benign lesions typically show at least focally retained basal cell layer

Prostatic Glands

(Left) Normal glands typically show an undulating epithelial/luminal interface. The secretory cells  have pale, sometimes vacuolated, cytoplasm. The basal cell layer  is prominent in this example. (Right) Normal prostate glands are typically of medium to large caliber and are aggregated into loose groups surrounded by fibromuscular stroma. Corpora amylacea  are common in the lumina of benign glands.



Glands and Stroma



Normal Glands

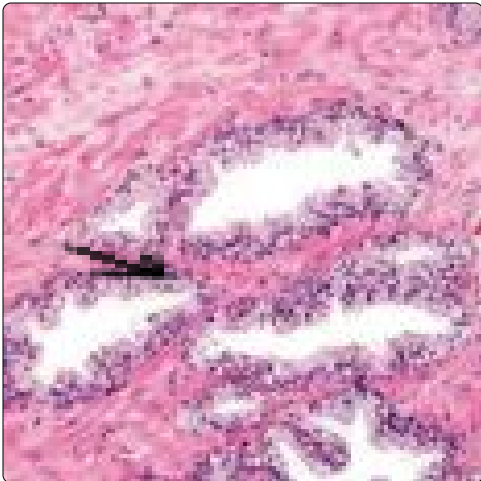


Architecture

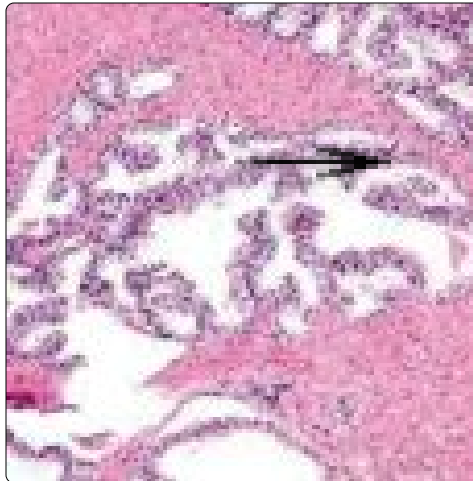


(Left) At low power, benign nonatrophic prostate glands are typically of similar caliber and are evenly distributed or arranged into vague aggregates. Glands may be rounded or elongated, arranged in a branching configuration. Elongated branching glands are sometimes referred to as "ducts." **(Right)** This benign gland demonstrates the typical undulating or tufted secretory cell layer. Glands of smaller caliber may be seen at the edge of larger glands due to tangential sectioning.

Undulating Contours

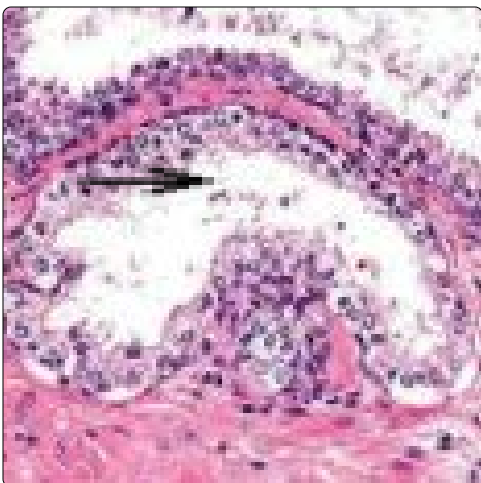


Papillary Excrescences

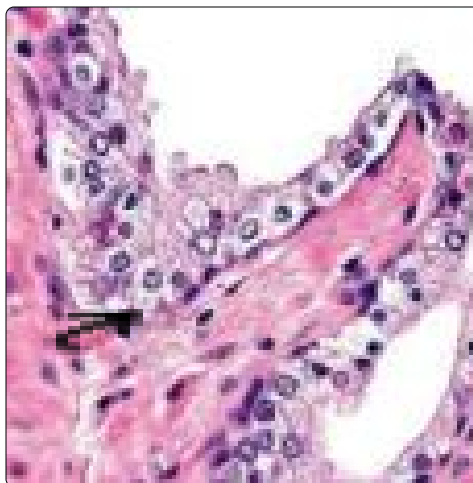


(Left) Some benign glands are of a smaller caliber and have less intraluminal complexity; however, the luminal surface still has some subtle undulation. In contrast, prostatic adenocarcinoma glands often have a very sharply defined luminal surface. **(Right)** Benign prostate glands may have larger intraluminal papillary excrescences, particularly in certain regions, such as the prostatic central zone. These large luminal infoldings are less common in carcinoma glands.

Nuclei



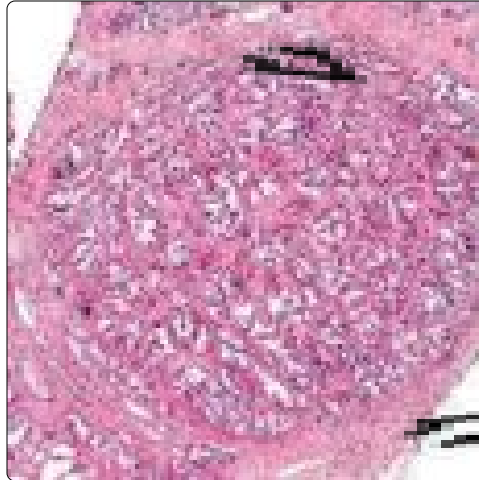
Secretory and Basal Cells



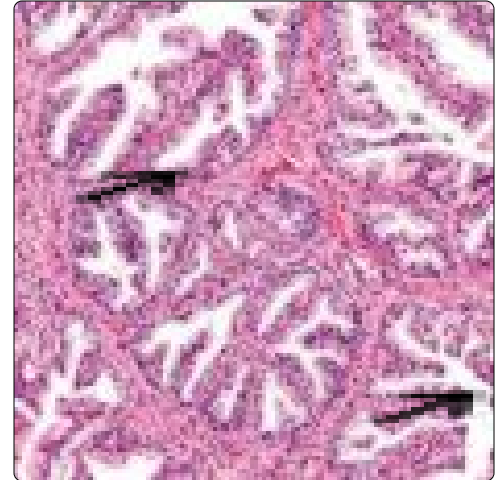
(Left) In normal glands, the distance between a nucleus and the basement membrane is variable. Many carcinomas are characterized by nuclei that are more uniformly basal in location. **(Right)** The pale to clear cytoplasm of the secretory cells in this gland is typical, as are the round nuclei that vary in distance from the basement membrane. A flattened basal cell layer is variably seen in benign glands, which may resemble adjacent stromal cells by routine H&E morphology.

BPH

(Left) In this transurethral prostate resection specimen, the nodular, circumscribed arrangement of these glands is characteristic of benign prostatic hyperplasia (BPH). Even on low-power examination, some intraluminal infolding within the glands is appreciable. **(Right)** This degree of intraluminal infolding with resultant papillary excrescences is typical of the benign prostatic glands often seen in glandular hyperplasia.

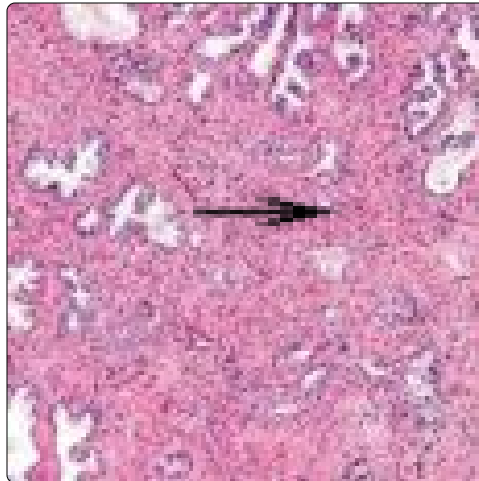


BPH: Intraluminal Infolding



BPH: Hyperplastic Glands and Stroma

(Left) In this example of BPH, there is an equal admixture of benign prostatic glands and prostatic stroma. One may see a varying admixture of glands and stroma within an individual specimen. **(Right)** This well-circumscribed nodule is characteristic of a pure hyperplastic stromal nodule, which is part of the spectrum of BPHs. These nodules are frequent incidental findings in radical prostatectomy specimens.

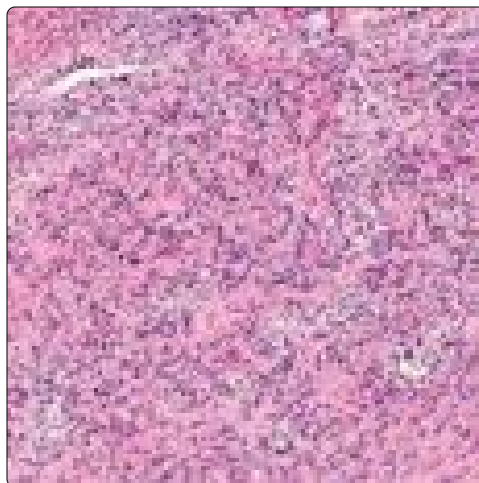


BPH: Nodularity

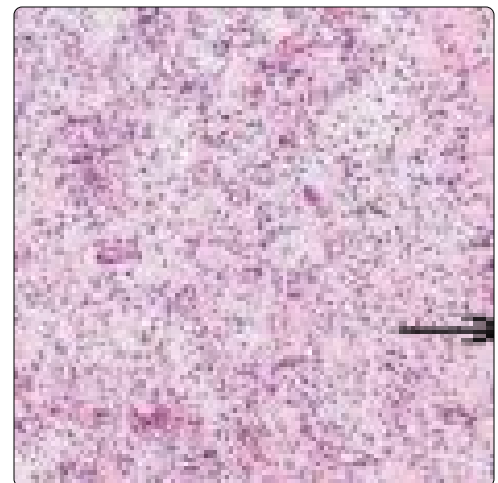


BPH: Stromal Hyperplasia

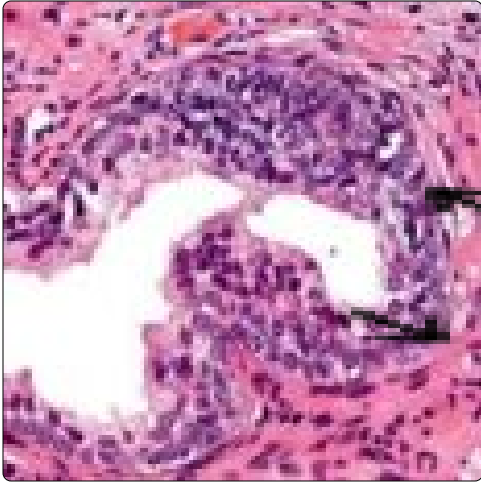
(Left) Cellular foci of pure prostatic stromal cells without glands are seen in areas of nodular stromal hyperplasia. The spindle cells are monomorphic without the degree of nuclear size variation or hyperchromasia often seen in a stromal neoplasm. **(Right)** Small-caliber blood vessels admixed with loose myxoid stroma are characteristic features of a benign hyperplastic stromal nodule.



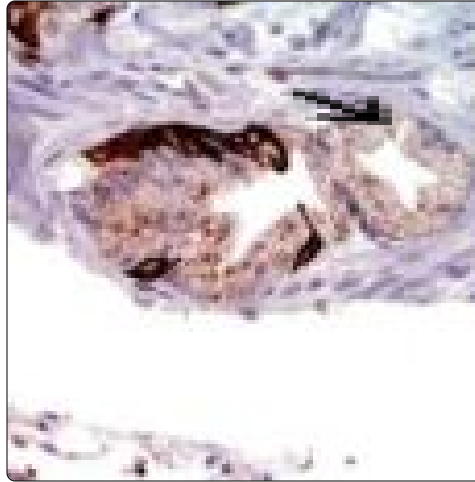
BPH: Small Stromal Blood Vessels



Basal Cell Hyperplasia

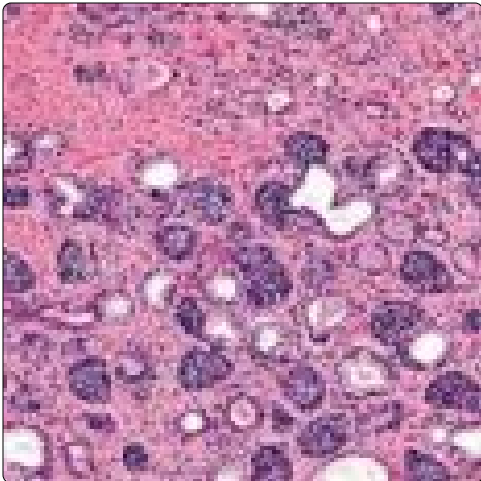


Prominent Basal Cells



(Left) This benign prostate gland with basal cell hyperplasia highlights the distinction between the basal cells and secretory cells. The secretory cells have more abundant cytoplasm and are lumenally oriented. The basal cells form nodular aggregates along the basement membrane side of the gland. **(Right)** The basal cells are highlighted by strong cytoplasmic immunoreactivity for high molecular weight cytokeratin. The secretory cell layer is nonreactive with this antibody.

Basal Cell Hyperplasia in BPH

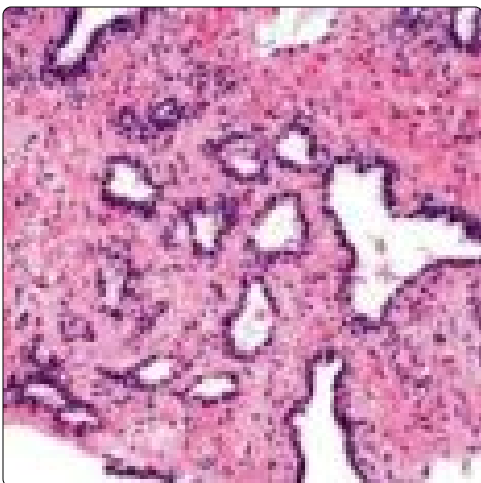


Simple Atrophy



(Left) Multiple glands contain peripheral nodules of basal cells or appear completely filled by dark basophilic cells with a high N:C ratio typical of basal cell hyperplasia. The presence of nests of basal cells within a population of otherwise normal glands can mimic an infiltrative process. **(Right)** In simple prostate gland atrophy, there is a marked loss of cytoplasm in the individual prostatic secretory cells. Frequently, the atrophic glands also have a more angulated shape.

Atrophic Glands



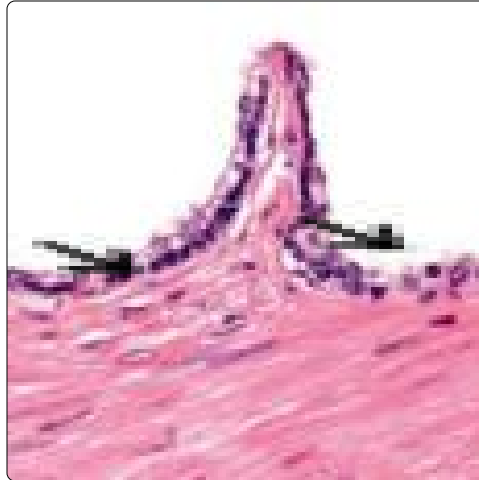
Cystic Atrophy



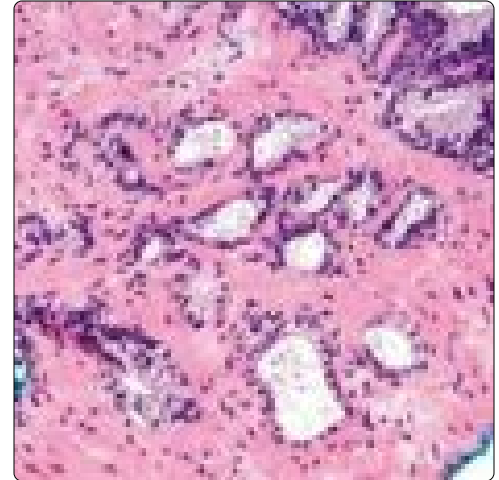
(Left) On higher power magnification, these atrophic glands have a uniform loss of cytoplasm that imparts a very basophilic appearance to the glands. When the benign atrophic glands are crowded, as in this example, they may mimic a low-grade adenocarcinoma. **(Right)** In cystic atrophy, the benign prostatic glands become markedly dilated and often have a rounded shape. Similar to the cells of simple atrophy, the individual prostatic secretory cells have a loss of luminal cytoplasm.

Cystic Atrophy: High Power

(Left) On evaluation of the lining epithelium in a large gland with cystic atrophy, the secretory cells contain no luminal cytoplasm, but there is some residual cytoplasm lateral to the nucleus. A distinct basal cell layer is evident in this example. **(Right)** The small glands of partial atrophy are typically crowded. There is loss of luminal cytoplasm and some gland angulation.

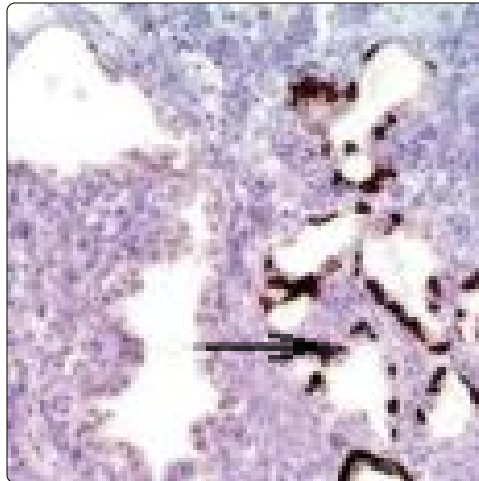


Partial Atrophy

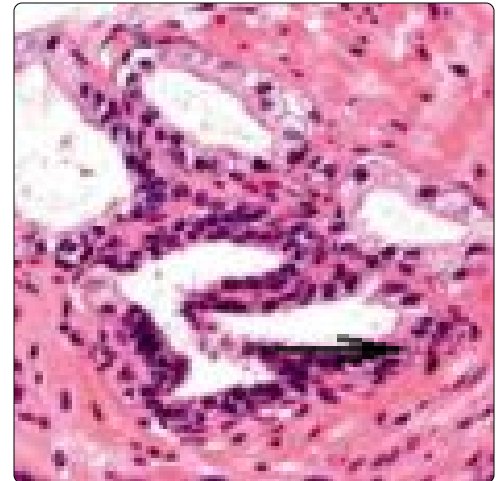


Partial Atrophy: Patchy Basal Cell Layer

(Left) In this high molecular weight cytokeratin antibody immunostain, a patchy discontinuous basal cell layer is highlighted in a subset of the glands, while some of the glands do not show a basal cell layer. This finding is typical of benign partial atrophy. **(Right)** On higher power magnification, the secretory cells in partial atrophy typically have variable areas of complete atrophy and areas with residual cytoplasm located lateral to the nucleus.

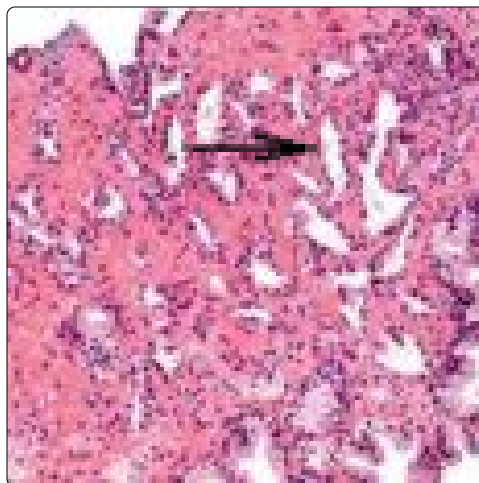


Partial Atrophy: High Power

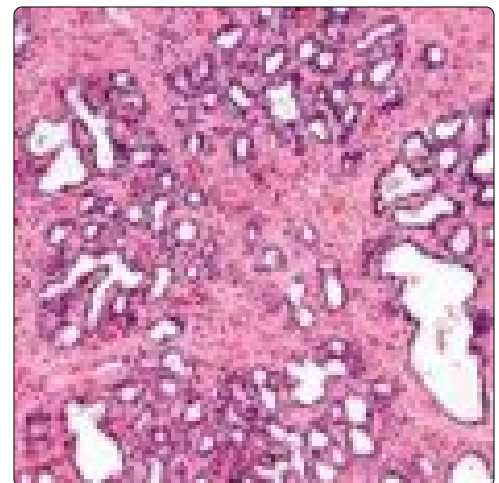


Partial Atrophy: Pseudoinfiltrative Appearance

(Left) In this example of partial atrophy, the glands have a more disorganized pseudoinfiltrative appearance. Atrophic glands often have a more angulated contour. In partial atrophy, the glands have variable amounts of luminal cytoplasm. **(Right)** Postatrophic hyperplasia is characterized by vaguely circumscribed collections of small rounded atrophic glands, often with associated larger caliber ducts/glands.



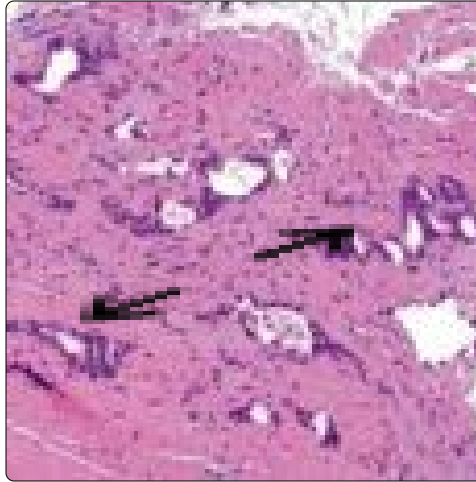
Postatrophic Hyperplasia



Postatrophic Hyperplasia: Prominent Basal Cell Layer

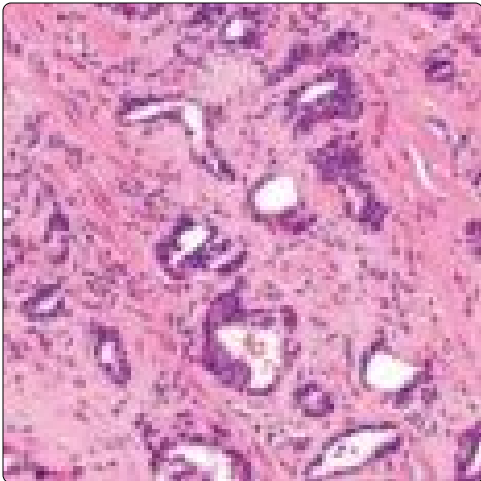


Sclerotic Atrophy

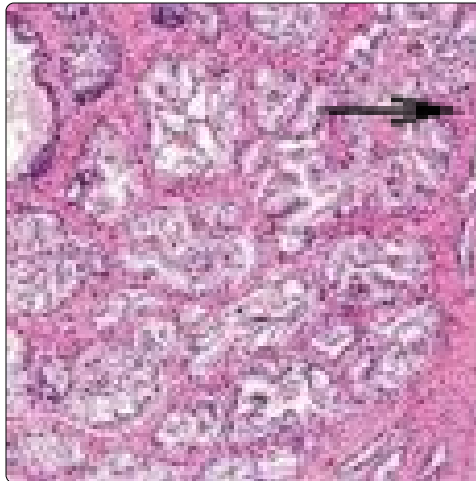


(Left) In contrast to partial atrophy, postatrophic hyperplasia has a very prominent basal cell population that can be easily demonstrated by immunohistochemical markers, such as high-molecular-weight cytokeratin (as shown here), cytokeratin 5/6, or p63. (Right) Sclerotic atrophy is a benign change that can closely mimic invasive carcinoma in other organ systems. The glands are often angulated with pointed ends , and the surrounding stroma appears more densely collagenized.

Stroma in Sclerotic Atrophy

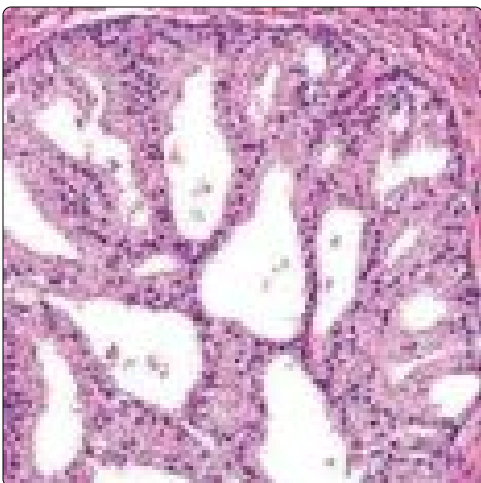


CCCH

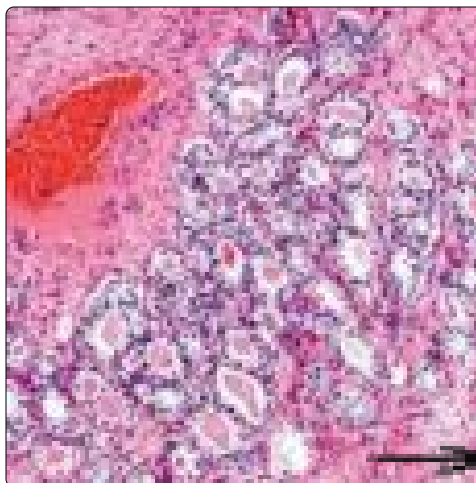


(Left) The surrounding changes seen in sclerotic atrophy resemble the stromal responses that may be seen in invasive carcinomas in other organs; however, this type of stromal response is rare in prostate cancer. (Right) In the transition zone, a benign cribriforming process may be encountered that is termed clear cell cribriform hyperplasia (CCCH). The epithelial cells may bridge the entire gland , a feature that mimics neoplastic intraductal lesions. (Courtesy G. Paner, MD.)

CCCH: Bland Cytologic Features



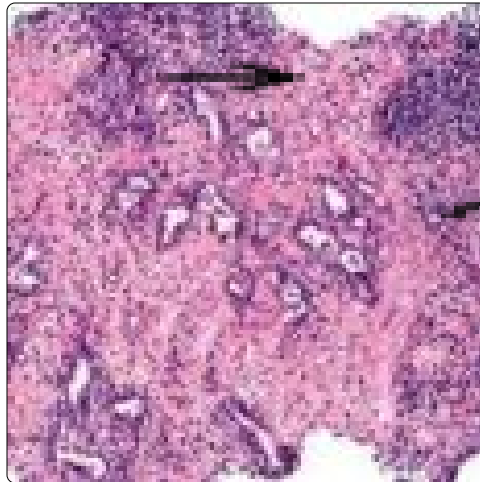
Mesonephric Remnants



(Left) CCCH, unlike intraductal carcinoma or cribriform high-grade prostatic intraepithelial neoplasia, has bland cytologic features without nucleomegaly &/or prominent nucleoli. (Courtesy G. Paner, MD.) (Right) Mesonephric remnants are identical to those commonly found in the uterine cervix. They are characterized by small glandular structures, which typically contain densely eosinophilic, rounded intraluminal concretions .

Inflammatory Changes

(Left) With associated inflammatory infiltrates (both acute  and chronic ) the benign prostatic glands may appear more crowded with a pseudoinfiltrative appearance and darker due to the loss of cytoplasm. **(Right)** In the presence of intraluminal neutrophils , benign prostatic glands may show some degree of nucleomegaly and may contain small nucleoli. These features should not be attributed to neoplasia in this inflammatory setting.

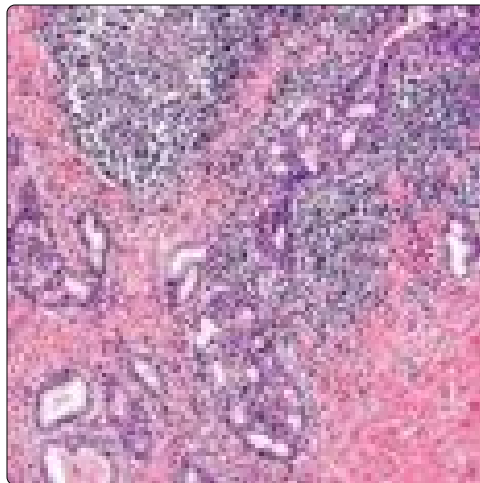


Reactive Glandular Changes

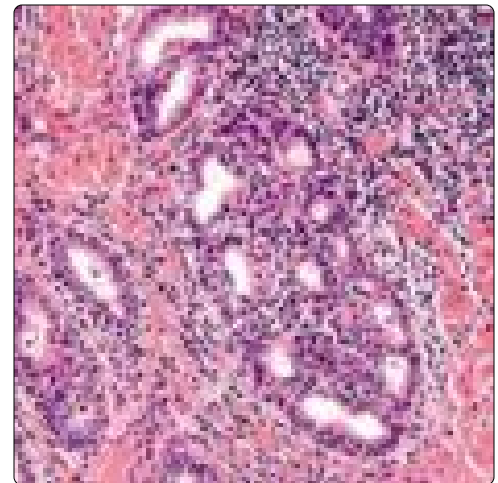


Pseudocribriform Pattern: Low Power

(Left) In prostates that contain a dense chronic inflammatory infiltrate, a pseudocribriform epithelial pattern  may be seen that can mimic a neoplastic process, such as cribriform adenocarcinoma. **(Right)** In this example of benign glands with associated chronic inflammation, the dense intraepithelial lymphocytic infiltrate imparts a pseudocribriform pattern with an increased basophilic appearance.

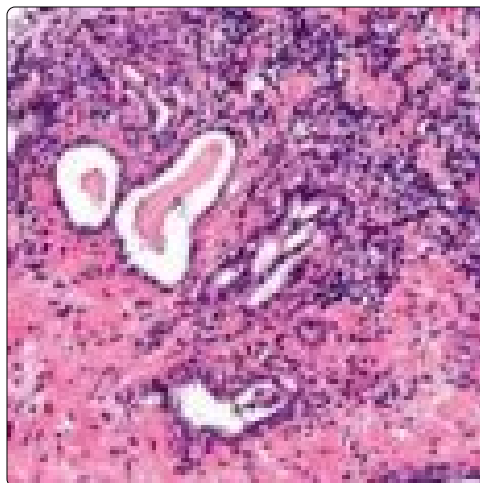


Pseudocribriform Pattern: High Power

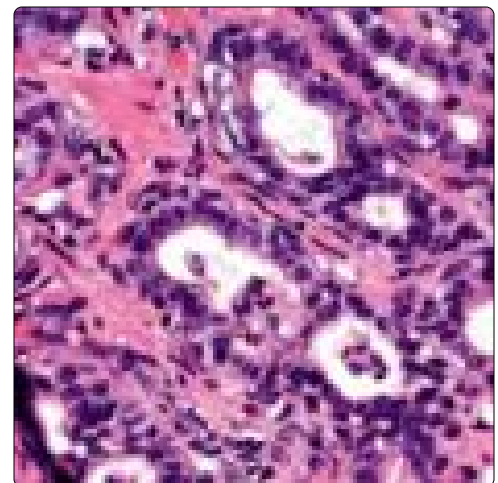


Epithelial Bridging

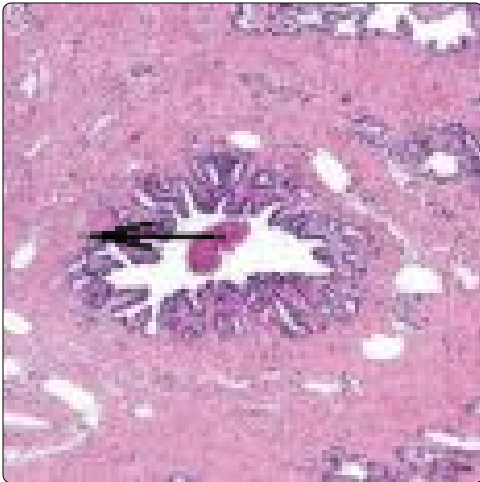
(Left) Benign prostatic glands may contain thin epithelial bridges in the setting of marked inflammatory infiltrates . This is one of the few benign cribriforming lesions. **(Right)** Inflamed benign prostate glands have a more basophilic appearance due to the loss of cytoplasm (higher nuclear:cytoplasmic ratio) and the increased number of cells per unit area. Cytologic features should be interpreted cautiously in the presence of intraluminal acute inflammation .



Inflamed Benign Glands



Ejaculatory Duct



Ejaculatory Duct: Stroma



(Left) In contrast to the dense smooth muscle of the seminal vesicle, the ejaculatory duct is surrounded by a loose collagenous stroma, which often contains delicate, thin-walled blood vessels. This structure is seen within the prostate gland, coursing toward the prostatic urethra. **(Right)** The loose fibrous stroma and dilated, thin-walled vessels in this biopsy tissue are characteristic of the soft tissue surrounding the ejaculatory duct.

Ejaculatory Duct: Features

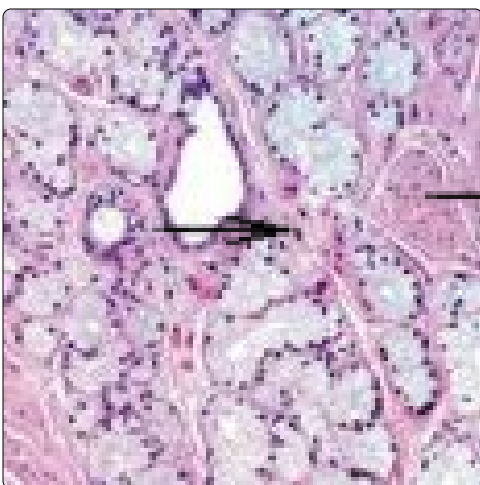


Cowper Gland

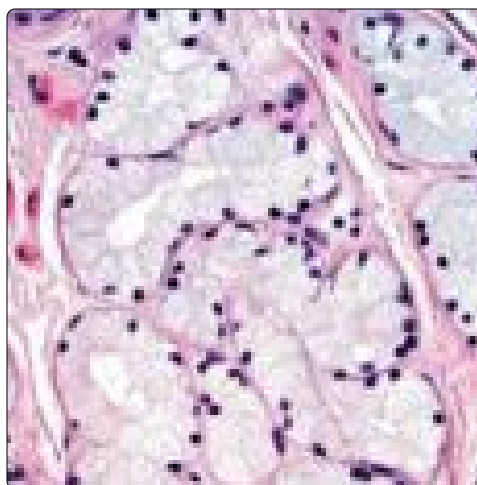


(Left) The epithelium that lines the ejaculatory duct is histologically indistinguishable from that of the seminal vesicle. It contains both the characteristic intracytoplasmic lipochrome pigment and scattered pleomorphic nuclei. **(Right)** The Cowper (or bulbourethral) glands are paired glandular structures located just distal to the prostate gland. They are typically intermixed with the surrounding skeletal muscle.

Cowper Gland: Histology



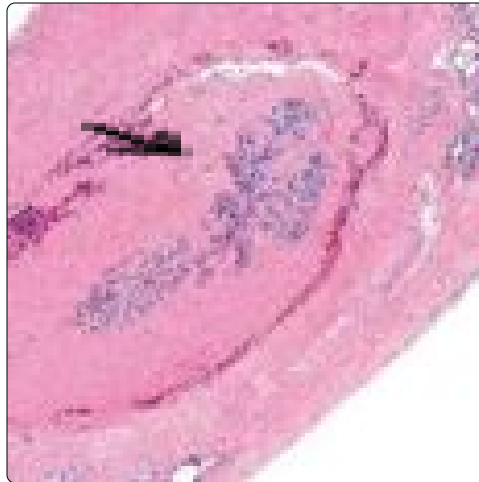
Cowper Gland: Cytology



(Left) The Cowper gland is characterized by a lobular arrangement with a central duct that is surrounded by acini that are filled with cells containing intracytoplasmic mucin. This appearance is similar to minor salivary glands. **(Right)** The small cytologically bland nuclei and the lobular architectural arrangement of a Cowper gland aid in its distinction from prostatic adenocarcinoma. In addition, the mucin associated with prostate cancer is commonly extracellular/intraluminal.

(Left) The seminal vesicles, which are paired structures, are located at the superior most aspect of the prostate gland and are attached posteriorly. Both have a compact layer of dense smooth muscle  that surrounds the luminal epithelium. Adjacent prostatic tissue  is often present in histologic sections. **(Right)** On high-power magnification, the dense smooth muscle wall  of the seminal vesicle is seen underlying the invaginations of epithelium .

Seminal Vesicle: Low Power



Seminal Vesicle: Smooth Muscle Wall

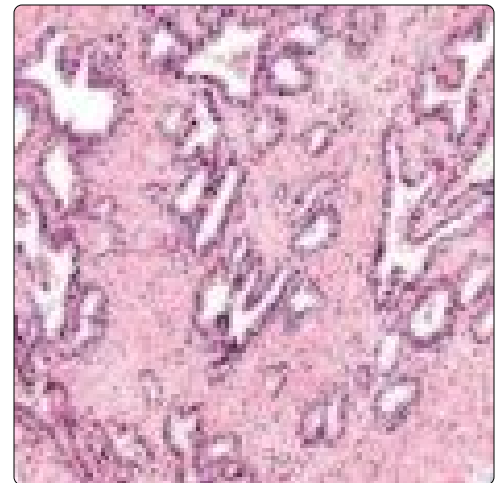


(Left) The seminal vesicle typically has a central lumen (right lower edge of the field)  surrounded by a layer of epithelium with complex invaginations into the surrounding stroma. **(Right)** At the periphery of the large central lumen of the seminal vesicle, the epithelium is frequently tangentially sectioned, imparting the appearance of infiltrative small glands. This appearance may closely mimic prostatic adenocarcinoma.

Seminal Vesicle: Features

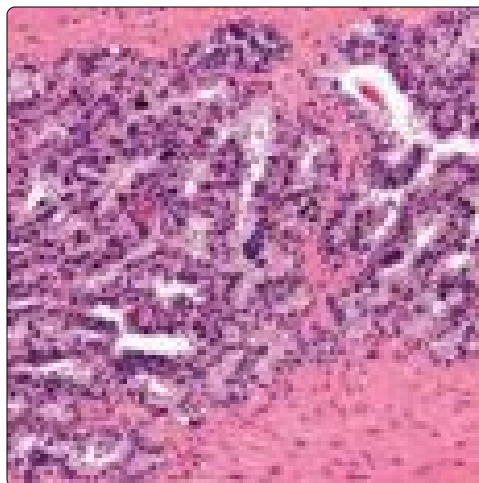


Seminal Vesicle: Pseudoinfiltrative Appearance

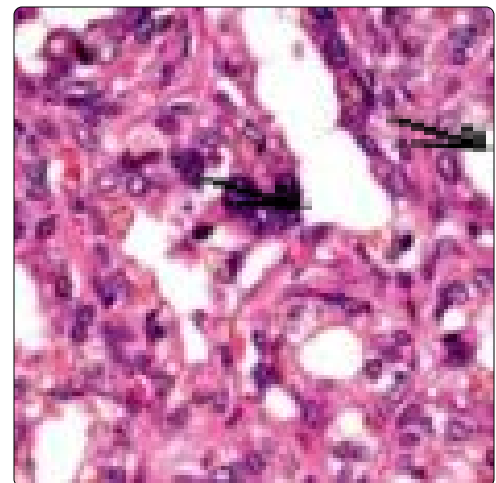


(Left) Seminal vesicle epithelium is characterized by cells with scattered pleomorphic nuclei . In contrast, prostatic adenocarcinoma is composed of cells with more monomorphic round nuclei. **(Right)** The combination of pleomorphic nuclei  and intracytoplasmic lipochrome pigment  helps to confirm seminal vesicle epithelium. In difficult cases, the nuclear expression of pax-8 is also characteristic of seminal vesicle epithelium.

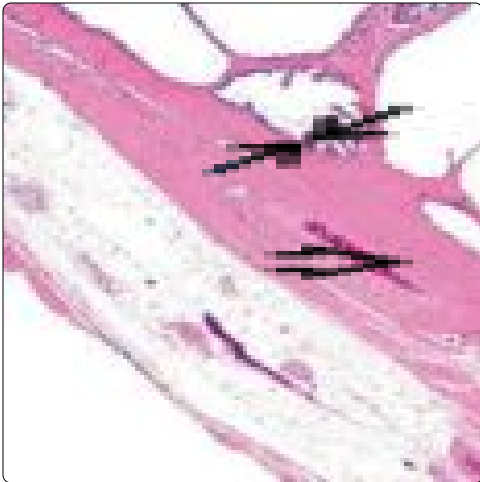
Seminal Vesicle: Cytology



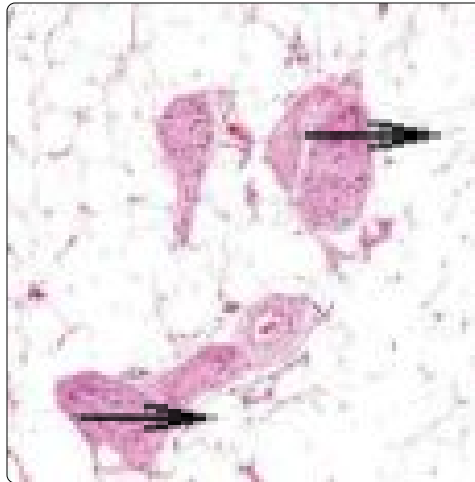
Seminal Vesicle: Intracytoplasmic Pigment



Extraprostatic Tissues

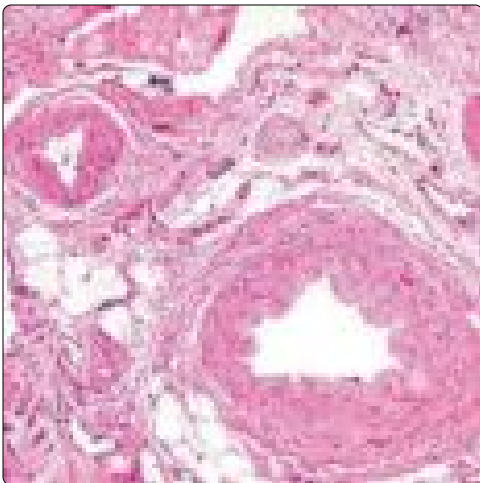


Extraprostatic Tissues: Peripheral Nerves

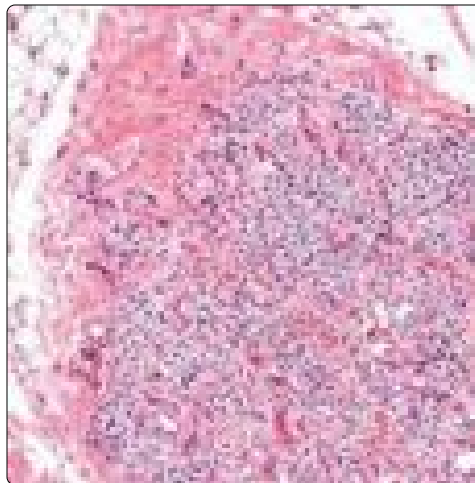


(Left) This section from a radical prostatectomy specimen highlights the extraprostatic tissue, defined as the soft tissue peripheral/external to the plane between the prostatic stroma and the adipose tissue. The plane is very distinct in this section, but it may be less well developed in other areas. **(Right)** Variably sized peripheral nerves are frequently encountered in the extraprostatic connective tissues.

Extraprostatic Tissues: Blood Vessels

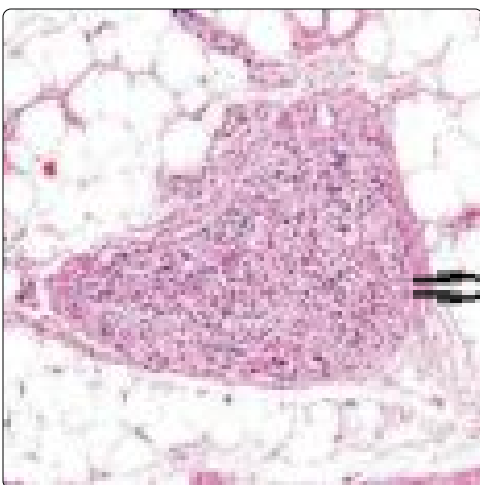


Extraprostatic Tissues: Paraganglia

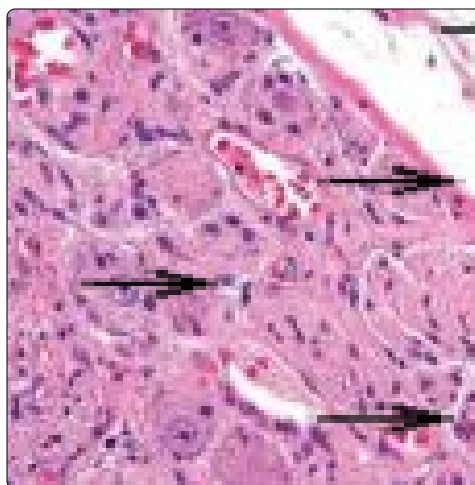


(Left) Blood vessels of varying caliber are common in the extraprostatic tissues. **(Right)** Normal paraganglia are occasionally found in extraprostatic tissues. The cells are often round and have clear to amphophilic cytoplasm. The cells may grow as small nests or have a more sheet-like arrangement. These paraganglia may easily be mistaken for high-grade prostatic adenocarcinoma. By immunohistochemistry, they are cytokeratin negative and synaptophysin/chromogranin positive.

Extraprostatic Tissues: Ganglion



Ganglion: High Power



(Left) Ganglia are also frequently found in the extraprostatic tissues around the prostate gland. They are composed of admixtures of peripheral nerve and collections of plump ganglion cells. **(Right)** At high-power evaluation, the ganglion cells are characterized by abundant eosinophilic cytoplasm, large round nuclei, and large prominent nucleoli. These cells are admixed with peripheral nerve.

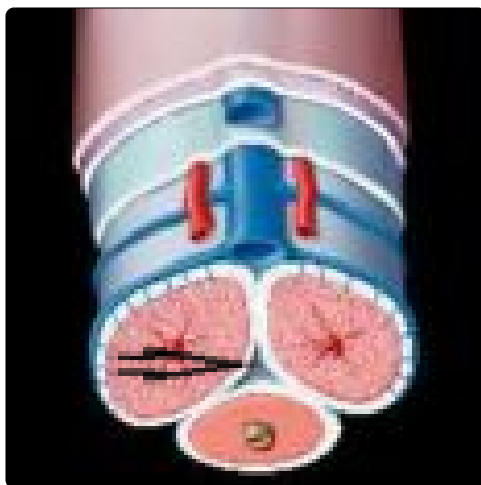
Penis

PENIS

Macroscopic Anatomy

- Shaft (proximal portion)
 - Predominantly composed of paired corpora cavernosa and ventrally situated corpus spongiosum
 - Shaft (distal portion)
 - Composed of glans, foreskin, and distal-most portions of corpus spongiosum and corpora cavernosa
 - Corpus spongiosum expands into cone-like shape at distal portion of penis to form majority of glans
 - Urethral meatus located centrally on tip of glans
 - External opening and termination of urethra
 - Corona
 - Elevated ridge of tissue present circumferentially at base of glans
 - Coronal sulcus
 - Indentation that lies adjacent and proximal to corona
 - On ventral-most aspect of shaft, corona and coronal sulcus replaced by frenulum (fold of tissue fixing foreskin to glans)
 - Foreskin
 - Also known as prepuce
 - Covers glans in uncircumcised persons
- ### Microscopic Anatomy
- Shaft (or body)
 - Paired corpora cavernosa and single corpus spongiosum
 - Composed of erectile tissue
 - Interconnecting networks of vascular spaces that fill with blood upon erection
 - Urethra is located centrally within corpus spongiosum
 - Tunica albuginea
 - Dense layer of fibrous tissue that surrounds each structure
 - Prominent around each corpus cavernosum
 - Thinner around corpus spongiosum
 - Median raphe
 - Point where tunica albuginea surrounding each corpus cavernosum meets and fuses at midline
 - Buck fascia
 - Layer of loose connective tissue rich in elastic fibers
 - Completely surrounds corpora cavernosa and corpus spongiosum and their associated tunics
 - Dartos muscle
 - Thin, irregular layer of smooth muscle bundles interspersed within loose connective tissue
 - Located around Buck fascia
 - Layer of skin (squamous epithelium) forms outermost layer of penile shaft
 - Glans
 - Outermost layer
 - Thin layer of stratified squamous epithelium that lacks adnexal structures and melanocytes
 - Lamina propria
 - Separates epithelial layer from corpus spongiosum
 - Consists of layer of loose connective tissue of varying thickness (1-2 mm)
 - Corpus spongiosum expanded at glans, forming cone-shaped structure composed of rich network of vascular spaces
 - In portion of population, distal-most parts of corpora cavernosa and tunica albuginea can extend into proximal portion of glans
 - Foreskin (prepuce)
 - Flap of tissue circumferentially covering glans
 - Composed of outer skin layer and inner mucosal layer
 - Outer epidermal layer
 - Consists of keratinizing stratified squamous epithelium and underlying dermis
 - Underlying dermis contains occasional associated adnexal structures
 - Inner mucosal layer
 - Histologically similar to that of glans and has underlying lamina propria
 - Dartos smooth muscle located centrally within foreskin; extension of smooth muscle layer found in shaft of penis

Penile Shaft

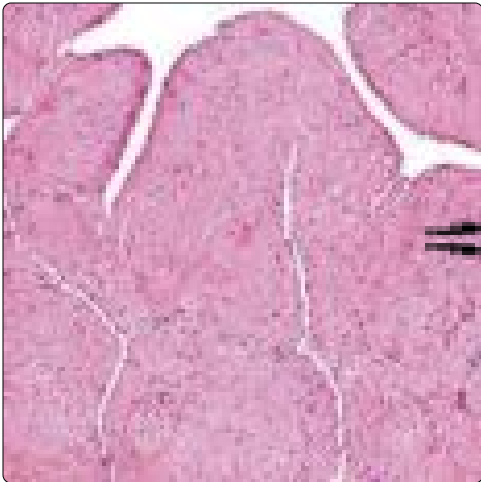


Penile Anatomy

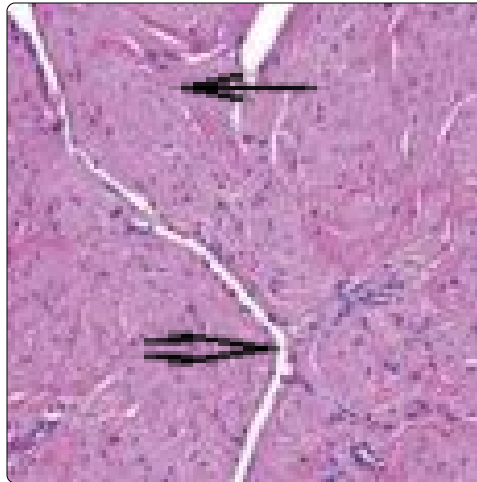


(Left) The shaft (or body) of the penis is composed predominantly of 3 elongated tubular structures of erectile tissue: The paired corpora cavernosa and the corpus spongiosum, which surrounds the urethra. **(Right)** The corpus spongiosum expands and acquires a cone-like shape at the distal-most portion of the penis, forming the majority of the glans. The foreskin can be seen surrounding the glans.

Corpora Cavernosa

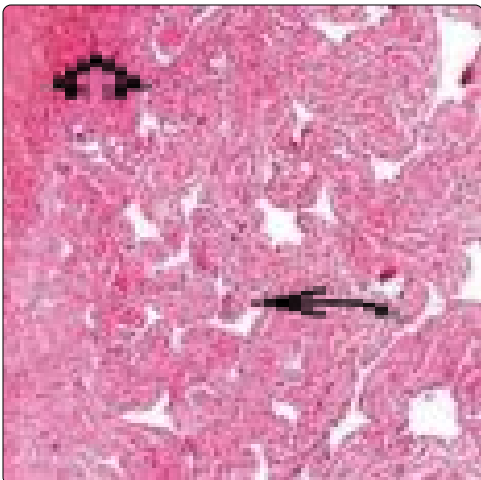


Corpora Cavernosa Trabeculae

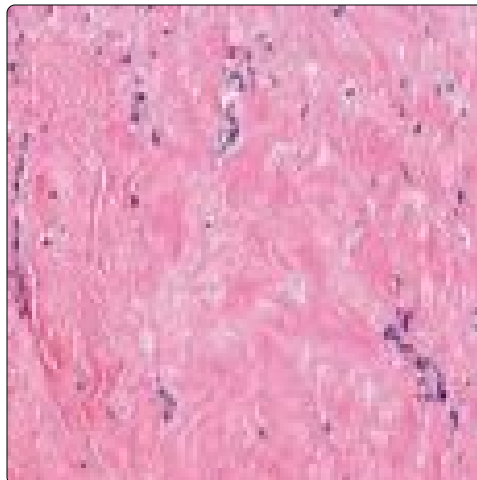


(Left) The corpora cavernosa are the paired main erectile tissues of the penis. They consist of an interanastomosing vascular network with intervening trabeculae composed of smooth muscle and fibrous tissue. The trabeculae of the corpora cavernosa tend to have a more predominant smooth muscle component than that seen in the corpus spongiosum. (Right) The trabeculae of the corpora cavernosa are lined by a single layer of endothelial cells.

Tunica Albuginea

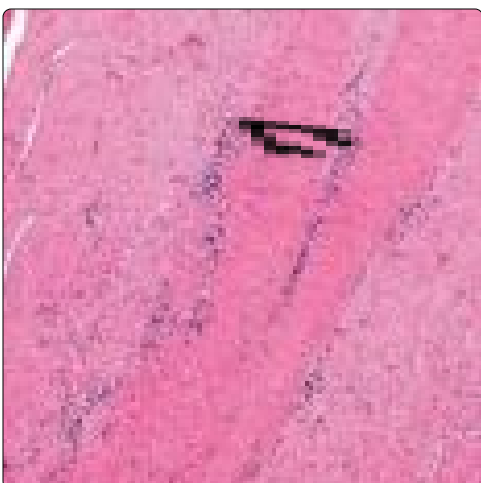


Tunica Albuginea Collagen

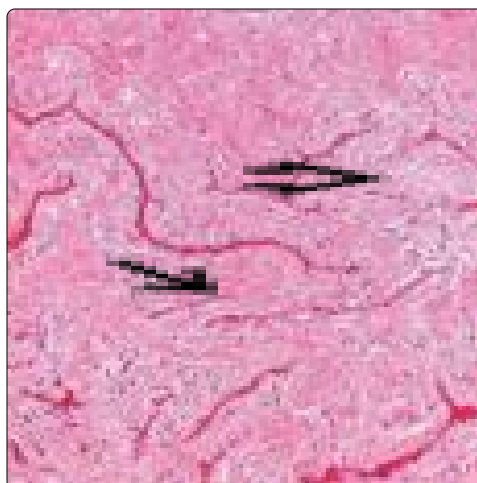


(Left) The corpora cavernosa are surrounded by a layer of thick connective tissue called the tunica albuginea. Note that the vascular trabeculae at the periphery are of a smaller caliber and appear less prominent than those seen within the central portions of the corpora cavernosa. (Right) The tunica albuginea consists of a poorly vascularized layer of fibrous connective tissue composed predominantly of thick collagen fibers.

Median Raphe



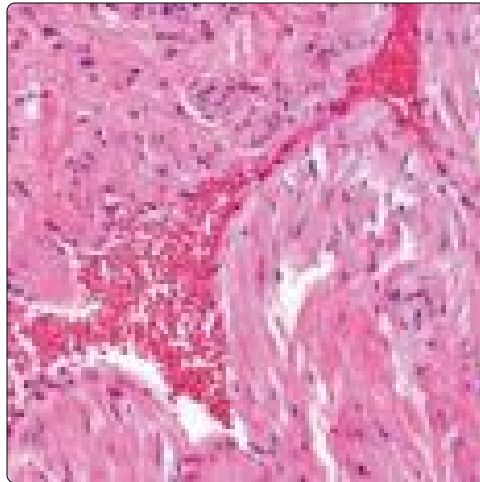
Corpus Spongiosum



(Left) The tunica albuginea encasing the 2 corpora cavernosa fuse at the midline, forming a fibrous septum. A complete septum is present in the more proximal portion of the shaft but becomes discontinuous within the distal portion. (Right) The endothelial-lined channels of the corpus spongiosum are surrounded by trabeculae composed of smooth muscle and fibrous tissue. The trabeculae of the corpus spongiosum tend to have less smooth muscle than those of the cavernosa.

Corpus Spongiosum Vasculature

(Left) The vascular spaces of the corpora spongiosum are lined by a layer of endothelial cells surrounded by the fibrous trabeculae. (Right) The urethra and its associated periurethral glands are located centrally within the corpus spongiosum. The vascular network and fibrous trabecula of the corpus spongiosum can be seen surrounding the urethra.

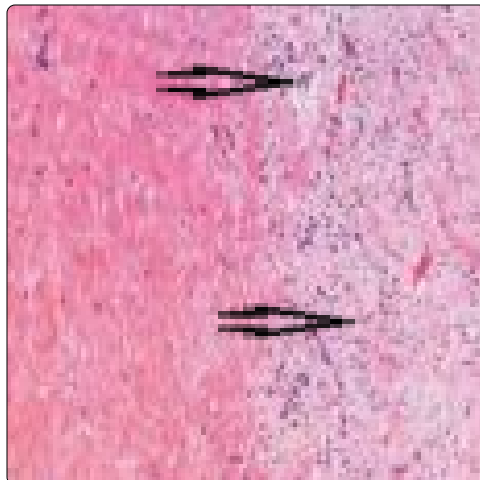


Corpus Spongiosum and Urethra



Tunica Albuginea of Corpus Spongiosum

(Left) The tunica albuginea of the corpus spongiosum is thinner than that of the corpora cavernosa. Unlike the corpora cavernosa, the vascular spaces of the corpus spongiosum do not tend to decrease in caliber toward the periphery of the structure and are of fairly uniform size throughout. (Right) The loose connective tissue of Buck fascia surrounds the dense layer of collagen fibers that make up the tunica albuginea.

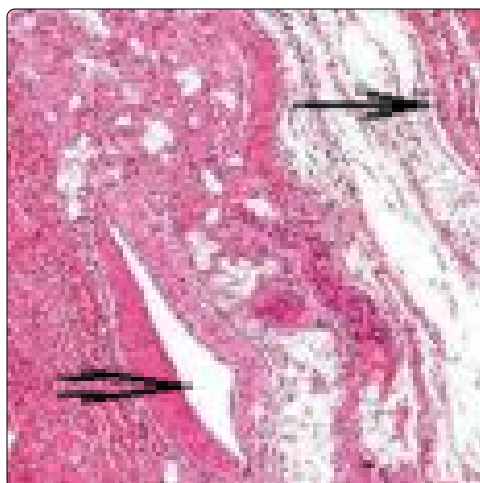


Buck Fascia

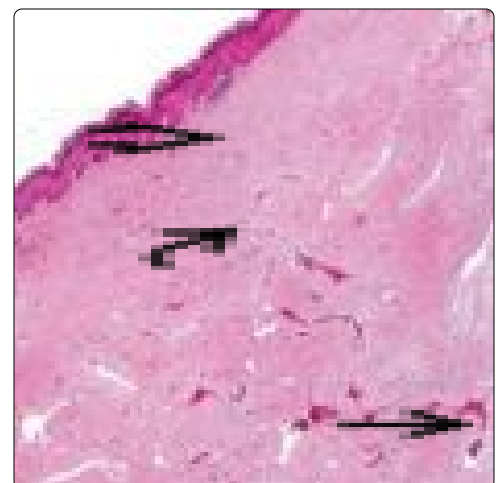


Buck Fascia Composition

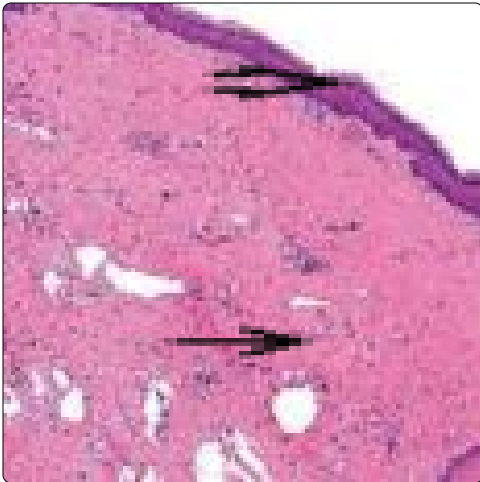
(Left) Buck fascia, seen here at higher power, is a layer of loose connective tissue with interspersed blood vessels and nerves that surrounds the thick layer of collagen that makes up the tunica albuginea. (Right) The squamous epithelium of the glans lacks adnexal structures, such as hair follicles or sebaceous glands. The lamina propria separates the glans from the network of vascular channels that make up the corpus spongiosum.



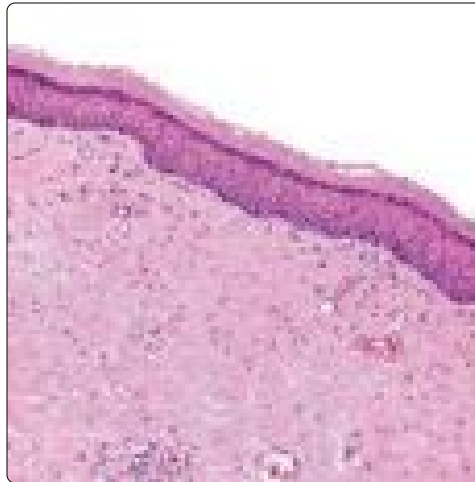
Glans Penis



Glans

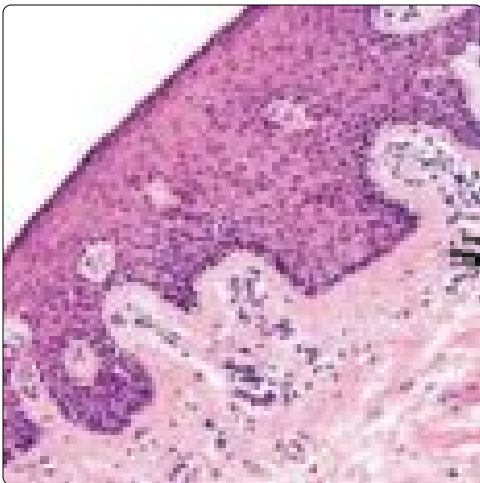


Glans Epithelium



(Left) The connective tissue layer comprising the lamina propria varies in thickness, and its border with the underlying corpus spongiosum is often ill-defined in histologic sections. (Right) The squamous epithelial layer of the glans may have a thin layer of keratinization in circumcised males. Note the lack of adnexal structures associated with the epithelium, as well as the lack of melanocytes within the basal layer of the squamous epithelium.

Foreskin

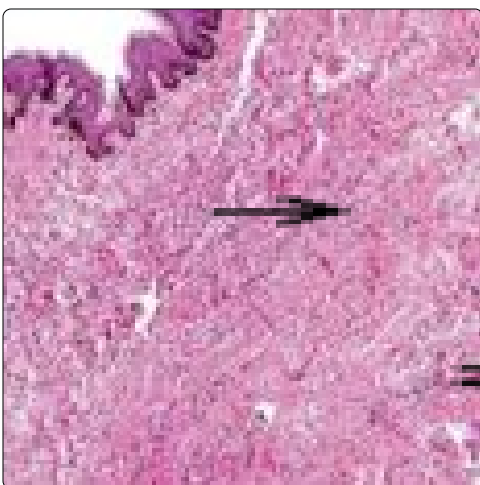


Foreskin Epithelium

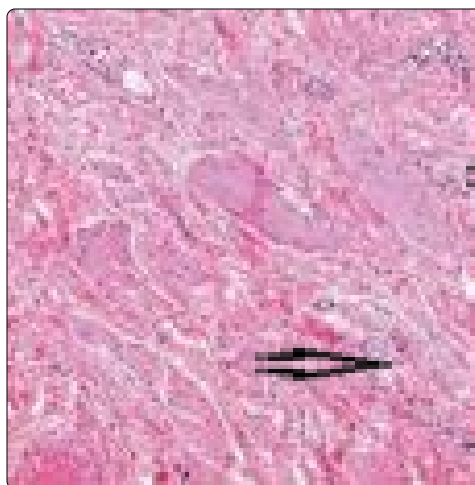


(Left) The outer epidermal layer of the foreskin is composed of keratinizing stratified squamous epithelium. Melanocytes and pigmentation can often be seen in the basal layer. (Right) The keratinizing squamous epithelium of the foreskin epidermis may have occasional adnexal structures identified within the underlying dermis. Note the pigmentation of the basal layer, which is in contrast with the squamous epithelium of the glans.

Dartos Muscle



Dartos Smooth Muscle



(Left) Underlying the dermis of the foreskin is the layer of dartos smooth muscle, which is located centrally within the foreskin, separating the cutaneous epidermis and dermis from the inner mucosal squamous epithelium and lamina propria. (Right) The dartos smooth muscle layer in the foreskin is continuous with that of the penile shaft and consists of bundles of smooth muscle interspersed within a fibrous connective tissue with associated nerves and blood vessels.

Testis and Associated Excretory Ducts

TESTIS

Macroscopic Anatomy

- Paired male gonads
 - Weigh ~ 15-20 g each
- Suspended within scrotal sac by spermatic cord
 - Spermatic cord contains vas deferens and blood vessels that supply testis
- Scrotum consists of 4 layers
 - Outer skin
 - Dartos muscle
 - Colles fascia
 - Parietal layer of tunica vaginalis

Microscopic Anatomy

- Outermost testicular capsule has 3 layers
 - Tunica vaginalis (visceral)
 - Outer layer lined by mesothelium
 - Tunica albuginea
 - Thick fibrous layer interior to tunica vaginalis
 - Tunica vasculosa
 - Loose connective tissue layer that divides testicular parenchyma into lobules via projects fibrous septa
- Interstitium
 - Composed of loose connective tissue that surrounds seminiferous tubules
 - Contains androgen-producing Leydig cells
 - Arranged singly or in clusters, often in association with nerve fibers
 - Abundant eosinophilic cytoplasm
 - May contain lipofuscin, Reinke crystalloids, lipid
 - May be present in other sites, including spermatic cord and epididymis
 - Also contains nerves, lymphatics, vessels
- Seminiferous tubules
 - Highly convoluted tubular structures that contain Sertoli cells and germ cells
 - Sertoli cells
 - Lie near basal aspect of seminiferous tubules and form blood-testis barrier

- Elongated cells with inconspicuous cytoplasm and round to irregular nuclei
- Germ cells
 - Comprise ~ 90% of cells within seminiferous tubules
 - Most immature cells (spermatogonia) form the basal-most layer
 - Most mature cells aggregate along luminal surface
- Germ cell maturation occurs in ~ 70 days and proceeds through orderly maturation process
 - Spermatogonia, primary spermatocytes, secondary spermatocytes, spermatids, mature spermatozoa

- Rete testis
 - Receives contents from seminiferous tubules
 - Characterized by its cavernous-appearing channels lined by flat to low columnar epithelium
- Efferent ductules
 - Receive contents of rete testis
 - Luminal surface has undulating configuration and is lined by columnar epithelium
- Epididymis
 - Composed predominantly of columnar cells with apical stereocilia, known as principal cells
- Vas deferens
 - Transports spermatozoa from epididymis to ejaculatory duct

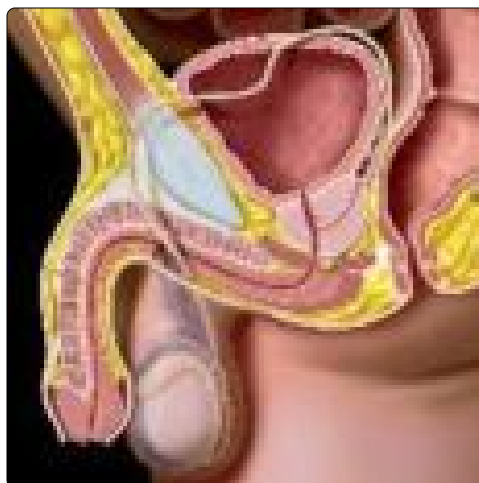
CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

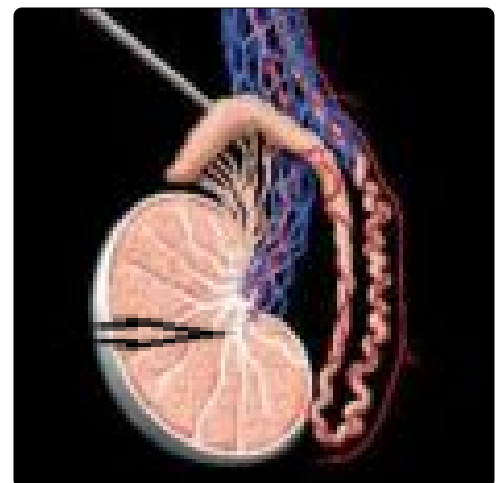
- At birth, testicle is composed of loose, immature mesenchymal tissue with small seminiferous tubules often lacking lumina
 - Majority of cells within seminiferous tubules of fetal and prepubertal testis are immature Sertoli cells, which have oval nuclei and inconspicuous nucleoli
 - Immature germ cells are only occasionally seen
- Decrease in spermatogenesis and fibrosis of seminiferous tubules are seen in elderly population

Testes

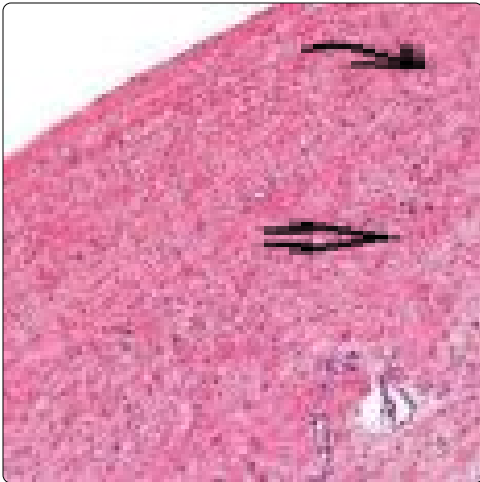
(Left) The testes are suspended within the scrotal sac by the spermatic cord. The cord contains the blood supply of the testis and the vas deferens, which transports the spermatozoa from the testis to the ejaculatory duct. (Right) Septa from the testicular capsule divide the testis into separate lobules. The epididymis, efferent ductules, and the rete testis are the tubular structures present at the mediastinum with adjacent blood vessels.



Testis and Excretory Ducts



Outer Surface

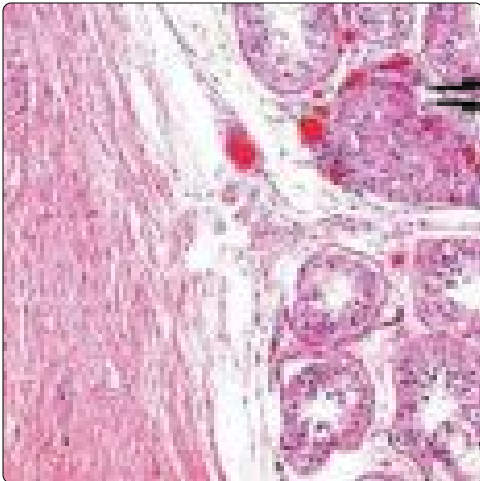


Tunica Vaginalis and Tunica Albuginea

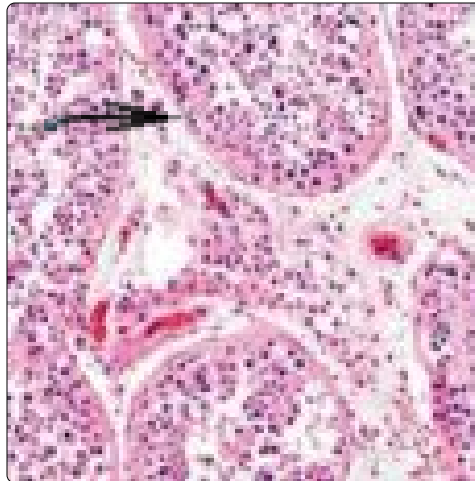


(Left) Beneath the flattened mesothelial layer of the tunica vaginalis lies the thick layer of dense collagen called the tunica albuginea, which forms a capsule surrounding the testis. (Right) The visceral portion of the tunica vaginalis consists of a single layer of flattened mesothelial cells that make up the outermost layer of the testicle, external to the tunica albuginea. The parietal counterpart of the tunica vaginalis lines the innermost layer of the scrotum.

Tunica Vasculosa

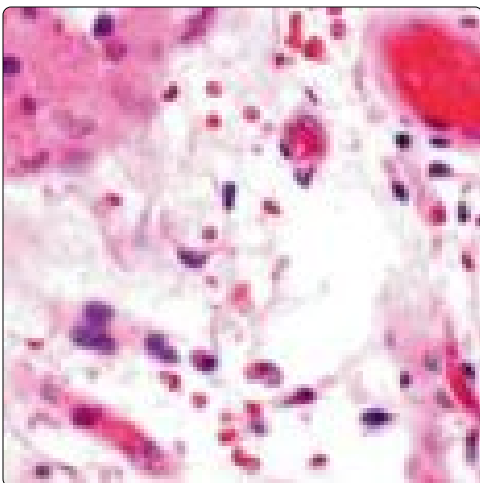


Interstitial

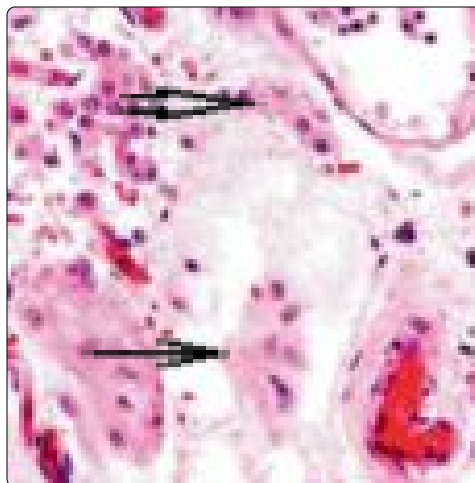


(Left) The tunica vasculosa is a layer of loose connective tissue containing numerous lymphatic spaces and blood vessels that lies beneath the tunica albuginea. The septa, which separate the testis into ~250 lobules, arise from the tunica vasculosa. (Right) The loose connective tissue between the seminiferous tubules is known as the interstitium. Interspersed within the interstitial tissue are blood vessels, mast cells, nerves, and Leydig cells.

Interstitial



Interstitial

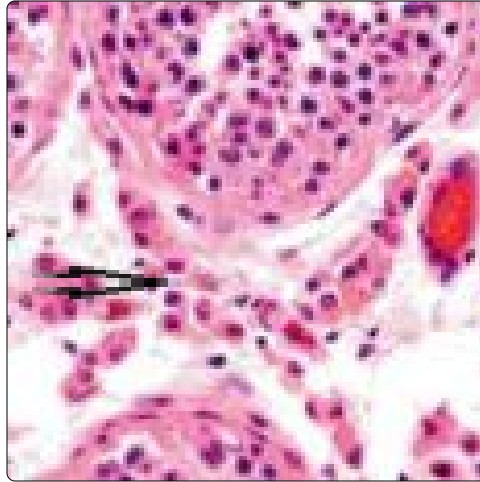


(Left) The loose connective tissue of the interstitium contains blood vessels and scattered mast cells. (Right) Along with blood vessels, nerves, and scattered mast cells are Leydig cells, which produce testosterone. Leydig cells are present as single cells or, more commonly, small clusters of cells with abundant eosinophilic cytoplasm and a round vesicular nucleus.

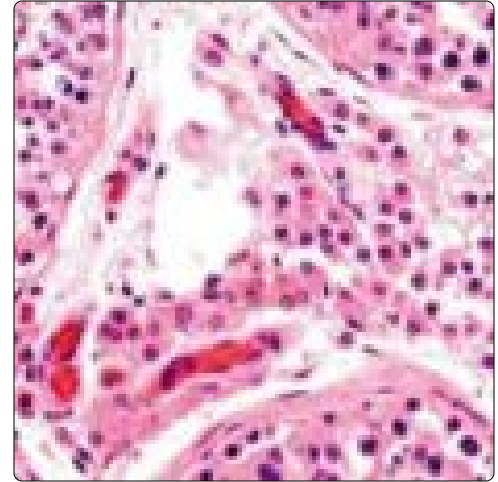
Testis and Associated Excretory Ducts

Leydig Cells

(Left) Leydig cells have eosinophilic cytoplasm, single vesicular nuclei, and often lie in close proximity to blood vessels and adjacent to the seminiferous tubules in the interstitium. Intracytoplasmic lipid droplets and lipofuscin pigment are commonly present in Leydig cells. (Right) Lipid droplets and brown pigment, known as lipofuscin, begin to accumulate within the brightly eosinophilic cytoplasm of the Leydig cells at puberty and increase with age.

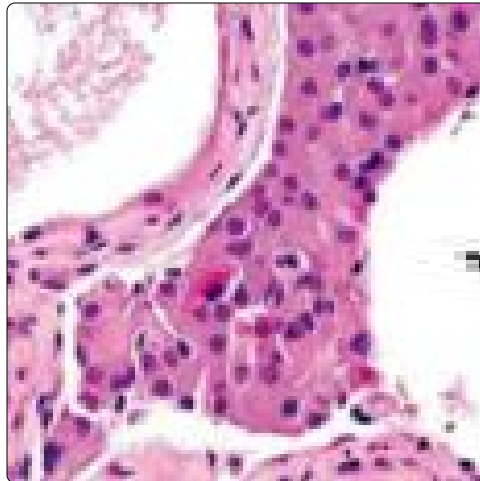


Leydig Cells

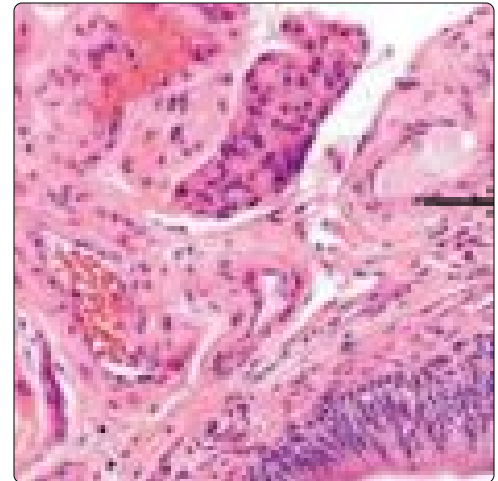


Reinke Crystals

(Left) Intracytoplasmic crystalloids, known as Reinke crystalloids, can occasionally be found within Leydig cells. The crystalloids are only identified in postpubertal testes. (Right) Leydig cells can be identified outside of the testicular interstitium, often within nerve fibers. The epididymis, spermatic cord, and hilar soft tissue are common sites where Leydig cells may be present. A cluster of Leydig cells is identified surrounding a nerve fiber adjacent to the epididymis.

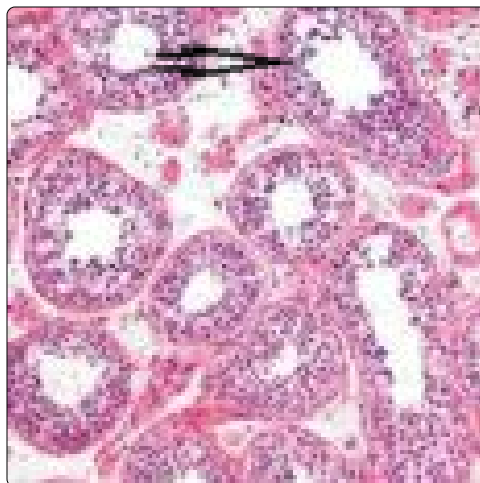


Leydig Cells

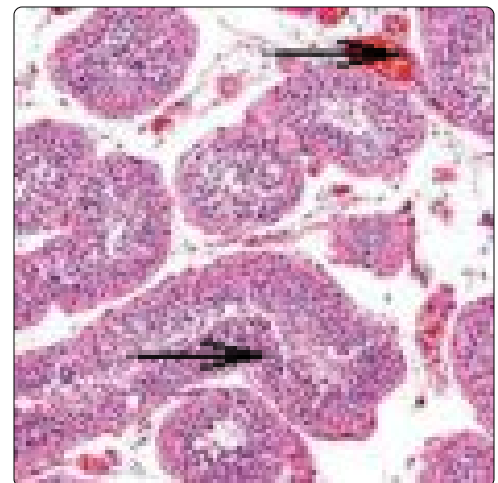


Seminiferous Tubules

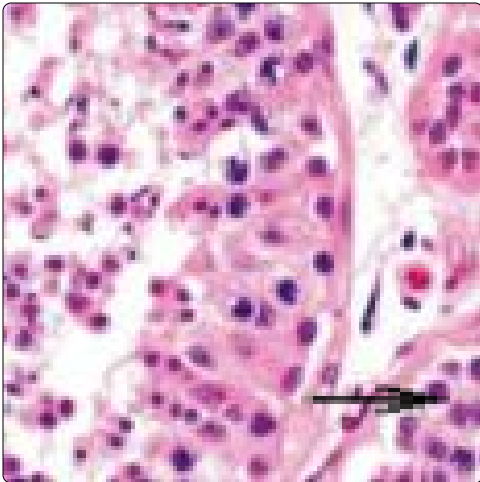
(Left) The seminiferous tubules are convoluted tubular structures containing the germ cells and Sertoli cells. A basement membrane and lamina propria surround each tubule, separating it from the adjacent interstitium of the testis. (Right) Each lobule of the testis houses between 1 and 4 long, convoluted seminiferous tubules, which contain the maturing germ cells within the testis before emptying their contents into the rete testis.



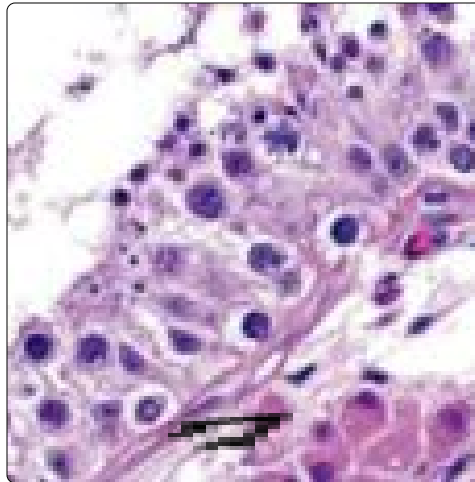
Seminiferous Tubules



Sertoli Cells

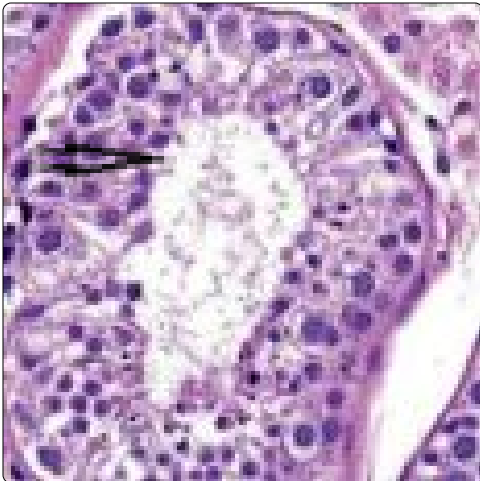


Sertoli Cells

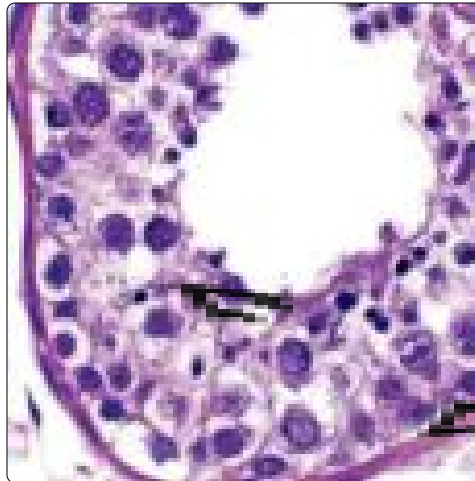


(Left) Sertoli cells are irregular columnar cells with inconspicuous cytoplasmic borders, round or slightly irregular nuclei, folded nuclear membranes, and prominent nucleoli. The cytoplasm forms extensions to support the adjacent germ cells. (Right) Sertoli cells lie between the spermatogonia and primary spermatocytes with the base of the cell attached to the underlying basement membrane. Tight junctions between Sertoli cells form the blood-testis barrier.

Sertoli Cells

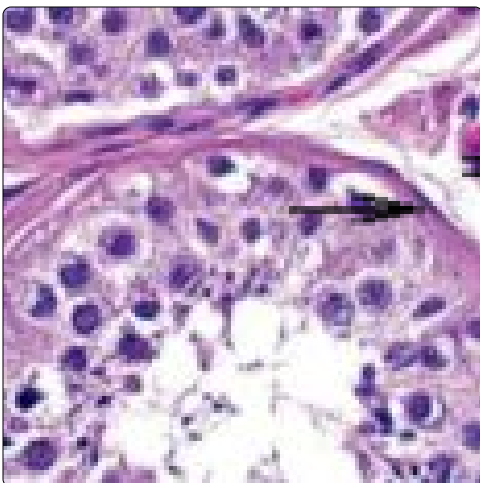


Spermatogonia

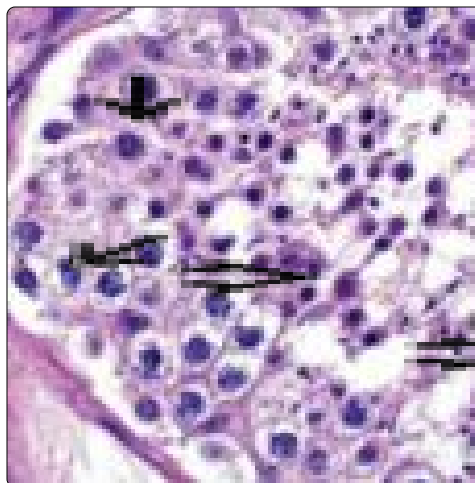


(Left) Sertoli cells support the surrounding germ cells and form the blood-testis barrier. It separates the tubular contents into a luminal compartment, which contains primary spermatocytes, secondary spermatocytes, spermatids, and an underlying compartment, which contains spermatogonia and primary spermatocytes. (Right) Spermatogonia are the most undifferentiated of the germ cells in the seminiferous tubules.

Spermatogonia



Primary Spermatocytes

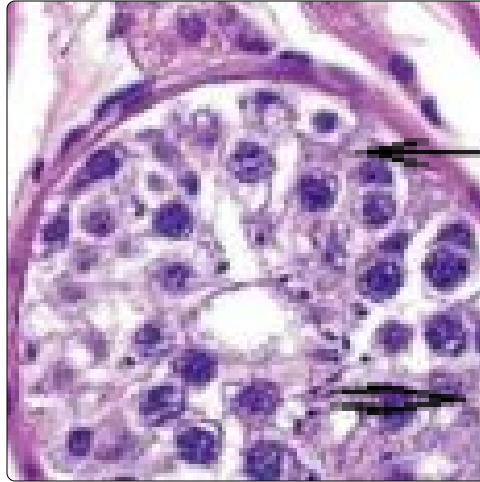


(Left) The small spermatogonia have round to ovoid nuclei and lie against the basement membrane of the tubule within the basal compartment. The spermatogonia proliferate both to replenish their population and to give rise to primary spermatocytes. Sertoli cells can be found adjacent to the spermatogonia. (Right) Primary spermatocytes have larger nuclei than the spermatogonia and occupy a more central location within the tubule.

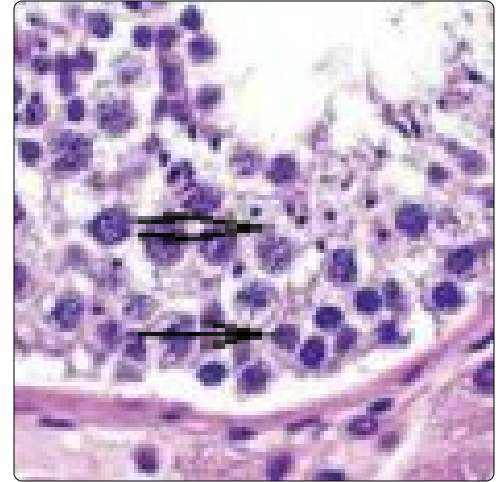
Testis and Associated Excretory Ducts

Primary Spermatocytes

(Left) Spermatogonia give rise to primary spermatocytes, which are larger and have characteristically coarse chromatin. The primary spermatocytes move from the basal compartment of the tubule to the luminal compartment soon after their maturation from spermatogonia. (Right) Nuclei of primary spermatocytes are large with coarse, clumped chromatin. Sertoli cells and spermatogonia occupy a more basal location in the tubule.

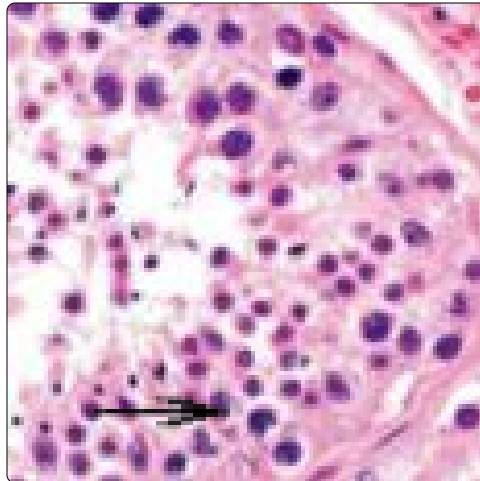


Primary Spermatocytes

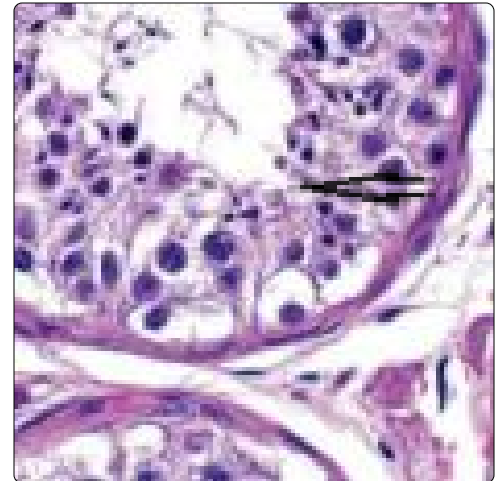


Spermatids

(Left) Primary spermatocytes mature into secondary spermatocytes, which quickly mature into spermatids. Secondary spermatocytes and early spermatids are morphologically similar with round nuclei much smaller than those of the primary spermatocytes. (Right) Spermatids vary morphologically as they mature into spermatozoa. Early forms have round nuclei with more abundant cytoplasm, whereas late spermatids have ovoid to elongated nuclei and much less cytoplasm.

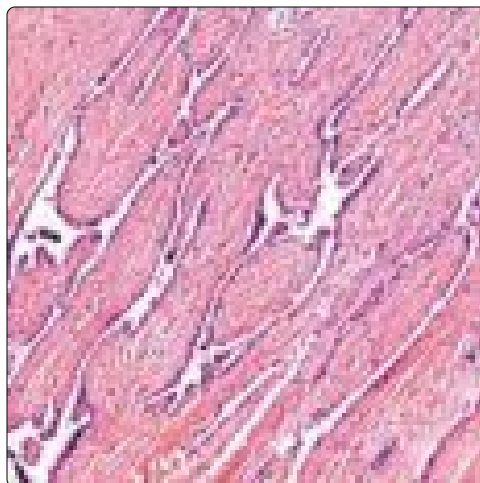


Spermatids

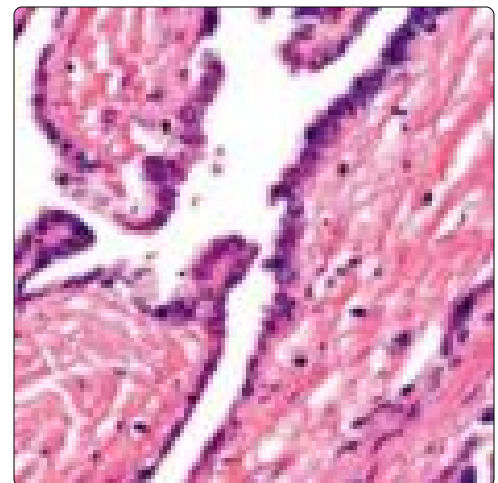


Rete Testis

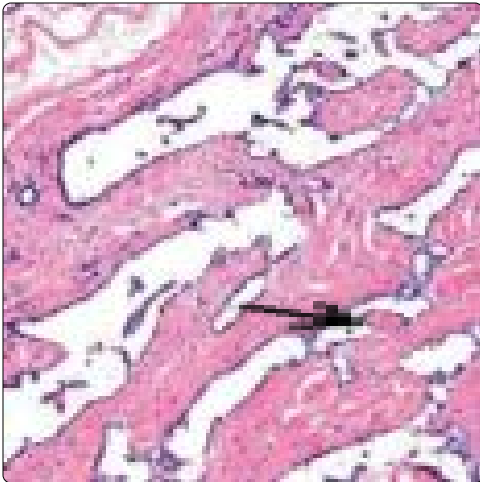
(Left) The seminiferous tubules empty their contents into the rete testis, a network of channels in the mediastinum of the testis. The tubules of the rete testis are surrounded by stroma containing collagen and elastic fibers with occasional myoid and fibroblastic cells. (Right) The flattened, low columnar epithelium of the rete testis has microvilli and a single flagellum on the luminal aspect, although these structures are not apparent on routine histologic sectioning.



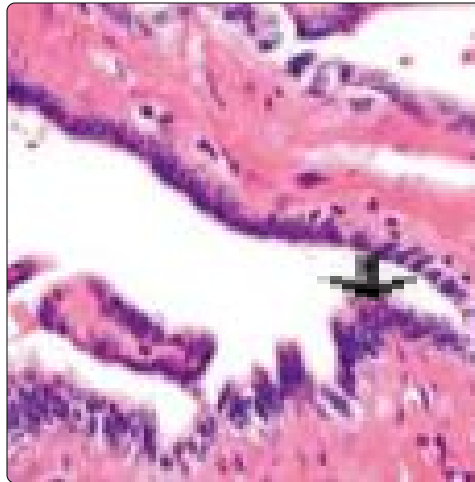
Rete Testis



Rete Testis

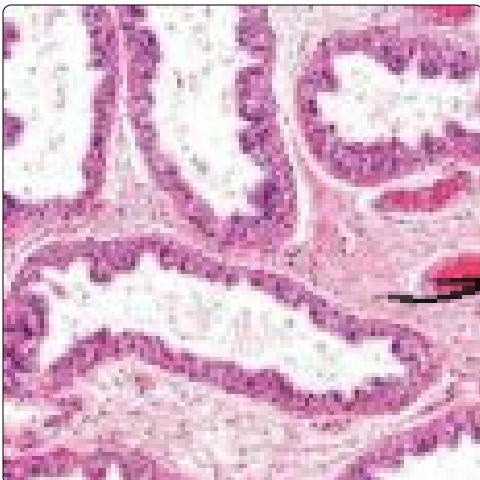


Rete Testis

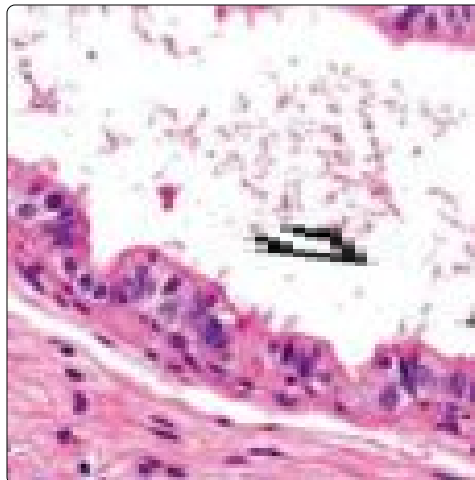


(Left) The chordae retis are columns of tissue that serve to connect the opposing walls of the rete testis. When viewed on cross sectioning, these structures appear as intraluminal islands of epithelial-lined fibrous tissue. (Right) The chordae retis are lined by the same flattened, low columnar epithelium as the walls of the rete testis. Thick collagen fibers with occasional myoid and fibroblastic cells can be seen in the surrounding stroma.

Efferent Ductules

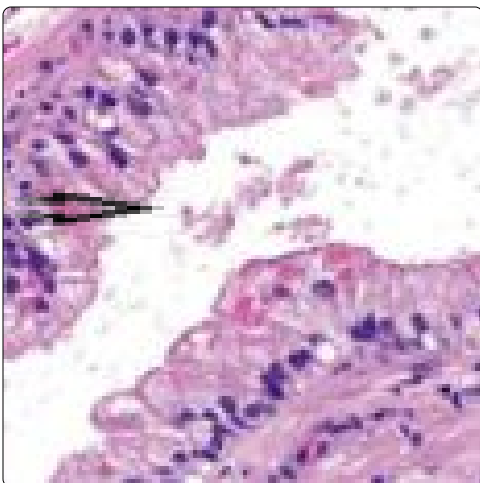


Efferent Ductules

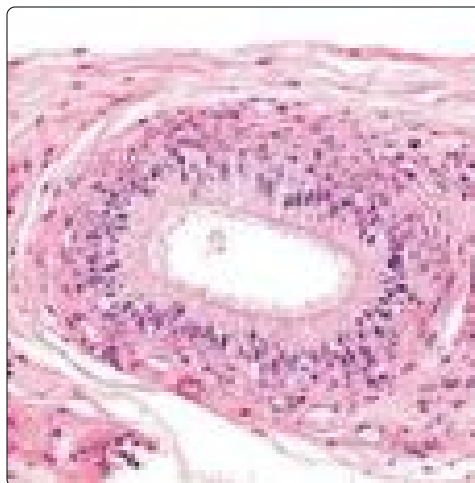


(Left) The efferent ductule luminal surface has an undulating architecture. The surrounding stroma contains smooth muscle and fibroblasts. (Right) The epithelium of the efferent ductules is composed of columnar cells, which may be ciliated, and basal cells. Lymphocytes may be identified within the epithelium, and the admixture of columnar and basal cells with lymphocytes results in an overall pseudostratified appearance to the epithelium.

Efferent Ductules



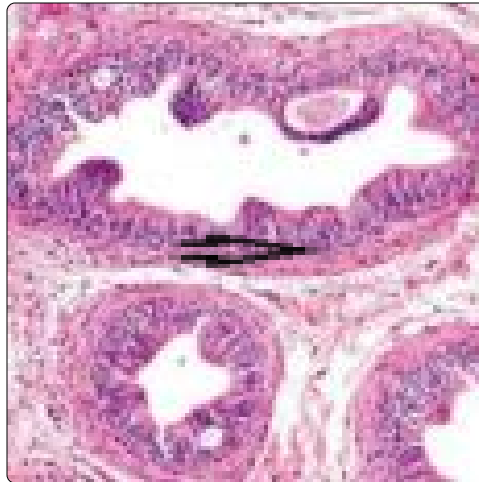
Epididymis



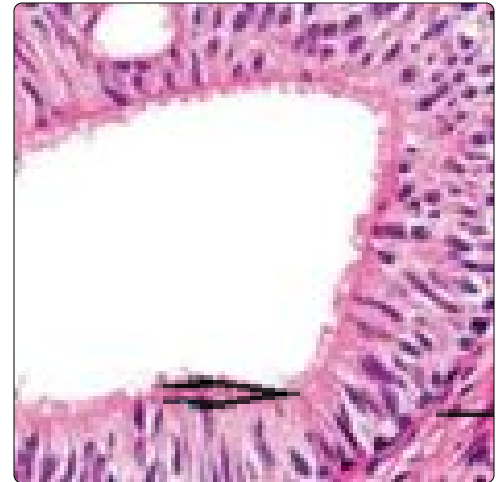
(Left) Cells with intracytoplasmic collections of brightly eosinophilic, PAS-positive globules are occasionally seen in the efferent ductules and, less commonly, in the epididymis. (Right) The tubules of the epididymis are more regular in shape and have a smoother, less undulating luminal surface when compared to the tubules of the efferent ductules. Surrounding the tubule is a well-defined layer of smooth muscle.

Epididymis

(Left) A cribriform pattern can be seen in the epididymis in ~ 1/2 of the population and should not be mistaken for a pathologic process. A well-defined layer of smooth muscle surrounds each tubule. (Right) The epithelium of the epididymis is composed of principal cells, basal cells, apical cells, and clear cells. The majority of the cells are the tall columnar principal cells, which have apical stereocilia. Lymphocytes are scattered throughout the epithelium.



Epididymis

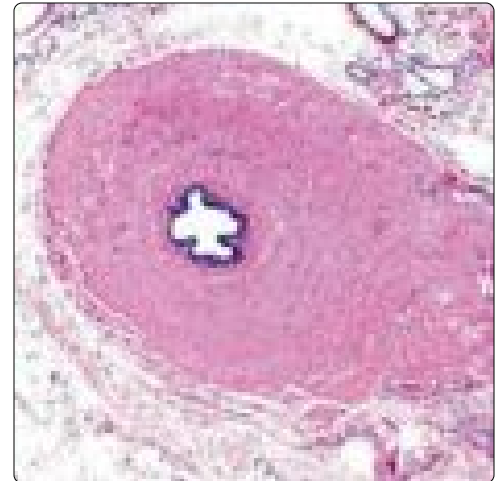


Epididymis

(Left) Eosinophilic inclusions can be identified within the nuclei of the tall columnar cells in the epididymis and in the cells of the vas deferens. Intracytoplasmic golden brown lipofuscin pigment can also be seen in the epididymis. (Right) The vas deferens is a long (~ 35 cm), muscular-coated structure that receives contents from the epididymis. The distal portion (ampulla) of the vas deferens joins with the seminal vesicle excretory duct to form the ejaculatory duct.

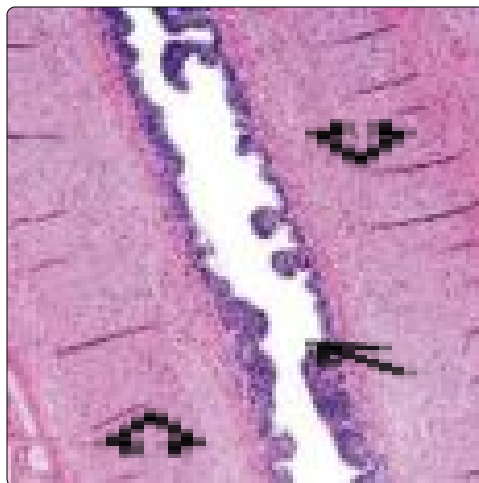


Vas Deferens



Vas Deferens

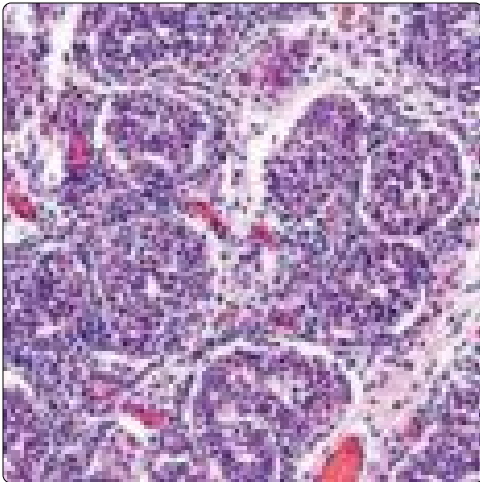
(Left) The thick muscular wall of the vas deferens is composed of 3 layers: An inner longitudinal layer, a middle circular layer, and an outer longitudinal layer. The epithelial surface may have a folded architecture, particularly in the more distal portion. (Right) The epithelium of the vas deferens is composed of columnar cells and basal cells. The columnar cells have stereocilia on the apical aspect, which can be seen on the luminal surface.



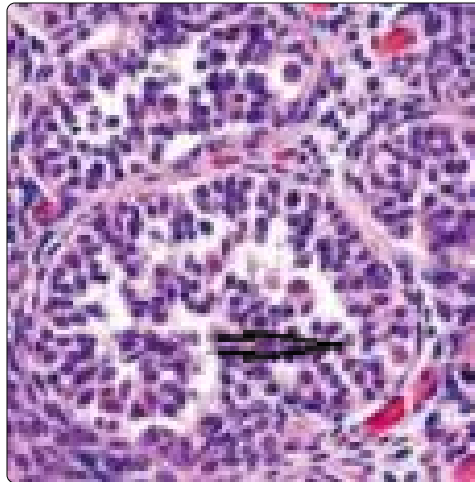
Vas Deferens



Prepubertal Testis

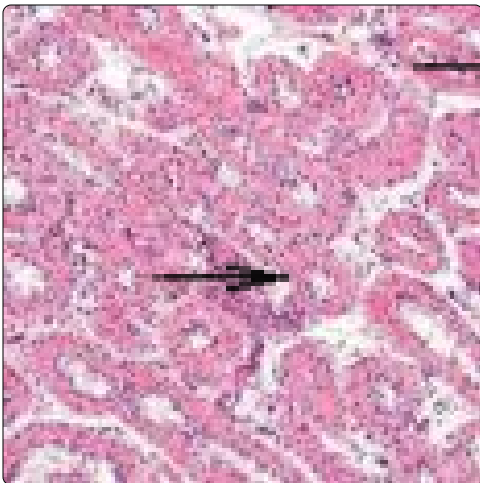


Prepubertal Testis



(Left) At birth, the testis is composed of loose, immature mesenchymal tissue with small seminiferous tubules, the majority of which lack lumina. (Right) At birth, the majority of the seminiferous tubules appear solid, lacking a central lumen. Immature Sertoli cells , with oval to elongated nuclei and inconspicuous nucleoli, comprise the majority of the cells within the tubules. Rare immature germ cells can be identified .

Peritubular Fibrosis

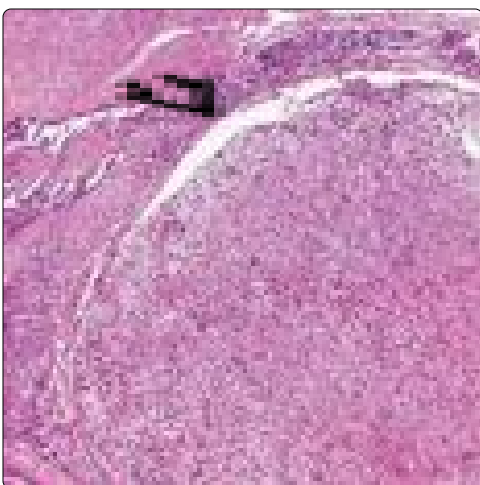


Testicular Appendages

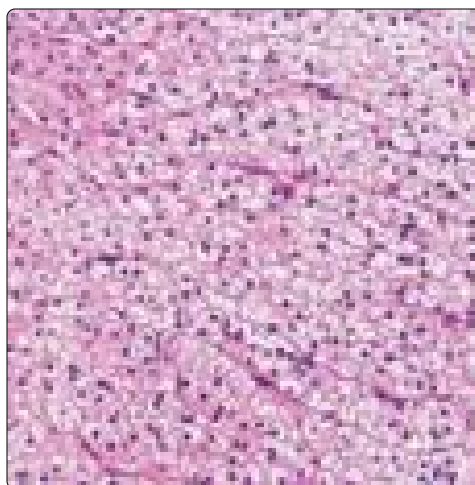


(Left) Thickening of the wall of the seminiferous tubules  may be identified in focal areas as men age. Widespread thickening and hyalinization of the seminiferous tubules can be seen in cryptorchid testicles, in which the testicles fail to completely descend from the abdomen. (Right) The testicular appendages are mesonephric or paramesonephric duct remnants. The appendix epididymis (pictured) is a remnant of the mesonephric duct and is lined by columnar epithelium, which may be ciliated.

Adrenal Cortical Rest



Adrenal Cortical Rest



(Left) Adrenal cortical rests  may be found in the spermatic cord or in the hilar soft tissue adjacent to the epididymis or rete testis. The adrenal tissue in this case is present within the hilar soft tissue with normal rete testis  identified in the upper left corner. (Right) Histologically unremarkable adrenal cortical tissue, with vacuolated, lipid-filled, lightly eosinophilic cytoplasm, is identified within adrenal cortical rests.

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SECTION 13

Female Genital Tract



Vulva	346
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Placenta	384

VULVA

Macroscopic Anatomy

- General anatomic landmarks
 - Female genitalia external to hymen
 - Anteriorly limited by (and including) mons pubis
 - Posteriorly delimited by anus
 - Laterally, extends to inguinal-gluteal folds
- Mons pubis
 - Prominence over pubic symphysis that is composed of adipose tissue
 - Anteriormost region of vulva
- Hymen
 - Border of distal-most extent of vagina and posterior aspect of vulvar vestibule
- Labia majora
 - Located lateral (and parallel) to labia minora
- Labia minora
 - Present lateral to vulvar vestibule and medial to labia majora
- Clitoris
 - Located anterior to frenulum at junction of labia minora
- Vulvar vestibule
 - Extends from exterior surface of hymen to clitoris anteriorly, fourchette posteriorly, labia minora anterolaterally, and Hart line posterolaterally
 - Includes vaginal opening and urethral orifice
 - Also includes openings for paired Skene ducts, vestibular glands, and Bartholin glands
- Bartholin glands
 - Paired mucin-secreting glands located posterolaterally in vulva
- Hart line
 - Inferior junction between vulvar vestibule and perineal skin
- Nonkeratinized in vestibule where it may be glycogenated
- Keratinization generally increases laterally toward surrounding skin
- Stroma
 - Stromal cell can be spindle, fusiform, or stellate and may have nuclear multilobation
- Adnexal structures
 - Labia majora contains eccrine glands, sebaceous glands, and hair follicles, similar to those found in skin at other locations
 - Labia minora is devoid of adnexal structures in most women
- Clitoris
 - Contains vascular erectile tissue
- Mammary-like anogenital glands
 - Located within sulcus between labia majora and minora
 - Lined by luminal cuboidal/columnar epithelial cells that have underlying myoepithelium
 - Luminal cells may have apocrine appearance
- Minor vestibular glands
 - Superficial glands that enter mucosal surface of vestibule
 - Mucinous-type epithelium that may be admixed with squamous epithelium
- Bartholin glands
 - Acini consist of mucinous cells
 - Main duct lined by transitional-type epithelium

CONSIDERATIONS AND VARIATIONS

Metaplasia

- Squamous metaplasia of underlying vestibular glands
 - May mimic invasive squamous cell carcinoma

Pitfalls and Artifacts

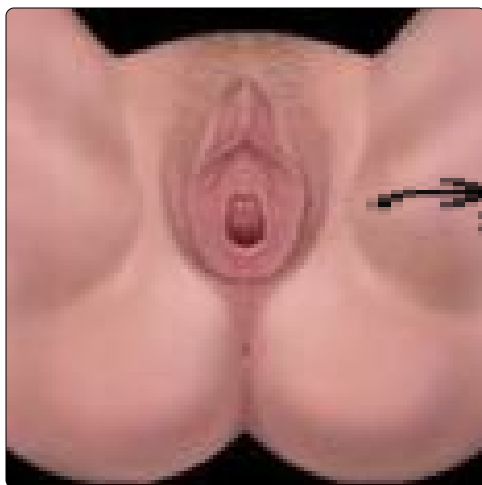
- Multinucleated atypia
 - Multinucleated epithelial cells without viral cytopathic effect
 - May mimic squamous dysplasia
- Vulvar stromal cells with atypia
 - Normal stromal cells may have large multilobated nuclei

Microscopic Anatomy

- Epithelium
 - Covered by stratified squamous epithelium
 - Keratinized in labia majora and minora

Anatomic Landmarks

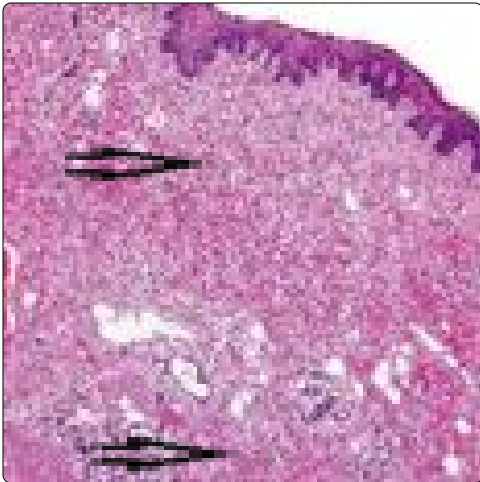
(Left) The major anatomic landmarks of the vulva include the mons pubis [A], clitoris [B], labia minora [C], labia majora [D], and vestibule [E]. The urethral meatus [F] is present within the vulvar vestibule. **(Right)** The vulva is covered by stratified squamous epithelium. The basal cell layer [G] has a higher nuclear:cytoplasmic ratio, and the cells gradually mature toward the surface.



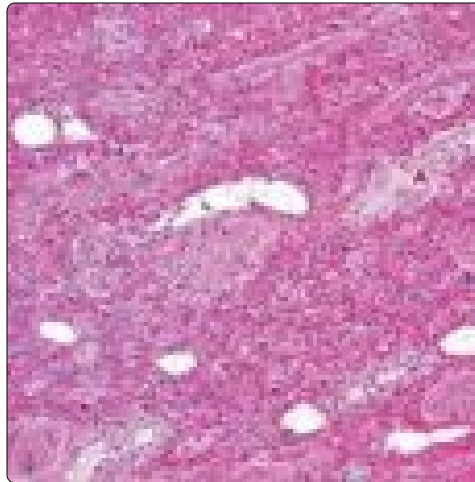
Epithelium



Labia Majora

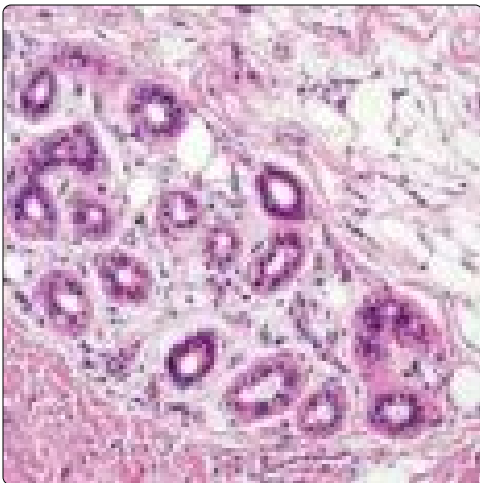


Labia Majora: Dermis

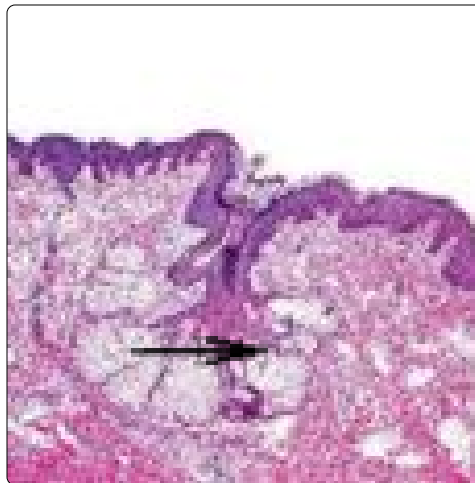


(Left) In the labia majora, the tissue underlying the epidermis contains collagen, adipose tissue, and adnexal structures. Adnexal structures and adipose tissue are typically absent in the labia minora. **(Right)** The dermis of the labia majora typically contains superficial adipose tissue and smooth muscle bundles as well as an admixture of cutaneous adnexal structures that extend to deeper layers.

Labia Majora: Eccrine Glands

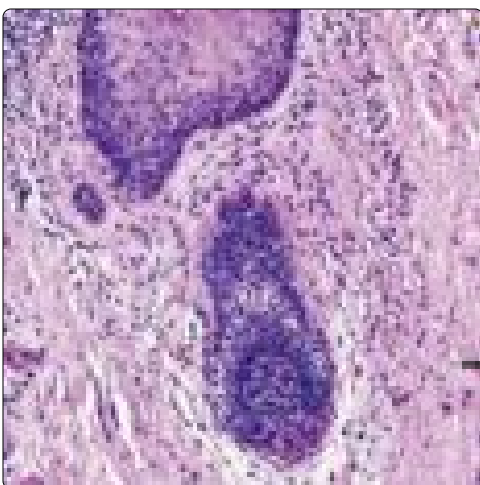


Labia Majora: Sebaceous Glands

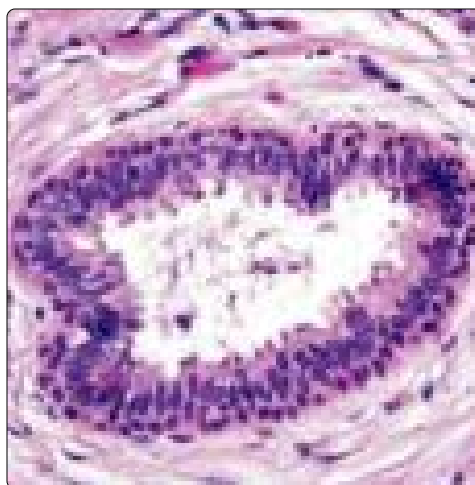


(Left) Eccrine-type glands are typically found in small circumscribed clusters. The secretory cells are a low cuboidal epithelium with scant eosinophilic cytoplasm. An indistinct surrounding myoepithelial layer is also present. **(Right)** Sebaceous glands, ± associated hair follicles, are commonly identified at various levels within the labia majora. As in other areas of the body, the sebaceous glands contain clear vacuolated cytoplasm.

Labia Majora: Hair Follicle



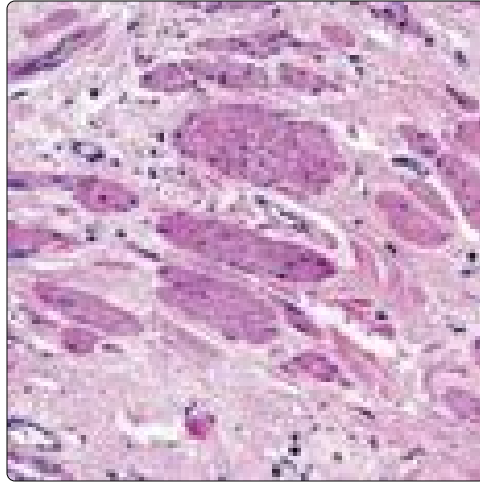
Labia Majora: Apocrine Glands



(Left) Hair follicles are also seen in the labia majora. This photomicrograph shows the base of the follicle (the bulb) with the invaginated papilla. **(Right)** Apocrine-type glands are also commonly identified within the labia majora. The secretory cells have more abundant eosinophilic cytoplasm with characteristic apical blebbing. A distinct myoepithelial layer is seen in this example.

Labia Majora: Smooth Muscle Bundles

(Left) Dense bundles of smooth muscle are also commonly identified in the labia majora, but they are not typically seen in the labia minora. **(Right)** The superficial stroma underlying the epithelium of the vulva may contain a focal zone with somewhat increased stromal cellularity. This stroma may contain a subset of larger atypical cells that are visible even on low-power magnification.

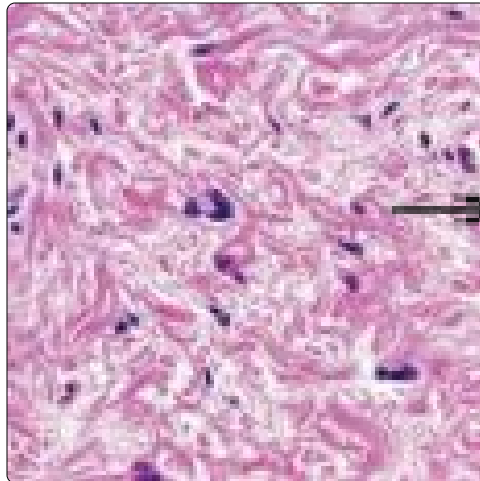


Vulvar Stromal Cells

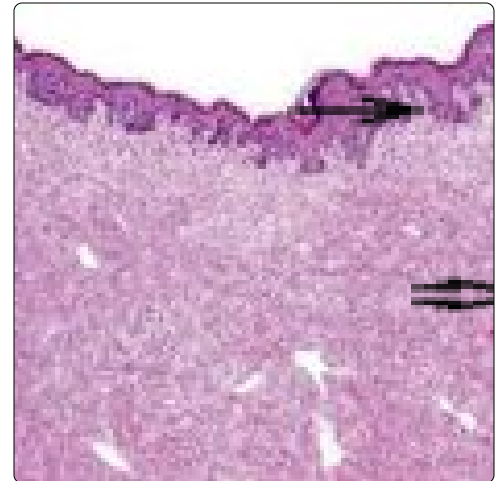


Multinucleated Stromal Cells

(Left) As in the vagina, the stromal cells of the vulva may have multilobated nuclei that can resemble a variety of neoplastic processes. These normal stromal cells are identical to those seen in fibroepithelial polyps. **(Right)** The squamous epithelium and the underlying stroma of the labia minora are demonstrated in this photomicrograph. The majority of the labia minora is devoid of adnexal structures and adipose tissue.

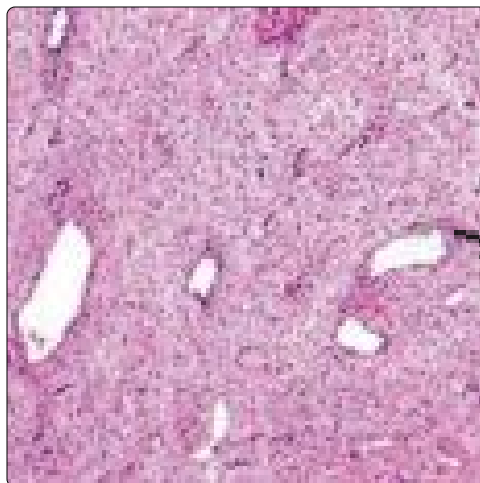


Labia Minora

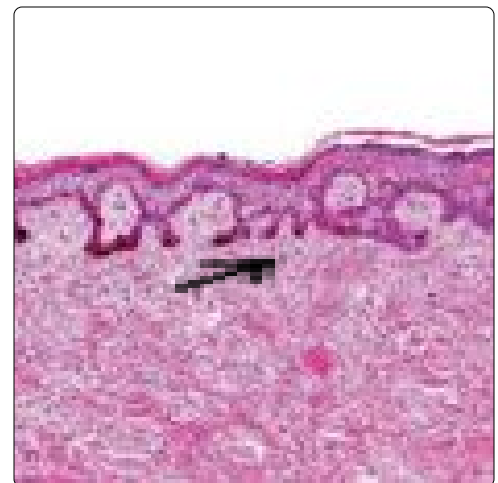


Labia Minora: Stroma

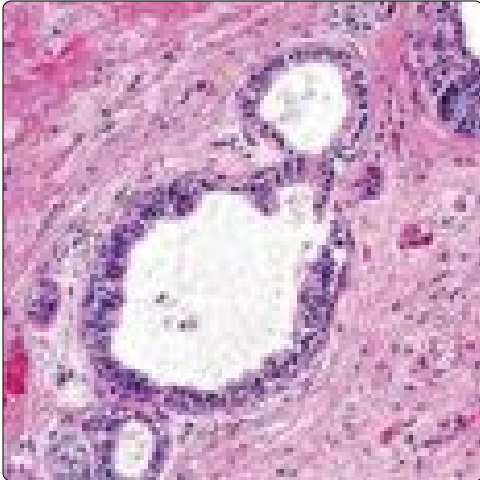
(Left) The stroma of the labia minora is rich in blood vessels and elastic fibers. Unlike the labia majora, the stroma of the labia minora does not usually contain admixed adipose tissue, smooth muscle, or adnexal structures. **(Right)** In the lateral and posterior regions of the labia minora, the epithelium may be pigmented. The epithelium of the labia majora in the region peripheral to the Hart line may also be pigmented.



Labia Minora: Epithelial Pigmentation



Mammary-Like Anogenital Glands



Mammary-Like Anogenital Glands: Cytology

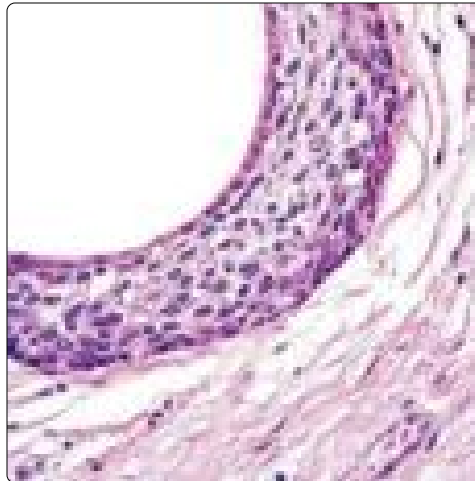


(Left) The sulcus between the labia majora and minora contains mammary-like anogenital glands. These glands may be the origin of breast-like lesions in the vulva, such as fibroadenoma and hidradenoma papilliferum. **(Right)** The mammary-like anogenital glands are morphologically similar to apocrine glands with eosinophilic cytoplasm and apical snouts. These glands are distinguished from vestibular, müllerian, or Bartholin glands by the absence of mucinous or ciliated epithelium.

Bartholin Duct

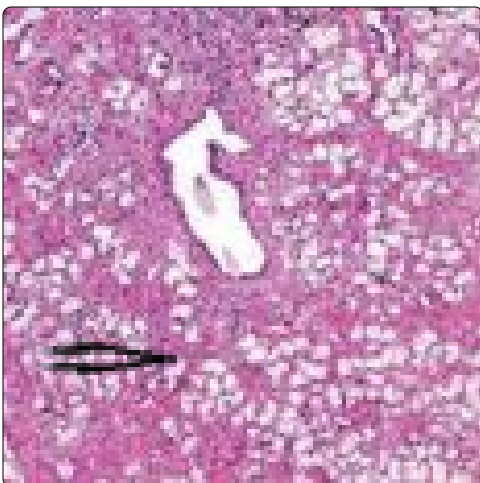


Bartholin Duct: Epithelium

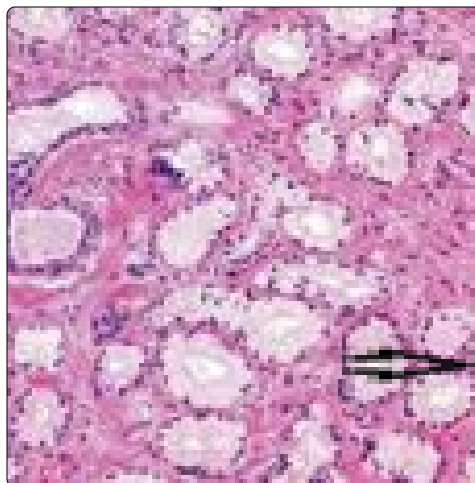


(Left) The Bartholin (bulbourethral) duct enters the vulvar vestibule posterolaterally, adjacent to the hymen. The duct often has a urothelial-type (transitional) epithelium with peripheral foci of mucinous-type glandular epithelium. **(Right)** The Bartholin (bulbourethral) duct epithelium has a histologic appearance similar to urothelium (transitional epithelium). Cystic change may occur in these ducts and may produce a clinically apparent mass lesion.

Bartholin Gland



Bartholin Gland



(Left) The ductal epithelium in the Bartholin (bulbourethral) gland ducts that is located near the main acini has a flattened appearance with scant cytoplasm, while the mucinous glands have abundant blue-gray cytoplasm. **(Right)** The mucinous epithelium of the Bartholin (bulbourethral) glands is characterized by small, cytologically bland, pyknotic nuclei with abundant pale cytoplasm. The intervening stroma is typically densely collagenized.

VAGINA

Macroscopic Anatomy

- Fibromuscular tube extending from vestibule of vulva, between labia minora, to uterine cervix
 - Located posterior (dorsal) to urinary bladder and anterior (ventral) to rectum
 - Anteriorly, vagina is separated from bladder by fibroadipose tissue; urethra enters vaginal wall distally
 - Posteriorly, upper 1/4 adjacent to rectouterine space
 - Middle portion is separated from rectum by rectovaginal septum, layer of fibroadipose tissue
 - Distal portion is separated from anal canal predominantly by sphincter musculature

Microscopic Anatomy

- Stratified nonkeratinizing squamous epithelium
 - Basal cell layer comprised of single columnar epithelial layer with high nuclear:cytoplasmic ratio
 - Parabasal layer comprised of ~ 2 cell layers, above basal layer, with higher nuclear:cytoplasmic ratio than more superficial layers
 - Intermediate cell layer is most prominent with more abundant cytoplasm that is sometimes glycogenated
 - Superficial cell layer appears flattened with cells showing pyknotic nuclei and dense eosinophilic cytoplasm with occasional keratohyalin granules
- Subepithelial stroma (lamina propria)
 - Loose connective tissue with elastic fibers
 - Scattered spindled to stellate stromal cells (nuclei may be multilobated)
 - Venous and lymphatic network
- Muscle
 - Outer longitudinal layer of smooth muscle is continuous with uterus
 - Thin inner circular smooth muscle layer
 - Fewer smooth muscle fibers anteriorly due to urethra
- Adventitia
 - Inner dense connective tissue layer adjacent to muscularis

- Outer loose connective tissue layer contains peripheral nerves, blood vessels, and lymphatics

CONSIDERATIONS AND VARIATIONS

Pitfalls and Artifacts

- Epithelial proliferation and maturation varies with menstrual cycle
 - Predominance of thick glycogenated squamous epithelium just prior to ovulation
 - Gradual decrease in cell layers (and cytoplasm) after ovulation until start of next cycle
- Atrophic epithelial changes
 - Fewer cell layers and loss of cytoplasm imparts basophilic appearance
 - Closely mimics high-grade squamous intraepithelial lesion
- Wolffian (mesonephric) duct remnants (most commonly in lateral walls)
 - Collection of round to elongated glands with central eosinophilic secretion ($\pm$ central duct)
 - Lined by short cuboidal layer of cytologically bland epithelial cells with little cytoplasm
 - May mimic adenocarcinoma, but remnants typically retain some lobular architecture
- Adenosis
 - Glandular epithelium located at stromal-epithelial interface, due to persistence of embryonic müllerian glandular tissue
 - Epithelium may be mucinous, tuboendometrioid, or embryonal type

Age-Related Variations

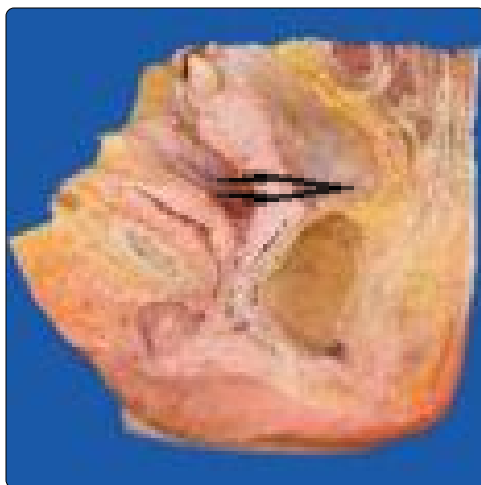
- After menopause, vaginal epithelium atrophies due to low estrogenic state
- In newborns, vaginal epithelium is often fully mature due to maternal estrogen

Metaplasia

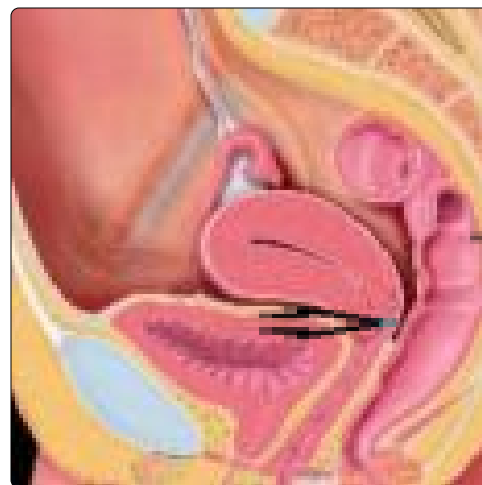
- Parakeratosis of epithelium may be seen with uterine prolapse

Location

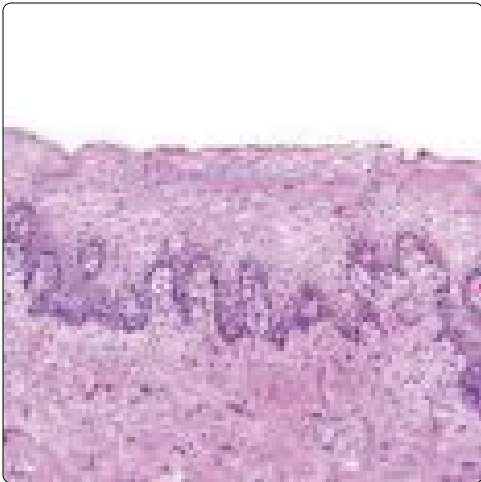
(Left) This sagittal dissection of the female pelvis (anterior on the right) shows the urinary bladder anterior to the uterus. The vagina is located posterior to the urethra. (Right) This schematic diagram shows the anatomic relationships between the urinary bladder, uterus, vagina, and rectum. The uterine cervix is present at the junction between the uterus and the vagina.



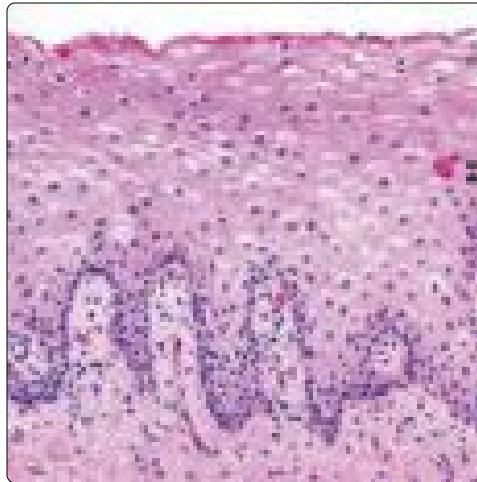
Regional Anatomy



Mucosa



Epithelium

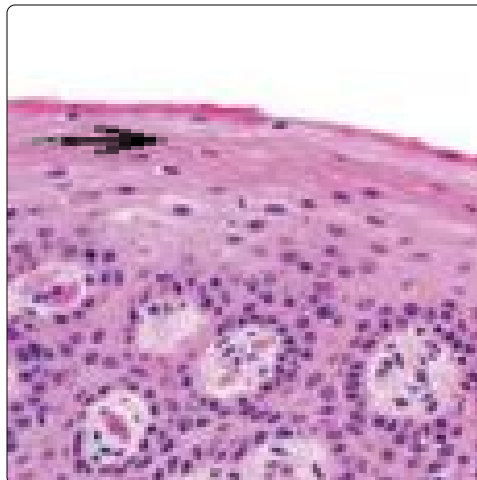


(Left) The vaginal mucosa consists of an epithelial layer and underlying lamina propria. (Right) The epithelium consists of 4 layers. The basal layer is a single layer of columnar cells with scant cytoplasm. The parabasal layer is ~ 2 cell layers thick with slightly more cytoplasm. Intermediate cells are the predominant cell type, and they have abundant eosinophilic cytoplasm. The superficial cells appear flattened, with pyknotic nuclei.

Basal, Parabasal, and Intermediate Cells

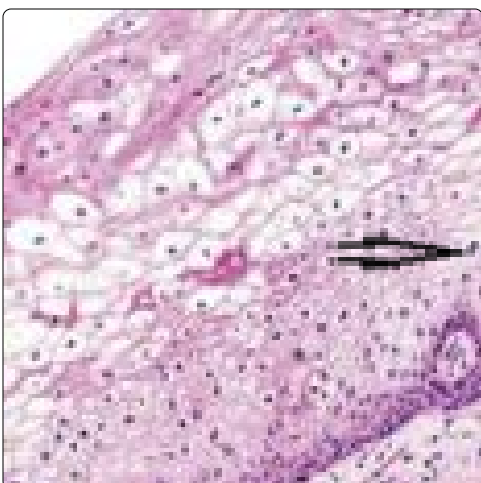


Superficial Cell Layer

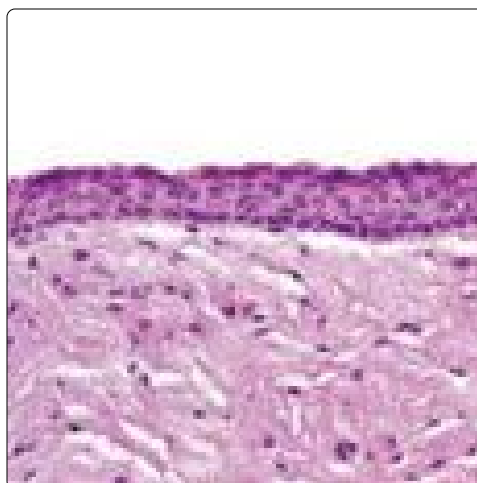


(Left) This section of epithelium shows a distinct basal cell layer at the epithelial-stromal junction. The parabasal cells have slightly more cytoplasm, and the intermediate cells typically have the most cytoplasm. (Right) This section of vaginal epithelium has a more prominent superficial cell layer, which is characterized by flattened cells with pyknotic nuclei arranged in a plane parallel to the basement membrane.

Epithelium (Glycogenated)



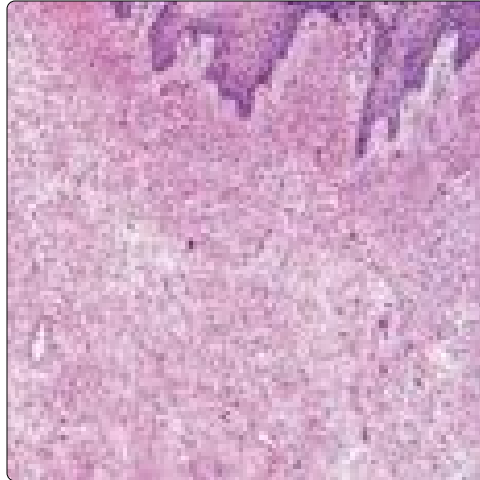
Epithelium (Atrophic)



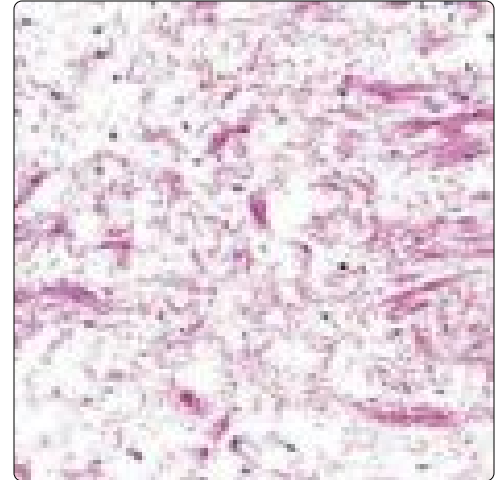
(Left) The intermediate cells may contain abundant intracytoplasmic glycogen that can potentially mimic the koilocytic changes of HPV infection. (Right) After menopause, reduced estrogen often leads to marked vaginal atrophy. Atrophic squamous epithelium has fewer cell layers, and the individual cells have less cytoplasm; these features impart a more basophilic appearance. These histologic changes may mimic a high-grade squamous intraepithelial lesion.

(Left) The lamina propria  lies directly beneath the squamous epithelium  and consists of collagen fibers of variable density, elastic fibers, sparse stromal cells, and varying numbers of venous and lymphatic channels. The lamina propria is typically more dense in deeper areas adjacent to the muscularis. **(Right)** The lamina propria of the vagina may have a very loose, hypocellular stroma with an edematous appearance.

Lamina Propria

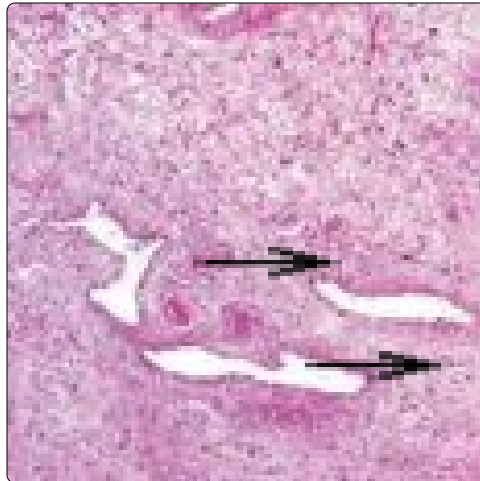


Lamina Propria: Composition

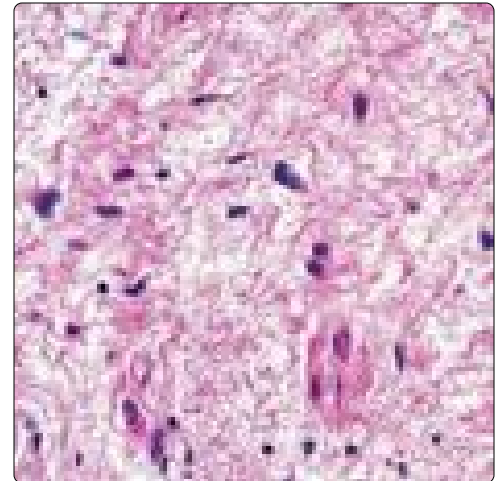


(Left) The lamina propria may also contain venous channels  that are similar to the erectile tissue seen in other anatomic sites. **(Right)** The vaginal lamina propria is typically very hypocellular. The individual stromal cells may be spindled, fusiform, or stellate. A subset of these stromal cells may have multilobated nuclei .

Lamina Propria: Vascular Channels

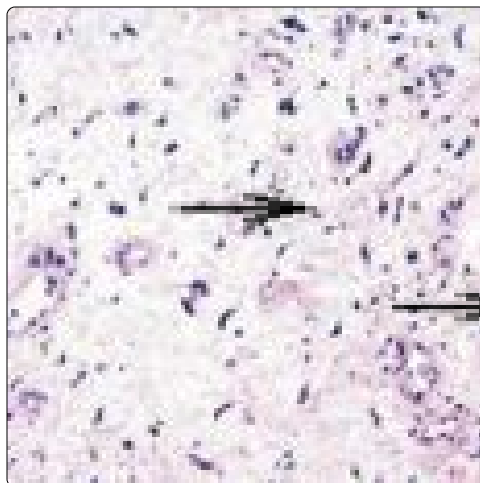


Lamina Propria: Cytologic Features

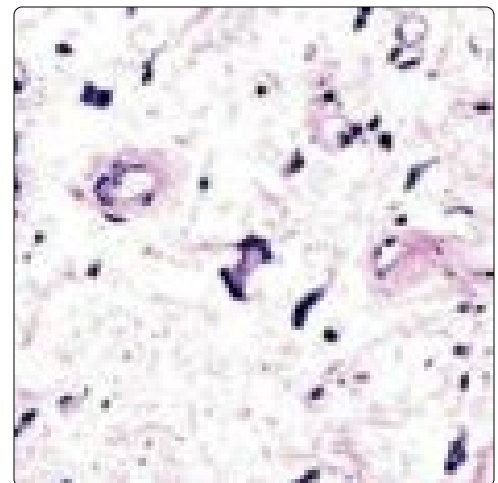


(Left) The multilobated stromal cells  of the vaginal lamina propria may be striking, even on low-power scanning magnification. Similar cells may also be seen in the cervix and vulva. **(Right)** The presence of these multilobated nuclei  in the vaginal lamina propria stromal cells may suggest the possibility of a neoplastic process, but this degree of atypia is within the spectrum of normal. These cells are identical to those seen in a fibroepithelial polyp.

Lamina Propria: Multilobated Stromal Cells



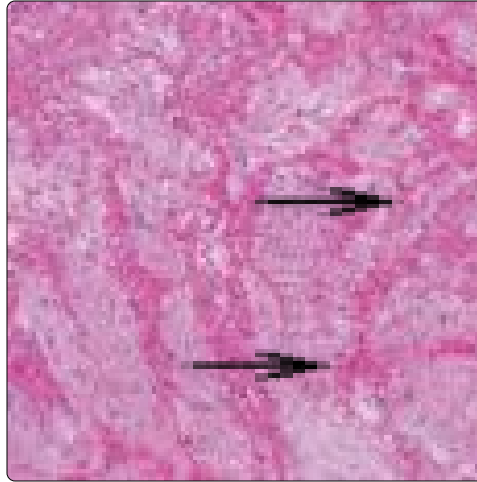
Multilobation



Vaginal Muscularis

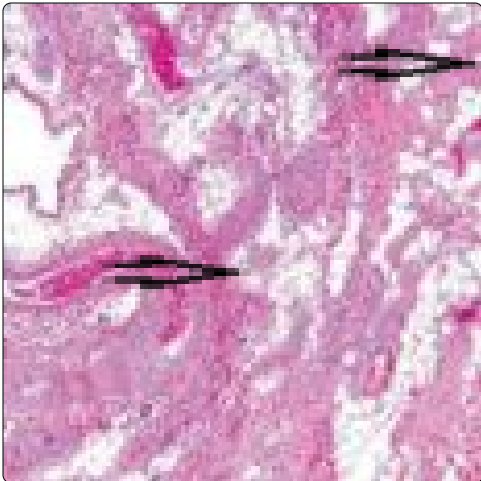


Muscularis: Smooth Muscle

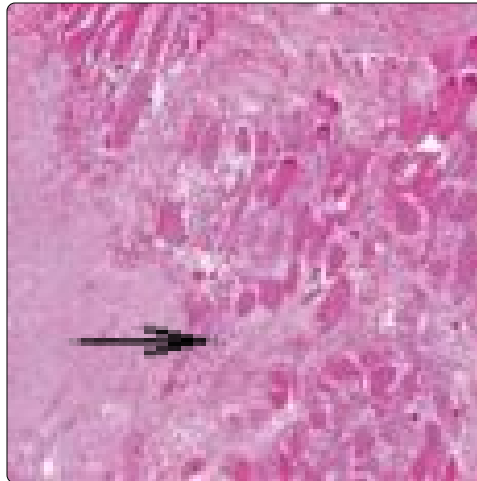


(Left) The muscularis of the vagina is composed of inner circular and outer longitudinal smooth muscle layers, but these 2 layers may be relatively indistinct. The outer longitudinal layer of the vaginal muscularis, which is thicker and more conspicuous, is continuous with the muscularis of the uterus. **(Right)** The muscularis is comprised of variably arranged fascicles of normal smooth muscle.

Rectovaginal Septum



Distal Vaginal Wall

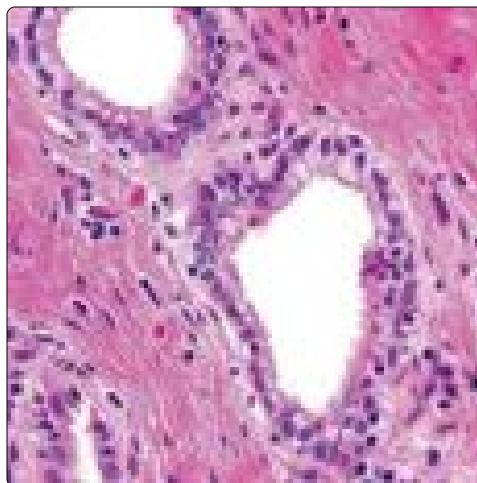


(Left) In the middle segment of the vagina, the rectum is separated from the posterior aspect of the vagina by a rim of fibroadipose tissue (the rectovaginal septum) that also contains blood vessels and small nerves. **(Right)** In the distal vagina near the introitus, the muscularis of the vaginal wall may merge and become admixed with the skeletal muscle fibers of the sphincter musculature (bulbospongiosus muscle).

Wolffian Duct Remnants



Müllerian Remnants



(Left) Wolffian duct (also called mesonephric or Gartner duct) remnants are not infrequently identified in the vagina. These glandular structures are typically lined by a cuboidal epithelial layer with little cytoplasm, and the lumen is often filled with eosinophilic secretions. **(Right)** Although rarely seen outside the setting of diethylstilbestrol exposure, remnants of müllerian ducts may occasionally be seen in the vagina. This example has a ciliated, tubal-type epithelium.

UTERUS

Macroscopic Anatomy

- Complex organ with multiple regions: Ectocervix, endocervix, lower uterine segment, endometrium, myometrium, and serosa
- Ectocervix: Distal end of uterus, typically covered by stratified squamous epithelium
- Endocervix: Tunnel-like lumen of uterine cervix, lined by mucin-secreting pits
- Lower uterine segment: Triangle-shaped transition from endocervix to uterine corpus
- Endometrium: Uterine mucosa composed of glands and stroma
- Myometrium: Muscular wall of uterus composed of layers of smooth muscle
- Serosa: Thin lining of body and fundus of uterus; covers portions of uterus that are not retroperitoneal

Microscopic Anatomy

- Ectocervix: Stratified squamous epithelium overlying fibrous stroma
- Transformation zone
 - Squamocolumnar junction that lies between ectocervix and endocervix
 - Zone identified by squamous metaplasia of endocervix; migrates during reproductive years
- Endocervix: Simple, columnar mucinous epithelial cells line endocervical pits that extend into fibrous stroma
- Lower uterine segment: Transition between endocervix and endometrium; admixtures of endometrial and endocervical glands and stroma may be seen in this region
- Endometrial glands
 - Columnar epithelial cells that vary from simple to pseudostratified
 - Epithelium may contain vacuoles (secretory) or be ciliated
 - Functionalis: Superficial endometrial layer, lost during menstruation
 - Basalis: Deep layer of more monomorphic glands that regenerates functionalis after shedding

- Endometrial stroma
 - Densely cellular supporting stroma composed of spindle cells with elongated nuclei admixed with blood vessels (spiral arterioles)
 - Stromal cells range from small and spindled (proliferative phase) to large, eosinophilic, and polygonal (secretory phase, pregnancy, hormone effect)
- Myometrium: Smooth muscle wall of uterus, composed of 3 layers (inner longitudinal, middle circular, and outer longitudinal)
 - Middle circular layer heavily vascularized and contains arcuate arteries
- Serosa: Thin mesothelial lining of uterine corpus and fundus

CONSIDERATIONS AND VARIATIONS

Age-Relation Variations

- During puberty, endocervical epithelium migrates toward ectocervix (ectropion) and then is slowly replaced by squamous epithelium during reproductive years
- Endometrial atrophy: Inactive, cystically dilated glands may be present after menopause

Metaplasia

- Squamous metaplasia: Occurs along transformation zone
- Ciliated (tubal) metaplasia: Common; occurs in endocervix and endometrial glands
- Mucinous metaplasia: Uncommon metaplasia of endometrium; typically consists of simple to stratified, columnar mucinous epithelial cells

Pitfalls and Artifacts

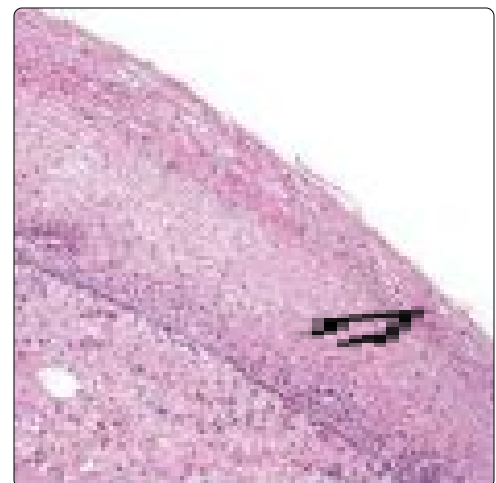
- Glandular compression, simulating hyperplasia, may be seen following biopsy
- Suction created by biopsy instruments may lead to "gland-in-gland" artifact
- Exfoliation artifact leads to epithelial shedding, which can mimic serous carcinoma
- Arias-Stella effect may closely resemble carcinoma due to marked atypia and glandular crowding

(Left) The uterus lies in the pelvis and comprises the ectocervix , endocervix , endometrium , and myometrium . (Right) The ectocervical mucosa consists of stratified, nonkeratinizing squamous epithelium composed of 3 layers: Basal/parabasal , stratum spongiosum , and superficial .

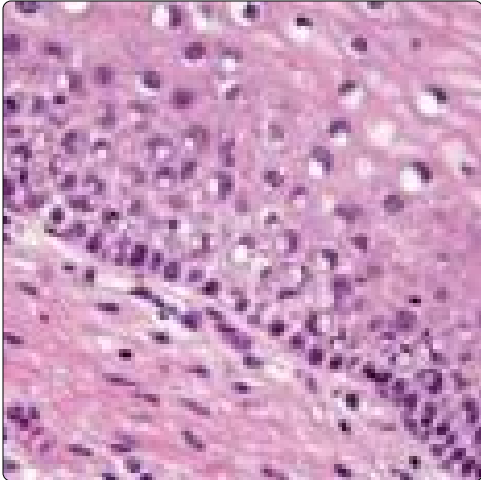
Uterus



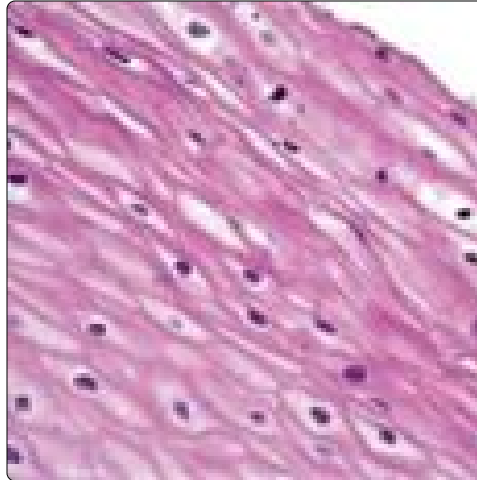
Ectocervix



Ectocervix: Basal Layer



Ectocervix: Superficial

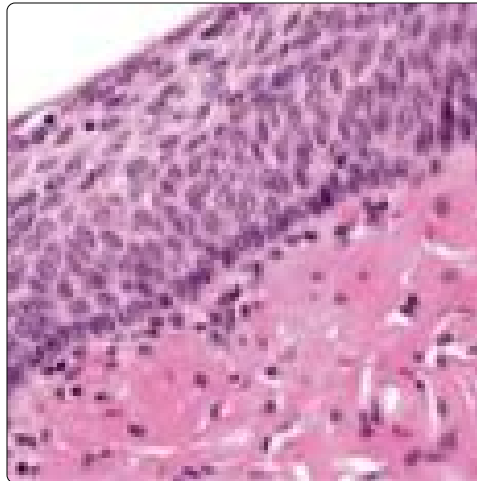


(Left) The basal/parabasal layers are composed of round to cuboidal cells. Basal cells [1] have scant cytoplasm and dark nuclei, whereas parabasal cells [2] are slightly larger and with more cytoplasm. Scattered mitotic figures may be seen. **(Right)** The stratum spongiosum and superficial layers compose the majority of the squamous epithelium. As cells mature, the nuclei become smaller and darker, while the cells become flat. Superficial cells may not have visible nuclei [3].

Ectocervix: Atrophy

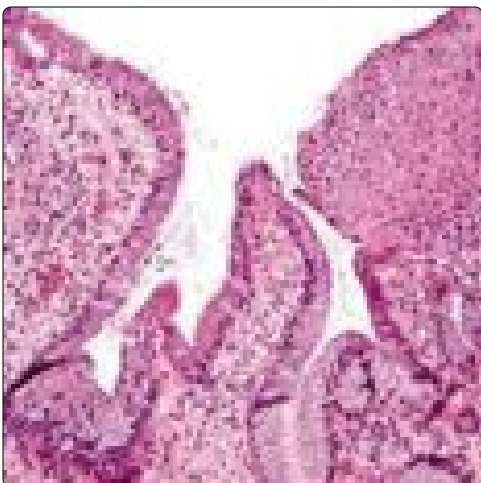


Ectocervix: Atrophy (Cytology)

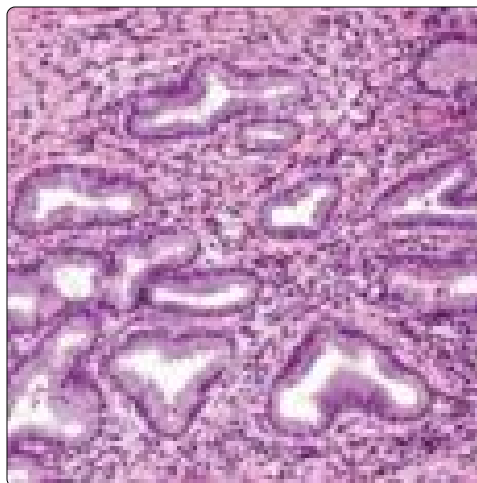


(Left) As patients age, the cervical epithelium may undergo atrophic changes. In atrophy, the epithelium is composed predominately of basal and parabasal cells. An easily apparent and orderly basal layer [1] is usually present. **(Right)** This photograph of atrophy shows the basal [2] and parabasal [3] layers along with a thinned stratum spongiosum and superficial layer. Loss of glycogen leads to decreased cytoplasm relative to the nucleus, which may mimic dysplasia. Mitotic figures should be absent.

Cervix: Transformation Zone



Endocervix: Endocervical Glands



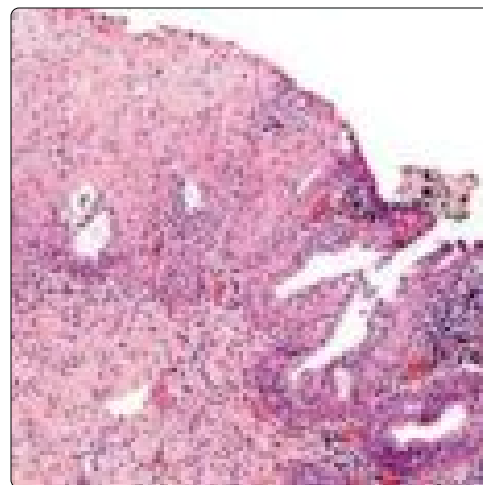
(Left) The transformation zone is an irregularly defined junction of squamous [1] and mucinous [2] epithelium. This squamocolumnar junction migrates to the ectocervix during puberty. As females age, squamous metaplasia causes the junction to migrate back into the endocervical canal. **(Right)** Endocervical "glands" (technically crypts, as they connect to the surface) are composed of a simple columnar mucinous epithelium [3] with basally oriented nuclei and amphophilic cytoplasm.

(Left) Cervical stroma is predominantly composed of fibrous tissue and blood vessels. Small wisps of smooth muscle may be present but are rarer than in the rest of the uterus. Scattered lymphocytes and plasma cells are normal. **(Right)** Squamous metaplasia occurs at the transformation zone and can extend along the surface of the endocervix. The metaplasia may involve glands, causing thickening of the epithelium and loss of normal mucinous cells.

Endocervix: Stroma

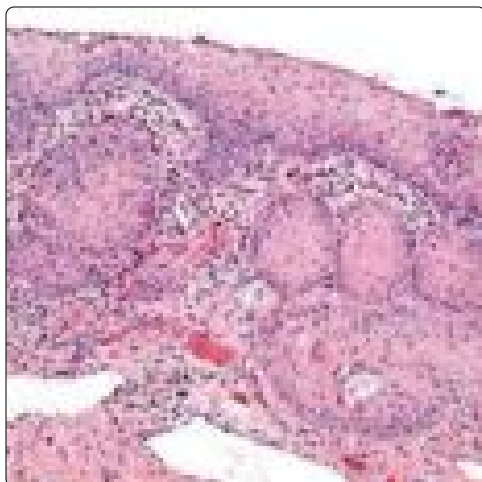


Endocervix: Squamous Metaplasia

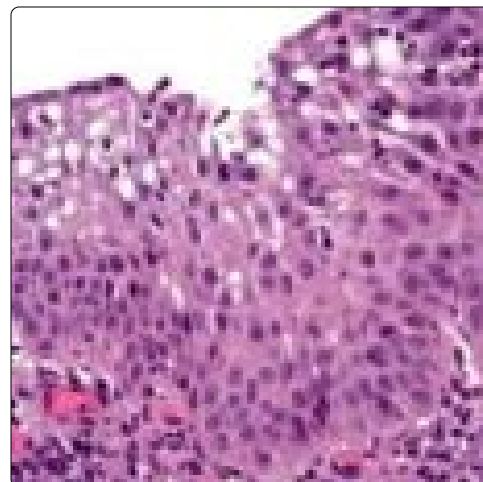


(Left) Squamous metaplasia can fill the endocervical glands, causing obliteration of the gland lumen. These glands typically appear as round nests of normal squamous epithelium. **(Right)** Immature squamous epithelium may occupy the transformation zone. This epithelium contains less glycogen than typical cervical epithelium and is composed of immature keratinocytes throughout all levels. These immature cells may mimic dysplasia; however, true atypia and atypical mitotic activity are absent.

Endocervix: Squamous Metaplasia (Nests)



Immature Squamous Metaplasia



(Left) The columnar mucinous cells of the endocervix may also undergo metaplasia to tubal epithelium. Involved glands may appear more hyperchromatic at low power, at times resembling endometrial glands or glands involved by adenocarcinoma in situ. **(Right)** High-power examination of areas of tubal metaplasia will reveal tubal-type epithelium composed of both ciliated and secretory cells. Note the lack of atypia, mitotic activity, and apoptotic debris seen in adenocarcinoma in situ.

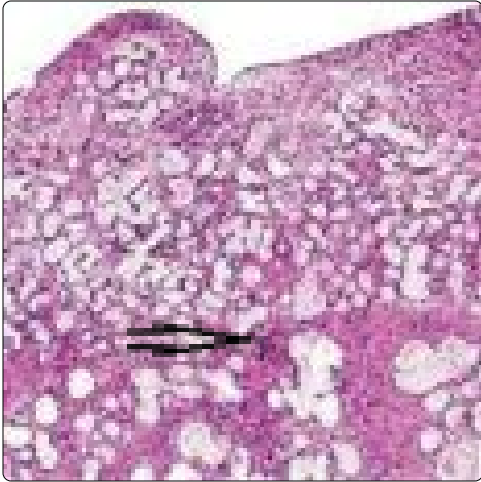
Endocervix: Tubal Metaplasia



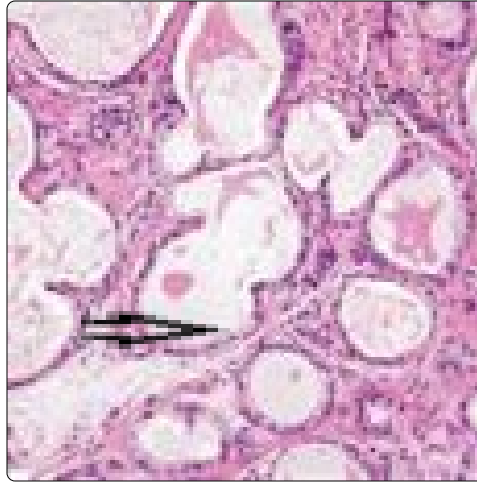
Endocervix: Tubal Metaplasia (Ciliation)



Endocervix: Tunnel Clusters

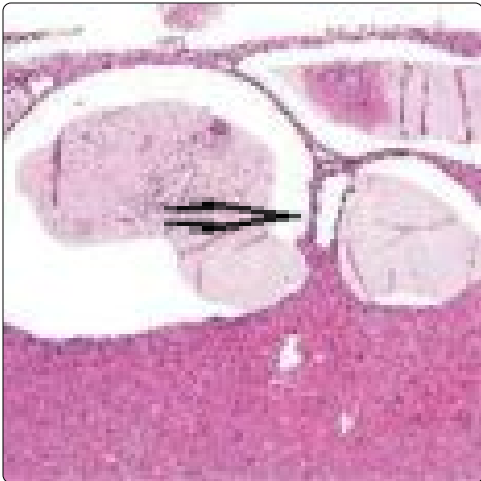


Endocervix: Tunnel Clusters (Cytology)

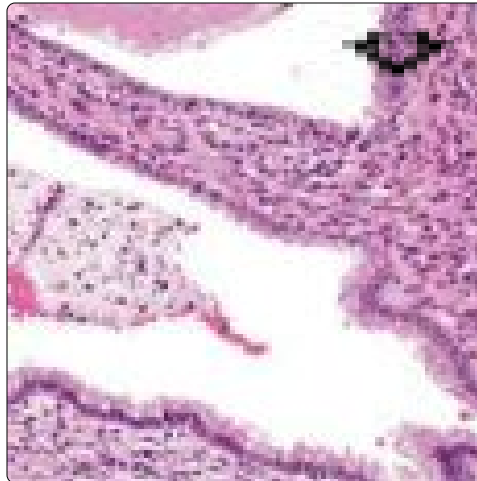


(Left) As squamous epithelium covers the openings to the endocervical glands, the glands become dilated and filled with mucinous material. Type A tunnel clusters are small, whereas type B tunnel clusters are dilated. Frequently, the 2 patterns are intermixed. (Right) Type A tunnel clusters have an attenuated simple cuboidal lining of mucinous cells and inspissated, eosinophilic, mucinous material in the lumina.

Endocervix: Nabothian Cyst

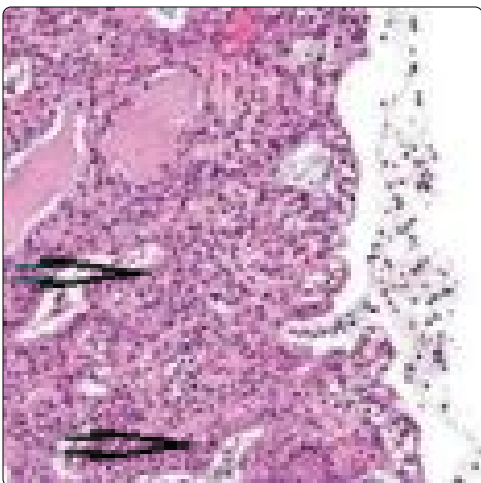


Endocervix: Nabothian Cyst (Lining)

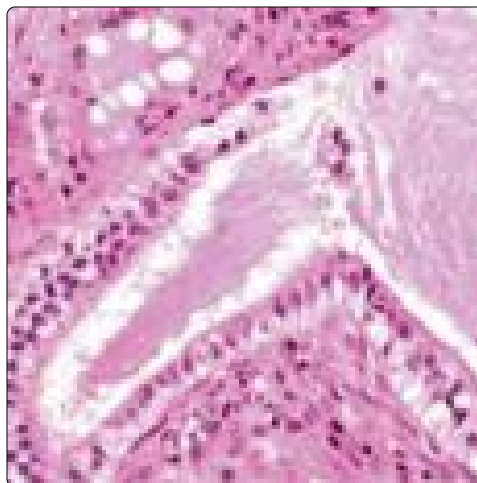


(Left) Occasionally, endocervical glands occluded by the process of squamous epithelialization will expand, forming cystic spaces filled with mucin. These cysts are known as nabothian cysts and may become large enough to form a cervical mass. (Right) The lining of a nabothian cyst is typically composed of endocervical columnar mucinous epithelium. The lining may become attenuated and assume a low cuboidal morphology.

Endocervix: Microglandular Hyperplasia



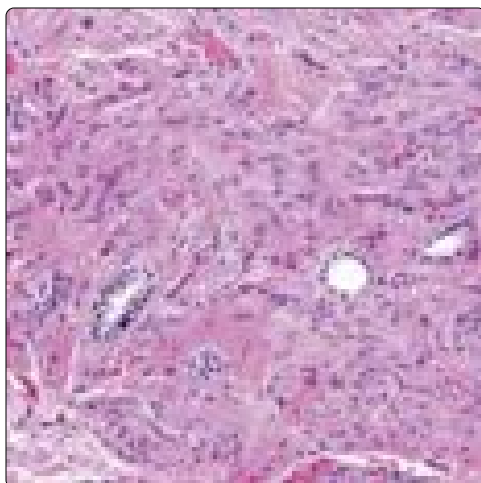
Endocervix: Microglandular Hyperplasia (Cytology)



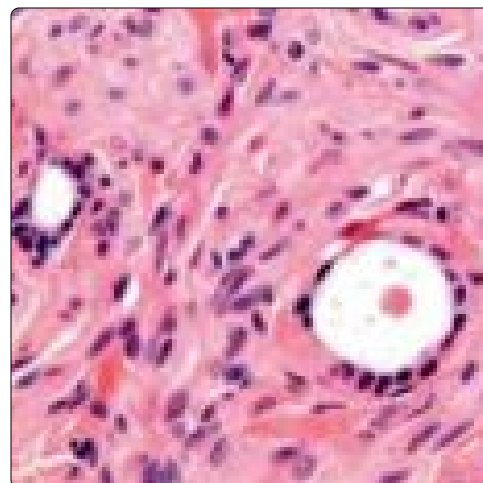
(Left) Microglandular hyperplasia of the cervix is composed of an increased number of small glands that proliferate in the endocervix. This phenomenon has been associated with oral contraceptive use. (Right) The glands that make up microglandular hyperplasia often have irregular contours, columnar mucinous epithelium with subnuclear vacuoles, and inspissated mucin. Scattered neutrophils can be seen in both the mucin and the epithelium.

Cervix: Mesonephric Remnant

(Left) Scattered small tubular glands may be seen deep within the cervical stroma. These glands are composed of low cuboidal epithelium and generally lack atypia and mitotic activity. Occasionally, they may be associated with a long cleft-like space. (Courtesy A. Cole, MD.) (Right) Mesonephric remnants are composed of bland, low cuboidal epithelium. Eosinophilic, PAS(+) secretions can commonly be identified within the lumen. (Courtesy A. Cole, MD.)

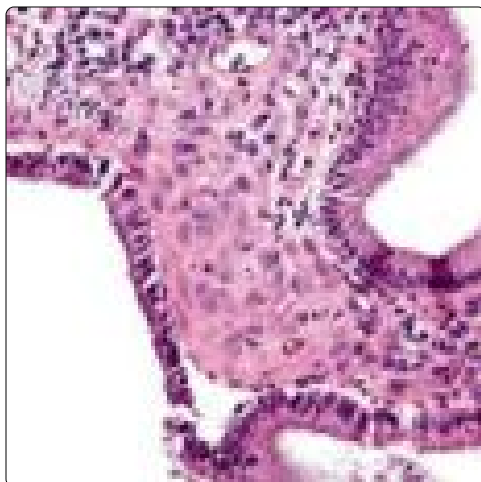


Cervix: Mesonephric Remnant (Cytology)

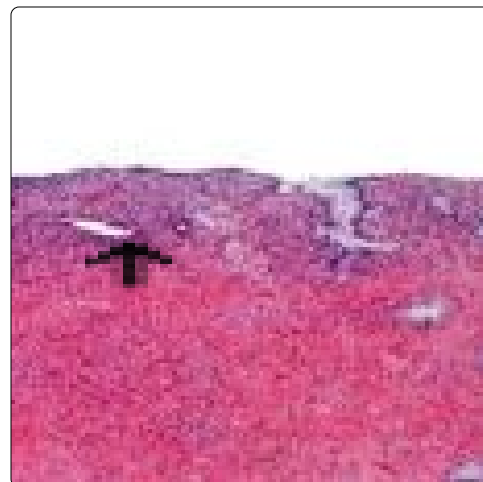


Cervix: Decidua

(Left) In pregnant patients, a stromal decidual reaction may occur. Ectopic decidua is composed of large, polygonal, eosinophilic cells with bland nuclei. Nests of these cells may mimic squamous metaplasia or invasive squamous cell carcinoma but can be distinguished from the latter by the lack of atypia. (Right) As the endocervix transitions to the lower uterine segment, the glands lose their basophilic mucin, and the stroma becomes more spindle and basophilic.

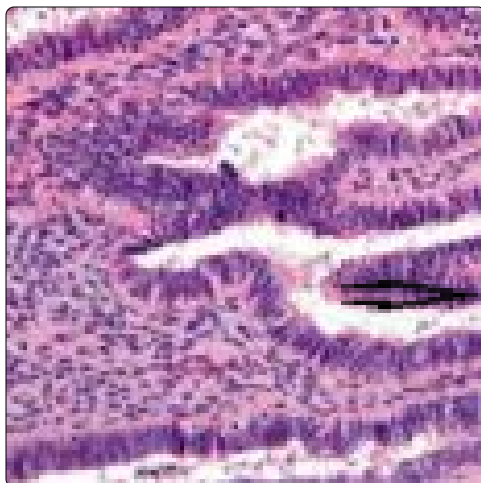


Cervix and Lower Uterine Segment

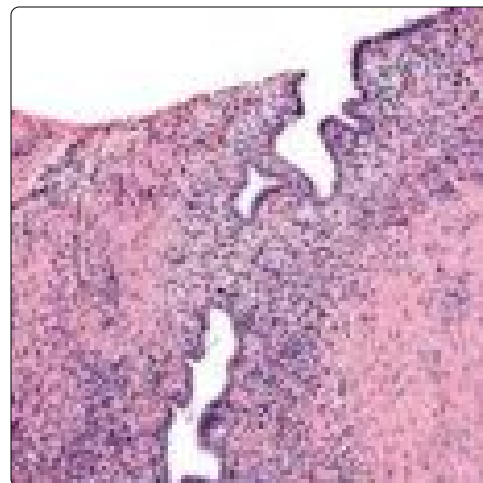


Lower Uterine Segment

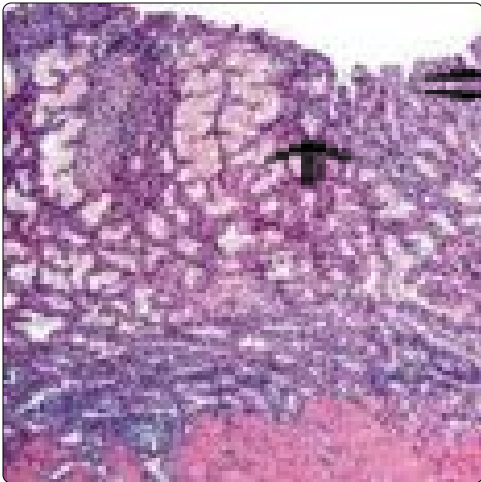
(Left) The lower uterine segment displays glandular and stromal changes that are intermediate between the endocervix and the endometrium. These endocervical glands are composed of columnar epithelium that lost the basophilic mucin seen in endocervical epithelium. (Right) The endometrial lining of the lower uterine segment is thinner than the uterine corpus endometrial lining. The stroma becomes more cellular and basophilic as it approaches the endometrium.



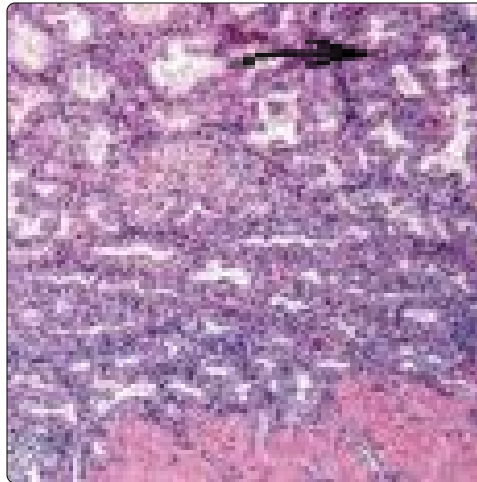
Lower Uterine Segment: Stroma



Endometrium

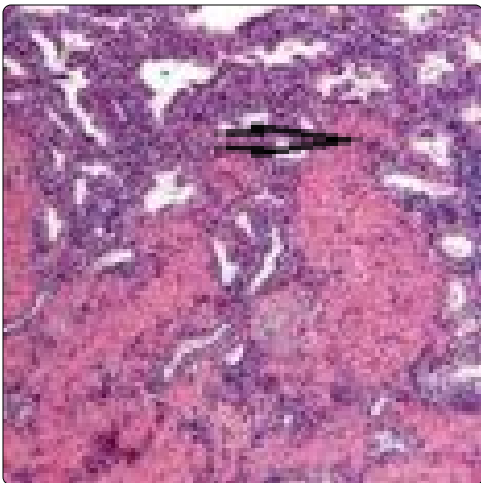


Endometrial Layers



(Left) The endometrial lining of the uterine corpus is much thicker than that of the lower uterine segment. It is composed of glands with intervening stroma. (Right) Two distinct layers of endometrium can be identified. The layer adjacent to the myometrium is known as the basalis. The basalis is the reserve cell layer of the endometrium that regenerates the functionalis after menses. The basalis, as opposed to the functionalis, is only slightly hormonally responsive.

Endomyometrium: Irregular Junction

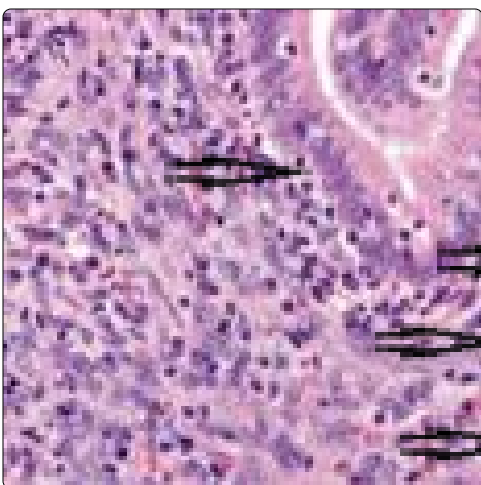


Endometrium: Lymphoid Aggregate

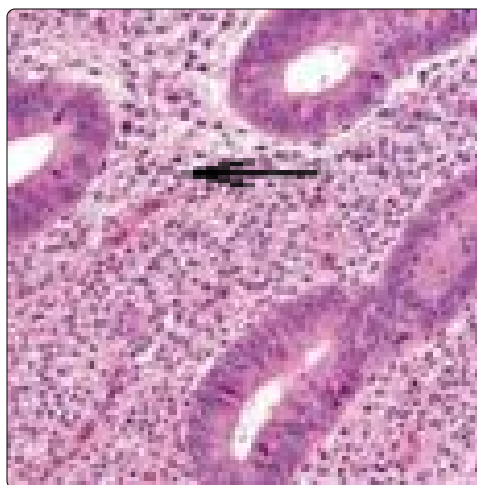


(Left) The interface between the endometrium and the myometrium may be irregular, which may cause difficulty in determining the depth of invasion in endometrial adenocarcinoma. The presence of endometrial stroma may help identify an irregular endometrial/myometrial interface involved by tumor as opposed to true invasion. (Right) Scattered lymphoid aggregates are normally seen within the endometrial stroma. These are a normal finding and not diagnostic of chronic endometritis.

Endometrium: Chronic Endometritis



Proliferative Endometrium



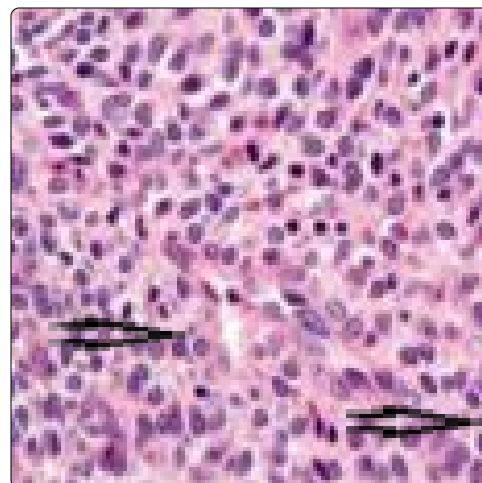
(Left) Plasma cell infiltrates are not a normal finding within the endometrium and are generally diagnostic of chronic endometritis. Endometrial polyps may contain rare plasma cells; however, this alone does not warrant a diagnosis of chronic endometritis. (Right) Proliferative endometrial glands have a rounded contour and are composed of pseudostratified columnar cells. Scattered mitotic figures should be readily apparent. Stromal mitoses may be identified.

Proliferative Gland

(Left) Proliferative glands are composed of pseudostratified columnar cells with mitotic figures [\[1\]](#). The nuclei are elongated (cigar-shaped) and contain vesicular chromatin and multiple small nucleoli [\[2\]](#). In the early proliferative phase, the glands are uncoiled, and as they grow, they become more convoluted. **(Right)** During the proliferative phase, the stromal cells have round to spindle nuclei [\[3\]](#) and scant cytoplasm. Mitotic figures are typically easy to identify [\[4\]](#).

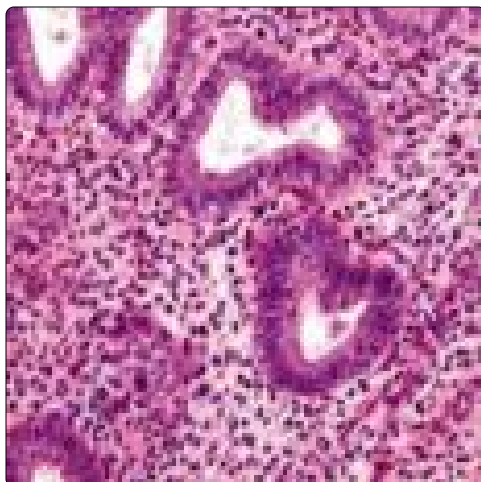


Proliferative Stroma

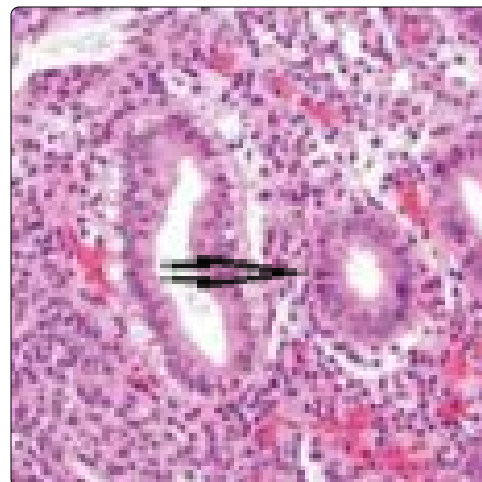


Interval Phase Endometrium

(Left) Around day 10 or 11 of the menstrual cycle, small subnuclear vacuoles [\[1\]](#) begin to form. These vacuoles will become more pronounced, peaking at day 17. The glandular contours become more irregular as the gland grows. Scattered proliferative-type glands and rare mitotic figures may be identified. **(Right)** At least 1/2 of the glands should contain subnuclear vacuoles [\[2\]](#) in early secretory endometrium (around day 16). (Courtesy A. Cole, MD.)

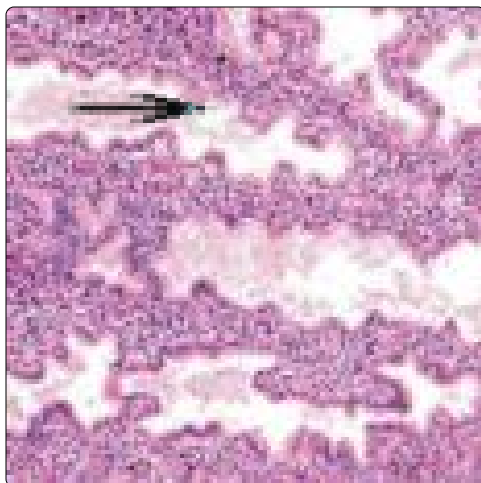


Early Secretory Endometrium

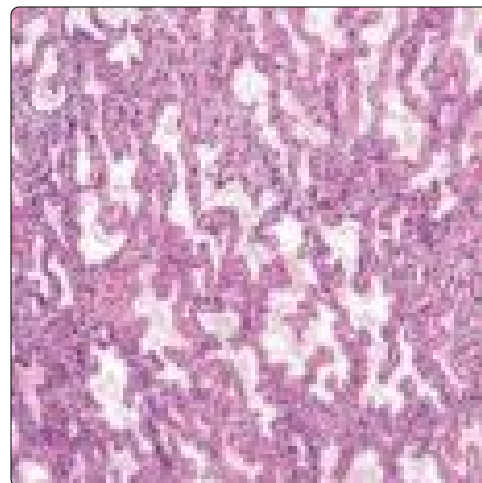


Secretory Endometrium

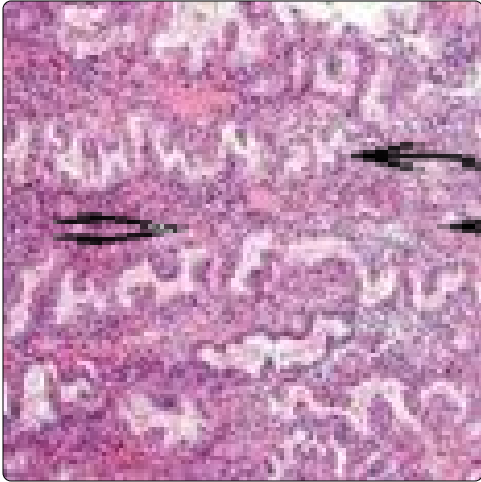
(Left) The middle of the secretory phase is marked by glands that lack cytoplasmic vacuoles but have abundant luminal secretions [\[1\]](#). The glandular epithelium is composed of a simple columnar epithelium with abundant eosinophilic cytoplasm [\[2\]](#). **(Right)** During the secretory phase, extensive glandular crowding may be present. This crowding must be distinguished from the glandular crowding seen in hyperplasia, which usually carries a proliferative morphologic profile.



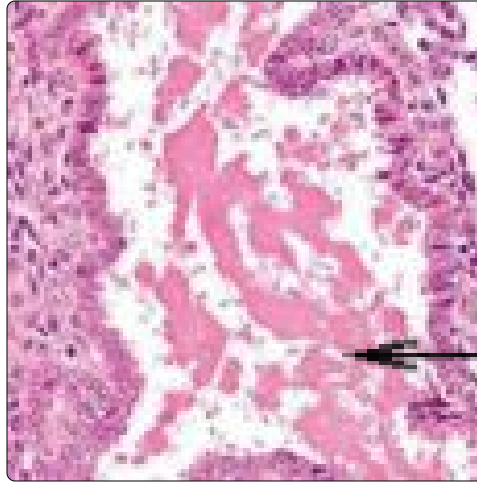
Secretory Endometrium: Crowding



Late Secretory Endometrium

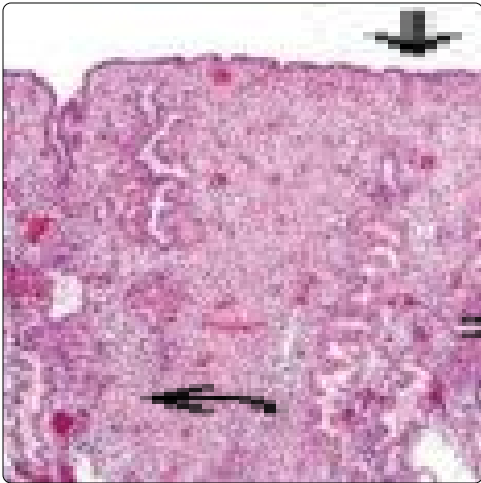


Late Secretory Endometrium: Eosinophilic Secretions

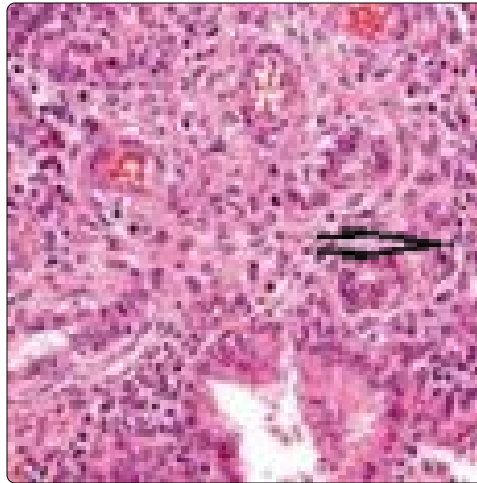


(Left) As the glands progress through the secretory phase, the glandular lumina become more convoluted (or "saw-toothed"). The spiral arterioles become more pronounced due to the presence of an early predecidual cuff surrounding the vessels. (Right) A late secretory gland with a dilated lumen contains inspissated eosinophilic secretions. Note the absence of vacuoles. The lining epithelium is columnar to cuboidal with rounded vesicular nuclei.

Secretory Endometrium: Surface

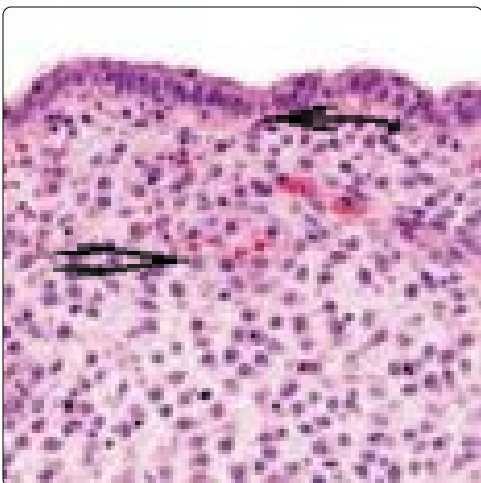


Spiral Arterioles

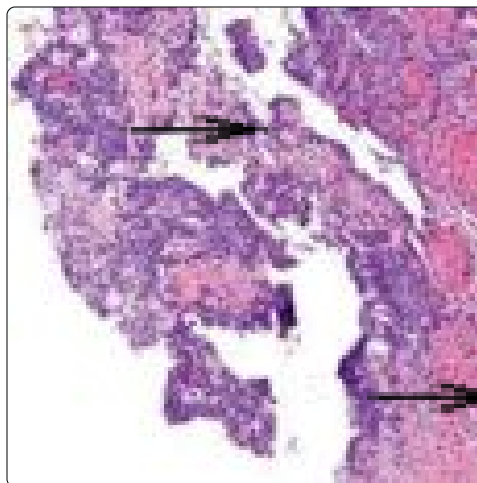


(Left) Stromal changes mark progression into the late secretory phase. These changes consist of the development of predecidua surrounding the superficial spiral arterioles. This predecidual change then expands and begins to coalesce beneath the endometrial surface. Exhausted, serrated secretory glands are present. (Right) These superficial spiral arterioles are surrounded by a cuff of relatively large, polygonal, eosinophilic predecidual cells.

Predecidua



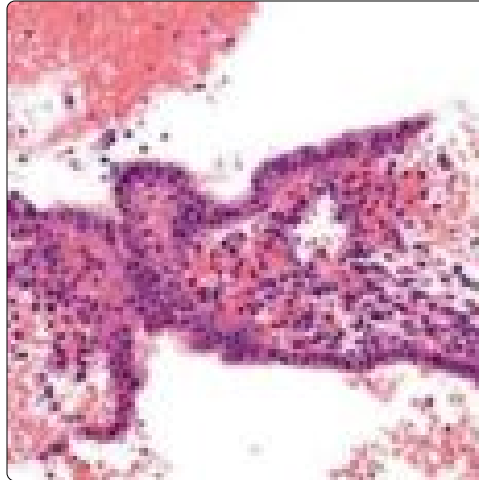
Menstrual Endometrium



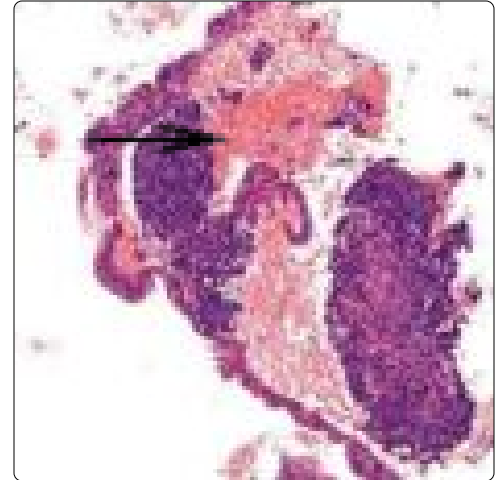
(Left) The predecidual cells coalesce beneath the endometrial surface. These cells contain abundant eosinophilic cytoplasm and are separated by increasing amounts of stromal edema. Scattered stromal inflammatory cells (granulocytes) are present and will increase in number as menstruation is approached. (Right) The superficial endometrium begins to fragment. Stromal hemorrhage and condensation (clumping) are present as well. (Courtesy A. Laury, MD.)

Menstrual Endometrium: Features

(Left) A fragment of superficial endometrium during menstruation shows marked eosinophilia. Stromal fragmentation and hemorrhage are apparent. (Right) As the endometrium breaks down, the stromal cells condense, forming rounded aggregates. Typically, a rim of eosinophilic epithelium can be seen adherent to the stromal fragments. This epithelium may contain nuclear atypia and should not be confused with malignancy.

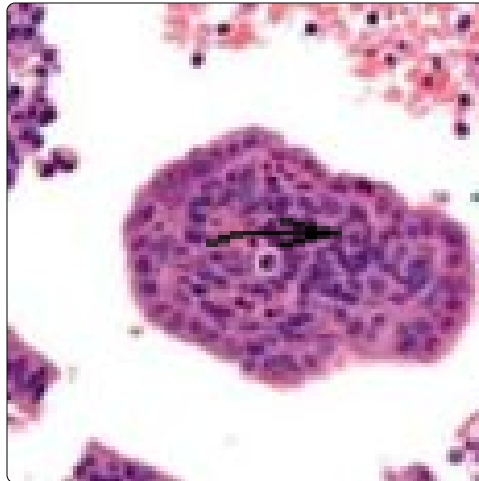


Endometrial Breakdown

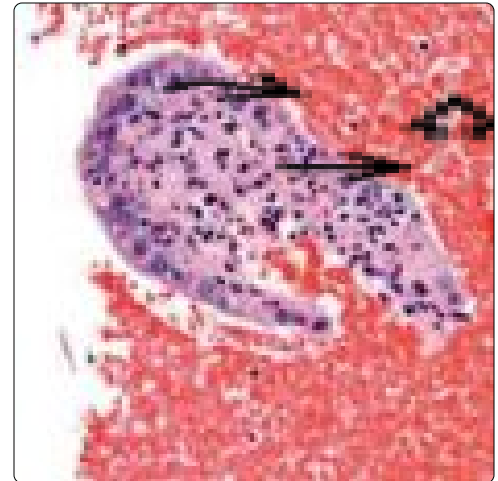


Endometrial Breakdown: "Blue Balls"

(Left) Stromal fragments, also known as "blue balls," are common in stromal breakdown. An apoptotic cell and scattered inflammatory cells can be seen within the stroma. Note the rim of eosinophilic glandular epithelium. (Right) This endometrial stromal fragment is extensively permeated by stromal granulocytes. The lining of epithelium has mild atypia consisting of variation of nuclear size and shape as well as variable nuclear chromasia.

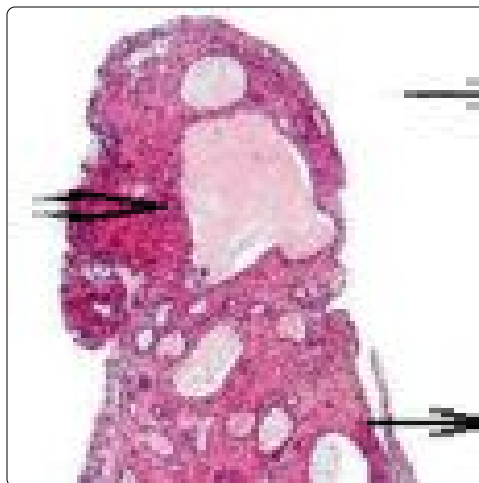


Endometrial Breakdown: Cytologic Features

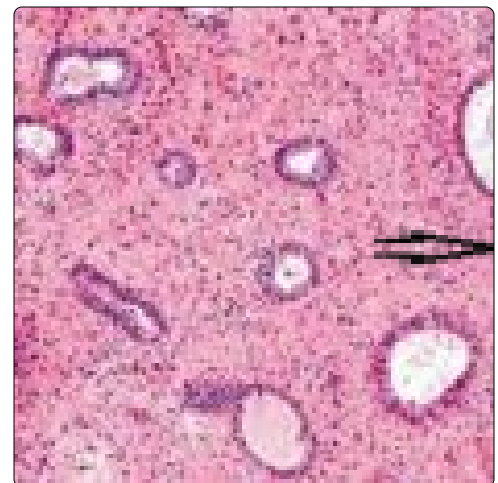


Endometrial Polyp

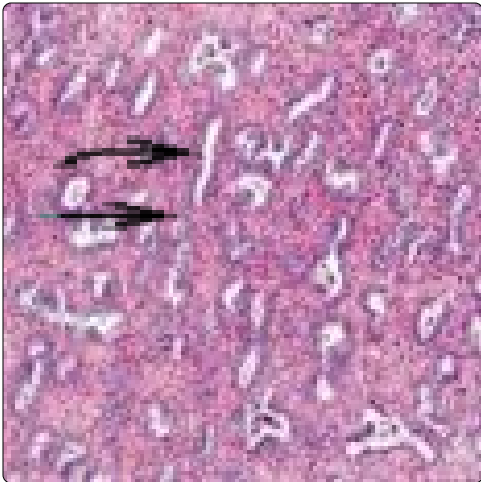
(Left) Endometrial polyps are common findings when examining uterine samples. In an intact specimen, one can appreciate the lining of the polyp along 3 sides by surface epithelium. Altered endometrial glands are another common finding. (Right) In addition to a polypoid shape and irregular endometrial glands, the stroma of the polyp differs from the stroma of the nonpolypoid endometrium. Typically, the stroma is less cellular and has a fibrotic, eosinophilic appearance.



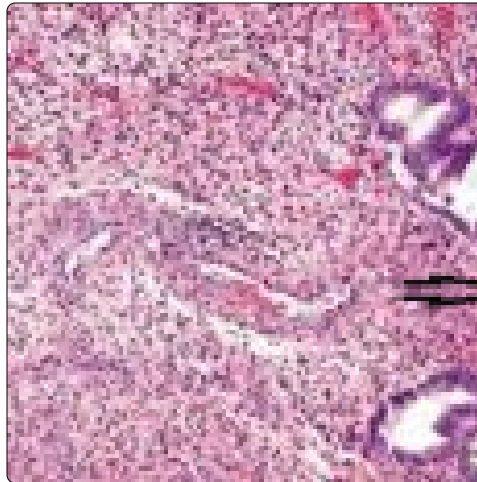
Endometrial Polyp: Stroma



Endometrial Polyp: Glands

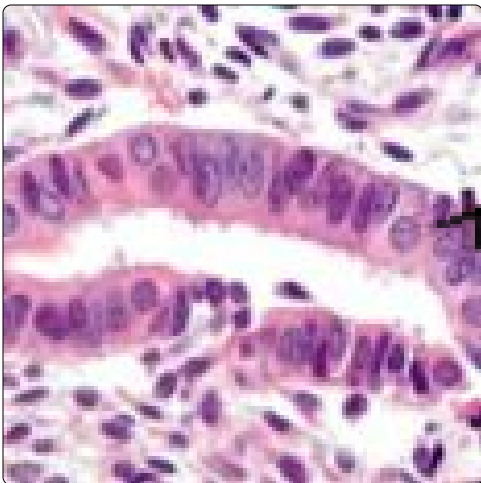


Endometrial Polyp: Blood Vessels

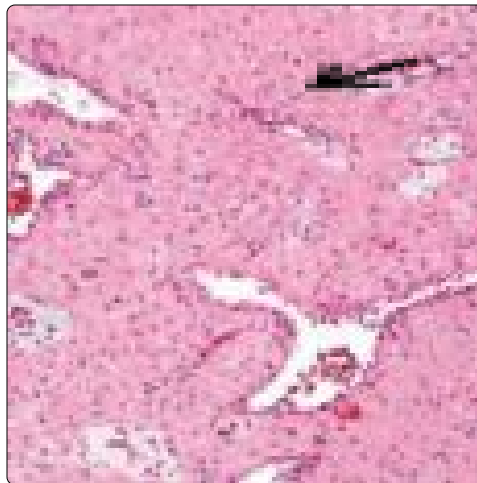


(Left) The morphology of endometrial glands in polyps is highly variable. The glands may be irregular or cystically dilated. The amount of stroma between the glands is variable as well, leading to glands that are widely spaced or slightly crowded, as seen in this example. (Right) Frequently, thick-walled blood vessels may be seen in the polyp stroma. Note the presence of relatively acellular stroma that is more eosinophilic than typical endometrial stroma.

Ciliated Cells

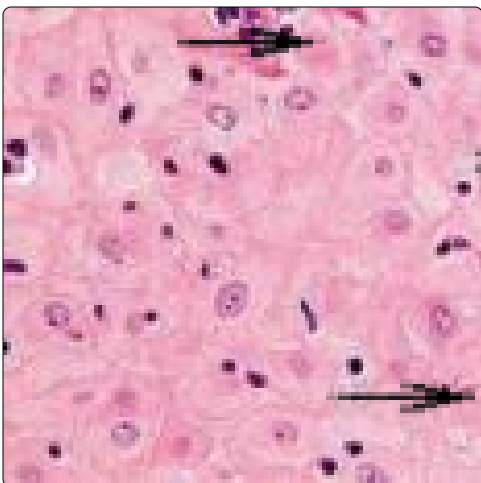


Decidualized Endometrium

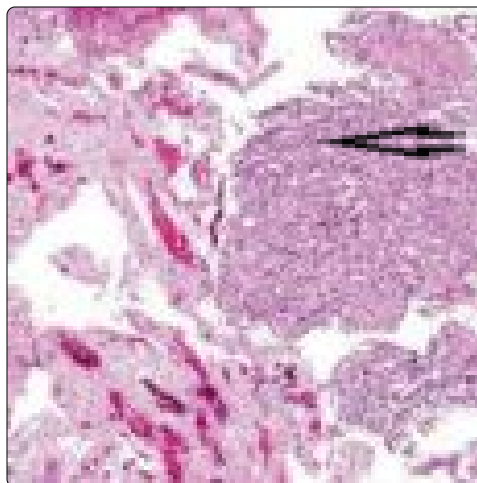


(Left) Ciliated cells can occasionally be found within endometrial glands. These cells are pear-shaped with round nuclei and abundant eosinophilic cytoplasm and apical cilia. (Right) During pregnancy, the endometrial glands and stroma become altered or decidualized. The glandular epithelium becomes thinned and either simple cuboidal or hobnailed. Endometrial stromal cells become enlarged and hypereosinophilic.

Decidualized Endometrium: Cytologic Features



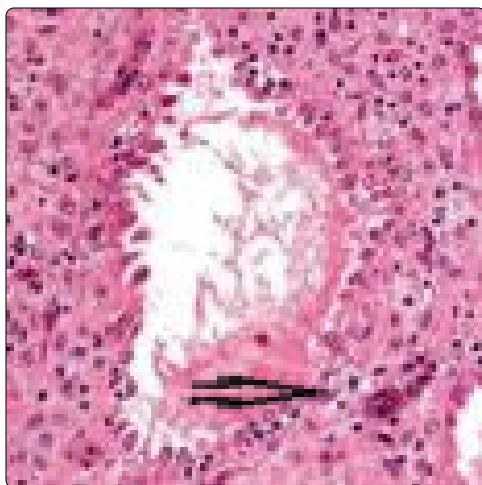
Decidualized and Secretory Endometrium



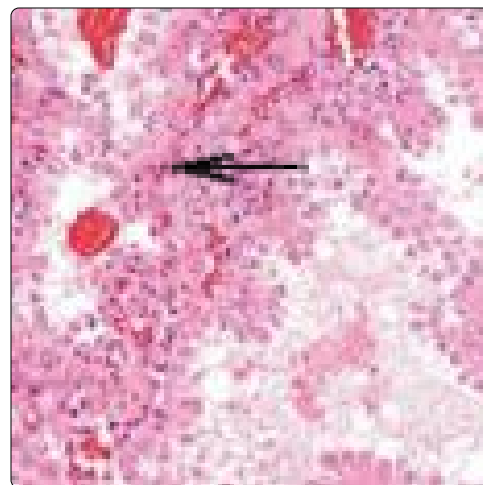
(Left) Decidualized stromal cells contain abundant eosinophilic cytoplasm and centrally placed, round nuclei with pinpoint nucleoli. The cells have a well-demarcated eosinophilic border. Scattered inflammatory cells (granulocytes) can be seen interspersed among the stromal cells. (Right) Decidualized endometrium with its abundant stroma can be seen on the left. Secretory endometrium with numerous, crowded glands occupies the right 1/2 of this image.

(Left) During pregnancy, hypersecretory changes may occur in the glandular epithelium. These changes have been termed Arias-Stella effect. The changes are highly variable and may be markedly atypical and confused with carcinoma. (Right) In Arias-Stella effect, the cells project into the endometrial glandular lumen (hobnailing) and have vesicular, optically clear nuclei. Only scant amounts of endometrial stroma may be present.

Arias-Stella Effect

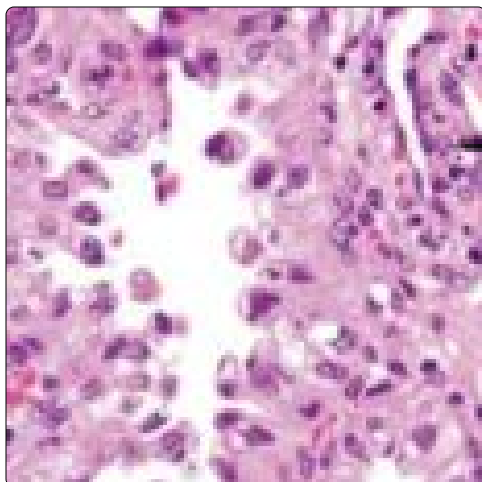


Arias-Stella Effect: Hobnail Cells

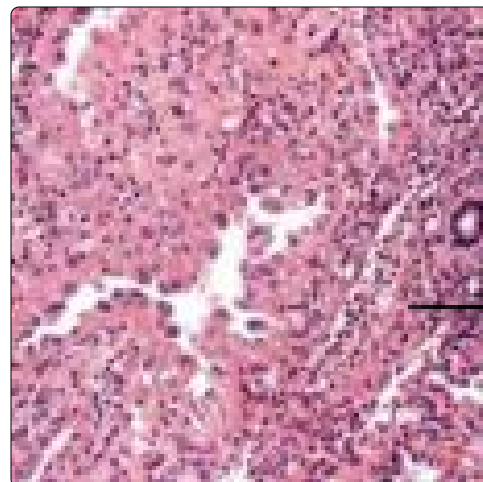


(Left) Glands involved by Arias-Stella effect may have markedly pleomorphic features. Enlarged cells can project and shed into the lumen. Additionally, the nuclei may become enlarged and hyperchromatic, having a glassy appearance. Mitotic figures should be essentially absent. (Courtesy A. Laury, MD.) (Right) A hypersecretory, enlarged gland involved by Arias-Stella effect is seen adjacent to a normal endometrial gland.

Arias-Stella Effect: Nuclear Pleomorphism

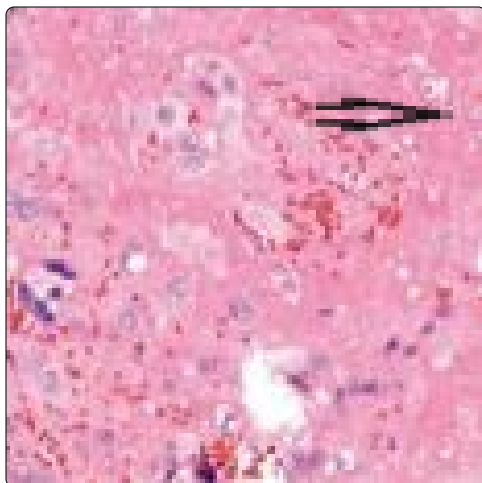


Arias-Stella Effect: Comparison to Normal Glands

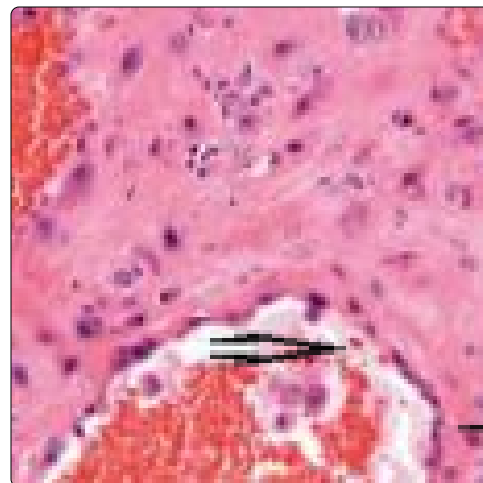


(Left) Examination of a uterine specimen from a pregnant patient may reveal the placental implantation site. This site is marked by the presence of extravillous, intermediate trophoblasts within a hypereosinophilic substance termed fibrinoid or Nitabuch fibrin. (Right) Intermediate trophoblasts remodel the decidual spiral arterioles, replacing the smooth muscle wall. This helps to facilitate oxygen transfer from the maternal to the fetal blood.

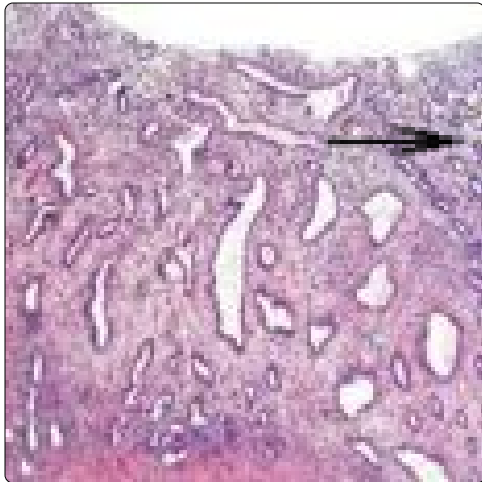
Implantation Site



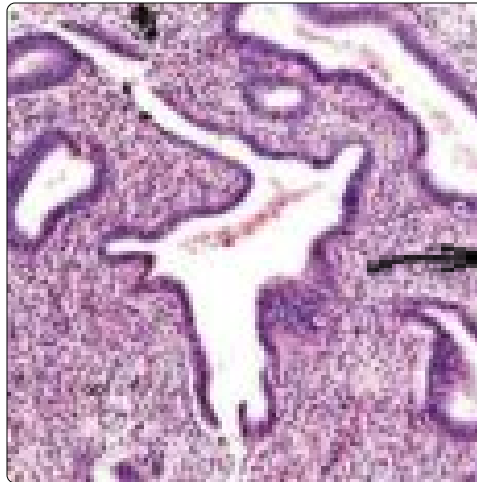
Implantation Site: Vessel



Disordered Proliferative Endometrium



Disordered Proliferative Endometrium: Glands

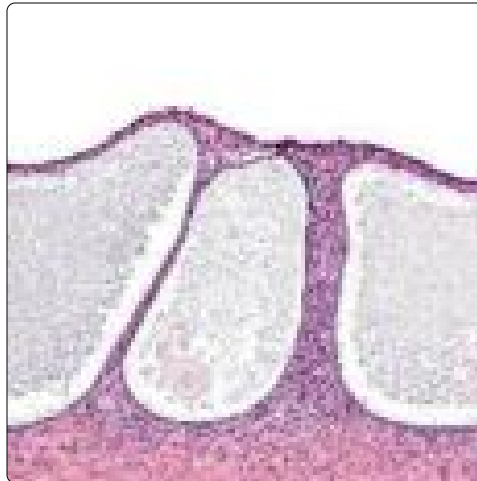


(Left) Disordered proliferative endometrium is common around perimenarche and perimenopausal years. It can be identified by the presence of scattered cystically dilated glands with irregular contours and outpouchings. This change is commonly attributed to unopposed estrogen. (Right) The glands in disordered proliferative endometrium may be normal or dilated. The stroma is similar to that seen in the proliferative phase. Mitotic figures and ciliated cells are usually present.

Anovulatory Endometrium



Atrophic Endometrium

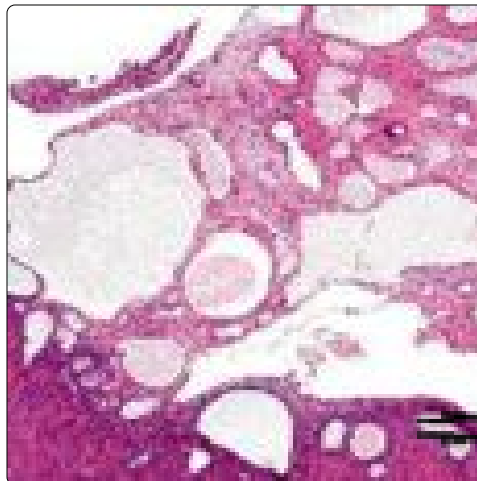


(Left) Anovulatory endometrium and disordered proliferative endometrium are interchangeable terms. In the presence of estrogen, the glands become cystically dilated. When estrogen is absent, dilation may be minimal; however, gland and stromal breakdown and tubal metaplasia may be present. (Right) Atrophic endometrium is composed of mitotically inactive, often cystically dilated glands lined by a simple cuboidal lining. Crowding should not be evaluated in the setting of atrophy.

Atrophic Polyp



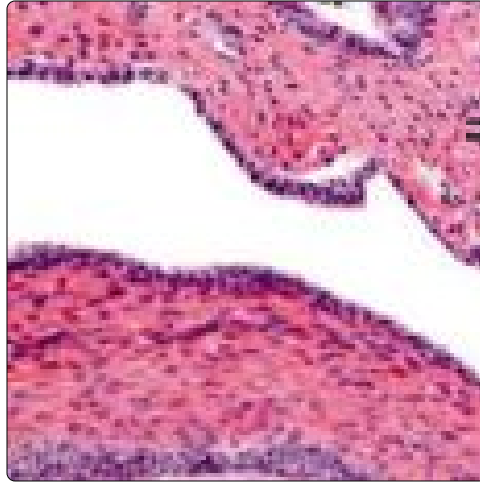
Atrophic Polyp: Cystically Dilated Glands



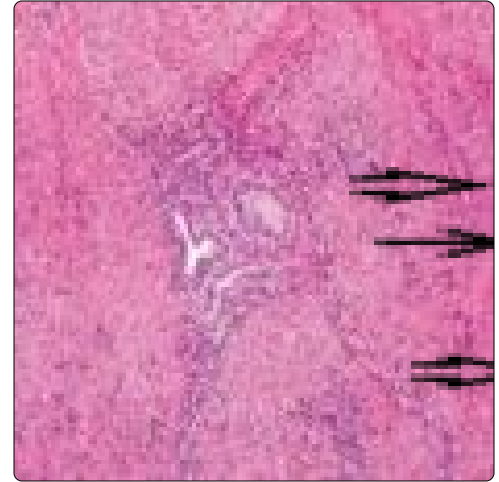
(Left) A mixture of atrophic small glands and cystic glands may be seen in cases of atrophy. Here, the glands are set in the fibrotic stroma of an atrophic endometrial polyp. (Right) Atrophic, cystically dilated glands can be seen both in an endometrial polyp and in the background endometrium. Although the area of glands relative to stroma is increased, this is not considered evidence of hyperplasia.

Atrophic Polyp: Glands

(Left) The lining of an atrophic endometrial gland is typically composed of a single lining of cuboidal to columnar epithelium with a high nuclear:cytoplasmic ratio. No mitotic activity should be present. (Right) Adenomyosis is defined as the presence of endometrial glands and stroma beyond the border of the endomyometrial junction. A minimum distance of 1-2 mm has been suggested to avoid overdiagnosis. Normal physiologic changes may occur in adenomyosis.

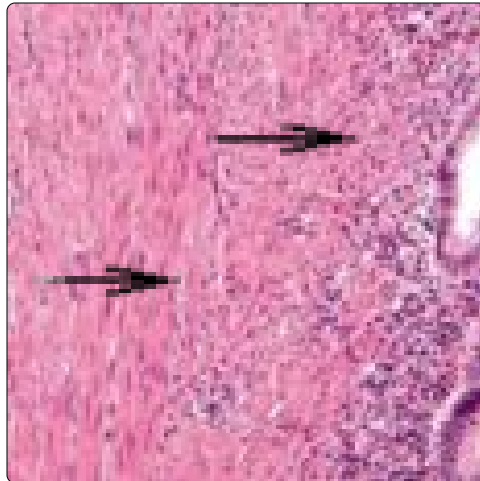


Adenomyosis

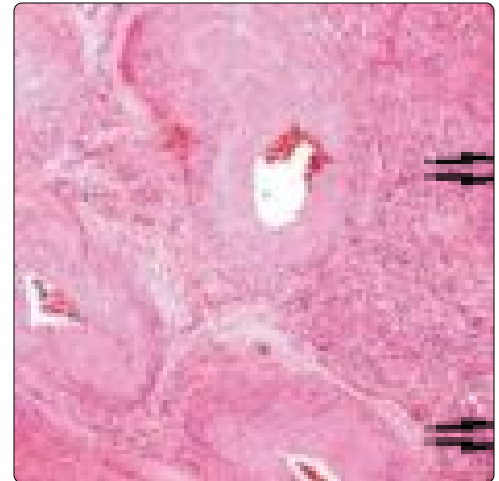


Superficial Myometrium

(Left) The myometrium is composed predominately of smooth muscle cells with elongated, cigar-shaped nuclei. Collagen and elastin fibers are present as well. The cells may greatly enlarge during pregnancy, and variable mitotic activity may be found, especially during the secretory phase. (Right) Deeper within the myometrium are the arcuate arteries. These arteries may be a helpful landmark when evaluating for possible myometrial invasion in cases of endometrial adenocarcinoma.



Arcuate Arteries

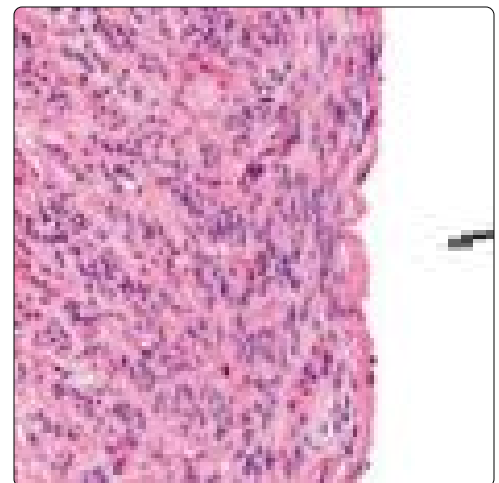


Arcuate Arteries: Features

(Left) The arcuate arteries may contain changes, such as atherosclerosis, medial calcification, or thrombosis. In multiparous patients, the uterine vasculature is more pronounced (as seen here) than that seen in a nulliparous patient. (Right) The deep myometrium is covered by a thin serosal lining. The serosa is composed of a thin layer of mesothelial cells with flattened nuclei.



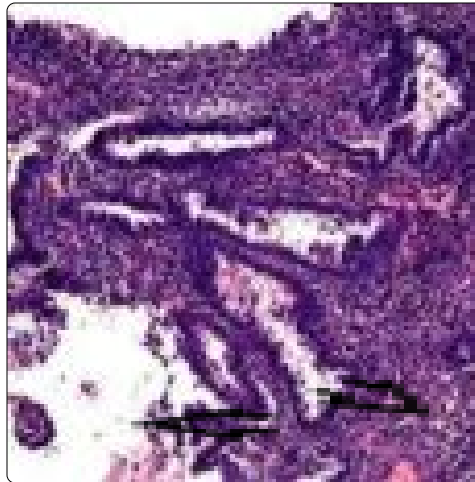
Uterine Serosa



Serosal Adhesion

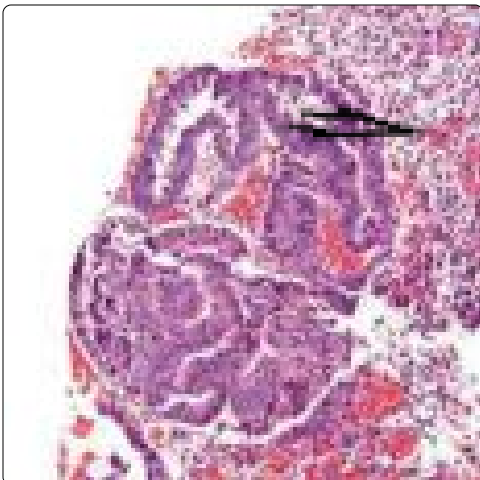


Exfoliation Artifact

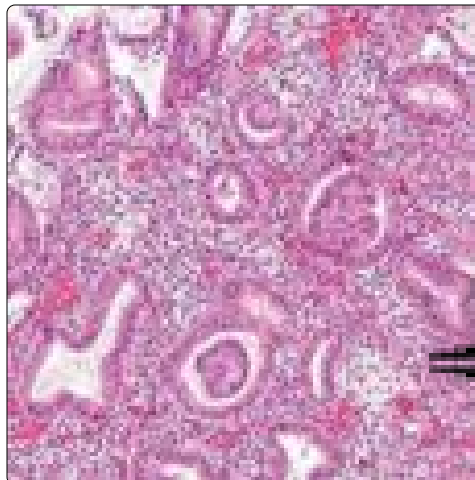


(Left) Fibrous adhesions may be present in patients who have had abdominal surgery or other causes of pelvic irritation. The presence of vascular growth in the adhesion is a helpful identifying feature. (Right) The instillation of fluids during hysteroscopy has a caustic effect on the endometrium. This is manifested by stromal fragmentation and hemorrhage and exfoliation of epithelial cells into the glandular lumen, particularly near the surface. (Courtesy C. Crum, MD.)

Fragmentation and Pseudocrowding

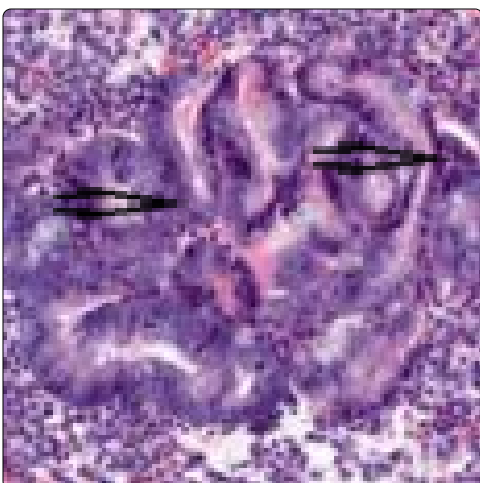


Telescoping Artifact

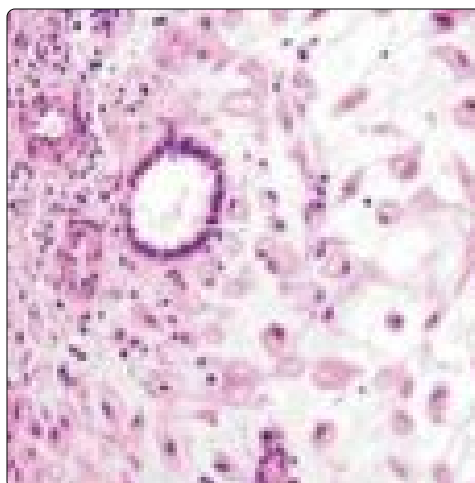


(Left) Fragmentation of endometrial tissue is often due to biopsy instruments. Strips of endometrial glandular epithelium without intervening stroma may appear markedly convoluted and should not be used to evaluate the presence of glandular crowding. (Right) Suction created by biopsy tools may lead to intussusception of glandular epithelium, leading to a "gland-in-gland" phenomenon that should be recognized as a benign artifact. (Courtesy A. Cole, MD.)

Severe Telescoping Artifact



Hormone Effects



(Left) Severe telescoping artifact may lead to the compression of a single gland tract, noted by a ribbon-like reduplication of the gland. This is another benign artifact that must not be interpreted as evidence of malignancy. (Courtesy A. Laury, MD.) (Right) Many cases of abnormal/dysfunctional bleeding are treated with progestin therapy. Progestins lead to small, tubular endometrial glands with an inactive lining as well as decidual change of the stroma cells.

Fallopian Tube

FALLOPIAN TUBE

Macroscopic Anatomy

- Composed of 3 regions
 - Isthmus
 - Extends from wall of uterus
 - Ampulla
 - Mid portion of fallopian tube that is more tortuous and dilated than isthmus
 - Fimbriated end
 - Composed of delicate, finger-like projections extending into pelvic cavity adjacent to ovary
- Broad ligament
 - Contains fallopian tube as well as blood vessels and supporting connective tissue
 - Predominately composed of fibroadipose tissue
 - Walthard rests, paratubal cysts, and endosalpingiosis frequently seen here

Microscopic Anatomy

- Epithelium
 - Varying admixture of secretory and ciliated cells
 - Number of ciliated cells believed to increase with increasing levels of estrogen
 - Regional variation in number of secretory and ciliated cells exists
- Stroma
 - Composed of smooth muscle and fibroconnective tissue
 - Thickest in isthmus; thins as fimbriated end is approached
 - 3 layers of smooth muscle are present in isthmus adjacent to uterine wall
 - Remainder of tube wall contains 2 layers of smooth muscle
- Outermost mesothelial cell layer
 - Reactive mesothelial cells may be present in inflammatory or malignant conditions
 - Reactive mesothelial cells identified by hobnail morphology, polygonal shape, and presence of nucleoli

- Regional microscopic variation
- Isthmus
 - Thick smooth muscle wall (3 layers) with muscle fibers in parallel and perpendicular orientation
 - Projections inside tube (plica) have minimal complexity
 - Secretory cells predominate
- Ampulla
 - Muscular wall thins and loses organization (2 layers)
 - Increase in number of plica and their complexity
- Fimbriated end
 - No muscular wall is present
 - Finger-like plica project into abdominal cavity
 - Ciliated cells predominate

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Atrophic changes can be seen after menopause

Hyperplasia

- Tubal epithelial hyperplasia is relatively unstudied and is evolving concept
- Hyperplasia most commonly involves secretory cells and likely leads to pelvic serous carcinogenesis

Metaplasia

- Squamous: Stratified, nonkeratinizing
- Mucinous: Columnar, mucin-containing cells
 - Rare phenomenon that may be associated with Peutz-Jeghers syndrome
- Transitional: Urothelial (bladder-like) epithelium
- Decidual change
 - Seen in pregnant or postpartum patients

Pitfalls and Artifacts

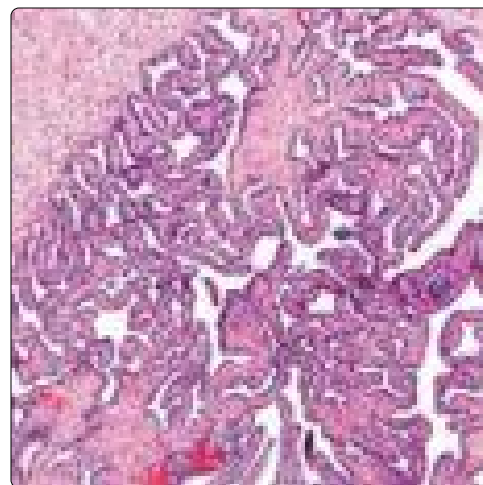
- Tangential sections through epithelium may mimic hyperplasia and malignancy
- Hydrosalpinx may mimic ovarian cyst, paratubal cyst, or endometriosis

(Left) The isthmus  is closest to the body of the uterus. The ampulla  comprises the mid portion of the tube, and the fimbriated end  is located at the opening of the tube into the pelvis. (Right) The ampulla is seen with its numerous plica, or finger-like projections, into the lumen of the tube. This area of the tube provides the environment that supports fertilization of the egg.

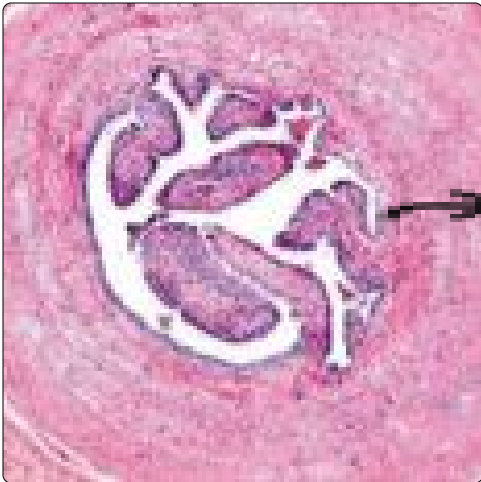
Fallopian Tube



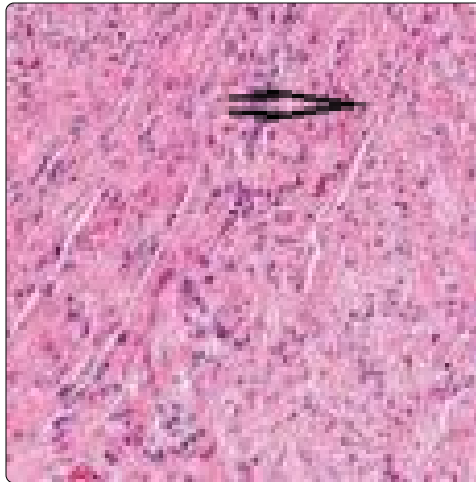
Ampulla



Isthmus

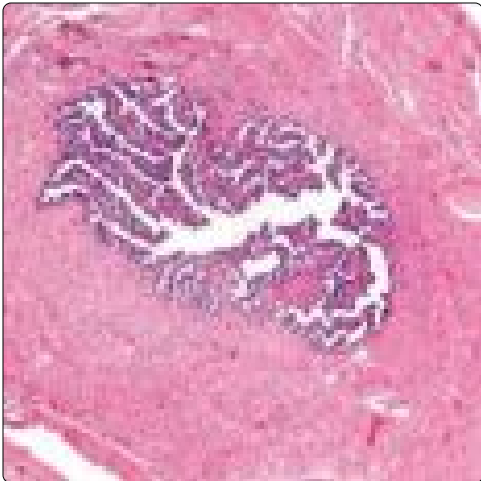


Isthmus: Smooth Muscle

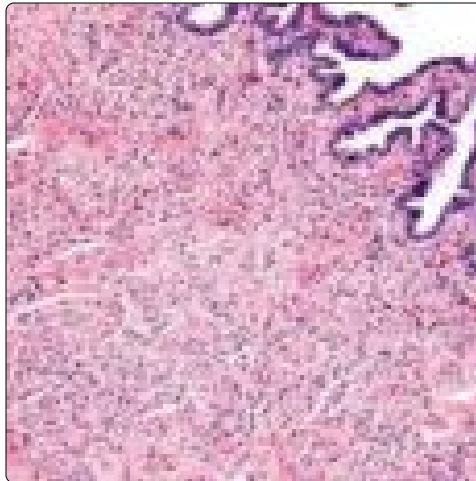


(Left) The isthmus has few plica and is predominately composed of a thick muscular wall composed of smooth muscle. (Right) The wall of the isthmus of the tube is composed of fascicles of smooth muscle that can be seen running perpendicularly and parallel to the long axis of the fallopian tube lumen.

Mid Portion of Tube

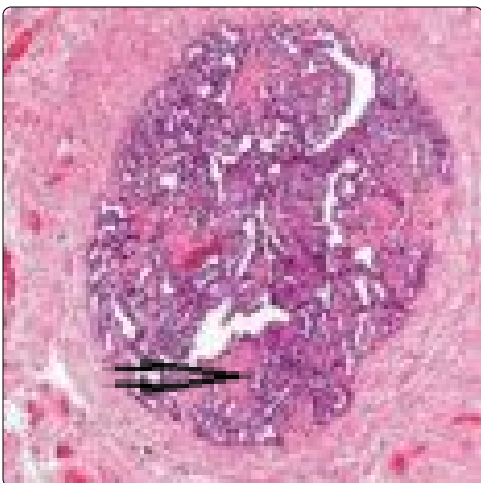


Mid Portion



(Left) Toward the fimbriated end of the tube, the muscular wall of the tube becomes more attenuated, and the number and complexity of the plica increases. This photo was taken in the mid portion between the isthmus and the ampulla. (Right) The muscular wall of the ampulla becomes thinner, and the organization of the smooth muscle cells decreases as one approaches the fimbriated end of the tube.

Ampulla



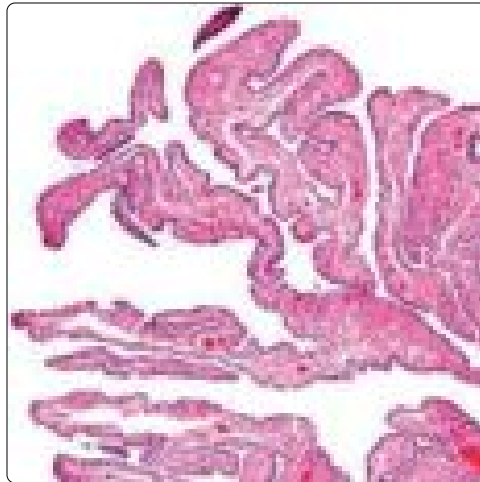
Ampulla: Smooth Muscle



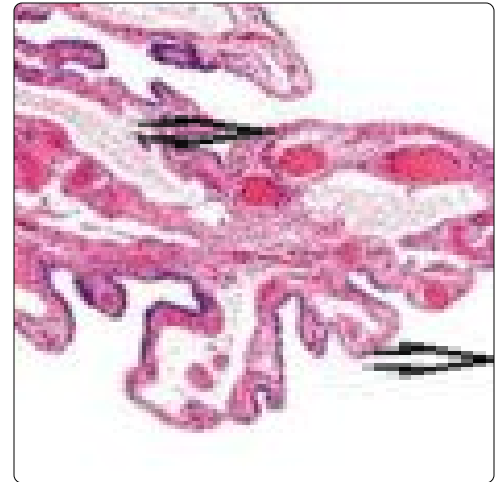
(Left) The ampulla of the fallopian tube has a much thinner muscular wall with numerous plica projecting into the tubal lumen. (Right) The wall of the ampulla is composed of wispy smooth muscle fibers that are no longer organized in large fascicles.

Fimbriated End

(Left) The fimbriated end of the fallopian tube consists of plica that project into the pelvic cavity. These plica often appear as islands of stroma that are lined by tubal epithelial cells. **(Right)** In this section of fallopian tube, numerous dilated lymphatic channels can be seen in the plical stroma.



Fimbriated End: Lymphatics

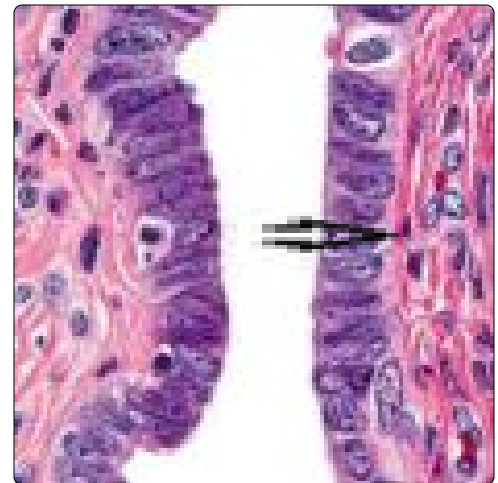


Epithelium

(Left) Tubal epithelium is composed of varying numbers of secretory cells and ciliated cells. The proportions of these cells vary in response to hormonal status. It is believed that the secretory cells can differentiate into ciliated cells. **(Right)** Occasionally, secretory cells are the predominant cell type, as seen here. Lymphocytes can be identified within the lining epithelial cells. Note the single ciliated cell. The intercalated cell (not pictured) is thought to represent a type of secretory cell.

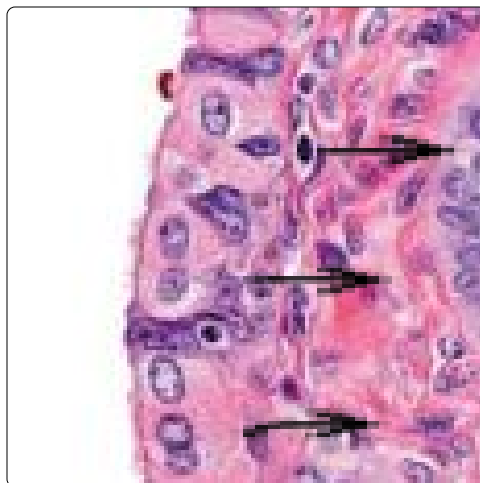


Epithelium: Secretory Cells



Epithelium: Ciliated Cells

(Left) Note the bright pink terminal bar that can be identified under the cilia. Ciliated cells are typically rounder and contain much more cytoplasm than secretory cells; they have been likened to fried eggs. **(Right)** Cysts may form at the periphery of the tube. These cysts may vary in size from microscopic to many centimeters in diameter. They are usually full of clear, thin fluid. This cyst has focal hemorrhage into the wall.



Paratubal Cyst



Paratubal Cyst: Epithelium

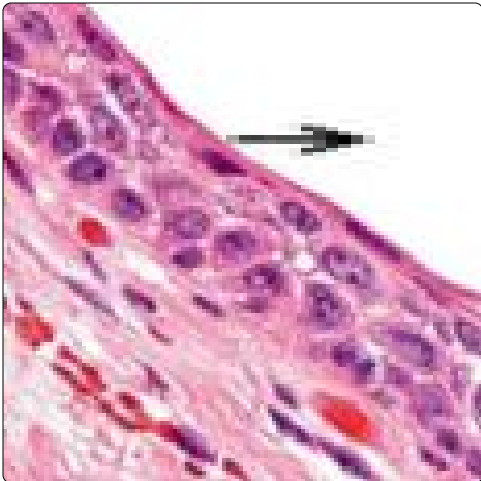


Paratubal Cyst: Eosinophilic Secretions

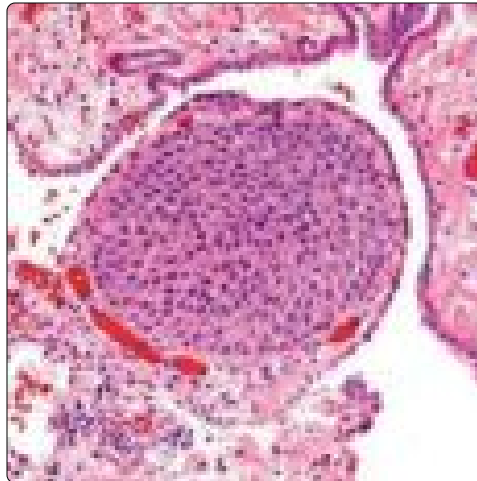


(Left) Paratubal cysts are typically lined by a simple, low cuboidal epithelium. The wall surrounding the cyst should be thin and devoid of muscle. (Right) This photomicrograph shows multiple thin-walled, paratubal cysts. Note the eosinophilic secretions that may accumulate within the cysts. The fallopian tube lumen and muscular wall can be identified in the bottom right corner. The lining epithelium may be so attenuated that it may be difficult to see.

Squamous Metaplasia



Walthard Rest

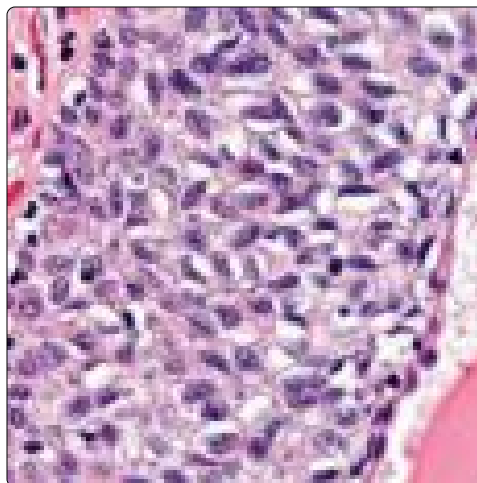


(Left) Squamous metaplasia of the tubal epithelium frequently occurs in the setting of inflammation or trauma. The normal lining is replaced by stratified squamous lining. Intercellular bridges (present as thin, pink lines) between the keratinocytes can be observed at high power. (Right) Walthard rests are a common finding in the fallopian tube. They are composed of transitional epithelium, similar to the epithelium found in the bladder.

Cystic Walthard Rest



Walthard Rest: Epithelium



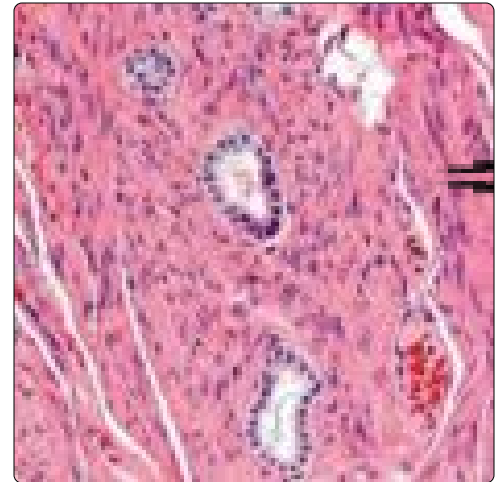
(Left) Walthard rests may become cystically dilated. The lining epithelium bears a resemblance to that seen in the bladder. Frequently, eosinophilic secretions may be seen in the lumen. (Right) Close examination of the lining of a Walthard rest shows elongated epithelial cells with scattered grooved nuclei. The nuclei are often found streaming perpendicularly to the lumen of the cyst.

Multiple Walthard Rests

(Left) This photomicrograph shows multiple cystic Walthard rests within the supporting connective tissue of the fallopian tube. (Right) Mesonephric remnants are commonly found in the tubal wall. They are composed of small, gland-like structures with a simple cuboidal lining. They are surrounded by a dense fibromuscular stroma.

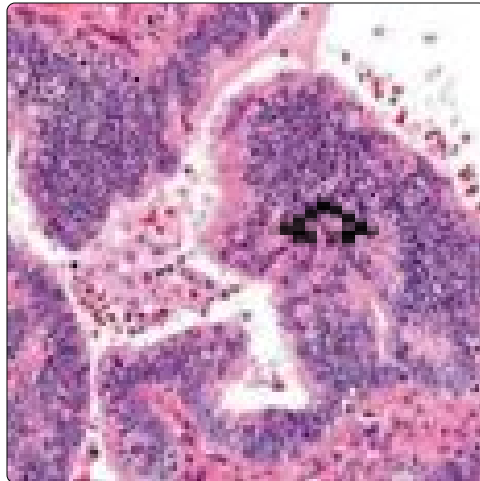


Mesonephric Remnant

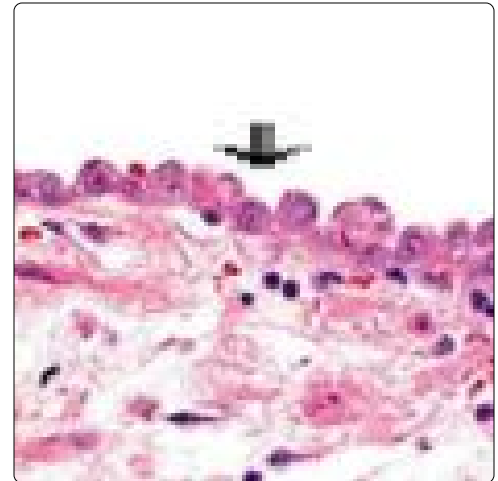


Artifact: Tangential Sectioning

(Left) Tangential sectioning of the epithelium of the fallopian tube may create areas that appear to be hyperplastic (so-called papillary hyperplasia of the fallopian tube). No significant nuclear atypia is seen. (Right) The outermost lining of the tube is composed of mesothelial cells that may become reactive. Reactive mesothelial cells are polygonal and often possess a hobnail morphology. Single, prominent nucleoli can be identified.

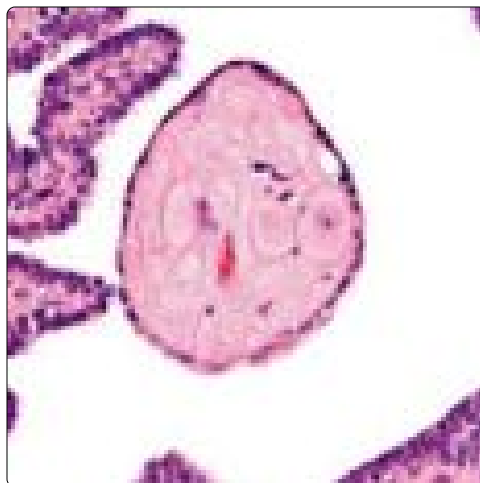


Reactive Mesothelium

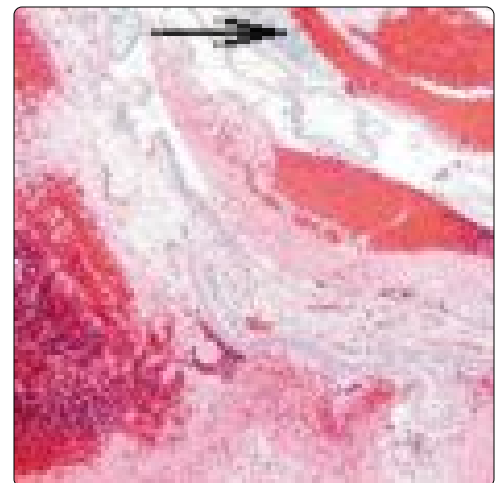


Decidual Change

(Left) Decidual change, marked by enlarged cells with abundant eosinophilic cytoplasm, may occur in patients who are pregnant or have recently been pregnant. (Right) Tubal pregnancies may contain placental villi adjacent to fallopian tube plica. There is often abundant hemorrhage within the tube (hematosalpinx). In advanced cases, rupture of the wall of the tube may occur.



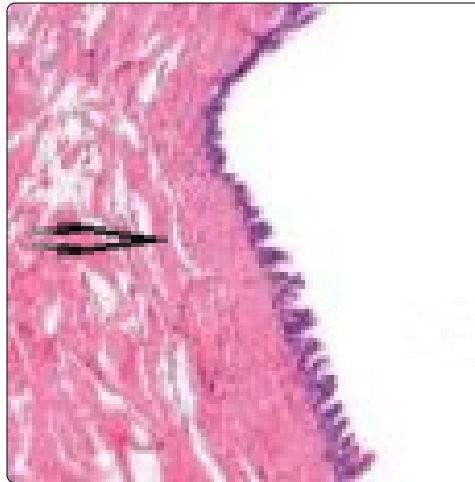
Tubal Pregnancy



Hydrosalpinx



Hydrosalpinx: Absence of Plica

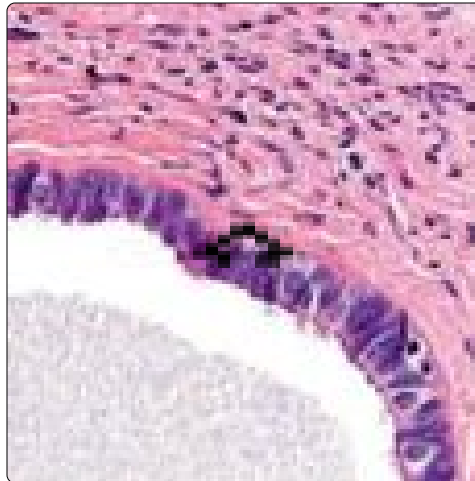


(Left) The tube may become cystically dilated, usually due to previous infection or inflammation; this is termed hydrosalpinx. In hydrosalpinx, the plica are often decreased in number or not present at all. (Right) In this example of hydrosalpinx, no plica are present. Note the muscular wall [X], which can help to differentiate a hydrosalpinx from a paratubal cyst or serous cyst microscopically.

Endosalpingiosis



Endosalpingiosis: Epithelium



(Left) Endosalpingiosis is the presence of cystic, normal tubal epithelium outside of the fallopian tube lumen. It may be seen in the wall of the fallopian tube and resemble a paratubal cyst or, in extreme cases, hydrosalpinx. In this figure, endosalpingiosis is seen involving the ovary [X]. (Right) The lining of an endosalpingiotic cyst is composed of normal tubal epithelium consisting of secretory [X] and ciliated cells [X].

Atrophic Changes



Atrophy



(Left) In postmenopausal patients, the fallopian tube undergoes atrophic changes. These changes consist of large, fibrous plica [X] with a simple lining of normal tubal epithelium. The number of plica and fimbria are often decreased. (Right) Atrophy of the fimbriated end leads to a decrease in number and complexity of the individual fimbria. There is a predominance of stroma compared to the epithelium.

OVARY**Macroscopic Anatomy**

- Cortex
 - Outer layer of ovary
 - Composed of stroma; contains developing follicles
- Medulla
 - Mid portion of ovary
 - Composed predominantly of stroma with blood vessels and nerves
- Hilum (a.k.a. hilus)
 - Where blood vessels and nerves enter parenchyma

Microscopic Anatomy

- Surface epithelium
 - Delicate single layer of cuboidal to columnar cells that cover outermost surface of ovary
- Cortex
 - Stroma composed of stromal cells (fibroblast-like cells) and developing follicles
 - Primordial follicle: Single layer of flat follicle cells surrounding oocyte
 - Primary follicle: 1-2 layers of follicular cells surrounding oocyte
 - Secondary (preantral) follicle: Multiple layers of granulosa cells as well as development of zona pellucida and beginning of fluid-filled cyst (antrum)
 - Mature (graafian) follicle: Eccentrically located oocyte and granulosa cells as well as well-developed antrum; established presence of theca interna and externa
- Medulla
 - Stroma composed of fibroblasts, collagen, and elastic fibers
 - Clusters of blood vessels transgressing from hilum
 - Occasionally, smooth muscle, fat, decidual, or luteinized cells may be identified
- Hilum
 - Connective tissue with large-caliber blood vessels and nerves

- Scattered nests of hilus cells, resembling Leydig cells of testis, may be present; Reinke crystals or cytoplasmic pigment may aid in their identification
- Rete ovarii
 - Tubules and slit-like spaces analogous to rete testis

CONSIDERATIONS AND VARIATIONS**Age-Related Variations**

- At birth, ovaries are much smaller than they are around beginning of puberty
- Ovaries increase 20-30x in size before onset of puberty
- At onset of puberty, ovaries have reached their full size
- Ovarian surface becomes more convoluted throughout reproductive life
- After menopause, ovaries lose ~ 50% of their size and weight

Hyperplasia

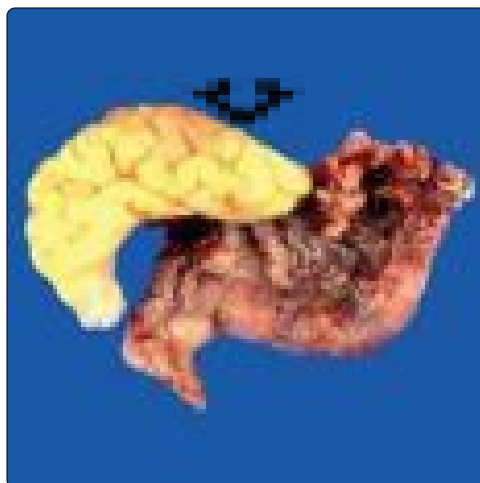
- Varying amounts of stromal hyperplasia may be seen in peri- and postmenopausal patients
- Cortical stromal hyperplasia may be diffuse or form discrete nodules

Metaplasia

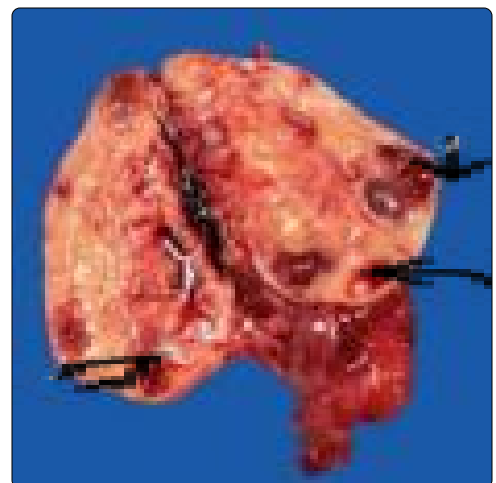
- Cortical inclusion glands or cysts may undergo metaplasia
 - Serous, mucinous, or transitional types
 - Walthard rests: Nests of transitional epithelium

Pitfalls and Artifacts

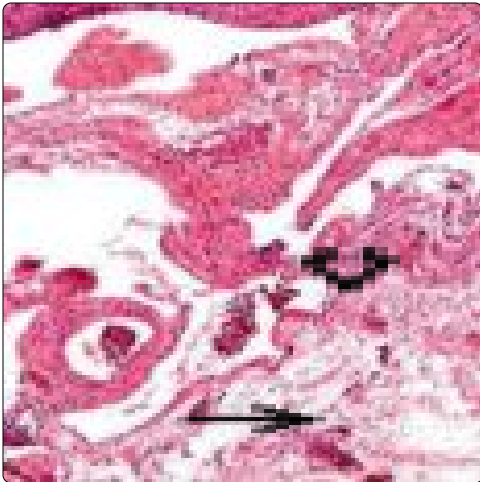
- Endosalpingiosis
 - Small cystic structures lined by tubal-type epithelium
- Benign ovarian cysts (corpus luteum of pregnancy or follicle cysts) may clinically resemble neoplasm
- Surface papillation may be confused with surface serous tumors (adenofibromas)
- Endometriosis, especially with reactive atypia, may simulate malignancy both clinically and histologically

Ovary

(Left) In this photograph of an ovary and the adjacent fallopian tube, the outer cortex of the ovary is visible. The convolution of the cortex increases with age. The fallopian tube has a fimbriated end. (Right) Small follicle cysts can be seen in the cortex on the cut surface of the ovary. The medulla lies under the cortex and is composed of stroma. The hilum comprises the most central portion.

Ovarian Cortex

Hilar Vessels

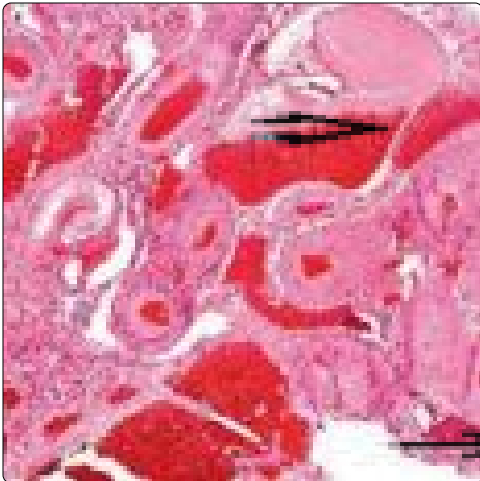


Medullary Vessels

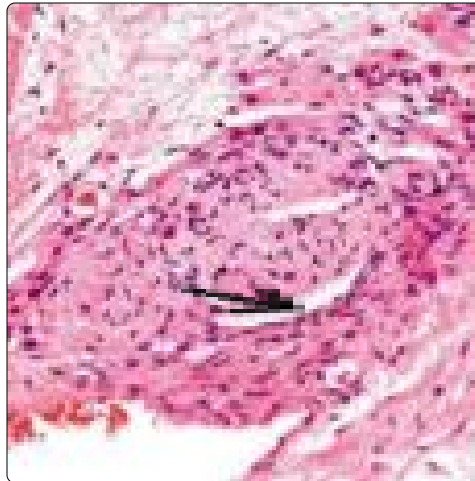


(Left) Numerous large blood vessels and abundant connective tissue are present at the hilum. The arteries and veins that supply the ovary enter here. (Right) Large medullary vessels become smaller and more coiled as they progress deeper into the medulla. In postmenopausal patients, the relative lack of stroma may give the false impression of an increase in the number of vessels, which may mimic a benign vascular tumor or malformation.

Medullary Arteries and Veins

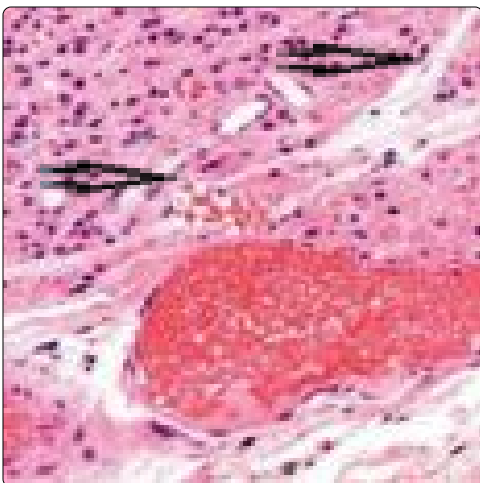


Hilum: Nerve

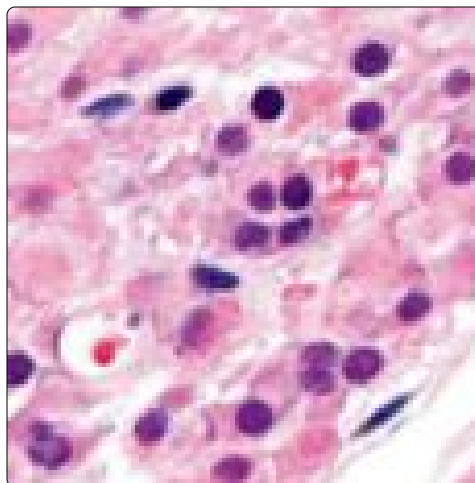


(Left) As the arteries and veins progress into the ovary, they become more convoluted. With age, changes, such as atherosclerosis, calcification, or amyloid deposition, may be seen in these vessels. (Right) Nerves of the sympathetic nervous system also enter the ovary at the hilum. Hilus cells, analogous to Leydig cells of the testis, may be seen in and around these nerves and around blood vessels.

Hilus Cells



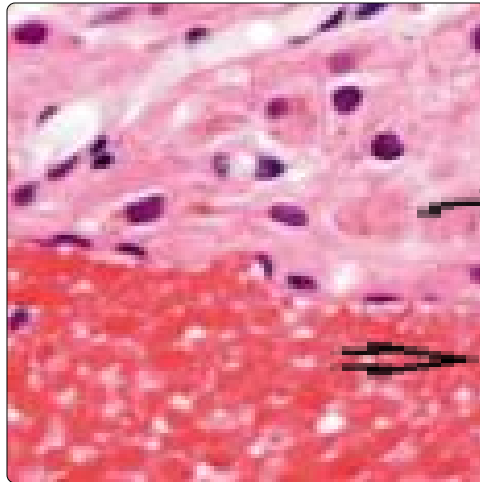
Hilus Cells: Reinke Crystal



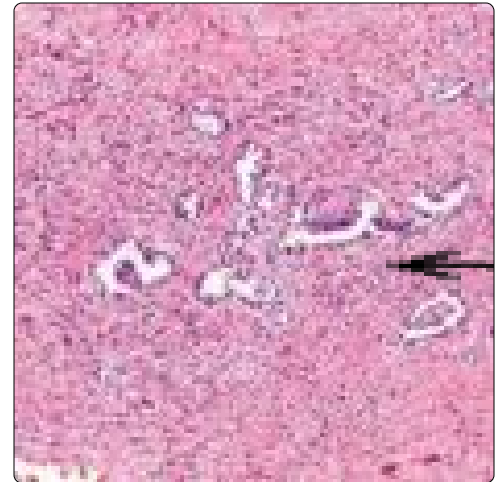
(Left) Hilus cells are polygonal; eosinophilic cells present within the ovarian hilum and abutting the medulla. These cells are absent during childhood; they begin to appear around puberty and increase in number throughout life. (Right) Eosinophilic crystals of Reinke may be identified within the cytoplasm of the ovarian hilus cells, although they are often absent. A Masson trichrome stain may aid in their detection.

Hilus Cells: Pigment

(Left) The presence of a fine, brown, granular pigment in the cytoplasm of hilus cells may aid in their identification. These hilus cells are adjacent to a blood vessel. (Right) The rete ovarii can be found in the hilum of the ovary. The rete ovarii are analogous to the rete testis and are composed of round to slit-like tubules with occasional luminal papillae. They are lined by epithelium that may appear flat to columnar.

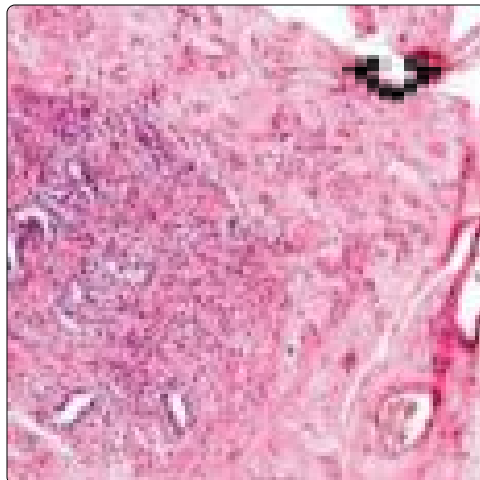


Rete Ovarii



Rete Ovarii and Hilus Cells

(Left) The rete ovarii is often surrounded by a well-demarcated cuff of spindled stroma. This stroma is similar to, but not contiguous with, the ovarian stroma. Clusters of hilus cells may be seen in association with the rete ovarii. (Right) Wolffian duct remnants may be identified throughout the gynecologic tract. They are most common in the vagina but can be identified in the hilum of the ovary. They should not be mistaken for metastatic carcinoma.

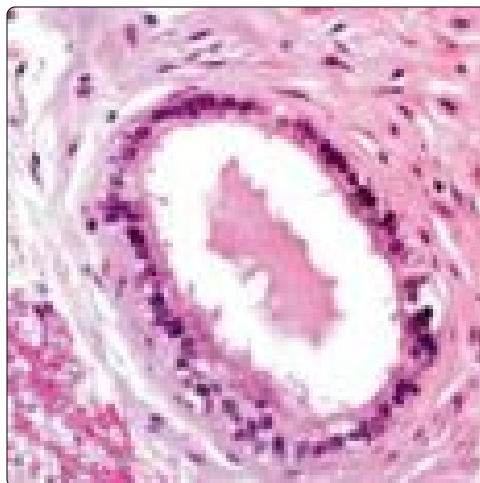


Wolffian Ducts

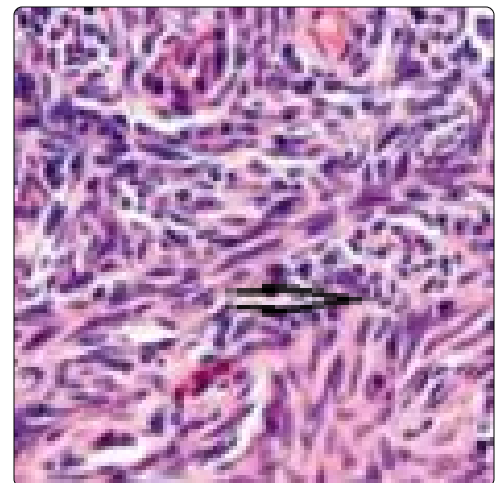


Wolffian Ducts: Eosinophilic Secretions

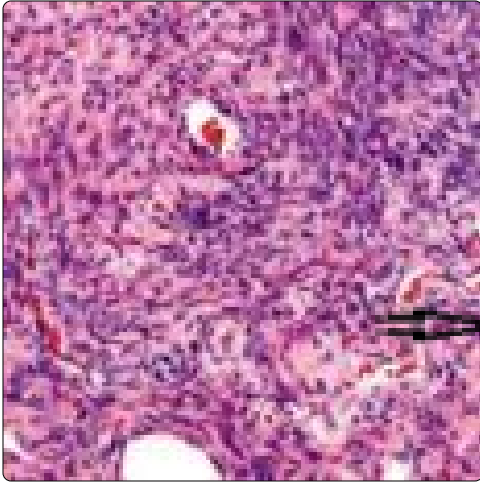
(Left) Wolffian ducts may be confused with the rete ovarii, but the presence of eosinophilic luminal secretions and the absence of a well-developed surrounding spindle cell stroma can aid in this distinction. The cells lining a wolffian duct are typically cuboidal with round nuclei. (Right) Spindled stromal cells are present within the medulla and the ovarian cortex. These cells have scant cytoplasm and elongated, tapered nuclei, often forming fascicles or whorls.



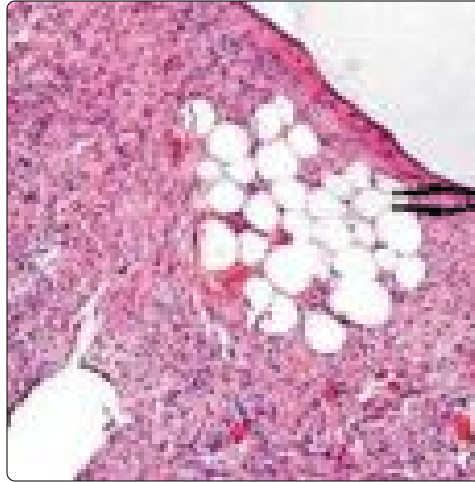
Ovarian Stroma



Stroma: Luteinized Cells

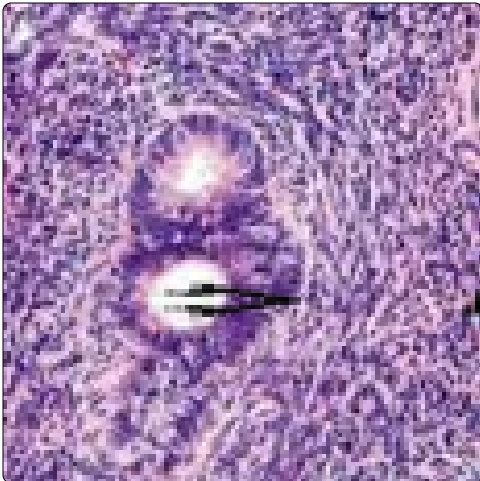


Stroma: Fat

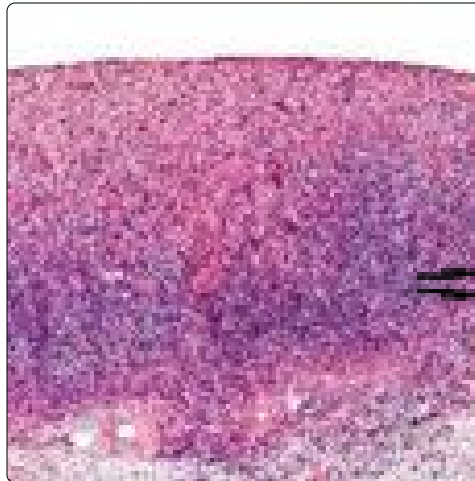


(Left) Luteinized cells may be seen interspersed within the medullary stroma. These cells can be recognized by their ample, eosinophilic cytoplasm and the presence of round, centrally placed nuclei. These cells increase in number throughout reproductive life. (Right) Heterotopic tissues may be identified within the stroma. Here, a collection of fat cells is present within the cortical stroma. Bundles of smooth muscle may be present as well.

Stroma: Endosalpingiosis

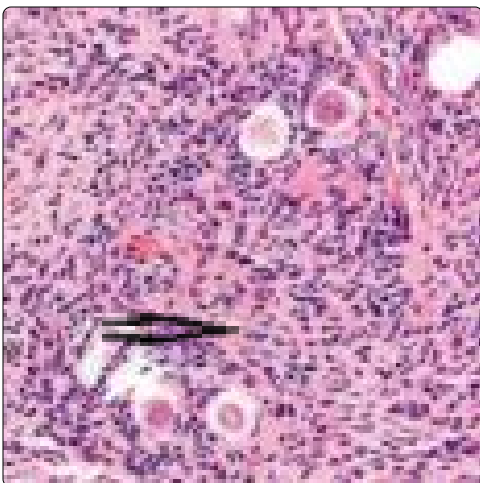


Corticomedullary Junction

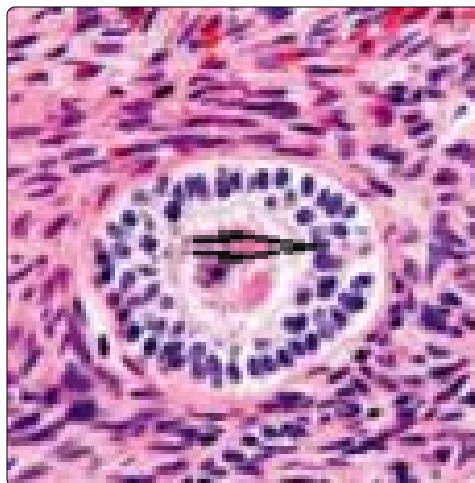


(Left) Within the stroma, scattered cortical inclusion cysts may be present. These cysts increase in number with age and are thought to be a result of ovulation. They may be lined by simple cuboidal epithelium or by tubal epithelium (endosalpingiosis). The presence of cilia can be helpful in identifying the latter. (Right) The junction between the medulla and the cortex is ill defined, and the 2 may blend and become indistinguishable.

Primordial Follicle



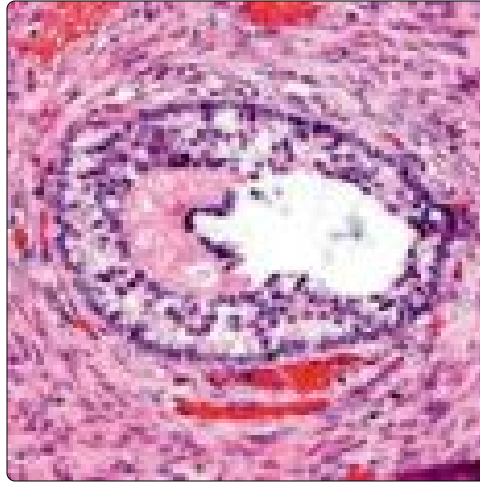
Primary Follicle



(Left) Primordial follicles are found in a linear distribution in the superficial cortex. They are lined by a single flattened layer of granulosa cells. In younger patients, oocytes may be identified within the center of the primordial follicle. (Right) As the follicle matures, the granulosa cells assume a cuboidal or columnar shape. In the primary follicle, 1-2 distinct layers of granulosa cells are present. The oocyte begins to enlarge as well.

Preantral Follicle

(Left) As follicles mature, they migrate deeper into the cortex. The preantral follicle is characterized by 3-5 layers of granulosa cells and the development of a thin, eosinophilic layer surrounding the oocyte (zona pellucida). (Right) Granulosa cells secrete fluid, eventually forming the antrum, or the portion filled with follicular "liquor." At this point, the oocyte is mature and surrounded by an increased number of granulosa cells, forming the cumulus oophorus.

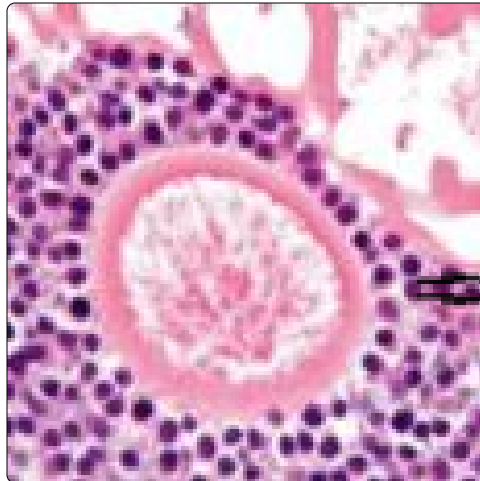


Mature Follicle

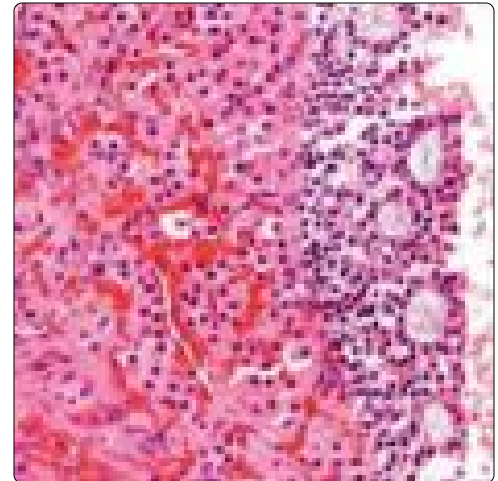


Mature Follicle: Oocyte

(Left) The mature oocyte, with its thickened zona pellucida, is surrounded by granulosa cells. (Right) As the maturing follicle progresses deeper into the cortex, the stromal cells differentiate into 2 specialized layers of theca cells. The granulosa cell layer abuts the inner of these 2 layers, the theca interna. Cystic Call-Exner bodies may be identified within the granulosa cell layer and recapitulate those found in granulosa cell tumors.

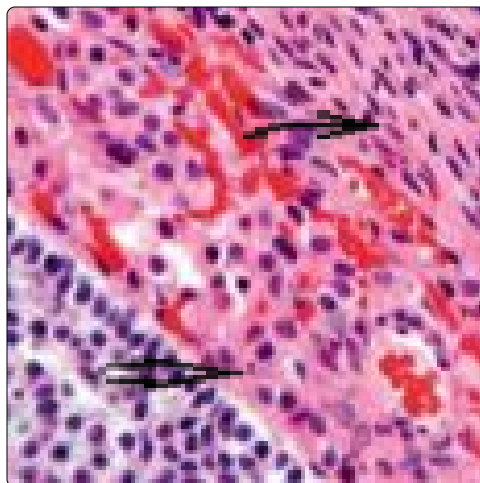


Mature Follicle: Granulosa and Theca Interna Layers

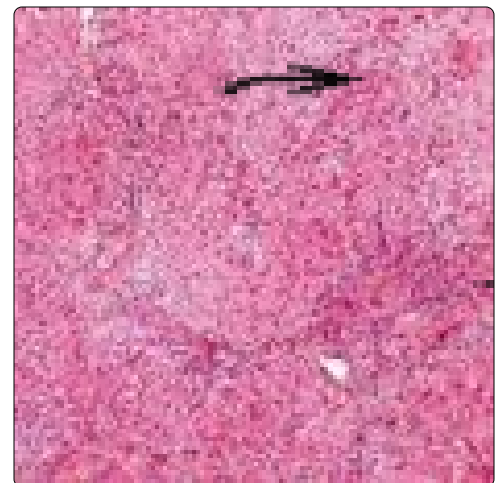


Theca Interna and Externa

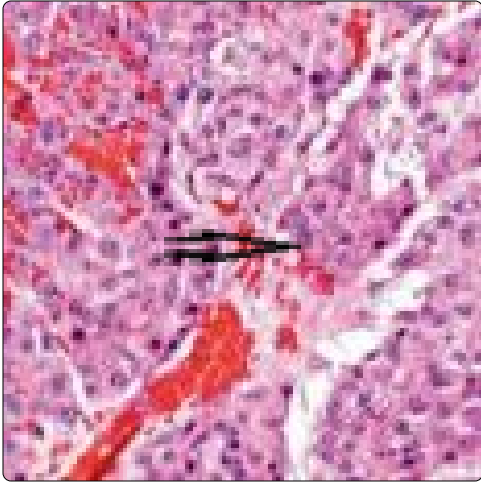
(Left) The theca externa layer comprises the outermost layer of the fully mature follicle. The theca interna is composed of round, eosinophilic cells with round nuclei, whereas the theca externa is composed of spindle, eosinophilic cells with elongated nuclei. Compare this to granulosa cells. (Right) After ovulation, the corpus luteum forms. The corpus luteum is composed of granulosa lutein cells and theca lutein cells, remnants of the granulosa and theca interna layers, respectively.



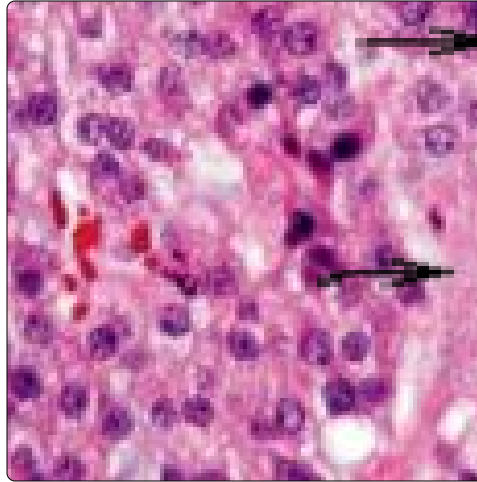
Corpus Luteum



Corpus Luteum: Menstruation

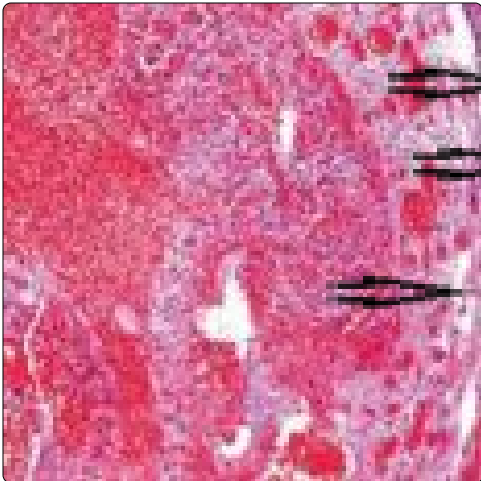


Corpus Luteum: Cytologic Features

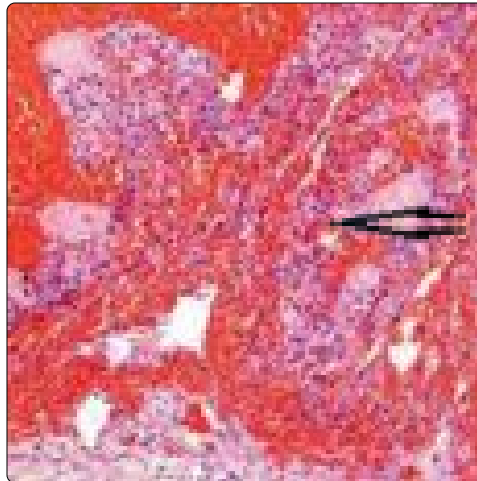


(Left) Within the corpus luteum, the granulosa cells become luteinized and are termed granulosa lutein cells. Granulosa lutein cells are enlarged, polygonal cells with ample eosinophilic cytoplasm and scattered lipid droplets. (Right) The nucleus of the granulosa lutein cell is round and vesicular with easily discernible nucleoli. Apoptotic debris and K cells, noted by their darkly staining cytoplasm and nuclei, may be identified as well.

Hemorrhagic Corpus Luteum

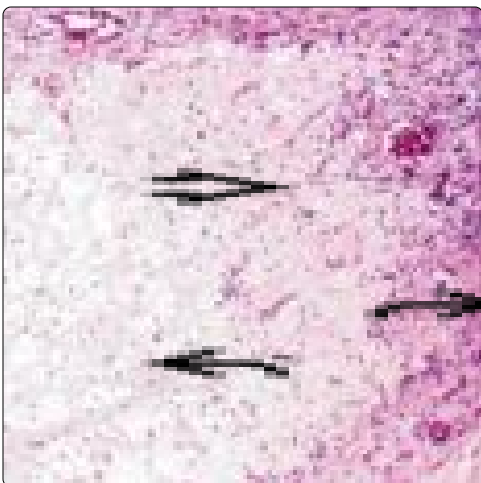


Hemorrhagic Corpus Luteum: Fibrin

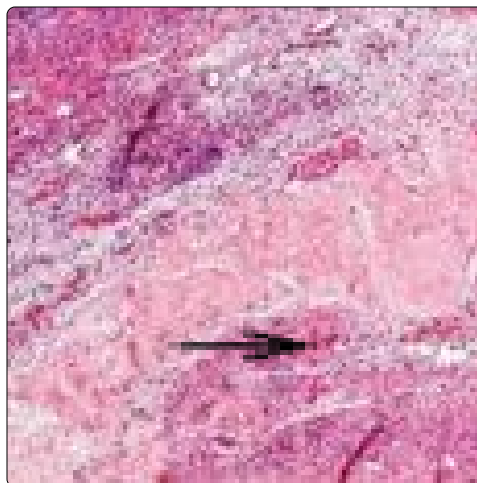


(Left) One of the most common complications associated with an otherwise normal corpus luteum is associated hemorrhage. Additionally, exceptionally large luteal cysts may mimic ovarian neoplasms and be excised. The undulating lining of this luteal cyst can be seen admixed with abundant hemorrhage. (Right) Fragments of granulosa lutein cells and fibrin thrombi can be helpful features when considering the differential diagnosis of a hemorrhagic ovarian cyst.

Involuting Corpus Luteum



Corpora Albicans



(Left) When a corpus luteum undergoes routine involution, the granulosa lutein cells accumulate abundant cytoplasmic lipid, which leads to pale staining. The cells become smaller and have pyknotic nuclei. (Right) The involuting corpus luteum is replaced by collagen and fibroblasts and eventually forms a scar known as a corpus albicans. Occasional macrophages may be identified in and around these structures. Corpora albicans may become resorbed and may disappear completely.

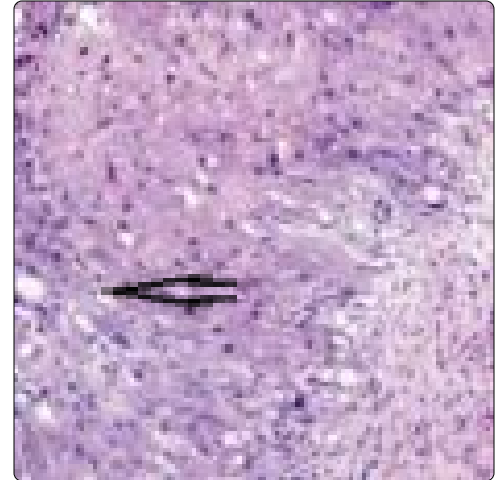
Ovary

Corpus Luteum: Pregnancy

(Left) During pregnancy, the corpus luteum may enlarge to form a variably sized cystic structure with a yellow lining. In striking cases, such as the one pictured, multiple, large cysts filled with fibrin and blood may form. (Right) During pregnancy, involution of the corpus luteum does not occur. The cells continue to enlarge, reaching a maximum size of around 55 μm . One histologic feature of pregnancy is the development of cytoplasmic vacuoles in the corpus luteum , which enlarge throughout pregnancy.

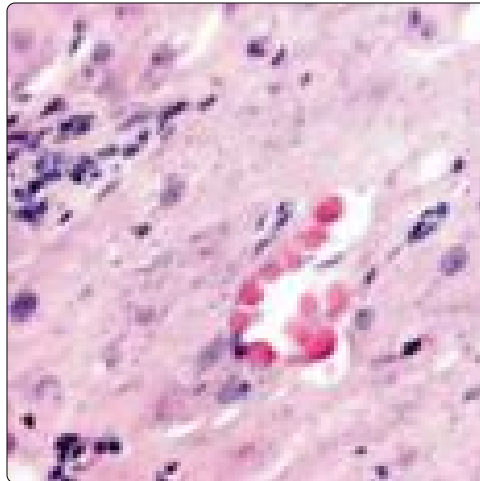


Corpus Luteum: Pregnancy (Noninvolution)

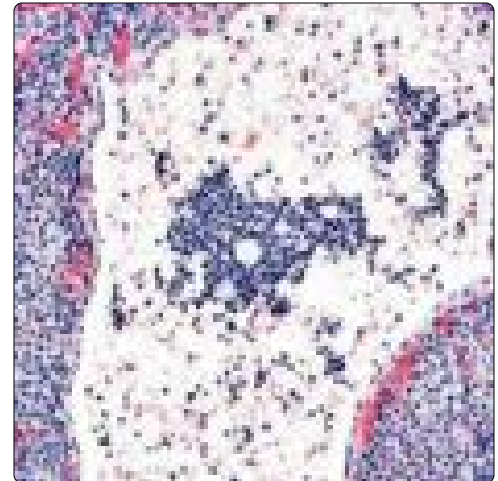


Corpus Luteum: Pregnancy (Hyaline Bodies)

(Left) Another histologic feature seen during pregnancy (and rarely in the corpus luteum of menstruation) are hyaline bodies . Hyaline bodies increase in number throughout gestation. (Right) The vast majority of follicles undergo atresia. Primordial and primary follicles involute completely and leave no evidence of their existence. Mature follicles (follicles that reach the antral stage) that involute are recognizable. The earliest sign is the degeneration of the oocyte .

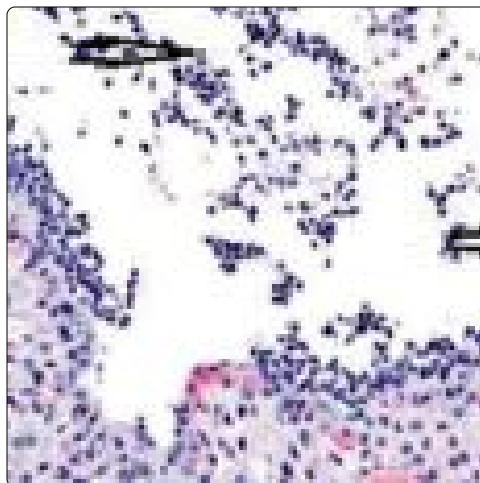


Involuting Follicle

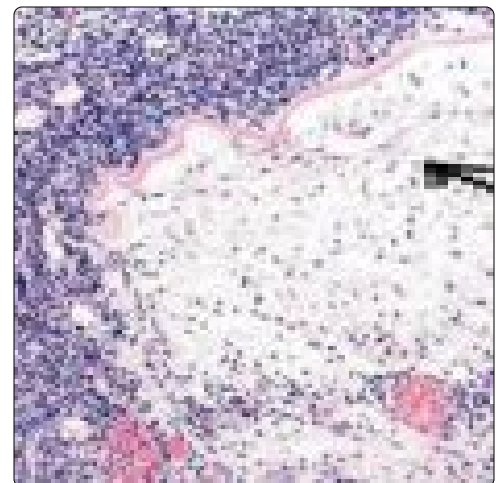


Involuting Follicle: Lining

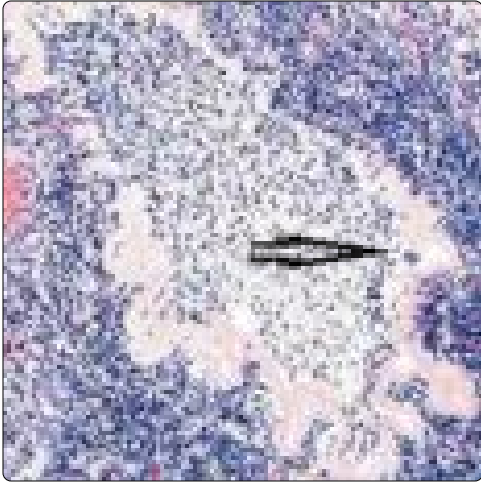
(Left) Following the degeneration of the oocyte, the granulosa cells become mitotically inert and decrease in size. This is followed by shedding of the granulosa cells  into the antral cavity. (Right) In the involuting follicle, the cystic cavity is eventually filled with loose, vascular stroma composed of connective tissue and fibroblasts . The undulating follicular basement membrane becomes thickened  and at this stage is known as the "glassy membrane."



Involuting Follicle: Stroma



Involuting Follicle: Late

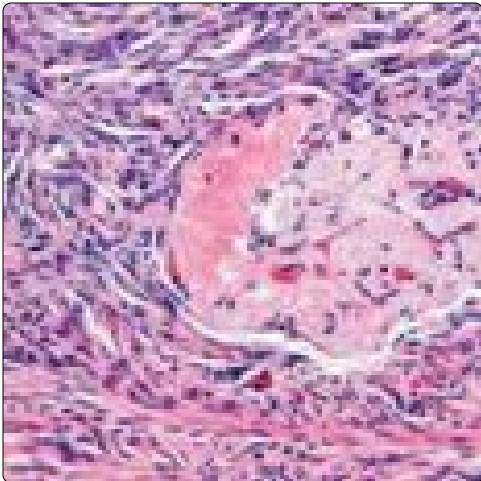


Follicle Cyst

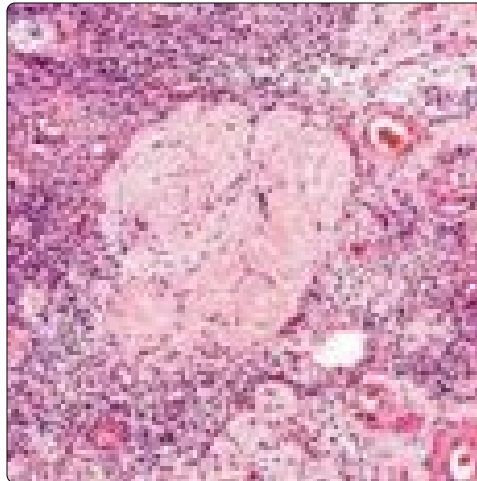


(Left) The "glassy membrane" represents the heretofore histologically inapparent basement membrane that lies between the granulosa cell layer and theca interna. Most mature follicles completely fill with fibrous connective tissue and form scars. (Right) Occasionally, the cystic portion of a follicle will persist. A thin lining of granulosa cells can usually be identified. When the cyst is > 3 cm in diameter, it is termed a follicle cyst.

Hyaline Scar

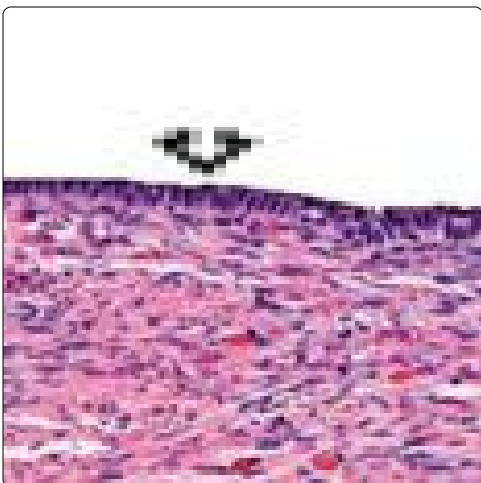


Corpora Fibrosa



(Left) The majority of follicles do not form cysts and become known as corpora fibrosa. As the involuting follicle shrinks, it becomes hyalinized, which can be seen occurring in the left portion of the pictured follicle. (Right) Once complete hyalinization has occurred, the resulting corpora fibrosa is formed. The corpora fibrosa is composed of an undulating band of hyaline tissue. Most of the resulting corpora fibrosa are eventually resorbed into the ovarian stroma.

Ovarian Surface



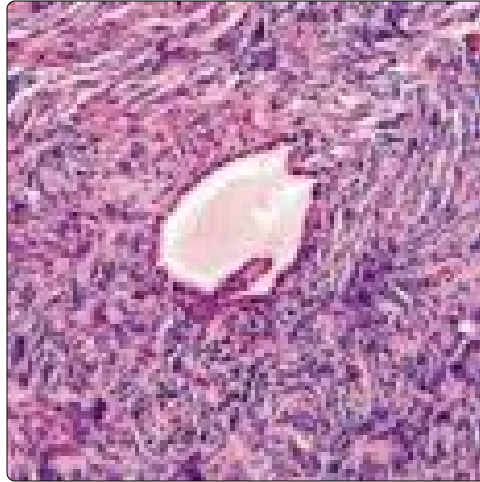
Cortical Inclusion Cyst



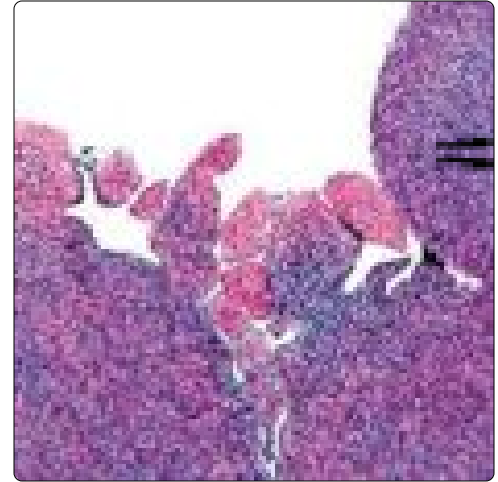
(Left) The surface epithelium of the ovary is a simple cuboidal to columnar lining with occasional areas of pseudostratification. This epithelium is very delicate and is typically avulsed during handling of the ovary during surgery or pathologic examination. (Right) Cortical inclusion cysts or glands are commonly found beneath the cortical surface. These glands (or cysts, if > 1 cm) represent invaginations of the surface epithelium that no longer communicate with the surface.

Cortical Inclusion Gland

(Left) Cortical inclusion cysts and glands may be found deeper within the cortical stroma. The lining is typically simple cuboidal to columnar epithelium, similar to that found on the ovarian surface. **(Right)** Surface papillation, or micropapillomatosis, is composed of finger-like projections of cortical stroma covered by otherwise unremarkable ovarian surface epithelium. The simple papillae and lack of epithelial atypia and stratification differentiate these from surface tumors.



Surface Papillation



Surface Adhesion

(Left) Delicate bands of fibrous tissue, termed adhesions, may be seen originating from the ovarian surface. Adhesions may be more numerous in patients with a history of abdominal surgery or pelvic trauma. They are associated with an increased risk of pelvic pain. **(Right)** Small collections of urothelial (transitional)-type epithelium may be seen within the ovary or broad ligament. These Walthard rests may contain small centrally located cysts.

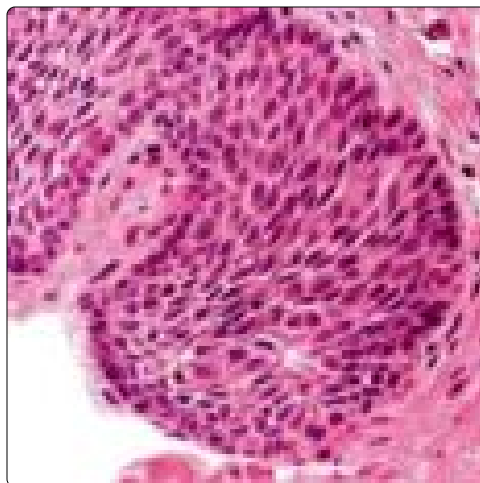


Walthard Rest

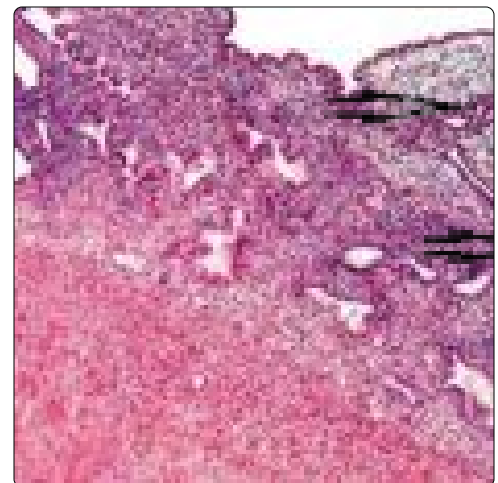


Walthard Rest: Epithelium

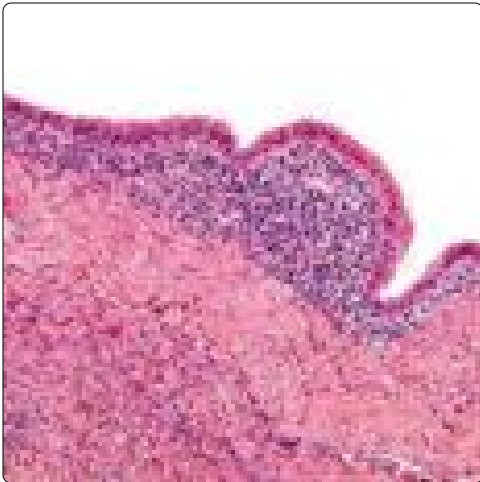
(Left) Walthard rests are composed of transitional epithelium that bears a striking similarity to bladder epithelium. A well-defined basal layer underlies a stratified epithelium that appears to "stream" toward the surface. **(Right)** A not-uncommon finding is the presence of otherwise unremarkable endometrial glands and stroma within the ovary, termed endometriosis. The glands or stroma may range in amount from abundant, as pictured, to completely absent.



Endometriosis



Endometriosis: Gland Poor

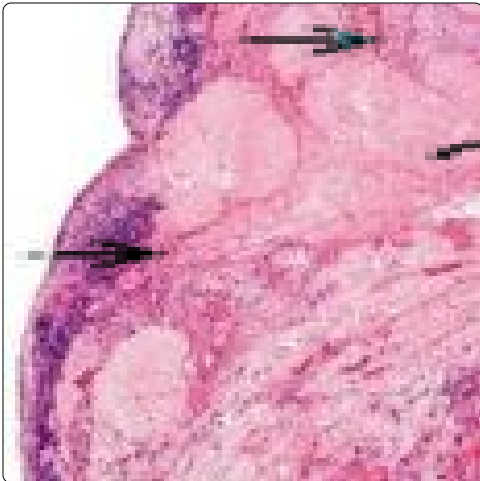


Endometriosis: Stromal Hemorrhage

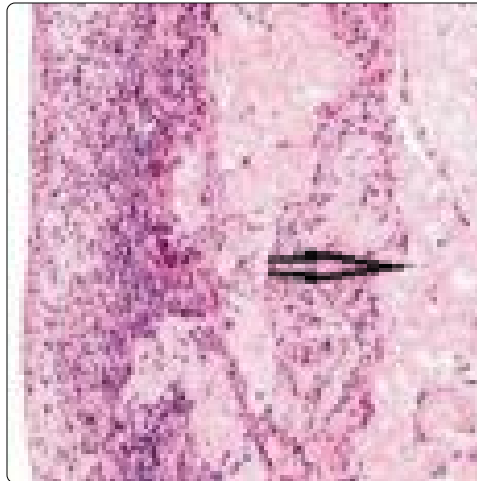


(Left) Some examples of endometriosis may lack endometrial glands and be composed of endometrial stroma. An epithelial lining may or may not be visible. (Right) Another helpful histologic feature of endometriosis is the presence of pigmented macrophages. These pigmented macrophages ingest the blood associated with endometriosis and may occupy the stromal compartment. In cases that lack glands or stroma, stromal hemorrhage may suggest the presence of endometriosis.

Atrophic Ovary

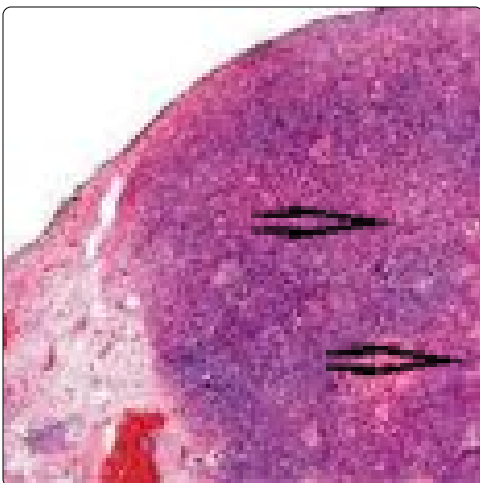


Atrophic Ovary: Cortex and Stroma

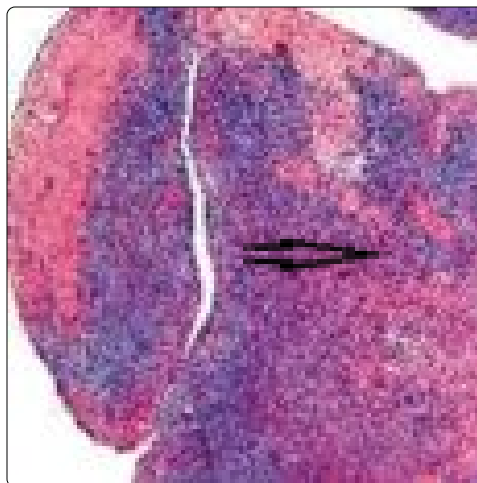


(Left) Peri- and postmenopausal patients' ovaries can display a range of atrophic changes. Examination typically reveals thinning of the cortex and scant medullary and hilar stroma. In some patients, numerous corpora albicantia may be present. (Right) This atrophic ovary displays a markedly thinned cortex with very scant stroma. Two corpora albicantia are present.

Stromal Hyperplasia: Nodular



Stromal Hyperplasia: Nonnodular



(Left) In some peri- and postmenopausal patients, there are varying degrees of stromal hyperplasia. In some patients, the stromal hyperplasia may have well-circumscribed borders and mimic sex cord-stromal neoplasms (fibroma). Occasionally, multiple nodules may be present and efface the normal ovarian architecture. (Right) This atrophic ovary displays a moderate amount of stromal hyperplasia. In this example, however, a discreet nodule is not formed.

PLACENTA

Macroscopic Anatomy

- Umbilical cord
 - Tan-white, partially translucent, rope-like structure
 - Houses 2 arteries and 1 vein
 - Contains Wharton jelly
- Amniotic (fetal) membranes
 - Sac-like structure
 - Contains fetus and amniotic fluid
 - 2 layers: Amnion and chorion
- Placental disc
 - Flat, disc-shaped organ
 - Composed of chorionic villi
 - Fetal surface (chorionic plate)
 - Maternal surface (cotyledons)

Microscopic Anatomy

- Umbilical cord
 - Outer surface lined by amniotic epithelium
 - Wharton jelly composed of mucopolysaccharides
 - Omphalomesenteric duct remnant may be present
 - Remnant connection of yolk sac to developing midgut
 - Tubular structure lined by columnar to cuboidal epithelium, $\pm$ mucin
 - Allantoic duct remnant
 - Represents involuted allantois
 - If present, located between umbilical arteries in proximal umbilical cord
 - Lined by various types of epithelia (e.g., transitional, simple cuboidal)
- Amnion
 - Single layer of cuboidal to flattened epithelial cells
 - Underlying basement membrane
 - Subamniotic connective tissue
- Chorion
 - Lies outside amnion
 - Composed predominately of trophoblastic cells (X cells) and stroma
- Stem villi

- Large villi extending from chorionic plate that give rise to secondary villi
- Secondary villi
 - Intermediate-sized villi arising from stem villi, which give rise to terminal villi
- Tertiary villi
 - Small villi that are rich with capillaries
 - Where maternal-fetal oxygen exchange takes place

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Immature villi
 - Large villi with abundant villous mesenchyme, centrally located vessels, and prominent trophoblastic linings
- Mature villi
 - Comparatively smaller and increased in number when compared to less mature villi
 - Vessels are more peripheral and trophoblasts are less pronounced

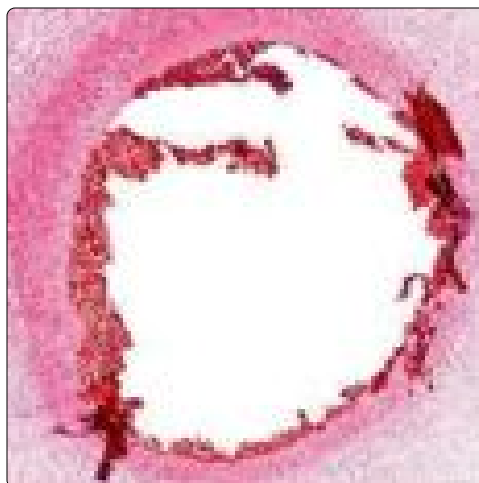
Metaplasia

- Squamous metaplasia of amniotic epithelium may occur normally or in presence of chorionic irritation

Pitfalls and Artifacts

- Vascular tortuosities in umbilical cord
 - Also described as false knots
 - Can lead to mistaken histologic impression of more than 3 vessels in cord
- Omphalomesenteric duct remnant variations
 - May show cystic changes with heterotopic tissues (e.g., gastric, pancreatic, bowel)
- Artifactual perivascular hemorrhage
 - Extravasated erythrocytes into Wharton jelly
 - Absence of hematoma clinically

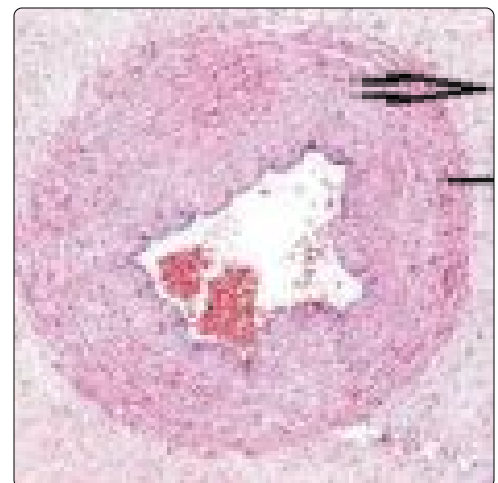
Umbilical Vein



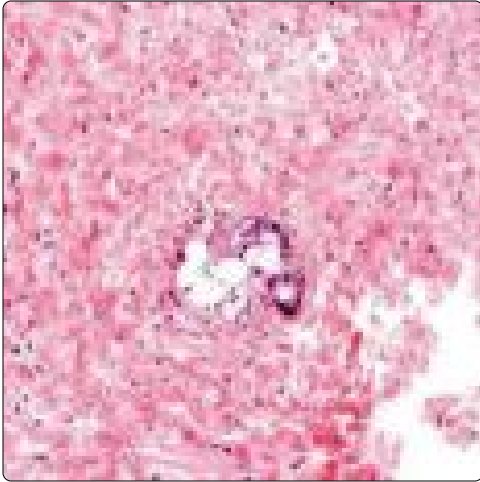
(Left) A normal umbilical cord contains 3 vessels: An umbilical vein, shown here, and 2 umbilical arteries.

(Right) The umbilical artery is composed of 2 muscular layers (inner  and outer ) and endothelium . All 3 vessels in the umbilical cord lack an adventitial layer, differentiating them from other blood vessels in the body. As the adventitia is absent, so too are the lymphatics and vasa vasorum that are typically present in this layer.

Umbilical Artery



Cord: Omphalomesenteric Duct Remnant

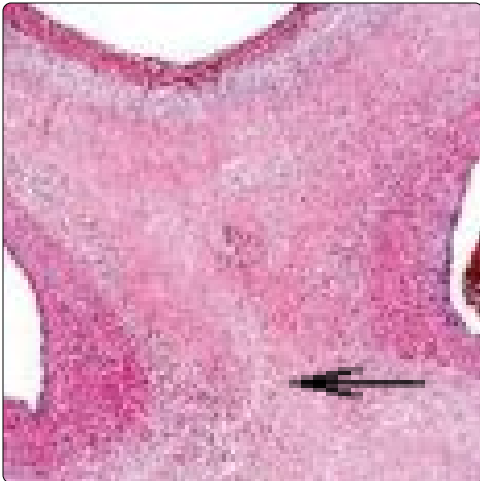


Cord: Omphalomesenteric Duct Remnant Lining

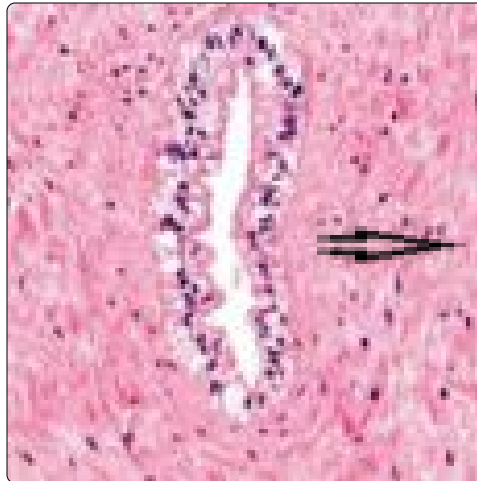


(Left) The omphalomesenteric duct represents the remnant connection of the yolk sac to the developing midgut. This structure is tubular and lined by columnar to cuboidal epithelium that may secrete mucin. Heterotopic tissue may be present, including gastric, pancreatic, or bowel. (Right) The lining epithelium of an omphalomesenteric may contain heterotopic tissue, such as mucinous epithelium with scattered goblet cells.

Cord: Allantoic Remnant



Cord: Allantoic Remnant Lining



(Left) The allantoic remnant (duct) represents the involuted allantois. This remnant can be encountered in the proximal portion of the umbilical cord, typically between the umbilical arteries. (Right) Various types of lining epithelium may be seen within allantoic remnants, including transitional epithelium, simple cuboidal epithelium (seen here with vacuolar change), and yolk sac-like cells. The allantoic remnant and its epithelial lining are rarely clinically significant.

Cord: Allantoic Remnant Location



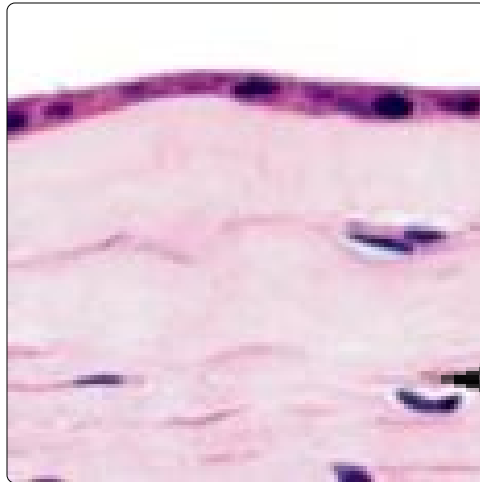
Cord: Hemorrhage



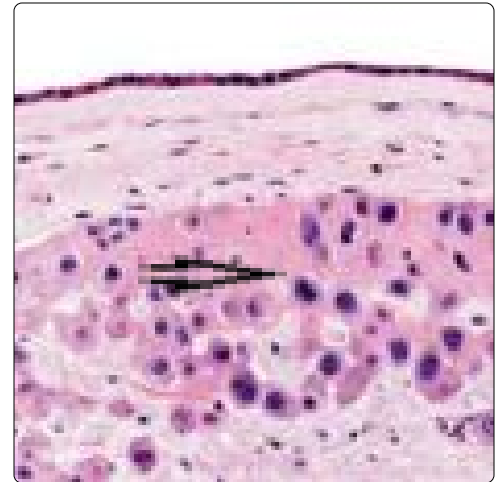
(Left) The lining epithelium of an allantoic remnant may not be present; however, they may be identified due to their location between the umbilical arteries. (Right) Occasionally, perivascular hemorrhage may be identified in the umbilical cord. If a grossly identified hematoma is not present, these areas likely represent artifactual extravasation of blood into the perivascular tissue (usually due to traction or cord clamps) and is of no clinical significance.

Amnion

(Left) The amniotic epithelium is usually composed of a single layer of cuboidal cells that may display squamoid features. The epithelium lies on a homogeneous, eosinophilic basement membrane. A layer of fibroblasts lies directly beneath the basement membrane. (Right) Chorionic "X cells" compose the 1st cellular layer of the chorion. These cells are typically seen admixed with the underlying maternal decidual cells.

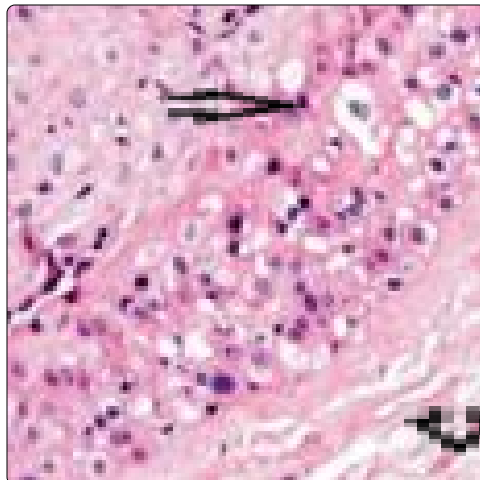


Amnion and Chorion

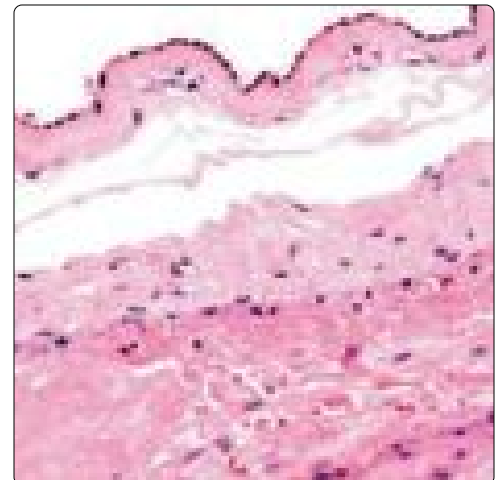


Chorion: Superficial Layers

(Left) The chorion consists of multiple layers. The spongy layer is relatively acellular and separates the amnion and chorion. Scattered macrophages and fibroblasts may be seen beneath the spongy layer. The next layer is the reticular zone, followed by a false basement membrane. Beneath this are the X cells and decidua. (Right) Artifactual subamniotic clefts and subchorionic fibrin may be present and have no clinical significance.



Subamniotic Clefts

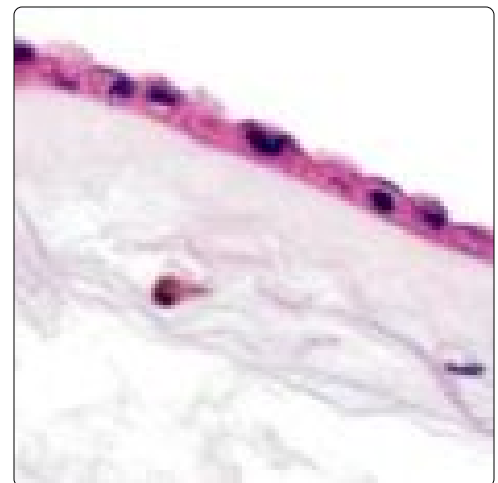


Membrane: Sclerotic Villi

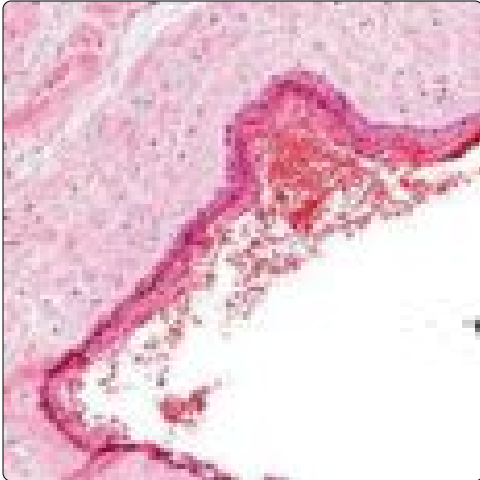
(Left) Scattered sclerotic villous remnants may be seen within the chorionic portion of the placental membranes. (Right) Scattered macrophages may be identified beneath the amnion, within the chorion or the artifactual cleft that may form. Here, a macrophage contains abundant globular brown pigment seen in meconium staining.



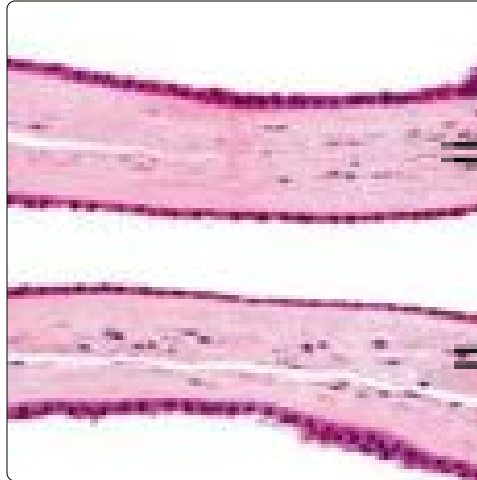
Subamniotic Macrophages



Amnion: Squamous Metaplasia

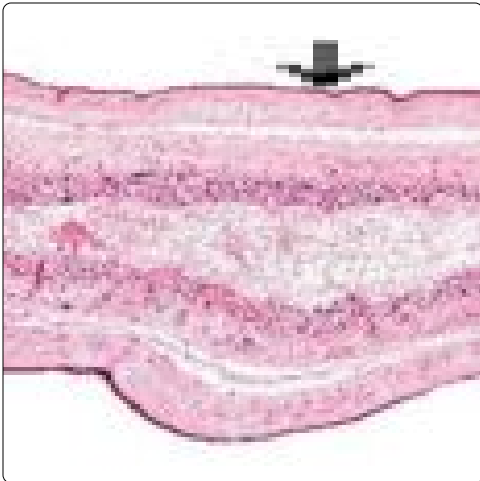


Diamniotic Monochorionic Membranes

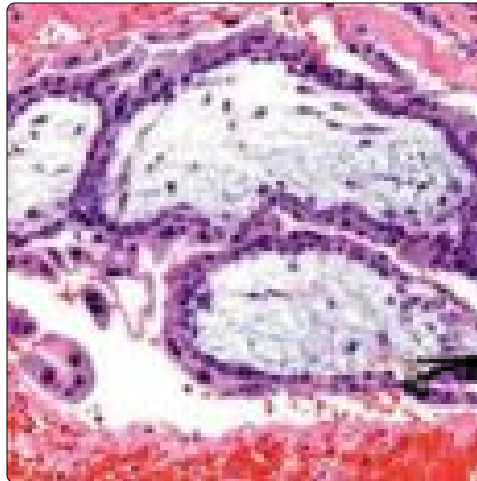


(Left) As can be seen in the umbilical cord, the amniocytes of the membranes and chorionic (fetal) surface may undergo squamous metaplasia. This finding has no clinical implications for the developing fetus. (Right) Several variations of membranes may be seen in multiple gestation pregnancies. The presence of 1 chorion, which is relatively attenuated and acellular, and 2 amniotic sacs leads to the diamniotic monochorionic membranes pictured here.

Diamniotic Dichorionic Membranes

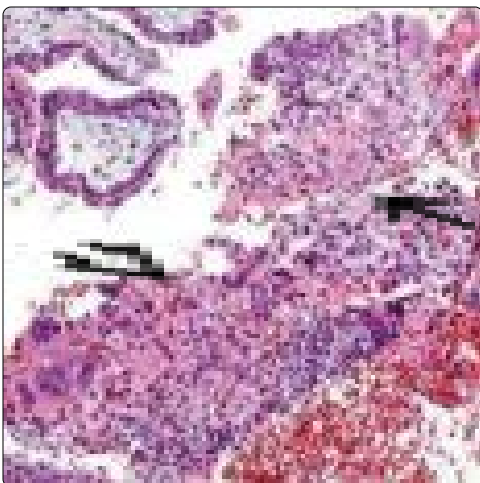


Immature Villi

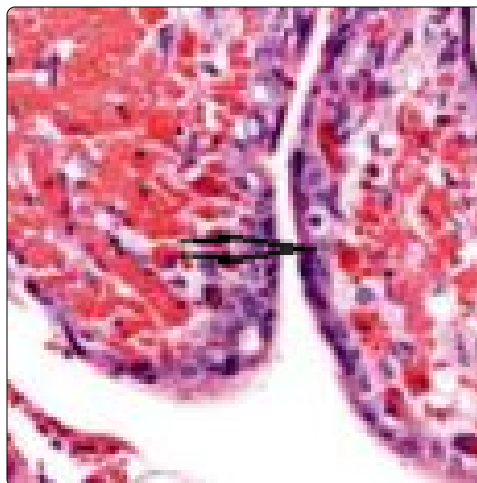


(Left) Twin pregnancies with 2 complete sets of amniotic membranes lead to the appearance of fused membranes (the membrane T zone). These "di-di" membranes are composed of 2 distinct chorionic and amniotic layers. (Right) Villi around 6 weeks of age are round to oval with basophilic stroma and a relative lack of stromal vessels. The cytotrophoblastic and syncytiotrophoblastic layers are distinct and highly cellular compared to later in gestation.

Immature Villi and Trophoblasts



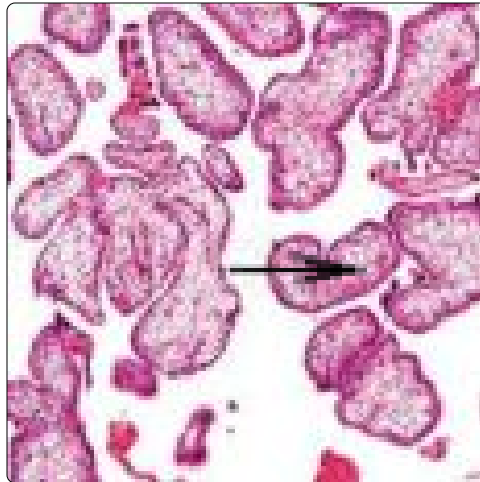
Fetal Blood Elements



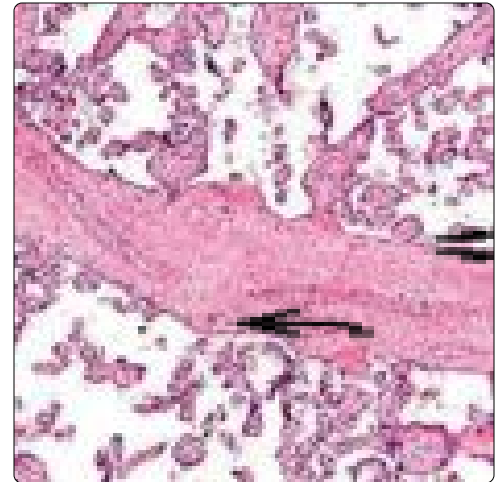
(Left) Extravillous trophoblasts (intermediate and syncytiotrophoblasts) may be relatively increased in number and clustered in close approximation. The presence of extremely immature villi should argue against a misdiagnosis of choriocarcinoma. (Right) Capillaries will begin to proliferate within the immature villi starting after the 6th week. Between this time and the 10th-12th weeks, nucleated fetal red blood cells may be identified in otherwise normal placentas.

Immature Villi

(Left) Immature villi are larger with a more cellular stroma than their mature counterparts. The vessels within the villi tend to be more centrally located. The villi become progressively smaller throughout pregnancy. **(Right)** Within the mature placenta, 3 types of villi may be identified. As the villi extend from the chorionic plate, the stem villi, which give rise to the secondary villi, will branch to form the tertiary villi.

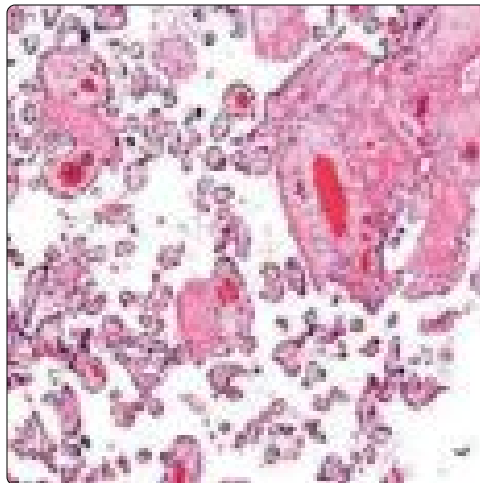


Placental Villi

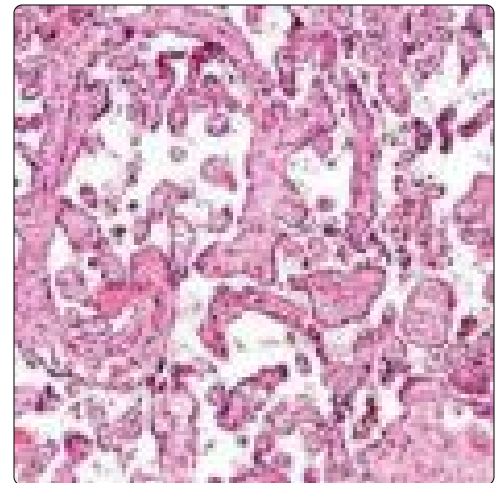


Placental Stem Villi

(Left) A large stem villus can be seen in the upper right corner. The stem villi contain the large caliber stem vessels, which coalesce in the chorionic plate and, eventually, the umbilical cord. **(Right)** Numerous secondary villi can be seen giving rise to scattered tertiary or terminal villi. These secondary villi can be differentiated from stem villi based on the lack of large, thick-walled arteries and veins.

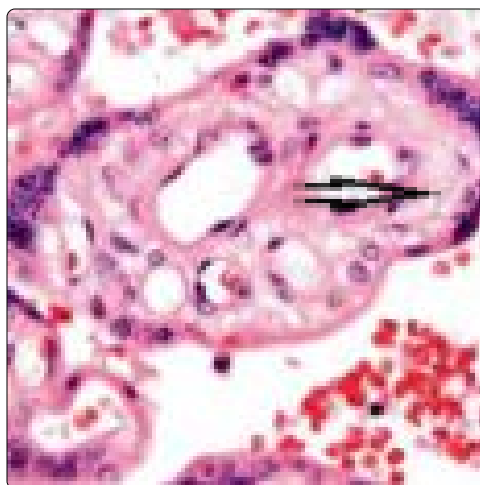


Placental Secondary Villi

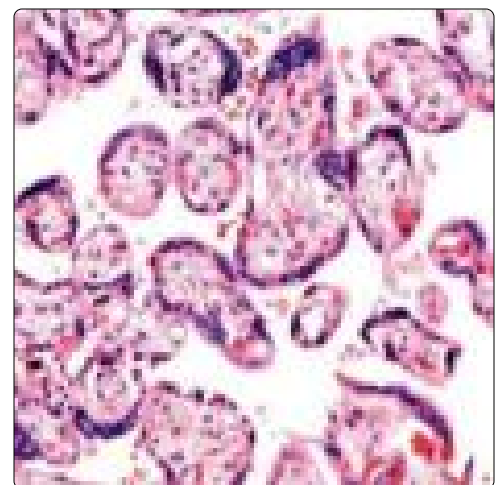


Secondary Villus

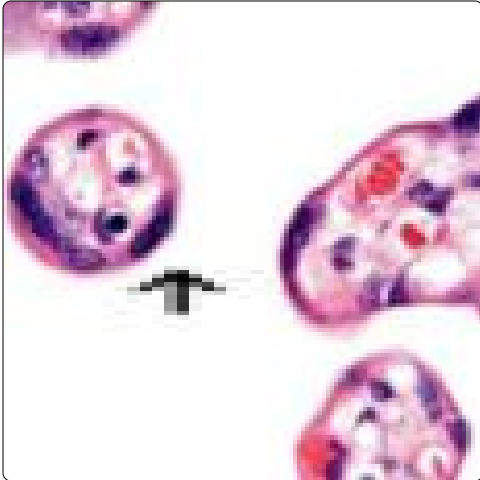
(Left) Secondary villi typically contain multiple capillaries that tend to be centrally located. In addition to having more vascular structures, the secondary villi are larger and contain more stroma when compared to their terminal counterparts. **(Right)** Tertiary villi vary in size but are generally smaller than secondary villi. The peripheral location of the villous capillaries helps to facilitate the transfer of oxygen from mother to fetus.



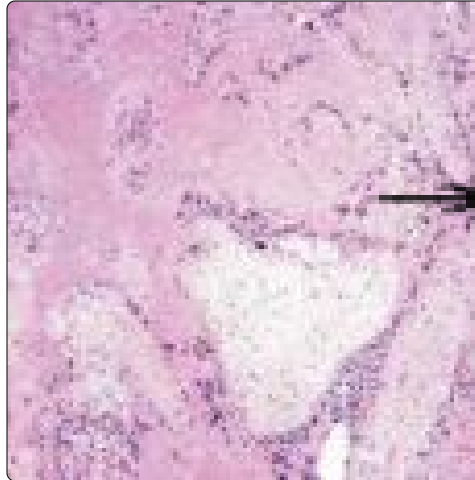
Placental Tertiary Villi



Tertiary Villi

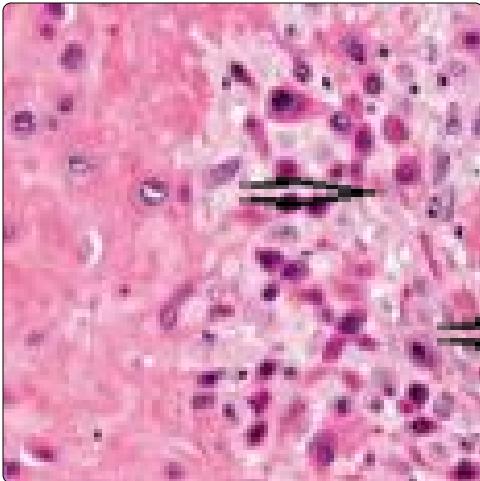


Perivillous Fibrin

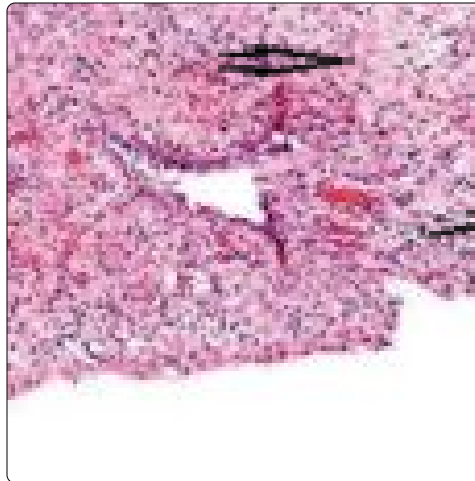


(Left) In mature tertiary villi, the cytotrophoblastic layer is typically not visible. The syncytiotrophoblastic layer is thin and irregular, and clusters of darkly staining syncytiotrophoblastic nuclei [X] may be the only evidence of its presence. Fetal blood [X] may be seen in the terminal villi capillaries. (Right) Scattered pockets of perivillous fibrin may be seen in otherwise normal placentas. The fibrin encases the villi [X], leading to their involution; however, trophoblasts [X] may persist.

Decidua

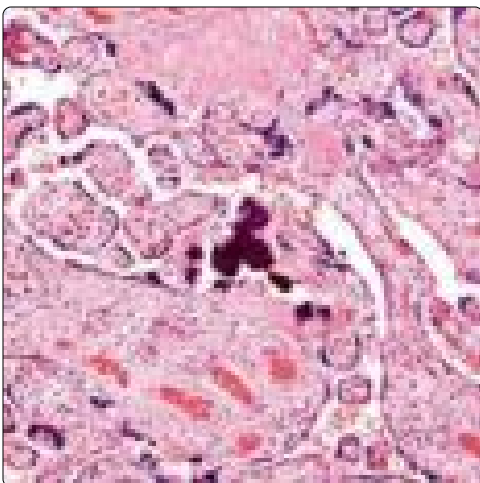


Decidualized Endometrium

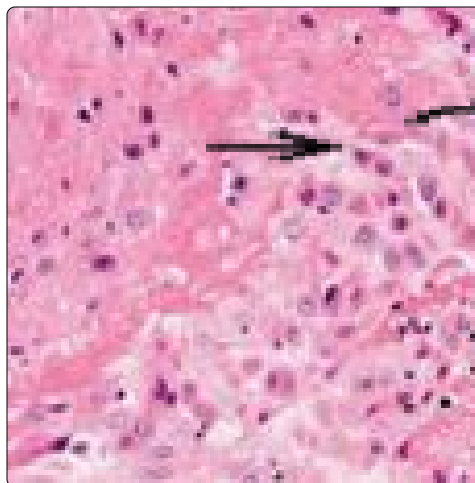


(Left) Endometrial stromal cells undergo a process of "decidualization," leading to the so-called decidua [X]. These cells can be identified by their abundant eosinophilic cytoplasm, centrally located vesicular nuclei, and well-defined cell borders. (Right) Occasionally, large fragments of accompanying decidualized endometrial glands [X] and stroma [X] may be present with the placenta. This may be the predominant component of so-called "products of conception" seen in cases of missed abortions or previable pregnancies.

Calcification



Implantation Site



(Left) Scattered calcifications [X] are commonly identified in or around placental villi in mature placentas. Calcification is often normal. It may be abnormal, however, if excessive for gestational age and may be a sign of ischemia. (Right) The implantation site is composed of maternal decidua [X], extravillous trophoblastic cells [X], and Nitabuch fibrinoid [X]. Identification of the implantation site may serve as proof of intrauterine pregnancy when villi are absent.

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SECTION 14

Endocrine



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Adrenal Gland

ADRENAL GLAND

Macroscopic Anatomy

- Located in retroperitoneum, superior and medial to kidneys
- Right adrenal gland is pyramidal; left adrenal gland is crescent-shaped
- On cut section, generally 3 zones are visible grossly
 - Outermost layer: Thick and bright yellow; mostly composed of zona fasciculata
 - Middle layer: Thin and brown, composed of zona reticularis
 - Innermost layer: Gray, composed of adrenal medulla
 - Present in head and body of gland but absent in tail

Microscopic Anatomy

- Fibrous capsule surrounds adrenal gland
- Adrenal cortex is histologically divided into 3 layers (from outermost to innermost)
 - Zona glomerulosa
 - Thin layer of cuboidal cells in packed clusters

- Zona fasciculata
 - Thick layer of highly vacuolated cells in columns
- Zona reticularis
 - Variably thick layer of deeply eosinophilic cells
- Adrenal medulla
 - Sheets of small to large, polygonal cells with basophilic to amphophilic granular cytoplasm

CONSIDERATIONS AND VARIATIONS

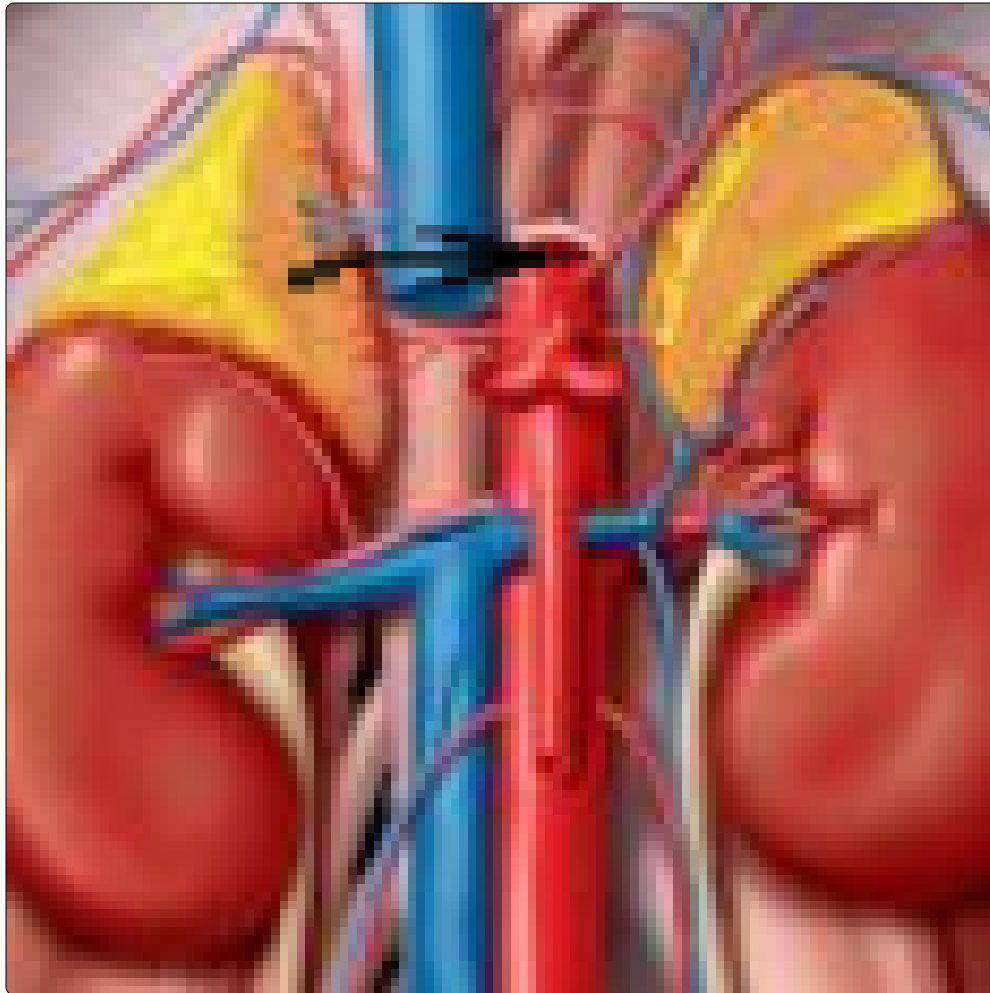
Age-Related Variations

- In neonates, majority of adrenal cortex is composed of provisional (or fetal) cortex with outer thin rim of definitive cortex
 - After birth, provisional cortex involutes, and definitive cortex becomes adult adrenal cortex

Hyperplasia

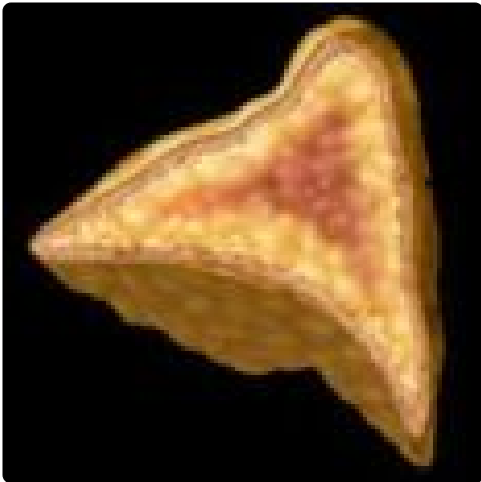
- Nodular adrenocortical hyperplasia consists of areas of hyperplastic adrenocortical cells that occur with increasing frequency with age

Adrenal Glands

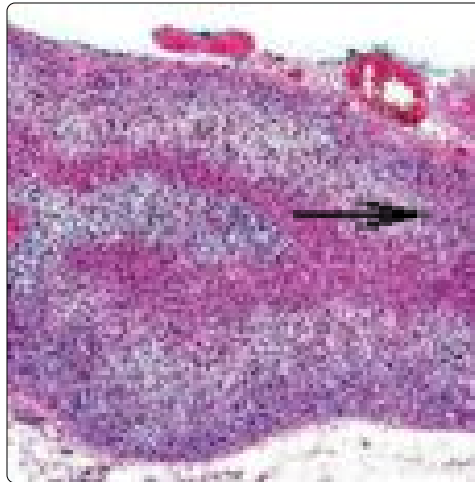


The adrenal glands  are located in the retroperitoneum, superior to the kidneys. The right adrenal gland is pyramidal in shape; the left adrenal has a crescent-shaped appearance. Each adrenal is supplied by the superior (arising from the inferior phrenic artery), middle (arising from the aorta), and inferior (arising from the renal artery) arteries.

Adrenal Gland Cross Section

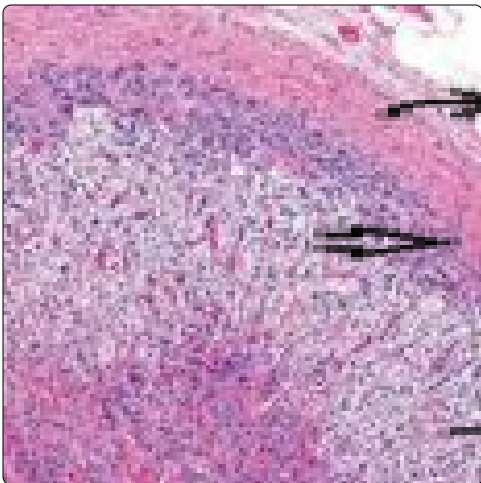


Adrenal Gland

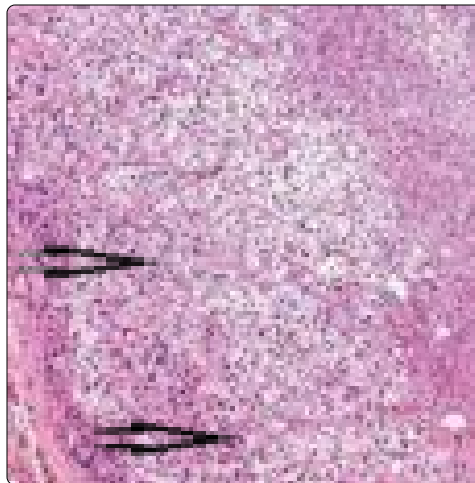


(Left) The innermost layer of the adrenal gland, the medulla, is composed of gray, glistening tissue and is surrounded by the outer bright yellow cortical tissue. (Right) At low magnification, the adrenal gland is composed of an outer cortex (composed of 3 distinct layers) and inner medulla. A thin fibrous capsule is present around the periphery.

Cortex

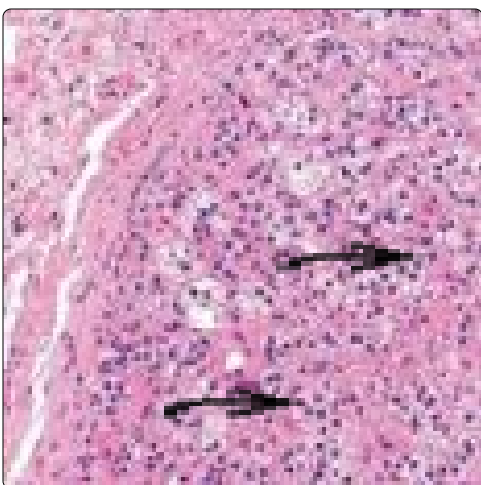


Layers

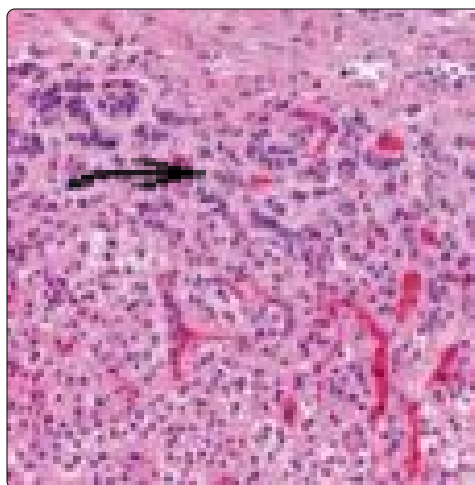


(Left) The adrenal cortex is composed of 3 layers. The zona glomerulosa lies directly beneath the capsule. The zona fasciculata forms the middle layer of the cortex. The zona reticularis forms the innermost layer. Note too the fibrous capsule. (Right) The thin, discontinuous layer of zona glomerulosa cells lies directly beneath the capsule. The deeply eosinophilic zona reticularis lies deepest. In between, the distinctive vacuolated cells of the zona fasciculata form the majority of the cortex.

Zona Glomerulosa



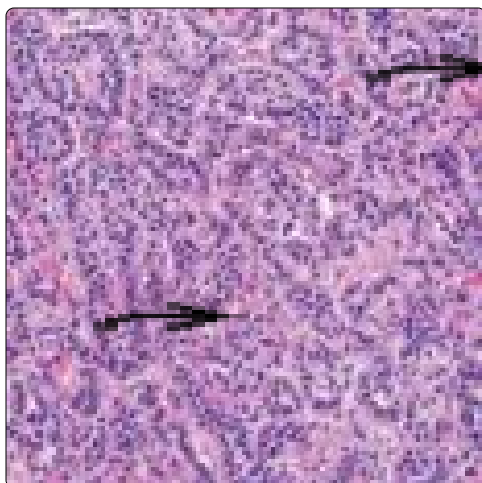
Zona Glomerulosa Cells



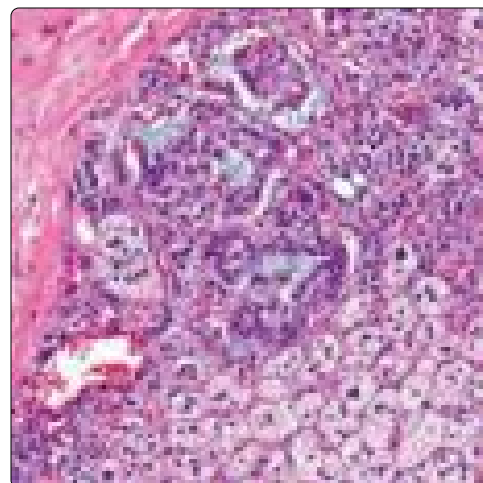
(Left) Zona glomerulosa cells are located directly beneath the adrenal capsule. This layer of the cortex is generally thin and often discontinuous, and in some areas, gaps allow for the cells of the zona fasciculata to lie directly against the adrenal capsule. (Right) The cells of the zona glomerulosa are arranged in small clusters and balls. They have slightly eosinophilic to amphophilic cytoplasm that is sparse compared to that of the underlying zona fasciculata cells.

Zona Glomerulosa Oblique Section

(Left) Although the zona glomerulosa layer is generally thin, oblique sectioning (as seen in this image) can give the appearance of more extensive growth. Note also that some of the cells can show a vague columnar morphology. (Right) Focal myxoid change may be seen in the outer adrenal cortex or in the adjacent fibroadipose tissue but is generally rare.

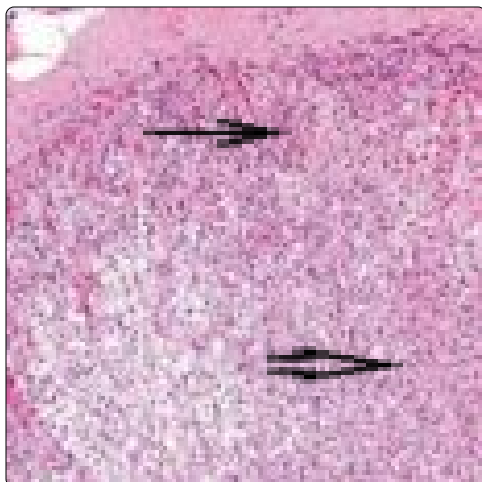


Rare Myxoid Change

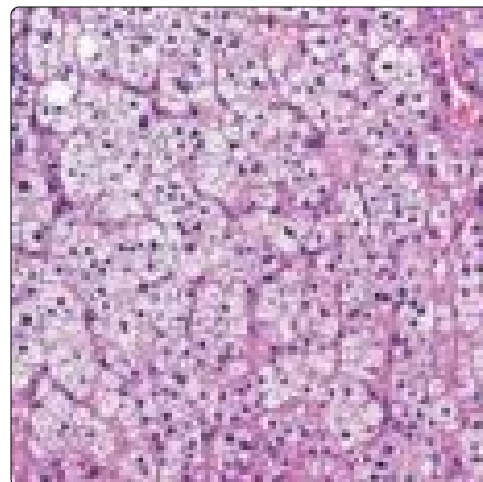


Zona Fasciculata

(Left) The zona fasciculata lies beneath the zona glomerulosa and accounts for the majority of the adrenal cortex. The cells have abundant lipid-laden cytoplasm, which produces the finely vacuolated appearance of the cells microscopically and the bright yellow color of the adrenal cortex macroscopically. (Right) This high-power photomicrograph illustrates the abundant, finely vacuolated cytoplasm of the cells of the zona fasciculata.

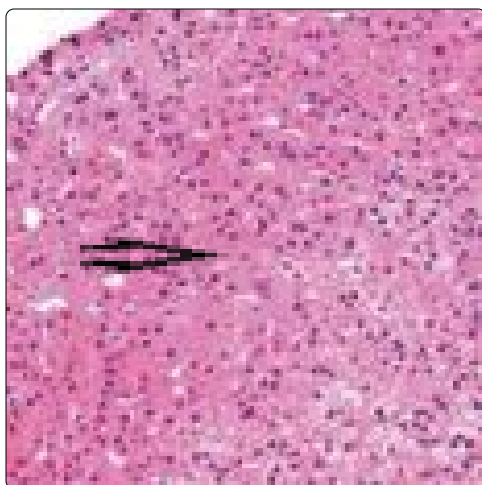


Characteristic Vacuolated Cytoplasm

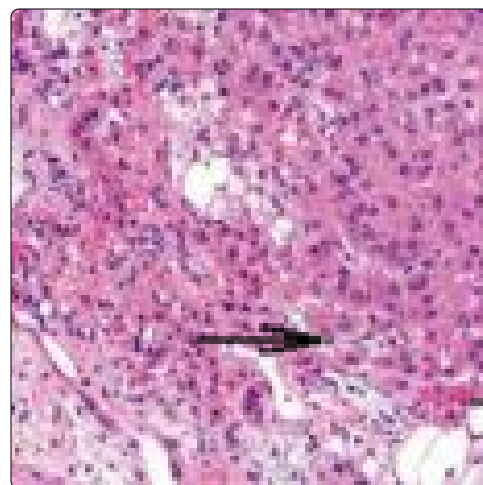


Zona Reticularis

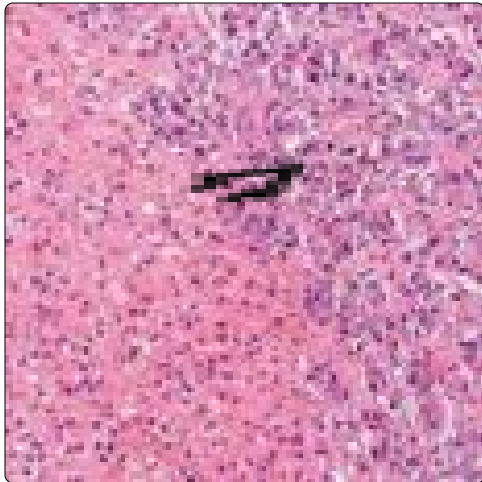
(Left) The zona reticularis forms the innermost layer of the adrenal cortex. The cytoplasm is more eosinophilic when compared to the cells of the adjacent zona fasciculata. (Right) The cells of the zona reticularis may occasionally contain intracytoplasmic accumulations of golden brown lipochrome pigment. This pigment is often most abundant within the deeper cells of the zona reticularis along the corticomedullary junction.



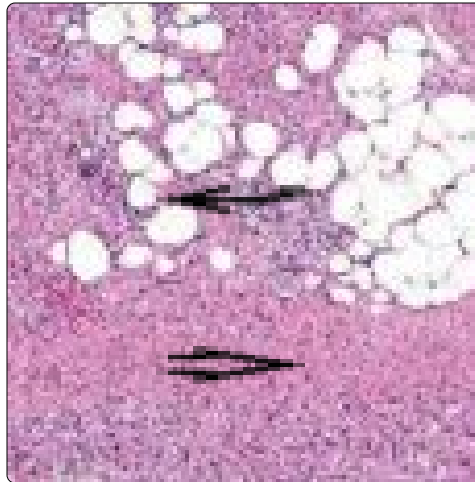
Lipochrome Pigment



Cortical-Medullary Junction



Adrenal Capsule

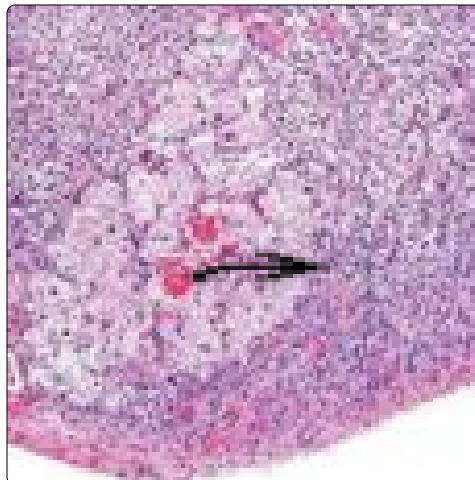


(Left) The junction of the zona reticularis layer of the adrenal cortex and the underlying adrenal medulla is irregular. The cytoplasm of the zona reticularis cells along the junction contain lipochrome pigment. (Right) Collections of adrenal cortical cells can be identified outside of the adrenal capsule within the periadrenal fat. These collections of benign cortical cells are normal and should not be confused with a neoplastic process.

Cortical Nodule

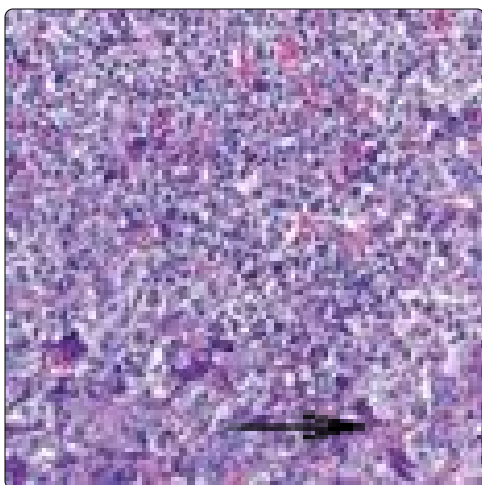


Cortical Nodule

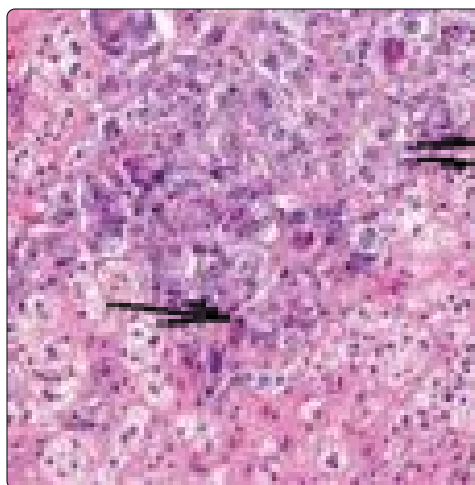


(Left) Occasionally, small microscopic nodules or hyperplastic aggregates may be encountered in the adrenal cortex, particularly within or associated with the capsule. Note also the darker adrenal medulla in this image. (Right) This vague cortical nodule is composed of zona fasciculata cells, as evidence by their abundant vacuolated and foamy cytomorphology. These localized collections are more common with increasing age.

Adrenal Medulla



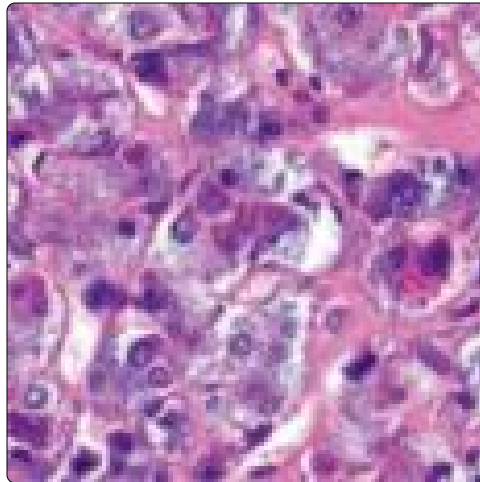
Adrenal Medulla Cells



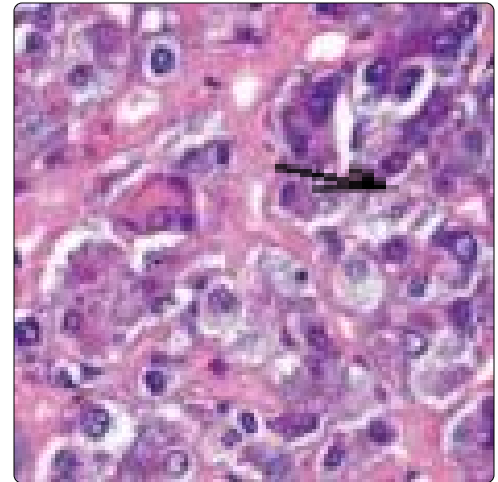
(Left) The cells of the adrenal medulla are arranged in nests and are surrounded by fibrovascular septa. The cytoplasm is basophilic to amphophilic and varies in size from small and round to large and polygonal and may show focal vacuolization. (Right) The cytoplasm of the adrenal medulla cells is basophilic and granular in appearance in contrast to those of the adrenal cortex.

Adrenal Medulla Cytology

(Left) The nuclei of the adrenal medulla cells appear vesicular with clumped chromatin and peripheral condensation along the nuclear membrane. A central nucleolus may be present, and the nuclei are often eccentrically located. Intracytoplasmic globules are sometimes identified within the cells. **(Right)** Rare ganglion cells can be seen interspersed among the adrenal medulla cells and can occur either singly or in small clusters.

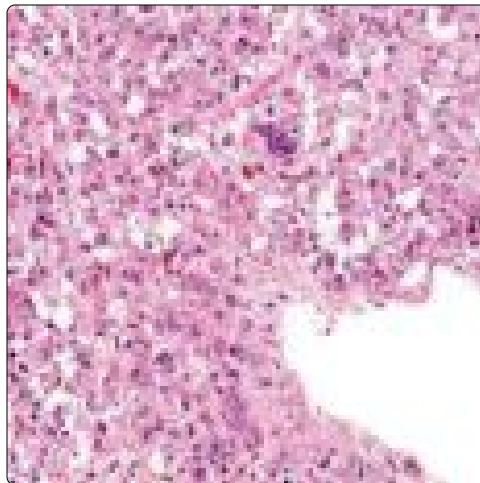


Rare Ganglion Cells

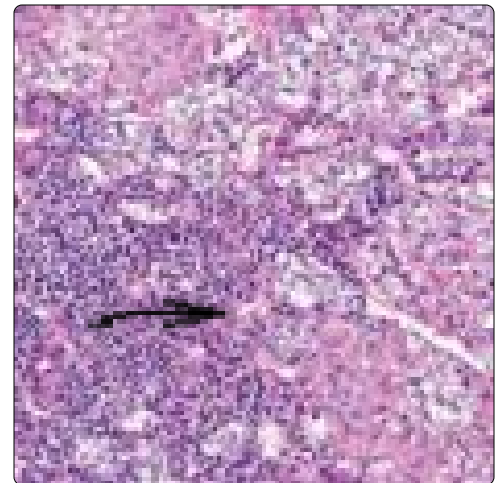


Focal Atypia

(Left) The cells of the adrenal medulla can appear atypical with nuclear enlargement and hyperchromasia. This atypia is normal and does not indicate a neoplastic process. **(Right)** Aggregates of chronic inflammatory cells, particularly lymphocytes and plasma cells, can be seen within the adrenal medulla or along the corticomedullary junction.

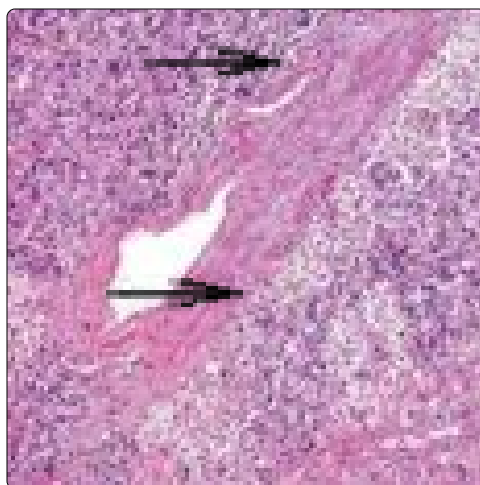


Inflammatory Cells

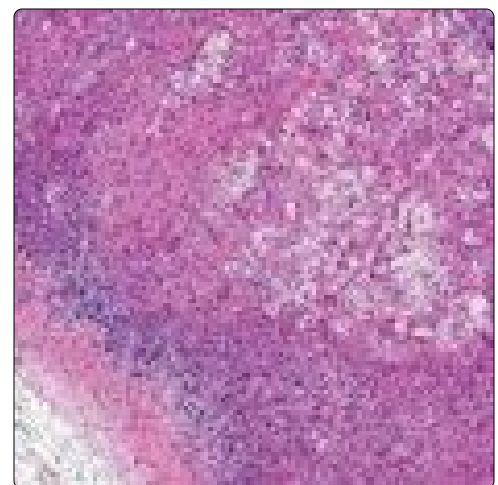


Central Adrenal Vein

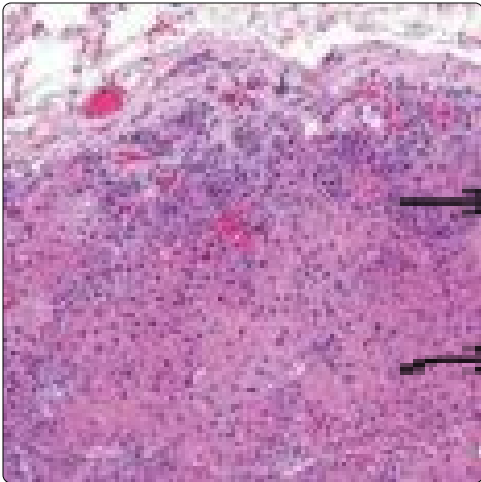
(Left) The central adrenal vein is centrally located within the medulla. It has an uneven muscular layer surrounding it, which may be completely absent in areas, allowing cells to be separated from the lumen by the intima only. Invaginated cortical cells can be seen surrounding the vein. **(Right)** Diffuse adrenocortical hyperplasia can be seen in a variety of disorders (such as Cushing syndrome) and often manifests as proliferation of either the zona fasciculata or zona reticularis or both.



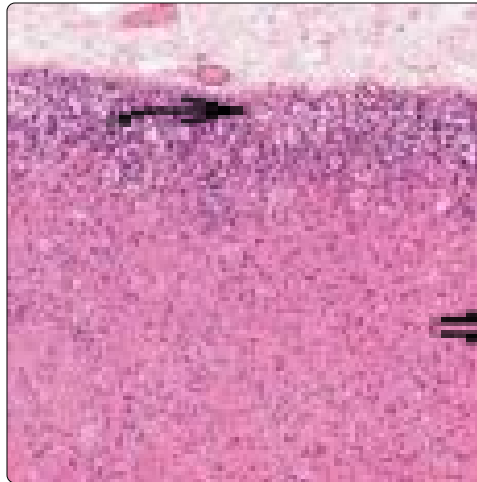
Diffuse Cortical Hyperplasia



Diffuse Cortical Hyperplasia

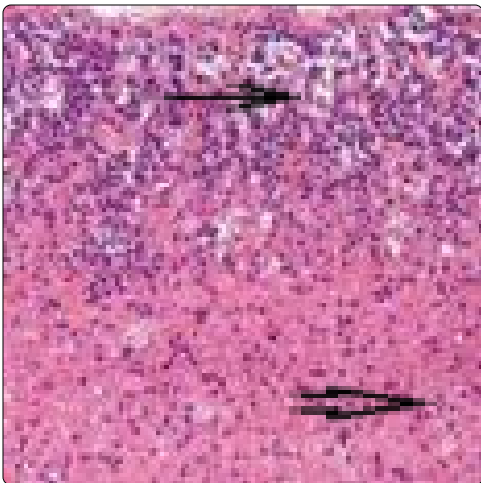


Adrenal Cortex, 1 Week After Birth

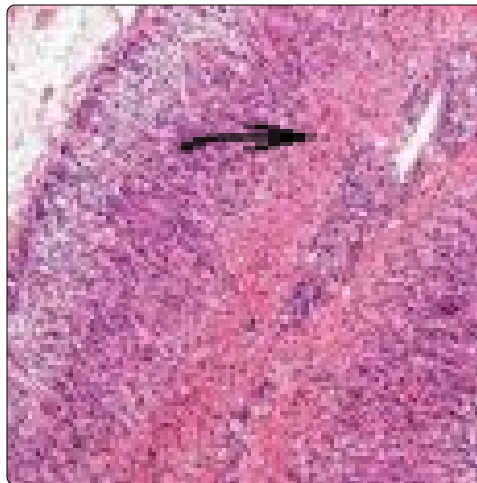


(Left) This H&E shows hyperplasia of zona reticularis cells [1] as evidenced by their voluminous, deeply eosinophilic cytoplasm. Note the smaller zona glomerulosa cells [2] for comparison. **(Right)** Directly beneath the adrenal capsule there is a thin band of small, hyperchromatic cells arranged in cords [3] that overlies the provisional (or fetal) cortex [4]. As the provisional cortex involutes, this outer layer will proliferate to become the definitive (or adult) adrenal cortex.

Adrenal Cortex, 1 Week After Birth

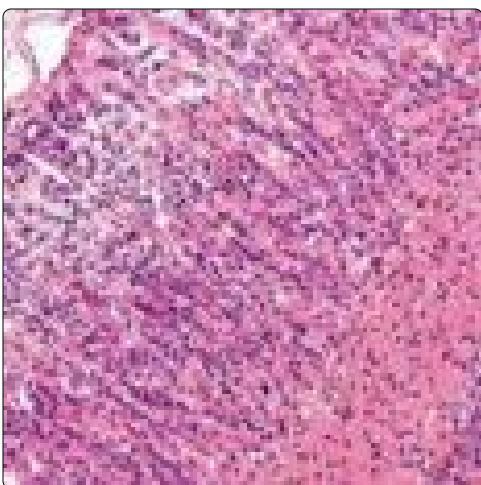


Adrenal Cortex, 3 Weeks After Birth

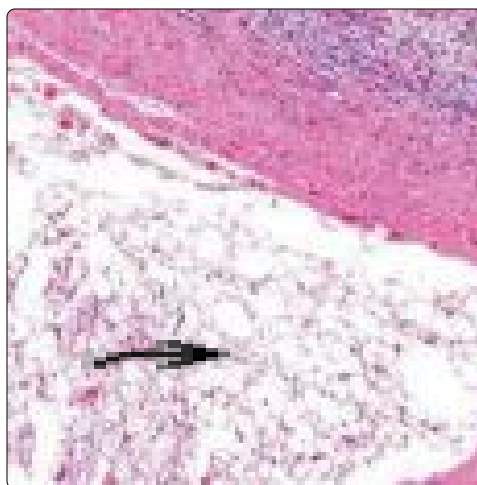


(Left) The provisional cortex [1] is composed of large cells with abundant eosinophilic cytoplasm and vesicular nuclei, as opposed to the smaller, hyperchromatic cells of the definitive cortex [2]. **(Right)** By 3 weeks after birth, the provisional cortex has largely involuted. The cells in the definitive cortex, which began as a thin subcapsular rim of cells, have now proliferated to form the adult adrenal cortex [3]. Within the central portion of the gland, the adrenal medulla [4] is visible.

Adrenal Cortex, 3 Weeks After Birth



Brown Fat



(Left) At 3 weeks after birth, the cells of the definitive adrenal cortex have expanded as the provisional cortex involuted and now form the cords and nests of the adult cortex. **(Right)** Normal brown fat [1] may be identified in the perirenal soft tissue. It is characterized by adipocytes featuring numerous small cytoplasmic vacuolizations, lending them a granular, clear to eosinophilic cytomorphology.

PARAGANGLIA

Macroscopic Anatomy

- Small, generally microscopic neuroendocrine organs associated with autonomic nervous system
 - Sympathetic paraganglia
 - Located along prevertebral and paravertebral sympathetic chains and in connective tissue surrounding pelvic organs
 - Parasympathetic paraganglia
 - Located along cervical and thoracic branches of glossopharyngeal and vagus nerves

Microscopic Anatomy

- Paraganglia contain 2 cell types
 - Neuroendocrine cells
 - Small round cells
 - Clear to amphophilic cytoplasm
 - Organized into nests (Zellballen) and cords
 - IHC: Synaptophysin (+), chromogranin (+), keratin (-)

- Supporting (sustentacular) cells
 - Usually flattened and inconspicuous
 - Partially or completely surround neuroendocrine cells
 - a.k.a. "satellite" or "type 2" cells
 - IHC: S100 protein (+), SOX10(+)

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Adrenal medulla and carotid bodies enlarge with age; other paraganglia decrease in size

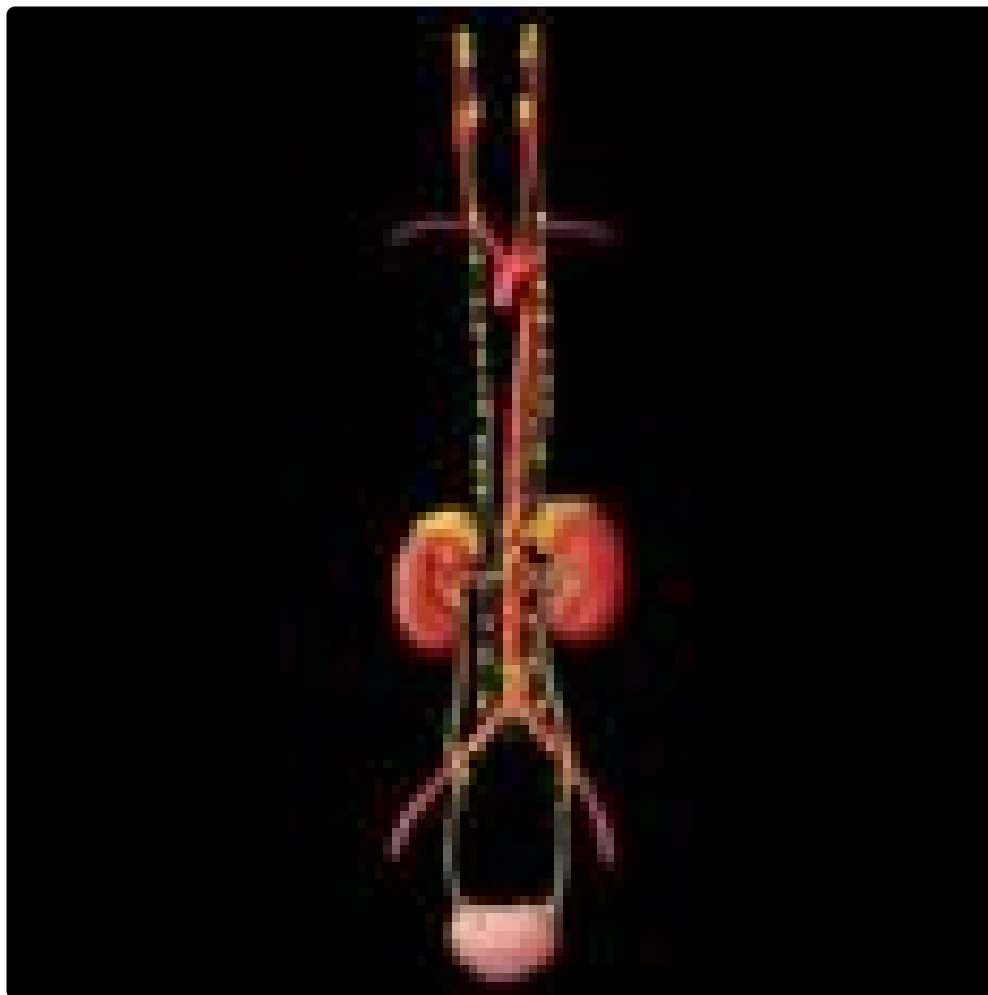
Hyperplasia

- Hyperplasia of carotid body may occur in some conditions, particularly hypoxic states (e.g., high altitude, pulmonary disorders, or hypertension)

Pitfalls and Artifacts

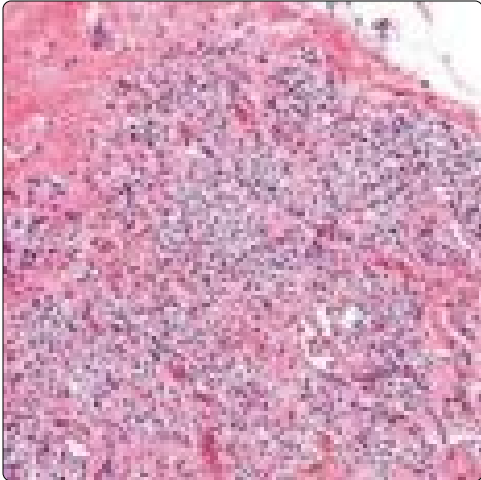
- Paraganglia, particularly when identified in organ resection specimens, may closely mimic nests of carcinoma

Paraganglia

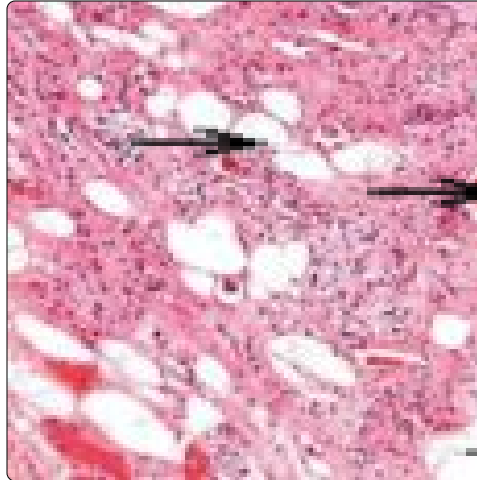


The sympathetic paraganglia are along the prevertebral and paravertebral sympathetic chains and include the adrenal medulla  and the organ of Zuckerkindl . They are numerous around pelvic organs, such as the bladder . The parasympathetic paraganglia are along the cranial and thoracic branches of the glossopharyngeal and vagus nerves and include the carotid bodies .

Paraganglia (Carotid Body)

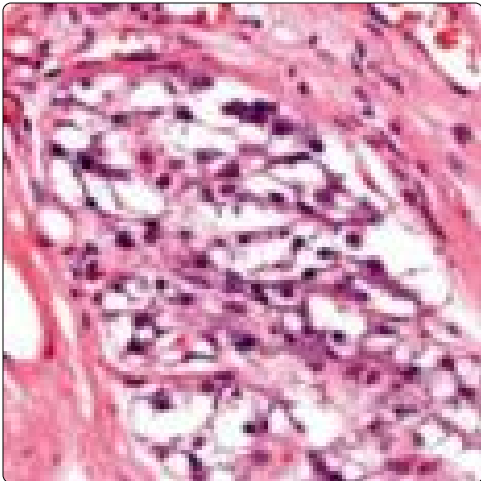


Scattered Nests

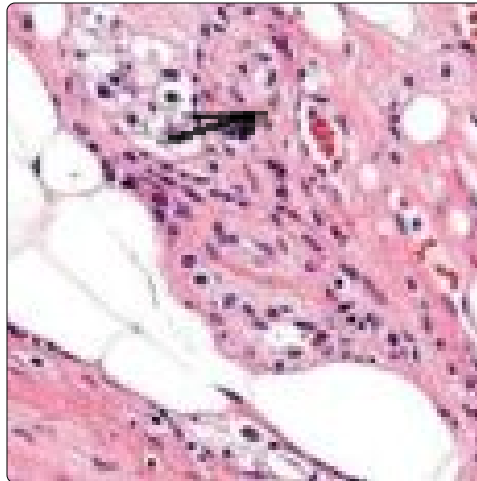


(Left) This example of a paraganglion (carotid body) is more cellular but is still comprised of tight nests of neuroendocrine cells surrounded by fibrous stroma. (Right) Some paraganglia, especially those in connective tissue adjacent to parenchymal organs, are comprised of very few nests scattered amongst fibrous tissue, nerves, and fat; therefore, they may be easily overlooked. In resection specimens, these paraganglia may be confused with nests of carcinoma cells.

Paraganglia Cytology

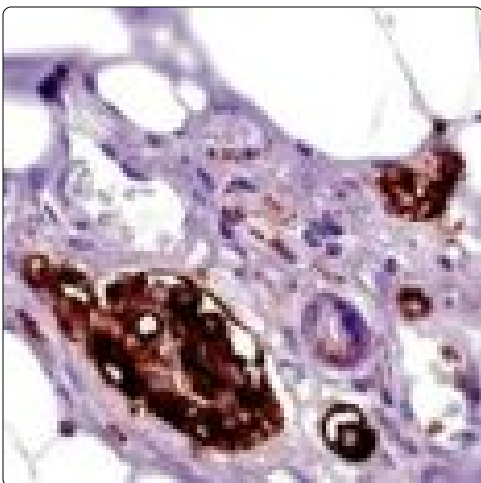


Clear Cytoplasm

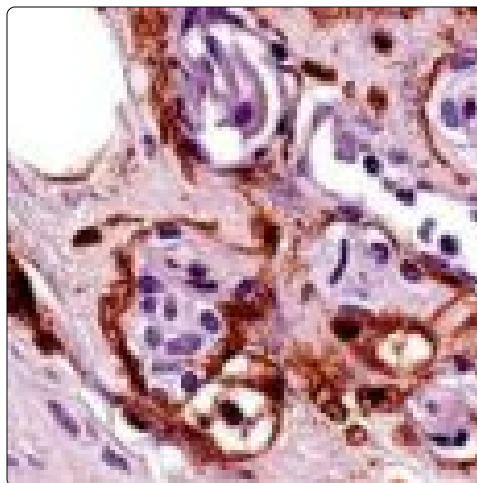


(Left) On high-power magnification, the paraganglia are characterized by nests of neuroendocrine cells with clear to amphophilic cytoplasm and round, uniform nuclei. The sustentacular cells are generally not identifiable. (Right) These typical nests of paraganglionic cells show predominantly clear cytoplasm in the neuroendocrine cells with the typical small uniform nuclei. The sustentacular cells are not conspicuous on routine H&E staining.

Paraganglia Immunophenotype



Paraganglia (S100 Protein)



(Left) The nests of paraganglionic cells are highlighted by immunostains for chromogranin (shown) and synaptophysin, both of which typically show strong cytoplasmic staining in the neuroendocrine cell component. Paraganglia cells are cytokeratin negative. (Right) Immunostains for S100 protein typically demonstrate strong cytoplasmic immunoreactivity in the flattened supporting (sustentacular) cells that surround the neuroendocrine cells. These cells are generally inconspicuous on H&E stain.

THYROID

Macroscopic Anatomy

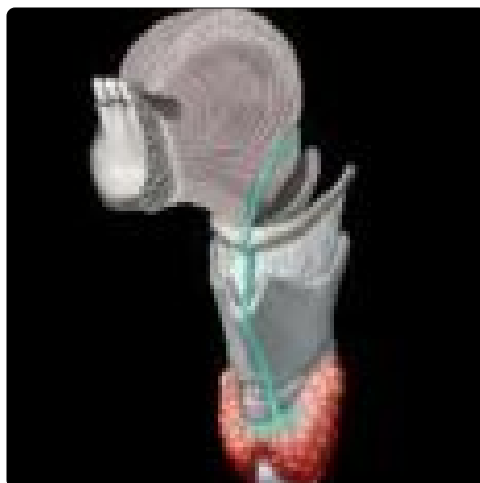
- H-shaped endocrine organ (15-25 g) composed of 2 lateral lobes (each ~ 5 cm long, 2.5 cm wide, 2 cm deep) connected by central isthmus
 - Located anterior to upper trachea, just below laryngeal cricoid cartilage
 - Lobes have pointed superior and blunt inferior poles
 - 40% have pyramidal lobe, which extends cephalad from isthmus
 - Size varies with stature, age, sex, iodine intake, hormonal status, and gland functional status
- Invested by delicate fibrous capsule that extends into gland dividing it into lobules (thyromeres)
- Cut surface is red-brown and firm
 - Nodules seen in 10% of nongoitrous euthyroid individuals
- Paired superior and inferior parathyroid glands are attached to posterior thyroid capsule
- Small lymph nodes can be seen around isthmus, including pretracheal (Delphian) lymph node
- Embryologically derived from median anlage and 2 lateral anlagen
 - Median anlage derived from 1st and 2nd branchial pouch endoderm at base of primitive pharynx during 4th week
 - Descends into neck, passing anterior to hyoid and larynx
 - Remains connected to foramen cecum by thyroglossal duct, which subsequently involutes
 - Failure of duct to involute can result in thyroglossal duct cysts
 - Lateral anlagen (ultimobranchial bodies) derive from 4th and 5th branchial pouches during 7th week and provide parafollicular C cells
 - Ultimobranchial bodies provide C cells, which disperse throughout mid to upper 1/3 of lateral thyroid lobes
 - Ultimobranchial body remnants, so-called solid cell nests, can be seen in lateral lobes of most thyroids

Microscopic Anatomy

- Functional unit of thyroid gland is follicle
 - Follicles are spherical cyst-like structures of variable size (0.2-1.0 mm on average) and shape
 - Follicles surrounded by extensive capillary network
 - Lined by monolayer of flat to columnar follicular epithelial cells (thyrocytes)
 - IHC: TTF-1(+), thyroglobulin (+), pax-8 (+), CK7(+)
 - Cell size varies by functional status of follicle
 - ◻ Flat cells are inactive, cuboidal cells secrete colloid, and columnar cells resorb colloid
- Colloid is eosinophilic to basophilic secretion composed mostly of thyroglobulin
 - Thyroglobulin
 - Iodinated glycoprotein and serves as inactive storage form of active thyroid hormones T3 and T4
 - Resorption vacuoles present in follicles resorbing colloid
 - Calcium oxalate crystals are often present in colloid
- Sanderson polsters
 - Small aggregates at 1 end of follicle that may have papillary or undulating appearance
 - Increased in hyperplastic conditions
- So-called palpation thyroiditis is common
 - Represents disrupted follicles replaced by macrophages, chronic inflammatory cells
- Focal areas of stromal adipose tissue can be seen
- Intrathyroidal skeletal muscle, cartilage, parathyroid glands, thymic tissue are occasionally present
- Parafollicular C cells secrete calcitonin
 - Cuboidal cells with pale cytoplasm with coarse chromatin
 - Found between follicular cells and basement membrane
 - Difficult to see in normal thyroids
 - Restricted to mid and superior lateral lobes
- Solid cell nests
 - Small (0.1- to 2.0-mm) multilobed clusters of oval basaloid cells interspersed among follicles
 - Small numbers of C cells present
 - Squamous metaplasia and cystic change not uncommon
 - Restricted to mid and superior lateral lobes

Thyroid Gland Embryology

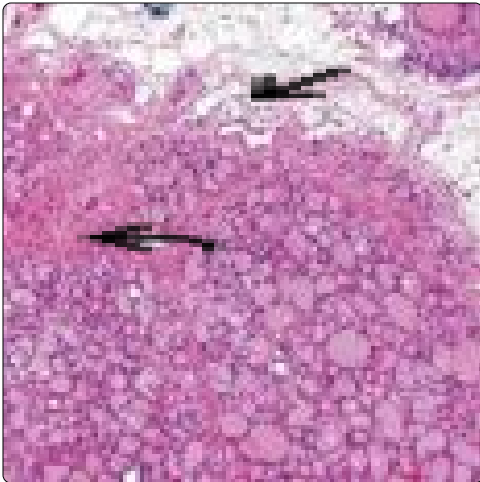
(Left) As the median thyroid anlage descends into the neck, it is attached to its origin at the foramen cecum by the thyroglossal duct. Failure to involute can result in thyroglossal duct cyst. **(Right)** Normal thyroid gland is composed of 2 lateral lobes connected by a central isthmus and invested by a thin fibrous capsule with numerous small surface vessels. Forty percent of thyroid glands will have a pyramidal lobe.



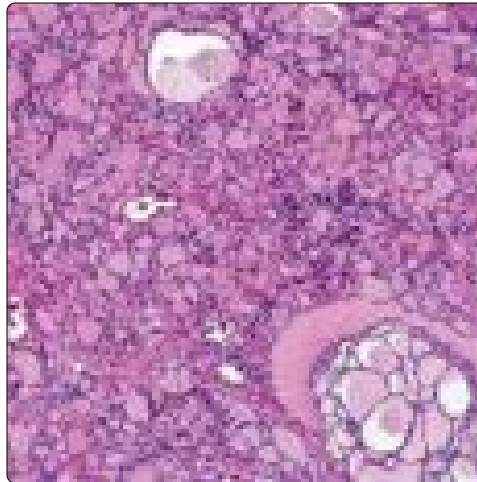
Thyroid Gland



Capsule and Extracapsular Thyroid Tissue

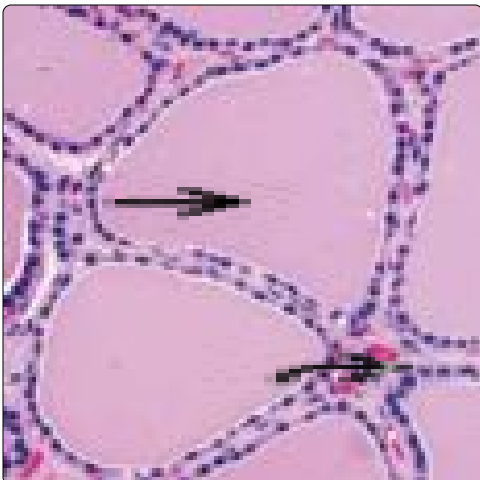


Incidental Nodules

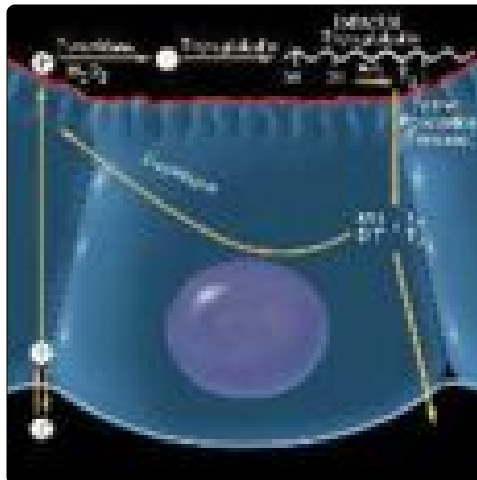


(Left) The thyroid gland is covered by a delicate fibrous capsule that extends into the parenchyma, dividing the gland into vague lobules (thyromeres). Small nodules of otherwise unremarkable thyroid tissue are often present outside of the capsule proper. (Right) Incidental nodules can be seen macroscopically in ~ 10% of nongitrous thyroid glands. Microscopic nodules and colloid cysts are also frequently seen in nongitrous and euthyroid individuals.

Follicular Epithelial Cells

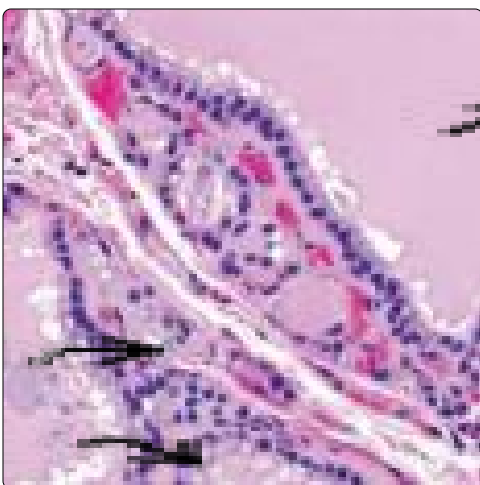


Thyroid Hormone Synthesis

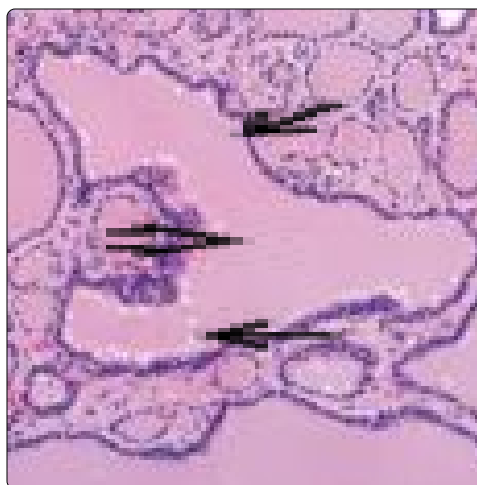


(Left) Follicles are lined by a monolayer of follicular cells surrounding a mass of colloid. Cuboidal follicular cells synthesize colloid. Flat follicular cells are inactive. (Right) Colloid consists mostly of thyroglobulin, an iodinated glycoprotein that serves as the storage form of the active thyroid hormones T3 and T4. In response to pituitary thyroid-stimulating hormone, follicular cells resorb thyroglobulin and convert it to T3 and T4, which are released into the perfollicular capillaries.

Resorption Vacuoles



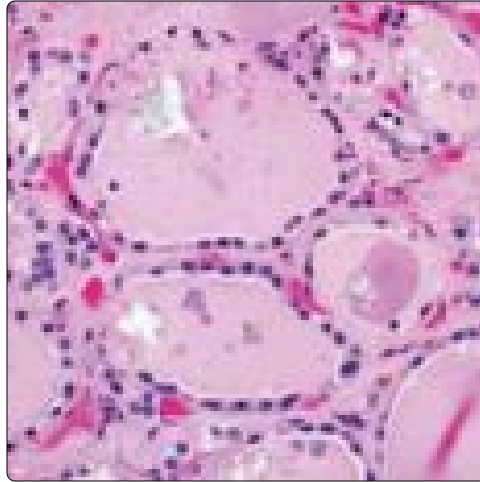
Sanderson Polster



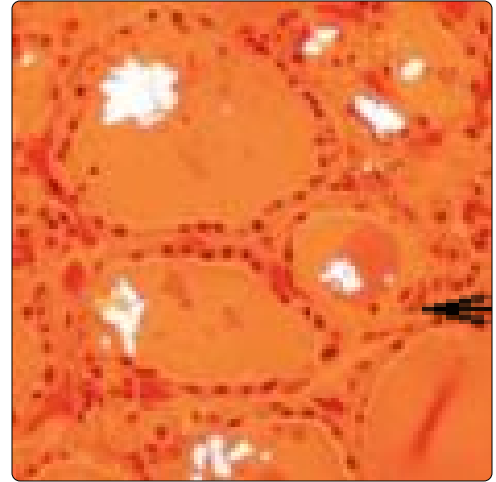
(Left) Follicles actively resorbing colloid are usually lined by more columnar follicular epithelial cells and demonstrate resorption vacuoles. These appear as a row of small vacuoles at the colloid and epithelial cell interface. (Right) Sanderson polsters are intraluminal collections of follicles at one end of a larger follicle and often have an undulating or papillary appearance. They are more common in active follicles, as evidenced by the resorption vacuoles.

Colloid Calcium Oxalate Crystals

(Left) Irregular, rhomboidal, anisotropic calcium oxalate crystals  can be seen within the colloid of most (~85% in autopsy studies) thyroid glands. They can be present in normal follicles but are more frequent with age, in renal failure patients on dialysis, and in inactive follicles. They are uncommon in cases of chronic thyroiditis. **(Right)** Although calcium oxalate crystals can be easily identified using routine light microscopy, they are more readily identified using polarized light .

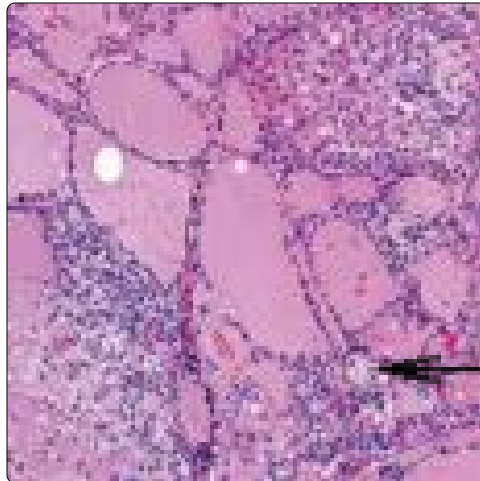


Calcium Oxalate Crystals, Polarized Light

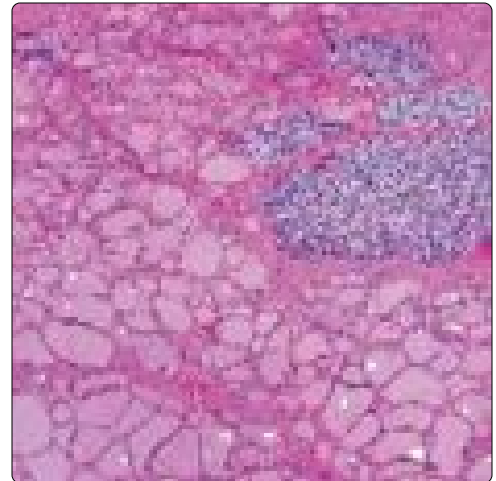


Palpation Thyroiditis

(Left) Ruptured follicles incite a granulomatous reaction referred to as palpation thyroiditis. These are seen in most resected thyroids and are characterized by macrophages and lymphocytes  replacing the damaged follicle to varying degrees. Although most cells are mononuclear macrophages, multinucleated cells can be seen as well. **(Right)** Intracapsular &/or intrathyroidal parathyroid tissue  or glands are not uncommonly seen in otherwise normal thyroid glands.

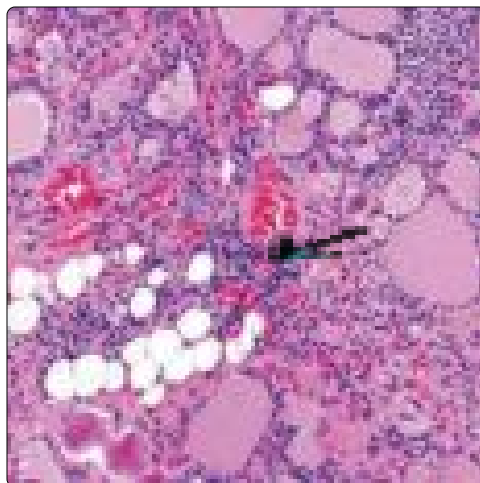


Intrathyroidal Parathyroid Tissue

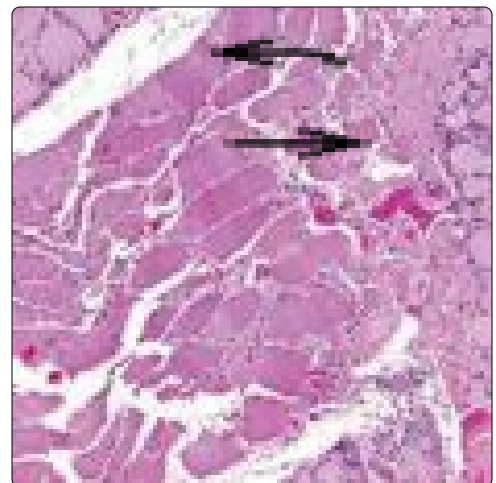


Stromal Adipose Tissue

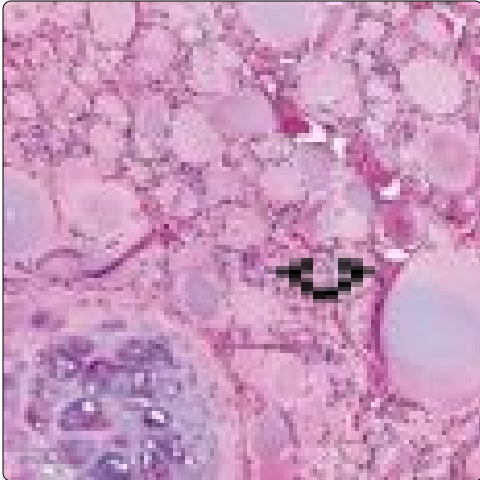
(Left) Thyroid glands generally have very little stroma. Small foci of mature adipose tissue can be seen between follicles in most thyroid glands. Often, it is in or near the capsule, but fat can also be present deep in the gland parenchyma . Small collections of chronic inflammatory cells  can be seen in many normal thyroid glands as well. **(Right)** Intrathyroidal  skeletal muscle  is common in the isthmus region. Follicles can also be seen in extrathyroidal skeletal muscle.



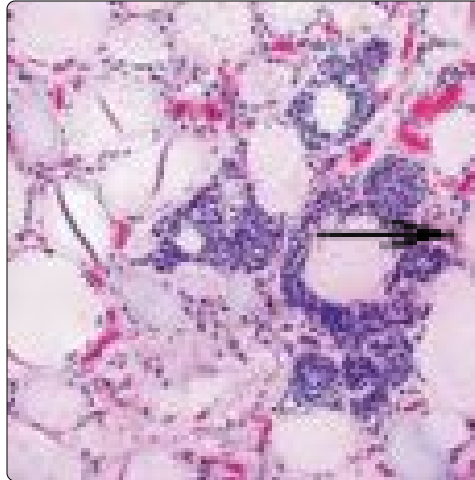
Intrathyroidal Skeletal Muscle



Heterotopic Cartilage

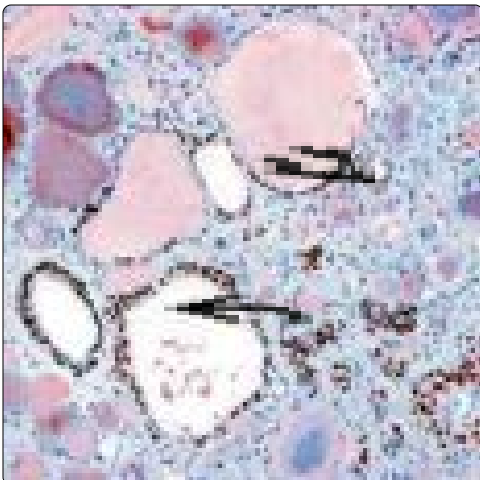


Solid Cell Nest

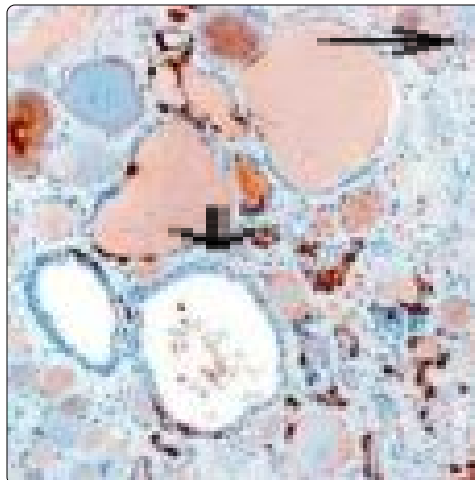


(Left) Intrathyroidal islands of heterotopic mature hyaline cartilage, presumably of branchial pouch derivation, can be seen in the stroma in ~10% of thyroid glands. (Right) Solid cell nests are common in thyroid glands and can be found in the mid and upper 1/3 of the lateral lobes, consistent with an ultimobranched body derivation. They are small multilobed structures composed of uniform basophilic cells. Focal squamous differentiation and cystic change are often seen.

Solid Cell Nest



Solid Cell Nest

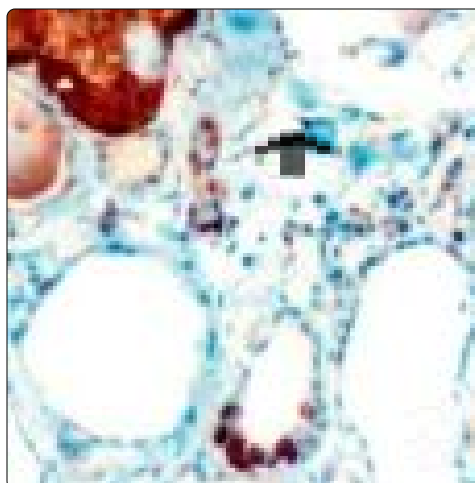


(Left) The basophilic cells are thought to be basal or stem cells. This is supported by strong nuclear reactivity for p63, a marker of basal/stem cells in stratified epithelia. Note the cystic areas and mixed follicles in this solid cell nest. The cystic areas can enlarge, resulting in clinically significant cysts. (Right) Some solid cell nests contain a small component of calcitonin-positive C cells, supporting the embryologic link of C cells and solid cell nests.

C-Cell Cluster



C-Cell Cluster



(Left) C cells are normally present in the mid and upper 1/3 of the lateral lobes. They are difficult to identify on routine H&E stains under normal conditions. They are easily identified under hyperplastic conditions, such as this C-cell nodule from a multiple endocrine neoplasia type 2 patient. The cells are larger than follicular cells and have abundant pale eosinophilic cytoplasm. (Right) Calcitonin stains can be used to identify C cells, such as those in this cluster and an adjacent follicle.

PARATHYROID

Macroscopic Anatomy

- Most people (90-95%) have 4 parathyroid glands
 - Paired superior and inferior glands
 - Number varies from 2-12, including supernumerary and ectopic glands
 - Intrathyroidal (3%) and intrathymic (10%) are common ectopic locations
- Small (4-6 mm long, 3-4 mm wide, 1-2 mm thick), yellow-tan, lentiform glands typically abutting posterior thyroid capsule
 - Each gland weighs ~ 30-40 mg but can weigh up to 80 mg; inferior glands slightly larger
 - Covered by delicate fibrous capsule with small surface vessels
- Location variable, but most (75%) superior glands lie posterior to midsuperior thyroid poles
 - Inferior gland locations more variable, but most (50%) lie lateral to lower thyroid poles
- Paired inferior glands originate from more cephalad 3rd branchial pouches along with thymus gland
 - Longer descent and comigration with thymus explain more variable location and incidence of intrathymic inferior parathyroid glands

Microscopic Anatomy

- Thinly encapsulated with fibrous septa extending into parenchyma dividing gland into vague lobules
- Stroma comprised of mature adipocytes and fibroconnective tissue with rich vascular supply
 - Most adult parathyroid glands contain 20-40% stromal adipose
 - Adipose:parenchymal cell ratio used to roughly correlate functional status; hyperfunctioning glands have less adipose
 - Proportion of adipose variably distributed throughout glands and not uniform or predictable
- Chief cell
 - Synthesize and secrete parathyroid hormone

- Small (6-8 μ m) polygonal cells with pale amphophilic, finely vacuolated cytoplasm and round central nuclei with coarse chromatin and small nucleoli
- Arranged in small nests and thin cords and separated by stromal adipose tissue
- Cells are intimately associated with delicate capillary network
- Some areas with acinar or follicular differentiation containing eosinophilic secretions can be seen
- IHC: Cytokeratin (+), chromogranin A (+), and parathyroid hormone (+)
 - Hyperfunctioning glands tend to have reduced chromogranin A and parathyroid hormone staining due to degranulation
- Contain glycogen and intracellular lipid
- Oxyphil (oncocyctic cell)
 - Larger (12-20 μ m) chief cells that are fewer in number
 - Abundant, granular eosinophilic cytoplasm; typically arranged in small nodules
 - Number increases with age, similar to oncocyctic change in some other organs (i.e., salivary gland)
 - Ultrastructurally, contain numerous irregular mitochondria similar to other oncocytes
- Clear cells are less common and represent chief cells with abundant cytoplasmic glycogen
- Unencapsulated parathyroid tissue can be seen in soft tissue adjacent to parathyroid glands as well as in skeletal muscle, thyroid gland, and thymus
- Intrathyroidal, intrathymic, and other ectopic areas can be sites of otherwise normal parathyroid glands
- Canals of Kürsteiner may be seen adjacent to some parathyroid glands and likely represent source for many parathyroid cysts

CONSIDERATIONS AND VARIATIONS

Age-Related Changes

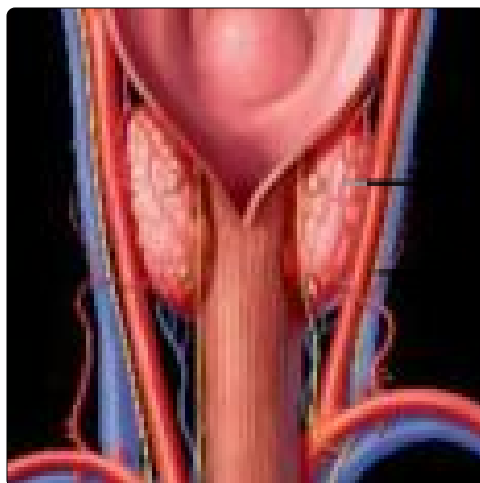
- Stromal adipose tissue increases with age

Pitfalls and Artifacts

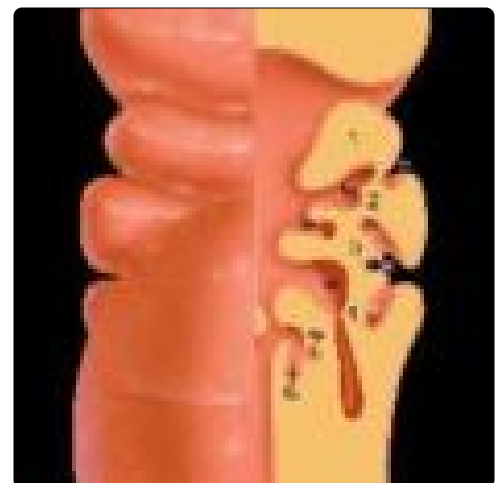
- Parathyroid with follicular change can mimic thyroid

Parathyroid Glands

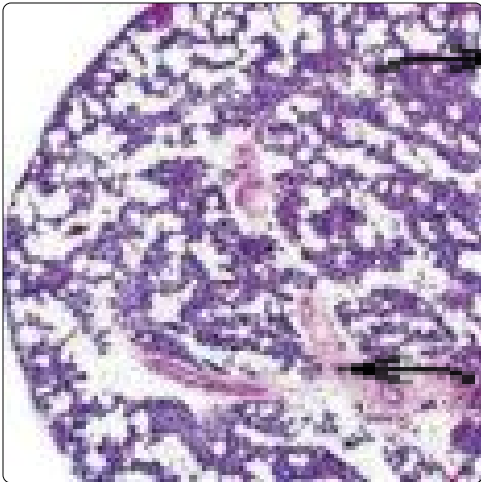
(Left) Most people (90-95%) have 4 parathyroid glands consisting of paired superior  and inferior glands . Most lie posterior to the thyroid gland poles but can also be found in several variant locations. **(Right)** Superior parathyroids arise from the 4th branchial pouch  along with the primordial lateral thyroids. Inferior parathyroids arise from the 3rd branchial pouch  along with the thymus .



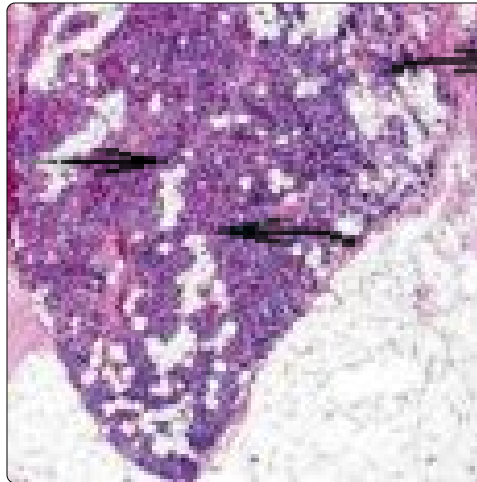
Anatomy



Stromal Adipose Tissue

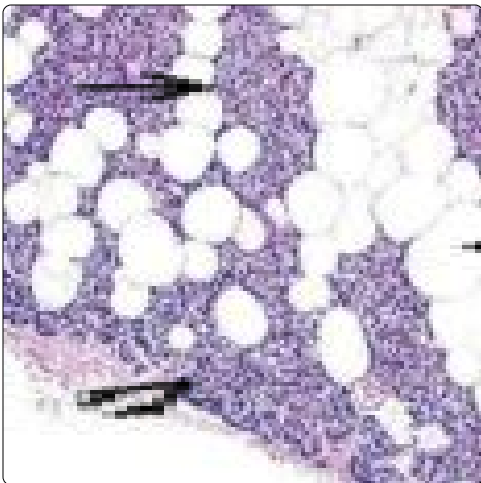


Stromal Adipose Tissue

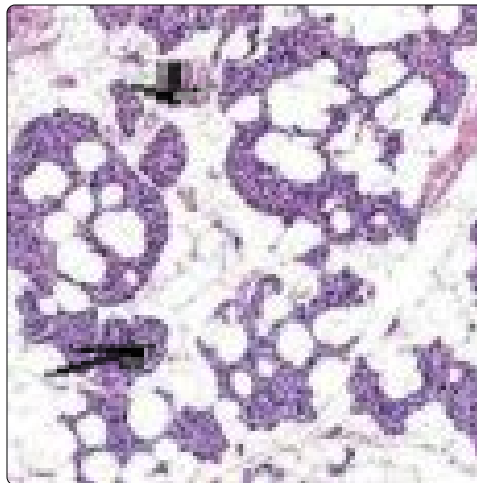


(Left) Stromal adipose increases with age and is sparse in children. Although variable, most normal adult parathyroid glands contain around 40% stromal adipose tissue [1]. Note the vascular pole region containing large arteries and veins extending into septa [2]. (Right) This normal parathyroid gland from a euparathyroid individual demonstrates less stromal adipose tissue (~ 20%) [3], illustrating the marked variation in chief cell cellularity [4] seen in normal parathyroid glands.

Capsule

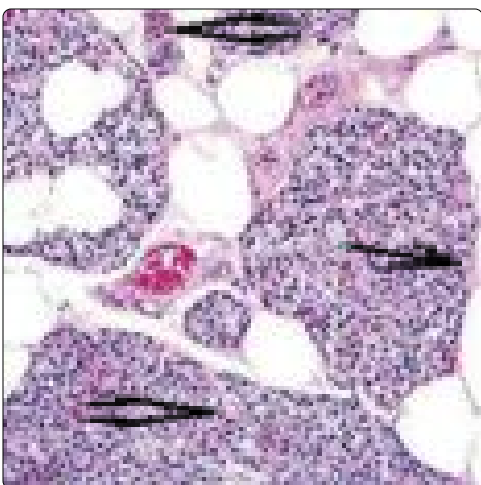


Lobules

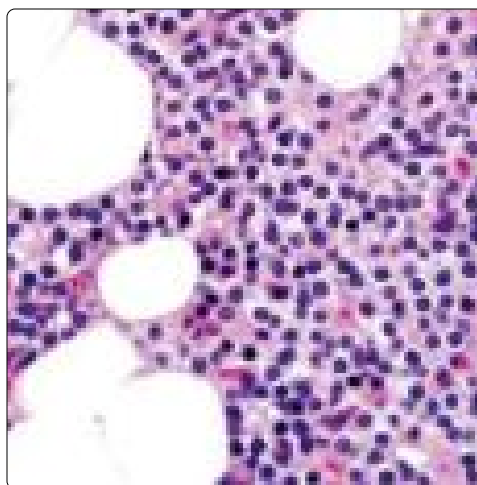


(Left) A delicate fibrous capsule [1] containing many small intracapsular vessels surrounds normal parathyroid glands. These surface vessels communicate with the vessels in the trabeculae, which receive blood from the hilar (vascular pole) vessels. The chief cells are arranged in nests and cords [2] and are separated by mature stromal adipocytes [3]. (Right) Although well-developed septa are uncommon, the vague lobularity [4] of the parathyroid can be appreciated at low-power magnification.

Vascular Stroma



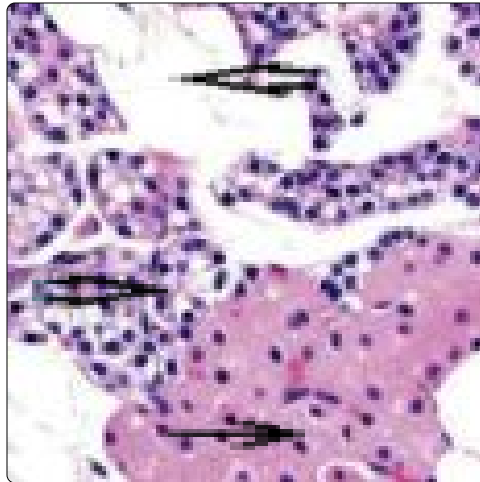
Chief Cells



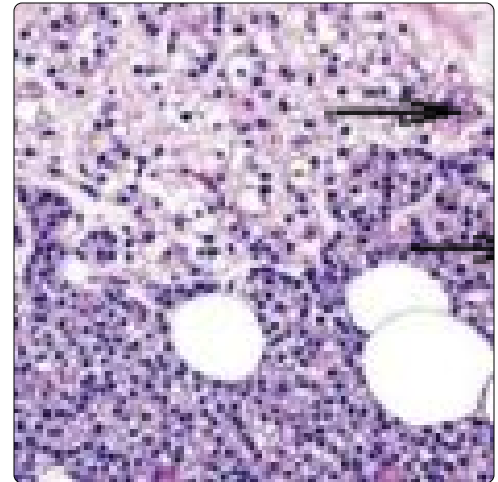
(Left) The fibrous septa contain the vascular supply, including small arterioles [1] and venules [2]. Numerous small capillaries [3] surround the chief cells within the lobules. (Right) Chief cells [4] are uniform small cells with pale amphophilic to clear vacuolated cytoplasm. The nuclei are round and small with coarse chromatin and inconspicuous nucleoli. These are the functional cells of the parathyroid, which synthesize and secrete parathyroid hormone in response to hypocalcemia.

Oxyphil (Oncocytic) Cells

(Left) Oxyphil cells are altered chief cells with abundant, granular, eosinophilic cytoplasm. They have less secretory activity and can be isolated or form small nodules, such as this focus [X]. Smaller transitional cells [X] with eosinophilic cytoplasm can be seen as well. (Right) Clear cells [X] are the least common cell type seen in adult parathyroid glands. They also represent altered chief cells, but their cytoplasm is clear due to accumulation of cytoplasmic glycogen.

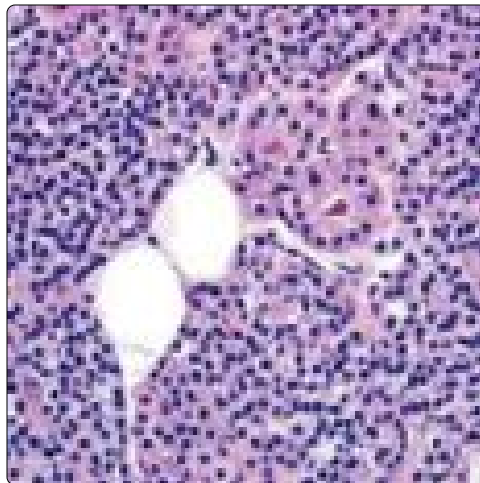


Clear Cells

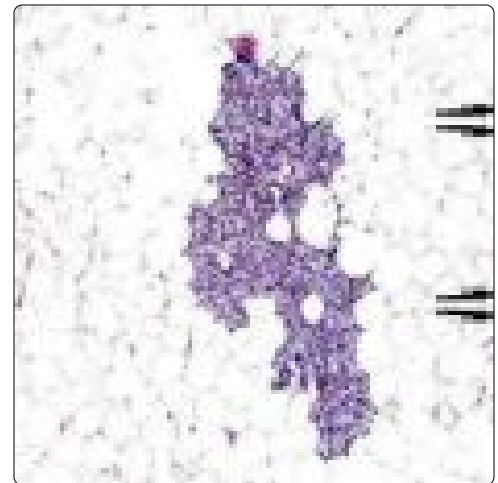


Follicular Change

(Left) Although uncommon, follicular (or acinar) change can be seen in both normal and hyperfunctioning glands. This is characterized by chief cells lining a lumen typically filled with dense eosinophilic secretion [X]. When extensive, this can mimic thyroid gland tissue. (Right) Unencapsulated ectopic parathyroid tissue [X] is commonly seen in tissues adjacent to parathyroid glands, such as this small focus within mature adipose tissue.

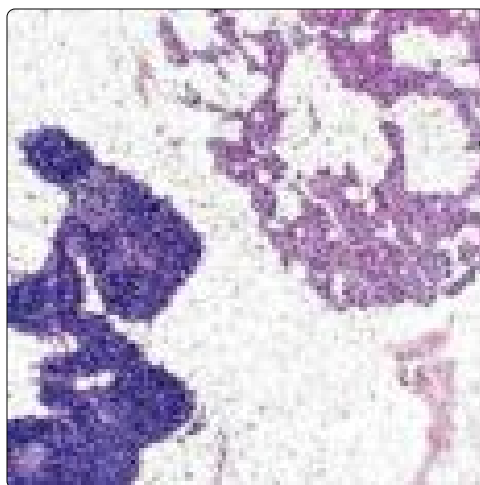


Ectopic Parathyroid Tissue

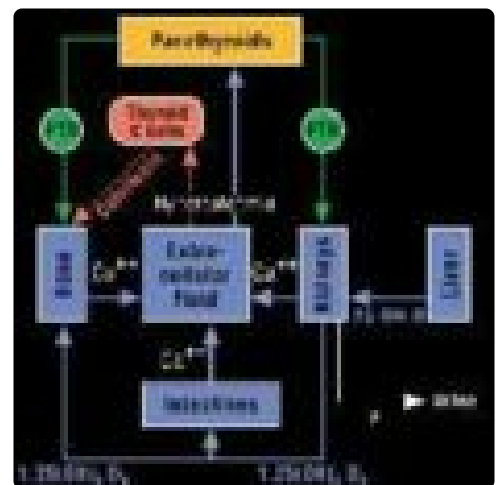


Intrathymic Parathyroid Gland

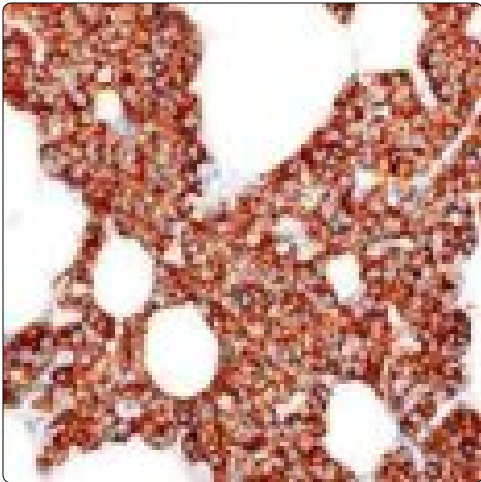
(Left) Ectopic &/or supernumerary inferior parathyroid glands [X] can be located in the mediastinal thymus [X] or in cervical thymic extensions. They can also be found higher up in the neck and pericardium. (Right) Parathyroid hormone (PTH) secretion is stimulated by hypocalcemia and acts to increase serum calcium by resorbing bone and increasing kidney reabsorption and intestinal absorption. PTH also stimulates production of active 1,25(OH)₂ vitamin D₃.



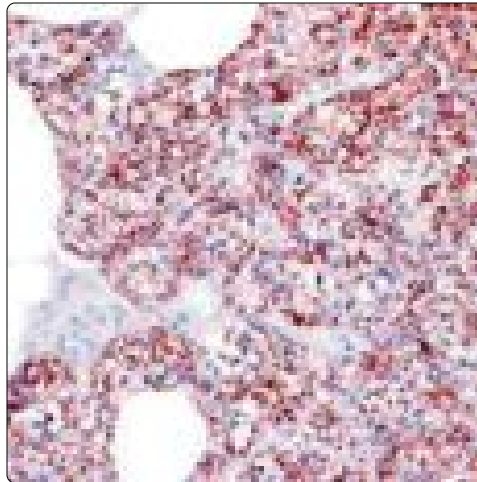
Parathyroid Hormone



Parathyroid Hormone

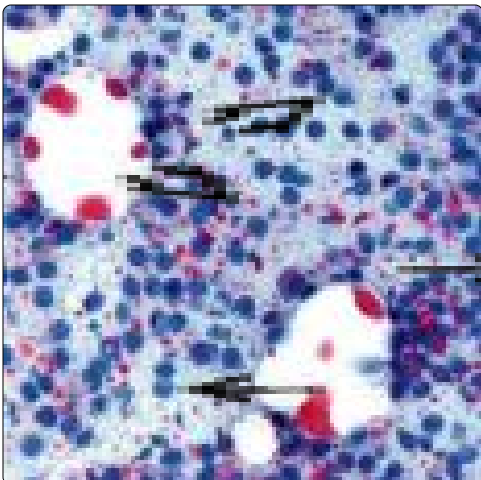


Chromogranin

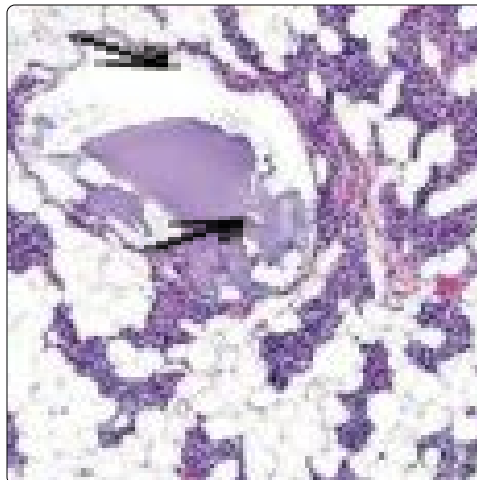


(Left) Chief cells demonstrate strong granular staining for PTH. The granular staining pattern is a result of the PTH protein being stored in cytoplasmic neurosecretory granules. Reduced intensity of PTH staining is seen in hyperfunctioning glands due to neurosecretory degranulation. **(Right)** Although not specific, chromogranin is also positive in chief cells with a granular pattern similar to that of PTH. Chromogranin is colocalized with PTH in the neurosecretory granules.

Intracellular Lipid

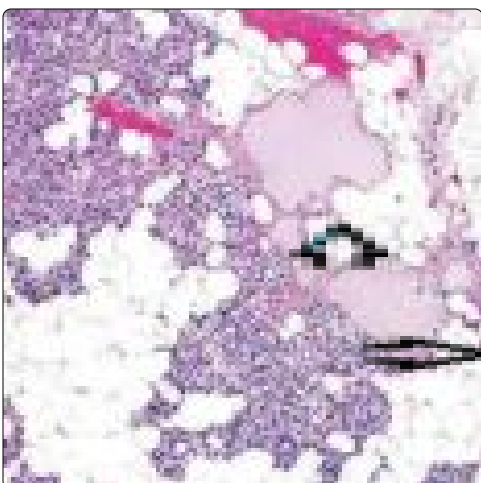


Parathyroid Microcyst

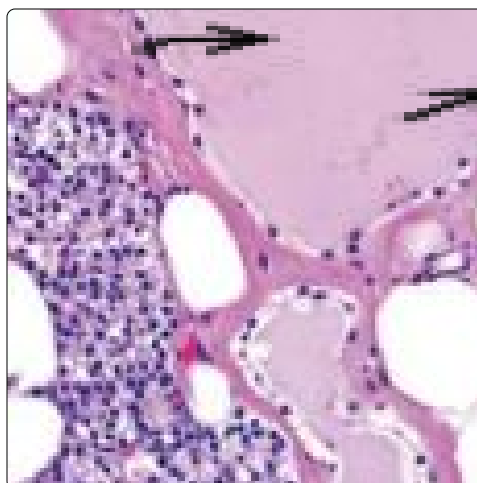


(Left) Chief cells contain intracytoplasmic lipid droplets, which can be highlighted by using fat stains on frozen sections, such as this oil red O stain. The amount of cytoplasmic lipid is roughly inversely proportional to secretory activity and is decreased in hyperparathyroid conditions. Note the mature fat cells with lipid droplets. **(Right)** Not uncommonly, small cysts lined by clear cuboidal cells can be seen within or adjacent to normal or abnormal parathyroid glands.

Canals of Kürsteiner



Canals of Kürsteiner



(Left) Canals of Kürsteiner are occasionally found adjacent to otherwise normal parathyroid glands. These likely represent rudimentary glandular structures derived from branchial endoderm. They account for incidental microcysts as well as most clinically evident parathyroid cysts. **(Right)** The canals are surrounded by sclerotic stroma and have lumina filled with eosinophilic secretions. They are lined by low cuboidal cells with clear cytoplasm, and they are PTH negative.

PINEAL GLAND

Macroscopic Anatomy

- Small, unpaired endocrine organ providing circadian and seasonal biorhythm control
- Midline location (quadrigeminal cistern) within loose connective tissue of velum interpositum
 - Stalk attached to diencephalon, posterior wall of 3rd ventricle
- Primary vascular supply is medial posterior choroidal artery (lacks blood-brain barrier)

Microscopic Anatomy

- Regular lobules of cellular cords/follicles, separated by connective tissue septa (leptomeninges)
- Pinealocytes
 - Epithelioid cells arranged in pinocytotic rosettes
 - Large lobated nuclei, sharply defined nucleoli
 - Photosensory and neuroendocrine derivation
 - IHC: Synaptophysin (+)

- Interstitium/stroma
 - Loose neuroglial stroma between cords of pinealocytes, in perivascular areas
 - Astrocytes have elongated nuclei and long, GFAP(+) cytoplasmic processes
 - Investing leptomeninges (vellum interpositum)
- Corpora arenacea
 - Calcium and phosphate salt concretions

CONSIDERATIONS AND VARIATIONS

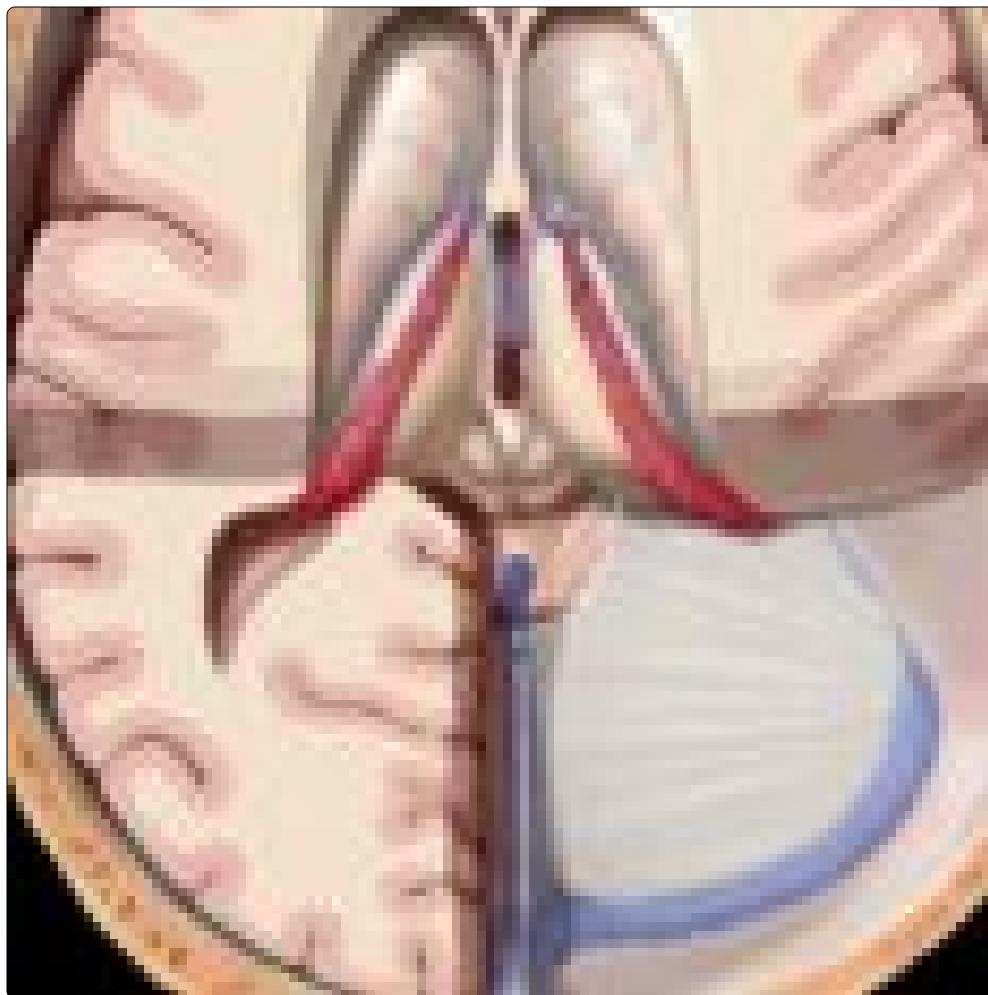
Age-Related Variations

- Gland enlarges until 1-2 years of age
- Gliosis, cystic change, increasing corpora arenacea occur with age

Pitfalls and Artifacts

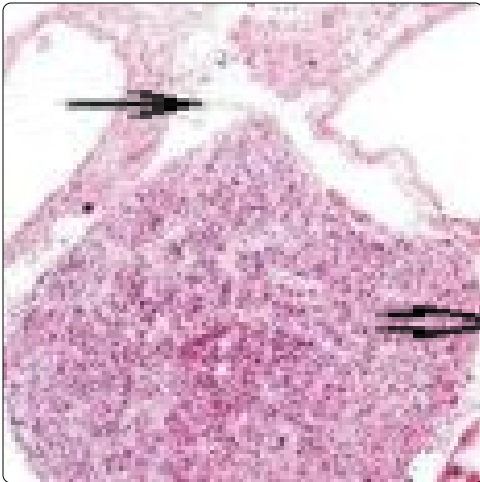
- Cellularity and gland-like histology may mimic neoplasm
- Difficult to cytologically distinguish normal from well-differentiated pineocytoma

Macroscopic Anatomy

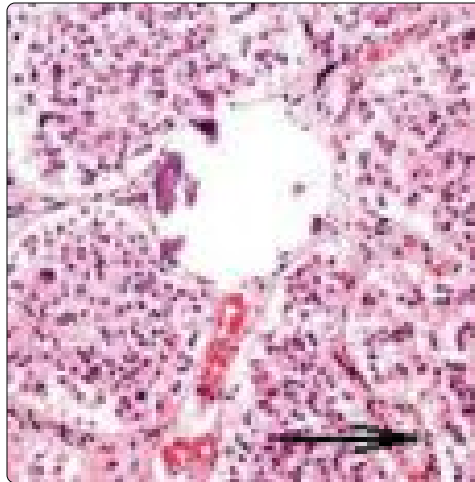


Graphic of a horizontal section through the pineal region viewed from above, corpus callosum removed, shows the pineal gland , superior colliculus , inferior colliculus , 3rd ventricle , internal cerebral veins , and vein of Galen .

Microscopic Anatomy

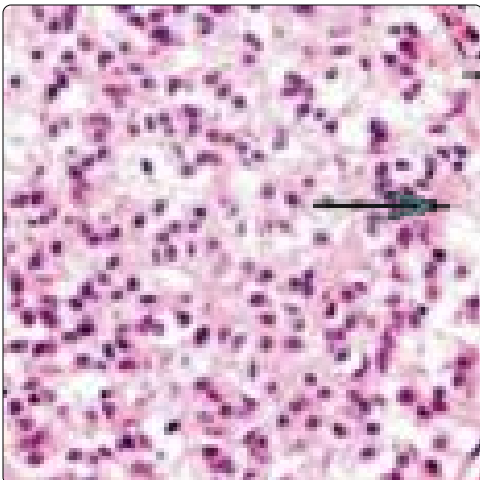


Lobular Architecture

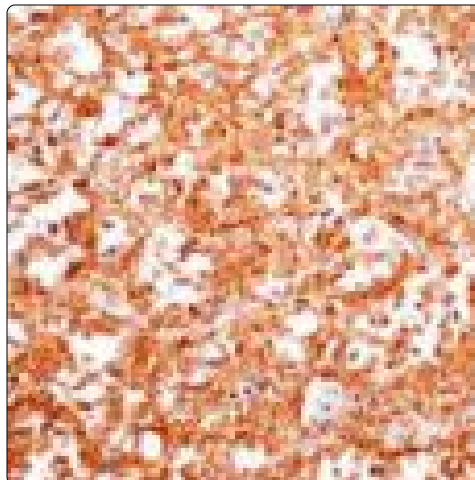


(Left) Low-power view shows the pineal gland and the paired internal cerebral veins located superiorly/dorsally. Anteriorly/rostrally is the pineal recess of the 3rd ventricle, which often contains choroid plexus. **(Right)** Regular lobules of cellular cords and follicles are separated by connective tissue septa formed by leptomeninges originating in the overlying pia mater. They contain blood vessels and unmyelinated postganglionic nerve fibers that will form connections with pinealocytes.

Pinealocytes

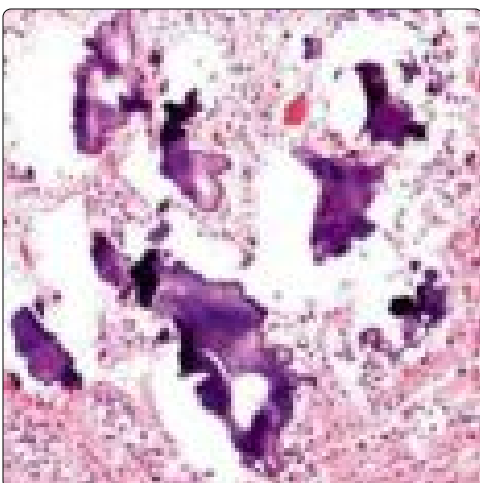


Synaptophysin

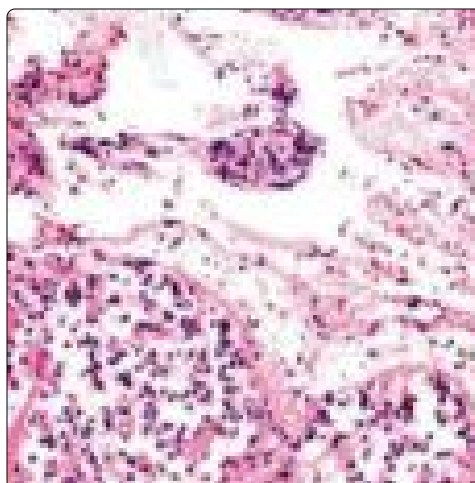


(Left) Epithelioid pinealocytes have large, lobated nuclei and prominent nucleoli. Loose neuroglial stroma composed of astrocytes is present in between the cords. **(Right)** Pinealocytes are strongly synaptophysin positive. An immunohistochemical stain for GFAP would stain the intervening neuroglial stroma and indigenous astrocytes, and would highlight the lobular architecture.

Corpora Arenacea



Leptomeninges



(Left) The large, purple, mineralized concretions in the pineal gland are termed corpora arenacea, acervuli cerebri, or brain sand. These calcium and phosphate salt precipitations increase after puberty and are almost ubiquitous with older age. **(Right)** High-power H&E section shows arachnoid cell nests in the investing leptomeninges (connective tissue or velum interpositum) that surround pineal gland lobules. These nests are the presumed source of pineal region meningiomas.

PITUITARY

Macroscopic Anatomy

- Small, bean-shaped organ measuring 1.0-1.5 cm and weighing 0.5-0.7 g
- Located centrally at base of brain, within saddle-shaped cavity of sphenoid bone (sella turcica)
 - Close to optic chiasm and cavernous sinuses
- Pituitary stalk connects it to hypothalamus and carries
 - Releasing/inhibitory hormones/factors (through portal vascular system) to adenohypophysis
 - Axonal processes of nerve cell bodies (from hypothalamic nuclei) to neurohypophysis
- Anterior pituitary lobe (adenohypophysis)
 - 80% of pituitary; composed of pars distalis (anterior), pars intermedia, and pars tuberalis (proximal)
 - Embryologically derived from Rathke pouch, which is cranial extension of oral cavity roof (ectoderm)
- Posterior pituitary lobe (neurohypophysis)
 - Embryologically derived from caudal outpouching of 3rd ventricle floor/diencephalon (neuroectoderm)
 - Composed of pars nervosa and infundibulum
 - In contrast to anterior lobe, supplied by arterial blood (superior and inferior hypophyseal arteries)

Microscopic Anatomy

- Anterior pituitary lobe (adenohypophysis)
 - Various cell types admixed, arranged in nests/acini
 - Each produces 1 hormone (except gonadotrophs)
 - Immunohistochemical stains for each hormone can help distinguish among cell types
 - Surrounded by stellate (sustentacular) cells and capillary-rich, fibroblast-derived reticulin network
 - Somatotrophs (50% of all cells, lateral location)
 - Acidophilic; produce growth hormone (GH)
 - Stimulation: GH-releasing hormone
 - Inhibition: GH-inhibiting hormone (somatostatin)
 - Lactotrophs (15-20%, posterolateral location)
 - Acidophilic/chromophobic; produce prolactin (PRL)
 - Inhibition: PRL-inhibitory factor (dopamine)

- Corticotrophs (15-20%, central mucoid wedge)
 - Basophilic, PAS positive; produce proopiomelanocortin, which is precursor of ACTH, MSH, endorphins/enkephalins, and β -lipotropins
 - Stimulation: Corticotropin-releasing hormone
- Thyrotrophs (5%, anterior central location)
 - Pale basophilic, PAS positive; produce thyroid-stimulating hormone
 - Stimulation: Thyrotropin-releasing hormone
- Gonadotrophs (10%, evenly distributed)
 - Basophilic, PAS positive; produce luteinizing hormone, follicle-stimulating hormone
 - Stimulation: Gonadotropin-releasing hormone
- Posterior pituitary lobe (neurohypophysis)
 - Unmyelinated axons and modified, GFAP-positive glial cells (pituicytes) in delicate stroma
 - Axons with granular terminal expansions (Herring bodies) store 2 hypothalamic peptide hormones
 - Antidiuretic hormone (vasopressin) and oxytocin
 - Released into systemic (venous) circulation

CONSIDERATIONS AND VARIATIONS

Age-Related Variations

- Corticotrophs in posterior lobe ("basophil invasion")
 - Surrounded by axons of neurohypophysis
- Interstitial fibrosis, amyloid deposits, lymphocytosis
- Granular cell nests/tumorlets in stalk or posterior lobe

Hyperplasia

- Anterior pituitary can enlarge 30% during pregnancy

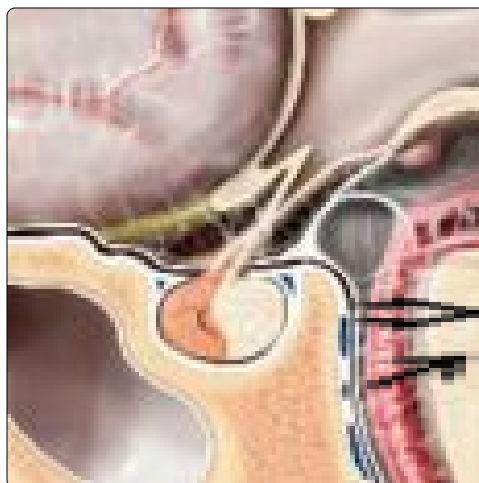
Metaplasia

- Squamous metaplasia, particularly in pars tuberalis

Pitfalls and Artifacts

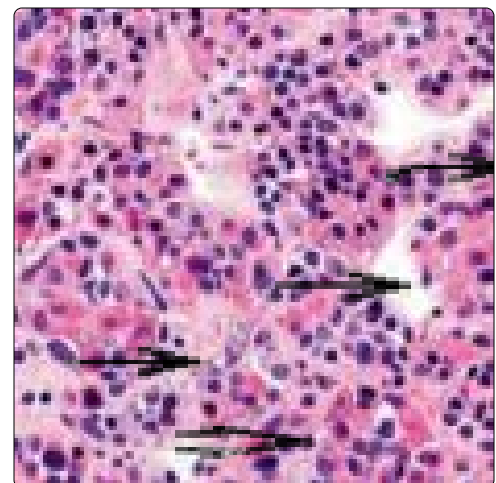
- Normal pituitary tissue may mimic adenoma, especially on frozen section or small specimens
- Salivary gland rests
- Incidental Rathke cleft remnant/cyst is common (30%)
- Crooke hyaline: Perinuclear keratin accumulation

Pituitary Gland

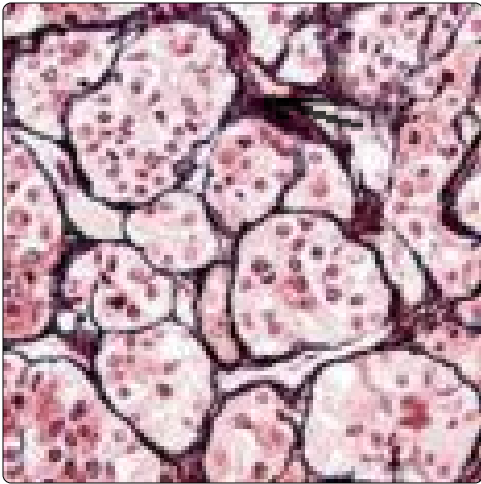


(Left) Lateral view shows the pituitary gland, consisting of anterior and posterior lobes, lying within the saddle-shaped cavity of sphenoid bone (sella turcica). The pituitary stalk connects the pituitary gland to the hypothalamus. (Right) The anterior lobe (adenohypophysis) contains admixed acidophilic, basophilic, and chromophobic cells arranged in acini. Capillaries and sustentacular cells make up the surrounding stroma.

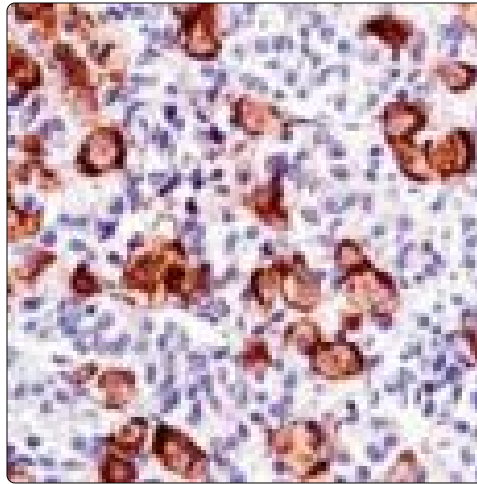
Anterior Pituitary Gland



Adenohypophysis Stroma

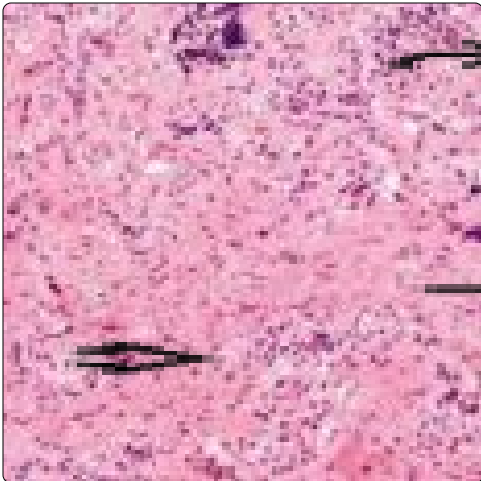


Corticotrophs of Adenohypophysis

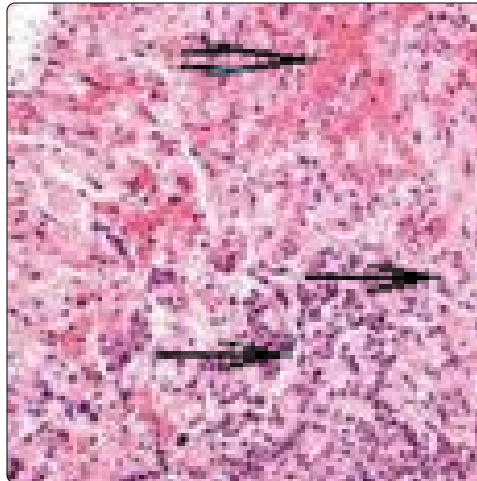


(Left) A reticulin stain highlights the intricate, capillary-rich reticulin stroma surrounding each acinus of the normal adenohypophysis, in contrast with pituitary adenomas, which generally lack this reticulin network. **(Right)** Corticotroph cells of the anterior pituitary stain with an immunohistochemical stain against ACTH. Each hormone-producing cell of the pituitary can be highlighted in this way with the corresponding antibody immunostain.

Fibrosis and Lymphocytosis

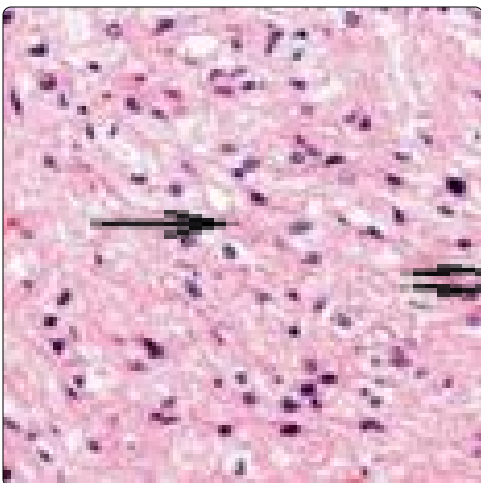


Basophil Invasion

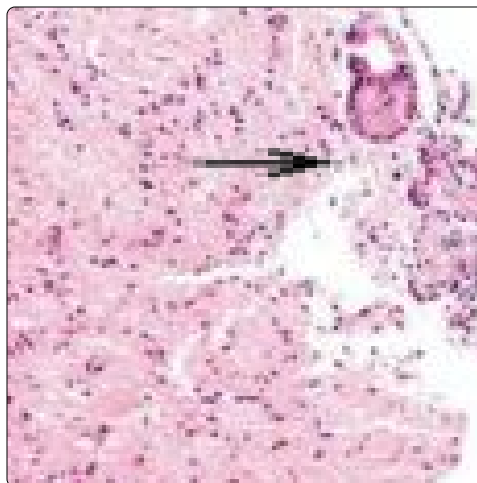


(Left) This H&E shows interstitial fibrosis of the stroma surrounding the acinar epithelium of anterior pituitary as well as collections of lymphocytes. Both of these changes are commonly seen in the pituitary gland with advancing age. **(Right)** This population of basophilic corticotroph cells from the anterior pituitary are infiltrating among the neuronal processes of the posterior lobe (neurohypophysis). This is not indicative of a neoplastic process.

Normal Neurohypophysis



Salivary Gland Rest



(Left) The posterior pituitary lobe is composed of neuropil (a network of unmyelinated axons of nerve cell bodies in hypothalamic nuclei) and pituicytes (modified glial cells with elongated nuclei that stain for GFAP). **(Right)** Salivary gland rests are common, especially at the junction of the neurohypophysis and pituitary stalk. They should not be confused with a neoplastic process.

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